Abstract Presentations from the AABB Annual Meeting and CTTXPO, Denver, CO, October 12-15, 2013

Scientific Section

Scientific Plenary Session

P1

Prospective Investigational Blood Donation Screening for *Babesia microti (Bm)*

E Moritz[1] (moritze@usa.redcross.org), S T Johnson[1], C Winton[1], R L Townsend[1], L Tonnetti[2], G A Foster[1], E M Brissette[3], V P Berardi[3], S L Stramer[1]. *[1]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [2]Jerome H. Holland Laboratory, American Red Cross, Rockville, MD, United States; [3]IMUGEN, Inc., Norwood, MA, United States*

Background/Case Studies: *Babesia microti (Bm)* is a transfusion-transmissible (TT) intraerythrocytic tick-borne parasite. The incidence of *Bm* infection is increasing in the US; no FDA-licensed screening test is available. We conduct blood donor screening under IND at selected drives in MA, CT, MN, and WI; data from June 2012-April 2013 are presented. **Study Design/Methods:** Arrayed fluorometric immunoassay (AFIA) and real-time polymerase chain reaction (PCR) (IMUGEN, Norwood, MA) for *B. microti* antibody and DNA are performed on EDTA-anticoagulated whole-blood samples. AFIA reactivity is defined as fluorescence at ≥1:128 dilution. Repeat AFIA-reactive (pos) samples are diluted to endpoint; confirmation includes IgM/IgG western blots. DNA is extracted using an automated membrane-based isolation/purification system. PCR primers/probes target the *B. microti* 18S ribosomal RNA gene; the 95% detection limit (LOD) is 66 piroplasms/mL. PCR reactive samples are retested x3; ≥1 reactive result is considered positive (pos). Initial PCR reactive samples retesting negative (neg) 3 times and samples exhibiting non-specific fluorescence are considered inconclusive (inc). Antibody-pos/PCR-neg/inc samples are retested using an enhanced sensitivity digital PCR (ePCR) with a 95% LOD of 10 piroplasms/mL. Quantitative PCR measures parasite load; hamster pairs are inoculated to determine infectivity. Reactive donors and recipients of implicated units are invited into a follow-up study for repeat testing and risk/demographic questioning. **Results/Findings:** Of 30, 435 donors screened, 155 (0.51%) tested pos/inc for *B. microti* antibodies and/or DNA, including 5 AFIA-neg/PCR-pos, 14 AFIA-pos/PCR-pos, and 133 AFIA-pos/PCR-neg; 2 were AFIA-inc/PCR-neg. Five PCR-neg and 1 PCR-inc (all AFIA pos at ≥256) tested ePCR pos. Parasite loads were 13-870,000 piroplasms/mL; 12/17 (70%) hamster pairs developed infections. At last follow-up (mean 124 days, range 17-310 days post-index), 68/70 (97%) of donors remained AFIA-pos (≥1:64). Among PCR-pos donors, 10/11 (91%) became PCR-neg but remained AFIA-pos at last follow-up (mean 147 days, range 39-219 days). Recipient tracing of transfused red cells from prior donations from pos donors revealed 2/13 AFIA-pos recipients; however, both had other risk factors. Since the initiation of screening, no reported cases of TT babesiosis (TTB) from screened donations occurred. However, 6 TTB cases have been identified in recipients receiving units from the same/adjacent locations but not included in screening; 5/6 implicated donors were PCR-pos 2-7 months post-index donation including 1 PCR-pos/AFIA-neg donor. Two additional TTB cases (*B. microti*) occurred in NJ and N CA. **Conclusion:** Blood donor screening for *Bm* in endemic areas is feasible and results in the removal of likely-infectious products from the blood supply. Followed donors appear to retain PCR positivity over several months but retain antibody reactivity for one year or longer.

Disclosure of Commercial Conflict of Interest

V. P. Berardi: IMUGEN Inc., Ownership or Partnership; E. M. Brissette: Nothing to disclose; G. A. Foster: Nothing to disclose; S. T. Johnson: Nothing to disclose; E. Moritz: Nothing to disclose; S. L. Stramer: Nothing to disclose; L. Tonnetti: Nothing to disclose; R. L. Townsend: Nothing to disclose; C. Winton: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. P. Berardi: Nothing to disclose; E. M. Brissette: Nothing to disclose; G. A. Foster: Nothing to disclose; S. T. Johnson: Nothing to disclose; E. Moritz: Nothing to disclose; S. L. Stramer: Nothing to disclose; L. Tonnetti: Terumo BCT, Grants or Research Support; R. L. Townsend: Nothing to disclose; C. Winton: Nothing to disclose

P2

Recombinant Blood Group Proteins Facilitate the Detection of Alloantibodies to High-Prevalence Antigens and Uncover Hidden Antibodies: Results of an International Study

A Seltsam[1] (axel.seltsam@bsd-nstob.de), F Wagner[1], M D Lambert[2], T Bullock[3], N Thornton[3], E A Scharberg[4], D Grueger[5], C Schneeweiss[5], R Blasczyk[6]. *[1]Institute Springe, German Red Cross Blood Service NSTOB, Springe, Germany; [2]Irish Blood Transfusion Service, National Blood Centre, Dublin, Ireland; [3]International Blood Group Reference Laboratory, NHSBT Filton Blood Centre, Bristol, United Kingdom; [4]Institute for Transfusion Medicine and Immunohematology, German Red Cross Blood Service Baden-Württemberg-Hessen, Baden-Baden, Germany; [5]Imusyn, Hannover, Germany; [6]Institute for Transfusion Medicine, Hannover Medical School, Hannover, Germany*

Background/Case Studies: Alloantibodies to high-prevalence red blood cell (RBC) antigens are not easily specified by routine antibody identification tests using standard test cell panels. Thus, occurrence of alloantibodies to high-prevalence antigens generally leads to a delay in antibody identification and a possible delay in patient care. Recombinant blood group proteins (rBGPs) are useful for identification of such antibodies. This multicenter study was conducted to test the use of rBGPs in regional and international RBC reference laboratories. **Study Design/Methods:** For a period of 6 months, single soluble rBGPs (Lu, Yt, Kn (CR1, LHR-D), JMH, Sc, Rg, Ch, Do, Cr) or cocktails of them were assessed for their ability to inhibit reactivity of antibodies to the corresponding antibody specificities. Four different national and international reference laboratories took part in the study. Initially, the use of rBGPs was validated by testing panels of well characterized patient serum samples containing antibodies to high-prevalence antigens in the inhibition assay. Then, the rBGPs were prospectively used in all participating laboratories for routine antibody identification and the results were compared to those obtained with RBCs-based diagnostics. **Results/Findings:** Panels of predefined antibodies to high-prevalence antigens (N = 40) were completely and specifically neutralized by the corresponding rBGP specificities. The rBGPs were used prospectively for antibody identification in a total of 81 cases. Antibodies to high-prevalence antigens were specifically inhibited by the corresponding rBGP specificities except for some CR1-related antibodies (N = 6) which may be directed to so far undefined epitopes not expressed on the truncated recombinant CR1. In five cases, more than 1 μg of rBGP was required to achieve full inhibition. In 13 cases, one (N = 5) or multiple (N = 8) additional clinically relevant alloantibodies could clearly be identified in the presence of an antibody to a high-prevalence antigen. In two laboratories, the rBGPs were successfully used for crossmatching to inhibit the reactivity of a clinically irrelevant antibody to a high-prevalence antigen, to determine compatibility between donor and recipient. **Conclusion:** The soluble rBGPs are stable and can be easily integrated into routine RBC serology. rBGP-based diagnostics enables the identification of antibodies to high-prevalence antigens without the need for rare RBC reagents which are often unavailable. The presence of underlying antibodies can be reliably detected and crossmatching results be validated in the presence of antibodies to high-prevalence antigens. Because of their capacity for single-step, direct RBC antibody determination, recombinant protein-based assays will greatly facilitate and accelerate the process of antibody identification, therefore resulting in more efficient blood supply for immunized patients.

Disclosure of Commercial Conflict of Interest

R. Blasczyk: Nothing to disclose; T. Bullock: Nothing to disclose; D. Grueger: No Answer; M. D. Lambert: Imusyn, Germany, Other; E. A. Scharberg: Nothing to disclose; C. Schneeweiss: Nothing to disclose; A. Seltsam: Nothing to disclose; N. Thornton: Nothing to disclose; F. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Blasczyk: Nothing to disclose; T. Bullock: Nothing to disclose; D. Grueger: No Answer; M. D. Lambert: Nothing to disclose; E. A. Scharberg: Nothing to disclose; C. Schneeweiss: Nothing to disclose; A. Seltsam: Macopharma, Grants or Research SupportImusyn, Grants or Research Support; N. Thornton: Nothing to disclose; F. Wagner: Red Cross Blood Service, Other

P3

Following Prolonged Refrigerator Storage, Younger Red Blood Cells (RBCs)are Preferentially Cleared in Mice Post-Transfusion

S Bandyopadhyay[1] (sheila.bandy@gmail.com), U Dholakia[2], K A Prestia[1,2], R O Francis[1], S L Spitalnik[1], E Hod[1]. *[1]Pathology and Cell Biology, Columbia University Medical Center, New York, NY, United States; [2]Comparative Medicine, Columbia University Medical Center, New York, NY, United States*

Background/Case Studies: Standard refrigerated storage of RBCs is associated with a storage lesion, with increased storage leading to increased hemolysis and decreased post-transfusion RBC recovery. By examining post-transfusion recovery of radiolabeled human RBCs stored in acid-citrate-dextrose (ACD) for 28 days, Mollison surmised that younger, less senescent RBCs are more damaged by storage and are, therefore, cleared preferentially after transfusion as compared to older, more senescent RBCs. We sought to confirm or refute this using a mouse model. **Study Design/Methods:** NHS-biotin injections biotinylated ~99% of the circulating RBCs in 25 Green Fluorescent Protein (GFP)-C57BL/6 mice. The ratio of biotinylated RBCs to newly-produced GFP-positive cells was tracked over time by flow cytometry until 50% of the RBCs remained biotinylated. Blood from these mice was pooled for a leukoreduced blood bank and stored in CPDA-1; it contained an older, senescent population of biotinylated and GFP-positive RBCs, and a younger population of GFP-positive RBCs. Blood was leukoreduced and fresh aliquots were transfused into 10 wild-type C57BL/6 mice for tracking by flow cytometry of the two transfused populations of differing RBC age. The blood was then refrigerator stored for 1 or 2 weeks and transfused into 10 other wild-type C57BL/6 mice. The ratios of the two RBC populations were tracked in the stored RBCs ex vivo and in the transfusion recipients in vivo. **Results/Findings:** The observed lifespan in vivo of GFP-C57BL/6 RBCs was 42 days. Leukoreduction did not affect the younger to older RBC ratio of fresh donor RBCs. Refrigerated storage led to a progressive loss of younger RBCs over time ex vivo (0.22% per storage day increase in the older RBC fraction; P = 0.01). Leukoreduced RBC transfusions after 0 or 7 days of storage resulted in clearance rates consistent with the process of natural senescence (ie, older, senescent RBCs were cleared first at the rate expected from the observed RBC lifespan). Transfusions of mouse RBCs stored for 2 weeks produced an initial clearance of storage-damaged RBCs in the first 2 hours post-transfusion. The ratio of biotinylated, older, senescent to younger RBCs increased from 1.0 to 1.2 in the first 2 hours post-transfusion, suggesting preferential clearance of younger RBCs. **Conclusion:** Prolonged refrigerated storage of mouse RBCs led to preferential loss of the younger, less senescent RBCs ex vivo during storage and during the initial in vivo clearance of storage-damaged RBCs. Thus, the data confirm Mollison's hypothesis and suggest that refrigerated RBC storage does not represent a process of accelerated natural RBC senescence.

Disclosure of Commercial Conflict of Interest

S. Bandyopadhyay: Nothing to disclose; U. Dholakia: Nothing to disclose; R. O. Francis: Nothing to disclose; E. Hod: Nothing to disclose; K. A. Prestia: Nothing to disclose; S. L. Spitalnik: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Bandyopadhyay: Nothing to disclose; U. Dholakia: Nothing to disclose; R. O. Francis: Nothing to disclose; E. Hod: Nothing to disclose; K. A. Prestia: Nothing to disclose; S. L. Spitalnik: Nothing to disclose

P4

Metabolomic Analysis of Stored Platelets and Correlations with Post-Transfusion Platelet Survivals

J C Zimring[1,3] (jzimring@psbc.org), J S Corson[1], M Jones[1], E Morin-Kensicki[2], R P Mohney[2], S J Slichter[1,4]. *[1]Puget Sound Blood Center, Seattle, WA, United States; [2]Research, Metabolon Inc., Research Triangle Park, NC, United States; [3]Laboratory Medicine, University of Washington, Seattle, WA, United States; [4]Medicine-Division of Hematology, University of Washington, Seattle, WA, United States*

Background/Case Studies: Platelets (PLTs) undergo changes during storage, which result in decreased viability. There is substantial donor-to-donor variation in how well platelets store, the biologic basis for which is unknown. Moreover, there are no in vitro assays that can reliably predict post-transfusion outcomes. **Study Design/Methods:** Leukoreduced apheresis PLTs were collected from healthy human subjects and stored for 8 days. Aliquots were taken on days 0, 1, 3, 5, and 8, frozen at -80°C, and subsequently analyzed by LC-MS/MS2 and GC-MS. After 5 days storage, a PLT sample was removed and a sample of fresh PLTs was obtained from the same donor. The stored and fresh PLTs were labeled with either 51Cr or 111In prior to transfusion. Post-transfusion survival was calculated as a ratio of stored to fresh PLT survivals. **Results/Findings:** 332 identified compounds and 86 Unelucidated Structures (US) were quantified. Consistent with existing literature, lactate increased over time while adenosine diphosphate (ADP) and serotonin decreased. Metabolomic analysis of stored PLTs showed: 1) Increased amino acids and organic osmolytes leading to potential osmotic stress; 2) Evidence of oxidative stress due to decreased cysteine and cysteine-glutathione disulfide and increased 2-aminobutyrate 5-oxoproline, markers of transulfuration pathway activity and glutathione turnover; 3) Lipid breakdown indicated by elevation of indicators of both lipolysis and phospholipid catabolism, suggesting dysregulation of energy metabolism; and 4) Accumulation of fatty acid beta-oxidation intermediates, acylcarnitines, and Krebs cycle intermediates, suggesting a Krebs cycle blockage and mitochondrial dysfunction. When the Krebs cycle's capacity to use pyruvate is overwhelmed, pyruvate is converted to lactate. Thus, these findings provide mechanistic insight into lactate accumulation and pH drop. Correlation coefficients (CC) were calculated for all analytes compared to post-transfusion survival. Significant correlations were observed between levels at time of collection, including adenine (CC = 0.95, p = .004), US1 (CC = 0.87, p = .02), 13-HODE/9-HODE (CC = –0.9, p = .016), US2 (CC = –0.86, p = .03), caprylate (CC = –0.86, p = .03). Laurate levels correlated (until 5 days of storage) to PLT survival (CC = -0.95, p = .004). Other compounds developed a significant correlation over the time course of storage, including C-glycosyltryptophan, US3, and US4. **Conclusion:** The current studies provide unique insights into the metabolic characteristics of platelets during storage, and may identify metabolites and pathways that predict PLT viability prior to transfusion. Due to the small sample size and the large number of analytes, the current studies represent preliminary data as part of an ongoing study that is actively enrolling new subjects.

Disclosure of Commercial Conflict of Interest

J. S. Corson: Nothing to disclose; M. Jones: Nothing to disclose; R. P. Mohney: Metabolon, Inc., Other; E. Morin-Kensicki: Nothing to disclose; S. J. Slichter: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. S. Corson: Terumo BCT, Grants or Research Support Genentech, Inc., Grants or Research Support, Fast Track Drugs & Biologics, LLC (DoD funds), Grants or Research Support; M. Jones: Nothing to disclose; R. P. Mohney: Nothing to disclose; E. Morin-Kensicki: Nothing to disclose; S. J. Slichter: Terumo BCT, Grants or Research Support; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium

P5

Hemoglobin Recovery after Blood Donation and the Effects of Iron Supplementation: The Hemoglobin and Iron Recovery Study (HEIRS)

J E Kiss[1] (jkiss@itxm.org), R G Cable[2], D Brambilla[3], G Simone[5], A E Mast[6], B R Spencer[4], M Stone[7], L H Tobler[7]. *[1]Institute for Transfusion Medicine, Pittsburgh, PA, United States; [2]American Red Cross, Farmington, CT, United States; [3]RTI, Rockville, MD, United States; [4]American Red Cross, Dedham, MA, United States; [5]National Heart, Lung, and Blood Institute, Bethesda, MD, United States; [6]BloodCenter of Wisconsin, Milwaukee, WI, United States; [7]Blood Systems Research Institute, San Francisco, CA, United States*

Background/Case Studies: Repeated blood donation causes iron deficiency in many donors, with consequent low hemoglobin deferral. Although donation is allowed every 56 days, the effects of donor iron status, gender, age, and use of iron supplements on the time to recovery of hemoglobin and iron stores are not well understood. **Study Design/Methods:** HEIRS randomized 214 blood donors (>1 donation in the past year) to take 38 mg of iron or no iron daily for 24 weeks following whole blood donation. Randomization was stratified by iron status [ferritin <26 (iron depleted) versus >26 ng/ml (iron replete)], gender, and age (<60 versus >60 years). Hemoglobin (Hgb) and ferritin were measured pre-donation and at seven post donation time points from 3-7 days to 24 weeks, and analyzed using repeated measures models. Reticulocyte counts were measured in a representative subset. Time to recovery of 80% of the postdonation decrease in Hgb (Hgb baseline—Hgb day 3) and % of donors with venous Hgb <12.5 g/dl at 56 days post donation were determined. **Results/Findings:** Mean enrollment ferritin was 56.3 ng/ml in the iron-replete group and 15.0 ng/ml in the iron-depleted group. Time to recovery of 80% of Hgb lost at donation was less than 56 days only in subjects randomized to receive iron, regardless of pre-donation ferritin, and was delayed well beyond 56 days in both

non-supplemented groups. These responses were further reflected in an increased reticulocyte response, and reduced percentage of subjects with Hgb < 12.5 g/dl at 56 days in those taking iron (Table). Even in the iron replete group, recovery of ferritin to baseline required about 90 days in those taking iron and >168 days in those not taking iron. **Conclusion:** This large-scale randomized clinical trial provides comprehensive quantitative data on the recovery of hemoglobin and iron stores following whole blood donation for diverse blood donor groups. Delayed recovery of Hgb and iron stores occurs frequently and is more pronounced in iron-depleted donors. Iron supplements hasten recovery in iron-depleted donors and enhance recovery even in donors with adequate pre-donation iron stores. The data suggest that implementation of iron supplementation programs will benefit donors by accelerating Hgb recovery, limiting the duration of iron deficiency, and reducing Hgb deferrals.

Disclosure of Commercial Conflict of Interest

D. Brambilla: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; A. E. Mast: Nothing to disclose; G. Simone: Nothing to disclose; B. R. Spencer: Nothing to disclose; M. Stone: Nothing to disclose; L. H. Tobler: No Answer

Disclosure of Grants Conflict of Interest

D. Brambilla: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; A. E. Mast: Novo Nordisk, Grants or Research Support; G. Simone: Nothing to disclose; B. R. Spencer: Nothing to disclose; M. Stone: Nothing to disclose; L. H. Tobler: No Answer

Ferritin (ng/ml)	Treatment	#w/Hgb < 12.5 g/dl on Day 56 (%) Women	Men	Hgb Recovery to 80% Baseline, Days (95%CI)
≤26	No iron	19/30 (63.3)	4/18 (27.8)	122 (94, >168)
	Iron 38 mg	10/33 (33.3)	1/18 (5.6)	29 (27, 39)
>26	No iron	3/33 (12.1)	0/20 (0)	78 (51, >168)
	Iron 38 mg	0/35 (0)	0/20 (0)	33 (25, 39)

P6

Identification of the Genetic Basis of PX2, a Recently Reported Glycolipid Blood Group Antigen

J S Westman[1] (Julia.Westman@med.lu.se), J R Storry[1,2], A K Hult[1,2], Hellberg[2], M L Olsson[1,2]. *[1]Dept. of Laboratory Medicine, Lund University, Lund, Sweden; [2]Clinical Immunology and Transfusion Medicine, University and Regional Laboratories, Region Skåne, Lund, Sweden*

Background/Case Studies: In 1982, Kannagi et al described a β3GalNAc-terminating glycolipid denoted x2. Unlike the P blood group antigen which has the same terminal sugar, the precursor structure for x2 is paragloboside and not the P^k antigen. A decade later, Thorn et al reported that x2 levels are elevated on RBCs of the P/P1/P^k-negative p phenotype. We recently encountered a patient with the P-deficient P_1^k phenotype and whose pre-transfusion test was incompatible with p RBCs. According to text books, such patients produce naturally occurring anti-P but should be compatible with p RBCs. Anti-P was confirmed and other known blood group antibodies were excluded. The same phenomenon was then also noted in other P-deficient donors. Consequently, ISBT recognised x2 as a new blood group antigen in the GLOB collection under the name PX2. We set out to identify its genetic basis. **Study Design/Methods:** In addition to standard RBC serology, the MEG-01 cell line was transfected with constructs containing variants of the *B3GALNT1* open reading frame (ORF) cloned into pEF1α-IRES-ZsGreen1. In addition to consensus, nonsense (202T) and missense (449G) mutants known to cause loss of P synthase activity were used, as well as a neutral missense mutant (376A) that does not affect P synthase activity. Empty vector was used as negative control. All experiments were performed in triplicate on 3 separate occasions. PX2 was detected by flow cytometry 48h post-transfection and on RBCs, using murine monoclonal anti-x2 (clone TH2) and PE-conjugated rat-anti-mouse anti-kappa. **Results/Findings:** Plasma from all P-deficient individuals was compatible with other P_1^k and P_2^k RBCs but not with p RBCs. We therefore hypothesized that the *B3GALNT1*-encoded P synthase, which is mutated in P deficiency, may also be responsible for PX2 synthesis. Cells transfected with the *B3GALNT1* ORF showed high PX2 expression (63% positive cells), compared to mock-transfected cells (1.8%) and rare null mutants (202T: 1.3% and 449G: 2%). A variant ORF (376A found in ~2.7% of alleles in the 1000 genomes project) neither affected PX2 (62%) nor P expression. Blocking studies showed human polyclonal anti-PX2 to inhibit TH2 binding to RBCs. Finally, the PX2 levels on RBCs were assessed by flow cytometry: p (strong) > Bombay (intermediate) > P_1 or P_2 (weak) > P^k (negative). **Conclusion:** Our data show that *B3GALNT1*, known to encode P synthase, is also able to encode PX2-synthesizing activity. Thus, in the absence of a functional enzyme neither P nor PX2 are formed. As a consequence, naturally occurring anti-P and anti-PX2 can be made. Based on this linkage, we propose that PX2 joins P as the second antigen in the GLOB system. Until the clinical significance of anti-PX2 is known, we also recommend that rare P_1^k or P_2^k RBC units are preferentially selected for transfusion to P^k patients since p RBCs may pose a risk for hemolytic transfusion reactions due to elevated PX2 levels.

Disclosure of Commercial Conflict of Interest

Hellberg: Nothing to disclose; A. K. Hult: Nothing to disclose; M. L. Olsson: Nothing to disclose; J. R. Storry: Nothing to disclose; J. S. Westman: Nothing to disclose

Disclosure of Grants Conflict of Interest

Hellberg: Nothing to disclose; A. K. Hult: Nothing to disclose; M. L. Olsson: Nothing to disclose; J. R. Storry: Nothing to disclose; J. S. Westman: Nothing to disclose

Red Blood Cells: Antibodies and Hemolysis

S1-010A

Female RBCs Have Less In Vitro Hemolysis and Greater In Vivo Post-Transfusion Recovery than Male Red Blood Cells (RBCs) in a Mouse Model

T Kanias[1] (tak77@pitt.edu), D Sinchar[1], H Waterman[2], K de Wolski[2], J C Zimring[2], M T Gladwin[1]. *[1]Medicine, University of Pittsburgh, Pittsburgh, PA, United States; [2]Puget Sound Blood Center Research Institute, University of Washington School of Medicine, Seattle, WA, United States*

Background/Case Studies: Donor genetic predisposition to hemolysis can significantly affect the recovery and survival of stored RBCs in the circulation. We have previously reported that female gender is associated with less hemolysis following routine RBC storage or in response to osmotic stress for human RBCs. The purpose of this study is to further characterize the correlation between genetic background and hemolysis in mouse models, and to test whether the observed gender differences in storage and osmotic fragility predict the survival of transfused RBCs in vivo. **Study Design/Methods:** Predisposition to hemolysis based upon gender was evaluated in 20 distinct inbred mouse strains, and in normal or ovariectomized (OVX) AKR/J mice. Washed RBCs were subjected to selected hemolytic assays including oxidative, osmotic, and mechanical hemolysis. RBCs from normal and OVX females were also subjected to 7-day storage in ADSOL, and calcium influx assays using Fluo-3AM. The effects of donor's gender on RBC survival after transfusion were studied in two mouse strains of low (C57BL/6J (B6)) or high (FVB/NJ) susceptibility to hemolysis. Mouse RBCs collected in CPDA-1 were filter leukoreduced and stored for 14 (B6) or 5 (FVB/NJ) days. Recovery of fresh or stored RBCs was evaluated by transfusing into B6xFVB F1 mice that were also transgenic for GFP such that all RBCs fluoresced green. Transfused RBCs were tracked by the GFP negative population. **Results/Findings:** The rate of hydrogen-peroxide-induced heme degradation was two-fold higher in BALB/c compared with B6 strains. Osmotic hemolysis (3 h pink test) was highest in AKR/J (48%) and lowest in BPH/2J (18%). Female gender demonstrated higher resistance to osmotic hemolysis, mechanical fragility, and AAPH oxidative hemolysis in 14, 15, and 17 of the 20 tested strains, respectively. Ovariectomy was associated with reduced intracellular calcium, and increased mechanical fragility, oxidative hemolysis, and storage hemolysis, but not osmotic fragility. Transfusion of stored B6 female RBCs had 80% 24-hour in vivo recovery versus 55% recovery of transfused male RBCs. A similar trend was observed with transfused FVB/NJ RBCs, although the overall 24h in vivo recovery rates were significantly lower than B6. **Conclusion:** These studies support the overreaching hypothesis that genetic variation, including gender, modulates hemolysis. Our gender studies in mice confirm that RBCs from reproductive females are less susceptible to storage or stress-induced hemolysis. Furthermore, we demonstrate that in vitro evaluation of RBC storage hemolysis and osmotic fragility correlates with post-transfusion recoveries in mice, and may have clinical relevance in terms of enhanced post-transfusion recovery and survival of female RBCs in humans. Our ongoing studies focus on the possible mechanisms linking sex hormones to hemolysis.

Disclosure of Commercial Conflict of Interest

K. de Wolski: Nothing to disclose; M. T. Gladwin: Nothing to disclose; T. Kanias: Nothing to disclose; D. Sinchar: Nothing to disclose; H. Waterman: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. de Wolski: Terumo, Grants or Research SupportImmucor, Grants or Research Support; M. T. Gladwin: Bayer Corp, Grants or Research Support, Gilead Sciences, Grants or Research Support, Bayer Corp, Consulting or Board of Director Fees; T. Kanias: Nothing to disclose; D. Sinchar: Nothing to disclose; H. Waterman: Immucor, Grants or Research Support, Terumo, Grants or Research Support; J. C. Zimring: Immucor, Grants or Research Support, Terumo, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo, Travel Support or Honorarium

S2-010A

Inhibition of Red Cell Alloimmunization Can Occur Independently from Red Cell Clearance in a Murine Model

H Yu[1,2] (yuhonghui.yhh@gmail.com), S R Stowell[4], J E Hendrickson[4], J C Zimring[5], M Uchikawa[6], A H Lazarus[1,3]. *[1]Canadian Blood Service, Toronto, ON, Canada; [2]Anesthesiology, Tongji Hospital, Huazhong University of Science and Technology, Wuhan, China; [3]Laboratory Medicine and the Keenan Research Centre, Li Ka Shing Knowledge Institute of St. Michael's Hospital, Toronto, ON, Canada; [4]Pathology and Laboratory Medicine, Emory University School of Medicine, Atlanta, GA, United States; [5]Puget Sound Blood Center Research Institute, Seattle, WA, United States; [6]Kanto-Koshinetsu Block Blood Center, Japanese Red Cross, Tokyo, Japan*

Background/Case Studies: One of the most effective interventions in clinical medicine has been the administration of Rh immune globulin (anti-D) to prevent immunization of the mother to the D antigen on fetal red blood cells (RBCs); a phenomenon termed "antibody mediated immune suppression" (AMIS). The mechanism of action of how such antibodies can prevent an immune response remains unclear but several theories have been proposed. The most commonly accepted theory to explain how the AMIS effect occurs with erythrocytes is by the rapid clearance of the antigen positive RBCs. Unfortunately, mouse models in current use have been unable to address this question because the RBC (commonly sheep RBC) are xenogeneic to the host, and the cells are rapidly cleared in the presence or absence of an opsonizing antibody. **Study Design/Methods:** To explore the role of RBC clearance in AMIS, we immunized naïve mice with murine transgenic RBCs expressing the HOD (Hen egg lysozyme [HEL], expressed in tandem sequence with Ovalbumin, and the complete human Duffy transmembrane protein) antigen in the presence versus absence of a polyclonal HEL-specific IgG as well as six different monoclonal antibodies to selected portions of the HOD molecule. The isotype, specificity, and ability to induce AMIS of these antibodies were compared with their ability to cause accelerated RBC clearance as well as sterically hinder the HOD antigen. **Results/Findings:** Mice immunized with HOD-RBC made IgM and IgG reactive with the HEL portion of the molecule only. HEL-specific IgM as well as IgG1, IgG2b and IgG3 were made in response to transfusion of HOD-RBCs. The dose of 10∧7 HOD-RBC (roughly equivalent to a 2-5 mL bleed in a human on a blood volume basis) was selected for the evaluation of AMIS induction. All six of the monoclonal antibodies tested could inhibit the response. The HEL-specific monoclonal antibodies (4B7, IgG1; GD7, IgG2b; 2F4, IgG1) did not accelerate the clearance of the HOD-RBC and displayed minimal steric hindrance properties. The three Duffy-specific antibodies (MIMA29, IgG2a, CBC-512, IgG1; K6, IgG1) all caused rapid clearance of HOD RBC and displayed no steric hindrance properties. **Conclusion:** This is the first demonstration of AMIS in a full murine model. The model clearly demonstrates that AMIS can occur normally in the absence of RBC clearance. The AMIS effect also appears to be independent of the IgG isotype and fine specificity of the AMIS inducing antibody.

Disclosure of Commercial Conflict of Interest

J. E. Hendrickson: Nothing to disclose; A. H. Lazarus: Nothing to disclose; S. R. Stowell: Nothing to disclose; M. Uchikawa: Nothing to disclose; H. Yu: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. E. Hendrickson: Terumo, Grants or Research Support; A. H. Lazarus: Nothing to disclose; S. R. Stowell: Nothing to disclose; M. Uchikawa: Nothing to disclose; H. Yu: Nothing to disclose; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium

S3-010A

Antibody-Mediated Immune Suppression by Fcγ Receptor Independent Mechanisms in a Fully Allogeneic Mouse Model of Red Blood Cell (RBC) Immunization

L Bernardo Reyes[1] (bernardoreyl@smh.ca), H Yu[2,1], A H Lazarus[2,1]. *[1]St Michael's Hospital, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada*

Background/Case Studies: The ability of IgG to suppress immune responses has been applied clinically in rhesus (Rh) prophylaxis. Rh-negative women may develop antibodies against Rh-positive erythrocytes acquired transplacentally from their fetuses, which can later damage fetal RBCs causing hemolytic disease of the fetus and newborn (HDFN). Anti-D immunoglobulin therapy has been highly successful in preventing HDFN; however, the immune inhibitory mechanism(s) remains poorly understood. This inhibitory effect of anti-D to fetal RBC immunization has been referred as AMIS. Two major theories behind this mechanism involve the binding of the IgG mediating the AMIS to activating or inhibitory Fcγ receptors (FcγR) which induce either erythrocyte clearance or B cell inhibition, respectively. In this work, the absolute role of activating and inhibitory Fcγ receptors in the AMIS mechanism was explored using a fully allogeneic mouse model of RBC immunization. HOD transgenic mice contain a RBC specific triple fusion recombinant protein comprised of hen egg lysozyme (HEL) in tandem sequence with sequences from ovalbumin and the complete human transmembrane Duffy antigen. HOD RBCs thus express a unique erythrocyte antigen that can be used to stimulate an immune response to the red cells. The immunodominant epitope in C57BL/6 mice is contained in the HEL portion of the molecule only. **Study Design/Methods:** HOD RBCs on the FVB background were injected into naïve C57BL/6 mice alone or together with anti-HEL antibodies (ie, AMIS). To assess the contribution of activating and inhibitory Fcγ receptors to the induction of an AMIS effect, normal mice vs Fcγ chain -/- mice or FcγRIIB -/- mice were used as recipients of HOD RBCs. F(ab')2 fragments of IgG were also produced and their ability to mediate AMIS effects were evaluated in C57BL/6 mice. The resulting antibody response to the HOD antigen was evaluated using HEL protein by ELISA. **Results/Findings:** Anti-HEL polyclonal antibodies successfully suppressed the antibody response to the HOD antigen in C57BL/6 mice. The same AMIS effect could also be observed in mice lacking the IgG binding receptors (activating as well as inhibitory) as compared to wild-type mice. In agreement with this, F(ab')2 fragments of the AMIS antibody, which cannot interact with either activating or inhibitory FcγRs, also reduced the antibody response nearly as efficiently as anti-HEL antibodies. **Conclusion:** Inhibition of in vivo antibody responses to HEL protein on HOD RBCs by polyclonal IgG can occur via an FcγR-independent pathway. These results may have implications for helping understand how anti-RBC antibodies, such as anti-D, may be able to attenuate the immune response to red cells.

Disclosure of Commercial Conflict of Interest

L. Bernardo Reyes: Nothing to disclose; A. H. Lazarus: Nothing to disclose; H. Yu: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Bernardo Reyes: Nothing to disclose; A. H. Lazarus: Nothing to disclose; H. Yu: Nothing to disclose

S4-010A

Development of an Improved Calculation for Estimating the Immunogenicity of Non-ABO Blood Group Antigens

G Stack[1,2] (gary.stack@va.gov), C A Tormey[1,2]. *[1]Pathology and Laboratory Medicine Service, VA Connecticut Healthcare System, West Haven, CT, United States; [2]Department of Laboratory Medicine, Yale University School of Medicine, New Haven, CT, United States*

Background/Case Studies: The immunogenicity of non-ABO blood group antigens is typically estimated from the relative number of antibodies of different specificities observed in a transfused population, with normalization for the probability that an antigen-negative recipient will be transfused an antigen-positive RBC unit. Such immunogenicity calculations have been based on the assumption that patients are transfused only one antigen-positive RBC unit each. Since the number of RBC units per transfused patient is typically higher, the calculations are predicted to overestimate the immunogenicity of low-frequency antigens and underestimate that of high-frequency antigens. The goals of this study were: 1) to assess the magnitude of this error; 2) to develop an immunogenicity calculation that corrects for RBC unit exposures; and 3) to further update the calculation by using an empirically-determined value for the immunogenicity of the reference antigen, K. **Study Design/Methods:** The number and specificity of blood

group alloantibodies at the study hospital were extracted from a transfusion service database (n = 205 antibodies). Naturally occurring and pregnancy-associated antibodies were excluded. The mean number of RBC units per transfused patient was determined by a review of randomly selected records (n = 600 patients). A literature review was conducted to find empirical values for the immunogenicity of K. **Results/Findings:** A revised immunogenicity calculation was developed by changing the term for the fractional probability of an antigen-positive donor from [1-Ag_xneg] to [1-(Ag_xneg)^{MRBC}], where Ag_xneg represents the expected fraction of donor units negative for the antigen, Ag_x, and MRBC is the mean number of RBC units per transfused patient. The revised term corresponds to the probability of exposure to at least one antigen-positive RBC unit from among the units transfused. Immunogenicity calculations were done on the 12 most commonly observed antibodies. A mean of 6.0 RBC units were transfused per patient, which when substituted into the revised equation, resulted in up to a 3.9-fold increase in immunogenicity over the uncorrected values. The higher an antigen's prevalence, the greater was the observed fold increase. Use of empirical data for K immunogenicity, instead of a traditionally-used estimate, resulted in a further 1.9-fold increase in all calculated immunogenicities. The corrected immunogenicity ranking was: K > Jk^a > Lu^a > E > c > P1 > M > Le^b > C > Le^a > Fy^a > S. **Conclusion:** A correction for the number of transfused RBC units and an updated value for K immunogenicity resulted in up to a 7-8-fold increase in the calculated immunogenicity of non-ABO blood group antigens. The corrected values should represent the most accurate immunogenicity estimates to date, and may lead to an improved understanding of the basis of blood group antigen immunogenicity.

Disclosure of Commercial Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Terumo, Inc., Grants or Research Support

S5-010A

Analysis of Red Blood Cell (RBC)-Bound Immunoglobulins (Ig) in "Coombs Negative" Hemolytic Anemia by Flow Cytometry (FC) and Capillary Tube (CT) Methods

W A Beres[1] (wendy.beres@redcross.org), S Singh[1], S J Nance[1].
[1]Technical Services, American Red Cross, Philadelphia, PA, United States

Background/Case Studies: Rarely, patients suspected of having autoimmune hemolytic anemia (AIHA) present with a negative direct antiglobulin test (DAT). It has been hypothesized that the possible reasons the RBC-bound IgG is undetectable are: the RBC-bound IgG is below the normal detection threshold; or RBC-bound IgA or IgM is present; or low-affinity IgG autoantibodies are present (*Immune Hemolytic Anemias*, 2nd ed, p. 208). Over a 7-month period, this rare phenomenon was suspected in 14 DAT-negative patients whose samples were referred to a national IRL for evaluation. **Study Design/Methods:** The routine battery of tests to detect low levels of Ig for DAT negative AIHA was performed on 14 samples: tube DAT; enzyme testing; direct polybrene test; polyethylene glycol; concentrated acid eluate; and anti-IgG, -IgA and -IgM (Jackson ImmunoResearch Lab, West Grove, PA) DATs by CT. For CT, optimal dilutions of anti-Ig in normal saline (0.9%) containing 6% bovine serum albumin (BSA) were used. Equal volumes (approximately 2 cm) of antisera or BSA (as a negative control) and 50% cell suspensions were used in the CT (0.4 × 75 mm). The CTs were sealed, incubated at a 60-degree angle at room temperature for 15 minutes, and observed for agglutination. The CTs were inverted and again observed for agglutination. Agglutination indicated RBC-bound Ig. The samples were tested by FC with fluorescein isothiocyanate (FITC)-labeled anti-human IgG, -IgA, -IgM (Life Technologies, Carlsbad, CA) or FITC-labeled anti-human IgA (Jackson ImmunoResearch Lab, West Grove, PA) at optimal dilutions in Dulbecco's PBS containing 0.6% BSA. 50,000 RBCs acquired from each sample by the Becton Dickinson FACSCaliburTM (San Jose, CA) FC were analyzed. Naïve ethylenediaminetetraacetic acid (EDTA)-anticoagulated samples and Ig coated RBCs were tested to determine FC settings and control for validity and cross-reactivity. Normal ranges for RBC-bound Ig were determined using the mean %M2 ±3 SD of normal donor samples. **Results/Findings:** Samples 1-9 yielded negative results in all test methods. Sample 10 was positive in Tube IgG-DAT on day 1, but negative by CT on day 6 and FC on day 10. Sample 11 was positive by CT with anti-IgA on day 12, but negative by FC on day 19. Samples 12 and 13 were positive by FC with anti-IgG, -IgA, -IgM, negative by CT and all other tests. Sample 14 was positive in FC with anti-IgG and polybrene testing, and not tested in CT. **Conclusion:** The DAT negative AIHA phenomenon is rare, but when a patient presents, detection of RBC-bound Ig can aid in treatment considerations. The results confirm the usefulness of a battery of tests in detection of low-level RBC-bound Ig, and that FC yielded the highest % positives (21%), confirming the work of Garratty (Semin Hematol 2005; 42:156-164), who found FC to be most productive in studying this patient population, with the same 21% positive by FC findings.

Disclosure of Commercial Conflict of Interest

W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; S. Singh: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; S. Singh: Nothing to disclose

S6-010A

Studies on Specificity of Selected Drug-Dependent Antibodies

P A Arndt[1] (patricia.arndt@redcross.org), J Revilla[1], R M Leger[1], G Garratty[1]. *[1]American Red Cross Blood Services, Pomona, CA, United States*

Background/Case Studies: Previous studies have described drug-dependent antibodies (DDAbs) with specificities in the following blood groups: Rh; I; P; Kell; Lutheran (more than one report for each); and Kidd; MNS; H; and Band 3 (one report each). Over the last 30 years, various specificity studies on DDAbs have also been performed. **Study Design/Methods:** Antibodies (n = 53) to 13 different drugs were tested in the presence of drug against RBCs of different phenotypes and/or RBCs treated with ficin or 0.2 M dithiothreitol (DTT). Some drug antibodies were diluted to achieve 1-2+ reactivity for testing. **Results/Findings:** See table. **Conclusion:** We tested 53 antibodies to 13 drugs (including the 3 currently most common causes of drug-induced immune hemolytic anemia: piperacillin, ceftriaxone and cefotetan). Only three of these drugs were previously reported to be associated with blood group specificities (diclofenac and piperacillin = Rh, sulfamethoxazole = H). In this study, only anti-piperacillin, anti-oxaliplatin and anti-sulfamethoxazole showed variation in reactivity with different RBCs. A large percentage of anti-piperacillin showed relative Rh specificity, especially when dilutions were tested (17/26 undiluted sera [65%] and 5/6 diluted sera [83%]) . We were surprised to see that 9/11 (82%) piperacillin antibodies showed relative specificity for Lutheran antigens and all nine also showed Rh specificity. At present, there is no explanation for piperacillin antibodies showing both Rh and Lutheran specificities as there is no relationship between these two blood group systems.

Disclosure of Commercial Conflict of Interest

P. A. Arndt: Nothing to disclose; G. Garratty: Nothing to disclose; R. M. Leger: Nothing to disclose; J. Revilla: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. A. Arndt: Nothing to disclose; G. Garratty: Nothing to disclose; R. M. Leger: Nothing to disclose; J. Revilla: Nothing to disclose

Results

Drug Antibody (# tested)	Diluted?	Non-reactive or weaker reactive RBCs	RBCs tested
Acetaminophen (1)	No	None	Adult I, adult i
Phenacetin (1)	No	None	
Probenecid (1)	No	None	
Quinidine (1)	No	None	
Teniposide (1)	No	None	
Cefotetan (4)	Yes	None	R1R1, R2R2, rr, Lu(a – b–), PP1Pk–, cord i, ficin-tt'd, DTT-tt'd; 2 also vs. Rhnull
Piperacillin (26)	No	none (9), R2R2 (n = 16), R1R1 (n = 1); Lu(a + b–) (n = 1/5); Lu(a + b–) and Lu(a – b–) (n = 4/5)	R1R1 & R2R2 (26); 5 also vs. Lu(a – b+), Lu(a + b–) & Lu(a – b–)
Cefotaxime (1)	Yes	None	R1R1, R2R2, rr (untreated and ficin-tt'd); DTT-tt'd, cord i, adult i, Oh, Rhnull, –D–, Lu(a – b–), Ko, PP1Pk–, Jk(a – b–), Vel–, U–, Ge: –2, –3 [5 piperacillin antibodies were also tested vs. Lu(a + b–)]
Cefotetan (6)	Yes	None	
Ceftriaxone (4)	Yes	None	
Cimetidine (1)	Yes	None	
Diclofenac (1)	Yes	None	
Oxaliplatin (2)	Yes	Ficin-tt'd RBCs (n = 1)	
Piperacillin (6*)	Yes	none (n = 1†), R2R2, Rhnull, –D–, Lu(a – b–) and Lu(a + b–) (n = 3); Rhnull, –D–, Lu(a – b–) and Lu(a + b–) (n = 1); R2R2, Rhnull, –D–, DTT-tt'd and ficin-tt'd (n = 1)	
Sulfamethoxazole (1)	Yes	DTT-tt'd, U–, Lu(a – b–)	

* 4 previously tested undiluted versus R1R1 and R2R2 RBCs (no specificity).

† This example of anti-piperacillin, tested at a 1/10 dilution, reacted only as a direct agglutinin (antiglobulin test was negative). All other examples of anti-piperacillin, tested at dilutions ranging from 1/2 to 1/4, reacted by the antiglobulin test (with or without an agglutinin).

Platelet and Leukocyte Biology

S7-010B

Novel Mechanism of Platelet Clearance in Immune-Mediated Thrombocytopenia (ITP): Roles of Platelet Desialylation in Diagnosis and Therapy of Anti-GPIb-mediated ITP

D E van der Wal[1,2] (diannevanderwal@gmail.com), J Li[1,4], G Zhu[1], I Yougbare[1], L Ma[1], B Vadasz[1,4], Q Zeng[3], V Leytin[1,2], J Freedman[1,4], H Ni[1,2]. *[1]Keenan Research Centre in the Li KaShing Knowledge Institute, St. Michael's Hospital, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada; [3]Department of Hematology, Anhui Medical University, Hefei, China; [4]Department of Laboratory Medicine and Pathobiology, Departments of Medicine and Physiology, University of Toronto, Toronto, ON, Canada*

Background/Case Studies: Immune thrombocytopenia (ITP) is an autoimmune disease characterized by autoantibodies against one's own platelet antigens, primarily glycoprotein (GP)IIbIIIa-integrin (70-80%) and GPIb-complex (20-40%). Although the current paradigm suggests clearance of opsonized platelets is through the reticuloendothelial system via Fcγ-receptors, our group and others have demonstrated that anti-GPIbα, but not anti-GPIIbIIIa, can induce thrombocytopenia via an Fc-independent pathway. This form of thrombocytopenia is shown here to be resistant to intravenous IgG (IVIG) therapy in murine ITP-models, a finding which is consistent with subsequent IVIG studies in human ITP patients. In addition, our recent retrospective study found that human anti-GPIb-mediated ITP patients seem more resistant to steroid therapy. Thus, binding of anti-GPIb antibodies may induce platelet clearance through a different mechanism which is currently unknown. **Study Design/Methods:** We have developed unique mouse anti-mouse monoclonal antibodies (mAbs) that are cross reactive to human platelets. Flow cytometry (FC) was used to evaluate antibody induction of platelet activation, apoptosis, desialylation, and Neu1. The link between anti-GPIbα induced platelet activation and desialylation was assessed with activation inhibitors and sialidase inhibitor N-acetyl-2, 3-dehydro-2-deoxy neuraminic acid (DANA). Fc-independent uptake of anti-GPIbα bound platelets was examined by flow in the presence of FcγR blockers, and the hepatic Ashwell-Morell receptor (AMR) blocker asialofetuin. **Results/Findings:** It was found that anti-GPIbα antibodies induced significantly more P-selectin, phosphatidylserine (PS)-exposure and, interestingly, desialylation, than anti-GPIIbIIIa antibodies both in vitro and in vivo. Importantly, incubations of human platelets with ITP-patient plasma showed similar effects. We also found platelet activation was through GPIbα which lies directly upstream of platelet desialylation, which further induces activation and apoptosis. Neu1 was identified as the candidate sialidase through increased surface expression. In mice, we found DANA was able to rescue platelet counts exclusively in anti-GPIbα mediated thrombocytopenia. We also found FcγR independent pathways, such as the AMR, are implicated in the clearance of anti-GPIbα bound platelets. **Conclusion:** We demonstrated for the first time that anti-GPIbα antibodies induce platelet activation, apoptosis and desialylation, leading to clearance via non-Fcγ receptors such as hepatic AMR. These findings may lead to novel therapeutic and diagnostic regimens, including the potential use of sialidase inhibitors as a solution for anti-GPIb-mediated ITP patients previously refractory to both steroid and IVIG therapies.

Disclosure of Commercial Conflict of Interest

J. Freedman: Nothing to disclose; V. Leytin: Nothing to disclose; J. Li: Nothing to disclose; L. Ma: Nothing to disclose; H. Ni: Nothing to disclose; B. Vadasz: Nothing to disclose; D. E. van der Wal: Nothing to disclose; I. Yougbare: Nothing to disclose; Q. Zeng: Nothing to disclose; G. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Freedman: Nothing to disclose; V. Leytin: Nothing to disclose; J. Li: Nothing to disclose; L. Ma: Nothing to disclose; H. Ni: Nothing to disclose; B. Vadasz: Nothing to disclose; D. E. van der Wal: Nothing to disclose; I. Yougbare: Nothing to disclose; Q. Zeng: Nothing to disclose; G. Zhu: Nothing to disclose

S8-010B

Responsiveness of Platelets to Adenospine Diphosphate (ADP) Requires Protein Synthesis: A Translational Inhibitor Study

P Schubert[1] (peter.schubert@blood.ca), B Culibrk[1], S Karwal[1], C A Lovatt[1], D Devine[1]. *[1]Centre for Blood Research, Canadian Blood Services, Vancouver, BC, Canada*

Background/Case Studies: The ability of enucleated platelet to carry out protein synthesis has become acknowledged; however, many questions are still unanswered including the detailed molecular mechanism. One way to address its involvement in platelet biology is to analyze the impact of protein translation on the platelet function. Two known translational inhibitors, rapamycin and puromycin, were used to monitor effects on platelet concentrate during storage. **Study Design/Methods:** Platelet concentrates produced by apheresis were split into small platelet storage bags and were spiked with 10 nM rapamycin, 1 mM puromycin, or kept untreated as storage control. Samples were drawn two hours after adding the inhibitors on day 1 as well as on days 2, 5, and 7 of storage. Routine platelet quality parameters such as platelet count, platelet activation as determined by P-selectin exposure on the platelet surface, glycoprotein IIbIIIa activation by PAC-1 binding, as well as glucose and lactate levels to monitor metabolic activity, were measured. Responsiveness to 10 μM ADP assessed as the increase of P-selectin expression and glycoprotein IIbIIIa activation by the agonist were determined in the same assay as well as by an extent of shape change assay. Platelet lysates were prepared using a Triton-based buffer and the expression levels of selected proteins were quantified by immuno blotting.

Results/Findings: Platelet count, platelet activation and platelet glycoprotein IIbIIIa activation as well as glucose and lactate levels were not affected by the presence of the translational inhibitors when compared to the untreated control. However, all assays targeting the responsiveness to ADP showed a marked reduction in the presence of puromycin, not seen in the presence of rapamycin which displayed a similar profile as the control samples. In order to place this observation into a biological context, the protein profile of RhoGDI, a regulator protein for the GTPase RhoA, which is known to be synthesized in platelets upon activation, was analyzed. The protein expression increased throughout storage and was not affected by rapamycin; however, the presence of puromycin triggered a strong decrease in the RhoGDI protein level, suggesting reduced protein synthesis. **Conclusion:** The significance of protein translation in platelets is still poorly understood. For the first time, the involvement of protein synthesis in the response mechanism of platelets to ADP was demonstrated here. Besides the observation that inhibition of RhoGDI protein synthesis is affecting degranulation, integrin activation, and actin cytoskeleton rearrangement, it was also discovered that puromycin might be involved in regulating translation of these proteins in platelets. This finding should further propel investigations of the significance of the ROCK/RhoA pathway to platelet protein translation.

Disclosure of Commercial Conflict of Interest

B. Culibrk: Terumo BCT, Grants or Research Support; D. Devine: Terumo BCT, Grants or Research Support; S. Karwal: Nothing to disclose; C. A. Lovatt: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

B. Culibrk: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support; Fresenius-Kabi, Grants or Research Support, MacoPharma, Grants or Research Support; S. Karwal: Nothing to disclose; C. A. Lovatt: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

S9-010B

Extended Storage of Blood Platelets in a Phase of Hibernation Induced by a Chemical Bioregulator

S Stienstra[1] (Stef.Stienstra@inter.nl.net). [1]BBB, Beek-Ubbergen, Netherlands

Background/Case Studies: A bioregulator, a chemical compound, is synthesised, which can bring mammalian cells in a phase of hibernation, such as the squirrel and Syrian hamster in winter. In hibernation, mammalian tissue and blood cells are protected against oxidative stress damage as the metabolism changes towards minimal need for oxygen and nutrition. It changes the metabolism so that the tissue and blood cells of hibernating animals can cool down to temperatures just above freezing point without cell damage or losing vitality. **Study Design/Methods:** The synthesised bioregulator dissolved in a standard cell storage buffer is added to a standard fresh platelet concentrate. Both standard platelet-rich plasma (PRP) platelet concentrates and apheresis platelet concentrates are used. The platelet concentrates were stored for 21 days without flat-bed shaking in standard blood bank bags in a refrigerator at 4C. After rewarming, ex-vivo vitality tests were done in an aggregometer, hematology counter, and with a flow cytometer, which indicated that the platelets have similar properties as fresh apheresis platelets for transfusion. The compound is tested for toxicity and can be washed out after it has been used to trigger the platelets into the hibernation phase, enabling the cold storage of platelets for an extended period without the conventional shaking on a flat-bed shaker. Animal tests are performed as well. **Results/Findings:** The developed compound initiates a phase of hibernation in human blood platelets, which enables the storage of platelets at 4C without suffering the so-called cold activation. In a German blood bank experiments were done which showed that it was possible to store platelets for several weeks in a phase of hibernation in a refrigerator without losing too much vitality. Platelets that have been stored for 21 days in standard blood bank bags without severe loss of functionality as tested with thrombin and ADP aggregation, TRAP, CD62p, and Annexin V analysis also gave results indicating functionality after transfusion. Remarkably, the swirling phenomenon was very strongly present for several weeks. In hibernation induced by the presence of the developed compounds, the platelets did not change shape or produce microvesicles. **Conclusion:** This new storage technology for platelets makes it possible to supply platelets for transfusion from blood banks to hospitals with the same logistics and shelf life as the distribution of packed red blood cells for transfusion.

Disclosure of Commercial Conflict of Interest

S. Stienstra: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Stienstra: Nothing to disclose

S10-010B

MiR-146a Mediates Intravenous immunoglobulin (IVIg) Anti-Inflammatory Effect in Lipopolysaccharides (LPS)-Treated Human Monocytes

L Loubaki[1] (lionel.loubaki@hema-quebec.qc.ca), D Chabot[1], I Paré[1], M Drouin[1], R Bazin[1]. *[1]R&D, Héma-Québec, Québec, QC, Canada*

Background/Case Studies: Intravenous immunoglobulin (IVIg) is a highly purified polyspecific immunoglobulin G preparation obtained from pooled plasma of several thousand healthy donors and is used to treat an increasing number of inflammatory and autoimmune diseases. The exact mechanisms involved in IVIg anti-inflammatory effects remain unclear; however, IVIg has been shown to abrogate pro-inflammatory cytokine secretion while increasing the expression of anti-inflammatory cytokines in monocytic cells. Monocytes are key players of the innate immune response and recent studies have shown that their differentiation and functions involve microRNAs which act through the inhibition of protein translation. Among them, miR-146a has been shown to supress NF-κB and IRF3 activity, two key transcription factors involved in TLR4 signaling and, consequently, to inhibit their target genes including INF-β, IL-6 and TNF-α. Therefore, miR-146a could lead to anti-inflammatory effects similar to those reported for IVIg. In this work, miR-146a was evaluated to see whether it is involved in the anti-inflammatory effects of IVIg following LPS stimulation of human monocytes. **Study Design/Methods:** Monocytes were obtained from the blood of healthy volunteers after informed consent and treated with LPS (1 μg/mL) or IVIg (15 mg/mL) alone or alternatively, pretreated with LPS followed by washing and addition of IVIg. The cells were incubated for 3, 6, and 12 hours prior to recovery and analysis of miR-146a, IL-6, IRF3, IFN-β, TGF-β1, and IL-1Ra expression by qPCR while IκBα, IRAK1, and TRAF6 expression at 24 hours was measured by Western blotting. **Results/Findings:** Addition of IVIg to resting and LPS-pretreated monocytes induced a significant up regulation of miR-146a expression (respectively 4.77 ± 1.18 and 6.34 ± 1.73-fold increase when compared to resting monocytes). This effect was associated with a significant reduction in the expression of miR-146a targets IRAK1 and TRAF6 and consequently of NF-κB activation. Furthermore, the mRNA expression of IRF3, another miR-146a target, and IFN-β was reduced by at least 4-fold in LPS-pretreated monocytes following treatment with IVIg, whereas TGF-β1 and IL-1Ra expression was increased by three-fold, thus skewing the balance between pro- and anti-inflammatory cytokine towards an anti-inflammatory condition. **Conclusion:** Our data suggest that miR-146a mediates some of the anti-inflammatory effects of IVIg in human monocytes. Our future work will address the possibility of using microRNA mimetics to replicate these effects, which could possibly lead to the development of synthetic IVIg substitutes.

Disclosure of Commercial Conflict of Interest

R. Bazin: Nothing to disclose; D. Chabot: No Answer; M. Drouin: Nothing to disclose; L. Loubaki: Nothing to disclose; I. Paré: No Answer

Disclosure of Grants Conflict of Interest

R. Bazin: Nothing to disclose; D. Chabot: No Answer; M. Drouin: Nothing to disclose; L. Loubaki: Nothing to disclose; I. Paré: No Answer

S11-010B

Mouse Erythrocytes Suppressed Inducible Isoform of NOS (iNOS) Expression in Activated RAW264.7 Macrophages

Q Li[1] (rhnullandrhmod@hotmail.com), G Lu[2], H Xiong[2], Z Zhu[1]. *[1]Shanghai Blood Center, Shanghai, China; [2]Immunology Institute, Mount Sinai School of Medicine, New York, NY, United States*

Background/Case Studies: Macrophages were important effector immune cells, which would be polarized into M1, M2, or M2-like subclasses in response to different stimuli. Lipopolysaccharides (LPS) and interferon-γ(IFN-γ) were the common stimulators of M1 macrophage differentiation and activated macrophages by either LPS or IFN-γ expressed high level of inducible isoform of NOS (iNOS), resulting in the production of nitric oxide (NO). Alblas et al found that soluble CD47 could activate rat macrophages to produce NO. CD47 was an important factor in the non-opsonic erythrophagocytosis system and expressed on erythrocyte membrane. The phenomenon suggested that erythrocytes might have the ability to regulate the macrophage polarization. The purpose was to analyze whether and how erythrocytes affect macrophage differentiation using mouse model. **Study**

Design/Methods: 4 × 10∧6 RAW 264.7 macrophages (from ATCC) were cultured in 6 cm dishes containing Dulbecco's modified Eagle's medium (DMEM) (Gibco) supplemented with 10 IU/ml penicillin, 10 μg/ml streptomycin, and 10% fetal bovine serum, and incubated in 5% CO2 at 37C. The mouse erythrocytes were obtained from 6-week-old C57BL/6 mice (Jackson Laboratories). Sodium citrate was used as anticoagulant. LPS (100 ng/ml) (090M4030, Sigma) or IFN-γ (10 ng/ml) (485-MI, R&D Systems) were added to RAW 264.7 co-cultured with mouse erythrocytes (1.2 × 10∧7). 24 hours later, NO production was detected by using Griess reagent (G2930, Promega), and iNOS expression was analyzed by Western blot with rabbit anti-mouse iNOS antibody (sc-650, SANTA CRUZ). **Results/Findings:** Erythrocytes themselves had no effect to trigger NO production; however, they significantly suppressed nitrite accumulation from RAW 264.7 activated with either LPS or IFN-γ (Table 1). In addition, Western blot showed iNOS expression was reduced in activated RA264.7 co-cultured with erythrocytes. **Conclusion:** iNOS and NO production were symbols for proinflammatory M1 macrophages. Mouse erythrocytes could inhibit LPS or IFN-γ-induced M1 macrophage polarization, which might have important biological significance since over activation of M1 macrophages could damage normal tissues by overproduction of proinflammatory cytokines. However, it was still not clear how erythrocytes contribute to this regulation. Future study would focus on the molecular mechanism for the regulation of macrophage activation by erythrocytes, and elucidated whether human erythrocytes had similar functions.

Disclosure of Commercial Conflict of Interest

Q. Li: Nothing to disclose; G. Lu: Nothing to disclose; H. Xiong: Nothing to disclose; Z. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

Q. Li: Nothing to disclose; G. Lu: Nothing to disclose; H. Xiong: Nothing to disclose; Z. Zhu: Nothing to disclose

TABLE 1. The production of NO induced by LPS, IFN-γ, or mouse erythrocytes

Group	NO production (μ m)	p value
RAW264. 7 control	3.8 ± 0.9	0.73
RAW264. 7 + mouse RBC	4.1 ± 1.2	
RAW264. 7 + iFN γ	28.4 ± 9.6	0.02
RAW 264. 7 + mouse RBC+IFN γ	9.4 ± 1.4	
RAW 264.7 + LPS	42.5 ± 11.9	0.01
RAW 264.7 + mouse RBC + LPS	15.2 ± 4.3	

S12-010B

Micro Ribonucleic Acid (miRNAs) as Potential Therapeutic Targets in Graft-versus-Host Disease (GVHD)

L Loubaki[1] (lionel.loubaki@hema-quebec.qc.ca), L Padet[1], R Bazin[1].
[1]R&D, Héma-Québec, Québec, QC, Canada

Background/Case Studies: Graft-versus-host disease (GVHD) is a common complication following an allogeneic transplantation in which recognition of histocompatibility determinants by allogeneic T cells together with the engagement of co-stimulatory molecules with their ligands is essential for the induction of T cell-mediated allograft rejection. Intravenous immunoglobulin (IVIg) is used as a replacement therapy in immunocompromised patients to prevent infections, but also improves the rate of graft survival in patients with high risk for rejection. Usually, immunosuppressive drugs are administered, typically corticosteroids along with cyclosporine or tacrolimus. However, responses to this steroid treatment are observed in only 50% of patients. Then, new strategies that can supplement standard treatment are required. MicroRNAs (miRNAs) that repress gene function through interaction with target miRNAs, have emerged to play critical roles in regulation of GVHD. Thus, the use of miRNA as therapeutic targets in GVHD could be a promising approach in the treatment GVHD. **Study Design/Methods:** The allogeneic mixed lymphocyte reaction (MLR) was used as an in vitro model of allograft rejection. Briefly, human peripheral blood mononuclear cells (PBMC) from two unrelated individuals were mixed together and incubated with or without 2 mg/ml of IVIg. Six hours after MLR initiation, T cells and monocytes were recovered and purified to allow the evaluation of miRNA expression by qPCR and the analysis of PD1 expression on T cells and PDL1 on monocytes by flow cytometry. A second MLR with cells from the same donors was performed in parallel to evaluate the extent of the reaction by measuring the secretion of IL-2, 24 hours following MLR initiation. **Results/Findings:** Preliminary data show that addition of IVIg to the MLR significantly reduces the strength of the reaction, resulting in a strong inhibition of IL-2 secretion that is associated with an 8-fold increased expression of miR-26 in T cells. A down-regulation by at least 4-fold of miR-513 accompanied by a 65% increase in the expression of PDL1 on monocytes is also observed in IVIg-supplemented MLR. **Conclusion:** Preliminary data identify miRNA-26 and miR-513 as potential therapeutic targets for GVHD treatment. In addition, this study opens the possibility of using miRNA-26 mimetic and miR-513 inhibitor to replicate these effects, which could possibly lead to the development of synthetic IVIg substitutes. Finally, this study contributes to a better understanding of the mechanisms by which IVIg may induce peripheral tolerance and improve graft survival in transplanted patients.

Disclosure of Commercial Conflict of Interest

R. Bazin: Nothing to disclose; L. Loubaki: Nothing to disclose; L. Padet: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Bazin: Nothing to disclose; L. Loubaki: Nothing to disclose; L. Padet: Nothing to disclose

Cellular Therapies

S13-010C

Identification of the Mechanisms Responsible for the Improved Platelet Recovery Achieved with Cord Blood(CB) Progenitors Expanded in Osteoblast (OST) Conditioned Medium

N Dumont[1] (nellie.dumont@hema-quebec.qc.ca), H Emond[1], B Celebi Saltik[1,2], M Drouin[1], R Bazin[1], D Mantovani[2], D Roy[3], N Pineault[1]. *[1]Research and Development, Hema-Québec, Québec, QC, Canada; [2]Laboratory for Biomaterials & Bioengineering, Département Min-Met-Materials Eng, Laval University, Québec, QC, Canada; [3]Centre de Recherche Hôpital Maisonneuve-Rosemont, Département Medicine, University of Montréal, Montreal, QC, Canada*

Background/Case Studies: Platelet recovery is significantly delayed in patients undergoing cord blood (CB) stem cell transplantation. To overcome this limitation, ex vivo expansion of hematopoietic stem and progenitor cell (HSPC) is investigated as a means to accelerate hematopoietic recovery and engraftment. We recently showed that osteoblasts (OST) derived from mesenchymal stromal cells (MSCs) represent a superior source of feeder cells for the expansion of HSPCs. This conclusion was supported by the increased expansion of myeloid progenitors in OST-based cultures, and by the robust engraftment of progenitors and platelets in mice transplanted with CB CD34+ cells expanded with OST-conditioned medium (CM). Herein, the mechanism by which CM improves the expansion of progenitors and subsequent recovery of platelets was investigated. **Study Design/Methods:** CB CD34+ cells were expanded in culture with MSC CM, OST CM or control medium. All cultures were supplemented with the cytokines SCF, FL and TPO. Human engraftment in NOD/Scid-gamma null (NSG) transplanted mice was analyzed by cytometry and progenitor assays. **Results/Findings:** First, the impact of CMs was measured on the expression of receptors known to be important for the homing of HSPCs to the marrow. Percentages of CD34 + LFA-1+ (48% vs 38%, $p > 0.05$), CD34 + CXCR4+ (33% vs 25%, $p < 0.05$) and LFA-1 + CXCR4+ (68.3% vs 49.8%, $p < 0.01$) CB cells were consistently greater in OST CM cultures vs control cultures. Next, the homing efficiency was measured of CB cells expanded in OST CM vs control medium in NSG mice. The average percentage (4.6% vs 5.3%, $p = 0.63$) and total number (3.1 vs 3.7 × 105, $p = 0.46$) of human CD45+ marrow cells in mice were similar 40 hours posttransplantation. In line with this result, human engraftment at 7 and 13 days post-transplant were also similar in mice transplanted with OST CM and control cultures, though OST's culture recipients consistently showed greater numbers of human myeloid (+28%) and megakaryocyte (+52%) progenitors ($p > 0.05$). Finally, the pattern of soluble factors released by osteoblasts to that of their parental MSCs by cytokine array was compared. Gro-alpha ($p < 0.01$), Rantes and IL-8 ($p = 0.08$) were upregulated in OST CM. However, addition of either factor at low (1-10 ng/mL) or high (50-100 ng/mL) concentrations failed to replicate the impact of OST CM on the expansion of CB CD34+ cells ($n = 3$). **Conclusion:** In summary, the data suggest that despite improved expression of CXCR4 and LFA-1, improved homing may not be the principal mechanism responsible for the robust platelet engraftment achieved with CB cells expanded with OST CM. The increased human progenitor content in the marrow of OST recipients suggest that OST CM might rather promote the maintenance of more primitive HSPCs in culture.

Disclosure of Commercial Conflict of Interest

R. Bazin: Nothing to disclose; B. Celebi Saltik: Nothing to disclose; M. Drouin: Nothing to disclose; N. Dumont: Nothing to disclose; H. Emond: Nothing to disclose; D. Mantovani: No Answer; N. Pineault: Nothing to disclose; D. Roy: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Bazin: Nothing to disclose; B. Celebi Saltik: Nothing to disclose; M. Drouin: Nothing to disclose; N. Dumont: Nothing to disclose; H. Emond: Nothing to disclose; D. Mantovani: No Answer; N. Pineault: Nothing to disclose; D. Roy: Kiadis Pharma, Grants or Research Support

S14-010C

Prediction of Poor Mobilization of Autologous CD34+ Cells in Multiple Myeloma (MM): Implications for Risk-Stratification

L J Costa[1] (costalj@musc.edu), E J Nista[1], F K Buadi[2], M Q Lacy[2], A Dispenzieri[2], C P Kramer[1], Y Kang[1], M A Gertz[2], R K Stuart[1], S K Kumar[2]. *[1]Medicine, Medical University of South Carolina, Charleston, SC, United States; [2]Hematology, Mayo Clinic College of Medicine, Rochester, MN, United States*

Background/Case Studies: Administration of hematopoietic growth-factors (GFs) is the simplest method to mobilize CD34+ cells for autologous transplantation in multiple myeloma (MM), but is limited by high risk of inadequate collection. Other strategies, chemotherapy + growth factor (GF) and GF + plerixafor, increase CD34+ yield but with more toxicity and/or cost. It is unknown if clinical characteristics can successfully predict poor mobilizers, so they can be assigned to mobilization with alternative strategies. **Study Design/Methods:** MM patients who underwent autologous CD34+ mobilization in one of the two participating institutions between 2000 and 2013 were retrospectively assigned into training and a validation cohort. Multiple regression analysis was applied to develop, from the training cohort, a predicting score (PS) based on clinical characteristics that would identify patients destined to be poor mobilizers (PM), defined as peripheral blood (PB)-CD34+ < 20/mm3, 4 days after initiation of GF. The performance of the PS was then tested in the validation cohort. **Results/Findings:** A complete dataset was available for 477 patients who were assigned to the training cohort (N = 239) or validation cohort (N = 238). Median age of patients was 59.4 years (IQR 52.7-65.4), median percentage of plasma cells in the bone marrow at mobilization was 5% (IQR 3%-15%), median time from diagnosis to mobilization was 6 months (IQR 4.8-8.5) and median platelet count was 223 × 103/mm3 (IQR 182-281). One hundred and thirty-four patients (28.1%) had prior exposure to lenalidomide, 393 (82.4%) had mobilization with filgrastim while 84 (17.6%) received mobilization with pegfilgrastim. Median PB-CD34+ count 4 days after initiation of GF was 16/mm3 (IQR 8-28). It was observed that only age, platelet count at time of mobilization, type of GF utilized, and duration of prior exposure to lenalidomide independently correlated with PB-CD34+ count and were therefore integrated in the PS. Overall 146 patients (61.6%) in the validation cohort were actual PM. The PS cut-off found in the training cohort to have 90% sensitivity for prediction of PM performed with 89.7% sensitivity but only 34.8% specificity in the validation cohort, and would have led to 10.3% of the actual PM not been assigned to alternative mobilization and mistakenly assigned 55.4% of the good mobilizers (GM) to alternative mobilization. Conversely, the PS cut-off developed to have 90% specificity for prediction of PM performed with 86.9% specificity but only 37% sensitivity in the validation cohort. **Conclusion:** Clinical characteristics identifiable prior to initiation of mobilization are inadequate predictors of poor mobilization and should not be used to stratify MM patients for different mobilization strategies. "Just-in-time" plerixafor is an attractive alternative to risk stratification.

Disclosure of Commercial Conflict of Interest

F. K. Buadi: Nothing to disclose; L. J. Costa: Sanofi, Travel Support or Honorarium; A. Dispenzieri: No Answer; M. A. Gertz: Nothing to disclose; Y. Kang: Nothing to disclose; C. P. Kramer: Nothing to disclose; S. K. Kumar: Nothing to disclose; M. Q. Lacy: Nothing to disclose; E. J. Nista: Nothing to disclose; R. K. Stuart: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. K. Buadi: Nothing to disclose; L. J. Costa: Nothing to disclose; A. Dispenzieri: No Answer; M. A. Gertz: Nothing to disclose; Y. Kang: Nothing to disclose; C. P. Kramer: Nothing to disclose; S. K. Kumar: Celgene, Grants or Research Support, Millennium, Grants or Research Support, Novartis, Grants or Research Support, Onyx, Grants or Research Support, Sanofi, Grants or Research Support; M. Q. Lacy: Nothing to disclose; E. J. Nista: Nothing to disclose; R. K. Stuart: Sunesis, Grants or Research Support, Millenium, Grants or Research Support, Sanofi-Aventis, Grants or Research Support, Novartis, Grants or Research Support, Celator, Grants or Research Support, Celtic Therapeutics, Grants or Research Support, Onconova, Grants or Research Support

S15-010C

Tolerogenic Dendritic Cells (tDCs) Modified By Tacrolimus Can Induce Immunological Tolerance and Inhibit Collagen-Induced Arthritis in Mice

Y Ren[1] (yana.ren021@hotmail.com). [1]Shanghai Blood Center, Shanghai, China

Background/Case Studies: Dendritic cells (DCs) exposed to various suppressive agents, as well as to immunosuppressive drugs, are endowed with tolerogenic functions termed "tolerogenic" DCs (tDCs). Due to their ability to dampen immune responses, tDCs have gained interest as a therapeutic strategy in autoimmune diseases such as rheumatoid arthritis (RA). Tacrolimus is a widely used immunosuppressive drug. There are a few reports showing that tacrolimus can inhibit the maturation of DCs and the ability to stimulate T cells. But the features of tDCs modified by tacrolimus and the therapeutic effects in arthritis are not clear. **Study Design/Methods:** tDCs derived from human monocytes were induced in vitro by GM-CSF, IL-4 with tacrolimus, and collected after cultivation of 7 days. The production of cytokine in tDCs modified by tacrolimus was analyzed. Then the functionality of these tDCs was determined via CFSE proliferation assay. To determine the therapeutic action of tDCs in vivo, the animal models of RA were established. In the model, arthritis was induced in DBA/1 mice after immunized with bovine type collagen on day 0 and day 21, termed collagen-induced arthritis (CIA). CIA mice received 3 intravenous injections of tDCs in a dose ranging from 10∧4-10∧6 cells. Arthritis severity was monitored for up to 4 weeks after treatment. Moreover, the effect of tDCs on T-cell populations was determined by intracellular cytokine staining (CS) and flow cytometry (FC). **Results/Findings:** These tDCs exhibit a typical tolerogenic phenotype of reduced costimulatory molecules HLA-DR, CD80, and CD86. The expression of mRNA encoding IL-10 and TGF-β in tDCs increased substantially after stimulating 18 hours by tacrolimus with the strongest responses for 24 hours. And the expression of them was further upregulated after tDCs treated with LPS and IFN-γ. In addition, tDCs secreted less TNF-α and more IL-10 comparing with immature DCs. tDCs had a reduced ability to activate allo-CD4 + T cells. Functionally these cells were suppressive towards allo-CD4 + T cells proliferation in suppression assay in a dose-dependent manner, producing less amounts of IL-12p70, TNF-a, and IFN-γ but increased levels of IL-10. tDCs generated with IL-10, TGF-β, dexamethasone, or cyclosporin A were also compared with tDCs modified by tacrolimus. tDCs induced by tacrolimus showed most powerful tolerogenic characteristics and exhibited the least T-cell stimulatory capacity. tDCs induced by tacrolimus show functional characteristics that make them best suited for tolerance inducing therapies. Treatment of arthritic mice with tDCs significantly inhibited disease severity and progression. The tDCs treatment could alter the proportion of Th1/Th17/Treg cell population. **Conclusion:** These results suggest that human tDCs induced by tacrolimus could promote T-cell tolerance and prevent autoimmune arthritis in animal models of CIA. The findings indicated the potential of therapeutic use of tDCs for the prevention or treatment of RA.

Disclosure of Commercial Conflict of Interest

Y. Ren: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Ren: Nothing to disclose

S16-010C

Separation of in Vitro-Derived Megakaryocytes and Platelets Using Spinning Membrane Filtration

A C Schlinker[1] (AlainaSchlinker2015@u.northwestern.edu), K Radwanski[2], C Wegener[2], W M Miller[1], K Min[2]. *[1]Department of Chemical and Biological Engineering, Northwestern University, Evanston, IL, United States; [2]Fresenius Kabi USA, Lake Zurich, IL, United States*

Background/Case Studies: In vitro platelet production would provide an alternative source of platelets while avoiding some of the current problems with donor platelets. Functional platelets have been generated in vitro using 2D culture systems. Subsequent production of transfusable platelet units would require separation of the platelets from other cells, including megakaryocytes. Although this separation can be performed via centrifugation, it

requires multiple steps and error-prone manual manipulation. The objective of this study was to evaluate a cell-washing device with spinning membrane filtration for rapid, one-step, automated separation of platelets from megakaryocytes. **Study Design/Methods:** Basal (2N and 4N) or phorbol 12-myristate 13-acetate (PMA)-stimulated (up to 32N) CHRF megakaryoblastic cells were suspended in PAS-V platelet additive solution. The CHRF cells were then mixed with apheresis platelets collected using the Amicus Separator. The mixture of CHRF cells and platelets was processed on a cell-washing device consisting of a polycarbonate spinning membrane with 4-μm cylindrical pores. Cells that passed through the membrane or were retained by the membrane were collected in separate containers. Samples from the appropriate fractions were analyzed to determine the concentrations of CHRF cells and platelets, CHRF cell mean ploidy, CHRF cell viability, and platelet CD62p expression. Preservation of CHRF cell and platelet function was assessed by proplatelet formation and CD62p level after adenosine diphosphate (ADP) stimulation, respectively. **Results/Findings:** >90% of platelets were recovered in the platelet fraction, which was virtually pure (~1 CHRF cell per 90,000 platelets), suggesting that CHRF cells were unable to pass through the membrane. The % of CD62p+ platelets was increased in the recovered vs the prewash fraction, but ADP stimulation resulted in nearly 100% CD62p+ platelets in both populations. Approximately 80% of CHRF cells were recovered in the CHRF cell fraction. Viability in the recovered CHRF cell fraction was unchanged when processing basal CHRF cells and decreased for processing PMA-stimulated CHRF cells, perhaps due to their fragility. The ploidy distribution of recovered CHRF cells was similar to that in the prewash fraction, and recovered CHRF cells were able to form proplatelets when transferred to a tissue-culture-treated surface. **Conclusion:** Membrane filtration is able to efficiently separate a mixture of platelets and megakaryocytes into individual components while maintaining cell viability and function.

Disclosure of Commercial Conflict of Interest

W. M. Miller: Nothing to disclose; K. Min: Nothing to disclose; K. Radwanski: Nothing to disclose; A. C. Schlinker: Nothing to disclose; C. Wegener: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. M. Miller: Fresenius Kabi, Grants or Research Support; K. Min: Fresenius, Grants or Research Support; K. Radwanski: Fresenius, Grants or Research Support; A. C. Schlinker: Fresenius Kabi, Grants or Research Support; C. Wegener: Fresenius, Grants or Research Support

		Platelets + Basal CHRF cells		Platelets + PMA-treated CHRF cells	
		Prewash	Recovered	Prewash	Recovered
Platelets	**% Recovery**	—	109.2	—	92.3
	% CD62P +, resting	43.6	45.4	42.9	73.7
	% CD62P +, ADP	N/A	M/A	98.3	96.6
CHRF cells	**% Recovery**	—	84.4	—	79.7
	% Viable	89.3	90.0	69.5	59.7
	% Apoptotic	2.1	1.6	11.2	17.5
	Mean ploidy	3.0	2.9	5.8	6.2

S17-010C

An Optimized Peripheral Blood Stem Cells (PBSC) Collection Protocol Based on Pre-Harvest CD34 Enumeration and Plerixafor Administration

S T Avecilla[1,2] (sca2002@med.cornell.edu), C A Pagan[1,2], C Goss[1,2], D Fridel[2], T M Mark[3], T Shore[3], K van Besien[3], M Cushing[1,2]. *[1]Pathology and Laboratory Medicine, Weill Cornell Medical College, New York, NY, United States; [2]Transfusion Medicine and Cellular Therapy, New York Presbyterian Hospital, New York, NY, United States; [3]Medicine, Div. Hematology and Oncology, Weill Cornell Medical College, New York, NY, United States*

Background/Case Studies: The collection of autologous peripheral blood stem cells (PBSC) can be challenging in patients with prior chemo and heme malignancies who do not mobilize well with standard granulocyte-colony stimulating factor (G-CSF) mobilization. Patients with poor mobilization often undergo large volume leukapheresis over multiple days to reach sufficient CD34 harvests which is not only unsafe, but also inefficient due to increases in procedure time, product processing time, and numerous low-yield products. The use of plerixafor, an alternative mobilizing agent, is an option to enhance harvest yield. However, there is no consensus as to when and under which conditions in a collection cycle it should be used. **Study Design/Methods:** A protocol was established for autologous PBSC donors which includes the measurement of CD34 counts on day 4 of mobilization (Table). Depending upon the day-4, CD34 count, the patient is administered plerixafor with GCSF. All autologous products collected from 12/2010-4/2011 (before protocol) and 12/2012-4/2013 (under protocol) were analyzed for patient and product specific parameters which included: patient weight; gender; age; diagnosis; nadir platelet count post collection; total CD34 harvest; TNC harvest; total blood volume processed; number of collection days; total product volume collected; and individual product volume and red blood cell (RBC) content. **Results/Findings:** 26 patients before and 24 patients after the initiation of the protocol were examined. Significant differences ($P < 0.05$ or lower) were found such that on protocol patients underwent less days of collection (1.25 vs 2.39days), lower total blood volume processed (25.9 vs 57.2 L), lower total product volume (324 vs 691 mL), lower RBC content per collection (9 vs 18 mL), and improved nadir platelet count postpheresis (86 vs 44 × 10^3/uL). There were no significant differences between the two groups with regard to patient weight, gender, age, diagnosis, baseline platelet count, total CD34, or CD34/kg harvested. **Conclusion:** The application of a consistent protocol that assesses a patient's response to GCSF PBSC mobilization a day prior to collection allows for preemptive administration of plerixafor to augment the mobilization. This strategy was found to decrease patient inconvenience by reducing days of collection and volume of blood processed in order to achieve the CD34 collection goals which remained unchanged before and after the protocol. It also resulted in safer collections with lower RBC contamination and avoidance of severe thrombocytopenia post collection. Overall this new protocol has enhanced patient care and simultaneously increased the efficiency of the apheresis and cellular therapy laboratory services.

Disclosure of Commercial Conflict of Interest

S. T. Avecilla: Nothing to disclose; M. Cushing: Nothing to disclose; D. Fridel: Nothing to disclose; C. Goss: Nothing to disclose; T. M. Mark: Nothing to disclose; C. A. Pagan: Nothing to disclose; T. Shore: Nothing to disclose; K. van Besien: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. T. Avecilla: Nothing to disclose; M. Cushing: Nothing to disclose; D. Fridel: Nothing to disclose; C. Goss: Nothing to disclose; T. M. Mark: Celgene, Travel Support or Honorarium, Celgene, Grants or Research Support, Celgene, Consulting or Board of Director Fees, Millennium, Travel Support or Honorarium; Onyx, Grants or Research SupportOnyx, Travel Support or Honorarium; C. A. Pagan: Nothing to disclose; T. Shore: Nothing to disclose; K. van Besien: Miltenyi, Grants or Research Support, Therakos, Grants or Research Support

Mobilization Protocol

Day 4 AM CD34	<10			10-40	>40
Day 4 Late PM Plerixafor	Yes			Yes	No
D5 AM CD34	Yes			No	No
D5 AM CD34	<8	8-20	>20	N/A	N/A
Volume of collection	No collection, GCSF + plerixafor and recheck CD 34 on D6	24 L	20 L	20 L	20 L

S18-010C

Hematopoietic Progenitor Cell (HPC) Mobilization in Patients with Chronic Granulomatous Disease (CGD) and Severe Combined Immunodeficiency

S R Panch[2] (sandhya.panch@nih.gov), Y C Yau[1], E Kang[3], H Malech[3], S F Leitman[1]. *[1]Transfusion Medicine, NIH, CC, Bethesda, MD, United States; [2]Hematology/Transfusion Medicine, NHLBI, NIH, Bethesda, MD, United States; [3]Laboratory of Host Defenses, NIAID, NIH, Bethesda, MD, United States*

Background/Case Studies: Granulocyte-colony stimulating factor (G-CSF) mobilized autologous hematopoietic progenitor cells (HPC) may be collected by apheresis of patients with chronic granulomatous disease (CGD) or severe combined immunodeficiency (SCID) for use in gene therapy trials. CD34+ mobilization responses have not been well characterized in this population. **Study Design/Methods:** CD34+ cell mobilization and collection in 73 consecutive CGD (n = 63) and SCID (n = 10) patients given G-CSF

10-16 ug/kg/d × 5 d were retrospectively evaluated, and in healthy allogeneic controls (n = 99) matched for age, weight, and G-CSF dose. **Results/Findings:** In subjects aged <20 years, day 5 peak pre-apheresis circulating CD34+ counts were significantly lower in CGD (n = 38) and SCID (n = 9) than in control subjects (n = 47); mean peak CD34 = 59, 64, and 81 cells/uL, respectively, p = 0.05. SCID patients had lower CD34 collection efficiency during apheresis than CGD and control subjects; mean efficiency 40%, 63% and 58%, respectively, p = 0.001. In subjects aged >20 years, CGD patients (n = 25) had significantly less robust CD34+ cell mobilization than controls (n = 52); mean peak CD34 41 and 113 cells/uL, respectively, p < 0.0001. In multivariate regression analysis of the CGD group, higher baseline platelet count and lower erythrocyte sedimentation rate (ESR) at time of mobilization were significantly correlated with CD34+ cell mobilization, p = 0.01 and 0.007, respectively. Mean ESR, in turn, was significantly higher among subjects with active infection and/or CGD-associated colitis at collection compared to those without these events (40 vs 16 mm/hr; p = 0.0001). In SCID but not CGD patients, CD34 collection efficiency was strongly correlated with higher red cell indices (MCV: R2 = 0.81, p = 0.002; MCH: R2 = 0.95, p < 0.0001; MCHC: R2 = 0.81, p < 0.002) but not hemoglobin. Mean ferritin was 20 vs 134 ug/L in CGD vs SCID patients. In 11 patients with poor CD34+ mobilization (<20 cells/uL) on day 4, a single dose of plerixafor 240 ug/kg 10 hrs prior to apheresis increased mean CD34 counts 5-fold, from 15 to 76 cells/uL (d4 to d5). **Conclusion:** CGD and SCID populations are characterized by significantly less robust CD34+ mobilization than healthy controls. This can be offset with use of plerixafor the day prior to apheresis. In CGD, the association of higher baseline platelet counts with better CD34+ mobilization may be related to common pathways of thrombopoiesis and HPC production; the presence of active infection or colitis, reflected by increased ESR, may negatively impact mobilization due to marrow suppressive effects. Among SCIDs, markedly reduced CD34 collection efficiencies were related to iron deficiency; variability in red cell size and density due to iron deficiency appeared to impair optimal optical control and separation of the MNC layer during apheresis. Prospective correction of iron deficiency may improve collection yields in these patients.

Disclosure of Commercial Conflict of Interest

E. Kang: Nothing to disclose; S. F. Leitman: Nothing to disclose; H. Malech: Nothing to disclose; S. R. Panch: Nothing to disclose; Y. C. Yau: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Kang: Nothing to disclose; S. F. Leitman: Nothing to disclose; H. Malech: Nothing to disclose; S. R. Panch: Nothing to disclose; Y. C. Yau: Nothing to disclose

Donor and Therapeutic Apheresis

S19-020A

Plasmapheresis Donation Using Saline Compensation: The Australian Experience

H Yang[1] (hyang@abmdr.org.au), J F Lee[2], B Bell[1], J M Pink[3], A J Keller[2]. *[1]Medical, Quality & Transplantation Services, Australian Red Cross Blood Service, Sydney, NSW, Australia; [2]Medical, Quality & Transplantation Services, Australian Red Cross Blood Service, Perth, WA, Australia; [3]Medical, Quality & Transplantation Services, Australian Red Cross Blood Service, Brisbane, QLD, Australia*

Background/Case Studies: Australian usage of intravenous immunoglobin (IVIg) has increased significantly over the past decade, and the Australian Red Cross Blood Service (ARCBS) has explored a number of strategies to increase local supply of source plasma while balancing donor safety. The infusion of saline solution as fluid replacement ("saline compensation") reduces the risk of vasovagal reaction (VVR) and also facilitates collection of higher plasma volumes. "End-donation" saline compensation, the infusion of saline after the final draw phase of an intermittent flow procedure, is already in widespread use around the world, but many VVRs occur during the procedure. Infusing saline earlier in the procedure should therefore prevent more VVRs, but with the risk of diluting the collected plasma. In 2009, ARCBS studied a "mid-donation" saline compensation protocol that delivered 250 mL after the first draw and 250 mL after the final draw, and found that while there was a dilutional effect, the protein content of collected plasma remained acceptable. Donors participating in the study tolerated large collection volumes—up to 840 mL. In 2010, ARCBS implemented an end-donation saline compensation protocol, and found that VVRs decreased significantly compared to plasmapheresis without saline compensation. This experience, coupled with the results of the earlier study, prompted ARCBS to implement a mid-donation protocol in 2011. **Study Design/Methods:** The Mid-Donation Saline Surveillance Study incorporated all ARCBS plasmapheresis collections in order to meaningfully compare VVR risk. The previously studied infusion protocol was used, with collection volume limited in Stage 1 to 16% of TBV, capped at 800 mL; and in Stage 2 to 18% of TBV, still capped at 800 mL. **Results/Findings:** (Table) **Conclusion:** Total protein levels in collected plasma remained above the European Pharmacopoeia standard of 50 g/L in both stages, while IgG showed comparatively less dilutional effect than total protein. The reported rate of VVRs was significantly lower in both Stage 1 and Stage 2 than under the preceding end-donation protocol (p < 0.001). It is concluded that a mid-donation saline compensation protocol reduces VVR risk compared to end-donation saline, and can also deliver acceptable protein content.

Disclosure of Commercial Conflict of Interest

B. Bell: Nothing to disclose; A. J. Keller: Nothing to disclose; J. F. Lee: Nothing to disclose; J. M. Pink: Nothing to disclose; H. Yang: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Bell: Nothing to disclose; A. J. Keller: Nothing to disclose; J. F. Lee: Nothing to disclose; J. M. Pink: Nothing to disclose; H. Yang: Nothing to disclose

Plasmapheresis collection protocol	No saline compensation: 13% of TBV, capped at 750 mL (1/19/10-7/4/10)	End-donation protocol: 16% of TBV, capped at 750 mL (8/2/10-1/30/11)	Mid-donation protocol Stage 1: 16% of TBV, capped at 800 mL (1/31/11-5/22/11)	Mid-donation protocol Stage 2: 18% of TBV, capped at 800 mL (5/23/11-10/31/11)
Total protein (g/L)	57.99	54.23	51.50	51.66
IgG (g/L)	7.47	7.26	7.04	7.14
Number of donations	90,494	150,930	101,351	145,035
Vasovagal reactions	1,032 1.14%	1,150 0.76%	568 0.56%	956 0.66%

S20-020A

Characterization of Respiratory Function in Myasthenia Gravis (MG) Patients after Plasma Exchange

J J O'Brien[1] (dr.jjobrien@gmail.com), G J Despotis[1]. *[1]Pathology & Immunology, Washington University School of Medicine, St. Louis, MO, United States*

Background/Case Studies: Myasthenia gravis (MG) is an autoimmune disease that leads to respiratory muscle weakness. MG can be treated by either immunosuppression and/or plasma exchange procedures (PEX). A Cochrane review found little benefit of acute management of MG with PEX and a 10-fold mortality increase when used alone that may be in part related to procedures during off-hours. The change in respiratory function (as reflected by reductions in vital capacity [VC] and negative inspiratory force [NIF]) in response to sequential PEX procedures in MG patients was evaluated. **Study Design/Methods:** Data were obtained from 33 patients who received multiple PEX from August 2011 to April 2013. Baseline VC and NIF were compared to post-PEX (PostP) values. Oneway, anova, linear, and

logistic regression analyses were performed with a statistical program (StataTM). We defined a responder (R) cohort as patients whose NIF or VC increased by 30% from baseline preprocedure values vs non-responder (NR) cohort. **Results/Findings:** Of the 33 patients, only 29 (88%) had pre- and post-PEX NIF and VC. Modest increases in NIF and VC (10% and 20%, respectively) were observed after first PEX. Substantially greater changes in NIF and VC were observed after second PEX (50 and 58%, respectively), and after fifth PEX (91 and 82%, respectively). Clinical response between cohorts is compared in the table. Despite a nominal overall response after the first PEX in majority of patients, clinically significant increases in mean NIF (–16 mm) and in VC (800 mL) were observed in a small percentage (15% and 24%, respectively). Baseline values for VC <1500 mL and NIF <–30 mm were statistically associated with response after first PEX using logistic regression (NIF ROC Area = 0.69; VC ROC Area = 0.82) as well as with other statistical methods. **Conclusion:** Since only a small subset of patients respond to the first PEX procedure, it is critical that clinicians identify these patients. Despite our preliminary findings of potential utility of NIF (<–30 mm) and VC (<1500 mL) cutoffs, further investigations that include other clinical variables (e.g. other therapies, concurrent lung disease, and vital signs) and outcome assessments are indicated.

Disclosure of Commercial Conflict of Interest

G. J. Despotis: Nothing to disclose; J. J. O'Brien: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. J. Despotis: Nothing to disclose; J. J. O'Brien: Nothing to disclose

Clinical Response in Respiratory Function Between Cohorts

	All Periods	PostP#1	PostP#2	PostP#3	PostP#4	PostP#5
NIF (mm Hg)	(n = 152)	(n = 33)	(n = 28)	(n = 22)	(n = 22)	(n = 14)
Responder (%)	38	15	50	59	68	71
Baseline (NR)	–44 ± 18	–43 ± 19	–49 ± 17	–51 ± 17	–58 ± 12	–60 ± 16
Baseline (R)	–47 ± 15	–21 ± 10*	–23 ± 11*	–26 ± 13*	–25 ± 12*	–23 ± 12*
Diff from BL (NR)	–2 ± 7	1 ± 5	–5 ± 9	–5 ± 20	–2 ± 17	–3 ± 18
Diff from BL (R)	–23 ± 12*	–16 ± 12*	–22 ± 12*	–23 ± 10*	–25 ± 12*	–24 ± 12*
VC (L)						
Responders (%)	32	24	43	36	59	57
Baseline (NR)	1.9 ± 1.0	2.1 ± 1.0	2.1 ± 1.0	1.6 ± 0.8	2.1 ± 0.7	2.0 ± 0.7
Baseline (R)	1.2 ± 0.8*	1.3 ± 0.7	1.1 ± 0.7*	1.1 ± 0.7	1.1 ± 0.7*	0.9 ± 0.6*
Diff from BL (NR)	0.01 ± 0.4	0.2 ± 0.5	0.04 ± 0.5	–0.04 ± 0.4	0.2 ± 0.3	0.2 ± 0.3
Diff from BL (R)	0.9 ± 0.5*	0.8 ± 0.4*	0.8 ± 0.7*	1.0 ± 0.6*	1.0 ± 0.6*	0.8 ± 0.2*

* $p < 0.01$ between R and NR

S21-020A

Initial Serum Ferritin Predicts Number of Therapeutic Phlebotomies to Iron Depletion in Secondary Iron Overload (IO)

S R Panch[2] (sandhya.panch@nih.gov), Y C Yau[1], T S Sweigart[1], K Diggs[1], S F Leitman[1]. *[1]Transfusion Medicine, NIH, CC, Bethesda, MD, United States; [2]Hematology/Transfusion Medicine, NHLBI, Bethesda, MD, United States*

Background/Case Studies: Phlebotomy therapy is known to reduce hyperferritinemia and mitigate systemic iron toxicity in patients with iron overload (IO) not due to genetic hemochromatosis. Laboratory response parameters and duration of therapy are less well characterized in this population. **Study Design/Methods:** Ninety-nine consecutive patients undergoing therapeutic phlebotomy were retrospectively evaluated for transfusional IO (TIO, n = 89). (Diagnoses included severe aplastic anemia [47%], sickle cell disease [14%, and myelodysplastic syndrome [14%; and 75% of this group had undergone hematopoietic stem cell transplant) and non-transfusional IO (NIO, n = 10). Diagnoses included porphyria cutanea tarda and anemias of ineffective erythropoiesis. Serum ferritin (SF), transferrin saturation (TS), hemoglobin, MCV, and liver function tests were measured at each visit. Phlebotomy volume ranged from 200-500 mL and was determined by weight and fingerstick hemoglobin. Phlebotomy goal was SF < 200 mcg/L or, in the presence of GVHD, infection, or steatosis where SF endpoints were not well-defined, an MCV decrease of at least 5% and/or TS <30% were used. **Results/Findings:** Mean SF levels prior to first phlebotomy among TIO and NIO subjects were 3093 and 477 mcg/L, respectively. Mean transfusion burden among TIO was 98 ± 116 (SD) (range 9-800) red cell units. Mean phlebotomy volume was 445 ± 73 mL and phlebotomy interval was 6 ± 7 wks. Two-thirds of subjects completed therapy, requiring a mean of 19 phlebotomies (median 12, range 4-89) to reach targeted endpoints. In these subjects, the overall volume removed was about 20% of that transfused. One third of subjects were lost to follow-up, 3% of whom had procedure intolerance. Overall, 15% reported mild side effects including headaches, nausea, and dizziness with most occurring with the first phlebotomy. Prior transfusion burden was poorly correlated with initial ferritin and total number of phlebotomies to target (NPT) in the TIO group. However, NPT was strongly correlated with initial SF ($R2 = 0.8$; $p < 0.0001$). ALT decreased significantly with serial phlebotomy (mean pre and post values, 68 and 38 U/L; $p = 0.002$). No significant changes were noted in mean TSH, free T4, or cardiac ejection fraction before and after treatment. **Conclusion:** Initial SF level but not transfusion burden was predictive of number of phlebotomy sessions to reach target. This may reflect individual variation in iron loading/off-loading kinetics. It may also be due to the variable amount of phlebotomy performed for diagnostic lab testing. Despite reasonable treatment tolerance, significant losses to follow-up were noted. It may be that providing patients with an estimated phlebotomy number and follow-up duration may improve compliance by providing a finite end point. In addition, hepatic function may improve with iron off-loading in this group.

Disclosure of Commercial Conflict of Interest

K. Diggs: Nothing to disclose; S. F. Leitman: Nothing to disclose; S. R. Panch: Nothing to disclose; T. S. Sweigart: Nothing to disclose; Y. C. Yau: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Diggs: Nothing to disclose; S. F. Leitman: Nothing to disclose; S. R. Panch: Nothing to disclose; T. S. Sweigart: Nothing to disclose; Y. C. Yau: Nothing to disclose

S22-020A

Absolute Immature Platelet Count Increment Predicts the Clinical Course of Thrombotic Thrombocytopenic Purpura (TTP) Patients

H Hong[1,2] (hong.hong@uhhospitals.org), L M Stempak[1,2], L M Sandhaus[1,2], R W Maitta[1,2]. *[1]Pathology, University Hospital Case Medical Center, Cleveland, OH, United States; [2]Case Western Reserve University School of Medicine, Cleveland, OH, United States*

Background/Case Studies: Measurement of the percent immature platelet fraction (%-IPF) based on ribonucleic acid (RNA) content and size using automated hematology analyzer has been shown to be a fast and reliable method to differentiate platelet consumption/destruction from hypoproduction. Increases in %-IPF have been described in patients with immune thrombocytopenic purpura (ITP) and acute thrombotic thrombocytopenic purpura (TTP). However, the dynamics of IPF during the course of TTP has not been studied. The purpose of our study is to investigate the changes in %-IPF and absolute immature platelet count (A-IPC) over the course of TTP treatment with therapeutic plasma exchange (TPE) and to determine if the %-IPF/ A-IPC may be useful in assessing treatment efficacy. **Study Design/Methods:** Seven TTP patients, confirmed by ADAMTS13 <5% were enrolled and included four males and three females ranging in age from 19 to 62 years old. All seven patients had been treated with TPE and the decision for the cessation of TPE was made when patient's platelet count was over 150,000 × 109/L. All patients had IPF values tested at initial diagnosis and at the end of their TPE regimen, and five patients had at least four IPF values obtained during their course of treatment. %-IPF and platelet counts were measured using the Sysmex XE-5000 Hematology Analyzer (Sysmex America Inc., Mundelein, IL). A-IPC was obtained by multiplying the %-IPF by the platelet count. Results were compared to the normal %-IPF range for 40 samples from healthy adults. A student t test (two-tailed) was performed

with a p <0.05 considered significant. **Results/Findings:** All patients had a significantly increased %-IPF (8.43 ± 3.33 % versus 2.87 ± 1.39 %, p = 4.20809E-09) and decreased A-IPC (4.32 ± 1.94 × 109/L versus 7.12 ± 3.22 × 109/L, p = 0.016) at initial diagnosis in comparison to normal adults. For the five patients who had four or more sequential values, an inverse relationship between the %-IPF and the platelet count was observed. There was a steady decline in the %-IPF as the platelet counts climbed towards normal. The A-IPC, however, increased to a peak value (10.18 ± 2.12 × 109/L) which preceded platelet count recovery and then declined. The time to reach A-IPC peak was 2.3 ± 1.4 days, and the time to reach normal platelet count was 3.0 ± 1.4 days. **Conclusion:** In our patient cohort there was a significantly increased %-IPF and reduced A-IPC level in TTP patients at initial diagnosis. The expected inverse relationship between %-IPF and platelet count was observed during the clinical course of TTP patients upon TPE treatment. The results suggest that the attainment of A-IPC peak may be a useful indicator for cessation of TPE in TTP patients.

Disclosure of Commercial Conflict of Interest

H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; L. M. Sandhaus: Nothing to disclose; L. M. Stempak: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; L. M. Sandhaus: Nothing to disclose; L. M. Stempak: Nothing to disclose

S23-020A

Double Red Cell Collection (DRCC): Comparison of Three Apheresis Systems

E A Burgstaler[1] (burgstaler.edwin@mayo.edu), K J Duffy[1], M Benike[1], S Borene[1], J M Ask[1], E L McIntire[1], N A Kramer[1], J R Stubbs[1], M J Gandhi[1]. *[1]Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: To increase the red blood cell inventory and achieve maximum benefit from low frequency donors, it was decided to pursue double red cell collection (DRCC). For this study, DRCC-only procedures were compared on the Terumo BCT Trima, Fenwal Alyx, and Haemonetics MCS+ 8150 (MCS+) instruments. We evaluated donor eligibility, times product met quality standards, donor time, technician (tech) time, donor tolerance, and donor acceptance. **Study Design/Methods:** Products from 40 consecutive complete donations per instrument were studied. Manufacturer recommendations were followed for machine operation. All procedures include addition of preservative to the product. Leukoreduction was part of the collection procedure for Alyx and Trima, while it was a separate process for MCS+. The red blood cell (RBC) target volumes were 360 ml for all three instruments and an additional 400 ml target for Alyx. Donor time included the time the donor was in the collection area. Tech time included time for machine set-up, collection, preservation addition, and filtration (if applicable). **Results/Findings:** Analysis of eligibility criteria for whole blood donors (N = 9046) over a 5-month period revealed that potentially 4998 donors would be eligible for MCS+, 4622 for Alyx, and 4213 for Trima. Results of the comparison of the 40 donor groups (80 products) are summarized in the table. Trima donors (102 kg) weighed significantly more than Alyx or MCS+ donors(91 kg) and were significantly taller (175 cm) than MCS+ (170 cm). The hematocrits were similar (43%). Mix of female/male donors was: 11/29 Trima and MCS+, 17/23 Alyx. All instruments met the standard that 95% of products contain >42.5 grams hemoglobin (Hgb). 100% of Trima, 96.3% Alyx, and 67.5% MCS+ products contained >51grams Hgb. All three instruments produced 100% of the products with <5 × 106 WBC; 100% of the Trima and Alyx products contained <1.0 × 106 WBC as did 93.8% of the MCS+ products. Frequencies of oral calcium given for tingling were 5% Trima, 2% Alyx, and 10% MCS+. All donors indicated they would be willing to donate DRCC again. **Conclusion:** All three instruments were capable of consistently collecting an acceptable leukoreduced double red cell product. However, the Alyx collected RBC products with significantly less Hgb than Trima, but more than MCS+, in significantly shorter donor time and technician time (if processing in the component area is considered for MCS+). Additionally, the Alyx could potentially collect from 409 more donors in 5 months than Trima, but 376 less than MCS+.

Disclosure of Commercial Conflict of Interest

J. M. Ask: Terumo BCT, Other, Haemonetics, Other, Fenwal, Other; M. Benike: Nothing to disclose; S. Borene: No Answer; E. A. Burgstaler: Fenwal, Travel Support or Honorarium; K. J. Duffy: Termuo BCT, Other, Fenwal, Other, Haemonetics, Other; M. J. Gandhi: Nothing to disclose; N. A. Kramer: Nothing to disclose; E. L. McIntire: Terumo BCT, Other, Haemonetic, Other, Fenwal, Other; J. R. Stubbs: Terumo BCT, Other, Fenwal, Alyx, Other, Haemonetics MCS + 8150, Other

Disclosure of Grants Conflict of Interest

J. M. Ask: Nothing to disclose; M. Benike: Nothing to disclose; S. Borene: No Answer; E. A. Burgstaler: Nothing to disclose; K. J. Duffy: Nothing to disclose; M. J. Gandhi: Nothing to disclose; N. A. Kramer: Nothing to disclose; E. L. McIntire: Nothing to disclose; J. R. Stubbs: Nothing to disclose

Hgb, Times, and Donor Tingling (Means)

	Product Hgb (gm)	Donor Time (min)	Tech Time (min)	Donor Tingling (%)
Trima	59.2	52	87	58§
Alyx	56.8*	45*	73*	30§
MCS+	51.5*‡	52‡	64*‡	18§

* sig different than Trima ‡ sig different than Alyx § sig not determined for Tingling

S24-020A

Prophylactic Red Blood Cell (RBC) Exchange for Minor ABO Donor-Recipient Incompatibility in Hematopoietic Transplant: A Single Center Experience

R A Cunard[2] (rcunard@ucsd.edu), I I Marquez[2], M Y Choi[2], T A Lane[1,2], E D Ball[2], A P Sanchez[2], D M Ward[2]. *[1]Pathology, University of California, San Diego, School of Medicine, San Diego, CA, United States; [2]Medicine, University of California, San Diego, School of Medicine, San Diego, CA, United States*

Background/Case Studies: Minor-side ABO incompatibility (ABOi) between allogeneic hematopoietic transplant (BMT) donor and recipient may cause severe hemolysis of recipient RBC due to the passenger lymphocyte syndrome (PLS). Following two severe hemolytic events in BMT patients due to PLS, one resulting in death despite early RBC exchange, prophylactic partial RBC exchange (PE) was initiated in selected BMT patients who had minor ABOi donors and who were deemed to be at high risk for severe postBMT RBC hemolysis due to PLS. Reported here is the incidence of severe PLS-related hemolysis and patient survival in minor ABOi BMT patients from January 2007-November 2010 (Control patients), and after initiating PE in selected patients from August 2010-November 2012 (Study patients). **Study Design/Methods:** Patient risk factors for severe PLS in ABOi BMT included: non-myeloablative BMT; no methotrexate; an HPC, Apheresis graft; pre-BMT Hb >8 gm/dL; and transfusion (tx) of <4 donor/pt compatible rbc one month before BMT. PEs were performed by continuous flow RBC exchange apheresis on day +2-+4 post BMT with 5-15 U of donor/patient-compatible RBC (mean-1.6 rbc volume). Severe hemolysis was defined as Hb decrease of at least 2 gm/dL in <24 hours, in the absence of other causes, with a new positive direct antiglobulin test (DAT) post BMT, and either a 50% increase in LDH or indirect bilirubin versus pre BMT. After Institutional Review Board (IRB) approval, patients with minor or bidirectional ABOi donors and patients who received PE were identified in the BMT and Apheresis Service databases. Medical records were reviewed by one to two MDs. **Results/Findings:** The 31 Control and 16 Study patients were well matched by diagnosis, comorbidities, age, and gender. There were more non-ablative BMT in Study patients (14/16; p < .005) vs Controls (10/31). Severe hemolysis occurred in 1/16 Study patients (6.3%) vs 6/31 Control patients (19.4%; NS) and at a median of 7.5 days (range 4-10) post-BMT. RBC tx post-BMT was greater in Study patients (16.5 units) versus Controls (6; p < .005), but apheresis platelet tx was similar in Study (5.5 units) versus (8) Controls; NS. Days to neutrophil (>500/uL) and platelet (>20 K/uL) engraftment were similar in Study (9.3 & 11.9) versus Control patients (10.9 & 13.9; NS). There was no difference in length of stay in Study (15 days) versus Control patients(16). PE-associated complications included hypotension requiring ICU stay (2 patients); tunnel infection, partial, and total internal jugular occlusion (1 patient each). Survival at one year was lower in Study patients (50%) versus Controls (81%; p < .02) by Kaplan Meier analysis. **Conclusion:** In this study, while there was a non-significant trend toward a lower incidence of severe hemolytic events in patients who received prophylactic RBC exchange, there was no difference in days of hospitalization, but overall RBC tx was greater in PE patients; there were several PE -related complications, and overall one year survival in the PE group was lower. This small study therefore failed to identify a clear role for selected prophylactic RBC exchange in patients who were deemed at risk for severe post-BMT PLS.

Disclosure of Commercial Conflict of Interest

E. D. Ball: Nothing to disclose; M. Y. Choi: Nothing to disclose; R. A. Cunard: Nothing to disclose; T. A. Lane: Nothing to disclose; I. I. Marquez: Nothing to disclose; A. P. Sanchez: Nothing to disclose; D. M. Ward: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. D. Ball: Cellgene, Grants or Research Support; M. Y. Choi: Nothing to disclose; R. A. Cunard: Nothing to disclose; T. A. Lane: Nothing to disclose; I. I. Marquez: Nothing to disclose; A. P. Sanchez: Nothing to disclose; D. M. Ward: Nothing to disclose

Transfusion-Transmitted Infectious Diseases: Viruses

S25-020B

Direct Dengue Virus-Platelet Binding: Replication of the Positive-Strand Ribonucleic acid (RNA) Viral Genome

A Y Simon[1] (syila@mail.ubc.ca), M R Sutherland[1], E L Pryzdial[1].
[1]Canadian Blood Services, R&D, Centre for Blood Research, University of British Columbia, Vancouver, BC, Canada

Background/Case Studies: Dengue, a positive strand RNA virus (DENV), infects ~50 million people annually. It usually manifests flu-like symptoms, although ~500,000 cases per annum escalate to life-threatening hemorrhagic fever and shock syndrome. Non-neutralizing antibodies likely cause these severe pathologies enhancing propagation by Fc-receptor-bearing cells. Platelet products are essential to treat severe DENV disease, as thrombocytopenia is a hallmark. Of possible concern in the transfusion lab, DENV binds to platelets, not only a probable basis for disease, but also suggesting that the virus may fractionate with platelets. An earlier study using semi-purified virus reported the interaction is Fc-dependent; implying platelets of only a small percentage of infected donors can bind DENV. Interestingly, platelets can replicate RNA and may conceivably turnover the DENV genome. Since >105 DENV/ml may exist in asymptomatic donor blood, the Fc-independent binding of purified DENV to purified platelets was reinvestigated and the ability of platelets to amplify the DENV genome was evaluated. **Study Design/Methods:** DENV serotypes 1-4 were propagated in Vero cells and purified on step-wise sucrose gradients. Freshly washed platelets were quantified and characterized for activation by flow cytometry and had no detectable leukocytes. The DENV-platelet interaction was measured after a 20 minute incubation. Bound virus was separated by differential centrifugation and quantified by qRT-PCR. Specific surface-bound virus was determined by trypsinization. Fc-receptors were blocked using Human TruStain FcX (Biolegend). **Results/Findings:** Direct binding of purified DENV1 and 2 to platelets was independent of Fc-receptors and saturable (~4 and 3 viruses/platelet at 37C, respectively). This further explains the basis of platelet dysfunction during infection, but also implies that the virus may partition with platelets during blood fractionation. Binding was confirmed antigenically. Thrombin treatment of platelets enhanced DENV binding, indicating stimulus-induced receptor expression. Furthermore, pretreatment of platelets with DENV1-4, increased P-selectin detection, suggesting that the virus can directly trigger platelet activation. At 25C, the temperature platelets are stored in the blood bank, the genome of DENV1-4 was replicated five- to seven-fold after 7 days. At 37C, only DENV2 and 3 genomes were replicated. **Conclusion:** For the first time, these data reveal direct DENV-platelet interactions and viral genome replication by platelets. These observations add to the mechanism explaining thrombocytopenia and dysfunctional platelets in DENV pathology. Consideration should be given to platelets obtained from asymptomatic donors, which may harbor and enhance DENV during storage.

Disclosure of Commercial Conflict of Interest

E. L. Pryzdial: Nothing to disclose; A. Y. Simon: Nothing to disclose; M. R. Sutherland: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. L. Pryzdial: Nothing to disclose; A. Y. Simon: Nothing to disclose; M. R. Sutherland: Nothing to disclose

S26-020B

New Genotype of West Nile Virus was Found in the 2012 US Epidemic Season

A Grinev[1] (andriyan.grinev@fda.hhs.gov), C Chancey[1], G Añez[1], D A Heisey[1], E Volkova[1], V Winkelman[2], P C Williamson[2], S L Stramer[3], M Rios[1]. *[1]CBER/FDA, Bethesda, MD, United States; [2]Creative Testing Solutions, Tempe, AZ, United States; [3]American Red Cross, Gaithersburg, MD, United States*

Background/Case Studies: Since its first identification in the US in 1999, West Nile virus (WNV) has become endemic, recurring for 14 consecutive years, and is estimated to have caused 4-5 million human infections, with 1505 lethal cases reported to CDC. Most WNV infections are asymptomatic, and it can be transmitted by transfusion and transplantation. Since 2003, donated blood has been routinely screened for WNV and thousands of transmissions have been prevented. In 2012, the WNV epidemic in the US was intense, and reports to the CDC account for 5387 disease cases, including 2734 neuroinvasive, 243 deaths and 597 presumptively viremic donors. The annual reoccurrence of outbreaks in the US differs from those observed in the old world suggesting viral adaptation. This study aimed to investigate genetic variability of WNV circulating in the US in 2012. **Study Design/Methods:** In this study, 24 viral isolates from blood donor specimens were produced and genetic analysis of the complete viral genome for ten isolates obtained from different geographical locations of the US have already been performed. Viral ribonucleic acid (RNAs) were isolated, sequenced, and compared with data existing in GenBank. Analyses of phylogenetic relationships were based on likelihood and Bayesian algorithms. **Results/Findings:** Viral isolates from human specimens collected in 2012 showed an average of 0.65% genetic variability when compared with the ancestor strain WNV-NY99. Most mutations were silent transitions (U↔C, A↔G); the number of nucleotide mutations ranged from 63 to 81 resulting in 5 to 8 amino acid substitutions. We found that six of ten completely sequenced isolates from North Dakota, South Dakota, Texas, Wyoming, and Mississippi are genetically close to each other. Those isolates shared 55 nucleotide mutations and four amino acid substitutions when compared with WNV-NY99. Phylogenetically, they are close to a 2011 isolate from Mississippi suggesting that the new WNV genotype might have emerged in 2011. Isolates from Arizona and California are different from this new genotype and phylogenetically belong to the local clades. **Conclusion:** Viral adaptation to domestic mosquitoes and birds played a major role in the spread of WNV in North America. The new genotype WN02 displaced the ancestor WN99 in 2002, followed by the appearances of genotype SW/WN03 in 2003 and subtype MW/WN06 in 2006, and overall genetic divergence continued to slowly increase until 2011. The finding of similar isolates across a broad geographic area in 2012 suggests that a new WNV genotype emerged in 2011 and spread over the Midwest bird flyways in 2012, coincident with the largest US WNV outbreak since 2003. The appearance of this genotype may mark the beginning of a new genetic shift of WNV after 10 years of steady drift.

Disclosure of Commercial Conflict of Interest

G. Añez: Nothing to disclose; C. Chancey: Nothing to disclose; A. Grinev: Nothing to disclose; D. A. Heisey: Nothing to disclose; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose; E. Volkova: Nothing to disclose; P. C. Williamson: Nothing to disclose; V. Winkelman: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Añez: Nothing to disclose; C. Chancey: Nothing to disclose; A. Grinev: Nothing to disclose; D. A. Heisey: Nothing to disclose; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose; E. Volkova: Nothing to disclose; P. C. Williamson: Nothing to disclose; V. Winkelman: Nothing to disclose

S27-020B

Impact of Testing the Blood Supply for Human T Lymphotropic Virus (HTLV)on Recipient Infections

S F O'Brien[1] (sheila.obrien@blood.ca), I R Dines[1], M Goldman[1], V Scalia[1], Q Yi[1], W Fan[1], M Fearon[1]. *[1]Canadian Blood Services, Ottawa, ON, Canada*

Background/Case Studies: HTLV-I and II are cell associated retroviruses which cause life-long infection, generally asymptomatic, but sometimes associated with adult T-cell leukemia and neurological diseases. This is an attempt to evaluate the frequency of recipient HTLV infections occurring before and after testing of the blood supply. **Study Design/Methods:** Beginning in 1990, blood donations were screened with the Abbott Anti-HTLV-I EIA assay, switching to the Organon Teknika Anti-HTLV-I/II assay in 1998, and the Abbott PRISM HTLV-I/II (Abbott Diagnostics Division, Wiesbaden, Germany) in 2003. Supplemental testing was performed with a radioimmunoprecipitation assay (RIPA) and later a Western blot. The annual HTLV rate was calculated from 1990 to 2012 for all donations. For HTLV-I/II positive donations, lookback involved identification of cellular components of all previous donations, and attempting to locate and test recipients. Traceback was initiated when transfusion transmission was queried for HTLV-I/II positive blood recipients. All donors of products that the recipient received were identified, with an attempt to locate and test them. **Results/Findings:** The HTLV supplemental assay positive rate decreased from 9.35 per 100,000 donations in 1990 to 1.01 in 2012. There were 76 % HTLV-I, 16% HTLV-II and 8% type undetermined from 2009-2012. The mean rate increased from 1.39 to 2.38 in1998 when the screening assay was upgraded to include HTLV-II ($p < 0.0001$). Among repeat donors the HTLV rate decreased from 9.4 per 100,000 donations in 1990 when testing was imple-

mented to 1.14 per 100,000 over the last 10 years. However among first time donors the rate has remained similar (8.63 per 100,000 donations in 2012). In the lookback study, of 109 HTLV-I/II positive donors, 508 components were transfused, of whom 147 recipients of cellular products were tested, and 18 (12%) were positive, all prior to the implementation of testing. Twenty-three traceback investigations were requested involving 324 transfused untested products. 219 (67.6%) of donors were tested and 13 (6%) were positive for HTLV. Seven involved red blood cell transfusions (from one to 11 days storage time), four platelet transfusions, one white blood cells and one product type unknown. Of four traceback requests involving products transfused after testing was implemented (two were after introduction of leukoreduction in 1999) from 17 donors associated, 13 donors were tested, and all were negative. **Conclusion:** Most HTLV-II infections were likely identified through cross-reactivity with the original HTLV-I assay. Transfusion transmission of HTLV-I/II occurred in the absence of testing and prior to universal leukoreduction, albeit infrequently, but there is no evidence of transmission in the 23 years since testing was implemented.

Disclosure of Commercial Conflict of Interest

I. R. Dines: Nothing to disclose; W. Fan: Nothing to disclose; M. Fearon: Nothing to disclose; M. Goldman: Nothing to disclose; S. F. O'Brien: Nothing to disclose; V. Scalia: Nothing to disclose; Q. Yi: No Answer

Disclosure of Grants Conflict of Interest

I. R. Dines: Nothing to disclose; W. Fan: Nothing to disclose; M. Fearon: Nothing to disclose; M. Goldman: Nothing to disclose; S. F. O'Brien: Nothing to disclose; V. Scalia: Nothing to disclose; Q. Yi: No Answer

S28-020B

Molecular, Virological, and Phylogenetic Characterization of Dengue Virus Types 1 and 4 in Blood Donors from Puerto Rico, 2012-2013

G Añez[1] (German.Anez@fda.hhs.gov), D A Heisey[1], L M Espina[1], S L Stramer[2], M Rios[1]. *[1]CBER, US Food and Drug Administration, Bethesda, MD, United States; [2]American Red Cross, Gaithersburg, MD, United States*

Background/Case Studies: Dengue viruses (DENV-1 to 4) are flaviviruses transmitted mainly by Aedes aegypti. DENV infection can be asymptomatic or range from mild disease to severe dengue. DENV has been detected in asymptomatic blood donors and transfusion-transmitted cases have been reported. DENV is endemic in Puerto Rico (PR). The aim of the study was 1) To compare DENV ribonucleic acid (RNA) concentrations in whole blood (WB) and the plasma component of samples collected from DENV-reactive donors. 2) To perform molecular epidemiology studies in DENV strains circulating in PR during 2012-2013. **Study Design/Methods:** The study included 123 WB and 121 plasma specimens from blood donors to the American Red Cross (ARC) from PR found reactive for DENV RNA by an investigational nucleic acid amplification test (NAT) assay during August 2012-March 2013. Since DENV-1 and -4 were reported circulating in the area, samples were first tested by type-specific TaqMan qRT-PCR assay for DENV-1 and those that produced negative results were tested for DENV-4. A total of 77 whole blood and 36 plasma samples that tested positive for DENV-1 or -4 were subjected to viral isolation in mosquito C6/36 cells. Cell culture supernatants were subjected to quantitation of viral loads by TaqMan and titration of infectious particles by focus-forming assay. Fourteen DENV-1 and two DENV-4 positive specimens had the complete envelope protein (E) gene amplified by RT-PCR and sequenced by the Sanger method. Viral sequences were subjected to phylogenetic analysis by the Maximum-likelihood method. **Results/Findings:** Positive TaqMan results were obtained in 22/121 (18.2%) plasma and in 60/123 (48.8%) WB specimens tested for DENV-1, while 14/121 (11.6%) plasma and 17/123 (13.8%) WB specimens tested positive for DENV-4. Comparison of DENV RNA titers found in plasma and WB shows equivalent viral titers for most samples, although some subjects had higher viral loads in either one of the tested components. In tissue culture, 17 plasma and WB samples that tested positive for DENV-1, and nine plasma and six WB samples that tested positive for DENV-4 were infectious as detected by both virological and molecular methods. DENV-1 E gene sequences were found to belong to the clade "c" of genotype V, clustering together with DENV-1 sequences from strains obtained from blood donors from PR during 2010. DENV-4 E gene sequences (genotype II) were closely related to strains from this DENV type circulating in blood donors in 2010. **Conclusion:** DENV-1 and DENV-4 continued to circulate in blood donors from PR during 2012 and 2013. NAT-reactive donations had infectious virus that could be isolated from both WB and plasma samples from viremic blood donors, as evidenced by viral isolation in C6/36 cells. DENV-1 strains from the clade "c" of genotype V and DENV-4 of genotype II continues to circulate in PR since 2010. The use of molecular methods for blood screening has been effective in detecting DENV-infected blood donations, enhancing blood safety.

Disclosure of Commercial Conflict of Interest

G. Añez: Nothing to disclose; L. M. Espina: No Answer; D. A. Heisey: Nothing to disclose; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Añez: Nothing to disclose; L. M. Espina: No Answer; D. A. Heisey: Nothing to disclose; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose

S29-020B

Comparison of Seven Hepatitis B Virus Nucleic Acid Testing (NAT) Assays in Selected Samples from American Blood Donors

J Allain[1] (jpa1000@cam.ac.uk), F Enjalbert[1], D E Krysztof[2], D Candotti[3], S L Stramer[2]. *[1]Haematology, University of Cambridge, Cambridge, United Kingdom; [2]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [3]Research & Development, National Health Service Blood and Transplant, Cambridge, United Kingdom*

Background/Case Studies: Sensitive nucleic acid testing (NAT) assays for hepatitus B virus (HBV), hepatitis C virus (HCV), and human immunodeficiency virus (HIV) nucleic acid detection are implemented for blood donation screening worldwide. These assays have variable ability to detect very low levels of HBV DNA in individual donations (ID). At borderline DNA detection levels, where Poisson distribution impacts results, distinguishing between true and false positive results is challenging. **Study Design/Methods:** The American Red Cross (ARC) selected 135 blood donor samples reactive by one or more routine NAT and/or serologic HBV markers that provided discrepant results when tested with four NATs: Ultrio; Ultrio Plus (both in ID and minipools [MP] of 16); MPX (ID and MP6); and NGI SuperQuant (ID). To further explore these discrepancies, three additional in-house NATs, either real-time polymerase chain reaction (PCR) after extraction from 0.5 or 5 ml of plasma (qPCR-0.5 & qPCR-5) or nested PCR from 5 ml of plasma, and sequencing were performed. **Results/Findings:** Among the 135 samples, 108 (80.6%) were reactive with at least two NATs out of seven assays and were considered positive, of which 67 (62.0%) were confirmed by sequence analysis. Ultrio Plus and MPX performed similarly (80.6% and 86.1% for sensitivity and 55.6% and 66.7% for specificity, respectively). The older (median 49 years), anti-HBc-positive donors carried predominantly genotype A (58%) with a high frequency of amino acid substitutions in the major hydrophilic region of the S protein. Younger (median 24 years) HBsAg-positive donors carried wild-type strains predominantly genotype B (32%) and E (24%), the latter in an apparent cluster. **Conclusion:** Highly sensitive NATs require the development of new confirmatory algorithms using different genomic regions and possibly yielding a sequence. The introduction of immigration-related HBV genotypes might considerably impact HBV epidemiology in the USA.

Disclosure of Commercial Conflict of Interest

J. Allain: Nothing to disclose; D. Candotti: Nothing to disclose; F. Enjalbert: Nothing to disclose; D. E. Krysztof: Nothing to disclose; S. L. Stramer: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Allain: Terumo BCT, Grants or Research Support; D. Candotti: Nothing to disclose; F. Enjalbert: Nothing to disclose; D. E. Krysztof: Nothing to disclose; S. L. Stramer: Nothing to disclose

Characteristics of Seven HBV DNA assays and relative performance on a selected panel of 135 samples reactive with HBV markers

	5 nested	Ultrio+	MPX	qPCR-5	Ultrio	qPCR-0.5	SuperQuant
Reactive of 108 positive 1	67	87	93	74	53	38	41/107
% sensitivity	62.0	80.6	86.1	68.5	35.2	49.1	38.1
Reactive of 67 confirmed 2	67	54	60	51	25	35	31
% sensitivity	100	80.6	80.6	76.1	37.3	52.2	46.3
Reactive of 27 negative 3	0	12	9	0	0	0	0
% specificity	100	55.6	66.7	100	100	100	100

1) The 108 samples considered 'positive' for HBV DNA were reactive with two or more of the seven assays or yielded a HBV DNA sequence. 2) The 67 'confirmed' samples all yielded an HBV DNA sequence. 3) The 27 'negative' samples were defined as either non-reactive with all seven HBV DNA assays or reactive with only one of seven assays.

S30-020B

Evaluation of Nucleic Acid Testing (NAT) and Serology for Hepatitis B Virus in a Donor Population

V Winkelman[1] (vwinkelman@mycts.org), S Cyrus[1], R Spizman[1], P C Williamson[1]. *[1]R&D Support, Creative Testing Solutions, Tempe, AZ, United States*

Background/Case Studies: In August 2010, a large regional blood donor testing facility implemented a Triplex (HIV-1/HCV/HBV) nucleic acid test (NAT). Initial screening on all donations included NAT, HBsAg, and Anti-HBc chemiluminescence (ChLIA) or EIA. Data was initially evaluated using correlation between HBsAg and NAT to evaluate the status of hepatitis B virus (HBV) for additional counseling information. The data was re-evaluated to include Anti-HBc in an effort to help determine if hepatitis B reactive donations could be identified using only NAT and Anti-HBc. **Study Design/ Methods:** All HBsAg repeat reactive samples tested after implementation of the Triplex test were selected for evaluation. Samples that are HBsAg confirmed positive (CP) and NAT NR are routinely routed for a correlation series which requires an individual discriminatory HBV NAT (dHBV) and an additional HBsAg test if the dHBV was NR. These samples were then evaluated based on Anti-HBc results. **Results/Findings:** A total of 5806 HBsAg repeat reactive samples were evaluated. Of these, 2199 (37.87%) were neutralization non-confirmed (NC), HBc NR, and NAT NR. Seventeen (0.29%) were neutralization NC, HBc R, and NAT NR. Five (0.09%) samples were neutralization NC, HBc NR, and NAT R of which four were non-discriminated and one had insufficient volume for discriminatory testing. Eight (0.14%) samples were neutralization NC, HBc R, and NAT R. 92 (1.58%) samples were neutralization confirmed positive (PC), HBc NR, and NAT NR. And 3485 (60.02%) samples were either neutralization CP with either Anti-HBc and/or NAT R, or testing was not completed due to other factors. The 92 samples that were neutralization CP, HBc NR, and NAT NR were further evaluated using the correlation series protocol. These samples were additionally tested using individual dHBV and an additional HBsAg test if the dHBV was NR. Of these samples, 22 (23.91%) were unable to be evaluated, one (1.09%) was dHBV R, and 69 (75%) were dHBV NR and HBsAg R upon retest. Donor history for the 1 dHBV R was unavailable for evaluation. Donor information was available on six of the 69 dHBV NR/HBsAg R donations. Of these, five donors had received Hepatitis B Vaccine at least 10 days prior and the other donor had received a flu shot eight days prior to donation. **Conclusion:** This data supports the need for integration of routine HBV NAT, HBsAg neutralization, and Anti-HBc, supplemented by additional testing and donor follow up, to properly classify HBV test reactive donors for donor counseling and management. Additional information would need to be obtained and analyzed for the 64 donors with no available history to determine their true hepatitis B status. Rigorous follow up for these types of donors is required to determine feasibility of dropping HBsAg given routine NAT plus Anti-HBc.

Disclosure of Commercial Conflict of Interest

S. Cyrus: Nothing to disclose; R. Spizman: Nothing to disclose; P. C. Williamson: Nothing to disclose; V. Winkelman: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Cyrus: Nothing to disclose; R. Spizman: Nothing to disclose; P. C. Williamson: Nothing to disclose; V. Winkelman: Nothing to disclose

Red Blood Cells: Molecular Basis of Blood Groups — Rh in RBC Immunohematology, Molecular Testing and Genetics

S31-020C

Characterization of the *RH* locus in African American Blood Donors

M A Keller[1] (margaret.keller@redcross.org), J A Crowley[1], T Horn[1], R Young[1], S Hondorp[1], D F Friedman[2], C M Westhoff[3], S T Chou[2]. *[1]National Molecular Laboratory, American Red Cross, Philadelphia, PA, United States; [2]Children's Hospital of Philadelphia, Philadelphia, PA, United States; [3]New York Blood Center, New York, NY, United States*

Background/Case Studies: Alloimmunization to Rh antigens is not uncommon in patients with sickle cell disease (SCD), despite transfusions matched for D, C, and E. It is known that African Americans have higher frequencies of *RH* variant alleles. This study aimed to characterize the *RH* alleles in a donor cohort to gauge the feasibility to match *RH* alleles in donors with those in patients on chronic transfusion regimens. Such patients could include those who are alloimmunized to Rh antigens as well as those for which *RH* genotype matching could prevent alloimmunization. **Study Design/ Methods:** A group of 217 self-declared African American blood donors were characterized for *RH* variants. Genotyping was performed using *RHCE* and *RHD* BeadChip™ platforms (BioArray Solutions, LLC) as well as polymerase chain reaction (PCR)-restriction fragment length polymorphism (RFLP) for *RHCE* nt254 and *RHD* nt1136 and hybrid Rhesus box PCR for detection of an *RHD* gene deletion. Inconclusive results were investigated using additional gel-based genotyping assays or genomic sequencing. **Results/ Findings:** Alleles could be unequivocally assigned in 88% of donors. Both conventional *RHCE* and *RHD* alleles were present in 20.3% of donors. Variant *RHCE* and *RHD* alleles were present in 65.9% and 43.3% of donors respectively, with 9.7% of donors carrying only variant alleles (Table). Approximately 10% of donors tested were compound heterozygotes for *RHCE* nt48G/C and nt733C/G, such that allele assignment was equivocal. The most common *RHCE* allele was *RHCE*ce48C*, carried by 35% of donors with definitive allele assignments. Detected by PCR-RFLP, 6% of donors carried *RHCE* nt254 variant associated with the hr^B- variant *RHCE*ceAG* and 21.2% of donors carried the marker of an *RHD*DAU0* allele. While 5.5% of donors were predicted to be D-, ~14% were predicted to express partial D+ and 2.8% an altered C antigen associated with an *RHD*D-CE-D* hybrid allele. A weakened e antigen encoded by *RHCE*ce48C* was predicted for nearly 20% of donors, partial c and partial e antigens in 12% of donors and an hr^B- or hr^B + w/- phenotype in 3% of donors. **Conclusion:** A cohort of African American blood donors were characterized for *RH* variants using combined array-based and gel-based genotyping. This screening highlights *RH* genetic diversity in African Americans, with approximately two-thirds of donors expressing variant Rh antigens. This study illustrates that ~10% of donors are ambiguous heterozygotes with equivocal allele assignments that would require cDNA analysis for resolution. In addition, the knowledge of *RHD* status in donors and patients could allow for matching for *RHD* variants, thus decreasing the demand for RhD negative blood.

Disclosure of Commercial Conflict of Interest

S. T. Chou: Nothing to disclose; J. A. Crowley: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; S. Hondorp: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research, SupportLife Technologies, Consulting or Board of Director Fees; C. M. Westhoff: Nothing to disclose; R. Young: No Answer

Disclosure of Grants Conflict of Interest

S. T. Chou: BioArray, Grants or Research Support; J. A. Crowley: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; S. Hondorp: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support; C. M. Westhoff: Nothing to disclose; R. Young: No Answer

***RH* Alleles in African American donors (N = 217)**

Alleles	RHCE (#)	RHCE (%)	RHD (#)	RHD (%)	RHCE & RHD (#)	RHCE & RHD (%)
conventional alleles only	51	23.5	118	54.4	44	20.2
conventional and variant alleles	101	46.5	60	27.6	126	58.1
variant alleles only	42	19.4	34	15.7	21	9.7
equivocal allele assignments	23	10.6	5	2.3	26	12.0

S32-020C

The Promiscuity of the Antibody Made By Hr- hr^S- People Homozygous for *RHCE*ceAR* Makes It a Valuable Tool to Screen for hr^S- Donors

C Lomas-Francis[1] (clomas-francis@nybloodcenter.org), S Vege[1], A Fuchisawa[1], Z Hu[1], C M Westhoff[1]. *[1]Immunohematology and Genomics, New York Blood Center, Long Island City, NY, United States*

Background/Case Studies: Several variant *RHCE*ce* alleles encode the hr^S- phenotype, and some, including *RHCE*ceAR*, *RHCE*EK* and *RHCE*BI* encode a hr^S- and Hr- phenotype. People with an hr^S-Hr- phenotype can make anti-hr^S and/or -Hr; depending on the specificity the reactivity may appear to be anti-e or anti-Rh17-like. Since 1998, plasma from a D + C – E – c + e+ hr^S – patient by albumin IAT has been used to screen thousands of samples from African Americans and latterly from Hispanics to find hr^S– donors. By albumin IAT the reactivity is anti-hr^S but by PEG IAT the reactivity is consistent with anti-Hr. As DNA analysis is becoming mainstream, the molecular basis of the antibody maker's hr^S– phenotype was recently determined. Additionally, for investigation of the specificity of the patient's

antibody, we reviewed the RH genotypes in donors identified using her plasma. **Study Design/Methods:** Antigen typing was performed by standard tube methods. Donor screening with anti-hr^S was performed by the IgG gel test (ID MTS) using the plasma diluted 1 in 10. Any nonreactive samples were retested with the neat plasma and submitted for DNA analysis. Cross-testing between antibody makers with the same or similar RHCE genotypes was performed. DNA was isolated from whole blood for genomic testing from available hr^S- donors. *RHCE* was analyzed by *RHCE* BeadChip array (Immucor) and manual molecular methods. For the patient, ribonucleic acid (RNA) was isolated from reticulocytes and Rh-specific cDNA was synthesized, amplified, and sequenced. **Results/Findings:** Testing revealed that the antibody maker is homozygous for the RHCE*ceAR allele that is predicted to encode partial c and partial e antigens and a hr^S–Hr–, hr^B+ phenotype. Record review showed that one of the samples compatible during the original 1998 antibody identification was also homozygous *RHCE*ceAR*. The donors identified by our screening include a fascinating combination of genotypes: *RHCE*ceAR/cE*, *RHCE*ceMO/ceBI*, *RHCE*ceMO/cE*, *RHCE*ceEK/cE*, encoding hr^S—and *RHCE*JAL/ce733G*; *RHCE*ce733G/cE* and *RHCE*ceS/cE*, encoding hr^B-. Patients with the *RHCE*ceAR/ceEK* or *RHCE*ceAR/ceAR* genotype were also compatible with the plasma. Cross-testing of patient antibodies showed that the antibody made by homozygous *RHCE*ceAR* people are compatible with *RHCE*ceAR/ceEK, RHCE*ceEK/cE907del*. Antibody made by a *RHCE*ceEK/cE907del* person was compatible with RBCs with *RHCE*ceAR* homozygous, *RHCE*ceMO* homozygous, *RHCE*ceMO/ceBI*. Antibody made by *RHCE*ceMO/ceMO, RHCE*ceCF/ceCF,* were not compatible with *RHCE*ceAR/ceAR*. **Conclusion:** The antibody made by homozygous *RHCE*ceAR* people is a valuable screening tool to detect multiple e variant donors, including hr^S- and hr^B-. Given the promiscuity of the anti-hr^S -Hr reactivity made by *RHCE*ceAR* people poses the question of whether or not a single epitope is altered or whether the changes in *RHCE*ce* exon 5 alter several epitopes by the induced change in conformation.

Disclosure of Commercial Conflict of Interest

A. Fuchisawa: Nothing to disclose; Z. Hu: No Answer; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Fuchisawa: Nothing to disclose; Z. Hu: No Answer; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

S33-020C

Detection of Altered *RHD* Alleles in D-Negative Pregnant Women

K B Torres[1] (kelyan_ff@hotmail.com), A C Gaspardi[1], D Machado[1], E Sippert[1], C R Fujita[2], J Pellegrino[1], L Castilho[1]. *[1]Hemocentro, UNICAMP, Campinas, Brazil; [2]Immunohematology, Fujisan, Fortaleza, Brazil*

Background/Case Studies: Anti-D is a frequent cause of hemolytic disease of the fetus and newborn. As a rule, immunization occurs in D negative pregnant women, but occasionally anti-D is also observed in carriers of D variants. Currently, maternal plasma analysis for determination of the fetal *RHD* status became an exciting new tool for the management of D-negative pregnant women, but one of the challenges in noninvasive fetal *RHD* genotyping is the presence of aberrant *RHD* alleles in the pregnant women. The aim of this study was to determine aberrant *RHD* alleles among D-negative pregnant women from a multiethnic population using a strategy involving serological and molecular methods. **Study Design/Methods:** Blood samples serologically identified as D-negative from 380 pregnant women (338 non-alloimmunized and 42 alloimmunized) were first tested by adsorption elution. Samples found to be positive by adsorption-elution were molecularly tested by using a Multiplex polymerase chain reaction (PCR), the RHD BeadChip (BioArray Solutions, Immucor), and sequencing of the *RHD*. **Results/Findings:** Among 380 samples, 17 (4.5%) were reactive by adsorption-elution. Molecular analyses confirmed altered *RHD* alleles in 14 (3.7%) samples. Eight were non- functional *RHD* (five *RHD* Ψ and three *RHD*CE(3-7)-D* hybrid gene). Six were functional *RHD* predicting partial D (one *RHD*DVI*), weak D (one *RHD*weak D type 2*, one *RHD*weak D type 4.3* and one *RHD*weak D type 38*) and DEL (one *RHD*K409K* and one *RHD*IVS3 + 1G > A*) phenotypes. The carriers for the *RHD*DVI* and *RHD*IVS3 + 1G > A* had anti-D and delivered a D-positive baby affected by a mild hemolytic disease (HDFN). In three samples the *RHD* was deleted, indicating a false-positive result in the adsorption elution test. **Conclusion:** This study showed altered *RHD* alleles in 3.7% of serologically D-negative pregnant women associated with a positive adsorption-elution test. This finding is important to design assays for noninvasive fetal *RHD* genotyping in this multiethnic group of pregnant women in order to avoid false results and also to manage the variants harbored in this population. The fact that the babies from the carriers of the partial D (one *RHD*DVI*) and DEL (*RHD*IVS3 + 1G > A*) phenotypes presented HDFN supports anti-D prophylaxis for such cases.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; C. R. Fujita: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; D. Machado: No Answer; J. Pellegrino: Nothing to disclose; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; C. R. Fujita: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; D. Machado: No Answer; J. Pellegrino: Nothing to disclose; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

S34-020C

Altered *RHD* and *RHCE* Alleles among African Brazilian Blood Donors

D Machado[1] (debora.machado@yahoo.com.br), E Sippert[1], A C Gaspardi[1], K B Torres[1], J Pellegrino Jr[1], L Castilho[1]. *[1]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: The *RH* locus consists of two homologous genes: the *RHD* encoding the D antigen and the *RHCE* encoding the Ce, ce, cE, and CE antigen forms. As a consequence of the homology and opposite orientation of the two *RH* genes, many rearrangements between these two genes occurred and resulted in hybrid genes. Over 200 *RHD* alleles and over 70 *RHCE* alleles have been reported, and new alleles are still being discovered. African Brazilians have been studied to determine these alleles in order to support chronically transfused patients with sickle cell disease (SCD) with a precisely genotype match donor blood units. **Study Design/Methods:** LDTs (PCR-SSP, Multiplex polymerase chain reaction (PCR) and sequencing exons) and *RHD* and *RHCE* BeadChip (BioArray Solutions, Immucor) were used to determine altered *RH* alleles among African Brazilian blood donors. **Results/Findings:** Of 570 blood donor samples sent to our laboratory with weak or discrepant results in the Rh serology typing, 67 had altered *RH* alleles. The table shows the distribution of *RH* variants found in African Brazilians. Compound heterozygote *RHCE* variants were found in 12 (18%) donors. The *RHD* and *RHCE* allele combinations found in the studied population were: *RHD*DAR* with *RHCE*ceAR*; *RHD*DAU0* with *RHCE*ceMO*; *RHD*CE(4-7)D* with *RHCE*ceS*; *RHD*DIIIa* with *RHD*CE(4-7)-D*; *RHD*DOL* with *RHCE*ceBI* and *RHD*DIVa-2* with *RHCE*ceTI*. **Conclusion:** The most common *RH* allele combinations found in this population were *RHD*DAR* with *RHCE*ceAR* and *RHD*D-CE(4-7)-D* with *RHCE*ceS*. Compound heterozygote *RHCE* altered alleles were common among African Brazilians. Knowledge of the prevalence of *RH* alleles in populations of African origin can be important to evaluate and minimize the risk of Rh alloimmunization in patients with Sickle Cell Disease.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; D. Machado: No Answer; J. Pellegrino Jr: No Answer; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; D. Machado: No Answer; J. Pellegrino Jr: No Answer; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

TABLE. Distribution of *RHD* and *RHCE* alleles among Brazilians

RHD alleles	n	*RHCE* alleles	n
*RHD*DAR*	9	*RHCE*ceAR*	5
*RHD*DAU0*	7	*RHCE*ceMO*	3
*RHD*DIIIa*	5	*RHCE*ce733G*	16
*RHD*DIVa-2*	2	*RHCE*ceTI*	2
*RHD*DOL*	3	*RHCE*ceBI*	3
*RHD*D-CE(4-7)-D*	6	*RHCE*ceS*	18
*RHD*DIIIa-CE(4-7)-D*	4	*RHCE*ce48C*	4
		*RHCE*ce48C,733G*	4

S35-020C

A Novel Allele with Nucleotide 307C > T (Pro103Ser) On *RHCE*ceAG* (254C > G) Encodes Robust C Antigen

S Vege[1] (svege@nybloodcenter.org), N C Johnson[2], R W Velliquette[1], C Lomas-Francis[1], C M Westhoff[1]. *[1]Immunohematology and Genomics, New York Blood Center, Long Island City, NY, United States; [2]Immunohematology Reference Laboratory, American Red Cross, Baltimore, MD, United States*

Background/Case Studies: The RH system is diverse at the genetic level, and as the number of alleles continues to increase it is important to determine which encode clinically significant Rh antigens and those important to detect in genotyping assays. RH was investigated in two African-American blood donors whose red blood cells (RBCs) typed strongly C+, but neither carried the *RHCE*Ce* gene as determined by multiple molecular methods including AS-PCR for nucleotide (nt) 307 C > T, the *Ce intron 2 insertion, and by an automated method. **Study Design/Methods:** Antigen typing was performed by standard tube methods. DNA was isolated from whole blood for genomic testing. HEA (BioArray/Immucor) was performed according to manufacture instructions. Ribonucleic acid (RNA) was isolated from reticulocytes and Rh-specific cDNA was synthesized, amplified, and sequenced. **Results/Findings:** The RBCs of Donor One typed as D + C+E-c + e+, and Donor Two as D + C+E + c+e+; no reduction in strength of reactivity compared to controls was observed. Typing of the RBCs with multiple reagents, and for one sample several monoclonal anti-C, demonstrated strong reactivity (Table). However, the predicted phenotype by HEA concluded both were C-, although the genotype report showed 103Pro/Ser with no intron 2 insertion. Manual multiplex AS-PCR (Singleton et al, *Blood*, 2000) also predicted the samples were C- as they lacked the *RHCE*Ce* intron 2 insertion and were positive only for 307C (*RH*c/c*). Molecular testing for other Rh antigens was consistent with the RBC phenotype, and interestingly, both samples were positive for the *RHD* pseudogene and both had one conventional *RHD in trans*, consistent with their D+ phenotype. Rh-cDNA sequencing revealed the presence of an altered *RHCE*ce* with a nucleotide 307C > T change, encoding Pro103Ser, which is the exofacial transmembrane residue on RhCe. A previously reported nt 254C > G change (Ala85Gly, designated RHCE*ceAG) was also present on the same allele. We tested four RBC samples with *RHCE*ceAG* alone or with conventional *RHCE*ce* and none reacted with anti-C. **Conclusion:** Robust C antigen expression in the absence of *RHCE*Ce* is reported. The results show that 103Ser on Rhce is associated with expression of C, in the absence of other C-specific amino acid residues (16Cys, 60Ile, 68Ser). This is the first report of C antigen expression on an altered Rhce protein (*RHCE*ce245G, 307T*) with RBCs strongly reactive with all commercial anti-C reagents.

Disclosure of Commercial Conflict of Interest

N. C. Johnson: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. C. Johnson: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Sample	RBC phenotype	Anti-C	RH genotype	
Donor 1	D + **C+**E-c + e+	3+/4+*	*RHCE*ce (254G, 307T) RHCE*ce(48C)*	RHD pseudogene RHD
Donor 2	D + **C+**E + c+e+	3+/4+	*RHCE*ce (254G,307T) RHCE*cE*	RHD pseudogene RHD

* With Immucor Gamma-clone, Ortho BioClone, BioRad Seraclone, monoclonal supernatants MS-24, MS-26, Mab S2, Mab C388

S36-020C

Genetic Linkage between *RHCE*ceHAR* and *RHD*DIVa.2* Alleles in People of African Descent

C Vrignaud[1] (cvrignaud@ints.fr), C Auxerre[1], M Roussel[1], S Ramelet[1], S Martin-Blanc[1], P Gallon[2], E Barre-Amirault[3], T Peyrard[1]. *[1]National Immunohematology Reference Laboratory, National Institute of Blood Transfusion, Paris, France; [2]EFS Ile de France, French National Blood Service, Paris, France; [3]EFS Pays de la Loire, French National Blood Service, Le Mans, France*

Background/Case Studies: Several studies showed in people of African descent the existence of a genetic linkage between variant *RHCE* and *RHD* alleles, eg *RHCE*ceBI* and *RHD*DOL*. *RHCE*ceHAR* is a rare allele found in Caucasians (mostly Germans), also called *RHCE*ce-D(5)-ce* (gene conversion with *RHD* exon 5). Despite being in cis with a *RHD*-deleted (*d*) allele ($R^{o\ Har}$ haplotype), it encodes a weak/partial D phenotype (DHAR), detectable only with a few monoclonal D typing reagents. It is shown here that *RHCE*ceHAR* may be encountered in Africans and then travels with *RHD*DIVa.2* (also known as *DIV type 1.0*), allele reported to encode a pseudo weak C reactivity with some C typing reagents. **Study Design/Methods:** Samples from eight unrelated African individuals were referred to the laboratory, either for a weak/discrepant C reactivity (n = 7) or anti-D in a D-positive type (n = 1). Red blood cells (RBCs) were investigated by standard hemagglutination techniques. *RHD* and *RHCE* were studied by RBC genotyping (wRHCE BeadChip, BioArray/Immucor) and DNA exon-sequencing. **Results/Findings:** In the seven individuals referred for C investigation, all were confirmed to be weakly reactive with anti-C MS-24 clone (≦2+ by gel technique) but negative with polyclonal anti-C, including adsorption/elution technique. Six subjects were D + C+^{w}E-c + e+ and one D + C+^{w}E + c+e + ^{w}. *RHCE* study showed no C polymorphism but a rare *RHCE*ceHAR* allele. This prompted the investigation of the *RHD* gene that revealed the systematic presence of a *RHD*DIVa.2* allele. Of note, three out of seven individuals were initially mistyped as *RHCE*Ce-D(5)-Ce* (*RHCE*CeVA*), since exon 2 was not sequenced at that time in our laboratory; this later appeared surprising as this allele had only been reported in Caucasians. The eighth individual showed a partial DIVa-2 type, consistent with alloanti-D in her serum. The C reactivity (MS-24 clone) was <1+. *RHCE* was investigated in order to search for a *RHCE*ceTI allele*, known to be usually in cis with *RHD*DIVa.2*; *RHCE*ceHAR* was unexpectedly detected in that case. All African people known with *RHCE*ceHAR* in our reference laboratory were found to carry *RHD*DIVa.2* (n = 8). Five of them were from Ivory Coast. Of note, three were *RHCE*ceHAR/RHCE*ce* and *RHD*DIVa.2/RHD*DIVa.2* or, much more likely, *RHD*DIVa.2/d*; all carried the c.48G > C change (with 2 at the homozygous state), usually found in C-negative African people. **Conclusion:** For the first time, it was shown that *RHCE*ceHAR* may be encountered in people of African descent. Unlike *RHCE*ceHAR* in Caucasians, this allele appears to carry c.48C in exon 1, which could justify the allocation of a specific name for the African allele. *RHCE*ceHAR* was also found to be genetically linked to *RHD*DIVa.2* in Africans. Lastly, it was confirmed that DIVa-2 is associated to a weak C reactivity with MS-24 clone, which turns out to be the major reason for referral in people with a *RHD*DIVa.2/RHCE*ceHAR* haplotype.

Disclosure of Commercial Conflict of Interest

C. Auxerre: Nothing to disclose; E. Barre-Amirault: Nothing to disclose; P. Gallon: Nothing to disclose; S. Martin-Blanc: Nothing to disclose; T. Peyrard: Nothing to disclose; S. Ramelet: Nothing to disclose; M. Roussel: Nothing to disclose; C. Vrignaud: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Auxerre: Nothing to disclose; E. Barre-Amirault: Nothing to disclose; P. Gallon: Nothing to disclose; S. Martin-Blanc: Nothing to disclose; T. Peyrard: Nothing to disclose; S. Ramelet: Nothing to disclose; M. Roussel: Nothing to disclose; C. Vrignaud: Nothing to disclose

Pediatric Transfusion Medicine

S37-020D

Hemorheological Aspects of Blood Transfusion in Sickle Cell Pediatric Patients

R Morandi[1] (riccardo.moranti@unipr.it), M Ballestri[3], F Fontana[3], M Maioli[3], G Palazzi[2], M Lodi[2], G Montagnani[1], P Paolucci[2], D Venturelli[1]. *[1]Transfusion Medicine Department, Azienda ospedaliero universitaria policlinico, Modena, Italy; [2]Department of Mother and Child, Azienda ospedaliero universitaria policlinico, Modena, Italy; [3]Department of Nephrology, Azienda ospedaliero universitaria policlinico, Modena, Italy*

Background/Case Studies: The hemorheological profile of Sickle Cell Disease (SCD) patients is not well characterized and changes in blood rheologic properties (ie blood viscosity, red blood cell [RBC] deformability) are thought to be important modulators of vascular response to transfusion and organ perfusion. Limited data exist in this field and almost all are in adult patients. The aim of this study was to evaluate the main hemorheological parameters in homozygous (HbSS) SCD pediatric patients before and after simple blood transfusion therapy. **Study Design/Methods:** Steady-state hemorheological profiles (plasmatic viscosity ηP, low shear rate viscosity ηS1, high shear rate viscosity ηS200, RBC aggregation index RAI, RBC elasticity (Taylor factor, Tk), blood mechanical spectrum G'), and hematological parameters were assessed in 24 healthy controls. The same parameters were analyzed in 36 SCD HbSS pediatric patients pre and post simple transfusion. We strictly followed the guidelines for international standardization in blood rheology measurements and interpretation (ICHS). All measurements were acquired through Haake Rotovisco RV20 (CV100 system Couette type, ZB15 sensor, Haake, Germany) at 37C. Hematocrit (Ht) was corrected to 40% in order to compare samples with different Ht. Patients were transfused with leukocyte-depleted RBC units following the Italian Oncoematology Pediatric Association (AIEOP) guidelines **Results/Findings:** SCD patients showed a significant increase in high shear rate viscosity and Tk values compared to the control group. No statistically significant modifications of the hemorheological parameters pre and post simple blood transfusion were detected. **Conclusion:** Abnormal blood rheology contributes to microcirculatory dysfunction due to increased RBC stiffness, increased extent and strength of RBC aggregation, and abnormal blood viscosity. SCD pediatric patients have a decreased RBC deformability which causes a higher microcirculatory viscosity. Hemorheological analyses of transfused patients suggest that no significant hemorheological variations would occur until 40% hematocrit is reached.

Disclosure of Commercial Conflict of Interest

M. Ballestri: No Answer; F. Fontana: No Answer; M. Lodi: Nothing to disclose; M. Maioli: No Answer; G. Montagnani: Nothing to disclose; R. Morandi: No Answer; G. Palazzi: No Answer; P. Paolucci: Nothing to disclose; D. Venturelli: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Ballestri: No Answer; F. Fontana: No Answer; M. Lodi: Nothing to disclose; M. Maioli: No Answer; G. Montagnani: Nothing to disclose; R. Morandi: No Answer; G. Palazzi: No Answer; P. Paolucci: Nothing to disclose; D. Venturelli: Nothing to disclose

Parameter	CTRL	SCD all	SCD-pre	SCD-post
Ht%	44.11 ± 4.38	28.39 ± 3.66	29.18 ± 1.11	34.48 ± 5.76
ηP (mPa*s)	1.4 ± 0.13	1.52 ± 0.28	1.43 ± 0.14	1.37 ± 0.16
ηS1C(mPa*s)	11.36 ± 4.8	21.11 ± 13.9	14.8 ± 7.5	11.7 ± 6.1
ηS200C (mPa*s)	3.61 ± 0.6	8.29 ± 2.9	6.47 ± 1.1	5.05 ± 1.6
RAIC	2.96 ± 0.95	2.62 ± 1.4	2.37 ± 1.1	2.72 ± 0.91
Tk	0.96 ± 0.1	1.18 ± 0.25	1.07 ± 0.09	0.95 ± 0.15
G'		>>	>>	>

Healthy controls (CTRL), sickle cell patients (SCD), pre transfusion (SCD-pre), post transfusion (SCD-post). CTRL vs SCD-all, ηS200C(mPa*s) $p < 0.0001$, Tk $p < 0.001$, SCD-pre vs SCD-post ηS200C(mPa*s) $p < 0.0001$. SCD-all vs CTRL G' $p = 0.03$, SCD-pre vs SCD-post G'$p = 0.03$.(Kruskal-Wallis or LSD/Waller-Duncan). Software SPSS19.0 IBM. Student's t distribution analysis R software.

S38-020D

Post-Transfusion Red Cell Survival of Neonatal Autologous and Adult Donor Biotin Labeled RBCs, Measured Concurrently, Are Not Different

R G Strauss[1] (ronald-strauss@uiowa.edu), D J Kuruvilla[2], D M Mock[3], N I Matthews[3], D Nalbant[4], R L Schmidt[4], P Veng-Pedersen[2], J A Widness[4]. *[1]Pathology & Pediatrics, University of Iowa College of Medicine, Iowa City, IA, United States; [2]Pharmaceutics, University of Iowa College of Pharmacy, Iowa City, IA, United States; [3]Biochemistry & Molecular Biology, University of Arkansas for Medical Sciences, Little Rock, AR, United States; [4]Pediatrics, University of Iowa College of Medicine, Iowa City, IA, United States*

Background/Case Studies: Determination of red cell survival (RCS) in infants is important for its clinical relevance to the evaluation of erythropoiesis in defining mechanisms of neonatal anemia. Multi-density biotin labeling of red blood cells (RBCs) (BioRBCs) in adults can be used to safely determine RCS of two or more RBC populations concurrently (Mock et al *Transfusion* 51:158, 2011). The objective was to compare RCS of neonatal autologous RBC with adult donor RBC in critically ill, very-low-birth-weight (VLBW) infants after clinical RBC transfusion. The hypothesis was that RCS of neonatal autologous RBCs, measured by BioRBCs + pharmacokinetic (PK) modeling, is substantially shorter than RCS of adult donor RBCs. **Study Design/Methods:** Separate populations of neonatal autologous and adult allogeneic RBCs were labeled at two discrete biotin densities. Both RBC populations were transfused into six premature infants (mean ± SD: birth wt: 808 ± 180 g, gest age: 25.7 ± 0.7 wk, study age: 2.5 ± 1.7 d). Serial samples (10 μL discarded blood) were analyzed for BioRBC enrichment by flow cytometry and RCS (ie, RBC life span) was calculated using PK modeling adjusted for confounders: laboratory phlebotomy, RBC transfusion, and growth (assuming constant blood volume per kg). This lifespan model assumes that, in the absence of phlebotomies, the disposition of transfused BioRBCs was due to senescence and did not vary over time. Transfused BioRBCs were assumed to have been produced under steady state. **Results/Findings:** Demographics of the study population of six premature infants and their RCS results are presented below (Table). RCS for neonatal and adult RBCs were similar: 62 ± 13 and 69 ± 14 d, respectively, $P = 0.17$. **Conclusion:** Contrary to the hypothesis and to previous published infant studies, RCS of adult and neonatal BioRBC, measured concurrently, were not different. This study demonstrates utility of concurrent multi-density BioRBC + PK modeling in determining in vivo RCS of multiple populations by flow cytometry (FC). (Supported by NIH HL046925; Thrasher Research Fund.)

Disclosure of Commercial Conflict of Interest

D. J. Kuruvilla: Nothing to disclose; N. I. Matthews: Nothing to disclose; D. M. Mock: Nothing to disclose; D. Nalbant: Nothing to disclose; R. L. Schmidt: Nothing to disclose; R. G. Strauss: Nothing to disclose; P. Veng-Pedersen: Nothing to disclose; J. A. Widness: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. J. Kuruvilla: Thrasher Research Fund, Grants or Research Support; N. I. Matthews: Nothing to disclose; D. M. Mock: Nothing to disclose; D. Nalbant: Nothing to disclose; R. L. Schmidt: Nothing to disclose; R. G. Strauss: Nothing to disclose; P. Veng-Pedersen: Nothing to disclose; J. A. Widness: Nothing to disclose

	Birth Weight (g)	GA (wk)	Age at Study (d)	Study Wt (g)	Survival of Autologous RBCs (d)	Survival of Allogeneic RBCs (d)	Initial Autologous BioRBC Enrichment	Initial Allogeneic BioRBC Enrichment	No. RBCTx During Study Period
Infant 1	721	25.7	1.3	721	78.9	81.1	0.9%	2.8%	5
Infant 2	708	26.1	5.4	746	65.4	65.4	1.2%	3.7%	6
Infant 3	914	25.7	2.7	775	61	86.9	2.2%	2.0%	5
Infant 4	813	26.0	0.5	813	67.7	75.1	3.1%	4.5%	4
Infant 5	1102	26.3	2.1	912	63.1	59.5	2.4%	3.4%	3
Infant 6	590	24.4	3.0	513	39.5	48.3	2.3%	2.6%	9
Mean	808	25.7	2.5	747	62.6	69.4	2.0%	3.2%	5.3
SD	180	0.7	1.7	133	12.9	14.4	0.8%	0.9%	2.1

S39-020D

Hemoglobin Variants Acquired Post Transfusion in Pediatric Sickle Cell Disease (SCD) Patients

S Pandey[1,2] (spandey@uams.edu), G Pesek[1,2]. *[1]Pathology, University of Arkansas for Medical Sciences, Little Rock, AR, United States; [2]Pathology, Arkansas Childrens' Hospital, Little Rock, AR, United States*

Background/Case Studies: Many SCD patients receive chronic transfusion support for disease related complications. At our pediatric institution, the goal of transfusion is to maintain a hematocrit at 25-30% and Hemoglobin S (HbS) concentration of less than 30%. Hb variants are prevalent in the general population; however, reports of transfusion-acquired Hb variants are rare. Three cases of acquired Hb variants in SCD patients seen at our institution are reported here. **Study Design/Methods:** SCD patients receive pre- and post-exchange transfusion hemoglobin electrophoresis (HBE). A retrospective review found three patients with new bands on post transfusion specimens during 2008-2013. Donor red cell segments were traced and HBE was performed to confirm the findings. **Results/Findings:** The first patient received four red cell units. Post transfusion HBE revealed a new band that corresponded with HbJ variant and constituted 6.5% (Table). An identical band was found on one donor unit comprising 52.5%. The other two patients received three and four red cell units respectively. Post transfusion specimens on both patients revealed a new band in the HbC region comprising 14.4% and 19.6% respectively (Table). Identical bands were found on donor segments comprising 52.5% and 74.4% respectively. None of the patients showed transfusion-associated adverse events. The donor center was contacted in these cases and the followup is unknown. **Conclusion:** There are at least 58 HbJ variants described with little to no clinical significance, but only one case of an acquired HbJ variant in the literature to date. There have been occasional reports of transfusion-acquired HbC, including one study where it is reported as the most common cause of transfusion-acquired hemoglobin variant. Heterozygous donors are asymptomatic and show no abnormalities during donor screening. Since HBE is not routinely performed on the donor specimen it may go unrecognized until the post transfusion recipient results pose diagnostic difficulties. There are no definitive guidelines on deferring these donors, hence one should be cognizant to prevent misdiagnosis. Additionally, donor centers need mechanisms to confirm the findings and counsel donors as needed. In a population where HbS negative blood is routinely requested, the effect of other Hb variants remains unknown. This is of special concern in the pediatric population where a single RBC unit can contribute a significant portion of the exchanged blood volume.

Disclosure of Commercial Conflict of Interest

S. Pandey: Nothing to disclose; G. Pesek: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Pandey: Nothing to disclose; G. Pesek: Nothing to disclose

Patients' pre and post transfusion hemoglobin electrophoresis results

	Patient 1		Patient 2		Patient 3	
Hemoglobin	Pre-transfusion (%)	Post-transfusion (%)	Pre-transfusion (%)	Post-transfusion (%)	Pre-transfusion (%)	Post-transfusion (%)
Hb-A	81.1	87.6	49.8	74.2	77.6	71.7
Hb-F	0.5	—	12.8	2.2	—	—
Hb-A2	2.7	2.5	3.0	2.7	3.1	3.3
Hb-S	15.7	3.4	34.4	6.5	19.3	5.4
Hb J variant	—	6.5	—	—	—	—
Hb-C	—	—	—	14.4	—	19.6

S40-020D

Carboplatin and Vincristine Induced Hemolytic Anemia in a Pediatric Patient with an Optic Pathway Glioma

M Betensky[2] (Betenskym@email.chop.edu), M J Fisher[2], S J Nance[3], M Weiss[2], C Witmer[2], D Sesok-Pizzini[1]. *[1]Pathology and Laboratory Medicine, The Children's Hospital of Philadelphia, Philadelphia, PA, United States; [2]Department of Pediatrics, The Children's Hospital of Philadelphia, Philadelphia, PA, United States; [3]Biomedical Services, IRL, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: Drug-induced immune hemolytic anemia (DIHA) is an uncommon event. We report a case of acute hemolysis associated with vincristine and carboplatin in a 35-month-old female with neurofibromatosis type I and an optic pathway glioma. The patient tolerated a year of chemotherapy with only minor allergic symptoms. During her last two cycles, she reported back and abdominal pain associated with episodes of dark urine days after her chemotherapy. After the last cycle, her hemoglobin fell from 12.9 to 3.9 g/dL over 12 days. The reticulocyte count peaked at 25%, LDH at 1130 U/L and haptoglobin was <10 mg/dL. Her peripheral blood smear revealed spherocytes. Serologic testing for DIHA was performed. **Study Design/Methods:** Serum antibody identification, acid eluate, and drug studies with serum and acid eluates were performed using carboplatin and vincristine. Two potential cross-reactive drugs, oxaliplatin and cisplatin, were also investigated. **Results/Findings:** An initial DAT was negative, but a DAT drawn 3 days later was 3+ with polyspecific-AGT, weak positive with monospecific anti-IgG, and 2+ with monospecific anti-C3d. No alloantibody to common red cell antigens were detected when the serum was tested against reagent red cell at the IS, RT, 37 degrees C-ALB and ALB-IgG-AGT phases. The patient's serum reacted with all reagent and autogeneic red cells tested at the Papain-IgG-AGT and PEG-IgG-AGT phases with no apparent specificity. No antibodies to oxaliplatin, cisplatin, or carboplatin were detected in the acid eluate method, but weak reactivity with vincristine using drug coated cells was shown. The patient's serum was also reactive with vincristine and carboplatin by the drug solution addition method, but only reactive with carboplatin using the drug coating method. Serum titers for carboplatin and vincristine were 2 and 0 at 37C phase and 256 and 8 at polyspecific-AGT, respectively. No antibodies to oxaliplatin or cisplatin were detected in the patient's serum. **Conclusion:** The patient experienced unexpected acute hemolysis associated with antibodies against carboplatin and vincristine. Carboplatin is reported to be associated with DIHA and detected by the "immune complex" or drug solution addition, drug adsorption, and autoantibody mechanisms. Despite the lower titers, vincristine-associated DIHA may be contributory in this case. The patient's hemolysis and positive direct antiglobulin test (DAT) abated with discontinuation of chemotherapy. Subsequently, her tumor increased in size and an alternative chemotherapy regimen was instituted. This case illustrates the importance of recognizing early the clinical signs of DIHA and the utility of serological testing to support the diagnosis to prevent possible life threatening hemolysis.

Disclosure of Commercial Conflict of Interest

M. Betensky: Nothing to disclose; M. J. Fisher: Nothing to disclose; S. J. Nance: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose; M. Weiss: Nothing to disclose; C. Witmer: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Betensky: Nothing to disclose; M. J. Fisher: Nothing to disclose; S. J. Nance: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose; M. Weiss: Nothing to disclose; C. Witmer: Nothing to disclose

S41-020D

Co-Infusion of Dextrose-Containing Solutions and Packed Red Blood Cells Does Not Adversely Affect in Vitro Red Blood Cell Quality

A Keir[1] (amy.keir@sickkids.ca), A Hansen[2], R Jankov[1], J Callum[3], J Acker[2,4]. *[1]Department of Paediatrics, University of Toronto, Toronto, ON, Canada; [2]Research and Development, Canadian Blood Services, Edmonton, AB, Canada; [3]Department of Laboratory Medicine and Pathobiology, University of Toronto, Toronto, ON, Canada; [4]Department of Laboratory Medicine and Pathology, University of Alberta, Edmonton, AB, Canada*

Background/Case Studies: Many neonatal intensive care units withhold milk feeds from preterm infants prior to, during, and post-transfusion, with the aim of reducing the risk of transfusion-associated necrotising enterocolitis. Additional intravenous points to infuse dextrose-containing solutions (DS) or heel-pricks to monitor blood glucose levels may be required to avoid hypoglycemia. Vascular access can be challenging and may lead to iatrogenic injury. Co-infusion of red blood cells (RBCs) with DS has the potential to significantly benefit this patient population, however, transfusion guidelines advise against co-infusing RBCs with solutions other than 0.9% saline. This study collected in vitro data to evaluate the impact of co-infusion with DS on RBC quality. **Study Design/Methods:** Two sets of nine pools of RBCs were stored for 5 days (n = 3), 14 days (n = 3), or 21 days (n = 3) and were irradiated ≤2 h before use. Phase 1: Samples from one set of pooled RBCs were infused through a neonatal transfusion apparatus alone or with treatment solutions: 5% DS; 10% DS 5% DS/0.2% saline; and 0.9% saline. Infusion pumps ran at 15 mL/h until a sample of 5 mL was obtained. Phase 2: Samples (5 mL) from the second set of pooled RBCs were placed in tubes alone or in a 1:1 ratio with treatment solutions. Tubes were incubated at room temperature and tested after 5, 30 and 180 min. Hemolysis, supernatant potassium, red blood cell (RBC) indices, morphology, and deformability were measured on all samples in both phases. **Results/Findings:** Phase 1: RBCs transfused alone through the apparatus had higher ($p < 0.01$) hematocrit, total hemoglobin, and supernatant potassium, and took approximately twice the time to collect the required sample volume compared to all other groups. No statistical differences were identified between groups transfused through the apparatus for hemolysis, mean cell volume (MCV), morphology, and deformability. Phase 2: MCV of samples containing DS increased with incubation length and were higher ($p < 0.01$) than RBCs incubated alone or with 0.9% saline after 30 and 180 min. RBCs incubated with DS for 30 and 180 min had lower ($p < 0.01$) maximum elongation indexes compared to RBCs incubated alone or with 0.9% saline. RBCs incubated with 5% DS and 5%/0.2% saline had higher ($p < 0.01$) hemolysis than RBCs incubated alone after 5, 30 and 180 min. Hemolysis and supernatant potassium in RBCs incubated with 5% DS and 5%DS/0.2% saline increased ($p < 0.01$) with incubation length to become higher ($p < 0.01$) than RBCs incubated with 0.9% saline after 180 min. **Conclusion:** In vitro quality of RBCs that were co-infused with 0.9% saline or 10% DS were not adversely impacted. Caution is warranted, however, when developing clinical guidelines as the co-infusion conditions (flow rate, incubation time and DS) can affect the quality of the transfused RBCs. Co-infusion of DS may be permissible under circumstances clinically relevant to the NICU.

Disclosure of Commercial Conflict of Interest

J. Acker: Fresenius Kabi, Grants or Research Support; J. Callum: Nothing to disclose; A. Hansen: Nothing to disclose; R. Jankov: Nothing to disclose; A. Keir: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Acker: Aquila Diagnostic Systems Inc., Ownership or Partnership, HemoCue, Travel Support or Honorarium, Fresenius Kabi, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support; J. Callum: Nothing to disclose; A. Hansen: HemoCue, Grants or Research Support, Haemonetics Corp., Grants or Research Support, Fenwal Inc., Grants or Research Support, Fresenius Kabi, Grants or Research Support; R. Jankov: Nothing to disclose; A. Keir: Nothing to disclose

S42-020D

A Study of Blood Utilization in Pediatric Cardiac Surgery

E Meyer[1,2] (emeyer5@emory.edu), C Josephson[1,2], S Pahz[2], P Jessup[2], N Guzzetta[3,4]. *[1]Pathology and Lab Medicine, Emory University School of Medicine, Atlanta, GA, United States; [2]Blood and Tissue Services, Children's Healthcare of Atlanta, Atlanta, GA, United States; [3]Anesthesiology, Emory University School of Medicine, Atlanta, GA, United States; [4]Anesthesiology, Children's Healthcare of Atlanta, Atlanta, GA, United States*

Background/Case Studies: Effective management of red blood cell (pRBC) inventory is a transfusion service priority. A crossmatch to transfusion ratio of 2:1 has been historically recommended by the College of American Pathologists to assist in blood utilization assessment. This recommendation is based on benchmarking in adult patients and does not include neonates <4 months of age who typically do not require a crossmatch prior to pRBC transfusion. Previous studies have suggested that a preparation to transfusion (P : T) ratio would more accurately reflect blood utilization in the pediatric population, though few benchmarking studies exist to suggest a target ratio (Schmotzer, et al *Transfusion*, 2010). To determine the efficiency of our blood inventory management, the P : T ratios and discard-volumes of pRBCs were evaluated intraoperatively and postoperatively for children and neonates undergoing cardiothoracic surgery at a large pediatric institution. It was hypothesized that the P : T ratios would be greater than 2:1 with a wastage percentage of 3%. **Study Design/Methods:** A retrospective review of all cardiac surgeries performed by three pediatric cardiac surgeons was undertaken from January-March 2012. For the purposes of this study, a single CPDA-, 1 leukocyte-reduced pRBC unit had an assumed volume of 250 mL. All crossmatches were either immediate spin or at Coombs phase. Data collected included patient sex, age, preoperative diagnosis, preoperative laboratory values (hemoglobin [Hgb], hematocrit [Hct], platelet count, INR), procedure performed, ABO/Rh status, volume of pRBC allocated/prepared/crossmatched, volume of pRBC transfused intraoperatively, and volume of pRBC transfused and discarded 24 hours prior to and after surgery. Wastage was calculated by pRBC median volume discarded divided by the median volume allocated. **Results/Findings:** Over a 2-month period, 73 children, 41% female, with a median age of 7 months (0 days-15 years) underwent 76 cardiac surgeries (88% on cardiopulmonary bypass pump). Preoperative laboratory values were available on 65 patients: median Hb 12.9 g/dL (9.3-16.4 g/dL), median platelet count 306 × 10e9/μl (83-635 × 10e9/μl), and median INR 1.0 (0.8-1.0). The median volume allocated/prepared/crossmatched was 1000 mL (45-1500 mL) with a median volume transfused of 250 mL (0-1000 mL). The P : T ratio of pRBCs for the 76 surgeries was 4:1. pRBC wastage was 10% with a median volume discarded of 100 mL (40-210 mL). **Conclusion:** The preliminary results of this retrospective review reveal a P : T ratio of 4:1 for children and neonates undergoing cardiac surgery, which is consistent with the hypothesis. Furthermore, although the wastage rate was higher than hypothesized, the total volume wasted was less than half of a CPDA-1 pRBC unit. Additional benchmarking studies will be necessary to determine optimal P : T ratios and acceptable wastage for common pediatric cardiac surgical procedures.

Disclosure of Commercial Conflict of Interest

N. Guzzetta: Nothing to disclose; P. Jessup: Nothing to disclose; C. Josephson: Nothing to disclose; E. Meyer: Nothing to disclose; S. Pahz: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Guzzetta: Nothing to disclose; P. Jessup: Nothing to disclose; C. Josephson: Immucor/Bioarray, Grants or Research Support, Octapharma, Consulting or Board of Director Fees; E. Meyer: Nothing to disclose; S. Pahz: Nothing to disclose

Donor Recruitment, Retention and Adverse Events

S43-030G

Development of a High-Throughput One-Step Method for the Identification of IgA Deficient Blood Donors

R Bazin[1] (renee.bazin@hema-quebec.qc.ca), T Tremblay[1], J Courtemanche[1], I Paré[1], P Rouleau[1], H Emond[1], M Drouin[1]. *[1]Héma-Québec, Québec, QC, Canada*

Background/Case Studies: A registry of IgA deficient donors has to be maintained to provide compatible blood components to IgA deficient patients, at the hospital's request. In this study, screening of blood donors for

IgA deficiency to compensate for the loss of active donors in the registry was recently reinitiated. Several screening methods mostly based on ELISA exist but, while being very reliable, they are also time-consuming. Typically, sample preparation and loading in microplates take about 2 and 7 hours for 1000 samples, respectively, for one technician. The ELISA technique itself requires about 4 hours to be completed. The objective is to improve the efficiency of screening. **Study Design/Methods:** A fluorescent-based competition assay was set up, requiring only one incubation step. The workload for sample distribution in microplates has been considerably reduced by using the Hamilton Microlab AT Plus 2 automated dispenser (located at our regulatory testing lab in Montreal) to collect plasma samples (50 ul) from the left-over tubes used for ABO/Rh testing, in IgA-coated 96-well microplates. The microplates are then frozen to prevent spillage and shipped to the R&D department in Quebec City. For testing, the microplates are first thawed and anti-human IgA-FITC conjugate is added directly to each well. The plates are incubated for one hour at 37C, washed, emptied, and read on a fluorometer. The anti-human IgA-FITC conjugate is neutralized by the IgA contained in normal plasma, resulting in background fluorescence. In contrast, IgA deficient plasma does not interfere with the binding of anti-human IgA FITC conjugate to IgA-coated wells, yielding high fluorescence and allowing the easy identification of IgA deficient plasma. **Results/Findings:** The overall procedure is completed within 2 hours, compared to more than 12 hours using a standard ELISA procedure for the same number of samples. Using this method, 10,763 samples from January 16 to January 31 2013 and identified 23 IgA deficient donors (frequency of 1/468). These samples were retested using our in-house cytometry-based assay for preliminary confirmation of the IgA deficiency status. All samples were IgA-negative, showing the specificity of our screening method. **Conclusion:** A cost-effective and efficient method for the screening of thousands of blood donors for IgA deficiency has been developed.

Disclosure of Commercial Conflict of Interest

R. Bazin: Nothing to disclose; J. Courtemanche: Nothing to disclose; M. Drouin: Nothing to disclose; H. Emond: Nothing to disclose; I. Paré: Nothing to disclose; P. Rouleau: Nothing to disclose; T. Tremblay: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Bazin: Nothing to disclose; J. Courtemanche: Nothing to disclose; M. Drouin: Nothing to disclose; H. Emond: Nothing to disclose; I. Paré: Nothing to disclose; P. Rouleau: Nothing to disclose; T. Tremblay: Nothing to disclose

S44-030G

Evaluation of Double Red Cell Donors with High Ferritin Levels

G Delage[1] (gilles.delage@hema-quebec.qc.ca), Y Grégoire[1], A Lebrun[1].
[1]Medical Affairs, Hema-Québec, Saint-Laurent, QC, Canada

Background/Case Studies: Upon introduction of double red cell collections (DRCC) at Héma-Québec (HQ), at the request of the regulatory authorities, a pilot ferritin monitoring program was put in place. This study reports the findings in donors with high ferritin levels. **Study Design/Methods:** All donors recruited to the double red cell apheresis program were tested for ferritin levels using the Modula E170 Immunochemical Analyser (Roche Diagnostics). Venous specimens were centrifuged within 2 hours of collection before being sent to an outside laboratory for analysis. All donors with high ferritin levels were solicited for a follow-up study. Upon receipt of signed informed consent, donors were seen by a trained nurse who completed a medical questionnaire, carried out a short physical examination, and drew blood for additional tests. The following battery of tests was done on each donor: CBC; sedimentation; C-reactive protein; hemoglobin and protein electrophoresis; haptoglobin; glucose; creatinine; ALT; AST; bilirubin; LDH; folic acid; B12; transfusion; ferritin; total serum iron and % saturation; and tests for C282Y and H63D mutation. They were then seen by one of the investigators (GD) with results for counselling purposes. This study was approved by the HQ Institutional Review Board. **Results/Findings:** By November 2012, 2546 donors had been recruited. Of these, 155 had elevated ferritin, of which 6 were known to suffer from hereditary hemochromatosis and were not investigated any further. 80 donors accepted to participate in the study. To date, genetic analysis results are available for 44 donors: C282Y homozygous (1); double C282Y/H63D heterozygote, (2); wild type/wild type, (24); C282Y heterozygote, (5); H63D heterozygote, (12). Upon retesting, 44 had normal, 3 had low, and 13 had elevated ferritin. Of those with repeatedly high ferritin levels, 2 had elevated iron levels with increased % saturation, and 11 had normal iron levels. The 2 with evidence of iron overload were C282Y homozygote and C282Y/H63D double heterozygote respectively. **Conclusion:** 155 of the 2542 donors had elevated ferritin. Of the 44 for whom genetic analysis was available at the time of preparation of this abstract, only 3 had mutations compatible with hemochromatosis: of these three, two were the only donors with biochemical evidence of iron overload. These data will be useful in informing policy on iron supplementation for blood donors.

Disclosure of Commercial Conflict of Interest

G. Delage: Nothing to disclose; Y. Grégoire: Nothing to disclose; A. Lebrun: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Delage: Nothing to disclose; Y. Grégoire: Nothing to disclose; A. Lebrun: Nothing to disclose

S45-030G

Factors Associated with Absent Iron Stores (AIS) in Male and Female Donors Tested for Ferritin

B Custer[1] (bcuster@bloodsystems.org), M D Bravo[2], P A Tomasulo[2], M P Busch[1], H Kamel[2]. *[1]Blood Systems Research Institute, San Francisco, CA, United States; [2]Medical Affairs, Blood Systems, Inc., Scottsdale, AZ, United States*

Background/Case Studies: To reduce iron deficiency in donors, a network of blood centers began screening donors for ferritin. In otherwise unselected donors, ferritin testing is conducted on donations from male donors with hemoglobin (Hb) of 12.5-13.4 g/dL and female donors with Hb of 12.5-12.9 g/dL. This study evaluates donor and donation factors associated with AIS (ferritin <12 mcg/L) among those tested for ferritin. **Study Design/Methods:** Data were analyzed from the first 4 months of screening. Proportions of AIS and multivariable analysis (MVA) logistic regression models were estimated for factors associated with AIS by gender for those screened for ferritin. Factors included in the analysis were: age; BMI; EBV; ABO blood type; education; pulse; systolic and diastolic blood pressure; counts of red blood cells (RBC) or plasma and/or platelet (PP) donation in the prior 2 years; and center. Odds Ratios (OR) and 95% Confidence Intervals (CI) are reported. **Results/Findings:** There were 226,470 allogeneic donations (53% male) during the study period. Of 120,208 donations from males, 7064 (6%) were tested for ferritin, leading to 3789 (3%) deferrals for low ferritin (LF, <30 mcg/L); of these, 48% had AIS. Of 106,262 donations from females, 16,658 (16%) were tested, leading to 8798 (8%) deferrals for LF (<20 mcg/L); of these, 59% had AIS. Factors associated with AIS were similar for males and females. MVA showed that the increasing number of RBC donations in the prior 2 years was the most significant factor associated with AIS. The magnitude of the effect was greatest for males (OR 22.3, 95%CI 17-30) who gave 10+ RBC in 2 years compared to males with no previous donation in the 2-year-period and for females (OR 6.3, 95%CI 4.8-8.1) with 10+ RBC donations compared to females with no previous donation in the 2-year-period. Independently, donors with ≥14 PP donations in the previous 2 years had significantly higher odds of AIS (OR 3.2, 95%CI 2.3-4.9 for males and 2.1, 1.6-2.7 for females) compared to no PP donations. Many other factors were also associated with increased odds of AIS, including younger age in females, and higher education in males. Factors associated with decreased OR for AIS were obese and extremely obese (BMI ≥30) male and female donors compared to overweight donors (BMI 25-29.9); and in males, low pulse (<65 bpm) compared to 65-90 bpm. Native American donors had higher ORs and African Americans had lower ORs compared to White race groups. **Conclusion:** While factors are similar for males and females, there are distinct differences in the magnitude of odds of AIS, which may reflect underlying differences in iron homeostasis by gender. The results indicate that management and policy decisions need to be gender-based. Furthermore, while increasing RBC donation in the previous 2 years is clearly associated with AIS, donors who frequently give PP are also at increased risk for AIS and should be advised appropriately.

Disclosure of Commercial Conflict of Interest

M. D. Bravo: Nothing to disclose; M. P. Busch: Nothing to disclose; B. Custer: Nothing to disclose; H. Kamel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. D. Bravo: Nothing to disclose; M. P. Busch: Novartis, Grants or Research Support Novartis, Travel Support or Honorarium; B. Custer: Novartis Vaccines and Diagnostics, Grants or Research Support; H. Kamel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

S46-030G

Limited Impact on Donor Iron Status from Longer Donation Intervals and Higher Male Donor Hemoglobin Cutoff: Results of Simulation Models Using REDS-II Data

B R Spencer[1] (bryan.spencer@redcross.org), D J Wright[2], G Simone[3], A E Mast[4], S Kleinman[5], J E Kiss[6], R J Benjamin[7], M P Busch[8], R G Cable[1]. *[1]American Red Cross, Dedham, MA, United States; [2]Westat, Rockville, MD, United States; [3]Division of Blood Diseases and Resources, National Heart, Lung and Blood Institute, Bethesda, MD, United States; [4]BloodCenter of Wisconsin, Milwaukee, WI, United States; [5]AABB, Bethesda, MD, United States; [6]Institute for Transfusion Medicine, Pittsburgh, PA, United States; [7]Holland Laboratories, American Red Cross, Rockville, MD, United States; [8]Blood Systems Research Institute, San Francisco, CA, United States*

Background/Case Studies: Depletion of iron stores is a well-documented outcome in frequent blood donors. Measures being considered to mitigate this risk, and to prevent the collection of blood from anemic male donors, include extending the minimum whole blood donation interval beyond 8 weeks and raising the minimum hemoglobin (Hb) level in male donors above 12.5 g/dL. Comprehensive estimates of the efficacy of these measures in reducing the prevalence of low iron in the blood donor population are lacking. **Study Design/Methods:** Six blood centers participating in Retrovirus Epidemiology Donor Study (REDS)-II systematically collected information on all blood donor visits from 2006-2009. Data on component type or, if deferred, cause for deferral, were compiled. Simulations were developed using a multi-stage approach that adequately reproduced the observed patterns of donor return, Hb and ferritin levels, and outcome of donor visit (successful donation of one or two red blood cell (RBC) components, deferral for Hb < 12.5, or other reason) seen in the REDS-II datasets. In the ferritin models, ferritin values were simulated using the model from the REDS-II Iron Evaluation Study RISE. Modified simulations were used to predict the potential impact on donor iron status and blood collections using different Hb cutoff and inter-donation interval qualification criteria. **Results/Findings:** Extrapolation of RISE inputs to the REDS-II donor population suggested that 13.1% of donors making a RBC donation have absent iron stores (AIS), defined as ferritin < 12 ng/mL. Increasing the minimum Hb cutoff in male donors by 0.5 or 1.0 g/dL was estimated to have no measurable impact on prevalence of AIS in those accepted for donation but would result in 1.1-4.0% fewer units collected. (Sim C and D in Table). In contrast, lengthened intervals between donations were estimated to lower the proportion of blood donors with AIS, by 1.5% to 2.3%. However, 80-90% of existing donors with AIS would remain iron deficient even at intervals of 12 to 16 weeks. These alternate intervals will impact blood availability with estimates of 7.7 to 13.3% fewer available units, respectively. **Conclusion:** These simulations suggest that higher Hb cutoffs in males and longer inter-donation intervals in both genders will have limited efficacy in preventing significant iron-deficiency in donors. More targeted management of high-risk donors, such as ferritin testing or providing iron supplements, might prove to be more effective measures with lower impact on blood availability.

Disclosure of Commercial Conflict of Interest

R. J. Benjamin: Nothing to disclose; M. P. Busch: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; S. Kleinman: Nothing to disclose; A. E. Mast: Nothing to disclose; G. Simone: Nothing to disclose; B. R. Spencer: Nothing to disclose; D. J. Wright: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. J. Benjamin: Nothing to disclose; M. P. Busch: Novartis, Grants or Research Support, Novartis, Travel Support or Honorarium; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; S. Kleinman: Nothing to disclose; A. E. Mast: Novo Nordisk, Grants or Research Support; G. Simone: Nothing to disclose; B. R. Spencer: Nothing to disclose; D. J. Wright: Nothing to disclose

Simulated Change in Donor Iron Depletion and Red Cell Collections Under Modified Policies

SimID	Hgb cutoff Female	Male	Donation Interval Female	Male	% RBC Donations With Ferr < 12	Red Cell Components Donated
Baseline	12.5	12.5	8	8	13.1%	2,112,497
Sim A	12.5	12.5	12	12	11.6%	Δ = –7.7%
Sim B	12.5	12.5	16	16	10.8%	Δ = –13.3%
Sim C	12.5	13.0	8	8	13.1%	Δ = –1.1%
Sim 6	12.5	13.5	8	8	13.1%	Δ = –4.0%
Sim E	12.5	13.5	12	12	11.6%	Δ = –10.0%
Sim F	12.5	13.5	16	16	10.8%	Δ = –14.8%

S47-030G

Quantification of Iron Stores and Iron Absorption in REDS-II Iron Evaluation Study (RISE) Donors

J E Kiss[1] (jkiss@itxm.org), R G Cable[2], W R Steele[3], D J Wright[4], J Bethel[4], R J Birch[4]. *[1]Institute for Transfusion Medicine, Pittsburgh, PA, United States; [2]American Red Cross, Farmington, CT, United States; [3]Transmissible Diseases, Holland Lab, American Red Cross, Washington, DC, United States; [4]Westat, Rockville, MD, United States*

Background/Case Studies: The amount of iron removed with each blood donation, ~230 mg, represents about 25% of the average body iron stores in men and nearly 75% in women. The National Health and Nutrition Examination Survey (NHANES) estimated iron stores in the general population to be 9.82 + 2.82 mg/kg in men (M) and 4.87 + 4.14 mg/kg in women (F). Using data from RISE, changes in blood donor iron status were analyzed in active donors to quantify the effects of repeat donation on iron stores and absorption. **Study Design/Methods:** RISE enrolled First-time/Reactivated (FT/RA) and Frequent (Frqt) blood donors in a15-24 month prospective cohort study. Iron stores were estimated from the known linear relationship between iron stores and the log ratio of serum transferrin receptor/ferritin: Iron Stores (mg/kg) = –[log(sTfR/ferritin)–2.8229]/0.1207 (Cook J. *Blood* 2003;101:3359); and red cell iron based on estimates of red cell volume and iron content of hemoglobin. These iron status measures were calculated at enrollment and during the final study visit, 15-24 months later. Absorbed iron per day was calculated as: (Iron lost from blood donation during study [calculated from red cell loss] + Obligate losses [estimated]—(Decrease in Iron Stores + Decrease in Red Cell Iron), then divided by days on study. **Results/Findings:** FT/RA donors at enrollment (Table) had comparable iron stores to the NHANES general population estimates seen above. During the 15-24 month follow-up period, Frqt donors had more study donations than FT/RA donors (Frqt M 7.5, F 6.2 donations v. FT/RA M 3.5, F 3.1 donations) and greater mean iron loss (Frqt M 2642 mg, F 2174 mg v. FT/RA M 2003 mg, F 1807 mg). However, Frqt donors had little change in iron stores, whereas the percentage of FT/RA donors with estimated iron stores ≤0 mg/kg increased between 4-5-fold. The estimated amount of iron absorbed per day on study for Frqt donors was higher compared to FT/RA donors (Table). **Conclusion:** This study is the first to provide quantitative data on iron absorption and changes in iron stores in a broad spectrum of blood donors. In contrast to ferritin measurement, the model can more precisely assess the degree of iron depletion, which can be analyzed as a continuous variable in determining iron deficits. Substantial differences in iron absorption among blood donors are present, which may reflect both the stress of repeated red cell loss, as well as individual differences in iron absorption that need further characterization. The model is particularly suited to assess the impact of mitigation measures such as iron supplementation.

Disclosure of Commercial Conflict of Interest

J. Bethel: Nothing to disclose; R. J. Birch: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; W. R. Steele: Nothing to disclose; D. J. Wright: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Bethel: Nothing to disclose; R. J. Birch: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: Nothing to disclose; W. R. Steele: Nothing to disclose; D. J. Wright: Nothing to disclose

	Plasma Ferritin (ng/mL) at Enrollment	Plasma Ferritin (ng/mL) at Final	Iron Stores at Enrollment (mg/kg)	Iron Stores at Final (mg/kg)	% with Iron Stores ≤0 mg/kg, Enrollment	% with Iron Stores ≤0 mg/kg, Final	Absorbed Iron on Study (mg/day)
FT/RA Females (n = 181)	43.0 (38.1, 48.6)	22.8 (20.3, 25.6)	6.7 (6.2, 7.1)	4.2 (3.7, 4.7)	2.2%	11.1%	2.76 (2.64, 2.87)
FT/RA Males (n = 143)	104 (92.9, 116)	42.6 (37.3, 48.7)	9.8 (9.4, 10.2)	6.6 (6.0, 7.1)	0.7%	2.8%	2.89 (2.75, 3.04)
Frqt Females (n = 478)	17.7 (16.7, 18.7)	18.7 (17.5, 20.0)	2.9 (2.6, 3.1)	3.3 (3.0, 3.6)	17.2%	15.5%	3.75 (3.67, 3.84)
Frqt Males (n = 506)	25.0 (23.4, 26.8)	24.1 (22.4, 25.9)	4.1 (3.8, 4.4)	4.2 (3.8, 4.5)	14.0%	14.2%	4.49 (4.41, 4.58)

Ferritin expressed as geometric mean; Other values expressed as mean (95%CI)

S48-030G

An Open, Randomised, Clinical Study of Oral Versus Intravenous Iron for Iron Substitution in Blood Donors

B E Ekermo[1] (Bengt.Ekermo@lio.se), P Forsberg[2], G Schedvin[3], G Berlin[1]. *[1]Dept. of Clinical Immunology & Transfusion Medicine, University Hospital, Linköping, Sweden; [2]Transfusion Medicine Wermland, Central Hospital, Karlstad, Sweden; [3]Dept. of Clinical Chemistry & Transfusion Medicine, Central Hospital, Växjö, Sweden*

Background/Case Studies: In a recent study,* a high frequency of iron deficiency and restless legs syndrome (RLS) was found in blood donors, and intravenous (IV) iron substituted iron loss more efficiently than oral iron. This study was performed to test the hypothesis that infusion of a greater amount of iron is more effective in maintaining hemoglobin (Hb) and iron levels in regular blood donors compared with a substitution regimen with oral iron. **Study Design/Methods:** 210 donors with ≥5 previous whole blood donations were randomised to one of two iron substitution strategies: 1) infusion of dextriferron 1000 mg after first donation in the study and then 200 mg after each donation; and 2) 20 tablets of 100 mg Fe2+ after each donation. Female donors were followed for four donations and male donors for five donations. The primary objective was to measure the difference in change in Hb between the two groups at visit 2, compared to baseline. Secondary objectives were: 1) to measure the difference in change in blood Hb between the two groups at the third, fourth, and fifth (males only) blood donation; and 2) to compare the two treatment groups regarding effects on iron status, health status (SF-36), International RLS Study Group Rating Scale score and fatigue status at each blood donation; and 3) to compare the two treatments regarding the incidence and severity of adverse events, tolerance, and compliance. **Results/Findings:** No difference was shown in change in Hb levels between the treatment groups at the second blood donation and throughout the trial. Both treatment groups had mean ferritin levels between 30 and 40 μg/L at baseline. There was a significantly greater increase in the levels of ferritin in group A at the second blood donation and throughout the study compared with group B. There were no significant differences between the two treatment groups regarding fatigue or the subsets of data concerning the SF-36 questionnaire from the first to the last blood donation. Analyses of serum hepcidin showed significant increase in group A compared with group B, and the levels of soluble transferrin receptor showed a significantly greater decrease in group A compared with group B. One third of the blood donors were iron deficient according to the PREFER criteria,† and in this subpopulation, treatment A led to a significant higher increase in Hb levels compared with group B, and also to improved female fatigue scores. The incidence of blood donors fulfilling the RLS criteria was 7%. No serious adverse reactions occurred. **Conclusion:** No difference was seen between the two treatment groups regarding changes in Hb. However, in the subgroup of donors with iron deficiency. IV iron increased hemoglobin and iron stores more efficiently than oral iron. Both drugs are safe to use for iron substitution in blood donors. * Birgegård et al, Vox Sang 2010; 99, 354-361 †Favrat B et al. Abstract 0405, European Hematology Association conference, Amsterdam, June 14-17, 2012

Disclosure of Commercial Conflict of Interest

G. Berlin: Vifor Pharma Nordiska AB, Grants or Research Support; B. E. Ekermo: Vifor Pharma Nordiska AB, Grants or Research Support; P. Forsberg: Vifor Pharma Ltd., Grants or Research Support; G. Schedvin: Vifor Pharma Nordiska AB, Grants or Research Support

Disclosure of Grants Conflict of Interest

G. Berlin: MacoPharma, Grants or Research Support; B. E. Ekermo: Nothing to disclose; P. Forsberg: Nothing to disclose; G. Schedvin: Nothing to disclose

Transfusion Practice

S49-030H

MELD Score and Preoperative INR-Based Stratification of Liver Transplantation Outcome

A F Brown[1] (browna6@uthscsa.edu), G Meny[1]. *[1]Pathology, University of Texas HSC, San Antonio, TX, United States*

Background/Case Studies: Organ allocation based on the model for end-stage liver disease (MELD) has been used to stratify patients on the waiting list for liver transplantation. Patients with a higher MELD score are prioritized for liver transplantation (Liver Transpl 2006; 12: 614). The impact of this model on perioperative management and graft outcome has not been completely assessed. The objective of this review was to assess correlation between MELD score and perioperative outcome. **Study Design/Methods:** The records of all adult liver transplants from 2008-2012 at a tertiary care university hospital were retrospectively reviewed. Patients were divided into groups as follows: low (</ = 20) and high MELD (>20) score; low (</ = 1.5) and high (>1.5) preoperative international normalized ratio (INR); and cases with or without concurrent hepatocellular carcinoma (HCC). Fresh frozen plasma (FFP) transfusion, total hospital stay, and graft-related mortality at 6 months were compared using paired T-test within each group. **Results/Findings:** A total of 227 cases were obtained which included 144 viral or alcohol-related, 27 cryptogenic, 24 autoimmune, 12 non-alcoholic steatohepatitis (NASH), and 20 other cases; 86 cases had HCC. The median intraoperative FFP transfusions for all cases was 11+/–11 units with significantly higher units transfused to patients with high MELD score (14+/–11 versus 6+/–4; p = <0.01) and high preoperative international normalized ratio (INR) (14+/–10 vs 8+/–10; p = <0.01). Additionally, high MELD score and high preoperative INR were associated with increased graft-related mortality (relative risk = 3.35 and 1.76, respectively) and length of hospital stay (6.5 and 4.5 additional days, respectively; p = <0.01). High MELD score had moderate direct correlation (r2 > 0.3) to length of hospital stay, preoperative INR, and FFP transfusions (r2 of 0.66, 0.39, and 0.37, respectively). Evaluation of cases with concurrent HCC (n = 86) revealed a generalized decrease in MELD score, preoperative INR, length of hospital stay, and FFP transfusion compared to the non-HCC group (mean decrease of 10.4, 0.6, 6.5, 3.2, respectively; p = <0.05). Graft-related mortality between HCC and non-HCC cases was not significantly different. **Conclusion:** Evaluation with MELD score and preoperative INR appears to provide significant predictive value to hospital course and FFP transfusion needs. Review of the HCC group reveals that transplant at lower MELD score and preoperative INR results in decreased FFP transfusion and shortened length of hospital stay. Lower MELD score (<20) appears to be associated with decreased graft-related mortality following liver transplant. Early transplant in patients with lower MELD score and preoperative INR should be considered due to a potential for improved outcome and increased cost savings due to shortened hospital stay. Decreasing hospital stay by 6 days would save at least $11,460 per transplant (Kaiser Family State Health facts, 2011).

Disclosure of Commercial Conflict of Interest

A. F. Brown: Nothing to disclose; G. Meny: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. F. Brown: Nothing to disclose; G. Meny: Nothing to disclose

S50-030H

Viscoelastic Measures of Acute Traumatic Coagulopathy: Utililizing ROTEM to Quantify Trauma Induced Coagulopathy and Predict Outcome

L Kornblith[1,2] (Lucy.Kornblith@ucsfmedctr.org), B J Redick[1,2], R F Vilardi[1,2], M F Nelson[1,2], M J Cohen[1,2]. *[1]University of California San Francisco, San Francisco, CA, United States; [2]Surgery, San Francisco General Hospital, San Francisco, CA, United States*

Background/Case Studies: Acute traumatic coagulopathy (ATC) is a well-defined entity present on admission in 20-40% of trauma patients presenting with injury and shock, and is associated with worse outcomes. ATC has classically been defined by plasma-based tests, but viscoelastic assays such as rotational thromboelastometry (ROTEM) are better suited to identify disordered clot formation and lysis associated with ATC. However, abnormalities of ROTEM associated with ATC have not been well characterized and viscoelastic definitions of ATC remain undefined. This study sought to understand the abnormalities of ROTEM parameters associated with ATC, and hypothesized that manufacturer 'normal ranges' for ROTEM do not adequately identify ATC. **Study Design/Methods:** Longitudinal plasma samples were prospectively collected from 255 critically-injured patients from 0-120 h. ROTEM, international normalized ratio/partial thromboplastin time (INR/PTT), and coagulation factors were measured and matched with demographics and outcomes. Coagulopathy by INR was defined as INR ≥1.3, and coagulopathy by ROTEM was defined by manufacturer 'normal range' cutoffs of CT > 79s, CFT > 159s, α-angle < 63°, or MCF < 50 mm. Massive transfusion (MT) was defined as >10 u of pRBCs in 6 h. **Results/Findings:** The 255 patients were a severely injured cohort with a median ISS of 22 and a mortality of 20.8%. 23.9% were coagulopathic by INR cutoff and 9.4% were coagulopathic by ROTEM 'normal range' cutoffs. Those coagulopathic by INR cutoff were severely injured, required more blood products, and had significantly worse outcomes than those not coagulopathic (ISS 30 versus 16; MT 23.0% versus 2.1%; mortality 42.6% versus 13.9%; all $p < 0.05$). Those coagulopathic by ROTEM 'normal range′ cutoffs had no significant differences in demographics or transfusion requirements. Factors II, V, VII, IX, X, ATIII, D-dimer, and PC were lower in those coagulopathic by INR (all $p < 0.05$), but had no significant differences in those coagulopathic by ROTEM 'normal range′ cutoffs. In multivariate regression, coagulopathy by INR cutoff was associated with 4.2× higher odds of MT (OR 1.1-16.2, $p < 0.05$) but coagulopathy by ROTEM 'normal range′ cutoffs had no statistically significant prediction of MT. Using median ROTEM values of patients requiring MT as cutoffs (CT > 77 s, CFT > 132 s, α-angle < 65°, and MCF < 54 mm) was better predictive of the need for MT and mortality at 24 h than both INR and ROTEM 'normal ranges' (all ROC AUC increased). Utilizing these new ROTEM cutoffs, 3.5% were additionally identified as coagulopathic and they had an associated 5.8× higher odds of MT (OR 1.2-26.9, $p < 0.05$). **Conclusion:** Given the utility of viscoelastic assays in identification of ATC, defining ROTEM cutoffs associated with traumatic coagulopathy is necessary for adequate prediction of transfusion needs and outcomes.

Disclosure of Commercial Conflict of Interest

M. J. Cohen: Nothing to disclose; L. Kornblith: Nothing to disclose; M. F. Nelson: Nothing to disclose; B. J. Redick: Nothing to disclose; R. F. Vilardi: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. J. Cohen: Nothing to disclose; L. Kornblith: Nothing to disclose; M. F. Nelson: Nothing to disclose; B. J. Redick: Nothing to disclose; R. F. Vilardi: Nothing to disclose

S51-030H

Centralized Recipient Databases Enhance the Serological Safety of Red Blood Cell (RBC) Transfusions for Sickle Cell Disease (SCD) Patients

S K Harm[1,2] (sharm@itxm.org), M H Yazer[1,2], G F Monis[3], M Delaney[3,4]. *[1]The Institute for Transfusion Medicine, Pittsburgh, PA, United States; [2]Department of Pathology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [3]Department of Laboratory Medicine, University of Washington, Seattle, WA, United States; [4]Puget Sound Blood Center, Seattle, WA, United States*

Background/Case Studies: Sickle cell disease (SCD) patients are often transfused with red blood cell (RBC) units and frequently produce RBC alloantibodies (Ab). Up to 37% of these RBC Abs may evanesce during the patient's lifetime, putting them in danger of experiencing a delayed hemolytic transfusion reaction (DHTR). Patients are usually unaware of their antibody status thereby rendering an accurate and accessible serological record critical for prevention of DHTRs. Each of the 35 hospitals serviced by two regional centralized transfusion services (CTS) are linked to a CTS electronic medical record (EMR) containing the serological data and diagnosis of all transfusion recipients. The goal of this study was to quantify the risk of DHTR by determining how often RBC Ab information obtained at one hospital would be critical to ensure a serologically safe transfusion at a different hospital as patients seek care over time and their Abs evanesce. **Study Design/Methods:** From the CTS EMRs, the records of SCD patients who were >18 years old and transfused with ≥1 RBC during 2012 (CTS 1), or during 2011 and 2012 (CTS 2), were reviewed. Records were abstracted to collect the number of all RBC transfusions since 1999 (CTS 1) and 1997 (CTS 2), the name and date(s) of the hospital(s) where the admission for transfusion occurred, and the patient's RBC Ab history. The index hospital was defined as the hospital at which an RBC Ab was initially detected. The standard of care at both CTSs is to prospectively provide Rh and K antigen matched RBCs to SCD recipients who have not produced an Ab, and more extensive antigen matching thereafter. **Results/Findings:** A total of 150 SCD patients were evaluated; 65/150 (43%) patients had ≥1 Ab. In 42/65 (65%) patients with Ab(s), ≥1 Ab evanesced. Of the 170 Abs in total that were identified, 85/170 (50%) evanesced. The median number of RBCs transfused and the number of hospitals visited for RBC transfusion by these 150 patients is shown in the Table. Of the 42 SCD patients with evanesced Abs, 12/42 (29%) patients had ≥20 admissions for transfusion at non-index hospitals after the Ab evanesced and were thus at the highest risk for DHTR; 14/42 (33%) patients had 5-19 admissions for transfusions at non-index hospitals, and 16/42 (38%) patients had ≤4 admissions for transfusions at non-index hospitals. **Conclusion:** Alloimmunized SCD patients with evanesced Ab(s) are at risk for DHTR. A centralized database containing the serological history of each patient can prevent DHTRs and provides a model for enhancing transfusion safety.

Disclosure of Commercial Conflict of Interest

M. Delaney: Nothing to disclose; S. K. Harm: Nothing to disclose; G. F. Monis: Nothing to disclose; M. H. Yazer: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Delaney: Nothing to disclose; S. K. Harm: Nothing to disclose; G. F. Monis: Nothing to disclose; M. H. Yazer: Nothing to disclose

RBC transfusion information for the SCD transfusion recipients.

	# of RBC units transfused per SCD recipient (n = 150)	# of hospital admissions during which all SCD patients (n = 150) received transfusion	# of hospital admissions during which SCD patients with Ab (n = 65) received transfusion	# of hospitals at which SCD patients with evanesced Ab (n = 42) received transfusion	# of non-index hospitals at which SCD patients with evanesced Ab (n = 42) received transfusion (Ab detected elsewhere, highest risk of DHTR)
Median	46	2	2	3	2
Range	1-1176	1-8	1-8	1-8	0-7

S52-030H

A Descriptive Analysis of Plasma use in the REDS-III Hospital Network

D Triulzi[1] (dtriulzi@itxm.org), J Gottschall[2], E L Murphy[4], E Fiebig[3], Y Wu[4], P M Ness[5], D Kor[6], D Chowdhury[7], D Fleischmann[7]. *[1]University of Pittsburgh and Institute for Transfusion Medicine, Pittsburgh, PA, United States; [2]Blood Center of Wisconsin, Milwaukee, WI, United States; [3]University of California San Francisco, San Francisco, CA, United States; [4]Yale New Haven Hospital, New Haven, CT, United States; [5]The Johns Hopkins School of Medicine, Baltimore, MD, United States; [6]Mayo Clinic Rochester, Rochester, MN, United States; [7]RTI International, Rockville, MD, United States*

Background/Case Studies: Over 4 million units of plasma are transfused annually in the US yet detailed information regarding its use is lacking. Such information is needed to identify opportunities for practice improvement and design of randomized clinical studies. **Study Design/Methods:** Ten hospitals participating in REDS-III collected detailed blood components, patient demographic, clinical and laboratory data from the blood bank and electronic medical records from October 1, 2010-September 30, 2011 for all adult patients transfused with plasma. A plasma dose was defined as the total number of units given within 4 hours of a prior unit. International normalized ratio (INR) was collected if available within 8 hours of the start of plasma transfusion and within 8 hours after the last unit of plasma. **Results/Findings:** 72,167 units of plasma were transfused in 19,596 doses to 9,269 patients (58% men, 42% women). Mean age was 63.2 ± 16.6 years. Most patients (71%) also received red cell transfusions (56,146 units) and 42% also received platelets (16,845 units). The median dose of plasma was 2.0 units (25-75% interquartile range 2 to 4, range 1-72); 15% of doses were 1 unit, and 45% were 2 units. When adjusted by patient body weight (kg), the median dose was 6.3 ml/kg (interquartile range 4.4 to 10.6). The pre-transfusion international normalized ratio (INR) was available on 69% of the doses and had a median value of 1.9 (interquartile range 1.6-2.6). Nearly a quarter (23.5%) of plasma transfusions were given for an INR of ≤1.5 and 48.5% for an INR ≥2.0. Among the 1, 2, 3 and 4 unit plasma doses the pre-transfusion median INR's were 1.7, 1.9, 2.0, and 2.2 respectively. The post-transfusion INR was available on 68% of doses and had a median value of 1.6 (interquartile range 1.4-2.0). Only 42% of plasma transfusions resulted in a post transfusion INR ≤1.5. The most commonly associated patient diagnoses were hypertension, atrial fibrillation, acute hemorrhage, acute kidney failure, and coronary artery disease. The most common issue locations were general ward (38%) and ICU (42%). The operating room and emergency department accounted for only 11% and 4% of issue locations respectively. **Conclusion:** The REDS-III plasma epidemiology database provides a large source of descriptive data on plasma utilization in the US. The data demonstrate substantial opportunity to improve transfusion practices for plasma, with evidence for both inadequate dosing (low plasma doses and inadequate correction of INR) and over-transfusion (plasma transfusion for INR ≤1.5). Measures to improve plasma transfusion practice should be directed at patients on medical and surgical wards and in the ICU where plasma is most commonly used.

Disclosure of Commercial Conflict of Interest

D. Chowdhury: Nothing to disclose; E. Fiebig: Nothing to disclose; D. Fleischmann: Nothing to disclose; J. Gottschall: Cerus, Other, Hemonetics, Other, Fenwal, Other, Macopharma, Other; D. Kor: No Answer; E. L. Murphy: Nothing to disclose; P. M. Ness: Nothing to disclose; D. Triulzi: Nothing to disclose; Y. Wu: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Chowdhury: Nothing to disclose; E. Fiebig: Nothing to disclose; D. Fleischmann: Nothing to disclose; J. Gottschall: Nothing to disclose; D. Kor: No Answer; E. L. Murphy: Nothing to disclose; P. M. Ness: Nothing to disclose; D. Triulzi: Nothing to disclose; Y. Wu: Nothing to disclose

S53-030H

The Proportion of Blood Products Administered to Individuals Who Die in the Hospital: An Observational Study of 244,013 Hospitalizations

N Shehata[1,2] (nshehata@mtsinai.on.ca), A Forster[3,4], N Lawrence[3], D M Rothwell[3], P C Hebert[3,4], K Wilson[3,4]. *[1]Medicine and Laboratory Medicine and Pathobiology, University of Toronto, Mount Sinai Hospital, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada; [3]Ottawa Hospital Research Institute, Ottawa, ON, Canada; [4]Medicine, University of Ottawa, Ottawa, ON, Canada*

Background/Case Studies: Transfusion therapy is a life-saving but also scarce and costly resource. Furthermore, the evidence for the use of blood products in many situations is not clear. Given the costs of transfusion and uncertainty over indications, this study sought to characterize patients who receive a transfusion and to determine the percentage of individuals who received transfusion in the hospital and died from all causes before hospital discharge. **Study Design/Methods:** This is a retrospective study of all hospital admissions from November 1, 2006-June 2012 using a large administrative database that serves as a data repository for three campuses of one university-affiliated hospital. All adult patients that were discharged during this interval were included. Two groups were identified: 1) the proportion of individuals who received a transfusion and died from all causes during their hospitalization; and 2) the proportion of blood products transfused to these patients. The baseline mortality risk at admission was calculated based on the model of Escobar and colleagues that was modified and validated using the Charlson comorbidity index as a measure of comorbidity. **Results/Findings:** 244,013 hospitalizations were identified of which 32,302 received a transfusion. Overall, 31.8% of patients hospitalized who received red cells or plasma had a higher burden of disease as measured by the Charlson comorbidity index compared to 13.2% of hospitalizations who did not receive blood. The observed and predicted mortality rates were similar for patients who received red blood cells, but there was a higher observed than predicted mortality rate for patients receiving plasma and platelets (a change of 7% for plasma and 6% for platelets). Twenty four percent of transfused blood products were administered to individuals who died during the hospitalization. The in-hospital mortality rate for patients not transfused was 4.1% compared to a mortality rate of 11.7% for individuals who received any transfusion. The in-hospital mortality rate for individuals who received plasma, red blood cells, and platelets, was 24.3%, 11.5%, and 18.8% respectively. Higher percentages (30.2%, 19.1%, and 25.9%) of transfused plasma, red blood cells, and platelets were given to individuals who died during the hospitalization. Almost half of plasma (45.6%) and platelet transfusion (45.3%) administered to patients admitted to the intensive care service and to the general medicine service (40.7% and 45.9%, respectively) were administered to patients who died during their hospitalization. **Conclusion:** A large percentage of blood products are administered to individuals who have a high burden of illness who ultimately succumb to their illness. Given the scarcity and cost of blood products, further research should be conducted to ensure effectiveness of transfusion therapy in individuals with high mortality risk.

Disclosure of Commercial Conflict of Interest

A. Forster: Nothing to disclose; P. C. Hebert: No Answer; N. Lawrence: Nothing to disclose; D. M. Rothwell: Nothing to disclose; N. Shehata: Nothing to disclose; K. Wilson: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Forster: Nothing to disclose; P. C. Hebert: No Answer; N. Lawrence: Nothing to disclose; D. M. Rothwell: Nothing to disclose; N. Shehata: Nothing to disclose; K. Wilson: Nothing to disclose

S54-030H

Platelet Dosing, Suspension Medium, and Inter-transfusion Interval

R Galloway-Haskins[1,2] (rgallowayh@nshs.edu), W Heaton[1,2], F N Boctor[2], N M Nikolis[2], A Stein[2]. *[1]Blood Component Research Laboratory, Feinstein Institute for Medical Research North Shore-LIJ Health System, Manhasset, NY, United States; [2]Blood Donor Transfusion Services, North Shore University Hospital, Manhasset, NY, United States*

Background/Case Studies: Since August 2010, reduced dose (1/2 single donor platelet [SDP] and VSDP) plasma platelet concentrates (PC) have been routinely supplied for patients requiring bleeding prophylaxis at North Shore University Hospital. Doses included: Standard SDP (~4×10^{11} platelets /product); Variable (VSDP) (~2.9×10^{11}); and 1/2 SDP (~ 1.9×10^{11}). In April, PAS 3 SDP were substituted for ~ 50% of routine full dose transfusions. Inter-transfusion intervals and reaction rates were monitored and reported

according to local procedure. **Study Design/Methods:** Routine SDP transfusion (Tx) products for those with pre-counts <12,000/m^3 were assigned by blood bank staff to prophylactic patients—defined as non-refractory without bleeding, sepsis, or consumption. Following transfusion, time to next transfusion was tracked with pre/post counts, reactions, and product category. Based on patient size, and for risk mitigation, full dose SDP were supplied for any patient with counts <8,000/m^3 and for those with Body Surface Area (BSA) > 2.3 m^2. As of Q2 2012, full dose PAS 3 SDP were available and transfusion records affirmatively monitored for reactions and correct product allocation. **Results/Findings:** The Table shows inter-Tx intervals sorted by product. Average platelet loads transfused per product averaged 3.84 ± 0.54 × 10^{11} platelets /SDP, 4.00 ± 0.53 × 10^{11} platelets /PAS 3 SDP, and 2.85 ± 0.20 × 10^{11} /VSDP, such that a VSDP averaged 74% of an SDP. Average transfused SDP plasma volumes were 242.34 ± 45.02 ml, PAS 3 SDP were 253.98 ± 59.77 ml, and VSDP were 218.67 ± 39.36 ml. Over a period associated with 27 reactions, none were associated with 1/2 SDP, two with VSDP, one with PAS 3 SDP, and 20 with plasma SDP. No level 4 bleeds were reported. **Conclusion:** Inter-Tx intervals generally paralleled platelet dose, and with a $p < 0.08182$ there was no significant difference between PAS 3 and plasma PC. Reduced dose SDP had less platelet reactions and both 1/2 dose and PAS 3 SDP units had significantly less plasma than full dose SDP. It is possible that the combination of reduced dose and PAS would decrease the reaction rates further, since during the study PAS 3 units were not dose reduced.

Disclosure of Commercial Conflict of Interest

F. N. Boctor: No Answer; R. Galloway-Haskins: Fenwal, Grants or Research Support; W. Heaton: Immunetics., Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. N. Boctor: No Answer; R. Galloway-Haskins: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, HemCon Medical Technologies, Grants or Research Support, Light Integra Technology, Grants or Research Support; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

TABLE 1. Prophylactic Inter-Tx Interval

	Q2/Q3 2012 Inter-Tx Interval Mean ± SD (Days)				
	April-May	June	July	August	Total
# Patients Tx	31	38	40	23	135
# Tx Events	119	185	162	46	512
SDP	1.4 ± 0.8	1.5 ± 0.9	1.4 ± 0.9	1.8 ± 1.0	1.5 ± 0.9
PAS 3 SDP	1.4 ± 1.0	1.5 ± 0.9	1.5 ± 0.9	1.4 ± 1.1	1.5 ± 0.9
VSDP	1.2 ± 0.7	1.1 ± 0.9	1.4 ± 1.0	1.6 ± 0.8	1.3 ± 0.9
1/2 SDP	0.7 ± 0.8	0.9 ± 0.4	0.8 ± 0.7	0.5 ± 0.3	0.7 ± 0.6

Red Blood Cells: Molecular Basis of Blood Groups – Non-Rh

S55-030I

New *FY*01* Allele with Weakened Expression of FY^a Antigen

P A Arndt[2] (patricia.arndt@redcross.org), G Garratty[2], T Horn[1], J Crowley[1], R Young[1], S M Heri[3], M A Keller[1]. *[1]National Molecular Laboratory, American Red Cross, Philadelphia, PA, United States; [2]Southern California Blood Services Region, American Red Cross Biomedical Services, Pomona, CA, United States; [3]Blood Bank, Children's Hospital Los Angeles, Los Angeles, CA, United States*

Background/Case Studies: The Duffy gene (*FY*) located on chromosome 1 consists of two exons. The reference allele is *FY*02 (FY*B)* and encodes the Fy^b antigen. An A > G single nucleotide polymorphism (SNP) at nucleotide (nt) 125 in exon 2 defines the *FY*01* (*FY*A*) allele and results in an aspartate to glycine change at residue 42 encoding Fy^a. In the *FY*02N.01* allele, a GATA box T > C SNP at nt-67 causes loss of *FY*02* expression on RBCs but not in tissues. In the *FY*02M.01* allele, a C > T SNP at nt265 (arginine 89 to cysteine) and G > A SNP at nt298 (alanine 100 to threonine). In the *FY*02M.02* allele, there is an additional G > T SNP at nt145 (alanine 49 to serine). The nt265C > T SNP is associated with weakening of Fy^b expression (Fy^x). This change has not been previously described in the *FY*01* allele. Flow cytometry (FC) is a useful method for detecting differences in antigen strength, eg due to zygosity, and has been used to study expression of Duffy antigens on RBCs with different genotypes (e.g., *FY*01/FY*01, FY*01/FY*02, FY*02/FY*02, FY*02M.01/FY*02M.01, FY*01/FY*02N.01, FY*02/FY*02N.01, FY*02N.01/FY*02N.01, FY*01/FY*02M.01, FY*02M.01/FY*02N.01*). Phenotype-matched units were desired for a fetus with alpha-thalassemia that had received multiple intrauterine transfusions. A fetal blood specimen was used for genotype determination. **Study Design/Methods:** Genotyping was performed by HEA BeadChip™, gel-based genotyping, and genomic sequence analysis. Antigen strength studies were performed by flow cytometry on RBCs after incubation with commercial anti-Fy^a and FITC Fab anti-human IgG. **Results/Findings:** The HEA BeadChip™ yielded indeterminate calls for the Duffy antigens. Genomic sequencing of FY exon 2 showed the sample was homozygous for 125G associated with *FY*01* and heterozygous for nt265C/T associated with the *FY*02M.01* allele. The nt145G > T SNP and the nt298G > A SNP associated with the known *FY*02M.01* and *FY*02M.02* alleles was not present. Blood samples were obtained from the parents and genotyping was performed. The mother's predicted phenotype by HEA BeadChip™ was Fy(a + b-) and was negative for the nt265C > T SNP. The father's sample, like that of the fetus, was homozygous for nt125G and heterozygous for nt265C/T. FC was performed using samples from the parents along with genotyped control cells. Median fluorescence intensity (MFI) were 70.4-98.2 for 5 Fy(a + b-) RBCs from *FY*01/FY*01* donors, 39.6-57.8 for 5 Fy(a + b+) RBCs, 40.3-49.6 for 3 Fy(a + b-) RBCs from *FY*01/FY*02N.01* donors, and 4.8-5.0 for 2 Fy(a-) RBCs. The mother's RBCs gave results of 63.2 MFI. The father's RBCs gave results of 44.3 MFI, clearly in the range for *FY*01* heterozygotes. **Conclusion:** The fetus and the father appear to share a novel *FY* allele containing both the nt125G and nt265C > T SNPs. This novel *FY*01* allele (provisional allele *FY*01M.01*) is associated with weakened expression of Fy^a antigen.

Disclosure of Commercial Conflict of Interest

P. A. Arndt: Nothing to disclose; J. Crowley: No Answer; G. Garratty: Nothing to disclose; S. M. Heri: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support, Life Technologies, Consulting or Board of Director Fees; R. Young: No Answer

Disclosure of Grants Conflict of Interest

P. A. Arndt: Nothing to disclose; J. Crowley: No Answer; G. Garratty: Nothing to disclose; S. M. Heri: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support; R. Young: No Answer

S56-030I

Serologic and Molecular Investigation of Novel Kidd System Alleles in African-Americans

A Burgos[1] (aburgos@nybloodcenter.org), S Vege[1], R W Velliquette[1], C Lomas-Francis[1], C M Westhoff[1]. *[1]Immunohematology and Genomics, New York Blood Center, New York, NY, United States*

Background/Case Studies: The Kidd system is more polymorphic than a single nucleotide (nt) change and the number of reported alleles continues to grow. JK in 3 persons was analyzed, whose RBC Jka/Jkb phenotype differed from the genotype—one who had made the corresponding antibody. **Study Design/Methods:** Red blood cell (RBC) typing was performed by standard tube method with polyclonal and monoclonal anti-Jka and -Jkb. DNA testing was by human erythrocyte antigen (HEA) (BioArray/Immucor). The coding exons of *JK* were amplified and sequenced. Adsorption elution was performed with polyclonal reagents. **Results/Findings:** Case Reports: DNA testing predicted a Jk(a + b+) phenotype in all 3 samples. Patient 1 was a multiply transfused 15-year-old African-American (AA) female with sickle cell disease (SCD) and anti-D,-S,-E,-Cob and anti-Jka in the plasma. Her pre-transfusion RBCs typed Jk(a-b+). Patient 2 was a 36-year old pregnant AA evaluated for a panreactive antibody whose RBCs typed Jk(a + b-). Sample 3 was from a AA male donor whose RBCs typed Jk(a-b+). All three samples were *JK*A/B* by HEA and confirmed by gene sequencing. Patient 1 was chronically transfused so no additional RBC testing was possible. Gene sequencing revealed a deletion of 24-bp in exon 4, predicted to eliminate 8 amino acids from the mature protein. Retyping Patient 2 RBCs confirmed they were non-reactive with Ortho BioClone and Immucor polyclonal anti-Jkb and adsorption elution was negative. *JK* sequencing found an insertion of a C at nt 157-159, encoding a frameshift (fs) at p.Pro53 with a stop at p.Asp60. Repeat testing of fresh RBCs from Sample 3 with anti-Jka gave 1 + w with Immucor, mi with Ortho, and no reactivity with BioRad. Reactivity diminished on storage but adsorption-elution recovered a robust anti-Jka. Sequencing found a c.130G > A change in exon 4, associated with altered Jka

expression, and insertion of nt A between 1062-1067 in exon 11, causing a fs at p.Lys355 and stop at p.Lys366. **Conclusion:** Three new JK alleles were reported, all in African-Americans; two with loss of Kidd antigen and the third with weak expression. *JK*A del27-50* was associated with production of anti-Jka indicating clinical significance for transfusion. *JK*B 157_159insC* caused a stop codon and testing is consistent with a null allele. *JK*A 130A, 1062_1067insA* encodes a stop 23 aa from the COOH terminus, but a small amount of protein reaches the membrane, which gives insight into trafficking and assembly of Kidd protein. As in other blood group systems, Kidd system genetic diversity may be increased in Blacks.

Disclosure of Commercial Conflict of Interest

A. Burgos: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Burgos: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Sample	Anti-Jka	Anti-Jkb	838G > A *JK*A/*B*	nt. change	amino acid change
Pt. 1	0	3+	*JK*A/*B*	*JK*A del 27-50*	del 10-17
Pt. 2	3+	0	*JK*A/*B*	*JK*B 157_159 insC*	Pro53fs + Asp60Stop
Donor 3	0/mi/1 + w	3+	*JK*A/*B*	*JK*A 130G > A, 1062_1067 insA*	Glu44Lys and Lys355fs + Lys366Stop

S57-030I

A New *KEL* Compound Heterozygous Background in a Patient with Anti-Ku and Unexpected Kp^b Reactivity Eluted from Her Red Blood Cells (RBCs)

R W Velliquette[1] (rvelliquette@nybloodcenter.org), S Vege[1], C Lomas-Francis[1], C Fong[1], C M Westhoff[1]. *[1]Laboratory of Immunohematology and Genomics, New York Blood Center, New York, NY, United States*

Background/Case Studies: An 84-year-old Caucasian woman presented with a GI bleed and an antibody in her plasma that reacted with all RBCs except her own. As her plasma did not react with RBCs treated with dithiothreitol (DTT), or with K_0 RBCs, an antibody to a Kell system antigen was suspected. A serological and molecular study was undertaken to define the patient's Kell phenotype and the corresponding antibody. **Study Design/Methods:** Antigen typing and antibody identification were performed by standard methods. Acid eluates were made with Gamma ELU-KIT II (Immucor). Plasma from the patient was tested by the indirect antiglobulin test (IAT) with phenotypically similar RBCs treated with papain, trypsin, α-chymotrypsin, and 200 mM DTT. Genomic DNA was isolated from white blood cells (WBCs). Human erythrocyte antigen (HEA) (BioArray/Immucor) testing was performed according to manufacturer's instructions. The 19 *KEL* exons, including flanking splice sites, were amplified and sequenced by standard methods. **Results/Findings:** The patient's plasma reacted 3+ in the IAT with all panel cells, and with papain, trypsin, or α-chymotrypsin treated RBCs, but was non-reactive with DTT-treated RBCs, K_0 RBCs and her own RBCs. Her plasma reacted 2+ with K + k-, K + k+Kp(a + b+), Kp(a + b-), or Js(b-) RBC samples. Her RBCs typed K-k- Kp(a-b-) Js(a-b-) and did not react with 2 examples of anti-Ku, suggesting her RBCs had a serologic Kell-null (K_0) phenotype. Based on the phenotype, her plasma likely contained anti-Ku. To rule out a K_{mod} phenotype, polyclonal anti-k was used in adsorption/elution studies with her RBCs and with k– and k+ controls. The patient's RBCs, recovered from liquid nitrogen storage, were very weakly reactive in the DAT due to bound IgG. Surprisingly, an eluate prepared from her naïve RBCs, and from her RBCs sensitized with anti-k, contained anti-Kp^b reactive (2+) in the IgG gel test. Kp(b-) or K_0 RBCs did not react with the eluates; her own RBCs reacted very weakly (+/–) equivalent to the DAT. By HEA, the patient's RBCs were predicted to be K-k+, Kp(a-b+), Js(a-b+). Sequencing of the KEL exons found she was heterozygous for changes on a *KEL*02* background: a nucleotide c.574C > T change encoding a p.Arg192Stop in exon 6 (ISBT *KEL*02N.07*), and a c.1664G > A change in exon 15 encoding a p.Gly555Glu change that has not been reported. **Conclusion:** A compound heterozygote was identified with a new *KEL*02*, c.1664G > A encoding p.Gly555Glu, in conjunction with a known null allele, *KEL*02N.07*, in a Caucasian female with an apparent K_0 phenotype and anti-Ku. The adsorption/elution studies suggest that the patient's RBCs may express low levels of Kell antigens. The surprising finding of anti-Kp^b reactivity in the eluate from her cells following adsorption of anti-k is reminiscent of previous observations of a K_0 patient who had made apparent anti-Kp^b [Transfusion 2012; 52:158A (Suppl)].

Disclosure of Commercial Conflict of Interest

C. Fong: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Fong: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

S58-030I

Serologic and Molecular Studies of the Vel-Phenotype in a Multi-ethnic Population

J R Storry[2] (jill.storry@med.lu.se), K Hue-Roye[1], R W Velliquette[1], S Vege[1], M L Olsson[3], C Lomas-Francis[1], C M Westhoff[1]. *[1]Immunohematology and Genomics, New York Blood Center, Long Island City, NY, United States; [2]Clinical Immunology and Transfusion Medicine, University Lund, Lund, Sweden; [3]University Hospital, Lund, Sweden*

Background/Case Studies: The carrier of the Vel antigen was reported in 2013 by three groups (Storry et al, *Nat Genet*. 45:537-41, Cvejic et al, *Nat Genet*. 45:542-5, Ballif et al, EMBO Mol Med.5:751-61). A 17-base pair deletion in *SMIM1*, which encodes a small transmembrane RBC protein, is the cause of the Vel- phenotype in all samples reported, with the exception of one heterozygote with a 17-bp deletion and p.Met51Lys. Weak expression was due to heterozygosity for the deletion, with the exception of one sample with p.Met51Arg. The goal of this study was to determine if the same 17-bp deletion causes the Vel- phenotype in a multi-ethnic population including African-Americans and Hispanics. **Study Design/Methods:** Forty samples were collected from donors and patients, including some originally identified as VEL:-1,-2 or VEL:-1, 2 (Issitt.Vox Sang 1968, 15:12) and 11 patients with anti-Vel. Samples were from Caucasians, African-Americans, and Hispanics. DNA was isolated from peripheral blood and additive solution-polymerase chain reaction (AS-PCR) was as described (Storry et al, Nat.Genet). Wild-type products of 178-bp were distinguished from 161-bp deletion products by gel electrophoresis. Amplification and sequencing of the *SM1M1* coding region was also performed. **Results/Findings:** 22 of 40 were homozygous for the 17-bp deletion (del/del), 14 were heterozygous (del/wt), and 4 were negative (wt/wt). Ten of the 11 patients with anti-Vel in their plasma were homozygous for the mutation and one was negative. Sequence analysis of the coding exons of *SMIM1* in the 4 samples without the deletion found no changes in three and heterozygosity for c.152T > G (p.Met51Arg) in one. Cross-testing of 4 anti-Vel (Table) with red blood cells (RBCs) from samples with two copies of the deletion (del/del) was negative, while RBCs from those with one copy (del/wt) were positive. **Conclusion:** Testing of 40 historically Vel- or Vel variant samples confirmed homozygosity for the 17-bp deletion as the cause of the Vel- phenotype in 22 samples and heterozygosity in 14 Vel + ^{w} samples, with one Vel + ^{w} due to p.Met51Arg. The 17-bp deletion was identified in samples from African-American and Hispanics alike, evidence that the molecular background is the same as that reported in Europeans. These results suggest that the so-called VEL:-1,-2 phenotype is associated with homozygosity for the *SMIM1* 17-bp deletion while the VEL:–1, 2 phenotype is associated with heterozygous carriers and weak expression of Vel. The underlying molecular basis of the non-deletion Vel- individual with anti-Vel and no changes in the coding sequence is under investigation.

Disclosure of Commercial Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; M. L. Olsson: Nothing to disclose; J. R. Storry: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; M. L. Olsson: Nothing to disclose; J. R. Storry: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

			Anti-Vel Samles			
Serology	Molecular	Predicted	V.S. 2006	Paul 1970	N.Q. 1982	D.W. 1996
VEL+ control	wt/wt	Vel+	4+	3+	2+	2+
VEL:–1, –2	del/del	Vel–	0	0	0	0
VEL:–1, –2	del/del	Vel–	0			
VEL:–1, 2	del/wt	Vel + ^{w}	3+	2+	1+	0
VEL:–1, 2	del/wt	Vel + ^{w}	3+	2+	1+	0
VEL:–1, 2	del/wt	Vel + ^{w}	3+	2+	1+	0

S59-030I

Serological and Molecular Studies of Vel Negative Donors: Further Evidence of Heterogeneity

J M Moulds[1] (jmmoulds@lifeshare.org), K L Billingsley[1], L Arnaud[2].
[1]Scientific Support, LifeShare Blood Centers, Shreveport, LA, United States; [2]INTS, National Institute of Blood Transfusion, Paris, France

Background/Case Studies: The first example of anti-Vel was published in 1952 and soon afterwards heterogeneity was noted between other examples of anti-Vel. Recently three independent studies reported that the Vel antigen was encoded by the *SMIM1* gene and that homozygosity for a 17 bp deletion in this gene was largely the predominant cause of the Vel negative phenotype. Cverjic et al, however, identified 5/69 (7.2%) Vel negative blood donors who were heterozygous for this deletion and hence were postulated to be Vel weak. **Study Design/Methods:** Fourteen Caucasian donors previously typed as Vel negative with two potent antisera were studied by both serology and DNA. A total of 11 anti-Vel sera were used at 24C, 37C and by antihuman globulin (AHG) to type the presumed Vel negative donors. Genotyping was initially performed by polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) analysis. For heterozygotes, sequencing of the *SMIM1* coding regions was performed. **Results/Findings:** Of the 14 serologically Vel negative donors, 6 (43%) were heterozygous for the 17 bp deletion. This is in contrast to 7/174 (4%) found in the three combined publications. A review of the donor records did not indicate any obvious familial patterns. Sequencing of the coding regions did not identify any additional polymorphism. Red cells were available from four of the six heterozygous donors and these were re-examined serologically. Three anti-Vel that reacted with antigen positive cells 1 to 2 + ^{w} were negative with all heterozygous samples. Of the five anti-Vel that reacted to 2 + ^{s} to 3+ with Vel+ cells, one was negative with all samples and four showed microscopic reactivity at AHG. However, three potent anti-Vel (titer >500) were tested and these results varied: one was negative with all samples; one was positive (2+) with all samples; and the last antibody was positive with two but negative with the other two donors. **Conclusion:** Data reported regarding the frequency of the 17 bp deletion in the *SMIM1* gene, has been predominantly from Europeans. It remains to be seen if the high frequency (43%) that was found is common to the US or rather is unique to our donor population. In general, weaker anti-Vel sera will phenotype the 17 bp deletion heterozygotes as Vel negative. Therefore, genotyping should be performed to identify 'true' Vel negatives, ie 17 bp deletion homozygotes. Even so, discordant results between serology and molecular remain and suggest further heterogeneity in the Vel negative phenotype that needs additional investigation.

Disclosure of Commercial Conflict of Interest

L. Arnaud: Nothing to disclose; K. L. Billingsley: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support

Disclosure of Grants Conflict of Interest

L. Arnaud: Nothing to disclose; K. L. Billingsley: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support, Grifols, Consulting or Board of Director Fees, Grifols, Travel Support or Honorarium

S60-030I

A New Molecular Basis (c.148G > T in *DAF*) for the Cromer-null Phenotype in a Yt(a-) MER2 (CROM) Proband with Anti-IFC

C Lomas-Francis[1] (clomas-francis@nybloodcenter.org), Y Wu[2], A Fuchisawa[1], S Vege[1], L A Williams[2], J Choate[2], Z M Neal[2], R Ross[2], C M Westhoff[1]. *[1]Immunohematology and Genomics, New York Blood Center, Long Island City, NY, United States; [2]Yale University, New Haven, CT, United States*

Background/Case Studies: The 18 antigens of the Cromer blood group system are carried on Decay Accelerating Factor (DAF) and encoded by *CROM*. Four molecular bases for silencing of *CROM* and the IFC- (or Inab) phenotype have been reported. A sample from a 26-year-old Pakistani woman, who recently delivered her first child (G3P1, with two prior spontaneous abortions of unknown etiology), was referred for identification of an antibody in her plasma that reacted with all red blood cells (RBCs) except her own. The antibody was first detected at her pre-op evaluation for recurrent cardiac myxoma 2 years prior. Testing revealed an antibody directed at a Cromer antigen and a serological and molecular investigation was initiated. **Study Design/Methods:** Antigen typing and antibody identification was performed by standard methods. Acid eluates were made with Gamma ELU-KIT II (Immucor). The patient's plasma was tested by the indirect antiglobulin test (IAT) with phenotypically similar RBCs treated with papain, trypsin, α-chymotrypsin, and 200 mM dithiothreitol (DTT). Her RBCs were tested with antibodies to high prevalence antigens and her plasma was tested with selected RBCs lacking high prevalence antigens. Genomic DNA was isolated from WBCs. *CROM (DAF)* exons 2 and 6 were amplified and sequenced by standard methods. DNA analysis was performed also on samples from her sister and daughter. Flow cytometric (FC) analysis was performed at the hospital clinical laboratory. **Results/Findings:** The patient's plasma reacted 2+ in the IAT, with polyethylene glycol (PEG) enhancement, with all panel cells, and with papain, trypsin, or DTT treated RBCs, but did not react with α-chymotrypsin treated RBCs, or with her own RBCs, suggesting an antibody to a Cromer antigen or possibly Yt^{a}. The patient's RBCs typed negative for Cr^{a}, Dr^{a}, Es^{a}, GUTI, WES^{b}, IFC, Do^{b}, Yt^{a} and MER2. Testing selected RBCs lacking high prevalence antigens revealed that only IFC-RBCs did not react indicating that the antibody is anti-IFC or directed at an unknown Cromer antigen. Adsorption/elution studies ruled out antibodies to Do^{b}, Yt^{a} and MER2 and to common antigens. DNA analysis revealed homozygosity for novel nucleotide changes c.147G > A (silent) and c.148G > T in exon 2 of *CROM (DAF)* predicted to encode a stop at amino acid 50 (p.Glu50Stop) in the protein. DNA from the patient's sister was consensus c.147G and c.148G in *CROM* exon 2, and that of her daughter was heterozygous c.147G/A and c.148G/T. All of the patient's RBCs are uniformly CD55- but are CD59+. **Conclusion:** A complex antibody investigation resulted in the identification of a patient with anti-IFC and a new molecular basis for the IFC- (Cromer null) phenotype, c.148G > T (p.Glu50Stop). Cromer antibodies have not been implicated in hemolytic disease of the fetus and newborn (HDFN) and only occasionally in transfusion reactions. Given the patient's complex antigen profile, autologous transfusion was recommended. Patient has not needed allogeneic transfusion since and her newborn had no signs of HDFN.

Disclosure of Commercial Conflict of Interest

J. Choate: Nothing to disclose; A. Fuchisawa: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; Z. M. Neal: Nothing to disclose; R. Ross: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose; L. A. Williams: Nothing to disclose; Y. Wu: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Choate: Nothing to disclose; A. Fuchisawa: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; Z. M. Neal: Nothing to disclose; R. Ross: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose; L. A. Williams: Nothing to disclose; Y. Wu: Nothing to disclose

Transfusion-Transmitted Infectious Diseases: Bacteria and Parasites

S61-030J

Long-Term Natural History Study of *Babesia microti (Bm)* in Connecticut (CT): Considerations for Blood Donor Re-entry Algorithms

L Tonnetti[1] (laura.tonnetti@redcross.org), S T Johnson[2], A M Thorp[1], E Van Tassell[2], D A Leiby[1]. *[1]Transmissible Diseases, American Red Cross, Rockville, MD, United States; [2]American Red Cross, Farmington, CT, United States*

Background/Case Studies: *Babesia microti (Bm)*, an intraerythrocytic parasite that causes human babesiosis, is the agent most commonly transmitted by blood transfusion in the US. Currently blood donors are asked if they ever had babesiosis, but most are unaware of carrying the pathogen since infections are often asymptomatic. An FDA-licensed test is not yet available, but investigational new drug (IND) protocols are exploring the use of antibody and/or nucleic acid tests to screen blood donors. With the possibility of a licensed screening assay(s) in the near term, considerations should be given to potential re-entry algorithms for donors deferred for *Bm*-infection. **Study Design/Methods:** Through a seroprevalence study previously conducted in *Bm*-endemic areas of Connecticut (CT), blood

donors who tested positive (≥1:64) by IFA for *Bm* antibodies, were indefinitely deferred and invited to enroll in a Natural History Study to follow the course of the infection. For the first three years, blood samples were collected and tested by immunoflouresence assay (IFA) and polymerase chain reaction (PCR) ~ every 1-2 months (Phase I). After completion of Phase I, consenting donors were enrolled in Phase II and tested by IFA and PCR once a year for up to 5 years. **Results/Findings:** A total of 180 donors were enrolled in the study for periods of times varying from 1 to 8 years. Donors could be divided into two categories: one group who sero-reverted to negative (n = 103, 57%); and one group who remained IFA positive throughout the study (n = 77, 43%). Among those who sero-reverted, 87% (n = 90) were antibody negative within the first two years of enrollment. This category also included: 14 PCR-positive donors (14%); 8 sero-reverted to negative within 1 year; 5 within 2 years; and 1 in 4 years. Of the donors who remain IFA positive, 14 (8%) maintained high antibody titers (≥1:256) throughout the study. In addition, 8 of the 14 donors (57%) were PCR positive at least one time during the study. Data suggesting possible re-infection were present in both groups. Among the donors who sero-revert to negative, 23 (22%) tested positive by IFA on a donation following the negative result. Likewise, 11 donors (14%) who were never antibody negative but reached low IFA titer of 1:64, spiked to IFA titer of ≥1:256 on their next donation. **Conclusion:** Through long-term surveillance of blood donors exposed to *Bm,* potential re-infection and the likelihood of chronically infected donors was observed. However, the study suggests that most donors sero-revert to negative antibody within 2 years of being infected, and could be considered for re-entry when coupled with a negative PCR result. Re-exposure to the parasite is fairly common in endemic areas like CT and may contribute to re-infection or elevated antibody titers. Donors with persistent high titer should be considered chronically infected and at-risk for transmitting the parasite; indefinite deferral may be a prudent option for these donors.

Disclosure of Commercial Conflict of Interest

S. T. Johnson: Nothing to disclose; D. A. Leiby: Nothing to disclose; A. M. Thorp: Nothing to disclose; L. Tonnetti: Nothing to disclose; E. Van Tassell: No Answer

Disclosure of Grants Conflict of Interest

S. T. Johnson: Nothing to disclose; D. A. Leiby: Terumo BCT, Grants or Research Support; A. M. Thorp: Terumo BCT, Grants or Research Support; L. Tonnetti: Terumo BCT, Grants or Research Support; E. Van Tassell: No Answer

S62-030J

Final Results of a Chagas Seroprevalence Study at Canadian Blood Services (CBS)

M Fearon[1] (margaret.fearon@blood.ca), S F O'Brien[2], I R Dines[1], M Huang[1], M Ndao[3], P Lagace-Wiens[4], V Scalia[2]. *[1]Medical Scientific and Research Affairs, Canadian Blood Services, Toronto, ON, Canada; [2]Medical Scientific and Research Affairs, Canadian Blood Services, Ottawa, ON, Canada; [3]National Reference Centre for Parasitology, McGill University, Montreal, QC, Canada; [4]Department of Medical Microbiology and Infectious Disease, University of Manitoba, Winnipeg, MB, Canada*

Background/Case Studies: Background: Chagas Disease (CD)is caused by a protozoan parasite, *Trypanosoma cruzi (T. cruzi)*, transmitted by triatomine bugs in Mexico, Central, and South America. Increasing immigration to North America from these countries has prompted the screening of blood donors for CD, and prior to implementation of testing there had been two cases of transfusion transmitted reported in Canada, both in Manitoba. In May 2010, Canadian Blood Services (CBS) began testing 'at risk' donors for Chagas antibody, identifying 13 positive donations in the first year. A seroprevalence study was set up in September 2010 to test a subset of 'no risk' donors. This was done primarily to assess the efficacy of the risk questions in identifying donors for Chagas testing. **Study Design/Methods:** A target of at least 60,000 donors from clinics in areas reflecting immigration from endemic countries and regions where positive results were found in selective testing was selected for Phase I and included donors from southern Ontario (ON), Manitoba (MB), Alberta (AB), and British Columbia (BC). Phase 2 targeted an additional 20,000 donors and was carried out in MB only. Donors who answered no to all three CD risk questions were tested for CD antibody using the Abbott PRISM Chagas assay. All repeat reactives were sent to the National Reference Centre for Parasitology (NRCP) at McGill University, Montreal, for testing by ELISA, immunoblot, and polymerase chain reaction (PCR). Samples were sent to Quest Laboratories (US)for the radioimmunoprecipitation assay (RIPA) and to Blood Systems Inc. laboratories for the Ortho Chagas assay. **Results/Findings:** From September 2010-October 2012, 84,078, donors were tested, with 62,117 tested in Phase I (26% of donations from these areas over the study) and 21,961 in Phase 2 (92% of Winnipeg donors tested during both phases). Only one confirmed positive donor was identified in Phase I. Further investigation revealed that this donor from MB was infected via vertical transmission from her mother who had been transfused prior to and around the time of the donor's birth. This donor would not have been detected by any risk questions. The recipient of the only product transfused from this donor tested negative for CD antibody. Phase II identified no additional confirmed positive donors. **Conclusion:** Based on the finding of only one confirmed positive donor out of more than 80,000 'no risk' donors tested, CBS has maintained the strategy of testing only at-risk donors, identified by questioning. This strategy represents a new paradigm in infectious disease testing at CBS.

Disclosure of Commercial Conflict of Interest

I. R. Dines: Nothing to disclose; M. Fearon: Nothing to disclose; M. Huang: Nothing to disclose; P. Lagace-Wiens: Nothing to disclose; M. Ndao: Nothing to disclose; S. F. O'Brien: Nothing to disclose; V. Scalia: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. R. Dines: Nothing to disclose; M. Fearon: Nothing to disclose; M. Huang: Nothing to disclose; P. Lagace-Wiens: Nothing to disclose; M. Ndao: Nothing to disclose; S. F. O'Brien: Nothing to disclose; V. Scalia: Nothing to disclose

S63-030J

Determination of Seroprevalence of *Babesia microti (Bm)* in Endemic and Non-Endemic Blood Donor Populations using an Investigational ELISA

A E Levin[1] (alevin@immunetics.com), P C Williamson[2], J L Erwin[1], S Cyrus[2], E Bloch[3], B H Shaz[4], D A Kessler[4], S R Telford[5], P J Krause[6], M P Busch[3]. *[1]Immunetics, Inc., Boston, MA, United States; [2]Creative Testing Solutions, Tempe, AZ, United States; [3]Blood Systems Research Institute, San Francisco, CA, United States; [4]New York Blood Center, New York, NY, United States; [5]Tufts University Cummings School of Veterinary Medicine, North Grafton, MA, United States; [6]Yale School of Public Health and Yale School of Medicine, New Haven, CT, United States*

Background/Case Studies: Transfusion-transmitted babesiosis, caused principally by *Babesia microti (Bm)*, has emerged as a significant risk to the blood supply in the US. Despite the acknowledged need for a mitigation strategy, there are still no licensed, commercially available Babesia assays suitable for high throughput blood screening. The aim of this study was to determine the prevalence of *Bm* antibodies in blood donors using an investigational ELISA to assess the feasibility of serological blood donor screening. **Study Design/Methods:** Blood donors were randomly selected from areas defined as high-risk (Suffolk County, NY), low-risk (Brooklyn and Manhattan, NY) and non-endemic (AZ) for *Bm* transmission. A total of 15,000 de-identified donor samples collected from August-October 2012 were tested using the investigational ELISA to detect *Bm* IgG and IgM antibodies. ELISA repeat reactive samples were further tested by *Bm* immunofluorescent assay (IFA), polymerase chain reaction (PCR), and peripheral blood smear examination. Samples from 72 clinically diagnosed and laboratory confirmed cases of babesiosis were also studied. **Results/Findings:** Rates of reactivity in donors and clinical cases are shown in Table 1 at the study EIA cut-off (S/CO = 1.0) and at a proposed higher cut-off (S/CO = 1.24). A single seropositive sample in the higher-risk endemic group was positive by PCR; all were blood smear negative. Overall concordance between ELISA and IFA among donor samples was 99.3%. At the higher cut-off, the frequency of enzyme immunoassay (EIA) positive/IFA non-reactive samples in the non-endemic region decreased by 65% with a 14.8% decrease in EIA positive/IFA positive donor samples in the high risk endemic region, all of which were PCR negative, and a 2.8% decrease in detection of clinical case samples. **Conclusion:** The seroprevalence of *Bm* in donors in a high risk area was about 1%, similar to that reported in other studies. ELISA and IFA results were highly concordant. The observation of a single PCR positive sample suggests that the overwhelming majority of seropositive donors show evidence of prior exposure to *Bm* rather than persistent active infection. A revised, higher cut-off significantly reduces the frequency of non-infected false positives with minimal impact on detection of true positives in donors and clinical cases, and is therefore proposed for prospective studies under investigational new drug (IND). This study has served to further validate and optimize the *Bm* ELISA and attests to the feasibility of this high throughput serological screening assay in blood donors.

Disclosure of Commercial Conflict of Interest

E. Bloch: Nothing to disclose; M. P. Busch: Nothing to disclose; S. Cyrus: Nothing to disclose; J. L. Erwin: Nothing to disclose; D. A. Kessler: Nothing to disclose; P. J. Krause: Nothing to disclose; A. E. Levin: Immunetics, Inc.,

Ownership or Partnership; B. H. Shaz: Haemonetics, no financial relationship, Other, Fenwal, Other; S. R. Telford: Nothing to disclose; P. C. Williamson: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Bloch: Nothing to disclose; M. P. Busch: Nothing to disclose; S. Cyrus: Nothing to disclose; J. L. Erwin: Nothing to disclose; D. A. Kessler: Immunetics, Grants or Research Support; P. J. Krause: Nothing to disclose; A. E. Levin: Nothing to disclose; B. H. Shaz: Nothing to disclose; S. R. Telford: Nothing to disclose; P. C. Williamson: Nothing to disclose

TABLE 1

	Study Cut-off (S/CO = 1.0)			Revised Cut-off (S/CO = 1.24)		
Region	EIA Repeat Reactive	IFA Reactive at 1:64	IFA Non-Reactive	EIA Repeat Reactive	IFA Reactive at 1:64	IFA Non-Reactive
High Risk Endemic (n = 5000)	54 (1.08%)	18 (0.36%)	36 (0.72%)	46 (0.92%)	17 (0.34%)	29 (0.58%)
Low Risk Endemic (n = 5000)	37 (0.74%)	3 (0.06%)	34 (0.68%)	27 (0.54%)	2 (0.04%)	25 (0.50%)
Non-Endemic (n = 5000)	21 (0.42%)	1 (0.02%)	20 (0.40%)	8 (0.16%)	1 (0.02%)	7 (0.14%)
Clinical Babesiosis Cases (n = 72)	71 (95.9%)	71 (95.9%)	1 (1.39%)	69 (93.2%)	69 (93.2%)	3 (4.17%)

S64-030J

***Trypanosoma Cruzi* Donor Screening in the US; Development of an Algorithm for Donor Reentry**

R L Townsend[1] (Rebecca.Townsend@redcross.org), D E Krysztof[1], G A Foster[1], B L Weixlmann[2], S D Muckenfuss[2], S L Stramer[1]. *[1]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [2]Confirmatory Laboratory, American Red Cross, Charlotte, NC, United States*

Background/Case Studies: In January 2007, universal (U) blood donation screening for *T. cruzi*, the parasite causing Chagas disease, began. It was replaced in August 2009 by selective testing (S) where each donor is qualified by a single nonreactive test. Chagas, endemic in Latin America, has been detected in US blood donors due to immigration of infected persons and competent parasite vectors in the US. While donor screening has been effective in removing antibody-positive donors, uninfected donor units are also lost and donors deferred. **Study Design/Methods:** Screening included: Ortho ELISA January 2007-August 2011 or Abbott PRISM September 2011-January 2013; confirmatory included radioimmunoprecipitation assay (RIPA) (Quest) until July 31, 2012 followed by Abbott enzyme strip immunoassay (ESA) resulting in three algorithms. For the Ortho/RIPA group, all repeat reactives (RRs) were RIPA tested; for the PRISM/RIPA group, only concordant RRs (PRISM + ELISA) were RIPA tested; for the PRISM/ESA group, all RRs were ELISA + ESA tested. To establish comparable data for donors screened by each algorithm, plasma unit and follow-up (f/u) samples from selected confirmed (P) and unconfirmed (N) donors having both sample types available were tested by PRISM, ELISA and ESA. Results were compared for accuracy and for a donor reentry algorithm. **Results/Findings:** Over a 6-year period (January 29, 2007-January 31, 2013), 22,887,268 donations were screened with 4143 RRs, 595 (14%) confirmed P; overall RR rate = 0.018%; specificity = 99.985%; positive predictive value (PPV) ranged from 6-20% by algorithm. The ELISA/RIPA group had the highest PPV and lowest RR rate (but combined two screening strategies, U + S). Less than 100% of RR/P samples repeated as RR when retested (plasma, f/u) indicating false positivity of each algorithm; the highest number remained RR/P in the PRISM/RIPA group where only PRISM + ELISA RRs were tested by RIPA. Less than 50% of RR/N samples continued as RR when retested (plasma or f/u), and none were P by RIPA or ESA. RR/N samples yielded 0-4% RR rates on alternate screens. Plasma samples for which three tests were performed from the subsets of RIPA N samples (221 + 202) were all ESA N indicating false positivity plus 174 retested ESA N from the PRISM/ESA group. (Table) **Conclusion:** Since no RR sample in any N group confirmed P from further testing, concordant screening test (ELISA + PRISM) nonreactivity in a f/u sample should be sufficient for reentry following an index ESA N test result.

Disclosure of Commercial Conflict of Interest

G. A. Foster: Nothing to disclose; D. E. Krysztof: Nothing to disclose; S. D. Muckenfuss: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose; B. L. Weixlmann: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. A. Foster: Nothing to disclose; D. E. Krysztof: Nothing to disclose; S. D. Muckenfuss: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose; B. L. Weixlmann: Nothing to disclose

		#(%) Plasma Unit RR		#(%) F/U RR	
Algorithm	#(%) RR	ELISA	PRISM	ELISA	PRISM
ELISA/RIPA	2497 (0.012) U + S #(%)P = 488 (20) #(%)N = 2009 (80)	399/481 (83)400/1937 (21)*	215/259 (83)7/347 (2)	210/245 (86)217/688 (32)	5/10 (50)1/23 (4)
PRISM/RIPA	1122 (0.094) S #(%)P = 64 (6) #(%)N = 1058 (94)	59/61 (97)0/481 (0)	61/64 (95)489/1026 (48)**	17/19 (89)0/96 (0)	19/20 (95)97/217 (45)
PRISM/ESA	524 (0.082) S #(%)P = 43 (8) #(%)N = 481 (92)	32/42 (76)1/172 (0.6)	42/43 (98)174/470 (37) ***	15/18 (83)1/36 (3)#	18/20 (90)27/82 (33)

* 221+ ** 202+ *** 174 ESA tested, none P # includes 9 PRISM N donors ELISA tested for an ESA post-marketing study, all ESA N

S65-030J

Bacterial Contamination of Transfused Platelet Products: A Six-Year Review of Fatalities Reported to the US Food and Drug Administration(FDA)

S Haddad[1] (salim.haddad@fda.hhs.gov), F Barash[1], R Espinola[1], O Illoh[1], S Cannon[1]. *[1]US Food and Drug Administration, Rockville, MD, United States*

Background/Case Studies: Platelets (PLTs) are particularly vulnerable to bacterial outgrowth due to their room temperature storage. The risk of PLT transfusion-associated septic reaction and related death from bacterial contamination is estimated, respectively, at 1 in 107,000 and 1 in 1 million transfused apheresis units. However these rates are underestimated due to the passive nature of the reporting mechanism. Fatalities reported to FDA were reviewed to assess the relationship, if any, between the PLT transfusion and the reported fatality. **Study Design/Methods:** A retrospective chart review of transfusion-related fatalities reported to FDA in fiscal years 2007 to 2012 was conducted by FDA medical officers to identify fatalities related to transfusion of bacterially contaminated PLTs. **Results/Findings:** 250 of 436 fatality reports were confirmed related to transfusion. Fatalities associated with PLT bacterial contamination were 16 of 250 (6.4%) for an overall average of 2.7/year. 11 cases were associated with apheresis PLTs, 1 with a pre-storage pool, and 4 with post-storage pools. The apheresis PLTs and the

pre-storage pool had all tested negative by the early bacterial screening culture. Implicated bacteria were 10 Gram positive (63%), mostly skin commensals, and 6 Gram negative organisms (37%), all enterics. There were no strict anaerobic organisms. Staphylococcus aureus accounted for 6 out of the 16 fatalities (38%). When reported (N = 7), days of transfusion were either day 4 or day 5. The fatalities occurred in nine males and seven females. Of seven fatalities in which the implicated apheresis PLT product had split components, non-fatal septic reactions were observed in three of the recipients of the split products. In 11 fatalities in which information on the donor was available, two cases were related to endogenous infections: the first (Streptococcus pneumoniae) was diagnosed shortly after the donation; the second (Klebsiella pneumoniae) led to the death of two neonates from a single contaminated product, the donor having harbored an occult infection diagnosed only subsequent to the fatal events. A unit contaminated with Morganella morganii was associated with the erroneous release of a contaminated unit (quarantine release error). **Conclusion:** Fatalities due to bacterial contamination of PLTs have been reported in both apheresis and pooled PLT products and are mostly associated with Gram-positive organisms and older platelets. Staphylococcus aureus was the most commonly identified organism. The findings emphasize the need for effective quarantine control systems and for prompt recall mechanisms of co-components of contaminated units. Sepsis-related fatalities persist despite the implementation of mitigating measures such as early culture, improvement in skin disinfection techniques, and diversion pouches. The residual risk of bacterial contamination in transfused PLTs remains a public heath concern.

Disclosure of Commercial Conflict of Interest

F. Barash: Nothing to disclose; S. Cannon: Nothing to disclose; R. Espinola: Nothing to disclose; S. Haddad: Nothing to disclose; O. Illoh: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Barash: Nothing to disclose; S. Cannon: Nothing to disclose; R. Espinola: Nothing to disclose; S. Haddad: Nothing to disclose; O. Illoh: Nothing to disclose

S66-030J

Risk of Transfusion-Transmitted Bacterial Infection: Data From The French National Haemovigilance Network, 2000-2012

B Lafeuillade[1] (brunolafeuillade@efs.sante.fr), F Eb[1], N Ounnoughene[1], R Petermann[1], J Decousser[1], B Pozzetto[1], P Weinbreck[1]. [1]Transfusion Transmitted Bacterial Infection Working Group, French Agency of Medical Safety of Health Products (Agence National de Sécurité du Médicament et des Produits de la Santé), Saint Denis, France

Background/Case Studies: Transfusion transmitted bacterial infection (TTBI) remains the most frequent residual infectious risk from transfusion. However, owing to heterogeneous collecting methods, the real incidence of TTBI is unknown. Based on the national hemovigilance network, this study determines and comments on the TTBI incidence in France over a 13-year period. **Study Design/Methods:** TTBI were reported by a network of local correspondent in each hospital and blood center; thereafter regional coordinators centralized them to the French Agency of Medical Safety of Health Products. A multidisciplinary group (called Transfusion Transmitted Bacterial Infection Working Group) consisting of specialists in transfusion, infectious diseases, and infection control analyzed retrospectively each TTBI with a standardized scale of imputability and severity. Only cases with likely or undisputable imputability were reported. Incidence per million of blood products and types of blood products (BP) were calculated for the 2000-2012 period. Incidence of severe (ie death or immediate life threatening) and death-associated TTBI were calculated per million of BP. Finally, the distribution of bacteria species responsible for TTBI was also reported globally and according to types of transfused blood products. **Results/Findings:** 36.9 million BP including 3.3 million platelet concentrated (PC), 29.7 million red blood cell (RBC), and 3.9 million fresh frozen plasma (FFP) were transfused over the 2000-2012 period. The incidence of TTBI was 2.11 per million of transfused BP, including 20.0 per million of PC and 0.4 per million of RBC. No TTBI was associated with FFP transfusion. In term of severity, the incidence of severe or death associated TTBI were 1.22 and 0.43 per million of BP, respectively. The top three bacteria responsible for TTBI were Staphylococcus aureus, coagulase negative staphylococci, and Escherichia coli. PC were responsible for 85% of TTBI; Gram negative bacteria represented 39% of PC related TTBI and Gram positive bacteria 61%, most of them belonging to the normal flora. For RBC, all but one TTBI were associated with Gram negative bacilli. The incidence for RBC was lower than the 1998-2000 Bacthem French study (0.4 vs of 5, 8 per million of RBC). **Conclusion:** The French haemovigilance system provides an accurate estimate of the TTBI incidence for the 2000-2012 period and documents a bacterial risk reduction for RBC comparing to the 1998-2000 period.

Disclosure of Commercial Conflict of Interest

F. Eb: No Answer; J. Decousser: Nothing to disclose; B. Lafeuillade: No Answer; N. Ounnoughene: No Answer; R. Petermann: Nothing to disclose; B. Pozzetto: No Answer; P. Weinbreck: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Eb: No Answer; J. Decousser: Nothing to disclose; B. Lafeuillade: No Answer; N. Ounnoughene: No Answer; R. Petermann: Nothing to disclose; B. Pozzetto: No Answer; P. Weinbreck: Nothing to disclose

Patient Blood Management

S67-030K

Lowering the Red Blood Cell (RBC) Transfusion Trigger: 2009-2012: Data from 2,441,361 Hospitalized Patients

R L Thurer[1] (rthurer@haemonetics.com), T Precopio[1], M A Popovsky[1]. [1]*Haemonetics, Braintree, MA, United States*

Background/Case Studies: Evidence-based indications for red blood cell (RBC) transfusion are an essential part of patient blood management (PBM). Since there is increasing information to support lowering of the RBC transfusion trigger, the study sought to determine whether that evidence is being adopted clinically. **Study Design/Methods:** Using data from a proprietary blood management business intelligence portal (IMPACT® Online, Haemonetics Corporation, Braintree, MA), the incidence of RBC transfusion related to the lowest hemoglobin level during hospitalization (nadir Hgb) was examined in 1,578,076 patients in 17 hospitals throughout the US during 2009-2012. These hospitals initiated a PBM program at varying times during the study period. We also reviewed the data of 863,285 patients at 6 additional hospitals (2011-2012) that had not yet begun a blood management initiative. **Results/Findings:** As seen in the Table, at hospitals with blood management programs the overall incidence of RBC transfusion as well as the transfusion rate for patients at all levels of nadir Hgb decreased during the study period. The preponderance of the reduction was in 2012 when all of the programs were fully established. Hospitals that had sufficient interest in PBM to subscribe to the data tool but had not yet started programs showed no reduction in the transfusion rates between 2011 and 2012. However, the transfusion rates for patients at those hospitals were already similar to the rates at the hospitals with programs. **Conclusion:** Clinicians adopt lower RBC transfusion triggers when evidence-based practice is introduced as part of a PBM program. However, for those hospitals for which data exist, it appears that clinicians in hospitals interested in PBM but without formal programs are already adopting lower hgb triggers for RBC transfusion. The majority of patients with hgb values of 8.0-8.9 g/dL are not being transfused. As evidence-based triggers continue to gain acceptance and multidimensional blood management programs are implemented, the incidence of RBC transfusion will most likely continue to decrease.

Disclosure of Commercial Conflict of Interest

M. A. Popovsky: Haemonetics Corporation, Other; T. Precopio: Haemonetics Corp., Other; R. L. Thurer: Haemonetics Corp., Other, Haemonetics Corp., Stocks or Bonds

Disclosure of Grants Conflict of Interest

M. A. Popovsky: Nothing to disclose; T. Precopio: Nothing to disclose; R. L. Thurer: Nothing to disclose

Transfusion Rate (%) at Patient Blood Management Hospitals

Nadir Hgb (g/dL)	<7.0	7.0-7.9	8.0-8.9	9.0-9.9	≥10.0	All Patients
2009	84.3	74.0	45.6	11.8	1.2	13.3
2010	83.7	72.3	40.5	10.6	1.2	12.9
2011	85.1	71.5	36.1	8.8	1.1	12.5
2012	80.6	65.2	29.0	7.2	0.9	11.1
Transfusion Rate (%) at Non Patient Blood Management Hospitals						
2011	81.6	66.1	29.5	6.7	0.8	13.5
2012	81.9	65.4	30.4	7.9	1.0	13.2

S68-030K

The Role of the Blood Bank in the Development of the Patient Blood Management Program over a 10-Year Period and a Comparison of Pre and Post Utilization Data

A E Gomensoro[1] (agomensoro@maimonidesmed.org), R Stroud[2], N M Ali[1]. *[1]Blood Bank/Donor Center, Maimonides Medical Center, Brooklyn, NY, United States; [2]Cardiothoracic Surgery, Maimonides Medical Center, Brooklyn, NY, United States*

Background/Case Studies: In the environment of escalating health-care costs, blood transfusion management must take into consideration the high cost of blood products. Even more importantly, a large and increasing body of literature points to a dose-dependent increase of mortality and adverse long-term outcomes associated with blood transfusion. In this communication, evidence is presented of the important role of developing a working protocol, which guides the transfusion of blood products in large metropolitan hospital. **Study Design/Methods:** A protocol was developed with different hospital departments to assure proper type and quantity of blood products for transfusion of asymptomatic patients. These guidelines include Hemoglobin (Hgb) <8 gm./dl for red cell transfusion, PT with INR of >1.5 or PTT > 50 seconds for fresh frozen plasma (FFP), fibrinogen <100 mg/ul for cryoprecipitate, <10,000/ul platelet for prophylactic platelet transfusion, and <50,000/ul platelet for therapeutic platelets transfusions. Specific protocol was developed for patients undergoing cardiovascular surgery. The orthopedic department participated in a multicenter, randomized, transfusion trial for patients undergoing hip fracture repair (FOCUS study). We compared the number of units of each product transfused from 2003-2007, pre protocol years, and in 2008-2012, post protocol years. The number of patient admissions for these two 5-year periods did not change significantly. **Results/Findings:** As a result of continuous efforts to improve transfusion practice through education of medical and nursing staff, and the cooperation and participation of different committees and working groups from the departments of medicine, surgery, and anesthesia, a significant decrease was observed in the transfused units of red blood cells (RBCs), FFP, and cryoprecipitate. Platelet transfusions were slightly increased due to the implementation of a new computer system in 2010 that did not include pre-transfusion platelet count. In March 2012, MIS corrected the problem, and the blood bank was immediately able to evaluate the appropriateness of requests for platelet transfusions. For RBCs there was a 17.27% decrease; for FFP a 23.89 % decrease; and for cryoprecipitate a 33% decrease. The savings from implementation of the PBM program at Maimonides Medical Center was $2,912,515, an average of $582,503 yearly. **Conclusion:** The present result emphasizes the importance of developing a protocol for blood transfusion and following up on reviewing blood orders before issuing blood components. It is equally important to keep the ordering physician informed.

Disclosure of Commercial Conflict of Interest

N. M. Ali: Nothing to disclose; A. E. Gomensoro: Nothing to disclose; R. Stroud: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. M. Ali: Nothing to disclose; A. E. Gomensoro: Nothing to disclose; R. Stroud: Nothing to disclose

Results and savings from implementation of the Patient Blood Management program at Maimonides Medical Center

Period	FFP Units/Years	Red Cells Units/% Years	Platelet Units/ Years	Cryoprecipitate Units/Years
2003-2007	27,965	70,692	5,592	3022
2008-2012	21,283	58,483	5,574	2005
Difference/Unit	6,682 × 50.00 = $334,100	12,209 × 215.00 = $2,624,935	-182 × 535.00 = - 97,370	1017 × 50.00 = $50,850
% Difference	23.89	17.27	-3.25	33.0

S69-030K

A One-Year Pilot Study Using a "LEAN Six Sigma" Approach and IMPACT® Online Significantly Reduced Hospital Blood Utilization and Expenses

S Patchin[1] (spatchinj123@cox.net), J Baumeister[1], M Powers[1]. *[1]Blood Bank, Mercy Hospital OKC, Oklahoma City, OK, United States*

Background/Case Studies: Hospital blood utilization (BU) efforts have been occurring nationwide to reduce patient risks, improve patient outcomes, and significantly reduce hospital expenses. This hospital collaborated with the local blood center to pilot a hospital BU project here. **Study Design/Methods:** Using the Define, Measure, Improve, Analyze, and Control (DMAIC) "LEAN Six Sigma" approach and the web-based software (IMPACT® Online [IOL], Haemonetics, Braintree, MA), BU was monitored for red blood cells (RBCs), plasma, platelets, and cryoprecipitate. During the Define phase, a project charter, business case, and stakeholder analysis were established. Also, critical to quality (CTQ) measurements were determined, key input/output variables (KPIV/KPOV) were defined, and suppliers, inputs, processes, outputs, and customers (SIPOC) were documented. In addition, a Pareto chart was generated to illustrate the top ten physicians responsible for transfusing the most blood products (BPs). Twelve months of baseline information was compiled and analyzed for: 1) Total units transfused (tx)—All BPs and RBCs only; 2) Total tx patients; 3) BU rates—All BPs and RBCs only; 4) % patients tx with a nadir hemoglobin (Hgb) >7 g/dL;and 5) % tx patients based on a final recorded Hgb >9 and 10 g/dL. Efforts to improve BU included: 1) Lowering the critical Hgb value to 7 g/dL; 2) Developing a computer blood product alert (BPA) message when physicians attempt to order BPs; 3) Posting flyers announcing the transfusion trigger has been lowered to 7 g/dL and encouraging physicians to adopt the philosophy "give one and reassess" before transfusing a second unit of RBCs; and 4) Offering multiple opportunities for physician education Additionally, root cause analysis exercises were performed, a solution prioritization matrixwas established, monthly feedback to physicians and nurses at the Blood Utilization Review Committee meetings was provided, periodic progress reports at the Clinical Safety Quality Council (CSQC) meetings was presented, and physician scorecards were prepared to provide them with feedback on their BU patterns and progress. **Results/Findings:** A cost and statistical data analysis were performed comparing baseline data to data obtained one year following the improvements listed above. Using the t-test analysis, there is a statistically significant improvement/decrease for the following parameters: 1) % patients tx when nadir Hgb > 7 g/dL = 25.3%; 2) Total # BPs tx = 16.09%; 3) Number of RBCs only tx = 22.56%; 4) BU rate—ALL BPs = 13.46%; 5) BU rate—RBCs only = 20.05%; 6) Cost of BPs = $488,255; and 7) Estimated Hospital Costs = $1,112,676. **Conclusion:** This one-year pilot study resulted in statistically significant reductions in BU and over $1.1 M in estimated hospital cost savings. Based on these significant reductions, this BU project will soon expand to six other Sisters of Mercy hospitals in two additional states.

Disclosure of Commercial Conflict of Interest

J. Baumeister: Haemonetics, Inc., Other; S. Patchin: Nothing to disclose; M. Powers: Haemonetics, Other

Disclosure of Grants Conflict of Interest

J. Baumeister: Nothing to disclose; S. Patchin: Nothing to disclose; M. Powers: Nothing to disclose

S70-030K

Reduction in Red Blood Cell (RBC) Transfusions is Associated with Fewer Complications and Shortened Length of Stay

R L Thurer[1] (rthurer@haemonetics.com), T Precopio[1], M A Popovsky[1]. *[1]Haemonetics, Braintree, MA, United States*

Background/Case Studies: Patients transfused with stored red blood cells (RBC) are thought to have poorer outcomes than similar patients who receive RBC transfusions. However, despite attempatients to match patients in the transfused and non-transfused groups, this primarily retrospective information may be biased by the tendency of clinicians to transfuse patients perceived to have a poorer prognosis. To better understand this conundrum, data was evaluated before and after the institution of patient blood management (PBM) programs at six hospitals in the US. **Study Design/Methods:** Using data from a proprietary blood management business intelligence portal (IMPACT® Online, Haemonetics Corp., Braintree, MA), the following was analyzed: the transfusion of allogeneic and preoperative autologous donor (PAD) RBC complication rates (the incidence of one or more cardiac,

respiratory, neurologic, or infectious complications) and length of hospital stay (LOS) for 2225 patients who had primary, unilateral total knee replacement during one year prior to the adoption of a PBM program. This was compared to data from 3157 patients at the same hospitals during one year following adoption of PBM. Additionally, changes were evaluated in the transfusion trigger by looking at the final hemoglobin prior to discharge of transfused patients as well as the transfusion rates for patients whose lowest recorded hemoglobin was 7.0-7.9 g/dL and 8.0-8.9 g/dL. **Results/Findings:** As shown in the Table, PBM was associated with lower RBC transfusion rates. This was in part due to a large reduction for patients whose lowest recorded hemoglobin was between 8.0 and 8.9 g/dL. The total number of allogeneic and PAD units transfused per patient was similarly reduced (allogeneic-0.46 to 0.20 and PAD—0.24 to 0.02) and the last recorded hemoglobin of transfused patients went from 10.02 g/dL during the baseline period to 9.34 g/dL in the PBM period reflecting a lower transfusion trigger. These changes in the pattern of RBC transfusion were associated with reductions in the complication rate and LOS. **Conclusion:** Adoption of PBM results in fewer stored blood transfusions. Clinicians are increasingly willing to reduce transfusions to patients with hemoglobin levels between 8.0 and 8.9 g/dL but are less inclined to avoid transfusion to patients whose hemoglobin is <8.0 g/dL. From this data, there is no evidence that this change results in poorer outcomes. Rather, the complication rates and LOS were reduced by 22% and 15% respectively, in a similar population following the adoption of PBM.

Disclosure of Commercial Conflict of Interest

M. A. Popovsky: Haemonetics Corp., Other; T. Precopio: Haemonetics Corp., Other; R. L. Thurer: Haemonetics Corp., Other, Haemonetics Corp., Stocks or Bonds

Disclosure of Grants Conflict of Interest

M. A. Popovsky: Nothing to disclose; T. Precopio: Nothing to disclose; R. L. Thurer: Nothing to disclose

	Allogeneic RBC Transfusion Rate (%)	PAD Transfusion Rate (%)	Complication Rate (% of Patients)	LOS (Days)	Transfusion Rate (%) Lowest Hgb 7.0-7.9 g/dL	Transfusion Rate (%) Lowest Hgb 8.0-8.9 g/dL
Baseline	25.8	19.1	2.3	3.53	88.8	66.6
PBM Program	12.4	1.3	1.8	3.00	82.5	39.1

S71-030K

Utility of Lead Screening of Whole Blood Tail Segments to Avoid Lead Exposure in Children

M Moumina[1] (mashaelmm@hotmail.com), V Criss[1], E Gourdine[1], N L Luban[1,2], E Wong[1,2]. *[1]Division of Laboratory Medicine, Children's National Medical Center, Washington, DC, United States; [2]Pediatrics and Pathology, George Washington School of Medicine and Health Sciences, Washington, DC, United States*

Background/Case Studies: Lead exposure remains an important environmental problem affecting children worldwide. Recent studies suggest that there is no lead concentration threshold for neurocognitive injury. We instituted a policy of avoiding red blood cell (RBC) transfusions to neonatal and pediatric patients depending on donor blood lead level (BLL). RBCs from donors with BLL <3.0 μg/dL are transfused to all patients, while RBCs from donors with BLL ≥3.0 and ≤10.0 μg/dL are transfused to patients ≥7 years of age. In contrast, RBCs from donors with BLL > 10.0 μg/dL are not transfused to any patient. However, not all transfusion medicine services have access to donor ethylenediaminetetraacetic acid (EDTA) whole blood (WB) samples for testing. In this study, we determined whether WB tail segments could be used to screen RBC units with potentially high lead levels instead of donor WB samples. **Study Design/Methods: A** three-month prospective laboratory study comparing lead levels in blood donors to lead levels from corresponding WB tail segments was carried out. The LeadCare® Blood Lead Testing System (ESA Inc., Chelsford MA) previously validated against this atomic absorption method and FDA approved for patient testing was used to evaluate lead levels from 177 EDTA WB samples and corresponding WB tail segments (PALL Leukotrap® WB System, Covina, CA). Linear regression analysis was performed between lead levels of donor WB and tail segments. Intra-assay precision was performed on tail segments in the 1.0 to 3.0 μg/dL range. Because RBCs contain the majority of lead in WB, the mean (and standard deviation) hematocrit (Hct) of WB donors (Hematastat®, Separation Technology, Sanford, FL) and corresponding tail segments (spun Hct) were compared, in order to estimate the influence of Hct on lead levels in the samples. **Results/Findings:** Lead levels of tail segments and WB samples highly correlated with each other. Linear regression equation between tail segment and WB lead levels was $y = 0.595\times + 0.534$ with $R = 0.732$, with slightly higher values for WB samples, reflecting the higher Hct of WB versus tail segment samples (41.9 versus 35.7%). A concordance of 97.7% (173/177) between WB and tail segment lead levels was demonstrated for samples <3.0 μg/dL. At the threshold of 3.0 μg/dL, there were 3 discordant cases. In two cases, the tail segment was ≥3.0 μg/dL while the WB value was <3.0 μg/dL (3.0 vs 1.3 and 3.0 versus 2.8, respectively). In one case, the tail segment was 2.6 μg/dL while the WB value was ≥3.0 (3.9) μg/dL. The remaining comparison showed a WB lead level versus tail segment (6.7 versus 4.5 μg/dL), respectively. Coefficient of variation was 4.3% and 10.8% (standard deviation of 0.10 and 0.12 μg/dL) at a mean tail segment lead value of 2.4 and 1.1 μg/dL, respectively. **Conclusion:** Screening of WB tail segments can be utilized to assess donor blood lead levels to avoid transfusion of RBC units with potentially elevated lead levels to children, especially neonates.

Disclosure of Commercial Conflict of Interest

V. Criss: Nothing to disclose; E. Gourdine: Nothing to disclose; N. L. Luban: Nothing to disclose; M. Moumina: Nothing to disclose; E. Wong: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. Criss: Nothing to disclose; E. Gourdine: Nothing to disclose; N. L. Luban: Nothing to disclose; M. Moumina: Nothing to disclose; E. Wong: Nothing to disclose

S72-030K

Storage Age of Blood Transfused in the US: Is It Changing?

R L Thurer[1] (rthurer@haemonetics.com), T Precopio[1], M A Popovsky[1]. *[1]Haemonetics, Braintree, MA, United States*

Background/Case Studies: Transfusion medicine specialists and clinicians are increasingly concerned about the blood storage lesion and the potential adverse effects associated with transfusion of stored blood. While this subject is still controversial, it was attempted here to understand the variability in the storage duration of transfused allogeneic red blood cells (RBC) that currently exists among hospitals in the US. It was also sought to determine whether the transfusion community is already responding to concerns about blood storage by altering inventory management practices and providing fresher blood. **Study Design/Methods:** Using a proprietary blood management business intelligence portal (IMPACT® Online, Haemonetics Corporation, Braintree, MA), 18 hospitals were identified (164,391 transfused units) in the US for which there is data regarding the storage duration of RBC transfused during 2012. For eight of the hospitals, there is data for a 4-year period from 2009-2012 (365,683 transfused units). **Results/Findings:** The mean age of transfused blood during 2012 was 20.9 ± 2.7 days and varied from 14.8 (median = 13) to 24.1 (median = 24) days. There was no substantive change in the duration of storage of transfused RBC in the hospitals studied over a 4-year period (Table). Approximately 25% of units continue to be transfused following more than 4 weeks of storage. Analysis of specific blood types also revealed no meaningful change except for O+ units. From 2009-2010, 50% of O+ units were transfused following less than 3 weeks storage. This increased to 60% during 2011-2012. **Conclusion:** There is considerable variability among hospitals in the storage age of transfused RBC. Despite concerns about the blood storage lesion, there has been no change in the age of blood transfused during the past 4 years. If the randomized trials currently in progress show that transfusions of blood stored for shorter durations result in better outcomes, the transfusion medicine community will face a significant challenge in responding to clinical demands for fresher transfusions.

Disclosure of Commercial Conflict of Interest

M. A. Popovsky: Haemonetics Corp., Other; T. Precopio: Haemonetics Corp., Other; R. L. Thurer: Haemonetics Corp., Other, Haemonetics Corp., Stocks or Bonds

Disclosure of Grants Conflict of Interest

M. A. Popovsky: Nothing to disclose; T. Precopio: Nothing to disclose; R. L. Thurer: Nothing to disclose

Storage Duration of Units Transfused 2009-2012

	# units (% of year's total)		
	0-20 days	21-28 days	≥29 days
2009	47,180 (47.5%)	27,123 (27.3%)	24,942 (25.1%)
2010	44,946 (48.1%)	24,812 (26.6%)	23,655 (25.3%)
2011	52,553 (56.6%)	21,703 (23.4%)	18,599 (20.0%)
2012	39,734 (49.6%)	20,986 (26.2%)	19,370 (24.2%)

Red Blood Cells: Blood Group Genes and Genotyping

S73-030L

Evidence That Microarray Genotyping is an Accurate Predictor of a Blood Group Phenotype

J M Moulds[1] (jmmoulds@lifeshare.org), G T Noumsi[1], J Hendrix[2], D McCaskill[1]. *[1]Scientific Support, LifeShare Blood Centers, Shreveport, LA, United States; [2]Immunohematology Reference Lab, Oklahoma Blood Institute, Oklahoma City, OK, United States*

Background/Case Studies: Currently the "gold standard" for blood group determination is based on serological reagents. However, the implementation of molecular methods for blood grouping has raised the question as to which is the most accurate method. During this initial microarray validation, it was noted that samples from ethnic minorities more frequently gave aberrant results. Thus, a group of African-American (A-A) or Native-American (N-A) blood donors was studied by both methods to determine their overall accuracy. **Study Design/Methods:** A total of 2627 samples were tested using FDA licensed antisera and by microarray (wHEA 3.3, BioArray Solutions). Antigens used for comparison included MN, Ss, Cc, Ee, K1, Fy^a/Fy^b, and Jk^a/Jk^b. Samples that initially appeared discordant (and were not due to data entry errors) were repeated with additional sources of licensed antisera and other molecular methods including DNA sequencing when appropriate. **Results/Findings:** A total of 12 samples (0.46%) were investigated due to conflicting results. The antigens that were discordant included: 4 S, 2 s, 2 E, 1 e, 1 c, 1 Jk^a and 1 Jk^b. Upon additional testing, 5 of the 12 samples gave results that were now concordant between serology and DNA. Of note in this group an Arg35His change in *GYPB* adversely affected the anti-s binding yielding a false negative with the initial antisera. A discrepant c typing, that was negative with the first antisera but positive with two other licensed antisera, was found to be due to a *RHCE*ceJAL* allele being paired with *RHCE*Ce*. Of the 7 samples (5 A-A, 2 N-A) that remained discordant upon repeat testing, various mechanisms were identified as the root cause. A S+ result by HEA, that was negative by serology, was shown to be due to a new *GYP(A-B)* hybrid allele that has the highest homology to *GYP-SAT*. A false negative genotype for a *GYPB*s* allele was due to two new SNPS in *GYPB* which adversely affected the primer binding. In the RH system, a *RHCE*ceMO* paired with a *RHCE*cE* resulted in a weakened e antigen that was detected with some but not all antisera. This sample gave an indeterminate call by HEA. In addition, the 2 N-A donors had the same 221G > A mutation which caused the silencing of their *RHCE*cE allele*. Other silenced alleles were found in the JK system including a *JK**A amorph due to a 866A > G substitution. **Conclusion:** Discordant results between serological and DNA based methods for blood grouping are very rare. More importantly, the study shows that contradictory results may be obtained even when using FDA licensed antisera for typing. The possibility of incorrectly typing a sample due to a new DNA mutation in the A-A and N-A populations was only 0.0002 and may be even lower in a Caucasian population.

Disclosure of Commercial Conflict of Interest

J. Hendrix: Nothing to disclose; D. McCaskill: No Answer; J. M. Moulds: BioArray Solutions, Grants or Research Support; G. T. Noumsi: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Hendrix: Nothing to disclose; D. McCaskill: No Answer; J. M. Moulds: BioArray Solutions, Grants or Research Support, Grifols, Consulting or Board of Director Fees, Grifols, Travel Support or Honorarium; G. T. Noumsi: Nothing to disclose

S74-030L

Let's Go Nano! Super Luminescent Nanoparticles for Bioimaging and Blood Genotyping Applications

D Brouard[1,2] (danny.brouard@hema-quebec.qc.ca), O Ratelle[2], D Boudreau[2], M St-Louis[1]. *[1]Research and Development, Héma-Québec, Québec, QC, Canada; [2]Département de chimie, Université Laval, Québec, QC, Canada*

Background/Case Studies: Noble metal nanoparticles (NPs) are attracting much attention lately and are increasingly integrated in innovative biosensing and bioimaging approaches. When metallic nanostructures are excited by an electric field, free electrons collectively oscillate in resonance with light, a phenomenon known as localized surface plasmon resonance (LSPR). The LSPR frequency of metal NPs can be tuned to interact positively with nearby fluorophores leading to an enhancement of their apparent quantum yield and photostability, a photophysical process called metal enhanced fluorescence (MEF). To exploit MEF, dyes can be encapsulated in an outer silica shell. This basic architecture also offers the opportunity to benefit from a flexible surface chemistry. NPs can be modified with a variety of functional groups facilitating the realization of crosslinking reactions with biomolecular probes or target biomolecules of interest. The objective of this work was to develop a fast and scalable synthesis for the preparation of these highly fluorescent NPs. **Study Design/Methods:** Luminescent silver/silica core-shell NPs were first applied to blood genotyping applications by flow cytometry (FC). Silver cores (40 nm) were prepared by the silver nitrate citrate reduction. Fluorophore molecules were incorporated in the silica shell (10 nm) using a silica precursor and a dye-silane coupling agent. Within 24 hours, large volumes (>500 mL) of fluorescent NPs solutions were obtained ([NPs] = 1E12 NPs/mL). The nanoparticlar architecture was confirmed by transmission electronic microscopy. First, single stranded 5'-NH2-oligonucleotide probes specific to DO*01 and DO*02 alleles were functionalized to a sulfhydryl modified surface using succinimidyl-4-(n-maleimidomethyl)cyclohexane-1-carboxylate. A cationic charged polymer was used to signale the hybridization reaction. Fluorescent signal was detected by FC. Second, mouse anti-human CD41 was grafted to NPs for cell imaging applications. **Results/Findings:** Specific single nucleotide polymorphism (SNP) detection in the picomolar range was achieved for oligonucleotide, amplicons or genomic human DNA used as targets. This translates to ~3E07 strands in a 50 µL hybridization volume. Because one recognition event is reported by thousands of MEF enhanced fluorophores, great detection contrast was observed for human platelets imaging using anti-CD41 NPs. **Conclusion:** Silver/silica core-shell nanoparticles were prepared by simple steps without any specific chemistry skills or apparatus needed. Versatile and affordable, they can be applied to a wide range of imaging and cell-tagging, or molecular detection applications to implement in blood banks and hospitals to perform fast and reliable PCR-free genotyping.

Disclosure of Commercial Conflict of Interest

D. Boudreau: Nothing to disclose; D. Brouard: Nothing to disclose; O. Ratelle: Nothing to disclose; M. St-Louis: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Boudreau: Nothing to disclose; D. Brouard: Nothing to disclose; O. Ratelle: Nothing to disclose; M. St-Louis: Nothing to disclose

S75-030L

Molecular Matching in Patients with Myelodisplastic Syndrome (MDS)

G A Guelsin[1] (gguelsin@gmail.com), F Traina[1], S Gilli[1], S T Olalla Saad[1], L Castilho[1]. *[1]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: Molecular DNA-based genetic methods have been provided an excellent tool for improving transfusion therapy in patients with Sickle Cell Disease (SCD), helping in decrease the risk of transfusion reactions, especially delayed transfusion reactions to existing alloantibodies, and in preventing alloimmunization. However, few studies are being performed in patients with myelodisplastic syndrome (MDS). Based on this, the usefulness of DNA methods was evaluated to provide a means to precisely genotype match donor blood units to the antigen-negative type of 34 Brazilian polytransfused patients(24 non alloimmunized and 10 alloimmunized) with MDS. **Study Design/Methods:** Antigen-match RBC units for ABO, D, C c E e, K, Fya, Fyb, Jka, Jkb, S, s, Doa, Dob, and Dia performing a molecular array (HEA BeadChip, BioArray Solutions) were investigated on the patient

samples and on the donor units serologically matched for them based on their ABO, Rh and K phenotypes, and presence of antibody(ies). **Results/Findings:** The most common extended phenotype predicted from the genotype found in the patients with MDS was D + C+c + E-e+ (R1r), K-, Fy(a + b+), Jk(a + b+), S-s+. Seventeen of 34 patients presented discrepancies or mismatches for multiple antigens between their genotype profile and the blood units antigen profile serologically matched for them. The main discrepancies or mismatches occurred in FY, JK, and MNS systems. We found discrepancies between the previous Rh phenotype and genotype-derived phenotypes in four alloimmunized patients chronically transfused. The majority of the alloimmunized patients had the *HLA-DRB1*04* allele with antibodies against the Rh antigens. Fifty percent of the polytransfused patients had autoantibodies. Although it was possible to find a better match for the patients in the genotype-typed units, matching for Rh, K, and S would be enough for most of the patients. **Conclusion:** Molecular analyses in patients with MDS are important to identify discrepancies in the serologic phenotype and the genotype-predicted phenotype, and for discrimination of allo versus autoantibodies. In the population, matching for C c E e, K showed to be cost effective for the treatment of chronically transfused patients. A possible association of *HLA-DRB1*04* allele and susceptibility to RBC alloimmunization in patients with MDS should be confirmed.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; S. Gilli: Nothing to disclose; G. A. Guelsin: Nothing to disclose; S. T. Olalla Saad: Nothing to disclose; F. Traina: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; S. Gilli: Nothing to disclose; G. A. Guelsin: Nothing to disclose; S. T. Olalla Saad: Nothing to disclose; F. Traina: Nothing to disclose

S76-030L

The Rare Dantu Phenotype: Serological Aspects and New Screening Approach through Red Blood Cell (RBC)Genotyping

J Nataf[1] (jnataf@ints.fr), S Poupel[1], D Gien[1], D Janvier[2], L Collet[5], G Brochier[3], J Antolini Gouvitsos[4], T Peyrard[1]. *[1]National Immunohematology Reference Laboratory, National Institute of Blood Transfusion, Paris, France; [2]EFS Ile de France, French National Blood Service, Paris, France; [3]EFS La Reunion, French National Blood Service, Saint Denis, France; [4]EFS Provence Alpes Cote d'Azur, French National Blood Service, Marseille, France; [5]Clinical Laboratory, Hospital of Mayotte, Mamoudzou, France*

Background/Case Studies: Dantu is a rare MNS variant type encoded by a hybrid *GYP(B-A)* glycophorin (GP) gene. The Dantu haplotype carries a normal *M-GYPA* gene, linked to a hybrid gene coding for the 39 N-terminal amino acids of GPBs fused to the 72-131 C-terminal residues of GPA. Three molecular types were reported: two in Africans (NE, Ph); and one in a Caucasian (MD). They produce a normal GPA-M, the same trypsin-resistant N and s GP(B-A) hybrid (with discrepant results for s reactivity depending on anti-s being used) and express the low-prevalence RBC antigen Dantu (MNS25). African Dantu types are considered U + ^{weak}, with an altered U antigen. Here we describe the study of 11 individuals (10 unrelated) referred to confirm anti-U immunization and/or a S-s- phenotype. **Study Design/Methods:** Patients' sera and red blood cells (RBCs) were investigated by standard hemagglutination techniques. Anti-S and anti-s typing reagents (polyclonal and monoclonal) were from Biorad and Diagast. U and Dantu antisera were previously collected. Genotyping was carried out with the HEA v1.2 BeadChip DNA bead-array technique (Immucor, BioArray Solutions). **Results/Findings:** 1) The first study was conducted on four subjects, native from Islands of the Indian Ocean (two pregnant women with anti-U, one relative and 1 S-s- donor), who were all found Dantu positive. They were U negative with anti-U made by U + ^{var} individuals but U positive with potent anti-U produced by U- patients. They typed s- with the monoclonal anti-s and s+ (4+) with the polyclonal anti-s. All were S negative with the 2 reagents used. When genotyped by the bead-array technique, all of them gave the same pattern: presence of the c.143T > C polymorphism (encoding the s antigen and named GPBS on BioArray report), associated with the failure to detect the point mutations c.230C > T (named GPB-230) and c.270 + 5g > t (named GPB-Int5). In brief, Dantu phenotype was found on the BioArray system to be GPBS BB, GPB-230 LS and GPB-Int5 LS (LS for "Low Signal"). 2) Since there were consistent results for the first four Dantu subjects, it was decided to extend the study to seven other cases with the same conflicting s and U typing results according to the reagents used. Six were pregnant women (three with anti-U) and one a blood donor. Five were from Islands of the Indian Ocean. Because of ABO incompatibility, their Dantu status could not be determined. Nevertheless, 7/7 showed the same BioArray profile, i.e. GPBS BB, GPB-230 LS and GPB-Int5 LS. **Conclusion:** A small cohort of S-s+/–U+ ^{weak} subjects with a singular RBC genotyping pattern is presented in this study, very likely corresponding to the rare Dantu type. Interestingly, 9/11 are native from Indian Ocean Islands (Malagasy, Reunion Island, Mayotte, Comoros) and exhibited a high rate of U alloimmunization (5/11). Further investigate the molecular background(s) of these Dantu people is planned.

Disclosure of Commercial Conflict of Interest

J. Antolini Gouvitsos: Nothing to disclose; G. Brochier: No Answer; L. Collet: No Answer; D. Gien: No Answer; D. Janvier: Nothing to disclose; J. Nataf: Nothing to disclose; T. Peyrard: Nothing to disclose; S. Poupel: No Answer

Disclosure of Grants Conflict of Interest

J. Antolini Gouvitsos: Nothing to disclose; G. Brochier: No Answer; L. Collet: No Answer; D. Gien: No Answer; D. Janvier: Nothing to disclose; J. Nataf: Nothing to disclose; T. Peyrard: Nothing to disclose; S. Poupel: No Answer

S77-030L

CD59 is a Candidate for a New Blood Group System

C Weinstock[1] (c.weinstock@blutspende.de), I von Zabern[1], B Hoechsmann[1], H Kyrieleis[2], C Dohna-Schwake[3], H Schrezenmeier[1], M Anliker[1]. *[1]Institute of Clinical Transfusion Medicine and Immunogenetics Ulm, German Red Cross Blood Service Baden-Wüerttemberg-Hessen, Institute of Transfusion Medicine, University of Ulm, Ulm, Germany; [2]Department of Pediatrics, Hospital Bethanien, Moers, Germany; [3]Department of Pediatrics, University Hospital of Essen, Essen, Germany*

Background/Case Studies: CD59 is a 20kDa cell surface glycoprotein that regulates complement-mediated cell lysis by inhibiting the incorporation of C9 into the final membrane attack complex. It is also known as membrane inhibitor of reactive lysis (MIRL) or as homologous restriction factor (HRF). Hereditary CD59 deficiency is very rare, these patients suffer from Coombs-negative hemolysis and severe neurological impairment. Investigating a young child with inherited CD59 deficiency, who was homozygous for the new CD59 allele c.146delA (Höchsmann B. et al 2013, submitted), an alloantibody against a high frequency erythrocyte antigen was found in her serum. It could be shown that this antibody was directed against CD59. **Study Design/Methods:** Serologic testing was done with a gel agglutination method (indirect antiglobulin test (IAT); DiaMed, Switzerland). Serum samples for control containing alloantibodies (anti-c, -C, -K, and -Fy^a) were from non-hematologic patients. For flow cytometry (FC), fluorescence-labelled anti-CD59 and anti-CD55 were used (Becton Dickinson, US). For inhibition studies polyhistidine-tagged recombinant soluble CD59 protein (sCD59) was used (Creative Bio Mart, US). **Results/Findings:** No CD59 expression was detected on the patient's RBC by flow cytometry. Her parents' RBC showed decreased expression of CD59, when compared to unselected donors. The patient serum agglutinated the RBC of several unselected donors (IAT; titer 4). After pre-incubation with soluble CD59 the serum reacted negative. For control, serum containing a different alloantibody was admixed to the patient serum. After pre-incubation with sCD59, only the patient's alloantibody was inhibited, but not the admixed alloantibody (anti-c, -C, -K, or -Fy^a, resp.). sCD59 also inhibited the monoclonal anti-CD59 used in FC, but not the monoclonal anti-CD55. Papain-treatment of RBC enhanced the reactivity of the patient serum, while dithiothreitol-treatment abolished it. **Conclusion:** Evidence is given that a CD59-deficient patient had formed an anti-CD59 antibody. Only immunoglobulins but no RBC had been administered when the child was 7 months old. However, one year later, when the first units of RBC had to be transfused after a hemolytic crisis, an antibody reacting with all donors and test cells was present (CD59 deficiency and, thus, the specificity of the antibody was not known at that time). No transfusion reactions were reported. Serologically the antibody specificity is easily identifiable by inhibition by recombinant sCD59. This is the first demonstration of an in vivo-produced antibody directed against the red cell membrane constituent CD59. Sequence analysis showed that both parents were heterozygous for the allele causing the $CD59_{null}$ phenotype in this child, indicating that the deficiency of the CD59 antigen was inherited. Therefore, CD59 is a candidate for a new blood group system.

Disclosure of Commercial Conflict of Interest

M. Anliker: Nothing to disclose; C. Dohna-Schwake: Nothing to disclose; B. Hoechsmann: Nothing to disclose; H. Kyrieleis: Nothing to disclose; H. Schrezenmeier: No Answer; I. von Zabern: Nothing to disclose; C. Weinstock: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Anliker: Nothing to disclose; C. Dohna-Schwake: Nothing to disclose; B. Hoechsmann: Alexion, Grants or Research Support, Alexion, Consulting or Board of Director Fees, Alexion, Travel Support or Honorarium; H. Kyrieleis: Nothing to disclose; H. Schrezenmeier: No Answer; I. von Zabern: Nothing to disclose; C. Weinstock: Nothing to disclose

S78-030L

Autoanti-Jk3 and Alloanti-Jka in a Patient with a Variant JK*A Gene

C Eckley[1] (ceckley@bloodsystems.org), D Figueroa[1], R Hoffman[1], D B Oliver[1], K Bielen[1], M S Irani[1], G T Noumsi[2], J M Moulds[2], S Vege[3], C M Westhoff[3]. *[1]Clinical Laboratories, Blood Systems, Inc., Tempe, AZ, United States; [2]Support Services Laboratory, Lifeshare Blood Centers, Shreveport, LA, United States; [3]Laboratory of Immunohematology and Genomics, New York Blood Center, Long Island City, NY, United States*

Background/Case Studies: A 58-year-old female of German descent with a diagnosis of metastatic ovarian cancer presented with dyspnea, anemia (hemoglobin 8.3 g/dL, hematocrit 25%) and a malignant pleural effusion. She was A, Rh-negative and last transfused in 1972. An antibody screen with autocontrol exhibited reactivity with all cells. PeG auto-adsorptions performed by the hospital laboratory failed to remove the reactivity. The patient's sample was submitted to the immunohematology reference laboratory for antibody identification. **Study Design/Methods:** The antibody screen, direct antiglobulin test (DAT) and eluate were performed using standard serologic tube methods. The patient's red blood cells (RBCs) were treated using the EGA kit (Immucor). Allogeneic adsorptions were performed using ficin-treated R1R1 Jk(a + b-) and rr Jk(a-b+) RBCs. Autologous adsorptions were performed using ZZAP-treated RBCs. To assess clinical significance of the antibodies, a two-stage complement sensitization monocyte monolayer assay (MMA) was performed. DNA was isolated by standard methods and genotyped by BioArray/Immucor HEA. JK gene sequencing was performed by exon-specific amplification and comparison to conventional sequences. **Results/Findings:** The screen was suggestive of a warm-reactive autoantibody. The patient's untreated RBCs were phenotyped as Jk(a-b-) using monoclonal antisera. The plasma and eluate were tested against selected RBCs and showed anti-Jk3 specificity. Autologous adsorptions removed the apparent anti-Jk3, while a strongly reactive anti-Jka remained. The DAT was strongly positive (4+) with Polyspecific, IgG, and C3 AHG. EGA treatments decreased the strength of the DAT (2+) and the patient's RBCs then typed as Jk(a + b+) with monoclonal antisera. The patient's sample was referred for MMA and DNA sequencing. The patient received one Rh+;Jk:-3 unit with no clinical sign of overt hemolysis. A pre transfusion MMA showed a monocyte index (MI) of 28.4% for the anti-Jka, and 6.3% for the anti-Jk3. However, a few months later the post-transfusion MIs increased to 36.2% and 8.9% respectively. The patient's genotype was JK*A/JK*B. JK exon- sequencing revealed a nucleotide change in exon 7, 486T > A (Ser162Arg), and the previously reported 588 A > G (silent) change. **Conclusion:** The patient presented with warm-reactive autoantibody with anti-Jk3 specificity with an underlying alloanti-Jka. The MMA indicated both the autoanti-Jk3 and anti-Jka were clinically significant. The finding of a novel nt 486A change in JK encoding an amino acid change Ser to Arg likely explains the production of alloanti-Jka in a Jk(a+) patient. The initial typing with monoclonal Jka and Jkb antisera gave negative results due to the heavy amount of autoanti-Jk3 coating the patient's RBCs and blocking the antigen sites. A biological explanation for the concurrent production of autoanti-JK3 remains elusive.

Disclosure of Commercial Conflict of Interest

K. Bielen: Nothing to disclose; C. Eckley: Nothing to disclose; D. Figueroa: Nothing to disclose; R. Hoffman: Nothing to disclose; M. S. Irani: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support, Novartis, Travel Support or Honorarium, Grifols, Consulting or Board of Director Fees; G. T. Noumsi: Nothing to disclose; D. B. Oliver: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Bielen: Nothing to disclose; C. Eckley: Nothing to disclose; D. Figueroa: Nothing to disclose; R. Hoffman: Nothing to disclose; M. S. Irani: Nothing to disclose; J. M. Moulds: Nothing to disclose; G. T. Noumsi: Nothing to disclose; D. B. Oliver: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Components and Component Processing

S79-040A

Baseline Variability of Whole Blood in Vitro Quality Parameters: An Analysis of 61 Donor-Matched Pairs

*S S Yonemura[1] (*susan.yonemura@Terumo BCT.com*)*, S Marschner[1], R P Goodrich[1]. *[1]Scientific Affairs, Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Evaluating new blood processing techniques involves in vitro tests to assess physical or biochemical changes in blood products resulting from investigational methods. In the US, FDA often recommends defining success in studies with at least 60 donor-matched pairs as ≤20% difference 95% of the time with a one-sided 95% confidence limit (95-95 rule), predicated on the idea that paired studies minimize the effect of donor-to-donor variation. This analysis examined the variability of baseline parameters measured in pairs of whole blood (WB) units collected ≥8 weeks apart. **Study Design/Methods:** Paired WB units were collected from 61 donors. WB was collected in CPD, targeting 500 mL ± 10% WB volume with 70 mL anticoagulant. WB samples were taken at 45-240 minutes following collection and kept neat or separated into red blood cell, platelet, and plasma fractions. In vitro measurements were compared using paired t-tests where the 95-95 rule was interpreted as a non-inferiority test with a 20% margin and a one-sided 95% confidence limit. Individual failures (pair variance ≥20%) were also tallied. **Results/Findings:** Seven parameters did not meet the non-inferiority criterion and many exhibited more than one failure (Table). Parameters with no failures were pH, RBC mass, Hct, total Hb, Na^+, K^+, bicarbonate, glucose, pCO_2, pO_2, PT, aPTT, and antithrombin. **Conclusion:** Substantial within-donor variability can be seen when in vitro quality parameters are measured in consecutive WB collections spaced at least 8 weeks apart. Given this variability, donor-matched controls may not be an appropriate comparison for all parameters, or more sophisticated statistical methods are needed to differentiate donor and treatment effects. Individual failures should be interpreted within the context of clinical relevance.

Disclosure of Commercial Conflict of Interest

R. P. Goodrich: Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other; S. S. Yonemura: Terumo BCT, Other

Disclosure of Grants Conflict of Interest

R. P. Goodrich: Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other; S. S. Yonemura: Terumo BCT, Other

Parameter	Difference (95% CI)	# Failures/N
Protein S	8.9% (10.4%)	2/61
HSR	8.5% (9.9%)	3/61
Plasma Hb	5.7% (10.8%)	5/61
Factor V	8.9% (10.2%)	5/61
ATP in RBC	9.5% (11.2%)	5/61
Plt Yield	9.9% (12.1%)	5/61
Fibrinogen	9.6% (11.8%)	6/61
vWF activity	11.3% (13.5%)	7/61
Residual WBC	11.5% (13.9%)	7/61
Protein C	13.5% (15.8%)	14/60
ATP in Plt	14.2% (17.0%)	14/61
ESC	14.1% (17.1%)	15/60
TAT *	32.6% (54.4%)	17/60
Factor XI	13.6% (16.0%)	18/61
2, 3-DPG *	18.7% (21.5%)	25/61
Factor VIIIc *	18.1% (21.1%)	26/61
Lactate *	22.1% (26.3%)	28/61
Factor VIIa *	27.2% (32.9%)	34/61
CD62P *	42.1% (51.2%)	36/52
Annexin V *	53.6% (67.2%)	53/59

* Failed non-inferiority criterion (20% falls within 95% confidence interval)

S80-040A

Donor-Specific Red Blood Cells (RBC) Fingerprints: Metabolomic Analysis Identifies a Subset of RBC Biomarkers with Storage Kinetics that Diverge between Donors and which Persist on Subsequent Donations

L J Dumont[1] (Larry.J.Dumont@hitchcock.org), J L Newman[2], S K Karatela[2], J C Zimring[3], J D Roback[2]. *[1]Department of Pathology, Geisel School of Medicine at Dartmouth University, Hanover, NH, United States; [2]Pathology and Laboratory Medicine, Emory University School of Medicine, Atlanta, GA, United States; [3]Puget Sound Blood Center Research Institute, Seattle, WA, United States*

Background/Case Studies: Quantification of chemical metabolites in red blood cells (RBCs) at the time of donation and during storage represents a promising approach to identify RBC units with suboptimal survival post-transfusion and/or the potential to cause adverse transfusion effects. Global quantification of RBC metabolites was previously performed over 42 days storage for RBCs collected from a cross-section of volunteer donors. New reports now show that the donor-specific metabolic fingerprints that were observed are largely reproducible and stable over multiple donations by the same donor. **Study Design/Methods:** Leukoreduced RBC components in AS-1 were prepared from three donors on two separate occasions each. The units were serially sampled over the 42 days of storage. Samples were snap-frozen on LN2 and later analyzed in parallel by GC/MS and LC/MS/MS. Biochemicals were identified by comparisons to a library of mass spectra derived from purified standards. **Results/Findings:** 221 and 264 defined metabolites were positively identified and quantitated in the first and second sets of RBC donations, respectively; of these biochemicals, 200 were found in both sets, and were further evaluated. These metabolites were divided into three groups. Ninety-three% (186 of 200) of these metabolites maintained similar expression patterns (longitudinal changes in levels as a function of storage time) after each donation. For 166 of these 186 consistent metabolites (83% of total; Group 1), the patterns for each donor were closely superimposed upon one another; metabolites illustrative of this pattern included 2, 3 DPG, lactate, and pyruvate. For the remaining 20 of these 186 metabolites (10% of total; Group 2), there were donor-specific expression patterns that significantly diverged from the RBC metabolomic patterns seen in the other donors (defined as persistent differences in metabolite levels >2-fold between donors); these divergent patterns were maintained in samples from subsequent donations. Examples included aspartate, catechol sulfate, pantothenate, and caffeine. For the remaining 7% of metabolites (14 of 200; Group 3), such as pipecolate and p-cresol sulfate, the patterns between the 3 donors was altered from the first collection to the second. **Conclusion:** For pretransfusion RBC metabolomic analysis to be clinically useful, metabolite kinetic patterns should be largely reproducible for individual donors over repeated donations. The present results show that the vast majority (93%) of RBC metabolites maintained identical (or very similar) expression kinetics during subsequent donations. Among these were 20 metabolites (Group 2) that showed significant variations in longitudinal patterns between donors, which persisted on subsequent donations. Thus, metabolites from Group 2 represent important targets for investigation to identify a "metabolic fingerprint" for pretransfusion testing to predict transfusion efficacy and safety.

Disclosure of Commercial Conflict of Interest

L. J. Dumont: CitraLab, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Advanced Preservation Technologies, Consulting or Board of Director Fees; S. K. Karatela: No Answer; J. L. Newman: Nothing to disclose; J. D. Roback: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. J. Dumont: CitraLab, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Fenwal, a Fresenius Kabi Company, Grants or Research Support, Haemonetics, Grants or Research Support, New Health Sciences, Grants or Research Support, Terumo BCT, Grants or Research Support, Advance Cell Technologies, Consulting or Board of Director Fees, Advanced Preservation Technologies, Consulting or Board of Director Fees, New Health Sciences, Consulting or Board of Director Fees, Terumo BCT, Consulting or Board of Director Fees; S. K. Karatela: No Answer; J. L. Newman: Nothing to disclose; J. D. Roback: Nothing to disclose; J. C. Zimring: Immucor, Grants or Research Support, Terumo, Grants or Research Support, Haemonetics,, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo, Travel Support or Honorarium

S81-040A

Depletion of Microparticles from the Supernatant of Stored Red Blood Cells (RBCs) Reduces Adhesion of Fresh Allogeneic Neutrophils to MCP-Treated Endothelial Cells under Continuous Flow Perfusion

W Xu[1] (wexu8277@hotmail.com), A K Sran[1], R L Sparrow[1]. *[1]Research and Development, Australian Red Cross Blood Service, Melbourne, VIC, Australia*

Background/Case Studies: For certain patient groups, red cell (RBC) transfusion has been linked to adverse immune responses, referred to as Transfusion-Related Immunomodulation (TRIM). Endothelial dysfunction may be implicated. Previously it was found that longer-stored RBCs promote the adhesion of fresh allogeneic neutrophils (PMN) to endothelial cells (EC) in an in vitro continuous flow perfusion model designed to simulate microvascular blood flow. Microparticles (MP) and cell-free hemoglobin (Hb) have been shown to contribute to immune responses. The aim of this study was to determine whether MP and Hb in the supernatant of stored RBC red cell survival (RCS) can modulate the interaction of PMN with EC using our in vitro continuous flow perfusion two-insult model of TRIM. **Study Design/Methods:** Leukoreduced RBCs (n = 7) were prepared and stored by standard Blood Service procedures. RCS was collected at 42 days. RCS was depleted of MP by centrifugation at 18,000x*g* (n = 7), depleted of Hb by treatment with a Hb-binding agent (HemogloBind) (n = 7) or depleted of both MP and Hb (n = 5). Pelleted MP (MP+) (n = 5) were resuspended in perfusion medium. Fresh allogeneic PMN (representing the patient's cells) were isolated from healthy volunteer blood (n = 7). Human umbilical vein endothelial cells (EC) (representing the patient's blood vessel wall) were cultured on multichannel flow perfusion slides. EC were stimulated with monocyte chemotactic protein-1 (10 ng/mL) for 4 h at 37°C (1st insult) and perfused for 2 min at 37C at a low shear stress of 0.5 dyne/cm^2 with one of medium alone (control), RCS, MP and/or Hb-depleted RCS or MP+ (all at a 5% final dilution) (2nd insult). Pretreated EC were perfused with PMN for 3 minutes. PMN adhesion to EC was recorded via a CCD camera fitted to an inverted microscope and analyzed with imaging software. Results are mean ±SD of 10 fields/sample. Paired t-tests determined significance. **Results/Findings:** Preperfusion of EC with RCS resulted in significantly increased adhesion of PMN compared to control (25 ± 4 vs 6 ± 2 adherent PMN/field; $p < 0.001$). In contrast, there was no significant difference between EC preperfused with control or with RCS that had been depleted of MP, Hb or both (6 ± 2, 8 ± 4, 6 ± 2, and 7 ± 5 adherent PMN/field, respectively). Preperfusion of EC with MP+ caused similar level of PMN adhesion as seen with RCS, which was significantly increased compared to control (23 ± 7 vs 6 ± 2 adherent PMN/field; $p = 0.002$). **Conclusion:** Depletion of MP or Hb from RCS reduced the adhesion of PMN to EC to levels comparable to preperfusion with medium alone. As a proportion of cell-free Hb is encapsulated in MP, further work is required to determine the relative contribution of MP versus Hb to PMN-EC adhesion. The results suggest that MP and Hb in RCS could play a bioactive role in the interaction of PMN and EC, which may be relevant to the biomechanism of TRIM.

Disclosure of Commercial Conflict of Interest

R. L. Sparrow: Nothing to disclose; A. K. Sran: Nothing to disclose; W. Xu: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. L. Sparrow: Nothing to disclose; A. K. Sran: Nothing to disclose; W. Xu: Nothing to disclose

S82-040A

Functional Inactivation of White Blood Cells by Riboflavin and Visible Light

B Zhang[1] (lotus_0411@hotmail.com), L Wang[2], Y Huang[1], Q Mo[1], X Wang[1], K Qian[1]. *[1]Institute of Transfusion, Shanghai Blood Center, Shanghai, China; [2]School of Life sciences, East China Normal University, Shanghai, China*

Background/Case Studies: Transfusion of allogeneic white blood cells (WBCs) may cause adverse reactions in immunocompromised recipients, including transfusion-associated graft-versus-host disease (TA-GVHD) which is often fatal and incurable. To reduce the risk of TA-GVHD, gamma irradiation was adopted for prevention. However, since the technical and clinical challenges associated with the use of gamma irradiators, pathogen reduction technologies were considered to be used to inactivate WBCs through photochemical effects. Although Mirasol and Intercept Blood System have been proved effective in WBC inactivation, ultraviolet light that both systems use for photosensitizer activation may cause extra damage to blood

components and operators. Thus, the in vitro inactivation effects of WBCs by riboflavin with visible light was evaluated here. **Study Design/Methods:** In this study, the in vitro effects of riboflavin with visible light (RB-Vis) treatment on lymphocyte proliferation and cytokine secretion were assessed. Peripheral blood mononuclear cells (PBMCs) collected from donors were suspended in autologous plasma with addition of 100 μM riboflavin and exposed to 420 nm visible light. The in vitro function of cells after RB-Vis treatment was examined and compared to cells obtained from untreated or gamma-irradiated cells by measuring cell proliferation, cytokine secretion and apoptotic analysis. **Results/Findings:** Our results demonstrated that RB-Vis treatment was more effective than gamma irradiation in inhibiting cell proliferation after phytohemagglutinin stimulation, with the average inhibition rate (n = 9) of 82.89% and 53.78%, respectively (Table). The secretion of IFN-γ and IL-10 were more effectively inhibited by RB-Vis treatment compared with irradiated cells. IL-6, IL-8 and TNF-α production were inhibited as effectively as gamma irradiation. The proportion of apoptotic and dead cells of RB-Vis and irradiation group are significantly higher than those of untreated group. **Conclusion:** RB-Vis treatment inhibits the ability of PBMCs proliferation and cytokine secretion as effectively as gamma irradiation, providing a potential alternative for TA-GVHD prevention.

Disclosure of Commercial Conflict of Interest

Y. Huang: No Answer; Q. Mo: No Answer; K. Qian: No Answer; L. Wang: No Answer; X. Wang: Nothing to disclose; B. Zhang: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Huang: No Answer; Q. Mo: No Answer; K. Qian: No Answer; L. Wang: No Answer; X. Wang: Nothing to disclose; B. Zhang: Nothing to disclose

Inhibition rate of lymphocytes proliferation upon PHA stimulation

	Control		Gamma irridiation			RB-Vis		
Sample No.	PHA–	PHA+	PHA–	PHA+	Inhibition rate of proliferation (%)	PHA–	PHA+	Inhibition rate of proliferation (%)
1	0.148	2.555	0.072	1.558	38.28	0.073	0.096	99.07
2	0.272	0.701	0.171	0.208	91.46	0.161	0.166	98.91
3	0.222	0.354	0.114	0.207	29.11	0.154	0.159	96.20
4	0.203	0.634	0.182	0.211	93.43	0.193	0.170	100
5	0.701	1.205	0.164	0.410	51.22	0.357	0.380	95.31
6	0.268	0.794	0.155	0.324	67.89	0.246	0.570	38.44
7	0.187	0.388	0.124	0.263	30.56	0.100	0.173	63.62
8	0.196	0.418	0.173	0.213	82.04	0.166	0.268	54.49
9	0.168	0.219	0.153	0.225	0.00	0.166	0.137	100

Cells were labeled with BrdU for 24 h. Values represent the O.D. measured by light absorbance at 450 nm. The inhibition rate of proliferation were calculated using the formula: [1-(AtPHA-At)/(AcPHA-Ac)]*100%.

S83-040A

Red Blood Cell (RBC) Storage in DINCH-PVC Storage Containers using Current and Next Generation Additive Solutions

K Radwanski[1] (katherine.radwanski@fenwalinc.com), K Min[1]. *[1]Fresenius Kabi USA, Lake Zurich, IL, United States*

Background/Case Studies: Over the last few years, the feasibility of using DINCH as an alternate plasticizer to DEHP in RBC storage containers has been evaluated using various commercially available additive solutions (eg AS-1, AS-3, SAGM) with substandard results in regards to spontaneous hemolysis during storage (approx. 0.5% on day 42). In previous work, it was found that E-Sol 5 (a next gen additive solution) reduces hemolysis levels in DINCH-stored RBCs down to or below levels observed in DEHP containers with AS-1. PAGGS-M is an alternative additive solution commercially available in the EU for storage of RBCs for up to 49 days, which has been shown to provide superior RBC storage when compared to SAGM. The objective of this study was to compare day 42 in vitro parameters for RBCs stored in DINCH-PVC containers with PAGGS-M or E-Sol 5 to control data from DEHP containers with AS-1 under worst case non-mixed storage conditions. It was hypothesized that RBCs stored in DINCH containers with PAGGS-M would have day 42 hemolysis levels that were comparable to levels in DEHP-AS-1 and DINCH-E-Sol 5 stored RBCs. **Study Design/Methods:** Paired, ABO matched whole blood (WB) units in CPD were collected into DEHP-PVC bags and pooled into non-plasticized bags within 15 minutes of collection. Pooled WB units were split into two containers and each unit was leukofiltered and processed into plasma and a red cell concentrate (RCC). Both units were stored in DINCH-PVC storage containers with either 105 mL of E-Sol 5 or 105 mL of PAGGS-M. Units were stored upright at 4C for 42 days with sampling on day 0 and 42 only. Data are compared to similarly prepared units stored with 110 mL of AS-1 in DEHP-PVC storage containers. **Results/Findings:** Total Hgb content was not significantly different among the three arms. ATP levels were above 2.7 (μmol/g Hb) on day 42 in all arms-3.2 ± 0.5, 3.5 ± 0.5, and 3.0 ± 0.4 for DINCH/PAGGSM, DINCH/E-Sol 5 and DEHP/AS-1 respectively. RBC metabolism was highest is DINCH/E-Sol 5 units, which also had the lowest levels of microparticles and hemolysis on day 42. Hemolysis and microparticle generation were comparable between DINCH/PAGGSM and DEHP/AS-1 units. **Conclusion:** RBCs stored in DINCH storage containers with PAGGS-M additive solution were comparable to RBCs in DEHP containers with AS-1 in regards to end of storage hemolysis levels, thereby showing feasibility of this storage container/additive solution combination. However, RBC storage in DINCH containers with E-Sol 5 additive solution was superior to both.

Disclosure of Commercial Conflict of Interest

K. Min: Nothing to disclose; K. Radwanski: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Min: Fresenius, Grants or Research Support; K. Radwanski: Fresenius, Grants or Research Support

D42 Parameters (Mean ± SD)	DINCH PAGGSM (n = 6)	DINCH E-Sol 5 (n = 6)	DEHP AS-1 (n = 6)
Lactate (mM)	29 ± 2*	35 ± 3*†	27 ± 2†
Microparticles (×10^3/μL)	36 ± 12*	19 ± 6*	46 ± 14
Hemolysis (%)	0.31 ± 0.15*	0.16 ± 0.05*†	0.27 ± 0.09†

*, † $p < 0.05$

S84-040A

Increased Cytokine Release from Platelet Concentrates Following Riboflavin/UV Light Treatment Accelerates Endothelial Cell Proliferation and Migration

P Schubert[1] (peter.schubert@blood.ca), B Culibrk[1], B Lautens[1], L Culibrk[1], E M Conway[1], R P Goodrich[2], D Devine[1]. *[1]Centre for Blood Research, Canadian Blood Services, Vancouver, BC, Canada; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Pathogen reduction technologies (PRT) seem to accelerate the development of platelet storage lesion, although prophylactic use of PRT platelets to treat thrombocytopenia is not associated with increased bleeding. During storage, platelets release granule proteins into the storage medium and upon transfusion; some of these proteins can interact with the blood vessel endothelium. Whether PRT alters the interaction between platelet factors and vessel endothelium is of interest in understanding the risk profile of PRT. **Study Design/Methods:** In a pool-and-split study, BC/PCs were either treated with riboflavin/UV light or untreated as control. Samples collected on day 1 (= day of production), days 2, 5 and 7 of storage were spun at 1500 rpm for 15 min to prepare the supernatant. Releasates from days 1 and 7 were screened with a cytokine antibody array and results were verified by ELISA. Supernatants collected throughout storage were incubated for 15 min with cultured Ea.hy926 cells, a HUVEC-adenocarcinoma immortalized cell hybrid. Treated endothelial cells were washed with PBS and lysed in a detergent-containing buffer. Cell lysates were analyzed for phosphorylation levels of kinases using immuno-blot.

Wound scratch assays were performed with and without exposure to the releasates and healing progression defined as a return of monolayer confluence as monitored by microscopy. **Results/Findings:** The cytokine antibody array revealed that both platelet-derived growth factor (PDGF) and epidermal growth factor (EGF) were elevated in the releasate of the PRT-treated samples by ~1.5-fold immediately after the treatment and by ~3-fold after 7 days of storage. Only small increases were seen in the untreated samples indicating that PRT triggered this release. Cells incubated with releasates collected on day 7 after PRT treatment showed the fastest cell migration resulting in scratch wound closure (monolayer confluence) within 43 hours. The control cells incubated with releasates prepared on day 7 without PRT treatment exhibited just 73 ± 4 % wound closure at that time point. To investigate the potential association with angiogenesis in this model, the phosphorylation status of kinases involved in cell migration was assessed. ERK and substrate proteins of PKC were phosphorylated by 3-fold and 2 to 5-fold, respectively by the releasate obtained from PRT-treated platelets; however, p38 and AKT phosphorylation levels were unchanged. **Conclusion:** This study identified accelerated growth of recipient vascular endothelial cells as a potential positive effect of PRT treatment of platelet concentrates. Although these studies were conducted in a model cell culture system, the findings suggest that released cytokines may contribute to improved vessel integrity following transfusion of platelets treated with riboflavin/UV light. This feature may overcome PRT-mediated platelet quality defects in prophylactic platelet transfusions.

Disclosure of Commercial Conflict of Interest

E. M. Conway: Nothing to disclose; B. Culibrk: Terumo BCT, Grants or Research Support; L. Culibrk: Terumo BCT, Grants or Research Support; D. Devine: Terumo BCT, Grants or Research Support; R. P. Goodrich: Terumo BCT, Other; B. Lautens: commercial products or services are discussed, Other; P. Schubert: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

E. M. Conway: Nothing to disclose; B. Culibrk: Nothing to disclose; L. Culibrk: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support, Fresenius Kabi, Grants or Research Support, MacoPharma, Grants or Research Support; R. P. Goodrich: Terumo BCT, Other; B. Lautens: Terumo BCT, Grants or Research Support; P. Schubert: Terumo BCT, Grants or Research Support

Recipient Non-Infectious Adverse Events

S85-040B

Red Blood Cells (RBC) Transfusions Modulate the Host Inflammatory Response and Prime the Endothelium for Subsequent Injury through the Receptor for Advanced Glycation End Products

N Mangalmurti[1] (nspatel@mail.med.upenn.edu), D Qing[1], D Siegel[2], J S Lee[3], G S Worthen[4], S M Albelda[1]. *[1]Medicine, University of Pennsylvania, Philadelphia, PA, United States; [2]Pathology, University of Pennsylvania, Philadelphia, PA, United States; [3]Medicine, University of Pittsburgh, Pittsburgh, PA, United States; [4]Pediatrics, University of Pennsylvania, Philadelphia, PA, United States*

Background/Case Studies: RBC transfusions are associated with acute respiratory distress syndrome, multi-system organ failure, and mortality in susceptible patients. Ligands for the pro-inflammatory "Receptor for Advanced Glycation End products" (RAGE) have previously been identified on stored RBCs and have shown that RBC transfusion increases lung endothelial RAGE expression. RAGE, as a pattern recognition receptor, sustains inflammatory responses through engagement of pathogen-derived and endogenously-derived inflammatory signals. Since RAGE ligands are elevated in multiple disease states such as sepsis, it was hypothesized that the upregulation of RAGE by RBC transfusions would prime the host to subsequent injury. **Study Design/Methods:** RBCs from C57/BL6 mice were leukoreduced and stored for 14 days. C57/BL6 mice or RAGE KO mice were transfused via tail vein with RBCs or sterile PBS (10 ul/g). Two hours following RBC transfusion, lypopolysaccharide (LPS) (1.5 mg/kg) was administered via tail vein. Mice were sacrificed 4 hours following LPS administration and lung and plasma chemokines were measured. In order to quantify endothelial injury, soluble VCAM-1 was measured by ELISA. **Results/Findings:** Although LPS led to a predictable increase in plasma and lung chemokines, RBC transfusion did not increase plasma or lung chemokines when compared with PBS-treated mice. However, pre-transfused LPS-treated mice demonstrated a marked increase in systemic chemokines when compared with non-transfused LPS-treated mice (MIP-2 12.0 ± 2.4 v 3.3 ± 0.6 ng/mL, p = 0.006; KC 612.2 ± 47.8 v 326.9 ± 70.2 ng/mL, p = 0.004; MCP-1 117.8 ± 5.6 v 72.4 ± 8.7 ng/mL, p = 0.001). Shedding of VCAM-1 was enhanced in pre-transfused LPS-treated mice when compared with non-transfused LPS-treated mice (1591.9 ± 86.3 v 1288.2 ± 97.1 ng/mL). RAGE KO pre-transfused LPS-treated mice demonstrated attenuated plasma sVCAM-1 levels when compared with WT-transfused LPS-treated mice (1097.8 ± 93.7 v 1430.6 ± 62.6 ng/mL). **Conclusion:** Transfusion with syngeneic RBCs enhances the inflammatory response to subsequent LPS. Furthermore, pre-transfusion with RBCs increased LPS-induced endothelial injury. These effects were dependent on RAGE as deletion of RAGE led to mitigation of endothelial injury. Thus one consequence of RBC transfusion in susceptible hosts is enhanced RAGE expression and susceptibility to subsequent injury. As RBC transfusions are common in various conditions such as trauma, cardiac surgery, and sepsis where endogenous and pathogen-derived inflammatory mediators have been detected, these observations may provide one plausible mechanism for the deleterious effects of RBC transfusions in these populations.

Disclosure of Commercial Conflict of Interest

S. M. Albelda: Nothing to disclose; J. S. Lee: Nothing to disclose; N. Mangalmurti: Nothing to disclose; D. Qing: Nothing to disclose; D. Siegel: Nothing to disclose; G. S. Worthen: No Answer

Disclosure of Grants Conflict of Interest

S. M. Albelda: Nothing to disclose; J. S. Lee: Nothing to disclose; N. Mangalmurti: Nothing to disclose; D. Qing: Nothing to disclose; D. Siegel: Nothing to disclose; G. S. Worthen: No Answer

S86-040B

The Impact of Platelet Additive Solution Apheresis Platelets on Allergic Transfusion Reactions and Corrected Count Increment

A Tobian[1] (atobian1@jhmi.edu), A K Fuller[1], K Uglik[1], P D Borge[2], D Tisch[3], R J Benjamin[2], P M Ness[1], K E King[1]. *[1]Pathology, Johns Hopkins University, Baltimore, MD, United States; [2]American Red Cross, Rockville, MD, United States; [3]Epidemiology and Biostatistics, Case Western Reserve University, Cleveland, OH, United States*

Background/Case Studies: The incidence of allergic transfusion reactions (ATRs) ranges from 1% to 3% of all transfusions and they are difficult to predict and prevent. Rates of ATRs appear to depend on the dose of plasma in the platelet product, and it has been previously shown that concentrating and washing apheresis platelets (APs) substantially reduce the number of ATRs. APs stored in platelet additive solution (PAS) comprise a mixture of 35% plasma and 65% InterSol (Fenwal, Lake Zurich, IL). The impact of PAS APs on the incidence of ATRs and the post-transfusion platelet increment was evaluated. **Study Design/Methods:** This retrospective study evaluated all ATRs among patients at a large university hospital that maintained a mixed inventory of PAS APs and non-PAS APs (standard plasma suspended platelets) for 6 months (June 1, 2012-December 4, 2012). ATRs were evaluated at the time of the reaction by transfusion medicine physicians who were blinded to the type of product transfused. Corrected count increments (CCI, [PLT count post—PLT count pre]*BSA/ number of equivalent units of platelets [5.5 × 10 e10] transfused) were calculated for AP transfusions of individuals who received both a PAS and non-PAS AP transfusion within a 7-day period. In addition, on the days of the two compared transfusion increments, the patient must have had similar bleeding status, similar maximum temperature, and post transfusion labs obtained at 1-4 hours and/or 12-24 hours. Hypothesis testing was performed with chi-square test for dichotomous variables (ATR rates) and Student's t-tests for continuous variables (CCIs). **Results/Findings:** The incidence of ATRs among the non-PAS APs was 1.85% (72 ATRs/3884 transfusions) and 1.01% (12 ATRs/1194 transfusions) for PAS APs (RR = 0.54, 95%CI = 0.30-0.99, p = 0.04). Compared to PAS AP products, ATR incidence was non-significantly lower for concentrated APs (incidence 0.64%, 5/779, RR = 0.64, 95%CI 0.23-1.81) and washed APs (incidence 0.46%, 2/434, RR = 0.46, 95%CI 0.10-2.04). However, there was no difference in the incidence of febrile non-hemolytic transfusion reactions between non-PAS APs (incidence 0.70%, 27/3884) compared to PAS APs (incidence 0.59%, 7/1194, p = 0.69). Among 223 individuals with paired non-PAS and PAS AP transfusions, the CCI at 1-4 hours after transfusion was 4932 (95%CI 4452-5412) for non-PAS APs and was lower for PAS APs, CCI = 3766 (95%CI 3375-4158, p = <0.001). However, there was no significant difference in CCI at 12-24 hours between non-PAS (CCI = 2135, 95%CI 1696-2573) and PAS APs (CCI = 1745, 95%CI 1272-2217, p = 0.14). **Conclusion:** PAS APs substantially reduce the number of ATRs. The CCI for PAS APs was lower immediately after transfusion, but there was no statistically significant difference at 12-24 hours.

Disclosure of Commercial Conflict of Interest

R. J. Benjamin: Nothing to disclose; P. D. Borge: Nothing to disclose; A. K. Fuller: Nothing to disclose; K. E. King: I have no relationship with Fenwal, Other; P. M. Ness: Nothing to disclose; D. Tisch: Nothing to disclose; A. Tobian: Fenwal, Grants or Research Support; K. Uglik: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. J. Benjamin: Nothing to disclose; P. D. Borge: Nothing to disclose; A. K. Fuller: Nothing to disclose; K. E. King: Terumo BCT, Grants or Research Support; P. M. Ness: Nothing to disclose; D. Tisch: Nothing to disclose; A. Tobian: Nothing to disclose; K. Uglik: Nothing to disclose

S87-040B

A Retrospective Review of Patient Factors, Transfusion Practices, and Outcomes in Patients with Transfusion Associated Circulatory Overload (TACO)

L Lieberman[1,2] (l.lieberman@utoronto.ca), C Maskens[3], C M Cserti-Gazdewich[4,1], M Hansen[6], Y Lin[5,1], J Pendergrast[4,1], Q Yi[7], J Callum[5,1]. *[1]Department of Laboratory Medicine & Pathobiology, University of Toronto, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada; [3]University of British Columbia, Vancouver, BC, Canada; [4]Department of Clinical Pathology, University Health Network, Toronto, ON, Canada; [5]Department of Clinical Pathology, Sunnybrook Health Sciences Centre, Toronto, ON, Canada; [6]Department of Cardiology, Sunnybrook Health Sciences Centre, Toronto, ON, Canada; [7]Canadian Blood Services, Ottawa, ON, Canada*

Background/Case Studies: Transfusion associated circulatory overload (TACO) is a common yet under-recognized and underreported complication of transfusions associated with significant morbidity and mortality. The objective of this study was to examine patient and transfusion characteristics in a cohort of TACO cases. **Study Design/Methods:** A retrospective chart review of 100 consecutive TACO cases reported at two academic centers was performed. Information related to demographics, medical history, radiologic and echocardiographic investigations, infusion practices, reaction features, management and outcome were collected. **Results/Findings:** Ninety-eight cases were accessible for review. A history of congestive heart failure (41%), renal dysfunction (44%), and age >70 (56%) were common in TACO patients. Suboptimal fluid status management and inappropriate infusion practices were commonly seen (such as verbal orders, double red cell transfusions, rapid infusion rates, and lack or improper timing of pre-emptive diuretics). The median volume of blood ordered was 500 mL and the median volume of crystalloid or colloid (preceding 24 hours) was 2200 mL. A physician order specifying the rate of infusion for the transfusion was documented only 50% of the time. Pre-emptive diuretics were ordered only 29% of the time, most commonly introduced mid-way or following the transfusion at a dose of 20 mg intravenously. Following TACO, 18% of the patients required transfer to the intensive care unit (ICU), 8% suffered a major complication and 5% died. **Conclusion:** Sub-optimal ordering and infusion practices may be contributing to the high incidence and severity of TACO. Research in TACO prevention strategies, such as slow rates of infusion and pre-emptive diuretics, is warranted.

Disclosure of Commercial Conflict of Interest

J. Callum: Nothing to disclose; C. M. Cserti-Gazdewich: Nothing to disclose; M. Hansen: Nothing to disclose; L. Lieberman: Nothing to disclose; Y. Lin: Nothing to disclose; C. Maskens: Nothing to disclose; J. Pendergrast: Nothing to disclose; Q. Yi: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Callum: Nothing to disclose; C. M. Cserti-Gazdewich: Nothing to disclose; M. Hansen: Nothing to disclose; L. Lieberman: Nothing to disclose; Y. Lin: Nothing to disclose; C. Maskens: Nothing to disclose; J. Pendergrast: Grifols, Grants or Research Support; Q. Yi: Nothing to disclose

S88-040B

Genome-Wide Association Study of HLA Allo-Antibody Formation and Persistence

M Seielstad[2] (Mark.Seielstad@ucsf.edu), G Page[3], N Gaddis[3], X Deng[1], T Lee[1], M C Lanteri[1], Y Wu[4], R Kakaiya[5], M P Busch[1]. *[1]Blood Systems Research Institute, San Francisco, CA, United States; [2]University of California San Francisco, San Francisco, CA, United States; [3]RTI International, Atlanta, GA, United States; [4]Yale University, New Haven, CT, United States; [5]Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: Alloimmunization is a clinical problem of paramount importance in transfusion medicine. Despite extensive investigation of the mechanisms underlying alloresponses in human and animal models, little is known concerning genetic factors that may influence the development of alloimmunity. A genome-wide association study was conducted to identify genetic factors predisposing to HLA alloimmunization arising from exposure to non-self HLA alleles during pregnancy. **Study Design/Methods:** Subjects were drawn from the REDS II Leukocyte Antibody Prevalence Study, a multi-center study designed to measure the prevalence of HLA and neutrophil antibodies in blood donors with and without a history of known allogeneic exposures through blood transfusion or pregnancy. Genotyping of 772 previously pregnant women with detectable HLA alloantibodies (cases) was attempted, matched with 772 women reporting >2 previous pregnancies without HLA alloantibodies (controls). A genotyping array was used containing probes for 1,008,624 SNPs. Data was obtained on 1542 samples. All women reported non-Hispanic white ancestry, and no differences between self-reported and genetic ancestry were noted. 1527 people had >99% call-rates. After excluding duplicates and related samples, the final sample size was 752 cases and 753 controls. Genotypes for 37,160,272 additional SNPs were imputed using IMPUTE2 and 1000 Genomes cosmopolitan reference haplotypes. Association analysis was performed for alloimmunization to either class I or class II molecules, or both, using logistic regression in ProbAble and adjusting for population substructure with the first two eigenvalues from EigenStrat. **Results/Findings:** A SNP in the NXPH2 gene surpassed genome-wide significance ($P = 2.06 \times 10\text{-}8$), with multiple adjacent markers exceeding P = 10-6, for women with anti-class I alloantibodies only. NXPH2 is a minimally characterized gene family member, first identified in neuronal tissue. Subsequent studies have implicated NXPH1, a gene family member, in the suppression of hematopoietic progenitor cells. SNPs in the E2F7 gene, a transcription factor related to cell-cycle control and cellular proliferation, also approached genome-wide significance. **Conclusion:** Common genetic variation appears to influence the risk of HLA alloimmunization induced by exposure to alloantigens during pregnancy. Further work to identify and characterize causal variants in NXPH2 and E2F7, and to understand their functions in the context of alloantibody formation is warranted. The absence of a prominent association with the HLA region, whether the result of limited sample size or a genuine absence of an association, will be investigated further.

Disclosure of Commercial Conflict of Interest

M. P. Busch: Nothing to disclose; X. Deng: Nothing to disclose; N. Gaddis: Nothing to disclose; R. Kakaiya: Nothing to disclose; M. C. Lanteri: Nothing to disclose; T. Lee: Nothing to disclose; G. Page: No Answer; M. Seielstad: Nothing to disclose; Y. Wu: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. P. Busch: Nothing to disclose; X. Deng: Nothing to disclose; N. Gaddis: Nothing to disclose; R. Kakaiya: Nothing to disclose; M. C. Lanteri: Nothing to disclose; T. Lee: Nothing to disclose; G. Page: No Answer; M. Seielstad: Nothing to disclose; Y. Wu: Nothing to disclose

S89-040B

A Novel Role for Complement C3 in Antibody-Induced Antigen-Modulation

S R Stowell[1] (srstowe@emory.edu), K Girard-Pierce[1], N Smith[1], J E Hendrickson[1], J C Zimring[2]. *[1]Pathology and Laboratory Medicine, Emory University, Decatur, GA, United States; [2]Transfusion Medicine Research, Puget Sound Blood Center, Seattle, WA, United States*

Background/Case Studies: Immune-mediated destruction of red blood cells (RBCs) may occur following incompatible transfusion, with some RBCs remaining in circulation despite the presence of RBC specific antibodies. The phenomenon of “antigen-modulation”, in which the ability to detect the target antigen recognized by antibodies diminishes, has previously been described to occur in multiple settings with different target antigens in humans. However, the mechanism(s) of this process remain unclear. With the goal of better understanding the mechanisms of incompatible RBC clearance, murine model was generated with RBC-specific expression of a clinically significant human antigen known to be involved in antigen-modulation (the KEL antigen). **Study Design/Methods:** C57BL/6 or KEL transgenic control recipient mice were transfused with KEL and C57BL/6 RBCs labeled with distinct lipophilic dyes to facilitate detection of cells following transfusion. In addition, C57BL/6, C3KO or Fcgamma R KO recipients were passively immunized with anti-KEL antibody followed by transfusion of labeled KEL and C57BL/6 RBCs. Post-transfusion survival and RBC bound antibody and complement levels were determined by flow cytometry (FC) at specific time points after transfusion utilizing the fluorescent lypophilic dyes, anti-IgM, anti-IgG, anti-C3, and anti-iC3b. Serum antibodies were also analyzed by FC, using KEL and control C57BL/6 RBCs as targets. **Results/Findings:**

C57BL/6 wild-type recipients transfused with transgenic KEL RBCs generated anti-KEL glycoprotein alloantibodies, which in both active and passive immunization settings, fixed complement, resulted in changes in detectable KEL antigen, and led to clearance of a portion of the incompatible RBCs. Mice genetically lacking complement had lesser degrees of KEL antigen-modulation and decreased incompatible RBC clearance rates, suggesting complement plays a role in both processes. Consistent with this, the accumulation of complement degradation products on KEL RBCs appeared to parallel the development of cellular resistance to antibody-mediated clearance. In contrast, mice genetically lacking Fcgamma receptors had similar degrees of KEL antigen-modulation as wild type recipients. **Conclusion:** These findings suggest that complement regulation may directly impact antigen-modulation and survival of RBCs following an incompatible transfusion. This KEL model may serve as a platform for the future development of strategies to minimize the dangers of RBC incompatibility, through intentional antigen- modulation or through alteration of clearance pathways involving C3.

Disclosure of Commercial Conflict of Interest

K. Girard-Pierce: Nothing to disclose; J. E. Hendrickson: No Answer; N. Smith: Nothing to disclose; S. R. Stowell: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Girard-Pierce: Nothing to disclose; J. E. Hendrickson: No Answer; N. Smith: Nothing to disclose; S. R. Stowell: Nothing to disclose; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium

S90-040B

Transfusion Reactions and the Critical Evaluation Period: Do the Guidelines Need Revisiting?

E A Gehrie[1] (eric.a.gehrie@vanderbilt.edu), P Young[1,2]. *[1]Pathology, Vanderbilt University Medical Center, Nashville, TN, United States; [2]Pathology, Nashville Veterans Affairs Hospital, Nashville, TN, United States*

Background/Case Studies: National and international guidelines on clinical transfusion practices warn that severe transfusion reactions commonly occur during the first 15 minutes of a transfusion. Accordingly, many institutions require a licensed transfusionist to remain with the transfusion recipient for the first 15 minutes. **Study Design/Methods:** All allergic, febrile, pulmonary (ie shortness of breath) and hypotensive transfusion reactions were evaluated, including those that resembled “anaphylactoid” type, over a 12-month period at Vanderbilt University Medical Center, which includes both an adult and pediatric tertiary care hospital. Demographics of these reactions, including component type, rate, volume transfused, and timing of the suspected reaction after initiation, were evaluated. **Results/Findings:** Out of 65,612 total transfusions, there were 276 (0.42%) reports of a suspected transfusion reaction. The median time from initiation of transfusion to beginning of a reaction was 24.5 minutes or longer (Table). There were five anaphylactoid reactions characterized by allergic symptoms with a component of airway compromise such as neck swelling or decreased oxygen saturation. On average, anaphylactoid reactions occurred after ~26 minutes of transfusion. The rate of transfusion in three of the five anaphylactoid reactions exceeded adult recommended transfusion rates listed in the Technical Manual. **Conclusion:** On average, symptoms of allergy or significant hypotension lead to cessation of transfusion after ~25 min. True anaphylactic reactions are rare (none reported in our institutions for 2011 or 2012 in ~120,000 reactions) and anaphylactoid reactions, on average, take ~26 minutes to materialize. Guidance on whether a full 15 minutes of transfusionist time should be allocated to closely monitor for acute transfusion reactions should be further evaluated. Clearer guidance to ensure an appropriate transfusion rate may be a more effective means of preventing harm to patients.

Disclosure of Commercial Conflict of Interest

E. A. Gehrie: Nothing to disclose; P. Young: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. A. Gehrie: Alexion Pharmaceuticals, Grants or Research Support; P. Young: Nothing to disclose

TABLE 1. Characteristics of Allergic, Febrile, Pulmonary and Hypotensive Transfusion Reactions

Characteristic	Allergic	Febrile	Pulmonary	Hypotensive
n	80	116	39	38
Median time, min (SD)	30 (33)	40 (49)	30 (49)	24.5 (46)
Median volume transfused, mL (SD)	195.5 (108)	100 (106)	223.5 (98)	100 (108)
Average Rate, mL/min	6.5	2.5	7.5	4
Most common component, adjusted	Platelets	Platelets	Platelets	Platelets

Blood Management Scientific

SP1

Encouraging Single Unit Red Cell Transfusion Prescription for in-Patients Can be Effective in Reducing Red Cell Usage

M F Tavares[1] (mtavares@rwmc.org), P DiQuattro[1], J D Sweeney[1]. *[1]Blood Bank, Roger Williams Hospital, Providence, RI, United States*

Background/Case Studies: The red cell transfusion incident rate in the US is approximately 50 units/1000 population/year, but in European countries, it is less, at 30-40 units/ 1000 population/year. This suggests that some red cell transfusions may be inappropriate (non-efficacious) or that a practice of excess dosing exists (two units vs single unit prescription). Two-unit transfusions are common in out-patient transfusion and such dosing may be reasonable in this context. Two-unit prescriptions for in-patients, however, may represent in some cases either unecessary transfusion or over-dosing. Intervening in such circumstances could result in cancellation or deferrment of the prescription or reduction of the prescription to a single unit. This has the potential to reduce overall red cell transfusion. **Study Design/Methods:** Since 2004 the technologists here have intervened to encourage single unit red cell transfusion in situations where the pretransfusion hemoglobin was above 8 g/dL and a requst for two units of red cells was received. The physicians were questioned regarding the presence of any bleeding and, if present, the extent of any bleeding. For non-bleeding patients or patients with minimum bleeding, single unit red cell transfusion was encouraged. Red cell transfusions were tabulated over a 15-year period (1998-2012) and in-patient red cell transfusions separated from out-patient red cell transfusion. The number of discharges per year were recorded and the data analyzed for each year as units RBC/1000 discharges. The 6-year period 1998-2003 (pre-intervention) was compared to the 9-year period 2004-2012 (post intervention). **Results/Findings:** As seen in the table, a progressive decrease in in-patient red cell usage is observed. For the period 1998-2003, the mean units transfused/1000 discharges was 509 ± 45 units/year and for 2004-2012, 396 ± 50, $p < 0.01$. **Conclusion:** This data shows a reduction in red cell usage of approximately 20% concurrent with a program to encourage a change in two units red cell prescriptions to a single unit prescription.

Disclosure of Commercial Conflict of Interest

P. DiQuattro: No Answer; J. D. Sweeney: Nothing to disclose; M. F. Tavares: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. DiQuattro: No Answer; J. D. Sweeney: Bio-Rad, Grants or Research Support; M. F. Tavares: Nothing to disclose

Year	Total In-Patient Red Cell Units Transfused	In-Patients RBC transfused: 1000 Discharges
1998	3199	440
1999	3943	507
2000	4536	571
2001	4183	535
2002	3936	485
2003	4137	518
2004	3850	465
2005	3618	461
2006	3582	438
2007	3256	406
2008	2907	377
2009	2841	377
2010	2468	343
2011	2882	378
2012	2462	321

SP2

Preoperative Anemia Screening: An Audit of Current Practices for Compliance with The Joint Commission (TJC) 2011 Patient Blood Management (PBM) Initiative

M Azenha[1] (mazenha@somerset-healthcare.com), S Rodriguez[1], M Kuriyan[1,2]. *[1]Somerset Medical Center, Somerville, NJ, United States; [2]Robert Wood Johnson University Hospital, New Brunswick, NJ, United States*

Background/Case Studies: Preoperative hemoglobin (Hgb) is shown to be a predictor of outcome for orthopedic surgery patients. Literature states that there is a known association of preoperative anemia with increased morbidity after surgery including exposure to allogeneic blood transfusions. The Joint Commission (TJC) recommends a Hgb test before the scheduled surgical procedure on patients for whom blood transfusion is a probability. This promotes early anemia recognition and enables treatment strategies to minimize a potential transfusion. **Study Design/Methods:** To evaluate the practice in our hospital for compliance with this recommendation, adult elective orthopedic surgery patients were selected. Monthly audits assessed if Hgb was done preoperatively. Documentation of preoperative anemia screening 14-45 days before anesthesia start date was checked (TJC recommendation) for one year (January 2012-December 2012). Each chart was reviewed for management of preoperative anemia in surgical patients with Hgb < 12 g/dL for females and <13 g/dL for males (TJC recommendation). Data analysis includes evaluation of preoperative anemia screening result. Percent compliance is calculated. The data was further analyzed based on physician or type of procedure and transfusions associated with lack of preoperative anemia screening. **Results/Findings:** A total of 303 orthopedic surgery patients were evaluated and 44 (15%) patients were transfused. Patients with preoperative Hgb done were 302 (99.7%) and patients with no Hgb done and no anemia treatment was 1 (<1%). There were 267 (88.1%) with normal Hgb and 26 (10%) were transfused. There were 35 (11.6%) patients with a low preoperative Hgb, of whom one was tested less than 14 days preoperatively, and 18 (51%) were transfused. From the low Hgb group, 27 (77%) had documented anemia treatment and 8 (23%) had no anemia treatment documented. From the low Hgb and documented anemia treatment group, 16 of 27 (59%) were transfused. Of the low Hgb with no anemia treatment documented group, two of eight (25%) were transfused during admission involving seven doctors, four knee replacements, and four hip replacements. **Conclusion:** There is a high level of compliance with TJC recommendation and there was no pattern by surgeon with noncompliance. The TJC Hgb levels are higher than that used to define anemia (Hgb 10 g/dL) locally. It could not be demonstrated that the high risk for a blood transfusion for patients that did not have their pre-operative anemia treatment completed prior to surgery was probably due to the low patient number.

Disclosure of Commercial Conflict of Interest

M. Azenha: Nothing to disclose; M. Kuriyan: Nothing to disclose; S. Rodriguez: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Azenha: Nothing to disclose; M. Kuriyan: Nothing to disclose; S. Rodriguez: Nothing to disclose

Data Summary of Audit Results

Type of Surgery Patient:	Number of Patients(%)	Number of Patients Transfused (%)
Total Evaluated	303	44 (15%)
No Hgb	1 (<1%)	0
Normal Hgb	267 (88.1%)	26 (10%)
Low Hgb	35 (11.6%)	18 (51%)
Low Hgb with Anemia Treatment	27 (77%)	16 (59%)
Low Hgb with No Anemia Treatment	8 (23%)	2 (25%)

SP3

Post-Operative Red Blood Cell (RBC) Transfusion: Liberal Versus Restrictive

M Hareuveni[1] (marah@tlvmc.gov.il), Y Sagy[2], A Meilik[3]. *[1]Blood Bank, TASMC, Tel Aviv, Israel; [2]Clinical performance research unit, Tel Aviv Souraski Medical Center (TASMC), Tel Aviv, Israel; [3]Clinical performance research unit, TASMC, Tel Aviv, Israel*

Background/Case Studies: Survival, length of hospital stay, morbidity, and other parameters have been looked at in the context of liberal vs restrictive RBC transfusion in a variety of conditions and populations. The restrictive approach has been shown to be at least non-inferior in critically ill infants as well as adults and in patients after orthopedic and cardiac surgery. Having set guidelines for transfusion in 1999 and revised them in 2005, adherence to these guidelines in the post operative period. was sought. **Study Design/Methods:** This is an observational retrospective single center cohort study. All patients who underwent a surgical procedure here (TASMC) and stayed in the hospital longer than 48 hours during 2012 were the starting population. Patients who were transfused during the operation or in the first hour after end of surgery were excluded. Patients who were not transfused intra-operatively were then grouped by their postoperative Hb level (above or below 9 g/dL) and by RBC transfusion receipt. Demography, ASA & Charlson scores, type of surgical procedure, peri-operative mortality, length of stay, ICU admission, infections, and laboratory parameters were examined. **Results/Findings:** 15,968 patients underwent one surgical procedure during 2012 and were hospitalized for >48 hours. Of 13,464 patients (84%) not transfused intra-operatively (or during the first post-op hour), 802 (5.9%) were transfused post-operatively. 642 (80%) had hemoglobin level <9 g/dL before transfusion. All transfused patients (regardless of Hb level) were older and had higher ASA and Charlson scores. Transfusion itself was found to be associated with an increased risk for outcomes in multivariate models: peri-hospitalization mortality, post-operative bacteremia, and prolonged post-operative hospitalization (>30 days), regardless of whether the Hb was lower or higher than 9 g/dL. The ASA score and the Norton score (referring to patient's functional status) added to the risk of post-operative bacteremia. Peri-hospitalization mortality was further associated with the Charlson co-morbidity score, and Hb drop (post-pre-operative). **Conclusion:** 90.8% of patients did not receive RBC transfusion post operatively. Among patients who were transfused, severity of comorbidities and impaired functional status were significant. Transfusion was found as an independent correlate for peri-hospitalization mortality, post-operative bacteremia, and prolonged post-operative hospitalization, regardless of Hb level.

Disclosure of Commercial Conflict of Interest

M. Hareuveni: Nothing to disclose; A. Meilik: Nothing to disclose; Y. Sagy: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Hareuveni: Nothing to disclose; A. Meilik: Nothing to disclose; Y. Sagy: Nothing to disclose

SP4

Implementing and Sustaining Patient Blood Management Outcome Improvements

J Miles[1] (Dr.Miles@prlnet.com), S Kerley[2], J E Menitove[3]. *[1]Pathology, Heartland Regional Medical Center, St. Joseph, MO, United States; [2]Pathology, St. Joseph Health Center, Kansas City, MO, United States; [3]Administration, Community Blood Center of Kansas City, Kansas City, MO, United States*

Background/Case Studies: The results of randomized, controlled trials show similar outcomes associated with restrictive and liberal transfusion strategies. It was assumed that adaptation of 7-8 g/dL transfusion threshold protocols would decrease the occurrence, at hospital discharge ("discharge hemoglobin"), of hemoglobin concentrations >10 g/dL among patients receiving transfusions. The "discharge hemoglobin" metric was used to estimate over-utilization, focus attention on improving transfusion practices, and implement computerized physician order entry (CPOE) protocols to sustain practice improvements. **Study Design/Methods:** The percentage of transfused patients with "discharge hemoglobin" level >10 g/dL at two community hospitals in 2009-2010 and, again, in 2012 was determined. Didactic educational programs and presentations focused on the physicians prescribing the majority of red cell transfusions. CPOE implementation included drop down menus consistent with approved order set indications for transfusion and required physicians to select one of these indications or "other." However, transfusions were not withheld on the basis of the stated indication. **Results/Findings:** At Hospital A, the percentage of patients with "discharge hemoglobin" >10 g/dL decreased from 48% in 2009 to 22% in ($\chi 2 = 55.76$, $p \leq 0.001$). Utilization decreased from 563 units per month to 417 (26%) and units transfused /patient from 3.0 to 2.7. Significant declines occurred among general surgery (60% "discharge hemoglobin" >10 g/dL to 25%) and internal medicine (52% to 23%) patients. A review of CPOE recorded indications showed >80% concordance between stated indications and laboratory results. At Hospital B, "discharge hemoglobin" results >10 g/dL dropped from 40% to 27% ($\chi 2 = 15.98$, $p \leq 0.001$). Utilization declined from 388 units/month to 278 (28%) while units transfused per patient increased from 1.4 to 3.3. In 2012, 17% of transfusion episodes involved single units. Normalizing the data in relationship to hospital census

showed a utilization decrease from 0.41 units/patient discharge to 0.36 units (12%). "Discharge hemoglobin" rates declined most significantly among hospitalist/internal medicine patients (44% to 29%). **Conclusion:** The "discharge hemoglobin" metric served as a global marker for estimating over-utilization and alerting the medical staff to re-evaluate transfusion practice guidelines. CPOE implementation provides a constant reminder of the new guidelines. Although it is anticipated CPOE will sustain the improvements, this requires on-going evaluation.

Disclosure of Commercial Conflict of Interest

S. Kerley: No Answer; J. E. Menitove: Nothing to disclose; J. Miles: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Kerley: No Answer; J. E. Menitove: Nothing to disclose; J. Miles: Nothing to disclose

SP5

No Benefit from Intensified Coagulation Factor Application in Cardiovascular Surgery and Solid Organ Transplantation

H G Heuft[1] (heuft.hans-gert@mh-hannover.de), S Tellkamp[2], D Scheinichen[2], R Blasczyk[1], A Tiede[3]. *[1]Transfusion Medicine, Hannover Medical School, Hannover, Germany; [2]Anaesthesiology and Intensive Care, Hannover Medical School, Hannover, Germany; [3]Haematology / Oncology, Haemostaseology, and Stem Cell Transplantation, Hannover Medical School, Hannover, Germany*

Background/Case Studies: Bleeding complications may contribute to major morbidity and mortality after surgery. Coagulation management, including fibrinogen (F. I), fibrogammin (F. XIII), prothrombin complex (PPSB), or antithrombin (AT), might avoid bleedings, reduce blood transfusions and could lead to a beneficial clinical outcome. The need for blood components and clinical outcome was analyzed in patients with major cardiovascular surgery or solid organ transplantation (TX) who were treated with increasing doses of coagulation factor concentrates. **Study Design/Methods:** In a retrospective comparative study, 795 patients of Hannover Medical School (2007, n = 312; 2009, n = 483) were investigated for the usage of blood components (red cells [RC], therapeutic plasma [TP], and platelets [PC]) and coagulation factors such as F.I, F.XIII, PPSB, and AT with regard to clinical factors (eg intensive care unit hospital stay [LOSICU], survival). Of the 795 patients, 611 underwent 5 categories of surgical interventions: Category I, combined heart/aortic surgery, n = 80, 2007; n = 98, 2009-in total 178 patients; Category II, heart arteries bypassing and/or heart valve surgery, partly combined n = 52, 2007; n = 164, 2009-in total 216 patients; Category III, heart or lung TX, partly combined, n = 33, 2007; n = 45, 2009-in total 78 patients; Category IV, artificial heart TX, n = 18, 2007; n = 31, 2009-in total 49 patients; and Category V, liver TX, n = 52, 2007; n = 38, 2009-in total 90 patients. **Results/Findings:** Selected results are shown in the table. From 2007-2009 the application of F.I, F.XIII, PPSB, and AT substantially increased by medians of 25% for AT to 68% for PPSB. In contrast, RC transfusions remained stable in all, TP and PC transfusions in 4 out of 5 surgical intervention categories. Regarding clinical outcome, LOSICU rose in 4 of 5 categories while survival did not significantly change in 2009 as compared to 2007. **Conclusion:** Overall, the sharp increase in the application of coagulation factors did not reduce blood transfusion needs nor improve clinical outcome.

Disclosure of Commercial Conflict of Interest

R. Blasczyk: Nothing to disclose; H. G. Heuft: CLS Behring, Other; D. Scheinichen: Nothing to disclose; S. Tellkamp: No Answer; A. Tiede: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Blasczyk: Nothing to disclose; H. G. Heuft: Nothing to disclose; D. Scheinichen: Nothing to disclose; S. Tellkamp: No Answer; A. Tiede: Baxter, Bayer, Behring, Biogen Idec, Novo Nordisk, Pfizer, Grants or Research Support, Baxter, Bayer, Behring, Biogen Idec, Novo Nordisk, Pfizer, Consulting or Board of Director Fees

	2007			2009		
	F. I	RC	Survival	F. I	RC	Surviva
	(g)	(n)	(%)	*(g,* p-value)	(n, p-value)	1 (%)
Cat I	6.7 ± 0.6	12 ± 1	90	9.5 ± 0.8; ≤0.007	12 ± 1; =0.80	78
Cat II	3.8 ± 0.5	9 ± 1	85	5.4 ± 0.4; <0.02	11 ± 1; =0.22	81
Cat III	8.6 ± 1.4	28 ± 4	70	8.2 ± 1.0; =0.79	21 ± 3; =0.15	78
Cat IV	6.1 ± 1.1	17 ± 3	89	8.9 ± 1.4; =0.18	14 ± 2; =0.37	90
Cat V	4.8 ± 0.6	21 ± 3	81	12.5 ± 2 0;≤0.000	21 ± 2; =0.61	77

RC = Red Cell Concentrates

SP6

Decreasing Inappropriate Two-Unit Red Blood Cell (RBC) Transfusions with a Restrictive Transfusion Policy

S Yerrabothala[1] (Swaroopa.Yerrabothala@Hitchcock.org), K P Desrosiers[2], Z M Szczepiorkowski[1,2], N M Dunbar[1,2]. *[1]Pathology, Dartmouth Hitchcock Medical Center, Lebanon, NH, United States; [2]Internal Medicine, Dartmouth Hitchcock Medical Center, Lebanon, NH, United States*

Background/Case Studies: The purpose of this study is to evaluate the impact of an evidence-based restrictive policy on red blood cell transfusion (RBC TXN) practice. The policy here states that all RBC TXNs in non-bleeding inpatients should be ordered as single units. Post-TXN hemoglobin (Hgb) should be obtained prior to ordering additional units. All medical and surgical inpatients (except the patients in the operating room) with Hgb ≤ 7 g/dL, cardiac patients with Hgb ≤ 8 g/dL, and septic patients with Hgb ≤ 10 g/dL are considered appropriate candidates for RBC TXN. **Study Design/Methods:** This is a retrospective study evaluating RBC TXN practice for 6 months after implementing this policy (Period 2: October 2012-March 2013) and comparing it to the same period in the year before implementing the policy (Period 1: October 2011-March 2012). All TXNs to the inpatients on medical and surgical floors, adult critical care units, hematology and oncology floor, emergency department, and obstetric floor were evaluated. From these data, all 2 unit TXNs were extracted. Each order for RBC TXN is considered one transfusion event. Chart reviews were performed to evaluate the appropriateness of 2-unit TXNs for Hgb trigger >6.3 g/dL. All 2-unit TXNs for Hgb ≤ 6.3 g/dL were considered appropriate for the purpose of this study. All TXNs with documented and/or laboratory evidence of active bleeding (drop in Hgb by ≥1 g/dL in 24 hours) or a documented need for a higher Hgb trigger (eg acute coronary syndrome or severe heart and/or lung disease) were considered clinically appropriate. The results were analyzed in a 2 × 2 contingency table using Fisher's exact test. **Results/Findings:** There was a statistically significant decrease in the percentage of 2-unit TXNs between Period 1 (47%) and Period 2 (15%). The number of 2-unit TXNs for Hgb trigger >6.3 g/dL decreased from 90% in Period 1 to 71% in Period 2. Based on chart review criteria, % of 2-unit TXNs deemed clinically inappropriate decreased from 59% in Period 1 to 38% in Period 2. **Conclusion:** Introduction of a single-unit transfusion policy with restrictive transfusion guidelines decreases the number of clinically inappropriate 2-unit RBC transfusions.

Disclosure of Commercial Conflict of Interest

K. P. Desrosiers: Nothing to disclose; N. M. Dunbar: Nothing to disclose; Z. M. Szczepiorkowski: Nothing to disclose; S. Yerrabothala: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. P. Desrosiers: Nothing to disclose; N. M. Dunbar: Nothing to disclose; Z. M. Szczepiorkowski: Fenwal, Inc., Consulting or Board of Director Fees, Terumo BCT, Grants or Research Support, Fenwal, Inc., Grants or Research Support; S. Yerrabothala: Nothing to disclose

	Period 1 10/2011-03/2012	Period 2 10/2012-03-2013
Total RBC TXN events	1905	1753
2 unit TXNs	849 (47%)	247 (15%)*
Total number of 2 unit TXNs	849	247
TXNs for Hgb ≤ 6.3 g/dL	86 (10%)	72 (29%)
TXNs for Hgb > 6.3 g/dL	763 (90%)	175 (71%)*
Total number of 2 unit TXNs for Hgb > 6.3 g.dL	763	175
Clinically appropriate	314 (41%)	108 (62%)
Clinically inappropriate	449 (59%)	67 (38%)*

* $p < 0.0001$

SP7

Early Intraoperative Blood Conservation Strategies Can Reduce Rates of Off-Pump Red Blood Cell (RBC) Transfusion in On-Pump Coronary Artery Bypass Graft Surgery

T Wilcox[1] (tanyaw@uci.edu), D M Lin[1,2], M Tran[1,3]. *[1]UC Irvine School of Medicine, Irvine, CA, United States; [2]Internal Medicine, UC Irvine Medical Center, Orange, CA, United States; [3]Department of Pathology and Laboratory Medicine, University of California Irvine Medical Center, Orange, CA, United States*

Background/Case Studies: The purpose of the study was to determine the effect of implementing two early intraoperative blood conservation strategies

(low prime perfusion circuit, autologous priming) on the rate of off-pump red blood cell (RBC) transfusion. Two late intraoperative strategies (cell salvage, hemoconcentration) were already well-established. **Study Design/ Methods:** An observational, before-after study of patients undergoing first time, on-pump coronary artery bypass graft (CABG) surgery was conducted at a single, low-volume, academic medical center over a 3-year period. The primary endpoint was the rate of off-pump RBC transfusion. Multivariate logistic regression analysis was performed with possible explanatory preoperative and intraoperative variables as determined by univariate analyses. Secondary endpoints included overall intraoperative blood component usage and postoperative complications. **Results/Findings:** The post-implementation (1-year period) and the historical (2-year period) groups were comprised of 38 and 72 patients, respectively. Compared to historical controls, the rate of off-pump RBC transfusion fell from 47.2% to 23.7% (p = 0.02), and the odds of receiving off-pump RBC transfusion are 65% lower (unadjusted OR = 0.35; 95% CI 0.14-0.81) during the post-implementation period. However, the post-implementation timing lost its explanatory value when adjusted for significant covariates (adjusted OR = 0.71; 95% CI 0.21-2.35). The last on-pump hematocrit level—which showed before-after differences (median Hct 26% [range 19-40] versus 28% [24-38], p = 0.002)—was the best explanatory variable (adjusted OR = 0.75; 95% CI 0.64-8.69). No before-after differences in the overall rates of intraoperative RBC (63.9% versus 55.3%, p = 0.38 and platelets (52.8% versus 60.5%, p = 0.44) were observed, except for cryoprecipitate (8.3% versus 26.3%, p = 0.01) and possibly FFP (34.7% versus 52.6%, p = 0.07). Lastly, no between-group differences in postoperative complication were observed, including mortality. **Conclusion:** Significant reductions in the rate of off-pump RBC transfusion during the post-implementation period were not readily explained by the post-implementation timing. Instead, the last on-pump hematocrit level appeared to be more explanatory, albeit limited by the wide confidence interval (CI) of the adjusted odds ratio. It is possible that the higher last on-pump hematocrit level observed in the post-implementation period may more directly reflect the two early intraoperative blood conservation strategies. In order to reduce overall intraoperative RBC transfusion and non-RBC component usage, the blood conservation program should develop and implement a comprehensive, multidisciplinary, goal-directed transfusion algorithm.

Disclosure of Commercial Conflict of Interest

D. M. Lin: Nothing to disclose; M. Tran: Nothing to disclose; T. Wilcox: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. M. Lin: Nothing to disclose; M. Tran: Nothing to disclose; T. Wilcox: Nothing to disclose

SP8

Use of ε-Aminocaproic Acid in Elective Total Hip Replacement to Reduce Transfusion Rate

K E Puca[1] (kathy.puca@bcw.edu), N L Fredrich[1], E Vermeulen[2], M Dahlgren[2], M J Anderson[3]. *[1]BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Aurora St. Luke's Medical Center, Milwaukee, WI, United States; [3]Aurora Advanced Healthcare, Mequon, WI, United States*

Background/Case Studies: Recently, the use of antifibinolytics has been shown to be safe and effective in reducing transfusion requirements in joint replacement surgeries. Tranexamic acid (TXA) and ε-aminocaproic acid (EACA) are both antifibrinoltyics with similar mode of action; however, more studies have reported on TXA in reducing allogeneic transfusions than EACA. While TXA is more potent than EACA, it is also more expensive than EACA in the US. Therefore it was decided to use EACA as a blood management strategy in joint replacement surgeries and report the effect of EACA in reducing transfusion rates in patients undergoing total hip replacement (THR). **Study Design/Methods:** Retrospective chart review as part of quality blood utilization for elective unilateral THR (procedure code 81.51) from April 2012-December 2012 at five hospital sites within a health care system was performed. Data collected included age, gender, severity of illness score, use of EACA and dose, RBC transfusion and number of units, and preadmission and discharge hemoglobin (Hb). EACA and RBC transfusions were ordered per surgeon's discretion. Dosing of EACA ranged from 5000 mg to 10,000 mg (based on patient's weight) and was administered IV over 20 minutes in surgery at or near time of incision. Patients who underwent presurgical autologous donation or intra or post-operative autologous cell salvage were excluded. Student's t-test was used for comparison of continuous variables and Chi-square test for comparison of categorical variables. Multivariate logistic regression to adjust for age, female gender, severity of illness, and admission Hb was performed. **Results/Findings:** Of the 469 patients who underwent unilateral THR involving 22 surgeons, 181 patients or 38.6% received EACA. Results are shown in the Table. Significantly fewer patients in the EACA group were transfused. Logistic regression model shows a significantly lower risk of transfusion in the patients who received EACA after adjusting for age, female gender, severity of illness, and admission Hb (Odds Ratio 0.154-0.546, p = 0.0002). **Conclusion:** THR patients who received EACA had a lower transfusion rate and fewer RBC units/patient compared to patients who did not receive antifbrinolytics. A significantly higher discharge Hb occurred in EACA patients despite fewer transfusions suggesting less postoperative blood loss.

Disclosure of Commercial Conflict of Interest

M. J. Anderson: Nothing to disclose; M. Dahlgren: Nothing to disclose; N. L. Fredrich: Nothing to disclose; K. E. Puca: Nothing to disclose; E. Vermeulen: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. J. Anderson: Nothing to disclose; M. Dahlgren: Nothing to disclose; N. L. Fredrich: Nothing to disclose; K. E. Puca: Nothing to disclose; E. Vermeulen: Nothing to disclose

	No Antifibrinolytics	EACA	*p*
n	288	181	
Age (years), mean(STD)	65.3 (11.76)	64.2 (11.39)	0.291
% Female, n(%)	182 (63.2%)	98 (54.1%)	0.052
Severity Score: 3 or 4, n(%)	42 (14.6%)	11 (6.1%)	0.005
Admission Hb, g/dL, mean(STD)	13.4 (1.42)	13.7 (1.30)	0.035
Patients transfused, n(%)	83 (28.8%)	18 (8.8%)	<0.0001
Total RBC units transfused, n	146	26	
RBC units per patient, mean(STD)	0.48 (0.91)	0.13 (0.50)	<0.0001
Discharge Hb, g/dL, mean(STD)	9.8 (1.16)	10.8 (1.17)	<0.0001

SP9

Use of ε-Aminocaproic Acid in Elective Total Knee Arthroplasty (TKA) to Reduce Transfusion Rate

K E Puca[1] (kathy.puca@bcw.edu), N L Fredrich[1], E Vermeulen[2], M Dahlgren[2], M J Anderson[3]. *[1]BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Aurora St. Luke's Medical Center, Milwaukee, WI, United States; [3]Aurora Advanced Healthcare, Mequon, WI, United States*

Background/Case Studies: Use of antifibrinolytics to reduce surgical blood loss and transfusions in orthopedic surgery is growing. Both tranexamic acid (TXA) and ε-aminocaproic acid (EACA) inhibit fibrinolysis by blocking the lysine-binding sites of plasmin(ogen) to fibrin thereby preventing clot dissolution. TXA and EACA differ in that TXA is 10 times more potent than EACA but in the US it is 100 times more expensive than EACA for an equivalent dose. While many published studies report the effectiveness of TXA in reducing blood loss and allogeneic transfusions in total knee arthroplasty (TKA), fewer studies have used EACA. This study presents using EACA as a blood management strategy for reducing transfusion requirement in TKA. **Study Design/Methods:** Retrospective chart review as part of quality blood utilization for elective unilateral TKA (procedure code 81.54) from April 2012-December 2012 at five hospital sites within a health care system was performed. Data collected included age, gender, severity of illness score, use of EACA and dose, RBC transfusion and number of units, preadmission and discharge hemoglobin (Hb). Patients who underwent presurgical autologous donation or postoperative autologous cell salvage were excluded. EACA and RBC transfusions were ordered per surgeon's discretion. Dosing of EACA ranged from 5,000 mg to 10,000 mg (based on patient's weight) and administered IV just prior to release of tourniquet. Student's t-test was used for comparison of continuous variables and Chi-square test for comparison of categorical variables. Multivariate logistic regression to adjust for age, female gender, severity of illness, and admission Hb was performed. **Results/Findings:** Of the 709 patients who underwent unilateral TKA involving 29 surgeons, 150 patients or 21.2% received EACA. Results are shown in the Table. Fewer patients were transfused in the EACA group. Logistic regression model shows a trend toward lower risk of transfusion in the patients who received EACA after adjusting for age, female gender, severity of illness, and admission Hb (Odds Ratio 0.161-1.054, p = 0.0881). **Conclusion:** TKA patients who received EACA had a lower transfusion rate and fewer RBC units/patient compared to patients who did not receive

antifbrinolytics. A significantly higher discharge Hb occurred in EACA patients despite fewer transfusions suggesting less postoperative blood loss.

Disclosure of Commercial Conflict of Interest

M. J. Anderson: Nothing to disclose; M. Dahlgren: Nothing to disclose; N. L. Fredrich: Nothing to disclose; K. E. Puca: Nothing to disclose; E. Vermeulen: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. J. Anderson: Nothing to disclose; M. Dahlgren: Nothing to disclose; N. L. Fredrich: Nothing to disclose; K. E. Puca: Nothing to disclose; E. Vermeulen: Nothing to disclose

	No Antifibrinolytic	EACA	*p*
n	559	150	
Age (years), mean(STD)	63.9 (11.63)	64.7 (10.01)	0.443
% Female, n(%)	341 (61.0%)	100 (66.7%)	0.204
Severity Score: 3 or 4, n(%)	88 (15.7%)	11 (7.3%)	0.008
Admission Hb, g/dL, mean(STD)	13.5 (1.51)	13.8 (1.24)	0.025
Patients transfused, n(%)	64 (11.4%)	6 (4.0%)	0.007
Total RBC units transfused, n	115	14	
RBC units per patient, mean(STD)	0.20 (0.65)	0.09 (0.50)	0.021
Discharge Hb, g/dL, mean(STD)	10.3 (1.36)	11.0 (1.18)	<0.0001

SP10

Intraoperative Coagulation-Based Hemotherapy Protocol Decreases Reoperation due to Postoperative Bleeding Complications with Left Ventricular Assist Device Implantation

B Castillo[1] (brian.s.castillo@uth.tmc.edu), E G Nedelcu[1], A Wahed[1], Y Bai[1], K L Klein[1], I Gregoric[1], M Patel[1], B Kar[1], P Loyalka[1], A Nguyen[1]. *[1]The University of Texas Health Science Center at Houston, Houston, TX, United States*

Background/Case Studies: Bleeding complications are significantly associated with left ventricular assist device (LVAD) implantation. One of the most significant postoperative complications is reoperation after implantation due to postoperative bleeding. In this study, the effectiveness of an intraoperative transfusion support protocol in reducing reoperation rate is evaluated. **Study Design/Methods:** The protocol includes sequential testing of functional hemostatic assays and transfusion triggers in three operative phases: baseline; after hemoconcentration; and post-heparin reversal. Transfusions were optimized using integrated analysis of laboratory data from each set of tests including CBC, DIC panel, antithrombin, VerifyNow-P2Y12, VerifyNow-ASA, and thromboelastography (TEG). Medical records of patients who underwent LVAD implantation at our center from May 2012-January 2013 and received intraoperative coagulation-based hemotherapy were reviewed. Patients undergoing reoperation for bleeding were further reviewed for detailed complications and risks. The reoperation rate is then compared to previously published data. **Results/Findings:** The coagulation-based hemotherapy protocol was applied to 24 patients (22 men, 2 women, mean age 55 ± 14 years) who underwent LVAD implantation over the study period. Of these patients, only two required reoperation. Of note, one of these two patients had LVAD implantation with concurrent mitral valve repair. The rate of reoperation here (8.3%) is found to be significantly lower than that published previously (23%). **Conclusion:** Coagulation-based hemotherapy protocol developed here contributes to decreasing reoperation rates due to bleeding complications for LVAD implants. A larger study with more data is needed to fully validate the results from this study.

Disclosure of Commercial Conflict of Interest

Y. Bai: Nothing to disclose; B. Castillo: Nothing to disclose; I. Gregoric: No Answer; B. Kar: No Answer; K. L. Klein: Nothing to disclose; P. Loyalka: No Answer; E. G. Nedelcu: Nothing to disclose; A. Nguyen: Nothing to disclose; M. Patel: No Answer; A. Wahed: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Bai: Nothing to disclose; B. Castillo: Nothing to disclose; I. Gregoric: No Answer; B. Kar: No Answer; K. L. Klein: Nothing to disclose; P. Loyalka: No Answer; E. G. Nedelcu: Nothing to disclose; A. Nguyen: Nothing to disclose; M. Patel: No Answer; A. Wahed: Nothing to disclose

SP11

More Effective Use of HLA-Matched and Crossmatched Platelet Products

J E Squires[1] (squiresj@musc.edu), K S Garner[1], M Riddle[1], S Moncada[1], M E Clayton[1]. *[1]Pathology, Medical University of South Carolina, Charleston, SC, United States*

Background/Case Studies: Numerous techniques have been used to reduce the unnecessary transfusion of blood products. Educational initiatives, personalized physician transfusion reports, and transfusion audits have reduced overall blood use here by approximately 30% since 2009 (based on transfusions per 1000 patient days). However, specialized platelet products (crossmatched and HLA-matched) continued to be poorly used. In order to reduce the inappropriate use and waste of these products, a revised platelet transfusion algorithm and a new management approach were introduced which has been successful in controlling the transfusion of these products. **Study Design/Methods:** Two complementary approaches were implemented to manage the ordering and use of HLA-matched and crossmatched platelets. An algorithm was designed in cooperation with the bone marrow transplant team (responsible for the use of over 90% of specialized platelet products in this hospital) so that bone marrow transplant physicians were aware and supportive of transfusion service ordering policies. In addition, pathology resident staff began to actively consult with bone marrow transplant staff to insure proper ordering of specialized platelet products as well as assuring that products received were appropriately used. **Results/Findings:** In 2010, the number of HLA-matched platelets used was 2.8/1000 patient days; by 2012 that number had dropped to 0.7/1000 patient days. More specifically, in 2010 our institution ordered a mean of 69 HLA-matched platelets per month (828 products annually), but by 2012 HLA-matched platelets had decreased to 21 per month (252 annually), a 69% decrease. One of the major benefits of this program, however, has been that HLA-matched platelets are more reliably used by the intended recipient. In 2010, approximately 23 HLA platelets per month were not used by the intended patient; by 2012 that number had decreased to an average of six products per month. **Conclusion:** Specialized platelet products are often needed for patients who are highly alloimmunized and hence refractory to platelet transfusion. In this institution, particularly in the adult bone marrow transplant program, large numbers of HLA-matched platelets were being ordered for these patients. However, many of these products were not being used for the intended recipient for a variety of reasons which, in turn, resulted in significant unbillable expense for the hospital. An algorithm was developed for the ordering of HLA-matched and crossmatched platelets. Additionally, pathology resident staff rotating in the blood bank manage the ordering and use of these products. This approach has led to a 69% decrease in the number of HLA-matched platelets ordered per month as well as a 74% decrease in the number of HLA-matched platelets that are not transfused to the intended, HLA-matched, recipient.

Disclosure of Commercial Conflict of Interest

M. E. Clayton: Nothing to disclose; K. S. Garner: Nothing to disclose; S. Moncada: Nothing to disclose; M. Riddle: Nothing to disclose; J. E. Squires: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. E. Clayton: Nothing to disclose; K. S. Garner: Nothing to disclose; S. Moncada: Nothing to disclose; M. Riddle: Nothing to disclose; J. E. Squires: Nothing to disclose

SP12

The Economic Impact of a Comprehensive Clinical and Surgical Blood Management Program for Primary Total Joint Arthroplasty in an Independent Community Hospital

R J Hakanson[1] (kneedoc@att.net). [1]Orthopaedic Surgery, Northern Hospital of Surry County, Mount Airy, NC, United States

Background/Case Studies: Low pre- and post-operative hematocrit levels in combination with blood loss associated with total joint arthroplasty (TJA) of the hip or knee can lead to various serious medical complications post-operatively. These patients often require allogenic blood transfusions, which include inherent risks to the patient, subsequent increased length of stay (LOS), and increased cost to the hospital and healthcare payer. The purpose of this study was to evaluate the influence of a comprehensive clinical and surgical blood management protocol on the post-operative transfusion rate in patients undergoing primary TJA within an independent community hospital. **Study Design/Methods:** A retrospective review of patient medical records was initiated with the intent to compare cases prior to and following the introduction of a comprehensive clinical and surgical blood management

program. All TJA cases performed for a one-year interval from 2010 into 2011 were reviewed and summarized as the "control" sample population. These cases included standard operative techniques and a traditional hematocrit target transfusion protocol. All TJA cases performed for a one year period beginning late 2011 through 2012 were reviewed and summarized as the "study" sample population. These cases included the use of a comprehensive pre-operative blood management program based on pre-operative hemoglobin (Hgb). Erythropoietin was not used in this program due to insurance constraints. Intra-operatively, a bi-polar sealing device was used to control intra- and subsequent post-operative blood loss. Post-operative transfusion criteria were based on symptomatic anemia. The primary endpoints for this study were change in Hgb values, transfusion rate, and patient's hospital LOS. **Results/Findings:** Being an independent community hospital, the majority of patients in this facility present with multiple severe comorbidities (diabetic, hypertensive, average BMI 34—range 21-60), and have a primary payer of Medicare/Medicaid health insurance. Overall, females made up 60% of the cases for review in both groups. There was a significant drop in Hgb post-operatively in all cases. The control transfusion rate was 32% for all TJA cases combined, which decreased to 1% for knees and 0% in the hips in the study cases. There was a significant decrease in LOS of almost one full day between groups. **Conclusion:** Our findings demonstrate a significant decrease in transfusion rates and LOS following TJA. Our comprehensive blood management protocol is safe, easy to implement, and yielded significant economic savings to the hospital. Further implementation of our program across all surgeons performing TJA has shown similar clinical and economic results.

Disclosure of Commercial Conflict of Interest

R. J. Hakanson: Medtronic Advanced Energy, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

R. J. Hakanson: Nothing to disclose

SP13

Impact of Blood Center Efforts to Support Hospital Patient Blood Management (PBM) through an Operational Analysis

B R Swanson[1] (bswanson@bloodsystems.org), M A Lieb[1], J A Sullivan[1], K Land[1]. *[1]Blood Systems Inc., Scottsdale, AZ, United States*

Background/Case Studies: As part of a large blood center's patient blood management program, a comprehensive hospital blood bank and transfusion practices assessment was developed. From 2010-2012, a clinical services team of physicians and quality/transfusion specialists performed 2-day onsite assessments of blood product management and utilization practices (aka Transfusion Dx ®) for 43 hospital transfusion services. Best practices and recommendations with several sample templates and brief how-to descriptions were provided along with appropriate references when available. **Study Design/Methods:** An anonymous followup survey was sent to assess the recommendation's perceived value. Hospitals were asked if they made or planned to make changes and why they had or had not made the changes for the following Transfusion Dx elements: inventory management; blood product ordering and utilization; preadmission testing; iatrogenic blood loss; emergency release; wastage; 72-hour hold; laboratory technologies; transfusion reactions; quality monitors; and various hospital based policies (blood review committee oversight, massive transfusion protocol, standardized transfusion guidelines, blood administration, pre-operative anemia management, and nursing procedures/ training). Questions about optional charge-master review and basic LEAN evaluation of work flow and equipment placement were also asked. **Results/Findings:** 13/43 (30%) of facilities responded. Most hospitals made or were likely to make changes regarding recommendations on transfusion practice guidelines, transfusion safety officers, inventory management, and blood product wastage, especially where examples were provided. Recommendations on pre-admission testing, analysis of blood inventory including wastage, eliminating the 72-hour hold policy, and the electronic crossmatch were considered most valuable. Benefits of the changes included financial savings, increased efficiency, and improved service to physicians and patients. The need to change the culture, recent staffing and management changes, and the lack of current benefit to operations were most noted as reasons for not making changes. It was noted in many closeout sessions that those hospitals that were already involved in PBM or systemwide patient safety activities began implementing recommendations during the assessment itself. Benefits of the recommendations were not always easily quantifiable by the hospitals, but many of the comments received showed evolving awareness, understanding, and commitment to PBM. **Conclusion:** This assessment is an excellent way to establish relationships with hospitals and blood center PBM experts. Blood centers should help raise the level of awareness and actively support PBM in their hospitals. Providing templates, objective examples, and references were useful. Helping hospitals quantify their PBM efforts including objective outcome measures is an important next step.

Disclosure of Commercial Conflict of Interest

K. Land: Nothing to disclose; M. A. Lieb: Nothing to disclose; J. A. Sullivan: Nothing to disclose; B. R. Swanson: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Land: Nothing to disclose; M. A. Lieb: Nothing to disclose; J. A. Sullivan: Nothing to disclose; B. R. Swanson: Nothing to disclose

SP14

Plasma and Platelet Audit in Preparation for an Institutional Blood Management Program

J Kreuter[1] (Kreuter.Justin@mayo.edu), K Schulte[1], K N Kuker[1], J R Stubbs[1], C M van Buskirk[1]. *[1]Mayo Clinic-Rochester, Rochester, MN, United States*

Background/Case Studies: On the verge of developing an institutional blood management program, it was important to understand the rate and common reasons for inappropriate plasma and platelet transfusion. In order to get a manageable yet unbiased subpopulation to study, an audit was performed of all reported transfusion reactions in 2011. Furthermore, cases of inappropriate transfusion that resulted in a transfusion reaction (ie patient harm) would be valuable later when educating medical staff on the importance of blood management for patient care. **Study Design/Methods:** Patient name, clinic number, product, and date for all plasma and platelet transfusion reactions in 2011 were obtained (n = 280). All relevant complete blood count (CBC) and coagulation testing results around the time of transfusion were collected, and a physician reviewed relevant clinical notes around the time of transfusion for information pertinent to the appropriateness of transfusion: diagnosis; additional relevant information; active bleeding; and planned procedure. Once the database was complete, a physician judged each case for appropriateness according to the Institutional Transfusion Policy. **Results/Findings:** Plasma units were implicated in 12% and platelet units were implicated in 30% of reported transfusion reactions in 2011. 30% of plasma and 12% of platelet transfusions were determined to be inappropriate by the institutional transfusion policy. Common reasons for inappropriate plasma transfusions were prophylactic plasma given greater than 24 hours prior to the intended procedure and prophylactic plasma given for minimal International Normalized Ratio (INR) elevation. The common reason for inappropriate platelet transfusion was prophylactic transfusion when platelet counts were sufficient. In addition, multiple examples of absent and contradictory documentation in the clinical notes were discovered. **Conclusion:** It was identified that a blood management program at the institution and education of the medical staff will need to emphasize the potential harm of prophylactic transfusions and the importance of accurate clinical documentation of decision making. Interestingly, only 12% of the platelet transfusions were judged as inappropriate; however, it was estimated this number would be significantly increased if multiple prophylactic transfusions in platelet refractory patients were addressed by the institutional transfusion policy. The current study clearly represents only the initial attempt towards implementation of safe and appropriate management of our blood products.

Disclosure of Commercial Conflict of Interest

J. Kreuter: Nothing to disclose; K. N. Kuker: Nothing to disclose; K. Schulte: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal, Alyx, Other, Haemonetics, Other; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Kreuter: Nothing to disclose; K. N. Kuker: Nothing to disclose; K. Schulte: Nothing to disclose; J. R. Stubbs: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

SP15

Impact of the Use of Structured, Data-Driven Donor Recruitment and Blood-Drive Planning on Blood Collections

S Lee[1] (leex4900@umn.edu), R Mwenda[2], E Nkya[2], N Espejo[1], D de Coning[3], J McCullough[1]. *[1]Laboratory Medicine and Pathology, University of Minnesota, Minneapolis, MN, United States; [2]Ministry of Health and Social Welfare, National Blood Transfusion Service, Dar es Salaam, United Republic of Tanzania*, *[3]American Association of Blood Banks, Bethesda, MD, United States*

Background/Case Studies: An important limiting factor of the blood supply in developing countries is the lack of organized, cohesive planning and

evaluation of blood drive(s). As a result, willing sponsors may not be utilized efficiently, donor recruitment staff are unable to learn from experience with previous drives and are unable to plan effectively, and collection staff are not used efficiently. **Study Design/Methods:** Staff was mentored regarding use of structured planning and best practices for regular donor recruitment. The PREP tool, an Excel-based system, was developed to record the number of collection staff, their availability based on location and travel, fixed site capacity, and history of collections by the host or location. Implementation of this comprehensive donor recruitment mentoring and structured planning including use of the PREP tool occurred over several months. This enabled structured, data driven planning for blood collections. **Results/Findings:** The number of units of blood collected during January, February, and March, 2013, was 36,000 compared with 26,000 units of blood during October, November, and December, 2012. During this time, there were no other major publicity activities that could have impacted blood collections and seasonal variations were not seen in the past. This approximate 38% increase in blood collections was primarily due to the use of this structured recruitment process and the PREP tool. **Conclusion:** A comprehensive blood donor recruitment program and structured blood drive planning can substantially increase blood collections. The PREP tool is a valuable part of this process which can assist in developing a structured and complete blood drive schedule. This comprehensive process has the potential to increase blood collection and improve utilization of collection staff enabling donor recruiters to focus on future blood drive hosts and optimal scheduling, thus further increasing the blood supply.

Disclosure of Commercial Conflict of Interest

D. de Coning: No Answer; N. Espejo: No Answer; S. Lee: No Answer; J. McCullough: Nothing to disclose; R. Mwenda: No Answer; E. Nkya: No Answer

Disclosure of Grants Conflict of Interest

D. de Coning: No Answer; N. Espejo: No Answer; S. Lee: No Answer; J. McCullough: Nothing to disclose; R. Mwenda: No Answer; E. Nkya: No Answer

SP16

Impact of Outpatient Red Cell Exchange in an Institution Patient Blood Management Program (PBM)

S Herring[1] (sharmstrong@uabmc.edu), M A Gibbs[1], J Adamski[1], M B Marques[1]. *[1]University of Alabama, Birmingham, Birmingham, AL, United States*

Background/Case Studies: In 2007 when a patient blood management (PBM) program was started, only hospitalized patients with hemoglobinopathies and life-threatening complications were receiving red cell exchanges (RBCX) in our institution. Since then, RBCX was started for secondary prevention of strokes in sickle cell anemia. It was sought to determine the effect of this change in practice in the overall red blood cell (RBC) utilization. **Study Design/Methods:** Apheresis records were reviewed from FY 2009 (when outpatient RBCX was started) to the present and recorded the numbers of patients, procedures and RBC units used during outpatient RBCX in comparison with the total number of units transfused for each time-period as well as those transfused in the outpatient setting but not for RBCX. **Results/Findings:** From a single patient who had RBCX in 2009, seven patients were treated so far in 2013 (Table). All RBCX were performed via subcutaneous ports inserted in the superior vena cava to provide vascular access. From 2009 to 2013, the number of procedures went from 3 per year to 26 in the first 6 months of 2013 (Table . In 2009, only 32 RBC units were transfused during outpatient RBCX, compared with 172 so far this year, for an annualized number of 344 (a 10-fold increase in 4 years). RBCX units accounted for 2.1% of all outpatient RBCs transfused in 2009, and 24.2% currently. Of the total units transfused in our institution, only 0.1% were given during outpatient RBCX in 2009, versus 1.2% in 2013. **Conclusion:** The PBM has successfully decreased the utilization of RBCs by 27% from 2007 to the present. Since 2009, there was a 10-fold increase in the requirement for transfusion for a single indication, namely secondary prevention of stroke in patients with sickle cell disease. This is the only indication for which the number of RBC units/patient increased significantly since the beginning of the PBM. It is suggested that other institutions working on a PBM that when analyzing the outcome of their program, it is essential to note if they have experienced a change in practice that led to an increase in demand for transfusion.

Disclosure of Commercial Conflict of Interest

J. Adamski: Nothing to disclose; M. A. Gibbs: Nothing to disclose; S. Herring: Nothing to disclose; M. B. Marques: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Adamski: Nothing to disclose; M. A. Gibbs: Nothing to disclose; S. Herring: Nothing to disclose; M. B. Marques: Nothing to disclose

fiscal year	Number of patients	Number of RBCX	RBC units used during RBCX	Percent of outpatient units used for RBCX	Percent of total units used for RBCX
2009	1	3	32	2.1	0.1
2010	3	20	214	11.6	0.7
2011	3	29	317	19.0	1.1
2012	4	35	281	17.3	0.9
2013*	7	26	172	24.2	1.2

RBCX: Red blood cell exchanges; * First 6 months. Impact of outpatient RBCX in RBC utilization

Cellular Therapies

SP17

Microbiological Control Validation of Stem Cell Preparations According to Regulators Recommendations

M Stoermer[1] (melanie.stoermer@uk-koeln.de), S Radojska[1], N J Hos[2], B S Gathof[1]. *[1]Transfusion Medicine, University Hospital of Cologne, , Cologne, Germany; [2]Institute of Medical Microbiology, Immunology and Hygiene, University Hospital of Cologne, Cologne, Germany*

Background/Case Studies: Microbial safety of cellular preparations presents a challenge for physicians, manufacturers, and regulators because contamination can occur at various points in processing. Automated culture systems represent major progress in microbiological control of cellular preparations compared with conventional culture methods. Therefore the Paul-Ehrlich-Institut (PEI, Federal Institute for Vaccines and Biomedicines, Germany) defined standardized conditions for the validation of the microbiological control of hematopoietic stem cell preparations (PEI, http://www.pei.de/SharedDocs/ *Downloads/blut/stammzellen-genehmigung-21a-amg/pei-stellungnahme-mikrobiologische-kontrolle-stammzellen.pdf*). In this study, the microbiological control of stem cell preparations was validated according to the recommendations of PEI. **Study Design/Methods:** Buffy coats (BC) served as surrogates to stem cell preparations. According to Ph. Eur. 2.6.27, 10-100 CFU of *B. subtilis, S. aureus, P. aeruginosa, P. acnes*, and *C. albicans* were spiked separately in duplicate into BCs or saline (positive control). Afterwards 3 ml of each sample were inoculated in duplicate into aerobic and anaerobic culture bottles and incubated in an automated culture system for 5 days followed by additional 9 days in the incubator at 35 C. In addition, one inoculated aerobic culture bottle was incubated in an incubator at 20 C. Baseline sterility of BCs was proven before spiking. Finally, all culture bottles were screened for bacteria at the end of incubation. Grown bacteria were identified by routine diagnostic tests and matrix-assisted laser ionisation time-of-flight analysis. **Results/Findings:** All bacterial strains could be detected starting with inoculums of 7-26 CFU per culture bottle. *P. acnes* could be detected only in the anaerobic culture bottles while *C. albicans* grew only under aerobic conditions. The other strains could be detected by using both aerobic and anaerobic cultivation with a shorter time to detection using aerobic culture bottles. All aerobic culture bottles incubated at 20 C were positive for bacteria 48 hours after spiking at the earliest due to the necessity of plating and additional over night incubation to generate visible colonies on the agar plate. **Conclusion:** In cellular products, where neither membrane filtration nor classical culture can be performed due to cells which inevitably cause culture media to become turbid, microbial safety can be improved by using automated culture systems. Moreover the use of two different temperatures as recommended by Ph. Eur. 2.6.1 for classical sterility testing would further improve bacterial safety due to some species that are not able to grow at higher temperature under the real-life condition of only an initial low bacterial count.

Disclosure of Commercial Conflict of Interest

B. S. Gathof: Nothing to disclose; N. J. Hos: Nothing to disclose; S. Radojska: Nothing to disclose; M. Stoermer: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. S. Gathof: AES Chemunex, Grants or Research Support, Light Integra, Grants or Research Support, Roche, Grants or Research Support; N. J. Hos: Nothing to disclose; S. Radojska: Nothing to disclose; M. Stoermer: Nothing to disclose

SP18

The Evaluation of Differences in Platelet Loss in Allogeneic Peripheral Blood Stem Cell Collections from Two Devices

E K Storch[1] (ekscns@gmail.com), J L Becker[2]. *[1]Pathology, University at Buffalo, Buffalo, NY, United States; [2]Pathology, Roswell Park Cancer Institute, Buffalo, NY, United States*

Background/Case Studies: Peripheral blood stem cells collected using apheresis are used to perform a majority of adult hematopoietic stem cell transplantations. Donor platelet loss represents an important factor to take into consideration when evaluating donor safety. As apheresis technology continues to advance, it is important to evaluate operating performance of current devices to provide for ongoing comparison. **Study Design/Methods:** The study is a retrospective evaluation of platelet losses from allogeneic peripheral blood stem cell collections performed using either the Terumo BCT Spectra (Denver, CO) or the Fenwal Amicus (Lake Zurich, IL). Collections were included for analysis when performed for 2 consecutive days utilizing the same device. Donors were assigned to a device for collection based upon the number of individuals collecting in a week and donor specific needs. Donors met the goals of collection in the two procedures. Procedures were excluded if additional collections were required. Data collected included daily donor platelet count, volume processed, product volume and product platelet count. Platelet loss per liter processed and change in donor count were calculated. **Results/Findings:** All collections occurred between January 1, 2010-December 31, 2012. A majority of collections were performed using peripheral access. Standard blood flow was 70 ml/min. The MNC program was used for the Spectra. The Amicus was used with manufacturer's cutoffs. The day 1 pre-platelet counts were not statistically different between the two devices. Pre-day 2 platelet counts on the Spectra were significantly lower than on the Amicus p = .001. Platelet loss as determined by the difference in pre-counts between day 1-2 was significantly greater with the Spectra p = .001. In the product the average platelet yield differed significantly, assessed both by the total product platelet ($408.57 \pm 228.71 \times 10^{9}$ for the Spectra vs $62.21 \pm 23.03 \times 10^{9}$ for the Amicus) and platelet loss per liter processed ($31.54 \pm 14.18 \times 10^{9}$ for the Spectra compared to $4.83 \pm 1.64 \times 10^{9}$ for the Amicus), p = .001. The percent decrease in platelet count from day 1 to 2 was $39.48 \pm 12.64\%$ for the Spectra, $11.90 \pm 9.89\%$ for the Amicus. The total CD34 counts were $338.42 \pm 220.66 \times 10^{6}$ for the Amicus and $387.26 \pm 300.40 \times 10^{6}$ for the Spectra. **Conclusion:** Platelet losses were lower in allogeneic stem cell collections using the Fenwal Amicus as compared to the Spectra. This difference can affect donor management and the assignment of donors to equipment when choices are available.

Disclosure of Commercial Conflict of Interest

J. L. Becker: Fenwal, Travel Support or Honorarium; E. K. Storch: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. L. Becker: Nothing to disclose; E. K. Storch: Nothing to disclose

	N	Cobe	Amicus	N	p-value
Pre-day 1 platelet count	33	242 ± 65	231 ± 61	25	
Pre-day 2 platelet count	33	145 ± 43	200 ± 42	25	.001
Change in Platelet count	33	98 ± 92	31 ± 29	25	.001
% Decrease	33	39 ± 41	12 ± 13	25	
Product platelet yield × 10^12	66	409 ± 319	62 ± 61	50	
Platelet loss/liter	66	32 ± 28	4.8 ± 4.7	50	.001

Values are average ± SD

SP19

Persistence of Donor-Derived HLA Antibodies in a Stem Cell Transplant Recipient

S E Hohensee[1] (samhohensee@gmail.com), M T Aubrey[2], J Gutman[3,2], B M Freed[2]. *[1]Pathology, University of Colorado, Aurora, CO, United States; [2]ClinImmune Labs, Aurora, CO, United States; [3]Medicine, University of Colorado, Aurora, CO, United States*

Background/Case Studies: Donor-specific HLA antibodies in recipients of hematopoietic stem cell transplants are a known risk factor for graft failure, but the importance of HLA antibodies in the donor are less well understood. It has been reported that donor HLA antibodies disappear within 60 days posttransplant. A single case is reported here of the recipient of an HLA-identical sibling transplant in which the donor had high levels of HLA antibodies due to multiple pregnancies. Since the antibodies were not directed against the recipient, there was no contraindication to transplant. **Study Design/Methods:** Serum samples were analyzed for HLA class I and class II HLA antibodies using the One Lambda Single Antigen Bead Kit on a Luminex instrument. **Results/Findings:** Antibody testing was performed on donor and recipient sera pretransplant and on recipient sera at 19, 73, 121, and 164 days posttransplant. Significant antibodies were defined as those with a mean florescence intensity (MFI) >2000. The recipient had seven HLA antibodies before transplant. Two of these overlapped with the donor's antibody profile and demonstrated increasing intensity throughout the posttransplant period. Of the remaining five preexisting antibodies, only one still has a significant MFI 164 days posttransplant. 100% of the donor's 81 HLA antibodies appeared in the recipient by 73 days after transplant (MFI range 2761-15163); 94% of the donor's HLA antibodies were still circulating 164 days posttransplant (MFI range 6410-17845). The remaining 6% were just below the intensity defined as significant. **Conclusion:** This patient passively acquired 100% of his donor's 81 class I and class II HLA antibodies and still has 94% of them 164 days after transplant. These observations suggest that donor-derived antibody-producing cells may persist in the recipient for many months, which has implications for chronic graft versus hot disease (GVHD). In addition, it raises the possibility that recipient antiviral immunity posttransplant could be boosted by immunizing the donor.

Disclosure of Commercial Conflict of Interest

M. T. Aubrey: Nothing to disclose; B. M. Freed: Nothing to disclose; J. Gutman: Nothing to disclose; S. E. Hohensee: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. T. Aubrey: Nothing to disclose; B. M. Freed: Nothing to disclose; J. Gutman: Nothing to disclose; S. E. Hohensee: Nothing to disclose

SP20

Protocol Development of Isolation, Cryopreservation, and Culture of Human Umbilical Cord Mesenchymal Stem Cells

V Kalodimou[1,2] (kalodimou@yahoo.gr), I Mitrou[2], M Kontogiorgi[3], A Papalois[3]. *[1]Flow Cytometry-Research & Regenerative Medicine, IASO-Maternity Hospital, Medstem Services-Cryobanks, Athens, Greece; [2]Cryobanks, IASO-Maternity Hospital, Medstem Services-Cryobanks, Athens, Greece; [3]Experimental-Research Center, ELPEN Pharmaceuticals, Athens, Greece*

Background/Case Studies: Mesenchymal cells (MSCs) have risen dramatically, as shown by the increasing number of research teams. MSCs have been described as being able to differentiate, to migrate to sites of tissue injury, and to have strong immunosuppressive properties that can be exploited for successful autologous as well as heterologous transplantations. Given the increasing promise of the clinical utility of MSCs it is of ongoing interest to establish safe isolation protocols. MSCs from Wharton Jelly are adherent cells in culture with common morphology; they express the same specific surface antigen and they can differentiate into mesodermal lineages in vitro. **Study Design/Methods:** The ideal way of umbilical cord dissection, homogenization, enzymatic digestion, subsequent cell retrieval, antibiotic treatment, yield, and viability of the isolated population was studied. Possible immunophenotypic changes before and after culture were sought by Fluorescence Activated Cell Sorting. Different cytokines and stimuli to the MSCs were tested as to their effect on the expression of molecules, which have been shown to enhance the migrating/engrafting capacity after transplantation. The FACS-based method used allowed the performance of gene expression profiling studies and in vitro differentiation studies on a pure population. **Results/Findings:** An optimized protocol was developed for MSCs isolation concerning the length of the umbilical cord, homogenization and maintenance, the use of antibiotics, the clearing of the vessels, the initial wash to reduce blood contamination, efficient chopping for reproducible, complete digestion, efficient filtration, cell death, viscosity of the suspension, and minimize cell loss by splitting in extensive dilution, and cryopreservation. Antigen expression studies were performed and the expression pattern was studied by flow cytometry (FC). A novel protocol for the isolation of MSCs co-expressing at least three different mesenchymal markers was developed. **Conclusion:** MSCs are able to migrate to sites of tissue injury and have strong immunosuppressive properties that can be exploited for successful autologous as well as heterologous transplantations. The elimination of non-MSCs in vitro expansion of total nucleated cells after isolation from a segment of the umbilical cord has been already been proposed. However, for practical and financial reasons the culture of MSCs on a day-to-day basis should be avoided. Additionally the MSC populations that can be expanded in vitro have not been described precisely. The ex vivo model for isolation/cryopreservation of MSCs must acquire the following characteristics: materials and methods must follow Good Manufacturing

Practice with the highest efficiency; the cryopreserved transplant must not be contaminated by bacteria or fungi; it must retain its characteristics long-term; and the whole procedure must be inexpensive.

Disclosure of Commercial Conflict of Interest

V. Kalodimou: Nothing to disclose; M. Kontogiorgi: No Answer; I. Mitrou: Nothing to disclose; A. Papalois: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. Kalodimou: Nothing to disclose; M. Kontogiorgi: No Answer; I. Mitrou: Nothing to disclose; A. Papalois: Nothing to disclose

SP21

Engraftment of Human Peripheral Blood Stem Cells in NOD/SCID/IL-2Rγ^{null} Mice Following Long-term Cryopreservation

K Hege[1,2] (kmfitzpa@iu.edu), C Groves[3], R Britton[3], E Skipworth[3], V Graves[3], S Srivastava[3,4], A Sinn[5,6], K E Pollok[5,6], K A Robertson[1,5], W Goebel[1,3]. *[1]Pediatric Hematology/Oncology/Stem Cell Transplant, Riley Hospital for Children at Indiana University Health, Indianapolis, IN, United States; [2]Morris Green Scholars Program, Indiana University School of Medicine, Indianapolis, IN, United States; [3]Cellular Therapy Lab, Indiana University Hospital at Indiana University Health, Indianapolis, IN, United States; [4]Adult Hematology/Oncology/Bone Marrow Transplant, Indiana University Hospital at Indiana University Health, Indianapolis, IN, United States; [5]Herman B. Wells Center for Pediatric Research, Indiana University School of Medicine, Indianapolis, IN, United States; [6]In Vivo Therapeutics Core, Indiana University Simon Cancer Center, Indianapolis, IN, United States*

Background/Case Studies: Peripheral blood stem cells (PBSC) are usually cryopreserved awaiting clinical use. Data is limited regarding how long PBSC can be cryopreserved and retain abilities to engraft. The recovery of PBSC following long-term cryopreservation was assessed as well their ability to engraft in NOD/SCID/IL-2Rγ^{null} (NSG) mice. **Study Design/Methods:** Seven discarded PBSC units, cryopreserved for 16 to 18 years (average 17.5 years), were thawed. Post-thaw cell recovery, viability and colony forming ability were assessed. To evaluate engraftment capabilities in vivo, two of the seven units were thawed, CD34+ cells selected by magnetic beads (quatroMACS; Miltenyi), and 2×10^5 CD34+ cells were infused into each of ten sublethally irradiated NSG mice (five mice per unit). Short-term engraftment was assessed at 4 weeks by detecting human CD45+ and CD34+ cells in murine peripheral blood by flow cytometry. **Results/Findings:** Average % cell recovery after thawing the 7 units was 92 ± 11.4% (mean ± SD) with average % post-thaw viability by trypan blue staining of 72 ± 19%. The 2 units for NSG mouse transplantation had an above-average post selection viability of 94 and 96%. Average colony counts from the seven thawed units were 3.4 ± 3 colony-forming unit-granulocyte-macrophage (CFU-GM) and 3.1 ± 3 burst-forming unit-erythroid (BFU-E) per plate. A single unit with 34% viability did not demonstrate any colony forming ability. Most significantly, at 4 weeks, transplanted NSG mice showed evidence of short-term engraftment with the presence of human CD34+ and CD45+ cells by flow cytometry. Five mice transplanted with unit A averaged 7.6 ± 1.5% CD45+ and 3.6 ± 1.3% CD34+ human cells, while five mice transplanted with unit B averaged 2.4 ± 0.5% CD45+ and 3.1 ± 1% CD34+ human cells. **Conclusion:** A highly efficient total cell recovery of 79 to 110% in all seven thawed units was demonstrated, and cell viability of 62 to 86% in six of the seven units. Most units showed growth of myeloid and erythroid colonies with the exception of one with the lowest post-thaw viability. More importantly, short-term engraftment was demonstrated at 4 weeks post-transplant in NSG mice. Fifteen additional mice have been transplanted using three more long-term cryopreserved PBSC units to confirm short-term (4 weeks) and long-term (12 weeks) engraftment. This study demonstrates PBSC can be cryopreserved for over 16 years and retain colony forming ability, engraft into NSG mice, and potentially be useful in clinical transplantation.

Disclosure of Commercial Conflict of Interest

R. Britton: Nothing to disclose; W. Goebel: Nothing to disclose; V. Graves: Nothing to disclose; C. Groves: Nothing to disclose; K. Hege: Nothing to disclose; K. E. Pollok: Nothing to disclose; K. A. Robertson: Nothing to disclose; A. Sinn: Nothing to disclose; E. Skipworth: Nothing to disclose; S. Srivastava: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Britton: Nothing to disclose; W. Goebel: Cook General BioTechnology, LLC, Consulting or Board of Director Fees; V. Graves: Nothing to disclose; C. Groves: Nothing to disclose; K. Hege: Nothing to disclose; K. E. Pollok: Nothing to disclose; K. A. Robertson: Novartis, provides drug for a clinical trial, Other; A. Sinn: Nothing to disclose; E. Skipworth: Nothing to disclose; S. Srivastava: Nothing to disclose

SP22

Successful Implementation of Routine Filtration of Cellular Therapy Products at Bedside

K Anjos[1] (kanjos@hsc.mb.ca), K Paulson[1], A Giftakis[1], M Tulloch[1], O Synova[1], C Duong[1], J Pitura[1], L Heinrichs[1], D Szwajcer[1], D Wall[1]. *[1]Cellular Therapy Laboratory, Manitoba Blood and Marrow Transplant Program, CancerCare Manitoba, Winnipeg, MB, Canada*

Background/Case Studies: Bedside filtration with a standard blood filter (170-260 uM pore size) is a standard part of transfusion medicine. Cellular therapy (CT) products, however, are not routinely filtered prior to infusion. Filtration might remove harmful macroaggregates which can lead to infusion complications. Concern about loss of hematopoietic progenitor cells (HPC) has prevented universal adoption of filtration. **Study Design/Methods:** This study consisted of two phases. The initial phase involved thawing and filtering cryopreserved HPC apheresis products indentified for discard (n = 15). Pre and post filter viability, CD34, CFU and TNC values were compared (Table). These filters were then dried, dissected, stained with Wrights/Giesma, and cellular debris composition examined microscopically. Based on the results of this initial phase; a multidisciplinary process change was implemented (LEAN implementation) to routinely filter all fresh and cryopreserved CT products (apheresis, marrow, cord blood, therapeutic cells) at the time of infusion. **Results/Findings:** There was no significant change in viability (TNC, CD34, or CFU) post filtering of the initial phase filtering the cryopreserved HPC apheresis products. Macroaggregates were routinely seen on all filters from both phases dissected from the infusion sets: fibrous strands, cell aggregates with neutrophils, platelets, and lymphocytes. No adverse events with the introduction of the routine filtration of all CT products prior to infusion were noted. Education of laboratory and clinical team members prior to the introduction of routine filtration relieved anxiety about this significant change in practice. Preliminary engraftment is consistent with institutional normal ranges. **Conclusion:** Routine filtration does not remove hematopoietic progenitors present in cryopreserved CT products but removes macroaggregates that may be harmful. The use of a defined process change management allowed for a safe and efficient transition of a major change within our transplant program. Filtration for macroaggregates should be considered for all CT products.

Disclosure of Commercial Conflict of Interest

K. Anjos: Nothing to disclose; C. Duong: No Answer; A. Giftakis: Nothing to disclose; L. Heinrichs: Nothing to disclose; K. Paulson: Nothing to disclose; J. Pitura: No Answer; O. Synova: Nothing to disclose; D. Szwajcer: Nothing to disclose; M. Tulloch: Nothing to disclose; D. Wall: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Anjos: Nothing to disclose; C. Duong: No Answer; A. Giftakis: Nothing to disclose; L. Heinrichs: Nothing to disclose; K. Paulson: Nothing to disclose; J. Pitura: No Answer; O. Synova: Nothing to disclose; D. Szwajcer: Jansen, Grants or Research Support; M. Tulloch: Nothing to disclose; D. Wall: Nothing to disclose

TABLE 1. Initial Phase of Cryopreserved HPC Apheresis Post Thaw/Post Filtration (n = 15)

	Post Thaw (A) mean +/– range	Post Filter (B) mean +/– range	Recovery (B/A)*100
TNC × 10e9	17.4 (8.0-33.1)	17.7 (8.5-33.3)	100%
Cell Viability % (trypan blue)	76.5 (64-84)	75.0 (60-85)	98%
Total CD34 × 10e8	3.9 (0.8-7.9)	4.1 (0.8-9.5)	100%
CFU-GM/10e5	556.7 (120-1370)	583.7 (120-1780)	100%

SP23

Autologous Platelet Gel (APG) Use in a Multihospital Health Care System

J S Raval[1] (jay_raval@med.unc.edu), R M Dyga[2], S K Harm[3], J H Waters[4]. *[1]Pathology and Laboratory Medicine, University of North Carolina, Chapel Hill, NC, United States; [2]Perfusion Services, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [3]Pathology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [4]Anesthesiology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States*

Background/Case Studies: Autologous platelet gel (APG) is the product of platelet-rich plasma combined with thrombin and calcium that creates a

coagulum. It is increasingly being used in a wide range of surgical cases for a variety of reasons, including improved wound healing, decreased rates of post-operative infections and excessive bleeding, and reduced post-operative pain. However, few studies have been performed which describe the spectrum of its use. The goal of this study was to characterize the use of APG in a multihospital health care system. **Study Design/Methods:** Patient and procedural data were gathered for all surgical cases that attempted APG collection from 2004-2012. There were three devices used during this time period: SmartPReP Platelet Concentrate System (Harvest Technologies Corp., Plymouth, MA, USA); Angel Whole Blood Separation System (Cytomedix, Inc., Gaithersburg, MD, USA); and Magellan Autologous Platelet Separator (Arteriocyte Medical Systems, Inc., Cleveland, OH, USA). The institutional quality improvement council approved this study. **Results/Findings:** There were 85 physicians at 11 hospitals who attempted APG collection in a total of 6317 procedures (range 27-3035 procedures per hospital). APG was most often used in cardiac surgery (3356/6317, 53.1%) and orthopedic surgery (2861/6317, 45.3%), followed by neurosurgery (59/6317, 0.9%), and vascular surgery (21/6317, 0.3%). Other uncommon users (five or fewer cases) included obstetrics/gynecology, oral/maxillofacial surgery, general surgery, thoracic surgery, plastic/reconstructive surgery, cardiology, and urology. Of all patients, the average age was 61 years and 56% were males. The average pre-operative complete blood count was within normal limits. The average blood processed per case was 71.5 mL (range 48.3-83.1 mL per hospital), and the average volume of platelet-rich plasma collected per case was 7.2 mL (range 3.9-9.7 mL per hospital). APG collection was combined with both intraoperative cell salvage and cardiopulmonary bypass in 33.3% (2104/6317) of all cases, and with intraoperative cell salvage only in 1.6% (104/6317) of all cases. Only 9 adverse events (0.1% of all cases) occurred in the entire system over the 8-year period, the most common reasons being device malfunction, inability to collect blood for processing, and clotting events in the device. **Conclusion:** In this multihospital health care system, APG is most frequently used in cardiac and orthopedic surgical procedures, and products are successfully collected in the vast majority of cases. However, there is a small group of physicians who also use APG in other types of surgical or interventional procedures.

Disclosure of Commercial Conflict of Interest

R. M. Dyga: Nothing to disclose; S. K. Harm: Nothing to disclose; J. S. Raval: Nothing to disclose; J. H. Waters: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. M. Dyga: Nothing to disclose; S. K. Harm: Nothing to disclose; J. S. Raval: Nothing to disclose; J. H. Waters: Haemonetics, Grants or Research Support, Coramed,Grants or Research Support

SP24

Impact of Anticoagulant on Cord Blood (CB) Processing Efficiency

K S Brown[1] (kbrown@cordblood.com), M L Selich[2], S Range[2], D Reavis[2], H L Brown[1]. *[1]Scientific and Medical Affairs, Cbr Systems, Inc., San Bruno, CA, United States; [2]Laboratory and Processing Facility, Cbr Systems, Inc., Tucson, AZ, United States*

Background/Case Studies: Umbilical cord blood (UCB) is a source of hematopoietic stem cells (HSCs) for use in transplant medicine and is being researched in emerging applications in regenerative medicine. The AXP AutoXpress Platform™ (Thermogenesis Corp.) is an automated, functionally-closed processing system which volume reduces UCB by plasma and red blood cell (RBC) reduction. During AXP processing of whole UCB, the total nucleated cell (TNC) population, including the mononuclear cell fraction and stem cell population, is delivered into a seamless blow-molded, multi-compartment storage bag, and the RBC and plasma fractions are distributed into their respective compartments. UCB collection bags contain an anticoagulant to prevent coagulation and maintain cell viability during transport to the processing facility. Although Citrate Phosphate Dextrose (CPD) has been widely used based on the commercial availability of prefilled collection bags, heparin is also routinely employed as an anticoagulant for processing HSCs, especially in bone marrow harvests. This study compares TNC distribution within the three compartments of the AXP processing bagset of UCB collected in lyophilized heparin (LH) or CPD. **Study Design/Methods:** UCB was collected from consenting mothers who elected to family bank or donate UCB. UCB was transported to Cord Blood Registry (CBR)s processing facility and processed on AXP. UCB collected in either LH (n = 11) or CPD (n = 10) were matched for age, collection volume (without anticoagulant), and pre-processing TNC content. Pre- and post-processing cell counts in each AXP compartment were determined by mass balance analysis. **Results/Findings:** There were no significant differences in age, collection volume, or pre-processing TNC count of UCB collected in LH or CPD. After processing, LH collected units had a greater mean TNC recovery in the product storage bag compared to CPD collected units (97.2% vs 80.0%; p =< 0.001). A mean of 12% of the original TNC content of CPD collected units was recovered in the RBC bag whereas <1% of the TNC of LH collected units was recovered from the RBC bag. While not statistically significant, combined TNC recovery across product, plasma and RBC bags was higher for LH collected units than for CPD collected units (98.0% and 94.6% respectively). **Conclusion:** Mass balance analysis indicates cell recovery during AXP processing is impacted by anticoagulant selection, with minimal TNC lost to non-product bags when UCB is collected in LH. This is critical as TNC content is a primary selection criterion for transplant physicians. It has previously been shown that anticoagulant choice also impacts cell concentration and viability during transportation of UCB to the processing facility. Therefore, it is important that the implications of anticoagulant choice be carefully considered to ensure that UCB unit quality is optimally maintained from the time of collection through processing.

Disclosure of Commercial Conflict of Interest

H. L. Brown: Nothing to disclose; K. S. Brown: Nothing to disclose; S. Range: Nothing to disclose; D. Reavis: Nothing to disclose; M. L. Selich: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. L. Brown: Nothing to disclose; K. S. Brown: Nothing to disclose; S. Range: Nothing to disclose; D. Reavis: Nothing to disclose; M. L. Selich: Nothing to disclose

SP25

Very Small, Embryonic-Like Stem Cells Isolated from Human Umbilical Cord Blood and Differentiated into Erythroid Cells in Vitro

F Wang[1,2] (wangfang81929@126.com), J Liu[1,2], J He[1,2], F Zhu[1,2], H Lv[1,2]. *[1]Blood Center of Zhejiang Province, Hangzhou, China; [2]Key Laboratory of Blood Safety Research, Ministry of Health, Hangzhou, China*

Background/Case Studies: Very small, embryonic-like (VSEL) stem cells are enriched among CD133+, CD34+ and CXCR4+ lineage negative (Lin-) CD45- cells, which express nuclear embryonic transcription factors Oct-4 and Nanog and are able to differentiate some tissues from all three germ layers in vitro. In this study, VSEL stem cells from human umbilical cord blood (UCB) were differentiated into erythroid cells in vitro were studied. **Study Design/Methods:** MNCs were isolated from human UCB using density gradient centrifugation. VSEL stem cells were prepared from MNCs using Lineage Cell Depletion Kit, CD45 MicroBeads, CD34 MicroBeads kit and CD133 MicroBeads kit successively. The sorted cells were characterized by flow cytometry (FC) with anti-CD34-FITC, anti-CD45-Percpcy5.5 and anti-CD133-PE. Expression of Oct4 and Nanog in nuclear of the cells were detected using polyclonal anti-Oct-4 and anti-Nanog by immunocytochemistry technique. The mRNA transcripatients amounts of Oct4 and Nanog were detected by RQ-PCR. The VSEL stem cells were cultured in serum-free erythroid differentiation medium in the presence of SCF, IL-3, Epo and LIF, and induced to differentiate erythroid cells, which were characterized by FC with anti-CD71-FITC, anti-CD235-PE and anti-CD240-PE. **Results/Findings:** Two groups cells (Lin-CD45-CD34+ cells and Lin-CD45-CD133+) were obtained. Oct-4 and Nanog were positive in the nuclei and the cytoplasm were positive with DAPI in these cells groups by immunocytochemistry technique. Oct-4 and Nanog transcripatients were identified in these cells. After being cultured, the differentiation cells were expressed CD71 on day 7,and expressed CD235 and CD240 on day 12. **Conclusion:** The Lin-CD45-CD34+ cells and Lin-CD45-CD133+ cells separated from human UCB are VSEL stem cells by this method, which can express embryonic transcription factors and differentiate into erythroid cells in vitro.

Disclosure of Commercial Conflict of Interest

J. He: No Answer; J. Liu: No Answer; H. Lv: No Answer; F. Wang: Nothing to disclose; F. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. He: No Answer; J. Liu: No Answer; H. Lv: No Answer; F. Wang: Nothing to disclose; F. Zhu: Nothing to disclose

SP26

Peripheral Blood CD34 Count as a Predictor of Successful Hematopietic Progenitor Cell Harvest

R Parameswaran[1] (vinod.parameswaran@avera.org), L Cooper[1], A Guzman[1], L Roberts[1], A Galal[1], K McCaul[1]. *[1]Avera McKennan Hospital and University Health Center, Sioux Falls, SD, United States*

Background/Case Studies: Peripheral blood hematopoietic progenitor cell (HPC) support is crucial to hematopoietic recovery after high dose therapy

followed by autologous or allogeneic transplantation. The timing of collection after cytokine priming is important, to ensure the collection of the minimum cell count required for engraftment. The efficiency of collection impacts the utilization of resources, material and human. Several strategies have been employed to initiate and optimize collection adequacy, including peripheral blood mononuclear cell counts, blood volume processed during apheresis and enumeration of peripheral blood HPC (PHPC) CD34 counts. A single institutional experience in the use of the PHPC counts is reported here to predict adequacy of peripheral blood HPC collection. **Study Design/Methods:** At the center, autologous transplants are performed routinely for high-risk hematological malignancies. Seventy-four consecutive autologous collections were analyzed. Granulocyte Colony Stimulating Factor (GCSF) and/or Plerixafor were used as mobilization agents. HPC collection usually commenced on day 5 of GCSF administration. A complete blood count, chemistry panel, and peripheral CD34 count were performed on each day of collection. Apheresis was performed to process 200 ml of blood volume /kg body weight with the premise that increasing the process volume would lead to increased recruitment of HPCs. Subsequently, in 2012 the HPC CD34 count was used additionally to make decisions regarding commencement of harvest and the use of Plerixafor. The minimum target HPC dose for collection was $2 \times 10\wedge 6$ CD34 cells/kg. Data was available on all 74 HPC collections (49 with Myeloma and 25 with Lymphoma). The Pearson's method was used to determine the correlation between the the PHPC CD34 counts, the process volume, and the product CD34 dose. **Results/Findings:** The collection target dose was reached in all donors. PHPC count ranged from 2.0-295 cells/uL (Median 21.0). The median product CD34 dose for all collections was $1.5 \times 10\wedge 6$ CD34 cells/kg (Range 0.12-27.12).There was a strong positive linear relationship ($r = 0.88$ with $p =$ value <0.0001) between the harvested CD34 dose/kg recipient weight and the PHPC CD34 count. In patients with myeloma the median PHPC count was 18 cells/uL (Range 2-60), the median product CD34 dose was $1.5 \times 10\wedge 6$ CD34 cells/kg (range 0.12-5.49) and the correlation coefficient was $r = 0.94$ with $p < 0.0001$. In patients with lymphoma, the median PHPC CD34 count was 20 cells/uL (Range 2-295), the product CD34 dose was $1.66 \times 10\wedge 6$ CD34 cells/kg (range 0.12-27.12), and the correlation coefficient was $r = 0.87$ with $p < 0.0001$. There was no significant linear correlation between volumes processed and the product CD34 dose ($p = 0.5$). **Conclusion:** Estimation of PHPC CD34 counts accurately predicts the adequacy of HPC collection, and may potentially help conserve resources and increase efficiency of HPC collection.

Disclosure of Commercial Conflict of Interest

L. Cooper: Nothing to disclose; A. Galal: Nothing to disclose; A. Guzman: Nothing to disclose; K. McCaul: No Answer; R. Parameswaran: Nothing to disclose; L. Roberts: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Cooper: Nothing to disclose; A. Galal: Nothing to disclose; A. Guzman: Nothing to disclose; K. McCaul: No Answer; R. Parameswaran: Nothing to disclose; L. Roberts: Nothing to disclose

SP27

Analysis of Critical Factors Influencing Primary-Cell Quality and Expansion Potential of ADSC

S Chen[2] (shyigen@ms26.hinet.net), P Shen[1], C Huang[1], P Tseng[1], W Lo[1]. *[1]Research and Development Department, MSC Laboratory, HealthBanks Biotech Co., Ltd., Taipei, Taiwan; [2]Division of Plastic and Reconstructive Surgery, Department of Surgery, Tri- Service General Hospital, National Defense Medical Center, Taipei, Taiwan*

Background/Case Studies: The feasible and abundant advantages of adult adipose-derived stem cells (ADSCs) obtained by liposuction have made ADSCs an attractive MSC source of autologous therapy for tissue engineering and regenerative medicine. **Study Design/Methods:** To optimize the condition for preparation of adipose tissue and manufacture of ADSC, we tried to identify the critical factors influencing primary cell quality and expansion potential of ADSC. **Results/Findings:** First of all, it was demonstrated that classical liposuction was the ideal collection technique to get superior stromal vascular fraction (SVF) isolation efficiency when comparing to water-jet liposuction (1.72 ± 0.3*E5 versus 0.3 ± 0.12*E5 cells/ml adipose tissue). After collection, the processed lipoaspirate (PLA) tissue should be storaged/transported at low temperature and set in subsequent process within 48 hours. Furthermore, tissue/cell viability evaluated by XTT assay was highly positively correlated to primary SVF isolation efficiency, primary ADSC isolation efficiency and expansion efficiency. We also manifested the donor age was inversely proportional to both the tissue/cell viability and the SVF/ADSC isolation efficiency. **Conclusion:** In conclusion, the results validated the effective method for evaluation and revealed the critical factors that influenced ADSC potential, which should be considered as the major issue in manufacture and quality control of ADSC product for clinical purpose.

Disclosure of Commercial Conflict of Interest

S. Chen: Nothing to disclose; C. Huang: Nothing to disclose; W. Lo: Nothing to disclose; P. Shen: Nothing to disclose; P. Tseng: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Chen: Nothing to disclose; C. Huang: Nothing to disclose; W. Lo: Nothing to disclose; P. Shen: Nothing to disclose; P. Tseng: Nothing to disclose

SP28

A Reproducible Potency Assay to Measure Differences in MSC-mediated T cell Suppression

D Bloom[1] (ddbloom@medicine.wisc.edu), N Bhatia[1], P Hanley[2], A Gee[2], M Armant[3], D H McKenna[4], J M Centanni[1], R Lindblad[5], D Hei[1], P Hematti[1]. *[1]University of Wisconsin-Madison, Madison, WI, United States; [2]Baylor College of Medicine, Houston, TX, United States; [3]Center for Human Cell Therapy, Boston, MA, United States; [4]University of Minnesota, Minneapolis, MN, United States; [5]The EMMES Corporation, Rockville, MD, United States*

Background/Case Studies: Bone marrow-derived mesenchymal stromal cells (BM-MSCs) culture-expanded ex vivo are able to down-modulate the immune activity of a wide range of leukocyte subsets, including T lymphocytes. As a result, they hold promise as a cellular therapeutic for immune diseases. However, over the past several years, BM-MSC immunotherapy has not met primary endpoints in clinical trials. The reasons for this clinical short-coming are not known. At the University of Wisconsin-Madison site for the Program Assistance for Cellular Therapy (PACT), an in vitro immune-potency assay (IPA) has been developed which measures the degree to which different MSC lines are able to suppress T cells in vitro. Such an assay could assist in the interpretation of in vivo efficacy. **Study Design/Methods:** This assay measures allogeneic MSC-mediated suppression of CD4+ T cell proliferation. Stimulatory antibodies specific for CD3 and CD28 induce the proliferation of CFSE-labeled TH cells over a wide range of PBMC : MSC ratios, from 1-1 to 1-0.05. The strength of T cell suppression is measured against an established MSC reference standard line to facilitate interpretation of relative potencies between test lines. A single value was calculated for each series of titrations of a test line and P values generated on repeat assays. The degree of suppression was compared to HLA-DR expression as well as MSC viability to examine whether any of these significantly influenced the inhibition of CD4+ T lymphocyte proliferation. **Results/Findings:** Eleven BM-MSC lines generated from four PACT sites were examined using this assay. There was significant variability in the degree of CD4+ T cell suppression among the lines tested, ranging from an IPA suppression value of 87% for the most suppressive line to 28% for the least suppressive. Viability post-thaw was greater than 74% for all MSC lines and did not correlate significantly with suppression of T cell proliferation ($p > 0.05$). MSC-mediated CD4+ T cell inhibition was not influenced by MSC expression of HLA-DR. However, the degree of suppression for all lines tested correlated with the level of IFN-gamma produced by PBMCs of distinct leukapheresis donors. **Conclusion:** The immune-potency assay developed at UW-PACT is practical and highly reproducible. The testing of BM-MSC lines from four PACT sites demonstrated extensive variability in CD4+ T cell suppression. Examination of more lines is required to define whether these differences are due to the MSC culture expansion technique utilized by each site or are the result of variability between bone marrow donors. It is proposed that the assay described here could be a key step to understanding the potential clinical efficacy of individual MSC lines for treatment of T cell-mediated disorders.

Disclosure of Commercial Conflict of Interest

M. Armant: Nothing to disclose; N. Bhatia: Nothing to disclose; D. Bloom: Nothing to disclose; J. M. Centanni: Nothing to disclose; A. Gee: Nothing to disclose; P. Hanley: Nothing to disclose; D. Hei: Nothing to disclose; P. Hematti: Nothing to disclose; R. Lindblad: Nothing to disclose; D. H. McKenna: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Armant: Nothing to disclose; N. Bhatia: Nothing to disclose; D. Bloom: Nothing to disclose; J. M. Centanni: Nothing to disclose; A. Gee: Nothing to disclose; P. Hanley: Nothing to disclose; D. Hei: Nothing to disclose; P. Hematti: Nothing to disclose; R. Lindblad: Nothing to disclose; D. H. McKenna: BEST Collaborative, Travel Support or Honorarium; Novartis, Grants or Research Support, ERYTECH Pharma, Consulting or Board of Director Fees

SP29

Comparative Predictive Value of Peripheral CD34 and Novel Parameters in Stem Cell Mobilization

B King[1] (bking3@partners.org), J Geary[2], O J Sturtevant[1], J Ritz[2], G Kao[2]. *[1]Cellular Therapies Quality Assurance, Dana-Farber Cancer Institute, Boston, MA, United States; [2]Cell Manipulation Core Facility, Dana-Farber Cancer Institute, Boston, MA, United States*

Background/Case Studies: Cellular therapy programs have endeavored to determine the appropriate time to initiate the collection of apheresis products to support autologous hematopoietic stem cell transplant patients and optimize CD34+ cell yield. Currently, peripheral blood CD34 + cell counts (pbCD34) are routinely used to determine initiation of apheresis. Sysmex has developed a hematology analyzer that can identify a population of immature myeloid cells in the peripheral blood (HPC) according to cell size, density and differential lysis resistance with very short turn around time. In this study, a comparison of these two methods (pbCD34 and HPC) to predict apheresis product CD34+ cell yield was performed to show that HPC counts can be used as an alternative method to determine initiation of collection. **Study Design/Methods:** The peripheral blood CD34 count and HPC was prospectively analyzed from a total of 170 adult autologous donors not mobilized with Plerixafor and recorded their same day apheresis product CD34+ cell yield. The pbCD34 and product CD34+ cell count were anayzed using single platform ISHAGE gating and the Beckman Coulter Stem-Kit™ on the Coulter FC500, and HPC analysis was performed on the Sysmex XE-2100. Receiver operating characteristics (ROC) curves were created for pbCD34 and HPC to predict a successful collection as collecting over $0.5 \times 10\wedge6$ CD34/kg in a one day. **Results/Findings:** The correlation between HPC and day one collected CD34/kg is 0.66 ($p < 0.001$), whereas the correlation for pbCD34 and CD34/kg on the same patients was 0.90 ($p < 0.001$). This correlation was unchanged for pbCD34 on subsequent days of collection but the correlation with the HPC value drops each day to approximately 0.3 by day 4. The ROC analysis to predict collection yield of over $0.5 \times 10\wedge6$ CD34/kg conclude that an HPC count of 20 has a sensitivity (true positve) and specificity (1-false positive) of 77% and 100 %, while pbCD34 of 10 yield a sensitivity and specificity of 92% and 100%. The area under the ROC curve for HPC is 0.81, and for pbCD34 it is 0.95. **Conclusion:** Same day peripheral blood HPC measurement with Sysmex XE-2100 can be used as an alternative method to pbCD34 to predict successful collection for autologous transplant patients not mobilized with Plerixafor. Compared to pbCD34 of 10, HPC of 20 was shown to have lower sensitivity but equivalent specificity to predict a successful collection. However, the substitution of HPC for pbCD34 will be important to cellular therapy programs that may not have immediate access to pbCD34 assay.

Disclosure of Commercial Conflict of Interest

J. Geary: Sysmex, Other; G. Kao: Nothing to disclose; B. King: Sysmex,Other; J. Ritz: Nothing to disclose; O. J. Sturtevant: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Geary: Nothing to disclose; G. Kao: Nothing to disclose; B. King: Nothing to disclose; J. Ritz: Nothing to disclose; O. J. Sturtevant: Nothing to disclose

SP30

Process Improvement for Double Cord Blood (CB) Transplant Patients

S C Sharpe[1] (susan.sharpe@northside.com), B C Alder[2], J L McCollum[2]. *[1]Transfusion Service & Tissue Bank, Northside Hospital, Atlanta, GA, United States; [2]Hematopoietic Stem Cell Laboratory, Northside Hospital, Atlanta, GA, United States*

Background/Case Studies: The Transfusion Service (TS) and Hematopoietic Stem Cell (HSC) Laboratory have always worked closely together to ensure that the TS provides the appropriate blood product support required of our stem cell transplant patients. Our institution approved a protocol for double CB transplant which presented a challenge for the TS to determine what type of blood products to issue for transfusion support of the recipient post transplant. A standardized process for the HSC Lab was implemented to notify the TS of the date that the transplant recipient was to receive their double CB transplant, and for the TS to document the donor CB types, and transfusion recommendations. **Study Design/Methods:** Initially, a process had been designed where the blood type and date of transplant was entered into the Lab Information System (LIS) for allogeneic stem cell or bone marrow transplant recipients. In order to document the two unique identifiers of the unrelated donor CB products used for one transplant recipient, we created a new test code. The test code was named Double Cord Blood Transplant (DCTX). The DCTX was developed to document the date of transplant, the two unique cord blood product identifiers and blood types, recipient blood type, and the transfusion recommendations of red cell and plasma products to use for the transplant recipient. A sample of each thawed CB product is given to the TS to perform a blood type on the day of transplant. The TS finalizes the results of the test code DCTX by entering the date of transplant, blood types, and transfusion recommendations. This information is placed in the recipient's electronic file. The TS personnel print the results of the DCTX and sign and date the form. This form is faxed to the HSC Lab. The HSC Lab signs and dates the faxed report which is then faxed back to the TS for their records. The HSC Lab maintains a copy of the signed DCTX report in the HSC Lab chart. **Results/Findings:** The advantage of the formalization of the process is ensuring the patient received the correct blood products at the time of transplant and for subsequent transfusions. All parties acknowledge the date of transplant and the new transfusion recommendations to ensure patient safety. **Conclusion:** Through the development of a formalized process, the TS and HSC Lab have developed and refined a process to ensure that all patients receiving a double cord blood transplant will receive the appropriate transfusion support.

Disclosure of Commercial Conflict of Interest

B. C. Alder: Nothing to disclose; J. L. McCollum: Nothing to disclose; S. C. Sharpe: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. C. Alder: Nothing to disclose; J. L. McCollum: Nothing to disclose; S. C. Sharpe: Nothing to disclose

SP31

Stem Cell Processing and Temperature Course

M Dijkstra-Tiekstra[1] (m.dijkstra@sanquin.nl), A Setroikromo[1], M Kraan[1], J de Wildt-Eggen[1]. *[1]Transfusion Monitoring, Sanquin Blood Supply, Groningen, Netherlands*

Background/Case Studies: Dimethyl sulfoxide (DMSO) is added to hematopoietic progenitor cells (HPC) to protect cells during cryopreservation. Before addition of DMSO to the HPC it will be diluted to a certain concentration in plasma. Different initial concentrations can be used. Adding of DMSO to HPC causes an exothermic reaction, exposing the cells to a higher temperature, which might affect the WBC viability after thawing. The aim was to study the effect of pre-cooling of 20% and 50% DMSO/plasma (D/P) and HPC simulant on temperature course and WBC viability. **Study Design/Methods:** Because HPC is not readily available for research purposes, a HPC simulant from buffy coats (BCs) and plasma with comparable cell counts but absence of $CD34^+$ or immature cells was used in this study. The HPC simulant was divided in six portions of 20 ml. Twelve ml plasma was added to products for 50% D/P to equalize the final volume to the 20% products. Under continuous agitation either 20 ml 20% D/P or 8 ml 50% D/P was added slowly. Finally, there were six study conditions varying for 20% D/P and 50% D/P, and for temperature of D/P and HPC simulant when adding D/P. For all conditions, the final DMSO concentration was 10% and the final WBC concentration was 100×10^9/l. The product was frozen using controlled rate freezing in about 62 minutes and placed into the vapour phase of liquid nitrogen. After storage for at least 1 week the product was thawed in a 37 C water bath and immediately 1:5 diluted in PBS and measured for WBC count and viability. Temperature was monitored for adding D/P to HPC simulant. **Results/Findings:** See table (mean ± SD, n = 4) Highest temperature was reached when D/P and HPC simulant were at RT. Increase in temperature was highest when D/P and HPC simulant were at <10 C, without influence of the 20% or 50% D/P. No influence of temperature of either D/P or HPC simulant or 20% or 50% D/P was found on WBC viability after thawing. **Conclusion:** Pre-cooling of this product or D/P seems not necessary and both 20% D/P and 50% D/P can be used in the freezing process of HPC (simulant). Additional testing using real HPC might be necessary.

Disclosure of Commercial Conflict of Interest

J. de Wildt-Eggen: Nothing to disclose; M. Dijkstra-Tiekstra: Nothing to disclose; M. Kraan: Nothing to disclose; A. Setroikromo: No Answer

Disclosure of Grants Conflict of Interest

J. de Wildt-Eggen: Nothing to disclose; M. Dijkstra-Tiekstra: Nothing to disclose; M. Kraan: Nothing to disclose; A. Setroikromo: No Answer

D/P HPC simulant	20% RT RT	20% <10°C RT	20% <10°C <10°C	50% RT RT	50% <10°C RT	50% <10°C <10°C
Temperature increase, °C	2.6 ± 0.7	3.4 ± 0.7	4.9 ± 1.7	2.2 ± 0.6[1]	1.9 ± 1.1[2]	5.8 ± 1.1[1,2]
Temperature max, °C	25.6 ± 1.0[3,4]	23.3 ± 1.4[3,5]	14.6 ± 1.2[4,5,6]	25.4 ± 0.9[7]	25.0 ± 0.3[8]	15.7 ± 0.9[6,7,8]
% WBC viability before D/P			93.3 ± 1.6			
% WBC viability with D/P	89.0 ± 4.9	87.9 ± 3.2	90.7 ± 2.0	87.1 ± 2.8[9]	88.7 ± 2.8	90.0 ± 2.1[9]
% WBC viability after thawing	61.9 ± 8.1	62.1 ± 6.1	63.9 ± 6.4	63.1 ± 6.9	61.6 ± 7.2	67.2 ± 4.8
% WBC recovery[11]	87.2 ± 2.5	88.6 ± 2.5[10]	87.0 ± 4.2	86.8 ± 1.8	85.3 ± 4.1[10]	87.4 ± 2.6

[1-10] $p < 0.05$ (ANOVA for repeated measurements followed by Tukey-Kramer, or paired t-test between 20% and 50% D/P) [11] after thawing compared to before adding D/P and contains both viable and non-viable cells

SP32

The Correlation between Total Nucleated Cell Count and Weight of Cord Blood Units (CBU) of Caucasian and Minority CB Units: A 12-Month Retrospective Study in 2012

A Lambert[1] (allambert@lifesouth.org), L Hernandez[2], M Arthur[1], J Merayo-Rodriguez[3]. *[1]Cellular Therapy, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States; [2]Quality Assurance, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States; [3]Medical Office, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States*

Background/Case Studies: Many factors are assessed when evaluating a collected cord blood unit (CBU) for processing and cryopreservation including donor eligibility, collection date, weight, volume, total nucleated cell count (TNC), and viability. This institution has established a cut-off value for weight, volume, and TNC when determining to process. This study will assess the correlation between the value of weight assessment and TNC of the unit when determining to process the unit for cryopreservation. The race and ethnicity of the units also determine whether the unit proceeds to processing and cryopreservation based upon TNC cut-off. **Study Design/Methods:** A retrospective study was performed to analyze the weight of each with the TNC for Caucasian and minority units to determine the correlation, if any, between weight and TNC for units collected during 2012. The correlation, if any, would provide a quicker means of assessing the unit for processing and cryopreservation and prevent any further testing and costly processing of the unit. **Results/Findings:** Table 2 provides the raw data that demonstrates the correlation between weight and TNC. For Caucasians and minorities, the minimum weight for assessment of TNC should be 130 g. At 120 g for Caucasians, the highest TNC value was 1.25 + E9, which falls in the evaluate range of the acceptable TNC criteria for our institution. At 130 g for Caucasians, the highest TNC value was 2.32 + E9, which is within the acceptable TNC range. At 120 g for minorities, the highest TNC value was 1.10 + E9, which falls in the evaluate range of the acceptable TNC range. At 130 g for minorities, the highest TNC value was 2.92 + E9, which is within the acceptable TNC range. **Conclusion:** Processing techniques prohibit post-processing TNC recoveries to be 100%. Because of the processing procedure, the average TNC recovery for processing is 84% (2013 processing TNC percent recovery average). For the CBUs with TNC values that fall in the evaluate range of the acceptable TNC range, the post-processing TNC would be considerably less which would provide a less desirable unit for transplantation. The recommendation would be that the weight criteria be set at 130 g, regardless of race or ethnicity of the CBU.

Disclosure of Commercial Conflict of Interest

M. Arthur: Nothing to disclose; L. Hernandez: Nothing to disclose; A. Lambert: Nothing to disclose; J. Merayo-Rodriguez: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Arthur: Nothing to disclose; L. Hernandez: Nothing to disclose; A. Lambert: Nothing to disclose; J. Merayo-Rodriguez: Nothing to disclose

SP33

Three-Year Study in a Cord Blood (CB) Bank to Confirm Quality, Evaluation, Operation, and Infection-Control Procedures

I Mitrou[2] (irenemitrou@yahoo.gr), V Kalodimou[1,2], N Panagiotopoulos[2], K Ntalis[2], A Vlachousis[2]. *[1]Flow Cytometry-Research & Regenerative Medicine, IASO-Maternity Hospital, Medstem Sservices-Cryobanks, Athens, Greece; [2]Cryobanks, IASO-Maternity Hospital, Medstem Services-Cryobanks, Athens, Greece*

Background/Case Studies: Stem cells have the remarkable potential to develop into many different cell types in the body. Serving as a sort of repair system for the body, they can theoretically divide without limit to replenish other cells as long as the person is still alive. When a stem cell divides, each new cell has the potential to either remain a stem cell or become another type of cell with a more specialized function, such as a muscle cell, or a red blood cell. Cord blood (CB) banking has a history that spans about 25 years. To date, about 8000 people have benefited from medical therapies using stem cells of the umbilical CB. Today CB is used in medical therapies similar to the way stem cells from bone marrow are used. To confirm continuing quality and evaluation efforts, standard operation procedures, and the infection control and safety, this current 3-year study was carried out. **Study Design/Methods:** A 3 year study (2008-2010) was reported of CB collection and storage concerning collection and storage of units according to standard operation procedures (SOPs) and infection control (serology, microbiology, and molecular diagnosis). **Results/Findings:** From a number of 4368 of units processed in 2008, 3644 in 2009, and 3099 in 2010, 0.69%, 0.93%, and 1.32% cultured specimens, respectively, were positive for the presence of bacterial /fungal pathogens after inoculation and culture by the BACT/Alert system. Caesarean collections, mostly, were free of contaminants. The strains in all the cases showed usual antibiotic sensitivities. From the total number of units processed in 2008, 2009, and 2010, 1.41%, 0.27%, and 0.26% specimens, respectively, had positive infectious disease markers which are HBsAg by immune control testing and HCV by nucleated acid test (NAT). **Conclusion:** About 11,482 CB units have been collected, processed, and stored. Infection control by blood culture and antibiotic testing showed that all of the isolated strains demonstrated usual antibiotic sensitivities. The contamination rates mentioned above, kind of bacteria and antibiotic resistances are negligible, confirming thus the capacities of the premises. Serological testing for viruses, syphilis, and NAT (nucleic acid test) for HIV and HCV are of major importance for the safety of transplantation. CD34+ cells present in the samples, in each time group, as well as their viability fluctuating with +/– 1, suggesting that as time passes the samples absolute counts and their viability remains stable. This conclusion is very promising for the future but more studies in longer periods of time must be done in order to make safe assumptions.

Disclosure of Commercial Conflict of Interest

V. Kalodimou: Nothing to disclose; I. Mitrou: Nothing to disclose; K. Ntalis: Nothing to disclose; N. Panagiotopoulos: Nothing to disclose; A. Vlachousis: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. Kalodimou: Nothing to disclose; I. Mitrou: Nothing to disclose; K. Ntalis: Nothing to disclose; N. Panagiotopoulos: Nothing to disclose; A. Vlachousis: Nothing to disclose

SP34

Stromal Cell Derived Factor-1 mRNA Expression of Peripheral Blood Stem Cells was Suppressed in a Dose Dependent Manner of Korean Red Ginseng

S G Kim[1,2] (sgkim@cu.ac.kr), A Lee[1], M Kim[1], H Jang[1], J H Lee[1,2], S Kim[3]. *[1]Laboratory Medicine, Daegu Catholic University Medical Center, Daegu, Republic of Korea; [2]Comprehensive Integrative Medicine Institute, Daegu, Republic of Korea; [3]Oriental Medicine, Daegu Haany University Medical Center, Daegu, Republic of Korea*

Background/Case Studies: Hematopoietic stem cell (HSC) transplantation is the treatment of choice for hematologic malignancies. For collecting peripheral blood stem cells, it is necessary to mobilize stem cells from bone marrow (BM). The interaction between chemokine receptor type 4 (CXCR4) and its ligand, stromal cell derived factor (SDF)-1, plays an important role in stem cell mobilization. There are still poor mobilizers who need to do more than 5-6 times of apheresis course to accomplish the target number of CD34

positive HSC. Korean red ginseng is one of the most popular medicinal herbs in Korea. Total saponin of panax ginseng may promote in vitro development of erythroid-lineage cells from CD34 positive cells. In this study, the effect of Korean red ginseng was sought on expression of SDF-1 and CXCR4 of peripheral blood stem cells exposed to doxorubicin. **Study Design/Methods:** Mononuclear cells (MNC) were isolated from apheresis products of allogeneous donor after obtaining informed consent. The cells were exposed to doxorubicin (600 ng/ml) for 24 hours and incubated in the medium with or without Korean red ginseng (25, 50, 100 ug/ml). The adherent cells and non-adherent cells which were regarded as hematopoietic stem cells were separated for the next experiments. Surface markers including CD34, CD44, CD106, and CD184 (CXCR4) on the non-adherent cells were measured by flow cytometry (FC). RT-PCR was done for detection of expression of mRNA of SDF-1 and CXCR4. **Results/Findings:** Surface expression of CD34, CD44, CD106, and CD184(CXCR4) of the non adherent cells were increased in dose dependent manner after being incubated with Korean red ginseng for 24 hours; % of positive cells were as follows: CD34, 24/27/36/38; CD44, 24/26/35/37; CD106, 27/34/38/40; and CD184, 33/39/44/46. Expression of SDF-1 mRNA of the doxorubicin treated non-adherent and adherent cells was decreased in dose dependent manner of Korean red ginseng, whereas CXCR4 expression was increased at the same manner. **Conclusion:** These results suggest that Korean red ginseng may play a synergistic role with CXCR4 inhibitor for stem cell mobilization when collecting peripheral blood stem cells by apheresis.

Disclosure of Commercial Conflict of Interest

H. Jang: Nothing to disclose; M. Kim: Nothing to disclose; S. G. Kim: Nothing to disclose; S. Kim: Nothing to disclose; A. Lee: Nothing to disclose; J. H. Lee: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Jang: Nothing to disclose; M. Kim: Nothing to disclose; S. G. Kim: Nothing to disclose; S. Kim: Nothing to disclose; A. Lee: Nothing to disclose; J. H. Lee: Nothing to disclose

SP35

Contribution of Cell Therapy with Stem Cells in Pulmonary Fibrosis in Rats

R G Félix[1] (rgfelix2005@hotmail.com), J C Vicentini-Oliveira[2], A T Fabro[3], O S Cotrim[4], J Ribeiro-Paes[5], M A Valerio[6], C Cunha[7], E Deffune[8].
[1]Pediatric, Faculty of Medicine—Botucatu—SP/BRAZIL, Botucatu-SP, Brazil; [2]Public Health, Veterinary Hospital—UNESP, Botucatu, Brazil; [3]Pathology, Faculty of Medicine—Botucatu/BRAZIL—UNESP, Botucatu, Brazil; [4]Blood Transfusion Center, Faculty of Medicine—Botucatu/BRAZIL—UNESP, Botucatu, Brazil; [5]Biotechnology, Faculty of Science—Assis- UNESPNESP, Assis, Brazil; [6]Blood Transfusion Center, Faculty of Science—Assis- UNESPNESP, Assis, Brazil; [7]Oncology, Clinical Hospital UNESP, Botucatu, Brazil; [8]Blood Transfusion Center, Faculty of Medicine- Botucatu/BRAZILI- UNESP, Botucatu, Brazil

Background/Case Studies: Idiopathic fulmonary fibrosis (IPF) is defined as a type of chronic fibrosing interstitial lung disease of unknown etiology, limited to the lungs, and that presents histological pattern of usual interstitial pneumonia. It is estimated about 5 million people are affected worldwide. The current clinical treatment is associated with transient and partial improvement with equivocal or unsatisfactory results. Cell therapy is presented as an alternative therapy with a potential application in IPF. This study aims to validate the model of IPF induced by bleomycin instillation and analysis of the contribution of cell therapy, using injection of mesenchymal stem cells, allogeneic, obtained from adipose tissue for the treatment of IPF. **Study Design/Methods:** A total of 60 male albino Wistar rats, weighing 250 g, aged 8 weeks, were formed into four experimental groups consisting of 15 animals in each. They were monitored for weight and saturation index in 3 stages: D0; D14; and D28. In D0, an instillation of bleomycin at a dose of 1.5 U/Kg was performed. In D14, an infusion of mesenchymal stem cells quantified and phenotyped by flow cytometry FP. In D28, animals were sacrificed for macro and microscopic analyzes. The mean adipose tissue removed was 3.35 g and after dissociation using collagenase type I cell count the was 5.3 $\times$ 104 lymphomononuclear cells/g of adipose tissue. These cells were expanded in culture for 21 days Knockout DMEM-F12 medium added with 10% fetal bovine serum. Three criteria were considered proof of MSC: adhesion to plastic, expression of CD90 by FC and the ability to differentiate into three lineages of mesodermal origin. **Results/Findings:** After expansion, the control flow cytometry showed 98% of cells expressing CD90. In D14, there was a pool of these MSC were infused intravenously into the tail vein at a concentration of 1 $\times$ 106 cells / animal in a volume of 200 µl saline. No adverse reactions or deaths were identified during the infusion or within 14 days. Histological analyzes were performed by two different experts in a double-blind study. Intrataqueal instillation of bleomycin at a dose of 1.5 U/Kg PF determined in 100% of animals whereas animals had 57.14% of PF in grade IV. The instillation of intratracheal bleomycin compromised the saturation index of animals and histological analyzes confirm the pattern of IPF. The parameters of weight gain, improved saturation index, recovery macroscopic, and microscopic lung tissue were fully achieved in the group treated with MSC-TA. **Conclusion:** It was concluded that from the standpoint of clinical and histological MSC-TA determined proven clinical and histological improvement in animals with pulmonary fibrosis of different degree and should be regarded as innovative therapeutic approach.

Disclosure of Commercial Conflict of Interest

O. S. Cotrim: No Answer; C. Cunha: No Answer; E. Deffune: No Answer; A. T. Fabro: Nothing to disclose; R. G. Félix: Nothing to disclose; J. Ribeiro-Paes: Nothing to disclose; M. A. Valerio: No Answer; J. C. Vicentini-Oliveira: Nothing to disclose

Disclosure of Grants Conflict of Interest

O. S. Cotrim: No Answer; C. Cunha: No Answer; E. Deffune: No Answer; A. T. Fabro: Nothing to disclose; R. G. Félix: Nothing to disclose; J. Ribeiro-Paes: Nothing to disclose; M. A. Valerio: No Answer; J. C. Vicentini-Oliveira: Nothing to disclose

SP36

Stem Cell Processing: Optimization for Added DMSO Concentration and Freezing Program

M Dijkstra-Tiekstra[1] (m.dijkstra@sanquin.nl), A Setroikromo[1], M Kraan[1], J de Wildt-Eggen[1]. *[1]Transfusion Monitoring, Sanquin Blood Supply, Groningen, Netherlands*

Background/Case Studies: To protect hematopoietic progenitor cells (HPC) during cryopreservation, dimethyl sulfoxide (DMSO) is added. In routine various concentrations of DMSO in plasma (D/P) are used for this, like 20% and 50%. Also the freezing rate might be important for HPC recovery and viability. The aim of this study was to investigate the effect of 20% versus 50% D/P and freezing program WBC viability and recovery. **Study Design/Methods:** Because HPC is not readily available for research purposes, a HPC simulant from buffy coats (BCs) and plasma with comparable cell counts but absence of CD34+ or immature cells was used in this study. The HPC simulant (<10 C) was divided in two portions of 20 ml. Twelve ml plasma was added to products for 50% D/P to equalize the final volume to the 20% products. Subsequently, either 20 ml 20% or 8 ml 50% D/P of <10 C was added slowly under continuous agitation, with a final 10% DMSO concentration. The product was frozen using controlled rate freezing, either program I (38 minutes; fast-rate) or program II (62 minutes; slow-rate) and placed into the vapour phase of liquid nitrogen. After storage for at least one week, the product was thawed in a 37 C water bath and immediately 1:5 diluted in PBS and measured for WBC count and viability. **Results/Findings:** See Table (mean $\pm$ SD, n = 4). No influence of 20% or 50% D/P was found on WBC viability, but WBC viability after thawing was higher for freezing program II (20% D/P p = 0.06; 50% D/P p = 0.03). However, WBC recovery was higher for freezing program I (20% D/P p = 0.11; 50% D/P p = 0.02). **Conclusion:** For freezing HPC simulant it was found that both 20% D/P and 50% D/P can be used. Further, the slow-rate freezing program II is preferred above the fast-rate freezing program I for freezing HPC (simulant). Additional testing using real HPC might be necessary.

Disclosure of Commercial Conflict of Interest

J. de Wildt-Eggen: Nothing to disclose; M. Dijkstra-Tiekstra: Nothing to disclose; M. Kraan: Nothing to disclose; A. Setroikromo: No Answer

Disclosure of Grants Conflict of Interest

J. de Wildt-Eggen: Nothing to disclose; M. Dijkstra-Tiekstra: Nothing to disclose; M. Kraan: Nothing to disclose; A. Setroikromo: No Answer

		Freezing program I		Freezing program II	
		20% D/P	50% D/P	20% D/P	50% D/P
Before adding D/P	WBC, $\times 10^9$/l	197 ± 1	196 ± 3	193 ± 91	193 ± 9[1]
	WBC viability, %	93.8 ± 2.7	92.6 ± 4.0	93.3 ± 1.6	93.3 ± 1.6
With D/P	WBC, $\times 10^9$/l	99 ± 2	98 ± 2	98 ± 2	97 ± 2
	WBC viability, %	88.5 ± 5.5	90.3 ± 2.6	90.7 ± 2.0	90.0 ± 2.1
After thawing	WBC viability, %	55.0 ± 4.1	56.3 ± 5.7[3]	63.9 ± 6.4	67.2 ± 4.8[3]
	WBC recovery, %[2]	91.7 ± 2.6	93.2 ± 2.4[4]	87.0 ± 4.2	87.4 ± 2.6[4]

[1] before adding plasma to equalize the volume [2] compared to before adding D/P and contains both viable and non-viable cells [3,4] $p < 0.05$ (t-test, paired for 20% vs 50% and unpaired for freezing method I versus II)

SP37

The Automated Hematology Analyzer in the Small Cell Therapy Laboratory

B Caudill[1] (bethany.caudill@osumc.edu), M Auker[1], J C Kelley[1], A Winterstein[1], H Bradbury[1], B Daneault[1], S Vasu[2], L O'Donnell[2]. *[1]The Ohio State University Comprehensive Cancer Center-Arthur G. James Cancer Hospital and Richard J. Solove Research Institute, Columbus, OH, United States; [2]The Ohio State University Department of Internal Medicine, Division of Hematology, Columbus, OH, United States*

Background/Case Studies: Hematopoietic stem cell products are tested for total nucleated cell content to calculate the CD34 and CD3 cells in the product, typically using a hematology analyzer. Our Cell Therapy Laboratory (CTL) uses the Beckman Coulter ACT diff. Due to specimen differences, low numbers of samples, and hours of operation the analyzer is not operated the same as in a hematology lab and requires more troubleshooting, maintenance, and service. The current CTL analyzer was installed in March 2012 and has had eight equipment malfunctions, an average of one malfunction every 7 weeks. With this high number of malfunctions the CTL evaluated the malfunctions and the best options for the lab. **Study Design/Methods:** An audit was performed of the equipment malfunctions and issues for the current analyzer since installation and the previous same analyzer for the 9 years it was in use. A poll of other cell therapy facilities was performed to research other analyzers used, how the instruments perform, and the methods used to prevent problems. **Results/Findings:** An overview of the equipment malfunctions for the CTL AcT analyzers were reviewed. Reults of polling indicated there were at least seven other types of hematology analyzers in use by other cell therapy facilities. All reported some type of extra maintenance being performed beyond manufacturer requirements. Two other laboratories that use an AcT perform shutdown and startup daily and clean the baths weekly. Based on manufacturer requirements, the CTL currently performs startup daily, shutdown every other day, and cleaning of the baths when needed. **Conclusion:** When comparing the two CTL AcT Diff instruments both had similar issues, but the new machine's occurred more frequently and quickly. There are a wide variety of other analyzers available for use. The type of instrument selected is based on previous local and peer experience, process validation requirements, capital and operational costs, and available services. Due to the uniqueness of the specimen type and usage, issues can be expected, making customer support and service an important consideration when choosing a hematology analyzer for a cell therapy facility.

Disclosure of Commercial Conflict of Interest

M. Auker: Nothing to disclose; H. Bradbury: Nothing to disclose; B. Caudill: Nothing to disclose; B. Daneault: Nothing to disclose; J. C. Kelley: Nothing to disclose; L. O'Donnell: Nothing to disclose; S. Vasu: Nothing to disclose; A. Winterstein: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Auker: Nothing to disclose; H. Bradbury: Nothing to disclose; B. Caudill: Nothing to disclose; B. Daneault: Nothing to disclose; J. C. Kelley: Nothing to disclose; L. O'Donnell: Nothing to disclose; S. Vasu: Nothing to disclose; A. Winterstein: Nothing to disclose

TABLE 1. Equimpment Malfunction Overview

	Cell Counter 2 (current)	Cell Counter 1 (previous)
Weeks In Use	56	448
Average Time Between Malfunctions	7 weeks	30 weeks
Total Number of Malfunctions	8	15
Symptom/Resolution	Number of Occurrences	Number of occurences
QC Failure/Syringe of Valve Replacement	1	4
Platelet Startup Failure/Decontamination	2	2
Vacuum Error/Vaccum Replaced	1	1
Reagent Card Failure/New Reagent Card	1	0
QC Failure/AVR Target Issue or Adjustment	2	2
Fatal Error/Wiring or Board Replacement	1	1
Spike in Histogram/Grounding Issue	0	1
Fatal Error/Probe Replacement	0	1
Other	0	3

SP38

Generation of Granulocytes and Platelets-Rich Plasma for the Treatment of Sepsis

M Fahrendorff[1] (melanie.fahrendorff@fresenius-kabi.com), D Preys[1], M Biehl[1], W F Boecker[1]. *[1]Fresenius Kabi Deutschland GmbH, Friedberg, Germany*

Background/Case Studies: Sepsis, with its extremely high mortality rate, is a severe medical condition that can be characterized as a whole-body inflammatory state. It is triggered by the immune system's response to a serious infection, most commonly caused by bacteria, fungi, viruses, or parasites in blood, urinary tract, lungs, skin, or other tissues. Sepsis can be thought of as falling within a continuum from infection to multiple organ dysfunction syndromes. Bacterial infections following implantation of biomaterials represent the most devastating complication in orthopedic and traumatology therapies. Over the last decade, systemic antibiotics have not provided an effective treatment against infections associated with implants. Today there is growing evidence that local activity of immuno-competent cells is able to reduce or even prevent both bacteria colonization of an implant and biofilm formation. A large body of data now exists, showing that leukocytes and particularly polymorphonuclear neutrophils (PMNs) play a significant role in fighting against implant-related infections, through a variety of mechanisms. Some authors conclude from their studies that lower local levels of PMNs in the early surgical wound are directly related to the subsequent occurrence of septic complications and higher early local leukocyte concentrations play a significant protective role against post-surgical infection. **Study Design/Methods:** Based on this background it is advisable and desirable to have a fast, reproducible and easy way for the production of platelet and granulocyte rich plasma (PGRP). That is the reason for the development of a Sequestration Procedure (SQ) in a cell washing device. This device works on the principle of a seal-less flow centrifuge comparable to continuous systems for hemapheresis which are widely used in blood banks for decades. The SQ procedure separates whole blood into packed red cells (PRC), plasma (PLS) as well as PGRP fractions. **Results/Findings:** The recovery reaches for granulocytes in the PGRP fraction approximately 50% and for platelets approximately 60%. **Conclusion:** The preparation of PGRP using an automated device is safe, easy, robust, fast and economic. Providing reproducible quality PGRP with this SQ procedure will allow designing broader clinical studies to show the benefit of the proposed therapy.

Disclosure of Commercial Conflict of Interest

M. Biehl: Fresenius, Ownership or Partnership; W. F. Boecker: Nothing to disclose; M. Fahrendorff: Fresenius Kabi, Ownership or Partnership; D. Preys: No Answer

Disclosure of Grants Conflict of Interest

M. Biehl: Nothing to disclose; W. F. Boecker: Nothing to disclose; M. Fahrendorff: Fresenius Kabi, Ownership or Partnership; D. Preys: No Answer

Components and Component Processing

SP39

Older Red Blood Cells RBCs Experience Enhanced Clearance Compared to Younger RBCs Following Simple Transfusion

S R Stowell[1] (srstowe@emory.edu), C M Arthur[1], N Smith[1], J C Zimring[2], J E Hendrickson[1]. *[1]Emory University, Decatur, GA, United States; [2]Transfusion Medicine Research, Puget Sound Blood Center, Seattle, WA, United States*

Background/Case Studies: Although many cells undergo turnover, RBCs display unique, age-related changes that target them for removal, a process commonly termed RBC senescence. While RBC senescence facilitates RBC removal during normal in vivo RBC turnover, it remains untested whether older RBCs also experience accelerated removal following collection, processing and transfusion. It was hypothesized that older RBCs would in fact be cleared preferentially over younger RBCs after transfusion, and tested this hypothesis in a murine model. **Study Design/Methods:** B6 or B6.HOD mice (with RBC specific expression of a triple fusion protein including the human Duffy antigen) were injected with N-hydroxysulfosuccinimide biotin, and their RBCs were evaluated at sequential points post-biotinylation. At approximately 1 or 35 days post-biotinylation, blood was harvested and RBCs were labeled with the lipophilic dye, DiI, to facilitate tracking post-transfusion. A subset of recipients was passively immunized with monoclonal anti-Duffy3 prior to HOD.B6 RBC transfusion. Following transfusion, the percentage of biotin + DiI + compared to biotin—DiI + RBCs was enumerated by flow cytometry (FC). **Results/Findings:** Immediately after whole mouse biotinylation, nearly 100% of donor RBCs were biotin positive. However, 35 days following biotinylation, approximately 20% of donor RBCs remained biotinyated. Biotinylated versus non-biotinylated RBCs at this time point will be referred to as older and younger RBCs, respectively. Following simple transfusion into naïve animals, syngeneic older RBCs displayed preferential removal compared to younger RBCs, with a 4- and 10-fold decrease in circulating older RBCs by days 3 and 7 post-transfusion, respectively. However, RBCs harvested only 1 day following biotinylation failed to displayed enhanced clearance as far out as 7 days following transfusion. In contrast, both older and younger HOD.B6 RBCs were equally sensitive to immediate clearance following incompatible transfusion into recipients passively immunized with anti-Duffy3. **Conclusion:** These results suggest older RBCs retain signals that result in preferential removal compared to younger RBCs following harvesting, processing and transfusion into a new recipient. Accelerated removal does not appear to be an artifact of biotin modification, as newly biotinylated donors failed to display enhanced clearance. Preferential clearance of older RBCs compared to younger RBCs likely reflects unique signals associated with RBC senescence, as both older and younger cells displayed equivalently rapid clearance following incompatible transfusion. In sum, this study suggests that RBCs retain senescence markers following transfusion, and provides a unique approach to study pathways responsible for regulating senescent RBC turnover in vivo.

Disclosure of Commercial Conflict of Interest

C. M. Arthur: No Answer; J. E. Hendrickson: No Answer; N. Smith: Nothing to disclose; S. R. Stowell: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. M. Arthur: No Answer; J. E. Hendrickson: No Answer; N. Smith: Nothing to disclose; S. R. Stowell: Nothing to disclose; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium

SP40

Profiling Phosphorylation of Protein Kinases during Platelet Storage: Novel Insights into Storage Lesion Development

P Schubert[1] (peter.schubert@blood.ca), S Karwal[1], B Culibrk[1], D Coupland[1], D Devine[1]. *[1]Centre for Blood Research, Canadian Blood Services, Vancouver, BC, Canada*

Background/Case Studies: The platelet storage lesion (PSL) describes all storage-related changes in platelets. Several proteomic studies have hinted at underlying molecular mechanisms of storage lesion development some of which correlate with routine in vitro quality parameters. These include impairment of platelet physiologic processes and ongoing protein translation during storage. Involvement of PI3-kinase-dependent Rap1 activation in PSL development highlighting a link to protein kinases has been previously demonstrated. Here changes are reported in the phosphorylation profile of selected kinases throughout storage of platelet concentrates in order to assess their contribution to features of the PSL. **Study Design/Methods:** Platelet concentrates prepared by the buffy coat (BC) method and suspended in plasma were sampled on day 1 (day of production), days 2, 5, and 7 during storage. Samples were analyzed for routine platelet quality parameters such as platelet count, platelet activation as determined by P-selectin exposure on the platelet surface, as well as glucose and lactate levels to monitor metabolic activity. Alterations in the phosphorylation status of 40 different kinases were determined using antibody-based arrays (RayBiotech, Norcross, GA, US) and results were verified by immuno blot analyses. **Results/Findings:** The changes in in vitro assay results throughout storage were in agreement with previous studies. Eleven kinases displayed changes greater than 1.5-fold in spot intensity comparing samples collected on day 1 and 7 of storage. The results of the array were confirmed by immunoblot analyses for several kinases. Among others, p38MAPK showed a 2-fold phosphorylation increase during storage, while phosphorylation of glucose synthesis kinase (GSK) and AMP-activated protein kinase (AMPK) decreased by 1.5- and 2-fold, respectively. The phosphorylation levels of p38MAPK and AMPK showed a correlation to the increase of platelet activation as monitored by P-selectin surface expression and to the decrease of the glucose concentration, respectively. **Conclusion:** PSL is a determinant of the quality of platelet concentrates. Kinases act as molecular switches in signal transduction by regulating protein function of their individual substrates. The p38MAPK is known to be linked to PSL development; however, GSK3 and AMPK have not previously been reported in association with PSL. The phosphorylated isoforms of these two kinases lack kinase function and the decreasing phosphorylation levels during storage suggest acceleration of their function. Because GSK3 is required for adequate glucose metabolism and AMPK is linked to ongoing protein synthesis, these changes are consistent with a process by which the storage lesion impairs platelet function while simultaneously driving protein synthesis. This is consistent with reported increases in concentration of specific proteins detected in proteomic profiling studies published to date.

Disclosure of Commercial Conflict of Interest

D. Coupland: Nothing to disclose; B. Culibrk: Terumo BCT, Grants or Research Support; D. Devine: Terumo BCT, Grants or Research Support; S. Karwal: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

D. Coupland: Nothing to disclose; B. Culibrk: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support, Fresenius-Kabi, Grants or Research Support, MacoPharma, Grants or Research Support; S. Karwal: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

SP41

Transglutaminase 2 is Not Required for Senescence of Red Blood Cells (RBCs) During Storage in a Mouse Model

K de Wolski[1] (KarenD@psbc.org), H Waterman[1], J C Zimring[1,2]. *[1]Puget Sound Blood Center, Seattle, WA, United States; [2]Laboratory Medicine, University of Washington, Seattle, WA, United States*

Background/Case Studies: RBCs have a defined life-span in any given individual, with an average of 120 days and 55 days in humans and mice, respectively. The mechanism(s) by which a senescent RBC is cleared from circulation is unclear, with multiple competing hypotheses. In the last decade, it has become clear that RBCs can undergo programmed cell death (called eryptosis). The mechanisms of eryptosis differ from apoptosis, as they use calpains instead of caspases, but the general theme is similar. Loss of RBC homeostasis leads to an influx of Ca2+, which leads to a number of downstream events, including activation of Ca2+ dependant proteases (calpains), kinases, phosphatases, scramblase, protein kinase C, and also a Ca2+ dependent transglutaminase (Tgm2). Tgm2 plays a special role in cellular degradation and eryptosis by catalyzing co-valent crosslinking of the free amino group of lysine residues and the gamma-carboxamide group of glutamine. Tgm2 activity results in crosslinking of RBC proteins, and inhibition of Tgm2 prevents breakdown of Band 3 during calcium influx. In the current study, it was hypothesized that Tgm2 was required for normal RBC senescence during blood storage. **Study Design/Methods:** Mice were obtained that have a targeted deletion of the Tgm2 locus, resulting in a null phenotype. Heterozygotes for the null locus were bred together, and littermates that were homozygous null or homozygous wild-type were identified by PCR. Western blot analysis of peripheral blood was used to confirm phenotype with genotype. Peripheral blood was collected from wild-type and

Tgm2 null mice in CPDA-1. The hematocrit was increased to 75% and blood was stored at 40 C for 14 days. The storage method was based upon established protocols previously shown to result in approximately 75% post-transfusion survival after 14 days storage. **Results/Findings:** The Tgm2 KO allele segregated with a frequency predicted by standard Mendalian genetics. Tgm2 protein was readily detectable in RBCs from wild-type mice, but there was no detectable Tgm2 in RBCs from homozygous Tgm2 KO littermates. There was no significant difference in post-transfusion survival between wild-type and Tgm2 KO RBCs when transfused fresh or after 14 days of storage. **Conclusion:** Tgm2 plays a clear biochemical role in eryptosis, resulting in the crosslinking of RBC proteins. However, the current studies demonstrate, in a murine system, that Tgm2 is not required for clearance of either freshly collected RBCs or RBCs stored as a model of banked blood. Such findings do not rule out a role for Tgm2, but they do reject the hypothesis that Tgm2 plays a crucial or necessary role in RBC clearance after storage.

Disclosure of Commercial Conflict of Interest

K. de Wolski: Nothing to disclose; H. Waterman: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. de Wolski: Terumo BCT, Grants or Research Support, Immucor, Grants or Research Support; H. Waterman: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium

SP42

Platelets Stored for 14 Days at 4 C are Comparable to Standard Room Temperature Storage for 5 Days

K Reddoch[1,2] (kreddoch@gmail.com), R Montgomery[2], P Nair[1], C Fedyk[2], H F Pidcoke[2], A Ramasubramanian[1], A P Cap[2]. *[1]Department of Biomedical Engineering, University of Texas, San Antonio, San Antonio, TX, United States; [2]US Army Institute of Surgical Research, Fort Sam Houston, TX, United States*

Background/Case Studies: Refrigerated platelets reduce the risk of bacterial contamination, promote faster control of bleeding, and, despite accelerated clearance, may improve outcomes after severe hemorrhage in the acute setting. Their hemostatic capacity is poorly characterized; this study assessed the hemostatic potential of platelets stored in cold and room temperature (RT) under standard blood banking conditions for up to 14 days. **Study Design/Methods:** Apheresis platelets (AP) collected from healthy donors (n = 10) were stored at room temperature (RT) with agitation or at 4 C without agitation. Measurements were performed on fresh, day 5, or day 14, and included: 1) platelet counts, mean platelet volumes, pH and metabolism; 2) impedance aggregometry in response to ADP, collagen, and TRAP; 3) clot rheometry; 4) expression of CD41a, CD42b, CD62P, CD154, and lactadherin using flow cytometry (FC); 5) microparticle release; and 6) release of soluble factors Thromboxane B2 (TxB2) and sCD40L (n = 5). The data is represented as mean ± SEM, and analyzed by repeated measures ANOVA. Differences from baseline were evaluated with Student's t test ($p < 0.05$). **Results/Findings:** Results are shown in Table 1. Metabolic parameters and aggregation were better or equal in 4 C platelets out to 14 days, with the single exception of TRAP-activated aggregometry. Fibrinogen and von Willebrand factor receptor expression levels were preserved in all groups. Cold samples were more activated than RT samples. Rheometry demonstrated increased clot stiffness in 4 C platelets compared to RT. Microparticle counts were similar to baseline in all groups. **Conclusion:** Chilled platelets were functionally and metabolically superior or equal to conventional RT stored platelets through day 14, suggesting the feasibility of longer platelet storage. While some markers of activation were present, microparticle formation did not increase. These data support the need to reevaluate the use of 4 C storage as an alternative to RT platelet storage for actively bleeding patients requiring immediate hemostasis.

Disclosure of Commercial Conflict of Interest

A. P. Cap: Nothing to disclose; C. Fedyk: APT Biotech, Other; R. Montgomery: No Answer; P. Nair: No Answer; H. F. Pidcoke: Advanced Preservation Technologies, Other; A. Ramasubramanian: Nothing to disclose; K. Reddoch: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. P. Cap: Terumo BCT, Grants or Research Support, Oxygen Biotherapeutics, Grants or Research Support; C. Fedyk: Nothing to disclose; R. Montgomery: No Answer; P. Nair: No Answer; H. F. Pidcoke: Oxygen Biotherapeutics, Inc., Grants or Research Support, Terumo BCT, Grants or Research Support; A. Ramasubramanian: Nothing to disclose; K. Reddoch: Nothing to disclose

	Day 1	Day5		Day 14
	Baseline	RT	4C	4C
Metabolic Parameters:				
pE	7.27 ± 0.12	7.03 ± 0.31	7.62 ± 0.09	7.51 ± 0.1
Glucose	332 ± 14	178 ± 61	300 ± 17	238 ± 24
Lactate	1.8 ± 1.2	16.0 ± 3.7	4.7 ± 1.3	11.0 ± 2.1
Aggregation.Response (AUC):				
ADP	13.4 ± 3.9	0.11 ± 0.11	13.3 ± 5.1	0.13 ± 0.13
Collagen	17.7 ± 5.3	0.3 ± 0.21	20.2 ± 3.1	6.1 ± 1.9
TRAP	80.2 ± 2.5	46.7 ± 9.4	50.4 ± 4.8	24.5 ± 5.0
Flow CytometryMarkers:				
CD42b GMF	3945.4 ± 140.3	3453.8 ± 170.9	3172.9 ± 101.5	2069.4 ± 203.5
CD62P(% Total)	34.8 ± 5.6	61.6 ± 6.3	68.9 ± 4.4	72.3 ± 3.1
Lactadherin (%Total)	3.8 ± 0.7	18.3 ± 4.0	19.4 ± 1.7	46.9 ± 4.5
Mcroparticles/ml ($\times l0^6$)	1.31 ± 0.67	0.84 ± 0.24	1.22 ± 0.32	1.10 ± 0.37
Soluble Marker Release (ng/ml):				
sCD40L	l.5 ± 0/1	2.7 ± 0.3	1.9 ± 0.3	ND
TxB_2	0.21 ± 0.05	2.98 ± 0.67	0.62 ± 0.15	ND

Functional response and soluble marker release of stored platelets. Bold = significant ($p < .05$); ND = no data.

SP43

Human Platelets, Stored in an Automated Temperature Cycler between 4.5 C and 37 C for 7 Days, Exhibited Improved In Vitro Properties if Stored with Intermittent Agitation

A Skripchenko[1] (Andrey.Skripchenko@redcross.org), A M Turgeon[1], D Thompson-Montgomery[1], H Awatefe[1], S J Wagner[1]. *[1]Blood Components, American Red Cross Holland Laboratory, Rockville, MD, United States*

Background/Case Studies: Platelets (PLTs) stored at cold temperatures more than 2 days show irreversible morphological and other in vitro changes which result in poor in vivo survival and recovery. Recent studies indicate that some of these decrements are reversible if PLTs are stored for shorter periods in the cold prior to re-warming. For example, PLTs, stored for 7 days with cycles of 12 hours at 4 C with subsequent incubation for 30 minutes at 37 C before returning back to 4 C had greater recovery than those of PLTs continuously stored for 7 days at room temperature or 4 C when transfused into SCID mice. In this study, the in vitro properties of PLTs stored in a commercially available automated incubator programmed to cycle for 12 hours at 4.5 C followed by 45 minutes at 37 C without agitation were compared to those with identical temperature cycling with 5 minutes of agitation

during warm up. **Study Design/Methods:** A single Trima apheresis PLT unit (n = 12) was collected under informed consent. Immediately after collection, PLTs were aliquoted into four 60 mL samples in CLX storage bags. One bag was stored at 20-24 C (Control-RT) with continuous agitation; a second bag was continuously stored at 4.5 C without agitation (Control-CT); a third bag was held in an automated temperature cycler (12 hrs at 4.5 C, 45 minutes at 37 C) without agitation (TC); a fourth bag was stored under the same conditions as third bag, but with 5 minutes of agitation during warm-up (TC-5). PLTs were sampled on Days 1, 5, and 7 for 17 in vitro parameters. Data were analyzed by ANOVA with repeated measures and Bonferroni corrections. **Results/Findings:** The rate of glucose consumption, morphology score, and HSR levels were increased; Annexin V binding and intracellular hydrogen peroxide production (IHP) were decreased in TC aliquots compared to those of Control-CT. TC-5 aliquots showed lower levels of intracellular superoxide accumulation (ISOA) and greater declines of IHP and Annexin V binding levels compared to those of TC relative to Control CT. PLTs stored in TC with intermittent agitation had greater CD42b expression, aggregation response and ESC levels compared to those of Control-CT. **Conclusion:** Several in vitro PLT properties were improved relative to those continuously stored in the cold if aliquots were held in an automated cycler (12 hours at 4.5 C, 45 minutes at 37 C) Introducing a 5-minute period of agitation further lessened some decrements caused by cold storage.

Disclosure of Commercial Conflict of Interest

H. Awatefe: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Awatefe: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; S. J. Wagner: Nothing to disclose

Day 7 Samples	Glucose consumption (fmol/PLT/day)	Morphology (% discoid PLTs)	HSR (%)	ESC (%)	Aggregation (%)	CD42b (Mean fluorescence)	Annexin V (%)	IHP (Mean fluorescence)	ISOA (%)
Control-RT	0.91 ± 0.21§	57.0 ± 8.3§	66.4 ± 9.1§	21.3 ± 3.5§	97.6 ± 21.6§	863 ± 135§	14.5 ± 3.3§	22.4 ± 7.1§	31.7 ± 3.9*
Control-CT	0.31 ± 0.13	2.5 ± 1.8	15.9 ± 7.1	4.0 ± 0.6	76.5 ± 12.4	629 ± 123	26.8 ± 4.8	45.7 ± 13.9	34.5 ± 4.0
TC	0.59 ± 0.17†	12.4 ± 4.3§	30.9 ± 6.4§	5.2 ± 1.4	78.8 ± 14.3	699 ± 137	21.4 ± 8.7*	35.7 ± 13.6*	32.3 ± 4.8
TC-5	0.60 ± 0.20§	14.9 ± 4.8§	33.6 ± 7.3§	6.5 ± 1.6*	90.3 ± 21.7*	738 ± 91*	18.2 ± 5.4§	27.3 ± 8.6§	31.1 ± 5.0†

Values represent average ± 1 SD. * $p < 0.05$, † $p < 0.01$, § $p < 0.001$, compared to Control-CT values.

SP44

Storage of Mirasol-Treated Platelets in Plasma or in Additive Solution

P F van der Meer[1] (p.vandermeer@sanquin.nl), B Daal[1], I J Bontekoe[1], J W Lagerberg[1], D de Korte[1]. *[1]Product and Process Development, Sanquin Blood Bank, Amsterdam, Netherlands*

Background/Case Studies: The use of pathogen reduction technologies (PRT) increases the safety of the blood supply. This is particularly true for platelet concentrates (PCs), which are stored at room temperature and are therefore prone to bacterial outgrowth. Previous studies have shown that PRT may be associated with elevated in vitro markers of platelet storage lesion, in part due to the use of ultra violet light. The aim was to investigate the effect of the Mirasol PRT system (Terumo BCT, Lakewood, CO) on in vitro quality of platelets in plasma. It was further studied whether the use of a platelet additive solution (PAS) is able to mitigate part of the effect of PRT on platelet quality. **Study Design/Methods:** Twenty buffy coats (BCs) were pooled, split in two equal parts; to one part, 2 units of plasma were added, to the other, 2 units of PAS (SSP+, 300 mL, MacoPharma, Tourcoing, France). Each part was then split again in two equal parts (to resemble pools of 5 BCs, which is the current routine). Of each pair of plasma units, one was Mirasol-treated 1), the other served as control 2). Similarly, one SSP+ unit was Mirasol-treated 3), and the other not 4). Units were stored for 8 days with sampling on day 1, 2, 6, and 8 for determination of in vitro measures. Eight paired experiments were performed. Data were compared using a repeated measures ANOVA. **Results/Findings:** Pairing of the units was successful, with no significant differences in volume or platelet count. A number of in vitro measures on day 8 of storage are summarized in the Table. The Mirasol treatment leads to elevated lactate production for PCs in plasma, leading to lower pH values. The use of PAS mostly abrogates this effect, and the Mirasol-treated PCs in SSP+ had only slightly higher lactate production and annexin V binding as control units in plasma. However, irrespective whether plasma or SSP+ was used, Mirasol treatment led to higher platelet activation as measured by CD62P expression and lower hypotonic shock response (HSR) scores. **Conclusion:** Both for platelets in plasma or in SSP+, Mirasol treatment leads to higher platelet activation and lower HSR scores. However, if SSP+ is used as storage solution for Mirasol-treated platelets, lactate metabolism and annexin V binding is lower, showing that PAS can partly mitigate the effect of PRT. The clinical relevance of this finding has yet to be demonstrated.

Disclosure of Commercial Conflict of Interest

I. J. Bontekoe: Nothing to disclose; B. Daal: Nothing to disclose; D. de Korte: Nothing to disclose; J. W. Lagerberg: Nothing to disclose; P. F. van der Meer: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. J. Bontekoe: Nothing to disclose; B. Daal: Nothing to disclose; D. de Korte: Terumo BCT, Grants or Research Support; J. W. Lagerberg: Nothing to disclose; P. F. van der Meer: Nothing to disclose

Day 8 values (mean ± SD)

	1	2	3	4	Statistics
pH	6.59 ± 0.03	6.91 ± 0.05	6.83 ± 0.04	7.09 ± 0.06	$p < 0.001$
CD62P expression, %	42 ± 3	16 ± 2	41 ± 4	13 ± 1	$p < 0.001$, except A vs C: n.s.; B vs D: $p < 0.05$
Annexin V binding, %	39 ± 4	16 ± 2	22 ± 4	13 ± 2	$p < 0.001$, except B vs D: n.s.
HSR, %	37 ± 4	64 ± 4	23 ± 4	50 ± 6	$p < 0.001$
Lactate production*	0.21 ± 0.03	0.12 ± 0.02	0.14 ± 0.02	0.09 ± 0.01	$p < 0.001$, except B vs. C: n.s.

* in mmol/10∧11 platelets/day

SP45

Addition of Sialidase or P38 MAPK Inhibitors Does Not Ameliorate Decrements in All In Vitro Platelet Storage Properties Caused By 4 C Storage

A Skripchenko[1] (Andrey.Skripchenko@redcross.org), H Awatefe[1], D Thompson-Montgomery[1], A M Turgeon[1], S J Wagner[1]. *[1]Blood Components, American Red Cross Holland Laboratory, Rockville, MD, United States*

Background/Case Studies: Bacterial proliferation is inhibited in platelets (PLTs) stored at refrigerated temperatures, but also dramatically decreases PLT in vivo survival. Recent studies have demonstrated that cold temperature (CT) stored PLTs secrete sialidases upon re-warming, removing sialic acid from the PLT surface, which may in turn be responsible for clustering of GP1bα and subsequent PLT clearance from circulation. Addition of a sialidase inhibitor greatly improved the recovery of transfused murine PLTs stored at CT for 2-48 hours. In this study, the in vitro properties of PLTs stored at 4 C with or without sialidase or with or without a p38 MAPK inhibitor were evaluated. **Study Design/Methods:** A single Trima apheresis PLT unit (n = 12) was collected under informed consent. Immediately after collection, PLTs were aliquoted into four 60-mL CLX storage bags. One bag was stored at 20-24 C (Control-RT) with continuous agitation; a second bag was stored

at 4 C without agitation (Control-CT); a third bag was held at 4 C without agitation with 1 mmol of sialidase inhibitor (N-Acetylneuraminic Acid, 2,3 Dehydro-2-deoxy-Sodium Salt) (DANA), a fourth bag was incubated in an identical manner as second bag with addition of 1 μM VX-702, a second generation p38 MAPK inhibitor (VX-702). All PLT aliquots were stored for 7 days and assayed on Days 1, 2, 4, and 7 for 17 in vitro parameters. Data were analyzed by ANOVA with repeated measures and Bonferroni corrections. **Results/Findings:** Beginning from day 1, all in vitro PLT parameters were adversely affected by CT compared to those of Control-RT, as previously reported by other investigators (day 1 data not shown). Similar in vitro storage properties were observed in CT PLT in the presence or absence of p38 MAPK or sialidase inhibitors. **Conclusion:** Addition of either sialidase or p38 MAPK inhibitors do not improve any in vitro parameters of PLTs stored at 4 C.

Disclosure of Commercial Conflict of Interest

H. Awatefe: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Awatefe: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; S. J. Wagner: Nothing to disclose

Day 7 Samples	pH	Glucose consumption rate (fmol/PLT/day)	pO_2 (mm Hg)	Morphology (% disks)	HSR (%)	ESC (%)	Aggregation Slope (%)	Annexin V (%)	CD42b (Mean fluorescence)
Control-RT	7.14 ± 0.10	0.90 ± 0.17	146 ± 8	55.8 ± 14.1	67.3 ± 8.1	19.3 ± 3.8	100 ± 23	9.8 ± 3.5	750 ± 290
Control-CT	7.47 ± 0.08*	0.31 ± 0.10*	218 ± 10*	0.1 ± 0.3*	16.5 ± 7.1*	3.4 ± 1.2*	75 ± 23*	20.1 ± 8.2*	580 ± 174*
DANA	7.45 ± 0.08*	0.33 ± 0.12*	217 ± 10*	0.5 ± 0.9*	17.7 ± 12.2*	3.4 ± 0.8*	73 ± 27*	22.6 ± 17.0§	557 ± 158*
VX-702	7.50 ± 0.09*	0.31 ± 0.12*	220 ± 8.3*	1.4 ± 1.7*	19.2 ± 8.4*	4.4 ± 1.0*	77 ± 22*	16.8 ± 6.2	617 ± 186§

Values represent average ± 1 SD; * $p < 0.001$, § $p < 0.01$ compared to Control-RT values

SP46

Human Platelets, Stored in an Automated Temperature Cycler Between 4.5 C and 37 C for 7 Days, Exhibited Improved In Vitro Properties and In Vivo Recovery in SCID Mice Compared to Continuous Cold Storage

A Skripchenko[1] (Andrey.Skripchenko@redcross.org), J G Vostal[2], M P Gelderman[2], D Thompson-Montgomery[1], H Awatefe[1], A M Turgeon[1], S J Wagner[1]. *[1]Blood Components, American Red Cross Holland Laboratory, Rockville, MD, United States; [2]Center for Biologics Evaluation and Research, FDA, Bethesda, MD, United States*

Background/Case Studies: Platelets (PLTs) stored at cold temperature for long periods have dramatically reduced bacterial growth but poor survival when infused. Although decrements in survival are irreversible with long periods of cold storage, results from the literature suggest that decrements from shorter periods of cold storage may be reversible with warming. For example, human PLTs manually cycled between 12 hours at 4 C followed by 30 minutes of 37 C had similar or greater in vivo recovery than those of PLTs stored at room temperature and greater recovery than those of human PLTs stored continuously at 4 C when transfused into SCID mice. In this study, the in vitro and in vivo properties of PLTs stored with different modes of agitation in an automated incubator programmed to cycle for 12 hours at 4.5 C followed by 1 hour at 37 C were evaluated. **Study Design/Methods:** A single Trima apheresis PLT unit (n = 12), was aliquoted into four 60 mL samples in CLX storage bags immediately after collection. One bag was stored at 20-24 C (Control-RT) with continuous agitation (CA); a second bag was stored at 4.5 C without agitation (Control-CT); a third bag was held in an automated temperature cycler (12 hours at 4.5 C, 1 hour at 37 C) (TC) with 5 minutes of agitation during warm-up (TC-5); a fourth bag was stored under the same conditions as the third bag, but with continuous agitation (TC-CA). PLTs were sampled on days 0, 5, and 7 for 17 in vitro parameters. On day 7, PLTs (n = 5) were infused into SCID mice (15/condition) and the in vivo recovery was determined at 20 minutes after transfusion. Data were analyzed by ANOVA with repeated measures and Bonferroni corrections. **Results/Findings:** The glucose consumption rate, morphology score, HSR levels, and aggregation were increased and oxygen levels with mitochondrial hydrogen peroxide accumulation was decreased in all TC aliquots compared to those stored at 4.5 C. pH levels and Annexin V binding of all TC PLT aliquots were comparable to those of Control-RT. Different modes of agitation did not alter either in vitro or in vivo PLT properties. After 7 days of storage, all TC PLTs had better recovery than 4.5 C PLTs and comparable to RT. **Conclusion:** PLTs, if stored with temperature cycling between 5 C and 37 C, have improved in vivo recovery compared to those continuously stored at cold temperature. The findings and conclusions in this abstract have not been formally disseminated by the FDA and should not be construed to represent any Agency determination or policy.

Disclosure of Commercial Conflict of Interest

H. Awatefe: Nothing to disclose; M. P. Gelderman: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; J. G. Vostal: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Awatefe: Nothing to disclose; M. P. Gelderman: Nothing to disclose; A. Skripchenko: Nothing to disclose; D. Thompson-Montgomery: Nothing to disclose; A. M. Turgeon: Nothing to disclose; J. G. Vostal: Nothing to disclose; S. J. Wagner: Nothing to disclose

Day 7 Samples	Glucose consumption (fmol/PLT/day)	Morphology (% discoid PLTs)	HSR (%)	pO_2 (mm Hg)	H_2O_2 (Mean fluorescence)	Aggregation (%)	pH	Annexin V (%)	PLT recovery (%)
Control-RT	0.91 ± 0.16§	59.9 ± 7.5§	63.0 ± 13.9§	152 ± 10§	35.9 ± 7.3*	108 ± 15§	7.2 ± 0.1§	10.2 ± 3.9†	90.2 ± 22.3§
Control-CT	0.31 ± 0.06	3.6 ± 1.2	16.5 ± 6.2	217 ± 7	68.5 ± 38.2	91 ± 12	7.4 ± 0.1	16.7 ± 6.3	49.6 ± 19.0
TC-5	0.63 ± 0.09§	16.3 ± 2.8§	34.3 ± 7.1†	176 ± 22§	42.6 ± 9.6	104 ± 12*	7.2 ± 0.1§	13.7 ± 9.2	82.9 ± 30.0§
TC-CA	0.62 ± 0.08§	20.8 ± 2.8§	36.2 ± 7.9§	181 ± 25§	34.8 ± 5.0*	102 ± 8*	7.3 ± 0.1§	14.7 ± 7.5	84.3 ± 22.1§

Values represent average ± 1 SD. * $p < 0.05$, † $p < 0.01$, § $p < 0.001$, compared to Control-CT values

SP47

Apheresis Platelets Collected from Fasting and Non-Fasting Donors Produce Components of Comparable Quality

S J Wagner[1] (Steven.Wagner2@redcross.org), S Seetharaman[1], J C Bolanos[1], J Kurtz[1]. *[1]Blood Components Dept., American Red Cross, Rockville, MD, United States*

Background/Case Studies: Some non-US blood centers collect apheresis platelets from fasting donors under the belief that these collection requirements lead to improved product quality. Evidence supporting this practice is lacking. A study with a cross-over design was conducted assessing the in vitro properties of apheresis platelets during 7 day storage collected from fasting and non-fasting donors. **Study Design/Methods:** Single units of apheresis platelets (Trima, Terumo BCT) were collected under informed consent from normal healthy donors (n = 12). Donors were randomized to a fasting and non-fasting donation group prior to their first collection. On the evening preceding donation, fasting donors were not permitted to eat any food or drink any beverage other than water from 8 PM until donation was complete the next morning. There were no food or beverage restrictions for the non-fasting group prior to donation. Donors who fasted prior to their first donation did not fast prior to their second donation; donors who had no food or beverage restrictions prior to their first donation fasted prior to their second donation. Units collected from fasting and non-fasting donors were stored at 20-24 C with continuous agitation and assayed for 15 in vitro platelet quality measures on days 1, 5, and 7. Assay results were compared by paired t-test, with a $p < 0.003$ considered significant due to the multiple

measures. **Results/Findings:** On day 1, platelets collected from fasting donors compared to non-fasting donors contained similar ($p > 0.003$) counts (1380 ± 164 versus 1368 ± 161 PLT/μL), pH (7.30 ± 0.05 versus 7.26 ± 0.04), morphology (78.4 ± 8.2 versus 78.4 ± 6.9 % disc.), aggregation amplitude (98.0 ± 10.0 versus 98.8 ± 8.0%), extent of shape change (ESC) (29.3 ± 4.7 versus 30.1 ± 3.9%) and CD62+ PLT (12.9 ± 18.0 versus 13.2 ± 17.1%) as well as all other measured in vitro properties. Comparable results ($p > 0.003$) between fasting and non fasting donor in vitro properties were also observed on day 7 (selected assays provided in table). **Conclusion:** Using a cross-over study design, donors providing apheresis platelets under fasting and non-fasting conditions yielded components of comparable in vitro quality during 7 days of storage.

Disclosure of Commercial Conflict of Interest

J. C. Bolanos: Nothing to disclose; J. Kurtz: Nothing to disclose; S. Seetharaman: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. C. Bolanos: Nothing to disclose; J. Kurtz: Nothing to disclose; S. Seetharaman: Nothing to disclose; S. J. Wagner: Nothing to disclose

Day 7 in vitro parameters (n = 12)

	Count (PLT/μL)	pH	Morphology (% disc.)	Aggregation (%)	ESC (%)	CD62 + PLT (%)
Fasting	1343 ± 128	7.17 ± 0.09	63.2 ± 13.0	90.1 ± 14.7	22.1 ± 5.1	22.2 ± 15.5
Non-fasting	1344 ± 138	7.14 ± 0.09	60.9 ± 11.1	89.0 ± 8.5	22.2 ± 4.6	22.2 ± 15.8

Values given in mean ± 1 SD

SP48

Lyophilized Plasma and Fresh Frozen Plasma are Equally Effective in Correcting Impaired Plasmatic Coagulation

U J Sachs[1,2] (ulrich.sachs@med.uni-giessen.de), R Marschall[2], G Dinges[3].
[1]Institute for Clinical Immunology and Transfusion Medicine, Justus Liebig University, Giessen, Germany; [2]Center for Transfusion Medicine and Hemotherapy, University Hospital Giessen and Marburg, Marburg, Germany; [3]Department of Anesthesiology and Intensive Care, University Hospital Giessen and Marburg, Marburg, Germany

Background/Case Studies: In massive blood loss, transfusion of blood components remains the mainstay of treatment, with some recent studies advocating an early and more aggressive use of plasma. In contrast to fresh frozen plasma (FFP), lyophilised (freeze-dried) plasma (LP) can be stored at room temperature and reconstitution for transfusion is faster than thawing FFP. However, only few data regarding the efficacy of LP are available in literature. **Study Design/Methods:** Citrated blood samples from 150 consecutive patients with prolonged prothrombin time (TP > 16.4 sec = INR > 1.5) were stored at –70 C until use. After thawing in a water bath at 37 C, samples were split and mixed in a 4:1 ratio with freshly thawed FFP or reconstituted LP, mimicking a transfusion of ~20 ml/kg body weight. Aliquots from each FFP (n = 50) and LP (n = 50) were used for three random patient samples. All samples were assessed for INR, protein S, and coagulation factor (F) V and VII activity, and mixed samples were also assessed for fibrinogen concentration using standard assays on a BCS-XP device (Siemens Health Care Diagnostics, Eschborn, Germany). Statistics were performed by Mann Whitney U test. **Results/Findings:** The following mean values were obtained in the patient samples: INR 2.7 (range, 1.5 to 5.9); protein S 50% (range, 2.7 to 122.8); FV activity 87.7% (range, 51 to 143); and FVII activity 33% (range, 7 to 78). After addition of FFP, the mean values were INR 1.6 (1.3 to 2.3), fibrinogen 3.8 g/L (1.4 to 8.1), protein S 60% (24 to 107), FV 91% (62 to147), and FVII 49% (24 to 83). After addition of LP, the mean values were INR 1.5 (1.2 to 2.1), fibrinogen 3.8 g/L (1.8 to 8.1), protein S 62% (30 to 113), FV 92% (69 to 139), and FVII 53% (23 to 87). There were no statistical differences for INR ($p = 0.35$), fibrinogen ($p = 0.92$), protein S ($p = 0.09$), FV ($p = 0.46$), or FVII ($p = 0.22$) between the two groups. **Conclusion:** This study demonstrates that lyophilised plasma is comparable to fresh frozen plasma in correcting impaired plasmatic coagulation in vitro. Since lyophilised plasma can be stored at room temperature and can be reconstituted rapidly, it is an attractive alternative to FFP, especially in emergency settings. In vivo studies are required to prove whether this is advantageous for our patients.

Disclosure of Commercial Conflict of Interest

G. Dinges: Nothing to disclose; R. Marschall: Nothing to disclose; U. J. Sachs: German Red Cross Blood Service West, Hagen, Germany, Grants or Research Support

Disclosure of Grants Conflict of Interest

G. Dinges: Nothing to disclose; R. Marschall: Nothing to disclose; U. J. Sachs: German Red Cross Blood Service West, Hagen, Germany, Grants or Research Support, Octapharma AG, Lachen, Switzerland, Consulting or Board of Director Fees

SP49

Visualization of Fluid Motion and Quantitation of Shear Stress in Platelet Storage Bags Subjected to Different Modes of Agitation

R Torres[1,2] (richard.torres@yale.edu), C A Tormey[1,2], G Stack[1,2].
[1]Pathology and Laboratory Medicine, VA Connecticut Healthcare, West Haven, CT, United States; [2]Laboratory Medicine, Yale University School of Medicine, New Haven, CT, United States

Background/Case Studies: Continuous agitation of blood bank-stored platelets, while necessary for adequate gas-exchange, subjects platelets to potentially deleterious shear stresses. Such shear stresses might contribute to platelet activation and granule release during storage, ultimately yielding a suboptimal product for transfusion. The degree to which various modes of agitation, eg flat-bed and rotational, contribute to shear stress and associated deterioration in platelet function during storage remains largely unknown. Earlier comparative analyses of modes of agitation, mostly performed more than 25 years ago, did not specifically examine the role of shear stress in the platelet storage lesion. Therefore, the goals of this study were to directly visualize fluid motion and determine shear stress generated by various geometries of platelet agitation. **Study Design/Methods:** Fluid motion was investigated in flatbed (72 cycles/minute), rotating circular (3 and 6 RPM), and tumbler (3 RPM) agitators. Density-matched, 1 mm polyethylene beads were suspended in 50 mL of water inside random-donor, CLX platelet storage bags. A video camera recorded bead motion during agitation, corresponding to fluid motion. Trajectory tracings of particles were obtained by analysis of recordings using particle tracking algorithms, from which fluid velocities were calculated. Estimations of relative wall shear stress and shear rates were made using the standard formula for shear stress in fluids and the calculated velocities. **Results/Findings:** Marked differences in particle motion and velocities were observed with different modes of agitation. As expected, fluid flow in rotating circular agitators followed a circular path, whereas both flatbed and tumbler agitation showed predominantly to-and-fro motion, parallel to the long axis of the bag. Remarkably, flatbed agitation resulted in average fluid velocities more than 25 times those observed with 3 RPM tumbler agitation and approximately 2.5 times those of 6 RPM rotating circular movement. Furthermore, calculated wall shear stress was highest with flatbed agitation, approximately 3- and 7-fold higher than rotating circular and tumbler motion, respectively. **Conclusion:** Despite being a popular form of platelet storage, flatbed platelet agitation induces significantly higher shear stresses than tumbler and rotating circular agitation. The findings indicate that tumbler agitation, which generated the least shear stress of all modalities, deserves re-examination. The precise effect of shear stress on the deterioration of platelet function during storage also deserves further study.

Disclosure of Commercial Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Nothing to disclose; R. Torres: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Terumo, Inc.,Grants or Research Support; R. Torres: Nothing to disclose

SP50

A Quantitative Proteomic Approach in Platelet Concentrates Led to the Identification of mRNAs Affected by the Riboflavin/UV Light Treatment

P Schubert[1] (peter.schubert@blood.ca), C Klein-Bosgoed[1], B Culibrk[1], S Karwal[1], R P Goodrich[2], D Devine[1]. *[1]Centre for Blood Research, Canadian Blood Services, Vancouver, BC, Canada; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Pathogen reduction technologies (PRT) were introduced into blood banking to further improve the safety of blood products. Although very effective on a variety of bacteria and viruses, PRT treatment accelerates the development of the platelet storage lesion. Based on the application of an in-depth proteomics approach to determine changes in the proteome of platelet concentrates subjected to riboflavin/UV treatment, mRNA levels of selected proteins changing in a proteomics approach, were analyzed. **Study Design/Methods:** In a two-arm pool-and-split study, apheresis platelet concentrates were either treated with riboflavin/UV (Mirasol technology) or kept untreated as control. Samples were drawn on day 0 (day of donation and treatment), days 1, 4, and 6 of storage and lysed for the proteomic analyses using the iTRAQ technology. Samples from the untreated arm on day 0 as well as from the treated arm on days 0, 1, and 6 were individually digested with trypsin and labeled with different isotope-coded covalent tags. The peptide mixture of all four samples was separated by SCX and analyzed by LC-MS/MS. Proteomic findings were verified by immune blot analyses. The RNA was extracted using the TRIzol method and total RNA and reverse Transcription Polymerase Chain Reaction (RT-PCR) were performed using specific primers to the target protein visualizing the product on agarose gels. **Results/Findings:** The proteomic results of three independent experiments showed a 93% overlap in protein identification and 87% agreement in the change of protein expression pattern (157 proteins) during storage both in the treated and the untreated study arm. Generally, individual changes in protein expression were small; however, thrombospondin (TSP) showed a 5-fold increase in expression throughout storage of untreated platelets, whereas a 4-fold decrease was observed in the PRT study arm. TSP level in the releasate was the same in both study arms. TSP is synthesized in platelets; its mRNA level was markedly reduced after PRT treatment but decayed slower in the control study arm. Other proteins, such as the GTPase Rac1 and actin, exhibited a decrease in protein expression only 7 days after PRT treatment with a concomitant reduction in mRNA levels. However, glyceraldehyde 3-phosphate dehydrogenase (GAPDH) was unaltered at both the protein and mRNA levels in both arms throughout storage. **Conclusion:** This quantitative proteomic approach revealed only subtle changes in the platelet proteome upon riboflavin/UV light treatment and subsequent storage confirming proteomic studies of other PRTs. Decreased protein levels throughout storage upon the PRT treatment are likely due to reduced mRNA levels of the individual protein at least in some cases. The identification of mRNAs affected by PRT treatment is necessary to further the understanding the impact of PRT on the functionality of platelets.

Disclosure of Commercial Conflict of Interest

B. Culibrk: Terumo BCT, Grants or Research Support; D. Devine: Terumo BCT, Grants or Research Support; R. P. Goodrich: Terumo BCT, Other; S. Karwal: Nothing to disclose; C. Klein-Bosgoed: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

B. Culibrk: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support, Fresenius-Kabi, Grants or Research Support, MacoPharma, Grants or Research Support; R. P. Goodrich: Terumo BCT, Other; S. Karwal: Nothing to disclose; C. Klein-Bosgoed: Nothing to disclose; P. Schubert: Terumo BCT, Grants or Research Support

SP51

Liquid Plasma can be a Suitable Alternative to Fresh Frozen Plasma in Management of Thrombotic Microangiopathies

S R Hui[1,2] (sxhui@texaschildrens.org), K T Nguyen[2], N Turner[3], V Kostousov[2], S K Hartman[1,2], K Eldin[1,2], B A Sapinsky[2], J Moake[3,4], J Teruya[1,2]. *[1]Pathology, Baylor College of Medicine, Houston, TX, United States; [2]Pathology, Texas Children's Hospital, Houston, TX, United States; [3]J.W. Cox Laboratory for Biomedical Engineering, Rice University, Houston, TX, United States; [4]Medicine, Baylor College of Medicine, Houston, TX, United States*

Background/Case Studies: Liquid plasma (LP) is used mainly for the purpose of emergency resuscitation during trauma or massive transfusion settings in the US, as fresh frozen plasma (FFP) requires additional time to prepare and thawed plasma (TP) may lead to increased wastage. It has recently been shown that LP may provide a superior hemostatic profile than TP, therefore, LP may become a more commonly available blood product in the future. To date, there has been no study to determine whether LP can be used in situations other than replacement therapy for coagulation factors. The aim of this study is to determine if LP contains and sustains adequate levels of ADAMTS13 (AT13) and Factor H (FH) during storage and therefore can possibly be used in the settings of thrombotic thrombocytopenic purpura (TTP) and atypical HUS (aHUS) as replacement products for simple transfusion or plasma exchange. **Study Design/Methods:** Six single donor LP units in CPD from whole blood collection (3 group O and 3 group A) were obtained from the Gulf Coast Regional Blood Center (Houston, Texas) and kept refrigerated at 1-6 C for 26 days until expiration. Five milliliter aliquots from each unit were collected on day 0-5, day 14, and day 26 and analyzed for PT/INR and PTT, functional levels of fibrinogen, Factors V and VIII, AT13, and antigenic level of FH. Data was analyzed using paired t-test and presented as mean ± SD with $p < 0.05$. **Results/Findings:** There were no significant differences in either FH or AT13 levels in LP during storage and these levels remained above normal limits (normal range for FH 242-759 μg/mL and AT13 >65%). In addition, PT, PTT, fibrinogen, and Factor VIII were also stable from day 0 to day 26. Only Factor V activity significantly decreased during storage. **Conclusion:** AT13, FH and the basic coagulation profile declined modestly, but not significantly in LP over 26 days at 1-6 C. These data demonstrated that LP can be considered as replacement product for transfusion or plasma exchange in TTP and aHUS. Utilizing LP may help reduce the wastage of plasma products and shorten turn-around-time in these clinical settings. Lastly, the no-thaw requirement and longer shelf-life of LP may increase availability of plasma products for TTP and aHUS patients in small community and rural hospitals.

Disclosure of Commercial Conflict of Interest

K. Eldin: Nothing to disclose; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose; V. Kostousov: Nothing to disclose; J. Moake: Nothing to disclose; K. T. Nguyen: Nothing to disclose; B. A. Sapinsky: Nothing to disclose; J. Teruya: No Answer; N. Turner: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Eldin: Nothing to disclose; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose; V. Kostousov: Nothing to disclose; J. Moake: Nothing to disclose; K. T. Nguyen: Nothing to disclose; B. A. Sapinsky: Nothing to disclose; J. Teruya: No Answer; N. Turner: Nothing to disclose

Laboratory test	Day 0-5	Day 14	Day 26
PT, sec (INR)	13.9 ± 0.7 (1.07 ± 0.07)	14.2 ± 1.5 (1.1 ± 0.1)	15.4 ± 2.4 (1.23 ± 0.3)
PTT, sec	33.4 ± 3.4	34 ± 3.9	34.7 ± 4.7
Fibrinogen, mg/dL	283 ± 31	262 ± 47	242 ± 96
FV, %	88 ± 18	71 ± 15*	53 ± 18**
FVIII, %	65 ± 22	62 ± 31	52 ± 26
ADAMTS13, %	160 ± 30	122 ± 47	99 ± 13
Factor H, μg/mL	457 ± 78	409 ± 64	371 ± 71

* $p = 0.006$ vs Day 0 ** $p = 0.001$ vs Day 14

SP52

Blood Bag Plasticizers Influence Microvesicle Size and Microvesiculation Rate in Red Blood Cell Concentrates

B Bicalho[1,2] (beatrizbicalho@gmail.com), J D Kurach[1], T R Turner[1], K Serrano[3,4], D Devine[3,4], J Acker[1,2]. *[1]Research and Development, Canadian Blood Services, Edmonton, AB, Canada; [2]Laboratory Medicine and Pathology, University of Alberta, Edmonton, AB, Canada; [3]Research and Development, Canadian Blood Services, Vancouver, BC, Canada; [4]Centre for Blood Research, University of British Columbia, Vancouver, BC, Canada*

Background/Case Studies: Diethylhexylphthalate (DEHP) is a plasticizer used in polyvinylchloride (PVC) blood collection packs that can improve red blood cell (RBC) stability and reduce RBC microvesiculation. DEHP and respective metabolites are potentially toxic; so the substitution of DEHP is of interest. This study examined the role that two DEHP-substitutes (diisononylestercyclohexane, DINCH; butyryltrihexylcitrate, CITRATE) have on the formation of RBC microvesiculation during hypothermic storage (HS). **Study Design/Methods:** A total of 15 pools of 2 RBC units were prepared. Each pool was divided into six pediatric bags: 2 PVC/DEHP; 2 PVC/DINCH; and 2 PVC/CITRATE. Units were stored at 1-6 C and tested on days 5, 21, 35, and 43. Flow cytometry (FC) and dynamic light scattering were used to

evaluate microvesiculation. **Results/Findings:** Supernatant Hb was higher ($p < 0.01$) in all the study units on day 21. However, in CITRATE units, the concentration of supernatant Hb was 2-fold higher ($p < 0.001$) than in DINCH or DEHP. This corresponded with a higher accumulation of microvesicles in CITRATE stored RBCs ($p = 0.05$). As HS progressed, CITRATE units developed the strongest correlation between supernatant Hb and microvesicles/μL ($R2 = 0.9$) when compared with DINCH or DEHP ($R2 = 0.8$). In CITRATE units, microvesicles increased by 3-fold at expiry, whereas in DINCH or in DEHP units, such increase was 0.5-fold ($p < 0.001$). Interesting, dynamic light scattering showed that DEHP associated microvesicles were smaller (45 ± 10 nm) in DINCH (80 ± 20 nm) or CITRATE (105 ± 25 nm) units on d5 ($p < 0.001$). At expiry, however, microvesicles were larger (100-200 nm), and DEHP-associated microvesicles had the highest (3-fold) increase in size. It is speculated that the observed changes in microvesicle size may indicate the relative incorporation of the plasticizer in the cell membrane. At expiry, % hemolysis in CITRATE units ($0.85 \pm 0.1\%$) was 2-fold higher than in DINCH or DEHP ($p < 0.001$). **Conclusion:** RBCs stored in DINCH and DEHP PVC containers are not different in terms of supernatant Hb, hemolysis and microvesicle quantity. CITRATE impaired RBC quality by favoring microvesiculation. Understanding the role that DEHP-substitutes have in stabilizing, or destabilizing the RBC membrane is an important factor in determining the safety and clinical efficacy of emerging RBC containers.

Disclosure of Commercial Conflict of Interest

J. Acker: Fresenius, Grants or Research Support; B. Bicalho: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support; J. D. Kurach: Fresenius, Grants or Research Support; K. Serrano: Nothing to disclose; T. R. Turner: Fresenius, Grants or Research Support

Disclosure of Grants Conflict of Interest

J. Acker: Aquila Diagnostic Systems Inc.,Ownership or Partnership, HemoCue, Travel Support or Honorarium, Fresenius, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support; B. Bicalho: Nothing to disclose; D. Devine: Terumo BCT, Grants or Research Support, Fresenius, Grants or Research Support, MacoPharma, Grants or Research Support; J. D. Kurach: HemoCue, Grants or Research Support, Haemonetics, Grants or Research Support, Fenwal, Grants or Research Support; K. Serrano: Fresenius, Grants or Research Support; T. R. Turner: HemoCue, Grants or Research Support, Haemonetics, Grants or Research Support, Fenwal Grants or Research Support

SP53

In Vitro Evaluation of a New In-Line Red Blood Cell (RBC) Leukoreduction Filter

M Einarson[1] (Mindy.Einarson@redcross.org), L Kline[2], J Gottschall[3], S Graminske[4], J G Nixon[1], D Ziegler[1], G George[2], M Dabay[2], S Heldke[4], J Skinner[5]. *[1]Research Department, American Red Cross, Penn-Jersey Region, Philadelphia, PA, United States; [2]Blood Components Department, American Red Cross Holland Laboratory, Rockville, MD, United States; [3]Medical Sciences Institute, BloodCenter of Wisconsin, Milwaukee, WI, United States; [4]Applied Research Laboratory, BloodCenter of Wisconsin, Milwaukee, WI, United States; [5]Regulatory Affairs, MacoPharma, Duluth, GA, United States*

Background/Case Studies: Red blood cells (RBC) are routinely leukoreduced before transfusion in order to minimize febrile nonhemolytic transfusion reactions, prevent HLA alloimmunization and reduce transmission of viruses such as cytomegalo virus (CMV). The objective of this study was to evaluate a new in-line RBC leukoreduction filter, the Leucoflex® CGP Leukocyte Reduction Filter System (MacoPharma USA) and compare it to an FDA-cleared product, the Leukotrap® RC System which has an in-line RBC (RC2D) filter (Pall Medical). The evaluation was sponsored by MacoPharma and performed at three sites. **Study Design/Methods:** Study subjects donated 500 ± 50 mL whole blood (WB) collected in CPD using the CGP system (Test) or in CP2D using the RC2D system (Control). WB was stored at 1-6C for 20-24 hours followed by separation of RBC using a hard spin at 1-6 C. Storage solution (AS-1 for Test, AS-3 for Control) was added to the RBC through the CGP or RC2D filter and the RBC were filtered at 1-6 C. Samples for in vitro tests were obtained from additive RBCs on days 1 (pre- and post-filtration), 28/29, 35/36, and 42/43 of storage. **Results/Findings:** A total of 153 evaluable units were collected (93 Test, 60 Control). Post-filtration RBC Recovery, calculated using total Hgb/unit, was ≥85% for all Test units compared to ≥85% for 54 of 60 Control units; 5 Control units had 84% and 1 had 83% Recovery. All filtered Test and Control units contained $<5 \times 10^6$ WBC/unit and had <1% hemolysis on Day 42/43. Statistical (t-test) comparisons showed no significant difference between Test and Control for WBC ($p = 0.14$), or Day 42/43 % Hemolysis ($p = 0.27$), Hgb ($p = 0.35$) or Plasma Hgb ($p = 0.52$); RBC Recovery and Day 42/43 Hct were significantly different ($P < 0.0001$). In vitro chemistry results (Table), are comparable given that the Test and Control RBC were stored in different additive solutions; statistical comparisons were not performed for these parameters. **Conclusion:** Both filters produced RBC units with residual WBC well below 5×10^6 WBC/unit and end of storage hemolysis below 1%. Post-filtration RBC Recovery was ≥85% in 93 of 93 Test units and 54 of 60 Control units. The RBC product produced by the new CGP filter set exceeds current AABB and FDA standards and is similar to the RBC product produced with the FDA approved Control set.

Disclosure of Commercial Conflict of Interest

M. Dabay: MacoPharma,Other; M. Einarson: MacoPharma, Grants or Research Support, OriGen Biomedical, Grants or Research Support; G. George: Becton Dickinson, Other, NanoEntek, Other; J. Gottschall: Cerus, Other, Haemonetics, Other, Fenwal, Other, MacoPharma, Other; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; Masimo, Grants or Research Support; S. Heldke: MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Masimo, Grants or Research Support, Cerus, Grants or Research Support, GenesisBPS, Grants or Research Support; L. Kline: MacoPharma, Grants or Research Support, Origen, Grants or Research Support, Masimo, Grants or Research Support; J. G. Nixon: MacoPharma, Grants or Research Support, Origen, Grants or Research Support; J. Skinner: Nothing to disclose; D. Ziegler: MacoPharma, Grants or Research Support, Origen, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. Dabay: Nothing to disclose; M. Einarson: Fenwal, Grants or Research Support; G. George: Nothing to disclose; J. Gottschall: Nothing to disclose; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus Corp., Grants or Research Support; S. Heldke: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus Corp., Grants or Research Support; L. Kline: Nothing to disclose; J. G. Nixon: Fenwal, Grants or Research Support; J. Skinner: Nothing to disclose; D. Ziegler: Fenwal, Grants or Research Support

Parameter (Mean ± SD)	Test (CGP) N = 93		Control (RC2D) N = 60	
	Day 1	Day 42/43	Day 1	Day 42/43
WBC (x10^6/unit)	0.18 ± 0.24	—	0.14 ± 0.14	—
Post-filter RBC recovery (%)	90.7 ± 0.02	—	87.0 ± 0.02	—
Hemolysis, end of storage (%)	—	0.21 ± 0.12	—	0.18 ± 0.13
ATP Day 42/43 (% of Day 1)	—	79.7 ± 10.4	—	*87.2 ± 11.9
Hemoglobin (g/dL)	18.8 ± 1.0	18.8 ± 1.0	18.7 ± 0.8	18.7 ± 0.8
Hematocrit (%)	56.2 ± 2.0	60.1 ± 2.5	56.1 ± 1.7	57.4 ± 2.2
Plasma Hemoglobin (mg/dL)	14.0 ± 18.0	90.6 ± 55.3	43.2 ± 42.3	84.5 ± 59.1
pH (37°C)	6.95 ± 0.05	6.43 ± 0.05	6.74 ± 0.04	6.41 ± 0.05
Glucose (mg/dL)	1002 ± 76	678 ± 61	747 ± 57	488 ± 51
Lactate (mM/L)	2.7 ± 0.7	29.6 ± 3.1	2.3 ± 0.5	25.5 ± 3.4
Potassium (mM/L)	2.3 ± 1.0	46.7 ± 3.9	2.2 ± 0.8	48.8 ± 4.1

* N = 59

SP54

Evaluation of Transport Time and Temperature on the Detection of Bacteria Using the eBDS Bacterial Detection System

M Dabay[1] (michelle.dabay@redcross.org), L Kline[1], T E Keegan[2], C A Hapip[1], N S Kaelber[1]. *[1]Holland Laboratory, American Red Cross, Rockville, MD, United States; [2]Biomedical Headquarters, American Red Cross, Washington, DC, United States*

Background/Case Studies: Centralization of bacterial testing for quality control (QC) of blood products would be desirable. The Haemonetics eBDS is the only approved bacterial detection technology for sampling of platelet product (PLT) into eBDS Sample Sets (SS) and transport of SS to a remote facility. This study was performed to ensure results are not affected by

transportation time and/or temperature. **Study Design/Methods:** Four bacteria reported in the package insert and associated with bacterial contamination of PLT were evaluated: *Bacillus cereus*, *Staphylococus epidermidis*, *Escherichia coli* and *Streptococcus agalactiae*. One leukoreduced apheresis PLT, for each bacterial organism, was collected and split between three bags. One bag was spiked with a bacterial suspension of 1-10 and one with 11-100 CFU/mL; one bag with PBS (Control). PLT samples from each bag were added to SS (three replicate SS/condition). The impact of transport time and temperature on O_2 level was evaluated by placing the SS immediately in the 35 C incubator (No Hold, N = 6) with agitation (per manufacturer's instructions), at 15-17 C, or at 23-25 C, for 9-11 hours (Hold, N = 12, 6/temp). A Pass/Fail result and O_2 level of the SS was measured after either 18 + 2 hours or 24 + 2 hours incubation at 35 C using the eBDS Oxygen Analyzer. Results were verified by plating a sample from each SS onto sheep blood agar (SBA) and checking for growth after 18-72 hours at 35 C incubation. **Results/Findings:** All SS inoculated with either the 1-10 or 11-100 CFU/mL of each of the four bacteria organisms were detected as Fail. All Control SS were detected as Pass. Pass and Fail results were consistent with O_2 levels measured by the analyzer, with lower O_2 levels measured in SS with bacteria compared to SS from the Controls in which much higher O_2 levels were detected, regardless of condition. SS held at either 16 C or 24 C for 10 hours had comparable O_2 measurements to SS incubated immediately at 35 C. There was also no difference in whether the SS were read at 18 or 24 hours for Control, 1-10 and 11-100 CFU/mL levels. Results obtained on SBA plates were consistent with eBDS analyzer. **Conclusion:** The eBDS accurately detected the presence of the four aerobic bacteria tested in this study and evaluated in SS with leukoreduced PLT held for 9-11 hr at 16 C and 24 C followed by incubation at 35 C for 18 to 24 hours. No false negatives in inoculated SS or false positives in Control samples were detected. Holding of SS for 9-11 hours at 16-24 C did not affect the detectability of bacterially contaminated PLT samples using eBDS, which would allow for the shipment of SS to a remote testing facility.

Disclosure of Commercial Conflict of Interest

M. Dabay: Haemonetics, Other; C. A. Hapip: Nothing to disclose; N. S. Kaelber: Nothing to disclose; T. E. Keegan: Haemonetics, Other, NanoEnTek, Other; L. Kline: MacoPharma, Grants or Research Support, Origen, Grants or Research Support, Masimo, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. Dabay: Nothing to disclose; C. A. Hapip: Nothing to disclose; N. S. Kaelber: Nothing to disclose; T. E. Keegan: Nothing to disclose; L. Kline: Nothing to disclose

Mean $\%O_2$ Range (Pass/Fail)

	Control	1-10 CFU/mL	11-100 CFU/mL
E. coli No Hold Hold[1]	18.04-18.83 (Pass) 17.28-19.76 (Pass)	0.07-0.47 (Fail) 0.00-0.49 (Fail)	0.06-0.50 (Fail) 0.02-0.55 (Fail)
S. epidermidis No Hold Hold[1]	18.02-18.60 (Pass) 17.91-19.69 (Pass)	0.85-4.07 (Fail) 0.65-3.31 (Fail)	0.66-1.94 (Fail) 0.72-1.69 (fail)
B. cereus No Hold Hold[1]	17.23-18.37 (Pass) 18.27-19.66 (Pass)	0.68-0.83 (Fail) 0.59-0.84 (Fail)	0.72-0.75 (Fail) 0.68-0.75 (Fail)
S. agalactiae No Hold Hold[1]	17.66-18.87 (Pass) 17.51-19.45 (Pass)	4.19-6.80 (Fail) 3.90-7.55 (Fail)	3.75-6.23 (Fail) 4.71-7.32 (Fail)

[1] 16°C + 24°C, 10 Hr

SP55

Bacterial Growth Inhibition by Cord Blood (CB) Constituents

M Cayer[1] (Marie-Pierre.Cayer@hema-quebec.qc.ca), J Pichette[1], D Fournier[2], G Delage[2], M Chevrier[2], L Thibault[1]. *[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada*

Background/Case Studies: Currently, there is no standard practice to perform sterility testing of cord blood (CB) samples. Some CB banks sampled the CB plasma. Others recover a portion of the residual volume remaining in the transfer bag or collect a small aliquot from the final product. Preliminary experiments have demonstrated that plasma extracted from several CB could inhibit the growth of several bacterial strains. In this work, we have studied the antimicrobial properties of plasma extracted from CB units. **Study Design/Methods:** The antimicrobial activity of plasma from 60 CB units and 20 adult whole blood samples (controls) was analyzed using agar disk diffusion method. Ten bacteria, known as frequent CB contaminants, were grown on blood agar plates on which 13 mm disks were placed. Thirty μL of plasma were next deposited onto disks. Zones of inhibition were measured after an overnight incubation at 37 C and compared to those of negative PBS controls. Plasma showing an antibacterial activity were heated to 56 C for 30 minutes to inhibit complement proteins or treated with a β-lactamase to abolish β-lactam antibiotic activity. Antibiotic susceptibilities (penicillin G, erythromycin, cefazolin, clindamycin and vancomycin) of all bacteria were determined by antibiograms. Zones of inhibition created by plasma were compared to antibiogram results. **Results/Findings:** Out of the 60 CB samples, 33% (20/60) showed a strong antimicrobial activity with at least one bacterium while none of the adult plasma inhibited bacteria growth. Only Gram-positive bacteria were inhibited by CB plasma samples. *M. luteus* was the most sensitive bacteria to CB plasma with an inhibition ring of up to 17 mm of radius. The inhibitory activity remained after heating at 56 C, indicating that complement proteins are not involved. However, treatment of inhibitory plasma with the β-lactamase completely abolished the inhibitory effect of 65% of these CB samples, confirming the presence of β-lactam antibiotics in those CB units. Zones of inhibition of plasma spiked with known amounts of penicillin G were compared to those of inhibitory CB samples with *S. aureus, S. epidermidis and M. luteus*. The concentration of Penicillin G in these samples was therefore estimated between 1 and 10 μg/mL even though CB units have been held for up to 48 hours at ambient temperature. **Conclusion:** In obstetric practice, antibiotics are commonly used to prevent the risk of contamination during childbirth. This study shows that antibiotics are present in CB units at concentrations that can inhibit the growth of several bacterial species. Additional work is required to determine their impact on methods of sample preparation for CB contamination testing.

Disclosure of Commercial Conflict of Interest

M. Cayer: Nothing to disclose; M. Chevrier: Nothing to disclose; G. Delage: Nothing to disclose; D. Fournier: Nothing to disclose; J. Pichette: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Cayer: Nothing to disclose; M. Chevrier: Nothing to disclose; G. Delage: Nothing to disclose; D. Fournier: Nothing to disclose; J. Pichette: Nothing to disclose; L. Thibault: Nothing to disclose

SP56

Reversal of Lesions Associated with 24-Hour Suboptimal Platelet Storage at 16 C by a p38MAPK Inhibitor, VX-702

S J Wagner[1] (Steven.Wagner2@redcross.org), S Seetharaman[1], J Kurtz[1], A Skripchenko[1]. *[1]Blood Components Department American Red Cross, Rockville, MD, United States*

Background/Case Studies: Infrequently in the winter months, platelets (PLT) are erroneously stored during transport in suboptimal conditions and must be discarded. Previous studies have demonstrated that many PLT in vitro storage properties are improved by addition of a second generation p38MAPK inhibitor, VX-702. A study was conducted to determine if PLT lesions associated with 24-hour storage at 16 C without agitation could be alleviated by VX-702. **Study Design/Methods:** Apheresis PLT (Trima, Terumo BCT) were collected under informed consent with a targeted yield of 4×10^{11} (n = 12). On day 0, 60 mL aliquots were transferred to 3 CLX containers: 1) a control that was held at room temperature (RT) with constant agitation; 2) a test unit, held at RT with agitation until day 2, when it was incubated at 16 C for 24hours without agitation and then returned to RT storage with agitation; 3) a test unit containing 1 μM VX-702 stored in an identical fashion as aliquot 2. Aliquots were stored for 7 days and assayed for 11 in vitro parameters. Measured values from assays of aliquots 2 and 3 were compared by two-tailed paired t-test, with p values < 0.004 considered significant because of the 11 comparisons. **Results/Findings:** Day 1 aliquots had a mean platelet count of 1330×10^3 PLT/μL. Storage at 16°C without agitation for 24 hours resulted in numerous changes in PLT metabolism, structure, activation and signs of apoptosis measured on day 7 (Table). Addition of VX-702 resulted in significantly improved measures in all assays except hydrogen peroxide accumulation, whose result was not significant ($p = 0.017$). **Conclusion:** Addition of the second generation p38MAPK inhibitor, VX-702 to PLT prevents most of the in vitro decrements observed on day 7 of PLT subjected to suboptimal storage conditions on days 2-3 (24-hour incubation at 16 C for 24hours).

Disclosure of Commercial Conflict of Interest

J. Kurtz: Nothing to disclose; S. Seetharaman: Nothing to disclose; A. Skripchenko: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Kurtz: Nothing to disclose; S. Seetharaman: Nothing to disclose; A. Skripchenko: Nothing to disclose; S. J. Wagner: Nothing to disclose

Day 7 in vitro PLT parameters

	Control (RT)	TEST	TEST + VX-702
Man PLT vol. (fL)	8.06 ± 0.81	8.72 ± 0.97	8.06 ± 0.79*
Morphology (% disc.)	63.6 ± 0.79	53.6 ± 9.2	62.2 ± 7.8*
Glucose (mg/dL)	212 ± 26	172 ± 36	225 ± 24*
Lactate (mM)	16.1 ± 2.7	19.6 ± 4.1	13.4 ± 2.3*
Bicarbonate (mM)	7.5 ± 1.5	5.1 ± 1.4	9.1 ± 1.5*
pH	7.05 ± 0.01	6.97 ± 0.13	7.13 ± 0.07*
CD62+ PLT (%)	57.6 ± 12.9	62.2 ± 12.1	28.6 ± 12.5±
CD42b (MFI)	737 ± 166	607 ± 158	791 ± 192*
HSR (%)	66.1 ± 10.4	56.9 ± 8.8	64.1 ± 8.1*
H_2O_2 (MFI)	49.5 ± 21.6	104.6 ± 88.1	31.7 ± 12.7
Annexin-V (%)	10.7 ± 3.4	14.7 ± 7.2	8.3 ± 4.0*

* $p < 0.004$, TEST versus TEST + VX-704

SP57

Efficacy and Safety of Pathogen Reduced Plasma in Plasma Therapy in Germany

S Rummler[1] (Silke.Rummler@med.uni-jena.de), C Weinigel[1], C Weise[1], D Barz[1]. *[1]Institute for Transfusion Medicine, University Hospital Jena, Jena, Germany*

Background/Case Studies: The institute received marketing authorisation for a leukocyte and platelet-depleted pathogen reduced single donor apheresis plasma (PIP) from the German authority Paul-Ehrlich-Institute in 2011. This plasma is processed via filtration with Plasmaflex PLAS4 and pathogen inactivation with the Intercept Blood System for Plasma. In March 2011 PIP was released for clinical use. Solvent detergent (SD) plasma and PIP were applied for plasma exchange therapy (TPE) without randomization depending on availability. **Study Design/Methods:** During the first 2 years of clinical application of PIP (03/2011-03/2013) altogether 11209 units were transfused. During TPE 1546 PIP-units were applied. 162 TPE procedures were performed on 45 patients. 793 units of SD plasma were transfused in 33 patients within 76 TPE. The main indications were transplant rejection episodes, chronic inflammatory neurological diseases and thrombocytopenia. Data were assessed retrospectively. **Results/Findings:** In total plasma transfusion was tolerated well. 3 transfusion reactions (0.027%) have been reported to our hemovigilance system. Plasma transfusion during TPE showed overall good tolerability with no thrombotic or bleeding episodes observed. 11 mild reactions were documented in patients' record of the PIP group (6.8%) versus 10 reactions in the SD group (13.2%). None of these reactions were reported to the hemovigilance system. Coagulation parameters (Quick, aPTT, Fibrinogen) were analysed before and after TPE. The Quick value decreased by 15% when using PIP versus 13% with SD, aPTT increased 2% with PIP versus. a 7.6% decrease under SD and fibrinogen decreased 31% (PIP) versus 26% (SD). **Conclusion:** The pathogen reduced plasma has shown a good tolerability in overall routine clinical use as well as for plasma exchange therapy. No unexpected transfusion reactions were observed. PIP has proven to be at least comparable to SD plasma. Possibly, further monitoring might show PIP being superior to SD plasma.

Disclosure of Commercial Conflict of Interest

D. Barz: Nothing to disclose; S. Rummler: Nothing to disclose; C. Weinigel: Nothing to disclose; C. Weise: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Barz: Nothing to disclose; S. Rummler: Nothing to disclose; C. Weinigel: Nothing to disclose; C. Weise: Nothing to disclose

SP58

In Vitro Evaluation of Stored Pediatric Apheresis Platelet Aliquots

R Dora[1] (Ross.Dora@bcw.edu), S Graminske[1], S Heldke[1], S Brooks[1], M de Arruda Indig[1], R Punzalan[2]. *[1]Applied Research Laboratory, BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Medical Sciences Institute, BloodCenter of Wisconsin, Milwaukee, WI, United States*

Background/Case Studies: Transfusing platelets to pediatric patients often times requires use of just a fraction of an apheresis platelet product. This adjustment is performed by transferring the desired volume to an aliquot bag. The platelet then must be transfused within 4 hours. The purpose of this study is to evaluate if platelet aliquots can be stored 1 day (24 hours) or longer in Pedi-Pak® bags (GenesisBPS). **Study Design/Methods:** Single dose platelets (SDP) (minimum 3×10^{11} platelets) were collected from healthy subjects into standard apheresis collection bags, stored overnight at 20-24 C, and sampled on day 1 for initial platelet concentration and BacT/Alert testing. After sampling on day 1, each product was split into 4 aliquots, each into a Pedi-Pak bag, in equal volumes, with some residual volume remaining in the collection bag. One of the aliquots was tested on day 1 of storage, a second on day 3, a third on day 6 and a fourth was kept as backup. In vitro laboratory testing included product volume, platelet concentration and yield, pH, pCO_2, pO_2, HCO_3, morphology score, P-selectin expression, extent of shape change (ESC), hypotonic shock response (HSR), glucose, lactate, and lactate dehydrogenase (LDH). **Results/Findings:** Six SDPs were collected with product volume range of 320-347 mL; platelet yield range was $4.0\text{-}5.9 \times 10^{11}$. Once split, 23 out of 24 aliquots maintained pH above 6.2. The only exception was the day 6 aliquot from the product that had the highest platelet concentration on day 1 (1695×10^3 plts/μL) and had virtually no glucose remaining at the end of storage. BacT/Alert testing was negative for all products. For all other parameters, there was minimal difference between day 1 and day 3 of storage, but greater changes were seen between day 3 and day 6 (Table). **Conclusion:** This feasibility study demonstrated that the Pedi-Pak may be used to store platelets for at least 3 days when the product is split on day 1 of storage. Further testing is needed to show if platelets aliquoted on day 2 will store adequately until day 5.

Disclosure of Commercial Conflict of Interest

S. Brooks: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; M. de Arruda Indig: GenesisBPS, Other; R. Dora: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; S. Heldke: MacoPharma, Grants or Research Support; Haemonetics, Grants or Research Support, Masimo, Grants or Research Support, Cerus, Grants or Research Support, GenesisBPS, Grants or Research Support; R. Punzalan: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Brooks: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; M. de Arruda Indig: Nothing to disclose; R. Dora: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; S. Heldke: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; R. Punzalan: GSK, Grants or Research Support, Baxter, Grants or Research Support

Platelet Aliquot Storage Data (Mean ± SD)

Parameter	Day 1 (N = 6)	Day 3 (N = 6)	Day 6 (N = 6)
Volume, Pre-Sample (mL)	80.8 ± 1.8	81.7 ± 1.3	79.8 ± 3.6
Platelet Yield ($x10^{11}$)	1.2 ± 0.1	1.2 ± 0.1	1.0 ± 0.1
Morphology Score (0-400)	326 ± 15	269 ± 32	203 ± 32
pH (37 °C)	7.2 ± 0.1	7.1 ± 0.1	6.7 ± 0.1*
pCO_2 (mmHg)	43 ± 10	45 ± 8	36 ± 9
pO_2 (mmHg)	148 ± 20	134 ± 17	151 ± 13
HCO_3 (mmol/L)	16.5 ± 0.8	11.7 ± 1.4	4.4 ± 0.9*
P-Selectin (%)	30.8 ± 16.1	32.2 ± 10.1	64.0 ± 14.3
HSR (%)	78.8 ± 6.0	76.7 ± 8.0	51.8 ± 20.0
ESC (%)	35.6 ± 6.4	34.9 ± 5.5	17.0 ± 11.5
Glucose (mg/dL)	297 ± 28	248 ± 19	123 ± 59
Lactate (mmol/L)	4.2 ± 0.9	9.7 ± 2.8	23.6 ± 8.7
LDH (U/L)	147 ± 52	152 ± 58	162 ± 49

* N = 5; one aliquot was not able to have pH and HCO_3 measured

SP59

The Immature Platelet Fraction and Immature Platelet Count Increase during Storage of Single-Donor Derived Platelet Units

H Hong[1,2] (hong.hong@uhhospitals.org), J Nofal[1], R W Maitta[1,2].
[1]Pathology, University Hospital Case Medical Center, Cleveland, OH, United States; [2]Case Western Reserve University School of Medicine, Cleveland, OH, United States

Background/Case Studies: Circulating immature platelets are larger and contain more nucleic acids especially RNA than mature platelets. The percent immature platelet fraction (%-IPF) in circulation reflects real-time thrombopoiesis and may correlate with platelet recovery in immune thrombocytopenic purpura and post-chemotherapy thrombocytopenia. However, the clinical significance of %-IPF in platelet transfusion is unknown. In the present study, we evaluated the dynamics of the %-IPF and the absolute immature platelet count (A-IPC) in single-donor derived platelet (SDPs) units during storage. **Study Design/Methods:** SDPs samples were obtained from local blood suppliers and stored using the same conditions of units used for patients, 22-24 C with continuous gentle agitation. 1-mL aliquots were taken from 5 units at day of donation (day 0) and from 10 units on days 3, 5, and 7 post donation. Platelet count, %-IPF, and A-IPC (obtained by multiplying the %-IPF by the platelet count) for all specimens were measured. In a separate experiment, five units were stored for up to 14 days and aliquots were taken at 3, 5, 7, 10, and 14 days post-donation. All aliquots were tested for %-IPF and platelet count within 1 hour of sampling using the automated hematology analyzer Sysmex XE-5000 (Sysmex Inc., Mundelein, IL). Statistical analysis was performed using a two-tailed student t test with significance set at $p < 0.05$. **Results/Findings:** %-IPF was 1.58 ± 0.3 % at day 0, similar to the %-IPF reference range (0.8-5.7 %) and A-IPC was $20.0 \pm 4.0 \times 109/l$. In storage, platelet counts were stable and did not significantly decrease until day 7 (day 0, $1290.2 \pm 125.5 \times 109/l$ versus day 7, $921.6 \pm 57.3 \times 109/l$; $p = 0.005$). %-IPF and A-IPC significantly increased by day 5 (4.5 ± 0.8 %, $p = 0.011$; and $57.7 \pm 12.5 \times 109/l$, $p = 0.025$ respectively) and even more by day 7 (6.2 ± 0.5 %, $p = 0.00001$; and $57.4 \pm 5.2 \times 109/l$, $p = 0.0002$ respectively). To test if the %-IPF increment in storage was due to A-IPC expansion ex-vivo, we analyzed the changes of %-IPF and A-IPC for 14 days of storage of SDPs from five different healthy donors. Compared to day 3 both %-IPF and A-IPC showed a significant increase by day 5 (2.7 ± 0.9 % versus 5.9 ± 1.5 %, $p = 0.007$; and $27.6 \pm 8.6 \times 109/l$ versus $63.1 \pm 15.5 \times 109/l$, $p = 0.006$ respectively) and remained high from day 7 to14. While %-IPF continued to increase from day 3 to14 (2.7 ± 0.9 % to 10.8 ± 1.9 %), A-IPC reached a plateau and remained high from day 7 to14 ($79.1 \pm 18.1 \times 109/l$ to $89.2 \pm 11.6 \times 109/l$). **Conclusion:** %-IPF and A-IPC continue to increase in SDPs during storage suggesting that A-IPC might be able to expand ex vivo. The findings suggest that A-IPC may also have a longer shelf life than its mature counterparts. Further characterization and analysis of the A-IPC may prove important in transfusion management and be of critical clinical importance.

Disclosure of Commercial Conflict of Interest

H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; J. Nofal: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; J. Nofal: Nothing to disclose

SP60

In Vitro Evaluation of Pediatric Erythrocyte Concentrates Stored in PAGGS-M in Four Different Storage Containers

J W Lagerberg[1] (j.lagerberg@sanquin.nl), D de Korte[1]. *[1]Sanquin Blood Supply, Amsterdam, Netherlands*

Background/Case Studies: The plasticizer di(2-ethylhexyl)-phthalate (DEHP) is a common component in blood collection systems. DEHP is non-covalently bound to the PVC polymer and can leach from PVC devices when the surface comes into contact with fluids. There are concerns that exposure to phthalates might induce developmental and reproductive toxicity. From previous studies it is known that the absence of DEHP has a detrimental effect on storage of red cells in SAG-M. Since with PAGGS-M this effect was less pronounced, in this study PAGGS-M was used as additive solution. The aim was to compare the in vitro quality of red cell concentrates in PAGGS-M, stored in pediatric red cell storage bags made from PVC with different plasticizers and different bag volume. **Study Design/Methods:** Leukodepleted red blood cell concentrate (RBC) in PAGGS-M (n = 7) was divided over four pediatric storage bags (60-75 mL/bag): DEHP-PVC (150 mL), BTHC-PVC (150 mL) and DINCH-PVC (150 and 600 mL). The RBCs were stored at 2-6 C for 42 days. A panel of in vitro red cell characteristics indicative for changes of viability and functionality of the red cells during cold storage were measured. **Results/Findings:** During storage the composition of the RBCs was very constant, with only minimal differences between the different storage bags. There were significant differences in hemolysis between the different storage bags. When stored in 150 mL bags, hemolysis was significantly higher in the BTHC bag, while hemolysis was comparable for the DEHP and DINCH bags. Hemolysis in the 600 mL DINCH bag was significantly higher than that in the 150 mL DEHP bag. No major differences in metabolic parameters during storage in the different storage bags were observed. **Conclusion:** During storage of the RBCs in PAGGSM in pediatric storage bags the normally occurring changes in quality parameters were observed. Storage in the BTHC-PVC container resulted in higher hemolysis as compared to the DEHP or DINCH containers. For the same plastic type, red cell in vitro quality during storage is better maintained in a smaller bag.

Parameter	Day 1 LD-RBC	Day 42 DEHP-PVC 150 mL	BTHC-PVC 150 mL	DINCH-PVC 150 mL	DINCH-PVC 600 mL
RBC x 10^12/L	0.73±0.39	3.69±0.33	3.65±0.34	3.68±0.3[illegible]	3.70±0.37
Ht (L/L)	0.63±0.02	0.64±0.02	0.64±0.02	0.65±0.02	0.64±0.03
MCV (fL)	98±3.7	97±4.2	97±4.6	97±4.5	96±4.4#
Hemolysis (%)	0.07±0.08	0.15±0.10	0.45±0.28#	0.20±0.13#	0.23±0.10#
ATP	5.4±0.18	4.3±0.79	4.3±0.6[illegible]	4.5±0.71	4.5±0.89

#: significantly different ($p<0.05$, paired t-test) from values obtained in DEHP-PVC, 150 mL
In vitro measures of pedi-RBCs stored in PAGGS-M (mean±SD, n=7)

Disclosure of Commercial Conflict of Interest

D. de Korte: Nothing to disclose; J. W. Lagerberg: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. de Korte: Terumo BCT, Grants or Research Support; J. W. Lagerberg: Nothing to disclose

SP61

In Vitro Properties of Platelets (Plts) Treated with UV Light in Open Air Compared to Mirasol PRT Treatment

D R Ambruso[1] (daniel.ambruso@ucdenver.edu), S Marschner[2], R Seewald[2], R Goodrich[2]. *[1]Pediatrics, University of Colorado Denver, Anschutz Medical Campus, Aurora, CO, United States; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Pathogen reduction technologies (PRTs) can induce changes in platelets (PLTs) which reduce recovery in the circulation. Studies have suggested a link to adverse events like transfusion related acute lung injury (TRALI). A two-event animal model of TRALI showed PLTs treated with UVB in air were associated with ALI. Mirasol PRT treatment is a closed system reducing exposure to air. The difference in PLTs treated with air compared to standard treatment in bags was evaluated. **Study Design/Methods:** Double apheresis PLT products were collected from healthy donors with TRIMA system under IRB approved protocol. One unit was treated with Mirasol standardly. Two aliquots (30 ml) were removed from the untreated unit and placed in open tissue culture flasks. One was exposed to 6.24 J/ml UV light and the other with 15.9 J/ml. Ten ml were removed from each sample, and Plts were washed in Tyrodes buffer with 0.4 μg/ml Prostacyclin. Samples were evaluated for PLT count and clumping/swirling by direct observation. Neutrophils (PMNs) were isolated from peripheral blood of healthy volunteers by standard techniques. Expression of P-selectin on Plt was detected using labeled antibodies to CD41a, CD62P, and isotype controls gated with a BD FACSCanto II; results were expressed as percent CD62P positive cells. After incubation of PMNs and Plts for 3 minutes at 370 C, PMN-platelet aggregates were identified by CD41a (platelets) and by forward and side scatter (PMNs) and were expressed as percent. **Results/Findings:** Untreated and Mirasol treated samples had normal swirl and similar platelet yields. Mirasol treated samples demonstrated increased P-selectin positive cells and PMN-Plt aggregates as shown previously.UV low and high/air exposed samples exhibited decreased counts, clumping, no swirl, and aincreased P-selectin positive cells and PMN-PLT aggregates. **Conclusion:** UV exposure of platelets in a system open to air results in more dramatic changes in platelets than those seen in the standard Mirasol process. This could account for ALI seen in the animal model and not documented in clinical studies with patients receiving Mirasol treated concentrates.

Disclosure of Commercial Conflict of Interest

D. R. Ambruso: Terumo BCT, Grants or Research Support; R. Goodrich: Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other; R. Seewald: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

D. R. Ambruso: Terumo BCT, Grants or Research Support; R. Goodrich: Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other; R. Seewald: Terumo BCT, Grants or Research Support

		Experiment 1		
	Count	Clumping/Swirl	Plts Positive for P-selectin (%)	PMNs Bound to Plts (%)
Untreated	1060 × 103/μl	No/Good Swirl	25.4	26.0
Standard Mirasol Treatment	927 × 103/μl	No/Good Swirl	42.3	32.7
6.24 J/ml Treated plus Air	788 × 103/μl	Moderate/no	34.7	30.0
2.4 J/cm2 (15.9 J/ml) Treated plus Air	530 × 103/μl	Severe/no	47.5	33.4
		Experiment 2		
	Count	Clumping/Swirl	Plts Positive for P-selectin (%)	PMNs Bound to Plts (%)
Untreated	1265 × 103/μl	No/Good Swirl	11.6	18.0
Standard Mirasol Treatment	1094 × 103/μl	No/Good Swirl	29.1	28.7
6.24 J/ml Treated plus Air	1015 × 103/μl	Moderate/no	22.4	26.0
2.4 J/cm2 (15.9 J/ml) Treated plus Air	478 × 103/μl	Severe/no	38.0	37.1

SP62

Validating the Quality of Non-Leukoreduced Single Units Recovered from Two Red Blood Cell (RBC) Procedures Using MCS®+8150

S Sawyer[1] (sherrie.sawyer@redcross.org), P H Whitley[1], M Wellington[1], S Hanley[1], L Y Maes[1], S Graminske[2], R Dora[2], J Gottschall[2], L W Yarmaloff[3], A Orr[3]. *[1]American Red Cross Mid Atlantic Research, Norfolk, VA, United States; [2]BloodCenter of Wisconsin, Milwaukee, WI, United States; [3]Clinical Research, Haemonetics Corp., Braintree, MA, United States*

Background/Case Studies: Two red blood cell (2 RBC) unit collections are an important contributor to the US blood supply. Infrequently, problems with venous access, donor reactions, and vein infiltration may cause the collection to be ended prematurely. This results in a loss of potentially transfusable blood products. The ability to recover one viable unit from discontinued 2 RBC collections would be of great value. The purpose of this study was to validate that single units of RBC's recovered from incomplete 2 RBC procedures meet the FDA in vitro requirements for non-leukoreduced RBC's within this study. **Study Design/Methods:** Non-leukoreduced CP2D/AS-3 single RBC products were collected at two clinical sites from 60 donors that qualified for a 2 RBC collection on the MCS®+8150 using LN832 disposable kits (Haemonetics, Braintree, MA). Targeted RBC volumes (180 to 210 mL) were determined by the MCS®+8150 based on height, weight, hematocrit, and gender. The 2 RBC procedure was manually stopped after the first RBC product was collected and before plasma or saline was returned to the donor to simulate typical single unit recoveries. Product weights were obtained to determine if the product volume met the acceptance criteria of ± 15% of the programmed target volume. In vitro testing on the single recovered unit was performed on days 0 and 42 of storage. Testing included hemolysis, ATP, CBC, glucose, lactate, potassium, and pH. **Results/Findings:** All red cell volumes were within 15% of the programmed target volume. Mean hemolysis at day 42 of storage was 0.35% (± 0.16) with no hemolysis greater that 1%. Glucose, lactate and potassium levels were similar to those seen in licensed red cell units. Mean ATP at day 42 was 3.19 (± 0.65) μmol/gHgb with a 73% (± 11) recovery from day 0 ATP. **Conclusion:** All units met the FDA acceptance criteria for this study which includes that all single recovered units had red cell volumes that were within 15% of the programmed target volume and had less than 1% hemolysis at end of storage. In vitro test results were comparable to those seen in licensed red cell units which would indicate that these units are acceptable for transfusion.

Disclosure of Commercial Conflict of Interest

R. Dora: Genesis BPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; J. Gottschall: Cerus, Other, Hemonetics, Other, Fenwal, Other, MacoPharma, Other; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; S. Hanley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. Y. Maes: Haemonetics, Grants or Research Support, Masimo, Grants or Research Support; A. Orr: Haemonetics, Stocks or Bonds; S. Sawyer: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; M. Wellington: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; P. H. Whitley: Massimo, Grants or Research Support, Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. W. Yarmaloff: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Dora: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; J. Gottschall: Nothing to disclose; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; S. Hanley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. Y. Maes: Haemonetics, Grants or Research Support; A. Orr: Haemonetics, Grants or Research Support; S. Sawyer: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; M. Wellington: Nothing to disclose; P. H. Whitley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. W. Yarmaloff: Nothing to disclose

Day 0 and Day 42 In Vitro Results

Mean (SD)	ATP (μmol/gHgb)	Hemolysis (%)	Glucose (mg/dl)	Lactate (mg/dl)	Potassium (mmol/l)
RBC Day 0	4.37 (0.62)	0.06 (0.02)	533 (40)	10.7 (8.7)	2.2 (0.3)
RBC Day 42	3.19 (0.65)	0.35 (0.16)	256 (45)	260.6 (50.4)	51.3 (4.0)

SP63

PRP-Depleted Erythrocyte Concentrates Stored in Different Additive Solutions in DEHP- and BTHC-PVC Storage Bags

J W Lagerberg[1] (j.lagerberg@sanquin.nl), D de Korte[1]. *[1]Sanquin Blood Supply, Amsterdam, Netherlands*

Background/Case Studies: The plasticizer di(2-ethylhexyl)-phthalate (DEHP) is a common component in PVC-based blood collection systems. There are concerns that exposure to phthalates might induce developmental and reproductive toxicity. For this, red cells are sometimes stored in BTHC-PVC containers. Absence of DEHP has detrimental effects on red cell stability which might be compensated for by alternative additive solutions.The aim was to compare the in vitro quality of red cell concentrates during storage in DEHP-PVC and BTHC-PVC containers and various additive solutions. **Study Design/Methods:** Overnight-held whole blood was processed according to the platelet rich plasma method. Red cells were diluted with SAGM, AS-1, AS-3 or PAGGS-M and leukodepleted by filtration. Red blood cell concentrates (RBC) were stored at 2-6 C for 42 days in 600 mL storage bags made from DEHP- or BTHC-PVC. A panel of in vitro red cell characteristics indicative for changes of viability and functionality of the red cells during cold storage were measured. **Results/Findings:** During storage the composition of the RBC was very constant, with only minimal differences between the different storage solutions and bags. The results obtained with PRP-depleted RBC in SAGM stored in DEHP-PVC correspond well with those of buffy-coat depleted RBC in SAGM. There were significant differences in hemolysis between the different additive solutions, with AS-1 and PAGGS-M giving less hemolysis than SAGM and AS-3. When stored in BTHC-PVC, hemolysis was higher as compared to DEHP-PVC. No major differences in metabolic parameters during storage in the different solutions

Parameter	Day 1	Day 42					
		DEHP SAGM	DEHP AS-1	DEHP AS-3	DEHP PAGGSM	BTHC SAGM	BTHC AS-1
RBC (10^{12}/L)	5.72±0.26	5.68±0.32	5.64±0.26	5.63±0.26	5.57±0.24	5.57±0.27	5.73±0.21
Hemolysis (%)	0.02±0.01	0.22±0.08	0.15±0.06	0.30±0.19	0.12±0.04*	0.38±0.28	0.33±0.13#
pH	6.93±0.04	6.39±0.03	6.36±0.07	6.36±0.02	6.35±0.02	6.42±0.03	6.4C±0.0C
lactate (mmol/L)	5.1±0.3	31±2.6	33±2.7	29±2.4	31±0.8	35±1.8	31±1.6
Annexin V+ cells (%)	n.c.	2.3±0.6	1.3±0.3*	3.0±1.4	1.7±0.6	1.9±2.2	2.6±0.6

*: p<0.05 compared to DEHP/SAGM
p<0.05 compared to DEHP/AS-1

In vitro measures of RBCs stored for 42 days (mean±SD, n=8)

	DHMC			USAISR			
	Baseline	5 days RT	11 days Xe	Baseline	5 days RT	11 days Xe	11 days Cold
TEG with Kaolin R (min)	8.3±1.3	8.0±0.8	8.2±0.9	6.4±0.8	6.8±1.0	6.4±0.7	5.8±1.0
TEG with Kaolin MA (mm)	76.7±2.3	74.4±1.7	67.6±4.6	69.7±5.4	63.3±4.9	66.9±5.7	68.5±3.4
TEG without Kaolin R (min)	26.8±7.2	28.6±7.1	20.4±7.6	13.9±2.6	15.8±5.4	15.6±2.6	12.2±3.6
TEG without Kaolin MA (mm)	74.4±1.0	71.3±4.1	61.3±7.0	61.1±5.9	56.3±12.2	65.8±4.3	64.4±1.5
Fibrinolysis at 30 min (%)	0.3±0.3	0.5±1.1	0.7±0.1	3.2±3.5	13.7±19.4	0.07±0.2	0

Summary TEG data for stored platelets (means ± SD)

and bags were observed. The number of Annexin V+ cells remained below 4%, with only minimal differences between the different bag and additive solution combinations. **Conclusion:** Hemolysis during storage depends on both additive solution and plasticizer. Storage in BTHC-PVC container resulted in higher hemolysis as compared to DEHP container, but maximal hemolysis was within regulatory requirements. AS-1 and PAGGS-M better maintained red cell integrity during storage than SAGM and AS-3.

Disclosure of Commercial Conflict of Interest

D. de Korte: Nothing to disclose; J. W. Lagerberg: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. de Korte: Terumo BCT, Grants or Research Support; J. W. Lagerberg: Nothing to disclose

SP64

Functionality of Chilled Platelets Stored under Hyperbaric Xenon

A P Cap[1] (andrew.p.cap.mil@mail.mil), K Reddoch[1,2], R Montgomery[1], H F Pidcoke[1], A Ramasubramanian[1], A Shumeev[3], I Ilyin[3], L Herschel[4], S L Waters[4], L J Dumont[4]. *[1]US Army Institute of Surgical Research, Fort Sam Houston, TX, United States; [2]Department of Biomedical Engineering, University of Texas, San Antonio, TX, United States; [3]Advanced Preservation Technologies, Boston, MA, United States; [4]Dartmouth-Hitchcock Medical Center, Hanover, NH, United States*

Background/Case Studies: Apheresis platelets (AP) are stored for up to 5 days at room temperature (22 C, RT). Refrigeration (4 C) would reduce the risk of bacterial overgrowth and allow longer storage; however, chilled platelets are more rapidly cleared from circulation following transfusion. A new technology was developed here for maintaining viability of chilled platelets for up to 14 days by refrigeration under hyperbaric xenon, which may stabilize platelet membranes and mitigate cold-mediated clearance. The aim of the present study is to investigate functional parameters of AP after 14 day storage under hyperbaric xenon. **Study Design/Methods:** Data are presented from three participating laboratories. Double AP units were collected from healthy volunteers using a Trima Accel (Terumo) Automated Blood Collection System and divided into experimental and control units. Experimental units were stored under 4 bars xenon at 4 C without agitation for 14 days (Xe). Control units were stored at 22 C with constant agitation for 5 days at 2 laboratories (RT) or under ambient air pressure at 4 C with no mixing for 14 days in one laboratory (Cold). Hemostatic functionality and platelet responsiveness were assessed by thromboelastography and light transmission aggregometry. **Results/Findings:** TEG results were within reference ranges for clot strength in Xe and Cold platelets. Fibrinolysis was out of reference range in RT but not for Xe or Cold platelets. Aggregation level on day 14 (ADP 10 μM + epinephrine 5 μM) is similar in the chilled samples (Cold 49.4 + −10.7 % (n = 6); Xe 46.6+ −14.7 % (n = 6)), but reduced compared to baseline (80.8 + −6.5 % n = 6). **Conclusion:** Xenon chilled samples preserve hemostatic function after 14 days of storage as indicated by the formation of an effective platelet clot in TEG and response to agonists. Clinical studies of recovery and survival of Xe platelets are indicated.

Disclosure of Commercial Conflict of Interest

A. P. Cap: Nothing to disclose; L. J. Dumont: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Advanced Preservation Technologies, Consulting or Board of Director Fees; L. Herschel: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support; I. Ilyin: Advanced Preservation Technologies, Ownership or Partnership; R. Montgomery: No Answer; H. F. Pidcoke: Advanced Preservation Technologies, Other; A. Ramasubramanian: Nothing to disclose; K. Reddoch: Nothing to disclose; A. Shumeev: Nothing to disclose; S. L. Waters: Citra Labs, Grants or Research Support

Disclosure of Grants Conflict of Interest

A. P. Cap: Terumo BCT, Grants or Research Support, Oxygen Biotherapeutics, Grants or Research Support; L. J. Dumont: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, New Health Sciences, Grants or Research Support, Terumo BCT, Grants or Research Support, Advance Cell Technologies, Consulting or Board of Director Fees, Advanced Preservation Technologies, Consulting or Board of Director Fees, New Health Sciences, Consulting or Board of Director Fees, Terumo BCT, Consulting or Board of Director Fees; L. Herschel: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Hemerus Medical, Grants or Research Support, New Health Sciences, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Terumo BCT, Grants or Research Support, SynGen, Grants or Research Support; I. Ilyin: Nothing to disclose; R. Montgomery: No Answer; H. F. Pidcoke: Oxygen Biotherapeutics, Inc., Grants or Research Support, Terumo BCT, Grants or Research Support; A. Ramasubramanian: Nothing to disclose; K. Reddoch: Nothing to disclose; A. Shumeev: Nothing to disclose; S. L. Waters: New Health Science, Grants or Research Support, Syn Gen, Grants or Research Support, Fenwall, Grants or Research Support, Haemonetics, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support

SP65

Citrate Buffer Addition Prevents Red Cell Hemolysis after Cryopreservation through a Non-Permeant Osmolarity Effect

J Cancelas[1] (jose.cancelas@uc.edu), N Rugg[1], E L Hartman[1], R Giulitto[1], M McAteer[2]. *[1]Hoxworth Blood Center, University of Cincinnati, Cincinnati, OH, United States; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Red blood cell (RBC) cryopreservation is a routine practice in hospitals and battlefield. Increased hemolysis of stored RBC is associated with increased morbi-mortality. Current standard practice for open system thawing of RBC uses saline/glucose solution for 24 h post-thaw storage. This study aimed to identify a translatable method to reduce post-thaw hemolysis. **Study Design/Methods:** Twelve CP2D WB units were processed into AS-3 RBC units, held for 21-26 days at 1-6 C and then frozen using the High Glycerol Method, and stored for 8-27 days at ≤ −65 C. At the end of storage, units were thawed and deglycerolized (COBE 2991). Seven 30 mL aliquots were prepared from each unit, resuspended in 15 mL of one of seven acidic solutions (A-G) and stored for 24 hours. Solution compositions were: 1) 150 mM NaCl/11 mM glucose (SG), pH = 5.0; 2) AS-3; 3) SG/22.8 mM Na3 citrate/2.2 mM citric acid, pH = 5.9; 4) SG/4.1 mM mannitol, pH = 5.3; 5) 135 mM NaCl/11 mM glucose/0.46 μM human serum albumin (HSA), pH = 6.8; 6) 135 mM NaCl/11 mM glucose/ 0.23 mM hydroxyethyl starch (HES), pH = 5.3; and, 7) SG/2 mM α-ketoglutarate (KG), pH = 3.8. While solutions C-G were ~ permeant isosmolar (281-311 mOsm/L) and exerted similar levels of oncotic pressure, solution A was isosmolar but exerted no significant oncotic pressure, and solution B had a reduced permeant osmolarity (210 mOsm/L). **Results/ Findings:** Citrate, mannitol, HSA, HES and KG showed significantly decreased hemolysis rates immediately after thaw (pre-storage) and at 24 h (post-storage) as compared to SG (Table). AS-3 was also cryoprotectant when assessed post-storage. However, addition of citrate buffer alone sufficed to diminish post-storage hemolysis ~35% more than AS-3 ($p < 0.001$). Addition of citrate buffer to SG thawing solution prevents hemolysis immediately after thawing and post-storage. Mannitol, albumin, HES and KG were also effective cryoprotectants. RBC adenosine 5′-triphosphate (ATP) levels at 24 hour post-storage were similar between the different solutions and did not correlate with the cryoprotectant effect (Table). **Conclusion:** Citrate protective effect seems to depend on maintenance of permeant iso-osmolarity and increased non-permeant osmotic pressure. These results

indicate that inclusion of citrate in the RBC resuspension buffer after thawing may be clinically relevant.

Disclosure of Commercial Conflict of Interest

J. Cancelas: Nothing to disclose; R. Giulitto: Nothing to disclose; E. L. Hartman: Terumo BCT, Grants or Research Support, Cerus, Grants or Research Support; M. McAteer: Terumo BCT, Other; N. Rugg: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Cancelas: Cerus, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Grants or Research Support, Therapure Inc., Grants or Research Support; R. Giulitto: Nothing to disclose; E. L. Hartman: Terumo BCT, Grants or Research Support, Cerus, Grants or Research Support; M. McAteer: Terumo BCT, Other; N. Rugg: Terumo BCT, Grants or Research Support

TABLE 1.

Mean ± SD (n = 12)	% Hemolysis		ATP (μmol/g Hb)
Solutions	Pre-Storage	Post-Storage	Post-Storage
A: SG	0.45 ± 0.12	1.36 ± 0.66	5.07 ± 0.91
B: AS-3	0.33 ± 0.08	0.63 ± 0.26*	4.90 ± 0.92
C: SG/Citrate	0.19 ± 0.06*	0.40 ± 0.12*	5.16 ± 0.95
D: SG/Mannitol	0.23 ± 0.10*	0.68 ± 0.22*	5.39 ± 0.96
E: SG/HSA	0.17 ± 0.05*	0.57 ± 0.18*	5.10 ± 0.98
F: SG/HES	0.17 ± 0.05*	0.48 ± 0.12*	5.16 ± 0.94
G: SG/KG	0.26 ± 0.10	0.77 ± 0.30*	5.43 ± 1.00

* $P < 0.01$ as compared to solution A

SP66

Paired 42-Day In Vitro Comparison of Red Cell Concentrates Produced from Whole Blood by Spinning Membrane Technology or Centrifugation

A Karpiel[1] (adrienne.karpiel@fresenius-kabi.com), B E Kusters[1], C Wegener[1], K Min[1]. *[1]Fresenius Kabi USA, Lake Zurich, IL, United States*

Background/Case Studies: Manual processing of whole blood (WB) units for production of red cell concentrates (RCC) currently relies on centrifugation for component separation. However, spinning membrane separation, which is commonly used in plasmapheresis, may also be employed for RCC production. In a previous study, Adsol-RCCs prepared from sodium citrate-WB using spinning membrane technology exhibited acceptable hemolysis (<0.8%) for up to 42 days. ATP levels, however, were reduced in comparison with historical data due to the use of a neutral pH anticoagulant. Since CPD anticoagulant is routinely used in WB collection, this study sought to directly compare the quality of red cell concentrates prepared from CPD- WB using automated spinning membrane technology to those prepared by manual processing with centrifugation. **Study Design/Methods:** Paired, ABO matched WB units in CPD were held at room temperature (22 C) for 4-6 hours. WB was then pooled, mixed, and split into 2 equivalent units. One unit was separated into plasma and Adsol-RCC product using a spinning membrane device. The second unit was separated by centrifugation and manual plasma expression using top and top processing; Adsol was then added to the resulting RCC. The Adsol-RCC pairs were leukoreduced by gravity filtration, and stored upright at 4 C for 42 days with day 7 mixing/sampling. Standard red cell in vitro parameters were assessed. **Results/Findings:** Total Hgb content of RCCs produced by either centrifugation or membrane separation was not significantly different. Day 0 spun hematocrit was higher in centrifuged versus membrane separated units (61 ± 1% versus 56 ± 2%, respectively $p < 0.05$). In vitro storage parameter results are summarized in table below. **Conclusion:** Automated separation of RCCs from CPD-WB by spinning membrane technology yielded RCCs with acceptable in vitro storage parameters out to day 42 with only small differences observed in comparison with centrifuged units. By collecting WB into CPD anticoagulant instead of sodium citrate, the expected improvement in red cell ATP maintenance was observed throughout the duration of storage.

Disclosure of Commercial Conflict of Interest

A. Karpiel: Nothing to disclose; B. E. Kusters: Nothing to disclose; K. Min: Nothing to disclose; C. Wegener: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Karpiel: Fresenius Kabi USA, Grants or Research Support; B. E. Kusters: Nothing to disclose; K. Min: Fresenius KABI USA, Grants or Research Support; C. Wegener: Fresenius Kabi USA, Grants or Research Support

Parameter, Mean ± SD (n = 10)	RBC Separation Method	Day 0	Day 21	Day 42
Hemolysis (%)	Membrane	0.13 ± 0.04*	0.20 ± 0.05*	0.30 ± 0.10*
	Centrifugation	0.08 ± 0.05*	0.12 ± 0.06*	0.21 ± 0.06*
Microparticles ($x10^3/\mu L$)	Membrane	10 ± 4	16 ± 5	26 ± 8
	Centrifugation	10 ± 5	18 ± 7	28 ± 6
Morphology (0-200)	Membrane	195 ± 3	156 ± 6*	137 ± 8*
	Centrifugation	196 ± 3	165 ± 8*	145 ± 7*
ATP (μmol/g Hb)	Membrane	5.1 ± 0.4	4.6 ± 0.5*	3.0 ± 0.4*
	Centrifugation	5.1 ± 0.6	5.2 ± 0.4*	3.4 ± 0.5*

Parameter, Mean ± SD (n = 10)	RBC Separation Method	Day 0	Day 7	Day 14
2,3 DPG (μmol/g Hb)	Membrane	9.6 ± 1.2*	3.9 ± 2.0*	Below LOD
	Centrifugation	10.1 ± 1.3*	4.6 ± 2.0*	Below LOD

* $p < 0.05$

SP67

Isohemagglutinin Titers in Platelet Additive Solution 3 (PAS 3) Platelets Versus Conventional Plasma Platelet Units

M Surowiecka[1] (suro003@umn.edu), R Dangerfield[2], J Welbig[2], S Pulkrabek[1], C S Cohn[1]. *[1]Department of Laboratory Medicine and Pathology, University of Minnesota Medical Center, Minneapolis, MN, United States; [2]Laboratory Medicine, University of Minnesota Medical Center, Fairview Hospital, Minneapolis, MN, United States*

Background/Case Studies: Many hospital transfusion services titer the isohemagglutinins in their type O platelet units, reserving those with elevated titers for type O patients. This institution uses a >1:200 titer as the cut-off value for elevated anti-A and/or -B levels. Titering is performed manually, and takes a significant amount of the technologist's time. Platelet units stored in additive solution 3 [(PAS 3) (InterSol, Fenwal, a Fresenius Kabi Co.)] contain 65% less plasma than conventional platelet units. This should give rise to lower isohemagglutinin levels, and less exposure for patients to out-of-type plasma. Type O units stored in PAS 3 versus plasma were compared to see if there are fewer PAS 3 units with titers >1:200. **Study Design/Methods:** A retrospective review of anti-A and/or -B titer results was undertaken for all type O platelet units received at University of Minnesota Medical Center from December 12, 2011-April 24, 2013. During this period a regular inventory of conventional and PAS 3 platelet units was maintained. The statistical significance of the data was analyzed using Fisher's exact test. **Results/Findings:** A total of 5471 platelet units were received during the study period. There were 3516 conventional plasma units and 1955 PAS 3 units. For conventional platelet units, 43/3516 (1.22%) had >1:200 titers versus 2/1955 (0.10%) for PAS 3 units ($p < 0.0001$). The two PAS 3 units with >1:200 titers were parts 1 and 2 from the same donor. **Conclusion:** Conventional plasma platelet units had an anti-A and/or -B titer >1:200 twelve times more often than PAS 3 units. This difference was found to be highly significant. These findings strongly suggest that a reduction in plasma leads to a concomitant reduction in isohemagglutinin levels.

Disclosure of Commercial Conflict of Interest

C. S. Cohn: Fenwal Inc., Grants or Research Support, Fenwal Inc., Travel Support or Honorarium; R. Dangerfield: Nothing to disclose; S. Pulkrabek: Fenwal Inc., Grants or Research Support; M. Surowiecka: No Answer; J. Welbig: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. S. Cohn: Fenwal Inc.,Travel, Support or Honorarium,Grants or Research Support; R. Dangerfield: Nothing to disclose; S. Pulkrabek: Nothing to disclose; M. Surowiecka: No Answer; J. Welbig: Nothing to disclose

Type O Platelet Unit Titers	PAS 3 Units N (%)	Conventional Plasma Units N (%)
>1:200 titer	2 (0.10%)	43 (1.22%)
<1:200 titer	1953	3473
Total	1955	3516

SP68

Large Animal Evaluation of the Safety and Efficacy of Whole Blood Treated with the Mirasol System in a Diffuse, Non-Surgical Bleeding Model

O T Okoye[1] (obi.okoye@usc.edu), H Reddy[2], M D Wong[1], S Doane[2], S Resnick[1], E Karamanos[1], D Skiada[1], R Goodrich[2], K Inaba[1]. *[1]Surgery, Los Angeles County + University of Southern California Medical Center, Los Angeles, CA, United States; [2]Blood Bank Technologies, Terumo BCT Biotechnologies, LLC, Lakewood, CO, United States*

Background/Case Studies: The resuscitation of critically ill trauma patients utilising Whole Blood (WB) can be a practical necessity for military combat care providers. As the understanding of the need for aggressive plasma and platelet replacement in addition to red cell transfusion increases, attention has been focused on the potential benefit of WB transfusion in the hospital setting as well. The Mirasol system has been demonstrated to effectively inactivate leucocytes and reduce pathogens in WB. The purpose of this study was to compare the safety and efficacy of Mirasol treated WB against untreated blood in an in vivo model of surgical bleeding. **Study Design/Methods:** A total of 18 anesthetized pigs (40 kg) underwent a 35% total blood volume bleed, followed by cooling to 35 C, and a standardized grade IV liver injury. Animals were randomized to resuscitation with either Mirasol treated or untreated WB (25% blood volume) over a 30-minute period. Treatment of porcine WB with the Mirasol System involved transfer of the WB to an illumination bag, addition of 35 mL of 500 μM riboflavin solution, and exposure to an 80 J/mL_{RBC} UV energy dose. All WB units (control and treated) were stored at 22 C-25 C for 24 hours prior to transfusion. Intraoperative bleeding was measured at 2 minutes, 15 minutes, and 30 minutes. Mortality, tissue oxygenation (Near-InfraRed Spectroscopy-NIRS), red cell parameters, platelets, leucocytes and coagulation indices were monitored up to day 14, after which the animals were sacrificed and tissues obtained for histopathological examination. Trends over time were monitored using repeated measures ANOVA. **Results/Findings:** Total intraoperative blood loss was similar in both arms (8.3 ± 3.2 versus 7.7 ± 3.9 ml/kg, $p = 0.72$). All animals survived up to day 14. Trended values over time did not show significant differences—NIRS readings ($p = 0.818$), hematocrit ($p = 0.429$), platelets ($p = 0.685$), leucocytes ($p = 0.545$), PT ($p = 0.688$), APTT (0.325), TEG-R ($p = 0.481$) or TEG-K ($p = 0.377$). Pathology samples did not show significant differences in either arm. **Conclusion:** Mirasol-treated Whole Blood did not impact survival, blood loss, tissue oxygen delivery, red cell indices, or coagulation parameters in a grade IV liver injury model. This data suggests that Mirasol-treated Whole Blood is both safe and efficacious in vivo.

Disclosure of Commercial Conflict of Interest

S. Doane: Terumo BCT, Other; R. Goodrich: Terumo BCT, Other; K. Inaba: Terumo BCT, Grants or Research Support; E. Karamanos: Terumo BCT, Grants or Research Support; O. T. Okoye: Terumo BCT, Grants or Research Support; H. Reddy: Terumo BCT, Other; S. Resnick: Terumo BCT, Grants or Research Support; D. Skiada: Terumo BCT, Grants or Research Support; M. D. Wong: Terumo BCT, Grants or Research Support

Disclosure of Grants Conflict of Interest

S. Doane: Nothing to disclose; R. Goodrich: Terumo BCT, Other; K. Inaba: Nothing to disclose; E. Karamanos: Nothing to disclose; O. T. Okoye: Nothing to disclose; H. Reddy: Nothing to disclose; S. Resnick: Nothing to disclose; D. Skiada: Nothing to disclose; M. D. Wong: Nothing to disclose

SP69

The Impact of Complying with FDA's 1-6 C Temperature Storage Requirement for Reissuance of Blood Products Issued within a Facility

J L Jacobson[1] (jessica.jacobson@nyumc.org), D Gajera[1]. *[1]Pathology, Bellevue Hospital/NYU School of Medicine, New York City, NY, United States*

Background/Case Studies: At the "Ask the FDA Session" at the 2008 AABB Meeting, the FDA stated that in order for blood issued and returned within a facility to be eligible for reissue it must have been kept between 1-6 C. This facility's policy permitted RBCs and plasmas to be returned and reissued to another patient if the product had been outside of the blood bank for less than 30 minutes. In 2011 each unit's temperature was measured upon return. Based on the number of recorded temperatures exceeding 7 C within 10-15 minutes post-issue, units were no longer accepted for reissue from areas without blood bank monitored refrigerators in March 2012. It was sought to evaluate the impact of this change on the waste/discard rates. **Study Design/Methods:** The numbers of discarded RBC and plasma units that had been issued to areas without a blood bank monitored refrigerator and were subsequently returned from April 1, 2012-March 31, 2013 were compared to the numbers from April1, 2011-March 31, 2012. The results were analyzed using the unpaired t test for statistical significance. **Results/Findings:** From April 1, 2011-March 31, 2012, 7964 RBCs and 2131 plasmas were issued, and 46 RBCs and 29 plasmas were returned and wasted. From April 1, 2012-March 31, 2013, 5329 RBCs and 1705 plasmas were issued, and 75 RBCs and 23 plasmas were returned and wasted. The average discard rate increased significantly from 0.6% to 1.2% for RBCs ($p < 0.05$). The average discard rate for plasma increased from 1.1% to 1.2%. **Conclusion:** By complying with the FDA requirement that blood products issued within an institution remain between 1-6 C in order to be eligible for reissue, our waste/discard rate for RBCs doubled. While the overall number of wasted RBCs remained relatively low and did not impact on our ability to provide RBCs, the decision to shifting from the "30 minute rule" to temperature maintenance of 1-6 C increased our RBC discard costs from $10,672 to $17,400.

Disclosure of Commercial Conflict of Interest

D. Gajera: Nothing to disclose; J. L. Jacobson: Fresenius-Kabi, Other, Terumo BCT, Other, Genesis BPS, Other, SMART Bio Tech Ltd., Other

Disclosure of Grants Conflict of Interest

D. Gajera: Nothing to disclose; J. L. Jacobson: G & W Laboratories Inc., Grants or Research Support

Blood Products Wasted and Transfused Per Quarter

	Product Type	1Q13	4Q12	3Q12	2Q12	1Q12	4Q11	3Q11	2Q12
Product Wasted	RBCs	13	5	34	23	9	8	21	8
	Plasmas	2	2	7	12	3	9	13	4
Products Transfused	RBCs	789	736	1854	1950	1961	2175	1998	1830
	Plasmas	245	201	464	795	429	576	772	354
Percent Wasted/Discarded	RBCs	1.6%	0.7%	1.8%	1.2%	0.5%	0.4%	1.1%	0.4%
	Plasmas	0.8%	1.0%	1.5%	0.7%	0.7%	1.6%	1.7%	1.1%

The hospital was closed to inpatient services following Hurricane Sandy from November 4, 2012-February 11, 2013 thus accounting for the decrease in the total number of products wasted and transfused during 4Q12 and 1Q13.

SP70

Application of a Real-Time PCR Assay for Bacterial Contamination Screening in Cord Blood (CB)

M Cayer[1] (Marie-Pierre.Cayer@hema-quebec.qc.ca), D Fournier[2], M Chevrier[2], L Thibault[1]. *[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada*

Background/Case Studies: Today, the gold standard method to detect microbial contamination of blood products relies on culture-based techniques which are time-consuming and above all require high inoculum volumes. Recently, Patel and coworkers (Transfusion, 2012) developed a new molecular strategy to detect bacterial contaminations in platelet concentrates. In this study, a qPCR assay was modified to detect the presence of bacterial contaminants in umbilical cord blood (UCB) samples. **Study Design/Methods:** UCB samples were first spiked with four bacterial strains (*Escherichia coli, Staphylococcus aureus, Klebsiella pneumoniae and Enterococcus sp.*) at 10, 100 and 500 colony-forming units (CFUs) ($n = 6$ per condition). One mL of spiked UCB was used to perform DNA extraction with

a modified protocol using a commercial kit designed for the targeted isolation from blood samples (Molzym, Bremen, Germany). PCR amplifications were done with a set of universal primers and a probe targeting a conserved region of the bacterial 16 S ribosomal DNA. In these experiments, PCR positive results were defined as a difference at least one C(t) value compared to negative control in the same PCR run. Reliability of the assay was monitored by amplifying a region of the human 12 S mitochondrial DNA. Ethidium-monoazide (EMA) treatment of the master mix was also assessed to eliminate unspecific signal. The qPCR detection limits was compared to the one of the BacT/ALERT system (bioMérieux, Marcy l'Etoile, France). **Results/Findings:** A poor reproducibility of the assay was observed with frequent contaminations in negative controls increasing detection limit values. Moreover, bacterial DNA extraction yield was affected by the presence of human DNA and inhibitors present in the UCB matrix. Treatment with EMA did not improve the detection of bacteria but rather affect the amplification performance. Sensitivity of the qPCR assay ranged at approximately 150 to 430 CFUs/mL and no differences were observed between the detection of Gram-negative and Gram-positive bacteria. In spite of these limitations, qPCR assay used only 1 mL of sample and is realized in less than four hours, which was much shorter than culture detection with the BacT/ALERT system. **Conclusion:** Current qPCR technology does not yet demonstrate the required sensitivity and precision for being used by cord blood banks to detect microbial contaminants in UCB samples. Nucleic acid amplification methods for the detection of cord blood contaminations still face several technical challenges including low levels of bacteria, small volume samples, and failure to differentiate live from dead cells and residual matrix components. Meanwhile, automated culture systems like the BacT/ALERT system remain more sensitive to detect contaminations in UCB samples.

Disclosure of Commercial Conflict of Interest

M. Cayer: Nothing to disclose; M. Chevrier: Nothing to disclose; D. Fournier: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Cayer: Nothing to disclose; M. Chevrier: Nothing to disclose; D. Fournier: Nothing to disclose; L. Thibault: Nothing to disclose

SP71

Optimizing the Washing Procedure of the Haemonetics ACP215 Automated System for Making Allogeneic IgA-deficient Red Blood Cell Units

A Jacques[1] (Annie.Jacques@hema-quebec.qc.ca), H Daoud[1], C Bédard[1], M Chevrier[2], L Thibault[1]. *[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada*

Background/Case Studies: In Canada, it is generally agreed that IgA-deficient patients should be transfused with washed red blood cells (RBCs) when blood products made from IgA-deficient blood donors are not available. According to current guidelines, RBC units should be washed with a minimum volume of 2 L of saline. The closed-system ACP 215 (Haemonetics) enables automated washing and 14 days of post-wash storage. In this work, the ability of the ACP 215 system for washing AS-3 and SAGM RBC units to comply with the CSA recommendation of less than 0.5 μg of IgA per mL was investigated. **Study Design/Methods:** For this work, the ACP 215 washer was programmed using a standard hematocrit of 60%. Only leukoreduced AS-3 and SAGM RBC units weighting ≥250 g were used. After washing, RBCs were stored in AS-3 at 4°C. Three washing protocols were evaluated to reduce IgA below 0.5 μg/mL: 1.84 L (control), 3.0 L and 2 consecutive procedures of 1.84 L (n = 6 per arm). Leukoreduced AS-3 and SAGM RBC units (450 mL) were washed with the ACP 215 system after 7, 14, 21 or 28 days of storage and stored post-wash for up to 14 days (n = 5 per arm). Washing efficiency, assessed by measuring the volume of residual plasma and IgA concentration, and in vitro quality parameters of RBCs washed at different storage times, were analyzed. **Results/Findings:** The IgA content of washed RBC units using 1.84 L and 3.0 L of saline remained above the acceptable limit of 0.5 μg of IgA per mL, at 1.4 ± 0.7 μg/mL and 1.3 ± 0.9 μg/mL, respectively. By performing two washes, the IgA content of RBC units was reduced from 202 ± 129 μg/mL to 0.007 ± 0.003 μg/mL. After two repeated procedures, the RBC recovery was 90 ± 4% for AS-3 units and 89 ± 4% for SAGM units. Measured after 7 days post-wash, the hemolysis rate in RBC units stored 14 days pre-wash was 0.5 ± 0.1% and 0.3 ± 0.0% in AS-3 and SAGM RBC units respectively Hemoglobin content in washed AS-3 RBC units was 53 ± 4 g/unit, and 48 ± 4 g/unit in SAGM RBC units. The residual volume of plasma was ≤ 2 μL in all units. **Conclusion:** This study shows that two sequential washes can reduce the IgA concentration of RBC units below the limit of 0.5 μg/mL, thereby complying with Canadian regulatory guidelines. Double-washed RBC units can be stored for up to 7 day post-wash when suspended in AS-3. A validation study is underway to confirm the routine use of the ACP 215 system to reduce the IgA content of RBC units.

Disclosure of Commercial Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; L. Thibault: Nothing to disclose

SP72

Preliminary Results for Post-Thaw Quality of SAGM and AS-3 Leukoreduced Red Blood Cells after Deglycerolization and Suspension in AS-3 with the ACP 215 Automated System

A Jacques[1] (Annie.Jacques@hema-quebec.qc.ca), H Daoud[1], C Bédard[1], A Laforce-Lavoie[1], M Chevrier[2], L Thibault[1]. *[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada*

Background/Case Studies: Currently, deglycerolized rare red blood cell (RBC) units are processed using COBE 2991 cell processors. Thawed RBC units that are suspended without nutrient solution have a limited shelf life of 24 hours at 1-6 C. The ACP 215 automated closed system from Haemonetics allows for prolonged postglycerolization storage. In this study, the post-thaw quality of SAGM and AS-3 RBC units previously frozen on day 14 was investigated on day 7 following deglycerolization and suspension in AS-3 solution. **Study Design/Methods:** Five leukoreduced AS-3 and 5 leukoreduced SAGM RBCs were frozen with 40% glycerol on day 14 using the automated ACP 215. After at least 48 hours of storage at –80 C, frozen RBC units were deglycerolized with the ACP 215 and suspended in about 100 mL of AS-3 additive solution. The instrument was set with a standard hematocrit of 0.60 L/L. Thawed RBC units were stored at 1 to 6 C. Samples were collected on day 7 to study the effects of deglycerolization on the quality of RBCs (hemoglobin, hematocrit, and hemolysis) and RBC metabolism (ATP, lactate and potassium). **Results/Findings:** After thawing, the percentage of recovery was comparable for both types of RBCs regardless of the nature of the nutrient solution used before freezing (85 ± 2 % versus 88 ± 5 % for AS-3 and SAGM RBC units, respectively). The hemoglobin content was 46 ± 2 g versus 39 ± 7 g for AS-3 and SAGM RBC units. The hematocrit of AS-3 and SAGM RBC units was 0.40 ± 0.02 L/L and 0.38 ± 0.03 L/L, respectively. After 7 days of storage, the mean level of hemolysis and ATP were comparable for both types of RBCs (0.4 ± 0.1% versus 0.5 ± 0.1% and 2.1 ± 0.5 μmol/g Hb versus 1.9 ± 0.5 μmol/g Hb for AS-3 and SAGM RBC units, respectively). After 7 days of storage, average potassium and lactate levels were slightly higher in AS-3 RBC than in SAGM RBC units (9.0 ± 2.6 mmol/L versus 7.9 ± 0.7 mmol/L for potassium and 5.8 ± 0.4 mmol/L versus 5.1 ± 0.9 mmol/L for lactate). No bacterial contamination was detected in these products. **Conclusion:** These results show that AS-3 and SAGM RBCs can be stored for up to 14 days before glycerolization and deglycerolization with the ACP 215 system. Once suspended in AS-3, packed red blood cells can be stored for up to 7 days to 1 to 6 C. A validation study has been undertaken to confirm these results.

Disclosure of Commercial Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; A. Laforce-Lavoie: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; A. Laforce-Lavoie: Nothing to disclose; L. Thibault: Nothing to disclose

SP73

Production of Standardized Labile Products from Single Whole Blood Donations: Evolution throughout Last Three Years

A Castrillo Fernández[1] (azucena.castrillo.fernandez@sergas.es), C Arcas Otero[1], M Rodríguez Calvo[1]. *[1]Components preparation, Centro de Transfusión de Galicia, Santiago de Compostela, Spain*

Background/Case Studies: Last year this transfusion center collected approximately 108,000 whole blood (WB) donations and 7,000 from

apheresis. WB is the raw material for production of labile blood products, necessary to cover the regional transfusion requirements. In 2012, 109,584 red blood concentrates (RBCC), 17,932 plasma units (inactivated by methylene blue), 6,515 platelet concentrates from apheresis, and 8,053 buffy-coat derived platelet concentrates (BCPC) were issued to hospitals. All platelet products are inactivated by the INTERCEPT®technology. All labile blood products (LBP) are leukodepleted. It was attempted to provide clinicians with standardized LBP, thus WB collection, blood components production, and preparation are checked and quality control were performed in different phases. **Study Design/Methods:** Two types of WB bag systems (MacoPharma) are used one, with an integrated filter for WB 1) and the other top and bottom configuration 2) with an integrated filter for leukodepletion of RBCC. In general WB from 1) is filtered and separated after 5-8 hours from donation. All WB from 2), more than 60% WB collected, was stored overnight in a temperature controlled room (22 ± 2 C). In the morning (15 to 18 hours from donation) WB is centrifuged at 4497 g for 18 minutes. After, WB is separated on Compomat G-4 (Fresenius). It was changed to Compomat G-5 last year. In this way, it is possible to obtain RBCC, plasma and buffy coat (BC) with a volume 52 ± 3 ml and a haematocrit of 37 ± 3%, these parameters of BC are critical to pooling BC (5BC) preparation with OrbiSac system (Terumo BCT). **Results/Findings:** From 2010-2012 the production of blood components from WB only was registered, number of units tested for quality control. The automation offered by some systems during component preparation eliminates the deviation caused by manual operations and increases productivity and yield of the platelet preparation/treatment process. **Conclusion:** Standardization of WB processing procedure allows the ability to obtain final blood components containing a reproducible quantity of active principles that meet the required standards. The results are consistent and continuous, thus facilitating therapeutic use of blood components and also making it easier to introduce new processes and technologies.

Disclosure of Commercial Conflict of Interest

C. Arcas Otero: Nothing to disclose; A. Castrillo Fernández: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arcas Otero: Nothing to disclose; A. Castrillo Fernández: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

	RBCC			BCPCs (Intercept®treated)			Plasma (MB treated)			
	Units tested	Hb (gr/u)	Hto (%)	Units tested	PQx $10^{11/unit}$	% Units $>3 \times 10^{11}$	Units tested	F.VIIIc (%)	F.V (%)	Fibrinogeno (mg/dl)
2010 RBCC: 108,269 BCPCs: 7,654 Plasma MB: 16,033	(a): 1,293 (b): 1,375	53.6 ± 6.8 52.1 ± 6.3	58.7 ± 6.5 60.6 ± 3.2	905	3.45 ± 0.4	88%	400	84.4 ± 26.8	71.5 ± 20.4	189 ± 44
2011 RBCC: 107,411 BCPCs: 7,919 Plasma MB: 14,727	(a): 1,287 (b): 1,410	53 ± 6.1 51.7 ± 5.7	58.2 ± 3.6 61.1 ± 2.4	1,116	3.42 ± 0.4	89%	423	86.7 ± 26.8	73.5 ± 19.8	199.5 ± 45.4
2012 RBCC: 105,861 BCPCs: 8,286 Plasma MB: 17,112	(a): 1,507 (b): 1,795	55 ± 6.3 51.7 ± 5.7	58.6 ± 3.9 61.1 ± 7.4	1,320	3.42 ± 0.4	88.4%	378	92.5 ± 27.7	76.1 ± 18.9	196.5 ± 43.3

SP74

Improving the Efficacy and Safety of Platelet Transfusion: Novel Naturally Occurring and Physiological Agents as Additives to Mitigate the Platelet Storage Lesion

S L Spinelli[1] (sherry_spinelli@urmc.rochester.edu), A E Casey[3], K L Lannan[2], C N Morrell[4], N Blumberg[1], R P Phipps[3]. *[1]Pathology and Laboratory Medicine, University of Rochester Medical Center, Rochester, NY, United States; [2]Microbiology and Immunology, University of Rochester Medical Center, Rochester, NY, United States; [3]Environmental Medicine, University of Rochester Medical Center, Rochester, NY, United States; [4]Medicine, University of Rochester Medical Center, Rochester, NY, United States*

Background/Case Studies: Platelet transfusion is the most widely used therapy for bleeding due to thrombocytopenia or platelet dysfunction. Poorly understood mechanisms that occur during platelet storage, termed the "platelet storage lesion", reduce platelet transfusion efficacy. Recent studies support the idea that platelet transfusions can be associated with morbidity and mortality, likely through proinflammatory and prothrombotic mechanisms. This laboratory discovered that platelets abundantly express transcription factors, especially peroxisome proliferator activated receptor gamma (PPARγ). Synthetic (*i.e.* thiazolidinediones) and natural (endogenously produced and metabolized) PPARγ ligands reversibly inhibit platelet activation in vitro and in vivo with no increase in bleeding. Many natural, dietarily ubiquitous molecules (*i.e.* omega fatty acids) are PPARγ ligands. The utility of using these dietary agents as storage additives to improve platelet function and potentially reduce morbidity due to transfusion was investigated. **Study Design/Methods:** Compounds were selected that have known beneficial properties with respect to inflammation, immune response, and in disease prevention. Human platelets for in vitro studies were collected from healthy, normal donors, leukoreduced, and stored compliant with the American Association of Blood Banks (AABB) and FDA standards, with and without test compounds. Platelet function was assessed using a variety of assays. Mouse platelets were collected, and stored similarly to human platelet concentrates. Stored mouse platelets were transfused to recipient mice and thrombus formation was measured. **Results/Findings:** A single addition of naturally occurring PPARγ ligands (ie α-linolenic acid) or structurally similar natural compounds (ie resveratrol) and synthetic PPARγ ligands (ie pioglitazone) reversibly inhibit platelet functions, such as mediator release (sCD40L, TXB_2, PF4) ($n = 3$, $p < 0.05$), surface marker expression (CD62P) ($n = 3$, $p < 0.05$), and platelet spreading on fibrinogen. Mouse platelets stored 24 hours increased thrombus formation, compared to fresh platelets ($n = 5$, $p < 0.03$), and this effect was mitigated after transfusion of resveratrol stored platelets. **Conclusion:** Additives of naturally occurring, physiologic PPARγ ligands can minimize activation during platelet storage and maximize platelet hemostatic function following transfusion. Maintaining platelets in a closer to normal state has the potential to enhance the benefits and reduce the risks of platelet transfusion therapy.

Disclosure of Commercial Conflict of Interest

N. Blumberg: Nothing to disclose; A. E. Casey: Nothing to disclose; K. L. Lannan: Nothing to disclose; C. N. Morrell: Nothing to disclose; R. P. Phipps: Nothing to disclose; S. L. Spinelli: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Blumberg: British American Tobacco Co., Grants or Research Support, Terumo BCT, Consulting or Board of Director Fees, Antek, Inc., Consulting or Board of Director Fees; A. E. Casey: BAT, Grants or Research Support; K. L. Lannan: Nothing to disclose; C. N. Morrell: Nothing to disclose; R. P. Phipps: Nothing to disclose; S. L. Spinelli: British American Tobacco Co., Grants or Research Support

SP75

Evaluation of Microparticle and Cytokine Production in Anaerobically Stored Packed Red Blood Cells Versus Standard Cold Storage

C M van Buskirk[1] (vanbuskirk.camille@mayo.edu), J Tarara[2], M Behrens[3], T Yoshida[4], W Jy[5]. *[1]Lab Medicine and Pathology, Division of Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Biochemistry and Molecular Biology, Mayo Clinic, Rochester, MN, United States; [3]Immunology, Mayo Clinic, Rochester, MN, United States; [4]Research and Development, Hemanext, New Health Sciences, Inc., Bethseda, MD, United States; [5]Wallace H. Coulter Platelet Laboratory, Department of Medicine, University of Miami School of Medicine, Miami, FL, United States*

Background/Case Studies: Controversy exists in the literature concerning potential negative clinical outcomes associated with transfusion of older blood. Packed red blood cell (PRBC) oxidative damage appears to play a major role in a series of events known as the "storage lesion." PRBC membrane changes, including red cell microparticle (RMP) formation with biochemical changes occur during storage, even in leukocyte reduced PRBCs.* **Study Design/Methods:** 15 units of whole blood were collected at

the Mayo Clinic Rochester Blood Donor Center from normal healthy volunteers utilizing a Pall Corporation Leukotrap® WB 500 mL collection bag set containing 70 mL CP2D (Pall Corporation, East Hills, NY). Within 1 hour the whole blood was passed through an in-line leucoreduction filter and centrifuged at 4563 × g for 5 minutes. Plasma was expressed. The unit was divided into 2 equal units in 600 mL storage bags. OFAS3 additive† was added to one unit (test), AS3 was added to the other (control). The set of test units were depleted of oxygen pre-storage using Ar gas exchange (in Ar-filled 600 mL bag, 10 minute agitation in a platelet mixer, for 6 exchanges) and stored under Ar atmosphere.‡ At weeks 0, 1, 2, 3, and 6, samples were removed from the PRBC units. RMPs were isolated from each sample by transferring 5 mL of stored PRBCs into 15 mL conical tubes. Samples were diluted with 10 mL isotonic saline (4 C). Samples were mixed for 30 seconds, then centrifuged at 250 × g for 10 minutes. Supernatants were transferred to 15 mL conical tubes for RMP tagging. Two fluorescent markers were utilized: Glycophorin A with phycoerythrin (PE) and Annexin V with fluorescein isothiocyanate (FITC). After a 20-minute incubation in the dark, analysis was performed using a FACS Canto Flow Cytometer. RMP data was captured as the number of events counted per 1 minute and normalized by total hemoglobin concentration of blood units. Plasma samples were frozen for single-batch testing for 22 cytokines using the Procarta Immunoassay Magnetic Bead kit. **Results/Findings:** Large variation was observed in RMP production rates among blood donors throughout the entire storage period. Generally, the anaerobic arm exhibited lower values compared to the conventional control arm during 2-6 weeks of storage (minimum of 80% of units in anaerobic arm had lower RMP). A statistical significance was achieved at 3 weeks of storage for GPA($p = 1.5 \times 10\text{-}6$ control—332 ± 168; test—160 ± 101 MPs events/min/g Hb), and Annexin ($p = 2 \times 10\text{-}5$, control—95.0 ± 31.8; test—47.0 ± 14.3). Cytokine data is currently under analysis. **Conclusion:** Anaerobic storage of PRBCs decreases the number of RMPs released during storage. This modification in the method of PRBC storage may improve clinical outcomes to transfusion of banked PRBCs.

*Transfusion.2009 Sep; 49(Suppl 3): 102A-103A
†Transfusion. 2008; 49: 458-64.
‡Vox Sang. 2007; 93: 184.

Disclosure of Commercial Conflict of Interest

M. Behrens: No Answer; W. Jy: Nothing to disclose; J. Tarara: Nothing to disclose; C. M. van Buskirk: Nothing to disclose; T. Yoshida: New Health Sciences Inc., Grants or Research Support, Other

Disclosure of Grants Conflict of Interest

M. Behrens: No Answer; W. Jy: Nothing to disclose; J. Tarara: Nothing to disclose; C. M. van Buskirk: Nothing to disclose; T. Yoshida: Nothing to disclose

SP76

Plasma Protein Analysis with Different Coagulation Analyzers

M Miklauz Hudziec[1] (meghan.hudziec@terumobct.com), J Mundt[1], E Hansen[1], S Marschner[1], S Lacomble[1], R Goodrich[1]. *[1]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Several Pathogen Reduction Technology (PRT) Systems, including the Mirasol System, are available to inactivate a wide range of pathogens in plasma products. In order to evaluate the effect of PRT on plasma quality, protein analysis is performed and results compared to applicable guidelines. A variety of coagulation (coag) analyzers are commercially available utilizing optical, mechanical, and chromogenic methodologies. Optical and chromogenic methods utilize different wavelengths of light for clot detection; mechanical methods detect change in sample turbidity as a clot is formed. The objective of this study was to evaluate variability between coag analyzers, by analyzing paired plasma samples at different sites. **Study Design/Methods:** Two studies were performed, each comparing three coag analyzers at different sites. Apheresis plasma samples were analyzed for Fibrinogen (FIB), and Factors V (FV), VIII (FVIII), and XI (FXI). The Siemens BCS (B; 340-570λ) and the Instrumentation Laboratory ACL TOP (A; 671λ) optical analyzers were evaluated. The Diagnostica Stago STA Compact (S) was used to mechanically quantify FIB, FV, and FXI, and to chromogenically (405λ) measure FVIIIc. Different sample dilutions and sample handling techniques were used at each site per standard protocols. Paired t-tests were performed on the data generated from each analyzer. **Results/Findings:** The data generated using paired samples for Study 1 and 2 is shown in the table below. Overall, the data generated was significantly different (SD) between analyzers. A single instrument or methodology did not consistently generate higher or lower protein concentrations. FVIII and FV show the most variability between coag analyzers, and FXI the least. **Conclusion:** Variability exists between plasma protein results obtained on different coag analyzers. Clotting factors FVIII and Fibrinogen must meet regional guidelines yet different equipment can generate results from a replicate sample ranging from 134-226 % activity and 294-366 mg/dL, respectively. This variability may be due to a combination of factors including differences in instrument methodologies, calibration curves (established for clinical sample analysis), reagents, sample dilutions, and sample handling. Although protein analysis is necessary to evaluate the impact of PRT on plasma quality, the effects of other assay variables need to be considered.

Disclosure of Commercial Conflict of Interest

R. Goodrich: Terumo BCT, Other; E. Hansen: Terumo BCT, Other; S. Lacomble: Nothing to disclose; S. Marschner: Employee of Terumo BCT, Other; M. Miklauz Hudziec: Terumo BCT, Ownership or Partnership; J. Mundt: Terumo BCT, Ownership or Partnership

Disclosure of Grants Conflict of Interest

R. Goodrich: Terumo BCT, Other; E. Hansen: Terumo BCT, Other; S. Lacomble: Nothing to disclose; S. Marschner: Employee of Terumo BCT, Other; M. Miklauz Hudziec: Nothing to disclose; J. Mundt: Terumo BCT, Ownership or Partnership

(N = 7) Control Plasma Protein Concentrations: Average(Min/Max)

Study	Instrument	FV	FVIII	FXI	FIB
			% Activity		mg/dL
1	A	92(83/105)^{S}	111(68/162)^{B}	107(77/128)	*
	B	*	96(62/134)$^{A/S}$	103(81/115)	279(259/305)
	S	84(75/93)^{A}	132(72/226)^{B}	120(87/153)	274(240/305)
2	A	90(70/112)$^{B/S}$	90(70/109)^{B}	100(89/117)	233(183/303)$^{B/S}$
	B	106(81/125)$^{A/S}$	113(94/135)$^{A/S}$	99(83/112)	282(220/365)^{A}
	S	95(76/111)$^{A/B}$	94(79/115)^{B}	98(81/108)	291(215/366)^{A}

* Not tested A/B/S -SD (p-value < 0.05)

SP77

In Vivo Evaluation of a New Red Blood Cell Leukoreduction Filter

S Heldke[1] (Sybil.Heldke@bcw.edu), S Graminske[1], R Dora[1], S Brooks[1], J Gottschall[2], M Einarson[3], R Vassallo[3], J G Nixon[3], D Ziegler[3], J Skinner[4]. *[1]Applied Research Laboratory, BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Medical Sciences Institute, BloodCenter of Wisconsin, Milwaukee, WI, United States; [3]Research, American Red Cross, Penn-Jersey Region, Philadelphia, PA, United States; [4]MacoPharma, Duluth, GA, United States*

Background/Case Studies: A new in-line filter system for leukoreduction of red blood cells (RBCs) was evaluated for leukoreduction and RBC recovery performance, as well as in vivo 24-hour autologous radiolabeled recovery of the RBCs after 42/43 days of 1-6 °C storage. The Leucoflex® System (MacoPharma USA) consists of a collection bag containing 70 mL of CPD, an in-line CGP filter, a storage bag containing AS-1 for RBC storage after filtration, and a bag for plasma storage. **Study Design/Methods:** The in vivo recovery study was conducted at two sites in the US. Each subject (N = 24) donated 500 ± 50 mL whole blood (WB) collected in the new system. WB was stored at 1-6 C for 20-24 hours before component processing and RBC filtration. Standard in vitro testing of these RBC units was completed both day 1 and day 42/43 of 1-6 C storage. At the end of storage, a sample was aseptically removed from each RBC unit, labeled with a radionuclide (^{51}Cr), and reinfused into the donor. Blood samples were obtained at specific intervals from 5-30 minutes after the infusion. Each donor returned at 24 hours ± 2 hours for a final blood sample to determine the 24-hour % recovery of ^{51}Cr-labeled RBCs. Day 43 infusions and testing were allowed in the event of a holiday or donor scheduling conflict. Five to ten days prior to each infusion, aseptically extracted samples were submitted for bacterial culture to verify lack of contamination. **Results/Findings:** The mean residual WBC of the 24 units was $0.17 \pm 0.16 \times 10^6$/unit, the post-filtration RBC recovery calculated by weight and hematocrit was 91.69% ± 0.02%, and the mean RBC 24-hour radiolabeled RBC recovery on Day 42/43 was 87% ± 6%. No units showed bacterial contamination. All other end of storage in vitro parameters were within accepted limits. **Conclusion:** All units met the FDA requirements for 24-hour in vivo recovery (≥75%, SD ≤9%, 95% lower confidence level for proportion of successful recoveries >70%), leukoreduction efficacy ($\leq 5 \times 10^6$ WBC/unit), and post-filtration recovery of RBCs (≥85%), and are therefore, suitable for transfusion.

Disclosure of Commercial Conflict of Interest

S. Brooks: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; R. Dora:

GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; M. Einarson: MacoPharma, Grants or Research Support, OriGen Biomedical, Grants or Research Support; J. Gottschall: Cerus, Other, Hemonetics, Other, Fenwal, Other, MacoPharma, Other; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; S. Heldke: MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Masimo, Grants or Research Support, Cerus, Grants or Research Support, GenesisBPS, Grants or Research Support; J. G. Nixon: MacoPharma,Grants or Research Support, Origen, Grants or Research Support; J. Skinner: Nothing to disclose; R. Vassallo: MacoPharma, Grants or Research Support; D. Ziegler: MacoPharma, Grants or Research Support, Origen, Grants or Research Support

Disclosure of Grants Conflict of Interest

S. Brooks: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Suppor, tCerus, Grants or Research Support; R. Dora: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; M. Einarson: Fenwal, Grants or Research Support; J. Gottschall: Nothing to disclose; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus Corporation, Grants or Research Support; S. Heldke: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; J. G. Nixon: Fenwal, Grants or Research Support; J. Skinner: Nothing to disclose; R. Vassallo: Fenwal, Grants or Research Support; D. Ziegler: Fenwal, Grants or Research Support

End of Storage Data (Day 42/43)

Parameter	Mean ± SD (N = 24)
Hemolysis (%)	0.21 ± 0.10
24-Hour Radiolabeled Recovery (%)	87 ± 6
pH (22 °C)	6.7 ± 0.1
Potassium (mmol/L)	45 ± 4
ATP (% of Day 1)	77.4 ± 8.6

SP78

Effect of Albumin and Fatty-Acids on Mechanical Fragility of Red Blood Cells

M Tarasev[1] (mtarasev@blazemedicaldevices.com), S Chakraborty[1], R Davenport[2]. *[1]Blaze Medical Device, LLC., Ann Arbor, MI, United States; [2]Pathology, University of Michigan, Ann Arbor, MI, United States*

Background/Case Studies: Red Blood Cell (RBC) ability to withstand mechanical stress without hemolysis is critical for cell survival and microcirculation. Such ability may be compromised by RBC storage, and may also be affected by intrinsic properties of donors as well as by those of patients receiving the transfusion. Mechanical fragility (MF) profiling provides a potentially useful way to test the impact of various factors, such as the cells' biochemical environment, on pRBC membrane fragility. **Study Design/Methods:** pRBC and plasma samples were from the University of Michigan Hospital Blood Bank. Essentially fatty-acid-free plasma was obtained by ultracentrifugation for 1 h at 100,000 g. pRBC stored in AS3 additive solution was diluted with compatible plasma or AS3 supplemented with albumin and polyethylene glycol, and subjected to mechanical stress via bead mill (oscillation 50 Hz; durations from 15 s to 60 m). MF profiles were plotted as percent hemolysis versus stress duration. **Results/Findings:** RBC MF significantly depended on albumin concentration in additive storage solution (AS), with minimum MF (maximum resistance to hemolysis) in the range of 2-10 g/dl. Increased MF at higher (40 g/dl) levels was partially due to potential osmotic and rheological effects as demonstrated by profiles collected in the presence PEG. The majority of albumin "protection" from hemolysis was likely due to protein-cell interaction, a conclusion supported by the difference in RBC MF response to changes in pH: while RBC MF measured in AS alone was not affected by such changes, in AS w/ albumin it exhibited marked dependence on pH and was linked to albumin conformation tracked through intrinsic albumin Trp/Tyr fluorescence. Overall, lower pH values corresponded with RBC higher resistance to mechanical stress. At physiological concentrations, albumin provided protection for RBC, reducing MF due to effects related to its presence in the media both prior to and during stress application. This protection was mostly negated in normal plasma, at least partially due to the presence of un-dissolved fatty acids. The negative impact of free fatty-acids on MF was significantly reduced if they were removed from plasma prior to the stressing. **Conclusion:** Blood composition and in particular its fatty-acid content can significantly affect RBC ability to withstand mechanical stress, and by extension survive in the bloodstream post-transfusion. It is proposed that the additional presence of cholesterol and triglycerides, also shown to negatively affect red cell fragility, accounted for higher MF of pRBC in plasma than in albumin. While further study is warranted, this data suggests a possible way to improve the quality of donated red cells and the survivability of transfused pRBC through monitoring and/or adjustment (eg pharmaceutically or dietarily) of fatty-acid content in donors and/or patients.

Disclosure of Commercial Conflict of Interest

S. Chakraborty: Nothing to disclose; R. Davenport: Nothing to disclose; M. Tarasev: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Chakraborty: Nothing to disclose; R. Davenport: Nothing to disclose; M. Tarasev: Blaze Medical Devices, LLC.,Ownership or Partnership

SP79

In Vitro and In Vivo Biochemical and Vasoactive Properties of Novel Hemoglobin-Loaded Synthetic Red Blood Cell-Like Hydrogel Particles

K Chen[1] (kaic@email.unc.edu), J Luft[1], J S Raval[2], J M DeSimone[1]. *[1]Chemistry, University of North Carolina, Chapel Hill, NC, United States; [2]Pathology and Laboratory Medicine, University of North Carolina, Chapel Hill, NC, United States*

Background/Case Studies: Red blood cells (RBCs) are a precious resource with small but measurable risks after donation and transfusion. To address the need for a safe, effective, shelf-stable, and non-immunogenic synthetic RBC substitute, the exacting fabrication technology of soft lithography has been adapted to mold RBC-like hydrogel particles (mimics) with the same size, shape, and elasticity of RBCs. These mimics have also been successfully loaded with bovine hemoglobin (Hb) in a reduced state. We sought to characterize various in vitro and in vivo biochemical and vasoactive properties of these novel mimics. **Study Design/Methods:** Hb-loaded mimics were evaluated for a variety of in vitro and in vivo (murine model) parameters compared against free Hb, including: stopped-flow spectrophotometric assessment of oxygen dissociation at 436 nm and nitric oxide dioxygenation at 419 nm; percent vasoconstriction as observed by fluorescent intravital microscopy; light microscopy evaluation by Prussian-blue staining for hepatic and renal iron deposition; hemolysis index via automated blood analyzer; lactate dehydrogenase (LDH); aspartate transaminase (AST); and creatinine (Cr). N = 3 for all in vivo data. **Results/Findings:** See Table. **Conclusion:** The novel Hb-loaded mimics have in vitro and in vivo characteristics which suggest that they are suitable for further exploration as a synthetic RBC substitute. Future in vivo studies evaluating safety and efficacy of oxygen delivery are planned and will further characterize these unique synthetic particles. The transfusion medicine community should continue to support research in artificial oxygen carriers, as they may benefit unique patient populations such as injured military personnel, individuals significantly allo-immunized against RBC antigens, and Jehovah's Witnesses.

Disclosure of Commercial Conflict of Interest

K. Chen: Liquidia Technologies, Grants or Research Support; J. M. DeSimone: Nothing to disclose; J. Luft: Nothing to disclose; J. S. Raval: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Chen: Liquidia Technologies, Grants or Research Support; J. M. DeSimone: Nothing to disclose; J. Luft: Liquidia Technologies, Grants or Research Support; J. S. Raval: Nothing to disclose

	Oxygen dissociation (1/sec)	Nitric oxide dioxygenation (1/μM*sec)	Percent vasoconstriction (%)	Hepatic iron	Renal iron	Hemolysis index	LDH (U/L)	AST (U/L)	Cr (mg/dL)
Hb-loaded mimics	16	7.4	3 ± 1	No increase vs. normal	No increase vs. normal	101 ± 42	1313 ± 169	68 ± 7	0.1 ± 0.0
Free Hb	23	18	25 ± 4	No increase vs. normal	No increase vs. normal	208 ± 49	2300 ± 363	122 ± 17	0.13 ± 0.06
P-value (T-test)	NA	NA	0.0008	NA	NA	0.10	0.03	0.02	0.42

SP80

Detecting Microparticles in Blood Components—A Potential Solution

A Labrie[2] (alabrie@lightintegra.com), J Leung[1], D Crettaz[3], M Prudent[3], N Lion[3], E Maurer-Spurej[1,2]. *[1]Pathology & Laboratory Medicine, University of British Columbia, Vancouver, BC, Canada; [2]LightIntegra Technology Inc., Vancouver, BC, Canada; [3]Service Régional Vaudois de Transfusion Sanguine, Epalinges, Switzerland*

Background/Case Studies: Many studies show that microparticles (MP) co-transfused with blood products result in adverse effects. Literature supports the negative effects of blood components containing microparticles. However, the factors contributing to MP formation and the role MPs play in recipient's reactions to transfusion are still unknown. Currently the most common methods to investigate microparticles are flow cytometry (FC) and proteomics. These are not always practical or feasible as they are time consuming and require invasive procedures. A major limitation of FC is that it fails to detect MPs smaller than 200 nm. Thus a non-invasive, routine test for MPs in blood components is needed to investigate their role in transfusion medicine. **Study Design/Methods:** ThromboLUX, a non-invasive test based on dynamic light scattering, was used to investigate MPs. Small, 100 μL samples of platelet concentrates, platelet-rich plasma or leukoreduced, SAGM-stored erythrocyte concentrates subjected to centrifugation and washing steps were loaded into capillaries. The Doppler shift of the light scattered by the suspended particles was analyzed to determine the MP size and content. **Results/Findings:** ThromboLUX was able to detect MP populations in a size range of 20 to 1100 nm and discriminate them from larger reference particles in the same sample, such as platelets or calibrated 1900 nm Polystyrene microspheres (Polysciences Inc, Warrington, PA). The average hydrodynamic radius R_H of erythrocyte-derived MPs was found to be 100 ± 16 nm (N = 29) which is smaller and more homogeneous than the previously reported size of platelet-derived MPs of R_H = 117 ± 78 nm. Chylomicrons in lipemic plasma coincided with both MP populations due to their size of R_H = 110 ± 20 nm. The presence of known concentrations of reference particles, such as platelets in platelet components, allowed the calculation of the MP concentration by intra-sample comparison. MP concentrations calculated from dynamic light scattering data correlated with concentrations determined by FC (Pearson's r = 0.94). **Conclusion:** Several studies showed that MPs in blood and blood components could be mediators or markers of disease and have prognostic potential. Transfusion of blood components with high MP loads (cell-derived or lipid) can cause severe complications in patients, however, measuring the MP load prior to transfusion has been challenging. ThromboLUX has the potential as a routine, non-invasive test to measure MPs in blood components.

Disclosure of Commercial Conflict of Interest

D. Crettaz: No Answer; A. Labrie: LightIntegra Technology, Other; J. Leung: LightIntegra Technology, Other; N. Lion: No Answer; E. Maurer-Spurej: LightIntegra Technology, Ownership or Partnership; M. Prudent: No Answer

Disclosure of Grants Conflict of Interest

D. Crettaz: No Answer; A. Labrie: LightIntegra Technology, Other; J. Leung: Nothing to disclose; N. Lion: No Answer; E. Maurer-Spurej: LightIntegra Technology,Ownership or Partnership; M. Prudent: No Answer

SP81

Improved Recovery of ATP and 2,3 DPG with Agitation during Incubation at 37 C in rejuvesol® Solution Treated CPD/AS-1 RBC

P H Whitley[1] (Pam.Whitley@redcross.org), S Sawyer[1], M Wellington[1], S Hanley[1], L Y Maes[1], L Herschel[2], A D Gray[3], L J Dumont[2]. *[1]American Red Cross Research, Mid-Atlantic Region, Norfolk, VA, United States; [2]Geisel School of Medicine at Dartmouth, Lebanon, NH, United States; [3]Citra Labs, LLC, Braintree, MA, United States*

Background/Case Studies: The addition of rejuvesol® Red Blood Cell (RBC) Processing Solution (a solution containing pyruvate, inosine, adenine and phosphate, Citra Labs, LLC, Braintree, MA) to RBCs, has been shown to increase ATP and 2,3 DPG levels to normal values. Although the study objective was to demonstrate that a manufacturing change had no effect on the rejuvesol® Solution, agitation of the rejuvenated RBC during incubation at 37 C appeared to improve recovery of ATP and 2,3 DPG. **Study Design/Methods:** Whole blood (WB) was collected (550 mL) from 60 healthy research subjects at two sites into the Fenwal Sepacell Flex-Excel System, hard spun, and processed at room temperature into CPD/AS-1 RBC. After 42 days (T42) of storage at 1-6 C, rejuvesol® Solution was added and incubated at 37 C in a circulating water bath for 1 hour; Site 1 used a Lindberg/Blue M Water Bath (no agitation) and Site 2 used a Helmer Water bath (with agitation). The RBC was glycerolized (40% w/v) and frozen (–80 C) in accordance with the Meryman method for at least 7 days prior to deglycerolization using the COBE Blood Cell Processor 2991. The RBC was then stored at 1-6 C in 0.9% saline 0.2% dextrose solution for an additional 24 hours. Testing was performed at day 0, T42, pre and post rejuvenation (RJV), post deglycerolization (DG) and after 24 hrs Post-DG (24 hr) for CBC, Na+, K+, supernatant hemoglobin, ATP, 2,3-DPG, osmolality and residual levels of hypoxanthine and inosine. **Results/Findings:** The mean in vitro % recovery values for DG RBCs were similar at both Site 1, 89.7%, and Site 2, 93.8%; and similar for CBC, Na+, K+, % hemolysis, osmolality, and residual levels of hypoxanthine and inosine. However, a significant difference ($p < 0.05$) was observed in the 2,3 DPG and ATP values immediately after rejuvenation, which was attributed to the agitation of the RBC/rejuvesol® Solution mixture during incubation process at Site 2; (Table). **Conclusion:** The rejuvenation process gave the expected results demonstrating excellent recovery of ATP, 2,3 DPG. These results indicate that RBC agitation during the incubation process may enhance the uptake and enzymatic conversion of rejuvesol® Solution, further improving ATP and 2,3 DPG levels.

Disclosure of Commercial Conflict of Interest

L. J. Dumont: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Consulting or Board of Director Fees; A. D. Gray: Citra Labs, Grants or Research Support; S. Hanley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. Herschel: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support; L. Y. Maes: Haemonetics, Grants or Research Support, Masimo, Grants or Research Support; S. Sawyer: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; M. Wellington: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; P. H. Whitley: Citra Labs, Grants or Research Support

Disclosure of Grants Conflict of Interest

L. J. Dumont: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, New Health Sciences, Grants or Research Support, Terumo BCT, Grants or Research Support, Advance Cell Technologies, Consulting or Board of Director Fees, Advanced Preservation Technologies, Consulting or Board of Director FeesNew , Health Sciences, Consulting or Board of Director Fees, Terumo BCT, Consulting or Board of Director Fees; A. D. Gray: Citra Labs, Grants or Research Support; S. Hanley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; L. Herschel: Citra Labs, Grants or Research Support, Advanced Preservation Technologies, Grants or Research Support, Hemerus Medical, Grants or Research Support, New Health Sciences, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Terumo BCT, Grants or Research Support, SynGen, Grants or Research Support; L. Y. Maes: Haemonetics, Grants or Research Support; S. Sawyer: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support; M. Wellington: Nothing to disclose; P. H. Whitley: Citra Labs, Grants or Research Support, Haemonetics, Grants or Research Support

TABLE 1. Mean (SD) ATP and 2,3 DPG for RJV RBC's (Site 1 = w/o Agitation, Site 2 = w/ Agitation)

	Day 0		Day 42		Post RJV		Post DG		24 hr	
TEST Mean (SD)	Site 1	Site 2	Site 1	Site 2	Site 1	Site 2	Site 1	Site 2	Site 1	Site 2
ATP (umol/gHb)	4.8* (0.5)	4.2 (0.7)	3.4* (0.6)	2.9 (0.5)	6.0 (0.9)	7.9** (1.4)	6.8 (0.7)	7.3** (1.4)	6.3 (0.8)	7.7** (1.1)
ATP (% of Day0)	100	100	70	69	126	193	143	177	131	188
2,3 DPG (umol/gHb)	13.1 (1.8)	13.4 (2.0)	0.7* (0.5)	0.4 (0.2)	6.8 (2.0)	10.6** (2.4)	15.7 (2.2)	16.2 (2.1)	14.7 (2.3)	18.5** (3.1)
2,3 DPG (% of Day0)	100	100	5.1	3.2	53	81	121	124	113	142

* Site 1 or ** Site 2 significantly higher ($p < 0.05$)

SP82

ALT-711, which Breaks Advanced Glycation End Products, Improves Murine RBC Storage

T Wang[1] (tina.wang@yale.edu), T Kim[1], T Padukkavidana[1], N Smith[2], S R Stowell[2], S Eisenbarth[3], J C Zimring[4], D A Spiegel[1], J E Hendrickson[2]. *[1]Chemistry, Yale University, New Haven, CT, United States; [2]Pediatrics and Pathology, Emory, Atlanta, GA, United States; [3]Immunobiology, Yale University, New Haven, CT, United States; [4]Puget Sound Blood Center, University of Washington, Seattle, WA, United States*

Background/Case Studies: Advanced glycation end products (AGEs), or amadori reaction products of glucose and proteins, have been found on both murine and human RBCs. This accumulation of AGEs has been shown to occur over the lifespan of RBCs in vivo, as well as during storage ex vivo in glucose containing anticoagulant preservative solutions. AGEs are known to be detrimental in non-transfusion settings, and it is hypothesized that AGE accumulation on RBCs may likewise be detrimental in transfusion settings. **Study Design/Methods:** Blood from transgenic murine blood donors expressing the HOD antigen on their RBCs was collected into CPDA-1 and leukoreduced. The AGE crosslink breaker ALT-711 or a placebo control compound were generated in the laboratory and added to the blood prior to storage, with analyses completed over a 14 day storage period. Pre-transfusion studies included analysis of percent hemolysis, AGEs by Western blot using polyclonal anti-AGE reagents, and AGE precursors by a flow cytometric dicarbonyl assay. Post-transfusion studies in recipients included flow analysis of RBC survival, serum cytokines, and anti-HOD alloimmunization. **Results/Findings:** In a compilation of five independent experiments, RBCs stored with ALT for 14 days had less free heme by cyanomethemoglobin assay (mean 2.2 fold less than control RBCs). Likewise, RBCs stored with ALT had improved post-transfusion recovery (mean 60% more transfused RBCs remained in circulation at 24 hours compared to control). In 2 of 3 experiments, recipients transfused with RBCs stored in ALT also had statistically significant decreases in serum KC and MCP-1 compared to those transfused with RBCs stored in the absence of ALT. Likewise, RBCs stored in ALT were less immunogenic, leading to decreased degrees of anti-HOD alloimmunization. AGEs and AGE precursors on HOD RBCs increased from the time of collection through day 14 of storage by both Western and dicarbonyl assays, with blood stored in ALT but not placebo having decreased detectable dicarbonyl AGE precursors. **Conclusion:** Analagous to what has been described for human RBCs, AGEs increase with storage duration on murine HOD.FVB RBCs. The AGE crosslink breaker ALT-711, which has recently been shown to minimize cardiovascular complications from diabetes, also improves RBC storage parameters in HOD.FVB RBCs. The decrease in hemolysis observed over storage duration is presumably due to ALT having an effect on AGE generation, given the decrease noted in detectable AGE dicarbonyl precursors and given that similar effects were not observed in the presence of RBCs stored with a functionally inactive placebo compound. However, existing reagents to measure AGEs are limited, and other mechanisms of action of ALT cannot be ruled out. Ongoing experiments are investigating whether the observed effects of ALT are generalizable to other murine strains and to human RBCs.

Disclosure of Commercial Conflict of Interest

S. Eisenbarth: Nothing to disclose; J. E. Hendrickson: Nothing to disclose; T. Kim: Nothing to disclose; T. Padukkavidana: Nothing to disclose; N. Smith: Nothing to disclose; D. A. Spiegel: No Answer; S. R. Stowell: Nothing to disclose; T. Wang: Nothing to disclose; J. C. Zimring: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Eisenbarth: Nothing to disclose; J. E. Hendrickson: Terumo BCT,Grants or Research Support; T. Kim: Nothing to disclose; T. Padukkavidana: Nothing to disclose; N. Smith: Nothing to disclose; D. A. Spiegel: No Answer; S. R. Stowell: Nothing to disclose; T. Wang: Nothing to disclose; J. C. Zimring: Immucor, Grants or Research Support, Terumo BCT, Grants or Research Support, Travel Support or Honorarium, Haemonetics, Consulting or Board of Director Fees, Cerus, Travel Support or Honorarium

		PC in plasma			PC in PASIII		
	Storage day	A	B	C	A	B	C
Platelets (10^3/uL)	1	359±33	359±32	356±3	317±22	317±19	327±20
Morphology score	1	237±15	288±0	290±1	313±18	313±19	316±18
	2, before hold	291±24	290±21	285±18	298±22	295±23	294±25
	2, after 6 h hold	258±24	243±13	n.d.	285±28	262±30	n.c.
	5, before hold	256±22	253±21	257±26	273±32	266±24	271±20
	5, after 6 h hold	n.d.	n.d	208±16	n.d	n.c.	251±9
	8	233±24	242±24	240±23	243±22	256±7	245±18

Table. In vitro measures of PCs with interrupted agitation (n=6, mean ± SD) n.d.: not determined

SP83

Platelet Concentrates Stored without Agitation For 6 Hours At 18 C

J W Lagerberg[1] (j.lagerberg@sanquin.nl), L de Laleijne-Liefting[1], P F van der Meer[1], D de Korte[1]. *[1]Sanquin Blood Supply, Amsterdam, Netherlands*

Background/Case Studies: Guidelines for blood transfusion prescribe a temperature 18-25 C for transport of platelet concentrates (PCs). In this blood center, the longest transportation time is less than 6 hours. To mimic worst-transport conditions, PCs were stored for 6 hours at 18 C without agitation on day 2 or day 5 of storage. As a comparison, PCs were stored without agitation on day 2 at 22 C. The aim of this study was to evaluate the effect of interruption of agitation on the in vitro quality of platelets during storage. **Study Design/Methods:** Whole blood (WB), overnight stored at ambient temperature, was processed using the buffy coat method. Pools from 5 BCs and 1 plasma unit or 280 mL of PASIII were used to prepare PCs. To allow for paired comparison, three PCs were pooled and split. PCs were stored on a flatbed shaker at 22-24 C for 8 days. Agitation was interrupted for 6 hours on: 1) day 2 at 22 C; 2) day 2 at 18 C; and 3) day 5 at 18 C. The PCs were sampled on day 1, 2, 5, and 8. Differences between groups were statistically tested using a paired Student's t-test. p-values < 0.05 were considered significant. **Results/Findings:** The results are summarized in the table. Pooling and splitting was successful, as no significant differences in platelet numbers were found on day 1. Interruption of agitation had no effect on hematologic and metabolic parameters or platelet activation. Interruption of agitation resulted in deterioration of morphology as measured by a decreased Kunicki score. While the decrease in morphology score was more pronounced at lower hold temperatures, the decrease was reversible. There were no major differences between storage in plasma and in PASIII. At day 8 of storage, platelet quality was comparable for all groups. **Conclusion:** Interruption of agitation of platelets for 6 hours at 18 C-22°C has a negative effect on platelet morphology. However, upon prolonged storage morphology is restored. Transport of platelets, without agitation for 6 hours and at the lowest allowed temperature (18 C), has no persistent negative effect on the in vitro quality of platelets.

Disclosure of Commercial Conflict of Interest

D. de Korte: Nothing to disclose; L. de Laleijne-Liefting: Nothing to disclose; J. W. Lagerberg: Nothing to disclose; P. F. van der Meer: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. de Korte: Terumo BCT, Grants or Research Support; L. de Laleijne-Liefting: Nothing to disclose; J. W. Lagerberg: Nothing to disclose; P. F. van der Meer: Nothing to disclose

SP84

Evaluation of an EVA Bag for the Storage of Frozen Red Blood Cells (RBCs)

M Dabay[1] (michelle.dabay@redcross.org), M Einarson[2], L Kline[1], J G Nixon[2], D Ziegler[2]. *[1]Holland Laboratory, American Red Cross, Rockville, MD, United States; [2]American Red Cross, Penn Jersey Region, Philadelphia, PA, United States*

Background/Case Studies: A study was performed to determine if the Cryostore™ EVA Freezing Bag (OriGen Biomedical) is an acceptable

alternative for storage of frozen RBC to the Fenwal Cryocyte Pack™. **Study Design/Methods:** Leak tests were performed by adding water to test bags at minimum and maximum freezing volumes (~745 and ~877 mL), placement into metal freezing canisters and exposure to multiple freeze-thaw cycles. For in vitro paired studies, two ABO/Rh compatible leukoreduced RBC were pooled and divided to create paired RBC. Paired units were frozen using glycerol as the cryoprotective agent in Cryostore (Test) or Cryocyte (Control) bags and frozen at ≤ –65 C for ≥ 1 week. Paired units were thawed and deglycerolized using the Cobe® 2991 cell processor and refrigerated overnight prior to testing. **Results/Findings:** The Test and Control bag fit well in metal freezing canisters; no leakage was observed following three freeze-thaw cycles. The table summarizes in vitro data collected (n = 12). Because of multiple parameters assayed in the study, $p < 0.007$ was considered significant. The post-deglycerolization mean RBC recovery in Test bags was 87% with a mean total hemoglobin of 49.1 g Hgb/unit; Control bags had a mean RBC recovery of 88% and mean total hemoglobin of 49.4 g Hgb/unit. The mean RBC recovery for Test and Control bags exceeded the acceptance criteria of ≥ 80% and total hemoglobin/unit of >42.5 g Hgb/unit for leukoreduced RBC. **Conclusion:** Results show RBC frozen in the Cyrostore™ EVA Freezing Bag, and subsequently deglycerolized, met the RBC recovery acceptance criteria of ≥80% of the preglycerolized RBC and a mean Total Hemoglobin of >42.5 grams/unit. Comparable quality parameters were obtained in the Cryostore bag compared to the Fenwal Cryocyte Pack™ Freezing Container.

Disclosure of Commercial Conflict of Interest

M. Dabay: Haemonetics, Other; M. Einarson: MacoPharma, Grants or Research Support, OriGen Biomedical, Grants or Research Support; L. Kline: MacoPharma, Grants or Research Support, Origen,Grants or Research Support, Masimo, Grants or Research Support; J. G. Nixon: MacoPharma, Grants or Research Support, Origen, Grants or Research Support; D. Ziegler: MacoPharma, Grants or Research Support, Origen, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. Dabay: Nothing to disclose; M. Einarson: Fenwal, Grants or Research Support; L. Kline: Nothing to disclose; J. G. Nixon: Fenwal, Grants or Research Support; D. Ziegler: Fenwal, Grants or Research Support

RBC Results—Mean ± 1 SD (Range)

Parameters	Test	Control	t-Test Paired
RBC Recovery (%)	86.9 ± 5.3 (77.7-93.0)	87.9 ± 4.1 (76.1-91.1)	0.54
Total Hgb/unit	49.0 ± 4.9 (41.1-58.6)	49.4 ± 3.3 (44.8-55.5)	0.73
Supernatant Hgb (mg/dL)	108.3 ± 32.1 (60.0-170.0)	112.5 ± 40.0 (60.0-180.0)	0.65
Hematocrit (%)	38.0 ± 5.4 (28.5-46.2)	37.9 ± 5.2 (30.6-46.7)	0.98
ATP (μmol/g Hgb)	5.4 ± 0.4 (4.8-6.0)	5.6 ± 0.5 (5.0-6.4)	0.02
Red Cell Morphology	89.1 ± 3.8 (80.6-95.2)	87.4 ± 5.6 (77.2-96.3)	0.15
pH (22°C)	6.79 ± 0.07 (6.64-6.92)	6.78 ± 0.06 (6.70-6.89)	0.75

N = 12

SP85

Improving Plasma Recovery by the Optimization of Whole Blood (WB) Centrifugation Process

L Gagné[2] (Louis-Philippe.Gagne@hema-quebec.qc.ca), E Fissette[2], A Jacques[1], J Dion[1], B Renaud[1], M de Grandmont[1], L Thibault[1].
[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada

Background/Case Studies: In recent years, Héma-Québec has introduced several changes in its blood processing operations. Automated extractors (MacoPharma) have now replaced manual extractors, thus improving the consistency and quality of blood products. The efficacy of these extractors is influenced by the settings applied during whole blood (WB) centrifugation prior to plasma extraction. Interestingly, our centrifugation settings have not been revised over the last 15 years. In this work, a new centrifugation time to improve plasma recovery has been revisedand validated. **Study Design/Methods:** For centrifugation settings verification, WB (450 mL) was collected in CP2D using the Pall Medical Leukotrap WB collection set. Blood from 3 ABO-compatible donors was pooled and split to obtain 3 identical WB units. After filtration, WB was centrifuged for 5, 10 or 15 min. at 5 147 × g (4 C). Quality parameters of blood products were measured (volume, yield and hemolysis) to determine the optimal centrifugation time. To validate a 10-minute centrifugation, 63 bags of whole blood were processed into blood products. Plasma extraction and AS-3 addition to red blood cell (RBC) units were achieved using automated extractors. The amount of residual plasma in RBC units was determined, based on the ratio of IgG in RBC supernatants and their respective plasma. Results were compared to blood products (n = 60) processed using our current centrifugation parameters (5 147 × g, 5 min.). **Results/Findings:** Increasing the centrifugation time from 5 minutes to 10 or 15 minutes increased plasma yield from 88% to 93% or 95%, respectively. After 42 days of storage, the hemolysis rates of RBC units were 0.2 %, 0.2 %, and 0.3 % for products centrifuged for 5, 10 and 15 minutes, respectively. By increasing centrifugation time from 5 to 10 minutes, the volume of plasma units (n = 63 per arm of the study) increased from 263 ± 23 mL to 283 ± 16 mL, for an improved yield of 20 mL per bag ($p < 0.05$). Conversely, increasing the centrifugation time has reduced the residual volume of plasma in RBCs from 39 ± 12 mL to 20 ± 5 mL. The percentage of hemolysis in RBCs complied with regulatory requirements in both centrifugation conditions (0.3% and 0.4% for 5-min. and 10-min. centrifugations). **Conclusion:** This work shows the importance of regularly reviewing the parameter settings applied to blood processing. An increase of only 5 minutes in the WB centrifugation time has little impact on operations, but significantly reduces the volume.

Disclosure of Commercial Conflict of Interest

M. de Grandmont: Nothing to disclose; J. Dion: Nothing to disclose; E. Fissette: Nothing to disclose; L. Gagné: Nothing to disclose; A. Jacques: Nothing to disclose; B. Renaud: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. de Grandmont: Nothing to disclose; J. Dion: Nothing to disclose; E. Fissette: Nothing to disclose; L. Gagné: Nothing to disclose; A. Jacques: Nothing to disclose; B. Renaud: Nothing to disclose; L. Thibault: Nothing to disclose

SP86

In Vitro Properties of Photochemically Pathogen-Reduced Treated Platelet Concentrates Suspended in SSP+ EVER for a 9-Day Storage Period

A Castrillo Fernández[1] (azucena.castrillo.fernandez@sergas.es), C Arcas Otero[1], A Díaz Pereira[1], M Adelantado[1], M Rodríguez Calvo[1].
[1]Components preparation, Centro de Transfusión de Galicia, Santiago de Compostela, Spain

Background/Case Studies: New platelet additive solutions (PAS) contain compounds that might improve the storage conditions for platelets. This study assessed in vitro function of platelet concentrate from buffy coat (BC-PC), treated with pathogen reduction technology (PRT) and stored in AS, PAS III M, or SSP+ (MacoPharma), that incorporates potassium and magnesium in relation to PAS III (InterSol). **Study Design/Methods:** Seventeen BC- PC were prepared using five standard BC products pooled together with SSP+, in the OrbiSac system (Terumo BCT). The ratio AS: plasma was 65:35 approximately.The units were treated with PRT (Intercept, Cerus Corp.). All products were stored for 9 days under standard blood banking conditions. BC-PCs were tested for platelet count, MPV (mean platelet volume), glucose, lactate, pH (at 22 C), CD62p expression, Annexin V binding and some biological response modifiers like sCD40L and sCD62p. The assessment of swirling was scored as 1, 2, and 3 (1 = no swirling; 2 = intermediate; 3 = good). **Results/Findings:** The volume and platelet per unit were 345 ± 7.8 mL and 3.5 ± 0.2 × 1011 respectively. pH was maintained more than 6.8 at day 9 and swirling remained at good level (score 3) for all units throughout storage. The residual leucocyte was determined on day 2 by flow cytometry (FC),and all BC-PCs had less than 1 × 106 WBC/unit. The results are shown on the included table. **Conclusion:** The BC-PCs treated with PRT and suspended in SSP+ can be stored for 9 days with acceptable cell quality like low glucose metabolism, low levels of spontaneous platelet activation and others metabolic and biological parameters are adequate.

Disclosure of Commercial Conflict of Interest

M. Adelantado: No Answer; C. Arcas Otero: Nothing to disclose; A. Castrillo Fernández: Nothing to disclose; A. Díaz Pereira: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Adelantado: No Answer; C. Arcas Otero: Nothing to disclose; A. Castrillo Fernández: Nothing to disclose; A. Díaz Pereira: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

Parameters	Day 2	Day 5	Day 7	Day 9
MPV (fL)	8.8 ± 0.5	8.8 ± 0.3	9 ± 0.3	9.1 ± 0.4
Glucose (mmol/L)	6.2 ± 0.6	4.9 ± 0.7	3.5 ± 0.2	1.9 ± 0.05
Lactate (mmol/L)	5.3 ± 0.1	7.9 ± 0.2	10.9 ± 0.2	14 ± 1.2
Swirling	3	3	3	3
pH (22°C)	7.13 ± 0.1	7.18 ± 0.1	7.18 ± 0.1	7.06 ± 0.1
CD62p (%)	9.3 ± 6	19.7 ± 5.9	17.2 ± 1.9	24.9 ± 10
AnnexinV (% binding)	1.8 ± 0.9	4 ± 0.8	3.1 ± 2.4	4 ± 1.5
sCD62p (ng/mL)	35.2 ± 7.2	48 ± 12	52 ± 15	78.3 ± 31
sCD40L (ng/mL)	5.6 ± 0.1	6.3 ± 3.4	9.1 ± 3.5	NT

data are given as mean ± SD for n = 17

SP87

Pooled Cryo Using Individual Cryo Units from Storage

S Wing[1] (swing@cbccts.org), J J Hulina[1], J Alexander[2], J Minneman[2]. *[1]Hospital and Laboratory Services, Community Blood Center, Dayton, OH, United States; [2]Administration, Community Blood Center, Dayton, OH, United States*

Background/Case Studies: Cryoprecipitated AHF (Cryo) is prepared by thawing 1st Stage Cryo between 1 C and 6 C and recovering the precipitate. The cold-insoluble precipitate is refrozen within 1 hour. Cryo contains coagulation factors VIII, XIII, fibrinogen, Von Willibrands Factor (vWF) and fibronectin. Single units of Cryo are being used less frequently by hospitals. Single units of Cryos are sometimes difficult and time-consuming to use by hospitals. "Fresh" cryo pools are currently being prepared and frozen by many blood centers. Preparing pooled Cryo from older, stored single cryo units will augment the pooled Cryo inventory and reduce the number of individual Cryo units that are likely to expire. **Study Design/Methods:** This study assessed the process of preparing frozen ten-unit pools of Cryos using individual units of Cryo stored in the walk-in freezer (WIF) not more than 10 months of age at the time of pool preparation. Individual units of Cryo should contain at least 80 IU of Factor VIII and at least 150 mg of fibrinogen in approximately 15 mL of plasma. Pools should contain multiples of these factors (e.g., a pool of 10 Cryo should contain at least 800 IU of Factor VIII and at least 1,500 mg of fibrinogen). Individual Cryo units were thawed and Pooled Cryo was prepared according to existing blood center procedures. Individual Cryo units were selected in groups of ten to range in age from 2 months to 10 months old. (Fewer than 10 individual Cryo units expiriing in May resulted in production of only nine pools of Cryo for this study.) All pools were refrozen within one hour of thawing the individual units and placed in storage in the WIF for one month prior to being sent to a reference lab for Factor VIII and Fibrinogen levels. **Results/Findings:** Each ten-unit pool was expected to contain ≥800 IU of Factor VIII and ≥1500 mg of Fibrinogen (Table). **Conclusion:** Cryo pools can be prepared from individual Cryo units in storage to augment a pooled Cryo inventory and reduce the number of individual Cryo units in inventory.

Disclosure of Commercial Conflict of Interest

J. Alexander: Nothing to disclose; J. J. Hulina: Nothing to disclose; J. Minneman: Nothing to disclose; S. Wing: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Alexander: Nothing to disclose; J. J. Hulina: Nothing to disclose; J. Minneman: Nothing to disclose; S. Wing: Nothing to disclose

TABLE 1.

Pool Number	Expiration Date	Factor VIII	Fibrinogen
W035412081109	11/01/13	1040	2260
W035412081110	10/22/13	1214	2417
W035412081113	09/01/13	1176	2164
W035412081114	08/17/13	1478	2451
W035412081117	07/10/13	1040	2247
W035412081127	06/06/13	1198	1980
W035412081132	04/02/13	1169	1900
W035412081137	03/03/13	1458	2152
W035412081135	02/07/13	1069	1704

SP88

In Vitro Evaluation of Platelets Stored in Three Different Additive Solutions

G C Leitner[1] (gerda.leitner@meduniwien.ac.at), J List[2], B Eichelberger[1], M Horvath[1], P Jilma[1]. *[1]Department for Bloodgroup Serology and Transfusion Medicine, Medical Uuniversity, General Hospita Vienna, Vienna, Austria; [2]Hematology, Hanusch Krankenhaus, Vienna, Austria*

Background/Case Studies: Usually platelets are stored in 100% plasma. Plasma contains all nutritive factors to store platelets for 5 days with minimal metabolic changes. In the 1980s, the first attempatients were made to replace plasma by platelet additive solutions (PAS) to gain plasma for other purposes. Since pathogen inactivation (PI) is an upcoming topic in transfusion medicine, additive solutions are again a matter of interest because PI methods require PAS with or without plasma as storage media. **Study Design/Methods:** The aim was to examine in a prospective analysis the impact of three different PAS on quality of platelets concentrates stored for up to 7 days. All donors (n = 12) were recruited from the routine multicomponent donor pool (3 females/9 males, median age 43 years; range 26-57) and fulfilled the requirements for triple platelet concentrate (PC) harvests. All procedures were performed with TrimaAcell® soft ware version 6.0 (Terumo BCT). Automatic PAS addition was switched off. The highly concentrated product was divided into three equal parts and the additive solutions (Composol® or SSP+® or Intersol™) were added to a final concentration of 56%. PCs were stored at 22 C (± 2 C) under continuous agitation until expiration. Samples were drawn under aseptic conditions on days 1, 5, and 7 to measure the extent of storage lesions by means of pH, O2, CO2, Glucose, LDH, lactate, MPV and aggregation response to thrombin receptor agonist peptide-6 (TRAP-6 50 μm). The study was performed in a cross over design. **Results/Findings:** The yield for each PC was mean 2.28 × 10^11 (SD 0.3). On day 1 there was no significant differences for any of the tested parameters. In all products minimal changes were measured for pH, MPV, O2, and CO2 during the 7 days of storage. Glucose decreased and LDH and lactate increased in all units over time. On day 7 glucose was significantly lower in Intersol PCs than in Composol and SSP+ PCs (12 mg% on day 7 versus 38 mg% in Composol and 36 mg% SSP+ products). Accordingly, LDH and lactate were significantly higher in Intersol units per day (224 U/L and 16.6 mmol/L) compared to Composol (100 U/L and 12.2 mmol/L) and SSP+ (143 U/L and 12.8 mmol/L). Interestingly there was no difference in platelet activation without TRAP during 5 days of storage between the three groups. However, SSP+ platelets showed on day 7 a significant lower response to TRAP-6 stimulation than platelets stored in Composol or Intersol. **Conclusion:** For 7 days of storage SSP+ and Composol are equal in maintaining glucose and keeping acid metabolites low. The clinical meaning of the low response to agonists on day 7 in SSP+ platelets has to be investigated.

Disclosure of Commercial Conflict of Interest

B. Eichelberger: Nothing to disclose; M. Horvath: No Answer; P. Jilma: Nothing to disclose; G. C. Leitner: Nothing to disclose; J. List: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Eichelberger: Nothing to disclose; M. Horvath: No Answer; P. Jilma: Nothing to disclose; G. C. Leitner: Nothing to disclose; J. List: Nothing to disclose

SP89

In Vitro Evaluation of Red Cell Concentrates Prepared by Automated Spinning Membrane Technology from Whole Blood Following an Overnight Cold Hold

A Karpiel[1] (adrienne.karpiel@fresenius-kabi.com), B E Kusters[1], C Wegener[1], K Min[1]. *[1]Fresenius Kabi USA, Lake Zurich, IL, United States*

Background/Case Studies: For operational flexibility, whole blood (WB) units may be stored at cold temperature overnight (<24 hrs, 2-8 C) prior to component separation if platelet production is not required. Separation of red cells concentrates (RCC) from whole blood (WB) is typically performed by centrifugation; however, spinning membrane technology can also be used for automated preparation of RCCs. Previously, we evaluated RCCs produced by spinning membrane technology from WB which was at or above ambient temperature. In this study, we sought to extend the evaluation to RCCs produced from WB which was held cold overnight prior to processing. **Study Design/Methods:** Paired, ABO matched WB units (500 mL) were collected in CPD and held overnight in a cooler on ice for an average of 18 hrs. WB units were transferred to a cold room (2-8 C) then pooled, mixed, and split into two equivalent units. Units were separated into plasma and RCC containing Adsol either by spinning membrane separation at room

temperature or cold centrifugation (4 C) with manual plasma expression. All RCC units were leukoreduced by gravity filtration under cold conditions. RCCs were stored upright at 4 C for 42 days with 7 day mixing/sampling beginning on day of processing (day 1). Standard red cell in vitro parameters were assessed. **Results/Findings:** Total Hgb content of RCCs was not significantly different between the two separation methods. Day 1 spun hematocrit for centrifuged versus membrane separated units was 60 ± 2% versus 55 ± 1%, respectively ($p < 0.05$). In vitro storage parameter results are summarized in table below. Membrane separated units had initially higher hemolysis levels on Day 1, which continued during storage, but there was no significant difference in the rate of hemolysis generation from Day 1-42 ($p > 0.05$). Moreover, all units were well below the upper hemolysis limit of 0.8% on Day 42. **Conclusion:** RCCs prepared from overnight cold hold WB using automated spinning membrane separation exhibit acceptable in vitro storage parameters for up to 42 days with only slight differences in comparison with manually processed RCCs.

Disclosure of Commercial Conflict of Interest

A. Karpiel: Nothing to disclose; B. E. Kusters: Nothing to disclose; K. Min: Nothing to disclose; C. Wegener: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Karpiel: Fresenius Kabi USA, Grants or Research Support; B. E. Kusters: Nothing to disclose; K. Min: Fresenius KABI, Grants or Research Support; C. Wegener: Fresenius Kabi, Grants or Research Support

Parameter, Mean ± SD (n = 10)	RBC Separation Method	Day 1	Day 21	Day 42
Hemolysis (%)	Membrane	0.13 ± 0.05*	0.21 ± 0.08*	0.31 ± 0.11*
	Centrifugation	0.06 ± 0.02*	0.12 ± 0.04*	0.21 ± 0.08*
Microparticles (x10^3/μL)	Membrane	7 ± 2	16 ± 5	35 ± 8
	Centrifugation	6 ± 2	15 ± 4	32 ± 8
Morphology (0-200)	Membrane	190 ± 5	159 ± 8*	139 ± 8*
	Centrifugation	193 ± 4	165 ± 7*	145 ± 10*
ATP (μmol/g Hb)	Membrane	4.1 ± 0.3*	4.5 ± 0.4*	2.7 ± 0.2*
	Centrifugation	4.3 ± 0.3*	5.0 ± 0.3*	3.1 ± 0.3*
Parameter, Mean ± SD (n = 10)	RBC Separation Method	Day 1	Day 7	Day 14
2,3 DPG (μmol/g Hb)	Membrane	13.3 ± 2.0*	5.1 ± 1.8*	Below LOD
	Centrifugation	13.9 ± 1.9*	5.4 ± 1.8*	Below LOD

* $p < 0.05$

SP90

Transfer of the S-303 Treatment System at Two US Blood Centers

J A Cancelas[1] (jose.cancelas@cchmc.org), J Gottschall[2], N Rugg[1], S Graminske[2], A Erickson[3], M Schott[3], N Mufti[3]. *[1]Hoxworth Blood Center, Cincinnati, OH, United States; [2]BloodCenter of Wisconsin, Milwaukee, WI, United States; [3]Cerus Corporation, Concord, CA, United States*

Background/Case Studies: Background: The INTERCEPT System is intended to inactivate pathogens and leukocytes in RBC components for transfusion, using S 303 to crosslink nucleic acids preventing replication of contaminating pathogens and residual leukocytes. In vitro studies were designed to transfer the S-303 treatment process at two US blood centers. For Study 1, the in vitro quality was assessed over 35 days (D) of storage in a paired study, comparing S-303 RBCs (T) to conventional AS-5 RBC (C). In Study 2, T units were evaluated post treatment and after 35 days. **Study Design/Methods:** Methods: Leukoreduced (LR) AS-5 RBCs were prepared from CPD whole blood units and stored at 1-6 C for 35 days. Study 1: ABO matched pairs of AS-5 RBC were combined and split on day 1. One unit per replicate (C) was placed at 1-6 C, the other (T) was treated with the S-303 process for RBC; T and C units were sampled for in vitro testing over 35 days of storage. Study 2: LR AS 5 RBCs were treated on day 1 with the S-303 process then stored at 1-6 C for 35 days and tested at both intervals. **Results/Findings:** Results: Study 1: At D35, all units had hemolysis <1%, and ATP >2 μmol/gHb. The post-storage, D35, K+, lactate, MCV, and pH were significantly lower in T than C. MCHC and morphology results differed across sites (Table 1). Study 2: Hb content was 67 ± 7 g/unit at Site 1 and 64 ± 6 g/unit at Site 2. Hb loss due to the S 303 process was 1 ± 1 g, post-treatment hemolysis ranged from 0.04-0.09%; on day 35, hemolysis was 0.09-0.63% and sterility testing was negative (Table). **Conclusion:** Conclusions: The S-303 Treatment System was successfully transferred to two US blood centers. The in vitro quality of S-303 treated RBCs was comparable to conventional RBCs. All S-303 treated RBC units met the US guideline for hemolysis for leukocyte reduced RBCs in additive solution at end of storage.

Disclosure of Commercial Conflict of Interest

J. A. Cancelas: Nothing to disclose; A. Erickson: No Answer; J. Gottschall: No Answer; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; N. Mufti: No Answer; N. Rugg: Cerus, Grants or Research Support, Terumo BCT, Grants or Research Support; M. Schott: No Answer

Disclosure of Grants Conflict of Interest

J. A. Cancelas: NIH, Grants or Research Support, US Dept of Defense, Grants or Research Support, Veteran Affairs Administration, Grants or Research Support, Cerus, Grants or Research Support, Hemerus LLC, Grants or Research Support, Terumo BCT, Grants or Research Support, Therapure Inc., Grants or Research Support, New Health Sciences Inc. , Grants or Research Support; A. Erickson: No Answer; J. Gottschall: No Answer; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; N. Mufti: No Answer; N. Rugg: Cerus, Grants or Research Support, Terumo BCT, Grants or Research Support; M. Schott: No Answer

Study 1 Day 35

Attribute (mean ± SD)	Site 1 (n=6) Test	Site 1 (n=6) Control	Site 2 (n=6) Test	Site 2 (n=6) Control
% Hemolysis	0.16 ± 0.08	0.17 ± 0.08	0.20 ± 0.04	0.19 ± 0.16
pH at 37°C	6.89 ± 0.08	6.67 ± 0.10[a]	6.96 ± 0.08	6.85 ± 0.03[a]
ATP (μmol/g Hb)	3.86 ± 0.96	3.33 ± 0.29[a]	4.61 ± 0.95	3.91 ± 0.66[a]
Extracellular K+ (mM)	47.2 ± 2.2	57.9 ± 3.7[a]	41.6 ± 3.7	41.1 ± 4.6
Extracellular glucose (mM)	14.5 ± 0.9	12.1 ± 0.8	18.3 ± 1.2	16.8 ± 1.3
Extracellular lactate (mM)	25.3 ± 1.3	35.0 ± 0.9[a]	17.8 ± 1.6	23.0 ± 1.4[a]
MCV (fL)	94 ± 3	96 ± 2	88 ± 1	90 ± 1
MCHC (g/dL)	31.3 ± 1.0	31.2 ± 0.5	34.9 ± 0.5	34.0 ± 0.3[a]
RBC morphology score	69.5 ± 7.0	75.1 ± 5.7[a,b]	75.6 ± 4.2	79.6 ± 4.6

Study 2

Attribute	Site 1 (n=12) mean ± SD	Site 1 (n=12) Criteria	Site 1 (n=12) Passed	Site 2 (n=16) mean ± SD	Site 2 (n=16) Criteria	Site 2 (n=16) Passed
Hemoglobin, g/unit	67±7	not applicable		64±6	not applicable	
Hemoglobin loss (g) (D0/1)	1 ±1	≤5	100%	1 ±1	≤5	100%
% Hemolysis (D35)	0.15 ±0.08	≤1.0	100%	0.3 ±0.1	≤1.0	100%
Sterility Testing (D35)	negative	negative	100%	negative	negative	100%

a. Student's paired t-test (2 tailed) used for non-repeated data.
b. $p < 0.05$

The INTERCEPT Blood System for Red Cells is in development and is not approved for sale.

SP91

Development of Free Plasma Hemoglobin Limit for Product Disposition

M Briel[1] (matthew_briel@bonfils.org), N A Stadele[1], J Steele[1], G Oliver[1], G Ramirez[1], F West[1], T Le[1]. *[1]Bonfils Blood Center, Denver, CO, United States*

Background/Case Studies: The quality control department currently utilizes a subjective and qualitative method of visual inspection to investigate RBC products for unacceptable levels of hemolysis. This study was conducted to establish baseline minimums for hematocrit (Hct) and hemoglobin (Hb) in stored RBC products. These lower bounds were then used to calculate limits for three product additive types (CPD, AS-3, and AS5) for use in a quantitative test for assessing free plasma hemoglobin as a measure of percent hemolysis. Current industry convention places the maximum level of acceptable percent hemolysis at 0.8%. **Study Design/Methods:** Hct and Hb

were measured using a differential hematology analyzer for 9 post-leukoreduction RBC units at 0 and 43 days to determine comparability. Measurements for threshold creation were taken from existing quality control runs using the same instrument. A maximum limit on free plasma Hb was determined by use of the following equation at a value of 0.8% hemolysis: % hemolysis = [(100-Hct) × free plasma Hb (g/dL)/total Hb (g/dL)]. **Results/Findings:** Hct values for day 0 ranged from 55.9-62.6%, and day 43 ranged from 56.1-62.5%. Hb values for day 0 ranged from 18.3-20.9 g/dL, and day 43 ranged from 18.7-21.2 g/dL. Both sets of results were analyzed using separate paired, two-tailed t-tests. The Hct t-test showed no statistically significant difference between days 0 and 43 ($p = .2724$); The Hb t-test showed an increase in Hb from day 0 to day 43 with high statistical significance ($p < 0.0001$). Existing quality control data for each product type was gathered and assessed for threshold creation (see table 1). **Conclusion:** Based on the t-test results, two assumptions were taken: a measurement of day 0 Hct is a reasonable estimate of the Hct at day 43, and a measurement of Hb taken at Day 0 will reflect the minimum (or at least, a consistent) Hb concentration of a unit over the storage duration. Assuming a normal distribution, we expected any unit of these types to fall within two standard deviations of their respective means (with 95% confidence). Therefore, $x\text{-}2\sigma$ will approximate the lower boundary for each measurement of any given unit for each additive type (Table). Maximum acceptable levels of free plasma hemoglobin (Table) reflecting a 0.8% hemolysis based on these lower bounds allowed quantitative, objective, and unbiased hemolysis assessment to be performed with a single measurement of free plasma hemoglobin.

Disclosure of Commercial Conflict of Interest

M. Briel: Nothing to disclose; T. Le: Nothing to disclose; G. Oliver: Nothing to disclose; G. Ramirez: Nothing to disclose; N. A. Stadele: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Briel: Nothing to disclose; T. Le: Nothing to disclose; G. Oliver: Nothing to disclose; G. Ramirez: Nothing to disclose; N. A. Stadele: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

TABLE 1.

	CPD		AS-5		AS-3	
	Hct (%)	Hb (g/dL)	Hct (%)	Hb (g/dL)	Hct (%)	Hb (g/dL)
n	22		105		87	
Mean	69.7	23.3	57.8	19.6	55.4	18.7
SD	2.17	1.07	2.78	1.12	2.79	1.09
Lower Limit at -2SD	65.4	21.2	52.3	17.3	49.8	16.5
Free Plasma Hb Limit (g/dL)	0.490		0.290		0.263	

SP92

Clinical Efficacy of Plasma Prepared with the INTERCEPT™ Blood System

V N Vilyaninov[1] (vilyaninov@mail.ru), V I Vachenko[1], S M Ramonenko[1]. *[1]Department of Blood Transfusion (Blood and Tissue), Kirovs Military Medical Academy, St Petersburg, Russian Federation*

Background/Case Studies: Improvements have been made in blood safety in the past two decades. Despite strict requirements for selection and acquisition of donors and the development of new methods to detect pathogens, plasma products still bear a high risk of disease transmission. Methods are now available modifying DNA and RNA, thus inactivating pathogens contaminating blood components with limited impact on the plasma proteins. Plasma treated with the INTERCEPT™ Blood System (IBS), was studied for its clinical efficacy. The aim of the study was to gather information about the influences the INTERCEPT™ pathogen inactivation (PI) treatment might have on different plasma components. **Study Design/Methods:** Plasma units (600 mL, n = 87) were collected from volunteer donors via apheresis (Autopheresis-C, Baxter, Round Lake, IL). PI was performed using the amotosalen/UVA-based INTERCEPT™ Blood System, (Cerus, Concord, CA). Prior to treatment the frozen units were thawed and kept at room temperature. Samples were taken before and after treatment. Hemostatic factors were measured with the STA Compact analyzer (STA Compact®, Roche, Mannheim, Germany). **Results/Findings:** A PI technology must demonstrate highest degree of retention of coagulation factors in the treated plasma unit. Here, the results are presented from a series of in vitro studies analyzing in great detail the function of plasma after treatment with the INTERCEPT™ Blood System. Only fibrinogen was reduced by more than 20% when compared with pretreatment values. The studied values are well within reference ranges and European standards. **Conclusion:** Encouraging results of clinical application show preserved therapeutic properties of PI plasma. Treatment of plasma with the INTERCEPT™ Blood System preserves proteins necessary for haemostasis without inappropriate activation of coagulation, fibrinolytic or complement pathways. The results are in accordance with those published including recent studies on the retention of coagulation factors in INTERCEPT™ treated plasma from apheresis or whole blood. These were also substantiated in a series of clinical trials and clinical routine use. Nevertheless, further clinical studies in special settings should be undertaken.

Disclosure of Commercial Conflict of Interest

S. M. Ramonenko: No Answer; V. I. Vachenko: No Answer; V. N. Vilyaninov: No Answer

Disclosure of Grants Conflict of Interest

S. M. Ramonenko: No Answer; V. I. Vachenko: No Answer; V. N. Vilyaninov: No Answer

Hemostatic properties of treated (PCT FFP) or untreated fresh frozen plasma (FFP).

Hemostatic parameters	FFP	PCT FFP
Partial Thrombin Time (sec)	15,9 ± 0,5	16,3 ± 0,4
Thrombin Time (sec)	16,6 ± 0,9	20,6 ± 0,7
Prothrombin Index (%)	71,6 ± 4,8	63,8 ± 3,1
Activated Partial Prothrombin time (sec)	34,2 ± 1,2	36,9 ± 1,2
International Normalized Ratio	1,32 ± 0,05	1,41 ± 0,04
Fibrinogen (g/L)	2,8 ± 0,1	2,21 ± 0,1*
Factor VIII (%)	91,4 ± 10,8	91,4 ± 10,8 84,8 ± 11,3

* P ≤ 0,01 compared to the untreated plasma.

SP93

Evaluation of Red Blood Cells Stored in CPP™ Bags

C Chen[1] (cyndi_chen@pall.com), J Davis[1], C Wang[1], J Fernandez[1], M A Popovsky[1], S Lucas[1]. *[1]Research and Development, Haemonetics Corporation, Covina, CA, United States*

Background/Case Studies: The plasticizer di(2-ethylhexyl) phthalate (DEHP) is a common component in medical plastic. DEHP is non-covalently bound and can leach from PVC material when in contact with blood. While controversial, concerns that exposure to phthalates may induce developmental and reproductive toxicity, makes non-DEHP alternatives desirable. This manufacturer currently utilizes the CPP™ bag, made from flexible PVC film using citrate type plasticizer, for plateletpheresis storage. Since the CPP™ bag is steam sterilizable and has good gas permeability, it has been suggested that it may be suitable as a universal storage bag for both platelets and red blood cells. This study is to evaluate and compare red cell storage quality of whole blood derived red cells stored in current DEHP-PVC bags compared to the proposed CPP™ bags. **Study Design/Methods:** Twenty-four whole blood (WB) units were collected and kept at room temperature (RT; 18-24 C) for 20-24 hours prior to processing. After hold, each pair of ABO matched WB units were pooled and split evenly. Control units had a standard DEHP-PVC red cell storage bag and the test units had the CPP™ red cell storage bag. Units were processed by standard centrifugation methods and then separated into individual components using the Compomat®G4. After separation, additive solution (SAG-M) was added to back-prime the filter and to mix with the packed red cells prior to leukocyte reduction. The leukocyte-depleted red cell concentrate (RCC) was stored at 4 C (1-6 C) and analyzed on day 1, 14, and 21 days of storage. **Results/Findings:** The mean red cell yield was $1.45 \times 10e12 \pm 0.09 \times 10e12$ in each split. Overall, the maintenance of red cell quality was satisfactory through 21 days of storage for the current DEHP-PVC bag. End of storage pH levels were greater than 6.6. Hemolysis throughout storage was < 1.0%. In comparison, the maintenance of red cell quality was not satisfactory through 21 days of storage for the proposed CPP™ storage bag. While pH levels were greater than 6.6, 1 of 12 units had hemolysis >1.0% at 21 days of storage. Results did not met AABB guidelines **Conclusion:** Red cells cannot be stored in the current CPP™ bag (flexible PVC using citrate type plasticizer) based on its current formulation. Optimization of the CPP™ bag formulation is required to develop a universal platelet and red blood cell storage bag.

Disclosure of Commercial Conflict of Interest

C. Chen: Nothing to disclose; J. Davis: No Answer; J. Fernandez: Nothing to disclose; S. Lucas: No Answer; M. A. Popovsky: Haemonetics, Other; C. Wang: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Chen: Nothing to disclose; J. Davis: No Answer; J. Fernandez: Nothing to disclose; S. Lucas: No Answer; M. A. Popovsky: Nothing to disclose; C. Wang: Nothing to disclose

TABLE 1. Red Cell Quality Assessment—Combined data (Mean ± SD; n = 12)

Parameters		Day 1	Day 14	Day 21
pH	DEHP-PVC	6.93 ± 0.03	6.75 ± 0.03	6.67 ± 0.035
(22°C)	CPP™	6.94 ± 0.03	6.78 ± 0.03	6.70 ± 0.037
Hemolysis (%)	DEHP-PVC	0.05 ± 0.01	0.10 ± 0.02	0.13 ± 0.03
	CPP™	0.05 ± 0.01	0.46 ± 0.11	0.72 ± 0.18

SP94

Direct Measurement of Residual Plasma in Red Cell Concentrates in Additive Solutions

T Cheves[1] (tcheves@lifespan.org), J D Sweeney[1]. *[1]Blood Bank, Roger Williams Hospital, Providence, RI, United States*

Background/Case Studies: Whole blood donations are typically separated into either two components (red cells in additive solution and plasma) or three components (red cells in additive solution, platelets, and plasma). The whole blood is subjected to a hard spin when two components are manufactured and an initial soft spin when three components are manufactured. It would be expected that the amount of residual plasma would be greater with an initial soft spin but comparative data are lacking. Furthermore, it is often stated by manufacturers that there is minimal residual plasma in the final red cell concentrate (15-35 mls), but there is little actual published data. This is of relevance since red cell concentrates in AS have been implicated in cases of transfusion related acute lung injury (TRALI) although the amount of residual plasma in such components is typically not known. **Study Design/Methods:** AS-1 RBCs and AS-3 RBCs from 500 ml collections were studied on day 42 (day of expiration). The AS-1 red cell concentrates were manufactured from whole blood donations (CPD) in which two components were produced and the AS-3 red cells from whole blood donations (CP2D) in which three components were produced. The total weight of each unit was measured and the weight of the plastic container tared. Using a specific gravity of 1.060, the volume of the unit was calculated. A sample was taken after mixing of the bag and the hematocrit determined, allowing calculation of the red cell and supernatant volume. 110 mls was subtracted from the supernatant volume to derive the volume of residual plasma. The data were summarized as descriptive statistics and analyzed using independent t tests. **Results/Findings:** A total of 19 AS-1 RBCs were studied and 26 AS-3 RBCs on day 42 (day of expiration). All but six were Group AB. Data are shown in the table as the mean +/– 1 SD. As expected, the AS-3 red cells contain more residual plasma. Moreover, the absolute amount of measured residual plasma for most units was greater than that which is commonly quoted (25-35 mls) . The median residual plasma volume in the AS-1 RBC's was 39 mls with 8/19 (35%) having greater than 40 mls of residual plasma. The median residual plasma volume in the AS-3 RBC's was 64 mls with 9/26 (35%) of the AS-3 RBCs containing 70 mls or greater. **Conclusion:** Residual plasma in AS-RBCs appears greater than is commonly quoted and may be relevant to RBC associated TRALI. Blood Centers should consider studying and optimizing centrifugational conditions to minimize residual plasma in AS-RBCs.

Disclosure of Commercial Conflict of Interest

T. Cheves: No Answer; J. D. Sweeney: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Cheves: No Answer; J. D. Sweeney: Bio-Rad, Grants or Research Support

	AS-1	AS-3	p
Volume RBC (mls)	192 ± 23	193 ± 21	0.08
Supernatant (mls)	148 ± 8	176 ± 13	<0.01
Volume plasma (mls)	38 ± 8	66 ± 13	<0.01
Minimum plasma (mls)	24	29	
Maximum plasma (mls)	57	98	

SP95

Does the 30-Minute Rule Always Apply?

K Jones[1] (Joneske18@comcast.net), T Ahlenius[1], T Berry[1], J Steele[1], F West[1], T Le[1]. *[1]Transfusion Medicine, Bonfils Blood Center, Denver, CO, United States*

Background/Case Studies: The 30 Minute Rule is an oft-questioned blood banking practice. This study was performed to determine the true elapsed time before blood exceeds 10 C. **Study Design/Methods:** The desired results represent the worst case scenario for time required for the core temperature of a standard packed red cell to reach 10 C. Thermocouples are most accurate when measuring core and center temperatures of individual units. In our study each RBC unit was divided into two equal volumes using an attached transfer bag with the thermocouple placed between the two units avoiding any labels on either bag. As volume impacts thermodynamics, the study consisted of four testing sets of 1, 5, 10, and 15 units. These sets are consistent with processing practices in both the Manufacturing and Reference Laboratories. Each total scenario was tested 3 times. The temperature probes were attached to the units while in the walk-in refrigerator to maintain storage temperature during preparation. Data collection began in the storage area; the units were then transported to the leukoreduction processing area. After approximately 4 hours, the units were then returned to a Manufacturing refrigerator for storage at 1-6 C for another 4 hours. Temperature of each unit and the associated ambient results were collected continuously for this 8-hour period. **Results/Findings:** The data was stratified to present all units in various scenarios including worst case outcomes defined as the units in the bin most exposed to the ambient environment. The results in Table 1 reflect the elapsed time (h : mm : ss) for a unit to exceed 10 C after removal from controlled storage, with tolerance bounds (p = 0.95) calculated. **Conclusion:** The time for an RBC unit to exceed 10 C is impacted by both the number of units being processed as well as the unit's location in relation to additional units. The 10 unit set results exceeded 10 C more quickly than expected. This is believed to be caused by environmental impacts such as air flows in that specific area in the laboratory. However, the results indicate that groups of red cell units can be exposed to ambient conditions for greater than 30 minutes while individual units may exceed 10 C at the core within 10 minutes of exposure.

Disclosure of Commercial Conflict of Interest

T. Ahlenius: Nothing to disclose; T. Berry: Nothing to disclose; K. Jones: Nothing to disclose; T. Le: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Ahlenius: Nothing to disclose; T. Berry: Nothing to disclose; K. Jones: Nothing to disclose; T. Le: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

Table 1	All Locations				Worst Case (4 Corners of Bin)			
Statistics	1 U	5 U	10 U	15 U	1 U	5 U	10 U	15 U
Average	0:34:40	1:46:43	2:03:52	2:29:34	34:40:30	1:24:48	1:21:12	1:37:41
SD	9:14:38	0:24:35	0:42:02	0:46:34	9:14:38	0:11:52	0:21:19	0:20:08
K Factor	2.57	2.57	2.22	2.13	2.57	3.71	2.74	2.74
95% Tolerance	0:10:55	0:43:21	0:30:33	0:50:23	10:55:07	0:40:47	0:22:47	0:42:31

U = Unit; time in hh : mm : ss; K factor from: Hahn, G. J. and Meeker, W. Q. (1991) Front Matter, in Statistical Intervals: A Guide for Practitioners, John Wiley & Sons, Inc., Hoboken, NJ, USA. doi: 10.1002/9780470316771.fmatter

SP96

Antioxidant Activity Studies of Vitamin E Emulsions in Red Blood Cells

C A da Siva[1] (calsbiologicas@gmail.com), C A Azevedo Filho[2], D C Silva[1], A S Nunes[1], M A Castro[3], A F de Almeida[3], S C Azevedo Lucena[4], B S Santos[2], M Barjas-Castro[5], A Fontes[1]. *[1]Departamento de Biofísica e Radiobiologia, Universidade Federal de Pernambuco, Recife, Brazil; [2]Departamento de Ciências Farmacêuticas, Universidade Federal de Pernambuco, Recife, Brazil; [3]Centro de Pesquisa Aggeu Magalhães—CPqAM, FIOCRUZ, Recife, Brazil; [4]Fundação Hemope, Recife, Brazil; [5]Hemocentro, Universidade Estadual de Campinas—Unicamp, Campinas, Brazil*

Background/Case Studies: During their storage, erythrocytes suffer progressive deterioration, which can compromise the post-transfusion integrity of red blood cells(RBCs) in circulation. Many of these storage lesions can be caused by oxidative damages on the RBCs membranes. Studies have reported that vitamin E is one of the most important antioxidants acting upon the lipids of cellular membranes. Thus, the present study aimed to evaluate whether the vitamin E antioxidant properties are capable of improving storage conditions of RBCs units. Due to their hydrophobic nature, vitamin E emulsions were prepared in different solvents (Phosphate Buffer Saline 1X—PBS, 0.9% saline and Sagmanitol-SAG-M). The emulsion characterizations confirmed the presence of vitamin E in the formulations and also showed the presence of particles in a nanometer range. **Study Design/Methods:** Nine CPD SAG-M RBCs units(Fresenius) from healthy donors were divided into two units. Three RBC units received emulsions of each kind of solvent (vitamin E-0.2 mg/mL). The other units received only the solvents (at the same volume used for the emulsions) and were used as control. Freshly prepared formulations were added aseptically to blood components and then the RBCs units were stored at 4 C. Quality control for RBCs units was performed by measuring hematocrit, hemoglobin and hemolysis at day 35 of storage. The reactive oxygen species (ROS) production was analyzed by cytometry using DCFH-DA(Invitrogen) at days 8, 21, and 35 of storage. Moreover, measurements of the elasticity of the RBCs (at least 20 cells/unit) were also performed at days 8, 21, and 35 using optical tweezers(Fontes, 2011). **Results/Findings:** All quality control parameters were within the normal range, except for the units with PBS that showed a degree of hemolysis higher than expected. Comparing vitamin E in saline formulation with control units no difference in ROS production was observed. However, the system with SAG-M and vitamin E emulsion showed a decay of average relative ROS production in at least 35% during storage period. As only the emulsion with vitamin E and SAG-M was able to reduce the ROS production, the experiment was continued with RBCs deformability only with this system and its respective control. These measurements showed that the RBC elasticity did not differ from control when stored with SAG-M and vitamin E emulsion. **Conclusion:** The data showed that the emulsion with vitamin E and SAG-M decreased ROS production significantly. Nevertheless, vitamin E was not capable of improving RBCs deformability. This vitamin acts against lipids oxidation whereas for preserving RBCs elasticity, the protein oxidation appears be more important. Studies with antioxidants acting upon RBCs membrane proteins deserve special attention as the gradual loss of elasticity is a relevant storage lesion.

Disclosure of Commercial Conflict of Interest

C. A. Azevedo Filho: Nothing to disclose; S. C. Azevedo Lucena: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; M. A. Castro: Nothing to disclose; C. A. da Siva: Nothing to disclose; A. F. de Almeida: Nothing to disclose; A. Fontes: Nothing to disclose; A. S. Nunes: Nothing to disclose; B. S. Santos: Nothing to disclose; D. C. Silva: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. A. Azevedo Filho: Nothing to disclose; S. C. Azevedo Lucena: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; M. A. Castro: Nothing to disclose; C. A. da Siva: Nothing to disclose; A. F. de Almeida: Nothing to disclose; A. Fontes: Nothing to disclose; A. S. Nunes: Nothing to disclose; B. S. Santos: Nothing to disclose; D. C. Silva: Nothing to disclose

SP97

Experience of Manufacturing Four versus Five Pooled Platelets using the AcrodoseTM Plus System

J Garbaini[1] (jennifer.garbaini@gmail.com), D Lal[1], T Corvino[1], D Strauss[1], B H Shaz[1]. *[1]New York Blood Center, New York, NY, United States*

Background/Case Studies: While whole blood donation is the primary source for red cell units in the United States, whole blood derived platelet pools account for about 16% of platelet units transfused. In an attempt to increase utilization of resource, the AcrodoseTM technology by Haemonetics Corporation to create a pooled, leukoreduced, ABO matched, bacteria tested platelet product that could be an alternate source of platelets was implemented. **Study Design/Methods:** From December 1, 2011-January 13, 2013, pools of four random-donor, ABO-matched platelets were made using the AcrodoseTM Plus System, Platelet Pool and Store Set with in-line leukoreduction filter and eBDS (Haemonetics Corp., Braintree, MA). After January 13, 2013, AcrodoseTM pools were made using 5 random-donor platelets, keeping all other aspects of processing and criteria for discarding the same. Pooling and filtration occurred within 24 to 48 hours after collection of whole blood units, according to manufacturers guidelines. Terumo sterile connecting device was used to make a sterile weld between the two pieces of tubing, maintaining a closed system. Each unit was required to meet specifications for volume [180 mL-420 mL], post filtration platelet yield [2.2-5.8 $\times$ 1011], and platelet concentration [$< 2 \times 106$ platelets/ul]. Other rejection criteria, according to SOP, include manufacturing issues, failing visual criteria, and infectious disease screening reactivity. **Results/Findings:** Despite continuous effort to improve system, the monthly rejection rate based on manufacturing criteria of four pool platelets was 10.9% (on average), range 4.5-19.2%. 95% of these failures were due to low platelet yield . Upon switching to five pool platelets, rejection rate dropped to 0.4% [range 0-0.7%] which were due to high platelet concentration and not platelet yield. **Conclusion:** Pools of four platelets resulted in >15% rejection of products despite review and optimization. Switching to pools of five platelets has decreasing rejection rates, without an increase in rejection rate of other criteria. Thus, it is recommended to use five rather than four pooled platelets when using this system.

Table 1: Comparison of 4 vs 5 Pool Platelets

	POOL4		POOL5		p-value
TOTAL	1541		721		
PASS	1219	80.4%	641	88.9%	p<0.001
REJECT	297	19.6%	80	11.1%	p<0.001
Platelet Yield	158	10.3%	0		p<0.001
Concentration	8	0.52%	3	0.42%	p 0.74
Volume	0		0		NA
Other Criteria*	105	6.93%	55	7.63%	p=0.51

*Other rejection criteria includes donor screening, infectious disease testing, hemoglobin sickle trait, and manufacturing problems.

Disclosure of Commercial Conflict of Interest

T. Corvino: Nothing to disclose; J. Garbaini: Nothing to disclose; D. Lal: Nothing to disclose; B. H. Shaz: Haemonetics, Other, Fewal, Other; D. Strauss: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Corvino: Nothing to disclose; J. Garbaini: Nothing to disclose; D. Lal: Nothing to disclose; B. H. Shaz: Nothing to disclose; D. Strauss: Nothing to disclose

SP98

Comparison of AS-3 and SAGM Red Blood Cell Units Washed with the Haemonetics ACP 215 Automated System and Stored in AS-3 Additive Solution

A Jacques[1] (Annie.Jacques@hema-quebec.qc.ca), H Daoud[1], C Bédard[1], M Chevrier[2], L Thibault[1]. *[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada*

Background/Case Studies: Currently, washed red blood cells (RBCs) are washed with COBE 2991 and stored in saline solution for up to 24 hours to reduce the risk of bacterial contamination. The closed-system ACP 215 (Haemonetics) enables automated washing and 14-day post-wash storage in SAGM solution. The quality parameters were studied of SAGM and AS-3 RBC units washed at various storage times and stored suspended in AS-3 solution. **Study Design/Methods:** Leukoreduced AS-3 and SAGM RBC units (450 mL) were washed using the ACP 215 system after 7, 14, 21, or 28 days of storage and stored post-wash for up to 14 days (n = 5 per arm). After washing, RBCs were stored in 80 mL of AS-3 solution. Product quality (volume, hematocrit, and hemolysis) and RBC metabolism (ATP, pH, lactate, potassium, and sodium) were analyzed before and after post-wash storage. Impact of a 25 Gy irradiation on their in vitro parameters was also studied. **Results/Findings:** Recovery was 96% for AS-3 RBCs and 90% for SAGM RBCs. Hemoglobin content in washed AS-3 RBC units was 55 $\pm$ 4 g/unit, and 44 $\pm$ 3 g/unit in SAGM RBC units. The hematocrit of AS-3 and SAGM RBC units was 0.54 $\pm$ 0.01 L/L and 0.49 $\pm$ 0.03 L/L, respectively. The % of hemolysis in RBC units washed after up to 14 days of storage and stored 7 days post-wash was approximately 0.5 % for both SAGM and AS-3 RBC

units. Immediately after washing, the average potassium level for both products was 1.0 ± 0.4 mmol/L, and increased to 10.4 ± 1.3 mmol/L after 7 days of post-wash storage. The residual volume of plasma was 0.40 ± 0.14 mL in AS-3 RBC units and 0.13 ± 0.02 mL in SAGM RBC units. Two days after irradiation, the hemolysis percent in washed RBC units was 0.4 ± 0.1% and 0.3 ± 0.1% in AS-3 RBC and SAGM RBC units. In these units, the average potassium level was comparable to fresh RBC units (7-day old) (13 ± 3 mmol/L vs, 16 ± 4 mmol/L in AS-3 and SAGM RBC units respectively). **Conclusion:** The data show that RBC units stored for up to 14 days before washing can efficiently be washed with the ACP 215 system and stored for up to 7 days post-wash when suspended in AS-3. The nutrient solution in which RBCs are stored before washing has little impact on their in vitro parameters. The quality of washed RBCs remains acceptable up to 48 hours post-irradiation. A validation study is underway to confirm the routine use of the ACP 215 system to wash blood for transfusion.

Disclosure of Commercial Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Bédard: Nothing to disclose; M. Chevrier: Nothing to disclose; H. Daoud: Nothing to disclose; A. Jacques: Nothing to disclose; L. Thibault: Nothing to disclose

SP99

The 30 Minute Rule Replacement

T Ahlenius[1] (trent_ahlenius@bonfils.org), K Jones[1], T Berry[1], J Steele[1], F West[1], T Le[1]. *[1]Bonfils Blood Center, Denver, CO, United States*

Background/Case Studies: Blood bankers have long used the historical and largely anecdotal 30 Minute Rule to assure product suitability of PRBC for transfusion. **Study Design/Methods:** This study examined the warming and cooling profiles of RBC units as an objective way to determine product suitability. Previous testing has shown thermocouples to accurately measure core temperature by using a center probe in a single unit. Therefore each RBC unit was divided in half using a transfer bag and the thermocouple was placed between the two units, not touching labels. Sets of 1, 5, 10, and 15 units were used (current practice in manufacturing and reference labs). Each set was tested three times. Refrigerated units (1-6 C) were removed to a leukoreduction processing area, where they remained at ambient temperature for 4 hours, then were returned to the refrigerator for 4 more hours. Warming and cooling profiles were determined for each 4 hour period based on a linear approximation of the temperature profile of individual units. Worst case results were noted in the area of the bin most exposed to the ambient environment; best case results were seen with units in the area of the bin surrounded by other units. **Results/Findings:** Linear approximations were stratified to represent various exposures and associated one sided tolerance bounds (p = 0.95) were calculated. Table results reflect warming and cooling profiles by number of RBC units and bin location. **Conclusion:** The number of units and bin location most strongly impacts RBC warming and cooling profiles, providing an objective basis for determining blood product suitability, compared to the classic 30 Minute Rule.

Disclosure of Commercial Conflict of Interest

T. Ahlenius: Nothing to disclose; T. Berry: Nothing to disclose; K. Jones: Nothing to disclose; T. Le: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Ahlenius: Nothing to disclose; T. Berry: Nothing to disclose; K. Jones: Nothing to disclose; T. Le: Nothing to disclose; J. Steele: Nothing to disclose; F. West: Nothing to disclose

WARMING			
# U, Bin Location	Avg. Deg.C /Hr	SD Deg.C/Hr	95% ToleranceDeg.C/Hr
1 U	3.81	0.90	6.12
5 U, C	2.96	0.18	3.61
5 U, S	2.45	0.21	3.07
5 U, A	2.64	0.31	3.43
10 U, C	2.76	0.21	3.35
10 U, S	2.13	0.21	2.66
10 U, A	2.38	0.38	3.22
15 U, C	2.63	0.21	3.21
15 U, S	2.13	0.20	2.62
15 U, M	1.70	0.17	2.21
15 U, A	2.18	0.38	2.98
COOLING			
1 U	–3.44	0.91	–1.11
5 U, C	–2.16	0.22	–1.34
5 U, S	–1.51	0.26	–0.73
5 U, A	–1.77	0.41	–0.72
10 U, C	–2.10	0.34	–1.18
10 U, S	–1.10	0.43	0.01
10 U, A	–1.50	0.63	–0.10
15 U, C	–1.94	0.41	–0.83
15 U, S	–0.99	0.35	–0.16
15 U, M	–0.50	0.23	0.19
15 U, A	–1.21	0.61	0.10

U = unit; C = corner; S = side; A = all; M = middle.

SP100

Temperature Devices for the Monitoring of Returned RBCs for Reissue

T M Gorey[1] (tgorey@partners.org), E Hart[3,1], A Finocchio[2]. *[1]Transfusion Service, Brigham and Women's Faulkner Hospital, Boston, MA, United States; [2]HVAC-Facilities, Brigham and Women's Faulkner Hospital, Boston, MA, United States; [3]Clinical Laboratory Science, University of Massachusetts Dartmouth, North Dartmouth, MA, United States*

Background/Case Studies: Regulatory agencies require red blood cells (RBCs) during transport to be maintained at 1-10 C to minimize the risk of an adverse event due to bacterial contamination. Once an RBC has been issued from the transfusion service (TS), it is extremely difficult to monitor the transport temperature in order to ensure the integrity of the RBC. It is essential the temperature not exceed 10 C in order for the TS to safely return the unit back into inventory for reissue at a later time. As an interdisciplinary approach, collaboration with the HVAC technician helped to evaluate the following devices: 1)TempCheck (TC) Rapid Response (Hampshire Controls Corp, Dover, NH), an electronic instrument, which incorporates a Type K thermocouple, insulated pad, and a digital readout that provides a non-invasive, quick reading and has a range of –40 to 40 C and an accuracy of ± 2 C; 2) Fluke 62 Mini Infrared (IR) (Fluke Corporation, Everett, WA), a portable non-contact IR thermometer that uses a customized two-channel infrared/contact thermometer mix signal application-specific integrated circuit with a range of –30 to 500°C and an accuracy of ± 1 C; and 3) HemoTemp® (HT) II Indicator (BioSynergy, Elk Grove Village, IL), a core correlated combination irreversible liquid crystal blood temperature indicator and a reversible liquid crystal thermometer. The reversible thermometer is calibrated to the internal core temperature of 1-9 C under ambient conditions. The irreversible portion (ie, "flower") warms when the core temperature exceeds approximately 6-10 C. After activation, the "flower" retains a blue color for at least 48 hours when blood is stored at 3 C. As blood warms, the "flower" changes from a blue to a brown color. **Study Design/Methods:** Ten expired RBCs were stored at 1-6 C for at least 10 hours prior to testing. The HT indicator was activated at 40-42 C and attached to the RBC unit according to the manufacturer's directions. After storage, a RBC with the attached HT indicator was removed from the refrigerator and kept at room temperature (RT). The temperature of the RBC and the RT was taken every 10 minutes for a total of 30 minutes for each method being evaluated. **Results/Findings:** All the units were within 1-10 C at 10 minutes for all methods evaluated: 1)HT: 9 units were at 7-9°C and 1 unit was ≥ 10°C at 20 minutes. All ten RBCs were ≥ 10 C at 30 minutes; 2) TC: 8 units were at 7-9 C and 2 were at 10-13 C at 10 minutes. One unit was at 7-9 C and 9 units were at 10-13 C at 30 minutes; and 3) IR: All ten RBCs were at 10-13 C at both 20 and 30 minutes. **Conclusion:** Although only a small number of units were

tested, we demonstrated that the best method for the TS to reliably monitor the temperature of returned RBCs back into inventory is the HemoTemp® II indicator. The TempCheck is a suitable alternative while the Fluke® IR thermometer is unsatisfactory because it reads the surface temperature up to 3 C higher.

Disclosure of Commercial Conflict of Interest

A. Finocchio: Nothing to disclose; T. M. Gorey: Nothing to disclose; E. Hart: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Finocchio: Nothing to disclose; T. M. Gorey: Nothing to disclose; E. Hart: Nothing to disclose

SP101

Optimize Platelet and Concurrent Plasma Collections using InterSol Platelet Additive Solution with Amicus

P McIlvain[1] (pmcilvain@bbd.org). [1]Donor Services, Blood Bank of Delmarva, Dover, DE, United States

Background/Case Studies: The Blood Bank of Delmarva (BBDelmarva) has recognized the increase in demand for plasma in the marketplace. As a response, in January 2012, the BBDelmarva took steps to optimize plasma collections at a satellite facility through the implementation of InterSol, Platelet Additive Solution, using the Fenwal Amicus Separator. When using InterSol, 65% of the plasma that would be collected for suspension/storage of the platelets is replaced by InterSol solution. **Study Design/Methods:** Platelet donors are currently restricted to a 28-day donation interval for concurrent plasma collection. In June 2012, the BBDelmarva applied for a variance to collect the plasma replaced by InterSol (PRBI) which will enable concurrent plasma collections to be optimized on more frequent donors while still meeting platelet inventory needs. With the alternate procedure approval the platelet and concurrent plasma donation interval will be the same as a platelet donation. While this variance is still pending; the collection of concurrent plasma is maximized since 65% of the plasma storage fluid is replaced by InterSol. A concurrent plasma product equal in volume to the plasma replaced by InterSol (PRBI) can be collected while still classifying the donor as a platelet donor for concurrent plasma donation interval. **Results/Findings:** The BBDelmarva experienced a 211.9% increase inconcurrent plasma collections and a 29.37% increase in recovered plasma sold in 2012, which was a direct result of implementation of InterSol at one facility. The whole blood plasma that would have been manufactured as FFP was able to be diverted to recovered plasma to increase revenue without compromising transfusion inventory needs. **Conclusion:** Using InterSol with Amicus platelet collections allows for additional plasma units to be collected. The BBDelmarva believes these procedures optimize the blood donors experience while increasing plasma revenues and improving Blood Center operations.

Disclosure of Commercial Conflict of Interest

P. McIlvain: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. McIlvain: Nothing to disclose

	Total Platelet Products Collected	Total Concurrent Plasma Products	Recovered Plasma Shipped
2010 w/o InterSol	2,214	1,459	12,585.040 L
2012 with InterSol	2,427	4,550	16,280.803 L
% Increase	9.6% Increase	211.9% Increase	29.37% Increase

SP102

Production of Standardized Labile Blood Products from a Single Whole Blood (WB) Donation at the EFS-Alsace Blood Center: Ten Years of Evolution and Impact of Inactivation Process

H Isola[1] (herve.isola@efs-alsace.fr), F Bigey[1], D Kientz[1], M Laforet[1], C Gachet[1], J Cazenave[1]. *[1]EFS Alsace, Strasbourg, France*

Background/Case Studies: The EFS-Alsace collects about 120,000 whole blood (WB) donations every year. This WB constitutes the raw material for production of the labile blood products (LBP) necessary to cover the regional and part of the national transfusion requirements. Thus, 120,000 red blood cell concentrates (RBCC), 13,000 buffy-coat derived platelet concentrates (BCPC) inactivated by the INTERCEPT® technology and 35,000 liters of plasma were produced in 2012. The particularity of this production resides in the wide variability of the biological parameters linked to the blood donors. The primary objective of the EFS-Alsace is to provide the clinicians with standardized LBP. Hence the organization of the WB collection and of the preparation and distribution of the LBP, as likewise the equipment and materials used for the entire production system, are adapted to the achievement of this objective. **Study Design/Methods:** On an average, 550 WB donations per day are collected, from 4 PM to 9 PM using a single type of WB collection system. This consists of quintuple bags (Fenwal), top and bottom, with integrated filters for leukodepletion of the RBCC and plasma. A unique target volume of WB is collected from all donors. The whole blood is stocked (15-18 hours) overnight in a temperature controlled room (21 ± 3 C). On day 1 the WB is centrifuged at 4000 rpm for 15 minutes (acceleration 9, break 5) using 6 Heraeus Cryofuge 6000i centrifuges (12 bags/cycle). The centrifuged products are then separated on 12 semi-automatic presses (Optipress II, Fenwal, mean separation time 110 ± 19 s) to obtain RBCC, BCs (volume = 49 ± 3 mL ; hematocrit = 43 ± 2 %) and plasma. The volume and hematocrit of both BC and pooling BC are the critical parameters of the procedure and must be efficiently controlled to obtain standardized LBP. **Results/Findings:** Since 2002, the mean volume of WB increased of 464 ± 14 mL to 481 ± 12 mL (n = 1,243,421) and the mean volume of the RBCC increased also to 258 ± 16 mL at 274 ± 18 mL (n = 1,212,559). The mean hemoglobin content increased to 50 ± 5 g at 54 ± 6 g. The mean volume of plasma increased to 282 ± 20 mL at 290 ± 19 mL (n = 1,217,250). The mean platelet content of a BCPC derived from 6 is 4.6 ± 0.5 10exp11 (n = 47,745) and 4.0 ± 0.5 10exp11 (n = 48,451) for a BCPC derived from 5. Since 2006, all BCPC were inactive by INTERCEPT system (n = 70,653). **Conclusion:** Standardization of the blood collection and processing procedure allows one to obtain final products containing a reproducible quantity of active substance from one unit to another. This reproducibility facilitates the prescription and therapeutic use of these LBP and also process modifications or integration of new process like reduction of pathogens in LBP.

Disclosure of Commercial Conflict of Interest

F. Bigey: Nothing to disclose; J. Cazenave: Cerus, Grants or Research Support, Travel Support or Honorarium, Consulting or Board of Director Fees; C. Gachet: Nothing to disclose; H. Isola: Nothing to disclose; D. Kientz: Nothing to disclose; M. Laforet: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Bigey: Nothing to disclose; J. Cazenave: Teruma BCT, Travel Support or Honorarium, Grants or Research Support, Fresenius, Travel Support or Honorarium, Cerus, Grants or Research Support, Consulting or Board of Director Fees, Haemonetics, Grants or Research Support, MacoPharma, Grants or Research Support; C. Gachet: Nothing to disclose; H. Isola: Nothing to disclose; D. Kientz: Nothing to disclose; M. Laforet: Nothing to disclose

SP103

Stability Of Selected Hematological Parameters in Blood Stored at 4 C for 72 Hours from Platelet Donors

A A Tendulkar[1] (anitatendulkar@gmail.com), S Gujral[2], M Tambe[1], R Kenjale[1], P Jain[1], B Ganesh[3]. *[1]Transfusion Medicine, Tata Memorial Hospital, Mumbai, India; [2]Hematopathology, Tata Memorial Hospital, Mumbai, India; [3]Medical Records Department, Tata Memorial Hospital, Mumbai, India*

Background/Case Studies: Apheresis platelets is a mainstay therapy in cancer patients. Approximately 3000 donors are tested annually in the tertiary cancer center here. Platelet drives are conducted to register willing donors after an informed consent. National guidelines for blood banks in India do not elaborate on the ideal storage time for blood samples to be tested for complete blood counts. Hence, testing of platelet donor samples is done on the day of collection which can be challenging because of issues such as manpower shortages, weekend hours, and a single cell counter in blood banks.The aim was to evaluate the stability of the results of specific hematologic parameters on a three part differential automated cell counter at time intervals of 4, 48, and 72 hours. **Study Design/Methods:** The study was approved by the Institutional Review Board (IRB) and funded by Department of Atomic Energy, India. Platelet donor preliminary screening (hemoglobin, weight, and medical history) was performed and an eligible donor's venous blood sample was collected in EDTA vacutainers for cell counts. The hemoglobin should be >12.5 g/dl and the platelet count >150 × 10e9/L for the donor to be eligible for a platelet donation. Blood samples were tested within 4 hours on a three-part differential cell counter (Humacount 30, Human,

Weisbaden,Germany) for hemoglobin, platelet and WBC count. The samples were stored at 4 C in a monitored blood bank refrigerator and were retested at 48 and 72 hours from the time of collection. The samples were allowed to reach room temperature (22-24 C) and mixed well before being introduced into the automated counter for testing. **Results/Findings:** 767 blood samples were tested for WBC count, Hemoglobin and platelet count at 4, 48, and 72 hours after collection. One way ANOVA was applied in SPSS (version 18). The mean WBC ct ($\times$ 10e9/L) at 4, 48, and 72 hours was 7.6,7.6,7.6 ($p = 0.76$). Mean hemoglobin (g/dl) at 4, 48, and 72 hours was 14.3,14.3,14.3 ($p = 0.53$). Mean platelet ct ($\times$ 10e9/L) at 4,48 and 72 hours was 313.4, 326.7, 330.5 ($p < 0.001$). Thus, there was no statistical difference seen in the mean values of WBC count and hemoglobin count at three time intervals. Even though difference in the mean platelet count at three time intervals was statistically significant ($p < 0.001$), it did not impact on donor acceptance criteria and was within the prescribed norms of 10% CV for platelet counts. Maximum difference in mean platelet counts at any time interval was 17 $\times$ 10e9/L. **Conclusion:** Blood samples for testing specific hematologic parameters of platelet donors were found to be stable at 4 C for up to 72 hours. This can facilitate increasing the platelet donor registry by organizing long distance and weekend camps. Hindrances due to manpower shortages, equipment breakdowns and weekend testing can be overcome with extended sample storage at 4 C. The study can be instrumental in formulating donor blood sample storage guidelines for blood banks.

Disclosure of Commercial Conflict of Interest

B. Ganesh: Nothing to disclose; S. Gujral: Nothing to disclose; P. Jain: Nothing to disclose; R. Kenjale: Nothing to disclose; M. Tambe: Nothing to disclose; A. A. Tendulkar: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Ganesh: Human GmbH, Grants or Research Support; S. Gujral: Human GmbH, Grants or Research Support; P. Jain: Human GmbH, Grants or Research Support; R. Kenjale: Human GmbH, Grants or Research Support; M. Tambe: Human GmbH, Grants or Research Support; A. A. Tendulkar: Human GmbH, Grants or Research Support

SP104

Reduced breakage of Frozen Blood Products with a New Design of Storage Casing

C Bédard[1] (Claudia.Bedard@hema-quebec.qc.ca), A Jacques[1], Ducas[1], N Dussault[1], P Lalonde[2], S Thibault[2], M Beaudin[2], L Thibault[1].
[1]Héma-Québec, Québec City, QC, Canada; [2]Héma-Québec, Montreal, QC, Canada

Background/Case Studies: Héma-Québec delivers about 90,000 frozen blood products to hospitals yearly. Frozen products are stored at <–18 C and distributed to hospitals on dry ice (–78 C). The transport and handling of these products at this low temperature increase the risk of breakage and cracking of plastic bags. Last year, nearly 2,000 frozen products were lost. The aim of this project was to develop a "green" packaging that would reduce the incidence of breakage by immobilizing the frozen product in its casing. **Study Design/Methods:** The new casing design must meet specific criteria: adaptable to all types of storage bags of frozen products, economical, reusable, recyclable, and have the least possible impact on our operations and those of hospitals. Vertical and horizontal drop tests from a height of 90 cm, a worst case scenario that simulates a fall from a work station, were conducted to evaluate prototype performance. The results were compared with our current casing. **Results/Findings:** The new design has been easily adapted to plasmapheresis units as well as whole blood derived frozen products. The casing, made of corrugated cardboard, includes an insert which firmly holds the bag in place to reduce its mobility. The lid has a large window to give access to the product label and is firmly held in place with Velcro type closures. During vertical drop tests, the new design reduced the breakage rate of the 500-mL fresh frozen plasma (FFP) and 250-mL frozen plasma (FP) from 70% and 60% respectively when compared to our current packaging. Despite a slight increase in breakage in horizontal drop tests with the 500-mL FFP, the new casing allowed a better protection of frozen products by reducing total breakage rate from 39% to 11%. **Conclusion:** The new casing design should reduce the rate of breakage and the loss of frozen blood products. The new casings will be put into operations shortly. However, it will take several months to assess their effectiveness in reducing the loss of frozen products.

Disclosure of Commercial Conflict of Interest

M. Beaudin: Nothing to disclose; C. Bédard: Nothing to disclose; . Ducas: Nothing to disclose; N. Dussault: Nothing to disclose; A. Jacques: Nothing to disclose; P. Lalonde: Nothing to disclose; L. Thibault: Nothing to disclose; S. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Beaudin: Nothing to disclose; C. Bédard: Nothing to disclose; . Ducas: Nothing to disclose; N. Dussault: Nothing to disclose; A. Jacques: Nothing to disclose; P. Lalonde: Nothing to disclose; L. Thibault: Nothing to disclose; S. Thibault: Nothing to disclose

Donor Recruitment, Retention, and Adverse Events: Quantitative Aspects; Donor Recruitment/Retention/Marketing and Donor (Suitability)

SP105

Motivation to Donate Among Granulocytes, Apheresis, and Whole Blood Donors in a Brazilian Blood Center

P C Rocha[2,1] (capuanirocha@gmail.com), E C Sabino[3], C D Oliveira[4], G M Patavino[1], L D Capuani[1], C de Almeida-Neto[1,2]. *[1]Fundação Pró-Sangue Hemocentro de São Paulo, São Paulo, Brazil; [2]Disciplina de Ciências Médicas, Faculdade de Medicina da Universidade de São Paulo, São Paulo, Brazil; [3]Departamento de Moléstias Infecciosas, Faculdade de Medicina da Universidade de São Paulo, São Paulo, Brazil; [4]Universidade Federal de São João del-Rei, Minas Gerais, Brazil*

Background/Case Studies: Transfusion of allogeneic granulocytes, collected by donor-apheresis, may be indicated to control life-threatening bacterial and fungal infection in neutropenic patients. Granulocyte donations are time-consuming and involve specific procedures, as administration of steroids and/or recombinant G-CSF to donors and collection by apheresis. In this center only community donors are eligible to donate granulocytes. It is important to understand motivations to donate among these donors to improve recruitment strategies. This study aim to describe the motivation factors characterized as altruism, self-interest, test-seeking and response to direct appeal among granulocyte donors comparing to platelet and whole blood donors. **Study Design/Methods:** A case-control study was conducted in which motivation to donate of 64 granulocytes donors cases were compared to 64 platelets apheresis and 68 first-time whole blood donors controls, from 2007-2011. Attributes of donor motivation measured in the survey included responses to altruism (four questions), direct appeal (four questions), self-interest (five questions), and test-seeking (nine questions). Frequencies of answers "Yes", "No," and "Don't know" were recorded and a p-value less than 0.05 was defined as significant. **Results/Findings:** Granulocyte donors are mainly male (56% versus. 41% versus. 39%; $p < 0.002$) when compared to controls, older (≥41 years, 34% versus. 6%; $p < 0.001$) and married (28% versus. 14%, $P < 0.0001$) when compared to whole blood donors. Different motivation to donate was not statically significant when comparing granulocytes to platelets donors. Altruism was not different among cases and controls. While granulocyte donors presented more attributes related to direct appeal as "I received a telephone call or a letter from the blood bank asking me to donate" (Yes answers, 90.6% versus. 54.4%, $p < 0.001$), whole blood donors answered "Yes" more frequently to questions related to test-seeking behaviour as "I think that blood donation is a good, fast, and anonymous way to get my blood tested" (29.9% versus. 20.3%, $p = 0.016$), "I like to donate to get my blood test results" (25% versus. 14.1%, $p = 0.031$), self-interest as "I want to get off work today" (39.7% versus. 17.5%, $p = 0.009$) and "I may need blood myself" (97% versus. 78.1%, $p = 0.004$). **Conclusion:** Altruism is a common attribute among different type of donors. Brazilian blood banks may be aware about the potential risks inherent in self-interest and test seeking behaviour among first-time whole blood donors. Direct appeal actions may be an effective strategy to recruit granulocyte donors.

Disclosure of Commercial Conflict of Interest

L. D. Capuani: Nothing to disclose; C. de Almeida-Neto: Nothing to disclose; C. D. Oliveira: Nothing to disclose; G. M. Patavino: Nothing to disclose; P. C. Rocha: Nothing to disclose; E. C. Sabino: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. D. Capuani: Nothing to disclose; C. de Almeida-Neto: Nothing to disclose; C. D. Oliveira: Nothing to disclose; G. M. Patavino: Nothing to disclose; P. C. Rocha: Nothing to disclose; E. C. Sabino: Nothing to disclose

SP106

Blood Donor Satisfaction and Reasons to Donate Blood: Experience in Argentina

M C Remesar[1] (mirtucris@yahoo.com.ar), V S Gallardo[1], S Kuperman[1], A E del Pozo[1]. *[1]Centro Regional de Hemoterapia, Hospital de Pediatria Garrahan, Ciudad Autonoma de Buenos Aires, Buenos Aires, Argentina*

Background/Case Studies: Surveys designed to evaluate satisfaction after blood donation (BD) and the reasons to donate blood, were already developed in other countries. The gathered information was useful to improve effectiveness on donor recruitment and retention. Implementing this survey in our scenario would be helpful to maintain the recently achieved 100% altruist BD in the Centro Regional Hemoterapia Garrahan (CRHG). **Study Design/Methods:** Cross-sectional study based on a pre-designed survey carried out by a social worker after the donor completed the BD. The survey was validated in situ and then administered in a non-compulsive, anonymous and confidential way. Donors were sampled by convenience. Answers were entered in a database (EpiInfo 2000, v 3.5.3). **Results/Findings:** A total of 743 donors answered the survey: 52% of them in extramural blood drives. 85% donated whole blood and 15% apheresis platelets. 23% of the surveyed people were first-time donors. From the rest, 55% donated blood at least four times in the last 5 years, and 45%, 2-3 times during the same period. Socio-demographic features: 52% men and 48% female. Age distribution: 16-19 years: 5.7%; 20-29: 27.5%; 30-39: 31.0%; 40-49: 22.4%; 50-59: 11.1% and ≥ 60: 2.4%. Education: Complete or incomplete university degrees: 61%; complete or incomplete high school: 30% and complete or incomplete elementary school: 9%. The 97% of surveyed blood donors rated the overall donation process at 8 to 10 on a scale of 10. The rate of reasons to donate, from "not important at all" to "very important", is shown in the table. **Conclusion:** Donors are highly satisfied with the BD process. Their selection of "helping others" and "being a frequent donor" as "very important" resulted a positive indicator of the developed recruitment campaigns at CRHG. Moreover, these concepatients should be reinforced in the promotion-recruitment program. It is relevant and of great value the low percentage of BD who considered "no important at all" to receive a recognition/gift for donating blood.

Disclosure of Commercial Conflict of Interest

A. E. del Pozo: Nothing to disclose; V. S. Gallardo: Nothing to disclose; S. Kuperman: Nothing to disclose; M. C. Remesar: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. E. del Pozo: Nothing to disclose; V. S. Gallardo: Nothing to disclose; S. Kuperman: Nothing to disclose; M. C. Remesar: Nothing to disclose

	To help others %	Being a frequent donor %	To feel good about myself %	Donating blood is a duty %	Receiving a gift or recognition %
Not important at all	0.9 (7/743)	6.9 (51/742)	20.1 (149/743)	46.4 (345/743)	86.7 (644/743)
Somewhat important	0.7 (5/743)	4.4 (33/742)	10.1 (75/743)	5.8 (43/743)	6.1 (45/743)
Indifferent	3.0 (22/743)	13.6 (101/742)	14 (104/743)	4.7 (35/743)	3.6 (27/743)
Important	18.2 (135/743)	32.9 (244/742)	21.9 (163/743)	14.8 (110/743)	2.3 (17/743)
Very important	77.3 (574/743)	42.2 (313/742)	33.9 (252/743)	28.3 (210/743)	1.3 (10/743)

SP107

SPatial Patterns of Blood Donor Deferrals

A B James[1] (abjames@emory.edu), L A Waller[2], B H Shaz[3], G B Schreiber[4], C Josephson[1], J D Roback[1]. *[1]Pathology and Laboratory Medicine, Emory University School of Medicine, Atlanta, GA, United States; [2]Biostatistics and Bioinformatics, Rollins School of Public Health, Emory University, Atlanta, GA, United States; [3]New York Blood Center, New York, NY, United States; [4]Epidemology Consultant, Bethesda, MD, United States*

Background/Case Studies: The objectives of this study are to describe spatial patterns of blood donor deferrals and to draw inferences in reference to the distribution of an observed set of locations and to detect whether these set of locations contain clusters of blood donor deferrals. The null hypothesis is a lack of clustering which describes that deferrals are likely to occur at any location within the study area, regardless of the locations of other deferrals. **Study Design/Methods:** Data were obtained from the American Red Cross Southern Region between 2004-2008. The study consisted of individuals (16-94 years old) who offered to donate blood in Georgia and were either temporarily or permanently deferred. Spatial analyses were conducted using R Statistical Software version 3.0 (R, 2012, Vienna, Austria) and SatScan (Computer program) ver. 9.1.1, (SatScan, Boston, MA). Specifically, observed spatial patterns of deferrals between men and women, race groups, and age groups were compared to identify unusual clusters of deferrals within the data. **Results/Findings:** The study population consisted of 104,292 blood donor deferrals throughout the state of Georgia. The majority of deferrals were females (71.0%), whites (59.2%) and younger adults (16-39 years), (60.9%). Among minorities, African Americans (23.1%) and Hispanics (2.3%) were deferred. The top five frequent reasons for deferrals were low hemoglobin (45.4%), blood pressure (8.7%), travel (8.2%), having a tattoo (5.3%), and infection (4.9%). The spatial analysis reveal significant local concentrations of male deferrals in the north central part of the state, significant local concentrations of African American deferrals across the middle of the state, significant local concentrations of Hispanic deferrals in urban areas, and significant concentrations of older deferrals in broad areas across the state. **Conclusion:** The spatial patterns reveal insight into deferral trends within population subgroups and offer opportunities for targeted intervention to decrease locally elevated deferral rates.

Disclosure of Commercial Conflict of Interest

A. B. James: Nothing to disclose; C. Josephson: Nothing to disclose; J. D. Roback: Nothing to disclose; G. B. Schreiber: Nothing to disclose; B. H. Shaz: Haemonetics, Other, Fewal, Other; L. A. Waller: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. B. James: Nothing to disclose; C. Josephson: Immucor/BioArray, Grants or Research Support, Octapharma, Consulting or Board of Director Fees; J. D. Roback: Nothing to disclose; G. B. Schreiber: Nothing to disclose; B. H. Shaz: Nothing to disclose; L. A. Waller: Nothing to disclose

SP108

Prevalence of Immunoglobulin A Deficiency (IgAD) in Shanghai Blood Donors and Efforts to Establish a Rare Blood Bank of IgAD in Shanghai

P Lu[1] (luping@sbc.org.cn), B Ling[1], R Li[1]. *[1]Shanghai Blood Center, Shanghai, China*

Background/Case Studies: Selective immunoglobulin A deficiency (IgAD) is the most common primary immunodeficiency with a frequency of 1 in 600 in Caucasians. The current definition, established by the Pan-American Group for Immunodeficiency and the European Society for Immunodeficiencies, defines the disorder as serum IgA levels <0.07 g/L with normal IgM and IgG levels in individuals ≥4 years of age. The prevalence in various ethnic groups ranges from 1:155 to 1:18,550. Individuals with IgAD may form specific antibodies against IgA, which can cause an anaphylactic response when the patient receives an IgA-containing blood transfusion. There is not enough accurate data in the Chinese population now. An attempt is made to calculate the prevalence of IgAD in a replicate cohort of the Chinese Han population in Shanghai by screening blood donors, and to estimate the risk of transfusion reactions induced by anti-IgA. **Study Design/Methods:** A sandwich enzyme-linked immunosorbent assay was performed to screen for IgAD and the other ELISA immunoassay used for confirmation. IgA antibodies were further detected by the ELISA test or by the DiaMed anti-IgA test in IgAD blood donors. **Results/Findings:** Of the total 61,625 healthy blood donors screened, 32 cases were confirmed as having IgA deficiency. Anti-IgA was identified in 3 of the 32 IgAD blood donors whose IgA levels were <0.07 g/L. Estimation of the theoretical risk for IgA anaphylactic transfusion reaction was 0.005%. **Conclusion:** The prevalence of IgAD in Chinese is low. However, potential risks exist in performing blood transfusion to IgAD persons, and measures should be taken to reduce IgA anaphylaxis. Meanwhile, its necessary to set up a Shanghai rare blood bank of IgAD donor for patient to meet the needs of IgA-poor transfusion.Further, the genetics of IgAD in Chinese blood donors should be analyzed and its molecular basis explored with special emphasis on the MHC region.

Disclosure of Commercial Conflict of Interest

R. Li: Nothing to disclose; B. Ling: Nothing to disclose; P. Lu: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Li: Nothing to disclose; B. Ling: Nothing to disclose; P. Lu: Nothing to disclose

SP109

High-Performance Liquid Chromatography for Screening of Abnormal Hemoglobins in Blood Donors Allows the Identification of Diabetic Individuals through the Detection of A1C Hemoglobin

R A Pereira[1] (ricardoap@einstein.br), M B Polite[1], S P Cabral[1], M T Farinha[1], S C Lira[1], M A Mota[1], J Levi[1], J M Kutner[1]. *[1]Hemotherapy and Cellular Therapy, Hospital Israelita Albert Einstein, São Paulo, Brazil*

Background/Case Studies: Approximately 5% of the population worldwide are healthy carriers of abnormal hemoglobin genes, particularly sickle cell anemia and thalassemia. In Brazil, screening for abnormal hemoglobins in blood donors is mandatory, and there must be a specific transfusion policy for blood units obtained from donors carrying the S trait (Hb AS). For this study, the agar-gel manual method was substituted for an automated equipment (G7—TOSOH Bioscience) for HPLC (High performance liquid chromatography), which identifies the main abnormal hemoglobins S, C and D. As HPLC can also be used to diagnose and follow-up diabetic individuals, it was involuntarily begun to identify donors with high HbA1c levels. The prevalence of diabetes in Brazil, as defined by the American Diabetes Association (Standards of Medical care in Diabetes) as HbA1c values >6.5, is around 7.9%. The aim of this study is to determine the percentage of individuals with high HbA1c among blood donors, and correlate these findings with clinical and demographic data, as well as to the donors' knowlegde of being diabetic at donation. **Study Design/Methods:** The new method was implemented on August 18, 2012. Data from blood donors tested between this date and February 27, 2013 were analyzed for gender, age, type of donation, height, if donor was aware of being diabetic at the screening interview before donation, renal and cardiac disease, and if the donor took any medication. **Results/Findings:** There were 6,989 donations in the studied period, of which 107 showed HbA1c values ≥ 6,5 (1.5%). Ten donors reported diabetes during the interview, and 18 took oral hypoglicemic agents, but did not report having diabetes at the time. Therefore, 28 donors were diabetic (26%) and 79 (74%) were not aware of that condition. The average body mass index (BMI) of the group with HbA1c <= 6.5 was 26, compared to 31 in the group with HbA1c >6.5. **Conclusion:** A considerably high percentage (1.5%) of blood donors with abnormal hemoglobin A1c content was identified. This could be anticipated, as the donors with controlled non-insulin dependent diabetes are allowed to donate blood. However, the majority of donors (74%) did not report that fact on the interview. The methodology used was beneficial in terms of automation and data traceability, but also for the identification of potential diabetic individuals. The introduction of this method raised the question on the most adequate approach towards such donors who are not aware of their possible diabetic condition.

Disclosure of Commercial Conflict of Interest

S. P. Cabral: No Answer; M. T. Farinha: No Answer; J. M. Kutner: Nothing to disclose; J. Levi: No Answer; S. C. Lira: No Answer; M. A. Mota: Nothing to disclose; R. A. Pereira: Nothing to disclose; M. B. Polite: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. P. Cabral: No Answer; M. T. Farinha: No Answer; J. M. Kutner: Nothing to disclose; J. Levi: No Answer; S. C. Lira: No Answer; M. A. Mota: Nothing to disclose; R. A. Pereira: Nothing to disclose; M. B. Polite: Nothing to disclose

SP110

100% Voluntary Donors: How to Get There

S Kuperman[1] (silvinakup@yahoo.com.ar), E F Ganza[1], A Montes[1], M C Remesar[1], M I Puppo[1], A E del Pozo[1]. *[1]Centro Regional de Hemoterapia, Hospital de Pediatria Garrahan, Ciudad de Buenos Aires, Buenos Aires, Argentina*

Background/Case Studies: Argentinean regulations establish that blood donation (BD) must be voluntary and non-remunerated. However in a significant part of the system, principally in the private sector, blood is collected from family members. The Centro Regional de Hemoterapia Garrahan (CRHG) , a high-complexity, 450-bed pediatric institution, decided to celebrate International Donor Day on June 14, 2011, by changing the BD system (from Voluntary BD + Family BD) to a fully Voluntary BD. The decision was taken based on a previous implementation in 2001 of an integral and sustainable voluntary and repeated BD promotion program.Aims: 1) To achieve a 100% of voluntary and repeated BD in order to improve transfusion safety; and 2) To maintain the blood bank's supply based on the existent program and the new proposed actions. **Study Design/Methods:** 1) To communicate the new policy to prescriber physicians, nurses and administrative personnel involved in the transfusion process. 2) To develope a message directed to each patient's family explaining the reasons and importance for the implemented changes, and the need for their collaboration in order to successfully consolidate the measure, as well as detailing the role they could play in the new scenario (participation as promoters, blood drives organization in their neighborhoods or institutions to which they belong). 3) To intensify recruitment of institutions for blood drives, favoring the repetition of institutions and donors. 4) To compare the TTI markers in 2011 (mixed system) with 2012 (100% Voluntary BD) in order to gauge the potential impact to transfusion safety. **Results/Findings:** The outcomes are shown in the table below. **Conclusion:** The radical change in the BD model was received with mixed feelings by patients' family and friends; the change was at odds with the predominant BD culture. They were afraid that their donation, albeit voluntary, would not be recorded as related to the patient. It was necessary to explain to them that if they wished to voluntarily donate they should not associate their name to the patient. The donors seen at the bank have diminished notoriously. The extramural donations constitute almost 50% of the supply of the period until December 2012. The TTI markers decreased, highlighting the importance of voluntary and fundamentally repeated donors.

Disclosure of Commercial Conflict of Interest

A. E. del Pozo: Nothing to disclose; E. F. Ganza: Nothing to disclose; S. Kuperman: Nothing to disclose; A. Montes: Nothing to disclose; M. I. Puppo: Nothing to disclose; M. C. Remesar: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. E. del Pozo: Nothing to disclose; E. F. Ganza: Nothing to disclose; S. Kuperman: Nothing to disclose; A. Montes: Nothing to disclose; M. I. Puppo: Nothing to disclose; M. C. Remesar: Nothing to disclose

Year	2011		2012	
Place of donation	CRHG (%)	Blood Drive (%)	CRHG (%)	Blood Drive (%)
Presenting donors	10.442 (75) (10442/14010)	3.568 (25) (3568/14010)	6.601 (52) (6601/12579)	5.978 (48) (5978/12579)
Collected units	11.054		11.641	
Prevalence T cruzi (%)	0.91 (101/11054)		0.44 (51/11641)	
Prevalence Syphilis (%)	1.00 (111/11054)		0.36 (42/11641)	
Prevalence HIV (%)	0.05 (6/11054)		0.026 (3/11641)	

SP111

Piloting an Instrument to Understand the Knowledge, Attitude, and Perceptions toward Blood Donation in The Democratic Republic of Congo

J Koster[1,3] (juliettekoster@gmail.com), S Y Ramazani[2,3]. *[1]Safe Blood For Africa Foundation, Johannesburg, South Africa; [2]National Blood Service, Democratic Republic of Congo, Kinshasa, The Democratic Republic of Congo; [3]Blood Safety, Centers for Disease Control, Atlanta, GA, United States*

Background/Case Studies: There is limited data on the knowledge, attitudes, and perceptions to blood donation in Africa. This is important given this abstracts aims to present a case study, which demonstrates the challenges of attempting to conduct a Knowledge Attitude and Perceptions (KAP) survey. Lessons learned from this particular experience in Kinshasa, Democratic Republic of Congo (DRC) in 2013 might assist future KAP initiatives. **Study Design/Methods:** Safe Blood For Africa Foundation (SBFA) used a newly designed survey instrument to understand the knowledge, attitudes and perceptions toward blood donation collection in Kinshasa, DRC. The study was conducted in collaboration with the National Blood Service (NBS) of the DRC. Local interviewers interviewed 290 people in either French or Lingala over a 12-day period in 2013. The subjects were selected using a convenience sampling approach. The interviews were conducted under the supervision of an SBFA programme officer. The KAP

survey instrument was approved by both the NBS of the DRC and the Centers for Disease Control prior to initiation of the study. The University of Kwazulu Natal in South Africa conducted the data analysis, using Structuring Equation Modelling. **Results/Findings:** While the level of comprehension of the KAP instrument met expectations, the instrument was too extensive, detracting from its practicality. The tool was designed to lead respondents to specific sections applicable to them. In practice, this failed and led to an overlap of responses complicating the analysis phase Furthermore, the interviewers had difficulties in understanding some survey questions, prompting biased interpretation of the subjects' responses. The convenience sampling introduced a noticeable sampling bias. **Conclusion:** This pilot study highlights the challenges of KAP studies where standardization is lacking. There are no standardised formats of a KAP study that are widely applicable. This has led to a spectrum of surveys under the KAP umbrella. A KAP model should take into account local differences and context and produces a design that allows this freedom and structures the basic requirements, which may assist researchers in addressing the population the most effective way. The absence of a streamlined KAP format may deter researchers from using the method and from using it correctly. While the KAP survey is a sophisticated tool that can inform rational approaches to blood donor recruitment and retention programs, it should not be regarded as a simple instrument; instead it relies on rigorous design in order to be effective.

Disclosure of Commercial Conflict of Interest

J. Koster: Nothing to disclose; S. Y. Ramazani: No Answer

Disclosure of Grants Conflict of Interest

J. Koster: Nothing to disclose; S. Y. Ramazani: No Answer

SP112

Correlating Red Blood Cell Unit Age with Incidence of Transfusion Related Acute Lung Injury

T Kieffer[1,2] (tkieffer@iupui.edu), J Cruz[2,1], D A Waxman[2,1]. *[1]Indiana University Health Pathology and Laboratory Medicine, Indiana University Health, Indianapolis, IN, United States; [2]Indiana Blood Center LLC, Indiana University Health, Indianapolis, IN, United States*

Background/Case Studies: Transfusion related acute lung injury (TRALI) is the leading cause of mortality among transfusions. The current theory consists of a priming event followed by exposure to a human leukocyte antigen (HLA) alloantibody from a transfused blood product. Yet a HLA alloantibody with cross reactivity is not always identified. This suggests an alternate pathology/mechanism. Recent discussions have called into question the safety of older red blood cell (RBC) units. Several large multi-facility trials are underway to evaluate this concern. To that end, the Indiana Blood Center (IBC), along with IU Health (IUH) Hospitals of Indianapolis, undertook this simple study to correlate the age of RBC units with the incidence of TRALI. **Study Design/Methods:** The medical director reviewed TRALI cases from all IBC cosignees before 2011. Cases with transfused RBC units and no causative units were selected. Unit age was calculated by subtracting transfusion from donation date. IUH was petitioned for total RBC units transfused for months when a TRALI case was reported. No other IBC consignees had data available. Unaltered RBC units (original expiration date) were selected and RBC unit ages were calculated by subtracting the difference between expiration and transfusion from 42. All unit ages were grouped by week and frequency was converted to relative %. A 1-prop z test was performed between the relative % of the same age range. **Results/Findings:** 39 TRALI cases met the study criteria involving 62 RBC units. 18 cases were from IUH, accounting for 29 RBC units. 148,056 RBC units were transfused at IUH over the requested dates, except 2001 and 2002 (not recorded). The frequency and relative % are shown in the Table. Both relative % appear to mirror each other. p values range from 1.0 to 0.25. **Conclusion:** Discussion:There is no apparent variance between transfused RBC units and TRALI cases suggesting no correlation between RBC unit age and TRALI reactions. It is assumed the RBC unit is causative, the transfusion practice of IUH extrapolate to other facilities, and reported TRALI cases are indeed TRALI cases. The primary limitation is the small TRALI sample size as reflected in the p values indicating a high probability of chance in the results. More data is required to elucidate any relationship between RBC unit age and adverse reactions.

Disclosure of Commercial Conflict of Interest

J. Cruz: Nothing to disclose; T. Kieffer: Nothing to disclose; D. A. Waxman: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Cruz: Nothing to disclose; T. Kieffer: Nothing to disclose; D. A. Waxman: Nothing to disclose

TABLE 1.

RBC Age Range in Weeks	IBC # TRALI cases (multifacility)	IUH # RBC transfused	Relative % TRALI Case	Relative % RBC transfused	p value
1	6	7684	9.6	5.1	1.0
2	15	30713	24.1	20.7	0.25
3	16	43240	25.8	29.2	0.72
4	10	31534	16.1	21.3	0.84
5	8	22060	12.9	14.9	0.67
6	7	12825	11.2	8.6	1.00
Total	62	148056			

SP113

Cause of "Low Hemoglobin" Deferrals in 500 First-Time Blood Donors (Previous Blood Donation or Not?)

B Newman[1] (bruce.newman@redcross.org), M Chbib[2], P Lemanski[3], S Shumunov[4]. *[1]American Red Cross, SE Michigan Region, Detroit, MI, United States; [2,3,4]Wayne State University, Detroit, MI, United States*

Background/Case Studies: Blood centers have measured hemoglobin concentrations in blood donors and in blood donors deferred for "low hemoglobin" in masse but have not purposely evaluated "low hemoglobin" deferrals (LHDs) in individual blood donors to determine if they were due to the previous blood donation or not. This relates to the benefit that might take place from iron replacement programs after successful blood donations. **Study Design/Methods:** 500 random first-time blood donors were chose who successfully donated between June 6, 2008-July 3, 2008 and then followed their whole-blood donations and LHDs for 3.75 years. The donor base was biased in that all blood donors' first-time donations were successful, only donors with at least two successful blood donations were included, and donors who were under 22 years old were not included. Blood donor LHDs that occurred within 2-6 months of a previous blood donation were counted as due to the previous donation while LHDs outside of that range were counted as not due to a previous donation. The purpose of the study was to determine if LHDs were due to the previous blood donation or not. **Results/Findings:** There were 252 women (50.4% of total group) and 248 men, who donated 1102 and 1048 units, respectively. The mean age, weight, and percentage of Caucasians in the women were 43 years, 167 lb, and 91%; and in the men, it was 42 years, 204 lb, and 81%. The "low hemoglobin" deferral rate was 4.4%, which is lower than the usual 7-8% observed in general blood donor populations. In women, 55 of the 85 (65%) LHDs were due to a previous blood donation; and in men, 8 of 12 (67%) LHDs were due to a previous blood donation. In total, 63 of the 97 (65%) LHDs were due to a previous blood donation. **Conclusion:** Based on applying the rule that LHDs within 2-6 months of a blood donation are due to the previous donation, most of the "low hemoglobin" blood donation deferrals (65%) in the studied database were due to a previous blood donation. An iron replacement program after successful blood donations would mitigate a portion of these LHDs.

Disclosure of Commercial Conflict of Interest

M. Chbib: Nothing to disclose; P. Lemanski: Nothing to disclose; B. Newman: Nothing to disclose; S. Shumunov: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Chbib: Nothing to disclose; P. Lemanski: Nothing to disclose; B. Newman: Nothing to disclose; S. Shumunov: Nothing to disclose

SP114

The Potential for an Iron Replacement Program after Successful Blood Donations to Affect "Low Hemoglobin" Deferrals

B Newman[1] (bruce.newman@redcross.org), M Chbib[2], P Lemanski[3], S Shumunov[4]. *[1]American Red Cross, SE Michigan Region, Detroit, MI, United States; [2,3,4]Wayne State University, Detroit, MI, United States*

Background/Case Studies: A question is how many "low hemoglobin" deferrals (LHDs) would be affected by an iron replacement program after successful blood donations? **Study Design/Methods:** Five-hundred first-time whole-blood donors, age >22 years, with successful first donations between June 6, 2008 July 7, 2008, and at least two successful blood donations, were followed for 3.75 years. LHDs that occurred within 2-6 months of a previous blood donation were defined as being due to the previous blood donation. The iron replacement program encouraged women and men to take iron pills / capsules when the women donated two or more times within a 12-month period or when men donated three or more times within a 12-month period. LHDs in the first two women-donations or first three men-donations were not included since the donors could not fulfill the criteria to take oral iron therapy. The number of times oral iron pills / capsules would have been recommended for the donation prior to the LHD, where the LHD was due to the donation, was evaluated. **Results/Findings:** There were 252 women (50.4% of total group) and 248 men, who donated 1102 and 1048 units, respectively. Of 95 LHDs, 63 (65%) were due to the previous blood donation and 32 (35%) were not. Of the 63 LHDs due to the previous donation, 15 could not be evaluated because of insufficient number of donations—12 in women and 3 in men. Of the 43 evaluable LHDs in women, iron would have been recommended in 39 of 43 (91%) donations prior to the LHD. Of the five evaluable LHDs in men, iron would have been recommended in 4 of 5 (80%) donations prior to the LHD. In total, iron would have been recommended in 43 of 48 (90%) donations prior to the LHD. When one includes the 32 LHDs that were not due to blood donation, the number of LHDs impacted would be 43 of 80 (54%) LHDs. **Conclusion:** The iron replacement program described above in the above database, with donor compliance, would have had an affect on 90% of the LHDs due to blood donation, and on 54% of all evaluable LHDs.

Disclosure of Commercial Conflict of Interest

M. Chbib: Nothing to disclose; P. Lemanski: Nothing to disclose; B. Newman: Nothing to disclose; S. Shumunov: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Chbib: Nothing to disclose; P. Lemanski: Nothing to disclose; B. Newman: Nothing to disclose; S. Shumunov: Nothing to disclose

SP115

Haemoglobin Pre-Donation Screening using Non-Invasive Method in Argentinean Blood Donors

S Oknaian[1] (sebaoknaian@yahoo.com.ar), S Kuperman[1], M I Puppo[1], J Atorino[1], A E del Pozo[1]. *[1]Blood Donor Center, Hospital de Pediatria Garrahan, Ciudad de Buenos Aires,Buenos Aires, Argentina*

Background/Case Studies: Hemoglobine (Hb) or hematocrit (Hct) tests are used routinely in blood banks for detection of anemia (pre-donation screening). Local national regulation establishes blood donor (BD) deferral when Hb or Hct are <12.5 g/dL or <38% respectively. Up to now all methods imply finger stick samples, leading to BD discomfort. A new available non-invasive method is expected to give a more comfortable donor care, which can contribute to increase repeated donations. The objective of this study was to evaluate the new non-invasive method comparing its results with a current invasive method and a reference method. **Study Design/Methods:** 308 whole blood donors and apheresis donors were screened for Hb by NBM-200 system (Osense Ltd., Nez Ziona, Israel) and by HemoCue Hb201+ (Hemocue AB, Angelholm, Sweden). Additionally venous blood sample were taken and tested by Cell Dyn 3200 (Abbott Lab. IL) as a reference method. All the personnel were trained before implementation of the study. Paired t-test was used to compare NBM-200 and HemoCue Hb201+ resuts to Cell Dyn 3200. **Results/Findings:** Finally 304 resuts (67.8% male, 32.2% female donors) were compared between NBM-200 system and Cell Dyn 3200. The mean Hb values were 14.59 g/dL for NBM-200 system and 14.75 g/dL for Cell Dyn 3200. The mean difference was −0.16 g/dL ($p = 0.007$). 303 resuts (68.3% male, 31.7% female donors) were compared between HemoCue Hb201+ and Cell Dyn 3200. The mean Hb values were 14.12 g/dL for HemoCue Hb201+ and 14.79 g/dL for Cell Dyn 3200. The mean difference was −0.67 g/dL ($p < 0.001$). **Conclusion:** Even though the p value showed statistic differences between each method to the reference method, the difference is not clinically significant. Findings indicate that Hb measurements using NBM-200 system among our BD population is at least, as accurate as measurements with the current in use invasive method. This non-invasive approach is a favourable alternative to perform the pre-donation screening avoiding finger stick.

Disclosure of Commercial Conflict of Interest

J. Atorino: OrSense, Other; A. E. del Pozo: Nothing to disclose; S. Kuperman: Nothing to disclose; S. Oknaian: Osense Ltd., Stocks or Bonds; M. I. Puppo: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Atorino: Nothing to disclose; A. E. del Pozo: Nothing to disclose; S. Kuperman: Nothing to disclose; S. Oknaian: Nothing to disclose; M. I. Puppo: Nothing to disclose

SP116

The Effect of Targeted Group O Double Red Cell Collections on Disproportionate Group O Issues and Collections

Y De Waal[1] (yvonne.dewaal@sanbs.org.za), S Fourie[3], S Prithvi Raj[2], T Scanes[2], V Reddy[2]. *[1]Operations, South African National Blood Service, Kimberley, South Africa; [2]Operations, South African National Blood Service, Johannesburg, South Africa; [3]Operations, South African National Blood Service, Bloemfontein, South Africa*

Background/Case Studies: For the period April 1, 2011-March 31, 2012 the South African National Blood Service (SANBS) blood banks issued 743,647 units of red cell products of which 387,579 (52.12%) were group O. This blood was collected from a donor population in which the group O distribution was 48.80% with 46% of the general population being group O. This disproportionate issue of group O blood to the general and donor population is of great concern to SANBS as it leads to chronic low blood stocks and acute blood shortages of, in particular, group O blood. The reason for this disproportionate issue of group O blood within SANBS can be attributed largely to the issue of non-crossmatched group O blood from unmanned emergency banks in remote rural areas. The ALYX Component Collection System allows the collection of two units of filtered red cell concentrates from donors, every 112 days. SANBS has run a pilot study to establish whether group O donor targeted double red cell collections, in rural areas, might alleviate the collections challenges resulting from the disproportionate issue of group O blood within SANBS. This presentation reports on the results of the pilot study carried out at Kimberley Branch of SANBS. **Study Design/Methods:** Double red cell collections were carried out on group O donors within the Kimberley Branch of SANBS for the period April 1, 2012-March 31, 2013. This was termed the Test Period. Whole blood and double red cells total and group O collections data for the Test period were compared to the Control period (April 1, 2011-March 31, 2012). Data regarding collections were obtained from the SANBS Business Intelligence system (BI). Statistical analysis of the difference between proportions was determined using a two tailed z-test. Statistical significance was evaluated at the 0.05 level. **Results/Findings:** During the control period 18712 units of whole blood were collected from 3352 group O whole blood donors resulting in a donation frequency (df) of 2.41 donations / year. For the test period, 2226 double red cell units were collected from 270 donors (df 3.0) during 970 procedures, and 15456 whole blood units were collected from 3004 donors at a df of 2.26. A significant increase in the % of group O donations ($z = 5.7571 : p < 0.05$) was observed following the introduction of group O targeted double red cell collections. This increase is probably due to a significantly higher donation frequency ($z = -5.1132 : p < 0.05$) in double red cell (3.00) over whole blood donors (2.26). **Conclusion:** The selective targeting of group O donors for double red cell collections appears to be effective in increasing group O red cell collections at the Kimberley Branch of SANBS.

Disclosure of Commercial Conflict of Interest

Y. De Waal: Nothing to disclose; S. Fourie: Nothing to disclose; S. Prithvi Raj: Fenwal, Other; V. Reddy: Nothing to disclose; T. Scanes: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. De Waal: Nothing to disclose; S. Fourie: Nothing to disclose; S. Prithvi Raj: Nothing to disclose; V. Reddy: Nothing to disclose; T. Scanes: Nothing to disclose

SP117

Giving Life: Motivators to Become a Volunteer Blood Donor

M De Maio[1,2] (marianademaio@ufl.edu), E F Ganza[2], S Kuperman[2].
[1]College of Journalism and Communications, University of Florida, Gainesville, FL, United States; [2]Centro Regional de Hemoterapia, Hospital de Pediatría Garrahan, Buenos Aires, Argentina

Background/Case Studies: Almost two years ago the Centro Regional de Hemoterapia Garrahan (CRHG) decided to change its Blood Donor (BD) recruitment policy to achieve a 100% voluntary goal, improving from the former 40% level. CRHG's policy change was risky because it required having in place a set of measures to guarantee blood supply before implementing the new system. The purpose of this study was to explore the motivators to become a volunteer blood donor (VBD) for the CRHG. Understanding the factors that motivate donors to give blood would facilitate improvements in recruitment and retention VBD programs. The theories framing this study are the transtheoretical model of behavior change and the theory of planned behavior. **Study Design/Methods:** Four homogeneous focus groups of first-time donors recruited through social media were conducted. Both genders and a range of age groups were represented. The people within those groups were unknown to the interviewer and each other. Additionally, 10 in-depth interviews were conducted with donors with similar characteristics. The main question asked was regarding their motivators to become a VBD for the CRHG. Furthermore, the importance of blood donation as a volunteer activity, deterrents, awareness of need, and effective recruitment messages were among other topics discussed. **Results/Findings:** The age of the participants ranged between 23 and 53 years old (male = 41%, female = 59%); the level of education complete or incomplete university degree (50%); complete or incomplete high school (37.5%); and complete or incomplete elementary school (12.5%). Participants believed that blood donation is an important volunteer activity. 92% were motivated to donate blood to the CRHG because it supplies a children's hospital. 97% donated blood at a different institution for a friend or family member under the replacement system before becoming VBD for the CRHG. Myths 87%, fear 89%, and inconvenience 85% were cited as major barriers to becoming a volunteer donor. They suggested that social media messages educating about myths and promoting extramural blood drives were key to recruiting more volunteer donors. The word "complicity" came out 65% as an important concept to be taken into consideration when drafting messages for social media. **Conclusion:** Blood banks should take into account that potential donors on social media value the act of donating blood. Recruitment messages should relate to those feelings. Moreover, providing convenient opportunities to donate should be at the top of the list for blood centers. Social media campaigns ought to include messages to demystify beliefs about blood donation and help donors overcome fears.

Disclosure of Commercial Conflict of Interest

M. De Maio: Nothing to disclose; E. F. Ganza: Nothing to disclose; S. Kuperman: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. De Maio: Nothing to disclose; E. F. Ganza: Nothing to disclose; S. Kuperman: Nothing to disclose

SP118

Donor Safety Improvements after Excessive Collections of Whole Blood Donations

R M Cusick[2,1] (rcusick@bloodsystems.org), M D Bravo[1], P L Jones[3], L Su[2,1]. *[1]Medical Affairs, Blood Systems, Inc, Scottsdale, AZ, United States; [2]Medical Affairs, United Blood Services, Scottsdale, AZ, United States; [3]Operations, Processes, and Procedures, Blood Systems, Inc, Scottsdale, AZ, United States*

Background/Case Studies: Inadvertent excessive collection (EC) of whole blood (WB) donors is a potential complication of blood donation resulting in increased red cell (RC) and iron losses; these can lead to anemia and/or concurrent adverse vasovagal reactions (VVR). The 2001 FDA Guidance for Industry Recommendations for Collecting Red Blood Cells by Automated Apheresis Methods stipulates a deferral of 112 days for RC losses equaling or exceeding 300 ml. A process to manage donors of EC with the goals of mitigating donor harm and better understanding causation of EC was implemented across a system of community blood centers. **Study Design/Methods:** An over-collection is defined as a WB collection exceeding 550 ml, but the process described here only intervenes in cases of EC. EC is defined for allogeneic collections as a unit with >570 ml (~706 g) collected in the primary bag, including WB from stripped tubing. RC loss was calculated using volume of blood lost, including 40 ml in the diversion pouch, and the donor hemoglobin. Donors of allogeneic WB donation (AWBD) resulting in a loss of >300 ml RC were deferred 112 days. Equipment was evaluated after each event and incidents were managed per deviation management standard operating procedures. Analysis of EC occurring with AWBD was performed using 7 months of data (07/23/12-02/22/13) collected after process implementation at 11 blood centers. Data from correlating VVR was analyzed for a 5 month subset of this data (07/23/12-12/31/12). **Results/Findings:** 294,681 AWBD were analyzed, and 136 EC were reported (4.6 per 10,000). Of these EC, 91/136 (67%) resulted in a RC loss of <300 ml and 45/136 (33%) in a loss of ≥300 ml RC. Total EC rates/10,000 and those resulting in a ≥300 ml RC loss among individual centers ranged from 1.4 to 10.1 and 0.0 to 3.72, respectively. Of 103 EC drawn in the 5 month data subset, 7 (7%) were associated with a reported VVR; for the same time period, 3414 VVR were reported for 223,285 total AWBD (1.5%). **Conclusion:** EC of AWBD is a potentially serious problem that merits tailored management of donors and evaluation of equipment and staff. EC may expose donors to potential harm by increasing the risk of VVR, and the industry standard of 56 day deferral may not allow donors of EC sufficient time to restore iron and/or RC levels. 45 AWBD resulting in a loss of ≥300 ml RC over 7 months were identified and thus received the benefit of an appropriate 112 day deferral. Rates of VVR for a 5 month subset of EC data were 7% compared to an overall rate of 1.5% for all AWBD during the same time period. Determination of the root cause of EC incidents, whether they are equipment- and/or staff-related, may enable decreased occurrence of this problem. The wide variation in the rate of EC among blood centers requires further elucidation and may help identify underlying factors contributing to EC.

Disclosure of Commercial Conflict of Interest

M. D. Bravo: Nothing to disclose; R. M. Cusick: Nothing to disclose; P. L. Jones: Nothing to disclose; L. Su: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. D. Bravo: Nothing to disclose; R. M. Cusick: Nothing to disclose; P. L. Jones: Nothing to disclose; L. Su: Nothing to disclose

SP119

Donor Retention after Deferral: Impact on Time to Return

J Crane[1] (jecrane@uic.edu), C Hoagland[2], W Gao[3], R Kakaiya[4], M Papari[4], S A Campbell Lee[1]. *[1]Pathology, University of Illinois Hospital and Health Sciences System, Chicago, IL, United States; [2]Information Technology, Institute for Transfusion Medicine, Pittsburgh, PA, United States; [3]Center for Clinical and Translational Science, University of Illinois at Chicago, Chicago, IL, United States; [4]LifeSource Blood Center, Rosemont, IL, United States*

Background/Case Studies: An aging donor population makes it important to study temporary deferral and its impact on donor retention. One question is does temporary deferral impact some groups more than others? Disproportionate deferral rates have been documented previously. Therefore, studies that measure deferrals and donor retention by ethnic groups are important to understand and address these differences. This study aims to look at the impact of donor deferral on different groups in a major metropolitan area and identify the impact of deferral on the interdonation interval (interval beyond ineligibility) of established donors. **Study Design/Methods:** In this retrospective study, allogeneic whole blood and red cell apheresis donors deferred over a 4-year period from January 2008-December 2011 were identified. For the Control Group (DS, N = 3321), all successful donors in the first week of January 2008 and any subsequent donations were identified. The Study Group (DF, N = 310) was composed of donors temporarily deferred in January and February 2008 and their subsequent visits. First-time (FT) and established (Est) donors were identified along with the date of donation, inter-donation interval, and donation site. Demographic data collected included sex, age, and self-reported race which was divided into African-American (AA), Asian-American (AsAm), Hispanic (His), White-Caucasian (C), and Other (O). Deferral criteria were per national guidelines. Calculations included Days Until Eligible (DUE), Actual Days to Next Donation (AD), and Interdonation Interval (IDI), calculated as the number of days past the eligibility date. In calculating return, it is assumed donors present with the intent to donate. **Results/Findings:** For statistical analysis, Chi-square/Fisher tests for categorical variable comparisons and T-test/ANOVA for continuous variables were used (SAS version 9.2). All significant findings had $p < 0.05$. AA and AsAm were deferred significantly more compared with C (18.2% vs 7.2%, 16.7% versus 7.2%). Differences with other races were not significant. Comparison of DF and DS found significantly less DUE in the DS group (62.8 vs 122.1), however no difference in AD or IDI. Further return analysis was performed after pooling the data. FT donors had more DUE,

(85.8 versus 64.2), more AD (265.3 vs 186.0), and longer IDI (197.9 vs 122.9) than Est donors. AsAm donors had significantly more DUE than C (27.1), while all except AA had significantly longer AD and IDI. Additional significant return findings are summarized in the table. **Conclusion:** Previous findings are supported while also pointing to the need for focused recruitment efforts targeting FT, AA, His, and AsAm donors. Targeted recruitment efforts may be useful in ensuring a stable blood supply.

Disclosure of Commercial Conflict of Interest

S. A. Campbell Lee: Nothing to disclose; J. Crane: Nothing to disclose; W. Gao: Nothing to disclose; C. Hoagland: Nothing to disclose; R. Kakaiya: Nothing to disclose; M. Papari: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. A. Campbell Lee: Nothing to disclose; J. Crane: Nothing to disclose; W. Gao: Nothing to disclose; C. Hoagland: Nothing to disclose; R. Kakaiya: Nothing to disclose; M. Papari: Nothing to disclose

Non-Pooled Data	DS > DF (74.0% vs 62.6%)
Pooled Data	Est > FT (81.5% vs 31.4%) C > AA, His (82.6% vs 61.6%, 82.6% vs 64.5%)

SP120

Investigating Enthusiasm for a "Donate Blood for Research" Program among Australian-Deferred Blood Donors: Examining the Attitudes of a Sample of Variant Creutzfeldt-Jakob Disease (vCJD) Risk Deferred Donors

D Waller[1] (dawaller@redcrossblood.org.au), A Garradd[1], S Cheah[1].
[1]Research and Development, Australian Red Cross Blood Service, Sydney, NSW, Australia

Background/Case Studies: The Australian Red Cross Blood Service (ARBCS), in addition to providing products for clinical purposes, also uses blood products in an extensive Research and Development program. Blood used for such projects must meet specific criteria and may be "clinical grade" and/or discarded product. Consequently, the availability of blood for research use may be limited in Australia during times of low supply. Elsewhere, Blood Services (eg Canada) run "Blood for Research" programs whereby donors deferred for specific reasons can donate blood product solely for research use. The ARedCBS is considering whether a similar program would be viable to ensure sufficient provision of blood for research. There are few scientific publications on perceptions and attitudes towards "Blood for Research" programs. Consequently, it is unknown whether attitudes towards the ARBCS would change if such a program were instituted. Additionally, the manner in which deferred donors might agree to participate is unknown. The aim of this research is to examine deferred donor attitudes to determine if a "Blood for Research" program would be a viable option in Australia. **Study Design/Methods:** A sample of 1449 donors previously deferred for variant Creutzfeldt-Jakob Disease (vCJD) risk were contacted via postal mail and asked to complete a short survey on their attitudes towards an Australian "Blood for Research" Program. This sample was chosen because there is a high rate of vCJD risk deferral in Australia and this number continues to grow with sustained migration from the United Kingdom. Survey responses were matched to demographic data from the Blood Service records for analyses. **Results/Findings:** Significant support for a "Blood for Research" program from the survey is presented with optimal methods of program implementation highlighted. Segmentation of survey results based on demographic data will be discussed with the purpose of identifying individuals likely to agree to donating blood for research use. Selected participants from the initial survey are also invited to join focus groups to gather further qualitative data on attitudes to donating blood for research use. **Conclusion:** This survey represents the initial phase of a series of consultations to understand community attitudes towards a "Blood for Research" program in Australia. Results will help to inform future evidence-based policy decisions at the Blood Service.

Disclosure of Commercial Conflict of Interest

S. Cheah: Nothing to disclose; A. Garradd: Nothing to disclose; D. Waller: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Cheah: Nothing to disclose; A. Garradd: Nothing to disclose; D. Waller: Nothing to disclose

SP121

A New Smartphone Application to Facilitate Young Donors Recruitment and Ease Blood Donations

E Shinar[1] (eilats@mda.org.il), I Abramovitch[2], R Tzabari[2], D Zimat[2], D Richter[1], E Yafe[2]. *[1]Blood Services Division, Magen David Adom in Israel, Ramat Gan, Israel; [2]Division of Training and Public Relation, Magen David Adom in Israel, Tel Aviv, Israel*

Background/Case Studies: Although the Israeli public is well aware of the importance of blood donations in saving lives and the need to preserve an adequate national blood inventory for emergencies, only 3.8% of the population donates blood regularly. An innovative smart phone application was developed, using an up-to-date technology, to facilitate recruitment of young donors, raise their awareness and ease their access to blood collection sites nationwide. **Study Design/Methods:** The application, which was designed in a friendly and easy-to-use technique, was made available on Apple's app-store and on Android's play store, and was accompanied by an explanatory short video on YouTube. It contains features such as: donor acceptance/deferral criteria; information regarding the nearest blood collection site, including a guided map through WAZE application; connection to social networks such as FaceBook and Twitter; and a possibility to communicate to friends the accessibility and/or waiting time at the donation site (using a simple red-yellow-green signal) so they can plan their arrival to the blood drive. This screen is also seen by the blood drive coordinator of the area, enabling the official to better plan and operate the drive. The "Contact Us" module is used to contact the blood services regarding questions, particular requests, and post donation information. A 24 hour post-donation "Thank You" message and a 90 days post-donation reminder are sent, inviting donors to come back for their next donation. The application also contains short videos on first aid and resuscitation instructions. **Results/Findings:** During the first month following the launching of the application, some 9,785 downloads were recorded, with 14,501 screen views. The average session duration was 47 seconds, with 1.94 screens /session. About 84.5% were returning users. The application was rated 4.6/5 by the users. **Conclusion:** This new application enables the blood services to reach a younger segment of the population, using modern tools. It helps raise public awareness regarding the importance of blood donations, assists donors to plan their blood donation, strengthens the contact with the public, and increases the blood services efficiency. The application can be downloaded by typing search words: "MDA", or "I donated blood", or "First Aid" on Apple's App Store and on Play store from Android.

Disclosure of Commercial Conflict of Interest

I. Abramovitch: No Answer; D. Richter: No Answer; E. Shinar: Nothing to disclose; R. Tzabari: No Answer; E. Yafe: No Answer; D. Zimat: No Answer

Disclosure of Grants Conflict of Interest

I. Abramovitch: No Answer; D. Richter: No Answer; E. Shinar: Nothing to disclose; R. Tzabari: No Answer; E. Yafe: No Answer; D. Zimat: No Answer

SP122

Blood Donation Registry: A Cost-Effective Mechanism of Regaining "Lost" Donors

D O Irving[1,2] (dirving@redcrossblood.org.au), R Slonim[2], E Garbarino[2], G Smith[1], D Waller[1], C Wang[3]. *[1]R&D, Australian Red Cross Blood Service, Sydney, NSW, Australia; [2]University of Sydney, Sydney, NSW, Australia; [3]Harvard Business School, Boston, MA, United States*

Background/Case Studies: Retaining an appropriately sized donor base to ensure adequacy of blood supply is a challenge to blood collection agencies throughout the world. In Australia, approximately 3.5% of the eligible donor population actively donate although a significantly larger proportion have donated at least once, often in response to advertising campaigns in times of supply shortage. In order to investigate methods to re-engage individuals who have fallen into the category of "lapsed donors," a pilot study was carried out where it was hypothesised that inviting individuals to make a moderate commitment, joining a registry in which participants would be contacted only in times of low inventory or specific need, would be more appealing to lapsed donors than making a larger commitment of returning to donate regularly during normal times. **Study Design/Methods:** Individuals who had previously donated on at least one occasion but who had not returned to donate in more than 28 months and were thus no longer being contacted by the Blood Service to donate were identified from archival Blood Service records. Contact scripatients were prepared and 2277 individuals (28-44 months lapsed, O or A type donors) were contacted and invited to join the donation registry. They were advised that by joining the registry, they would only be contacted in times of need, such as when inventory was less

than three days, and would be contacted no more than three times in a year. Responses were compared with business as usual approaches (routine advertising and marketing campaigns). **Results/Findings:** Contact Center operators reported enthusiastic support from those called, with most expressing strong interest in joining such a registry. Of the 2277 individuals contacted, 1856 (81.5%) joined the registry. During a period of inventory shortage, 832 of these registry donors were contacted, advised of the shortage, and asked to donate. A 16% conversion rate to appointments was achieved (percentage of appointments made per call) with 81 blood donations collected. Thus, roughly 10% of calls made to registry participants were converted to blood donations. Compared with standard operating practice, this is approximately twice the response rate obtained from a control group of otherwise identical donors. **Conclusion:** Results from this pilot study indicate that a higher than expected proportion of long-term lapsed donors, who would otherwise be lost to the Blood Service, are willing to join a registry. Furthermore, those who join that registry are more likely to donate blood during times of need than donors who are regularly contacted. Further studies are underway to try to understand the motivators driving these responses. This blood donation registry will enable the Blood Service to keep such individuals engaged, coordinate volunt.

Disclosure of Commercial Conflict of Interest

E. Garbarino: Australain Red Cross Blood Service, Grants or Research Support; D. O. Irving: Nothing to disclose; R. Slonim: Nothing to disclose; G. Smith: Nothing to disclose; D. Waller: Nothing to disclose; C. Wang: No Answer

Disclosure of Grants Conflict of Interest

E. Garbarino: Australian Red Cross Blood Service, Grants or Research Support; D. O. Irving: Nothing to disclose; R. Slonim: Nothing to disclose; G. Smith: Nothing to disclose; D. Waller: Nothing to disclose; C. Wang: No Answer

SP123

Non-Invasive Detection of Anemia by Occlusion Spectroscopy

A Weinstein[1] (aharon.weinstein@orsense.com), O Herzenstein[1], A Kononenko[1], E Gabis[1], A Korenberg[2], M Hod[3]. *[1]OrSense Ltd., Petah Tikva, Israel; [2]Department of Hematology and Blood Bank, Assaf Harofeh Medical Center, Zerifin, Israel; [3]Helen Schneider Hospital for Women, Rabin Medical Center, Petah Tikva, Israel*

Background/Case Studies: Hemoglobin (Hb) measurements allow the detection of anemia and hemorrhage, and are widely used in blood banks, for pre-donation screening, and in hospital wards such as ICU and ER. The introduction of non-invasive Hb measurements can improve donor recruitment by eliminating pain, reducing the need for trained personnel and minimizing the infection risk for donors and health providers. The purpose of this multi-center study is to test the sensitivity of the NBM-200 system (OrSense Ltd.) in screening of anemia, and to evaluate its performance in a broad range of Hb values. The NBM-200 is based on Occlusion Spectroscopy technology in the red/near-infrared range. The measurement is performed using an annular, multi-wavelength probe with pneumatically operated cuffs, with which an over-systolic pressure is produced at the finger base. The monitor is targeted for use by blood donation centers, caregivers in hospitals, OB/GYN, and large population screening. **Study Design/Methods:** The clinical study was conducted on patients in the Hematology and Blood Bank Department of Assaf Harofeh Medical Center (Zerifin, Israel) and on pregnant women in the Women Hospital of the Rabin Medical Center (Petah Tikva, Israel). Upon receipt of informed consent, two non-invasive Hgb measurements were performed on the right and left thumbs of the subject. Reference Hb values were obtained from venous blood samples and evaluated on a Beckman Coulter LH 750 blood analyzer. **Results/Findings:** A total of 1255 volunteers (478 male, 777 female) in the ages 19-95 years participated in the trial, yielding 2395 data pairs. Reference hgb values were in the range 5.8-17.9 g/dL, with an average of 11.9 ± 1.9 g/dL, and the values obtained by the NBM-200 device were in the range 5.0-17.6 g/dL with an average of 12 ± 1.7 g/dL. The mean error (bias) of the NBM-200 readings comparing to the reference was 0.1 g/dL, and the mean absolute error was 0.8 g/dL. The correlation was 0.86 and the accuracy, defined as the standard deviation of error, was 0.97 g/dL. For repeating tests the precision was 0.45 g/dL with CV of 3.7%. Of the values obtained, 1462 (61%) were under 12.5 g/dL, the exclusion border lines commonly used in blood donation centers. The NBM detected these values with sensitivity of 86% and specificity of 82%. Focusing on lower Hb values, there were 768 (32%) points that fell under 11 g/dL, and the NBM identified 98% of them as under the border line, thus excluding almost all the significant anemic values. **Conclusion:** The non-invasive NBM-200 showed accurate performance compared with the invasive reference values and showed high sensitivity to anemia. The operating staff found the non-invasive system easy to use with minimal training, safe and highly appreciated by the subjects. The findings support the use of the device as a simple and accurate measurement technique for measurement of hemoglobin and anemia detection.

Disclosure of Commercial Conflict of Interest

E. Gabis: No Answer; O. Herzenstein: No Answer; M. Hod: OrSense, Grants or Research Support; A. Kononenko: No Answer; A. Korenberg: OrSense, Grants or Research Support; A. Weinstein: OrSense, Grants or Research Support, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

E. Gabis: No Answer; O. Herzenstein: No Answer; M. Hod: OrSense, Grants or Research Support; A. Kononenko: No Answer; A. Korenberg: OrSense, Grants or Research Support; A. Weinstein: OrSense, Grants or Research Support, Consulting or Board of Director Fees

SP124

An Unusual Donor Reaction Case Report: Dystonic Conversion Reaction

J W Smith[1] (jsmith@obi.org), P Latschar[1], J Armitage[1]. *[1]Medical Administration, Oklahoma Blood Institute, Oklahoma City, OK, United States*

Background/Case Studies: Vaso-vagal and hypovolemic reactions are some of the more common reactions associated with whole blood donation. Signs, symptoms, and treatments are dealt with regularly by blood collection staff. In this case study, a reaction is reported which began with commonplace symptoms but progressed in severity to resemble stroke-like symptoms. **Study Design/Methods:** A 17 year old female gave her second whole blood donation at a mobile-based high school blood drive. She was noted to be 5′ 10″ tall, approximately 150-175 pounds, with hematocrit of 38%. Predonation vital signs included a blood pressure of 112/60 with pulse of 95. The donation was completed successfully and uneventfully at 9:28 AM. Subsequently she developed lightheadedness, weakness, nausea, and difficulty breathing. Usual treatment steps were initiated, but the donor was noted to have "tetany," described by staff as rigidity of arms and legs, and "loss of consciousness," noted as inability to respond to questions or events. Blood pressure remained stable throughout (last BP by staff 118/76 at 11:38 AM). Pulse rate did not decrease and was noted to be 98 at 11:38 AM. Due to the donor's inability to respond to questions and persisting stroke-like rigidity of left side, she was transported by ambulance to a local hospital. **Results/Findings:** Initial emergency department assessment was tetany of hands and feet and non-verbal status. Her guardian provided history of one similar occurrence in the past. Blood and urine tests revealed no significant abnormalities, including negative drug toxicity panel and pregnancy assessment. The donor was able to walk upon leaving the emergency department at 13:55 PM that afternoon, but returned the next morning complaining of pain and tightness of left arm and leg, and amnesia of events of the prior day. Subsequent work-up by MRI, CT, and EEG revealed no abnormalities. The neurologist noted slightly decreased strength of left arm and leg but no other significant neurologic abnormalities. A history of traumatic childhood with history of sexual abuse was acknowledged, and this post-donation reaction was diagnosed as a dystonic conversion reaction. A conversion reaction refers to a condition in which motor or sensory symptoms reflect psychological conflict. Dystonia, or altered muscle tone, was one manifestation in this case. She was referred for further psychotherapy. **Conclusion:** Blood collection staff deal regularly with donor reactions, many of which relate to vaso-vagal or hypotensive states. Treatment is administered and the donor routinely stabilizes. This case documents an example of an unusual psychiatric condition that may present with signs and symptoms very similar to vaso-vagal reactions. It is important to review all aspects of a case and be alert when the presentation seems out of proportion to the documented physical findings.

Disclosure of Commercial Conflict of Interest

J. Armitage: Nothing to disclose; P. Latschar: Nothing to disclose; J. W. Smith: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Armitage: Nothing to disclose; P. Latschar: Nothing to disclose; J. W. Smith: Nothing to disclose

SP125

Iron Supplement Use and Iron Deficiency Awareness among Active Donors: Baseline Survey Results

W R Steele[1] (whitney.steele@redcross.org), E P Notari[1], A F Eder[2].
[1]Transmissible Diseases—Holland Laboratory, American Red Cross—Biomedical Services, Rockville, MD, United States; [2]Medical Office, American Red Cross—Biomedical Services, Rockville, MD, United States

Background/Case Studies: Individuals who donate whole blood or double red cells are at risk of iron depletion and deficiency. The blood system here recently developed new materials to educate donors about the risk of iron deficiency and the need for frequent donors to replace the iron lost with each blood donation. Before releasing the new materials, a baseline survey of active blood donors was conducted to assess their knowledge about blood donation and iron deficiency and their current use of dietary iron supplements. **Study Design/Methods:** Whole blood or double red cell donors who gave in February 2013 were randomly selected for the survey if they had ≥2 donations in the last year and a valid email address. Donors (n = 21,837) were emailed an invitation letter and a link to an online survey. Simple frequencies and percentages were calculated after removing missing values and X^2 analysis performed (SAS® 9.2). **Results/Findings:** A total of 6700 (30.7%) responses were received (56% female, 93% Caucasian, median age 51 years, 64% had ≥3 donations in the last 12 months) and 6330 with complete responses were included in this analysis. Only 24% of donors correctly identified that the fingerstick test only measures hemoglobin (Hb) while 63% said that it measured iron and 7% said that it measured both iron and Hb. Although the pre- and post-donation written materials contain statements about blood donation removing iron from the body and possibly causing or aggravating iron-deficiency anemia, only 31% of respondents said they had read this before and/or after donation. Over 96% of donors agreed with the statement "Donors who are accepted to give blood must have normal iron levels". More than half of donors reported taking a multivitamin and of those 63% said it contained iron, 25% said it did not and 12% said they didn't know. Nine percent of donors said they took a separate iron supplement. Overall 38% of donors reported taking some form of iron (29.1% multi-vitamin with iron, 5.3% separate iron supplement, 3.8% both). Significantly more women than men reported taking iron (45.5% versus 29.5%, $p < 0.0001$). Donors who gave ≥3 times in the last year were more likely to report taking iron than donors giving less frequently (41% versus 32%, $p < 0.0001$). Of the donors taking a multi-vitamin with iron, 21% said they started taking it because they donate blood and of those taking a separate iron supplement, 53.6% said the same. **Conclusion:** Awareness of the possibility of becoming iron-deficient from blood donation is quite low even in this population of active donors. Many donors have misconceptions about the Hb test measuring iron and donor qualification. Almost half of the women and nearly a third of the men who are active donors already report taking iron replacement. The survey establishes a baseline for assessing if the new educational materials increase awareness of iron deficiency and encourage iron supplementation use after blood donation.

Disclosure of Commercial Conflict of Interest

A. F. Eder: Nothing to disclose; E. P. Notari: Nothing to disclose; W. R. Steele: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. F. Eder: Nothing to disclose; E. P. Notari: Nothing to disclose; W. R. Steele: Nothing to disclose

SP126

Prediction and Control of the Anti-D Donor Cohort in the Netherlands

M Janssen[1] (m.p.janssen@umcutrecht.nl), M A Berkowska[2], O Verhagen[2], L Della Valle[2], H Vrielink[3], E van der Schoot[2]. *[1]Transfusion Technology Assessment, UMC Utrecht, Utrecht, Netherlands; [2]Department of Experimental Immunohematology, Sanquin, Amsterdam, Netherlands; [3]Sanquin Blood supply, Amsterdam, Netherlands*

Background/Case Studies: Presence of the maternal anti-D antibodies is the main cause of the haemolytic disease of the fetus and newborn. Thus, to prevent perinatal immunization, RhD-negative women pregnant with the RhD-positive child receive allogenic anti-D immunoglobulins. The main source of these antibodies is the plasma obtained from healthy female volunteers who were primarily immunized during pregnancy, and receive additional antigen boosters to maintain high antibody titers in the post-reproductive age. The ageing of the active anti-D donor population in combination with the success of the anti-D prophylaxis, which results in fewer primary immunizations, may result in a shortage of anti-D plasma in the future. Therefore, an analysis of the dynamics of the change in size of the anti-D donor cohort is required in order to predict the future size of the current donor population while accounting for donor age. **Study Design/Methods:** The length donor careers was studied with survival analyses using recorded donor visits in the period from 1995-2012. An age dependent anti-D donor retention model was developed in accordance with findings from the survival analyses, and various Monte-Carlo simulations were performed to predict the size and variability of the future donor population. **Results/Findings:** The number of active anti-D donors reduced from 300 in 1996 to 200 in 2012. At the same time the median donor age increased from 50 years in 1996 to 60 years in 2008 (0.8 years per year). Survival analyses show that after the first year dropout among donors is random and hardly influenced by age (except for the maximum donation age). Contrarily, the probability of donors stopping after their first donation is strongly depending on donor age. With a simple model it was possible to accurately predict the size, age distribution and proportion of forced retirements in the donor population over a period of 15 years (1997-2012). From 2009 onwards an increase in dropout rate among new donors is observed. For various dropout rates and donor recruitment scenarios, which differ in the number of new donors per year and the age distribution of these donors, the long-term donor population size was determined. The simulations show that for example with a dropout rate of 10% one would need 40 new donors per year to maintain a donor population of 200 donors, with a dropout rate of 15% one would require 55 new donors annually. **Conclusion:** Contrary to current perception, ageing of the donor population is no threat to the anti-D antibody supply. The recently observed reduction in the size of the anti-D donor population is rather caused by the increased dropout rate among new donors. What is causing this change needs to be further studied. To determine the number of new donors to be recruited annually for a sufficient anti-D supply, the developed model can be used with the observed dropout rate and the number of active donors required as inputs.

Disclosure of Commercial Conflict of Interest

M. A. Berkowska: Nothing to disclose; L. Della Valle: Nothing to disclose; M. Janssen: Nothing to disclose; E. van der Schoot: Nothing to disclose; O. Verhagen: Nothing to disclose; H. Vrielink: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. A. Berkowska: Nothing to disclose; L. Della Valle: Nothing to disclose; M. Janssen: Sanquin Blood Supply Foundation, Other; E. van der Schoot: Nothing to disclose; O. Verhagen: Nothing to disclose; H. Vrielink: Nothing to disclose

SP127

Adverse Events during Healthy Donor and Autologous Peripheral Blood Stem Cell Collections

S M Morgan[1] (jagus005@umn.edu), A D Johnson[1], S Shaikh[2], C S Cohn[1], N D Zantek[1]. *[1]Laboratory Medicine and Pathology, University of Minnesota, Minneapolis, MN, United States; [2]American Red Cross, Mid-America Blood Services Division, St. Paul, MN, United States*

Background/Case Studies: Apheresis procedures are generally safe and well tolerated; however, adverse events (AEs) can occur. AEs range from mild to severe and include a variety of mechanisms such as hypocalcemia/citrate toxicity, neurologic, allergic, and vascular access issues. There is limited data comparing the AE rates in peripheral blood stem cell (PBSC) collections of healthy (allogeneic) donors versus autologous patients. **Study Design/Methods:** A single center retrospective review of PBSC collections from a 12-month period was conducted. Procedures were performed in an apheresis center and inpatient and outpatient care units. Adult collections were performed on the Amicus (Fresenious Kabi, Germany) blood cell separator and pediatric collections were performed on the Cobe Spectra (Terumo BCT, Lakewood, CO). Anticoagulant citrate dextrose -A (ACD-A) was used for anticoagulation. PBSC collections were 5 hours duration unless the patient weighed 12 kilograms or less in which case the duration was decreased to 4 hours. Prophylactic intravenous 10% calcium gluconate infusion was given at a rate of 20 mg/kg/hr throughout the procedure and, if a blood prime was required, the rate was increased to 40 mg/kg/hr. If the patient experienced symptoms of hypocalcemia/citrate toxicity (paresthesias/tingling, nausea, or muscle cramps), the 10% calcium gluconate infusion was increased to a maximum of 60 mg/kg/hr. AEs were recorded as they occurred. **Results/Findings:** A total of 1470 adult and pediatric apheresis procedures were performed. Of these, 360 were PBSC collections including 265 autologous collections and 95 healthy donor collections. Overall, eight patients experienced a total of eight (2.2%) AEs. Three (1.1%) AEs occurred in three autologous donors: two hypocalcemia/citrate toxicity reactions and one event of stridor and decreasing oxygen saturation. Five

(5.3%) AEs occurred in five healthy donors: three hypocalcemia/citrate toxicity reactions, one vascular access failure resulting in an incomplete procedure, and one requirement for multiple venipunctures to obtain vascular access. The rate of reactions was significantly different (P = 0.03) between autologous and healthy donors. The reactions were all classified as mild to moderate in severity. **Conclusion:** Mild to moderate AEs occur not infrequently in PBSC collections. Healthy donors experienced more AEs during PBSC collection than autologous patients. Health care workers involved in the collection process must be aware and prepared for this increased risk in order to facilitate the appropriate care in a timely manner.

Disclosure of Commercial Conflict of Interest

C. S. Cohn: Fenwal Inc., Grants or Research Support,Travel Support or Honorarium; A. D. Johnson: Nothing to disclose; S. M. Morgan: Nothing to disclose; S. Shaikh: Nothing to disclose; N. D. Zantek: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. S. Cohn: Fenwal Inc., Travel Support or Honorarium,Grants or Research Support; A. D. Johnson: Nothing to disclose; S. M. Morgan: Nothing to disclose; S. Shaikh: Nothing to disclose; N. D. Zantek: Nothing to disclose

SP128

Iron Depletion and Replacement in Blood Donors: Testing Feasibility of large Scale Ferritin Screening and Iron Replacement at Blood Centers

J B Gorlin[1] (jed@mbc.org), K Doeden[1], D Elsmore[1], A Hove[1], L Katz[2], K Kirbach[2]. *[1]Physician Services, Memorial Blood Centers, St. Paul, MN, United States; [2]Medical Services, Mississippi Valley Regional Blood Center, Davenport, IA, United States*

Background/Case Studies: Frequent blood donations lead to iron depletion. A whole blood donation of 450-500 mL contains 55-70 g of hemoglobin (Hb) and consequently 187 mg to 238 mg of iron. A 2011 FDA Workshop concluded that blood collection centers need to mitigate development of iron deficiency among blood donors. As an alternative to raising Hb eligibility thresholds or lengthening donor intervals, this study is investigating measuring iron stores and replacing iron in donors with low ferritin. **Study Design/Methods:** Two thousand blood donors at higher risk for iron depletion (Hb values of 12.0-13.0 g/dL for females and 12.5-13.5 g/dL for males) are being enrolled following donor consent and IRB approval. A ferritin test level is performed on each donor to screen for iron depletion, as well as a complete blood count (CBC). For females with hemoglobin 12.0-12.4 only samples are drawn. On all other study subjects ferritin is measured from a sample obtained during routine donor testing. The threshold for low ferritin is defined as ≤20 ng/mL for females and ≤30 ng/mL for males. Zinc protoporphyrin point of care testing will also be performed on a limited subset of donors and values compared to ferritin values subsequently obtained to test for feasibility of using a point of care iron determination. Participants below the ferritin threshold are deferred 112 days and offered iron supplements. The participants above the ferritin threshold are deferred the standard 56 days from blood donation (no iron supplements offered). Both groups are asked to return after the deferral period for a repeat ferritin test, CBC, and blood donation (if eligible). **Results/Findings:** As of May 3, 2013, 41 donors have been enrolled in the study. 90% of participants are below the threshold for low ferritin. The average initial ferritin of all participants is 10.6 ng/mL with a standard deviation of 8. To date only four participants have returned for follow-up, and returned above the ferritin threshold. The average ferritin improvement (defined as the final ferritin value minus initial ferritin value) is 11.ng/mL. MCV also increased for those presenting with low MCV. Three% of participants offered iron supplements refused. A total of 18 participants have had a midway inquiry to check tolerance and adherence to the supplementation regiment. 67% are taking the supplements daily and 89% of these participants are taking the supplements at least 5 days a week. **Conclusion:** The vast majority (90%) of donors who present to fixed sites with hemoglobins near the lower threshold have ferritin levels below the threshold for low ferritin. Although small numbers of returned donors preclude establishing statistical significance, initial findings suggest clear improvement in blood donors' ferritin levels with iron supplementation and longer deferral period between blood donations in an operationally feasible iron supplementation program.

Disclosure of Commercial Conflict of Interest

K. Doeden: No Answer; D. Elsmore: Nothing to disclose; J. B. Gorlin: Ortho, Grants or Research Support, CVS-Caremark, Grants or Research Support; A. Hove: No Answer; L. Katz: Nothing to disclose; K. Kirbach: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Doeden: No Answer; D. Elsmore: Nothing to disclose; J. B. Gorlin: Roche, Grants or Research Support, Diasorin, Grants or Research Support, Beckman-Coulter, Grants or Research Support; A. Hove: No Answer; L. Katz: Nothing to disclose; K. Kirbach: Nothing to disclose

SP129

Hypertension Screening among Healthy Young Adult Volunteer Blood Donors

S Eason[1,3] (season@carterbloodcare.org), S Goudar[1], M Sayers[1,2]. *[1]Carter BloodCare, Bedford, TX, United States; [2]University of Texas Southwest Medical Center, Dallas, TX, United States; [3]Center for Community Health, University of North Texas Health Science Center, Fort Worth, TX, United States*

Background/Case Studies: Hypertension is a major contributor to cardiovascular disease, which is the leading cause of death in the US. Since hypertension can progress asymptomatically over a period of time, early screening and identification of individuals at risk are essential. The volunteer blood donor setting offers such an opportunity. It was decided to use this setting to compare the prevalence of hypertension in volunteer blood donors from 2010-2012 according to their age, gender, and ethnicity. **Study Design/Methods:** Blood pressure was measured using automated equipment (Welch Allyn, Spot Vital Signs LXi) on 75,303 volunteer blood donors age <20. Blood pressure levels were defined as Normal <120/80, Prehypertension 120-139/80-89, and Hypertension ≥140 and/or ≥90, systolic, diastolic respectively. Gender was evenly distributed 45.4% (34,166) male and 54.6% (41,137) female. The four ethnic groups included (with males and females, respectively, in parenthesis): African Americans, 6,675 (3,195, 3,480); Asian, 1,801 (924, 877); Caucasian, 39,197 (17,934, 21,263); Hispanic, 17,212 (7,519, 9,963); and those not identifying an ethnicity were categorized as "other," 10,418 (4,594, 5,824). **Results/Findings:** Prehypertension and hypertension occurred more often in males, 51%, 33.4%; and in females, 14.1%, 3.3% respectively. **Conclusion:** Blood donor screening provides the opportunity to identify hypertension in an ostensibly healthy subset of the population. Although prehypertension and hypertension was detected across all ethnicities and both genders, early identification of hypertension at the prehypertension level could be of particular value to this younger group, who are the least likely to have regular screening.

Disclosure of Commercial Conflict of Interest

S. Eason: Nothing to disclose; S. Goudar: Nothing to disclose; M. Sayers: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Eason: Nothing to disclose; S. Goudar: Nothing to disclose; M. Sayers: Nothing to disclose

Percentage of blood donors, under age 20, within defined blood pressure levels

	African American		Asian		Caucasian		Hispanic		Other	
BP Levels Systolic/Diastolic	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Normal <120/80	32.0%	56.3%	41.0%	72.1%	35.3%	62.6%	34.1%	66.6%	36.4%	62.9%
Prehypertension 120-139/80-89	51.0%	39.3%	48.5%	26.1%	50.7%	34.1%	51.8%	30.5%	50.6%	33.6%
Hypertension ≥140/90	17.1%	4.4%	10.5%	1.8%	14.0%	3.3%	14.1%	2.9%	13.1%	3.5%

SP130

Evaluation of Blood Donor Ferritin Testing

J Dunn Williams[1] (jdunnwilliams@mycts.org), M Dixon[2], G F Robertson[1], M D Bravo[4], R C Williams[3], G F Leparc[1], S Caglioti[1], P C Williamson[1].
[1]Phoenix, Creative Testing Solutions, Phoenix, AZ, United States; [2]Dallas, Creative Testing Solutions, Phoenix, AZ, United States; [3]Phoenix Epidemiology and Clinical Research Branch, National Institutes of Health, Phoenix, AZ, United States; [4]Blood Systems, Inc., Scottsdale, AZ, United States

Background/Case Studies: AABB recommends that blood centers implement strategies to monitor iron deficiency in blood donors. Therefore, ferritin testing was implemented. This study reports early results of ferritin screening from two donor testing laboratories. **Study Design/Methods:** Ferritin testing was performed on preselected blood donors (males with hemoglobin [Hb] of 12.5-13.4 g/dL; females with Hb of 12.5-12.9 g/dL). Male donors with ferritin levels <30 ng/mL and female donors with ferritin levels <20 ng/mL were deferred by the blood center for 24 weeks. Results within the instrument linearity (8-450 ng/mL) were considered valid and results outside that range were considered invalid. **Results/Findings:** From November-April 2013, 30,219 blood donations (71% female; 29% male) were selected for ferritin testing across 17-blood centers. A total of 26,968 donations (89.2%) had valid results and 3,251 (10.8%) were invalid. Within the invalid results: 3011 (92.6%) had levels <8 ng/mL; 72 (2.2%) had levels >450 ng/mL and 168 (5.2%) were not tested. Within valid results the mean for females was 32.3 ng/mL [95%CL: 31.75-32.77] and for males 54 ng/mL [95%CL: 52.55-55.38]. Descriptive statistics were performed using standard 2×2 and 2×4 contingency table analyses to evaluate independent reciprocal events based on gender and age for results outside instrument linearity, as well as blood center donor deferral for low test results within instrument linearity. 1)Results outside instrument linearity: for 3083 test results (a) gender comparison with values <8 ng/mL, 77.3% were females versus 22.7% males; values >450 ng/mL, 15.3% were females vs 84.7% males; $p < 0.0001$; (b) Age comparison (separated into 4-quartiles (Q), young to old) for values <8 ng/mL: Q-1: 36%, Q-2: 29%, Q-3: 23%,Q-4 : 12%; for values >450 ng/mL: Q-1: 0%, Q-2: 29%, Q-3: 36%,Q-4: 35%. The majority of out of range values <8 ng/mL was in Q-1/Q-2 at 65.5%; while the majority out of range >450 ng/mL were in Q-3/Q-4 at 70.8%; $p < 0.0001$. 2) Results within instrument linearity: for 26,968 donations tested, 12,826 (47.6%) were deferred versus 14,142 (52.4%) not deferred. Within the deferrals by gender, there were 8860 females (69%) and 3966 males (31%), $p < 0.0002$. Within all age categories more than 50% of donors were not deferred with the largest difference shown in Q-2 with 54.9% not deferred, 45.1% deferred, $p < 0.0001$. **Conclusion:** Donors selected for testing had low Hb; therefore low ferritin was expected, not individuals with iron overload. The laboratory's inability to provide test result values occurs more frequently due to low ferritin levels which are found more often in females than males and more often in younger blood donors. Within valid results the blood centers' deferral occurs more often in females than males. Donor testing laboratories that provide ferritin levels allow blood centers to evaluate and properly manage donor safety for iron deficiency anemia.

Disclosure of Commercial Conflict of Interest

M. D. Bravo: Nothing to disclose; S. Caglioti: Nothing to disclose; M. Dixon: No Answer; J. Dunn Williams: Nothing to disclose; G. F. Leparc: Nothing to disclose; G. F. Robertson: Nothing to disclose; R. C. Williams: Nothing to disclose; P. C. Williamson: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. D. Bravo: Nothing to disclose; S. Caglioti: Nothing to disclose; M. Dixon: No Answer; J. Dunn Williams: Nothing to disclose; G. F. Leparc: Nothing to disclose; G. F. Robertson: Nothing to disclose; R. C. Williams: Nothing to disclose; P. C. Williamson: Nothing to disclose

SP131

Assessment of Noninvasive Hemoglobin Testing for Blood Donation Screening

S Graminske[1] (sharon.graminske@bcw.edu), L Kline[2], P H Whitley[3], M Corradino[4]. *[1]BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]American Red Cross Holland Laboratory, Rockville, MD, United States; [3]American Red Cross, Norfolk, VA, United States; [4]Clinical Research, Masimo Corp, Irvine, CA, United States*

Background/Case Studies: The Pronto-7 is a noninvasive, handheld, spectrophotometric device that uses a multiple-wavelength optical finger sensor to estimate hemoglobin concentration (SpHb). Because hemoglobin testing with SpHb is fast and painless it may offer advantages over invasive methods in the blood donation setting. The aim of this study was to compare the accuracy, sensitivity and specificity for donor qualification of SpHb and three invasive hemoglobin (Hb) test methods to the analysis of venous blood with the cyanmethemoglobin assay (HiCN) in prospective blood donors. **Study Design/Methods:** Prospective blood donors from five blood donation facilities were consented. To enrich the sample of deferrable subjects, prospective donors likely to be near the hb level of 12.5 g/dL were preferentially recruited. Following noninvasive measurements with SpHb (Pronto-7 sw ver 2.2.16, with rainbow 4D sensor, rev F, Masimo Corp.) capillary blood samples were obtained and measured with HemoCue 301 devices (HQc, HemoCue, Inc.) and by the copper sulfate method (CuSO4). Lastly, venous blood samples were collected and analyzed by a laboratory hematology analyzer (Coulter, Beckman Coulter Model LH 500,) and HiCN, the gold standard for hemoglobin measurement. Bias and precision (1 SD) of the three quantitative methods (SpHb, HQc and Coulter) were calculated. Additionally, sensitivity (ability to detect hemoglobin <12.5 g/dL) and specificity (ability to detect hemoglobin ≥12.5 g/dL) of each method were calculated, followed by a McNemar test to determine significant differences between tests ($p \leq 0.05$). **Results/Findings:** Data were collected from 429 subjects. Based on HiCN, 107 subjects would have been deferred (25%) and 322 (75%) accepted. Bias and precision were 0.1 ± 1.1 g/dL for SpHb, 0.4 ± 0.7 g/dL for HQc and -0.2 ± 0.3 g/dL for Coulter. Sensitivity and specificity (95% confidence intervals) for each test method are shown in Table 1. For SpHb, sensitivity was not significantly different from HQc ($p = 0.06$) and specificity was not different from CuSO4 ($p = 0.70$). **Conclusion:** SpHb displayed clinically acceptable accuracy for screening hemoglobin. Additionally, in our sample of prospective blood donors with a higher deferral rate, SpHb had similar sensitivity and specificity as other invasive capillary tests cleared for use in blood donation. SpHb has advantages over invasive methods of reducing sharps injuries and blood exposure to clinical staff while offering a painless qualification method that may improve the donor experience.

Disclosure of Commercial Conflict of Interest

M. Corradino: Masimo, Other; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; L. Kline: MacoPharma, Grants or Research Support, Origen, Grants or Research Support, Masimo, Grants or Research Support; P. H. Whitley: Massimo, Grants or Research Support, Citra, Grants or Research Support, Haemonetics, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. Corradino: Masimo, Grants or Research Support; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support; L. Kline: Nothing to disclose; P. H. Whitley: Citra, Grants or Research Support, Haemonetics,Grants or Research Support

TABLE 1. Methods compared to HiCN

n = 429	Sensitivity, (CI) (n = 107) %	Specificity (CI) (n = 322) %
SpHb	53 (44-62)	88 (84-92)
HQc	65 (56-74)	95 (93-97)
Coulter	96 (92-100)	96 (94-98)
CuSO4	85 (78-92)	87 (83-91)

SP132

Attention to and Usefulness of the Written Post-Donation Instructions for Repeat Donors

E P Notari[1] (ed.notari@redcross.org), A F Eder[2], W R Steele[1]. *[1]Transmissible Diseases—Holland Laboratory, American Red Cross-Biomedical Services, Rockville, MD, United States; [2]Medical Office, American Red Cros-Biomedical Services, Rockville, MD, United States*

Background/Case Studies: Blood centers must provide written post-donation instructions to all donors about possible adverse events after giving blood and what to do if they occur. However, few centers have evaluated the effectiveness of their educational efforts. A survey was conducted of repeat blood donors to assess their baseline knowledge and recollection of the post-donation written instructions. **Study Design/Methods:** Individuals who had donated blood were randomly selected for the survey if they had ≥2 donations in the previous year and a valid email address. The online survey included a series of questions about the one-page, double-sided post-donation written instructions. To prompt donors' recall, an image of the actual information sheet was provided. Simple frequencies and percentages were

calculated and chi-square analysis performed (SAS® 9.2). **Results/Findings:** Over 21,000 donors were emailed invitations and 6700 responded (response rate—31%). Among all the respondents, more than half (56%) of were female; their median age was 51; and over 90% were Caucasian. Most donors said they had read all or most of the instruction sheet after donation (all—26%, most—27%, about half—11%, less than half—17%, none—19%). Of those who said they did not read it or did not read all of it, the vast majority (87%—none and 96%—about half or less) said it was because they were a repeat donor and had read it before. What they remembered varied significantly overall and by how much of the sheet they had read (Table). All comparisons were statistically significant ($p < 0.002$). **Conclusion:** Most repeat donors read all or at least part of the instruction sheet. The predominant reason given for not reading all of it was the expected familiarity with its content as a repeat donor. The material was apparently successful in conveying instructions about post-donation self-care, possible reactions and when to contact the blood center, but fewer donors adequately recalled adverse reaction care instructions or how to replace iron after blood donation. These findings suggest that blood centers should develop ways to alert seasoned donors to new educational messages and materials.

Disclosure of Commercial Conflict of Interest

A. F. Eder: Nothing to disclose; E. P. Notari: Nothing to disclose; W. R. Steele: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. F. Eder: Nothing to disclose; E. P. Notari: Nothing to disclose; W. R. Steele: Nothing to disclose

Information that repeat donors remember reading on the post-donation information instruction sheet overall and by how much of the instruction sheet they read (n = 5397*)

What do you remember reading on the instruction sheet? (Check ALL that you remember)	Total	All or Most	Half or Less
That you should drink extra liquids/avoid heavy exercise/keep the bandage on after you give blood	96.3%	97.1%	94.8%
That you should call the Red Cross if you have questions or problems after donation	87.2%	88.8%	84.2%
That you should call the Red Cross if you have additional health information to tell them	81.2%	83.2%	77.3%
What reactions you may have after donating (eg fainting, bruises)	74.3%	77.9%	67.3%
What to do if you feel dizzy or faint	49.7%	53.3%	42.8%
How to care for bruises	32.8%	36.6%	25.5%
How you can restore(replace) your iron after donation	27.4%	30.2%	22.1%
What you should do if you are injured after fainting	11.4%	13.5%	7.4%
No responses	1.5%	1.1%	2.3%

* Does not include donors reporting not reading the information sheet.

SP133

Predictors of Venous Hemoglobin (Hb) in the Week Following Whole Blood Donation

R G Cable[1] (Ritchard.Cable@redcross.org), D Brambilla[2], G Simone[3], J E Kiss[4]. *[1]American Red Cross, Farmington, CT, United States; [2]RTI International, Rockville, MD, United States; [3]National Heart, Lung, and Blood Institute, Bethesda, MD, United States; [4]Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: Immediate post-donation venous hemoglobin (Hb) has been used to derive pre-donation (baseline) venous Hb in research studies, but post-donation Hb levels after blood volume equilibration has occurred have not been studied. These data may inform policy considerations and assist in interpretation of blood counts performed after blood donation. **Study Design/Methods:** The Hemoglobin and Iron Recovery Study (HEIRS) collected venous Hb (g/dL) and ferritin (ng/mL) following blood donation, on 296 consenting whole blood donors (137 M; 159 F): pre-donation (baseline) and 3-7 days after donation. Reticulocyte counts were available on a representative subset. The 3-7 day interval was assumed to follow restoration of blood volume. Height, weight, and gender were recorded and baseline estimated blood volume (EBV) calculated. The percentage drop in hemoglobin from baseline was calculated and a linear regression model of post-donation Hb as a function of gender, EBV, baseline ferritin, baseline Hb, and days since donation (3-7) was developed. The percentage drop in venous Hb from baseline was also compared with the ratio of the estimated whole blood draw (535 mL including samples) to EBV. **Results/Findings:** Post-donation Hb dropped 10% from baseline (8% M and 12% F). The percentage drop in Hb for individual subjects was consistent with that expected from the loss of whole blood as a proportion of EBV. Parameter estimates for the linear regression model of post-donation Hb are listed in the Table. Post-donation Hb varied with gender, baseline Hb, EBV, and days since donation, but not with baseline ferritin. Hemoglobin increased from days 3-7. The mean reticulocyte count increased (from 46,700/μL to 62,200/μL) consistent with the rising Hb. The drop in Hb post-donation was 0.4 g/dL greater in females than in males even after adjusting for days since donation, EBV and the lower expected baseline Hb in women. **Conclusion:** Hb in the immediate post-donation period is associated with pre-donation Hb and EBV, and increases from days 3-7 after donation, likely due to early release of reticulocytes from the bone marrow and/or delayed plasma volume equilibration. Females have an unexplained larger drop in Hb than males after correcting for days since donation, EBV and baseline Hb. These data will be useful in considering changes in the Hb requirements for blood donation, for clinical Hb assessments conducted in the week after blood donation, and for interpreting recovery from whole blood donation in studies such as HEIRS.

Disclosure of Commercial Conflict of Interest

D. Brambilla: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: No Answer; G. Simone: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Brambilla: Nothing to disclose; R. G. Cable: Nothing to disclose; J. E. Kiss: No Answer; G. Simone: Nothing to disclose

Predictors of Post-Donation Hemoglobin (g/dL)

Predictor	Parameter Estimate	SE	p
Days since donation	0.0959	0.0224	<0.0001
Gender (female)	–0.4403	0.1038	<0.0001
Baseline ferritin (ng/mL)	0.0014	0.0009	0.14
Baseline hemoglobin (g/dL)	0.7414	0.0338	<0.0001
EBV (L)	0.1553	0.0539	0.0043
Intercept	1.2172	0.5671	

SP134

Predicting First Lifetime Plasma Donation among Whole Blood Donors

G Godin[1] (gaston.godin@fsi.ulaval.ca), M Germain[2]. *[1]Faculty of Nursing, Laval University, Québec City, QC, Canada; [2]Medical Affairs, Hema-Québec, Québec City, QC, Canada*

Background/Case Studies: No publication has investigated the motivation of whole blood donors to become plasma donors. Given the increasing importance of recruiting new plasma donors, the aim of the present study was to understand and predict first lifetime plasma donation among a population of whole blood donors. **Study Design/Methods:** The sample of 252 whole blood donors (blood types: O+, B+, AB+ and AB–), who never donated plasma, recruited at 33 different blood drives located within 45 minutes of the plasma center, were recruited during the resting period after the donation. A questionnaire was given to assess the following Theory of Planned Behavior variables: intention, attitudes, subjective norm and perceived behavioral control. The dependent variables were: 1) Intention to give plasma within the next 6 months; 2) Completing the registration form to become a plasma donor; 3) First lifetime plasma donation; and 4) Total number of plasma donations at 6-months follow-up. Behavioral information was collected either on site (intention and signed registration form) or extracted from the Progesa database (first plasma donation and frequency of donation) at six-month follow-up. **Results/Findings:** A total of 116 donors registered to become plasma donors and agreed to be phoned when eligible to donate. At 6 months, 57 individuals had made 98 plasma donations (range of 1 to 6 donations per donor). Completing the form to become a plasma donor was predicted by intention (OR = 7.3, CI95%: 2.81-19.39) and donor status (repeat vs first-time donor, OR = 2.71, CI95%: 1.05-6.97). First lifetime plasma donation was predicted by intention (OR = 4.86, CI95%: 1.38-17.15) and age (50 + years, OR = 2.4, CI95%: 1.10-5.26). Number of plasma

donations was predicted by intention ($\beta = 0.33$, $p < .01$), age (increase with age, $\beta = 0.21$, $p < .001$) and gender (being a man, $\beta = 0.13$, $p < .05$). Finally, intention was determined by attitudes ($\beta = 0.34$, $p < .001$) and perceived behavioral control ($\beta = 0.60$, $p < .001$). **Conclusion:** To recruit new plasma donors, blood agencies should develop promotion strategies aimed at men aged 50 years and more who are repeat blood donors. Such promotion strategies should be coupled with specific actions to eliminate/lower the perceived barriers and highlight the perceived advantages to become a plasma donor.

Disclosure of Commercial Conflict of Interest

M. Germain: Nothing to disclose; G. Godin: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Germain: Nothing to disclose; G. Godin: Nothing to disclose

SP135

The Promotion of Plasma Donation: A Randomized Trial of Recruitment Strategies among Whole Blood Donors

G Godin[1] (gaston.godin@fsi.ulaval.ca), M Germain[2]. *[1]Faculty of Nursing, Laval University, Québec City, QC, Canada; [2]Medical Affairs, Hema-Québec, Québec City, QC, Canada*

Background/Case Studies: The literature in the field of blood donation has mainly focused on identifying the motives of blood donors to give blood again. However, much less attention has been devoted to plasma donation and to test interventions aimed at promoting starting giving plasma among whole blood donors. **Study Design/Methods:** A sample of 924 whole blood donors (blood types: O+, B+, AB+ and AB–), recruited at 33 different blood drives located within 45 minutes of a plasma center, were randomized to one of three conditions: control (C); information only by nurse (I by RN); and information plus a self-positive image message by nurse (I + SPI by RN). Participants in the C condition were given only a file containing a leaflet defining and explaining plasma donation and the procedure: when, where, and how. Participants in the I by RN condition received the same leaflet, but its content was explained face-to-face by a nurse. Participants in the I + SPI by RN condition also reviewed the document face-to-face with a nurse, but the leaflet also contained specific references to feelings of self-positive image as expressed by regular plasma donors. The dependent variables were whether the donor gave her/his first lifetime plasma donation and the number of plasma donations during the 6-month follow-up period. Behavioral information was extracted from the Progesa database. **Results/Findings:** Overall, 141 (15.3%) donors gave plasma at least once during the 6-month follow-up period. In total, they made 403 plasma donations. The intention-to-treat analysis (GENMOD procedure, Binomial distribution) showed a main effect on the proportion of blood donors in the experimental conditions who gave plasma at least once compared to the control condition ($\chi 2$ (1, N = 924) = 31.97, $p < .001$); however, the two experimental conditions did not differ. For the frequency of plasma donation (GENMOD procedure, Poisson distribution), at 6-month follow-up, the I by RN ($\chi 2(1, N = 924) = 25.22$, $p < .001$, $d = 0.26$) and I + SPI by RN conditions ($\chi 2(1, N = 924) = 52.56$, $p < .001$; $d = 0.32$) showed significantly greater frequency of plasma donations compared to the control condition. Moreover, the I + SPI by RN intervention significantly outperformed the I by RN condition (($\chi 2(1, N = 924) = 5.04$, $p < .05$, $d = 0.13$). **Conclusion:** The two interventions were quite successful in promoting first lifetime plasma donation among whole blood donors. Nonetheless, the results also suggest that adding references to feelings of self-positive image such as feeling good, being proud, and that giving plasma is a rewarding personal experience, favor a higher frequency of plasma donation among individuals who decide to become plasma donors.

Disclosure of Commercial Conflict of Interest

M. Germain: Nothing to disclose; G. Godin: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Germain: Nothing to disclose; G. Godin: Nothing to disclose

SP136

Presence of Combined Hypertension and Hypercholesterolemia in Adolescent Blood Donors

S Eason[1,3] (season@carterbloodcare.org), S Goudar[1], M Sayers[1,2]. *[1]Carter BloodCare, Bedford, TX, United States; [2]University of Texas Southwest Medical Center, Dallas, TX, United States; [3]Center for Community Health, University of North Texas Health Science Center, Fort Worth, TX, United States*

Background/Case Studies: Elevated blood pressure and hypercholesterolemia are associated with increased risk for future cardiovascular disease which can progress, asymptomatically, over a period of time. Consequently, early screening and identification, especially of younger populations, is essential. Since the volunteer blood donor setting offers such an opportunity, it was decided to compare the prevalence of combined hypertension and hypercholesterolemia in adolescent volunteer blood donors, according to their gender, and ethnicity. **Study Design/Methods:** Participants included unduplicated donors between 2010-2012 who were aged 20 years and younger. Blood pressure was measured using automated equipment (Welch Allyn, Spot Vital Signs LXi). Hypertension was defined as a systolic pressure ≥140 mmHg and/or diastolic pressure ≥90 mmHg. Total nonfasting cholesterol was determined (Beckman Coulter AU680), and the results ≥200 mg/dL were regarded as abnormal. Donors were invited to declare their ethnicity at the time of donation. **Results/Findings:** There were 49,202 in the study (22,392 male and 26,810 female). 85% declared their ethnicity as African American, Asian, Caucasian, or Hispanic, and the remaining were categorized as "other." Males had the highest risk of hypertension in all ethnicities. Females had the highest risk of hypercholesterolemia in all ethnicities. However, males had the greatest risk of having both hypertension and hypercholesterolemia. **Conclusion:** The blood donor screening provides an opportunity to identify two risk factors, hypertension and hypercholesterolemia, in an ostensibly healthy subset of the population. The fact that this scrutiny does identify individuals with multiple risk factors suggests the need for future studies to explore effective communication strategies to build healthier behaviors.

Disclosure of Commercial Conflict of Interest

S. Eason: Nothing to disclose; S. Goudar: Nothing to disclose; M. Sayers: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Eason: Nothing to disclose; S. Goudar: Nothing to disclose; M. Sayers: Nothing to disclose

Ethnicity n = (M/F)	African American (2,168/2,393)	Asian (610/564)	Caucasian (11,202/13,120)	Hispanic (5,070/6,442)	Other (3,324/4,291)
%Hypertensive M/F	17.16%/3.93%	10.0%/2.48%	14.85%/3.42%	14.39%/2.86%	12.83%/3.71%
%Cholesterol ≥200 mg/dL M/F	4.985/7.90%	9.67%/10.46%	4.41%/7.88%	6.27%6.44%	6.10%/7.64%
%Hypertensive & Cholesterol ≥200 mg/dL M/F	.96%/.63%	1.47%/.71%	1.08%/.46%	1.53%/.22%	1.40%/.37%

SP137

Hemoglobin (Hb) Level, Safe Enough to Qualify Donors?

A Lebrun[1] (andre.lebrun@hema-quebec.qc.ca), N Ceneston[1], J Constanzo-Yanez[1], M Chevrier[1], G Delage[1]. *[1]Héma-Québec, Montréal, QC, Canada*

Background/Case Studies: Due to increasing needs of compatible blood for sickle cell anemia patients, a program was launched in 2009 to inform members of our black community about this disease and encourage them to increase their blood donations. But, although registered black women equaled men in number, they were deferred 15 times more often than men due to low Hb. **Study Design/Methods:** This observation raised the following question: to what extent it could be linked with a high prevalence of iron deficiency among the black women donor population? In order to clarify this point, some of them were asked to participate in a study. They belonged to two groups: those who were recalled because they had been deferred previously due to low Hb and did not give blood after; and those who presented at a blood drive on their own. After their acceptance, they were submitted to fingertip puncture for Hb and venous puncture for ferritin. In both groups, Hb was determined by HemoCue and ferritin in the setting of an hospital analyser. **Results/Findings:** As shown in Table 1, the combined analysis of Hb vs ferritin found in black women included in the study indicates a prevalence of iron deficiency (<12 ug/L) of 18% among those who qualified for blood donation with the 125 g criteria. Globally, 25% of black

women who registered were found iron deficient, a level much higher than the 10% observed among Caucasian women of childbearing age. **Conclusion:** 25% of female black donors presented absent iron store, including 18% of those who qualify for blood donation based on a Hb level of ≥125 g/L. Iron deficiency may have a negative impact on health, and blood donations may aggravate it. Options for management include increasing the inter-donation interval, ferritin testing and iron supplementation. Of the three, the most pragmatic, and logical measure is the offer of iron supplements for these donors, considering their importance for reaching the objective of the program.

Disclosure of Commercial Conflict of Interest

N. Ceneston: Nothing to disclose; M. Chevrier: Nothing to disclose; J. Constanzo-Yanez: Nothing to disclose; G. Delage: Nothing to disclose; A. Lebrun: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Ceneston: Nothing to disclose; M. Chevrier: Nothing to disclose; J. Constanzo-Yanez: Nothing to disclose; G. Delage: Nothing to disclose; A. Lebrun: Nothing to disclose

Donor Status	Ferritin ug/l	Hb g/l	
		≥125	<125
Recalled	≥12 : 40	21	19
	<12 : 14	5	9
Presented at Blood Drive	≥12 : 26	15	11
	<12 : 8	3	5
Total	≥12 : 66	36	30
	<12 : 22 (25%)	8 (18%)	14
	88	44	

SP138

Phlebotomy Duration and Vasovagal Reaction Rates in 5.4 Million Donations

B Newman[1] (bruce.newman@redcross.org), B Siegfried[2]. *[1]American Red Cross, SE Michigan Region, Detroit, MI, United States; [2]American Red Cross, Great Lakes Region, Lansing, MI, United States*

Background/Case Studies: The relationships between phlebotomy duration and vasovagal reaction rates and phlebotomy duration and syncope rates are not well characterized. **Study Design/Methods:** Phlebotomy durations versus vasovagal reaction rates and syncope rates were evaluated in 5.8 million whole-blood donations in 2006, using 4-13-minute durations; 5.4 million or 94% of all donations were within those durations. The same was done for donations by 17-year-old donors because of their high reaction rate and for subsets of these young donors defined by sex and donation history (first time or repeat) because of their known relations to the reaction rate. Reaction and syncope rates in the 4-8-minute durations were compared with the rates for 9-13-minute durations using a Fisher Exact Test at $p <$ the 0.05 significance level. **Results/Findings:** The table shows that for all donations, the vasovagal reaction and syncope rates were almost twice as high in the higher phlebotomy durations than in the lower ones. Vasovagal reaction rates also increased in each of the four 17-year-old populations with increasing phlebotomy durations. Syncope rates increased with increasing phlebotomy durations in first-time 17-year-old donations but not in repeat 17-year-old donations. **Conclusion:** Phlebotomy duration is directly related to vasovagal reaction rates in the general donor population and in 17-year-old donors in particular. Phlebotomy duration is also directly related to syncope rates in the general donor population, in 17-year-old first-time donors, but not in 17-year-old repeat donors. The data suggest that decreasing the phlebotomy duration might decrease vasovagal reaction rates.

Disclosure of Commercial Conflict of Interest

B. Newman: Nothing to disclose; B. Siegfried: No Answer

Disclosure of Grants Conflict of Interest

B. Newman: Nothing to disclose; B. Siegfried: No Answer

Table. Reaction Rates

Age (years), Donation History, Sex	Number	Vasovagal Reaction Type	Phlebotomy Duration (min) 4-8	9-13	p
All donors	5,448,357	N & S	2.0%	3.8%	<0.0001
All donors	5,448,357	S	0.08%	0.14%	<0.0001
17,FT,F	116,906	N & S	10.4%	12.0%	<0.0001
17,FT,M	110,869	N & S	5.9%	7.7%	<0.0001
17,R,F	51,070	N & S	8.2%	9.9%	<0.0001
17,R,M	42,295	N & S	4.2%	5.4%	<0.0001
17,FT,M&F	227,775	S	0.33%	0.42%	<0.0004
17,R,M&F	93,365	S	0.25%	0.28%	0.23

FT = first-time; R = repeat; M = male; F = female; S = syncopal reaction; N = non-syncopal reaction

SP139

Which Repeat Blood Donors Respond to Emergency Appeals for Blood?

J Rios[1] (Jorge.Rios@redcross.org), I M Khandaker[1], A Crumpton[1]. *[1]American Red Cross Blood Services, New England Region, Dedham, MA, United States*

Background/Case Studies: Recruitment of voluntary non-remunerated blood donors during an emergency appeal (due to a low blood inventory) has historically been bolstered by investigations of donor habits and motivations. In contrast to this cognitive model approach, we assess donors' civic engagement and other related pro-social characteristics to build an actionable framework for emergency appeal planning. **Study Design/Methods:** During late March 2013 all presenting allogeneic whole blood and apheresis double red blood cells donors donating at selected non-high school, non-college blood drives in three counties of a large metropolitan area in the Northeast US were given a survey. The survey contained 10 questions–including questions about the donors' educational level and voting participation at major elections–with some questions with no, yes, and not sure answers. Donors were asked to complete the survey before undergoing their health history screens and mini-physical exams. Data from first-time donors and from deferred donors was excluded from the analysis. **Results/Findings:** 638 unique repeat donors presented to donate whole blood and apheresis double red blood cells at the blood drives where the survey was administered. 429 (67%) repeat donors (232 M; 197 F) agreed to participate in the study. The overwhelming majority of the study donors voted in the 2012 US presidential election (92%) . Male gender and knowing a person who received a blood transfusion were related to responding to emergency appeals for blood donations (see Table). Education, knowledge of a political affair fact, and prior registration to become an organ, bone marrow, or stem cell donor were not related to the responses of the repeat donors to emergency appeals for blood ($p >$ 0.05 with a Chi-square test for independence.) **Conclusion:** Male gender and knowing a person who received a blood transfusion are characteristics of repeat donors who have responded to emergency appeals for blood. A large number of blood donors have engaged in civic activities such as voting at major elections. Appeals for blood donations that make references to stories about blood transfusion recipients and to the civic nature of donating blood might improve the response of repeat donors for blood donations during emergencies.

Disclosure of Commercial Conflict of Interest

A. Crumpton: Nothing to disclose; I. M. Khandaker: Nothing to disclose; J. Rios: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Crumpton: Nothing to disclose; I. M. Khandaker: Nothing to disclose; J. Rios: Nothing to disclose

		Response to an emergency appeal for blood donations					
		No	Yes	Not sure	No answer	Total	p value*
Gender	Male	89	114	27	2	232	0.01
	Female	90	69	38	0	197	
	Total	**179**	**183**	**65**	**12**	**429**	
Know a person who received a blood transfusion	No	81	54	15	1	151	0.04
	Yes	85	109	39	1	234	
	Not sure	12	20	10	0	42	
	No answer	0	0	1	0	1	
	More than 1 answer	1	0	0	0	1	
	Total	**179**	**183**	**65**	**2**	**429**	

* = Chi-square test for independence.

SP140

Racial Differences in Syncope Rates among South African Blood Donors

K Van den Berg[1] (karin.vandenberg@sanbs.org.za), E Bloch[2,4], N Moleli[1], C Ingram[1], R Swanevelder[1], T Muthivhi[1], J Langer[3], B Custer[2,4], E L Murphy[4,2]. *[1]South African National Blood Service (SANBS), San Francisco, CA, United States; [2]Blood Systems Research Institute (BSRI), San Francisco, CA, United States; [3]RTI International, Rockville, MD, United States; [4]University of California San Francisco (UCSF), San Francisco, CA, United States*

Background/Case Studies: Adverse reactions during blood donation negatively affect return rates and retention of blood donors and counteract donor recruitment efforts. The South African National Blood Service (SANBS) is dependent on young donors to meet transfusion demand and is steadily increasing the proportion of Black donors to make the donor pool more representative of the population. **Study Design/Methods:** All adverse donation events that were reported in blood donors at SANBS over a 6-month time period (January-June 2012)were analyzed. Standardized operating procedures govern reporting and recording of adverse events into the database. Adverse events were predefined as the International Haemovigilance Network categories covering syncopal and pre-syncopal reactions: Immediate fainting (defined as syncope in the donation area) and delayed fainting (defined as syncope after leaving the donation area), each with and without complication or injury. Incidence of syncopal and pre-syncopal reactions was calculated per 10,000 donation attempatients stratified by demographics and comparisons between incidence rates were tested by Fisher's Exact test. In South Africa, "Asian" race includes mostly people of south Asian descent while "Colored" includes people of mixed African and other descent. **Results/Findings:** In 6 months, 315,275 blood donors made 490,470 donation attempatients and 2793 adverse events were recorded including syncopal reactions, hematoma, nerve injury, arm pain, etc. There were 1175 syncope and pre-syncope (35% of recorded reactions, a rate of 24 per 10,000 donation attempatients), with significant ($p < 0.01$) differences by age, sex and race (see Table). Pre- and syncope incidence was highest in females aged under 18 years: 92 per 10,000 in Whites; 23 per 10,000 in Blacks, 205 per 10,000 in Asians and 42 per 10,000 in Colored. **Conclusion:** Syncope and pre-syncope incidence appears to be lower in South African than US blood donors (Wiltbank et al *Transfusion* 2008). Although rare, syncope occurs disproportionately in the young, female and Asian and White blood donors. The inability in the available data to control for donor weight is a limitation. These data support the presence of an underlying physiological basis for the difference in risk of syncope by race.

		Number of donations	N Syncope (per 10,000)
All donors		490,470	1175 (24)
Race:	White	289,075	612 (21)
	Black	129,637	141 (11)
	Asian	38,557	304 (79)
	Colored	24,664	39 (16)
Age:	<18 years	64,559	392 (61)
	18-30 years	156,630	462 (29)
	>30 years	269,281	321 (12)
Sex:	Female	223,355	681 (30)
	Male	267,113	494 (18)

Disclosure of Commercial Conflict of Interest

E. Bloch: Nothing to disclose; B. Custer: Novartis Vaccines and Diagnositics, Grants or Research Support; C. Ingram: Nothing to disclose; J. Langer: No Answer; N. Moleli: No Answer; E. L. Murphy: Nothing to disclose; T. Muthivhi: Nothing to disclose; R. Swanevelder: Nothing to disclose; K. Van den Berg: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Bloch: Nothing to disclose; B. Custer: Novartis, Vaccines and Diagnostics, Grants or Research Support; C. Ingram: Nothing to disclose; J. Langer: No Answer; N. Moleli: No Answer; E. L. Murphy: Nothing to disclose; T. Muthivhi: Nothing to disclose; R. Swanevelder: Nothing to disclose; K. Van den Berg: Nothing to disclose

SP141

Incidence of Low Iron Stores in Blood Donors with Acceptable Hemoglobin Levels: A Single Center Experience

K J Duffy[1] (duffy.kimberly@mayo.edu), S C Bryant[2], M Benike[1], J R Stubbs[1], M J Gandhi[1]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Biomedical Statistics and Informatics, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Frequent blood donations may result in depletion of iron stores with normal hemoglobin (Hb). Regulatory agencies are encouraging blood donor centers to evaluate options to mitigate this risk. The option is explored here to increase inter-donation interval. To evaluate the iron stores in the blood donors, ferritin levels in a subset of the blood donors was measured. **Study Design/Methods:** After getting the institutional review board approval, over a 5 month period (August-December 2012), plasma samples from 1000 random whole blood donors (7% of donor pool) were tested for ferritin levels at the time of donation. In addition, donors were queried regarding iron supplement and red meat intake. Each donor's past donation frequency in the preceding 12 months and time of last donation was captured. Ferritin levels were evaluated using a two-site immunoenzymatic ("sandwich") assay on the Beckman Coulter UniCel D × I 800. Low ferritin was defined as less than 24 ng/mL for males, and less than 11 ng/mL for females. Data was analyzed using the Wilcoxon rank sum test and Kruskal-Wallis test, as appropriate, using SAS 9.3 software, with significance defined as p-value < 0.05. **Results/Findings:** Median age of the blood donor was 53 years (range: 17-90 years) with equal gender ratio. 96% of the donors were Caucasian. 361 (36%) donors had low ferritin. Median ferritin levels for female donors was 19 ng/mL (range: 3-228 ng/mL) and for male donors was 24 ng/mL (range: 4-324 ng/mL) . Donor iron supplement, meat intake, donation frequency, and time since last donation are summarized in the table. Of the donors with low ferritin, those that took iron supplements donated more frequently than those who did not take iron supplements; however, this was not statistically significant ($p = 0.66$). **Conclusion:** 36% of the eligible donors with normal hemoglobin that donated had decreased iron stores. Low iron stores correlate with frequency of previous donations and time from last donation. Based on these findings, the whole blood inter-donation interval will be increased from 8 to 12 weeks. Role of iron supplementation to mitigate depleting iron stores needs additional investigation.

Disclosure of Commercial Conflict of Interest

M. Benike: Nothing to disclose; S. C. Bryant: Nothing to disclose; K. J. Duffy: Termuo BCT, Other, Fenwal, Other, Haemonetics, Other; M. J. Gandhi:

Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal, Other, Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

M. Benike: Nothing to disclose; S. C. Bryant: Nothing to disclose; K. J. Duffy: Nothing to disclose; M. J. Gandhi: Nothing to disclose; J. R. Stubbs: Nothing to disclose

	Low Ferritin	Normal Ferritin	p-value
Take iron supplements?, N(%)			<0.0001
No	284 (78.89%)	403 (63.27%)	
Yes	76 (21.11%)	234 (36.73%)	
Meat eating Frequency, N(%)			0.91
Never/Less than 1/week	41 (11.36%)	71 (11.11%)	
1-2/week	169 (46.81%)	290 (45.39%)	
3 or more/week	151 (41.83%)	278 (43.51%)	
Donations within past year, median (range)			
Not taking iron supplements	4 (1,7)	3 (1, 7)	<0.0001
Taking iron supplements	5 (1,7)	3 (1, 7)	<0.0001
Time since last donation, weeks, median (range)	10.9 (8.0, 299.1)	15.0 (8.0, 585.1)	<0.0001

SP142

Effect of Increasing Pulse Range on Allogeneic Donor Eligibility and Reaction Rates

L J Sutor[1,2] (lsutor@carterbloodcare.org), J M Centilli[1]. *[1]Carter BloodCare, Bedford, TX, United States; [2]Pathology, University of Texas, Southwestern, Dallas, TX, United States*

Background/Case Studies: The 26th edition of *AABB Standards*, effective November 1, 2009, dropped the requirement for facilities to measure and record a donor's pulse for eligibility to donate allogeneic blood. The acceptable range for donation prior to that time was between 50 and 100 beats per minute (bpm), without irregularity. Donors with lower pulses could be accepted if the donor was considered athletic and otherwise healthy. This blood center continued the measurement of pulse, but extended the range. The practice of noting irregularities of pulse was discontinued. The effect this change had on donor deferrals and also donor reaction rates was recorded here. **Study Design/Methods:** On December 15, 2009 this blood center changed its allogeneic donor eligibility requirement for pulse from a range of 50-100 to 45-110 bpm. Staff was instructed to no longer feel for pulse irregularities. For consistency in measurement, deferral data were retrospectively compared only during the time that pulse and blood pressure were measured by automated instrument (starting December 30, 2008). Vasovagal reaction comparison was done with donors from 2010-2012. **Results/Findings:** In the 12 months prior to the change, 415,161 donors presented and 6303 were deferred for pulse issues, for a deferral rate of 1.52%. After the change in pulse eligibility rules, over the next 12 month period, 2204 donors were deferred for pulse from 403,597 donors presenting, for a deferral rate of 0.55%. Vasovagal reaction rates for female donors with pulse greater than 100 were noted to be somewhat higher than donors with pulses in the previous range but in a linear manner (Table). No difference was noted in reaction rates for male donors. **Conclusion:** Because of medical concerns, it was decided not to completely drop the measurement of pulse. However, it was shown that the number of acceptable donors could be increased by widening the pulse range and no longer attempting to identify irregularities of pulse. Vasovagal reaction rates were slightly higher in the group of new female donors.

Disclosure of Commercial Conflict of Interest

J. M. Centilli: Nothing to disclose; L. J. Sutor: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. M. Centilli: Nothing to disclose; L. J. Sutor: Nothing to disclose

Vasovagal Reactions by Gender and Pulse (Percent)

Pulse (bpm)	45	55	65	75	85	95	100	105	110
Females	2.65	1.12	1.82	2.06	2.72	3.27	3.47	3.71	4.28
Males	0.75	0.75	0.79	1.00	1.18	1.31	1.60	1.55	1.44

Donor and Therapeutic Apheresis

SP143

Factors Affecting Platelet Yield during Plateletpheresis Donations

C S Rao[1] (srinu_2741@yahoo.co.in), R R Sharma[1], N Marwaha[1]. *[1]Transfusion Medicine, Post Graduate Institute of Medical and Educational Research, Chandigarh, India*

Background/Case Studies: High-yield plateletpheresis enables the collection of multiple units from the same donor in a single donation and may have the potential to reduce donor exposure, risks inherent with allogenic transfusion, effective management of platelet inventory, and cost effectiveness of the procedure. **Study Design/Methods:** A total of 500 plateletpheresis donors were included in this prospective study. All procedures were performed using a fully automated platform of plateletpheresis, Trima Accel version 5.1 (Gambro BCT), Amicus Crescendo version 3.2 (Baxter Healthcare Corp.).The parameters assessed were donor physical characteristics, donor hematological profile (pre donation) and the procedural variables with the platelet yield of the product harvested and divided into two groups Group-I ($\leq 4 \times 10^{11}$/unit, n = 318) and Group-II ($>4 \times 10^{11}$/unit, n = 182). **Results/Findings:** A total of 500 prospective plateletpheresis donors (age range 18-56 years) were enrolled. In the study, actual platelet yield was positively correlating with the weight (r = 0.155) of the donors (Group-I [72.47 ± 9.72] and Group-II [75.37 ± 9.98]), and height (r = 0.129) of the donors (Group-I [170.31 ± 4.29] and Group-II [171.46 ± 5.14]) and is statistically significant ($p < 0.01$). Actual platelet yield harvested is negatively correlating with the Hb (r = –0.15) and Hct(r = –0.59) but was not statistically significant ($p > 0.05$). (Group-I [14.53 ± 0.98 g/dl] and [43.60 ± 2.74%] and in Group-II [14.49 ± 0.98 g/dl] and [43.22 ± 2.64%]). Actual platelet yield was positively correlating with the pre donation platelet count (r = 0.624) of the donors and was statistically significant ($p < 0.01$). (Group-I [2.19 ± 0.436X 10^5/µl] Group-II [2.86 ± 0.505X 10^5/µl]). Procedural parameter, blood volume processed (r = 0.706) (Group-I [2486.73 ± 389.698 ml], Group-II [3320.10 ± 502.87 ml]), was highly correlating with the the actual platelet yield harvested and was statistically significant (p value < 0.01). **Conclusion:** Pre donation platelet count and blood volume processed are the main predictable's of platelet yield. Hence donors with pre procedure platelet counts (≥2 $\times 10^5$/µL) with higher body surface area can be considered for higher yield platelet products.

Disclosure of Commercial Conflict of Interest

N. Marwaha: No Answer; C. S. Rao: Nothing to disclose; R. R. Sharma: No Answer

Disclosure of Grants Conflict of Interest

N. Marwaha: No Answer; C. S. Rao: Nothing to disclose; R. R. Sharma: No Answer

Factors Affecting Platelet Yield

Variables	Group I ($\leq 4 \times 10^{11}$/unit, n = 318)	Group II ($>4 \times 10^{11}$/unit, n = 182)
Weight (in kgs)	72.47 ± 9.72	75.37 ± 9.98
Height (in cms)	170.31 ± 4.29	171.46 ± 5.14
Hemoglobin(g/dl)	14.53 ± 0.98	14.49 ± 0.98
Hematocrit (%)	43.60 ± 2.74	43.22 ± 2.64
Platelet count (/µl)	2.19 ± 0.436 X 10^5	2.86 ± 0.505 × 10^5
Blood volume processed (ml)	2486.73 ± 389.698	3320.10 ± 502.87

SP144

Developing Screening for Anti-A and Anti-B Titers in Type-O Apheresis Platelets

G Bachowski[1] (Gary.Bachowski@redcross.org), D C Mair[1], R S Lose[1], S Shaikh[1]. *[1]American Red Cross, Saint Paul, MN, United States*

Background/Case Studies: Hemolytic reactions due to ABO-incompatible plasma from apheresis platelets remain a concern when ABO unmatched products are transfused. Currently in the US, almost all apheresis platelets are stored in plasma which can pose risk for hemolysis. That risk is assumed to be proportional to anti-A,-B or –AB quantity, though no guidelines or safety recommendations exist for transfusions involving this incompatibility. Variation in lab technique makes comparison of titers and correlation with clinical results difficult. The goal here was to sample RBC and platelet donors and decide on a standard method to screen titer levels so that platelets reduced

in risk to cause hemolytic reactions could potentially be provided. data are presented from titer testing of both random RBC and platelet donors, and operational trial for routine apherersis platelet screening. **Study Design/Methods:** Blood samples from 50 random type-O apheresis platelet donors were gathered in an unlinked manner with no associated demographic data or donor identity. Serial titers of anti-A and anti-B for each EDTA blood sample and corresponding platelet rich plasma (PRP) were determined by manual tube methods and observed in immediate spin (IS), room temperature (RT), and indirect antiglobulin test (IAT) phases. Based on these results, a test run of these 50 samples at set dilutions in IS phase (1 : 50, 1 : 100, 1 : 200), and hospital customer requests, an operational trial was begun on daily apheresis collections of O-platelets that were diluted 1 : 200 and screened at IS phase. These data were also compared to results previously reported for 203 random RBC donors from our lab (2011 AABB Meeting Abstract). **Results/Findings:** See Table. Results for the 50 platelet donor and product samples were very similar to those for 203 whole blood donors tested in the previous phase of the study by the same methods. EDTA blood samples and platelet rich plasma gave similar results. The operational trial screening test of O-platelet products/EDTA donor tubes diluted 1 : 200 found 2.9% and 1.6% of the first 520 products to be positive for anti-A and anti-B, respectively. **Conclusion:** Literature review shows varied methods and recommendations that even include suggestions the majority of our O-donors have ABO titers posing hemolytic risk to recipients. As a step to decrease risk, we designed an operational trial to check either EDTA blood samples or platelet rich plasma in IS phase at a 1 : 200 dilution. By these critera only a small fraction (3-4%) of O-platelets would be excluded based on positive anti-A or B tests.

Disclosure of Commercial Conflict of Interest

G. Bachowski: Nothing to disclose; R. S. Lose: Nothing to disclose; D. C. Mair: Nothing to disclose; S. Shaikh: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Bachowski: Nothing to disclose; R. S. Lose: Nothing to disclose; D. C. Mair: Fenwal, Travel Support or Honorarium; S. Shaikh: Nothing to disclose

Median Anti-A and Anti-B Titers of 50 Apheresis Platelet Donors and Products

	Platelet Donor Blood (EDTA tube)			Platelet Rich Plasma		
Phase	IS	RT	IAT	IS	RT	IAT
Anti-A Titer	1:32	1:64	1:128	1:32	1:64	1:128
Anti-B Titer	1:16	1:32	1:64	1:16	1:32	1:64

SP145

The Automatization in Granulocyte Collection using the Optia Device Eases the Procedure and Provides Concentrates of High Quality

G C Leitner[1] (gerda.leitner@meduniwien.ac.at), V Kolovratova[1], M Horvath[1], N Worel[1]. *[1]Department for Bloodgroup Serology and Transfusion Medicine, Medical University, General Hospital Vienna, Vienna, Austria*

Background/Case Studies: The therapeutic effect of granulocyte transfusions in leukopenic patients with severe infections refractory against antibiotics or antimycotics is generally accepted. The challenge is the ability to collect sufficient numbers of granulocytes to induce eradication of pathogens. In recent decades several approaches were initiated to enable the collection of a therapeutic dose (4 1010 granulocytes per day) of neutrophiles (ANC). The last innovation in this field is the fully automated Optia® device / Cobe (COBE BCT, Lakewood, CO). In a prospective analysis it was attempted to show that the fully automated system of Optia/Cobe provides granulocyte concentrates not inferior in terms of yield, purity and donor comfort to the current devices. **Study Design/Methods:** 26 donors, mobilized with prednisolon (P), were recruited from the routine multicomponent donor pool (2 females, 24 males, median age 42 years, range 27-63). These volunteers donated one concentrate. Fourteen collections were obtained from six donors, all relatives of the recipients, (1 female, 5 males, median age 30 yrs, range 24-59) who were mobilized with G-CSF at a dose of 5 μg/kg bodyweight (BW). All collections were performed with Optia® device according to the instructions of the manufacturer. A mean of 6.1 L (SD 0.3) of donor's blood volume was processed within 120 minutes. Anticoagulation was done with a mixture of Hetastarch 6% and sodium citrate 46.7%. **Results/Findings:** None of the donors experienced any side effects. The gain of BW was in mean 0.4 kg for the routine (P)—and 0.2 kg for the G-CSF group. A mean of 449 mL (SD 61) was collected with a purity of in mean 82% (SD 6.1) ANC for both mobilization groups. As expected, a significant difference was found in the absolute yield of ANC in the products, 5.9 (SD 3.1) × 1010 in the G-CSF group vs 1.9 (SD 0.5) × 1010 in the P group. Compared with historical data the product volume as well as the absolute ANC yield in the P group was significantly higher in Optia than in collections done with Spectra® /Cobe: 449 mL vs 225 mL and 1.9 × 1010 (SD 0.5) vs 1.3 × 1010 (SD 0.6) respectively, $p < 0.5$. **Conclusion:** Granulocyte collections with the fully automated Optia® device are safe for donors. The procedure requires less man-power than Spectra® as a continuous visual control of the collection-line is negligible accompanied by high quality of the products.

Disclosure of Commercial Conflict of Interest

M. Horvath: No Answer; V. Kolovratova: No Answer; G. C. Leitner: Nothing to disclose; N. Worel: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Horvath: No Answer; V. Kolovratova: No Answer; G. C. Leitner: Nothing to disclose; N. Worel: Nothing to disclose

SP146

Obstetric Outcomes in Sickle Cell Disease Patients Receiving Chronic Red Blood Cell Exchange

B Chang[1] (brian.n.chang@uth.tmc.edu), Y Bai[1]. *[1]Pathology and Laboratory Medicine, University of Texas Health Science Center, Houston, TX, United States*

Background/Case Studies: Sickle cell disease (SCD) is associated with many complications during pregnancy and treatment frequently involves blood transfusions. Although a few studies have examined whether prophylactic transfusions reduces obstetric complications (eg pre-eclampsia, preterm delivery, intrauterine growth restriction, intrauterine fetal death) versus transfusion only in acute situations, controversy still remains. However, there are increasing numbers of SCD patients that begin chronic red blood cell exchange (RBCx) at a young age who later become pregnant. This group is the focus of our nascent study into the outcomes and impact of chronic RBCx on pregnancy. **Study Design/Methods:** A brief review of RBCx cases from 2006-2013 identified two pregnant SCD patients (both HbSS) on a chronic exchange protocol secondary to stroke and hemochromatosis, respectively. Four additional pregnant SCD patients who received transfusions only for acute complications were included for comparison. **Results/Findings:** A 22-year-old G2P0101 patient on chronic RBCx for stroke prophylaxis had an otherwise uncomplicated pregnancy until developing pre-eclampsia at 35 weeks gestation and therefore underwent emergent c-section. Her prior pregnancy also resulted in C-section at 36 weeks also due to pre-eclampsia. The other chronic RBCx patient was a 39-year-old G2P1001 with evidence of intrauterine growth restriction. Induction of labor and vaginal delivery occurred at 37 weeks 1 day without further complication. The non-chronic RBCx patients received treatments for a variety of acute conditions including stroke, acute chest syndrome, and severe pain crises. Three patients had a prior history of one or more 1st trimester spontaneous abortions or 2nd trimester intrauterine fetal demise. One developed pre-eclampsia but delivered at term whereas another developed atypical pre-eclampsia necessitating pre-mature delivery at 28 weeks 1 day. A third patient delivered by C-section at 34 weeks secondary to non-reassuring fetal heart tones. **Conclusion:** Every patient reviewed did have some form of obstetrical complication during the course of pregnancy. However, in the chronic RBCx cohort they appeared to be fewer in number and of less severity since neither patient had any history of spontaneous abortions nor intrauterine fetal demise and they carried all pregnancies into at least the 35th week of gestation. One distinguishing feature in these patients versus those in other studies is the fact that RBCx was performed regularly prior to conception and throughout fetal development in the critical 1st trimester and onwards. Under this regimen Hgb S can be maintained at lower levels and for longer duration. Although the small number of patients examined to date precludes definitive conclusions regarding pregnancy outcomes in SCD patients receiving chronic RBCx, a larger future study would provide the statistical power to do so.

Disclosure of Commercial Conflict of Interest

Y. Bai: Nothing to disclose; B. Chang: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Bai: Nothing to disclose; B. Chang: Nothing to disclose

SP147

Plasmapheresis as an Adjunct Therapy for Hyperhemolysis in a Patient with Sickle Cell Disease (SCD)

C L Gonzalez[1] (CGonzalez@Stanford.edu), P Lavinghousez[4], J A Hughes[2], J McNaughton[2], S A Galel[1], M R Jeng[3], M K Viele[2], M Fontaine[2]. *[1]Stanford Blood Center, Stanford University School of Medicine, Palo Alto, CA, United States; [2]Pathology, Stanford University School of Medicine, Stanford, CA, United States; [3]Department of Pediatrics, Stanford University School of Medicine, Stanford, CA, United States; [4]Department of Pediatrics, Dell Children's Medical Center and University of Texas Southwestern Medical School, Austin, TX, United States*

Background/Case Studies: Hyperhemolysis is a known transfusion complication in patients with sickle cell disease (SCD). Current treatment strategies include withholding further transfusion, steroids, IVIG, and rituximab but do not show consistently positive results. Aa severe case of hyperhemolysis with a dramatic rise in hemoglobin (Hgb) in response to combined therapy including plasmapheresis is reported here. **Study Design/Methods:** Chart Review. **Results/Findings:** A 21 year-old, O+ male with SCD and known history of allo/auto-immunization (anti-hrB, K, S, Fy3, Cob, He, warm auto, +DAT), osteonecrosis, and restrictive lung and heart disease was admitted in pain crisis with a Hgb of 4.8 (g/d). During a prior admission 6 days earlier he incremented adequately with one RBC unit; however; in this admission his Hgb initially fell to 2.5 despite transfusion of 3 RBC units (see table). He was transferred to the ICU, treated with IVIG and plasma exchange (1.5 volume, plasma frozen within 24 hours) and transfused with four more predominantly antigen negative units. However, his Hgb further declined to a nadir of 1.0. He was intubated because of obtundation and cardiovascular instability. On hospital day 8, two days after plasmapheresis and steroids were started, and three days after IVIG was initiated, one dose of rituximab was given. His Hgb began to improve and rose to a high of 7.8 over the next three days. An attempt was made to extubate the patient but he developed seizures. He was determined to have irreversible severe hypoxic brain injury and placed on comfort care. One dose of IVIG was given but plasmapheresis was discontinued and steroids tapered. His Hgb fell to 2.4 over the next week and he expired. **Conclusion:** In this case of SCD with complicated immunologic and therapeutic features the addition of plasma exchange to steroids, IVIG, and rituximab may have helped ameliorate the hyperhemolysis. Plasmapheresis may be a helpful adjunct in the treatment of hyperhemolysis.

Disclosure of Commercial Conflict of Interest

M. Fontaine: Nothing to disclose; S. A. Galel: Nothing to disclose; C. L. Gonzalez: Nothing to disclose; J. A. Hughes: Nothing to disclose; M. R. Jeng: Nothing to disclose; P. Lavinghousez: No Answer; J. McNaughton: Nothing to disclose; M. K. Viele: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Fontaine: Nothing to disclose; S. A. Galel: Nothing to disclose; C. L. Gonzalez: Nothing to disclose; J. A. Hughes: Nothing to disclose; M. R. Jeng: Eli Lily, Grants or Research Support, Grifols, CSL Behring, Biotest, Talecris, Grants or Research Support; P. Lavinghousez: No Answer; J. McNaughton: Nothing to disclose; M. K. Viele: Nothing to disclose

Clinical Course

Day	RBC units Given	Hgb g/dl	Methyl-prednisolone g	IVIG g	Rituxibmab mg	Plasma exchange
1 admitted	0	4.8				
2	1 neg for: C,e,Fya,Fyb,K,S	4.2				
3	1 neg for: C,e,Fya,Fyb,K,S	3.9				
4	1 neg for: C,Fya,Fyb,K,S	3.4				
5 admitted to ICU	0	2.5		24 × 2		
6 steroids plasma exchange started	1 neg for: C,e,Fya,Fyb,K,S	1.8 am 1.5 pm	1.0			yes
7 intubated	3 neg for: C,E,Fya,Fyb,K,S C,e,Fya,Fyb,K,S,M C,Fya,Fyb,K,S	1.3 am 1.0 pm	1.0	19		yes
8 rituximab	1 neg for: C,e,Fya,Fyb,K,S	1.4 am 1.9 pm	1.0	19	560	yes
9	1 neg for: Fya,Fyb,K, S	3.3	1.0	20		yes
10	1 neg for: C,e,Fya,Fyb,K,S	4.0	1.0			yes
11	1 neg for: C,e,Fya,Fyb,K,S	7.8	1.0			yes
12 extubated seizure	0	7.7 am 6.5 pm	1.0			
13 comfort care	0	5.1	1.0			
14	0	5.0	0.75			
16	0	3.6	0.50			
17	0	2.5	0.25	20		
18	0	2.4	0.10			
19 expires	0					

SP148

Plasma Exchange in Patients with Sickle Cell Crisis Complicated by Thrombotic Thrombcytopenic Purpura (TTP)

A T Al Bahrani[1] (atalbahrani@hotmail.com), K Haque[1]. *[1]Blood Bank and Transfusion Medicine, King Fahad Specialist Hospital-Dammam, Dammam, Saudi Arabia*

Background/Case Studies: Sickle cell disease (SCD) is one of the major health problems in Saudi Arabia. Rarely, sickle cell crisis patient presents features like thrombotic thrombocytopenic purpura (TTP). Two cases of sickle cell crisis who also presented the clinical features of TTP improved by plasma exchange (PLEX) are reported here. **Study Design/Methods:** Patient I (P-I), Saudi female, 28 years) and Patient-II (P-II), Saudi male, 38 years, are included in this report. Sickle cell crisis is alleviated by red blood cell exchange (RBCX). Post-RBCX HbS in P-I is 28% and P-II 20%. Prior to PLEX, both patients present normal head magnetic resonance imaging, high lactate dehydrogenase (LDH), 1+ schistocytes, normal ADAMST13 protease activity, thrombocytopenia, headache, mental confusion, and are considered as cases of SCD-TTD. An automated Spectra Optia Apheresis, version 7.0, (Terumo.BCT, Lakewood, CO) is used to perform all PLEX. Consent is obtained for PLEX. Based on patient's gender, height, weight and hematocrit level, Spectra Optia computes anticipated volume of replacement fluid. ACD-A solution is used as anticoagulant with a whole blood ratio is 10 : 1 to 13 : 1. Fresh frozen plasma is used as replacement fluid. 1.0-1.5 volume PLEX is performed daily over the course of 8-10 days. **Results/Findings:** In P-I, LDH level (normal: 62-115 U/L) decreased from 3904 to 737 on day 2, and PLT count (normal: 150-400 109/L) increased from 34 to 173 on day-5 of PLEX. In P-II, LDH decreased from 1729 to 787 on day 5, and PLT count increased from 18 to 77 on day 8 of PLEX; 2 weeks later PLT count reached normal whereas his higher creatinine level (normal: 62-115 mmol/L) range is 398-499 which indicates underlying renal condition. The number of schistocytes declined from 1+ to occasional and later cleared following PLEX. The nucleated RBC count is markedly elevated initially, but decreased over time. Hemoglobin levels are relatively stable in both patients during the course of PLEX. **Conclusion:** When SCD and TTP conditions present in parallel a clear-cut diagnosis is not easy. If clinical and pathological features suggest TTP-SCD, the PLEX should be the first choice of treatment despite normal ADAMST13 activity. It is suggested that PLEX reduces plasma cytokines and inflammatory mediators that can avoid red cell destruction and organ damages in sickle cell crisis. Moreover, PLEX effectively lessens vasoocclusion in SCD when RBCX alone is not helpful. Both SCD-TTP patients did not show severe thrombocytopenia like TTP alone as seen in functional hyposplenism during sickle cell crisis. LDH levels are observed as a little higher than normal level even after cessation of PLEX courses; this is not unexpected in SCD. This should be followed up on more cases like SCD-TTP in the future; particularly in a sickle cell prevalent (17%) area as here.

Disclosure of Commercial Conflict of Interest

A. T. Al Bahrani: Nothing to disclose; K. Haque: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. T. Al Bahrani: Nothing to disclose; K. Haque: Nothing to disclose

SP149

Re-examination of 30-day Survival and Relapse Rates in Patients with Thrombotic Thrombocytopenic Purpura (TTP)-Hemolytic Uremic Syndrome (HUS)

C E Bittencourt[1,2] (cassi_euzebio@hotmail.com), J P Ha[1,2], R W Maitta[1,2]. *[1]Pathology, University Hospitals Case Medical Center, Cleveland, OH, United States; [2]Pathology, Case Western University School of Medicine, Cleveland, OH, United States*

Background/Case Studies: Thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS) are characterized by microangiopathic hemolytic anemia and thrombocytopenia. The survival of patients with TTP-HUS has continued to improve with therapeutic plasma exchange (TPE); however, before the advent of TPE only 10% of patients survived, and markedly different survival rates have been reported since. The mortality rate of TTP-HUS patients who received TPE here was studied and compared to published data. **Study Design/Methods:** This retrospective study analyzed the overall 30-day mortality rate attributed to TTP-HUS and risk of relapse at University Hospitals Case Medical Center from 01/01/2008 to 12/31/2012, and compared it to two previous published studies looking at TTP-HUS reporting responses to therapy and survival. Studies looking at different patient populations (TTP patients only) or describing a different mortality time interval were excluded. Statistical significance was established using a two-tailed Fisher test with a $p < 0.05$ set for significance. **Results/Findings:** A total of 59 patients were analyzed; 11 patients were excluded due to testing that ruled out the initial diagnosis of TTP-HUS, either by laboratory testing or clinical reassessment found a different disease etiology. The remaining 48 patients were diagnosed with TTP-HUS and received a series of TPEs. Idiopathic causes were diagnosed in 36 patients (75%) and secondary causes in 12 patients. Patients received a mean of 13 TPEs (range 1-42), complete response was obtained in 79% of patients (n = 38) (platelet count $> 1 \times 10^{11}$ /L); partial response in 8% of patients (n = 4) and 13% had no response (N = 6). 40% of the patients obtained platelet counts $> 1.5 \times 10^{11}$ /L. Overall 30-day mortality attributed to TTP-HUS was 2% (1/48). Comparison of the mortality rate to combined data of two published studies of 19% (42/219) showed a statistically significant difference, $p = 0.002$. The relapse rate here was 23% (n = 11), which was statistically similar to previous published reports of 18-36%. **Conclusion:** This study shows that the 30-day mortality rate attributed to TTP-HUS was significantly lower than published reports. The wide distribution in the observed mortality among institutions/studies may be due to the presence of other confounding factors in the patient populations under study. Co-morbidities, promptness of TPE initiation, and delay in diagnosis may begin to explain the observed discrepancies among studies. Additionally, this institution is a referral center for other institutions in the region and patients may begin adequate treatment prior to transfer or may not have survived their disease prior to transfer.

Disclosure of Commercial Conflict of Interest

C. E. Bittencourt: Nothing to disclose; J. P. Ha: Nothing to disclose; R. W. Maitta: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. E. Bittencourt: Nothing to disclose; J. P. Ha: Nothing to disclose; R. W. Maitta: Nothing to disclose

SP150

Increased Predictability of Platelet Split Levels using a Hematology Analyzer for Donor Qualification

C Nare[1] (cnare@bbd.org), M Healy[1], S Hall[1], R Roper[1]. *[1]Blood Bank of Delmarva, Newark, DE, United States*

Background/Case Studies: Preparation and collection of apheresis platelets (APLTS) generally requires presetting the apheresis equipment with donor specific data prior to donation. Without an accurate assessment of the donor's current platelet count, the final product may not meet the expected concentration of platelets for splitting or may even necessitate product [destruction or wastage] if the concentration does not meet established minimal levels. The cost of inaccurate collections can be significant not only in lost revenue for partial units, but also for the time and materials of collections that must be discarded. **Study Design/Methods:** This study evaluated the predictability of using a small Hematology analyzer (HA) to pre-screen donors in the donor center and determine the appropriate setting for APLTS versus using a "historical average." The parameters analyzed included the number of procedures where the actual count missed the targeted count for single, double, and triple units, and the overall yield of predictability. Data was collected at two donor collection facilities and then pooled. **Results/Findings:** While the predictability of single units was very similar with either method, the predictability of accurately targeting double or triple units was markedly improved using the hematology analyzer. Similar results were obtained at both donor collection sites, and the pooled numbers are summarized here. The historical average accurately targeted double units only 72% of the time, whereas pre-qualification with the HA was predictive in 86%. For the triple units, there was a similar, but smaller effect, with a historical average yield predictability of 59% and an HA yield predictability of 70%. Overall, across the three categories the accuracy of donation targets increased by 6.9%. **Conclusion:** The implementation of the HA to pre-screen platelet donors for acceptability and to determine the collection settings on the apheresis machine resulted in significant savings to the blood bank. Each unusable or lost unit costs the blood center $300 in time, materials, and unsellable product, while each incorrectly targeted unit results in a $600 loss when it is sold as a partial. It is estimated the implementation of this technology will result in an annual benefit of $21,576 in additional revenue.

Disclosure of Commercial Conflict of Interest

S. Hall: Nothing to disclose; M. Healy: Nothing to disclose; C. Nare: Nothing to disclose; R. Roper: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Hall: Nothing to disclose; M. Healy: Nothing to disclose; C. Nare: Nothing to disclose; R. Roper: Nothing to disclose

SP151

Therapeutic Plasma Exchange (TPE) in a Rare Paraneoplastic Syndrome (PNS)

H Shafi[1] (heidishafi@hotmail.com), H M Mason[1], D J Chaffin[1], E Klapper[1]. *[1]Department of Pathology and Laboratory Medicine, Cedars-Sinai Medical Center, Los Angeles, CA, United States*

Background/Case Studies: Background: PNS affect approximately 1% of patients with underlying malignancies. These syndromes may affect any part of the nervous system from cerebral cortex to neuromuscular junction and muscle. Pathogenesis of PNS is not well understood; however immunologic factors are believed to be important as antibody and T-cell responses against nervous system antigens have been described for many of these disorders. American Society for Apheresis has categorized PNS as a category III indication for therapeutic apheresis and its use in this setting is controversial. A case is reported here where TPE was successfully used to treat neurological manifestations of a rare neoplastic syndrome due to infiltrating, mucinous duct carcinoma of the breast. A previously healthy 56-year-old female presented with new onset neurological abnormalities including jaw clenching, facial droop, and diplopia. A workup including LP and brain MRI was non-contributory. A presumptive diagnosis of rhombencephalitis (RE) was made based on clinical symptoms. She was treated with steroids and discharged home. She was readmitted after 1 month with progressive dysphagia, involuntary tongue biting due to extreme jaw clenching, and left facial twitching. Soon after admission, she became poorly responsive, unable to protect her airway, and was intubated. Repeat brain MRI, LP, bacterial, viral and fungal cultures, and screens for heavy metal intoxication were all negative. A complete paraneoplastic antibody (PNAb) workup including Neuronal Nuclear Ab, Purkinje Cell Cytoplasmic Ab, Anti-Hu, Anti-Ri, Anti-Yo, anti-NMDA receptor, and anti-Glutamic Acid Decarboxylase was negative. The patient's jaw clenching worsened causing loss of portions of her tongue, necessitating extraction of all her teeth. Since all common infectious etiologies had been ruled out, therapeutic plasma exchange (TPE) to treat possible immune-mediated RE was initiated. After 5 TPE the bruxism improved, however, diplopia and nystagmus persisted. An additional course of 5 TPE was requested. Initial partial response to TPE prompted further investigation for evidence of occult malignancy, despite negative PNAb panel. Ultrasound of the left breast demonstrated a small lump; needle biopsy and axillary dissection revealed metastatic infiltrating ductal carcinoma with mucinous features. She continued to show improvement with TPE, but symptoms persisted. Following discharge she received outpatient chemotherapy and her neurologic symptoms significantly improved. **Study Design/Methods:** NA. **Results/Findings:** NA. **Conclusion:** This case demonstrates the utility of TPE in the evaluation and treatment of rare PNS; in the face of negative

PNAb, response to TPE may indicate an immune mediated process and the need to pursue occult malignancies. While response to TPE can be variable, this case illustrates the potential benefit a short course may offer.

Disclosure of Commercial Conflict of Interest

D. J. Chaffin: Nothing to disclose; E. Klapper: Nothing to disclose; H. M. Mason: Nothing to disclose; H. Shafi: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. J. Chaffin: Nothing to disclose; E. Klapper: Nothing to disclose; H. M. Mason: Nothing to disclose; H. Shafi: Nothing to disclose

SP152

New Thrombotic Thrombocytopenic Purpura (TTP) Guidelines Helps to Improve Recovery: A Look Back at the Introduction of New Guidelines in the Treatment of TTP in One Regional Treatment Center

H E Stone[1] (helen.stone@hotmail.co.uk), D Potok[1]. *[1]Specialist Therapeutic Services Oxford, National Health Service Blood and Transplant Service, Headington, Oxford, United Kingdom*

Background/Case Studies: Thrombotic Thrombocytopenic Purpura (TTP) is a rare blood coagulation disorder, which in the United Kingdom has a reported incidence of six cases per million per year. Specialist Therapeutic Services (STS), which is a function of the National Health Service Blood and Transplant (NHSBT), provides a specialist apheresis service to the Oxford University Hospitals (OUH). Patients who are referred with suspected TTP are admitted under the care of the OUH and then referred to STS for plasma exchange. If left untreated TTP has a mortality rate of around 90% and early commencement of plasma exchange is essential. If plasma exchange is commenced within 24 hours the survival rate is over 90%. **Study Design/ Methods:** A reflective and observational review was undertaken in this regional treatment center of the impact of early diagnosis and intervention following new guidelines for TTP. The review looked at all cases treated as recommended by the British Committee for the Standards in Haematology guidelines, on the diagnosis and management of thrombotic thrombocytopenic purpura and other thrombotic microangiopathies. **Results/Findings:** 1.5 plasma volume plasma exchanges using solvent detergent washed fresh frozen plasma (S/D FFP), are recommended for at least 3 days until laboratory tests stabilise, followed by one plasma volume S/D FFP exchanges daily, for at least 2 days after reaching a platelet count of ≥150; this new recommendation has been introduced in OUH. Previously 1.5 plasma volumes were considered only if the patients' condition had deteriorated and the monoclonal antibody rituximab was used only in refractory TTP. In OUH, a course of rituximab is now commenced earlier as recommended by the new guidelines when there is evidence of neurological or cardiac involvement, and it had been found that patients' platelet and LDH levels improve and become within normal range earlier. **Conclusion:** It is well recorded in research and through observation that due to the high risk of death in the first 24 hours following onset of acute symptoms, early recognition of TTP and commencement of treatment is essential, and that having access to specialist expertise and PEX is an advantageous resource. Although the longer procedures of 1.5 plasma volumes for at least 3 days, along with increased use of rituximab as suggested by the new guidelines appears more aggressive than the 1 plasma volume exchanges which were undertaken previously, unless the patients condition deteriorated, it has been observed that platelet counts increase and LDH return to within their normal ranges more quickly and in turn the patients have a shorter stay in the hospital.

Disclosure of Commercial Conflict of Interest

D. Potok: Nothing to disclose; H. E. Stone: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Potok: Nothing to disclose; H. E. Stone: Nothing to disclose

SP153

Clinical Effect of Therapeutic Plasma Exhange (TPE) with Mirasol PRT-Treated Fresh Frozen Plasma in Patients with Acquired Thrombotic Thrombocytopenic Purpura (TTP)

M Letowska[1] (letowska@ihit.waw.pl), J Windyga[1], R Poglod[1], A Rosiek[1], L Rzymkiewicz[1], K Bykowska[1], E Lachert[1]. *[1]Institute of Hematology and Transfusion Medicine, Warsaw, Poland*

Background/Case Studies: Thrombotic thrombocytopenic purpura (TTP), a rare life-threatening disease first described by Elli Moschovitz (1925), is a thrombotic microcoangiopathy characterized by acute thrombocytopenia, microangiopathic hemolytic anemia, and organ failure from microvascular occlusion. Congenital and acquired TTP (aTTP) are induced by von Willebrand factor-cleaving protease ADAMTS13 (A Disintegrin And Metalloproteinase with ThromboSpondin-1 motifs) deficiency usually severe (<10% of normal activity);in aTTP it results from ADAMTS13 directed autoantibodies. The most common form of aTTP (4-6 cases/million) is acute idiopatic TTP, an autoimmune disease characterized by antibodies (usually IgG) directed against ADAMTS 13. Secondary TTP may result from pregnancy, HIV infection, or malignancy. Therapeutic plasma exchange (TPE) is standard treatment for aTTP. Following British Guidelines TTP diagnosis should be a medical emergency based on clinical history, examination and laboratory parameters including blood film review. TPE treatment should be initiated immediately (preferably within 4-8 hours of thrombocytopenia and hemolytic anemia diagnosis). In the Institute of Hematology and Transfusion Medicine (IHTM) Mirasol Pathogen Reduction Technology (PRT)- treated FFP was used for management of aTTP patients. **Study Design/Methods:** Evaluation of TPE treatment with Mirasol-treated FFP in IHTM aTTP patients. **Results/Findings:** Between April 2011-pril 2013, aTTP was diagnosed in seven patients (six females and one male), median age 49 (24-68); two with acute idiopathic TTP, two with secondary TTP (drug- and malignancy- related). Diagnosis was based on clinical and laboratory findings: hemolytic anemia with schistocytes in blood smear, thrombocytopenia, neurological symptoms,and elevated LDH. In this group 150 TPE procedures were performed, median per patient 13 (4-87) with 1508 units of Mirasol-treated FFP, median per patient 142 (38-890). Steroids were administered simultaneously. TPE and steroids proved ineffective in one case and cyclophosphamide and rituximab were introduced. In six out of seven patients ADAMTS 13 activity and antibody titer was determined prior to first TPE. In five of them ADAMTS 13 activity was <10%. In one malignancy—related TTP case ADAMTS 13 activity was 44%. In six patients the antibody titer was 6 IU/ml-130,9 IU/ml. In all patients TPE was started with 1,5 PV exchanges daily. . In one aTTP patient TPE and steroid treatment was discontinued following prostatic cancer diagnosis. **Conclusion:** Treatment proved successful in six out of seven aTTP patients. It was discontinued in one patient following prostatic cancer diagnosis. No serious transfusion-related adverse reactions were reported.

Disclosure of Commercial Conflict of Interest

K. Bykowska: No Answer; E. Lachert: No Answer; M. Letowska: Nothing to disclose; R. Poglod: No Answer; A. Rosiek: No Answer; L. Rzymkiewicz: No Answer; J. Windyga: No Answer

Disclosure of Grants Conflict of Interest

K. Bykowska: No Answer; E. Lachert: No Answer; M. Letowska: Nothing to disclose; R. Poglod: No Answer; A. Rosiek: No Answer; L. Rzymkiewicz: No Answer; J. Windyga: No Answer

SP154

A Study of How Eculizumab has had an Impact on the Use of Plasma Exchange in Paediatric Patients with Haemolytic Uraemic Syndrome (HUS)

A Chauhan[1] (alka.chauhan@nhsbt.nhs.uk), S Douglas[1]. *[1]Specialist Therapeutic Services, National Blood and Transplant Service, Bristol, United Kingdom*

Background/Case Studies: Haemolytic Uraemic Syndrome (HUS) predominantly but not exclusively affects children. HUS is a disease characterised by haemolytic anaemia, caused by the destruction of red blood cells, causing acute renal failure and a low platelet count. There are two main types of HUS: Infection induced-diarrhoea associated HUS—(D + HUS); or Typical HUS. D + HUS occurs 2-10 days after abrupt onset of bloody diarrhea due to verocytotoxin (Shiga-like toxin) primarily E-Coli 0157: H7. This type of HUS, according to the American Association for Apheresis (ASFA) 2010, accounts for between 85-90% of all HUS. Non stx-HUS or HUS not associated by infection-atypical or aHUS is less common. This form of HUS may either be sporadic or familial. Over the past 4 years, a number of patients with both aHUS and typical HUS have been treated in this study. The majority of patients have been pediatrics ranging in age from 8 months to 4 years old, with a few adults presenting with typical HUS. It was noted that there was a significant decline in referrals of pediatric patients presenting with HUS that needed plasma exchange, therefore a review was conducted to show whether the use of Eculizumab as a first line treatment had influenced this change. **Study Design/Methods:** Observational review of number of children treated with HUS in 2009-2012, and the impact of the new drug Eculizimab and how it has caused a decline in the use of Plasma Exchange (PE) as a first line treatment for HUS. **Results/Findings:** During

the period from 2009-2012, a total of five paediatric patients presenting with HUS were being treated with plasma exchange. Of the five patients treated, four no longer required plasma exchange treatment due to beginning on the drug Eculizumab. **Conclusion:** It was observed that over the course of 4 years (2009-2012) the use of PE as a first line treatment for patients with HUS has been superseded by the use of the drug Eculizmab. This impact has led to the decline in the number of patients treated with PE here.

Disclosure of Commercial Conflict of Interest

A. Chauhan: Nothing to disclose; S. Douglas: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Chauhan: Nothing to disclose; S. Douglas: Nothing to disclose

SP155

Comparisons of Adverse Reactions During Therapeutic Apheresis: A 5 Year Experience at an Academic Medical Center

S Datla[1,2] (drdatlamd@gmail.com), D Hendrix[2], K A Downes[2,3], V Griffin[2], R W Maitta[2,3]. *[1]Transfusion Medicine, Northern Ohio American Red Cross, Cleveland, OH, United States; [2]Pathology, Transfusion Medicine, University Hospitals Case Medical Center, Cleveland, OH, United States; [3]Pathology, Case Western Reserve University School of Medicine, Cleveland, OH, United States*

Background/Case Studies: Therapeutic apheresis (TA) has been used since 1960; however, there is limited literature examining the incidence of adverse events (AEs) during TA. Available studies lack uniformity in data collection and no common database is available to report to AEs in the US. The incidence of AEs was studied and compared to published data from the US and Europe. **Study Design/Methods:** This retrospective study looked at AEs incidence for 60 months (January 1, 2007-December 31, 2011) that occurred during or immediately after TA procedures performed here. Apheresis nurses performed all procedures using the COBE® Spectra Apheresis System, Software versions 4.7 and 6.1 (Terumo BCT, Lakewood, CO), and reported associated AEs. AEs' severity was stratified according to Table 1. Data were compared to published European and other US studies that reported the total number of procedures and percentage of adverse events/procedures. A Chi square statistical analysis with Yates correction (two-tailed) was performed with a $p < 0.05$ considered significant. **Results/Findings:** A total of 2862 therapeutic apheresis procedures were performed on 966 patients at UHCMC with an overall AE incidence of 8.1% (233). Of the 233 adverse events during the study period the severity grade incidence was: mild 114 (3.9%); moderate 74 (2.5%), severe 45 (1.5%), and fatal 0 (0%). Comparisons of these AEs to total procedures (233/2862) to both European (N = 6, 1997/14927) and US studies (N = 3, 1183/5858) showed a statistically significant difference, $p < 0.0001$. Similarly, the lower AEs incidence of European studies was statistically significant compared to US studies, $p < 0.0001$. Overall AEs/procedures in US studies was (20.2%), European studies (13.4%) and at UHCMC (8.1%). **Conclusion:** The study shows that the incidence of AEs was significantly lower than current published data. The difference in the incidence of AEs stresses the need for uniform reporting and stratification of AEs and development of a common database to report complications due to therapeutic apheresis.

Disclosure of Commercial Conflict of Interest

S. Datla: Nothing to disclose; K. A. Downes: Nothing to disclose; V. Griffin: Nothing to disclose; D. Hendrix: No Answer; R. W. Maitta: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Datla: Nothing to disclose; K. A. Downes: Nothing to disclose; V. Griffin: Nothing to disclose; D. Hendrix: No Answer; R. W. Maitta: Nothing to disclose

TABLE I: Adverse Events Grading

Grade	Quality	Intervention
I	Mild	No intervention required
II	Moderate	Intervention required; Treatment completed
III	Severe	Procedure Interrupted or abandoned
IV	Fatal	Patient Expired

SP156

Using Oral Calcium Supplementation Successfully for Therapeutic Plasma Exchange: Our Experience

A Das[1] (anneddas@gmail.com), S McGregor[1], R Knoebel[1], B W Baron[1], A Treml[1]. *[1]University of Chicago Medicine, Chicago, IL, United States*

Background/Case Studies: Severe hypocalcemia secondary to citrate in anticoagulant solution used in therapeutic plasma exchange (TPE) is uncommon. Symptoms of mild to moderate hypocalcemia can be significant. To avoid this during TPE, albumin used as replacement fluid is supplemented with calcium gluconate. If symptoms develop, they are often manageable with oral calcium carbonate. In patients with low ionized calcium levels, calcium may be replenished with intravenous (IV) calcium gluconate prior to or during TPE. The recent national shortage of calcium forced providers of TPEs to adopt alternative calcium supplementation strategies. To our knowledge there are no published descriptions of oral calcium use for prevention of hypocalcemia in the setting of TPE. **Study Design/Methods:** Since late March 2013, due to severe shortage of IV calcium gluconate here, calcium gluconate was no longer added to 5% albumin bottles. Calcium chloride was not an option. Working with a pharmacy, a plan was devised to pre-medicate patients with oral calcium carbonate. Patients were evaluated during and after each TPE for symptoms of hypocalcemia, and also chart reviews were also performed to assess ionized calcium levels prior to TPE and the next morning. **Results/Findings:** A total of 25 TPEs were performed on seven patients (three male, four female; mean age 51 years). During 19 TPEs, five patients received 2500 mg oral calcium carbonate prior to TPE and had a prn order for additional 2500 mg oral calcium carbonate during TPE if symptomatic. For 6 TPEs, there were two patients unable to take oral calcium who received IV calcium gluconate prior to the TPE. For patients who received oral calcium carbonate, ionized calcium levels (reference range 4.60-5.40 mg/dL) ranged from 3.56-4.40 mg/dL (pre) to 3.23-4.44 mg/dL (post). For the two patients who received IV calcium gluconate, ionized calcium levels ranged from 4.28-4.68 mg/dL (pre) to 4.20-5.08 mg/dL (post). Only one patient reported perioral tingling, but did not request additional oral calcium carbonate during TPE. No patients received additional calcium supplementation during TPE. **Conclusion:** Use of IV calcium has been the standard of care for prevention of hypocalcemia associated with TPEs. This data and recent experience support the possibility that oral calcium carbonate may be a suitable alternative to IV calcium even when there is no limitation of availability. These findings must be validated in a larger population before being adopted for routine use in TPE.

Disclosure of Commercial Conflict of Interest

B. W. Baron: Nothing to disclose; A. Das: Nothing to disclose; R. Knoebel: Nothing to disclose; S. McGregor: Nothing to disclose; A. Treml: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. W. Baron: Nothing to disclose; A. Das: Nothing to disclose; R. Knoebel: Nothing to disclose; S. McGregor: Nothing to disclose; A. Treml: Nothing to disclose

SP157

Therapeutic Plasma Exchange on The Treatment of Recurrent Focal Segmental Glomerulosclerosis after Pediatric Kidney Transplantation

J Luzzi[1] (djrluzzi@gmail.com), C C Borba[1], E H Goto[1], J Rosa[1], P O Messias[1], D B Jesuino[1], M B Fonseca[2], M C Camargo[2], P K Nogueira[2]. *[1]Hospital Samaritano Blood Bank, Transfusion and Cell Therapy Service, São Paulo, Brazil; [2]Renal Transplantation and Pediatric Nephrology Service, Hospital Samaritano of São Paulo, São Paulo, Brazil*

Background/Case Studies: Recurrence of focal segmental glomerulosclerosis (FSGS) occurs in 25-53% of patients with this glomerulopathy who receive kidney transplant and the risk of recurrence appears to be higher in pediatric recipients. Recurrent disease often leads to graft loss. The pathophysiology of disease is yet to be elucidated but some studies indicate that glomerular injury is associated with a low weight permeability factor released by lymphocytes that affects podocytes and filtration architecture. Therapeutic plasma exchange (TPE) is an extracorporeal blood purification technique designed to remove plasma-borne pathogenic substances and has been accepted as standard treatment option in a variety of diseases. **Study Design/Methods:** This work presents a series of seven cases transplanted or received after first renal transplantation with recurrent FSGS and treated with TPE. Data from patients were retrospectively reviewed from medical records from the past 4 years. Recurrence of FSGS was determined based on the presence of nephrotic-range proteinuria in the absence of another cause. Nephrotic-range was defined by urine protein to creatinine range

(UP/C) > 2 g/L or by 24h-urine proteinuria (UP24) > 3,5 g. Graft loss was determined by renal biopsy. Response to treatment was defined as a UP/C < 2 g/L or UP24 < 3.5 g. TPE was performed according to standard protocols with mean 1.0 ± 0.1 total blood volume exchange. **Results/Findings:** All cases were children with mean age of 12 ± 5 years. White ethnicity represented 43% while 57% were Afro-descendent. In all cases, patients were immunosupressed with either metilprednisolone pulse or prednisone. The first two cases did not receive immunoglobulin (Ig) immunosupression, four out of five received 500 mg/Kg Ig, and one case received 1000 mg/Kg Ig. Patients were treated with rituximab (RTX) (mean of 3.3 ± 1.2 doses of 375 mg/m2) + TPE or TPE-only. TPE procedures were started at 34.73 ± 31.84 days after FSGS recurrence (Range: 02-120). Mean TPE procedures were 25.29 ± 16.18 (Range: 04-49). Treatment was effective for five of seven cases (PreUP/C range: 20.9-149.9; PostUP/C range: 0.00-0.66), from which four were from RTX + TPE group and 01 from TPE-only group. One case had a partial response with more than 35% decrease in UP24 (PreUP24: 13.90; PostUP24: 8.81) but lost graft 04 months after relapse. This case received neither Ig nor RTX. One case has still not responded (PreUP/C: 5.10; PostUP/C: 6.00).This case had an adverse event to RTX and was from TPE-only group. The only case that did not respond to treatment (PreUP/C: 8.59; PostUP/C: 12.12),has lost graft after 18 months, and has not received Ig immunosupression. **Conclusion:** Although further investigations are needed, the data suggest that association of immunoglobulin and corticosteroid immunosupression with rituximab and TPE are beneficial for the treatment of recurrent FSGS in pediatric patients.

Disclosure of Commercial Conflict of Interest

C. C. Borba: Nothing to disclose; M. C. Camargo: Nothing to disclose; M. B. Fonseca: Nothing to disclose; E. H. Goto: Nothing to disclose; D. B. Jesuino: Nothing to disclose; J. Luzzi: Nothing to disclose; P. O. Messias: Nothing to disclose; P. K. Nogueira: Nothing to disclose; J. Rosa: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. C. Borba: Nothing to disclose; M. C. Camargo: Nothing to disclose; M. B. Fonseca: Nothing to disclose; E. H. Goto: Nothing to disclose; D. B. Jesuino: Nothing to disclose; J. Luzzi: Nothing to disclose; P. O. Messias: Nothing to disclose; P. K. Nogueira: Nothing to disclose; J. Rosa: Nothing to disclose

SP158

Potential Role of Therapeutic Plasma Exchange (TPE) in the Management of Primary Focal Segmental Glomerulosclerosis (FSGS)

A Shameli[1,2] (afshin.shameli@uhhospitals.org), K M Dell[3,4], B A Vogt[3,4], R W Maitta[1,2]. *[1]Pathology, University Hospitals Case Medical Center, Cleveland, OH, United States; [2]Pathology, Case Western Reserve University, Cleveland, OH, United States; [3]Pediatrics, Rainbow Babies and Children's Hospital, Cleveland, OH, United States; [4]Pediatrics, Case Western Reserve University, Cleveland, OH, United States*

Background/Case Studies: Therapeutic plasma exchange (TPE) plays a critical role in treatment of a range of kidney disorders such as glomerulonephritis, renal vasculitis, cryoglobulinemia, and myeloma cast nephropathy. Role of TPE in the treatment of focal segmental glomerulosclerosis (FSGS) is limited to the recurrent disease in kidney transplants, and is believed to be effective by removing circulating glomerular permeability factors. Notably, only few studies have shown effectiveness of TPE in treatment of FSGS in native kidneys. A patient study is presented here with refractory nephrotic syndrome, resistant to immunosuppresants, who was treated with TPE. **Study Design/Methods:** A 4-year-old male presented with progressively worsening edema, hypertension, hematuria, proteinuria, and oliguria. Initial renal biopsy was reported as mesangial proliferative glomerulonephritis with rare glomerular scarring. Patient was treated with corticosteroids, cyclophosphamide, and rituximab without response. He required hemodialysis for fluid removal and maintenance of electrolyte balance. A repeat biopsy 4 months later showed changes consistent with focal segmental glomerulosclerosis but without pronounced scarring. Five months after initial presentation, in light of the biopsy results suggesting a recoverable lesion, the decision was made to initiate a trial of TPE. A total of nine TPE procedures, exchanging one plasma volume were performed over a period of three weeks. Albumin or a combination of albumin and plasma were used as replacement fluid depending on the level of fibrinogen and coagulation tests. A red cell prime was performed prior to the initiation of TPE. Calcium gluconate at a rate of 12 mL/L of replacement fluid was used. **Results/Findings:** Creatinine values decreased during the first 10 days after initiation of TPE, as compared to 15 days prior to starting the procedures (1.34 ± 0.20 versus 2.05 ± 0.60). Urine protein to creatinine ratio (mg/mg) also decreased and this effect was more notable during the third week of TPE (from 47 after the second procedure to 14 after the eighth procedure). Clinically, there was increased urine output, especially during the first week of TPE. After initiation of tacrolimus therapy, the TPE was discontinued. Unfortunately, the effects of TPE were not sustained and the patient's hypervolemia, serum creatinine, and proteinuria returned to pre-TPE values. **Conclusion:** These findings suggest that TPE may be a useful therapeutic modality in managing primary FSGS disease in addition to the current use in recurrent disease post renal transplant. However, the effects may not be sustained and further investigation is required to study the efficacy of TPE in combination with immunosuppressive interventions in primary FSGS.

Disclosure of Commercial Conflict of Interest

K. M. Dell: Nothing to disclose; R. W. Maitta: Nothing to disclose; A. Shameli: Nothing to disclose; B. A. Vogt: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. M. Dell: Amgen, Grants or Research Support, Abbott, Grants or Research Support; R. W. Maitta: Nothing to disclose; A. Shameli: Nothing to disclose; B. A. Vogt: Nothing to disclose

SP159

Use of Plasmapheresis in Heparin-Induced Thrombocytopenia (HIT) with Thrombosis and Acute Hemorrhage

B Castillo[1] (brian.s.castillo@uth.tmc.edu), A Nguyen[1], Y Bai[1]. *[1]The University of Texas Health Science Center at Houston, Houston, TX, United States*

Background/Case Studies: Heparin induced thrombocytopenia (HIT) is an acquired prothrombotic disorder associated with heparin exposure mediated by antibodies to platelet factor 4 (PF4)/heparin complexes. Management of HIT includes immediate discontinuation of heparin and use of non-heparin alternative for anticoagulation. Patients with HIT with thrombosis (HITT) complicated by acute bleeding presents a clinical dilemma in managing HITT. A case of HITT complicated by acute intracranial hemorrhage managed by plasmapheresis is presented here. **Study Design/Methods:** A 64-year-old female with a history of atrial fibrillation presented with a T12 fracture. She was started on deep venous thrombosis (DVT) prophylaxis with subcutaneous heparin. Five days after starting heparin she had a high 4Ts score for HIT confirmed by positive immunoassay for anti-PF4/heparin antibodies and serotonin release assay (SRA). In addition she was found to have new cerebral infarcts, bilateral adrenal hemorrhage (secondary to venous thrombosis) and intracranial hemorrhagic lesion secondary to thrombocytopenia. Discontinuation of heparin only led to slight improvement of platelet count (48 K to 76 K). Given the risks of severe complications associated with either platelet transfusion or the use of anticoagulation, the patient was managed with daily plasma exchange and laboratory monitoring for platelet count as well as daily pre- and post-procedure heparin induced platelet aggregation (HIPA) and HIT antibody testing. **Results/Findings:** A total of four apheresis procedures were performed. After two procedures, the patient had a dramatic increase in platelet count (78 K to 148 K) with negative post procedure HIPA assay. After three procedures the patient had returned to her initial baseline platelet count (188 K) and was found to have both negative post procedure HIPA and immunoassay. On the fourth procedure the patient had negative pre and post HIPA and immunoassay with continued trending upward platelet count and resolution of ongoing intracranial hemorrhage based on CT imaging and clinical presentation. The patient was then started on a non-heparin alternative anticoagulant with no further sequelae. **Conclusion:** This case presents a unique clinical scenario of HITT complicated by intracranial bleeding. It demonstrates that plasma exchange is a useful alternative in such clinical presentation. As shown by the functional and immunoassays in this series the success of plasma exchange is achieved by removal of the PF4/heparin complexes, thus preventing on going thrombosis and allowing platelet recovery. This deters any further bleeding complications allowing for implementation of appropriate anticoagulation.

Disclosure of Commercial Conflict of Interest

Y. Bai: Nothing to disclose; B. Castillo: Nothing to disclose; A. Nguyen: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Bai: Nothing to disclose; B. Castillo: Nothing to disclose; A. Nguyen: Nothing to disclose

SP160

The Use of Rituximab in Patients Receiving Apheresis Procedures for Myasthenia Gravis (MG) at a Single Institution

M Losos[1] (Michael.Losos@osumc.edu), S Scrape[1]. *[1]Pathology, The Ohio State University Medical Center, Columbus, OH, United States*

Background/Case Studies: Myasthenia Gravis (MG) is an autoimmune neuromuscular disease which can cause chronic skeletal muscle weakness of varying degrees. At this institution, in addition to using traditional anticholinesterase agents and immunosuppressive drugs for treatment of MG, patients may be referred to the apheresis center for outpatient therapeutic plasma exchange procedures (PEX). PEX can provide significant relief from disease symptoms for many patients but may require a patient to live with a tunneled apheresis catheter. Many of our patients with chronic mild to moderate MG symptoms have required outpatient PEX procedures on an uninterrupted schedule for years. Seven patients are reported on here who were diagnosed with MG receiving PEX to help manage disease symptoms,and who, in addition to their standard drug therapy, also received rituximab. Rituximab is a chimeric monoclonal antibody against CD20 positive B-cells used to treat several types of hematologic malignancies and autoimmune diseases. **Study Design/Methods:** A retrospective chart review was performed for all MG patients undergoing outpatient PEX at our apheresis center between January 2009-December 2012 who also received rituximab during that period. For each patient, the start of their outpatient PEX procedures was recorded, number of PEX received, other medications administered for MG, changes to the patient's procedure schedule, and any side effects or health concerns presenting after rituximab infusion. **Results/Findings:** During the time frame reviewed, seven patients received rituximab as treatment for MG. All had received at least 24 months of weekly, outpatient PEX prior to receiving their first dose of rituximab and none had received a diagnosis of thymoma. Dosing of rituximab and the schedule of infusions were not consistent between patients. After all patients received 1 gram initial infusions, the maintenance dosing schedule varied considerably between patients with some patients not receiving subsequent infusions and others requiring regular maintenance therapy. After the first set of infusions, six of the seven patients were able to stop PEX altogether within several weeks (range 3-6). The seventh patient showed temporary symptomatic improvement which lasted approximately 8 weeks but she eventually needed to resume her pre-infusion PEX schedule and has responded similarly to subsequent infusions. All patients have also been able to reduce, at least to some degree, the dosage or frequency of anticholinesterase agents and immunosuppressive drugs taken for MG. To date, there have been no significant side effects or health concerns related to infusions in these patients. **Conclusion:** A significant decrease in the number of PEX required to support chronic MG patients who have received rituximab has been observed here. These patients have also been able to reduce dosages and/or administration frequency of their traditional MG medications.

Disclosure of Commercial Conflict of Interest

M. Losos: Nothing to disclose; S. Scrape: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Losos: Nothing to disclose; S. Scrape: Nothing to disclose

SP161

Therapeutic Plasma Exchange (TPE) and Intravenous Immunoglobulin (IVIG) as Primary Therapy for Rh (D) Alloimmunization in Pregnancy Precludes the Need for Intrauterine Transfusion (IUT)

M Bellone[1,2] (mbellone@gmail.com), F N Boctor[1], L E Logdberg[1], W Heaton[1]. *[1]North Shore University Hospital and Hofstra North Shore-LIJ School of Medicine, Manhasset, NY, United States; [2]New York Blood Center, New York, NY, United States*

Background/Case Studies: Patients diagnosed with severe maternal Rh (D) alloimmunization early in gestation without a history of hemolytic disease of the fetus and newborn (HDFN) need aggressive intervention in order to prevent severe anemia and hydrops fetalis. An intrauterine transfusion (IUT) is indicated to prevent fetal death once severe fetal anemia has been detected, but is associated with risk. Immunomodulatory treatments may obviate the need for IUT. Protocols combining therapeutic plasma exchange (TPE) and intravenous immunoglobulin (IVIG) have been described, but they usually bridge to IUT. A common schedule is three TPE treatments in a single week followed by weekly IVIG until week 20 of gestation. This report demonstrates that increased interval TPE and prolonged IVIG therapy can serve as primary therapy for fetal protection. **Study Design/Methods:** See Results. **Results/Findings:** A 27-year-old G4, P0-1-2-0 Caucasian female with a history of ruptured ectopic pregnancy presented at 12 weeks of gestation with a very high anti-Rh (D) titer (2048). TPE was performed during that week, and twice more in the following week, with a fourth, final exchange during week 14. A loading dose of IVIG (2 g/kg) was administered over two days following the third TPE and thereafter weekly at 1 g/kg until week 28 (total, 14 doses). The antibody titer decreased to 256 by the beginning of week 15 of gestation and remained stable at that level for the remainder of the pregnancy. Doppler ultrasonographic measurement of the fetal middle cerebral artery peak flow velocity (MCA-PSV) performed throughout gestation consistently found the PSV multiple of medians to be under 1.5. A healthy male infant (APGAR score 9/9) was delivered at 37 weeks of gestation. Laboratory evidence of immune-mediated hemolysis was detected at birth, including a positive direct antiglobulin test with anti-IgG, a positive acid elution test for anti-D, a 3-hour post-delivery hematocrit of 39.8%, and a 10-hour post-delivery bilirubin level 13.7 mg/dl. By 19-hours post-delivery the bilirubin level peaked at 18 mg/dl. The infant underwent successful treatment with an IVIG dose of 750 mg/Kg and a single exchange transfusion. **Conclusion:** The TPE-IVIG protocol, a four TPE 3-week regimen complemented by prolonging IVIG therapy, was successful at preventing the onset of severe fetal anemia as assessed by MCA-PSV. In contrast, previously described TPE-IVIG schedules typically necessitated IUT. Since IUT may be fatal, this approach offers a safer, less-invasive treatment regime that can adequately sustain a fetus until term.

Disclosure of Commercial Conflict of Interest

M. Bellone: Nothing to disclose; F. N. Boctor: No Answer; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; L. E. Logdberg: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Bellone: Nothing to disclose; F. N. Boctor: No Answer; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; L. E. Logdberg: Nothing to disclose

SP162

Newly Licensed Solvent Detergent Treated Plasma in Therapeutic Plasma Exchange: A Single Center Experience

K Pavenski[1,2] (pavenskik@smh.ca), E Krok[1], M F Anderson[1]. *[1]Laboratory Medicine, St. Michael's Hospital, Toronto, ON, Canada; [2]Laboratory Medicine and Pathobiology, University of Toronto, Toronto, ON, Canada*

Background/Case Studies: Solvent detergent treated plasma SDP (Octaplasma) was licensed in Canada in 2012 and in the US in 2013. In addition to pathogen inactivation, other advantages of SDP may include reduced risk of TRALI and allergic reactions. A significant proportion (30-70%) of patients undergoing therapeutic plasma exchange (TPE) for thrombotic thrombocytopenic purpura (TTP) experienced allergic reactions (Reutter 2001; Rizvi 2000). The experience with SDP in TPE of patients with thrombotic microangiopathies and a history of allergic reaction to plasma was examined. **Study Design/Methods:** A retrospective chart review was performed of all patients who underwent TPE with SDP from March 1, 2012-April 30, 2013. **Results/Findings:** To date, about 1600 units of SDP were transfused to five patients: four patients with TTP, and one patient with atypical hemolytic uremic syndrome. The cohort included two males and three females, and the mean age of patients was 42 years. All patients started TPE with FP (frozen plasma) or CSP (cryosupernatant plasma) and were switched to SDP following an allergic reaction. Allergic reactions included urticaria (four patients), dyspnea (two patients), angioedema (one patient), anaphylaxis (one patient), and nausea/vomiting (one patient). SDP was well tolerated and no patients experienced recrudescence of allergic reactions. One patient developed symptomatic unilateral distal popliteal vein thrombus 7 days post last treatment with SDP. At the time of the event, patient's platelet count was 250 $\times$ 10E9/L; unfortunately, protein S level was not measured. One patient was switched back to CSP because of concern about lack of response and a suspicion of a venous thrombotic event. Repeat imaging however did not confirm the presence of a thrombus. All patients with TTP received corticosteroids; three out of four TTP patients also received Rituximab for refractory or relapsed disease. In TTP patients, the mean total number of TPE treatments with SDP was 26 as compared to an institutional average of 18 TPE treatments when CSP or FP was used. Limitations of our study include very small numbers of patients; influence of unique patient factors; availability and tolerability of adjunct TTP therapies. **Conclusion:** SDP was well tolerated in patients who needed TPE with

plasma, and may be of particular value in patients with allergic reactions. One patient experienced a symptomatic thrombus. With regards to efficacy, it is premature to draw a conclusion and data collection will continue.

Disclosure of Commercial Conflict of Interest

M. F. Anderson: Nothing to disclose; E. Krok: Nothing to disclose; K. Pavenski: Octapharma, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. F. Anderson: Nothing to disclose; E. Krok: Nothing to disclose; K. Pavenski: CSL Behring Inc., Grants or Research Support, Octapharma, Grants or Research Support, Baxter Inc., Grants or Research Support, CSL Behring Inc., Travel Support or Honorarium, Alexion Pharmaceuticals, Travel Support or Honorarium

SP163

Intra-Operative Red Blood Cell (RBC) Exchange for Sickle Cell Disease (SCD) Patients Undergoing Cardiothoracic Surgery

J Louie[1] (jlouie@nshs.edu), K A Bocchieri[2], J Chiang[1], M Graver[2].
[1]Pathology, North Shore LIJ Health System, New Hyde Park, NY, United States; [2]Cardiothoracic Surgery, North Shore LIJ Health System, New Hyde Park, NY, United States

Background/Case Studies: Patients with Sickle Cell Disease (SCD) undergoing cardiothoracic surgery (CTS) require a reduction of their Hgb S level to 30% to 40%. This is usually accomplished by several simple transfusions and/or RBC exchange preoperatively and will involve extra preoperative resources, perhaps requiring extra patient days preoperatively. However, a modified cardiopulmonary bypass (CPB) system can do an RBC exchange intra-operatively and obviates the need for preoperative resources. Three SCD patients were studied: one pediatric with ventricular septal defect; and two adults with damaged mitral valves, who underwent CTS with intra-operative RBC exchange via the CPB system. All patients had satisfactory surgical outcomes. **Study Design/Methods:** One blood volume mixture of washed PRBC, FFP, albumin, and PlasmaLyte A at a 30% Hct. was used to prime the CPB circuit. The CPB circuit was modified by placing a shunt on the venous drainage line connected to a reservoir to receive the patient's entire blood volume while blood pressure was maintained at a minimum of 50-60 mm Hg with isotonic fluids and pressors. After the one blood volume was withdrawn, the CPB system was started and arterial blood flow initiated. The draw and return cycle took approximately two minutes. The autologous blood was processed to recover platelets and plasma for later use. The operating room time was not significantly lengthened. **Results/Findings:** A one blood volume exchange was needed to achieve post exchange Fraction of Cell Remaining (FCR) in the 8%-15% range and Hgb S levels in the 4% 15% range. All surgical procedures were concluded as planned. All patients recovered without significant post-op complications. **Conclusion:** Intra-Operative RBC exchange using the CPB system for SCD patients undergoing Cardiothoracic Surgery (CTS) yields very good results for exchange efficiency, reduction of preoperative resources, and patient outcome. Since almost all the patient's blood can be removed without simultaneous infusion of blood, less blood is needed than in a conventional RBC exchange using an apheresis cell separator. One blood volume replacement was used in these cases but lower amounts of blood can probably be used to achieve an Hgb S level in the 30% range. There is no need to bring in the patient prior to surgery for simple transfusion or RBC exchange since the exchange is done in the operating room. There is no significant increase in operating room time or need for special personnel. Finally, CPB RBC exchange can reduce risk for SCD patients requiring immediate emergency surgery when conventional RBC exchange is not possible.

Disclosure of Commercial Conflict of Interest

K. A. Bocchieri: Nothing to disclose; J. Chiang: Nothing to disclose; M. Graver: Nothing to disclose; J. Louie: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. A. Bocchieri: Nothing to disclose; J. Chiang: Nothing to disclose; M. Graver: Nothing to disclose; J. Louie: Nothing to disclose

SP164

Efficiency and Safety of Plasma Exchange using Three Apheresis Systems on the Same Patients

J Cid[1] (jcid@clinic.ub.es), M Acosta[1], A Faúndez[1], M Lozano[1].
[1]Hemotherapy and Hemostasis, Hospital Clínic, Barcelona, Spain

Background/Case Studies: A new protocol for the AMICUS separator (Fenwal, Inc., Lake Zurich, IL) enables the device to perform plasma exchange (PE). The aim of the study was to compare the plasma removal efficiency (PRE) and safety of Amicus, Optia (Terumo BCT, Leuven, Belgium), and Spectra (Terumo BCT) when performing PE on the same patients. **Study Design/Methods:** Patients who received PEs on the three apheresis systems were retrospectively reviewed. Tthe total amount of plasma removedwas recorded, which included plasma and anticoagulant, as well as the total amount of plasma processed. PRE was calculated as the fraction of processed plasma that is actually removed from the patient. Laboratory assays were performed from patient's blood sampled immediately before starting and immediately after finishing each PE procedure (complete blood count, prothrombin time and fibrinogen measurements, acid/base equilibrium parameters, and ionic parameters (total calcium, magnesium, sodium, and potassium)). Any adverse event was also noted. **Results/Findings:** The series comprises five men and two women with a median age of 46 years (range 23-72). Patients suffered from antibody-mediated rejection after kidney transplantation (three cases), HLA-incompatible kidney transplantation (two cases), acute disseminated encephalomyelitis, and Goodpasture's syndrome. Overall, the patients received 76 PEs. We performed 18, 44, and 14 PEs with Amicus, Optia, and Spectra, respectively. The median order of PEs performed was 6th with Amicus (range: 3rd-13th), 7th with Optia (1st-21st), and 6th with Spectra (1st-18th). The median order of PE procedures was not different when comparing the three apheresis systems ($p = 0.94$; Kruskal-Wallis test). The mean ± SD of PRE with Amicus, Optia, and Spectra was 79% ± 8%, 82% ± 5%, and 70% ± 8%, respectively. The mean PRE was lower with Spectra when compared with Optia ($p < 0.001$), and with Amicus ($p = 0.001$). The absolute change of ionic Ca (mmoL/L) was higher with Spectra (0.19 ± 0.16) when compared with Amicus (0.04 ± 0.13; $p = 0.01$) and with Optia (0.05 ± 0.12; $p = 0.002$); the absolute change of Na (mEq/L) was higher with Spectra (3.71 ± 1.98) when compared with Amicus (1.18 ± 1.97; $p = 0.002$) and with Optia (1.02 ± 1.93; $p < 0.001$). No differences were observed among the three apheresis systems in the other laboratory assays performed from patient's blood sampled immediately before starting and immediately after finishing each PE procedure. No secondary effects were observed after PE procedures with the three apheresis systems. **Conclusion:** The three apheresis systems were efficient and safe for performing PE procedures. PRE was similar with Amicus and Optia, and superior to Spectra.

Disclosure of Commercial Conflict of Interest

M. Acosta: No Answer; J. Cid: Fenwal, Travel Support or Honorarium, Terumo BCT, Travel Support or Honorarium; A. Faúndez: Nothing to disclose; M. Lozano: Terumo BCT, Grants or Research Support,Travel Support or Honorarium

Disclosure of Grants Conflict of Interest

M. Acosta: No Answer; J. Cid: Nothing to disclose; A. Faúndez: Nothing to disclose; M. Lozano: Nothing to disclose

SP165

Successful Treatment of Anti-D and Anti-K Alloimmunization in Pregnancy with Combination of Therapeutic Plasma Exchange (TPE), Intravenous Immunoglobulin (IVIG) and Intrauterine Transfusion (IUT)

M Teruya[1] (mteruya@gmail.com), N Alouch[2], K Moise[2], Y Bai[2]. *[1]The Methodist Hospital, Houston, TX, United States; [2]University of Texas Health Science Center, Houston, TX, United States*

Background/Case Studies: Alloimmunization in pregnancy can lead to hemolytic disease of the fetus and newborn (HDFN), a severe fetal anemia that can result in fetal hydrops and eventual in utero death. Different immunotherapy modalities have been used for HDFN, including therapeutic plasma exchange (TPE), and intravenous immune globulin (IVIG). A combined strategy of TPE and IVIG in early gestation along with multiple intrauterine transfusions (IUTs) is presented here to manage a woman with high titer of red blood cell (RBC) alloantibodies and a history of a previous intrauterine fetal demise. **Study Design/Methods:** A 28-year-old woman with history of anti-D and anti-K alloimmunization diagnosed during her first pregnancy 3 years ago, presented at 10 weeks gestational age. Her first pregnancy resulted in a baby with hyperbilirubinemia who required three simple transfusions, but was otherwise healthy. Subsequently, she had two more pregnancies; one ended at 21 wks gestation with an intrauterine demise due to hydrops fetalis and the other ended in a spontaneous abortion at 5 weeks gestation. In the pregnancy of interest, she presented at 9 weeks gestation and was found to be type O, RhD negative with high anti-D antibody titer of 128 and anti-K antibody titer of 512. She was treated with 3 TPEs in week 11 of gestation every other day followed by weekly IVIG infusions until 20 weeks. **Results/Findings:** An elevated middle cerebral artery Doppler at 22 wks gestation suggested fetal anemia, so IUTs were

initiated. The fetus was found to be D-antigen and K-antigen positive. In addition to anti-D and anti-K antibodies, the patient was found to have non-specific IgG. It was not possible to rule out anti-Jk(b) antibody. Although these changes in her antibody workup were likely due to the IVIG, blood that fully matched the maternal phenotype was issued. After seven total IUTs, she successfully delivered a healthy baby at 38 weeks. The infant did develop anemia at 3 weeks of life and had to be readmitted for a top-transfusion. **Conclusion:** A combined immunotherapy of TPE and IVIG leads to a successful treatment of severe maternal red cell antibody alloimmunization. Fully phenotypically-matched RBCs may help in preventing further immunization in complex HDFN cases.

Disclosure of Commercial Conflict of Interest

N. Alouch: Nothing to disclose; Y. Bai: Nothing to disclose; K. Moise: Nothing to disclose; M. Teruya: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Alouch: Nothing to disclose; Y. Bai: Nothing to disclose; K. Moise: Nothing to disclose; M. Teruya: Nothing to disclose

Gestational Age	11 wks (pre-TPE)	12 wks (post-TPE)	16 wks	22 wks	32 wks
Anti-D Titer	32-128	32	64	128	256
Anti-K Titer	256	128	256	1024	512

SP166

Comparative Effectiveness of Plasma Prepared with Pathogen Inactivation (PI) (Amotosalen-UVA) and Quarantine for Treatment of Acquired Immune Thrombotic Thrombocytopenia Purpura (TTP)

R Herbrecht[3] (raoul.herbrecht@chru-strasbourg.fr), M Ojeda-Uribe[2], J Cazenave[4], D Kientz[4], A Bohbot[3], O Hinschberger[2], K Liu[3], E Remy[3], C Ernst[1], L Corash[1]. *[1]Medical Affairs, Cerus Corp., Concord, CA, United States; [2]Hématologie Clinique, Hôpital E Muller, Mulhouse, France; [3]Oncology and Hematology, Hôpital Hautpierre, Strasbourg, France; [4]Etablissement Francais du Sang, Alsace, Strasbourg, France*

Background/Case Studies: Acquired immune TTP is a rare microangiopathy (~ 4 cases/10^6 persons/yr) due to ADAMTS13 deficiency. Untreated, mortality is >90%, but with therapeutic plasma exchange (TPE) remission is ~ 80% and mortality is reduced to <20%. TPE requires exposure to large volumes of plasma with increased risk of transfusion-transmitted infection (TTI). PI of plasma with Amotosalen-UVA (INTERCEPT Blood System for Plasma™ [IFFP], Cerus, Concord, CA) may mitigate TTI risk. A randomized controlled trial was conducted for the clinical indication of IFFP for TTP (Transfusion 2006; 46 : 1693), but included only 35 patients due to the infrequency of TTP. Universal implementation of IFFP by the Établissement Français du Sang Alsace (EFS-Alsace) in 9/2007 to replace quarantine plasma (QFFP) provided an expanded experience with IFFP for TPE of TTP. **Study Design/Methods:** The objective was to compare the efficacy and safety of IFFP to QFFP for acquired immune TTP. A retrospective, 2-period cohort study was conducted. EFS Alsace is the sole provider of FFP to regional TTP treatment centers (CHRU Strasbourg and CH Mulhouse), and administers an active hemovigilance program to assess patient outcomes. All patients treated with TPE for suspected TTP with an ADAMTS13 assay from 1/1998 to 3/2013 were reviewed, and patients with the following baseline diagnostic criteria were included: platelet count <100 $\times$ 10^9/L; microangiopathic anemia; creatinine <354 μmol/L; and ADAMTS13 < 10% without non-immune TTP. The primary efficacy outcome was the % of patients in remission 30 days after initiation of TPE. Secondary measures included the % in remission at day 60, number of TPEs, volume of plasma used, and relapse rates. The primary safety measure was the % of patients with treatment related serious adverse events (SAE) in the Cardiac System Organ Class (SOC). Secondary safety outcomes were: proportion of days with Grade 3 or 4 hypokalemia and mortality. Median values were presented for continuous variables. **Results/Findings:** 49 patients were screened and 31 qualified for review (Cohort 1: QFFP = 13; Cohort 2: IFFP = 18). Baseline demographics and disease severity between cohorts were comparable. IFFP was comparable to QFFP for efficacy and safety outcomes (Table). **Conclusion:** In this retrospective cohort study over 15 years, IFFP demonstrated comparable efficacy and safety to QFFP for TPE of acquired immune TTP.

Disclosure of Commercial Conflict of Interest

A. Bohbot: Nothing to disclose; J. Cazenave: Cerus, Grants or Research Support, Travel Support or Honorarium, Consulting or Board of Director Fees; L. Corash: Cerus, Ownership or Partnership; C. Ernst: Nothing to disclose; R. Herbrecht: Cerus, Grants or Research Support; O. Hinschberger: No Answer; D. Kientz: Nothing to disclose; K. Liu: Nothing to disclose; M. Ojeda-Uribe: Cerus, Grants or Research Support; E. Remy: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Bohbot: Nothing to disclose; J. Cazenave: Terumo BCT, Travel Support or Honorarium, Fresenius, Travel Support or Honorarium, Cerus, Grants or Research Support, Terumo BCT, Grants or Research Support, Haemonetics, Grants or Research Support, MacoPharma, Grants or Research Support, Cerus, Consulting or Board of Director Fees; L. Corash: Nothing to disclose; C. Ernst: Nothing to disclose; R. Herbrecht: Cerus, Grants or Research Support, Pfizer, Grants or Research Support, Travel Support or Honorarium, Astellas, Travel Support or Honorarium, Basilea, Travel Support or Honorarium, MSD, Travel Support or Honorarium, Gilead, Travel Support or Honorarium, Schering-Plough, Travel Support or Honorarium, Cell Therapeutics, Travel Support or Honorarium; O. Hinschberger: No Answer; D. Kientz: Nothing to disclose; K. Liu: Nothing to disclose; M. Ojeda-Uribe: Nothing to disclose; E. Remy: Nothing to disclose

Outcome Measure	IFFP (n = 18)	QFFP (n = 13)	P
Remission day 30 (%)	61.1	61.5	1.00
Remission day 60 (%)	77.8	76.9	1.00
Remission with Relapse (%)	7.1	30.0	0.27
TPE Procedures (n)	7.1	9.8	0.60
Volume Plasma (L)	23.1	26.1	1.00
Related Cardiac SAE (%)	0.0	7.7	0.42
Grade 3, 4 Hypokalemia	0.0	0.0	1.00
All-Cause Mortality (%)	16.7	23.1	0.68

Pediatric Transfusion Medicine

SP167

An Anti-hrB-like Antibody in a Sickle Cell Patient with a Variant e Confirmed by Molecular Testing

V Bhoj[2] (vijay.bhoj@uphs.upenn.edu), P Blair[1], D F Friedman[3], M A Keller[4], S J Nance[5], C C Norris[3], D Sesok-Pizzini[1]. *[1]Department of Pathology and Laboratory Medicine, Children's Hospital of Philadelphia, Philadelphia, PA, United States; [2]Department of Pathology and Laboratory Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA, United States; [3]Department of Pediatrics, Children's Hospital of Philadelphia, Philadelphia, PA, United States; [4]National Molecular Laboratory, American Red Cross, Philadelphia, PA, United States; [5]Testing Laboratory, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: The high incidence red cell antigen, hrB, has a serological relationship with the e antigen; the hrB-negative phenotype is typically identified in African Americans (AA) with e variants. Anti-hrB-like antibodies are rare and may be a complication in patients requiring frequent transfusions. Although transfusing hrB- blood may be achieved by using R2R2 (e-, hrB-) blood, transfusing HrB negative blood requires the use of the American Rare Donor Program. A case of a sickle cell patient with anti-hrB-like who presented for chronic transfusion for stroke prevention and the molecular typing results that confirmed the safe use of R2R2 blood is reported here. **Study Design/Methods:** Patient typing and antibody workup was performed by gel and tube methods. Molecular testing was performed by polymerase chain reaction (PCR), restriction fragment length polymorphism (RFLP), and RHCE BeadChip™. **Results/Findings:** A 7 year-old AA female with sickle cell disease (SCD)was presented for prophylactic transfusion. She was found to be A+ with a positive antibody screen and a pan-reactive antibody panel. A DAT was positive at IgG and C3 and an eluate was pan-reactive. Adsorption of her plasma at 37C with R1, R2 or rr cells demonstrated an allo anti-hrB-like with PEG-IgG-AGT in the R2 adsorbed plasma, which was not removed with further adsorptions with e- red cells. Alloantibodies to other common antigens were ruled out. Two packed RBC (pRBC) units were requested before completion of molecular testing. A serological phenotype on the patient could not be performed due to recent transfusions and protocols for SCD routinely require E, C, and K negative blood. However, due to concerns of an anti-hrB-like antibody and the rarity of hrB– units (sibling donors were not available), crossmatch compatible e-units (predicted to be hrB–) were transfused successfully. Molecular testing revealed a RHCE*cE/RHCE*ce733G genotype corresponding to the predicted phenotype: C– E+ c+ partial e+ VS+ V+ hrB + w/–. The patient later required additional pRBC and was successfully transfused crossmatch compatible e-units. In the event the patient was RHCE*ce733G homozygous, a search of the American Rare Donor Program confirmed only 41 compatible

donors in the US. **Conclusion:** Serologic testing demonstrated an allo anti-hrB-like antibody. Genotypic analysis predicted one normal E and a variant e allele, that latter of which is considered by some to lack hrB, supporting the decision for safe transfusion of e- units (R2R2). The molecular data in this case supported the safe use of R2R2 units, avoiding the need for time and resources to find rare matched donors. This case highlights the contribution of molecular genotyping to pRBC selection and emphasizes the importance of clinical judgment when confronted with an antibody to a high-incidence antigen with work-up in progress.

Disclosure of Commercial Conflict of Interest

V. Bhoj: Nothing to disclose; P. Blair: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; M. A. Keller: BioArray, Grants or Research Support, Life Technologies, Consulting or Board of Director Fees; S. J. Nance: Nothing to disclose; C. C. Norris: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. Bhoj: Nothing to disclose; P. Blair: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; M. A. Keller: BioArray, Grants or Research Support; S. J. Nance: Nothing to disclose; C. C. Norris: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose

SP168

Seven Day Storage of Apheresis Derived Split Platelets in ELX HP (Polyolefin) Bags

C Chen[1] (cyndi_chen@pall.com), Y Garcia-Quezada[1], M A Popovsky[1], S Lucas[1]. *[1]Research and Development, Haemonetics Corporation, Braintree, MA, United States*

Background/Case Studies: Currently, apheresis derived adult platelet doses can be split into four paediatric packs for up to 5 days of platelet storage. These split units are stored in CLX® bags which is suitable for 5 days of storage. The ELX HP bag (polyolefin) has been proven for 7 days of storage of platelets derived from random donors, buffy coat, and apheresis platelets. It would be desirable to store split units for extended periods, providing logistical and clinical advantages to transfusion services. The objective was to evaluate the platelet storage quality of apheresis derived platelet splits stored in ELX HP bag. **Study Design/Methods:** On the day after collection, 11 apheresis units (HemaCare, Van Nuys, CA) were processed and each unit was split evenly into four paediatric doses and stored in ELX HP storage bags. Each apheresis unit had a least $3 \times 10e11$ platelets prior to four-way spilt. Platelets were sampled on days 2, 5, 7, and 9 days of storage for quality assessment. All units were stored on a platelet horizontal agitator at 22C. **Results/Findings:** The mean platelet yield was $0.8 \times 10e11 \pm 0.1 \times 10e11$ in each split. Overall, the maintenance of % extent shape change (ESC), % hypotonic shock recovery (HSC), and % p-selectin were satisfactory. End of storage pH levels were greater than 6.4. No statistically significant difference was found between the four-way splits through 9 days of storage. Results met European directive and guidelines. **Conclusion:** Apheresis derived adult platelet doses can be split into four pediatric doses and stored for up to 7 days of storage when stored in ELX HP (polyelefin) bags. Such storage provides advantages for transfusion services.

Disclosure of Commercial Conflict of Interest

C. Chen: Nothing to disclose; Y. Garcia-Quezada: No Answer; S. Lucas: No Answer; M. A. Popovsky: Haemonetics, Other

Disclosure of Grants Conflict of Interest

C. Chen: Nothing to disclose; Y. Garcia-Quezada: No Answer; S. Lucas: No Answer; M. A. Popovsky: Nothing to disclose

TABLE 1. Platelet Quality Assessment—Combined data (Mean ± SD; n = 11)

Parameters		Day 2	Day 5	Day 7	Day 9
pH (22°C)	Split 1	7.47 ± 0.08	7.26 ± 0.10	7.05 ± 0.18	6.73 ± 0.28
	Split 2	7.48 ± 0.06	7.31 ± 0.08	7.12 ± 0.14	6.80 ± 0.28
	Split 3	7.48 ± 0.07	7.28 ± 0.10	7.06 ± 0.16	6.76 ± 0.25
	Split 4	7.49 ± 0.06	7.34 ± 0.07	7.16 ± 0.12	6.83 ± 0.22
ESC (%)	Split 1	31.0 ± 3.4	26.6 ± 1.8	22.9 ± 2.2	16.5 ± 5.6
	Split 2	31.1 ± 2.6	26.6 ± 1.9	23.7 ± 2.8	18.3 ± 4.2
	Split 3	30.1 ± 3.2	26.5 ± 3.4	22.6 ± 2.0	16.0 ± 5.2
	Split 4	31.8 ± 3.7	26.2 ± 2.9	24.1 ± 3.3	19.2 ± 3.5
HSR (%)	Split 1	83.0 ± 10.3	70.8 ± 10.9	64.3 ± 10.2	48.3 ± 22.4
	Split 2	83.3 ± 6.1	73.7 ± 12.1	66.4 ± 9.3	52.5 ± 17.6
	Split 3	83.2 ± 7.0	72.3 ± 11.8	67.4 ± 11.0	52.7 ± 20.4
	Split 4	84.1 ± 8.4	71.8 ± 11.3	71.1 ± 9.2	58.3 ± 13.8
P-selectin (%)	Split 1	10.8 ± 3.8	16.8 ± 3.4	21.8 ± 6.1	37.9 ± 13.4
	Split 2	10.8 ± 5.2	15.0 ± 3.6	21.4 ± 7.8	34.5 ± 9.3
	Split 3	10.6 ± 4.0	16.1 ± 4.0	22.9 ± 7.5	38.9 ± 11.4
	Split 4	10.9 ± 7.7	14.0 ± 3.9	20.4 ± 7.7	35.2 ± 10.0

SP169

A Novel Low-Frequency Antibody Causing Hemolytic Disease of the Fetus and Newborn (HDFN)

C Melland[1] (cami_melland@bonfils.org), M LaSarre[1], A Deering[1], K Maschmann[1], M McCollom[1], C C Silliman[2], F West[1], T Le[1]. *[1]Reference Laboratory, Bonfils Blood Center, Denver, CO, United States; [2]Pediatrics, University of Colorado, Aurora, CO, United States*

Background/Case Studies: Clinically significant low-frequency (or, low-prevalence) antibodies occurring in pregnant individuals are problematic because they are often undetectable by routine antibody screening techniques. These antibodies are not usually suspected until the effects of hemolysis are seen in the fetus or newborn. An interesting case is reported here in which a term child died of the sequelae of HDFN; of note, the mother previously had multiple spontaneous abortions. **Study Design/Methods:** A cord blood sample was received with a request for a DAT investigation. The IgG DAT was 4+ and the eluate was non-reactive. A low-frequency antibody was suspected and the eluate was run against 9 cells known to be positive for low-frequency antigens.One V+ cell came up very weakly positive, but anti-V was ruled out on a different V+ cell. Several months later, the RL received a blood sample from the mother for antibody identification, as part of the workup for her history of unexplained spontaneous abortions. Twenty-four red cells bearing low-frequency antigens were tested with PEG enhancement. All cells were negative except one Cx+ cell. Anti-Cx was ruled out on a different Cx+ cell. The reaction of the mother's plasma against the father's cells was 4+ with PEG enhancement. Blood from the mother and the father was tested by molecular methods for the standard array of antigens; there were no unusual findings. **Results/Findings:** The mother's antigen(s) to the low frequency antibody were not affected by ficin or DTT treatment. The antigen integrity, the severity of the HDFN and the 2 positive reactions occurring on Cx+ and V+ cells indicates that the antibody may be specific to an antigen in the Rh system. The father's sample was sent for RH gene sequencing; significant findings will be published. **Conclusion:** Regardless of the blood group system to which the antigen belongs, once a clinically significant antibody has been formed, the treatment then focuses on the fetus' well being . The current or any subsequent pregnancy should be monitored closely for the onset of fetal anemia by intracranial Doppler or fetal blood sampling and once the onset of HDFN occurs, possible treatments may include intrauterine transfusion, plasmapheresis to lessen the antibody load and/or IVIG therapy for the mother, induced delivery and possible exchange transfusion after birth.

Disclosure of Commercial Conflict of Interest

A. Deering: Nothing to disclose; M. LaSarre: Nothing to disclose; T. Le: Nothing to disclose; K. Maschmann: No Answer; M. McCollom: Nothing to disclose; C. Melland: Nothing to disclose; C. C. Silliman: Nothing to disclose; F. West: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Deering: Nothing to disclose; M. LaSarre: Nothing to disclose; T. Le: Nothing to disclose; K. Maschmann: No Answer; M. McCollom: Nothing to disclose; C. Melland: Nothing to disclose; C. C. Silliman: Pall Corp/Haemonetics, Grants or Research Support; F. West: Nothing to disclose

SP170

Massive Transfusion as a Predictor of Mortality in Pediatric Patients Receiving Extracorporeal Life Support

A M Winkler[1] (awinkl2@emory.edu), M L Paden[2], C Josephson[1,3].
[1]Department of Pathology and Laboratory Medicine, Emory University, Atlanta, GA, United States; [2]Department of Pediatrics, Division of Critical Care, Emory University, Atlanta, GA, United States; [3]Department of Pediatrics, Emory University, Atlanta, GA, United States

Background/Case Studies: Extracorporeal life support (ECLS) is a technique that has been shown to be effective in supporting patients of all ages with life-threatening cardio-respiratory failure of varying etiologies, and has become the standard therapy for neonatal, pediatric and adult patients with acute cardiac or respiratory failure unresponsive to traditional management. As a result of added extracorporeal volume, acquired hemostatic changes, and occurrence of hematologic complications, transfusion requirements associated with ECLS are high and patients are commonly massively transfused; however, the association between massive transfusion (MT) and mortality has not been formally investigated. **Study Design/Methods:** A cohort study of all ECLS patients less than 18 years of age was conducted at a single institution from January 1, 2009-December 31, 2011 to examine the association between MT, defined as RBC transfusion of greater than one blood volume (<10 kg 80 ml/kg, >10 kg 70 ml/kg) plus circuit volume (<10 kg 450 ml, >10 kg 750 ml), and all-cause mortality. **Results/Findings:** A total of 165 patients received ECLS support and baseline characteristics are summarized in Table 1. MT, which occurred in 32.7% of the entire cohort and 36.3% of neonatal subjects, was associated with increased all-cause mortality, and this effect was stronger for neonates (OR 6.45, CI 1.81-22.9, p = 0.004). Kaplan-Meier estimates for the probability of death did significantly differ between subjects who were massively transfused versus those who were not (Log rank < 0.0001) and the effect of MT was time-dependent. Subjects who were massively transfused with an ECLS duration greater than 4 hours had an increased hazard of death (HR 3.38, CI1.59-7.20, p = 0.002) when controlling for ECLS mode. **Conclusion:** MT in ECLS is independently associated with an increased risk for all-cause mortality, similar to investigations linking RBC transfusion to increased morbidity and mortality in non-ECLS critically ill patients.

Disclosure of Commercial Conflict of Interest

C. Josephson: Nothing to disclose; M. L. Paden: Nothing to disclose; A. M. Winkler: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Josephson: Immucor/BioArray, Grants or Research Support, Octapharma, Consulting or Board of Director Fees; M. L. Paden: Nothing to disclose; A. M. Winkler: Nothing to disclose

	MT (n = 54)	Not MT (m = 111)	p-value
Age (days)	4 (3-10.5)	85 (1-2207)	<0.0001
Weight (kg)	4.76 (3.2-23.9)	3.8 (2.94-4.6)	<0.0001
Male Sex (%)	34 (63%)	66 (59.5%)	0.667
Raoc			0.73
White	22 (42.3%)	37 (34.9%)	
Black	23 (44.2%)	47 (44.3%)	
Hispanic	3 (5.8%)	13 (12.3%)	
Asian	2 (3.9%)	5 (4.7%)	
Other	2 (3.9%)	4 (3.8%)	
ECLS Mode			
VV	10 (18.5%)	71 (64%)	<0.0001
VA	44 (81.5%)	40 (36%)	
Pre-ECLS PEL.OD	20.3 (6.5)	21.2 (7.16)	0.44
Pre-ECLS Hemoglobin (g/dl)	12.3 (2.7)	12.9 (2.8)	0.14
Pre-ECLS INR	1.6 (0.4)	1.6 (0.6)	0.72
Pre-ECLS APTT (see)	74.7 (70.0)	52.3 (38.7)	0.03
PRE-ECLS Fibrinogen (mg/dl)	249 (129)	278 (174)	0.39
Pre-ECLS Platelets (per nl)	203 (133)	238 (178)	0.25
ECLS Dmation (hr)	157 (106.6)	208 (183.5)	0.06
Mortality	29 (53.7%)	31 (27.9%)	0.001

SP171

Are Short-Term Intensive Care Unit Admissions for Monitoring during the First Transfusion in Pediatric Patients Necessary?

A Berg[1] (bergan2@upmc.edu), R Courtney[2], L Krishnamurti[2], D J Triulzi[3,1], M H Yazer[3,1]. *[1]Pathology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [2]Hematology and Oncology, Children's Hospital of Pittsburgh, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [3]The Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: Some pediatric patients, typically those that are very young or felt to be especially sick, are, at the discretion of their attending physicians, temporarily admitted to the intensive care unit (ICU) for close observation during their first transfusion. These patients do not require ICU admission for treatment of their underlying disease, but are admitted simply for close monitoring during their first transfusion. If a significant reaction that requires ICU management does not occur, at the completion of the transfusion these patients are transferred to a regular ward where future blood products are administered as necessary. The aim of this project was to determine if temporary ICU admissions for the duration of the first transfusion are warranted. **Study Design/Methods:** From the blood bank records of a tertiary care specialist pediatric hospital, a list of patients who experienced a transfusion reaction from 2007-2012, the nature of the reaction, and whether it occurred on the patient's first recorded transfusion was documented. The hospital location where the transfusion that caused the reaction occurred, and whether the patient was evaluated by the ICU team or transferred to the ICU for management of the reaction, was determined from the patient's electronic medical record. Due to limitations in the information technology system it was not possible to determine how many patients were temporarily admitted to the ICU for observation during transfusion. **Results/Findings:** During the study period this pediatric hospital transfused an average of 7724 +/– 571 RBCs, 13,712 +/– 2461 whole blood platelet equivalents and 3786 +/– 755 plasma units per year. There were 232 reactions reported in 204 patients. Overall, the most common reactions were febrile non-hemolytic (87/232, 38%), and mild allergic (82/232, 35%). Of these 204 patients, 13 (6.4%) different patients experienced a reaction during their first recorded transfusion. All 13 reactions to the first transfusion were mild (8 febrile non-hemolytic, 5 mild allergic) and did not require ICU management. None of these 13 patients had been temporarily admitted to the ICU for close monitoring during their first transfusion. During the study period there were 6/232 (2.6%) reactions involving significant hypotension and/or hypoxia that required ICU consultation and/or management. **Conclusion:** Given the small number of reactions to the first transfusion, and the infrequent occurrence of severe reactions that required ICU management, the practice of temporarily admitting pediatric patients to the ICU for monitoring during the first transfusion seems unnecessary.

Disclosure of Commercial Conflict of Interest

A. Berg: Nothing to disclose; R. Courtney: Nothing to disclose; L. Krishnamurti: No Answer; D. J. Triulzi: Nothing to disclose; M. H. Yazer: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Berg: Nothing to disclose; R. Courtney: Nothing to disclose; L. Krishnamurti: No Answer; D. J. Triulzi: Nothing to disclose; M. H. Yazer: Nothing to disclose

SP172

Fifteen Years Review of Solvent/Detergent (S/D) Plasma in a Pediatric Population

V Witt[1] (volker.witt@stanna.at), L Weidner[1], H Pichler[1], R Ritter[1], M Wurth[1], W Holter[1]. *[1]St. Anna Children Hospital, Vienna, Austria*

Background/Case Studies: Since 1996, St. Anna Children Hospital in Vienna has transfused S/D plasma (Octaplas®, Octapharma AG) as its primary source of fresh frozen plasma (FFP) for their pediatric patients. A review of transfusion records from 1996-2011 was performed to explore the side effect profile of S/D plasma . Altough S/D plasma is used world-wide in pediatric patients, this is the first report describing the experience of S/D plasma in a pediatric population. **Study Design/Methods:** Transfusion records from 1996 2012 were reviewed. Data from therapeutic plasma exchange (TPE) were collected retrospectively from patient records. Thirty one TPE patients were identified Transfusion data were collected from our internal hemovigilance system. TPE patients were consent to allow reporting their data for retrospective analysis and entering these data in national and international registries. **Results/Findings:** From 2008-2012, 282 TPE in 30 patients were performed. The median age was 13.8 years (min 0.94—max

22.7 y) and a median bodyweight of 40 kg (min 9.9—max.80 kg). A median of 6 TPE were performed per patient (1-40) with a median of 1940 mL S/D plasma exchanged (min 371—max 5270 mL) corresponding to 48.2 mL/kg (min 13—max 99 mL/kg). In total, 2244 units of S/D plasma were used for TPE in children and young adults representing 70% (2244/3178) of the S/D plasma units transfused in that time period. The replacement fluid was mainly S/D plasma alone (n = 185), or human albumin and for the last third S/D plasma (n = 73) and rarely human albumin alone (n = 17). INR, aPTT, fibrinogen and anti-thrombin III were routinely measured in these patients. When S/D plasma alone was transfused or in combination with human albumin, a normalization of the parameters was observed after TPE even if they were increased, normal or decreased before the procedure. Using human albumin alone, it was observed, for example, an increase of the INR (1 to 1.8), the aPTT (<40 s before, ≥40 s after TPE). The differences for coagulation parameters in the albumin only group versus the S/D only and S/D in combination with albumin were statistically significant. In single transfusions, no side effects were reported. No transfusion related acute lung injury (TRALI) was observed. In TPE, one patient developed an allergic reaction and was switched to human albumin as replacement fluid, and one catheter occlusion occurred after TPE, where it was unclear if this was due to the replacement fluid or due to the Broviac catheter. **Conclusion:** S/D plasma, used as the main source of FFP at this pediatric institution, has shown to provide satisfactory hemostatic properties with no associated transfusion reactions commonly observed with FFP.

Disclosure of Commercial Conflict of Interest

W. Holter: No Answer; H. Pichler: No Answer; R. Ritter: No Answer; L. Weidner: No Answer; V. Witt: Octapharma, Consulting or Board of Director Fees; M. Wurth: No Answer

Disclosure of Grants Conflict of Interest

W. Holter: No Answer; H. Pichler: No Answer; R. Ritter: No Answer; L. Weidner: No Answer; V. Witt: Nothing to disclose; M. Wurth: No Answer

SP173

Evaluation of Three Sterile Tubing Welding Devices to Weld PVC Blood Bags to DEHP-Free Pediatric Syringe Sets

J L Jacobson[1] (jessica.jacobson@nyumc.org), D Gajera[1]. *[1]Pathology, Bellevue Hospital/NYU School of Medicine, New York City, NY, United States*

Background/Case Studies: Bis(2-ethylexyl) phthalate (DEHP) is a chemical that has been used to as a plasticizer in many medical devices including blood bags and transfusion tubing. Under the high temperatures required for processing polyvinyl chloride (PVC) products, DEHP can leach out of the blood bag into the blood product and has been associated with many clinically significant side effects. Consequently, the FDA and the American Academy of Pediatrics advises clinicians to use DEHP-free products where available. The blood bank issues pediatric aliquots for transfusion to neonates in the neonatal intensive care unit (NICU). The NICU would like the blood bank to issue ISBT-128 labeled blood filled DEHP-free syringes instead of pouches. All FDA approved sterile tubing welders are approved only for welding PVC to PVC. The sterile tubing welder options were investigated for filling DEHP-free syringes made from blood products stored in PVC bags for the NICU. **Study Design/Methods:** The Fresenius Kabi CompoDock, the Terumo SCDII Sterile Tubing Welder, and the Genesis BPS TCD Sterile Tube Welder were evaluated for their ability to weld PCV blood bags to PVC tubing and PVC blood bag to DEHP-free syringe sets manufactured by Charter Medical and Genesis BPS. **Results/Findings:** All three devices were able to effectively weld PVC to PVC. The CompoDock was able to weld 0 of 20 PVC blood bags to DEHP-free syringe set tubing. Both the SCDII and TCD completed ten of ten welds of PVC blood bags to DEHP-free tubing. The SCDII and TCD were also able to weld five of five DEHP-free tubing to DEHP-free tubing. **Conclusion:** Although neither the Terumo SCDII Sterile Tubing Welder nor the Genesis PBS TCD Sterile Tube Welder was approved by FDA to weld PVC to DEHP-free plastic, both are capable of welding PVC blood bags to DEHP-free syringe sets. The Fresenius Kabi CompoDock cannot be used to produce DEHP-free syringe sets for neonatal transfusions. Given that blood is stored for much longer periods of time in DEHP containing PVC blood bags than it is in the transfusion syringe sets, the potential added safety of using DEHP-free syringe sets may not justify the added expense of purchasing a new sterile tube welding device if the lab already has the CompoDock.

Disclosure of Commercial Conflict of Interest

D. Gajera: Nothing to disclose; J. L. Jacobson: Fresenius, Other, Terumo BCT, Other, Genesis BPS, Other, SMART Bio Tech Ltd., Other

Disclosure of Grants Conflict of Interest

D. Gajera: Nothing to disclose; J. L. Jacobson: G & W Laboratories Inc., Grants or Research Support

Comparison of Sterile Tube Welding Devices

Device	Weld PVC to PVC	Weld PVC to DEHP-Free Charter Medical Tubing	Wedl PVC to DEHP = Free & BPA-Free Genesis Tubing	Weld DEHP-Free to DEHP-Free
Fresenius Kabi CompoDock	20 of 20	0 of 20	0 of 20	0 of 20
Terumo SCDII Sterile Tubing Welder	5 of 5	5 of 5	5 of 5	5 of 5
Genesis BPS TCD Sterile Tube Welder	5 of 5	5 of 5	5 of 5	5 of 5

SP174

Product Usage in Massive Transfusion at a Tertiary Pediatric Hospital

E I Schindler[1] (eschindler@path.wustl.edu), J Vosler[2], D Baker[2], R Jackups[1,3]. *[1]Pathology & Immunology, Washington University School of Medicine, St. Louis, MO, United States; [2]St. Louis Children's Hospital, St. Louis, MO, United States; [3]Pediatrics, Washington University School of Medicine, St. Louis, MO, United States*

Background/Case Studies: Although massive transfusion has been extensively studied in the adult setting, minimal data exist on the optimal ratio and amount of blood components in the pediatric setting. In 2011, a pediatric massive transfusion protocol (MTP) was implemented here. This retrospective study examines blood component usage under the MTP, specifically whether products have been transfused in an appropriate ratio, the ratio of products transfused to products issued, and the final disposition of returned blood products. **Study Design/Methods:** This is a retrospective study of all MTP activations from January 2012-March 2013 at a single tertiary pediatric hospital. Product usage data was analyzed using SPSS Statistical Package, v.20. **Results/Findings:** The massive transfusion protocol was initiated 25 times during the study period for 23 patients. The age range for patients was 0-18 years, except for one adult patient (age 66). The median and range number of products issued was 4 units plasma (2-11), 4 units RBCs (2-12), 1 unit SDP (0-3), and 0 units cryoprecipitate (0-4). Plasma and RBCs were transfused in a 1:1 ratio in 15/20 cases. No product was transfused in five cases. The median and range percentage of product transfused per amount of product issued for each component type were: plasma 30% (0-80), RBCs 30% (0-100), SDP 40% (0-100), and cryoprecipitate 0% (0-50). Three patients died during MTP. Protocol activation occurred in the emergency room (six times), operating room (11 times), ICU (seven times) and cardiac catheterization lab (once). The percentage of plasma transfused per amount of plasma issued was significantly increased in the emergency room compared to all other hospital sites (P = 0.021). Aside from this, there was no significant difference in product usage and final disposition across all locations. **Conclusion:** This study characterizes product usage during massive transfusion at this institution, with the majority of physicians transfusing within the 1:1 recommended plasma to RBC ratio. There was low usage of cryoprecipitate, which is only partially explained by the possible lag time in delivery due to thawing. In the five cases in which no product was transfused, all patients lived at least another 12 hours beyond the MTP activation. Three of these cases were associated with a traumatic head injury. The presence of a single older adult in this study highlights the fact that MTP protocols at children's hospitals must be prepared to address MTP in older patients receiving specialty treatment.

Disclosure of Commercial Conflict of Interest

D. Baker: Nothing to disclose; R. Jackups: Nothing to disclose; E. I. Schindler: Nothing to disclose; J. Vosler: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Baker: Nothing to disclose; R. Jackups: Nothing to disclose; E. I. Schindler: Nothing to disclose; J. Vosler: Nothing to disclose

SP175

Description of Effectiveness of Permanent Implantable Ports in Routine Erythrocytapheresis Procedures in Pediatric Patients with Sickle Cell Disease (SCD) in a 5-year Case Series

J Webb[1] (JennW1009@gmail.com), D F Friedman[1], J H Holmes[2].
[1]Division of Hematology, Children's Hospital of Philadelphia, Philadelphia, PA, United States; [2]Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, United States

Background/Case Studies: Adequate venous access is a limiting factor for chronic erythrocytapheresis in pediatric patients with sickle cell disease (SCD). Though several studies have evaluated the short and long-term complication rates for central lines, they have not evaluated the efficacy of central line placement in achieving hematologic goals. **Study Design/Methods:** Records were reviewed for all pediatric patients with SCD receiving routine erythrocytapheresis with goal percent hemoglobin S of <30% between January 1, 2005-December 31, 2010 at this institution. Characteristics were compared between those with central access and those without. All subjects with central access had implanted ports due to institutional preference. For those procedures performed using a port, the percentage of successfully completed procedures, steady-state hemoglobin, steady-state hematocrit, and percent hemoglobin S prior to procedures were compared. **Results/Findings:** The cohort included 73 pediatric patients with SCD receiving routine apheresis during this period for a total of 3955 erythrocytapheresis procedures. Thirteen subjects had implanted ports at cohort entry. Of these, three subjects had their ports removed during the observation period and three additional subjects had ports placed during the observation period. Those with central lines tended to be older (mean age 14.5 years versus 12.8 years) and have a higher body mass index (mean body mass index 23.3 kg/m2 versus 18.9 kg/m2). The population with ports had a higher percentage of females (50% versus 42%). Those with ports also tended to have been receiving routine erythrocytapheresis for a shorter duration of time (176 days versus 633 days). The average duration of a port was 581 days during this window of observation. The percentage of successfully completed procedures was slightly lower in those with ports (97.6% versus 98.1%). The mean percent hemoglobin S for procedures performed with a port was 31.6% versus 30.5% in those with peripheral access. The mean steady-state hemoglobin and hematocrit were nearly identical for procedures performed with and without central lines. **Conclusion:** Adequate venous access is necessary for erythrocytapheresis. In this cohort, the percentage of successfully completed procedures, hemoglobin, hematocrit and percent hemoglobin S at the time of the procedure were comparable between the two groups. This suggests that implanted ports serve an important role in achieving the hematologic goals of erythrocytapheresis in pediatric patients with SCD. Further research is necessary to determine if an individual's probability of successfully achieving hematologic outcomes improves with placement of a central line.

Disclosure of Commercial Conflict of Interest

D. F. Friedman: BioArray, Grants or Research Support; J. H. Holmes: Nothing to disclose; J. Webb: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. F. Friedman: BioArray, Grants or Research Support; J. H. Holmes: Nothing to disclose; J. Webb: Nothing to disclose

SP176

Irradiated Blood Products and Their Effect on Transfused Neonates

J Lee[1] (kaming.lee@uch.edu). [1]University of Colorado Hospital, Aurora, CO, United States

Background/Case Studies: It is a common practice to provide irradiated cellular components when transfusing neonates due to concerns of transfusion associated graft versus host disease (TA-GVHD). On the other hand, once irradiated, red blood cells have a shorter life and increased leakage of potassium into the extracellular space. Each institution varies in their guidelines but the general consensus is to provide fresh blood. This is particularly seen when providing small volume transfusions to the neonate population. A change of practice was recently implemented here to address these concerns. As of September 2011, all cellular components prepared for neonatal transfusions were irradiated. To address concerns of elevated potassium levels, red blood cells were not used past 14 days from the date of irradiation of the parent unit. Platelets, with a shelf life of 5 days, were used up to the expiration of the unit. **Study Design/Methods:** A retrospective study was performed on data collected between April 1, 2011-March 31, 2012 reflecting 6 months prior to and 6 months after implementing the irradiation policy. Among the data collected were pre- and post-transfusion potassium levels, presence of TA-GVHD related symptoms, diagnosis, birth weight, age at transfusion, and whether the blood product was irradiated or not. The data took into account prepared aliquots and excluded exchange and intrauterine transfusions. Statistical analysis was calculated for the average, mean, minimum, maximum, and sum of each data set. The significance of irradiated versus non-irradiated blood products was calculated using the non-parametric Wilcoxan-Rank-Sum test where if P >= 0.05, it is not significantly different. **Results/Findings:** A total of 65 infants were transfused within the specified year with RBC aliquots totaling 227 and platelet aliquots totaling 26. Of the 227 RBC aliquots, 149 were non-irradiated and 75 were irradiated. Of the 26 platelet aliquots, 20 were non-irradiated and 6 were irradiated. Of these 65 infants, five died after three received non-irradiated blood and two received irradiated blood. Presence of TA-GVHD associated symptoms such as rash, GI bleed, liver dysfunction, and fever was absent in all infants. Sixty-one of the infants were pre-term. The average change in potassium level was zero although the highest rise was 3.2 mmol/L and the lowest decrease was 2.2 mmol/L. The P values were 0.54 and 0.24, respectively for the RBC and platelet aliquots. Thus, there was no significant difference between irradiated and non-irradiated blood products. **Conclusion:** A significant difference in potassium levels was not demonstrated either clinically or statistically in the group of 65 patients observed during the year. Due to the limited quantity of patients, there may not have been enough to demonstrate a difference. Limitations of the retrospective study in terms of data quality may have an influence as well.

Disclosure of Commercial Conflict of Interest

J. Lee: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Lee: Nothing to disclose

SP177

Two Cases of Asymptomatic Massive Fetomaternal Hemorrhage (FMH)

A R Peedin[1] (apeedin@unch.unc.edu), J S Raval[1], Y A Park[1], E T Weimer[1], J Schmitz[1], M Mazepa[1]. *[1]Pathology & Laboratory Medicine, University of North Carolina, Chapel Hill, NC, United States*

Background/Case Studies: Evaluation of fetomaternal hemorrhage (FMH) in the immediate postpartum period is critical for the timely administration of Rh immunoglobulin (RhIG) prophylaxis to minimize the risk of alloimmunization in D-negative mothers of D-positive newborns. S series of two clinically unsuspected cases of massive FMHs (209.5 mL and 75 mL) were identified and reported on. **Study Design/Methods:** Two cases of massive, asymptomatic FMH identified within the past 12 months were retrospectively reviewed. After positive fetal bleed screens via rosette testing, flow cytometry (FC) for hemoglobin F was performed to quantify the volume of hemorrhage. **Results/Findings:** Patient A: A 27-year-old G3P2002, O-negative woman delivered a term infant by normal spontaneous vaginal delivery. In the immediate post-partum setting, she was administered one dose of RhIG. After a positive fetal bleed screen, flow cytometry identified that 4.19% of the patient's whole blood represented fetal cells, corresponding to 209.5 mL of fetal blood in maternal circulation. Because of this larger than expected volume, FC for D antigen was performed to rule out hereditary persistence of fetal hemoglobin; 3% of the patient's whole blood was D-positive, confirming a fetal origin. These findings resulted in the administration of 8 total doses of RhIG. Patient B: A 25-year-old G3P2002, O-negative woman delivered a term infant by C section. In the immediate post-partum setting, she was administered one dose of RhIG. After a positive fetal bleed screen, flow cytometry identified that 1.5% of the patient's whole blood represented fetal cells, corresponding to 75 mL of fetal blood in maternal circulation. This volume resulted in the administration of four total doses of RhIG. Both infants had APGAR scores of eight or above at 1 and 5 minutes of life, indicating that they were not adversely affected by the hemorrhages. **Conclusion:** These cases represent examples of massive FMH that were clinically silent. Despite the high volumes of fetal blood loss, neither bleed appears to have resulted in adverse outcomes in the infants, raising the issue of possible chronic fetal bleeding. These D-negative patients required doses of immunoprophylaxis well in excess of the standard postpartum RhIG dose. FC is a more sensitive, precise and accurate assay than the acid elution test for calculation of RhIG dosing. The consequences of under-dosing D-negative patients after FMH are serious, potentially resulting in future hemolytic disease of the fetus and newborn or hemolytic transfusion reactions. These cases highlight the importance of rapidly obtaining accurate measurements of fetal blood loss in cases of FMH, even in clinically silent cases, to ensure timely administration of adequate immunoprophylaxis to D-negative patients.

Disclosure of Commercial Conflict of Interest

M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; A. R. Peedin: Nothing to disclose; J. S. Raval: Nothing to disclose; J. Schmitz: Nothing to disclose; E. T. Weimer: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; A. R. Peedin: Nothing to disclose; J. S. Raval: Nothing to disclose; J. Schmitz: Nothing to disclose; E. T. Weimer: Nothing to disclose

SP178

Blood Level Levels in Transfused Neonatal and Pediatric Patients: A Retrospective Analysis

D Thornton[1] (david.thornton@nationwidechildrens.org), S V Rudmann[2], T Bannerman[2], S Townsend[1], K Nicol[1]. *[1]Pathology & Lab Medicine, Nationwide Children's Hospital, Columbus, OH, United States; [2]Medical Laboratory Science Division, The Ohio State University College of Medicine, Columbus, OH, United States*

Background/Case Studies: Lead is a well established heavy metal toxin and the adverse health affects associated with elevated blood lead levels (BLL) have been widely studied and documented. New concern has been given to the research demonstrating the effect low blood lead levels on IQ and academic performance; however, a threshold level for the effects of lead has yet to be found. Recent studies regarding the concentration of lead in donor units transfused to children prompted this study to examine the patient population here. **Study Design/Methods:** The laboratory data base was queried for patients having both a blood lead level and a blood bank encounter between January 1, 2009–June 20, 2012. Twenty six hundred sixty nine (2669) patients were identified, and patients were excluded that had lead level testing performed more than one year after their latest transfusion, leaving 553 patients. The patient age of those transfused and their blood lead levels were examined. **Results/Findings:** Of the 553 transfused children found to have blood lead level testing, 282 were neonates (<120 days old) (51%). Seventy four percent (208/282) of the neonates had lead levels of <2 μg/dL, while 17% had levels of 2 μg/dL. The median number of days from transfusion to testing was 314 days. Of the older transfused children (mean age 1.8 years), 67% (181/271) had lead levels of <2 ug/dL and 20% (55/271) had lead levels of 2 μg/dL. The median number of days between transfusion and testing was 163 days. The mean blood lead level of the transfused neonates was 1.4 μg/dL and the mean blood lead level of those older was 1.9 μg/dL. **Conclusion:** Despite reports of premature infants being exposed to elevated lead levels through blood transfusions, this review of transfused pediatric patients having blood lead level testing revealed that 89% had levels </= 2 μg/dl and 96% had levels </=5 μg/dL. This value is similar to the percentage of children aged 1-5 years with BLLs at or above the upper reference interval value of 5 μg/dL calculated using the 2007-2010 NHANES cycle which was 2.6%, and lower than those reported to the CDC in 2011 which was 5.8%. There is no evidence in our data to suggest a relationship between transfusion history and blood lead levels in neonatal and pediatric recipients. However, given the limitations of this retrospective review, additional research is warranted.

Disclosure of Commercial Conflict of Interest

T. Bannerman: Nothing to disclose; K. Nicol: Nothing to disclose; S. V. Rudmann: Nothing to disclose; D. Thornton: Nothing to disclose; S. Townsend: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Bannerman: Nothing to disclose; K. Nicol: Nothing to disclose; S. V. Rudmann: Nothing to disclose; D. Thornton: Nothing to disclose; S. Townsend: Nothing to disclose

SP179

Transfusion Requirements and Complications in Massively-Transfused Pediatric Patients Receiving Extracorporeal Life Support (ECLS)

A M Winkler[1] (awinkl2@emory.edu), M L Paden[2], C Josephson[1,3]. *[1]Department of Pathology and Laboratory Medicine, Emory University, Atlanta, GA, United States; [2]Department of Pediatrics, Division of Critical Care, Emory University, Atlanta, GA, United States; [3]Department of Pediatrics, Emory University, Atlanta, GA, United States*

Background/Case Studies: Extracorporeal life support (ECLS) is a technique that has been shown to be effective in supporting patients of all ages with life-threatening cardio-respiratory failure of varying etiologies and has become the standard therapy for neonatal, pediatric and adult patients with acute cardiac or respiratory failure that is unresponsive to traditional management. However, hematologic complications, including both hemorrhage and thrombosis, occur in 2-17% of ECLS patients and are associated with increased mortality. Moreover, if a patient develops a hemorrhagic complication, transfusion and factor replacement therapy are critical to the management and survival of the ECLS patient. As a result, transfusion requirements associated with ECLS are high and patients are commonly massively transfused. **Study Design/Methods:** A cohort study of all ECLS patients less than 18 years of age was conducted at a single institution from January 1, 2009–December 31, 2011 examined the association between massive transfusion (MT), defined as RBC transfusion of greater than one blood volume (<10 kg 80 ml/kg, >10 kg 70 ml/kg) plus circuit volume (<10 kg 450 ml, >10 kg 750 ml), and both mechanical and hemorrhagic complications. **Results/Findings:** A total of 165 patients received ECLS support, transfusion and factor replacement; data are summarized in Table 1. Subjects who were MT received transfusion of higher volumes of RBC, FFP, and PLT, although the ratio of RBC: FFP was greater than 1.0 for both groups. While MT was associated with an increased risk of all-cause mortality, there was no difference in reported mechanical or hemorrhagic complications. **Conclusion:** MT is independently associated with an increased risk for all-cause mortality in ECLS; however, mortality does not correlate with the occurrence of mechanical or hemorrhagic complications and the mechanism linking massive transfusion and mortality remains elusive. As a result, prospective research must continue to investigate the mechanism(s) by which transfusion contributes to morbidity and mortality in ECLS patients.

Disclosure of Commercial Conflict of Interest

C. Josephson: Nothing to disclose; M. L. Paden: Nothing to disclose; A. M. Winkler: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Josephson: Immucor/BioArray, Grants or Research Support, Octapharma, Consulting or Board of Director Fees; M. L. Paden: Nothing to disclose; A. M. Winkler: Nothing to disclose

	MT (n = 54)	Not-MT (n = 111)	p-value
First 24 Hours of ECLS			
RBC (ml/kg)	342.5 (178.6)	121.9 (69.6)	<0.0001
FFP (ml/kg)	55.0 (76.0)	10.3 (12.2)	<0.0001
PLT (ml/kg)	62.9 (34.3)	29.5 (46.4)	<0.0001
Cryo (luuls)	3.4 (9.7)	0.1 (0.8)	0.0003
rFVIIa (mcg/kg)	5.4 (24.1)	0.81 (0.85)	0.08
Duration of ECLS			
RBC (ml/kg)	305.5 (327.1)	150.6 (275.2)	0.002
FFP (ml/kg)	83.0 (147.9)	24.8 (51.7)	0.0003
PLT (ml/ kg)	235.2 (390.7)	163.9 (227.6)	0.14
Cryo (unils)	1.7 (3.0)	1.0 (3.5)	0.23
rFVIIt (mcg/kg)	10.6 (48.8)	1.63 (13.1)	0.07
Antifibrinolytic Use	17 (31.5%)	26 (23.4%)	0.269
Mcchanical Complications	8 (14.8%)	28 (25.2%)	0.129
Hemonhagic Complications	22 (40.7%)	48 (43.2%)	0.760

Platelet and Leukocyte Biology

SP180

Immature Platelet Counts May Predict which Cases of Suspected Heparin-Induced Thrombocytopenia (HIT) Yield a Positive anti-PF4 Immunoassay

J P Ha[1,2] (hajennifer@gmail.com), H Hong[1,2], R W Maitta[1,2]. *[1]Pathology, University Hospitals Case Medical Center, Cleveland, OH, United States; [2]Pathology, Case Western Reserve University School of Medicine, Cleveland, OH, United States*

Background/Case Studies: Heparin prophylaxis is established therapy for the prevention of venous thromboembolism. Type II heparin-induced thrombocytopenia (HIT) is an immune mediated hypercoagulable disorder mediated by antibodies to platelet factor 4-heparin complexes (anti-PF4) that can be elicited by heparin therapy. Testing for anti-PF4 antibodies is sensitive, has a high negative predictive value, and moderate specificity. The percent immature platelet fraction (%-IPF) has been suggested as useful in the diagnosis of idiopathic thrombocytopenia purpura and acute thrombo-

cytopenic purpura. In the present study, the utility of %-IPF and absolute immature platelet count (A-IPC) in the diagnosis of HIT was evaluated. **Study Design/Methods:** A case-control study was undertaken here following 14 thrombocytopenic patients (platelet counts < 1.5×10^{11}/L) tested for anti-PF4 by ELISA assay; 7 patients had a positive anti-PF4 test and 7 patients had a negative test. %-IPF and platelet counts on the same specimens or one obtained within 24 hours of HIT testing for each patient were measured using the Sysmex XE-5000 Hematology Analyzer (Sysmex Inc. Mundelein, IL). A-IPC was calculated by multiplying %-IPF by the patient's platelet count. Statistical analysis was performed using a two tailed student t-test with a $p < 0.05$ set for significance. **Results/Findings:** %-IPF for patients with anti-PF4 testing that was positive and negative were 9.43 ± 6.23% and 4.74 ± 4.01% respectively. Statistical analysis of the A-IPC results for patients with a positive anti-PF4 assay compared to the normal range showed no statistical difference ($8.19 \pm 2.64 \times 10^9$/L versus $7.12 \pm 3.22 \times 10^9$/L respectively; $p = 0.37$.) A similar comparison of the A-IPC of patients who had a negative anti-PF4 test to the normal range showed a significant difference with minimal variability among patients ($4.01 \pm 1.76 \times 10^9$/L versus $7.12 \pm 3.22 \times 10^9$/L respectively; $p = 0.024$). Comparison of the A-IPC of patients with a negative anti-PF-4 to those with a positive test showed a highly significant difference ($4.01 \pm 1.76 \times 10^9$/L versus $8.19 \pm 2.64 \times 10^9$/L respectively; $p = 0.0055$). **Conclusion:** Patients testing positive for anti-PF4 do not differ in A-IPC from the normal range. There was a significant decrease in A-IPC for patients with a negative anti-PF4 compared to the reference range and those testing positive for anti-PF4. The findings suggest that patients with negative anti-PF4 have thrombocytopenia due to destruction or hypoproduction of mature and immature platelets; whereas in HIT (positive anti-PF4), mature platelets are mainly targeted by pathology. A decrease of A-IPC may be used to avoid unnecessary HIT testing in patients and reserve testing for thrombocytopenic patients with a normal A-IPC.

Disclosure of Commercial Conflict of Interest

J. P. Ha: Nothing to disclose; H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. P. Ha: Nothing to disclose; H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose

SP181

Comparison of A-antigen Expression in Single Donor and Pooled Platelet Concentrates

L Cooling[1] (lcooling@med.umich.edu), A Hickey[1], R D Davenport[1].
[1]Pathology, University of Michigan, Ann Arbor, MI, United States

Background/Case Studies: ABO-incompatible platelet (PLT) transfusion can be associated with positive PLT crossmatches and PLT transfusion failures, especially in group O patients with high-titer ABO antibodies. Donor-specific differences in PLT ABO expression also contribute, with group A2 PLT having little/no A-expression and highly variable A-expression in A1 donors, including a high expressor phenotype in 5-7% donors (HXP, %A > 70-75%) . It was hypothesized that mean A-expression in group A prepooled PLT (PP-PLT) was equal or less than single donor platelets (SDP), and with less inter-unit variability. A-antigen expression in SDP and PP-PLT were compared, in parallel, with three anti-A reagents by 2-color flow cytometry. **Study Design/Methods:** A-antigen expression was examined in 150 in-date PLTs (11-group O, 139 A) including 60 SDP, 65 PP-PLT and 14 "virtual pool" PLT (VP-PLT) prepared from five SDP. PLT (5 × 10∧6) were stained with anti-CD61-APC, Helix pomatia-AF488 (HPA-488; 0.1 μg, 0.5 μg/mL), FITC-labeled monoclonal anti-A (mAb NaM87, 1 μg) and Dolichos biflorus (DBA-FITC, 10 μg/mL). Controls included RBC, group O PLT, IgG-APC, IgG-FITC, and Sambucus nigra lectin (SNA-FITC, 5 μg/ml). Results were reported as %A-positive PLT (%A+, × ± SD) by flow cytometry. Statistical analysis included 2-tail and paired t-test, chi square, coefficient of variation [CV], and correlation coefficient [R]. **Results/Findings:** As reported, SDP displayed a wide range in A-expression between donors (%A+ = 2-92%). DBA-FITC showed weak staining relative to other reagents (1-47%, Table 1; R = 0.33-57). There was a good correlation between HPA-488 and mAb NaM87 staining (R = 0.87); however, MAb NaM87 was overly sensitive with 50% SDP and 17% of PP-PLT typing as HXP (%A+ > 70%, p = 0.0001). HPA-488 (0.1 μg/ml) was most consistent with previous studies, with 5.2% SDP typing HXP (Table 1). In PP-PLT, the mean %A+ was equivalent to SDP but with less variability (CV = 30% vs 70%). A mean %A+ = 30-40% was observed in 55% of PP-PLT units but only 10% of SDP (p = 0.0008; HPA-488). The average %A+ in pooled PLT reflected the SDP constituents. In VP-PLT, the measured %A+ was within 5 ± 4% of predicted value (R = 0.97). Half of VP-PLT contained at least one A2-SDP, with a sharp drop in %A with VP-PLT containing > two A2-SDP (R = 0.70). **Conclusion:** PP-PLT had less inter-unit variability, with %A+ averaging 30-40% in 55% of units and no units showing an HXP-phenotype with HPA-488. For PLT screening, HPA-488 was superior to DBA and mAb NaM87 for identifying A2 and HXP donors/units.

Disclosure of Commercial Conflict of Interest

L. Cooling: Nothing to disclose; R. D. Davenport: Nothing to disclose; A. Hickey: No Answer

Disclosure of Grants Conflict of Interest

L. Cooling: Nothing to disclose; R. D. Davenport: Nothing to disclose; A. Hickey: No Answer

TABLE 1

	Percent SDP units		Percent Positive Platelets (X ± SD)				
Reagent	%A < 5%	%A > 70%	SDP (n-60)	PP-PLT (n = 65)	VP-PLT (n = 14)	Pool Total (n = 79)	P*
DBA-FITC	31%	0%	11 ± 9	10 ± 3	7 ± 3	9 ± 5	0.10
HPA-488 (0.1 ug)	26%	5.2%	36 ± 25	40 ± 12	41 ± 12	40 ± 12	0.38
HPA-488 (0.5 ug)	26%	15.8%	43 ± 8	48 ± 15	47 ± 17	48 ± 16	0.46
NaM87-FITC	10%	50%	57 ± 33	54 ± 16	59 ± 20	53 ± 17	0.48
SNA-FITC	—	—	>99	>99	>99	>99	1.0

* SDP vs Pool PLT, t-test

SP182

Defining an Appropriate Leucoreduction Strategy by Serial Assessment of Cytokine Levels in Platelet Concentrates Prepared by Different Methods

D K Bains[1] (drdaljit@outlook.com), R R Sharma[1], N Marwaha[1].
[1]Department of Transfusion Medicine, Post Graduate Institute of Medical and Educational Research, Chandigarh, India

Background/Case Studies: Different methods of platelet concentrate preparations leave behind certain number of residual leukocytes, accounting for most of the febrile non-hemolytic transfusion reactions, especially in multitransfused patients. Various inflammatory cytokines, such as Tumor Necrosis Factor-α (TNF-α), Interleukin-1β (IL-1β), Interleukin-6 (IL-6) are generated during storage and have been implicated for these adverse effects. **Study Design/Methods:** A total of 50 platelet concentrate(PC) units which met the national guidelines for quality criteria were included in the study, of which 20 were platelets prepared from Platelet rich plasma(PRP-PC), 20 Buffy Coat derived(BC-PC) and 10 were apheresis units. Five pools of platelet rich plasma (PRP-PC) and buffy-coat (BC-PC) each were prepared and divided into two halves. One half of the pool was leucofiltered using third generation leucofilter whereas the other half was stored as such. All the platelet units were stored in a horizontal platelet agitator and were assessed for residual leucocyte content (using nageotte chamber for leucoreduced PCs) and cytokine levels(IL-1β,IL-6 and TNF- α) with commercially available assays on day0, 3, and 5. **Results/Findings:** Comparative analysis of various cytokine levels in different platelet preparations (pre and post leucofiltration) was done . There was a statistically significant rise in cytokine levels (IL-1β, IL-6 and TNF-α) in non-leucofiltered(NLF) random donor platelet concentrates (PRP-PC and BC-PC) during storage (day 3 and 5) whereas leucofiltered(LF) random donor platelet concentrates (PRP-PC and BC-PC) and apheresis platelet concentrates did not show any significant rise in cytokine levels (on days 3 and 5) over the baseline values at day 0. Platelets with higher leucocyte content ($\times 10^8$ and $\times 10^7$) tend to have higher cytokine levels(IL-1β andIL-6)accumulation as compared to units with lower

WBC content ($\times 10^6$) on days 3 and 5 of storage. The cytokine levels had positive correlation with the residual number of leucocytes and the length of storage of the product. The platelet units with leucocyte content below ($\times 10^6$) consistently had minimal or no detectable cytokines during storage. Among all cytokines assessed, IL-6 levels rose maximum in different platelet preparations. **Conclusion:** This data suggests that although AP-PCs are superior to non-filtered PRP-PC and BC-PC in terms of in vitro quality control parameters and cytokine generation during storage, BC-PC mode of platelet preparation followed by prestorage leucofiltration is the best method to store platelets and minimise the cytokine accumulation and are best suited for transfusion in multitransfused hemato-oncologic patients.

Disclosure of Commercial Conflict of Interest

D. K. Bains: Nothing to disclose; N. Marwaha: No Answer; R. R. Sharma: No Answer

Disclosure of Grants Conflict of Interest

D. K. Bains: Nothing to disclose; N. Marwaha: No Answer; R. R. Sharma: No Answer

SP183

CD8+ CD25+ Regulatory T Cells: Unveiling New Mechanisms of Pathogenesis and Treatment of ITP

L Ma[1] (mali@smh.ca), J Li[1,4], E Simpson[1], L Baker[1,3], G Zhu[1], P Chen[1,2], J Freedman[1,4], H Ni[1,2]. *[1]Keenan Research Centre in the Li Ka Shing Knowledge Institute, St. Michael's Hospital, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada; [3]Department of Microbiology and Immunology, McGill University, Montréal, QC, Canada; [4]Department of Laboratory Medicine and Pathobiology, Departments of Medicine and Physiology, University of Toronto, Toronto, ON, Canada*

Background/Case Studies: Immune thrombocytopenia (ITP) is an autoimmune disease targeting one's own platelets primarily via platelet surface antigens Glycoprotein (GP) IIbIIIa (70-80%) and GPIb complex (20-40%). Currently, it is thought that ITP is predominantly an antibody mediated disorder; however, antibodies can be detected only in 50-70% of patients. Thus, the prevailing view is that T-lymphocytic cytoxic effects may be important in platelet destruction. Studies have demonstrated that CD8+ T-cells are capable of platelet killing. However, their contribution to platelet consumption in ITP is controversial. The disparity between the vastly larger platelet number and the minor CD8+ T-cell number reduces the significance of contact-dependent killing of platelets. Recently, preliminary data from our lab suggests a putative regulatory role for CD8+ T cells in ITP. Although CD8+ regulatory T cells have been well studied in other autoimmune diseases, their function in the context of ITP has never been examined. **Study Design/Methods:** Using an active murine model of ITP, wild-type (WT) Balb/C mice were transfused with splenocytes from WT platelet immunized β3–/– or GPIb–/– mice to induce thrombocytopenia. This encompasses both antibody and cell-mediated ITP. **Results/Findings:** Significantly increased CD8 + CD25 + Foxp3 + Treg percentages in the blood, thymus and spleen in immunized β3–/– mice were found. WT Balb/c mice engrafted with non-depleted β3–/– splenocytes had a higher platelet count than those engrafted with CD8+ T cell depleted splenocytes. Further, injection of CD8+ T-cells from immunized β3–/– mice into β3–/– splenocyte engrafted WT mice was able to rescue platelet counts whereas injection of CD8+ T cells from non-immunized mice could not. Interestingly, none of the CD8+ regulatory effects were seen in the anti-GPIb-active-model. Preliminary in vitro data suggests that CD8+ T-cells from immunized β3–/– mice significantly enhanced IL-10 secretion, inhibited lymphocyte proliferation, platelet apoptosis, and platelet associated IgG production. Lastly, it was found that CD8+ T cell depleted splenocyte engrafted mice were less responsive to dexamethasone treatment compared to non-depleted engrafted mice. **Conclusion:** It is believed that this is the first time a CD8+ regulatory role has been demonstrated in ITP and steroid treatment. Immunized β3–/– mice but not GPIb–/– had significant increases of CD8 + CD25 + Foxp3 + Treg which imparts a global protective role rather than cytotoxic in the entire CD8+ population. The data has not only important implications in understanding the immunopathogenesis between anti-GPIb and anti-GPIIbIIIa mediated ITP, but also in designing therapies which target those who are refractory to standard treatment (eg steroids).

Disclosure of Commercial Conflict of Interest

L. Baker: Nothing to disclose; P. Chen: Nothing to disclose; J. Freedman: Nothing to disclose; J. Li: Nothing to disclose; L. Ma: Nothing to disclose; H. Ni: Nothing to disclose; E. Simpson: Nothing to disclose; G. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Baker: Nothing to disclose; P. Chen: Nothing to disclose; J. Freedman: Nothing to disclose; J. Li: Nothing to disclose; L. Ma: Nothing to disclose; H. Ni: Nothing to disclose; E. Simpson: Nothing to disclose; G. Zhu: Nothing to disclose

SP184

Full Length Nucleotide Sequence of *SLC44A2* Alleles Encoding Human Neutrophil Antigen-3 (HNA-3) in African Americans and Caucasians

Q Chen[1,2] (qngchen@gmail.com), S C Ardinski[1], K Lam[1], M J Huvard[1], P Schmid[1], W A Flegel[1]. *[1]Department Transfusion Medicine, NIH Clinical Center, Bethesda, MD, United States; [2]Jiangsu Province Blood Center, Nanjing, China*

Background/Case Studies: Human neutrophil antigen-3 (HNA-3) alloantibodies are a major cause of severe transfusion-related acute lung injury (TRALI) and are involved in neonatal alloimmune neutropenia (NAIN). The frequencies of single nucleotide polymorphisms (SNPs) indicative of clinically relevant *SLC44A2* alleles are known for Caucasian, Asian and African American populations. This study screened for other SNPs in all 22 exons encoding HNA-3. **Study Design/Methods:** Genomic DNA from 245 African Americans, 103 Caucasians, 4 Asians, and 2 Hispanics were collected. A method was devised and applied to determine the full length coding and adjacent intron sequences for all 22 *SLC44A2* exons from genomic DNA by using 8 PCR amplifications only. **Results/Findings:** Eight SNPs were detected in the *SLC44A2* coding sequence, of which 1 was novel. 27 SNPs were detected in the *SLC44A2* non-coding sequence (1 in 5′UTR; 20 in the introns; and 6 in the 3′UTR), of which 5 were novel. **Conclusion:** A total of 35 SNPs, including 6 novel SNPs, were found in a random survey of the *SLC44A2* gene covering all coding and some adjacent nucleotide sequences. Population frequencies of all SNPs were established. The presence of null alleles was excluded corroborating previously published HNA-3 variant frequencies in African Americans and Caucasians.

Disclosure of Commercial Conflict of Interest

S. C. Ardinski: Nothing to disclose; Q. Chen: Nothing to disclose; W. A. Flegel: RHD Genotyping Patents, Other; M. J. Huvard: Nothing to disclose; K. Lam: Nothing to disclose; P. Schmid: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. C. Ardinski: Nothing to disclose; Q. Chen: Nothing to disclose; W. A. Flegel: Nothing to disclose; M. J. Huvard: Nothing to disclose; K. Lam: Nothing to disclose; P. Schmid: Bayer Schweiz AG, Grants or Research Support,Consulting or Board of Director Fees, CSL Behring, Grants or Research Support, Shire Human Genetic Therapies, Grants or Research Support

SP185

The EpHLA System for Virtual Platelet Crossmatching: A New Tool for Platelet Transfusion Efficiency

C B Bub[1,3] (carolinabub@gmail.com), B M Martinelli[1], T M Avelino[1], A C Gonçalez[1], H L Sales-Filho[2], M A Torres[3], M L Barjas-Castro[1], S J do Monte[2], V Castro[1]. *[1]Hematology and Hemotherapy Center-Instituto Nacional de Tecnologia do Sangue, State University of Campinas-UNICAMP, Campinas, Brazil; [2]Immunogenetics and Molecular Biology Laboratory, Federal Unversity of Piaui, Teresina, Brazil; [3]Histocompatibility Laboratory, Hospital Israelita Albert Einstein, São Paulo, Brazil*

Background/Case Studies: Immune platelet refractoriness (iPR) is mainly due to HLA alloimmunization. Transfusion to support iPR patients depends on compatible platelets (HLA fullmatch or partial mismatch -PM- without a corresponding mismatch antibody). HLA system is highly polymorphic leading to the frequent use of PM platelets. Donor selection process is complex and the availability of tools that could make it easier is desirable. Indeed, the HLAMatchMaker (HLAMM) system, which uses different "triplets" or "eplets" encoding HLA antigens has great utility for this purpose, but is still is a laborious tool for donor selection. Recently a new system—automated EpHLA system (EpHLA)- has been developed which enhances the automatization of HLA matched donor recognition, however the efficiency of this system in iPR support has not yet been evaluated. Thus, the objective of the present study was to evaluate the efficacy of EpHLA in iPR support. **Study Design/Methods:** A group of HLA antibodies positive iPR hematological patients followed in a reference hospital during the period of

2008-2012 were enrolled in the study. HLA antibodies were detected using single antigen Luminex (MFI cutoff: 500). HLA compatible platelet transfusions for iRP patients selected by HLAMM triplets were reanalyzed using EpHLA (virtual crossmatch—V × M). This analysis was compared to flow cytometry crossmatch (FC × M) routinely performed for those patients. The concordance between V × M and FC × M was evaluated, and platelet transfusion response (efficiency = CCI ≥ 5000) analyzed when possible. **Results/Findings:** Forty three plateletpheresis selected by HLAMM and FC × M transfused for 14 patients were evaluated. In 19 of these transfusions, efficiency could be evaluated by CCI. V × M presented a negative predictive value (NPV) of 100% (35/35) and a specificity of 94.6% (35/37) compared to FC × M. The sensibility of the method was 100% (6/6) with a positive predictive value (PPV) of 75% (6/8)—p < 0.001. The median number of distinct eplets in transfusions was 4 (range: 0-24). In the CCI subgroup, CCI median was 12.500 (range: −10.000 to 24.000) and number of distinct eplets in transfusions evaluated was 4 (range: 0-16). **Conclusion:** The results obtained in the present study showed that virtual platelet xmatch using EpHLA presented high NPV, easy software management and automation of the donor selection, leading to a high performance in iPR patients' clinical support. The high PPV as well as a good CCI median reinforce the EpHLA system as a useful tool to crossmatch platelets for transfusion support for patients with refractoriness. Nevertheless, FC × M remains an important method to identify any possible new alloimmunization against the remaining mismatched antigens (HLA or HPA).

Disclosure of Commercial Conflict of Interest

T. M. Avelino: Nothing to disclose; M. L. Barjas-Castro: Nothing to disclose; C. B. Bub: Nothing to disclose; V. Castro: Nothing to disclose; S. J. do Monte: Nothing to disclose; A. C. Gonçalez: No Answer; B. M. Martinelli: Nothing to disclose; H. L. Sales-Filho: Nothing to disclose; M. A. Torres: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. M. Avelino: Nothing to disclose; M. L. Barjas-Castro: Nothing to disclose; C. B. Bub: Nothing to disclose; V. Castro: Nothing to disclose; S. J. do Monte: Shire Inc., Grants or Research Support; A. C. Gonçalez: No Answer; B. M. Martinelli: Nothing to disclose; H. L. Sales-Filho: Nothing to disclose; M. A. Torres: Nothing to disclose

V × M and FC × M data obtained in the study

N	FC × M positive	FC × M negative
V × M positive	6	2
V × M negative	0	35

P < 0.001

SP186

Laboratory Evaluation and Management of Platelet Refractoriness, Using Both Crossmatching and HLA Matching

G Ramsey[1,2] (g-ramsey@northwestern.edu), R D Sumugod[2], P F Lindholm[1,2]. *[1]Pathology, Northwestern University, Chicago, IL, United States; [2]Blood Bank, Northwestern Memorial Hospital, Chicago, IL, United States*

Background/Case Studies: Despite widely used leukoreduction, platelet refractoriness (PR) remains an important issue. **Study Design/Methods:** Suspected PR cases were tested with 1) solid-phase red cell adherence (SPRCA) test for platelet antibodies (Capture-P®, Immucor, Norcross, GA) and 2) HLA Class I lymphocytotoxicity (LCT) screen, reported as Panel Reactive Antibody (PRA) before and after (IgG) serum treatment with dithiothreitol (DTT). PRAs which were <20% or mainly IgM (<20% after DTT) were considered negative. When the SPRCA screen was positive, crossmatch-compatible (XM) platelets were sought initially (Capture-P®). HLA-matched (HLA) platelets were sought in LCT+ patients when either the SPRCA screen was negative, the XM testing was broadly incompatible, or XM platelets were ineffective. Selected patients had platelet-specific antibody studies. One year of PR testing and the pre- and post-transfusion platelet counts for XM and HLA platelets was retrospectively reviewed. Transfusion responses for XM or HLA platelets were compared for achieving therapeutic (rising to ≥40/nL) or prophylactic (rising to ≥10/nL) levels. The median post-transfusion platelet count interval was 3 hours (range 0.5-24 hours, interquartiles 2 and 6 hours). Fisher's exact test was used for statistical frequency comparisons. **Results/Findings:** In 2012, 50 patients had 52 completed PR studies; 22 studies (42%) were negative in both tests, 9 (17%) were SPRCA+, 7 (13%) were HLA LCT+, and 14 (27%) were antibody-positive (Ab+) in both tests. Three cases (10% of 30 Ab+ sera) were found to have anti-HPA-1b (anti-PL^{A2}). Twenty patients received evaluable special platelets: 7 XM, 6 HLA, and 7 both. The Table shows the best responses in each patient for HLA platelets, XM platelets, and overall. HLA platelets tended to give better responses than XM platelets, but the distribution was not statistically different. In the seven patients receiving both XM and HLA platelets, four achieved a higher best-response category with HLA, one with XM, and two the same with both. **Conclusion:** Using both platelet SPRCA testing and HLA LCT, 58% of studies for suspected PR were Ab+. The SPRCA test was negative in 33% (7/21) of sera with HLA antibodies. However, 27% (9/30) of Ab+ sera were reactive only in SPRCA, suggesting that it adds benefit. HLA platelets tended to give better responses than XM platelets, but XM platelets can be available faster and are the main option in SPRCA+, LCT- patients. When either XM or HLA platelets were given based on dual testing, 80% (16/20) of immune-refractory patients had therapeutic or prophylactic responses to selected platelets.

Disclosure of Commercial Conflict of Interest

P. F. Lindholm: Nothing to disclose; G. Ramsey: Nothing to disclose; R. D. Sumugod: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. F. Lindholm: Nothing to disclose; G. Ramsey: Nothing to disclose; R. D. Sumugod: Nothing to disclose

Best Responses	Rose to ≥40/nL	Rose to ≥10/nL	Stayed 10-39/nL	Stayed <10/nL
XM (14 patients)	3	6	0	5
HLA (13 patients)	7	4	2	0
Overall (20 patients)	9	7	0	4

SP187

HLA Alloimmunization is Not Always Predictive of Platelet Transfusion Refractoriness

W S Crews[1] (wscrews@gmail.com), J K Kay[1], J H Herman[1]. *[1]Transfusion Medicine, Thomas Jefferson University Hospital, Philadelphia, PA, United States*

Background/Case Studies: HLA Class I alloimmunization (ALLO) defined by lymphocytotoxic antibodies (LCT) is known to cause platelet transfusion (TX) refractoriness (REFR). LCT present in a high degree, defined by percent reactive antibody (PRA), usually causes rapid failure of unmatched TX, especially pooled platelets (PPL). ALLO methodology has largely migrated away from LCT to a platform using fluorescence and single antigen beads (BEAD) in an effort to improve antibody identification. It has been noticed that BEAD testing seems overly sensitive and has poor correlation with TX REFR. It is now recommended to wait to observe REFR before ordering matched platelets because of this, regardless of PRA. A retrospective review of patients (PATIENTS) with ALLO by BEAD was performed to see whether it was as predictive of TX REFR as LCT had been. **Study Design/Methods:** Forty PATIENTS with BEAD PRA >20% who received >2 TX and had informative pre- and post-TX platelet counts were retrospectively identified dating to the point at which preemptive matching had been discontinued. Analysis was confined only to PATIENTS who had PPL TX to avoid inadvertent matched apheresis platelets (SDP). Twenty-six of the 40 fulfilled this criterion. CCI was calculated for intervals of 0-4, 4-24, and >24 hours post-TX. In the absence of the actual dose of the TX, the median content obtained from our supplier (3.6 × 10e11) was used. PPL have less variability of actual content from the median compared to SDP. Clinical risk factors for REFR were not assessed. **Results/Findings:** Fourteen of the 26 PATIENTS had no CCI >4500, and were suspected of having immune REFR. The CCI results for the remaining 12 are in the Table. A surprisingly high number of PATIENTS failed to demonstrate REFR, with poor correlation to PRA by BEAD. This study was unable to assess whether confounding variables such as splenomegaly, consumptive coagulopathy, auto or drug antibodies and/or sepsis may have contributed to the CCI results. However, this would skew the results in favor of suspected immune REFR. **Conclusion:** Despite having BEAD PRA, many PATIENTS may be able to continue receiving unmatched PPL and avoid the logistic difficulties and expense of matched TX. A better method to predict immune REFR would be helpful, in order to target the resources needed for matched TX.

Disclosure of Commercial Conflict of Interest

W. S. Crews: Nothing to disclose; J. H. Herman: Nothing to disclose; J. K. Kay: No Answer

Disclosure of Grants Conflict of Interest

W. S. Crews: Nothing to disclose; J. H. Herman: Nothing to disclose; J. K. Kay: No Answer

CCI Results

PT#	PRA %	CCI 0-4 (#TX)	CCI 4-24 (#TX)	CCI >24 (#TX)
1	94	17.4 (2)	10.3 (6)	
2	49	21.5 (5)	6.4 (1)	
3	80	6.6 (9)	9.4 (19)	
4	26	17.4 (3)	5.1 (3)	
5	36	2.2 (1)	6.5 (2)	
6	43	19.2 (2)	15.2 (8)	1.7 (1)
7	80		6.2 (3)	
8	57	7.5 (1)	5.7 (3)	
9	48	5.1 (1)	3.4 (4)	
10	26	6.1 (8)	1.2 (18)	
11	26	5.2 (3)	3.4 (5)	
12	86	43.1 (3)		

SP188

Silica Beads are Superior to Polystyrene for Sizing Cellular Microparticles

B K Parida[1] (bijaya.k.parida.vol@mail.mil), H Garrastazu[1], A P Cap[1], S J McFaul[1]. *[1]Blood Research, US Army Institute of Surgical Research, San Antonio, TX, United States*

Background/Case Studies: Cellular microparticles (MP) are submicron vesicles found in biological fluids. They are studied for their potential role in pathological conditions. Flow cytometry (FC) is used to quantify MP, and to determine MP cellular origin and expression of specific molecules on their membranes. Often, 1 μm polystyrene (PoSt) beads are used to distinguish MP from cells by setting a MP gate in a forward scatter (FSC) versus side scatter (SSC) plot. PoSt beads have a higher refractive index than cellular material and exhibit higher FSC and SSC than cells and MP of equal size. Consequently, some platelets are included within the MP gate (<1 μm), which incorrectly increases the reported percentage of platelet-derived MP. Silica (Si) beads have a lower refractive index than the PoSt beads so their light scattering is closer to that of cellular vesicles. In this study the use of Si beads was demonstrated as a particle standard to enumerate and size cellular MP by FC. **Study Design/Methods:** Blood was collected from two donors into citrated tubes according to an IRB-approved protocol. Platelet rich plasma (PRP) was prepared by centrifugation at 200 × g for 10 minutes, and platelet poor plasma (PPP) was prepared by centrifugation at 3,000 × g for 10 minutes followed by a second centrifugation at 3,000 × g for 10 minutes. Platelets and MP in PRP and PPP, respectively, were sized using a Canto II (BD Biosciences) flow cytometer calibrated using PoSt and Si beads (201-2000 nm), and compared to sizes determined by electrozone (displacement of electrolyte) analyses using Elzone II (platelet sizes) and qNano (MP sizes) particle analyzers. The Canto II was equipped with a FSC PMT detector that improved resolution of small particles. **Results/Findings:** Log(FSC) increased linearly as a function of size of PoSt and Si beads (R2 = 0.98 for both). FSC values were 1.4-14 times higher for PoSt beads than for Si beads between 200-1000 nm. As a result, calculated sizes of platelets and MP were smaller using PoSt bead standards (Table I) and deviated greatly from sizes determined by the electrozone and qNano methods (50% deviation for platelets and 35% for MP). Setting a MP gate at 1 μm with PoSt beads yielded 10.5% and 18.5% platelet contamination within the MP gate in two PRP samples, but no platelet contamination when the 1 μm gate was set with Si beads. **Conclusion:** Calibration of FSC using Si beads of different sizes enables determination of MP sizes, and accurate establishment of MP gates that minimizes platelet contamination of MP populations.

Disclosure of Commercial Conflict of Interest

A. P. Cap: No Answer; H. Garrastazu: Nothing to disclose; S. J. McFaul: Nothing to disclose; B. K. Parida: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. P. Cap: No Answer; H. Garrastazu: Nothing to disclose; S. J. McFaul: Nothing to disclose; B. K. Parida: Nothing to disclose

Mean Sizes (nm) of Platelets and MP

	Flow Cytometry		Electrozone	
Sample	Polystryrene	Silica	Elzone	qNano
Platelets	1135	1875	2334	
MP (200-500 nm)	209	304		320
MP (501-1000 nm)	391	641		599

SP189

Genotype Frequency of Human Neutrophil Antigen-3 (HNA-3) Polymorphism in the Han Population of Jiangsu Province, China

Q Chen[1] (qngchen@gmail.com), J Xiao[1], Z Liu[2], W A Flegel[3]. *[1]Jiangsu Province Blood Center, Nanjing, China; [2]Institute of Blood Transfusion, Chinese Academy of Medical Science and Peking Union Medical College, Chengdu, China; [3]Department Transfusion Medicine, NIH Clinical Center, Bethesda, MD, United States*

Background/Case Studies: Human neutrophil antigen-3 (HNA-3) alloantibodies are a major cause of severe transfusion-related acute lung injury (TRALI) and are involved in neonatal alloimmune neutropenia (NAIN). The frequencies of single nucleotide polymorphisms (SNPs) indicative of *SLC44A2* alleles encoding HNA-3 vary among different populations. The frequencies of *SLC44A2* alleles in the Jiangsu province were unknown. **Study Design/Methods:** Genomic DNA was extracted from 163 Han Chinese blood donors in the Jiangsu province. To detect SNPs indicative of clinically relevant HNA-3 variants, previously established methods were applied for PCR-SSP and nucleotide sequencing of exon 5 to exon 7 and adjacent introns. **Results/Findings:** 73 and 20 donors were homozygous for the known alleles encoding the HNA-3a and HNA-3b variants, respectively, while 70 were heterozygous. The frequency of the HNA-3a allele was 0.66 and HNA-3b 0.34. Nucleotide sequencing confirmed PCR-SSP results; no C457T mutation was detected. **Conclusion:** In this random survey of Han Chinese population in the Jiangsu province, the *SLC44A2*(R154Q) allele frequency was shown to be similar to previous studies in Shanghai and Nanning, but higher than in Guangzhou. The molecular screening identified blood donors at risk of harboring anti-HNA-3a. Although TRALI is rarely reported in China, the incidence of developing anti-HNA-3a may be higher than in Caucasian and African American populations.

Disclosure of Commercial Conflict of Interest

Q. Chen: Nothing to disclose; W. A. Flegel: RHD Genotyping Patents, Other; Z. Liu: Nothing to disclose; J. Xiao: Nothing to disclose

Disclosure of Grants Conflict of Interest

Q. Chen: Nothing to disclose; W. A. Flegel: Nothing to disclose; Z. Liu: Nothing to disclose; J. Xiao: Nothing to disclose

SP190

Allele Frequencies of HPA-1a, HPA-1b, HPA-2a, HPA-2b, HPA-15a, and HPA-15b in Brazilian Blood Donors of Different Ethnicities

S L Bonifácio[1] (bonifaciosilvia@ig.com.br), A Mendrone[1], C L Dinardo[1]. *[1]Immunohematology, Fundação Pró-Sangue, São Paulo, Brazil*

Background/Case Studies: HPA 1, 2, and 15 are platelet specific glycoproteins eventually involved in cases of immune platelet refractoriness and neonatal immune thrombocytopenia. Frequencies of different types of HPA vary amongst ethnically diverse populations, especially between Amerindians and Caucasians. Knowing the Brazilian population, it was thought that HPA alleles frequencies may facilitate the provision of HPA-matched platelet units to alloimmunized patients. Considering the extreme ethnic admixture of Brazilian population, it is not known if the frequency of HPA alleles still varies according to self-declared ancestry and if selecting platelet apheresis donors based on this criterion is worth when searching for a specific phenotype. The purpose of this study was to determine HPA-1a; HPA-1b; HPA-2a; HPA-2b; HPA-15a, and HPA-15b allele frequencies among Brazilian blood donors of diverse self-declared ethnic groups. **Study Design/Methods:** 175 blood donors were randomly recruited by a non-therapeutic apheresis team in a one-year-time period. DNA was extracted from 100 peripheral blood samples using a commercial DNA extraction kit (QIAamp DNA, QIAGEN, Germany). Detection of HPA-1a/1b; HPA-2a/2b and HPA-15a/15b polymorphisms was performed using specific primers by ASP-PCR. **Results/Findings:** Prevalence of HPA-1a allele was 0.86 in Brazilian self-declared Caucasians and 0.819 in African descendants (p = 0.86), whereas prevalence of HPA-1b was 0.14 and 0.181, respectively (p = 0.52). The frequencies of HPA-1b were higher than previously described for Brazilians (0.075-Caucasians and 0.097-African descendants, Castro 1999) and were similar to those presented by European population (0.169, Randem 2003). Prevalence found for HPA-2a was 0.79 in Caucasians and 0.76 in African descendants (p = 0.81) and for HPA-2b was 0.21 and 0.24, respectively (p = 0.5716). The prevalence of the HPA-15a allele was 0.53 in Caucasians and 0.51 in African descendants (p = 0.9121) and HPA-15b was 0.47 and 0.49, respectively (p = 0.988). Those frequencies (HPA-2 and HPA-15) were similar to those presented by earlier studies held in different countries. **Conclusion:** Allelic frequencies of HPA-1a, HPA-1b, HPA-2a, HPA-2b, HPA-15a, and HPA-15b did not differ between self-declared Caucasians and

African descendants in a sample of Brazilian blood donors (São Paulo). A relatively higher frequency of HPA-1b allele was found in our sample than in previous Brazilian studies, probably due to regional differences related to the immigration process. Considering the admixed population, selecting donors based on self-declared ancestry may not be a worthwhile strategy for finding one specific HPA phenotype.

Disclosure of Commercial Conflict of Interest

S. L. Bonifácio: No Answer; C. L. Dinardo: Nothing to disclose; A. Mendrone: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. L. Bonifácio: No Answer; C. L. Dinardo: Nothing to disclose; A. Mendrone: Nothing to disclose

SP191

Low Level HLA Antibodies Do Not Correlate with Failed Platelet Transfusion

M B Pagano[1] (mpagano3@jhmi.edu), R Vega[3], A K Fuller[1], K E King[1], M S Leffell[2]. *[1]Pathology, Johns Hopkins Medical Institutions, Baltimore, MD, United States; [2]Medicine, Johns Hopkins Medical Institutions, Baltimore, MD, United States; [3]Immunogenetics, Johns Hopkins Medical Institutions, Baltimore, MD, United States*

Background/Case Studies: HLA antibody (HLA-ab) detection methods have evolved with the development of more sensitive assays, such as flow cytometry (FC) and single antigen (SA) beads. Although the presence of donor specific HLA-abs in the recipient is known to result in decreased platelet increment, it is not known if differences in antibody levels have different impact on platelet increment. The purpose of this study is to determine if low level HLA-abs correlate with platelet transfusion failures. **Study Design/Methods:** Retrospective analysis of platelet transfusions in 13 chronically transfused patients with HLA-abs was performed. Apheresis, leukoreduced, irradiated platelets of known HLA type were transfused. Recipient HLA-abs were analyzed on multianalyte bead assays (Lifecodes Immuncor; One Lambda, Inc) and were classified based on their relative strength in three categories, Complement Dependent Cytotoxicity (CDC), FC, and Low level (Table). Risk factors for platelet refractoriness were analyzed, including strength of HLA-abs, fever (T ≥ 38.3C), splenomegaly, patient/platelet ABO mismatch, and platelet age. The corrected count increment (CCI) was calculated as (PLT count post—PLT count pre)*BSA/number of equivalent units of platelets (5.5 × 10e10) transfused, and was assessed between 1 and 11 hours post transfusion. Transfusions with a CCI < 5000 at 1 hour, or <4500 at 5 hours, were considered failed transfusions. **Results/Findings:** A total of 13 patients with 238 platelets transfusions were analyzed. Of the 238 platelet transfusions, 147 (62%) were HLA compatible and 91 (38%) were HLA incompatible. Failed transfusions were observed in 65 (71%) incompatible transfusions, and 64 (44%) compatible transfusions. The majority of incompatible and failed transfusions had CDC (85%) or FC (84%) HLA-ab levels, and approximately half (53%) had Low antibody levels. Independent of the antibody levels, 100% of patients with two or more refractory risk factors other than donor specific HLA-abs had failed platelet transfusions. A logistic regression showed that failed platelet transfusions were associated with CDC levels (p = 0.004), FC (p = 0.026) and Low levels (p = 0.782) of donor specific antibodies, ABO patient/platelet mismatch (p = 0.000), fever (p = 0.020), splenomegaly (p = 0.468), and older platelets (p = 0.012). **Conclusion:** Low level donor specific HLA-ab did not correlate with failed transfusions. From the laboratory perspective, reporting low level donor specific antibodies could exclude patients from receiving platelet units that could result in successful increments. MBP and RV contributed equally to this project.

Disclosure of Commercial Conflict of Interest

A. K. Fuller: Nothing to disclose; K. E. King: Nothing to disclose; M. S. Leffell: Nothing to disclose; M. B. Pagano: Nothing to disclose; R. Vega: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. K. Fuller: Nothing to disclose; K. E. King: Terumo BCT, Grants or Research Support; M. S. Leffell: Nothing to disclose; M. B. Pagano: Nothing to disclose; R. Vega: Nothing to disclose

Classification of HLA- Ab Level

	Luminex Phenotype (Mean Fluorescent Intensity, MFI)	Luminex SA (MFI)
CDC	>10,000	>17,000
Flow	2,500-9,999	7,000-16,999
Low Level	>500-2,400	>1,000-6,999

SP192

Transfusion Related Acute Lung Infection (TRALI) Caused by Transfusion of Red Blood Cells Containing HNA-1b and HLA Leukocyte Antibodies

B R Curtis[1,2] (brian.curtis@bcw.edu), W Bottner[4], S Grady[1], J G McFarland[1,3], J Gottschall[3,2]. *[1]Platelet & Neutrophil Immunology Lab, BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Blood Research Institute, BloodCenter of Wisconsin, Milwaukee, WI, United States; [3]Medical Sciences Institute, BloodCenter of Wisconsin, Milwaukee, WI, United States; [4]Department of Transfusion Medicine, Gunderson Lutheran Medical Center, La Crosse, WI, United States*

Background/Case Studies: Transfusion Related Acute Lung Infection (TRALI) continues to be the leading cause of transfusion-related death despite institution of several risk reduction measures. These measures have significantly lowered the frequency of leukocyte antibody mediated events involving high volume plasma products, but non-antibody TRALI and TRALI associated with red blood cells (RBCs) remains a significant risk. A case of TRALI was reported associated with transfusion of packed RBCs (prbcs) containing neutrophil antibodies specific for HNA-1b and class I HLA-A11. An 89-year-old male being treated for anemia secondary to traumatic hematoma of the thigh received transfusion of two leuko-reduced pRBCs from two different blood donors. One to two hours after transfusion of the second prbc (4 hours after the first), he developed severe dyspnea and tachycardia. BP increased from 87/51 to 121/68, pulse 85 to 115, respirations 20 to 34. Chest x-ray showed bilateral pulmonary infiltrates, and SpO2 was 87%. The patient was transferred to the ICU where he was stabilized and recovered. **Study Design/Methods:** Serum tests for class I and II HLA antibodies were performed using mixed and single antigen Luminex bead assays. Neutrophil antibody testing was performed by flow cytometry and monoclonal immobilzation of granulocyte anitgens (MAIGA) assay. HNA genotyping was by allele-specific PCR, and HLA typing by low resolution sequencing. **Results/Findings:** Donors of the associated blood products were both female, but only the donor of the first unit was available for additional lab studies. She was G4P0 and had previously tested positive for HLA antibodies allowing only her pRBCs to be used for transfusion. Serum testing showed class I HLA-A11, multi-specific class II HLA-DR, and strong IgG HNA-1b neutrophil antibodies. The donor typed HNA-1b negative, and the patient typed HNA-1b/1b and HLA-A11 but negative for all HLA-DR cognate antigens. **Conclusion:** The patient's symptomology, timing of the reaction with respect to start of transfusion, and laboratory confirmation of both HLA and HNA specific antibodies in the donor's serum matching cognate antigens on the patient's leukocytes is highly suggestive of TRALI. This would see to be only the second report of TRALI due to HNA-1b, although it is not certain if the HLA-A11, HNA-1b or both antibodies were responsible. A look back study for this donor is planned. It is also interesting that the implicated blood product was a pRBC, which contained < 50 ml of plasma, and the donor was known to have HLA antibodies in her serum from previous donor screening. This case demonstrates that despite recent risk reduction measures, 1) transfusion of plasma from mostly males, 2) donor HLA antibody screening, and 3) reduction in overall blood use, antibody mediated TRALI can still occur, even with pRBCs, and indicates that routine screening of blood donors for HNA antibodies should perhaps be considered.

Disclosure of Commercial Conflict of Interest

W. Bottner: No Answer; B. R. Curtis: Nothing to disclose; J. Gottschall: No Answer; S. Grady: Nothing to disclose; J. G. McFarland: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Bottner: No Answer; B. R. Curtis: Nothing to disclose; J. Gottschall: No Answer; S. Grady: Nothing to disclose; J. G. McFarland: Nothing to disclose

SP193

Establishment of Platelet Donor Registry Improves the Treatment of Platelet Transfusion Refractoriness in Shanghai

P Lu[1] (luping@sbc.org.cn), Z Lu[2], B Ling[1], R Li[1]. *[1]Immunohematology Platelet Lab, Shanghai Blood Center, Shanghai, China; [2]Department of Blood Transfusion, Tongji Hospital Affiliated to Shanghai Tongji University, Shanghai, China*

Background/Case Studies: Transfusion platelet matches with antigenic similarity would evoke less allorecognition and immune activation . Strategies have been based on the theory that selection for HLA-A and HLA-B cross-reactive groups (CREGs), compatible donor as well as ABO/HPA-matched donor, will predict good increments in platelet corrected count after platelet transfusion. **Study Design/Methods:** To establish platelet donor registry in Shanghai, 1729 repeat donors were typed for HLA-A, -B and

HPA-1,-2, -3, -4, -5, -6, and -15 using the standard PCR-SSP method. Fifteen patients were studied with hematologic or oncologic diseases with refractoriness to platelet transfusions from random donors who are receiving 31 units of apheresis platelet products transfusion. **Results/Findings:** Fifteen patients (7 males, 8 females) were studied showing platelet refractoriness to random donor platelets (1 hour corrected count increment [CCI] <7500/ml/m2, % of platelet recovery[PPR] <20%) before. Patients were phenotyped for both HLA-A, -B and HPA-1,-2, -3, -4, -5, -6, and -15. Apheresis platelets from donor registry in Shanghai matched to patients ABO, HPA and HLA-A and HLA-B cross-reactive groups (CREGs) are transfused. Nine patients (9/15, 60%) show 1 hour CCI > 10000; 12 patients (12/15, 80%) show 24 hours CCI > 4500; 10 patients (10/15, 66.7%) show 1 hour PPR > 30%; eight patients (8/15, 53.3%) show 1 hour PPR > 60%; and 10 patients (10/15, 66.7%) show 24 hours PPR > 20%. The mean 1 hour, 24 hr CCI and PPR values from the best donors were significantly higher than those from random donors they transfused before. **Conclusion:** The use of HLA-A,-B, and HPA, ABO compatible platelets improves posttransfusion 1 hour, 24 hours CCI values and percentage of platelet recovery in refractory patients. Transfusion with HLA-A,-B and HPA, ABO-matched platelets is mandatory to reduce the risk of bleeding in PTR patients. Refractoriness to platelet transfusions developed in at least 50% of the patients observed. Establish large-sized platelet donor registry with HLA class I, HPA, ABO-typed may be needed to circumvent platelet-specific antibodies of unknown specificity in all chronically transfused patients.

Disclosure of Commercial Conflict of Interest

R. Li: Nothing to disclose; B. Ling: Nothing to disclose; P. Lu: Nothing to disclose; Z. Lu: No Answer

Disclosure of Grants Conflict of Interest

R. Li: Nothing to disclose; B. Ling: Nothing to disclose; P. Lu: Nothing to disclose; Z. Lu: No Answer

SP194

The Allele Frequencies of Human Platelet Antigen (HPA)-21w in Chinese

Y Nie[1] (manieyongmei@gmail.com), J Ma[2]. *[1]Department of QM, Guangzhou Blood Center, Guangzhou, China; [2]Division of Gastrointestinal & Pancreatic Surgery, The First Affiliated Hospital of Sun Yat-sen University, Guangzhou, China*

Background/Case Studies: Human platelet antigen (HPA)-21bw was newly reported to be associated with severe neonatal alloimmune thrombocytopenia (NAIT). HPA-21bw was detected to be a low-frequency human platelet antigen in some populations. However, the allele frequencies of HPA-21w are still unknown in Chinese population. In this study it was sought to investigate the allele frequencies of HPA-21w by employing polymerase chain reaction-sequence specific primers (PCR-SSP) method in the Chinese population. **Study Design/Methods:** A total of 820 unrelated apheresis platelet donors in Guangzhou were involved in this study. The oligonucleotide primer sequences for HPA-21w were designed according the DNA sequences from genebank. A pair of internal positive control primers 5′- TGC CTT CCC AAC CAT TCC CTT A-3′ and 5′- CCA CTC ACG GAT TTC TGT TGT GTT TC-3′ were included in PCR to amplify a product from human growth hormone gene. The samples resulted from PCR-SSP with HPA-21w allele were verified by DNA sequencing. **Results/Findings:** The sequence specific primers had their specific 204bp-fragment amplification products. A 434bp-PCR product from human growth hormone gene was also amplified as the internal control. The high coincidence between the allele frequencies for HPA-21bw from PCR-SSP and DNA sequencing was observed in the study. The allele frequencies of HPA-21a and HPA-21bw were 0.9945 (1631/1640) and 0.0055 (9/1640) respectively. **Conclusion:** The study indicated that HPA-21bw occurs at a low but appreciable frequency in Chinese population. More attention should be paid to diagnosis and treatment of severe NAIT.

Disclosure of Commercial Conflict of Interest

J. Ma: Nothing to disclose; Y. Nie: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Ma: Nothing to disclose; Y. Nie: Nothing to disclose

Table 1.

	Pre-Transfusion			Pre-IVIG			Post-IVIG		
Platelet count (10(9)/L)	182			2			39		
Serum platelet antibody testing	Result	OD*	Negative Cutoff	Result	OD*	Negative Cutoff	Result	OD*	Negative Cutoff
HPA-1a/1a,3a/3a,4a	Negative (Neg)	0.079	0.229	Positive (Pos)	0.425	0.266	Neg	0.085	0.266
HPA-1b/1b,3b/3b,4a	Neg	0.066	0.219	Pos	0.329	0.234	Neg	0.063	0.234
HPA-1a/1a,3b/3b,4a	Neg	0.054	0.22	Pos	0.298	0.218	Neg	0.057	0.218
HPA-5b/5b	Neg	0.064	0.235	Pos	0.362	0.21	Neg	0.063	0.21
HPA-5a/5a	Neg	0.065	0.23	Pos	0.312	0.213	Neg	0.062	0.213
GPIb/IX	Neg	0.044	0.081	Pos	0.098	0.096	Neg	0.036	0.096
HLA	Neg	0.068	0.178	Pos	0.249	0.194	Neg	0.07	0.194

*OD= optical density

SP195

A Challenging Case of Immune-Mediated Thrombocytopenia

H M Reeves[1] (reeves.hollie@mayo.edu), A J Wood[1], J Kreuter[1], D Gastineau[1,2], J L Winters[1], M J Gandhi[1]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Hematology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Post-transfusion purpura (PTP) is characterized by a sudden, dramatic decrease in platelet count 5 to 10 days following a blood transfusion. Drug-induced thrombocytopenia often presents as a sudden decrease in platelet count that is usually transient and resolves with discontinuation of the offending drug. This is a case of a 63 year-old male with sudden onset thrombocytopenia following recent transfusion while also receiving antibiotic therapy. **Study Design/Methods:** A retrospective chart review was performed. **Results/Findings:** On April 24, 2013, a 63-year-old male was admitted to the intensive care unit (ICU) for respiratory failure. He was started on antibiotic therapy. On admission his platelet count was 182 (×10(9)/L). He received four units of fresh frozen plasma on April 25, 2013. On April 28, 2013 his platelet count was 132. Subsequent platelet count 24 hours later revealed a platelet count of 1. Autoimmune thrombocytopenia was a concern and a cell bound platelet antibody assay was weakly positive. The patient was started on high dose steroids and suspect drugs were discontinued, including vancomycin, piperacillin/tazobactam, and levofloxacin. Given the history of recent transfusion and the severe, acute drop in platelet count, PTP was suspected. Serum platelet antibody testing was performed. Intravenous immunoglobulin (IVIG) was started on April 30, 2013 and the following day platelet count had increased to 39. (Table). Platelet antigen genotyping results show the patient is heterozygous at HPA-1 locus. Specific drug antibody testing is pending. **Conclusion:** Due to the timing and severity of the thrombocytopenia, one should suspect drug-induced antibodies, as well as PTP and autoimmune thrombocytopenia. Laboratory testing may not be readily available and reactivity with serum platelet antibody testing should be interpreted cautiously. Transfusion services should be aware of the separate entities as transfusion management will differ.

Disclosure of Commercial Conflict of Interest

M. J. Gandhi: Nothing to disclose; D. Gastineau: Nothing to disclose; J. Kreuter: Nothing to disclose; H. M. Reeves: Nothing to disclose; J. L. Winters: Nothing to disclose; A. J. Wood: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. J. Gandhi: Nothing to disclose; D. Gastineau: Nothing to disclose; J. Kreuter: Nothing to disclose; H. M. Reeves: Nothing to disclose; J. L. Winters: Nothing to disclose; A. J. Wood: Nothing to disclose

SP196

Distribution of HLA-C Alleles in Shandong Han Ethnic Population of China

H Long[1] (sdhla@yahoo.cn), J Liu[1], Y Zhu[1], C Zhu[1], Y Zhang[1], Y Song[1]. *[1]Blood Center of Shandong Province, Jinan, China*

Background/Case Studies: The high diversity of HLA alleles and haplotypes has made it a very useful genetic marker for anthropological study and disease association. The previous studies on polymorphism of HLA of the Han population in Shandong province of China included only HLA-A, HLA-B, and HLA-DRB1 loci (Zhang Yi). The analysis of HLA-C alleles in high-resolution level is presented here. **Study Design/Methods:**

One hundred and fifty potential donors were selected from the Bone Marrow Donor Program. All were self-identified as being of Han ethnic extraction from Shandong province. HLA-C typing was performed using PCR-SBT over exons 2, 3, and 4. Ambiguities with the relevant exons were resolved using additional sequencing. Alleles were generally given in four-digit form. Alleles with identical sequence within exons 2, 3, and 4 were characterized through the letter "g"(for group) that was appended to the fist possible allele. Allele frequencies were derived from allele numbers counting. **Results/Findings:** A total of 26 alleles of HLA-C were observed and the frequencies were indicated as follows: C*06 : 02 (14.67%); C*07 : 02 (13.33%); C*04 : 01 g (8.67%); C*03 : 04 (8.67%); C*03 : 03 (8.33%); C*01 : 02 (7.33%); C*03 : 02 (5.33%); C*15 : 02 (4.67%); C*08 : 01 g (4.67%); C*14 : 03 (4.33%); C*14 : 02 (4.00%); C*12 : 02 (3.33%); C*12 : 03 (2.67%); C*07 : 01 g (2.33%); C*02 : 02 (1.67%); C*08 : 03 (1.33%); C*15 : 05 (0.67%); C*08 : 02 (0.67%); C*07 : 04 (0.67%); C*05 : 01 (0.67%); C*04 : 03 (0.67%); C*17 : 01 g (0.33%); C*16 : 04(0.33%); C*15 : 13(0.33%); and C*15 : 04(0.33%). The accumulated frequency of the most common six alleles was up to 61%. And 17 alleles with the frequency over 1% comprised 95.33% of the total alleles. A comparison of HLA-C distribution with different populations showed that few differences were observed between Shandong Han population(Northern Chinese population) and Sichuan Han population(Southern Chinese population) (xia ling), whereas there were significant differences compared with German (Alexander H. Schmit) and African American (Tu B). **Conclusion:** The distribution of HLA-C alleles in the Shandong Han population shows very useful information to increase understanding of the genetics of HLA in Shandong Han population and will be helpful in further studies.

Disclosure of Commercial Conflict of Interest

J. Liu: No Answer; H. Long: Nothing to disclose; Y. Song: No Answer; Y. Zhang: No Answer; C. Zhu: No Answer; Y. Zhu: No Answer

Disclosure of Grants Conflict of Interest

J. Liu: No Answer; H. Long: Nothing to disclose; Y. Song: No Answer; Y. Zhang: No Answer; C. Zhu: No Answer; Y. Zhu: No Answer

SP197

The Phenotype and Genetype of CD36 Deficiency in Guangzhou Blood Donors

X Ye[1] (huang_yexin@yahoo.com.cn), W Xia[1], J Liu[1], H Ding[1], X Xu[1], J Deng[1], Y Chen[1], Y Chen[1]. *[1]Institute of Blood Transfusion, Guangzhou Blood Center, Guangzhou, China*

Background/Case Studies: CD36 (also known as GPIV) deficiency is known to be responsible for the production of anti-Naka antibodies in different clinical settings such as fetal/neonatal alloimmune thrombocytopenia (FNAIT), platelet transfusion refractoriness (PTR) and post-transfusion purpura (PTP). But the related mutation frequency in China is not known.This study is to analyze the frequency for the phenotype and genetye of CD36 deficiency in Guangzhou blood. **Study Design/Methods:** 998 healthy blood donors were typed for CD36 deficiency on platelets and monocytes using flow cytometry (FC). PCR-SBT analysis was performed to identify the molecular basis underlying the CD36 deficiency. Antibody screening was performed using a PLT antibody solid-phase kit (PAKPLUS, GTI Diagnostics). **Results/Findings:** By analysis of 998 healthy blood donors in China, 1.8%(18/998) individuals failed to express CD36 on platelet surface. In 5/12 individuals no CD36 expression was detected both on platelets and monocytes. The frequencies of type I and II deficient are approximately 0.5 and 1.3%. By PCR-SBT analysis, 1228-1239 delATTGTGCCTATT and 329-330 delAC appeared to be the most common mutations related to type I CD36 deficiency in China. A case of GPIV antibodies in type I CD36 deficient donor (female) was found. Four new mutations of CD36 gene were found: 1200-5inv49bp, 429 + 4insg, exon4delgCAAGTT, and C220T; all are type I CD36 deficient. **Conclusion:** This study is the first to describe the phenotype and genetype of CD36 deficiency by PCR-SBT and FC in Guangzhou and will help to fill the gaps of platelet related CD36 genetic mutation.The basic information provided by this study is useful to clinical prognosis and diagnosis, will be useful in future applications, and possibly lead to the direct discovery of a new allelic gene.

Disclosure of Commercial Conflict of Interest

Y. Chen: No Answer; Y. Chen: No Answer; J. Deng: No Answer; H. Ding: No Answer; J. Liu: No Answer; W. Xia: No Answer; X. Xu: No Answer; X. Ye: Nothing to disclose

Table C: CD36 mutations found 18 Chinese blood donors (§ new mutations)

Sample No.	Sex	CD36 Ph/Mo	Type	Mutations of CD36 gene	GTI PAKPLUS			
					GP IIb/IIIa	GP Ia/IIa	GP IIIb	GPIV
GZ-2771	F	neg/pos	II	Normal	neg	neg	neg	neg
GZ-2902	M	pos/pos	II	1228-1239delATTGTGCCTATT	neg	neg	neg	neg
GZ-2970	F	pos/pos	I	1200-5inv49bp§ + 429+4insg§	neg	neg	neg	neg
GZ-3004	F	neg/neg	I	1228-1239delATTGTGCCTATT + exon4delgCAAGTT§	neg	neg	neg	neg
GZ-3075	M	neg/pos	II	329-330delAC + C156T	neg	neg	neg	neg
GZ-3183	F	neg/neg	I	329-330delAC + C220T§	neg	neg	neg	FUS
GZ-3200	M	neg/pos	II	-Normal	neg	neg	neg	neg
GZ-3264	M	neg/pos	II	-Normal	neg	neg	neg	neg
GZ 3425	M	neg/pos	II	Normal	neg	neg	neg	neg
GZ 3429	M	neg/neg	I	1228-1239delATTGTGCCTATT + 329-330delAC	neg	neg	neg	neg
GZ-3441	M	neg/pos	II	-Normal	neg	neg	neg	neg
GZ-3450	M	pos/pos	II	-Normal	neg	neg	neg	neg
GZ-3527	M	neg/pos	II	- Normal				
GZ-3561	30/M	neg/pos	II	429+4insg§	neg	neg	neg	neg
GZ-3565	20/F	neg/pos	II	- Normal	neg	neg	neg	neg
GZ-3566	23/F	neg/neg	I	1228-1239delATTGTGCCTATT + 329-330delAC	neg	neg	neg	neg
GZ-3647	25/M	neg/pos	II	- Normal	neg	neg	neg	neg
GZ-3660	35/F	neg/pos	II	1228-1239delATTGTGCCTATT	neg	neg	neg	neg

Disclosure of Grants Conflict of Interest

Y. Chen: No Answer; Y. Chen: No Answer; J. Deng: No Answer; H. Ding: No Answer; J. Liu: No Answer; W. Xia: No Answer; X. Xu: No Answer; X. Ye: Nothing to disclose

SP198

A Novel HLA-A Allele, A*24 : 191, was Identified by Sequence-Based Typing in a Chinese Donor

H Long[1] (sdhla@yahoo.cn), C Zhu[1], Y Zhu[1], Y Song[1], Y Zhang[1]. *[1]Blood Center of Shandong Province, Jinan, China*

Background/Case Studies: Human leukocyte antigen (HLA) is one of the most polymorphic genetic systems in human genome. In a study ending January 2013, 8949 HLA alleles including 2132 HLA-A alleles have been reported according to IMGT/HLA database. A novel HLA-A allele, A*24 : 191, has been identified in a registered donor of China Bone Marrow Donor Program. **Study Design/Methods:** The sample was initially detected during a routine HLA tying using SBT technique (ROSE-HLA-SBT commercial kit; ROSE Europe GmbH, Frankfurt, Germany). An unexplainable nucleotide mismatch between the A*02 and the A*24 was obtained which strongly suggested the possible existence of a novel allele of A locus. To clarify this phenomena of HLA-A in the sample, Protrans S4 HLA-A Single Allele-Specific Sequencing Set (Protrans GmbH, Ketsch, Germany) was used to retype it again. The A*02 and the A*24 group allele were amplified and separated by the allele-group-specific primers. Then sequencing reactions of exons 2, 3, and 4 in both directions and exons 1 and 5 in one direction were performed and electrophoresed on an ABI 3730 DNA Analyzer. **Results/Findings:** The repeated sequencing result showed that the A locus consists of A*02 : 01 : 01 : 01 and an A*24 : 156 related novel allele on which the nucleotide difference occurs. The sequence of the novel allele differs from the closest matching allele A*24 : 156 by one nucleotide substitution at nt 256 where G → C (codon 62 GAG → CAG) resulting in one amino acid change, 62 Glu → Gln. **Conclusion:** The sequence of the new A*24 allele was submitted to GenBank in July 2011 with the accession number JN209958. The name HLA-A*24 : 191 has been officially assigned by the World Health Organization Nomenclature Committee for Factors of the HLA System in January 2012.

Disclosure of Commercial Conflict of Interest

H. Long: Nothing to disclose; Y. Song: No Answer; Y. Zhang: No Answer; C. Zhu: No Answer; Y. Zhu: No Answer

Disclosure of Grants Conflict of Interest

H. Long: Nothing to disclose; Y. Song: No Answer; Y. Zhang: No Answer; C. Zhu: No Answer; Y. Zhu: No Answer

SP199

A Study of Neonatal Alloimmune Thrombocytopenia (NAIT) Caused by Parent Platelet Antigens Incompatibility

W Xia[1] (xiawenjie1977@yahoo.com.cn), X Ye[1]. *[1]Guangzhou Blood Center, Guangzhou, China*

Background/Case Studies: Neonatal alloimmune thrombocytopenia (NAIT) is a disease that affects fetuses and newborns, in which the platelet count is decreased. Genetic differences between the fetus and mother may result in the expression of certain antigens by fetal platelets, not expressed by the mother. Fetomaternal transfusions result in the recognition of these antigens by the mother's immune system as non-self, with the subsequent generation of allo-reactive antibodies which cross the placenta. **Study Design/Methods:** To investigate the correlation between platelet glycoprotein specific antibodies/HLA antibodies and the neonatal alloimmune thrombocytopenia 22 patients with neonatal immune thrombocytopenia were selected for this study. The patients were genotyped for HLA-A/B as well as HPA systems by standard PCR SSP assays. The platelet glycoprotein specific antibodies and HLA antibodies in serum and platelet elution were tested with a solid phase ELISA. Then the specificity of antibodies was analyzed according to the corresponding antigens. **Results/Findings:** Among the 22 NAIT patients: 14 were alloimmune; 6 patients (27.3%) had HLA antibodies; 3 patients, (13.6%) had HPA antibodies; 5 patients, (22.7%) had both HLA and HPA antibodies; 2 patients (9.1%) had passive immunity factors; and 6 patients, (27.3%) were caused by auto-antibodies. The common alloimmune antibodies were anti-HLA-A2, 19, anti-B40, and anti-HPA-3a. **Conclusion:** The main reason for neonatal immune thrombocytopenia was parent platelet antigens incompatibility. The outcome of this study is of great significance in the prevention of clinical prenatal diagnosis and postnatal treatment.

Disclosure of Commercial Conflict of Interest

W. Xia: Nothing to disclose; X. Ye: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Xia: Nothing to disclose; X. Ye: Nothing to disclose

Table1.The analysis of six cases of neonatal alloimmune thrombocytopenia caused by HLA mismatching

No		HLA-A	HLA-B	HLA-DR	Anti-HLA
1	patient	11,11	13, 56(22)	12,16	-B22
	mother	02,11	13,46	14,16	
	father	02,11	35, 56	04,12	
2	patient	11,11	13,15	12,12	-A2
	mother	11,11	13,40	12,16	
3	patient	02,11	15,40	14,15	-A2, -B40
	mother	01,11	15,37	10,15	
4	patient	32,33	44,58	03,07	-B44
	mother	11,32	35,58	03,15	
5	patient	02,02	46, 56(22)	04,09	-A11, -B22
	mother	02,11	40(60),46	04,09	
	father	02,11	15(75), 56	04,12	
6	patient	02, 33	38,58	01,17	-A2
	mother	11, 33	27,58	12,17	
	father	02, 31	38,57	01,07	

Table2. The analysis of those cases of neonatal alloimmune thrombocytopenia caused by HPA incompatibility

No		HPA1-17	Anti-HPA (PAK-FLOW)	
7	patient	3a b, 15ab	GPIIb/IIIa	Anti-3a
	mother	2a b, 3bb	GPIIb/IIIa	
	father	15ab		
8	patient	3a b, 6a b, 15bb	GPIIb/IIIa	Anti-3a
	mother	3bb, 6a b, 15bb	GPIIb/IIIa	
	father	15bb		
9	patient	5a b		
	mother	15ab	GPIa/IIa	Anti-5b
	father	3a b, 5a b		

SP200

Transfusion of Human Platelets Treated with Mirasol Pathogen Reduction Technology Does Not Induce Acute Lung Injury in Mice

A Caudrillier[1] (axelle.caudrillier@ucsf.edu), B Mallavia-Onaindia[1], L Rouse[2,1], S Marschner[2,1], M R Looney[1]. *[1]Medicine and Laboratory Medicine, University of California, San Francisco, San Francisco, CA, United States; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Pathogen reduction technology (PRT) has been developed in an attempt to make the blood supply safer, but there is controversy as to whether it may induce structural or functional changes to platelets that could lead to acute lung injury after transfusion. In this study, a commercial PRT system and treatment of human platelets was used to determine if acute lung injury develops after transfusion of PRT-treated platelets into immunodeficient mice. **Study Design/Methods:** Human platelets were obtained from normal healthy volunteers, split into 2 samples, and either not treated (control) or treated with Mirasol Pathogen Reduction Technology (PRT) on day 1 of storage. Platelets (10^8 or 10^9) on day 1 or day 5 of storage were transfused into LPS primed (0.1 mg/kg i.p., 24 h prior) NOD/SCID mice and acute lung injury measurements were obtained at 4 hours post-transfusion using the gravimetric method to determine extravascular lung water and the extravasation of 125 I-labeled albumin to determine lung vascular permeability. Platelet P-selectin expression was determined by flow cytometry. **Results/Findings:** P-selectin expression was higher in the Mirasol PRT-treated platelets compared to control platelets on storage day 1 (8% versus 2%, $p < 0.05$) and day 5 (19% versus 11%, $p < 0.05$). Transfusion of control versus Mirasol PRT-treated platelets (day 5 of storage,10∧9 platelets per mouse) did not result in increased extravascular lung water (6 µl versus. 10 µl, respectively; $p = 0.19$) or increased lung vascular permeability (24 µl versus. 27 µl, respectively; $p = 0.47$). Transfusion of day 1 platelets did not produce lung injury in either group. Whereas in a model of transfusion-related acute lung injury, NOD/SCID mice challenged with H-2Kd monoclonal antibody (1 mg/kg, i.v.) versus isotype control antibody had increased extravascular lung water (32 µl versus 10 µl, respectively; $p < 0.05$) and increased lung vascular permeability (55 µl versus 24 µl, respectively; $p < 0.05$). Using lung intravital microscopy and CFSE-labeled platelets, a difference was not detected in the sequestration of platelets in the lung circulation in the control versus Mirasol PRT-treated groups. **Conclusion:** Mirasol-PRT produced an increase in platelet P-selectin expression, but transfusion of human platelets treated with Mirasol PRT into immunodeficient mice did not result in greater platelet sequestration in the lungs or the development of acute lung injury.

Disclosure of Commercial Conflict of Interest

A. Caudrillier: Terumo BCT, Grants or Research Support; M. R. Looney: Terumo BCT, Grants or Research Support; B. Mallavia-Onaindia: Terumo BCT, Grants or Research Support; S. Marschner: Employee of Terumo BCT, Other; L. Rouse: Terumo BCT, Other

Disclosure of Grants Conflict of Interest

A. Caudrillier: Terumo BCT, Grants or Research Support; M. R. Looney: Terumo BCT, Grants or Research Support; B. Mallavia-Onaindia: Terumo BCT, Grants or Research Support; S. Marschner: Employee of Terumo BCT, Other; L. Rouse: US Deparment of Defense, Stocks or Bonds

Recipient Non-Infectious Adverse Events

SP201

Hemovigilance Data Reported to the National Healthcare Safety Network, 2010-2012

A Harvey[1] (fom7@cdc.gov), K Chung[1], M J Kuehnert[1]. *[1]Office of Blood, Organ, and Other Tissue Safety, Centers for Disease Control and Prevention, Atlanta, GA, United States*

Background/Case Studies: The National Healthcare Safety Network (NHSN) Hemovigilance Module is a surveillance system operated by the CDC used by US healthcare facilities to report transfusion-related adverse events. Participation is voluntary. **Study Design/Methods:** Facilities enrolled before 2013 that reported complete data for a minimum of 3 months, including denominators and adverse events, were included in analysis. **Results/Findings:** In 2010-2012, 63 facilities contributed 1175 months of adverse reaction and denominator data. Facilities ranged from 10-1100 beds. The total number of red blood cells, platelets, plasma, and

cryoprecipitate transfused was 1,973,355 units; red blood cells accounted for 57.3% of products transfused. Facilities reported collecting 1,699,655 samples for type and crossmatch. A total of 5056 CDC-defined adverse reactions were reported; associated product information was available for 4806 (Table). Allergic and febrile non-hemolytic reactions occurred most frequently, but were reported as severe in only 175 of 3,962 cases (4.4%). The least frequently reported reactions (sepsis, lung injury, and acute hemolytic reactions), were reported as severe in 42 of 60 cases (70%). Forty-three facilities reported 35,915 incidents, including 38 associated with adverse reactions. Incidents related to sample collection, handling, or testing totaled 24,614, occurring at an overall rate of 14 incidents/1000 samples collected. Incidents related to product storage, modification, or transfusion totaled 11,301, occurring at an overall rate of 7 incidents/1000 products transfused. **Conclusion:** Minor allergic and febrile reactions were reported most frequently. Sepsis, lung injury, and acute hemolytic reactions were rarely reported but were often severe compared with other reactions (eg allergic). Reporting requirements were reduced in January 2013 to increase enrollment and focus on events most likely to result in negative patient outcomes.

Disclosure of Commercial Conflict of Interest

K. Chung: Nothing to disclose; A. Harvey: Nothing to disclose; M. J. Kuehnert: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Chung: Nothing to disclose; A. Harvey: Nothing to disclose; M. J. Kuehnert: Nothing to disclose

TABLE 1. Reactions Associated with Red Blood Cells, Platelets, or Plasma

Reaction	Total	Rate per Units Transfused
Post-Transfusion Sepsis	7	1:50,876¥
Delayed Serologic	313	1:4,749§
Delayed Hemolytic	87	1:17,086§
Acute Hemolytic	24	1:61,938§
Allergic	2,237	1:829‡
Serious Allergic	157	1:11,817‡
Febrile Non-Hemolytic	1,725	1:1,076‡
Circulatory Overload	194	1:9,563‡
Hypotension	122	1:15,207‡
Dyspnea	68	1:27,283‡
Lung Injury	25	1:74,210‡
All Reactions	**4,806**	**1:386‡**

¥ Platelets transfused (n = 356,130) § Platelets/RBCs transfused (n = 1,486,513) ‡ Platelets/RBCs/plasma transfused (n = 1,855,244)

SP202

Perioperative Transfusion-Related Acute Lung Injury (TRALI): The Canadian Blood Services Experience

A Q Alam[1,3] (asim.q.alam@gmail.com), M Huang[3], Q Yi[2,3], B Hannach[3]. *[1]Department of Anesthesia & Pain Medicine, The Hospital for Sick Children, The University of Toronto, Faculty of Medicine, Toronto, ON, Canada; [2]Dalla Lana School of Public Health, University of Toronto, Toronto, ON, Canada; [3]Canadian Blood Services, Toronto/Ottawa, ON, Canada*

Background/Case Studies: Transfusion-related acute lung injury (TRALI) is a devastating transfusion-associated adverse event reported after transfusion. There is a paucity of data on the incidence and characteristics of TRALI cases that occur perioperatively. Suspected perioperative TRALI cases reported to Canadian Blood Services (CBS)between 2001 and 2012 were classified and compared to non-perioperative cases to elucidate specific surgical factors that make them particularly at risk for developing TRALI. **Study Design/Methods:** Using Canadian Consensus Conference definitions, all suspected TRALI cases reported to CBS since 2001 were reviewed by two experts or, from 2006-2012, an expert TRALI Medical Review Group (TMRG); they were then classified and detailed in a database. Two additional reviewers further categorized them as occurring within 72 hours from the onset of surgery (perioperative) or not in that period (non-perioperative). Various demographic and characteristic variables of each case were collected and compared between the groups. **Results/Findings:** Between 2001-2012, a total of 469 suspected TRALI were reported to CBS; 303 were determined to be within the TRALI diagnosis spectrum. Of those, (112) (38%) were identified as occurring during the perioperative period. Patients who underwent cardiac surgery requiring cardiopulmonary bypass (25.0%), general surgery (18.0%), and orthopedic patients (12.5%), represented the three largest surgical groups. Perioperative TRALI cases comprised of more men (53.6 versus 41.4, p = 0.0395) than non-perioperative patients. More perioperative TRALI patients required supplemental O2 (14.3% versus 3.1%, p = 0.0003), required mechanical ventilation (18.8% versus 3.1%), or were from the ICU (14.3% versus 3.7%, p = 0.0043), than non-perioperative TRALI patients before their operations. The surgical patients were transfused approximately 673 products and on average were transfused more products than non-perioperative patients (6.0[SD = 8.3] versus 3.6[5.2], p = 0.0002). Perioperative TRALI patients were transfused more frozen and fresh frozen plasma and cryoprecipitate than non-TRALI patients. Non-perioperative patients utilized more cryosupernatant plasma. There was no difference between donor antibody test results between the groups. **Conclusion:** CBS data has provided insight into the nature of TRALI cases that occur perioperatively; this group represents a large proportion of TRALI cases. Perioperative transfusion should be carefully considered and monitored, especially in patients with at-risk characteristics. Further research will be required to delineate the exact mechanisms behind perioperative TRALI.

Disclosure of Commercial Conflict of Interest

A. Q. Alam: Nothing to disclose; B. Hannach: Nothing to disclose; M. Huang: Nothing to disclose; Q. Yi: No Answer

Disclosure of Grants Conflict of Interest

A. Q. Alam: Nothing to disclose; B. Hannach: Nothing to disclose; M. Huang: Nothing to disclose; Q. Yi: No Answer

SP203

Hospitalized Patients Transfused with Storage-Aged (but Not Fresh) Red Blood Cells (RBCs) Demonstrate Impaired Endothelial Function and Significant Reductions in Circulating Nitrite Levels

R Neuman[1] (rneuman@emory.edu), S K Karatela[2], V Menon[1], J L Newman[2], D Polhemus[3], S Sher[1], D J Lefer[3], A A Quyyumi[1], J D Roback[2]. *[1]Cardiovascular Diseases, Emory University, Atlanta, GA, United States; [2]Department of Pathology and Laboratory Medicine, Emory University, Atlanta, GA, United States; [3]Surgery, Emory University, Atlanta, GA, United States*

Background/Case Studies: Transfusion of storage-aged (SA) packed red blood cells (RBCs) has been associated with worse clinical outcomes, but the mechanisms underlying clinical manifestations of the storage lesion remain unknown. It was hypothesized that these adverse effects may be due to disruption of nitric oxide (NO) signaling and endothelial function by SA-RBCs. To investigate these possible physiologic effects of RBC storage age on ill and anemic patients, randomized hospitalized patients were randomized with orders for RBC transfusion to receive either fresh-RBCs or SA-RBCs. Prior to, during, and after transfusion, the patients were monitored for endothelial function, plasma nitrite (NO_2), 2,3-diphosphoglycerate (DPG), and plasma free hemoglobin (FHb). **Study Design/Methods:** 44 hospital inpatients (aged 59 ± 14 years, 46% female) receiving crossmatched RBC transfusions for clinical indications were randomized to receive either fresh RBC (aged <10 days, n = 20) or SA-RBC (aged >21 days, n = 24). Vascular endothelial function was assessed using flow-mediated dilation (FMD) of the brachial artery before, during, and following (1 hour and 24 hours) transfusion. 2,3-DPG, FHb, and NO_2 were measured before and at 1 and 24 hours after transfusion. Data were analyzed using a linear mixed effects model. **Results/Findings:** There were no differences in demographics, cardiovascular disease (CVD) risk factors, known CVD, or amount of RBCs transfused in either group. The mean (±SD) age of RBC units was 9.1 ± 3.1 days and 29.8 ± 5.7 days in the fresh and SA groups, respectively. Patients receiving fresh RBCs had no significant change in FMD. However, FMD significantly decreased by 2.4% (p = 0.037) 24 hours after transfusion of SA-RBCs. Patients receiving fresh-RBCs had a 0.191 ± 0.438 and 0.069 ± 0.242 μM increase in NO_2 at 1 and 24 hours after transfusion, respectively (p = ns, for pre versus post). In contrast, patients receiving SA-RBC showed a decrease in NO_2 of 0.055 ± 0.227 and 0.079 ± 0.226 μM at 1 and 24 hours, respectively (p = ns, pre versus. post). The difference in NO_2 in recipients of fresh-RBC vs SA-RBC was significant at 1 hr (p = 0.027), and approached significance at 24 hrs (p = 0.052). Baseline 2,3-DPG was 50.57 ± 37.44 and 41.44 ± 24.03 μM/gHb in the fresh and SA groups respectively. There was no significant change at 1 or 24 hours post transfusion in either group. Although FHb prior to transfusion was higher in the fresh group (0.166 ± 0.12 g/dL) than the SA group (0.109 ± 0.042 g/dL, p = 0.043), there was no significant change following transfusion in either group. **Conclusion:** In ill and anemic hospitalized patients, transfusion of SA-RBCs (but not fresh-RBCs) was associated with post-transfusion

endothelial dysfunction manifest as a decreased FMD response and reduced NO2 levels. These effects were not accompanied by changes in 2,3-DPG or FHb. Further studies to identify and correct the causes of endothelial dysfunction and reduced NO2 levels after SA-RBC transfusions appear warranted.

Disclosure of Commercial Conflict of Interest

S. K. Karatela: No Answer; D. J. Lefer: Nothing to disclose; V. Menon: Nothing to disclose; R. Neuman: Nothing to disclose; J. L. Newman: Nothing to disclose; D. Polhemus: Nothing to disclose; A. A. Quyyumi: No Answer; J. D. Roback: Nothing to disclose; S. Sher: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. K. Karatela: No Answer; D. J. Lefer: Nothing to disclose; V. Menon: Nothing to disclose; R. Neuman: Nothing to disclose; J. L. Newman: Nothing to disclose; D. Polhemus: Nothing to disclose; A. A. Quyyumi: No Answer; J. D. Roback: Nothing to disclose; S. Sher: Nothing to disclose

SP204

The Fluidic Angioplasty Theory in Transfusion-Associated Circulatory Overload (TACO) Events Marked by Fever: Is It the Age of the Patient or the Age of the Product?

C M Cserti-Gazdewich[1] (Christine.Cserti@uhn.ca), L D'Onofrio[1], J Pendergrast[1]. *[1]Transfusion Medicine Laboratory, University Health Network/University of Toronto, Toronto, ON, Canada*

Background/Case Studies: Transfusion-Associated Circulatory Overload (TACO) has been noted to exhibit an unusually high prevalence of concomitant fever. This may or may not be the consequence of the nature of reaction review, with TACO investigations secondarily discovering the coincidence of pyrexia-prone active comorbidities, or transfusion-associated fever investigations secondarily discovering unreported cardiorespiratory evidence of circulatory overloading. On the other hand, cases of febrile TACO may be a real entity, with cases simply reflecting the use of longer-stored (and thus more pro-inflammatory) products. Alternatively, a "fluidic angioplasty" or "cumulative barotraumas stress" hypothesis (Andrzejewski et al, *Transfusion* 2012: 2310-20) proposes that volume overloading of the circulation promotes atherosclerotic plaque destabilization and release of pyrogenic mediators (bioactive lipids, etc). In the "causual conjecture," febrile TACO may therefore be marked by higher product age, and/or higher recipient age (as a surrogate for disruptable atheroma burden). **Study Design/Methods:** Hemovigilance data from a tri-hospital university-affiliated network (~1000 beds) were reviewed for the period between January 2010-December 2012 to assess cases of TACO with and without fever (TACO + F vs TACO-F). Recipient age distributions and a simplified product age distribution were compared. (To address the complexity of analyzing unit ages in instances of multi-component exposure, only TACO cases stemming from red cell transfusions [pRBC] with a uniform analyzable age [same-/similar-aged [±3 d] units or a solitary unit] were studied.) Data were assessed and compared using medians, interquartile ranges (IQR), and Mann-Whitney U test (2-tailed, for significance level at $p < 0.05$). **Results/Findings:** Over a 3-year-period, the hospital transfusion service received 972 reaction reports (among 106,733 pRBC, 25,121 adult-dose platelet concentrates, and 35,271 plasma units transfused). TACO was noted in 107 (11%); 7 definite/54 likely/47 possible. TACO + F occurred in 44 (41%), versus 63 lacking febrile features (Tmax <38°C, no rigors/chills). TACO + F included 10 who were at/near 38C (±0.5) at baseline, leaving 34 non-pre-existing fever cases, among which 16 were marked by new post-transfusion fever (47%) as the dominant or referral-based disturbance. Patient ages in TACO + F (69 y, IQR 58-78, range 19-93) were not different from TACO-F (64 years, IQR 53-75, range 23-88), $p = 0.2$. Among analyzable pRBC recipients (19 TACO + F, 32 TACO-F), product ages were nearly identical in distribution (22 d, IQR 19-25, range 7-37; 22 d, IQR 18-28, range 11-40; $p = 0.9$). **Conclusion:** Notwithstanding possible beta error, TACO ± F did not appear to be a function of product age or older (and thus presumably more atheromatous) patients, but was instead circumstantially ascertained by secondary disturbance discoveries.

Disclosure of Commercial Conflict of Interest

C. M. Cserti-Gazdewich: Nothing to disclose; L. D'Onofrio: Nothing to disclose; J. Pendergrast: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. M. Cserti-Gazdewich: Nothing to disclose; L. D'Onofrio: Nothing to disclose; J. Pendergrast: Grifols, Grants or Research Support

SP205

A Comparison of Adverse Reaction Rates for Platelet Additive Solution 3 (PAS 3) Versus Plasma Platelet Units

C S Cohn[1] (cscohn@umn.edu), J R Stubbs[5], J Schwartz[3], M Cushing[2], C Goss[2], B H Shaz[6], D C Mair[7], W Heaton[4]. *[1]Laboratory Medicine and Pathology, University of Minnesota Medical Center, Minneapolis, MN, United States; [2]Pathology and Laboratory Medicine, New York Presbyterian Hospital, Weill Cornell Medical College, New York, NY, United States; [3]Pathology and Cell Biology, New York Presbyterian Hospital, Columbia University, New York, NY, United States; [4]Transfusion Medicine, North Shore LIJ Medical Group, Manhasset, NY, United States; [5]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States; [6]New York Blood Center, New York, NY, United States; [7]American Red Cross, St. Paul, MN, United States*

Background/Case Studies: Apheresis platelet transfusion (txn) reactions range from mild urticaria to severe events. Recent publications have suggested a linkage to the plasma content. Plasma reduction is known to reduce adverse reactions (AR), but volume reduction/washing compromises product quality. Platelets stored in additive solution (PAS) have shown a reduction in ARs compared to plasma-based units (PP). PAS 3 (InterSol, Fenwal, a Fresenius Kabi Co.) has been FDA approved for use with AMICUS-derived apheresis platelets. PAS 3 platelets are stored in 65% PAS/35% plasma. A multicenter study comparing ARs in PAS 3 platelets versus PP is reported. **Study Design/Methods:** Study design was open label, non-randomized retrospective review assuming parallel product availability in routine practice. A total of 14,005 txns from six study sites were included with 9845 PP txns being given to 2202 patients and 4160 PAS 3 to 1444 patients. Non-inferiority (NI) was shown if the 97.5% Upper Confidence Limit (UCL) of the Relative Risk of Reaction (RR) was less than 2.0. If NI was proven, an analysis for superiority of PAS 3 over PP was undertaken. ARs were categorized using the Biovigilance Component of the National Healthcare Safety Network. Active surveillance was used to detect ARs. An independent clinical events committee (CEC) adjudicated the reported events in a manner blinded to platelet type and study site. **Results/Findings:** The study ran from January 29, 2012-September 30, 2012. 165 CEC-adjudicated ARs were reported (the CEC eliminated 2 AR from the PP arm and changed the category of 2 PAS 3 and 2 PP): 142 from PP and 23 from PAS 3 units. The % of txns with ARs were: 1.37% for PP; 0.55% for PAS 3 ($p < 0.0001$) and 1.13% overall. The RR for PAS 3 versus PP was calculated as 0.403 (UCL 0.663). The ARs with the highest rates were allergic (ATR) and febrile non-hemolytic (FNHTR), at 0.66% and 0.40% of the total txns reported, respectively. The RR for ATR and FNHTR, respectively, was 0.350 (UCL 0.686) and 0.336 (UCL 0.827). Further analysis showed that PAS 3 was statistically superior to PP. No TRALI was reported in either arm. The percentage of patients with an AR per total patients receiving PP was 5.22%, and from PAS 3 was 1.52%. There were 7 PP and 0 PAS 3 txns that led to more than one AR. **Conclusion:** This study found that NI and superiority of PAS 3 was statistically substantiated compared to PP. PAS 3 NI and superiority were also shown for ATR and FNHTR, separately.

Disclosure of Commercial Conflict of Interest

C. S. Cohn: Fenwal, Grants or Research Support,Travel Support or Honorarium; M. Cushing: Nothing to disclose; C. Goss: Nothing to disclose; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium Fenwal, Grants or Research Support; D. C. Mair: Nothing to disclose; J. Schwartz: Nothing to disclose; B. H. Shaz: Haemonetics, no financial relationship, Other, Fenwal, no financial relationship, Other; J. R. Stubbs: Terumo BCT, Other, Fenwal, Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

C. S. Cohn: Fenwal, Travel Support or Honorarium, Grants or Research Support; M. Cushing: Nothing to disclose; C. Goss: Nothing to disclose; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; D. C. Mair: Fenwal, Travel Support or Honorarium; J. Schwartz: Nothing to disclose; B. H. Shaz: Nothing to disclose; J. R. Stubbs: Nothing to disclose

Site	PAS 3 Transfusions	Plasma Transfusions	Reaction Type	PAS 3 (N = 4160) Transfusions n (%)	Plasma (N = 9845)
A	1340	3125	ATR	12 (.29)	81 (.82)
B	1002	3053	FNHTR	7 (.17)	49 (.50)
C	319	719	Hypotensive TR	1 (.02)	2 (.02)
D	420	639	TACO	2 (.05)	6 (.06)
E	444	933	Transfusion Associated Dyspnea	0 (.00)	2 (.02)
F	635	1376	Other	1 (.02)	2 (.02)
			Total	23 (.55)	135* (1.37)

* 135 PP led to 142 ARs

SP206

Allergic Transfusion Reactions (ATR) in Patients with IgA Deficiency

W Anani[1] (ananiwq@upmc.edu), D J Triulzi[2,1], L Qu[2,1]. *[1]Pathology, University of Pittsburgh, Pittsburgh, PA, United States; [2]The Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: IgA deficiency is the most common immunodeficiency estimated to affect one in 700 individuals. Many of these individuals are asymptomatic and testing for IgA occurs in a non-transfusion related setting. It was sought to determine the value of IgA level in predicting allergic transfusion reaction by evaluating the temporal relationship between documented IgA levels and blood product transfusions. **Study Design/Methods:** All IgA measurements performed at a university hospital laboratory between January 2008-December 2012 were retrospectively reviewed. Patients with IgA levels <7 mg/dL (the institution's laboratory cut-off for IgA reporting) were reviewed for history blood product transfusions and transfusion reactions. **Results/Findings:** IgA level was measured in 15,521 patients during the 5-year observation period, with 140 (0.9%) patients with IgA levels <7 mg/dL. 41 of the 140 (29%) patients who received blood product transfusions. Severe allergic transfusion reactions were reported in 5 of the 41 (12%) patients. One patient, who had an IgA level measured for clinical indications prior to any blood transfusions (unknown to the blood bank at the time of blood product issue), developed a severe allergic reaction with hypotension, tachycardia, and hypoxia during the first transfusion. The other four patients (two with urticaria and dyspnea; two with hypoxia) whose IgA levels <7 mg/dL were detected as part of a blood bank transfusion reaction investigation after the patients had developed symptoms of allergic reactions. The remaining 36 patients had no adverse transfusion reactions documented despite multiple blood product transfusions. No long-term morbidity or mortality was reported as a result of the allergic reaction. **Conclusion:** There is low likelihood (1 out of 37 or 2.7%) of an anaphylactic transfusion reaction in patients with no prior history of transfusion reactions and whose IgA level was incidentally found to be <7 mg/dL.

Disclosure of Commercial Conflict of Interest

W. Anani: Nothing to disclose; L. Qu: Nothing to disclose; D. J. Triulzi: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Anani: Nothing to disclose; L. Qu: Nothing to disclose; D. J. Triulzi: Nothing to disclose

SP207

Acute Transfusion Reactions in Pediatric versus Adult Patients

P Young[1] (pampee.young@vanderbilt.edu), F Oakley[2]. *[1,2]Pathology, Vanderbilt University, Nashville, TN, United States*

Background/Case Studies: The majority of reports on transfusion reactions address adult patients. Less is known about the types, incidence, and other clinical details of acute transfusion reactions in the pediatric populations. Furthermore,there have been no previous reports, as far as is known, directly comparing these aspects between adults and pediatric patient populations to assess if there are differences. **Study Design/Methods:** From January 1, 2011-February 1, 2013, all reported adult and pediatric transfusion reactions here were evaluated by Transfusion Medicine clinical service. The information was subsequently shared with the Hemovigilance data base. Data provided to Hemovigilance included age, sex, blood product associated with the reaction, severity of the reaction, and the type of transfusion reaction. The transfusion reaction data for this period was subsequently collated and obtained for analysis. **Results/Findings:** A total of 133,671 transfusions were performed here during the study period including 20,179 platelet transfusions, 31,605 plasma transfusions, 79,933 RBC transfusions, and 2,154 cryoprecipitate transfusions. Over the same period 108 pediatric and 277 adult transfusion reactions were recorded. This corresponds to an incidence of 0.62 transfusions reactions per 100 patients within the pediatric population and an incidence of 0.24 transfusion reactions per 100 patients within the adult population. In both the adult and pediatric population, transfusion reactions were most commonly associated with platelet transfusions, followed by RBCs and then plasma transfusions. No transfusion reactions were observed in either population following cryoprecipitate transfusion. Within the pediatric population subcomponent analysis identified multiple differences when compared to the adult population, including an increased incidence of allergic transfusion reactions (0.27/100 versus 0.11/100), febrile non-hemolytic transfusion reactions (0.19/100 versus 0.047/100) and hypotensive transfusion reactions (0.029/100 versus 0.0078/100). as well. **Conclusion:** This is the first study, as far as is known, to provide detailed comparisons of acute transfusion reactions to all blood products between pediatric versus adult populations at a single institution. This comparison between these two patient populations is reasonable as the clinical transfusionists at both the pediatric and adult hospitals undergo similar training and are supported by a single transfusion service and culture. Collectively these data provide insight into the pediatric transfusion reactions demonstrating a general increase in the incidence of transfusion reactions within the pediatric population as well as a number of subgroup and reaction specific differences.

Disclosure of Commercial Conflict of Interest

F. Oakley: No Answer; P. Young: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Oakley: No Answer; P. Young: Nothing to disclose

SP208

Passive Reporting Greatly Underestimates the Rate of Transfusion Associated Circulatory Overload after Platelet Transfusion

J S Raval[1] (jay_raval@med.unc.edu), M Mazepa[1], S L Russell[1], C C Immel[1], H C Whinna[1], Y A Park[1]. *[1]Pathology and Laboratory Medicine, University of North Carolina, Chapel Hill, NC, United States*

Background/Case Studies: Transfusion associated circulatory overload (TACO) is the second leading cause of transfusion-related fatalities reported to the US Food and Drug Administration (FDA). While its rate has been previously investigated after transfusion with red blood cells and plasma, no data is available regarding the frequency of this reaction after platelet transfusion. Our goal was to determine the rate of TACO after platelet transfusion at our university medical center. **Study Design/Methods:** This study was comprised of both a retrospective analysis and a prospective analysis. The retrospective analysis of the electronic transfusion reaction records was conducted to determine the historical rate by passive reporting of TACO after platelet transfusion from 2000-2012. The prospective analysis was performed over a 30-day period and included active surveillance of all non-emergently issued, non-operative inpatients and outpatients ≥16 years of age who had received platelets as their only blood component transfusion in the previous 6 hours. Data collected included patient demographics, vital signs pre- and post-transfusion, previous 24 hour fluid inputs/outputs, use of supplemental oxygen, any subjective reports of dyspnea, and infusion rates. For the prospective analysis, all variables were collected within 24 hours of transfusion from the electronic medical record and, when necessary, interviews with the transfusing care providers and/or patients. Data is presented as mean ± SD unless otherwise stated. **Results/Findings:** In the 13-year retrospective analysis, 366 reactions to platelets were reported to the Transfusion Medicine Service, and six of these reactions (1.6%) were TACO. The historical rate of TACO was 1 : 5997 transfused platelet units. During the prospective 30-day active surveillance period, 225 eligible patients received a total of 334 units of platelets. The average platelet transfusion volume was 261 ± 26 mL, and the average rate of infusion was 391 ± 198 mL/hr. During this time, two unreported TACO reactions in two different patients were discovered, characterized by new-onset hypertension, crackles on lung auscultation, dyspnea, hypoxia, and supplemental oxygen requirements; both patients had complete resolution of all of these signs and symptoms with diuresis. Both patients were oncology inpatients, >50 years of age, >1000 mL fluid positive over 24 hours, and had platelet infusion rates of >500 mL/hr. The rate of TACO during this active surveillance period was 1 : 167 transfused platelet units. **Conclusion:** Similar to other blood components, TACO due to platelet transfusions is greatly underestimated

by passive reporting. Additional efforts must be made to facilitate the detection and prevention of these reactions through physician and nurse education.

Disclosure of Commercial Conflict of Interest

C. C. Immel: Nothing to disclose; M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; J. S. Raval: Nothing to disclose; S. L. Russell: Nothing to disclose; H. C. Whinna: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. C. Immel: Nothing to disclose; M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; J. S. Raval: Nothing to disclose; S. L. Russell: Nothing to disclose; H. C. Whinna: Nothing to disclose

SP209

AABB Validation Study of the CDC National Healthcare Safety Network's Hemovigilance Module Adverse Events Definitions Protocol

J P AuBuchon[1] (jima@psbc.org), B I Whitaker[2], J Malasky[2], M K Fung[3]. *[1]Puget Sound Blood Center, Seattle, WA, United States; [2]Center for Data and Special Programs, AABB, Bethesda, MD, United States; [3]Pathology, Fletcher Allen Health Care, Burlington, VT, United States*

Background/Case Studies: Utility of any hemovigilance system is dependent upon appropriate and reproducible application of the system's definitions. This becomes even more important in systems where submissions are not reviewed by an adjudicating body. It was sought to determine how staff coded adverse reactions at institutions that had or had not received training on the application of definitions used in the CDC's National Healthcare Safety Network (NHSN) Hemovigilance Module (HV). **Study Design/Methods:** Facilities that were (11) or were not (11) currently submitting adverse reaction data to the HV were recruited to review 36 hypothetical cases containing elements of 37 case definitions from among 13 different diagnostic classifications. Respondents were required to assign a case definition (diagnosis), severity, and imputability using the January 2013 Surveillance Protocol definitions of the HV. Those submitting HV data would have had access to an instructional slide set prepared by CDC using similar hypothetical cases. Respondents were those who would usually be preparing cases for HV submission and/or reviewing transfusion reaction work-ups. Concordance of responses was tallied for all respondents and each group and was then compared via chi-square analysis. A panel of experts provided the expected responses. **Results/Findings:** Result forms were received from 22 institutions. The frequency of agreement with the expert assessment was not different according to prior training exposure, and results were tallied across both groups. The consensus (mode) case definition agreed with the expert assessment in 34 of 37 diagnoses (92%) (Table) Discordance in the "no reaction" category was noted in an apparent febrile reaction case where the temperature elevation did not meet the case definition requirement and in a complex case that had elements of autoimmune hemolysis. There was 76% concordance on severity and 71% on imputability scoring. **Conclusion:** Despite delineated definitions, considerable variability in responses was seen, and this was not reduced by the available training. This degree of error could degrade the utility of comparative reports. More effective training with emphasis on the details and application of the surveillance definitions may be helpful.

Disclosure of Commercial Conflict of Interest

J. P. AuBuchon: Nothing to disclose; M. K. Fung: Nothing to disclose; J. Malasky: Nothing to disclose; B. I. Whitaker: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. P. AuBuchon: Nothing to disclose; M. K. Fung: Nothing to disclose; J. Malasky: Nothing to disclose; B. I. Whitaker: Nothing to disclose

Reaction Type	Number of Cases	Concordance with Expert
Allergic + Febrile	6	6 (80.2% of all responses)
Pulmonary	8	7 (71.6%)
Hemolysis	8	8 (79.0%)
HTR	3	3 (66.7%)
TTI	2	2 (72.7%)
PTP	2	2 (93.2%)
TAGVHD	1	1 (90.9%)
Other (hyperkalemia)	1	1 (40.9%)
No reaction	6	4 (41.1%)
OVERALL	37	70.3%

(See NHSN HV Surveillance Protocol for abbreviations.)

SP210

Post-Transfusion Purpura (PTP) Occurrence among the Inpatient Elderly Medicare Beneficiaries as Recorded in Large Administrative Databases During 2011

M Menis[1] (Mikhail.Menis@fda.hhs.gov), S A Anderson[1], R Forshee[1], S McKean[2], R Gondalia[2], R Warnock[2], B Golding[1], J A Kelman[3], R Ball[1], H S Izurieta[1]. *[1]FDA/CBER, Rockville, MD, United States; [2]Acumen LLC, Burlingame, CA, United States; [3]CMS, Baltimore, MD, United States*

Background/Case Studies: Post-Transfusion Purpura (PTP) is a serious transfusion complication resulting in sudden thrombocytopenia with mucous membrane hemorrhage as well as gastrointestinal and urinary tract bleeding. There are no known published population-based studies evaluating PTP among the US elderly. The study objective was to assess PTP occurrence and potential risk factors among elderly Medicare beneficiaries, age 65 and older, transfused in the inpatient setting in 2011. **Study Design/Methods:** This retrospective, claims-based study utilized large Medicare databases for 2011 in coordination with the Centers for Medicare and Medicaid Services (CMS) and under CBER's Safe Blood Program within the SafeRx Project. Transfusions of blood and blood components were identified by recorded ICD-9-CM procedure codes and revenue center codes. PTP was ascertained via ICD-9-CM diagnosis code 287.41 titled "Posttransfusion purpura". Revenue center units were used to quantify blood use. This study evaluated PTP rates (per 100,000 inpatient transfusion stays) among elderly Medicare beneficiaries, overall and by age, gender, race, number of units and blood components transfused. Chi-squared tests were performed to compare PTP rates and ascertain PTP occurrence trends by age and number of units transfused. **Results/Findings:** Among 2,147,038 inpatient transfusion stays for elderly in 2011, 40 had PTP diagnosis code recorded, an overall rate of 1.9 per 100,000 stays. PTP rates (per 100,000 inpatient transfusion stays) varied by blood components, number of units, age, gender, and race. PTP rates by blood component groups were as follows: 1.5 for RBCs only; 1.4 for plasma only; 19.9 for platelets only; 20.8 for platelets and plasma; 1.2 for RBCs and plasma; 15.1 for RBCs and platelets; 14.2 for RBCs, plasma and platelets. PTP rates by number of units transfused were: 1.4 for 1 unit, 1.1 for 2-4 units, 3.7 for 5-9 units, and 6.5 for >9 units transfused ($p < 0.0001$). PTP rates for age categories 65-69, 70-74, 75-79, 80-84, 85 and over were 2.8, 3.2, 1.2, 1.0, and 1.4, respectively ($p = 0.017$). Females and males had PTP rates of 2.0 and 1.7, respectively; whites and non-whites had PTP rates of 2.0 and 1.4, respectively. **Conclusion:** The population-based study suggests substantially higher PTP rates for stays with platelets transfused either by themselves or in combination with RBCs and/or plasma. The study also suggests an increase in PTP rates with number of units transfused and a potential reduction in PTP occurrence with advancing age. The study was based on claims data and thus limitations include potential under- or misrecording of PTP diagnosis code, transfusion procedures, and units transfused, as well as lack of clinical details to validate the recorded PTP diagnosis. Multivariate regression analyses are planned to further assess potential risk factors for PTP occurrence and control for confounders.

Disclosure of Commercial Conflict of Interest

S. A. Anderson: Nothing to disclose; R. Ball: Nothing to disclose; R. Forshee: Nothing to disclose; B. Golding: Nothing to disclose; R. Gondalia: Nothing to disclose; H. S. Izurieta: Nothing to disclose; J. A. Kelman: Nothing to disclose; S. McKean: Nothing to disclose; M. Menis: Nothing to disclose; R. Warnock: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. A. Anderson: Nothing to disclose; R. Ball: Nothing to disclose; R. Forshee: Nothing to disclose; B. Golding: Nothing to disclose; R. Gondalia: Nothing to disclose; H. S. Izurieta: Nothing to disclose; J. A. Kelman: Nothing to disclose; S. McKean: Nothing to disclose; M. Menis: Nothing to disclose; R. Warnock: Nothing to disclose

SP211

Intravenous Immunoglobular (IVIG)-Associated Adverse Outcomes: Comparison to Monograph Rates and Product Type

S Nahirniak[1,2] (Susan.Nahirniak@capitalhealth.ca), H Blain[1], G Clarke[1,2]. *[1,2]Laboratory Medicine & Pathology, Alberta Health Services, Edmonton, AB, Canada*

Background/Case Studies: Intravenous immunoglobulin (IVIG) is a plasma protein product used in immune deficiency as immune replacement therapy or to modulate the immune response in auto-antibody mediated disease. Statements in product monographs indicate that adverse events may be seen in up to 16% of IVIG infusions (non specific, fever, chills, nausea, headache/migraine, and allergic symptoms) but that aseptic meningitis and

IVIG hemolysis should occur at a frequency of <0.01% of IVIG infusions. **Study Design/Methods:** As part of the IVIG utilization program, all patients receiving IVIG have data collected on, and stored in, an Access database, for the indication, dose, duration of therapy, specific product/manufacturer, objective outcome measures and any adverse outcomes associated with IVIG therapy. **Results/Findings:** A total of 19,041 IVIG infusions occurring over a 6-year time period (2007-2012) were reviewed. This accounted for 2468 patients and 98,0169 grams of IVIG. In total, 274 suspect reactions were reported to the transfusion service. Thirty five of these were deemed to be unrelated to IVIG infusion. The overall reaction rate to IVIG was determined to be 1.26% of infusion episodes or 0.024% per gram (g) infused. Only one case of TRALI and one of anaphylaxis were identified in the entire cohort. Other reaction rates are broken down by product type in the table below. The frequency of aseptic meningitis (n = 3) is consistent with monograph rates. However, the rates of allergic and febrile reaction were lower than expected with some product differences identified. The hemolysis rates were higher than the monograph projections for our three main IVIG products and also demonstrate differences between the products. **Conclusion:** The lower overall reactions as well as the lower allergic and febrile events may reflect differences in reporting in a research trial versus clinical setting. Despite this, our data suggests that there may be product specific differences in the rates of adverse events associated with IVIG therapy but whether this is truly related to product type or dosing protocols and other patient specific factors needs to be further evaluated.

Disclosure of Commercial Conflict of Interest

H. Blain: CSL Behring, Grifols, Bayer, Other; G. Clarke: CSL, Behring, Other, Grifols,Other, Baxter, Other, Talecris, Other, Bayer, Other; S. Nahirniak: CSL Behring, Other, Baxter, Other, Grifols, Other, Bayer, Other

Disclosure of Grants Conflict of Interest

H. Blain: Nothing to disclose; G. Clarke: Nothing to disclose; S. Nahirniak: Nothing to disclose

Comparison of IVIG adverse events by product

	Monograph	Product A	Product B	Product C	Product D	Product E	Overall
# of infusions		2654	12473	2714	963	237	19041
Overall AE	up to 16%	2.3%	0.94%	1.95%	0.62%	0.84%	1.26%
Aseptic meningitis	<0.01%	0.07%	0.008%	0.00%	0.00%	0.00%	0.015%
Hemolysis	<0.01%	0.68%	0.38%	0.59%	0.00%	0.00%	0.00%
Allergic	0.50-4.00%	0.26%	0.26%	0.11%	0.10%	0.42%	0.24%
Febrile	1.10-24.60%	0.19%	0.03%	0.15%	0.10%	0.42%	0.07%

SP212

The Novel Use of Cetirizine in Place of Diphenhydramine for Prevention of Allergic Transfusion Reactions (ATR): Excessive Sedation is Reduced Without Increasing Breakthrough Reactions

M Mazepa[1] (mmazepa@unch.unc.edu), Y A Park[1], J S Raval[1]. *[1]Pathology and Laboratory Medicine, University of North Carolina, Chapel Hill, NC, United States*

Background/Case Studies: Diphenhydramine is commonly used for secondary prophylaxis of Allergic Transfusion Reactions (ATR) in chronically transfused patients despite a lack of evidence that the risk for ATR is reduced by this therapy. The perception that this is a low risk therapy likely contributes to this practice. However, particularly with IV administration, the potential for excessive sedation and risk for abuse exists. Cetirizine, a second generation oral antihistamine, is an attractive alternative, but has not been evaluated for treatment or prevention of ATR. Cetirizine is rapidly absorbed orally, with onset of antihistamine effect and peak plasma concentrations in <1 hour, while lacking the adverse CNS toxicity profile of diphenhydramine. **Study Design/Methods:** In this retrospective case crossover study, 12 subjects with Sickle Cell Anemia on chronic red blood cell exchange regimens with a history of ATRs were crossed-over from diphenhydramine to cetirizine for secondary prophylaxis of ATR. Retrospectively, apheresis charts were reviewed for the 6 month period of diphenhydramine therapy prior to crossover, and then for the 6 month period following crossing over to cetirizine. The presence of breakthrough ATR and excessive sedation were recorded for each procedure. Breakthrough ATR was defined as pruritus or urticaria requiring administration of additional antihistamine and excessive sedation was defined as any RN notation of excessive somnolence that occurred during the exchange transfusion (excluding symptoms that occurred after use of additional antihistamine for breakthrough symptoms). **Results/Findings:** In the 6 month diphenhydramine period, 34 procedures were complicated by four ATRs and eight notes of excessive sedation. In comparison, in the 6 month cetirizine period, 42 procedures were complicated by six ATRs and one note of excessive sedation. Breakthrough ATRs were uncommon in both period, but did not significantly increase following crossover to cetirizine (Odds ratio 1.25; 95% CI 0.323-4.84). However, excessive sedation was greatly diminished following crossing over to cetirizine (Odd ratio 12.6; 95% CI 1.48-106.7). **Conclusion:** This exploratory study of cetirizine for the secondary prophylaxis of ATRs in chronically transfused patients provides the first evidence that cetirizine likely offers the same degree of secondary prophylaxis for ATRs while greatly diminishing the risk for excessive sedation and likely reduces the risk for abuse. Overall it appears that cetirizine may have a better risk-to-benefit profile than the current standard of diphenhydramine. A larger randomized case crossover study is warranted to further investigate the use of cetirizine for secondary prevention of ATRs in chronically transfused patients with a history of ATRs.

Disclosure of Commercial Conflict of Interest

M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; J. S. Raval: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Mazepa: Nothing to disclose; Y. A. Park: Nothing to disclose; J. S. Raval: Nothing to disclose

SP213

Product and Diagnosis Specific Rates of Intravenous Immune Globulin (IVIG)-Associated Hemolysis

S Nahirniak[1,2] (Susan.Nahirniak@capitalhealth.ca), H Blain[2], T E McKelvey[2], G Clarke[1,2]. *[1]University of Alberta, Edmonton, AB, Canada; [2]Laboratory Medicine & Pathology, Alberta Health Services, Edmonton, AB, Canada*

Background/Case Studies: Intravenous immune globulin (IVIG) is a plasma protein product used in immune replacement therapy or to modulate the immune response in autoantibody mediated disease. The literature states that IVIG associated hemolytic events are rare. This data indicates that the rate of hemolysis associated with IVIG is 3% of treated patients based on prospective monitoring of doses >1 g/kg of IVIG. **Study Design/Methods:** This IVIG utilization program records data regarding indication, dose, duration of therapy, IVIG product/manufacturer, objective outcome measures, and adverse outcomes on all patients receiving IVIG. Data is stored in an Access database. Hemolysis data was evaluated over a 6-year time frame from 2007-2012. **Results/Findings:** 55 episodes of IVIG associated hemolysis, occuring between 2007-August 2011, were reviewed. A significant difference in hemolysis rate by IVIG brand was identified. Product C had a 10-fold higher rate at 16 : 86050 (0.0002) hemolytic reactions per grams infused compared to Product A at 1: 56585 (0.00002) and Product B at 40 : 519887 (0.00008). Local policies and procedures were changed to limit the use of Product C in non-hematology patients, to restrict dosing to less than 2 g/kg whenever possible and to increase the use of Product A as well as discussions with Product C manufacturers about potential role of processing. In 2013, follow up was performed to evaluate the impact of these changes. This included an additional 26 hemolytic events (24 patients) and 273, 474 grams of IVIG. Product A now has a rate of 18 reactions over 124132 grams infused (0.0001); Product B had 47 : 675219 (0.00007) and Product C has dropped to 16 : 136645 (0.0001). Seven patients had repeat hemolysis on subsequent exposure, with 3/7 hemolyzing despite change of product type. In the entire cohort, the majority of implicated patients were blood group A (81%) and received >2 g/kg of IVIG. There was no significant difference in gender (53% female, 47% male) but slight variability in rates of hemolysis per patient diagnosis were seen. Idiopathic thrombocytopenic purpura (ITP), transplant rejection, and chronic idiopathic demyelinating

polyneuropathy (CIDP) were the top three implicated diagnoses at 21%, 13%, and 13% of cases, respectively but these diagnoses represent 27%, 7% and 3% of all patients in the database. **Conclusion:** This data demonstrates that there are product specific differences in the rates of hemolysis. The data also suggests possible differences due to diagnosis with a relatively higher rate in CIDP patients . However the impact of dosing protocols, manufacturing processes/changes, and other patient specific factors on the hemolysis rates still requires further investigation.

Disclosure of Commercial Conflict of Interest

H. Blain: CSL Behring, Grifols, Bayer, Other; G. Clarke: Baxter,Other, Grifols, Other, CSL Behring, Other; T. E. McKelvey: Nothing to disclose; S. Nahirniak: CSL Behring, Other, Baxter, Other, Grifols, Other, Bayer, Other

Disclosure of Grants Conflict of Interest

H. Blain: Nothing to disclose; G. Clarke: Nothing to disclose; T. E. McKelvey: Nothing to disclose; S. Nahirniak: Nothing to disclose

SP214

Febrile Non-Hemolytic Transfusion Reaction (FNHTR) Occurrence among Inpatient Elderly Medicare Beneficiaries as Recorded in Large Administrative Databases in 2011

M Menis[1] (Mikhail.Menis@fda.hhs.gov), S A Anderson[1], R Forshee[1], S McKean[2], R Gondalia[2], R Warnock[2], B Golding[1], J A Kelman[3], R Ball[1], H S Izurieta[1]. *[1]FDA/CBER, Rockville, MD, United States; [2]Acumen LLC, Burlingame, CA, United States; [3]CMS, Baltimore, MD, United States*

Background/Case Studies: Febrile Non-Hemolytic Transfusion Reaction (FNHTR) is acute transfusion complication resulting in fever, chills, rigors, headache, nausea, and vomiting. The study objective was to assess FNHTR occurrence and potential risk factors among elderly Medicare beneficiaries, ages 65 and older, transfused in the inpatient setting in 2011. **Study Design/Methods:** This retrospective claims-based study utilized large Medicare administrative databases for calendar year 2011 in coordination with Centers for Medicare & Medicaid Services (CMS) and under Center for Biologics Evaluation and Research (CBER)'s Safe Blood Program within the SafeRx Project. Transfusions were identified by recorded procedure and revenue center codes, while FNHTR was ascertained via ICD-9-CM diagnosis code. Revenue center units were used to ascertain number of blood units transfused. The study evaluated FNHTR rates (per 100,000 inpatient transfusion stays) among elderly Medicare beneficiaries, overall and by age, gender, race, number of units and blood components transfused. Chi-squared tests were performed to compare FNHTR rates and assess FNHTR occurrence trends by age and units transfused. **Results/Findings:** Among 2,147,038 inpatient transfusion stays for elderly in 2011, 1,243 had FNHTR diagnosis code recorded, an overall rate of 57.9 per 100,000 stays. FNHTR rates (per 100,000 inpatient transfusion stays) varied by age, gender, race, number of units, and blood components transfused. FNHTR rates for age categories 65-69, 70-74, 75-79, 80-84, and 85 and over were 65.3, 66.5, 63.7, 53.9, and 43.5, respectively ($p < 0.0001$). Females and males had FNHTR rates of 60.9 and 54.0, respectively; whites and non-whites had FNHTR rates of 57.2 and 61.5, respectively. FNHTR rates by number of units transfused were: 43.1 for 1 unit, 54.5 for 2-4 units, 77.5 for 5-9 units, and 90.7 for >9 units transfused ($p < 0.0001$). FNHTR rates by blood component groups were: 24.3 for plasma only; 68.9 for RBCs only; 67.5 for platelets only; 62.3 for platelets and plasma; 67.4 for RBCs and plasma; 193.7 for RBCs and platelets; 109.2 for RBCs, plasma and platelets, and 12.5 for autologous transfusion only. **Conclusion:** This population-based study among the US elderly suggests a decline in FNHTR rates with age and an increase in FNHTR occurrence with greater number of units transfused. The study showed substantially elevated FNHTR rates for stays with RBCs transfused in combination with platelets, and suggests the lowest FNHTR rates for plasma only and autologous transfusion only stays. The study was based on claims data and thus limitations include potential under- or mis-recording of FNHTR diagnosis code, transfusion procedures, and units transfused as well as lack of clinical details to validate the recorded FNHTR diagnosis. Multivariate regression analyses are planned to identify potential risk factors for FNHTR occurrence.

Disclosure of Commercial Conflict of Interest

S. A. Anderson: Nothing to disclose; R. Ball: Nothing to disclose; R. Forshee: Nothing to disclose; B. Golding: Nothing to disclose; R. Gondalia: Nothing to disclose; H. S. Izurieta: Nothing to disclose; J. A. Kelman: Nothing to disclose; S. McKean: Nothing to disclose; M. Menis: Nothing to disclose; R. Warnock: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. A. Anderson: Nothing to disclose; R. Ball: Nothing to disclose; R. Forshee: Nothing to disclose; B. Golding: Nothing to disclose; R. Gondalia: Nothing to disclose; H. S. Izurieta: Nothing to disclose; J. A. Kelman: Nothing to disclose; S. McKean: Nothing to disclose; M. Menis: Nothing to disclose; R. Warnock: Nothing to disclose

SP215

Repeated Anaphylaxis with Platelets due to an Anti-Rg

M B Elkins[1] (matelkin@med.umich.edu), L R Dake[1], R Davenport[1], L Cooling[1]. *[1]Pathology, University of Michigan, Ann Arbor, MI, United States*

Background/Case Studies: Chido (Ch) and Rodgers (Rg) are high incidence antigens (98.3% Ch, 97% Rg) expressed on complement C4. Alloantibodies to Ch/Rg may cause positive cross-matches, but are not a cause of hemolytic transfusion reactions. There have been reports of anti-Ch/Rg causing anaphylactic transfusion reactions. This study reports on the diagnosis and management of a Ch+/Rg– patient who developed an anti-Rg alloantibody causing repeated anaphylaxis to platelet transfusions and positive RBC cross-matches. **Study Design/Methods:** The patient was a 67-year-old woman with a past medical history significant for rheumatoid arthritis, an abdominal aortic aneurysm (AAA) status-post repair in 2011, and myelodysplastic syndrome (MDS with isolated deletion 5). She was admitted to an outside hospital in 2012 for elective resection of perianal Bowen's disease. Her transfusion history was significant for 1 unit of platelets (PLT) in 1988 and 1 unit of PLT during her AAA repair without incidents; however, 1 month after her AAA repair, she received 1 unit PLT with an anaphylactic reaction. Preoperatively, the patient was transfused with 3 unit RBC (Adsol) for HGB = 6.7 with good response (HGB 6.7->8.9 gm/dL) and no adverse reaction. After premedication with Solumedrol and Benadryl, the patient was transfused with 1 unit PLT for PLT = $8 \times 10\text{^}11$/L. She developed anaphylaxis with severe hypotension (BP 100/51->62/-), tachycardia (HR 68->120/min), dyspnea (RR 20->40/min), hypoxia (%O2 100->94%), rigors, throat swelling, and confusion: there was no rash or hives noted. A post-transfusion chest x-ray showed mild interstitial edema. She was treated with additional Solumedrol, Benadryl, fluid boluses, dopamine, epinephrine, and supplemental oxygen. After stabilization, she was transferred to a university hospital for evaluation. **Results/Findings:** Initial type and screens at both hospitals were negative. A repeat type and screen 4 weeks later was reactive (2/2 cells) on IAT; DAT was negative (polyspecific, IgG). Antibody panels showed a weak panagglutinin reactive with all cells (2+) except the autocontrol. Antibody reactivity was sensitive to Ficin and pooled plasma neutralization (5 donors). The patient's plasma reacted strongly with Ch-/Rg+ RBC but not with Ch+/Rg– RBC. The patient's RBC typed as Ch+/Rg– with human polyclonal antisera. Additional testing showed a serum IgA = 158 mg/dL, haptoglobin = 87 ng/dL, and normal IgG subclass concentrations. The patient was subsequently diagnosed with acute erythroleukemia. During induction chemotherapy, she was supported with washed blood products (45 PLT, 39 RBC) without additional allergic reactions. **Conclusion:** This is the second report of repeated anaphylaxis due to allo-anti-Rg. Similar to one reported case of anaphylaxis due to allo-anti-Ch case, the patient was clinically supported through chemotherapy-induced cytopenias with washed blood components without anaphylaxis.

Disclosure of Commercial Conflict of Interest

L. Cooling: Nothing to disclose; L. R. Dake: Nothing to disclose; R. Davenport: Nothing to disclose; M. B. Elkins: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Cooling: Nothing to disclose; L. R. Dake: Nothing to disclose; R. Davenport: Nothing to disclose; M. B. Elkins: Nothing to disclose

SP216

Activation and Function of Platelets Post Fire Burn

D de Korte[1] (D.DEKORTE@SANQUIN.NL), H Korsten[1], R Marck[2,3], J W Lagerberg[1], P F van der Meer[1], K Gardien[2,3]. *[1]Sanquin Blood Bank, Amsterdam, Netherlands; [2]Red Cross Hospital, Beverwijk, Netherlands; [3]Association of Dutch Burn Centres, Beverwijk, Netherlands*

Background/Case Studies: Burn injuries have, besides disruption of integrity of the skin, a severe impact on the physiologic homeostasis of the human body. This is a complex process about which a lot remains unknown. Platelet (PLT) counts show a distinct course post-burn, with a nadir around day 3 followed by a thrombocythemic period with a peak around day 15. However, it is unknown how burn injury affects PLT function. In trauma patients, PLTs

are activated for at least 72 hours after injury and have a hyper reactivity in the first 48 hours, but their functionality is diminished in non-survivors. Although PLTs are involved in many processes in the recovery after burn injury, such as hemostasis, wound healing, and immune response, little is known about the activation and functionality of PLTs post fire burn (PFB). PLT dysfunction might affect bleeding risk, hence the timing of surgery, or the susceptibility to infection or wound healing problems. It was the aim to study here PLT activation and function during the post fire burn period. **Study Design/Methods:** Five patients were studied up to 24 days PFB. Blood samples of the burn patient were collected in CPD tubes on day 0-1, days 3-4, days 8-9, days 12-16, and days 20-24 PFB and analyzed within 4 h. Functional tests (PLT Function Analyzer (PFA-100) and Thrombo Elasto Graph (TEG)) were performed. PLT Rich Plasma (PRP) was made, and surface markers (CD62P; CD63; CD31; PAC-1; Annexin-V) of PLTs were tested. In the residual plasma Fibrinogen, activated partial thromboplastin time (APTT) and prothrombin time (PT) were analyzed. **Results/Findings:** PLT counts showed PFB the course as described above, but in functional tests patients were aberrant from controls. In the TEG, PBF samples from all 5 patients had a faster reaction time (coagulation), a larger angle (polymerization velocity) and larger maximal amplitude (clot strength). Between day 3 and day 9 PFB the PFA-100 showed closure times below the lower limit. Between days 9 and 24 some recovery to normal values was found for both TEG and PFA, as well as for PLT counts. Thus, functional tests showed that PFB PLTs are hyper reactive, independent of the actual PLT count. There were some differences in surface markers indicating activation of PLTs, with a peak between day 3 and day 9 PFB and a slow recovery to normal values after day 9. Fibrinogen was increased between days 3 and 9 and over the whole PFB period PT was normal, while APTT was around or above the upper normal limit. **Conclusion:** TEG and PFA analysis showed that initially PFB PLTs from burn patients are hyper reactive, but functional. Surface marker analysis showed that PFB PLTs are activated, with a recovery to normal during longer follow up. Fibrinogen was increased, but a depletion of clotting factor(s) involved in the intrinsic coagulation pathway. Because autologous PRP is sometimes used to support skin transplantation, the amount of growth factors in PLTs PFB will be studied.

Disclosure of Commercial Conflict of Interest

D. de Korte: Nothing to disclose; K. Gardien: No Answer; H. Korsten: Nothing to disclose; J. W. Lagerberg: Nothing to disclose; R. Marck: No Answer; P. F. van der Meer: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. de Korte: Terumo BCT,Grants or Research Support; K. Gardien: No Answer; H. Korsten: Nothing to disclose; J. W. Lagerberg: Nothing to disclose; R. Marck: No Answer; P. F. van der Meer: Nothing to disclose

SP217

Time to Onset of Acute Transfusion Reactions after Transfusion Initiation

N Lau[1] (nlau@dhs.lacounty.gov), M D Le[1], R H Finck[1]. *[1]Pathology and Laboratory Medicine, Harbor-University of California-Los Angeles Medical Center, Torrance, CA, United States*

Background/Case Studies: The Standards for Blood Banks and Transfusion Services mandates that patients who receive blood transfusions must be observed for potential adverse events during the transfusion and for an appropriate time thereafter, but does not specify the optimal frequency of clinical assessment or the length of post-transfusion monitoring. Determining when transfusion reactions (tx rxns) are most likely to occur can help to optimize guidelines for the extent of routine clinical monitoring needed for patients during and after transfusions. **Study Design/Methods:** All tx rxn investigations performed by the transfusion medicine service were retrospectively reviewed here between January 1, 2010-December 31, 2012. Classification of each tx rxn was performed at the time of laboratory investigation. For each event, we calculated the time period between the initiation of the transfusion and the development of the reaction. **Results/Findings:** 122 transfusion reactions were investigated during the study period; 15 were excluded due to incomplete record keeping. Of the remaining 107, 45 reactions not classified as acute tx rxns were excluded from analysis ("not a tx rxn" [6], "symptoms due to underlying medical condition" [28], and "delayed hemolytic/serologic tx rxn" [11]). Sixty-two acute tx rxns with complete data remained for analysis. Number of reactions and time periods for the development of the reaction after the initiation of the transfusion are shown (table 1). **Conclusion:** The majority of tx rxns occur within the hour following the initiation of the transfusion. There was only 1 tx rxn that occurred more than 3 hours after the beginning of the transfusion episode. This information may be useful when determining policies for monitoring patients both during and after the transfusion.

Disclosure of Commercial Conflict of Interest

R. H. Finck: Nothing to disclose; N. Lau: Nothing to disclose; M. D. Le: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. H. Finck: Nothing to disclose; N. Lau: Nothing to disclose; M. D. Le: Nothing to disclose

TABLE 1. Transfusion reaction episodes per specified time interval of onset

	Time increment (in minutes)*					
	≤15	15-60	61-120	121-180	181≤	Total
Mild allergic	7	18	3	3		31
Severe allergic	1	1				2
Anaphylaxis		1				1
FNTR	6	3	5	6	1	21
Chills	2		1			3
TACO		1				1
TRALI		1				1
Acute hypotensive		1				1
Pain		1				1
Total (%)	16 (25.8)	27 (43.5)	9 (14.5)	9 (14.5)	1 (1.6)	62

* Time increment = Timeat start of transfusion-Timeat of transfusion reaction per nursing record

SP218

Painful and Red Vein Syndrome Associated with Transfusion of Red Blood Cells (RBCs)

P Robillard[1] (pierre.robillard@hema-quebec.qc.ca), A Jacques[2], Deschênes[2], C Bédard[2], L Thibault[2]. *[1]Medical Affairs, Hema Québec, Montréal, QC, Canada; [2]R&D, Hema Québec, Québec, QC, Canada*

Background/Case Studies: In November 2011, Héma-Québec was notified by a hospital transfusion safety officer of three recent cases of transfusion reactions characterized by exquisite pain and/or redness at injection site or along the trajectory of vein in red blood cell (RBC) recipients. Upon further investigation, 10 such cases had occurred during the previous year at that hospital and 14 others were uncovered in other Québec hospitals during the period of July 2009-November 2011. **Study Design/Methods:** An investigation was conducted on the characteristics of these reactions, the types of products, the materials involved in collection and processing, the age of products, the recipients' diagnoses, concomitant medications, and blood donor's characteristics. Na, K, glucose, lactate and bradykinin (BK) levels were measured in the two types of RBC units implicated in these adverse reactions: whole blood (WB)-derived leukoreduced AS-3 RBCs (n = 30) and apheresis leukoreduced AS-3 RBCs (n = 30) produced with Trima Accel instruments (Terumo BCT. Lakewood, CO). K, Na, glucose and lactate were measured using a GEM 3000 analyzer on fresh (F) (days 2, 3, or 4) and expired (Exp) (days 42-47) RBC units. BK levels were measured at expiration using a validated ELISA. **Results/Findings:** A total of 24 recipients experienced: pain at injection site 10 (41.7%), pain at site + along vein 5 (20.8%), redness at site 1 (4.2%), redness at site + along vein 2 (8.3%), redness + pain at site 5 (20.8%), or redness + pain at site + along vein 1 (4.2%). Pain was described as intense burning sensation that disappeared when transfusion was stopped. In three cases, pain and/or redness returned when transfusion was resumed. In all cases, no clinical signs were observed with the infusion of another RBC unit. Half of the patients were treated for cancer. One patient was on active chemotherapy and 3 received oral K. None received I.V. K. 50% of transfused RBC units were collected by apheresis although this product represents only 3.4% of the RBC deliveries. The mean age of units was 35 ± 9 days (37.5% were >35 d). Results for biochemical parameters were: (Table). **Conclusion:** This investigation could not unveil any specific cause for the transfusion reactions observed. However these results suggest that apheresis RBC processing could trigger the generation of BK in some units. BK has been shown to elicit inflammatory reactions and pain at injection sites.

Disclosure of Commercial Conflict of Interest

C. Bédard: Nothing to disclose; . Deschênes: Nothing to disclose; A. Jacques: Nothing to disclose; P. Robillard: Nothing to disclose; L. Thibault: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Bédard: Nothing to disclose; Deschênes: Nothing to disclose; A. Jacques: Nothing to disclose; P. Robillard: Nothing to disclose; L. Thibault: Nothing to disclose

Parameter (mean ± SD)	WB-derived RBC		Apheresis RBC	
	F	Exp	F	Exp
Na, mmol/L	132 ± 2	118 ± 3	129 ± 2	111 ± 4
K, mmol/L	6 ± 1	25 ± 2	6 ± 1	26 ± 3
Glucose, mmol/L	32 ± 3	24 ± 2	24 ± 2	12 ± 2
Lactate, mmol/L	5 ± 1	17 ± 2	8 ± 2	17 ± 2
BK, pg/mL	—	1119 ± 1157	—	4734 ± 21,929

Red Blood Cells: ABO Blood Group

SP219

Detection of ABO Incompatibility in the Extended IgG Gel Crossmatch

R M Barrett[1,2] (Rachelle.Barrett@salemhospital.org), T Kasper[1]. *[1]Blood Bank, Salem Health Laboratory, Salem, OR, United States; [2]SBB Program, University of Texas Medical Branch, Galveston, TX, United States*

Background/Case Studies: AABB Standard 5.15.1 states: "The crossmatch shall use methods that demonstrate ABO incompatibility and clinically significant antibodies to red cell antigens and shall include an antiglobulin test." When a clinically significant antibody other than ABO is detected during pretransfusion testing, the crossmatch is performed using the same method used in detection. Many extended crossmatch methods do not have an immediate spin reaction. The testing facility for this study uses MTS IgG Gel cards for detection of clinically significant antibodies other than ABO in extended crossmatch. To meet this standard, an immediate spin crossmatch is required in addition to this extended crossmatch. If it could be proven ABO incompatibility is detectable even by methods not designed for such purposes, the extra step of completing an immediate spin crossmatch in the case of the extended crossmatch could be eliminated for the sake of efficiency. **Study Design/Methods:** Twenty randomly selected samples of each ABO blood type were crossmatched with known ABO incompatible donors at tube immediate spin and on MTS IgG gel cards. Known ABO compatible crossmatch donors were used as controls. Randomly selected samples were also tested for complement activity using commercial anti-c3b-c3d in tube. **Results/Findings:** ABO incompatible crossmatch was apparent 100% of the time at immediate spin for all samples tested. Type O and A samples demonstrated incompatibility in 100% of samples tested on MTS IgG gel. Type B samples gave negative reactions on IgG gel in 15% of samples tested. Furthermore, type B samples had significantly lower grade reactions on gel as compared with Type O and A samples. Samples that were negative on gel were also negative for complement activity. Positive gel reactions correlated with positive complement in the few samples tested. **Conclusion:** The clinical significance of the ABO antibodies made by the % of patients testing negative in gel is questionable. If the ability to fix complement is diminished for these antibodies, are these patients then protected from hemolytic transfusion reactions when exposed to type incompatible blood? Until further study determines the clinical significance of these differences in antibody activity, this study suggests completing an immediate spin crossmatch in addition to extended crossmatch methods when extended crossmatch testing methods do not already include an immediate spin reaction.

Disclosure of Commercial Conflict of Interest

R. M. Barrett: Nothing to disclose; T. Kasper: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. M. Barrett: Nothing to disclose; T. Kasper: Nothing to disclose

SP220

Successful ABO Incompatible Kidney Transplantation: Using BioRad Column Agglutination Technology to Assess the Risk

D G Bruce[1] (david.bruce@nhsbt.nhs.uk), M Williams[1], W M Howell[2], M Maley[1], V Carter[2]. *[1]Red Cell Immuohaematology, NHS Blood and Transplant, Newcastle-Upon-Tyne, United Kingdom; [2]Histocompatibility and Immunogenetics, NHS Blood and Transplant, Newcastle-Upon-Tyne, United Kingdom*

Background/Case Studies: Determining the antibody titer to mismatched ABO blood group antigens is critical in assessing the immunological risk associated with ABO-incompatible (ABOi) transplantation. (Tydén G, Kumlien G, Fehrman, I, *Transplantation*. 2003 Aug 27; 76(4):730-1) demonstrated that successful kidney transplantation could be performed when a pre-transplant isoagglutinin titre of eight or lower was achieved. Pilot UK National External Quality Assessment Scheme exercises for titration of ABO antibodies show poor interlaboratory reproducibility. For one sample, tested in 31 laboratories, IgG titers ranged from 8-2048 with a mean of 128. This variation raises questions about the safety of relying on a standard level when assessing suitability for transplant and demonstrates there is a need to provide more consistent reproducible results. a series of 28 ABOi live donor kidney transplants are reported on here where pre- and post-transplant IgM and IgG ABO antibody titers were measured using BioRad column agglutination technology (CAT). **Study Design/Methods:** Patients were divided into two groups:- those who had starting titers ≤8 pre-transplant and those with starting titers >8. The titer endpoint was used to guide number of pre transplant treatments. Patients with titer >8 had rituximab and antibody removal (immunoabsorbtion or plasmapheresis) pre-transplant, whilst those with titer ≤8 received standard immunosuppression and no antibody removal. Manual serial two-fold dilutions of the patient samples were made using phosphate buffered saline. These were tested against a pool of either A1, A2, or B cells depending on the donor's ABO group. 50 mL of 0.8% cells were pipetted into BioRad CAT microtubes and 25 mL of test sample was added. IgG levels were determined using BioRad I.D. gel "LISS/Coombs" IAT cards. The room temperature (RT) direct agglutination titer end point was determined using BioRad I.D. neutral cards. Each ID-Card was incubated for 15 minutes (the IAT test cards at 37C and the neutral cards at RT). **Results/Findings:** Twenty-eight ABOi kidneys have been transplanted to date using this protocol. Fourteen required antibody removal and 14 did not. There was no rejection or early immunological events recorded in either group, and medium/long term outcome in this cohort of ABOi transplanted patients is comparable to those receiving ABO compatible grafts. **Conclusion:** It has been shown that manual titers using BioRad CAT can be used to successfully facilitate ABOi transplantation. Providing accurate antibody titer levels helps quantify the associated risk and determines the need for treatment. It has been shown that ABOi live donor transplantation is safe when pre-transplant ABO titers determined by our method are ≤8.

Disclosure of Commercial Conflict of Interest

D. G. Bruce: Nothing to disclose; V. Carter: Nothing to disclose; W. M. Howell: Nothing to disclose; M. Maley: Nothing to disclose; M. Williams: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. G. Bruce: Nothing to disclose; V. Carter: Nothing to disclose; W. M. Howell: Nothing to disclose; M. Maley: Nothing to disclose; M. Williams: Nothing to disclose

SP221

ABO Blood Type and Risk of Cancer among Blood Donors

F Vahidnia[1] (fvahidnia@bloodsystems.org), N V Hirschler[2], M Agapova[4], A Chinn[2], M P Busch[1,3], B Custer[1,3]. *[1]Blood Systems Research Institute, San Francisco, CA, United States; [2]Blood Centers of the Pacific, San Francisco, CA, United States; [3]Laboratory Medicine, University of California, San Francisco, CA, United States; [4]University of Washington, Seattle, WA, United States*

Background/Case Studies: Studies have demonstrated a relationship between ABO blood type and risk of cancer. Non-O blood type has been associated with higher risk of pancreatic cancer among different populations. **Study Design/Methods:** A linked database was used here, including data from blood donors who participated in the Retroviral Epidemiology Donor Study (REDS, 1991-2002) from Blood Centers of the Pacific (BCP), San Francisco, CA, and California Cancer Registry (CCR, 1991-2010). ABO blood type was measured at the blood center at time of donation. Primary cancers diagnosed after first donation to repository were identified. Frequency of different cancers was compared by anatomic site among donors

with blood type O and non-O (A, B or AB). Adjusted hazard ratio (HR) were calculated for all primary cancers using Cox regression models among all donors as well as donors who developed any cancer, smoking related cancers (lung, bladder, kidney, pancreas, oral cavity and pharynx), and cancers of digestive system. **Results/Findings:** Among 55,812 blood donors, representing 870,160 person-years of follow-up, 4, 899 cases of primary cancers were identified. 49% of donors who developed cancer had blood type O, versus 48% of those who were not diagnosed with a cancer. There were no significant differences in the frequency of cancers diagnosed among donors with blood type O versus non-O by anatomic site, except for 67 cases of pancreatic cancer (0.9% versus 1.8%, $p < 0.01$). Compared to blood type O, the adjusted HRs for pancreatic cancer in donors with types B, A, and AB were 1.99 (95% CI, 0.94-4.24), 2.04 (95% CI, 1.17-3.53), and 2.29 (95% CI, 0.79-6.67), respectively. In the multivariable analysis, compared with donors with blood type O, donors with non-O blood type had more than two times the risk of developing pancreatic cancer. Among donors who developed a smoking-related cancer or a cancer of digestive system the risk of pancreatic cancer remained elevated in non-O blood type donors. **Conclusion:** This prospective study of blood donors demonstrates an association between non-O blood type and long-term risk of developing pancreatic cancer. Further studies are suggested to investigate the role of ABO blood type antigens in development of pancreatic cancer.

Disclosure of Commercial Conflict of Interest

M. Agapova: Nothing to disclose; M. P. Busch: Nothing to disclose; A. Chinn: No Answer; B. Custer: Novartis, Grants or Research Support; N. V. Hirschler: Nothing to disclose; F. Vahidnia: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Agapova: GE Healthcare, Grants or Research Support; M. P. Busch: Novartis, Grants or Research Support, Novartis, Travel Support or Honorarium; A. Chinn: No Answer; B. Custer: Novartis, Grants or Research Support; N. V. Hirschler: Nothing to disclose; F. Vahidnia: Nothing to disclose

Adjusted hazard ratios (HR) for pancreatic cancer in donors with non-O blood type versus O, Blood Centers of the Pacific, 1991-2010*

		Blood Type		
		A, B, or AB		
Cohort	person-years	Adjusted HR	(95% CI)	p-value
All donors	870,160	2.05	(1.22-3.43)	0.007
Donors with all types of cancer	46,649	2.26	(1.34-3.79)	0.002
Donors with smoking-related cancers**	6,916.6	2.20	(1.31-3.70)	0.003
Donors with cancers of digestive system$	5,257.5	1.97	(1.17-3.33)	0.011

* HR adjusted for age, race and sex; pancreatic cancer cases n = 67
** Cancers of lung, bladder, kidney, pancreas, oral cavity and pharynx
$ Cancers of esophagus, stomach, small intestine, colon & rectum, liver and bile ducts and pancreas.

SP222
Weak ABO Subgroups: Should We Do More to Identify These Donors?
K Bowman[1] (kelly_bowman@bonfils.org), C Hintz[1], C Kreutzer[1], C Melland[1], F West[1], T Le[1]. *[1]Bonfils Blood Center, Denver, CO, United States*

Background/Case Studies: Multiple donors at this blood center have a history of blood donation as unremarkable group O donors, only to be later revealed as weak A subgroups with strong anti-A1 antibody. One particular donor donated 93 units over 25 years' time before the discrepancy was recognized. Routine blood center testing may indicate that a donor is group O, however investigation of the donor's true blood type may be initiated when a hospital reports a discrepancy in the course of their ABO confirmation, or when reactions between donor plasma and reagent A1 cells are noted to progressively weaken over time. Three such donors are investigated in this study. **Study Design/Methods:** These donor samples were tested with reagent red cells and antisera from multiple manufacturers, including anti-A, anti-B, anti-A,B, A1 lectin, A1 cells, A2 cells, and B cells. Adsorption with pooled anti-A,B from random donors with negative antibody screens, followed by elution studies were performed for the confirmation of weakly expressed A antigen. Finally, samples were referred for molecular ABO genotyping. **Results/Findings:** Three donors previously thought to be group O reacted weakly with anti-A,B purchased from one commercial manufacturer, but did not react with anti-A,B from two other manufacturers. Adsorption and elution studies confirmed the presence of A antigen on all three donor's cells. Molecular typing indicated the presence of Aw01 or Aw02 alleles, either of which may alter the efficacy of the enzyme glycosyltransferase to produce A antigen. **Conclusion:** The findings raise the question: do current donor ABO typing procedures allow the opportunity to detect individuals with weak subgroups? Furthermore, what is the value of revealing these weak subgroups? Moreover, once a weak subgroup is detected, the donor is deferred by the blood center to avoid possible typing discrepancies and delays in patient care by the transfusion service. Utilizing A2 cells or anti-A,B from multiple manufacturers or implementing molecular ABO typing are currently cost prohibitive options when broadly applied; however these tests may be required on a case-by-case basis. Lookback protocols gave no indication that the transfusion of these very weak A subgroups to type O patients caused any adverse reactions. Further clinical correlation with additional donors who are serologically group O, but may actually have weakened expression of A or B antigen, is required to determine the importance of routinely identifying donors with these rare ABO types.

Disclosure of Commercial Conflict of Interest

K. Bowman: Nothing to disclose; C. Hintz: Nothing to disclose; C. Kreutzer: Nothing to disclose; T. Le: Nothing to disclose; C. Melland: Nothing to disclose; F. West: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Bowman: Nothing to disclose; C. Hintz: Nothing to disclose; C. Kreutzer: Nothing to disclose; T. Le: Nothing to disclose; C. Melland: Nothing to disclose; F. West: Nothing to disclose

SP223
Blood Group A Sub Typing
M T Aguilucho[1] (maguilucho@njsharingnetwork.org), D S King[1], P Abes[1], J Szafran[1], P Rao[1]. *[1]Laboratory, NJ Sharing Network, New Providence, NJ, United States*

Background/Case Studies: ABO Group A sub typing is not routinely tested or reported in the blood bank industry. For the purpose of organ transplantation Blood Group A subgroup is a vital result. As ABO incompatible transplants become more common, new challenges arise in obtaining a clear A subgroup. Depending on the patient's history, reagents used and methodology, discrepancies occur. The way to differentiate subgroup A1 from weaker subgroups such as A2, is to type cells using an Anti-A1 lectin. Inconsistent results and interpretation with A1 lectin lead to discrepancies between laboratories and even different samples from the same patient. **Study Design/Methods:** Weak reactions with A1 lectin on certain patient samples prompted further testing using A1 lectins from various vendors, lectins with different specificities and even different methodologies. Patients were tested with Anti-A1 lectin, Anti-H lectin and finally by molecular technology (performed at NYBC). **Results/Findings:** Results are displayed in the table below. All samples were tested with Anti-A1 lectin. Not all samples had testing performed with Anti-H lectin. All samples had molecular confirmation. Results were as expected. Samples with weak Anti-A1 reactions suspected of being A2, were confirmed as A2 by molecular testing. **Conclusion:** Based on the results presented in the table below, we conclude that weak positive reactions with A1 lectin, by tube methodology, are present with certain patients that are phenotypically A2 or A2B. In some instances a rare allele or A intermediate allele may be present. These "false +" reactions are not typically present when testing with A1 lectin is performed by the gel method (not supported by vendor). Adding an additional lectin, Anti-H, will help to differentiate A1 vs A2 cells since Anti-H is expected to react strongly with A2 cells and very weakly or not at all with A1 cells.

Disclosure of Commercial Conflict of Interest

P. Abes: Nothing to disclose; M. T. Aguilucho: Nothing to disclose; D. S. King: Nothing to disclose; P. Rao: Nothing to disclose; J. Szafran: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. Abes: Nothing to disclose; M. T. Aguilucho: Nothing to disclose; D. S. King: Nothing to disclose; P. Rao: Nothing to disclose; J. Szafran: Nothing to disclose

Results

	Anti-A1 Lectin (tube method)	Anti-A1 Lectin (gel method)	Anti-H Lectin (tube method)	Anti-H Lectin (gel method)	ABO Molecular Genotype	Predicted Molecular Phenotype
68261	Weak +	Neg	4+	4+	*A2/*O1	A2
66838	Weak +	Neg	NT	NT	*A2/*A2	A2
66844	Neg*	Neg	3+	NT	*A201/*O09	A2
66087	Neg*	Neg	NT	NT	A2/O1	A2
60752	Weak +	Neg	NT	NT	A2/B101	A2B
57390	1+	mf	NT	NT	*A2/*O1	A2
68876	Neg*	Neg	3+	4+	A2/O1	A2
68263	1+	NT	3+	NT	*A2/*O1	A2

*Follow up was performed due to discrepancies with another laboratory. NT = Not Tested

SP224

Two New *B* Alleles, *ABO*B(926A 930G)* and *ABO*B(502G)*, are Involved in ABO Typing Discrepancies

S Vege[1] (svege@nybloodcenter.org), K Hue-Roye[1], P Patel[1], P M Santos[2], C Ewing[2], C Lomas-Francis[1], C M Westhoff[1]. *[1]Genomics Laboratory, New York Blood Center, Long Island City, NY, United States; [2]Blood Bank, JFK Medical Center, Edison, NJ, United States*

Background/Case Studies: While traditional serology techniques are the standard for characterizing the common ABO antigens, molecular genotyping is often needed for resolving ABO typing discrepancies. As over 100 A and B alleles have been reported, ABO gene sequencing is essential for predicting the presence of a specific subgroup. Alleles encoding A subgroups outnumber the B subgroup alleles reported to date. The ABO alleles in a Hispanic donor are investigated here whose ABO RBC typing was discrepant with the plasma reactivity, and in an Asian mother-child pair presenting with a maternal type of Group AB whilst her baby appeared as Group O. **Study Design/Methods:** Serologic testing was performed by standard tube agglutination. Genomic DNA from WBCs was used for duplex polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) (Vox Sang 69: 242, 1995). Sequence analysis was performed on amplified products of *ABO* exons 6 and 7. **Results/Findings:** For the donor sample, RBCs were non-reactive with Gamma-clone (Immucor) anti-A, -yB, and -A,B and very weakly reactive with Ortho BioClone anti-B and -A,B. The serum was reactive with A_1 and A_2 cells and non-reactive with B cells, suggesting a B subgroup. By PCR-RFLP assay, B and O alleles were present. Sequencing of exons 6 and 7 confirmed a deletion O allele, specifically *O11*, and identified a B allele with a novel change, c.926G > A, predicted to encode a p.Tyr309Cys amino acid change. Additionally this allele did not have the characteristic B nucleotide change c.930G > A, which is silent and does not encode an amino acid change. For the mother-child pair, the mother's RBCs were strongly reactive with anti-A, and -A,B, and moderately reactive with anti-B. Her plasma was non-reactive with A_1, A_2, and B cells. The baby's untreated RBCs were non-reactive with anti-A, -B, and -A,B whereas the baby's papain treated RBCs reacted very weakly with the anti-B and -A,B reagents. By PCR-RFLP, the maternal sample carried A^2/B alleles and the baby had B/O^1 alleles. Gene sequencing confirmed the A^2 allele in the maternal sample and the O1 allele in the baby. Both mother and child were found to carry a new B allele with c.502C > G, predicted to encode p.Arg168Gly. **Conclusion:** Two novel B alleles were identified: one with c.926G > A (p.Tyr309Cys) and lacking the silent c.930G > A change in a Hispanic donor associated with very weak B antigen expression (*ABO*B[926A, 930G*]); and another with c.502C > G (p.Arg168Gly) in an Asian mother and her child (*ABO*B(502G).* The very weak B antigen was not readily detected on the newborn's RBCs, as ABO antigens are not fully expressed at birth. This case illustrates the application of ABO molecular testing to resolve situations such as apparent maternal exclusion.

Disclosure of Commercial Conflict of Interest

C. Ewing: Nothing to disclose; K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; P. Patel: Nothing to disclose; P. M. Santos: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Ewing: Nothing to disclose; K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; P. Patel: Nothing to disclose; P. M. Santos: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

SP225

A Novel *B* Allele Causes a Weak B Phenotype Identified with a Single Source of Anti-B Monoclonal Reagent

G A Denomme[1] (greg.denomme@bcw.edu), K M Annen[1], M Schanen[1], D Berchanskiy[1], M Kalvelage[1], T Pugh[1], N Pasyak[1], M H Yazer[2,3]. *[1]Immunohematology, BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Department of Pathology, University of Pittsburgh, Pittsburgh, PA, United States; [3]The Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: Weak hemagglutination observed with murine monoclonal anti-A and anti-B typing reagents is the hallmark presentation of a subgroup allele. Often, the weak expression of A or B is observed when the RBCs are tested with a variety of murine monoclonal reagents. Studied here is a 43-year-old, female, Caucasian group B blood donor whose RBCs reacted strongly with three FDA-licensed reagents but weakly with one anti-B murine monoclonal reagent. **Study Design/Methods:** The donor's RBCs and plasma were tested on the PK7300 automated instrument and by manual tube methods after discovery of weak agglutination. The reactivity of the plasma isoagglutinins was tested by standard tube method. Genomic DNA and BigDye termination dideoxynucleotide sequencing was used to determine the nucleotide sequence of *ABO* exons 6 and 7 and flanking intronic regions. **Results/Findings:** The RBCs and plasma of a donor with a history of 60 lifetime donations demonstrated normal serological reactivity on routine testing using the PK7300 instrument. However, weak (w+) reactivity was observed with an FDA-licensed murine monoclonal anti-B during ABO group confirmation before the unit entered the inventory of a local hospital's transfusion service. The unit was sent to the reference laboratory for analysis. Weak reactivity with immediate spin testing was confirmed and 3+ hemagglutination was observed after a 10 minute room temperature incubation and re-centrifugation. The red cells were tested with three other sources of murine monoclonal anti-B; two produced strong (4+) immediate spin reactivity and one produced 3+ mixed field hemagglutination. *ABO* exon 6 sequencing revealed a 261ΔG, 297A > G, and a novel 361T > C (Phe121Leu). *ABO* exon 7 had SNPs consistent with the presence of *O01* and *B101* alleles. **Conclusion:** The 361T > C SNP (Phe121Leu) characterized in this donor represents a previously unreported *B* variant allele. Phe121 lies within a beta sheet proximal to the Manganese ion. Both phenylalanine and leucine are non-polar amino acids. The phenylalanine to leucine amino acid substitution is highly conserved, and is therefore not expected to alter the enzyme's tertiary structure. The requirement for incubation to obtain strong hemagglutination with one of four murine monoclonal antibodies, and 3+ mixed field in another, suggests that the Group B glycosyltransferase has subtle but altered enzymatic activity. This work underscores that careful hemagglutination studies of A and B expression can identify variant ABO glycosyltransferases.

Disclosure of Commercial Conflict of Interest

K. M. Annen: Nothing to disclose; D. Berchanskiy: Nothing to disclose; G. A. Denomme: Nothing to disclose; M. Kalvelage: Nothing to disclose; N. Pasyak: Nothing to disclose; T. Pugh: Nothing to disclose; M. Schanen: Nothing to disclose; M. H. Yazer: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. M. Annen: Nothing to disclose; D. Berchanskiy: Nothing to disclose; G. A. Denomme: Nothing to disclose; M. Kalvelage: Nothing to disclose; N. Pasyak: Nothing to disclose; T. Pugh: Nothing to disclose; M. Schanen: Nothing to disclose; M. H. Yazer: Nothing to disclose

SP226

_B303_ is the Most Frequent Allele among Taiwanese B_3/AB_3 Donors

M Yang[1] (menghua.tp@blood.org.tw), M Chen[1], Y Hung[1], C Hung[1]. *[1]Taipei Blood Center, Taiwan Blood Services Foundation, Taipei, Taiwan*

Background/Case Studies: B_3 / AB_3 phenotypes were found over 67% among ABO subgroups (including para-Bambay) in Taiwan. B_3 phenotype was first reported in 1972 and it was characterized by mixed filed agglutination with anti-B and anti-A,B, and absence of anti-B in the serum. The agglutinated and unagglutinated cells of mixed-filed reaction can be observed through microscope and column agglutination technique. *B303* and *B304* alleles were published among Taiwanese, which *B303* harbored an [IVS3] + 5G --> A mutation and *B304* allele harbored a 247G --> T mutation. Thirty-nine cases of B_3 phenotype and 26 cases of AB_3 phenotype were randomly collected for this study to analyze the phenotypic characters and the variation of their *ABO* gene. **Study Design/Methods:** A total of 65 cases of B_3 / AB_3 phenotypes were collected. Serological phenotyping were done by commercially available monoclonal antisera and RBC panels. Agglutinations with monoclonal anti-B were observed microscopically. Column agglutination technique was utilized to detect the mixed-field reaction. Sequence analysis of *ABO* gene was done as well. **Results/Findings:** In the 65 cases of B_3 / AB_3 phenotypes, mixed-field agglutinations with monoclonal anti-B were observed. Weak anti-B antibodies in serum were found in 13 cases. All cases were genotyped as *B/O* or *A/B* heterozygotes with common *O01*, *O02* or *A102* alleles. Analyzing the *B* alleles, 49 of them have *B303* allele, five have *Bw12* allele (278C>T), four have *Bw11* allele (695T>C), one has *Bw03* allele (721C>T, which was also shown in *Aw04* allele) and one has *Bw05* alleles (539G>A, which was also shown in *A304* allele). In addition, one case of *B111* allele (306C>T) and four cases of *B101* allele were found in AB_3 cases. **Conclusion:** Column agglutination technique is a good way to detect mixed-field agglutination of B_3 / AB_3. Sequence analysis is required for detailed sub-grouping. From this study, *B303* is the most frequent allele, which is about 75% (49/65) of B_3 / AB_3 cases. Eleven cases are found to have *Bw* alleles, including *Bw03*, *Bw05*, *Bw11*, *Bw12* alleles. Weak anti-B occurring rate is 20% (13/65). Ten of *B303* cases and three of *Bw11* cases have weak anti-B in sera but not continuously shown in every donation of each donor. Notably, four AB_3 cases have common *B101* allele and one has *B111* allele.

Disclosure of Commercial Conflict of Interest

M. Chen: Nothing to disclose; C. Hung: Nothing to disclose; Y. Hung: Nothing to disclose; M. Yang: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Chen: Nothing to disclose; C. Hung: Nothing to disclose; Y. Hung: Nothing to disclose; M. Yang: Nothing to disclose

Red Blood Cells: Alloimmune and Autoimmune Hemolysis

SP227

Life Threatening Anti-Kpb-Induced Severe Delayed Hemolytic Transfusion Reaction

E A Fadeyi[1] (efadeyi@wakehealth.edu), P Persad[1], J H Simmons[1], M Jones[1], G J Pomper[1]. *[1]Pathology, Wake Forest University School of Medicine, Winston-Salem, NC, United States*

Background/Case Studies: Kpb is a high-frequency red blood cell antigen occurring in 100% of all populations and Kpb antibody presents in less than one in 5000 Caucasians. Anti-Kpb is a member of the subgroup of Kell antigens and has been reported to produce only mild hemolysis; others however have noted that the Kpb antibody must be taken seriously, since marked hemolysis may occur. The experience here with this case would support the latter conclusion. Since the antigen is universally prevalent there is no reported case of a delayed hemolytic transfusion reaction (DHTR). There have been cases reported of autoantibodies to Kell related blood group antigens. A case of a severe DHTR attributed to anti-Kpb after multiple red blood cell transfusions is being reported. **Study Design/Methods:** A 66-year-old Caucasian woman presented with a history of multiple transfusions for recurrent lower gastrointestinal hemorrhages. Three years prior to presentation, the patient had received 18 units of packed red blood cells for lower gastrointestinal bleeding with several negative antibody screens. Six months later, she received six additional red cell units for anemia, with a negative antibody screen. One month prior to presentation, the patient was again admitted for anemia, received six additional units and still demonstrated a negative antibody screen. Following the most recent treatment for anemia, the patient presented with a life threatening hemolytic anemia requiring blood transfusions. **Results/Findings:** Upon presentation, the laboratory evidence of hemolysis included anemia (hemoglobin of 5.3 g/dL), an elevated LDH, and low haptoglobin. Shortly after admission, her hemoglobin further decreased to a nadir 5.0 g/dL. The serology demonstrated a positive antibody screen consistent with anti-K and anti-Kpb in plasma, and a positive direct antiglobulin test for IgG and complement (C3d). Further serologic work up showed only anti-Kpb in the eluate consistent with a DHTR. Because the identification of an antibody to a high frequency antigen is complicated serologically, the patient's blood was sent to a reference laboratory for a red blood cell genotype. The red cell genotyping demonstrated a predicted Kp (a+b-) phenotype consistent with alloimmunization to anti-Kpb. A nationwide search produced 8 units of Kpb negative, K negative red blood cells compatible with the patient. They were transfused over the course of two weeks without incident and her hemoglobin stabilized at 7.3 g/dL prior to discharge. **Conclusion:** This case demonstrates the clinical importance of an antibody to a high frequency Kell subgroup antigen. Although extremely rare, the antibody can occur in the setting of a DHTR. The difficulty of obtaining compatible blood would involve a nationwide search of the rare donor registry.

Disclosure of Commercial Conflict of Interest

E. A. Fadeyi: Nothing to disclose; M. Jones: Nothing to disclose; P. Persad: Nothing to disclose; G. J. Pomper: Nothing to disclose; J. H. Simmons: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. A. Fadeyi: Nothing to disclose; M. Jones: Nothing to disclose; P. Persad: Nothing to disclose; G. J. Pomper: Nothing to disclose; J. H. Simmons: Nothing to disclose

SP228

A Suspected Case of Delayed Hemolytic Transfusion Reaction (DHTR) Caused by Anti-Fy^b in a Patient with Sickle Cell Disease (SCD)

M A Mota[1] (marizamota@einstein.br), L D Santos[1], E P Bastos[1], L F Almeida[1], J M Furtado[1], F Rocha[1], M G Aravechia[1], L Castilho[2], J M Kutner[1]. *[1]Hospital Israelita Albert Einstein, Sao Paulo, Brazil; [2]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: Fy^b is known to be a poor immunogen estimated to be 20 times less immunogenic than Fy^a . Patients of African origin with the Fy(b-) phenotype, in special those who have Fy(a-b-) red blood cells (RBCs) may possess an allele encoding RBC Fy^b antigen silenced by a GATA box mutation *(FY*67t>c)* and therefore, they would not be expected to make anti-Fy^b . An usual case of a delayed hemolytic transfusion reaction (DHTR) caused by anti-Fy^b in a patient with sickle cell disease (SCD) phenotyped as Fy(a+b-) is reported on here. A 22-year-old female chronically transfused patient with SCD alloimmunized to C and E antigens receiving phenotype-matched RBCs for Rh and K presented 8 days after the last transfusion with severe anemia and signs and symptoms resembling vaso-occlusive crisis. Her RBCs typed O, D+C-E-c+e+, K-, Fy(a+b-), Jk(a+b+). Antibody screening showed an additional antibody in her serum and an anti-Fy^b was identified. **Study Design/Methods:** All serologic tests were performed in gel. Antibody screening and identification were performed with LISS/Coombs and Enzyme gel cards (BioRad and Grifols) using commercially prepared panel cells. FY/GATA genotyping was performed in the patient's DNA sample using the HEA BeadChip (BioArray Solutions, Immucor). The patient had received O R_or, K-, Fy(a-b+) RBC units in her last transfusion. Serologic features of anti-Fy^b was analyzed and the clinical significance of the antibody was evaluated by analysis of the hemoglobin (hgb) levels pre and post-transfusion and determined by change from baseline pre-transfusion hgb and percentage (%) of hgb S. **Results/Findings:** The serum of the patient was reactive with all C- E- Fy(b+) RBCs and negative with all C-E- Fy(b-) RBCs. The patient's RBCs were Fy(a+b-) and her FY genotyping showed that she was *FY*A/FY*A -67t*. On the basis of clinical and laboratory findings, DHTR was strongly suspected. **Conclusion:** In this report, it was demonstrated that anti-Fy^b in SCD patients can be clinically significant. Although, the occurrence of anti-Fy^b is rare in patients with SCD it is important to perform genotyping before making the assumption that a patient with the Fy(b-) phenotype will not make anti-Fy^b if transfused with Fy(b+) blood.

Disclosure of Commercial Conflict of Interest

L. F. Almeida: No Answer; M. G. Aravechia: Nothing to disclose; E. P. Bastos: No Answer; L. Castilho: Nothing to disclose; J. M. Furtado: No Answer;

J. M. Kutner: Nothing to disclose; M. A. Mota: Nothing to disclose; F. Rocha: No Answer; L. D. Santos: No Answer

Disclosure of Grants Conflict of Interest

L. F. Almeida: No Answer; M. G. Aravechia: Nothing to disclose; E. P. Bastos: No Answer; L. Castilho: Nothing to disclose; J. M. Furtado: No Answer; J. M. Kutner: Nothing to disclose; M. A. Mota: Nothing to disclose; F. Rocha: No Answer; L. D. Santos: No Answer

SP229

Platelet Additive Solution Reduces ABO Antibody Titers

M Bellone[1] (mbellone@gmail.com), S Paradiso[1], K C Greco[1], B J Greco[1], B H Shaz[1]. *[1]New York Blood Center, New York, NY, United States*

Background/Case Studies: While it is always preferable to transfuse patients with ABO-compatible single donor platelets (SDPs), inventory limitations result in transfusion services issuing SDPs with minor-ABO incompatibility. Such units contain a large volume (200-400 ml) of incompatible donor plasma that contains a naturally occurring antibody against the A and/or B antigen the donor lacks. This puts recipients, especially those with smaller blood volumes, at risk of developing acute passive alloimmune intravascular hemolysis. In order to prevent this, all group O SDP donations at our institution undergo anti-A titer testing and those with titers less than 100 are labeled as such. Hospital transfusion services are therefore alerted that such units should be transfused to group O recipients. This study was undertaken in order to determine if platelets stored in platelet additive solution (PAS) (65% additive solution/35% plasma) are low titer. **Study Design/Methods:** The isohemagglutinin titer was determined using conventional tube testing (CTT). A 1:100 dilution of 100 donor samples was prepared collected in EDTA-containing vacutainer tubes (ie pre-PAS) and 100 samples from corresponding SDP-PAS units. Agglutination of reverse typing cells was assessed immediately following centrifugation and gentle resuspension at room temperature. Donors consisted of 33 group O, 44 group A, 18 group B, and 5 group AB. Anti-A titration routinely performed on 2193 group O SDPs in a 4 month period was used to determine the baseline incidence of a high titer anti-A in our donor population. **Results/Findings:** A total of 11 pre-PAS samples had titers greater than 100, including eight group O donors and three group B donors. The high titer group O donors consisted of four males and five females, while all three high titer group B donors were male. None of the corresponding SDP-PAS units were found to have titers greater than 100. A total of 211 group O SDP units were found in a 4-month period to have an anti-A titer greater than 100, yielding a monthly average baseline incidence of 9.6%. **Conclusion:** The results indicate that PAS effectively reduces donor isohemagglutinin titers to below 100. It is likely a result of removing 65% of plasma as well as diluting residual antibodies by PAS. This presumably mitigates the risk of acute passive alloimmune hemolysis while also enabling transfusion services to more effectively manage their SDP inventory. If the results are confirmed by larger studies, SDP-PAS can serve as a universal SDP that can be safely transfused in all patients regardless of ABO group.

Disclosure of Commercial Conflict of Interest

M. Bellone: Nothing to disclose; B. J. Greco: Nothing to disclose; K. C. Greco: Nothing to disclose; S. Paradiso: Nothing to disclose; B. H. Shaz: Haemonetics, no financial relationship, Other, Fenwal, no financial relationship, Other

Disclosure of Grants Conflict of Interest

M. Bellone: Nothing to disclose; B. J. Greco: Nothing to disclose; K. C. Greco: Nothing to disclose; S. Paradiso: Nothing to disclose; B. H. Shaz: Nothing to disclose

SP230

A Case of Tk-activation

C Yamada[1] (yamadac@umich.edu), N Shaller[1], M B Elkins[1], R D Davenport[1]. *[1]University of Michigan, Ann Arbor, MI, United States*

Background/Case Studies: The most common cause of polyagglutination is exposure of antigenic determinants, which are part of the structure of the normal red cell membrane, by the action of bacterial enzymes. Several determinants have been discovered, but even the most common one among them (T-activation) is rare in adults, and the mechanism of hemolysis is not fully understood. A case of extravascular hemolysis with Tk-activation with negative polyagglutination is reported here. **Study Design/Methods:** A 50-year-old female with a history of stage IV high-grade papillary serous ovarian cancer status post resection in 2008 presented with increasing right lower quadrant abdominal pain, clear vaginal discharge for 2 weeks, malaise, nausea, vomiting, fever, and chills for one day, and recent swelling of her lower legs up to the upper thigh. She received multiple courses of chemotherapy for extensive metastasis, most recently completed etoposide one month prior to presentation. Physical examination demonstrated known masses on lower abdomen and vaginal wall. Laboratory values showed only mild anemia (Hgb 10.3g/dL) and mild neutrophilia (87.2%). CXR and CT showed known pulmonary and pelvic metastases. Piperacillin/tazobactam and vancomycin were started and the patient developed hemolytic anemia on next day with Hgb of 7.6g/dL, LDH of 1135IU/L and haptoglobin of <10mg/dL. At this point, she had mild renal failure (Cr 1.6mg/dL) with AST of 60IU/L. Peripheral smear showed numerous spherocytes, burr cells, and large platelets. Blood bank tests were all negative except positivity with Arachis hypogea lectin. Tk-activation was suspected and levofloxacin started. Her hemolysis subsided in 2 days and the patient was discharged after 9 days in stable condition. **Results/Findings:** Spherocytes in peripheral smear and lab data supported extravascular hemolysis. Antibody detection tests were negative including screening and PEG enhancement test. Eluate, DAT, enhanced DAT with ficin treated cells, and cold panel were also negative. Tests for polyagglutination with both normal adult and cord blood sera were negative; however, a lectin panel showed positivity with Arachis hypogea suggesting Tk-activation. Microbiology tests on feces showed few vancomycin resistant Enterococcus species, but blood/urine cultures were negative. High fibrinogen and D-dimer with negative DVT scan of legs suggested possibility of DIC. **Conclusion:** This patient had extravascular hemolysis most likely due to Tk-activation. Prompt response to antibiotic treatment also supports the diagnosis. In this case, the mechanism of hemolysis from Tk-activation is probably due to shortened RBC survival by altered RBC membrane sialic acid content rather than more common complement activation by antigen-antibody reaction.

Disclosure of Commercial Conflict of Interest

R. D. Davenport: Nothing to disclose; M. B. Elkins: Nothing to disclose; N. Shaller: Nothing to disclose; C. Yamada: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. D. Davenport: Nothing to disclose; M. B. Elkins: Nothing to disclose; N. Shaller: Nothing to disclose; C. Yamada: Nothing to disclose

SP231

Clinical and Reference Lab Characteristics of Patients with Suspected Direct Antiglobulin Test (DAT) Negative Immune Hemolytic Anemias

M S Karafin[1,2] (matthew.karafin@bcw.edu), G A Denomme[1], M Schanen[1], J L Gottschall[1,2]. *[1]BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Department of Pathology, Medical College of Wisconsin, Milwaukee, WI, United States*

Background/Case Studies: Clinical evidence of warm autoimmune hemolytic anemia is present in 2-11% of patients whose direct antiglobulin test (DAT) is negative. The clinical underpinnings associated with DAT negative immune hemolysis is poorly understood, and the current study aims to further define the clinical characteristics associated with this form of immune hemolytic anemia. **Study Design/Methods:** A 19-question survey was sent to all referring labs that sent patient samples for an enhanced DAT during 2011-2012. The enhanced DAT evaluation involves Gel, PEG, and 4C LISS wash DAT enhancement methods. The survey requested additional clinical information about each patient, including nadir hemoglobin, bilirubin, LDH, haptoglobin, current diagnoses, treatment modalities, and patient outcomes. **Results/Findings:** 309 samples were received for an enhanced DAT evaluation. Of these, 30 (9.7%) had a positive enhanced DAT (17 [5.5%] had a positive Gel, 1 [0.3%] had a positive PEG, and 16 [5.2%] had a positive 4C LISS), and 16 (5.2%) had a positive eluate. Detailed clinical and laboratory information was available on 24 patients with the enhanced investigation. Of these, 9 (37.5%) were found to have a positive DAT. Together, the reported mean nadir hemoglobin was 7.1 (range 3.2-11.4) and the LDH was 918 (range 200-4562). More than half (14) presented with a haptoglobin <10, and 13 (54.2%) reported spherocytes on peripheral smear. Respondents reported warm autoimmune hemolytic anemia as part of the initial working diagnosis prior to the enhanced DAT in 13 cases. Also, 17 (70.8%) reported using steroids, 17 (70.8%) did not change their treatment plan, and 13 did not change their diagnosis as a result of the enhanced DAT findings. Lastly, the survey revealed 15 patients had their hemolysis resolve within 2.5 weeks to 7 months or became stable with treatment, and only one died as a result of the reported hemolytic anemia. **Conclusion:** It was found that immune hemolysis detected only by enhanced methods is uncommon, and occurs in about 10% of evaluated cases. The survey revealed that DAT negative patients present with anemia, increased LDH and low haptoglobin, and these patients are generally treated as having an immune hemolytic anemia.

Respondents generally did not change their treatment plan as a result of the enhanced results, and the majority of DAT negative anemias resolved or became stable with treatment.

Disclosure of Commercial Conflict of Interest

G. A. Denomme: Nothing to disclose; J. L. Gottschall: No Answer; M. S. Karafin: Nothing to disclose; M. Schanen: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. A. Denomme: Nothing to disclose; J. L. Gottschall: No Answer; M. S. Karafin: Nothing to disclose; M. Schanen: Nothing to disclose

SP232

Autoimmune Hemolytic Anemia with Low Affinity Autoantibodies Presenting as DAT Negative Hemolysis Caused by a Left Ventricular Assist Device (LVAD)

E A Fadeyi[1] (efadeyi@wakehealth.edu), P Persad[1], J H Simmons[1], M Jones[1], G J Pomper[1]. *[1]Pathology, Wake Forest University School of Medicine, Winston-Salem, NC, United States*

Background/Case Studies: Left Ventricular Assist Devices (LVAD) are indicated in cardiac failure as a bridge to myocardial recovery, or heart transplantation. Hemolysis is a complication of several LVAD types. Destruction of red blood cells (RBCs) is the result of wall shear stress, flow acceleration and interaction with artificial surfaces. Hemolysis is also a valuable parameter to assess the proper functioning of the LVAD. An unexpected increase of hemolysis can indicate a thrombus, shift or malpositioning of the LVAD inflow cannula. Distinguishing between mechanical (LVAD) and immune mediated hemolysis can be difficult. **Study Design/Methods:** A 44-year-old female with nonischemic cardiomyopathy presented with progressive anemia secondary to hemolysis from her LVAD. Six months after receiving her first LVAD implantation, a replacement LVAD was needed secondary to pump thrombosis. She continued to have occasional schistocytes on her peripheral smear suggestive of ongoing hemolysis despite replacement of her second LVAD. Mechanical hemolysis was ruled out by repeated high resolution ultrasound showing a normal functioning LVAD. Re-evaluation of her anemia revealed a persistent mildly elevated LDH with low haptoglobin, negative direct antiglobulin test (DAT) and antibody screen. Over 10 months, she had received a total of 81 RBCs for chronic anemia of which 36 RBCs were given following LVAD replacement. Her antibody screen and DAT remained negative throughout. She was treated with iron sucrose, IVIG, methylprednisolone, and rituximab with no improvement in her anemia. Her LDH continued to be mildly elevated. **Results/Findings:** Additional reference serologic testing demonstrates a positive DAT when using a Gel enhancement technique and a cold LISS wash; however, the eluate was negative. A low affinity IgG autoantibody was detected in the patient's serum when using a 4C LISS wash in the indirect antiglobulin test which may cause severe hemolytic anemia. There were no alloantibodies identified. Both DAT and antibody screen for IgA and IgM were negative by serology and by flow cytometry. A splenectomy stabilized her hemoglobin; however, she remained anemic. The patient eventually developed an LVAD related thrombus, progressively declined, and expired. **Conclusion:** DAT negative hemolytic anemia may require additional testing using non-routine methods to rule out low affinity warm autoantibodies. This case demonstrates how low affinity IgG autoantibodies can lead to a severe refractory autoimmune hemolytic anemia and how difficult it is to detect serologically. A DAT negative hemolytic anemia needs a thorough serologic evaluation to rule out clinically significant low affinity warm autoantibodies.

Disclosure of Commercial Conflict of Interest

E. A. Fadeyi: Nothing to disclose; M. Jones: Nothing to disclose; P. Persad: Nothing to disclose; G. J. Pomper: Nothing to disclose; J. H. Simmons: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. A. Fadeyi: Nothing to disclose; M. Jones: Nothing to disclose; P. Persad: Nothing to disclose; G. J. Pomper: Nothing to disclose; J. H. Simmons: Nothing to disclose

SP233

Auto-Anti-Kp^b in a Lymphoma Patient

J Crane[1] (jecrane@uic.edu), C M Howard-Menk[1], H Patel[2], M Papari[1]. *[1]LifeSource Blood Center, Rosemont, IL, United States; [2]Pathology, John H. Stroger, Jr. Hospital of Cook County, Chicago, IL, United States*

Background/Case Studies: Warm autoimmune hemolytic anemia (WAIHA) is most commonly due to IgG autoantibodies directed against Rh proteins and the Wr^b antigen. Sporadic reports of other specificities have also been reported. A case of auto-anti-Kp^b in a lymphoma patient is reported here. **Study Design/Methods:** The patient is a 37-year-old male with Stage IV Hodgkin lymphoma, initially diagnosed in December 2009. He is status post 6 cycles of chemotherapy. He presented to the hospital in December 2012 and was found to have a significant drop in hemoglobin (15.3 g/dL on 2/19/12→8.2 g/dL on 11/15/12→5.3 g/dL on 12/14/12). Direct anti-globulin test (DAT) was positive for both IgG and complement, indirect bilirubin was elevated, and haptoglobin was decreased, with reticulocyte count 14%, suggestive of a hemolytic anemia. Antibody screen was positive and a sample was sent to the blood center for identification. **Results/Findings:** A 3+ anti-Kp^b antibody was detected in the eluate. As the patient stated he had never been transfused, and had a red cell phenotype positive for Kp^b, this antibody was defined as an autoantibody with Kp^b specificity. He was transfused two units of Kp^b -negative packed red blood cells (PRBC) on December 16th, followed by a third unit December 17th. Hemoglobin improved to 9.5 g/dL after the third unit. **Conclusion:** Most antibodies in the Kell system arise secondary to a sensitizing event, such as pregnancy or transfusion. In patients with Kell-related autoimmunity, 75% is specific for anti-K. Reports of other specificities in the Kell system are documented. Auto-anti-Kp^b, as shown in this patient, was first documented in 1972, and presented with an acquired depression of Kell antigens. The cause of the depression was unknown however other reports suggest it may be environmentally-related. Antigen typing of our patient demonstrated a 1-2+ Kp^b compared to a control value of 2+, suggestive of weakened expression. Auto-anti-Kp^b with associated depression of Kp^b has also been reported in infancy. Auto-anti-Kp^b without associated depression of Kell antigens has also been seen. Regarding transfusion support, units negative for Kp^b were available for this patient. This may not always be the case as the antigen has an almost 100% frequency. As with other autoimmune specificities (ex. e), it is not clear if antigen-positive cells will have less survival. Case reports have been published advocating either viewpoint of antigen-positive or antigen-negative red cell transfusion support. Given the prevalence of AIHA in leukemia/lymphoma patients, it is likely there are other cases of rare presentations of AIHA. Reports of rare cases are important to further understanding in their treatment.

Disclosure of Commercial Conflict of Interest

J. Crane: Nothing to disclose; C. M. Howard-Menk: Nothing to disclose; M. Papari: Nothing to disclose; H. Patel: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Crane: Nothing to disclose; C. M. Howard-Menk: Nothing to disclose; M. Papari: Nothing to disclose; H. Patel: Nothing to disclose

SP234

Autoimmune Hemolytic Anemia and Its Relationship With Hemophagocytic Lymphohistiocytosis (HLH)

J Atallah[2] (jeanpaul_atalla@yahoo.com), M Dhar[2], M N Siddique[2], M T Hurford[1], M Asgari[1], E Isaak[1]. *[1]Transfusion Medicine, Staten Island University Hospital, Staten Island, NY, United States; [2]Hematology/Oncology, Staten Island University Hospital, Staten Island, NY, United States*

Background/Case Studies: Hemophagocytic lymphohistiocytosis (HLH) is a rare disorder usually seen in children. In cases without well-known genetic mutations, the diagnosis is based on a constellation of clinical criteria. One criterion is erythrocytopenia, although the mechanism of anemia has not been well characterized. Hemophagocytosis, which increases as disease progresses, causes decreased red cell production, but autoimmune mediated hemolysis and its incidence in HLH have not been described. **Study Design/Methods:** The study is a retrospective case report. **Results/Findings:** A 46-year-old man with history of beta thalassemia minor, of bariatric surgery, and of idiopathic thrombocytopenic purpura in remission was admitted for generalized weakness and jaundice. His physical exam showed icterus and splenomegaly. His initial hemoglobin was 10.6 g/dl and his platelet count was 309,000/microliter. His total bilirubin was 15.9 mg/dl with a direct bilirubin of 1.6 mg/dl. Subsequently, his hemoglobin fell to 4.1 g/dl despite transfusion of 26 units of PRBC. His platelet count also fell to 96,000/microliter. The direct antiglobulin test (DAT) was positive by cold wash LISS technique, and autoimmune hemolytic anemia was diagnosed. Studies for infectious etiologies were all negative. The patient was initially placed on rituximab with a small improvement in hemoglobin level over 2 days even though his overall clinical condition deteriorated .His total bilirubin peaked at 86 mg/dl during that time with both direct and indirect components. The initial ferritin was 1,544 ng/ml, but increased to 17,000 ng/ml. A liver biopsy showed cholestasis, and the first bone marrow biopsy showed a

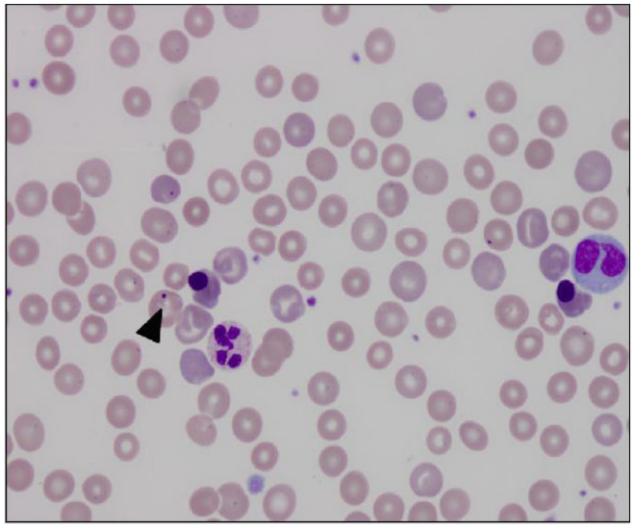
few definite foci of erythrophagocytosis. A repeat bone marrow biopsy revealed extensive erythrophagocytosis. The triglyceride level increased to 330 mg/dl, the DAT increased in strength, and he fulfilled six of the eight criteria for HLH. He had hyperbilirubinemia, splenomegaly, thrombocytopenia, elevated soluble CD25, marked hemophagocytosis, hyponatremia, and hypertriglyceridemia. He received etoposide, dexamethasone, and rituximab. An HLH mutational screen was negative. The patient received nine cycles of dexamethasone and etoposide and four weeks of rituximab with normalization of his soluble CD25, a decrease in ferritin and an increase in the hemoglobin level to 9g/dl without transfusional requirement for 4 weeks. **Conclusion:** HLH includes erythrocytopenia as a diagnostic criterion, but the etiology and features are not well characterized. An immune mediated hemolysis may be an important feature. The first case is described here of an HLH patient with erythrocytopenia due to both erythrophagocytosis and a documented autoimmune hemolytic component which increased in strength as the clinical picture deteriorated.

Disclosure of Commercial Conflict of Interest

M. Asgari: Nothing to disclose; J. Atallah: Nothing to disclose; M. Dhar: Nothing to disclose; M. T. Hurford: Nothing to disclose; E. Isaak: Nothing to disclose; M. N. Siddique: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Asgari: Nothing to disclose; J. Atallah: Nothing to disclose; M. Dhar: Nothing to disclose; M. T. Hurford: Nothing to disclose; E. Isaak: Nothing to disclose; M. N. Siddique: Nothing to disclose



SP235

A Clinical Conundrum: Evans Syndrome in an Asplenic Patient with *Babesiosis*

J J Shatzel[1] (joseph.j.shatzel@hitchcock.org), N Chu[2], G Garratty[3], K Mody[4], E Bengtson[4], N M Dunbar[5,1]. *[1]Internal Medicine, Dartmouth Hitchcock Mecial Center, Lebanon, NH, United States; [2]Geisel School of Medicine at Dartmouth, Lebanon, NH, United States; [3]American Red Cross Blood Services Southern California Region, Pomona, CA, United States; [4]Hematology-Oncology, Dartmouth Hitchcock Medical Center, Lebanon, NH, United States; [5]Pathology, Dartmouth Hitchcock Medical Center, Lebanon, NH, United States*

Background/Case Studies: Evans syndrome is characterized by the simultaneous or sequential presence of multiple autoimmune cytopenias. It is often secondary to rheumatologic disorders or lymphoid malignancies, but has not previously been associated with babesiosis. A case of Evans syndrome in an asplenic patient precipitated by active babesiosis is presented here. **Study Design/Methods:** Case report and review of the literature. **Results/Findings:** The patient's complicated past medical history made the initial work up challenging and included: a history of primary autoimmune hemolytic anemia treated by splenectomy that presented 12 years prior and required analysis from an outside reference laboratory for detection of the offending antibody; a history of Hodgkin lymphoma 4 years prior, currently in remission; and recent *babesia* infection diagnosed and treated 1 month prior. His presentation was complicated by hemolysis which again required investigation by the same outside reference laboratory, as well as large new lymphadenopathy on imaging which was initially concerning for a relapse of lymphoma. **Conclusion:** Asplenic patients are predisposed to more severe illness during babesiosis infection, and clinically significant AIHA in asplenic patients, as well as other autoimmune sequelae have been reported prior. This patient's initially negative DAT was perplexing, but Immune hemolytic anemia with negative routine serology does occur, with DAT negative AIHAs thought to represent 5%-10% of all AIHA. Autoimmune cytopenias are strongly associated with lymphoproliferative disorders. Our patient's benign lymphadnopathy was unexpected. Lastly, Evans syndrome is more than simple coincidental cytopenias but instead represents a state of severe immune deregulation. Evans syndrome may be a multiple step process and the treatment of an initial cytopenia may not prevent a second.

Disclosure of Commercial Conflict of Interest

E. Bengtson: No Answer; N. Chu: Nothing to disclose; N. M. Dunbar: Nothing to disclose; G. Garratty: Nothing to disclose; K. Mody: Nothing to disclose; J. J. Shatzel: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Bengtson: No Answer; N. Chu: Nothing to disclose; N. M. Dunbar: Nothing to disclose; G. Garratty: Nothing to disclose; K. Mody: Nothing to disclose; J. J. Shatzel: Nothing to disclose

SP236

Low-Titer Cold Agglutinin Disease: A Case Series

J M Petersen[1] (jpetersen@mednet.ucla.edu), D Dumani[1], D Goldfinger[1], Q Lu[1], A Ziman[1]. *[1]University of California-Los Angeles, Los Angeles, CA, United States*

Background/Case Studies: Cold agglutinin disease (CAD) is typically characterized by a relatively high titer (usually higher than 1000) of cold agglutinins consisting of IgM antibodies directed against the polysaccharaide antigens of the red blood cell. In normal individuals, cold agglutinins are commonly present in low titers (less than 64) without apparent hemolysis. Rare case reports exist of patients with low titer CAD (as low as 16), but to the author's knowledge, a case series of multiple patients with low titer CAD has not yet been well documented in the literature, and some of the cases in this series have titers lower than previously reported (as low as two). **Study Design/Methods:** A search for patients with documented CAD and low cold agglutinin titers (defined as ≤512) was performed in Meditech for the period of January 1, 2010 to present. Additional clinical information was obtained from a medical chart review. **Results/Findings:** Nine patients were identified with low titer cold agglutinins and accompanying hemolysis; only one patient (Patient 4) had a concurrent warm-reactive IgG autoantibody and all other causes of hemolysis were excluded (Table). The patient population consisted of one six-year-old male child and eight adults (age range of 63-90 years; five males and three females). The titers of the patients ranged from 2 to 256. **Conclusion:** While CAD is commonly seen in the context of high titer cold agglutinins (defined as >1000), in rare cases, CAD can have clinically significant hemolysis in the presence of low titers. However, in such cases the thermal amplitude may help solidify the diagnosis of CAD in the setting of hemolysis.

Disclosure of Commercial Conflict of Interest

D. Dumani: Nothing to disclose; D. Goldfinger: Nothing to disclose; Q. Lu: Nothing to disclose; J. M. Petersen: Nothing to disclose; A. Ziman: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Dumani: Nothing to disclose; D. Goldfinger: Nothing to disclose; Q. Lu: Nothing to disclose; J. M. Petersen: Nothing to disclose; A. Ziman: Nothing to disclose

Patient	Age	Sex	Titer	Thermal Amplitude	Lowest Hemoglobin ‡ (g/dl)	Direct antiglobulin test (IgG/C3)	Highest LDH* (U/l)	Haptoglobin (mg/dL)‡	Bilirubin (mg/dL)	
									Total	Direct
Patient 1	65	M	1:2	30°C	4.5	0-/2+	2101	<8	21.5	12.0
Patient 2	73	M	1:8	37°C	7.9	0-/4+	255	<8	2.9	0.9
Patient 3	70	F	1:16	30°C	5.4	0-/2+	1050	<8	4.1	0.8
Patient 4	0	M	1:16	37°C	Hemolyzed†	1+/2+	920	<8	1.5	0.5
Patient 5	72	M	1:32	30°C	Hemolyzed†	0-/3+	670	<8	1.7	NP•
Patient 6	69	M	1:32	37°C	8.9	0-/3+	491	<8	8.8	3.8
Patient 7	50	F	1:256	22-25°C	9.2	0-/4+	333	<8	3.3	NP•
Patient 8	48	F	1:256	22-25°C	10.3	0-/3+	541	<8	1.8	0.3
Patient 9	63	M	1:256	37°C	6.0	0-/3+	845	<8	47.2	23.6

† Hemoglobin analysis not performed due to hemolyzed sample. Hematocrit performed three days prior to cold agglutinin screen was 21.8% for patient 3. Hematocrit performed on day of sample for patient 4 was 25.8%.

‡ Within 30 days of the date of the cold agglutinin screen.

• NP = Not performed

* LD = Lactate dehydrogenase

SP237

A Case of Syphilitic Paroxysmal Cold Hemoglobinuria with Unusual Clinical and Serological Findings

R M Leger[1] (regina.leger@redcross.org), J L Allen[2], K Grima[2,3], G Garratty[1]. *[1]American Red Cross Blood Services, Pomona, CA, United States; [2]Blood Bank, The Brooklyn Hospital Center, Brooklyn, NY, United States; [3]American Red Cross Donor Client Support Center, Philadelphia, PA, United States*

Background/Case Studies: Classical chronic syphilitic and non-syphilitic paroxysmal cold hemoglobinuria (PCH) brought on by exposure to cold is extremely rare these days; more commonly PCH is seen as acute anemia and hemoglobinuria, without exposure to cold, in young children after a viral infection. The diagnostic test for PCH, the Donath-Landsteiner (D-L) test, demonstrates the biphasic hemolysin which most often has anti-P specificity. The direct antiglobulin test (DAT) is usually only positive for C3, and an eluate and antibody screen are usually nonreactive. A 91-year-old woman had gangrene of digits and tip of her nose, hemolytic anemia, a history of syphilis, and a positive D-L test. Gangrene has been described in PCH in early literature but is not typical, so further characterization of the D-L antibody was pursued. **Study Design/Methods:** The D-L test was performed in the presence of fresh complement with P+ and PP1Pk- RBCs; DAT and indirect antiglobulin tests (IAT) were performed and acid eluates tested. The serum, plasma and eluate were treated with dithiothreitol (DTT). Because DTT is anti-complementary, DTT-treated serum was dialyzed against phosphate buffered saline to remove the DTT, and tested with the addition of fresh complement in the D-L test. **Results/Findings:** The negative control for the routine DAT was positive; 37C washed RBCs reacted 2+ with anti-IgG and anti-C3, and were nonreactive with anti-IgM, anti-IgA and the 6% albumin control. The eluate was nonreactive with saline-suspended RBCs, but reacted 3½+ when polyethylene glycol (PEG) was added. The D-L antibody had anti-P specificity (3+); no monophasic lysin was detected. The D-L test on the eluate + complement was positive (2+). The plasma/serum was nonreactive by saline-IAT, but reacted by PEG-IAT. For both eluate and plasma, PEG caused direct agglutination of RBCs, ie, agglutinates were seen after washing for the IAT and prior to adding anti-IgG. By PEG-IAT the plasma was nonreactive with p and P1k RBCs, and the eluate was nonreactive with p RBCs (P1k RBCs not tested); thus the PEG-dependent agglutinin was also anti-P. DTT- treated plasma/eluate + PEG was nonreactive indicating the PEG-dependent agglutinin was IgM. A D-L test of dialyzed-DTT-treated serum + complement was positive, confirming the D-L biphasic hemolysin was IgG. **Conclusion:** This patient with syphilitic PCH also had gangrene of digits and nose. The DAT was positive for both IgG and C3. A PEG-dependent IgM directly agglutinating anti-P was demonstrated in this patient's plasma and eluate in addition to the IgG anti-P biphasic hemolysin. Anti-P agglutinins can be observed in syphilitic PCH in addition to the D-L antibody, but usually are not reactive above 20C and are not detected in routine tests. Reactivity of this patient's plasma and eluate by PEG-IAT could have been incorrectly interpreted as an IgG antibody except for careful observation of the RBCs during the washing phase.

Disclosure of Commercial Conflict of Interest

J. L. Allen: Nothing to disclose; G. Garratty: Nothing to disclose; K. Grima: Nothing to disclose; R. M. Leger: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. L. Allen: Nothing to disclose; G. Garratty: Nothing to disclose; K. Grima: Nothing to disclose; R. M. Leger: Nothing to disclose

SP238

Passenger Lymphocyte Syndrome (PLS) after Liver Transplantation

N Carpio[1] (carpiojarafuel@gmail.com), P Solves Alcaina[1], F Moscardo[1], A Lancharro[1], I Cano[1], M A Sanz[1]. *[1]Blood Bank, Hospital Universitari i Politècnic La Fe, Valencia, Spain*

Background/Case Studies: The passenger lymphocyte syndrome (PLS) consists of the production of antibodies by the donor passenger lymphocytes infused with the organ, against the recipient red blood cell (RBC) antigens. PLS is mostly detected in solid organ transplantation with minor ABO mismatch. Antibodies against the Rh and ABO antigens are mostly involved. The objective was to review the incidence of PLS occurred in patients underwent liver transplantation (LT) at The Hospital Univeristari I Politècnic La Fe from Valencia. **Study Design/Methods:** The LPS detected in the transfusion service for a 7–year .period (2006-2012) was reviewed. During first 3 months after LT, a direct antiglobulin test (DAT) was systematically performed in each transfusion for all patients. The DAT was performed using polyspecific antihuman globulin according to standard methods. Red blood cell (RBC) eluates were prepared and tested by acid elution (Elukitt II). Eluates from patients with positive screening tests were tested for specificity using RBC panels with known antigens. Data were analysed using SPSS software. **Results/Findings:** A total of 745 LT were performed in herel in the analysed period. Liver from minor incompatible ABO donors was transplanted to 40 patients. Non identical but compatible Rh transplants were performed to 98 patients. A total of 11 cases of PLS were detected by means by positive DAT. Results are shown in the table. Patients were nine men and two women; mean age was 42.1 ± 11.2 years. Moderate hemolysis was evidenced in nine cases. At diagnoses of PLS, mean hemoglobin and bilirrubin level were 6.8 ± 1.47 g/dl and 10.68 ± 8.65 mg/dl respectively. Treatment consisted of compatible ABO donor-receptor transfusions. With a mean follow-up of 33.3 months, eight patients are alive and three dead. **Conclusion:** In this experience, PLS occurs in 25 % of LT with minor ABO incompatibility. Most cases were due to antibodies against ABO and hemolysis was moderate or mild. This event must be suspected in ABO minor incompatibility transplants that had a positive DAT and/or hemolysis during first weeks after transplantation.

Disclosure of Commercial Conflict of Interest

I. Cano: Nothing to disclose; N. Carpio: Nothing to disclose; A. Lancharro: Nothing to disclose; F. Moscardo: Nothing to disclose; M. A. Sanz: No Answer; P. Solves Alcaina: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. Cano: Nothing to disclose; N. Carpio: Nothing to disclose; A. Lancharro: Nothing to disclose; F. Moscardo: Genzyme-Sanofi,Grants or Research Support; M. A. Sanz: No Answer; P. Solves Alcaina: Nothing to disclose

Case	ABO Receptor	Donor ABO	Days after LT	Antibody in eluate	Hemolysis
1	A+	O+	10	Anti A	No
2	AB+	B+	12	Anti-A	Yes
3	A+	O+	18	Anti-A	Yes
4	A+	O-	23	Anti-A	Yes
5	B+	O+	18	Anti-B	Yes
6	A+	A-	12	Anti-D	Yes
7	A+	O+	38	Anti-A	Yes
8	A-	O+	16	Anti-A	Yes
9	AB+	O+	10	Anti-B	Yes
10	B+	O+	12	Anti B	Yes
11	A+	O+	13	Anti-A	Uncertain

Red Blood Cells: Blood Group Alloimmunization

SP239

Modeling Alloantibody Formation to High Frequency Red Cell Antigens in Immune Responders Using Genotypic Data

R S Shirey[1] (sshirey@jhmi.edu), K Ravindran[1], P A Brunker[1]. *[1]Transfusion Medicine, Johns Hopkins Hospital, Baltimore, MD, United States*

Background/Case Studies: Alloimmunization to blood group antigens is unpredictable and poorly understood. With the advent of RBC antigen genotyping, patients can now be identified who are negative for high frequency antigens (HFA) and at risk for developing the corresponding antibodies. The prevalence and risk factors for HFA alloimmunization in patients at risk was examined, based on RBC antigen genotypes. **Study Design/Methods:** Genotyping for human erythrocyte antigens using the HEA BeadChip™ technology (BioArray Solutions, Immucor, Warren, NJ) was performed on all patient specimens referred for molecular testing due to a history of alloimmunization or diagnosis of sickle cell disease (SCD). All serologic testing was performed according to methods outlined in the *AABB Technical Manual*. Patients predicted to be negative for an HFA (ie, k-, Kp(b-), Js(b-), U-, Lu(b-), Di(b-), Co(a-), Jo(a-), Hy-, LW(a-), and Sc:-1) were retrospecitively identified, using molecular laboratory data from June 2009-March 2013. Serologic data (eg the number and specificities of antibodies) and clinical data (eg transfusion history, oncologic diagnosis, or diagnosis of SCD) were analyzed. Simple and multiple logistic regression and exact methods were used to model the risk factors for alloimmunization to an HFA in this susceptible and highly transfused group. **Results/Findings:** Of 3210 patients genotyped at loci in the HEA kit, 32 (0.1%) were homozygous for at least one variant predicting absence of an HFA (Table). Prior transfusion or pregnancy history was available for 29 (91%). Four susceptible patients made an antibody to an HFA (12.5% of all, 13.8% of those with a documented exposure). Two of these four patients (50%) had made an alloantibody to another alloantigen. The odds of forming an antibody to an HFA were not related to the total number of transfusions ($p = 0.47$), the total number of alloantibodies found ($p = 0.61$), or the diagnosis of SCD ($p = 0.77$). Adjustment in a multiple logistic regression was also not significant for each variable ($p = 0.6$, $p = 0.7$, and $p = 0.7$, respectively). **Conclusion:** Although they had a known exposure to allogeneic blood through transfusion or pregnancy, 86.2% of patients at risk for alloantibody formation to an HFA in fact did not mount an alloimmune response to that antigen. Part of this effect may relate to the differential immunogenicity of these antigens. Possible risk factors including the number of transfused units or the total number of alloantibodies made were not predictors of making an alloantibody to an HFA in our sample. The results suggest that other patient-specific risk factors for alloimmunization exist.

Disclosure of Commercial Conflict of Interest

P. A. Brunker: Nothing to disclose; K. Ravindran: Nothing to disclose; R. S. Shirey: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. A. Brunker: Nothing to disclose; K. Ravindran: Nothing to disclose; R. S. Shirey: Nothing to disclose

TABLE 1. Alloantibodies in HFA-negative patients(Patients)

HFA Genotype	k-	Js(b-)	U-	U-,Jo(a-)	Co(a-)	Lu(b-)	Hy-,Jo(a-)	Jo(a-)	TOTAL
No. of Patients	1	8	5	2	2	1	6	7	32
No. of Patients with Antibody to HFA	0	1	2	1	0	0	0	0	4

SP240

Pre-Existing Non-ABO Blood Group Alloantibodies Identify a "Responder" Subgroup Prone to Subsequent Red Cell Alloimmunization

C A Tormey[1,2] (christopher.tormey@yale.edu), G Stack[1,2]. *[1]Pathology & Laboratory Medicine, VA Connecticut Healthcare System, West Haven, CT, United States; [2]Laboratory Medicine, Yale University School of Medicine, New Haven, CT, United States*

Background/Case Studies: Some evidence suggests that a subset of patients is predisposed to blood group antigen alloimmunization ("responders"), and that these individuals develop alloantibodies at higher rates than others. However, the existence of this responder subpopulation has not been proven. Therefore, to further test the responder hypothesis, the alloimmunization rate was investigated of patients with pre-existing antibodies, ie antibodies present on first testing at our facility. Their alloimmunization rates following transfusion were compared to that of a control group consisting of patients with no prior antibodies. Given that patients with pre-existing antibodies are responders, they should have a higher subsequent alloimmunization rate, if the responder hypothesis is correct. Important aspects of the study were: 1) the inclusion of a control group; 2) the exclusion of anamnestic responses and naturally-occurring antibodies; and 3) a requirement for follow-up testing after transfusion in order to accurately quantitate new alloimmunization. **Study Design/Methods:** A retrospective chart review was performed on patients who had type-and-screen testing at a general hospital. Patients included in the pre-existing antibody arm were those: 1) whose first antibody screen was reactive at 37C and/or antiglobulin phase with an antibody definitely identified by panel; 2) who received RBC transfusion at this facility; and 3) who had one or more antibody screens ≥28 days after their first RBC transfusion. Antibody specificities and dates/times of antibody screens and RBC transfusions were recorded. Data were also collected on a control group that did not have pre-existing antibodies, but met all other inclusion criteria. **Results/Findings:** Of the patients with pre-existing antibodies, 15.9% (11/69) became alloimmunized following subsequent transfusion(s). By comparison, 2.4% (5/207) of control patients were subsequently alloimmunized, which was a significantly lower rate ($P = 0.00015$; Fisher exact test). This difference was unlikely to be due to unequal antibody detection, since the mean number of antibody screens in the two groups was not significantly different (6.5 [pre-existing] versus 5.3 [controls]; $P = 0.16$; Mann-Whitney U test). Alloimmunization in the responders of the control group was similar to that of responders with pre-existing antibodies, as evidenced by identical numbers of antibodies per alloimmunized patient (1.6) and a similar rate of multiple alloimmunization (45.5% [pre-existing] versus 40.0% [controls]). **Conclusion:** Patients with pre-existing non-ABO blood group antibodies developed additional transfusion-related alloantibodies at a higher rate than controls. These findings support the existence of a responder subpopulation.

Disclosure of Commercial Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Stack: Nothing to disclose; C. A. Tormey: Terumo BCT, Grants or Research Support

SP241

Immunogenicity of Blood Group Antigens and Extended Phenotype Matching for Blood Transfusion

C H Sørensen[1] (claess.Soerensen@rsyd.dk), U Sprogøe[1], L Espensen[1]. *[1]Clinical Immunology, Odense University Hospital, Odense, Denmark*

Background/Case Studies: Blood group antigens are by definition immunogenic. However, the immunogenicity varies significantly between specificities. With the increasing availability of high throughput red cell pheno- or genotyping methods and computerized inventory blood management, it might be feasible to routinely match for antigens other than A,B and RhD. **Study Design/Methods:** In order to identify the antigens most suited for extended routine matching, the immunsation rates and the immunogenicity of clinically important blood group antigens were calculated. This was done by retrieving information on the results of antibody identification from the data system, ProSang, for the years 2008-2012 (both included) for the County of Funen. Data corresponded to an approximate total of 160,000 transfusions (equivalent to 5,500 antibody identification procedures). Each patient was counted only once for each antibody specificity. Immunogenicity/potency for each antigen was calculated as an estimate of the frequency of anti-X per 100 X-antigen incompatible transfusions: 100 * number of patients with anti-X/ (frequency of X-antigen positive donors* frequency of X-antigen negative patients * 80,000). 80,000 representing total number of transfusions/2 to correct for the fact that an estimated 50% of all transfusions

were given to a patient after identification of any given antibody. Caucasian antigen frequencies were obtained from the literature (Blood Group Antigen Facts Book, 2nd ed.). **Results/Findings:** A total of 1615 red cell antibodies were identified. All antibodies with specificities of little clinical relevance were excluded (eg A1, Lea, Leb,Ch1, P1, Kna, Doa,and Yta) and also anti-D (n = 246, 16%). The prevalence of the specificities of the remaining 1306 antibodies were ranked as (% of total): K(22), E(21), M(12), Fya(9.7), Cw(7.4), C(6.7), c(5.8), Lua(4.7), Jka(3.1), S(2.5), e(1.5), Kpa(1.4), Jkb(1.1), Wra(1.0), and Fyb(0.5). Calculation of immunogenicity yielded the following ranking of specificities (immunogenicity): Cw(6.2), K(4.3), E(1.7), e(1.2), Kpa(1.2), M(1.1), Lua(1.1), Fya(0.7), c(0.6), C(0.5), Jka(0.3), S(0.17), Jkb(0.1) and Fyb(0.1). **Conclusion:** In accordance with the literature, anti-K was found to be the most prevalent antibody and K to be one of the most potent antigens. The high rankings of the potencies of Cw and E are probably overestimations, since high proportions of anti-Cw, and anti-E are naturally occuring. Following the potency of K with some distance e, Kpa, M, Lua and Fya all had potencies close to 1%. Taking into account the clinical importance of the corresponding antibody the antigens K, Cw, Kpa, and Fya may be seleted as priority specificities to match in a setting of extended phenotype or genotype matched transfusions.

Disclosure of Commercial Conflict of Interest

L. Espensen: Nothing to disclose; U. Sprogøe: BioRad,Other; C. H. Sørensen: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Espensen: Nothing to disclose; U. Sprogøe: BioRad,Other; C. H. Sørensen: Nothing to disclose

SP242

The Persistence of Diego Blood Group Alloantibodies

F A Moretto[1] (fefa@UNICAMP.br), A T Mascarenhas[1], P R Azevedo[1], M Locatelli[1], V Castro[1], M Barjas-Castro[1]. *[1]Hemocentro UNICAMP, Campinas, Brazil*

Background/Case Studies: The Diego blood group system was first described in 1955 in a Venezuelan family. It soon became apparent that Dia antigen, though relatively common in South American Indians, is rare in most other populations, including Caucasians (<0.01%). Antibodies anti-Dia and -Dib present clinical significance and can cause transfusion reaction and hemolytic disease of the newborn. RBC alloantibodies may disappear over time, predisposing patients to the risk of receiving incompatible RBC transfusion. Despite some inter individual differences, the evanescence of alloantibodies was found to vary by antigenic specificity of blood groups. **Study Design/Methods:** The aim of this study was to analyze the disappearance of alloantibodies anti- Dia and anti-Dib after transfusion and /or pregnancy alloimmunization in patients and blood donors. Transfusion service records were screened to identify donors (8) and patients (11) with anti-Dia or -Dib and those who had been screened at least once after initial antibody detection. Transfusion and pregnancy history and data of first and subsequent alloantibody screenings were analyzed. Donors were asked whether they accepted to be submitted to new sample collections and a medical interview regarding transfusions and pregnancies. Alloantidody detection was carried out using commercial 3-cell for antibody screening (Dia antigen in at least 1-cell) and patient/donor serum or plasma. Antibody identification was performed using 11-cells panel in gel (BioRad or Grifols) and tube tests. LISS (low-ionic-strength saline), PEG (polyetileneglycol) and Papain enzyme were used to enhance antigen-antibody interactions. These techniques were used to identify alloantibodies from 1997-2013; methods used before 1997 were not documented. **Results/Findings:** The age media of patients and donors was 41.6 years (17-68y),and included 12 women and 7 men. In all cases the studied antibodies (18 anti-Dia and 1 -Dib) were detectable in the last screening, with time of follow up testing between 1-29 years (10 < 5y and 9 ≥ 5y). In 6/19 cases, the interval between initial positive screen and their last document antibody identification was over 8 years. This group included one blood donor with anti-Dib (phenotype DiaDia), who had history of two pregnancies (29 and 27 years ago) with no transfusions. **Conclusion:** This is the first publication concerning Diego alloantibody evanescence and the data suggested that anti-Dia and -Dib are very stable with a long persistence over time. The patients and donors studied had no multiple antigenic exposures that could justify the long rate of antibody persistence.

Disclosure of Commercial Conflict of Interest

P. R. Azevedo: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; V. Castro: Nothing to disclose; M. Locatelli: Nothing to disclose; A. T. Mascarenhas: Nothing to disclose; F. A. Moretto: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. R. Azevedo: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; V. Castro: Nothing to disclose; M. Locatelli: Nothing to disclose; A. T. Mascarenhas: Nothing to disclose; F. A. Moretto: Nothing to disclose

SP243

A Follow-Up Study of RhD neg Patients Deliberately Receiving RhD pos Red Cell Units

B Persson[1] (barbro.persson@akademiska.se), Englund[1], A Joelsson Saxner[1], J Säfwenberg[1]. *[1]Department of Clinical Immunology and Transfusion Medicine, Uppsala University Hospital, Uppsala, Sweden*

Background/Case Studies: The blood center here has adopted a policy to ensure the availability of RhD neg units for RhD neg women of childbearing age (< 50 years) and for anti-D immunized patients. If the stock of RhD neg blood is below a certain level, RhD pos red cell units are issued to selected RhD neg patients who are likely to need several units. This study presents the consequences of the policy. **Study Design/Methods:** From the database, containing information from the 1960s onwards, data was retrospecitively compiled on: RhD neg patients receiving RhD pos red cell units during a 2-year period (2005-2007), the number of transfused units during a 5-day period, if a posttransfusion sample was analyzed, and if the patient was immunized. The diagnosis at the time of transfusion was obtained from the medical records. The time period was chosen for the opportunity to receive a posttransfusion sample. The patients included had never, according to the database, received any RhD pos erythrocyte units before, and all had a negative pretransfusion test (red cell antibody screen). **Results/Findings:** During the study period 110 RhD neg patients received RhD pos units. From 59 patients a posttransfusion sample was received 2 weeks up to 6 years later. 37 patients had a negative posttransfusion test and 22 patients were immunized with an anti-D; nine of these patients had additional Rh system antibodies. During the 5-day period, 11 patients died. Of 17 patients receiving 8 or more units, posttransfusion samples were received from 11 patients, and 6 were immunized with an anti-D. Of 59 patients receiving 1-2 units, samples from 28 patients were received, showing 8 with an anti-D. 34 patients received 2-8 units. 75 of the patients were men, 9 of them under the age of 50; and 35 patients were women including one woman 49.5 years old. Only 5 were trauma patients. 42 patients had a diagnosis of malignancy; of 23 posttransfusion samples 9 showed an anti-D. 42 were surgical patients, and 24 posttransfusion samples showing 11 with an anti-D. 21 patients had other diagnoses. 568 RhD pos units were transfused to RhD neg patients during the study period compared with the total of approximately 24 500 red cell units annually transfused here. **Conclusion:** All patients in need of blood were transfused without delay. One woman received blood against the policy but with permission from the doctor on duty. Of the 59 RhD neg patients receiving RhD pos units and with posttransfusion samples, 22 were immunized with an anti-D. A malignancy diagnosis did not decrease the degree of immunization. The risk of having an anti-D in selected patients is acceptable as it enables reduced consumption of RhD neg units. The number of patients is low in relation to overall use of red cell units. The data will continue to be collected for several more years.

Disclosure of Commercial Conflict of Interest

Englund: Nothing to disclose; A. Joelsson Saxner: Nothing to disclose; B. Persson: Nothing to disclose; J. Säfwenberg: Nothing to disclose

Disclosure of Grants Conflict of Interest

Englund: Nothing to disclose; A. Joelsson Saxner: Nothing to disclose; B. Persson: Nothing to disclose; J. Säfwenberg: Nothing to disclose

SP244

An Insight into the Kinetics of Prophylactic Rh Immune Globulin Using an In-Vitro Model of Maternal Red Cells with Weakly Reacting D Antigens

M Shabani-Rad[1] (mtshrad@hotmail.com), J L Hendry[1], J Luider[1], I Auer[1], A Mansoor[1]. *[1]Foothills Medical Center, Calgary, AB, Canada*

Background/Case Studies: Despite worldwide Rh Immune Globulin (RhIG) prophylaxis, the rate of anti-D alloimmunization has increased even in developed countries more than would expected due to only partial and variant weak D phenotypes. Flow cytometry (FC) and serology (Anti-Globulin Test) were used to assess the effectiveness of RhIG prophylaxis among weakly reacting D cases. **Study Design/Methods:** The study was performed using three experiments. First, red cells (RCs) from weak D cases (5), partial D

cases (2), and Rh negative and positive control adult O RCs were exposed to polyclonal RhIG (WinRho) at different dilutions, and after wash incubated with fluorochrome labelled Anti-human IgG, IgA, and IgM following FC protocols. All cases were analysed by FC to determine the density of D antigens on RCs based on bound RhIG. In the second experiment, all RCs were incubated with RhIG, and after wash mixed and incubated with Rh positive fetal O red cells (final ratio of 1%) to study the kinetics of RhIG among maternal and fetal red cells. And finally, maternal blood samples from weak D and partial D cases were collected 3-4 weeks after RhIG administration to study the presence of bound and free RhIG on red cells and maternal plasma. **Results/Findings:** Adult and fetal O+ red cells expressed a Medium fluorescent intensity (MFI) of 40-50 based on bound RhIG. The MFI of Weak D cases for bound RhIG were 2.5-5.0. Partial D cases showed MFI of 7 and 8. FC analysis showed that fetal red cells mixed with maternal red cells (all weak D and partial D cases) had higher affinity to absorb the bound RhIG from maternal red cells. FC analysis did not show an adequate sensitivity to measure the quantity of bound and free RhIG on maternal red cells and within the plasma, therefore indirect AGT was used to confirm the presence of free RhIG within the plasma of partial and weak D cases (3-4 weeks post RhIG blood samples). Bound RhIG was not detected on either weak or partial D maternal red cells. **Conclusion:** Density of RhIG bound D antigens measured by FC confirmed the binding of polyclonal RhIG to the D epitopes of both weak and partial D cases. This experiment also shows that fetal RCs will compete with maternal RCs by preferential shifting of bound RhIG antibodies on maternal RCs to fetal red cells. In addition, this study supports the ongoing presence of RhIG in maternal plasma despite the low ratio of RhIG to total RC binding capacity in weak and partial D cases as well as higher sensitivity of AGT for evaluation of RhIG kinetics in maternal plasma. Given the complexities of the D antigens, it seems RhIG prophylaxis is the safest and cost effective alternative to prevent the anti-D alloimmunization in the absence of a 100% specific and sensitive diagnostic test.

Disclosure of Commercial Conflict of Interest

I. Auer: No Answer; J. L. Hendry: Nothing to disclose; J. Luider: Nothing to disclose; A. Mansoor: No Answer; M. Shabani-Rad: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. Auer: No Answer; J. L. Hendry: Nothing to disclose; J. Luider: Nothing to disclose; A. Mansoor: No Answer; M. Shabani-Rad: Nothing to disclose

SP245

Effect of Rh Immune Globulin on Non-D Alloimmunization in Pregnancy

R Kansara[1] (roopesh03@yahoo.ca), C Desrochers[1], R Fallis[2], A Ponnampalam[1]. *[1]University of Manitoba, Winnipeg, MB, Canada; [2]Canadian Blood Services, Winnipeg, MB, Canada*

Background/Case Studies: Prophylactic administration of Rh immune globulin (RhIg) has dramatically reduced the incidence of D antigen alloimmunization. The mechanism of action remains unproven, but current proposals include enhanced macrophage clearance of anti-D coated fetal cells and nonspecific immune suppression. Observational data suggest prophylactic RhIg suppresses formation of HLA antibodies in prenatal women. It is hypothesized here that routine prophylactic use of RhIg leads to lower rates of alloimmunization against a broader (non-D) group of red cell antigens. **Study Design/Methods:** The Canadian Blood Services (CBS) Perinatal Program routinely tests all prenatal samples for ABO and RhD type, antibody screen, antibody identification, and antibody titre. Between January 2011-December 2012, the electronic registry in Manitoba was retroactively evaluated to identify the number of individuals screened for antibodies, number of positive antibody screens, and alloantibody specificity. RhD negative women, who received RhIg prophylaxis, formed the intervention group to quantify the rate of non-D alloimmunization. RhD positive women, who did not receive routine RhIg prophylaxis, formed the control group, with which to compare the difference in alloimmunization rates. **Results/Findings:** During the 24 month observation period, 40,691 individuals were screened as part of their prenatal assessment. Of this cohort, 35,524 (87%) were RhD positive and 5,167 (13%) were RhD negative. In the RhD positive group, 312 (0.88%) individuals were found to have a positive antibody screen with antibody specificity identified. The three most-common antibodies, in descending frequency, were anti-E, anti-K, and anti-c. In the RhD negative group, 51 individuals had a positive antibody screen with antibody specificity identified. The three most-common antibodies in descending frequency were anti-D, anti-C, and anti-K. Since the primary hypothesis was to assess the rate of non-D antibody formation, 28 of the 51 RhD negative individuals who developed anti-D were excluded from the anaylysis. After exclusion, the rates of alloimmunization against non-D red cell antigens were 0.45% for the RhD negative cohort and 0.88% for the RhD positive cohort; relative risk 0.51, 95% C.I. 0.33 to 0.78, p = 0.0018. **Conclusion:** By comparing the rate of alloimmunization against non-D red cell antigens in two groups of prenatal women, a significant reduction in antibody formation in women that receive RhIg prophylaxis was shown. It is proposed that this reduction in antibody formation is due to RhIg-induced nonspecific immune suppression. Alternate possibilities include antibody suppression from impurities in RhIg formulations or lower predisposition for antibody formation in RhD negative individuals. If the finding is replicable in other institutions, further study would be warranted to find novel means to prevent hemolytic disease of the newborn caused by non-D related antibodies.

Disclosure of Commercial Conflict of Interest

C. Desrochers: Nothing to disclose; R. Fallis: Nothing to disclose; R. Kansara: Nothing to disclose; A. Ponnampalam: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Desrochers: Nothing to disclose; R. Fallis: Nothing to disclose; R. Kansara: Nothing to disclose; A. Ponnampalam: Nothing to disclose

SP246

Clinical Significance of Rh Antibodies Produced by Sickle Cell Disease (SCD) Patients with *RH* Variants

E Sippert[1] (emiliasippert@hotmail.com), G A Guelsin[1], D Machado[1], A C Gaspardi[1], C R Fujita[2], J Pellegrino Jr[1], S Gilli[1], L Castilho[1]. *[1]Hemocentro, UNICAMP, Campinas, Brazil; [2]Immunohematology, Fujisan, Fortaleza, Brazil*

Background/Case Studies: Alloimmunization is a major complication in patients with Sickle Cell Disease (SCD) receiving red blood cell (RBC) transfusions and despite provision of Rh phenotyped RBC units, Rh antibodies still occur. Those antibodies in antigen positive patients for corresponding Rh antigen are considered autoantibodies in many cases but variant *RH* alleles found in SCD patients can also contribute to Rh immunization. In this study, variant *RH* alleles were characterized in 48 SCD patients who made antibodies to Rh antigens despite antigen-positive status and evaluated the clinical significance of the antibodies produced. **Study Design/Methods:** RHD and RHCE BeadChip (BioArray Solutions, Warren, NJ), and/or amplification and sequencing of exons were used to identify the *RH* variants. Serologic features of all Rh antibodies in antigen positive patients were analyzed and the clinical significance of the antibodies was evaluated by retrospective analysis of the hemoglobin (hgb) levels pre and post-transfusion and determined by change from baseline pre-transfusion hgb and percentage (%) of hgb S. **Results/Findings:** Variant *RH* alleles in 34/48 (71%) of SCD patients with Rh antibodies (21 anti-D, 17 anti-e, and 10 anti-C) were identified. Molecular analyses revealed the presence of partial *RHD* alleles in 17/21 (81%) patients with anti-D. Variant *RHCE*ce* alleles associated with altered e antigen were found in 12/17 (71%) patients with anti-e. Of the 10 patients with anti-C, 8 (80%) had a partial C produced by *(C)ceS*. Retrospective analysis showed that 58% of antibodies produced by the patients with *RH* variants were involved on delayed hemolytic transfusion reactions (DHTR) or decreased transfused RBC survival. **Conclusion:** In this study, it was demonstrated that most Rh antibodies in SCD patients with *RH* variants can be clinically significant and therefore matching patients based on *RH* variants should be considered.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; C. R. Fujita: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; S. Gilli: Nothing to disclose; G. A. Guelsin: Nothing to disclose; D. Machado: No Answer; J. Pellegrino Jr: No Answer; E. Sippert: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; C. R. Fujita: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; S. Gilli: Nothing to disclose; G. A. Guelsin: Nothing to disclose; D. Machado: No Answer; J. Pellegrino Jr: No Answer; E. Sippert: Nothing to disclose

SP247

Percentage of West-African Ancestry Predicts Red Cell Alloimmunization in Transfused African American Sickle Cell Disease (SCD) Patients

S Campbell Lee[1] (scampbe@uic.edu), R Machado[5], N A Ellis[3], K Batai[4], J Garcia[5], R A Kittles[2]. *[1]Pathology, University of Illinois at Chicago, Chicago, IL, United States; [2]Medicine, Institute of Human Genetics, University of Illinois at Chicago, Chicago, IL, United States; [3]Pediatrics, Institute of Human Genetics, University of Illinois at Chicago, Chicago, IL, United States; [4]School of Public Health, University of Illinois at Chicago, Chicago, IL, United States; [5]Medicine, University of Illinois at Chicago, Chicago, IL, United States*

Background/Case Studies: Red cell alloimmunization occurs in approximately 30% of transfused sickle cell disease (SCD) patients. Because red blood cell (RBC) transfusion is an important part of therapy, the need for additional antigen matching once alloimmunization occurs is problematic and leads to therapeutic limitations. Thus, identification of risk factors for alloimmunization would benefit this patient population. A genome-wide association study (GWAS) was performed to identify possible genes associated with RBC alloimmunization. **Study Design/Methods:** The transfusion records were reviewed of 256 sickle cell disease patients from whom DNAs were available. 33.6% of the patients had RBC alloantibodies; the most frequent were anti-C, E, and K1. Patients were genotyped for 2,217,402 single nucleotide polymorphisms (SNPs) using the Axiom™ Genome-Wide PanAFR Array (Affymetrix). This array offers high genomic coverage (>85%) for alleles with frequencies >2% in populations with West African ancestry. SNPs with a call rate <90%, Hardy Weinberg equilibrium P < 0.0001, or having a minor allele frequency <1% were excluded from the analysis. In total, 2,113,177 SNPs were analyzed in 256 individuals. Population structure was examined by principal components analysis. No outliers were identified and no samples were excluded. P values, odds ratios, and 95% confidence intervals were calculated using logistic regression models, and using age, gender, and West African ancestry as covariates. To account for multiple testing, a genome-wide significance threshold was set at P < 2.4 x 10-8. **Results/Findings:** Logistic regression analyses revealed that female gender (P = 0.003) and percent of African ancestry (P = 0.03) were associated with RBC alloimmunization, but age (P = 0.90) and number of transfusions (P = 0.20) were not significant. No SNPs reached genome-wide significance for the association with RBC alloimmunization. The three smallest P values were with the SNPs, rs1507094 (RALYL), rs4345649 (POLR1E), and rs10941424 (FYB) (P = 1.31 x 10-6; 2.82 x 10-6; and 3.21 x 10-6, respectively). Of note, the FYB gene on 5p13.1 is a binding protein involved in platelet activation and expression of IL-2. **Conclusion:** In this cohort, individuals with higher African genetic ancestry and females are more likely to develop RBC alloantibodies. A larger sample size would be necessary to find SNPs significantly associated with RBC alloimmunization.

Disclosure of Commercial Conflict of Interest

K. Batai: Nothing to disclose; S. Campbell Lee: Nothing to disclose; N. A. Ellis: Nothing to disclose; J. Garcia: No Answer; R. A. Kittles: Nothing to disclose; R. Machado: No Answer

Disclosure of Grants Conflict of Interest

K. Batai: Nothing to disclose; S. Campbell Lee: Nothing to disclose; N. A. Ellis: Nothing to disclose; J. Garcia: No Answer; R. A. Kittles: Nothing to disclose; R. Machado: No Answer

SP248

Red Blood Cell (RBC) Alloimmunization in Patients with Sickle Cell Cisease (SCD): A Brazilian Institutional Experience

J G Muniz[1] (janagmuniz@yahoo.com.br), C Arnoni[1], R Person[1], D Gazito[1], F Latini[1], J Barreto[1], L Castilho[2]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: Red blood cell (RBC) transfusion remains the main treatment in sickle cell disease (SCD) patients. Complications of transfusions, such as alloimmunization, can create challenges in the management of patients with SCD. Alloimmunization is associated with hemolytic transfusion reactions, autoantibody formation, difficulties in RBC antigen matching and delays provision of safe transfusion. The alloimmunization rate in SCD patients ranges considerably, dependent primarily on the extent of RBC antigen matching and exposure frequency. Thus, the aim of this study was to determine the frequency of red cell alloantibodies among SCD patients who received regular transfusions and evaluate the antigen disparity between the patients and donors of highly diverse ancestry. **Study Design/Methods:** A total of 80 chronically transfused patients with SCD and 162 blood donors were included in this study. The presence of alloantibodies in the patients was investigated by performing antibody screening and identification in gel cards. Rh, K/k, Fya/b, Jka/b and Ss phenotyping in patients and donors was performed by hemmaglutination in tube and gel test. RBC antigen frequencies were calculated to determine the most common antigen-negative phenotypes. **Results/Findings:** From 80 patients, 39% were alloimmunized and more than 50% of them presented more than one alloantibody. The main antibodies identified were anti-C (16.2%), anti-E (12.5%), anti-K (8.7%), anti-D (6.25%), anti-Jka (3.7%), anti-c (2.5%), anti-Fya (1.2%), anti-e (1.2%), anti-Jkb (1.2%), anti-S (1.2%), anti-Dia (1.2%), anti-Cw (1.2%), and anti-Fya (1.2%) anti-Fy3 (1.2%). The most common antigen-negative phenotypes for patients were 10% C-E-K-Jk(b-); 8.8% C-E-K-Fy(a-); 5% C-E-K-Fy(a-)Fy(b-); 2.5% C-E-K-Fy(a-)Jk(b-); and 2.5% C-E-K-Fy(a-)Fy(b-)Jk(b-). Analysis of the small pool of donors showed respectively 5.4%, 8%, 3.3%, 0.7%, and 0% of compatible donors for the most prevalent antigen-negative phenotypes found in the SCD patients. **Conclusion:** The results confirmed that the rate of alloimmunization in SCD patients who receive regular transfusions is high and that at least the use of C, E, and K RBC phenotyped-matched transfusions could significantly reduce the incidence of alloantibody production in SCD. The need to increase our pool of phenotyped donors was also verified, mainly donors C-E-K-Fy(a-)Jk(b-), to improve transfusion practice by avoiding alloimmunization. High throughput genotyping can be an excellent tool for typing large numbers of donors in order to increase the extended antigen profiles of the donors.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP249

Alloimmunization with Uncommon Alloantibodies in Sickle Cell Disease (SCD) Patients

A Pathare[1] (avp16@hotmail.com), S Alkindi[1,2], A Z Al Riyami[1], S Al-Mahrooqi[1], F Wasim[1], V Panjwani[1], Z Jabeen[1], S Al Hosni[1], S Al Hinai[1], D J Dennison[1]. *[1]Department of Haematology, Sultan Qaboos University Hospital, Muscat, Oman; [2]College of Medicine & Health Sciences, Sultan Qaboos University, Muscat, Oman*

Background/Case Studies: Blood transfusion forms an integral part of the supportive care in patients with Sickle Cell Disease (SCD). Red cell alloimmunization remains a frequent complication, especially in patients who are managed with regular exchange transfusions as part of their current standard of care. **Study Design/Methods:** A retrospective analysis was performed on the records of SCD patients in the database obtained over the past 21 years. **Results/Findings:** Out of a total of 1356 SCD patients followed at this center since 1992, 612(45.1%) have been transfused in contrast to 744(54.9%) patients who have never been transfused. Among the 612 patients who received at least one unit red blood cell transfusion, 118(19.3%) SCD patients have demonstrated alloimmunization. Single alloantibody was seen in 71 patients (63%), of whom 67.9% were against Rh antigen; whereas, an additional 25.1% were against Kell antigen. Uncommon alloantibodies encountered were anti-Yta, anti-Jra, anti-P, and anti-Jsb. **Conclusion:** Conclusion: The cumulative prevalence of alloimmunization was 19.3% over the past 21 years. Alloantibodies against Rh antigens occurred in about 68% cases; whereas, approximately 93% of the patients developed alloimmunization against both Rh and Kell antigens. Anti-P is an extremely rare antibody and may cause severe haemolytic transfusion reaction. Anti-Jra and Anti-Jsb antibodies may cause transfusion reactions whereas Anti-Yta is not reported to cause transfusion reactions. Importantly, as these are sickle cell patients, providing them with antigen negative blood is at times next to impossible and hence preventive measures need to be emphasized.

Disclosure of Commercial Conflict of Interest

S. Al Hinai: Nothing to disclose; S. Al Hosni: Nothing to disclose; A. Z. Al Riyami: Nothing to disclose; S. Al-Mahrooqi: Nothing to disclose; S. Alkindi: No Answer; D. J. Dennison: Nothing to disclose; Z. Jabeen: Nothing to disclose; V. Panjwani: Nothing to disclose; A. Pathare: Nothing to disclose; F. Wasim: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Al Hinai: Nothing to disclose; S. Al Hosni: Nothing to disclose; A. Z. Al Riyami: Nothing to disclose; S. Al-Mahrooqi: Nothing to disclose; S. Alkindi: No Answer; D. J. Dennison: Nothing to disclose; Z. Jabeen: Nothing to disclose; V. Panjwani: Nothing to disclose; A. Pathare: Nothing to disclose; F. Wasim: Nothing to disclose

SP250

Economic Evaluation of a Hypothetical Screening Test for Alloimmunization Risk among Transfused Sickle Cell Disease (SCD) Patients

S Kacker[1] (skacker1@jhmi.edu), P M Ness[1], W Savage[2], K D Frick[1], R S Shirey[1], K E King[1], A Tobian[1]. *[1]Johns Hopkins University, Baltimore, MD, United States; [2]Harvard University, Boston, MA, United States*

Background/Case Studies: Approximately 30% of chronically transfused sickle cell disease (SCD) patients become alloimmunized, potentially resulting in delayed hemolytic or serologic reactions and difficulty finding compatible units. Transfusion services are currently unable to identify at-risk patients, and thus either provide prophylactically antigen matched blood to all SCD patients or only to patients who have previously developed antibodies. Prophylactic antigen matching has been shown to cost between $369,482 and $769,284 to prevent a single alloimmunization event. However, if an assay to identify potential responders were to become available, the results could help determine which patients should receive prophylactically matched blood. **Study Design/Methods:** A Markov-based decision tree was constructed to evaluate the characteristics of a hypothetical screening test (cost, sensitivity, and specificity) which would identify patients at risk of alloimmunization. The model simulated a cohort of initially transfusion naïve pediatric and adult SCD patients who received either simple or exchange chronic transfusion over 15 years. Alloimmunization events and direct medical costs, including any transfusion or alloimmunization-related expenses and pre- or post-transfusion testing, were estimated under four strategies: prophylactic antigen matching for all transfused patients; antigen matching only for patients with previously developed alloantibodies (history-based); prophylactic antigen matching for patients screening positive using a perfect screening test with 100% sensitivity and 100% specificity; and prophylactic antigen matching for patients screening positive using an imperfect test with variable sensitivity and specificity. Antigen matching under all strategies was limited to antigens C, E, and K. Costs were expressed in 2012 US$, and a hospital perspective was used. Costs and alloimmunization events were discounted at 3%. **Results/Findings:** Under base-case assumptions with an assay cost of $250, prophylactic antigen matching based on a perfect screening test is expected to cost $7,439 per patient more than history-based antigen matching over 15 years, but result in 38% fewer alloimmunization events. It would cost $29,756 to prevent a single alloimmunization event. However, antigen matching using a perfect screening test would be cost saving compared to a strategy of antigen matching for all patients while the cost of the test is less than $10,000. An imperfect test priced at $125, with 75% sensitivity and 75% specificity, is expected to decrease alloimmunization events by 27%, at an additional cost of $33,986 per patient over a 15 year period. **Conclusion:** Development of a screening test to identify patients likely to become alloimmunized, even if imperfect, may be a cost-effective method to reduce alloimmunization among chronically transfused SCD patients.

Disclosure of Commercial Conflict of Interest

K. D. Frick: Nothing to disclose; S. Kacker: Nothing to disclose; K. E. King: I have no relationship with Fenwal, Other; P. M. Ness: Nothing to disclose; W. Savage: Nothing to disclose; R. S. Shirey: Nothing to disclose; A. Tobian: Fenwal, Grants or Research Support

Disclosure of Grants Conflict of Interest

K. D. Frick: Nothing to disclose; S. Kacker: Nothing to disclose; K. E. King: Terumo BCT, Grants or Research Support; P. M. Ness: Nothing to disclose; W. Savage: Nothing to disclose; R. S. Shirey: Nothing to disclose; A. Tobian: Nothing to disclose

SP251

Benefits of Extended Phenotype Matching in Patients Undergoing Chronic Transfusion and Evaluation of Donor Availability

M A Mota[1] (marizamota@einstein.br), R O Cruz[1], P B Freire[1], V B Bezerra[1], L Visciglia[1], J C Silva[1], M B Pereira[1], M G Aravechia[1], L Castilho[2], J M Kutner[1]. *[1]Hemoterapia e Terapia Celular, Hospital Israelita Albert Einstein, São Paulo, Brazil; [2]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: Extended phenotype matching is an effective strategy to reduce alloimmunization to RBC antigens in patients who are in need of chronic transfusion support. In this institution, patients who require multiple transfusions are placed on chronic prophylactic transfusion protocols in order to provide blood that will survive maximally. These protocols range from providing limited phenotype-matched RBCs for only Rh and K antigens to providing extended phenotype-matched RBCs for Rh, K, S, Fya, and Jka antigens prior to transfusion. The effectiveness of these protocols in reducing alloimmunization was assessed and evaluated to see if the antigen diversity of the donor pool meets the demands and the antigen diversity of the patients undergoing chronic transfusions. **Study Design/Methods:** Age, sex, number of transfusions, and the rate of alloimmunization were determined during the course of 1 year in patients who received units Rh and K-matched and in patients receiving extended phenotype-matched blood transfusions. The prevalence of the most common phenotypes found in these patients were also determined in a year donor inventory. **Results/Findings:** A total of 235 chronically transfused patients (16 hemoglobinopathies, 31 anemias, and 188 hematologic malignancies), 137 males and 98 females, were enrolled. Mean age was 50 years. All patients receiving phenotype-matched RBC units had received at least 10 transfusions of RBCs before the study. The rate of alloimmunization in the patients receiving extended phenotype-matched RBC was 2.6% while in the patients receiving blood units Rh and K-matched only, alloimmunization was 8.4 %. The most prevalent Rh phenotype found in these patients was R1r (44%). The most prevalent antigen-negative profiles were 70% E- K-, 20% E- K- Fy(a-), Jk(a-), S- and 15% E- K-, Fy(a-),and Jk(b-). Comparison of the most prevalent extended antigen-negative phenotypes in patients with donors revealed 6% donors were antigen-negative matches. In 15% of patients, extended matching could be improved by increasing the phenotyped/genotyped RBC units in the inventory and the recruitment of donors focusing on extended antigen profiles of the patients. **Conclusion:** Extended phenotype matching would have prevented alloimmunization in 15 % of patients who formed alloantibodies. Analysis of antigen-negative profiles in patients undergoing chronic transfusions and in the donor pool is important to manage the inventory and to establish a successful phenotype/genotype extended matched protocol in order to avoid transfusion delays and alloimmunization. Independent of the approach used, it is important that the RBC units be readily available in order to supply the patients.

Disclosure of Commercial Conflict of Interest

M. G. Aravechia: Nothing to disclose; V. B. Bezerra: No Answer; L. Castilho: Nothing to disclose; R. O. Cruz: Nothing to disclose; P. B. Freire: No Answer; J. M. Kutner: Nothing to disclose; M. A. Mota: Nothing to disclose; M. B. Pereira: No Answer; J. C. Silva: No Answer; L. Visciglia: No Answer

Disclosure of Grants Conflict of Interest

M. G. Aravechia: Nothing to disclose; V. B. Bezerra: No Answer; L. Castilho: Nothing to disclose; R. O. Cruz: Nothing to disclose; P. B. Freire: No Answer; J. M. Kutner: Nothing to disclose; M. A. Mota: Nothing to disclose; M. B. Pereira: No Answer; J. C. Silva: No Answer; L. Visciglia: No Answer

SP252

Different Red Cell Alloimmunization Rates among Transfusion-Dependent Benign Red Cell Disorders: A Comparison between Pure Red Cell Aplasia, Thalassemia Intermedia, and Thalassemia Major

A Ang[1,2] (ang_ai_leen@hsa.gov.sg), Y Lee[2], C Tan[2], A Ho[2], Y Chen[2], T Hein[2], W Ng[1]. *[1]Blood Services Group, Health Sciences Authority, Singapore, Singapore; [2]Hematology, Singapore General Hospital, Singapore, Singapore*

Background/Case Studies: Pure red cell aplasia (PRCA), thalassemia intermedia (TI) and thalassemia major (TM) are common transfusion-dependent benign red cell (RC) disorders in this adult Hematology unit. Leucodepleted but not antigen-matched (except ABO- and D-matched) RC are transfused to these patients, unless there are clinically significant RC alloantibodies (alloAbs). These patients likely share similar RC antigen frequencies as blood donors due to similar ethnic composition. This study compares RC alloimmunization between PRCA, TI, and TM and may guide the role of prophylactic RC antigen matching in these patients. **Study Design/Methods:** PRCA, TI and TM patients with ≥2 transfusions of at least 1 month apart were included. Their demographic, clinical, transfusion, and RC alloimmunization data were retrospectively recorded and analyzed. Only clinically significant alloAbs to non-ABO and -D antigens were analyzed. **Results/Findings:** There were 37 males and 35 females with a median age of 35(19-83) years, comprising of 12 PRCA, 13 deletional HbH disease (delHBH), 6 HbH Constant Spring disease (HCS), 14 β TI, and 27 TM patients (Table). Despite TM having significantly more RC transfusions than other disorders($P = 0.00$), it was β TI and HCS which had the highest

likelihood of RC alloimmunization. Collectively, this was signifcantly higher than PRCA (OR11.0, 95%CI[1.2,100], P = 0.02) and approached significance on comparing with delHBH (OR5.5, 95%CI[1.0,31.0], P = 0.07) and TM (OR3.5, 95%CI[1.0,12.4], P = 0.07). At 30 transfused RC units, β TI and HCS collectively had higher cumulative alloimmunization incidence than PRCA(P = 0.00), TM(P = 0.00) and delHBH(P = 0.14). PRCA had the lowest % of alloimmunized patients though not significantly different from TM and delHBH(P>0.4). No TM patients developed >1 RC alloAb and were significantly less likely to do so than β TI and HCS(P = 0.03). One HCS patient had symptomatic delayed hemolytic transfusion reaction and one β TI patient developed delayed hyperhemolysis with RC alloimmunization. Most alloAbs are specific for Rh (55%) and Kidd (14%) antigens and only 3% are against K. In females, the RC alloimmunization was temporally unrelated to pregnancy except for one who developed anti-Mia when transfused during pregnancy. **Conclusion:** β TI and HCS patients have the greatest rate and likelihood of RC alloimmunization, and possibly are at greater risk of complications from RC alloimmunization due to its disease nature. Prophylactic RC antigen matching will benefit them the most and matching for Rh and Kidd antigens will be most suitable for these patients. More analysis on PRCA is needed to confirm if prophylactic RC antigen matching can be omitted due to its low alloimmunization rate (likely because patients usually receive immunosuppressants).

Disclosure of Commercial Conflict of Interest

A. Ang: Nothing to disclose; Y. Chen: Nothing to disclose; T. Hein: Nothing to disclose; A. Ho: Nothing to disclose; Y. Lee: Nothing to disclose; W. Ng: No Answer; C. Tan: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Ang: Nothing to disclose; Y. Chen: Nothing to disclose; T. Hein: Nothing to disclose; A. Ho: Nothing to disclose; Y. Lee: Nothing to disclose; W. Ng: No Answer; C. Tan: Nothing to disclose

	PRCA(N = 12)	delHBH (N = 13)	HCS (N = 6)	β TI (N = 14)	TM (N = 27)
Median(range): Age of initiating regular transfusion (years) Number of transfused RC units	63 (0.1-82) 31 (4-1403)	51 (5-77) 20 (4-96)	23 (3-53) 25(4-162)	16 (2-33) 32 (2-495)	1 (0.25-7) 1083 (449-2028)
RC alloimmunization Proportion with ≥ 1 alloAb Proportion with ≥ 2 alloAbs Cumulative alloimmunization incidence at 30 transfused RC units	8% 8% 0%	15% 8% 8%	50% 17% 60%	50% 21% 50%	22% 0% 0%
RC AlloAb specificity	K, C	C, E, Jka, Fyb, Mia	C, c, E	C, E, f, S, Dia	C c E e, Jka, S, Mia

SP253

Prevention of Alloimmunization in Patients with Warm Autoantibodies using Extended Human Erythrocyte Antigen Typing

H Shafi[1] (heidishafi@hotmail.com), M R Edmunds[2], J Seo[1], I A Abumuhor[1], E C Vamvakas[3], E Klapper[1]. *[1]Department of Pathology and Laboratory Medicine, Cedars-Sinai Medical Center, Los Angeles, CA, United States; [2]Blood Centers of the Pacific, San Francisco, CA, United States; [3]Los Angeles, CA, United States*

Background/Case Studies: Patients with warm autoantibodies (WAA) that have been transfused have a 32% prevalence of alloimmunization, compared to 1-2% of transfused patients in general. Alloimmunization may increase the risk of transfusion reactions associated with future transfusion therapy, lead to delays in provision of blood, and increase cost and complexity of serological workups. Therefore, many transfusion services that care for patients who are high immune responders, including WAA patients who are likely to have ongoing transfusion needs, have implemented strategies to reduce the risk of alloimmunization. One such method of reducing the risk of alloimmunization is the provision of extended matched blood, whereby patients are transfused blood lacking clinically significant antigens that they themselves lack. In a previous study, the feasibility was reported of providing extended matched blood by DNA typing methods for 74 patients with WAA. The outcome this transfusion strategy has had on rates of alloimmunization over a 30-month follow-up period can now be reported. **Study Design/Methods:** Between October 1, 2010-September 30, 2011 (1 year), all patients with WAA undergoing pre-transfusion work up at the Cedars Sinai Medical Center blood bank were prospectively included in the study. Extended human erythrocyte antigen typing (x-HEA) for C c E e, K, k, Jka, Jkb, Fya, Fyb, S, and s was performed using DNA microarray technology (BioArray). Patients received blood that was maximally matched for these antigens when x-HEA results were available. Patients were followed for up to 30 months post-transfusion for the development of alloantibodies. **Results/Findings:** A total of 74 patients with WAA were genotyped. Thirty (41% of) patients were transfused extended-matched RBCs to the extent possible based on their x-HEA results. Of these, 24 (80%) had at least one subsequent antibody screen performed from 1-30 months after receiving extended-matched blood. No new alloantibodies were detected in any of these patients. A total of 1049 units were transfused to these 30 patients, of which 93% (977 units) were matched for all clinically significant x-HEA antigens. **Conclusion:** The results support the efficacy of the employed transfusion algorithm for the prevention of alloimmunization in patients with warm autoantibodies. Genotyping methods have previously been shown to accurately predict serologic phenotype, and high throughput, semi- automated platforms can be employed for comprehensive red cell typing in a hospital transfusion service. Once a patient's x-HEA type is determined, the results may be used to improve transfusion practice by facilitating transfusion of more highly matched blood for patients known to be at risk for alloimmunization.

Disclosure of Commercial Conflict of Interest

I. A. Abumuhor: Nothing to disclose; M. R. Edmunds: Nothing to disclose; E. Klapper: Nothing to disclose; J. Seo: Nothing to disclose; H. Shafi: Nothing to disclose; E. C. Vamvakas: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. A. Abumuhor: Nothing to disclose; M. R. Edmunds: Nothing to disclose; E. Klapper: Nothing to disclose; J. Seo: Nothing to disclose; H. Shafi: Nothing to disclose; E. C. Vamvakas: Nothing to disclose

SP254

Prevalence and Characterization of Erythrocyte Antibodies in Cardiovascular Surgery Patients

O Peñuela[1] (oscar_penuela@hotmail.com), L Gómez[3,1], J Perea[2], B Espitia[2]. *[1]Investigación y Desarrollo, Banco de Sangre Higuera Escalante, Floridablanca, Colombia; [2]Laboratorio Clínico, Fundación Clínica Shaio, Bogotá, Colombia; [3]Grupo de Investigación en Ciencias Biomédicas, Universidad Antonio Nariño, Bogotá, Colombia*

Background/Case Studies: Development of new red blood cell (RBC) alloantibodies (alloimmunization) is one of the most frequent adverse reactions after an RBC transfusion. Several irregular red blood cell alloantibodies, produced by alloimmunization of antigens in transfusions or pregnancies, have clinical importance because they could cause hemolysis in the fetus, newborn or in transfused patients. The aim of this study was to establish the prevalence and characterization of alloantibodies in a population of cardiovascular surgery patients. **Study Design/Methods:** Serological studies were done here in 1904 cardiovascular surgery patients receiving 4613 RBC units, during a period of 1 year in 2012. Before transfusion, all samples were submitted to irregular antibody testing and, when positive, to the identification and serial titration of alloantibodies. The following data were collected from each patient: gender, age, ABO/Rh blood group, type of acute disease or medical emergency, and the number of transfusions, as well as the results of screening and identification of irregular antibodies. **Results/Findings:** Of 1904 patients tested, 38 (2%) have irregular RBC antibodies. Fourteen (37%) individuals do not have a recent transfusion history. Among the alloinmmunized individuals, 30 (79%) were women and 8 (21%) men. Mean age was 66 years, and 3.9 units were administered per patient (range: 1-35). Antibodies characterization was: anti-c: 2 (5.3%); anti-E: 12 (31.6%); anti-K: 7 (18,4%); anti-Fya: 2 (5.3%); anti-e: 1 (2.6%); anti-Jka: 3 (7,9%); anti-D: 6 (15,8%); and anti-C: 2 (5.3%). Autoantibodies characterization was: 3 (7,9%). ABO/Rh distribution was: O positive: 57.9%; O negative: 7.9%; A

positive: 18.4%, A negative: 5.3%; B positive: 5.3%; B negative: 5.3%; no AB individuals were identified. **Conclusion:** The present study shows that of 1904 cardiovascular surgery patients treated in 2012, 38 (2%) have irregular RBC alloantibodies before transfusion. The frequency of alloimmunization was according to antibodies prevalence in general hospital population (0.2%-2%). The higher occurrence of alloantibodies against antigens of Rh (60.6%) and Kell (18.4%) systems is in agreement with the literature; both systems have highly immunogenic antigens. There was a significantly higher rate in women suggesting that the risk of alloimmunization might be influenced by the gender of the recipient, in particular due to gestations. It is suggested that phenotyping and compatibilization for the C c E e (Rh system), and K (Kell system) antigens should be extended, when possible, to all individuals with programmed surgeries and acute clinical events that do not need emergency transfusions.

Disclosure of Commercial Conflict of Interest

B. Espitia: No Answer; L. Gómez: No Answer; J. Perea: No Answer; O. Peñuela: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Espitia: No Answer; L. Gómez: No Answer; J. Perea: No Answer; O. Peñuela: Nothing to disclose

SP255

A Case of a Gerbich Antibody with an Unusual Reactivity Pattern Associated with Severe Hemolytic Disease of the Newborn and Prominent Suppression of Erythropoiesis

R C Skeate[1] (robert.skeate@blood.ca), C Tomlinson[2], D Louis[2], D Nicholls[2], O Elkhateeb[2], W Lau[2], J M Cote[3], C Lomas-Francis[4], R W Velliquette[4], S Vege[4]. *[1]Canadian Blood Services, Toronto, ON, Canada; [2]Hospital for Sick Children, Toronto, ON, Canada; [3]Canadian Blood Services, Ottawa, ON, Canada; [4]New York Blood Center, New York, NY, United States*

Background/Case Studies: Alloantibodies to Gerbich (Ge) antigens are rare. Anti-Ge3 has been associated with hemolytic transfusion reaction (HTR) and hemolytic disease of the fetus and newborn (HDFN) with suppression of erythropoiesis, while anti-Ge2 only with HTR. The case patient's mother is a blood group A, G3P2 27-year-old of Honduran descent whose second child had neonatal jaundice not requiring therapy. Her third pregnancy was complicated only by a difficult to characterize Ge antibody identified on a 32-week prenatal sample. At term delivery, the blood group A, 3.8 kg male was healthy with hemoglobin (HGB) 148g/L and total bilirubin (TB) 66µmol/L, but DAT was positive and he was HGB 133g/L and TB 89µmol/L within a few hours. He was managed with phototherapy, hydration, and IVIG 1g/kg with TB peaking at 273µmol/L on day 4. He was discharged day 6 but readmitted day 10 with HGB 57g/L, retic 3.9%, absolute retic count (RC) 64 x10e9/L, and TB 59µmol/L. Crossmatch compatible thawed GE:-2,-3,4 RBCs were transfused (HGB to 121g/L) and he was discharged. Over several weeks HGB dropped slowly, TB did not rise and RC remained inappropriately low. At 6 weeks HGB was 104g/L and RC was 171x 10e9/L without further transfusion. **Study Design/Methods:** Antigen typing, antibody identification, and DAT were performed by standard hemagglutination methods. The mother's plasma was tested by IAT with phenotypically similar RBCs treated with papain, trypsin, α-chymotrypsin, and 200mM dithiothreitol (DTT). Genomic DNA was isolated from WBCs. PCR assays for GE exons 2, 3 and 4 were performed and exons 2 and 4 were sequenced by standard methods. **Results/Findings:** The mother's plasma was reactive (4+) by saline or PEG IAT with all RBCs tested except her own. It did not react with GE:-2-3 or Ge:-2,-3,-4 RBCs by saline or PEG-IAT, but reacted with GE:-2,3 RBCs, consistent with anti-Ge3. It reacted with α-chymotrypsin or 0.200mMDTT treated RBCs, was non-reactive with papain treated RBCs and was non-reactive or reacted only very weakly with trypsin-treated red cells. This pattern of reactivity with enzyme or DTT treated RBCs is more consistent with anti-Ge2 than anti-Ge3. Genomic DNA analysis revealed a deletion of GE exon 3 consistent with her serologically determined GE:-2,-3, 4 phenotype. Sequencing of exons 2 and 4 found no changes. **Conclusion:** A clinically significant Gerbich antibody with unusual specificity is presented here. It reacted with GE:-2,3 RBCs but not with GE:-2,-3 RBCs; this is consistent with anti-Ge3; however the reactivity pattern with enzyme or DTT treated RBCs was more consistent with anti-Ge2. GE:-2,-3 RBCs were compatible and showed good clinical effect on transfusion. This case further supports previous observations that antibodies with Ge reactivity, such as reported for anti-Ge3, can be associated with severe HDFN with prominent suppression of erythropoiesis.

Disclosure of Commercial Conflict of Interest

J. M. Cote: Nothing to disclose; O. Elkhateeb: Nothing to disclose; W. Lau: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; D. Louis: Nothing to disclose; D. Nicholls: Nothing to disclose; R. C. Skeate: Nothing to disclose; C. Tomlinson: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. M. Cote: Nothing to disclose; O. Elkhateeb: Nothing to disclose; W. Lau: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; D. Louis: Nothing to disclose; D. Nicholls: Nothing to disclose; R. C. Skeate: Nothing to disclose; C. Tomlinson: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose

SP256

Absence of Hemolytic Transfusion Reaction in a Patient with Anti-Joa Chronically Transfused with Jo(a+) Red Blood Cells

S D Whelchel[1] (swhelchel@indianablood.org), G T Noumsi[2], J M Moulds[2], J K Slayten[1], D A Waxman[1], J Cruz[1]. *[1]Immunohematology Reference Lab, Indiana Blood Center, Indianapolis, IN, United States; [2]Scientific Support Services, LifeShare Blood Centers, Shreveport, LA, United States*

Background/Case Studies: A 42-year-old patient with sickle cell disease (SCD) presented with an acute respiratory distress syndrome (ARDS). Following administration of 12 crossmatch compatible RBCs units, the patient developed signs of a delayed hemolytic transfusion reaction. A sample was submitted to the Immuno-Hematology Reference Laboratory for investigation. **Study Design/Methods:** A cell separation was performed in order to determine the patient phenotype. DNA was isolated by standard methods and genotyped by BioArray/Immucor HEA. The antibody identification was performed using a standard PEG-AHG tube method. In order to access the clinical significance of the antibody, a monocyte monolayer assay (MMA) was performed, and RBCs units crossmatched using both serology and MMA. **Results/Findings:** On receipt, the initial sample was brown in color and very turbid; with a 1+mf DAT. The patient was phenotyped as rr, K-, Fy(a-b-), S-s+ using antisera, and Jo(a-) using genotyping. Anti-Fya and anti-Joa were identified in the eluate. MMA using segments from six of the previously transfused units showed that the anti-Fya was primarily responsible for the observed transfusion reaction with a MI = 5.85%, as compared to a MI = 0.57% for the anti-Joa. Two Fy(a-), Jo(a+) MMA crossmatched RBC units were transfused within 2 days, and the hemoglobin level went from 5.5g/dl to 9.0g/dl. Over a month period, the patient was seen in the hospital two more times; an additional anti-Jkb was identified; and five Fy(a-), Jk(b-), Jo(a+) MMA crossmatched RBC units were transfused. Two days after the last transfusion, the total bilirubin and the haptoglobin levels were respectively 3.3mg/dl and 28.4mg/dl. In addition, the anti-Joa never showed an increase in titer, as proven by the weakly positive reactivity at AGT in subsequent post-transfusion samples. Six days following the last hospital stay, a new antibody evaluation was performed and clearly demonstrated anti-Fya, anti-S and anti-Joa. As a result for this patient, future management of transfusion needs will require MMA crossmatch for providing RBC units offering the best survival advantage. **Conclusion:** This is an example of a patient with anti-Joa in addition to anti-Fya, anti-Jkb and anti-S, who was chronically transfused with Jo(a+) red cells units, with no signs of an overt hemolytic transfusion reaction.

Disclosure of Commercial Conflict of Interest

J. Cruz: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support, Novartis, Travel Support or Honorarium, Grifols, Consulting or Board of Director Fees; G. T. Noumsi: Nothing to disclose; J. K. Slayten: Nothing to disclose; D. A. Waxman: Nothing to disclose; S. D. Whelchel: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Cruz: Nothing to disclose; J. M. Moulds: Nothing to disclose; G. T. Noumsi: Nothing to disclose; J. K. Slayten: Nothing to disclose; D. A. Waxman: Nothing to disclose; S. D. Whelchel: Nothing to disclose

SP257

First Report of the Langereis-Phenotype in a Saudi Patient with Anti-Lan

F Anwar[1] (farzala@hotmail.com), M A Abdelaal[1], E M Mensinger[1], D J Lewis[1]. *[1]Pathology & Laboratory Medicine, King Abdulaziz Medical City, Jeddah, Saudi Arabia*

Background/Case Studies: The Lan (Langereis) red cell antigen is of high-incidence with Lan- phenotype occurrence in approximately 1 in 20,000 random population to 1 in 50,000 in Japanese population. Anti-Lan can

cause severe hemolytic transfusion reaction and cause hemolytic disease of fetus and new born (HDFN). A case of a Saudi patient with Lan- phenotype with anti-Lan is reported on here. **Study Design/Methods:** A 37-year-old Saudi female gravida 7, para 5 with one set of twins who died in utero, was admitted for medical termination of pregnancy. Her hemoglobin on admission was 11.3 g/dl. Pretransfusion testing yielded an ABO result of O positive and a positive antibody screen with all three screening cells demonstrating 2+ reactivity. The plasma was reactive 2+ with all 11 cells of the reagent red cell panel. The bromelin treated panel red cell enhanced the antibody reactivity to 4+. The auto-control using untreated and papain pretreated RBCs was negative. A direct antiglobulin test with poly AHG was negative. The sample was referred to the International Blood Group Reference Laboratory (Bristol, United Kingdom) for further testing and antibody identification. **Results/Findings:** Based upon serological presentation, an antibody to an unknown high-frequency red cell antigen/antigens was suspected. The reference laboratory identified anti-Lan in the plasma with no concomitant alloantibodies present. The patient's RBCs were confirmed as Lan antigen negative. **Conclusion:** This is the first report of the Lan- phenotype with anti-Lan in a Saudi national and there are no published reports of this rare phenotype occurring in neighboring Arab countries. Further familial studies are required to establish the pedigree of the propositus and potentially identify a readily available donor pool. Based upon the occurrence of consanguinity in Saudi Arabia, it may prove fruitful to screen the blood donor population for the Lan antigen relative to its autosomal recessive mode of inheritance.

Disclosure of Commercial Conflict of Interest

M. A. Abdelaal: Nothing to disclose; F. Anwar: Nothing to disclose; D. J. Lewis: Nothing to disclose; E. M. Mensinger: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. A. Abdelaal: Nothing to disclose; F. Anwar: Nothing to disclose; D. J. Lewis: Nothing to disclose; E. M. Mensinger: Nothing to disclose

SP258

Anti-G: A Case of an Underestimated Alloantibody In Pregnancy

S Valsami[1] (serenavalsami@yahoo.com), M Papakonstantinou[1], G Papadopoulos[1], M Dadakaridou[2], G Kanavos[2], E Desylla[1], I Flesiopoulou[1], L Foudoulaki-Paparizos[1]. *[1]Blood Transfusion Service, General Hospital of Nikea, Piraeus, Greece; [2]Obstetrics Department, REA Obstetrics-Gynecology Hospital, Athens, Greece*

Background/Case Studies: Anti-G antibody is an antibody directed against the G antigen of the Rh system, which is expressed in almost all red cells that carry D or C antigen or both. It has been reported to cause hemolytic disease of the fetus and newborn. Anti-G often masquerades as anti-D and anti-C (anti-D+C) leading to no prophylactic Rh immune globulin (RhIG) administration with possible severe consequences in subsequent pregnancies. A rare case of anti-G alone formation during pregnancy is reported here. **Study Design/Methods:** A 29-year-old pregnant woman, who was first presented to the department at 7 weeks of gestation for routine prenatal work-up, was alloimmunized to G antigen during pregnancy. **Results/Findings:** She was typed as blood group O D-, C-, E-, c+ e+. The pregnant woman had a recent past history of a miscarriage followed by RhIG prophylaxis as appropriate, 6 months before admission. She had no history of past blood transfusion. The Indirect Antiglobulin Test was negative at booking and became positive at 21st week of gestation. The antibody identification, using commercially available test erythrocytes (Identisera Diana/Extended—Grifols S.A. Spain) showed that the patient's serum reacted with all but ccddee (rr) and ccddEe (r' 'r) test RBCs, suggesting the presence of anti-D+C. However, due to the pregnant woman's and father's Rh phenotype (father: blood group A D+, C-c+E+e+), the presence of anti-G with or without anti-D was strongly suspected. Absorption/elution tests using group OCcddee (r'r) test RBCs, confirmed the presence of anti-G without anti-D in the pregnant's serum. The antibody titer, against group O CCDee (R1R1) test RBCs was determined at several time points and was always found below 8. The pregnancy was closely monitored. Middle cerebral arteria-peak systolic velocity, MCA-PSV was normal at 22 and 33 weeks. At 37 weeks an elective C section was performed and a healthy male neonate was born. The neonate's blood group was A D+, C-c+E+e+. At birth the infant's Direct Antiglobulin Test (DAT) was positive (DAT = +2), serum bilirubin was 3,5 mg/dl and hemoglobin was 16,5 gr/dl. At 3 days the bilirubin peaked at 15 mg/dl and the newborn received phototherapy. No transfusion support was required and the infant was discharged in good health on day 5. The mother received a full dose of RhIG at 28 weeks of gestation and a second dose postpartum in order to prevent D alloimmunization. **Conclusion:** A rare case of anti-G identification alone during pregnancy that was possibly attributed to the mild hemolysis of the newborn is reported here. Anti-G cases often can be misidentified as anti-D+C. Cases of pregnant women carrying anti-G, with or without anti-C, but not anti-D are not infrequent and should be identified, in order to receive adequate RhIG prophylaxis to prevent alloimmunization to D.

Disclosure of Commercial Conflict of Interest

M. Dadakaridou: No Answer; E. Desylla: Nothing to disclose; I. Flesiopoulou: Nothing to disclose; L. Foudoulaki-Paparizos: Nothing to disclose; G. Kanavos: Nothing to disclose; G. Papadopoulos: Nothing to disclose; M. Papakonstantinou: Nothing to disclose; S. Valsami: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Dadakaridou: No Answer; E. Desylla: Nothing to disclose; I. Flesiopoulou: Nothing to disclose; L. Foudoulaki-Paparizos: Nothing to disclose; G. Kanavos: Nothing to disclose; G. Papadopoulos: Nothing to disclose; M. Papakonstantinou: Nothing to disclose; S. Valsami: Nothing to disclose

SP259

An Example of Apparent Allo Anti-E in a Patient who is E+

J Spahic[1] (jspahic@indianablood.org), S M Davisson[1], K Evans[1], J K Slayten[1], J Cruz[1]. *[1]Immunohematology Reference Lab, Indiana Blood Center, Indianapolis, IN, United States*

Background/Case Studies: The subject is a 60-year-old female with no prior history of antibody development. Three months prior, the patient had been transfused with red blood cells after knee replacement surgery. The antibody identification (ABID) was ordered on the group O, D Positive patient before an additional knee surgery. The pre-surgical screening showed the patient demonstrated a positive antibody screen with a panel exhibiting a pattern of anti-E. However, the Rh typing revealed that the patient is E+e+.These unexpected findings were further investigated by the Immunohematology Reference Lab (IRL). **Study Design/Methods:** DAT testing was performed using a tube method with Polyspecific AHG, Anti IgG, and anti-C3b,-C3d (Immucor, Norcross, GA). ABID panels were tested by PEG-AHG, Ficin-AHG (Immucor, Norcross, GA) and Gel (IgG Card, Ortho, Raritan, NJ). A ZZAP treated autoadsorption was completed by a standard method. Red cell antigen typing was completed with two different FDA licensed antisera (Immucor, Norcross, GA) and with two additional human derived anti-E. The sample was subsequently submitted for RHCE Genotyping (RhCE ver. 2.0 microarray BioArray Solutions, Immucor, Norcross,GA). **Results/Findings:** PEG-AHG and Ficin-AHG panels showed a simple pattern of anti-E. The DAT and autocontrol were negative but an auto-anti-E was suspected since the patient typed as E+. An auto adsorption was performed to investigate if the antibody is allo or auto in nature. The auto adsorption was incomplete and failed to remove the pattern of anti-E. The E antigen typing with FDA licensed and human derived antisera determined the patient to be E+. Genotyping by two molecular platforms confirmed that the patient has normal E+e+ alleles. **Conclusion:** While the patient has normal E+ expression, the serologic reactivity in panels, DAT, and auto control demonstrate what appears to be an unexpected allo anti-E. Since the serology demonstrated anti-E, it was suggested for this patient to receive E antigen negative units at this time. This case demonstrated an unexpected pattern of an anti-E in a patient verified by serologic and molecular methods to be E+.

Disclosure of Commercial Conflict of Interest

J. Cruz: Nothing to disclose; S. M. Davisson: Nothing to disclose; K. Evans: Nothing to disclose; J. K. Slayten: Nothing to disclose; J. Spahic: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Cruz: Nothing to disclose; S. M. Davisson: Nothing to disclose; K. Evans: Nothing to disclose; J. K. Slayten: Nothing to disclose; J. Spahic: Nothing to disclose

SP260

Two Cases of Naturally Occurring Anti-Fy^b

T Rees[1] (terry.rees@imail.org). [1]Immunohematology Reference Lab, American Red Cross Blood Services, Salt Lake City, UT, United States

Background/Case Studies: Anti-Fy^b is usually IgG and occurs as a result of prior red cell stimulation. The Fy^b antigen is not a very immunogenic antigen and is found 20 times less often than its counterpart anti-Fy^a. There is only one official documented case of a naturally occurring anti-Fy^b reported in Poland in 2000. **Study Design/Methods:** A 27-year-old male presented with

bleeding tonsils 1 week post tonsillectomy. A type and screen was ordered. The patient's antibody screen was positive and a sample was forwarded to an Immunohematology Reference Laboratory for antibody identification. The hospital verified with the patient that he had never been transfused. Initial antibody screening was performed using a three cell screen in both LISS: AHG and PEG:IgG. Two out of three screening cells were reactive in both LISS and PEG at IAT only. A 10-cell panel of reagent red cells was tested with PEG:IgG and six out of ten0 cells reacted strongly in PEG:IgG. **Results/Findings:** The results of the panel confirmed the presence of an anti-Fy^b. Since the formation of a naturally occurring anti-Fy^b is extremely rare, the accuracy of the patient's transfusion history was questioned. The hospital was asked to re-verify the patient's transfusion history. An interview of the patient's parents and a thorough search of the patient's medical records confirmed that he had never received a transfusion. Further investigation of the patient's life revealed that he had been exposed to animal blood through the butchering of his own game. Within 3 months of the discovery of this anti-Fy^b, the presence of an anti-Fy^b in the serum of a male whole-blood donor was also confirmed. Suspicious of the strength of the anti-Fy^b after 12 years, the donor was contacted to obtain a transfusion history. The donor confirmed he had never received a transfusion. He also reported one incidence of preparing game about 15-20 years ago. **Conclusion:** The formation of a naturally occurring anti-Fy^b is very rare. The 2000 Polish case cited that the man was a butcher of both domestic and game animals. The common exposure scenario of these three individuals to animal blood could explain their sensitization to Fy^b. Since the Fy^b antigen is known to be found on the red cells of some apes, one could speculate that the Fy^b antigen may also be present on the red cells of certain game such as deer or elk. Or, it could be that these three examples are all cases of a heteroagglutinin. Perhaps certain game harbor a bacteria that is antigenically similar to the Fy^b antigen thus causing the stimulation of an anti-Fy^b heteroagglutinin. Further studies would need to be performed to identify the specific cause of the stimulation of anti-Fy^b formation in non-transfused individuals.

Disclosure of Commercial Conflict of Interest

T. Rees: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Rees: Nothing to disclose

SP261

Generation of Multiple Red Cell Alloantibodies in a Post-Splenectomy Trauma Patient

A W Moritz[1] (MoritzA@uthscsa.edu), G Meny[1]. *[1]Pathology, University of Texas Health Science Center-San Antonio, San Antonio, TX, United States*

Background/Case Studies: The rate of red cell (RBC) alloantibody formation following uncrossmatched emergency release (UNC) in trauma has been described as 3%, with delayed hemolytic transfusion reaction at approximately 0.02% (Mulay et al., *Transfusion* 2012, Oct 15). A case is reported here of a trauma patient who developed multiple alloantibodies post-splenectomy and was managed with crossmatch and antigen incompatible blood. **Study Design/Methods:** Antibody screening was performed using solid phase technique on the BioRad TANGO and standard gel and tube techniques. Antigen typing was performed using licensed antisera or laboratory developed molecular methods. **Results/Findings:** A 50-year-old diabetic gravida 0 Black female suffered multiple injuries, including splenic lacerations, from a motor vehicle crash. She was on hemodialysis and had received prior RBCs. Initial antibody screen was positive for anti-Fya; she typed as group O, D positive. Eight RBC units (four Fya antigen positive), two fresh frozen plasma (FFP) units and one single donor platelet (SDP) were given emergently. Splenectomy was performed. On day 7 the direct antiglobulin test (DAT) was +w (anti-IgG) only, with anti-Fya in the eluate. The patient had no demonstrable significant hemolysis (Table). Subsequent downtrending hemoglobin was believed to be due to trauma, surgery, and chronic disease. On day 10, a sample reacted with all panel cells, with 2+ IgG DAT, with anti-Fya again in the eluate. No compatible RBCs were identified. Patient samples were referred to a reference laboratory. Four Fy (a-b-) and K- crossmatch incompatible RBCs were issued. At 30 days, the DAT was 3+ (anti IgG) and +w (anti-C3d); an eluate contained anti-Fya and anti-D. Reference laboratory results demonstrated the presence of anti-D, -C, -S, -Fya, -Jkb, and antibodies with HTLA characteristics. An RHD genotype was performed which showed a probable phenotype of weak partial RHD type 4.0/RHDΨ. **Conclusion:** There may be many factors for the rapid generation of multiple alloantibodies. The patient may be an allo-responder, given the history of prior transfusions. It is also probable that the patient received non-matched blood, exposing the patient to multiple incompatible antigens (uncommon blood type with a predicted weak partial RHD 4.0). It is unclear of the role of splenectomy in stimulating the generation of alloantibodies or in reducing the rate of hemolysis when incompatible blood was administered. Incompatible blood was administered with no apparent adverse clinical outcome. However, multiple alloantibodies were generated.

Disclosure of Commercial Conflict of Interest

G. Meny: Nothing to disclose; A. W. Moritz: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Meny: Nothing to disclose; A. W. Moritz: Nothing to disclose

Hemolysis Evaluation Laboratory Values

Days after admission	0	1	3	6	13	16	18	34
Total Serum Bilirubin (mg/dL)	0.3	0.8	0.5	1.0	0.7	0.6	0.5	0.4
Direct Serum Bilirubin (mg/dL)		0.3		0.6			0.2	
Hemoglobin (g/dL)	8.2	12.6	9.7	7.7	8.5	7.9	7.0	6.7
Haptoglobin (mg/dL)								198

Red Blood Cells: Blood Group Genes and Genotyping: General Issues

SP262

Pilot Study for Performance Evaluation of ID-CORE XT™: A Blood Group Systems Genotyping Tool

A Molano-Bilbao[1] (amolano@progenika.com), D García-Crespo[1], I Apraiz[1], M Szczypiorska[1], E Olano[1], L Jofre[1], D Tejedor[1], M Lopez[1]. *[1]Progenika Biopharma S.A., Derio, Spain*

Background/Case Studies: ID-CORE XT™ (Progenika; Derio, Spain) is a genetic test based on Luminex xMAP® technology for the simultaneous identification of multiple alleles of the following blood group systems: Rh, Kell, Kidd, Duffy, MNS, Diego, Dombrock, Colton, Cartwright, and Lutheran. The assay analyzes 29 polymorphisms determining 37 antigens of the aforementioned blood group systems. The objective of this pilot study is a small scale preliminary evaluation of the performance of ID-CORE XT™. **Study Design/Methods:** The performance evaluation of the ID-CORE XT™ was established through evaluation of 100 genomic DNA samples previously typed by BLOODchip® Reference v4.0, ID-CORE™ and bi-directional sequencing. The samples were selected as follows: 57 patients and 43 donors with an ethnic origin of 54 Caucasian and 46 Afro-American. Genomic DNA extracted from whole blood was amplified by a single multiplex PCR. PCR products were hybridized, labeled with a fluorescent conjugate and detected with a Luminex® System. The raw data were processed with the property ID-CORE XT™ Analysis Software to obtain a genotype and predict a phenotype for blood groups. **Results/Findings:** No experimental errors or no call results occurred. No discrepancies with reference method were found, being the accuracy among SNPs, genotypes and predicted phenotypes 100%. The comparison of ID-CORE XT™ predicted phenotypes (presence or absence of antigens) are summarized in the Table below. **Conclusion:** ID-CORE XT™ test is an accurate tool for the determination of the mentioned Blood Group Systems in this pilot study of donors and patients of two ethnic group populations.

Disclosure of Commercial Conflict of Interest

I. Apraiz: Nothing to disclose; D. García-Crespo: Progenika Biopharma, Ownership or Partnership; L. Jofre: Progenika Biopharma, Other; M. Lopez: Nothing to disclose; A. Molano-Bilbao: Progenika Biopharma, Ownership or Partnership; E. Olano: Nothing to disclose; M. Szczypiorska: Nothing to disclose; D. Tejedor: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. Apraiz: Nothing to disclose; D. García-Crespo: Nothing to disclose; L. Jofre: Nothing to disclose; M. Lopez: Nothing to disclose; A. Molano-Bilbao: Nothing to disclose; E. Olano: Nothing to disclose; M. Szczypiorska: Nothing to disclose; D. Tejedor: Nothing to disclose

Blood group system	Antigens (ISBT phenotype)	Antigen presence		Antigen absence	
		Number of samples tested per antigen	% Accuracy	Number of samples tested per antigen	% Accuracy
Rh	C (RH:2)	70	100	30	100
	E (RH:3)	13	100	87	100
	c (RH:4)	99	100	1	100
	e (RH:5)	99	100	1	100
	CW (RH:8)	0	100	100	100
	V (RH:10)	23	100*	77	100*
	hrS (RH:19)	100	100*	0	100
	VS (RH:20)	52	100	48	100
	hrB (RH:31)	82	100*	18	100*
Kell	K (KEL:1)	6	100	94	100
	k (KEL:2)	100	100	0	100
	Kpa (KEL:3)	0	100	100	100
	Kpb (KEL:4)	100	100	0	100
	Jsa (KEL:6)	12	100	88	100
	Jsb (KEL:7)	100	100	0	100
Kidd	Jka (JK:1)	84	100	16	100
	Jkb (JK:2)	52	100	48	100
Duffy	Fya (FY:1)	43	100	57	100
	Fyb (FY:2)	50	100	50	100
MNS	M (MNS:1)	77	100	23	100
	N (MNS:2)	73	100	27	100
	S (MNS:3)	43	100	57	100
	s (MNS:4)	88	100	12	100
	U (MNS:5)	100	100	0	100
Diego	M a (MNS:7)	0	100	100	100
	Dia (DI:1)	0	100	100	100
	Dib (DI:2)	100	100	0	100
Dombrock	Doa (DO:1)	61	100	39	100
	Dob (DO:2)	90	100	10	100
	Hy (DO:4)	100	100*	0	100
	Joa (DO:5)	99	100*	1	100*
Colton	Coa (CO:1)	100	100	0	100
	Cob (CO:2)	3	100	97	100
Cartwright	Yta (YT:1)	100	100*	0	100
	Ytb (YT:2)	7	100*	93	100*
Lutheran	Lua (LU:1)	8	100	92	100
	Lub (LU:2)	100	100	0	100

*Reference method only available for 46 samples

SP263

Routine Use of Molecular Donor Typing Results: Five Years of Experience with Pooled Capillary Electrophoresis

F Wagner[1] (fwagner@bsd-nstob.de), A Doescher[2], R Bittner[1], T H Müller[1].
[1]Institut Springe, DRK Blutspendedienst NSTOB, Springe, Germany;
[2]Institut Bremen-Oldenburg, DRK Blutspendedienst NSTOB, Oldenburg, Germany

Background/Case Studies: At this blood service, high throughput molecular blood group prediction of donors by pooled capillary electrophoresis was started in 2007. An IT solution allowing guidance of units to a laboratory inventory according to the results was introduced in January 2009. The increasing number of typed donors may lead to changes in the workflow. The development since 2009 was compiled. **Study Design/Methods:** Extended blood group antigen profiles for 16 antigens (Fy^a, Fy^b, Jk^a, Jk^b, M, N, S, s, Co^a, Co^b, Yt^a, Yt^b, Lu^a, Lu^b, Kp^a,and Kp^b) were obtained by pooled capillary electrophoresis and stored in the donor database. Donations complying specific rules were diverted to laboratory inventories and released to the general supply only after a certain hold time. Number of typed donors, number of diverted units per month, and distribution of antigens were evaluated from the IT system. Mismatches of predicted and serologically confirmed phenotype were analyzed. **Results/Findings:** Molecular typing started with 1066 donors per month and is currently continued at 615 donors per month. Total number of typed donors is 62,574; in 47,121 donors, the full set of antigens was determined. In 2012, 16.6% of all donations were from molecularly typed donors corresponding to 8928 units per month, of which 1831 were diverted to the laboratory inventory (2009: 798). Considering the most frequently demanded antigens Fy^a, Jk^a, S, and M, quadruple-negative units per month increased from 35 in 2009 (3.5%) to 64 in 2012 (4.4%); triple-negative from 183 (23%) to 472 (26%). Considering antigens Fy^a, Fy^b, Jk^a, Jk^b, M, N, S, and s, quadruple-negative units increased from 115 (14%) to 279 (15%), triple-negative from 274 (34%) to 730 (40%): Units negative for Yt^a, Co^a, Lu^b, or Kp^b increased from 15 (1.9%) to 37 (2%). The percentage of rr units decreased from 47% to 40%, those of O rr from 33% to 22%. In 2012, 43 discrepancies of molecular prediction and serologic re-testing were noted; these were due to technical failures (n = 16), manual overinterpretation of low quality results (n = 15), incorrect serologic results (n = 11) and misrecognized alleles (n = 1). **Conclusion:** The increasing number of typed donors mainly led to increasing numbers of units diverted to the laboratory inventory, while the "quality" of units as scored by number of negative antigens increased only slightly. Future strategies should focus on optimized selection of units for the laboratory inventory. Most prediction failures were related to technical problems, therefore adding additional single nucleotide polymorphisms to the current detection system will not improve prediction accuracy.

Disclosure of Commercial Conflict of Interest

R. Bittner: Nothing to disclose; A. Doescher: Nothing to disclose; T. H. Müller: Nothing to disclose; F. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Bittner: Nothing to disclose; A. Doescher: Nothing to disclose; T. H. Müller: MacoPharma, Grants or Research Support; F. Wagner: Red Cross Blood Service, Other

SP264

Comparison of Molecular Biology Commercial Kits for Donor Red Blood Cell (RBC) and Platelet (PLT) Typing of the Lombardy Rare-Donor Program

S Scognamiglio[1] (serena.scognamiglio@policlinico.mi.it), M Villa[1], C Paccapelo[1], M Manera[1], R Parisi[1], A Manazza[1], G Portella[1], N Revelli[1], M Marconi[1]. *[1]Immunohaematology Reference Laboratory and Immunohaematology Transfusion Centre, Milano, Italy*

Background/Case Studies: During September-November 2012, new molecular biology commercial kits were compared based on the Luminex xMAP® method (IDHPA and IDCore+, Progenika Biopharma, Derio, Spain) with those used in our laboratory. As recommended by international guidelines, a validation plan was designed to evaluate sensitivity and specificity. **Study Design/Methods:** Methods: During the validation tests DNA was used that was extracted from EDTA whole blood of selected blood donors by automatic instrument QIAcube (QIAGEN, Valencia, CA) and stored at + 4C for a week and then at -40C. A total of 144 donor samples, previously typed by HEA BeadChip™ kit (BioArray Solutions, Warren, NJ) and/or serology by a high-productivity system (NeoGalileo, Immucor, Norcross, GA) were tested by ID Core+ kit. Moreover, 230 samples, previously typed by HPA BeadChip™ kit (BioArray Solutions) and/or THROMBOTYPE® (GTI Diagnostics,) and/or HPA-Type (BAGene, BAG Health Care, Lichtenstein, Germany), were tested by ID HPA kit. **Results/Findings:** Results: Unlike the HEA BeadChip™ kit, the ID Core+ kit allows to type also for Yta/Ytb antigens. In this regard, a rare donor known as Coa negative resulted also Yta negative. Further tests were carried out by RARE ID kit (GTI Diagnostics), which confirmed the negativity of both antigens. Eight of the 144 samples tested with ID Core+ showed discrepant results between serology and molecular biology. Seven of the 230 samples tested with ID HPA were discrepant by HPA BeadChip™ kit only for the HPA 2b antigen. In particular, two of seven samples resulted HPA 2(a+b+) by BeadChip™ and HPA 2(a+b-) by ID HPA kit. The tests were repeated with both methods and also with the THROMBOTYPE® kit (GTI Diagnostics) which confirmed the BeadChip™ result. Sequencing analysis on the 15 discrepancies was performed by BLOODChip Service (Progenika Biopharma). In four samples the sequencing analysis confirmed molecular results and not serology, in nine samples instead revealed some polymorphisms in the primers region that prevent the alleles amplification. One sample was coincident between ID-Core+ and serology, not with BeadChip™, one sample was inconclusive. **Conclusion:** Conclusion: This study confirms that RBC and PLT typing should always be done with several different methods, at least one in serology and another in molecular biology. In fact, the molecular methods can give false positive phenotype results if splicing or frameshift mutations or deletions are present, false negative results if the amplification and/or the hybridation does not occur or reveal serological mistyped typing. For this reason, it is very important for Immunohematology Reference Laboratories to use commercial kits based on different technologies and to develop agreements and partnerships with other laboratories capable of performing sequencing studies when necessary.

Disclosure of Commercial Conflict of Interest

A. Manazza: Nothing to disclose; M. Manera: Nothing to disclose; M. Marconi: Nothing to disclose; C. Paccapelo: Nothing to disclose; R. Parisi: Nothing to disclose; G. Portella: Nothing to disclose; N. Revelli: Nothing to disclose; S. Scognamiglio: Nothing to disclose; M. Villa: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Manazza: Nothing to disclose; M. Manera: Nothing to disclose; M. Marconi: Nothing to disclose; C. Paccapelo: Nothing to disclose; R. Parisi: Nothing to disclose; G. Portella: Nothing to disclose; N. Revelli: Nothing to disclose; S. Scognamiglio: Nothing to disclose; M. Villa: Nothing to disclose

SP265

Search of Antigen-Negative Blood Donors using a SNaPshot Protocol to Genotype 16 Alleles Responsible for Expression of Clinically Significant Antigens from RH, KELL, DUFFY, KIDD, and MNS Blood Group Systems

D Gazito[1] (dianagazito@gmail.com), C Arnoni[1], J G Muniz[1], R Person[1], J Barreto[1], L Castilho[2], F Latini[1]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: Molecular analyses are being implemented in transfusion medicine as they permit the simultaneous determination of multiple alleles encoding red blood cell (RBC) antigens in donors and patients and can facilitate the provision of more extensively matched blood for patients by allowing a better selection of donors for given recipients on the basis of the respective antigen repertoires.In this study, It was hoped to develop a SNaPshot protocol that identifies nine SNPs responsible for expression of 12 clinical significant antigens (C c E e, K, k, Fya, Fyb, Jka, Jkb, S, and s) in order to search for antigen-negative donors to facilitate the provision of matched blood for multi-transfused patients. **Study Design/Methods:** DNA samples from 120 blood donors were submitted to a multiplex PCR that amplifies all regions containing selected SNPs. PCR products were evaluated in 3% agarose gels, purified and used in an internal reaction containing probe primers that were designed to discriminate each SNP. This internal reaction was purified, denatured and analyzed in a 3500xL Genetic Analyzer (Applied Biosystem). Samples were also genotyped using PCR-RFLP and phenotyped using conventional serological techniques. **Results/Findings:** The analyses showed that the SNaPshot protocol herein developed successfully genotyped all analyzed samples for RHCE*C/RHCE*c, RHCE*E/RHCE*e, FY*01/FY*02, FY*01N.01/FY*02N.01, KEL*01/KEL*02, JK*01/JK*02, GYPB*03/GYPB*04, GYPB*03N.01, and GYPB*03N.03. PCR-RFLP genotyping and phenotyping results corroborated with SNaPshot in all analyzed samples. **Conclusion:** This SNaPshot protocol allows a medium throughput screening of most clinically significant antigens and a continuous search of blood donors with specific predicted phenotypes. Besides, this protocol may be an option to other molecular strategies, as it is reliable and presents reasonable hands-on-time and costs. The presented SNaPshot protocol may be helpful in situations where phenotyping is impaired, can be a tool to provide fast genotyping results, and might be part of a strategy to improve quality of transfusion by matching clinically significant antigens to reduce alloimunization.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP266

Applying Molecular Immunohematology to Regularly Transfused Thalassemic Patients in Thailand

P Rujirojindakul[1] (rupairay@hotmail.com), W A Flegel[2]. *[1]Pathology, Prince of Songkla University, Hat Yai, Thailand; [2]Transfusion Medicine, National Institutes of Health, Bethesda, MD, United States*

Background/Case Studies: Red cell transfusion is the principal therapy in patients with severe thalassemias and hemoglobinopathies, which are prevalent in Thailand. Serologic red cell typing is confounded by chronic RBC transfusion, because of circulating donor red cells. The concordance was evaluated of the serologic phenotypes between a routine and a reference laboratory and with red cell genotyping. **Study Design/Methods:** Ten consecutive Thai patients with β-thalassemia major who received regular transfusions were enrolled. Phenotypes were tested serologically at Songklanagarind Hospital and at the NIH. Red cell genotyping was performed with commercially available kits and a platform. **Results/Findings:** In only three patients, the red cell genotyping was concordant with the serologic RBC phenotypes for five antithetical antigen pairs in four blood group systems at the two institutions (Table). At NIH, 32 of the 100 serologic tests yielded invalid or discrepant results. The positive predictive value of serology did not reach one for any blood group system at either of the two institutions in this set of ten patients. **Conclusion:** Within this small study, numerous discrepancies were observed between serologic RBC phenotypes at the two hospitals; red cell genotyping enabled determination of the blood group, when serology failed due to transfused red cells. The utility of serologic tests is therefore questioned in regularly transfused pediatric patients. It is proposed to rely solely on red cell genotyping, which requires training for laboratory personnel and physicians. Red cell genotyping outperformed red cell serology by an order of magnitude in regularly transfused patients.

Table 1. Concordance between serologic and DNA-based results in 10 regularly transfused patients with thalassemia

	Test method (hospital) and serologic result *					
	Tube or gel (Songkhla)			Tube (NIH)		
Red cell genotyping result †	+/-	+/+	-/+	+/-	+/+	-/+
Rh system	C+c-	C+c+	C-c+	C+c-	C+c+	C-c+
*RHCE*C/ RHCE*C*	3	0	0	3	0	0
*RHCE*C/ RHCE*c*	2	3	0	0	5	0
*RHCE*c/ RHCE*c*	1	1	0	0	2	0
Rh system	E+e-	E+e+	E-e+	E+e-	E+e+	E-e+
*RHCE*E/ RHCE*E*	0	0	1	0	0	1
*RHCE*E/ RHCE*e*	0	3	2	0	5	0
*RHCE*e/ RHCE*e*	0	0	4	0	0	4
Duffy system	a+b-	a+b+	a-b+	a+b-	a+b+	a-b+
*FY*A/FY*A*	2	7	0	5	4	0
*FY*A/FY*B*	0	1	0	0	1	0
*FY*B/FY*B*	0	0	0	0	0	0
Kidd system	a+b-	a+b+	a-b+	a+b-	a+b+	a-b+
*JK*A/JK*A*	0	3	0	1	2	0
*JK*A/JK*B*	0	1	0	0	1	0
*JK*B/JK*B*	0	5	1	0	2	4
MNS system	S+s-	S+s+	S-s+	S+s-	S+s+	S-s+
*GYPB*S/ GYPB*S*	0	0	0	0	0	0
*GYPB*S/ GYPB*s*	0	1	0	0	1	0
*GYPB*s/ GYPB*s*	0	6	3	0	3	6

* The shaded cells indicate concordances between serologic reactivity and red cell genotyping (predicted phenotype) results. All results outside of the shaded cells represent discordances.

† allele designations as published[43]

Disclosure of Commercial Conflict of Interest

W. A. Flegel: RHD Genotyping Patents, Other; P. Rujirojindakul: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. A. Flegel: Nothing to disclose; P. Rujirojindakul: Nothing to disclose

SP267

Semi-Automation of BeadChip™ Genotyping Using the QIAgility

C J Colvin[1] (carrie.colvin@redcross.org), K Entwistle[2], J C Fusco[2], M A Keller[1]. *[1]National Molecular Laboratory, American Red Cross Biomedical Services, Philadelphia, PA, United States; [2]QIAGEN, Inc., Hilden, Germany*

Background/Case Studies: HEA BeadChip™ platform (BioArray Solutions, Warren, NJ) predicts the phenotype of 38 blood group antigens by genotyping 24 polymorphisms. The platform is available in a 96 sample plate or 8 sample strip configuration. The genotyping process involves four steps with multiple pipette manipulations. There is currently no automated solution for this process; all procedures are performed manually. This workflow presents a challenge, both in terms of process controls and scalability. Technicians use manual pipettors to manipulate DNA samples and reaction mixes, creating the potential for mistakes including sample mix-up and sample-to-sample contamination, as well as inconsistencies in pipetting techniques. Integration of an automated liquid handling platform would achieve improved process controls. **Study Design/Methods:** The QIAgility (Qiagen) was selected due to its compact size, cost and range of

capabilities. Four programs were developed using the QIAgility QIAsoft software to perform the different steps of the assay (PCR setup, Post-PCR Cleanup, Lambda Exonuclease, and Elongation). Each step was tested extensively to determine optimal master mix volumes, deck layout and pipetting accuracy. The HEA BeadChip™ assay was performed in plate and strip format on over 100 DNA samples that had been previously tested using the standard (manual) method and the results compared. With the exception of the final step, in which the reaction mix is loaded onto the BeadChip carrier plate, the entire process was completed using the QIAgility. **Results/Findings:** There was complete concordance of genotypes between samples in which pipetting steps were performed manually or using the QIAgility. There was no contamination, assay failure or sample mix-up using the QIAgility. Reaction volumes were consistent across all samples on all programs. The entire process requires approximately 2 hours of liquid handler time, and all steps of the assay can be completed to reflect the manufacturer's instructions. Hands-on time for the automated method was about 40 minutes, compared to about 1 hour and 40 minutes for the manual method, when setting up a 96 sample plate. **Conclusion:** The QIAgility successfully performed the four protocols of the BeadChip™ process in both plate and strip format. There was no contamination or sample mix-up. The process eliminates pipetting inconsistencies or errors that are common in manual procedures. The process requires more time to set up using the QIAgility robot as compared to the manual method. An affordable, commercially-available liquid handler can be customized to automate pipetting steps involved in BeadChip™ genotyping.

Disclosure of Commercial Conflict of Interest

C. J. Colvin: Nothing to disclose; K. Entwistle: No Answer; J. C. Fusco: Nothing to disclose; M. A. Keller: BioArray Solutions, Grants or Research Support, Life Technologies, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

C. J. Colvin: Nothing to disclose; K. Entwistle: No Answer; J. C. Fusco: Nothing to disclose; M. A. Keller: BioArray Solutions, Grants or Research Support

SP268

Highly Multiplexed Genotyping of Human Erythrocyte Antigens using Leukoreduced Blood

X Lin[1] (xlin@immucor.com), J Patel[1], J Yang[1]. *[1]BioArray Solutions, Warren, NJ, United States*

Background/Case Studies: Leukoreduction (LR) is a process by which leukocytes are filtered from blood or blood components. The LR process significantly reduces the risk of alloimmunization to leukocyte antigens, which may cause febrile nonhemolytic transfusion reactions. Current transfusion practice requires that, at minimum, blood selected for transfusion to a patient be phenotyped to be antigen negative to the existing alloantibodies in the patient's serum. Recently, DNA analysis has emerged as a powerful, versatile and cost effective method for blood group antigen phenotype determination. DNA analysis from blood relies on the fact that only white blood cells (WBC) have genomic DNA. With LR blood having ≤ 10 WBC/μL, it is difficult to carry out DNA assays using standard extraction protocols. Here, we describe a novel method for determining phenotypes of the human erythrocyte antigens (HEA) in LR blood. **Study Design/Methods:** The prototype HEA LR BeadChip™ Assay presented here allows for multiplexed determination of 38 red cell antigens and phenotypic variants plus one mutation for hemoglobin S. The key assay steps are DNA extraction, multiplexed PCR amplification, and Beadchip detection using the automated Array Imaging System. Performance of the HEA LR test was verified by using 84 whole blood (WB) and 233 LR blood samples obtained from several clinical institutions in the US. For a subset of 39 samples, paired blood samples from both before and after leukoreduction were available. Genomic DNA from LR and WB blood samples were extracted with the InviGenius automated extractor and the Giga blood DNA Kit (Stratec Molecular GmbH, Berlin, Germany). To evaluate performance of the HEA LR Beadchip Kit, the extracted DNA samples from before and after leukoreduction were analyzed. In parallel, WB DNA samples were further analyzed by using the Research Use Only (RUO) HEA 1.2 BeadChip Kit. Wherever available, comparisons were also made to serological typing. **Results/Findings:** The HEA LR BeadChip assay had a 96.1% (224/233) sample success rate when using LR blood in the internal verification study. Nine samples with extremely low DNA concentration (0.001-0.0047 ng/uL) failed to produce successful results with the HEA LR assay. The HEA LR BeadChip assay had a 100% (84/84) success rate when used with whole blood. There was 99.9% concordance between the 39 pairs of WB and LR DNA by using the prototype LR assay. WB DNA test results in the HEA LR and the RUO HEA 1.2 BeadChip assays were 100% concordant. In addition, the typing results from HEA LR test and available serology typing were 96.2% concordant. For the samples that were discordant with serology, further investigation using Sequence Specific Primer (SSP)-PCR and sequencing gave results that were concordant with the HEA LR typing. **Conclusion:** The prototype HEA LR BeadChip assay has demonstrated the ability to determine the erythrocyte antigens in the leukoreduced blood.

Disclosure of Commercial Conflict of Interest

X. Lin: Nothing to disclose; J. Patel: BioArray Solutions, Other; J. Yang: Employer, BioArray Solutions, Other

Disclosure of Grants Conflict of Interest

X. Lin: Nothing to disclose; J. Patel: Nothing to disclose; J. Yang: Nothing to disclose

SP269

Feasibility Study of HEA BeadChip Genotyping With DNA Extracted from Saliva Samples

B I Sodowich[1] (bsodowich@immucor.com), J Patel[1], E Enriquez[1], G Ishtiaq[1], U Naik[1], D Garg[1]. *[1]Research and Development, BioArray Solutions, Warren, NJ, United States*

Background/Case Studies: Human Erythrocyte Antigen (HEA) genotyping using genetic microarray analysis allows the determination of many blood group antigens in a single test. It is demonstrated here that genomic DNA extracted from saliva samples can be used for HEA genotyping and produce identical results when compared with DNA extracted from blood or buccal swabs. **Study Design/Methods:** The HEA BeadChip™ kit (for Research Use Only, BioArray Solutions, Immucor, Warren, NJ) is a multiplexed molecular test for simultaneous detection of 38 antigens and phenotypic variants. Twelve healthy donors who have been previously typed using the HEA BeadChip kit, self-collected saliva using commercially available collection kits (DNA Genotek, Ottawa, ON, Canada). DNA was extracted from two 0.5mL aliquots of each saliva sample using either the prepIT®*L2P manual DNA purification protocol or prepIT®*C2D Genomic DNA MiniPrep Kit (DNA Genotek, Ottawa, ON, Canada). Purified DNA from all samples was measured both with the PicoGeen® assay (Life Technologies, Carlsbad, CA) and 260/280 absorbance. Genotypes were determined using the HEA BeadChip assay and compared to the results from the previous typing. **Results/Findings:** The manual extraction yielded 3.4-fold more DNA than the miniprep extraction per sample. Both procedures produced similar sample purity based on the A260/A280 ratio. Paired samples produced results in 100% agreement with each other and reference HEA typing. **Conclusion:** Genomic DNA extracted from donor saliva samples produced successful HEA typing results. These results present the potential to produce valid HEA typing analysis of blood transfusion donors and recipients from a sample collection method that is both non-invasive and self-collecting.
*For research use only. Not for use in diagnostic procedures.

Disclosure of Commercial Conflict of Interest

E. Enriquez: BioArray Solutions, Other; D. Garg: BioArray Solutions, Other; G. Ishtiaq: BioArray Solutions, Ownership or Partnership; U. Naik: BioArray Solutions, Other; J. Patel: BioArray Solutions, Other; B. I. Sodowich: BioArray Solutions, Other

Disclosure of Grants Conflict of Interest

E. Enriquez: Nothing to disclose; D. Garg: Nothing to disclose; G. Ishtiaq: Nothing to disclose; U. Naik: Nothing to disclose; J. Patel: Nothing to disclose; B. I. Sodowich: BioArray Solutions, Other

		Saliva Manual Extraction		Saliva MiniPrep Extraction		*HEA BeadChip Result	
Sample	Reference Source	dsDNA ng/μL	A260/A280	dsDNA ng/μL	A260/A280	Saliva Extraction Agreement	Reference Source Agreement
BASS01	Blood	92.9	1.90	40.1	1.90	100%	100%
BASS02	Blood	166.2	1.90	83.6	1.89	100%	100%
BASS03	Blood	28.7	1.72	3.7	1.94	100%	100%
BASS04	Blood	114.2	1.85	52.3	1.89	100%	100%
BASS05	Blood	48.2	1.85	5.9	1.90	100%	100%
BASS06	Blood	118.0	1.92	3.4	1.96	100%	100%
BASS08	Buccal Swab	40.3	1.89	1.9	1.86	100%	100%
BASS11	Blood	34.9	1.88	9.4	1.89	100%	100%
BASS19	Blood	42.4	1.78	<1.0	2.05	100%	100%
BASS20	Buccal Swab	21.6	1.77	6.1	1.85	100%	100%
BASS21	Blood	133.6	1.83	44.6	1.86	100%	100%
BASS22	Blood	34.9	1.76	6.6	1.86	100%	100%
Average	N/A	73.0	1.84	21.5	1.90	N/A	N/A
Std. Dev.	N/A	49.2	0.06	27.0	0.06	N/A	N/A

*The HEA BeadChip detects of 38 antigens and phenotypic variants

SP270

Development of Genomic DNA Reference Panels for Blood Group Genotyping Tests

M Liu[1] (meihong.liu@fda.hhs.gov), T C Mercado[2], Z Liu[1]. *[1]Laboratory of Immunohematology and Molecular Testing, CBER/OBRR/DBA/DRB, FDA, Kensington, MD, United States; [2]CBER/OBRR/DBA/DRB, FDA, Kensington, MD, United States*

Background/Case Studies: High-throughput multiplex molecular methods have been evaluated as an alternative or an adjunct to conventional blood group serology methods. Comprehensive and well-validated DNA reference panels are necessary to ensure the quality of the molecular methods. Reference panels are also essential for the development and validation of new typing kits, lot release and monitoring of test kit performance. Currently, very limited reference materials are available for blood group molecular typing. The laboratory initiated a project to develop comprehensive genomic DNA reference panels for blood group genotyping tests. **Study Design/Methods:** Genetic variants were selected that were associated with the expression of 19 clinically significant antigens from 17 blood group systems as targets for the reference panels. Donors with genetic variants for these blood group antigens either as heterozygous and homozygous genotypes were identified. Whole blood was collected from the consenting donors. For each genetic target, two independent molecular assays will be used to initially confirm donor's genotypes in the laboratory. Immortalized cells lines were generated from the mononuclear cells isolated from whole blood. **Results/Findings:** Blood from donors who carry rare genotypes/phenotypes, for example Coa-Cob+ and Kpa+Kpb-, was recruited and collected first. Genomic DNA, red blood cells and PBMCs were banked in aliquots for future blood group typing if necessary. More than ten cell lines from these donors were established. 23 custom assays were developed and 15 predesigned TaqMan SNP assays (Applied Biosystems) were tested to detect these genetic variants. A few assays, including K/k, Joa, S silence (nt230C>T), and DIa/DIb, initially failed to make acceptable genotype calls. In addition, challenges were encountered in designing assays for the 37 base pair (bp) insertion (RHDψ), the 109 bp insertion (RhC), and nt 307T>C (Rhc) substitution detection. Bi-directional sequencing will also be used to confirm the genotypes. PCR amplification and sequencing primer sets were designed for all targets and optimized amplification conditions. **Conclusion:** Molecular assays for most target blood group sequences were able to be successfully designed. However, additional validation of the assay is needed using more pre-typed samples with three potential genotypes (homozygous allele one, homozygous allele two, and heterozygous). More cell lines will be created from additional donors to develop comprehensive reference panels. The lyophilized DNA from cultured cells will be tested to determine its suitability as reference panel material in a collaborative study. It is intended to provide these panels to manufacturers to support their kit development processes and for future lot release once blood group DNA tests are approved.

Disclosure of Commercial Conflict of Interest

M. Liu: Nothing to disclose; Z. Liu: Nothing to disclose; T. C. Mercado: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Liu: Nothing to disclose; Z. Liu: Nothing to disclose; T. C. Mercado: Nothing to disclose

SP271

Cell Line DNA Reference Panel for HEA BeadChip Based Red Blood Cell (RBC) Phenotyping

H Huang[1] (hhuang@immucor.com), E Enriquez[1], U Naik[1], J Yang[1]. *[1]BioArray Solutions Ltd., Warren, NJ, United States*

Background/Case Studies: Human erythrocyte antigens (HEA), located on the membrane of the red blood cells, are polymorphic inherited markers. Alloimmunization to HEA may produce severe immune response in some transfusion recipients. The Research Use Only (RUO) HEA BeadChip™ assay is a time and cost effective multiplexed genotype test that allows for the determination of 38 RBC antigens and phenotypic variants (from 11 blood groups) simultaneously for multiple samples in a single assay. Given, there is a worldwide paucity of DNA reference materials for multiplexed RBC genotyping, it is desirable to establish a library of reference DNA samples that covers the genetic variants for the RBC antigens. Here we present the results of an effort to create a cell line DNA reference library representing both "A" and "B" alleles for all SNPs detectable in the HEA BeadChip assay. **Study Design/Methods:** Over 200 cell line DNAs were selected (Coriell Cell Repositories, Camden, NJ) for screening using the HEA 1.2 BeadChip assay. The criteria for the sample selection were based on the ethnic distribution of the HEA variants to maximize the potential to identify samples with rare types of antigens such as heterozygous LU and LW. **Results/Findings:** After screening, 11 cell line DNAs were identified as the minimum panel that covers both "A" and "B" alleles for all SNPs detectable in the existing HEA BeadChip assay. The genotypes of these eleven samples have been confirmed by bi-directional sequencing using the Sanger method. **Conclusion:** A panel of well characterized cell line DNAs has been established as the reference standard for the RUO HEA BeadChip assay, which can be used to evaluate performance of the assay.

Disclosure of Commercial Conflict of Interest

E. Enriquez: BioArray Solutions, Other; H. Huang: BioArray Solutions, Other; U. Naik: BioArray Solutions, Other; J. Yang: Employer, BioArray Solutions, Other

Disclosure of Grants Conflict of Interest

E. Enriquez: Nothing to disclose; H. Huang: Nothing to disclose; U. Naik: Nothing to disclose; J. Yang: Nothing to disclose

SP272

Development and Validation of Genotyping Panel for Blood Group Antigens

C J Colvin[1] (carrie.colvin@redcross.org), T Horn[1], J Crowley[1], M Laig[2], E Grigorenko[3], M A Keller[1]. *[1]National Molecular Laboratory, American Red Cross Biomedical Services, Philadelphia, PA, United States; [2]Life Technologies, Foster City, CA, United States; [3]Diatherix Laboratories, Inc., Huntsville, AL, United States*

Background/Case Studies: Many red blood cell (RBC) antigens are the result of single nucleotide polymorphisms (SNPs) in the gene encoding the protein. Thus, SNP genotyping can be used to determine the RBC profile of blood donors. There are several genotyping tests currently available as research use only products. They vary by blood group and analyte content as well as the testing platform and ease of automation. None allow the user to add content to the panel or to customize the configuration of the panel. It was sought to create a blood group genotyping panel based on the TaqMan® OpenArray® platform (Life Technologies, Foster City, CA) that allowed efficient RBC genotyping with minimal hands-on time. **Study Design/Methods:** Multiple TaqMan® SNP Genotyping assays were designed that amplified and discriminated between alleles based on SNPs associated with blood group antigens. Arrays were loaded using an OpenArray® Accufill™ System (Life Technologies). Initial studies used the ABI GeneAmp® 9700 flat block thermal cycler for PCR amplification and the OpenArray NT cycler for detection. Later development combined both processes on the QuantStudio™ 12K Flex Real-Time PCR System (Life Technologies). AlleleTyper Software was developed to predict antigen phenotypes from genotypes. **Results/Findings:** TaqMan® assays discriminated between alleles of 28 SNPs associated with 40 antigens in 13 blood groups (YT, CO, CR, DI, DO, FY, KEL, JK, KN, LW, LU, MNS, SC, and RH). The RH markers predict C/c, E/e, V, and VS status. For predicting the C antigen, an intronic insertion/deletion polymorphism was also interrogated. Failure to amplify three *GYPB* markers was used to predict of *GYPB* deletion associated with a U-negative status. More than 100 DNA samples of known genotype were tested and used to select the optimal assay for each SNP. DNA extracted using the QIAcube and Universal BioRobot® (Qiagen) was tested. Universal BioRobot extracted DNA in which TopElute was utilized

showed an increased SNP assay failure rate when compared to samples isolated without the use of TopElute. For SNPs in highly homologous regions, several assays were designed, and the assay with the best discrimination between genotypes was selected for the panel. The panel was printed on OpenArray plates and used for genotyping. When four OpenArray plates are processed simultaneously, 380 samples can be genotyped in 4 hours, with about 1 hour of additional hands-on time. **Conclusion:** A comprehensive RBC panel has been developed and validated. The panel predicts 40 antigens in 13 blood group systems with minimal hands-on time and custom software generates an antigen report. Such a system is well-suited for large blood centers performing blood donor genotyping.

Disclosure of Commercial Conflict of Interest

C. J. Colvin: Nothing to disclose; J. Crowley: No Answer; E. Grigorenko: No Answer; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research, SupportLife Technologies, Consulting or Board of Director Fees; M. Laig: Life Technologies, Other

Disclosure of Grants Conflict of Interest

C. J. Colvin: Nothing to disclose; J. Crowley: No Answer; E. Grigorenko: No Answer; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support; M. Laig: Life Technologies, Other

Red Blood Cells: Blood Group Genes and Genotyping: Non-Rh

SP273

One KEL*841C_842A (Kpc) Homozygous Case in a Caucasian Blood Donor

A Molano-Bilbao[1] (amolano@progenika.com), M Chudy[2], J Kress[2], N Rapun[1], L Jofre[1], D Tejedor[1], M Lopez[1]. *[1]Progenika Biopharma S.A., Derio, Spain; [2]Molecular Virology IVD Testing Laboratory, Paul-Ehrlich-Institut, Langen, Germany*

Background/Case Studies: Besides Rh epitopes, KEL gene polymorphic sites constitute the second most immunogenic blood group antigen. Currently, 60 mutations at KEL locus are recognized, which translate into 34 polymorphic antigens of KEL glycoprotein. KEL*842G>A is one of them and involves an aminoacidic change at position 281 from an Arginine in the most common allele (Kpb antigen) to a Glutamine (Kpc antigen). KEL*841C_842G (Kpb) is a high frequency genotype in all populations, KEL*841T_842G (Kpa) is a low frequency genotype in Caucasian population and KEL*841C_842A (Kpc) has been described as a low frequency genotype in the Japanese population (up to 0.32%) and has not been reported in the Caucasian population. **Study Design/Methods:** A genetic study of Kell blood group in a German anonymous blood donor for testing purposes was performed. KEL gene exon 8 was sequenced by Sanger method in ABI PRISM® 3100 Genetic Analyzer. The German blood donor is a Caucasian adult female. **Results/Findings:** The sequencing revealed that the sample genotype is KEL*841C_842A homozygous. This genotype corresponds to the third allele described for codon 281 (Kpc). The predicted phenotype for this sample would be K-k+; Kp(a-b-c+); Jsa-Jsb+. **Conclusion:** The Kp(a-b-c+) is a very relevant phenotype in transfusion medicine (both for donor and patients). As far as is known, this is the first case of KEL*841C_842A (Kpc) homozygous genotype reported in a Caucasian sample. This is an example of a genotyping application for the management of patients and donors in transfusion.

Bibliography

Lee S, **Wu X**, **Son S**, **Naime D**, **Reid M**, **Okubo Y**, **Sistonen P**, **Redman C**. Transfusion. 1996 Jun;36(6):490-4. Point mutations characterize KEL10, the KEL3, KEL4, and KEL21 alleles, and the KEL17 and KEL11 alleles.
Geoff Daniels. 2002, 2nd edition. Blackwell Science. Human Blood groups: 298-302.

Disclosure of Commercial Conflict of Interest

M. Chudy: Nothing to disclose; L. Jofre: Progenika Biopharma, Other; J. Kress: Nothing to disclose; M. Lopez: Nothing to disclose; A. Molano-Bilbao: Progenika Biopharma, Ownership or Partnership; N. Rapun: Nothing to disclose; D. Tejedor: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Chudy: Nothing to disclose; L. Jofre: Nothing to disclose; J. Kress: Nothing to disclose; M. Lopez: Nothing to disclose; A. Molano-Bilbao: Nothing to disclose; N. Rapun: Nothing to disclose; D. Tejedor: Nothing to disclose

SP274

A New KEL*71A>G Change Associated with K0 Phenotype in a Brazilian Patient Presenting with anti-Ku

C Arnoni[1] (carine.arnoni@colsan.org.br), D Gazito[1], J G Muniz[1], R Person[1], J Barreto[1], F Latini[1], L Castilho[2]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: A 61-year-old Brazilian woman with hepatic encephalopathy and hemoglobin of 4.4 g/dL presented an antibody reactive with all RBCs except her own. The direct antiglobulin test (DAT) and autocontrol were negative. Two RBC units were requested for her. Patient RBC was K- k- Js(a-) Js(b-) Kp(a-) and Kp(b-). The antibody was nonreactive with with K0 RBCs. Serological and molecular studies were undertaken to define the patient's Kell phenotype and the corresponding antibody. **Study Design/Methods:** Antibody identification and antigen typing of RBCs were performed by hemagglutination in gel cards. Genomic DNA was used to amplify and sequence all KEL and XK exons. **Results/Findings:** The patient's serum reacted with all panel cells, and did not react with K0 RBCs. The presence of anti-Ku was confirmed and other underlying antibodies were ruled out. Sequence analysis revealed the patient was homozygous for a new change in KEL exon 2, 71G>A encoding Trp24Stop. No other changes were identified. **Conclusion:** The serological results are consistent with the patient's RBCs expressing the K0 phenotype and anti-Ku in the serum. DNA sequencing identified a new change, 71G>A leading to a premature stop codon affecting the expression of the Kell protein in the membrane.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP275

JK nt130G>A Found on Both *JK*01* and *JK*02* Alleles in US and Brazilian Populations

G A Guelsin[1] (gguelsin@gmail.com), T Horn[2], J Crowley[2], A C Gaspardi[1], L Castilho[1], M A Keller[2]. *[1]Hemocentro, UNICAMP, Campinas, Brazil; [2]National Molecular Laboratory, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: The Kidd blood group system includes the antithetical antigens Jk^a and Jk^b resulting from single nucleotide polymorphism (SNP) at nucleotide (nt) 838 in exon 9 encoding alleles *JK*01* and *JK*02* in the *JK* gene. A G>A SNP at nt130 in exon 4 was previously reported on the *JK*01* allele (*JK*01W.01*) and is associated with weakening of the Jk^a antigen. It was sought to determine the frequency of this allele in US and Brazilian populations and to examine its association with *JK*01* and *JK*02*. **Study Design/Methods:** An allele-specific PCR assay was implemented to determine the *JK* nt130 genotype. Philadelphia blood donors and Brazilian donors and patients were genotyped for the nt130 SNP using this assay. Samples were also genotyped for the nt838 using HEA BeadChip™ (BioArray Solutions) or PCR-RFLP and in some cases genomic sequence analysis. Jk antigen typing was performed in a subset of samples using standard serologic methods. **Results/Findings:** US Caucasian, African-American and US Hispanic blood donors from the Philadelphia region and blood donors and sickle cell disease (SCD) patients from Brazil were genotyped. Allele and genotype frequencies are listed in the table below. Two US Caucasian donors, three US Hispanic donors, one Brazilian sickle cell patient, and one Brazilian donor were homozygous for *JK*01W.01*. One African American donor who was heterozygous for the *JK* nt130 SNP was homozygous for *JK*02*. A fraction of donors in each group were compound heterozygotes for nt130G/A and nt838A/G (1.5% US Caucasian donors, 13% African American donors, 19.4% US Hispanics, 5% of Brazilian donors and 10.7% Brazilian SCD patients). A subset of the Brazilian donors (N = 129) were phenotyped; 53% typed Jk(a+b-) and, of these, 4% (N = 3) were compound heterozygous for the *JK* nt130 and nt838 SNPs. **Conclusion:** The *JK* nt130 SNP is a common variant in African American, US Hispanic, and Brazilian populations. Though initially reported on a *JK*01* allele, it has been identified in this study on a *JK*02* allele in an African American blood donor. Further, serologic testing of Brazilians supports the presence of the *JK* 130G>A SNP on the *JK*02* allele in this population as well. The risk of alloimmunization to this newly-discovered weakened Jk^b has yet to be determined. These results suggest that when there are discordant Jk^b typings or anti-Jk^b is detected, the presence of this allele should be considered.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; J. Crowley: No Answer; A. C. Gaspardi: Nothing to disclose; G. A. Guelsin: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research, SupportLife Technologies, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; J. Crowley: No Answer; A. C. Gaspardi: Nothing to disclose; G. A. Guelsin: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support

Allele and Genotype Frequencies of *JK* nt130 SNP

Allele Frequency	US Cauc (131)	US AA (103)	US Hispanic (108)	Br Donor (279)	Br SCD Patients (112)
G	0.95	0.83	0.77	0.89	0.86
A	0.05	0.17	0.23	0.11	0.14
Genotype Frequency					
GG	0.92	0.66	0.59	0.79	0.73
AG	0.07	0.34	0.35	0.20	0.26
AA	0.01	0	0.06	0.01	0.01

SP276

A Novel JK*A Variant Detected Only by Solid-Phase Testing

W S Crews[1] (wscrews@gmail.com), J M Gould[1], J Crowley[2], M A Keller[2], J H Herman[1]. *[1]Transfusion Medicine, Thomas Jefferson University Hospital, Philadelphia, PA, United States; [2]National Molecular Laboratory, American Red Cross Biomedical Services, Philadelphia, PA, United States*

Background/Case Studies: The Kidd blood group system is encoded by the JK or SLC14A1 gene. The antithetical antigens Jk(a) and Jk(b) encoded by JK*A (JK*01) and JK*B (JK*02) differ by a single nucleotide polymorphism at nucleotide 838G>A that changes Aspartic acid 280 to Asparagine. Alleles with SNPs that weaken Jk(a) (N = 2) or Jk(b) (N = 1) have been identified. Null alleles have been reported that silence Jk(a) (N = 6) or Jk(b) (N = 7) by introducing premature stop codons, splice site mutations that result in exon skipping or nonconservative amino acid substitutions. Antibodies to Jk antigens have been shown to have clinical significance. **Study Design/Methods:** A 68-year-old male required 13 units of red cells beginning upon admission for ongoing GI bleeding. Twenty days after admission, he developed a newly positive antibody screen by solid-phase testing (Immucor ECHO, Norcross, GA). Anti-Jk(a) was identified. The pre-transfusion sample was phenotyped as Jk(a) negative. A sample five days later was non-reactive on a different solid-phase platform (Immucor Galileo), but four days after that a new sample again showed anti-Jk(a) on the ECHO. The original panels varied reactivity: 1-3+ to 13 of 19 Jk(a) homozygous and only 2 of 14 heterozygous cells. The autocontrol and DAT were negative. The second reactive sample had similar variable reactivity. The four units transfused 4 days before the anti-Jk(a) appeared were Jk(a) positive. Since reactivity was variable, the original sample was tested by PEG tube and gel (Ortho Diagnostics, Raritan NJ) methods. The screens and panels were negative by both methods. Molecular genotyping was requested to resolve the discrepant serologic findings. **Results/Findings:** HEA BeadChip (BioArray Solutions, Warren, NJ) genotyping was performed and predicted the patient to be Jk(a+b+). Genomic DNA sequence analysis of all coding exons (4 through 11) of the JK gene was performed. Heterozygosity for the nonsynonymous single nucleotide polymorphism (SNP) at nt838A>G responsible for the JK*A/JK*B alleles confirmed BeadChip results. A single base deletion at nucleotide (nt)723 was identified as well as synonymous SNPs at nt549 and nt588. PCR amplification of a gene fragment encompassing exon 8, intron 8 and exon 9 was ligated into a plasmid and cloned. DNA sequence analysis showed that the nt723delA and the nt838A (JK*A) were co-localized. **Conclusion:** The nt723delA variant has been found previously on a JKB allele and was associated with a Jk(b-) phenotype. This patient carried the nt723delA on a JK*A allele. The deletion is predicted to result in a frameshift and truncated protein (I262X). This novel null allele has been assigned the provisional allele name JK*01N.07. This case shows how molecular analysis helps resolve confusing serologic results. Perhaps the single-method variable reactivity of the anti-Jk(a) made by this variant individual requires epitopic expression dependent upon the ECHO solid-phase platform.

Disclosure of Commercial Conflict of Interest

W. S. Crews: Nothing to disclose; J. Crowley: No Answer; J. M. Gould: Nothing to disclose; J. H. Herman: Nothing to disclose; M. A. Keller: BioArray Solutions, Grants or Research, SupportLife Technologies, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

W. S. Crews: Nothing to disclose; J. Crowley: No Answer; J. M. Gould: Nothing to disclose; J. H. Herman: Nothing to disclose; M. A. Keller: BioArray Solutions, Grants or Research Support

SP277

Characterization and Prevalence of Kidd System Genotype/Phenotype Discrepancies in Minority Blood Donors

S Vege[1] (svege@nybloodcenter.org), K Hue-Roye[1], R W Velliquette[1], C Lomas-Francis[1], D Charles-Pierre[1], C M Westhoff[1]. *[1]Immunohematology and Genomics Laboratory, New York Blood Center, Long Island City, NY, United States*

Background/Case Studies: JK silencing alleles are not uncommon and while originally associated with Polynesians, Asians, and Finns, altered *JK* have now been found in many ethnic groups. *JK* was investigated in five donors with discordant Kidd serologic and molecular results when testing 3,500 African American (AA) and Hispanic donors. **Study Design/Methods:** Standard methods were used for RBC typing with commercial reagents and for adsorption-elution with polyclonal anti-Jk^b (Immucor). DNA was isolated from WBCs and HEA (BioArray/Immucor) was used to test 2000 AA and 1500 Hispanic donors from 2011-2013. Amplification and sequencing of the coding region of *JK* was performed by standard molecular methods. **Results/Findings:** When comparing serologic and HEA predicted results, five discrepancies (two Jk^a, and three Jk^b) were found; all were in AAs. *JKA discrepancies*: one donor's RBCs were initially typed as Jk(a-b-), but HEA predicted a Jk(a+b–) phenotype. Repeat typing of the RBCs indicated the RBCs were very weakly (microscopic) Jk(a+), much less reactive than expected for an apparent double dose Jk^a. *JK* sequencing confirmed *JK*A/*A*, but both alleles carried the c.130G>A change (p.Glu44Lys) associated with weak, altered Jk^a expression (ISBT *JK*01W.01*). RBCs of the other donor typed as Jk(a-b+), but HEA predicted Jk(a+b+). *JK* sequencing confirmed *JK*A/B*, but *JK*A* had a c.956C>T change (p.Thr319Met) associated with loss of Jk^a expression (ISBT *JK*01N.04*). *JKB discrepancies*: the RBCs of all three typed as Jk(a+b-) with Immucor and Ortho reagents, but HEA predicted Jk(a+b+). The RBCs failed to adsorb and elute anti-Jk^b. *JK* sequencing confirmed the JK*A/B genotype and identified a c.191G>A change in *JK*B* encoding p.Arg64Gln in all three (ISBT *JK*02N.09*). **Conclusion:** Two Jk^a and three Jk^b phenotype/genotype discrepancies were found in 2000 AA donors, a prevalence of 0.1% for Jk^a, 0.15% for Jk^b and 0.25% for JK discrepancies overall. No JK discrepancies were identified in 1500 Hispanic donors. Jk^b discrepancies were all associated with c.191G>A and this allele was reported in four AA samples (prevalence of 1:400 or 0.25%, [Billingsley et al. 2011 *Transfusion* SP284]). It was associated with a Jk(b-) phenotype based on testing with a single monoclonal, and it was not clear if Jk^b was weakly expressed. In testing three examples, the RBCs did not react with multiple reagents and did not adsorb and elute polyclonal anti-Jk^b, supporting the classification as a serologic null. It is reported here that RBCs from samples homozygous for *JK*A/A* with c.130A/A have low levels of antigen expression and can appear to be a Jk(a-b-), Jk null.

Disclosure of Commercial Conflict of Interest

D. Charles-Pierre: Nothing to disclose; K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Charles-Pierre: Nothing to disclose; K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

SP278

A New Duffy Allele, *FY*A* 395G>A (p.Gly132Asp), Associated with Silencing Fy^a Expression

S Vege[1] (svege@nybloodcenter.org), K Hue-Roye[1], R W Velliquette[1], C Lomas-Francis[1], C M Westhoff[1]. *[1]Genomics Laboratory, New York Blood Center, Long Island City, NY, United States*

Background/Case Studies: The common antigens, Fy^a and Fy^b, in the Duffy system are encoded by a single nucleotide change, c.125G>A, in exon 2 of *FY*. To date, five *FY*A* and two *FY*B* silenced or null alleles, and two *FY*B* encoding weak antigen expression have been reported. All are rare with the notable exception of the -67t>c GATA promoter mutation common in African Black ethnic groups, and the c.265C>T allele called *FY*X*, thought to

be more prevalent in Caucasians. We investigated RBC Fy antigen expression and the *FY* gene in a African-American donor whose RBCs were historically Fy(a-b-), but by molecular testing the sample was *FY*A/*B* with a single copy of the silencing -67t>c GATA mutation, predicting a single dose Fy(a+) and Fy(b-) RBC phenotype. **Study Design/Methods:** Standard tube typing methods were used for RBC testing with Immucor, Ortho, and Quotient (ALBAsera) polyclonal anti-Fy^a reagents, and adsorption-elution studies were done with Immucor polyclonal anti-Fy^a. DNA was isolated from WBCs by standard methods. BioArray/Immucor HEA beadchip testing was according to manufacturer's instructions. Amplification and sequencing of the promoter and exon coding regions of *FY* was performed. Cloning and sequencing of amplified products of *FY* exon 2 was done by standard molecular methods **Results/Findings:** HEA testing predicted the sample was Fy(a+b-), but the RBCs typed as Fy (a-b-). In additional testing with multiple polyclonal anti-Fy^a (3 reagents), no reactivity was detected. The RBCs failed to adsorb and elute anti-Fy^a. Investigation of *FY* by sequencing confirmed the presence of the -67t/c promoter change and the presence of both *FY*A* and *FY*B*. Sequencing of exon 2 identified a nucleotide c.395G>A change, predicted to encode an amino acid change p.Gly132Asp. No other changes were found. Cloning and sequencing of *FY* exon 2 products was performed to confirm the specific allele carrying the mutation. The nt c.395A was found with nt c.125G (*FY*A*). **Conclusion:** A novel allele was identified, *FY*A* with c.395G>A, encoding p.Gly132Asp, in a African-American whose RBCs were Fy(a-) with multiple reagents and did not adsorb and elute anti-Fy^a. This allele was the only Fy^a discrepancy found when phenotype and genotype were compared in 4500 minority Black and Hispanic donors, suggesting it may be uncommon or rare. With the exception of the -67t/c promoter mutation, the five other silenced *FY*A* or *FY*B* reported to date are all associated with premature stop codons. This is the first silenced FY allele encoding a single amino acid change in the protein.

Disclosure of Commercial Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; S. Vege: Nothing to disclose; R. W. Velliquette: Nothing to disclose; C. M. Westhoff: Nothing to disclose

SP279

Two New *FY* Variant Alleles Responsible for a Weakened Expression of the Fy^b Antigen

E Gauthier[1] (egauthier@ints.fr), F Pecquet[1], M Hennion[2], C Vrignaud[1], C Narboux[2], T Peyrard[1]. *[1]National Immunohematology Reference Laboratory, National Institute of Blood Transfusion, Paris, France; [2]EFS Nord de France, French National Blood Service, Lille, France*

Background/Case Studies: Duffy (FY) antigens reside on a seven-transmembrane protein, the Duffy Antigen/Receptor for Chemokines (DARC), expressed on erythrocytes and non-erythroid cells. DARC glycoprotein plays a role in malaria infection and as a receptor for chemokines. FY antibodies are considered clinically significant. Fy^a and Fy^b are antithetical antigens encoded by 2 codominant alleles, *FY*A* and *FY*B*. There are three classical phenotypes in the FY system: Fy(a+b-), Fy(a-b+), and Fy(a+b+). Two atypical phenotypes are also encountered: Fy(a-b-) and Fy^x. Fy(a-b-) corresponds to the Duffy null phenotype, very frequent in people of African ancestry. It is due to a c.-67t>c mutation in the GATA promoter region upstream of the *FY*B* allele (*FY*B Null*). Fy^x corresponds to a very weak expression of Fy^b, mainly encountered in Caucasians, and caused by the c.265C>T point mutation in *FY*B* gene exon 2, which encodes a p.Arg89Cys amino acid substitution. Of note, the p.Ala100Thr polymorphism was systematically found to be associated. Fy^x seems quite frequent in Caucasians, as it was reported that approximately 3% of people being typed as Fy(a+b-) were actually *FY*A/FY*X*. **Study Design/Methods:** The present study aimed to characterize the molecular basis of a Fy(a+b+^{w}) phenotype in two unrelated Caucasian donors. RBCs were investigated by standard hemagglutination techniques. Fy^b typing was done with two polyclonal reagents (Biotest, DiaMed). *FY* genotyping was carried out by real-time PCR (Light Cycler, Roche Diagnostics) and *FY* gene amplification/genomic DNA sequencing was performed by using specific primers. **Results/Findings:** The RBCs of the two probands were confirmed to show a strong decrease of Fy^b expression. *FY* genotyping showed in both donors the presence of *FY*A* and *FY*B* alleles and the absence of the two mutations c.-67t>c (no *FY*B Null* allele) and c.265C>T (no *FY*X* allele). Interestingly, the real-time PCR for *FY*X* genotyping revealed in the first donor a slight melting curve shift, which seemed to lead to suspecting a molecular alteration very likely close to position 265. Indeed, *FY* gene sequencing analysis revealed a mutation at position 266 (c.266G>A), predicted to encode a p.Arg89His amino acid substitution, which affects the same critical amino acid as that encoded by the *FY*X* allele (p.Arg89Cys). The *FY* gene of the second donor exhibited a c.901C>T change in exon 2, leading to the substitution of a proline to an alanine at position 303. This mutation on a *FY*B* allele probably alters the FY protein conformational properties and dramatically reduces Fy^b antigen expression on the red blood cell surface. **Conclusion:** This study reports two novel alterations of the *FY* gene associated with a very weakened expression of the Fy^b antigen in the Caucasian population. Systematic *FY* sequencing turns out to be of interest when Fy^b is weakly expressed and no *FY*X* or *FY*B Null* alleles could be detected by standard RBC genotyping.

Disclosure of Commercial Conflict of Interest

E. Gauthier: Nothing to disclose; M. Hennion: Nothing to disclose; C. Narboux: Nothing to disclose; F. Pecquet: Nothing to disclose; T. Peyrard: Nothing to disclose; C. Vrignaud: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Gauthier: Nothing to disclose; M. Hennion: Nothing to disclose; C. Narboux: Nothing to disclose; F. Pecquet: Nothing to disclose; T. Peyrard: Nothing to disclose; C. Vrignaud: Nothing to disclose

SP280

A New Dombrock Allele, *DO*B* with 791G>A, and Discrepant Results with Two Molecular Methods

S Vege[1] (svege@nybloodcenter.org), K Hue-Roye[1], P Patel[1], C Lomas-Francis[1], C M Westhoff[1]. *[1]Genomics Laboratory, New York Blood Center, Long Island City, NY, United States*

Background/Case Studies: The Dombrock blood group system consists of two antithetical antigens, Do^a and Do^b, and six high prevalence antigens including Hy, Jo^a and Gy^a. There are no licensed antisera for typing RBCs for Dombrock antigens, and in-house and/or human source antisera are not always reliable. Consequently molecular testing for the single nucleotide polymorphism (SNP) at c.793A>G in *DO* has become the standard for typing for Doa/b. An unusual discrepancy was investigated here that presented when *DO* testing was performed by PCR-RFLP and by automated HEA (Human Erythrocyte Antigen) beadchip assay. **Study Design/Methods:** Genomic DNA was isolated from the WBCs with standard methods. *DO*A/*B* determination was performed with a PCR-RFLP with *Bse*RI targeting nucleotide (nt) c.793A>G (p.Asn265Asp). HEA (BioArray/Immucor) testing was performed according to manufacturer's instructions. Amplification and sequencing of *DO* exon 2 was performed by standard molecular methods. **Results/Findings:** A multiply transfused African-American female with sickle cell disease was submitted for genotyping for *FY*, *GYB*S*, and *DO* in 2007. By PCR-RFLP testing, her predicted phenotype was Fy(a-b-), S-s+, and Do(**a**+b+). In 2013 her sample was submitted for HEA testing and the assay predicted the RBCs were Do(**a**-b+). Testing of the 2013 sample by PCR-RFLP with *Bse*RI resulted in a fragment pattern consistent with the 2007 results of Do(**a**+b+). To investigate the cause for the discrepant results, *DO* exon 2 was amplified and sequenced. The sample was homozygous for the c.793A>G change (p.793G/G) associated with *DO*B/*B*, consistent with the HEA results. However, an additional nucleotide change, c.791G>A, predicted to encode an amino acid change p.Gly264Glu, was present (c.791G/A). This new polymorphism is also located in the *Bse*RI restriction site (GAGGAG) and ablates cutting of the novel *DO*B*. The effect of this change on expression of Do^b, could not be tested as the sample was not informative due to the presence of on conventional *DO*B in trans* to *DO*B 791G>A*. **Conclusion:** A new *DO*B* was identified with c.791G>A change encoding p.Gly264Glu. This allele was revealed when investigating discrepancy between *DO* prediction by manual PCR-RFLP and by HEA assay. The c.791G>A change is only two bases from the c.793A>G associated with *DO*A/*B*, and results in the loss of the *Bse*RI restriction site which is a hallmark of *DO*A*. This results in a false positive *DO*A* determination by PCR-RFLP with *Bse*RI. Additional studies are needed to determine the prevalence of this allele.

Disclosure of Commercial Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; P. Patel: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Hue-Roye: Nothing to disclose; C. Lomas-Francis: Nothing to disclose; P. Patel: Nothing to disclose; S. Vege: Nothing to disclose; C. M. Westhoff: Nothing to disclose

Table 1

ID	RHD Variant	Predicted Effect	Allelic Description	Serology
1.1	c.8C>G	Ser3Cys	*RHD*8G*	Weak D
1.2	c.742T>G	Leu94Arg	*RHD*742G*	Del
1.3	c.667T>G	Phe223Val	*RHD*667C,819A*	Partial D
	c.819G>A	Silent		
1.4	IVS6+1G>A	Splice site	*RHD*IVS6+1A*	RhD-
1.5	c.915delC	Frameshift 357Stp	*RHD*915delC*	RhD
1.6	c.1179G>A	Trp393Stp	*RHD*1179A*	RhD-
2.1	IVS1+1G>T	Splice site	*RHD*IVS1+1T*	Del
2.2	c.993C>G	Ile331Met	*RHD*993G*	Weak D
2.3	c.1187C>G	Pro396Arg	*RHD*1187G*	Weak D
3.1	c.1010T>G	Leu337Arg	*RHD*1010G*	RhD (1)
3.2	IVS9-1G>A	Splice site	*RHD*IVS9-1A*	RhD-
4.1	c.916G>A	Val306Ile	*RHD*916A,932G,1154C*	Weak D
	c.932A>G	Tyr311Cys		
	c.1154G>C	Gly385Ala		
4.2	c.1157T>A	Leu386Stp	*RHD*1157A*	RhD-
5.1	IVS4-2A>C	Splice site	*RHD*IVS4-2C*	Partial D
5.2	c.1065C>T	Silent	*RHD*1065T*	Weak D
6.1	c.201G>A	Silent	*RHD*201A,203A,1136T*	RhD+ (2)
	c.203G>A	Ser68Asn		
	c.1136C>T	Thr379Met		

(1) Del phenotype not excluded.
(2) Partial D phenotype not excluded.

Red Blood Cells: Blood Group Genes and Genotyping: Rh

SP281

Identification of Novel *RHD* Variants

F Garcia[1] (fgarcia.trans@salud.madrid.org), M Goldman[2], F Puente[3], M Rodríguez Calvo[4], E Muniz-Diaz[5], M Azcarate[6], H Alshatti[7], G Ochoa[8]. *[1]Centro de Transfusion de Madrid, Madrid, Spain; [2]Canadian Blood Services, Ottawa, ON, Canada; [3]Banco de Sangre y Tejidos de Aragon, Zaragoza, Spain; [4]Centro de Transfusion de Galicia, Santiago de Compostela, Spain; [5]Banc de Sang i Teixits, Barcelona, Spain; [6]Centro Vasco de Transfusion y Tejidos Humanos, Galdakao, Spain; [7]Kuwait Central Blood Bank, Jabriya, Kuwait; [8]Progenika Inc, Medford, MA, United States*

Background/Case Studies: Resolution of discrepancies between serology and genotyping microarrays by DNA sequencing and extended serology often reveals the presence of genetic variants. This report describes 16 previously unreported *RHD* alleles identified in donor and patient samples during discrepancy resolution. **Study Design/Methods:** Phenotyping was performed by standard serology methods. Genotyping was performed on BloodChip v4, an oligonucleotide microarray that interrogates 72 polymorphic positions in the *RHD* gene. DNA sequencing of *RHD* exons 1-10 and flanking intronic regions was performed by the Sanger dideoxy method. **Results/Findings:** Table 1 shows the *RHD* variants identified by sequencing, the predicted effect on gene products, and the serology. The variant alleles, present in hemizygosity or homozygosity, span the spectrum of RhD phenotypes. Two alleles associate with Partial D phenotypes (Samples 1.3 and 5.1), two with Del phenotypes (Samples 1.2 and 2.1), five with Weak D phenotypes (Samples 1.1, 2.2, 2.3, 4.1, and 5.2), and five with D negative phenotypes (Samples 1.4, 1.5, 1.6, 3.2, and 4.2). The phenotype of allele *RHD*667G,819A* is possibly identical to that of DFV, since c.819G>A is a silent SNP. The Weak D phenotype associated with allele *RHD*1065T* is possibly due to the formation of a new splice site, since c.1065C>T is a silent SNP. Sample 3.1 was not tested by adsorption/elution, so a Del phenotype has not been excluded for allele *RHD*1010G*. Allele *RHD*201A,203A,1136T* (sample 6.1) includes c.1136C>T, a nucleotide substitution shared by D positive and Partial D (DAU) variants. Since this sample was tested with an anti-D MoAb blend, a Partial D phenotype has not been excluded. **Conclusion:** Discrepancies between serology and genotyping help identify new *RHD* variants. Identification of variants, in turn, helps improve the accuracy of genotyping microarrays.

Disclosure of Commercial Conflict of Interest

H. Alshatti: Nothing to disclose; M. Azcarate: Nothing to disclose; F. Garcia: Nothing to disclose; M. Goldman: Nothing to disclose; E. Muniz-Diaz: Nothing to disclose; G. Ochoa: Nothing to disclose; F. Puente: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. Alshatti: Nothing to disclose; M. Azcarate: Nothing to disclose; F. Garcia: Nothing to disclose; M. Goldman: Nothing to disclose; E. Muniz-Diaz: Nothing to disclose; G. Ochoa: Nothing to disclose; F. Puente: Nothing to disclose; M. Rodríguez Calvo: Nothing to disclose

SP282

A Novel Molecular Background for the First Example of D•• (Evans Positive) Rare Phenoype in a Proband of African Origin

M Roussel[1] (mroussel@ints.fr), G Juszczak[1], D Gien[1], C Vrignaud[1], R Mortelecque[2], T Peyrard[1]. *[1]National Immunohematology Reference Laboratory, National Institute of Blood Transfusion, Paris, France; [2]EFS Ile de France, French National Blood Service, Paris, France*

Background/Case Studies: Rh antigens are encoded by two highly homologous and closely linked genes, *RHD* and *RHCE*. Their genomic organization facilitates genetic material exchange during meiosis, especially through the so-called "gene conversion" mechanism, which may result either in functional or non-functional *RHD-CE-D* and *RHCE-D-CE* hybrid alleles. *RHCE-D-CE* alleles, at homozygous or compound heterozygous state, may lead to exceptional phenotypes with abolished or extremely weak expression of the RHCE protein. This is the case for the Rh-deletion haplotypes D- -, Dc-, DIV(C) -, DC^w- and D••, for which a significant part of *RHCE* is replaced by its *RHD* counterpart. Unlike D- -, D•• expresses the low-prevalence Evans antigen (RH37). D•• turns out to be exceptional and was only reported in three Caucasian families, with distinct *RHCE* molecular backgrounds: *RHCE-D(2-6)-CE* (AT), *RHD(1-6)-CE* (HD) and *RHD(1-part of 6)-CE* (JD). As a result, Evans was suspected to be likely encoded by the 5' end of *RHD* exon 6 and 3' end of *RHCE* exon 7. The first example of D•• in a proband of African origin is described here, encoded by a novel molecular background. **Study Design/Methods:** Blood samples from Mrs. DS, native from Algeria (North Africa), were referred to the laboratory to confirm a D+C-E-c-e- type and investigate an antibody to a high-prevalence antigen. Serum and RBCs were investigated by standard hemagglutination techniques. Anti-Evans was obtained from our own collection and SCARF exchange program. *RHCE* was studied by genomic DNA sequencing and with a RBC genotyping device (wRHCE BeadChip, BioArray/Immucor). *RHD* was investigated by the wRHD BeadChip technique. **Results/Findings:** Mrs. DS was confirmed to be D+C-E-c-e- with standard reagents. Adsorption/elution of polyclonal C, E, c and e antisera showed a weak C reactivity. The proband was found to be Evans+ with two different reagent sources and also typed C^w-, C^x-, G+. The serum contained a potent antibody that was concluded to be anti-Hr_o (anti-RH17). RBCs of Mrs. DS were reactive with a high-titer anti-Hr_o made by a D-—patient. Study of *RHCE* gene revealed the presence of a hybrid *RHCE-D(4-6)-CE* allele, at the homozygous state. *RHCE* exons 1 and 2 carried the changes responsible for the C polymorphism, what may explain the very weak C reactivity in this patient. *RHD* investigation was consistent with a normal D antigen. **Conclusion:** The very first example of D•• in a proband of African origin is reported here. This exceptional phenotype is encoded by a novel molecular background, involving a hybrid *RHCE-D(4-6)-CE* homozygous allele. Since Mrs. DS carries the same *D(6)-CE*(7) hybrid sequence than the formerly reported D•• AT and HD (Mrs. Helen Dav.) probands, this study provides new arguments for the molecular basis of the low-prevalence Evans antigen, as being indeed the likely result of a unique amino acid sequence and conformational change within the RHCE protein at the RHD/RHCE junction.

Disclosure of Commercial Conflict of Interest

D. Gien: No Answer; G. Juszczak: No Answer; R. Mortelecque: Nothing to disclose; T. Peyrard: Nothing to disclose; M. Roussel: Nothing to disclose; C. Vrignaud: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Gien: No Answer; G. Juszczak: No Answer; R. Mortelecque: Nothing to disclose; T. Peyrard: Nothing to disclose; M. Roussel: Nothing to disclose; C. Vrignaud: Nothing to disclose

SP283

Identification of a Mutation in the *RHAG* Gene of Japanese with Weak D Phenotype

H Tsuneyama[1] (h-tsuneyama@ktks.bbc.jrc.or.jp), K Isa[2], K Ogasawara[2], R Yabe[1], M Uchikawa[1,2], M Minami[1]. *[1]Japanese Red Cross Kanto-Koshinnetsu Block Blood Center, Tokyo, Japan; [2]Japanese Red Cross Central Institute, Tokyo, Japan*

Background/Case Studies: Weak D phenotype is typically caused by amino acid substitutions RhD polypeptide. RhD and RhCcEe polypeptides are known to be expressed as a complex with Rh associated glycoprotein (RhAG) on red cell membrane, and hence, deletion and/or the marked decrease of RhAG results in the phenotypes of Rh_{null} and Rh_{mod}. The missense mutation (572G>A) on *RHAG* gene is reported here in five Japanese blood donors who are considered to relate to the decrease of RhD antigen expression. **Study Design/Methods:** Hemagglutination were examined using commercial polyclonal and monoclonal reagents. Flow cytometric (FC) analysis was used FITC-conjugated monoclonal anti-D (HIRO-3). Genomic DNA from peripheral blood was used for *RHAG*- and *RH*-specific exon sequencing. **Results/Findings:** The D antigen of the five individuals was weaker than R_1r RBCs by direct agglutination test with commercial reagents. Furthermore, agglutination titers were weaker than R_1r RBCs by indirect antiglobulin test. FC analysis revealed that the amount of D antigen on their RBCs was decreased by 4-12% compared with R_1r RBCs. The strength of reactivity of the RBCs with anti-Rh17 and -RhAG that react with their are similar to the RBCs with normal Rh phenotype. In all of the individuals, *RHAG* gene sequence analysis revealed a single base change (572G>A) in exon 4, predicting a 191Arg>Gln substitution in the intracellular domain of the RhAG protein. All of the individuals had normal *RHD* and *RHCE* gene. **Conclusion:** In weak D, the RhD proteins have one or more amino acid substitutions within either the membrane-spanning domains or the cytoplasmic loops of the protein, whereas the five weak D individuals in this report appear to be caused by RhAG missense mutation.

Disclosure of Commercial Conflict of Interest

K. Isa: Nothing to disclose; M. Minami: No Answer; K. Ogasawara: No Answer; H. Tsuneyama: Nothing to disclose; M. Uchikawa: Nothing to disclose; R. Yabe: No Answer

Disclosure of Grants Conflict of Interest

K. Isa: Nothing to disclose; M. Minami: No Answer; K. Ogasawara: No Answer; H. Tsuneyama: Nothing to disclose; M. Uchikawa: Nothing to disclose; R. Yabe: No Answer

SP284

Molecular Characterization of RhD Variants in Zhejiang Han Population, China

Y Ying[1,2] (yingyanling@126.com), X Xu[1,2], X Hong[1,2], S Chen[1,2], K Ma[1,2], X Lan[1,2], J He[1,2], F Zhu[1,2], H Lv[1,2]. *[1]Blood Center of Zhejiang Province, Hangzhou, China; [2]Key Laboratory of Blood Safety Research, MOH, Hanzhou, China*

Background/Case Studies: The frequency of the RhD variant phenotype is varied in the different ethnic populations and their partial molecular basis is reported. A comprehensive study of RhD variants was investigated in the Zhejiang Han population of China. **Study Design/Methods:** RhD antigen was typed with a blended IgM and IgG monoclonal anti-D using a tube protocol. Samples that agglutinated weakly by the tube method or only reacted by indirect antiglobulin test were classified as D variants. The full coding regions of RHD gene of RhD variants were amplified with polymerase chain reaction (PCR)and then directly sequenced. RHD zygosity test was performed based on the hybrid Rhesus box with PCR-SSP. **Results/Findings:** A total of 37 individuals were typed as RhD variants. According to the results of the serology and sequencing, 19 individuals were identified as weak D, including 11 individuals with weak D type 15, one with weak D type 33, three with heterozygote of weak D type 15 and 17, weak D type 15 and RHD(K409K), new weak D allele 95C>A and RHD(K409K), and the other four individuals with three novel weak D alleles(101A>G in exon 1 and 670A>G, 779A>G in exon 5 respectively). Seven individuals were confirmed as partial RhD, including three with DVI Type 3, two with DVI Type 2, one with DCS-1 allele, and one with a new 3-bp deletionin exon 1 (130-132 del CTC). All of six DELs were possessed the RHD(K409K) allele and five samples were not found any mutations in the sequencing region. **Conclusion:** The molecular basis of RhD variants in the Zhejiang Han population was obtained and five novel RHD alleles were identified. These data may help to type RhD phenotype and make the transfusion strategy of the RhD variants.

Disclosure of Commercial Conflict of Interest

S. Chen: No Answer; J. He: No Answer; X. Hong: No Answer; X. Lan: No Answer; H. Lv: No Answer; K. Ma: No Answer; X. Xu: No Answer; Y. Ying: Nothing to disclose; F. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Chen: No Answer; J. He: No Answer; X. Hong: No Answer; X. Lan: No Answer; H. Lv: No Answer; K. Ma: No Answer; X. Xu: No Answer; Y. Ying: Nothing to disclose; F. Zhu: Nothing to disclose

SP285

Weak D Type 54 Cases Exhibit Similar Topology and Distribution of Epitopes of RhD Compared with the DVII Category

L Ye[1] (ye_luyi@hotmail.com), W Zhang[1], W Gao[1], Z Zhu[1]. *[1]Shanghai Institute of Blood Transfusion, Shanghai Blood Center, Shanghai, China*

Background/Case Studies: The purpose is to study the RhD epitopes and potential impact on the topology of the RhD protein of weak D type 54. **Study Design/Methods:** Routine reagents were used for determination of Rh types for three weak D type 54 samples. D-screen was used to analyze the RhD epitopes. All the *RHD* exons were sequenced, and *RHD* zygosity was determined. Robetta was used for modeling. **Results/Findings:** Only 365C>T mutation was identified in two samples, whereas 365C>T heterozygous mutation and 1227G>A heterozygous mutation were simultaneously found in another sample. Two samples tested with all monoclonal anti-Ds in D-screen showed the same D epitope profiles as DVII, which belongs to the partial D category. Moreover, another sample with incomplete D-screen results showed similar epitope profiles. Modeling analysis showed that the intracellular mutation S122L of weak D type 54 might have influence on extracellular protein structure of the RhD protein. **Conclusion:** Weak D type 54 was first reported in Chinese people and characterized as partial D. When exposed to RhD positive RBCs, it still remains to be determined whether people with weak D type 54 would be alloimmunized. DISCLOSURES: The authors have no conflict of interest.

Disclosure of Commercial Conflict of Interest

W. Gao: No Answer; L. Ye: Nothing to disclose; W. Zhang: No Answer; Z. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Gao: No Answer; L. Ye: Nothing to disclose; W. Zhang: No Answer; Z. Zhu: Nothing to disclose

SP286

DEL-Associated Alleles, Including Novel RHD DEL Alleles, Detected in 2.17% of RhD Negative Blood Donors with RhC and/or RhE Haplotypes

S Scott[1] (sscott@redcrossbloo.org.au), L Tilley[2], R McBean[1], R L Flower[1], C A Hyland[1]. *[1]R&D, Australian Red Cross Blood Service, Kelvin Grove, QLD, Australia; [2]Bristol Institute for Transfusion Sciences, National Blood Service, Bristol, United Kingdom*

Background/Case Studies: Exposure to red cells from donors carrying the RHD*1227G>A allele, associated with the DEL phenotype, has been implicated in primary and secondary immunization in RhD negative recipients. This RHD allele is found in approximately one third of East Asian population groups who present with the RhD negative phenotype on routine serology. The aim of this study was to define the distribution of DEL RHD alleles in an Australian donor population. **Study Design/Methods:** RhD negative donors with C and/or E haplotype were genotyped for RHD exons 4, 5 and 10 by qPCR. RHD genotypes were defined further by SNP microarray testing on BLOODchip Reference v4 (Progenika) and samples presenting with a "no call" or apparent RHD positive genotype were sequenced. Extended serological testing included DEL phenotyping using monoclonal anti-D 6 antibody (MCAD6) and the D-SCREEN monoclonal anti-D antibody panel (DIAMED). **Results/Findings:** DEL associated RHD alleles were detected in 38/2027 donors. These comprised: RHD*1227G>A, (0.79%; 95% CI:0.4-1.17%); RHD*885G>T (0.49%; 0.19-0.8%); RHD*IVS3+1G>A (0.3%; 0.06-0.53%); RHD*94insT (0.2%: 0-0.39) and RHD*IVS1+1G>A (0.1%: 0-0.24%) alleles. A further eight donors carried RHD alleles defined by sequencing, (Table). One carried the RHD*1177T>C allele, associated with weak D phenotype, and seven carried a putative novel RHD allele. DEL phenotyping detected D antigen expression for six of the eight donors (donor 1 with predicted weak D expression and donors 2, 3, 6, 7 and 8), bringing the number of donors with DEL/weak D predicted or observed to 44/2027. **Conclusion:** DEL

associated or weak RHD alleles were defined in 2.17% of donors tested, 44/2027 (95% CI: 1.54-2.81). These rates are similar to European studies with the exception that the RHD*1227G>A allele is more prevalent in this present study, consistent with the dynamic demographics in Australia. The detection of novel RHD alleles amongst 11.4% (5/44) of the DEL positive cohort further demonstrates the diversity at the RH locus.

Disclosure of Commercial Conflict of Interest

R. L. Flower: Nothing to disclose; C. A. Hyland: Nothing to disclose; R. McBean: Nothing to disclose; S. Scott: No Answer; L. Tilley: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. L. Flower: Nothing to disclose; C. A. Hyland: Nothing to disclose; R. McBean: Nothing to disclose; S. Scott: No Answer; L. Tilley: Nothing to disclose

TABLE 1

Donor	Allele Desription	Predicted Effect	Del Phenotype
1	RHD*1177T>C 1	Known weak D	Positive
2	RHD*IVS1+5C 1	Splice site	Positive
3	RHD*IVS2-2delA 1	Splice site	Positive
4	RHD*IVS6+2A 1	Splice Site	Negative 2
5	RHD*297del23 1	Frame shift;stop codon	Negative 2
6	D exon 8 deleted or CE exchange	Unknown	Positive
7	D exon 8 deleted or CE exchange	Unknown	Positive
8	D exon 9 deleted or CE exchange	Unknown	Positive

1 Sequencing kindly performed by Progenika 2 DEL phenotyping limited to one monoclonal antibody and therefore not conclusive

SP287

RHCE Polymorphisms in Brazilian Patients with Sickle Cell Disease (SCD)

J G Muniz[1] (janagmuniz@yahoo.com.br), C Arnoni[1], R Person[1], D Gazito[1], J Barreto[1], F Latini[1], L Castilho[2]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: Rh epitopes are highly conformational and single amino acid changes in the protein can change the expression of Rh antigens. Single Nucleotide Polymorphism (SNP) and genetic exchange are primarily responsible for the large number of variations on Rh antigens expression. The molecular background of Rh variants in people of African descent is predominantly described in the RHCE*ce allelic form. The association with production of clinically significant antibodies against high-incidence antigens has also been reported. Based on this, the aim of this study was to determine the polymorphisms related to RHCE variants in Brazilian patients with sickle cell disease (SCD) presenting the RHCE*ce haplotype. **Study Design/Methods:** DNA samples of 100 SCD patients were analyzed. Red blood cells of all patients were phenotyped by serological methods using gel test to D, C, E, c, and e. A total of 40 patients with the RHCE*ce haplotype were selected to molecular analysis, that included PCR-RFLP to 48G>C and 1025C>T SNPs, AS-PCR to 1006G>T SNP, and sequencing of RHCE exon 5. **Results/Findings:** From 40 patients with the RHCE*ce haplotype, 18 samples presented variant alleles as shown in table 1. Four samples were homozygous for altered RHCE alleles, 1 sample was compound heterozygous and 13 samples had one normal allele. **Conclusion:** The results show that 45% of the Brazilian SCD patients studied presented RHCE*ce variant alleles. The most common altered allele found in this population of SCD patients was RHCE*ce48C. In this population, only five patients could potentially develop antibody (four homozygous and one compound heterozygous). This finding explains the low rate of alloimmunization to RhCE variants found in this mix population of SCD patients.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

TABLE 1. *RHCE* alleles frequency in SCD patients

N	Genotype
6	***RHCE*ceVS.01/ce***
4	***RHCE*ce48C/ce***
3	***RHCE*ce48C/ce48C***
2	***RHCE*ceMO/ce***
1	***RHCE*ceVS.01/ceVS.01***
1	***RHCE*ceAR/ce***
1	***RHCE*ceAR/ce48C***

SP288

Correlation Between Serology and Genotyping of RhD variants in Egyptians

E Hussein[1] (eimanhussein@ymail.com), J Teruya[2]. *[1]Clinical Pathology, CairoUniversity, Cairo, Egypt; [2]Pathology & Immunology, Pediatrics, and Medicine, Baylor College of Medicine and Texas Children's Hospital, Houston, TX, United States*

Background/Case Studies: Patients with weak D types 1, 2, 3, and 4.1 can be safely considered D positive. Anti D alloimmunization has been documented for patients with partial D and weak D types 4.0, 4.2 (DAR), 11 and 15. In the era of changing technology with the production of new methods and sensitive monoclonal reagents, RhD typing can be problematic. Discrepancies caused by various reactivity of D variants can only be resolved by the use of RhD genotyping. However, this strategy cannot be applied routinely due to the cost and feasibility. It has been documented that D variants are demonstrated among individuals with positivity for at least C or E. It is considered to be affordable for some countries to test D negative donors who are C or E positive for D variants. Samples with weak D and partial D were investigated for RhCE phenotype. The reactivity of different serological techniques with D variants was also evaluated. It was proposed that an algorithm could be found based on serology that matches the Egyptian genetic frequency data, and correctly assigns donors and patients, using the lowest possible expense. **Study Design/Methods:** Weak D was identified by BAGene DNA-SSP kits, while partial D was categorized with the BIO-RAD advanced partial D typing kit. Routine D typing by immediate spin (IS) tube method was performed in parallel with an automated gel test(DiaClon ABD-Confirmation for donors, Bio-RAD). The weak D test was performed by indirect antiglobulin test (IAT). Results obtained by different serological techniques were compared. RhCE phenotype was performed by IS. **Results/Findings:** Among 31 D variants, only 5 were C or E positive (16.1 %). R0r phenotype was associated with the remaining 26 samples (83.9%) and constituted weak D types 4.2 (38.5%), and 4.0/4.1 (11.5%), partial DIII (34.6%), and partial DV (15.4%). Gel reacted strongly with partial DIII and DV. Ten samples with DIII and DV typed as D positive with IS. All weak D were positive by IAT, while all partial D were positive by gel and IAT. **Conclusion:** RhD policies should be adjusted to match population data. Detection of D variants among C or E positive donors may not be an optimal strategy for Egyptians. In Egyptians, serology cannot discriminate weak D from partial D, but may provide a clue about the probable weak D or partial D to be tested molecularly with the appropriate kit. Patients with Partial DIII and DV may be typed as D positive at IS phase. Reagent selection is important to appropriately assign donors and patients with the DIII and DV types.

Disclosure of Commercial Conflict of Interest

E. Hussein: Nothing to disclose; J. Teruya: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Hussein: Nothing to disclose; J. Teruya: Nothing to disclose

RhCE phenotype and reaction strength of various D variants with different serological techniques

RhD variant	Number	RhCE	IS (Number)	Automated Gel test	IAT
Partial DIII (N = 9)	5 4	ce ce	- (3) 1-2+ (5) 3+ (1)	2-3+ 2-3+ 3+	3-4+ 3-4+ 3+
Partial DV (N = 6)	2 4	Ce ce	- 1-2+	2-3+ 2-3+	2-3+ 2-3+
Weak D type 4.2 (N = 10)	10	ce	-	-	1-2+
Weak D type 4.0/4.1 (No = 5)	3 1 1	ce Ce Ce	—	—	1-2+ 2+ 3 +
Weak D type 15 (N = 1)	1	cE	-	-	2+

SP289

Analysis of Partial D and Weak D Blood Donors in Taiwan

S Lin[1] (katrina.ks@blood.org.tw), H Lin[1], C Wu[1], K Lin[1], C Hung[1], T Lin[2]. *[1]Laboratory, Kaohsiung Blood Center, Taiwan Blood Services Foundation, Kaohsiung, Taiwan; [2]Department of Laboratory Medicine, E-DA Hospita I-SHOU University, Kaohsiung, Taiwan*

Background/Case Studies: The RHD gene is highly polymorphic and the existence of a large number of alleles results in RhD variant phenotypes. Monoclonal antibodies (MoAb) are being developed to identify D variants such as partial D and weak D; however, molecular analysis offers a tool for further classification of weak and partial D. The purpose of this study was to investigate the molecular and serologic characterization of donor samples with partial D and weak D type. **Study Design/Methods:** From 2006-2012, 71 samples demonstrated D negative in tube methods but positive reactions at AHG phase (Du test) were categorized as weak D. The samples were identified for the partial D phenotype by Partial RhD Typing Kit. Of them, 17 genomic DNA of donors were collected to study by using commercial genotyping kits and zygosity test. In addition, nucleotide sequencing of RHD was performed to confirm the discrepancy of the phenotype and the genotype. **Results/Findings:** According to the serologic reactions, 71 samples can be divided to DFR (n = 35), DVI (n = 16), DOL (n = 1), weak D (n = 13), indeterminate (n = 6). The genotyping results show that the DFR phenotype represent weak D type 15 and some cases contain the alleles of weak D type 15 and RHD 1227A. The group of DVI phenotype was categorized as DVI Type 3 allele. Two cases of indeterminate results show the same triplet deletion (686-688delGAA) at exon 5 of RHD and their serologic reactions are similar to DVL-1. **Conclusion:** Majority of weak D phenotypes in Taiwan population are identified to be partial D type (DVI type 3) and weak D type 15.

Disclosure of Commercial Conflict of Interest

C. Hung: Nothing to disclose; H. Lin: Nothing to disclose; K. Lin: Nothing to disclose; S. Lin: Nothing to disclose; T. Lin: No Answer; C. Wu: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Hung: Nothing to disclose; H. Lin: Nothing to disclose; K. Lin: Nothing to disclose; S. Lin: Nothing to disclose; T. Lin: No Answer; C. Wu: Nothing to disclose

SP290

Investigation of R*HCE*ceJAL* in a Southern US Black Population

J M Moulds[1] (jmmoulds@lifeshare.org), K L Billingsley[1], G T Noumsi[1], P Gowland[2]. *[1]Scientific Support, LifeShare Blood Centers, Shreveport, LA, United States; [2]Blood Transfusion Service SRC, Berne, Switzerland*

Background/Case Studies: JAL is a low frequency RH antigen found in Whites, Blacks and Asians. In Blacks, the most common gene, *RHCE*ceJAL*, produces weakened expression of c, e, f, V, VS, hr^B and hr^S. It has also been associated with the partial D coded by *RHD*DAU0*. Recently two donors phenotyped as R2R2 but genotyped as R2r. It was found that both carried the *RHCE*ceJAL* allele and the weak expression of little e was not detected with the routine antisera. Therefore, the Black donor and patient populations was investigated to determine if there were other cases of incorrect phenotypes due to *RHCE*ceJAL*. **Study Design/Methods:** Because of the reported associations, we purposely enriched our test population by selecting samples that were positive for DAU, V and VS. Genomic DNA samples were screened for V/VS using microarray assays (BioArray & Progenika) and testing for DAU was performed using several methods including a laboratory developed SSP-PCR, AS-PCR (BAG) and a microarray (BioArray). Selected samples were further tested for RHCE*ceJAL using SSP-PCR or microarray. **Results/Findings:** Of 9296 donors tested for V/VS, a total of 2128 (23%) were positive for both antigens. To estimate the frequency of DAU alleles, a group of 180 random Black donors were tested and 47 (26%) were DAU+. The most frequent DAU subtypes were *RHD*DAU3* and *RHD*DAU5*. From the DAU+ and V/V+ groups, three subsets were selected for further JAL testing including: 25 positive only for DAU (group 1); 132 positive only for V/VS (group 2); and 11 positive for both DAU and V/VS (group 3). No JAL+ were found in either groups 1 or 2; however, one heterozygous *RHCE*ceJAL* was found in group 3. Since his other allele had produced a normal e, there was no typing discrepancy. **Conclusion:** Although DAU alleles occur in a high frequency in the Black population studied here, few were of the DAU0 type. Thus it appears that the association with the *RHCE*ceJAL* allele is specific to DAU0 and not the DAU group as a whole. Even though *RHCE*ceJAL* is of low frequency in the Black population, it should still be considered when there is an apparent RH typing discrepancy.

Disclosure of Commercial Conflict of Interest

K. L. Billingsley: Nothing to disclose; P. Gowland: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support; G. T. Noumsi: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. L. Billingsley: Nothing to disclose; P. Gowland: Nothing to disclose; J. M. Moulds: BioArray Solutions, Grants or Research Support, Grifols, Consulting or Board of Director Fees, Travel Support or Honorarium; G. T. Noumsi: Nothing to disclose

SP291

Clinical Relevance of Variant *RHD* and *RHCE* Genes in a Sickle Cell Disease (SCD) Population

J M Moulds[1] (jmmoulds@lifeshare.org), K L Billingsley[1], G T Noumsi[1], D McCaskill[1]. *[1]Scientific Support, LifeShare Blood Centers, Shreveport, LA, United States*

Background/Case Studies: Even though many institutions now provide Cc, Ee, K1 matched blood for their sickle cell disease (SCD) patients, alloimmunization remains a problem in this patient population. Serological typing may be unable to detect RH variants in Blacks, thus SCD patients may be at risk for alloimmunization to Rh antigens. Over the past few years, blood centers have adopted microarray genotyping of both patient and donors in order to provide the best matched blood for transfusion. The genotyping data was analyzed from a large cohort of SCD patients (n = 416) to determine if more complete RH genotyping was needed. **Study Design/Methods:** Genomic DNA was prepared from samples obtained from both pediatric and adult SCD patients submitted blood group genotype. The initial genotype was performed using the HEA microarray (BioArray Solutions). Samples reported as possible r's and those with RH antibodies were studied further using the RHD and RHCE microarrays. Serological typing for the C antigen was done using licensed reagents containing the MS24 clone (Immucor, Ortho). **Results/Findings:** Of the initial cohort, 185 patients required RHD and RHCE genotyping. Of this group, 15 had produced anti-D and 8 of these were D variants that included: *RHD*DAU-3*, *DAU-5*, *DIIIa,* and *DIV*. For the C antigen typing, 35 (8.4%) were reported as (0*) or possible r's. Further investigation showed that 23/416 (5.5%) actually had the r's allele which is a RHD-CE-D hybrid producing a partial C antigen. However, all 23 samples reacted as expected with the licensed anti-C reagents and no weakening of C was observed but 7 of the patients had made allo anti-C. Of the remaining 12 (0*) samples, 9 (26%) were associated with *RHD*DIIIa* and 3 (8%) with other *RHD* alleles. At the *RHCE* locus, 13 patients were identified as being hr^B negative and 4 (31%) of these had produced anti-hr^B. Two patients that typed as hr^S negative had produced anti-hr^S and both genotype homozygous *RHCE*ceAR*. In addition, 18 patients carried alleles that produced partial e; 6 had *RHCE*ceMO* and 12 had *RHCE*ceTI*. **Conclusion:** HEA can detect the RHD-CE-D hybrid that produces a partial C. Even though these patients serologically type as C+ they should receive C antigen negative blood. However, the data demonstrate that Rh typing by serology often cannot detect the variant D and CE types found in SCD patients leaving these patients at risk for RH alloimmunization.

Disclosure of Commercial Conflict of Interest

K. L. Billingsley: Nothing to disclose; D. McCaskill: No Answer; J. M. Moulds: BioArray Solutions, Grants or Research Support; G. T. Noumsi: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. L. Billingsley: Nothing to disclose; D. McCaskill: No Answer; J. M. Moulds: BioArray Solutions, Grants or Research Support. Grifols, Consulting or Board of Director Fees,Travel Support or Honorarium; G. T. Noumsi: Nothing to disclose

SP292

Partial C Phenotype and the Risk for Alloimmunization; The Case for Red Cell Genotyping as a Screening Tool in the Detection of Variant Rh Alleles

R Shafi[1] (rabias@bcm.edu), B Sapinsky[1,2], S K Hartman[1,2], S R Hui[1,2], K Eldin[1,2], J Teruya[1,3]. *[1]Pathology & Immunology, Baylor College of Medicine, Houston, TX, United States; [2]Pathology & Immunology, Texas Children's Hospital, Houston, TX, United States; [3]Medicine and Pediatrics, Baylor College of Medicine, Houston, TX, United States*

Background/Case Studies: The highly complex Rh antigen system encodes many variant alleles, the majority of which are largely found in the African American (AA) population. Variant Rh alleles form an important trigger to the development of alloimmunization in sickle cell disease (SSD) patients, particularly those of AA ethnicity. We present a case of alloimmunization against the C antigen in a sickle cell patient of AA descent who was found to be C antigen positive by serological phenotyping. **Study Design/Methods:** Red cell genotyping was performed using genomic DNA and PCR-hybridization probes to determine the presence or absence of nucleotides associated with the expression of variant Rh alleles. **Results/Findings:** A 15-year-old AA female was presented with HbSS disease who had multiple prior hospital admissions secondary to vaso-occlusive crisis. Patient was typed as positive for C, c and e antigens per the serologic phenotype method.The patient was re-admitted in vaso-occlusive crisis after receiving two units of packed red blood cells (pRBC). Hemoglobin (Hb) was initially stable at 9.6 gm/dL but dropped to a nadir of 5.2 gm/dL with the appearance of red coloured urine. Laboratory values obtained at this time included: LDH of 24,639 IU/liter, plasma Hb of 122.4 mg/dL and total and indirect bilirubin of 6.0 and 2.3 gm/dL respectively. DAT was positive with poly, IgG and C3. With a positive antibody screen, an antibody panel identified anti C antibody and weak warm autoantibody with all other clinically significant alloantibodies ruled out. Eluate was positive for anti-C. At this time, a red blood cell genotyping panel was performed for the detection of variant Rh alleles. A hybrid RHD-CE-D gene was identified with two heterozygous alleles; RHCE*ces and RHCE*ce48C which are associated with the partial big C and partial little e phenotype respectively. pRBCS transfused prior to the development of anti-C were retrospectively found to be C positive. **Conclusion:** Published data available indicates a higher rate of alloimmunization to C antigen in patients with a partial versus normal C phenotype. With the increased frequency of Rh variants in the AA population, there is a failure of routine serological methods to detect many of these variants and the subsequent risk for alloimmunization. It is proposed that the use of red blood cell genotyping be used as a screening tool in the detection of the partial C phenotype as a means to prevent alloimmunization to anti-C in AA SSD patients, especially for female patients due to the concern of future pregnancy. The provision of C negative pRBC units to such patients could ultimately help prevent polyimmunization which is a major cause for transfusion failure in SSD patients.

Disclosure of Commercial Conflict of Interest

K. Eldin: Nothing to disclose; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose; B. Sapinsky: No Answer; R. Shafi: Nothing to disclose; J. Teruya: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Eldin: Nothing to disclose; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose; B. Sapinsky: No Answer; R. Shafi: Nothing to disclose; J. Teruya: Nothing to disclose

SP293

Resolution of Maternal D Typing Using Serology and Genotyping

P Berardi[1] (pberardi@toh.on.ca), J Hannon[2], G Clarke[2], T Alport[3], G H Growe[4], D Lane[5], R Fallis[5], J M Cote[6], G Ochoa[7], M Goldman[6]. *[1]Department of Pathology and Laboratory Medicine, The Ottawa Hospital, Ottawa, ON, Canada; [2]Prenatal Testing Laboratory, Canadian Blood Services, Edmonton, AB, Canada; [3]Prenatal Testing Laboratory, Canadian Blood Services, Regina, SK, Canada; [4]Prenatal Testing Laboratory, Canadian Blood Services, Vancouver, BC, Canada; [5]Prenatal Testing Laboratory, Canadian Blood Services, Winnipeg, MB, Canada; [6]National Immunohematology Reference Laboratory, Canadian Blood Services, Ottawa, ON, Canada; [7]R&D, Progenika Inc., Medford, MA, United States*

Background/Case Studies: All prenatal patients undergo serologic screening for RhD to determine if Rh immune globulin (RhIG) should be administered at 28 weeks. For patients with low expression of D, an algorithm incorporating molecular biology can assist in the classification of D variants. **Study Design/Methods:** Samples that gave weak (≤1+) or inconclusive results using automated solid phase testing (Immucor Galileo) with Series 4 (S4) and Series 5 (S5) monoclonal anti-D reagents, were tested using a manual tube method, S4, S5 reagents and Immucor Novaclone anti-D. Samples with weakly positive macroscopic results on immediate spin or after a 5-minute RT incubation with one or more reagents were sent for genotyping using an automated platform (Progenika BLOODchip). **Results/Findings:** From April 2011-April 2013, approximately 284,000 patients were tested; 37,127 were classified as D negative by serology. Results of the 102 sent for genotyping are shown (Table), along with policy for RhIG administration. There were six samples that required subsequent interrogation by gene sequencing, two compound heterozygotes (RHD*DAR, RHD*weak D type1and RHD*DAR, RHD Psi) and two novel D alleles not listed in the Rhesus Base or dbRBC database at the time of detection; one variant found in three samples was localised to exon 9 by BLOODchip, while the other found in a patient with anti-D gave a "no call" result. **Conclusion:** Use of the algorithm resulted in better classification of prenatal patients, with avoidance of RhIG administration in 72% of genotyped cases.

Disclosure of Commercial Conflict of Interest

T. Alport: Nothing to disclose; P. Berardi: Nothing to disclose; G. Clarke: Baxter, Other, Grifols, Other, CSL Behring, Other; J. M. Cote: Nothing to disclose; R. Fallis: Nothing to disclose; M. Goldman: Nothing to disclose; G. H. Growe: Nothing to disclose; J. Hannon: Nothing to disclose; D. Lane: Nothing to disclose; G. Ochoa: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Alport: Nothing to disclose; P. Berardi: Nothing to disclose; G. Clarke: Nothing to disclose; J. M. Cote: Nothing to disclose; R. Fallis: Nothing to disclose; M. Goldman: Nothing to disclose; G. H. Growe: Nothing to disclose; J. Hannon: Nothing to disclose; D. Lane: Nothing to disclose; G. Ochoa: Nothing to disclose

Genotype	n	RhiG
KHD-Positive	11	No
RHD Negative	1	Yes
Weak D—Types 1, 2, 3	56	No
Weak D—Types 4.0, 4.1	6	No
Weak D—Type 42	2	Yes
Partial D—DAK	8	Yes
Partial D—Others	12	Yes
Compound Heterozygotes	2	Yes
New variant, exon 9	*3*	Yes
New variant, exon 1	1	No (anti-D present)
TOTAL	102	2S

SP294

Validation of a Protocol for Fetal *RHD* Genotyping from Maternal Plasma in a Multi-Ethnic Population

M A Mota[1] (marizamota@einstein.br), M R Dezan[1], R O Cruz[1], T H Costa[1], F M Conti[1], M G Aravechia[1], J E Levi[1], L Castilho[2], J M Kutner[1]. *[1]Hospital Israelita Albert Einstein, Sao Paulo, Brazil; [2]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: Maternal plasma analysis for determination of the fetal *RHD* status is an exciting tool for the management of RhD-negative pregnancies, especially in alloimunized women. Few assays are designed in such a way that the fetus can be typed in mothers with non-functional *RHD* or presenting *RHD* variants. Based on this, was validated of a fetal *RHD* genotyping assay in maternal plasma and assessed its accuracy by analysis of gDNA from individuals with non-functional *RHD* and *RHD* variants. **Study Design/Methods:** Taqman technology was used to detect by real-time PCR, exons 4, 5, 7, and 10 of *RHD* . As internal control primers/probes were used set to CCR5. The assay was validated with fetal DNA extracted from maternal plasma of 11 RhD-negative pregnant women varying from 11-28 weeks of gestation. Peripheral or umbilical cord blood from respective neonates were collected during delivery and hemagglutination was performed. In order to assess the accuracy of the assay in our population gDNA samples were analyzed from three individuals presenting non-functional *RHD* (1 *RHDψ*, 2 *RHD-CE-D*) and from 20 individuals with *RHD* variants: *RHD*DVI* (n = 4); *RHD*DFR*(n = 3); *RHD*DIV* (n = 2); *RHD*DV* (n = 2); *RHD*DAR* (n = 2); *RHD*DIII* (n = 2); *RHD*DAU* (n = 1); *RHD*DVII* (n = 1); *RHD*Del* (n = 1); *RHD*DAR/RHD*DIII* (n = 1); and *RHD*IV/RHDψ* (n = 1). gDNA samples from individuals typed as RhD+ and RhD- were also tested. The population

studied was from southeastern Brazil with highly mixed ethnic backgrounds. **Results/Findings:** Among the 11 fetal DNA samples extracted from maternal plasma, 9 were genotyped as *RHD* + and 2 as *RHD* -. In this group, all results agreed with the neonatal typing. The *RHDψ* gDNA sample was exon 5 negative, as expected, and the 2 *RHD-CE-D* hybrid gene gDNAs were positive only for exon 10. The four *RHD*DVI* samples were negative for exons 4 and 5, two *RHD*DIV* were exon 7 negative, one *RHD*DFR* was exon 4 negative while one *RHD*DV* was exon 5 negative. The results were discrepant in three *RHD* variant samples: 2 *RHD*DFR* and 1 *RHD*DV* . All RhD+ samples were positive for all exons and all RhD- were negative for all exons. **Conclusion:** A test accuracy of 100% was demonstrated for prediction of the RhD phenotype in fetus of RhD-negative pregnant women on DNA from maternal plasma. The findings showed that *RHD* (exons 4, 5, 7 and 10) protocol can accurately be adopted for fetal *RHD* genotyping in a multi-ethnic population.

Disclosure of Commercial Conflict of Interest

M. G. Aravechia: Nothing to disclose; L. Castilho: Nothing to disclose; F. M. Conti: Nothing to disclose; T. H. Costa: Nothing to disclose; R. O. Cruz: Nothing to disclose; M. R. Dezan: Nothing to disclose; J. M. Kutner: Nothing to disclose; J. E. Levi: Nothing to disclose; M. A. Mota: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. G. Aravechia: Nothing to disclose; L. Castilho: Nothing to disclose; F. M. Conti: Nothing to disclose; T. H. Costa: Nothing to disclose; R. O. Cruz: Nothing to disclose; M. R. Dezan: Nothing to disclose; J. M. Kutner: Nothing to disclose; J. E. Levi: Novartis, Grants or Research Support; M. A. Mota: Nothing to disclose

SP295

Application of the Bio-Rad Free Fetal DNA KIT® RHD in a Semi-Automated Set Up

M A Jakobsen[1] (marianne.jakobsen@rsyd.dk), S B Mortensen[1], U Sprogøe[1]. *[1]Department of Clinical Immunology, Odense University Hospital, Odense, Denmark*

Background/Case Studies: Noninvasive determination of fetal RHD status before RhIG administration is mandatory in Denmark. The CE approved fetal RHD kit from Bio-Rad targeting RHD exon 5, 7 and 10 is based on manual DNA purification and PCR setup. We applied the kit in a semi automated setup and compared the results with those obtained by a standard in-house assay targeting RHD exon 5 and 10. **Study Design/Methods:** Collection tubes were centrifuged and 1000 µl plasma was transferred to sample cartridge by a Xiril pipetting robot. External DNA control was added manually. DNA was purified using MagNA Pure TNA-LV kit with MagNA Pure LC 2.0 (Roche) and DNA was eluted in 50 µl buffer. Finally, PCR was set up on MagNA Pure LC2.0 according to the instruction from Bio-Rad, and real-time analysis was run on LightCycler® 480 (Roche). Unique sample ID barcodes were read by Xiril and electronically transferred to MagNA Pure LC2.0 and LightCycler® 480, thus ensuring patient traceability during the complete process. **Results/Findings:** A total of 137 samples were run with both assays. In 83% (114/137) of cases, the results were concordant (Table). Sixteen samples rated as inconclusive with the in-house method were assigned as RhD neg by Bio-Rad kit, a discrepancy probably caused by weak unspecific signals in the in-house method. One of the inconclusive and two of the positive samples (in-house method) were assigned as D variants by the Bio-Rad kit due to missing exon 5/7 signal. For one of these samples, we obtained strong exon 5 signal (in-house method), suggesting that Bio-Rad kit PCR failed for exon 5 in that particular sample. Similarly, four samples rated positive with the in-house method were inconclusive using the Bio-Rad kit due to missing exon 10 signal. Results on umbilical cord blood serological RhD typing were not available. **Conclusion:** The Bio-Rad Free Fetal DNA kit® RHD may be applied in a semi automated setup using Xiril/MagNA Pure LC 2.0/LightCycler® 480. Compared to a manual procedure, this setup carries the benefit of full patient ID traceability. The performance of the kit compares favorably to our in-house method, although for at least five samples failure of the Bio-Rad kit was documented. Final conclusion on kit performance awaits results on serological cord blood RhD typing.

Disclosure of Commercial Conflict of Interest

M. A. Jakobsen: Labex (Denmark), Grants or Research Support; S. B. Mortensen: Nothing to disclose; U. Sprogøe: BioRad, Other

Disclosure of Grants Conflict of Interest

M. A. Jakobsen: Nothing to disclose; S. B. Mortensen: Nothing to disclose; U. Sprogøe: BioRad, Other

TABLE 1: *RHD* results using the two different methods

		BIORAD			
		RHD pos	*RHD* neg	D variant	Incondusive
IN HOUSE	*RHD pos*	*78*	0	*2*	4
	RHD neg	0	36	0	0
	Incondusive	0	16	1	0

SP296

Molecular Genotyping Provides Guidance in a Case of Maternal Anti-hr^s

K M Annen[1] (kyle.annen@bcw.edu), S K Hartman[2], G A Denomme[1], J Gottschall[1], S R Hui[2]. *[1]BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Pathology, Texas Children's Hospital, Baylor College of Medicine, Houston, TX, United States*

Background/Case Studies: Alloimmunization in pregnancy is most commonly associated with antibodies to D, c, E, and K. Hemolytic disease of the fetus and newborn (HDFN) due to antibodies of unknown specificity pose diagnostic and therapeutic challenges. This is a case of a 24-year-old, G2P1, African American woman who presented just prior to delivery at 37 weeks of pregnancy. Initial workup showed a mixed-strength pan-positive screen with negative DAT and autocontrol, suspicious for an antibody to a high prevalence antigen with underlying anti-C and anti-E. She delivered before the antibody could be identified. **Study Design/Methods:** Maternal and cord blood samples were submitted for red cell genotyping. Genotyping included evaluation for: 1) common Rh antigen variant single nucleotide polymorphisms (SNPs) of C, e, and c antigens; 2) common partial D alleles; and 3) RHD zygosity. **Results/Findings:** Investigation revealed maternal antibodies with anti-E, anti-C and anti-e-like specificity. The mother was homozygous for the following SNPs: RHCE*ce48C; 733G; and *712G, indicating RHCE*ceAR variant homozygosity. The predicted Rhce phenotype (C-, E-, c+, partial e+ and hr^s-) is consistent with anti-C, anti-E, and anti-hr^s alloimmunization. By molecular partial D and zygosity analyses, the patient carries the DAR allele and is hemizygous for RHD. Thus, the mother's RH haplotypes are DAR-RHCE*ceAR/RHCE*ceAR. The baby's cord blood analyses confirmed the infant's inheritance of a DAR allele and an unspecified RHD. Presence of the RHCE*ceAR allele in the infant was indicated by SNPs RHCE*ce48C, 676C/G, 733C/G, 712A/G; also present was RHCE*IVS2+109ins. The infant's most probable alleles were RHCE*CE/RHCE*ceAR with predicted Rh phenotype: D+, C+, E+, c+, e+, hr^s-. The infant was delivered uneventfully, but had mild hemolysis requiring phototherapy. The infant's DAT was 4+ IgG with 3+ reactivity to a 3-cell eluate screen. **Conclusion:** This case was complicated by an unusual maternal RHD type combined with an antibody to a high-prevalence antigen of unknown significance at the time of delivery. By serology, the mother was Group A+ with anti-E, anti-C and an unknown high-prevalence antibody. By genotype, she was homozygous for RHCE*ceAR variant and hemizygous for DAR (partial D). Individuals with DAR can develop anti-D if exposed to wild-type D antigen; therefore RhIG prophylaxis was warranted. Resolution of infant inheritance by molecular genotyping determined maternal partial-e and paternal D and E antigens. Thus, the infant lacks the hr^s antigen and HDFN was attributed to anti-E and C. An earlier prenatal screen would have allowed adequate time for identification of anti-E versus anti-hr^s on fetal cells and for determining the need for RhIG prophylaxis, as well as plan for transfusion management of the mother and infant.

Disclosure of Commercial Conflict of Interest

K. M. Annen: Nothing to disclose; G. A. Denomme: Nothing to disclose; J. Gottschall: Cerus, Other, Hemonetics, Other, Fenwal, Other, MacoPharma, Other; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. M. Annen: Nothing to disclose; G. A. Denomme: Nothing to disclose; J. Gottschall: Nothing to disclose; S. K. Hartman: Nothing to disclose; S. R. Hui: Nothing to disclose

SP297

D Variants Classification Following Discrepant Results after Different Blood Donations

C Arnoni[1] (carine.arnoni@colsan.org.br), J G Muniz[1], R Person[1], D Gazito[1], L Castilho[2], J Barreto[1], F Latini[1]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: D variants can have a broad spectrum of D antigen quantity on the red blood cell surface. Some of them may be

erroneously typed as D–negative by serology in the donor routine and may cause anti-D immunization when transfused to D-negative patients. Discrepancies were detected in the routine here in the D typing in some donors who were formerly typed as D-negative in the previous donations, and typed as D-positive in the current donation. Based on this, the aim of this study was to classify the D variants associated with these discrepant D typing results. **Study Design/Methods:** D typing on donor samples was performed by manual microplates and gel test in the previous donations and by Neo® (Immucor) in the current donation. Fifteen donors typed as D-positive by Neo®, who were previously phenotyped as D-negative, were selected for this study. DNA was extracted and RHD genotyping was performed on those samples using multiplex PCR, PCR-RFLP and sequencing. **Results/Findings:** The number of blood donations among these donors varied from 1 to 10. Nine out of 15 donors were genotyped as weak D type 38 in a total of 45 blood units transfused. Besides, three donors, related to five blood units, were genotyped as weak D type 2; 2 donors carrying weak D type 4.2.2, related to 8 previous donations; and one donor, with one prior donation, identified as weak D type 11; were earlier considered as D-negative. **Conclusion:** This analysis shows that D variants with low D antigen expression require more sensitive serological methods to be identified. To avoid this misclassification as D-negative, genotyping using pools of D-negative DNA could also be a good alternative.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP298

***RHCE* Characterization requires both commercial genotyping arrays and lab-developed tests**

J A Crowley[1] (jessica.crowley@redcross.org), T Horn[1], M A Keller[1].
[1]National Molecular Laboratory, American Red Cross Biomedical Services, Philadelphia, PA, United States

Background/Case Studies: Altered phenotypes of the Rh blood group system arise from genetic variation of the *RHD* and *RHCE* genes. Variant RH alleles can be due to the presence of one or more genetic variants and molecular analyses can detect alleles associated with weakened or partial antigens as well as with the loss of high prevalence antigens such as hr^B. Genotyping to identify *RHCE* variants is valuable as serologic methods are hampered by reagent limitations, can miss weak antigens and cannot distinguish partial antigens. BioArray Solutions *RHCE* BeadChip™ includes 25 genetic markers associated with the RhCE antigens. A software algorithm assigns predicted phenotypes based on the pattern of single nucleotide polymorphisms (SNPs) detected and allele assignments take into account what alleles have been reported in the literature and their allele frequency. Lab-developed tests used in conjunction with this commercial genotyping array are able to characterize patient or donor samples more precisely than the array alone. Two examples are provided to illustrate this point. **Study Design/Methods:** Patient samples submitted for *RHCE* characterization were genotyped using the *RHCE* BeadChip™ kit in conjunction with multiplex PCR, allele-specific (AS) PCR and PCR-RFLP assays, exon-specific sequencing and in some cases cDNA analysis. **Results/Findings:** Case 1 involves a 55-year-old female with sickle cell disease (SCD) who was demonstrating a possible anti-e antibody. *RHCE* BeadChip™ found the sample to be heterozygous for *RHCE* nucleotide nt48G>C and nt733C>G. The *RHCE* BeadChip™ software assigned the genotype ce/ce^S with the predicted phenotype of C- E- c+ e+. This allele assignment assumes that the variants are in cis (*RHCE*ce/ RHCE* ce48C,733G*). cDNA analysis was performed and identified two variant alleles *RHCE*ce48C* and *RHCE*ce733G*. The *RHCE*ce48C* allele encodes a weakened e antigen, while the *RHCE*ce733G* allele encodes a partial e with an hr^B+w/- phenotype. Though the risk of alloimmunization to e antigen is unclear, inheritence of these two variant alleles puts this patient at risk of production of anti-f(ce). Case 2 involves a 25-year-old male with SCD who was demonstrating a possible anti-hr^B antibody. *RHCE* BeadChip™ detected no variant markers and predicted the phenotype to be C- E+ c+ e+ which would be expected to be hr^B+. However, a PCR-RFLP found the sample to be heterozygous for *RHCE* nt254C/G, a marker not included on the *RHCE* BeadChip™. Thus, the probable alleles were *RHCE*cE* and *RHCE*ceAG*. The *RHCE*ceAG* allele encodes a partial e with an hr^B- phenotype. With these *RHCE* alleles, the patient is predicted to be hr^B- and would be at risk to produce anti-hr^B. **Conclusion:** These cases illustrate the benefit of pairing commercial genotyping arrays with laboratory-developed tests and sequence analysis for a comprehensive approach to characterizing *RHCE* alleles in donors and patients.

Disclosure of Commercial Conflict of Interest

J. A. Crowley: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support, Life Technologies, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

J. A. Crowley: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support

SP299

Comparative Analysis of Molecular BeadChip Assays and Serology for C c E and e, and VS and V Antigens

G Ishtiaq[1] (gishtiaq@immucor.com), Y Song[1], T Hailemariam[1], E Enriquez[1], U Naik[1], D Garg[1]. *[1]BioArray Solutions Ltd., Warren, NJ, United States*

Background/Case Studies: Molecular methods are becoming increasingly useful in enhancing immunohematological typing. They overcome the potential inherent limitations of serological methods thereby reducing the risk of alloimmuizations. Variants associated with altered expression of C c E and e antigen are a result of molecular changes in one or more exons of the RHCE gene. Serological methods may not always be able to identify these variants from wild type. This study involved a comparative analysis of BeadChip assays HEA 1.2 and RHCE (Both for Research Use Only in US) with serology to identify such variants to overall improve transfusion outcomes. **Study Design/Methods:** A total of 98 DNA samples were evaluated. Eighty nine whole blood samples were obtained from Immucor (Norcross; GA) and DNA was extracted using the QIAGEN Qiamp DSP Blood Mini kit (Qiagen cat# 61104). Serology results for these samples were made available from Immucor (Norcross, GA). Nine additional DNA samples were acquired from multiple other sources. HEA 1.2 and RHCE Bead Chip assays (BioArray Solutions Ltd., Warren NJ) were performed on all 98 samples. The HEA1.2 assay identifies 38 minor blood group antigens and phenotypic variants and the RHCE BeadChip identifies more than 30 major antigens (C/c, E/e) and phenotypic variants. **Results/Findings:** No variants were detected in 64/98 samples tested and were found to be in agreement between HEA 1.2, RHCE and serological assays for C c E e antigens. RHCE variants were identified in 34 samples. 25/34 samples with variants were also concordant between the molecular and serological assays for the above antigens. 9/34 samples were concordant between the molecular methods (HEA 1.2 and RHCE) but discordant with serology. Among the discordant samples, the 48C polymorphism was identified in 3/9 samples, 2/9 samples carried the ceMO variant, and one sample each of ceTI and ceS (340) variants was found. In 2/9 samples the cause for discordance between molecular and serological methods could not be explained. The molecular results were verified using various methods and found to be concordant. Serology on these samples however could not be repeated due to lack of sample availability. Hence rendering 98 % concordance rate with serology result in detection of major antigens. **Conclusion:** Extended antigen typing by DNA analysis using BeadChip based molecular methods is useful in identifying variants of the RHCE gene that result in expression of weak and partial RhCE antigens that cause discrepancies in serological typing.

Disclosure of Commercial Conflict of Interest

E. Enriquez: BioArray Solutions, Other; D. Garg: BioArray Solutions, Other; T. Hailemariam: BioArray Solutions, Employer, Other; G. Ishtiaq: BioArray Solutions, Ownership or Partnership; U. Naik: BioArray Solutions, Other; Y. Song: BioArray Solutions, Ownership or Partnership

Disclosure of Grants Conflict of Interest

E. Enriquez: Nothing to disclose; D. Garg: Nothing to disclose; T. Hailemariam: BioArray, Employer, Other; G. Ishtiaq: Nothing to disclose; U. Naik: Nothing to disclose; Y. Song: Nothing to disclose

SP300

Feasibility of the *RHD* BeadChip™ Kit to Identify and Characterize *RHD* Variant Alleles

T Horn[1] (Trina.Horn@redcross.org), J A Crowley[1], M A Keller[1]. *[1]National Molecular Laboratory, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: The *RHD* gene is highly polymorphic and genetic changes within this region can result in a partial, weak or null

phenotype of the RhD antigen. Current methodologies for *RHD* molecular characterization include commercial and lab-developed tests that interrogate individual genetic variants. BioArray Solutions *RHD* BeadChip™ simultaneously detects 35 genetic markers in the *RHD* gene in an 8-test format. Important to note, the prevalent *DAU0* allele characterized by nt1136C>T cannot be detected due to the lack of this marker on the platform. A set of samples of known *RH* genotypes were analyzed with the *RHD* BeadChip™ to evaluate its effectiveness to detect complex *RHD* variant alleles. **Study Design/Methods:** Donor and patient samples were selected to encompass a variety of *RHD* alleles with intent to cover as many analytes as possible. *RHD* BeadChip™ testing was performed according to manufacturer's directions. Twenty-two samples of known genotype were tested in duplicate covering 20 of the 35 analytes included on the BeadChip™. Genotype results were compared, and samples showing discordant or indeterminate findings were investigated further, utilizing alternative molecular methods. Six additional known samples were added to the sample set to investigate reproducibility of analytes or alleles which did not yield expected results. After implementation, unexpected genotype calls in multiple samples also were investigated using alternative methods. **Results/Findings:** Four of the 22 samples yielded unexpected genotype results. The nt697G>C was erroneously detected in the two samples with *RHD*DAU4* and *RHD*DV.5* alleles. Exon-specific sequencing confirmed the variant was absent. A sample with a *RHD*DNB* allele which carries the nt1063G>A change showed weakened signals in two neighboring analytes resulting in "xx" calls. The same was true in a sample carrying the nt667T>G change. Several samples were observed where *RHD*DAR* homozygotes or hemizygotes yielded LS calls for exon 5 analytes and samples with *RHD*DAR* alleles *in trans* to another *RHD* allele failed to call the variant at nt667T>G. Additionally, five samples yielded low signal (LS) calls for exon 7 or exon 9 analytes and the phenotypical variant was reported as indeterminate. DNA sequence analysis demonstrated that the reference sequence was present in these regions in all but one sample. **Conclusion:** The *RHD* BeadChip™ test was validated with 28 samples of known *RH* genotype. Under several circumstances, some analytes yield LS or xx calls not described in the manufacturer's limitations; these required further investigation. This work demonstrates that the genotype at nt697 may require confirmation. Comprehensive genotype results should be obtained by combining *RHD* BeadChip™ testing with other molecular tests to address limitations of this commercial platform.

Disclosure of Commercial Conflict of Interest

J. A. Crowley: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support, Life Technologies, Consulting or Board of Director Fees

Disclosure of Grants Conflict of Interest

J. A. Crowley: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support

SP301

Evaluation of the BioArray Solutions RHD Beadchip™ Test in Comparison to Established Methods for RhD Investigation

A Matteocci[1] (antonella.matteocci@virgilio.it), C Spaccino[1], B Pham[2], S Kappler-Gratias[2], S P Grimsley[3], K Finning[3], P Bailly[4], M Silvy[4]. *[1]UOC-SIMT, S. Camillo Hospital, Rome, Italy; [2]CNRGS, INTS, Paris, France; [3]IBGRL, NHSBT, Filton (Bristol), United Kingdom; [4]EFS, Marseille, France*

Background/Case Studies: An altered expression of the RhD antigen can be found in people of different ethnic origins with variable frequencies and variable reactivity in serological tests with commercially available anti-D sera. In recent years, DNA-based tests have allowed a better characterization of many RHD genetic variants. The new RHD BeadChip™ assay (Immucor-BioArray Solutions), a test based on the allele-specific elongation of synthetic probes that are linked to immobilized color-coded beads, covers 35 polymorphic sites of the RHD gene in a single-tube analysis, thus allowing determination of the presence or absence of the selected alleles associated with a given phenotype. **Study Design/Methods:** A collaborative study involving four European sites (Rome-S.Camillo, Paris-CNRGS, Bristol-IBGRL, Marseille-EFS) was started in 2012 with the aim to evaluate the performance of the RHD BeadChip™ test for the investigation of RhD variants in comparison to established serological and DNA-based methods. Four-hundred and ninety-two samples were specifically selected to achieve maximum coverage of variants targeted by the RHD BeadChip™ assay, so that D-pos and D-neg samples accounted for a minority (98 samples). All samples were tested using the RHD BeadChip™ kit and other reference methods including serology and/or molecular tests (PCR-SSP and DNA-sequencing). In order to evaluate the assay reproducibility and repeatability, six blind coded DNA samples were divided into aliquots that were distributed and tested three times on three different days at all sites. **Results/Findings:** The RHD BeadChip™ test had the following levels of concordance with established methods: 98.3% (121/123 samples) versus serology, with two samples that were RHD gene positive but antigen negative due to a mutation not detected by the RHD BeadChip™ test; 94.6% (212/224 samples) versus PCR-SSP, with 2.6% error rates with both SSP and RHD BeadChip™ methods in calling the right variants; 95% (135/142 samples) versus DNA-sequencing. Sensitivity (Snty) and specificity (Spty) were as follows:

detection of D positive, Snty = 100%, Spty = 90%;
detection of Partial D, Snty = 93.5%, Spty = 100%;
detection of Weak D, Snty = 98.5%, Spty = 100%; and
detection of D negative, Snty = 92.6%, Spty = 96.1%.

Blind coded samples (n = 6) produced results with 100% agreement with reference typing on all runs/days. **Conclusion:** The BioArray Solutions RHD BeadChip™ kit complements conventional serological assays by providing an innovative tool to help resolve ambiguous typings in patients and aid in the identification of compatible donors with DNA analysis of RHD variants. The high levels of sensitivity and specificity demonstrate its value in determining variants of RhD antigens, although its use is recommended as an adjunct to routine methods utilized for determination of RhD status.

Disclosure of Commercial Conflict of Interest

P. Bailly: Nothing to disclose; K. Finning: Nothing to disclose; S. P. Grimsley: Nothing to disclose; S. Kappler-Gratias: No Answer; A. Matteocci: Immucor-Gamma, Other; B. Pham: No Answer; M. Silvy: Nothing to disclose; C. Spaccino: No Answer

Disclosure of Grants Conflict of Interest

P. Bailly: Nothing to disclose; K. Finning: Nothing to disclose; S. P. Grimsley: Nothing to disclose; S. Kappler-Gratias: No Answer; A. Matteocci: Immucor-Gamma, Grants or Research Support; B. Pham: No Answer; M. Silvy: Nothing to disclose; C. Spaccino: No Answer

SP302

Evaluation of Molecular Methods to Determine D Variants

A C Gaspardi[1] (karolgaspardi@yahoo.com.br), D Machado[1], K B Torres[1], E Sippert[1], G A Guelsin[1], L Castilho[1]. *[1]Hemocentro, UNICAMP, Campinas, Brazil*

Background/Case Studies: The D antigen is highly immunogenic, requiring only a small quantity of transfused red blood cells (RBCs) to cause alloimmunization in D- recipients. Rh discrepancies are becoming a problem during routine testing due to D variants. The identification of these variants by serology is not easy and therefore, *RHD* genotyping is been used for a better characterization of partial D and weak D types. Based on this, it was sought to evaluate different molecular methods to determine D variants. **Study Design/Methods:** Samples from 42 Brazilian donors with an atypical D expression, including discrepancies or reactivity weaker than 3+ in D typing were selected. Molecular methods used to investigate D variants included LDTs (PCR-SSP, Multiplex-PCR) and the RHD BeadChipTM Kit (BioArray Solutions, Immucor) performed according to the manufactures' instructions. Sequencing exons was used as a confirmatory test. **Results/Findings:** Molecular analyses allowed the identification of D variants in all samples analyzed. The most common variants in the studied population were weak D type 4.0 followed by DVI, weak D type 4.2.2, weak D type 3, weak D type 2, DIVa and DAU4. The results among the methods used were concordant in 36 samples (86%), while discordant results were observed in 6 (14%) as shown in table 1. **Conclusion:** RHD BeadChip was concordant with sequencing in all discrepant results. As common SNPs are present in several D variants it is important to evaluate the molecular methods used in order to better characterize the samples with altered *RHD* alleles in each population.

Disclosure of Commercial Conflict of Interest

L. Castilho: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; G. A. Guelsin: Nothing to disclose; D. Machado: No Answer; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

Disclosure of Grants Conflict of Interest

L. Castilho: Nothing to disclose; A. C. Gaspardi: Nothing to disclose; G. A. Guelsin: Nothing to disclose; D. Machado: No Answer; E. Sippert: Nothing to disclose; K. B. Torres: No Answer

TABLE 1. Samples with discordant results

Samples (n)	PCR-SSP	Multiplex-PCR	RHD BeadChip	Sequencing exons
4	Weak D type 4.0	DVI	Weak D type 4.0	Weak D type 4.0
1	D positive	D positive	DAU-4	DAU-4
1	D positive	D positive	DV type 5	DV type 5
2	Weak D type 4.0	DVI	Weak D type 4.2.2	Weak D type 4.2.2
1	Weak D type 3	DVI	DAR/DIIIa-CE(4-7)-D	DAR/DIIIa-CE(4-7)-D
1	Weak D type 4.0	D positive	Weak D type 4.0/RHDΨ	Weak D type 4.0/RHDΨ

SP303

RH Genotype and Variant Screening using Cell Line DNA using the RH BeadChip

Y Song[1] (ysong@immucor.com), T Hailemariam[1], E Enriquez[1], G Ishtiaq[1], U Naik[1], D Garg[1]. *[1]R&D, BioArray Solutions Ltd., Warren, NJ, United States*

Background/Case Studies: The RH locus consists of two highly homologous genes, RHD and RHCE that produce highly variable and immunogenic RBC blood group antigens. The human Rh system encompasses five main antigens "D", "C", "c", "E" and "e" that are the most commonly identified and the most significant in blood transfusion. Therefore Rh type screening is important for transfusion safety and prevention of haemolytic disease of newborn. RH BeadChip kits (For research use only) provide a rapid and reliable method to identify a wide range of clinically significant RH genotype and complex phenotypic variants including partial, weak alleles, hybrids, deletions and C c E e. Many of the phenotypic variants are rare and population specific, hence availability of reference DNA samples to evaluate assay performance in the BeadChip assay and other molecular methods is limited due to inconsistent supply of well characterized samples. It is hence imperative to build a reference DNA library covering the maximum number of variants. This study evaluated cell line samples as a source of continual supply of well characterized DNA samples for RH genotyping. **Study Design/Methods:** Cell line DNA from different ethnic groups, healthy donors and donors with genetic disorders (Coriell Institute Camden, NJ), were utilized in this study. A total of 274 cell line DNA samples were tested with RHCE and RHD BeadChip kits (BioArray Solutions, Warren, NJ) (For Research Use Only). Cell lines of interest were subjected to bidirectional Sanger sequencing to confirm the presence of the variant. (GENEWIZ South Plainfield, NJ). **Results/Findings:** 216/274 (78.8%) samples were determined to be RHD positive. 35/274 (12.8%) were partial D and 4/274(1.5%) were weak D samples. On the RHCE panel, 75/274 (27.4%) samples were with RHCE type-Ccee; 94 /274 (34.3%) samples were RHCE type ccee; 42/274 (15.3%) samples were RHCE type ccEe; 22/274 (8%) samples were RHCE type CcEe; 36/274 (13.1%) were RHCE type CCee; 2/274 (0.7%) samples were RHCE type CCEe; and 3/274 (1.1%) samples were RHCE type ccEE. In addition some rare RHD allele combinations (eg DAR/DAU5 or DV type 1 or DBS2 [2E-14]) and RHCE allele combinations (eg ceS/Cw) were also found. **Conclusion:** Cell lines can be used as continual source of DNA in order to build a reference DNA library for RH genotyping. Research use only, not for use in diagnostic procedures.

Disclosure of Commercial Conflict of Interest

E. Enriquez: BioArray Solutions, Other; D. Garg: BioArray Solutions, Other; T. Hailemariam: BioArray Solutions, Employer,Other; G. Ishtiaq: BioArray Solutions, Ownership or Partnership; U. Naik: BioArray Solutions, Other; Y. Song: BioArray Solutions, Ownership or Partnership

Disclosure of Grants Conflict of Interest

E. Enriquez: Nothing to disclose; D. Garg: Nothing to disclose; T. Hailemariam: BioArray Solutions, Employer, Other; G. Ishtiaq: Nothing to disclose; U. Naik: Nothing to disclose; Y. Song: Nothing to disclose

SP304

Genotype Compatibility Tables for Matching Patients and Donors for Rh Variants

M A Keller[1] (margaret.keller@redcross.org), T Horn[1], J Crowley[1], S J Nance[2], G Meny[3], C Flickinger[2]. *[1]National Molecular Laboratory, American Red Cross Biomedical Services, Philadelphia, PA, United States; [2]National Reference Laboratory for Blood Group Serology, American Red Cross Biomedical Services, Philadelphia, PA, United States; [3]Department of Pathology, University of Texas Health Science Center at San Antonio, San Antonio, TX, United States*

Background/Case Studies: Genetic variation in *RHCE* and *RHD* is common in individuals of African descent. In the sickle cell disease (SCD) patient population, alloimmunization to Rh antigens is not uncommon, even in clinical programs with C, c, E and e matching. When blood donors who carry Rh variants are available, matching donors to patients with like variants may decrease alloimmunization and, in the case of patients with *RHD* variants, preserve D- units. An algorithm to assist in donor selection through rare donor registries such as the American Rare Donor Program and to offer clinicians transfusion alternatives was developed. The algorithm delineates all *RH* allele combinations that represent potentially-compatible donors for a patient based on their *RH* alleles. **Study Design/Methods:** Genotyping of *RHCE* and *RHD* was performed by multiplex PCR, PCR-RFLP, allele-specific PCR, DNA sequence analysis, and/or *RHD* and *RHCE* BeadChip (BioArray Solutions, Warren, NJ). Alleles were assigned based on the genotype findings using ISBT Blood Group Allele terminology. RH genotype compatibility tables were constructed to specify potential matches based on matching *RHCE* and *RHD* alleles individually. Donors who have non-functional *RHD* (via gene deletion or pseudogene) are included in these tables. A database of donors for which *RH* alleles have been characterized was evaluated for potential matches to patients based on *RH* alleles. **Results/Findings:** Genotype compatibility tables of *RHCE* and *RHD* alleles were created for patients with nine distinct *RH* allele combinations. Of these, two were for patients with E- hr^B- phenotypes associated with homozygosity for *RHCE*ce48C, 733G, 1006T* while two were for patients with one *RHCE* allele associated with a hr^B- phenotype paired with *RHCE*ce733G* that is associated with a hr^B+w/- phenotype. Patients carried conventional, partial or hybrid *RHD* genes. In one example, for a patient with a r'^S allele and a *RHCE*ce733G* allele, donors homozygous for the r'^S allele as well as those homozygous for *RHCE*ce733G* allele or compound heterozygotes were rated as level 1 matches. Donors who carried an *RHCE* allele with an additional variant such as *RHCE*ce48C, 733G* would represent a level two match. In this case, r'^S homozygous donor units were identified and transfused with no untoward transfusion responses reported. In other cases where genotype compatibility tables were used, though level two match donors were identified, units from such donors were unavailable. **Conclusion:** Genotype compatibility tables of *RHCE* and *RHD* alleles were established that specify *RHD* and *RHCE* genotypes of donors representing potential matches for a patient with specific *RH* variant alleles. These tools can aid in elucidating variables that determine risk of alloimmunization to Rh antigens. Further, such tools highlight the *RH* genotype of donors yet to be identified, recruited, and retained.

Disclosure of Commercial Conflict of Interest

J. Crowley: No Answer; C. Flickinger: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research, SupportLife Technologies, Consulting or Board of Director Fees; G. Meny: Nothing to disclose; S. J. Nance: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Crowley: No Answer; C. Flickinger: Nothing to disclose; T. Horn: No Answer; M. A. Keller: BioArray Solutions, Grants or Research Support; G. Meny: Nothing to disclose; S. J. Nance: Nothing to disclose

SP305

***RHD* Alleles in RhD: Blood Donors Identified through Electronic Interfacing of Red Cell Genotype and Phenotype**

G A Denomme[1] (greg.denomme@bcw.edu), M Schanen[1], K Bensing[1], C Piefer[1], R Udani[2], D B Bellissimo[2]. *[1]Immunohematology & Transfusion Services, BloodCenter of Wisconsin, Milwaukee, WI, United States; [2]Molecular Diagnostics Laboratory, BloodCenter of Wisconsin, Milwaukee, WI, United States*

Background/Case Studies: Weak, Partial, and Del D phenotypes are due to a number of uncommon and rare *RHD* alleles, the expression of which can be missed by routine D antigen phenotyping methods. Molecular screening approaches using DNA extracted from whole blood pools of RhD– donors makes it possible to evaluate large numbers of donors for the presence of *RHD*. However, creating these pools, extracting DNA, and resolving *RHD*+ samples from the pools is an off-line reflex test that can be a time-consuming

process. Individual RhD– donors were screened for an *RHD*-specific nucleotide and developed an electronic interface coupled to the RhD phenotype to flag donors with an *RHD*+/RhD– discrepancy. **Study Design/Methods:** High throughput red cell genotyping (RCG) is used currently to evaluate blood donors for *RHD*DIII/IV*[455A>C], hybrid *RHDIII-CE-D*[410G>A], *RHCE*C*[IVS2+109ins], *RHCEC/c*[307T>C], and *RHCE*E/e*[676C>G]. Since the *RHD*[455A] is present in the wild-type and many other *RHD* alleles, an electronic interface was designed to compare *RHD*[455A] output with RhD phenotype after each run and to generate an exception report of donors with an *RHD*+/RhD– discrepancy. Discrepancies were grouped according to the predicted C/c, E/e phenotype and donor-declared ethnicity. **Results/Findings:** A total of 138 of 6,637 RhD– donors (2.1%) were positive for the *RHD*[455A] nucleotide. Among 440 RhD– African Americans (AA) donors, 81 rr (cde/cde) had an *RHDΨ*; two of these donors were confirmed *RHDΨ*+ by an *RHDΨ* gel electrophoresis method, and another 22 (N = 101) donors remain to be confirmed. The remaining rr (cde/cde) donors included 12 Hispanic (HI), 7 Caucasian (CA), 1 Native American (NA), and 2 donors of undeclared ethnicity. Two of 12 HI donors were confirmed *RHDΨ*+ by gel electrophoresis. A total of seven discrepancies were observed among r′r (Cde/cde), r″r (cdE/cde), and r′r″ (Cde/cdE) donors. All discrepancies are summarized in the table. **Conclusion:** The integration of an RCG platform and data handling makes individual *RHD* testing of RhD– donors feasible. This process removes the need to make pools of blood and to separate them to identify *RHD*+ samples. When tested in real-time, individual donor units are temporarily quarantined until the RhD status can be resolved by reflex serological and molecular testing.

Disclosure of Commercial Conflict of Interest

D. B. Bellissimo: Nothing to disclose; K. Bensing: Nothing to disclose; G. A. Denomme: Nothing to disclose; C. Piefer: Nothing to disclose; M. Schanen: Nothing to disclose; R. Udani: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. B. Bellissimo: Nothing to disclose; K. Bensing: Nothing to disclose; G. A. Denomme: Nothing to disclose; C. Piefer: Nothing to disclose; M. Schanen: Nothing to disclose; R. Udani: Nothing to disclose

***RHD*+/RhD– Discrepancies Grouped by Rh Penotype and Ethnicity**

	Ethnicity					
Rh Phenotype	CA	AA	HI	AS	OT	# Confirmed
C-E-c+e+	7	101	12		2	83
C+e-c+e+	3	2	1	2		3
C-E+c+e+	6	1				3
C+E+c+e+				1		1
RhD–	5793	440	295	38	71	6637
% Discrepant	0.12	23	4.4	7.9	2.9	

Red Blood Cells: Blood Group Serologic Testing

SP306

Correlation of Flow Cytometric (FC) and Automated Serological Analyses of D Typing

C Arnoni[1] (carine.arnoni@colsan.org.br), J G Muniz[1], R Person[1], D Gazito[1], L Castilho[2], F Latini[1], J Barreto[1]. *[1]COLSAN, São Paulo, Brazil; [2]INCTs, UNICAMP, Campinas, Brazil*

Background/Case Studies: D typing is a constant challenge in the donor routine, since it may vary according to methods used. Reagents with higher sensitivity and specificity are required to avoid false D-negative results in donors harboring weak D variants. Although characterized molecularly, different weak D variants may present a broad spectrum of RhD amount in the red blood cell (RBC) surface. It was sought to determine the relation between the score result obtained in the D typing performed by Neo® (Immucor), and the quantification of D antigens performed by flow cytometry. **Study Design/Methods:** A total of 26 samples, including samples typed as D-negative and D-positive with different levels of agglutination, were selected to the study. The samples were analyzed in Neo® (Immucor) using an anti-D IgM (RUM-1) and by flow cytometry (FC) according to previous study (WA Flegel, *Transfusion Clinique et Biologique*, 2002—Spreadsheet Version1.1). Results obtained in Neo® were expressed in a score from 0-99, while those obtained by FC were expressed in % that means the relative epitopes densities. Molecular techniques, including multiplex PCR, PCR-RFLP and sequencing were used to identify RhD variants. **Results/Findings:** Neo Immucor-detected samples with very low antigen density, quantified by flow cytometry as 8.71% of epitopes density. After molecular characterization, D variant samples were identified as weak D type 3, weak D type 4.0, and weak D type 4.2.2 that presented epitopes density from 4.08 to 18.60% and a Neo score from 58 to 90. Additionally, a sample was classified as DMH, a partial D, that presented 31.5% of epitopes density and a score of 95 in Neo®. Samples with relative epitopes density >31% were detected by Neo® with the highest range of score varying from 95 to 99. **Conclusion:** The results showed that the most common variants of the analyzed samples were successfully detected by Neo®. There was a good correlation with epitopes density until approximately 31%, when the score reported by Neo® reached a plateau. The results showed that Neo® has high sensitivity to detect weak D antigens and the score results are comparable to those obtained by FC.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; L. Castilho: Nothing to disclose; D. Gazito: Nothing to disclose; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP307

Top versus Bottom Red Blood Cell (RBC) Sample Drawing for Blood Group Typing: Analysis of Mixed Fields in Solid Phase and Column Agglutination

S Flommersfeld[2] (flommers@med.uni-marburg.de), B Ertl[2], G Bein[1,2], U J Sachs[1,2]. *[1]Institute for Clinical Immunology and Transfusion Medicine, Justus Liebig University, Giessen, Germany; [2]Center for Transfusion Medicine and Hemotherapy, University Hospital Giessen and Marburg, Marburg, Germany*

Background/Case Studies: The recognition of mixed fields (MF) may hamper proper blood group typing in patients with a history of recent RBC transfusions. Several case reports have claimed that in automated blood group typing, mixed fields may occur more frequently, since in automatic devices, RBCs are pipetted from the bottom of the test tube. Transfused RBCs are supposed to be preferably present at the bottom because they gain weight during shelf storage and may thus settle faster than autologous cells. **Study Design/Methods:** During a time period of 4 weeks, 22 patients were identified who were transfused with RBCs incompatible for CcEe or K due to shortage of fully compatible RBCs. CcEe and K typing was performed on the latest available post-transfusion blood sample manually (using column agglutination from BioRad, Dreieich, Germany) or automatically (using the solid phase system GalileoNEO from Immucor, Rödermark, Germany). RBCs were collected from the top and bottom parts of the test tube. **Results/Findings:** Mixed fields were identified in 20/22 patients (28/37 antigen discrepancies) by column agglutination (CA) and in 14/22 patients (17/37 antigen discrepancies) in the automated solid phase (SP) assay. Findings between top and bottom samples differed in 6/22 patients in both systems (antigen discrepancies, 12/37 for CA and 8/37 for SP). CA detected four antigen discrepancies in the top and eight in the bottom sample; SP detected five antigen discrepancies in the top and three in the bottom sample. The mean age of transfused RBCs in samples with MF in the top sample was 11 days, but was 19 days for those with MF in the bottom sample ($p = 0.004$). **Conclusion:** The SP assay is less sensitive to recent transfusions compared to CA. Since all samples in which mixed field were not detected by SP were typed correctly, this could be an advantage of the SP assay in clinical routine. If, however, identification of previous transfusions is warranted, CA should be used preferably (sensitivity approx. 90% versus 64%). In contrast to previous reports, transfused RBCs are not regularly found on the bottom of the test tube. Evidence was found that this phenomenon might increase with the age of the transfused RBC. Again, SP was more robust than CA in typing CcEe and K from these samples.

Disclosure of Commercial Conflict of Interest

G. Bein: Nothing to disclose; B. Ertl: No Answer; S. Flommersfeld: Nothing to disclose; U. J. Sachs: German Red Cross Blood Service West, Hagen, Germany, Grants or Research Support

Disclosure of Grants Conflict of Interest

G. Bein: Nothing to disclose; B. Ertl: No Answer; S. Flommersfeld: Nothing to disclose; U. J. Sachs: German Red Cross Blood Service West, Hagen, Germany, Grants or Research Support, Octapharma AG, Lachen, Switzerland, Consulting or Board of Director Fees

SP308

Kell Antigen Blocking Phenomenon in a Case of Haemolytic Disease of the Fetus and Newborn (HDFN) in a Woman with Multiple Red Blood Cell (RBC) Alloantibodies

S Manfroi[1] (silvia.manfroi@aosp.bo.it), P Pagliaro[1]. *[1]Immunohematology and Transfusion Service, Policlinico S.Orsola-Malpighi, Bologna, Italy*

Background/Case Studies: A case is studied here of false-negative K typing caused by blocking of K antigen sites by maternal IgG anti-K in a severe case of hemolytic disease of the fetus and newborn (HDFN). The mother was a 29-year-old Italian with multiple RBC alloantibodies: anti-K (titre 1024), anti-s (titre 1) and anti-Lua (titre 1); she was grouped as O RhD positive and phenotyped as CCDee, kk, S+s-, Lu(a-b+). **Study Design/ Methods:** Red blood cell (RBC) serology typing was performed with commercially available antisera (Immucor, Germany) and confirmed by gel technique (DiaMed, Switzerland). Direct antiglobulin test (DAT) on cord cells was performed using DiaMed DC-ScreeningII card (DiaMed); antibody detection, identification, and titration was done using commercially available panel cells by gel technique (DiaMed). Elution was performed by Lui freeze-thaw procedure. Genomic DNA isolated from cord buffy coat sample was genotyped using BLOODchip® IDCore+ kit (Progenika Biopharma/Grifols, Spain). **Results/Findings:** During screening for HDFN by weekly ultrasound measurement of fetal middle cerebral artery peak systolic velocity (MCA-PSV), at 21 weeks a severe fetal anemia was suspected due to MCA-PSV >1.5 MoM. Obstetricians obtained foetal blood sample by cordocentesis and severe anaemia was confirmed with haemogobin (Hb) 5.4 g/dL in the sample before intrauterine blood transfusion (IUT). By conventional tube testing the fetal sample was typed as O RhD positive and phenotyped as CCDee and kk. The DAT gave strong positive results (4+), indirect antiglobulin test was positive and antibody was identified as anti-K (titre 2). Kell phenotype was repeated with tube and gel technique but results unchanged. In the suspicion of blocking K phenomenon we eluted maternal IgG antibodies from cord RBCs. DAT performed on eluted RBCs gave a 1+ reaction (from initial 4+) and foetal RBC could be phenotyped by gel technique as Kk. The eluate specificity was anti-K. The pregnant woman's anti-K was capable of blocking the K antigen also on blood donor K-positive RBC. Genotyping performed on DNA confirmed foetal serological typing as CCDee, Kk, S+s+, Lu(a+b+). Foetus received IUT of 30 ml RBC and Hb concentration raised to 15.2 g/dL; other two IUT were performed and K blocking phenomenon repeated. **Conclusion:** Blood grouping failures are uncommon using modern monoclonal antisera. However this phenomenon has been reported with high antibody titres (>256) and should be suspected in presence of severe foetal anaemia, high antibody titre, DAT 4+ and absence of antigens on foetal RBC. Genotyping procedures are helpful to solve these cases.

Disclosure of Commercial Conflict of Interest

S. Manfroi: Nothing to disclose; P. Pagliaro: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Manfroi: Nothing to disclose; P. Pagliaro: Nothing to disclose

SP309

Requests for Phenotyped Red Blood Cells (RBCs) in a Large Multi-State Blood Supplier

D Figueroa[1] (dfigueroa@bloodsystems.org), T Glanzer[3], M S Irani[2,1], F Nizzi[2,1], P A Tomasulo[2], B A Lenes[2]. *[1]Clinical Laboratories, Blood Systems, Inc., Tempe, AZ, United States; [2]Medical Affairs, Blood Systems, Inc., Scottsdale, AZ, United States; [3]Blood Center Divison, Blood Systems, Inc., Scottsdale, AZ, United States*

Background/Case Studies: Transfusion of phenotyped units to patients with antibodies is common. Very little data exists about the phenotypes requested, the frequency of requests, and their geographic distribution. A program to provide the same turnaround time for patients with antibodies as for patients without antibodies was instituted in a large multistate regional blood supplier with a 20 state geographic footprint and great ethnic diversity in its donor base. The following data were observed. **Study Design/ Methods:** ABO/RhD, number of phenotyped units requested, urgency, location, and specific phenotypes were determined and entered in a spreadsheet and the data was analyzed. The requests were classified based on types and number of antigens including C,c,E,e,K,Fya,Fyb,Jka, Jkb,S,s. The categories were as follows: Common (negative for C,c,E,e and K only, or other antigens excluding Fya,Fyb,Jka, Jkb,S,s), Uncommon (C,c,E,e,K and one to two of the following antigens Fya,Fyb,Jka, Jkb, Jkb,S,s), Rare (C,c,E,e,K, and three or more of the following antigens Fya,Fyb,Jka, Jkb,S,s), and Very Rare (high frequency antigen negative units). **Results/Findings:** The mean number of RBC units collected per month in 2012 was 80,357 and of those 32,081 (39.9%) were previously phenotyped at least for RhCcEe and K. An average of 9.7% of non-previously phenotyped donors were screened for RhCcEe and K each month. Out of 918,126 total RBC's distributed 33,194 units were phenotyped (3.6%). The most frequently requested phenotyped units (14144 [43%]) were in the RH (CEce) and KEL (K) systems only. The common requests as defined above consisted of 15031 units and made up 45% of the total phenotyped units. Table 1 summarizes the frequency of requests by classification and geography. **Conclusion:** Of all requests for phenotyped RBC units, 43% can be met by blood centers typing and labeling all units for RhCcEe and K. Geographic variation of phenotype requests may require systems to pre-position phenotyped products to better meet current and historical local needs.

Table 1 Phenotyped RBC Units by State

State	Common	Uncommon	Rare	Very Rare	Total Distribution
CA	3819 (1.75%)	5829 (2.5%)	1180 (0.5%)	14	230984
AZ	5500 (3.1%)	4256 (2.4%)	99 (0.1%)	9	178118
IX	1342 (0.9%)	1423 (1.0%)	40 (0.0%)	0	147389
IA, MS	1369 (1.0%)	1347 (1.0%)	320 (0.2%)	21	132032
NV	1344 (1.8%)	1485 (1.9%)	62 (0.1%)	25	76229
ND, SD	712 (1.2%)	675 (1.1%)	5 (0.0%)	3	60810
NM	343 (0.6%)	832 (1.5%)	55 (0.1%)	0	57031
MT, WY	602 (1.7%)	471 (1.3%)	12 (0.0%)	0	35533
Total	**15031 (1.6%)**	**16318 (1.8%)**	**1773 (0.2%)**	**72**	**918126**

Disclosure of Commercial Conflict of Interest

D. Figueroa: Nothing to disclose; T. Glanzer: No Answer; M. S. Irani: Nothing to disclose; B. A. Lenes: Nothing to disclose; F. Nizzi: Mediware, Other, Blood Systems, Other; P. A. Tomasulo: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Figueroa: Nothing to disclose; T. Glanzer: No Answer; M. S. Irani: Nothing to disclose; B. A. Lenes: Nothing to disclose; F. Nizzi: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

SP310

The Frequency of Clinically-Significant Red Cell Alloantibody Formation during a 14-Day Interval in Patients who were Recently Transfused

J Garbaini[2] (jennifer.garbaini@gmail.com), S T Avecilla[1,2], C Goss[1,2], D T Lo[2], D Degtyaryova[2], D W Chang[3], M Cushing[1,2]. *[1]Pathology and Laboratory Medicine, Weill Cornell Medical College, New York, NY, United States; [2]Transfusion Medicine and Cellular Therapy, NewYork Presbyterian Hospital, New York, NY, United States; [3]The Wharton School, University of Pennsylvania, Philadelphia, PA, United States*

Background/Case Studies: New alloantibody formation in patients who have a history of red cell alloimmunization and recent transfusion is a very real risk and steps must be taken to ensure that they are detected in a timely manner to maintain the utmost of patient safety. Unlike antibody screening, there is no currently accepted universal policy as to the appropriate frequency of performing red cell antibody identification testing (ABI). **Study Design/Methods:** From September 2010-September 2012, all antibody identification entries in the WCMC-NYP laboratory database were screened for the development of new clinically significant red cell alloantibodies in previously alloimmunized patients for whom a type and screen was ordered. More in-depth analysis was conducted on the screening candidate cases and a consensus decision amongst four physicians was achieved as to the validity of new alloantibody formation within a 14 day (or less) testing interval. **Results/Findings:** 8948 antibody identification entries were examined which correlate to 2792 unique patients. 30 patients were found to have formed at least one new alloantibody within the 14 day interval. 33 antibodies were identified; however only 13 antibodies (0.15% of all ABI performed or 0.47% of all patients examined) were deemed to be unambiguous, newly formed clinically significant alloantibodies for which a change in transfusion practice would occur: α-C (1), α-E (1), α-Fya (3), α-Fyb (1), α-Jka (5), α-Jkb (1), α-K (1). A change in policy to perform ABI every 14 days instead of every 72 hours in recently transfused/pregnant patients would avoid 505 ABI over a 2-year period in this institution, an estimated savings of $60,000 per year (including FTE salary and reagent costs). **Conclusion:** Retrospective

analysis of antibody identification entries over a 2-year period revealed that only 0.15% of all ABI identified newly formed clinically significant antibodies within a 14 day or less observation window which correlates to 0.47% of patients. There would certainly be resource-conservation advantages to changing the interval from every 3 days to 14 days; however the increased risk of hemolytic transfusion reactions must be considered.

Disclosure of Commercial Conflict of Interest

S. T. Avecilla: Nothing to disclose; D. W. Chang: Nothing to disclose; M. Cushing: Nothing to disclose; D. Degtyaryova: Nothing to disclose; J. Garbaini: Nothing to disclose; C. Goss: Nothing to disclose; D. T. Lo: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. T. Avecilla: Nothing to disclose; D. W. Chang: Nothing to disclose; M. Cushing: Nothing to disclose; D. Degtyaryova: Nothing to disclose; J. Garbaini: Nothing to disclose; C. Goss: Nothing to disclose; D. T. Lo: Nothing to disclose

SP311

A Case of Masquerading Alloantibodies: The Value of a Multi-Technique Approach

P M Wennersten[1] (PWennersten@carterbloodcare.org), L J Sutor[1,2]. *[1]Reference and Transfusion, Carter BloodCare, Bedford, TX, United States; [2]University of Texas Southwestern Medical School, Dallas, TX, United States*

Background/Case Studies: In an immunohematology reference lab, samples received for antibody identification react in many different ways that require a variety of approaches. Sometimes, the clues from initial testing can lead to faulty assumptions and misdirection. Fortunately, a well supplied reference lab will have access to a variety of techniques and reagents that used together can reveal the true identity of the antibodies involved. **Study Design/Methods:** Patient plasma and cells submitted for antibody identification were tested by standard tube and gel techniques to determine ABORh, DAT, and antibody screen results. **Results/Findings:** A sample was received from a 65-year-old male. It was reported by the referring hospital that his antibody screen was negative by solid phase 3 weeks prior and an unknown number of red blood cell (RBC) units had been transfused. The initial testing showed strong reactivity with all cells of the forward and reverse ABO/Rh as well as a three cell antibody screen tested at LISS 37C and IAT, PeG IAT, and gel IAT. An albumin control and DAT with polyspecific anti-human globulin were negative. The sample also reacted strongly at IS with all cells including cord blood and I- panel cells, but the autocontrol was strongly positive only at 4C. The initial assumption was that the plasma contained a cold autoantibody. The antibody screen was repeated with a prewarm technique, but still reacted strongly at 37C and IAT, including a saline, no enhancement technique. Subsequent testing, including the use of ficin treated cells and antisera for phenotyping, showed the apparent cold autoantibody to be a red herring. The patient was M- so the ABO reverse type was repeated with M- reverse cells, resolving the ABO discrepancy but not explaining all of the IS antibody screen reactions. Further gel testing performed with M- cells yielded reactions of varying strengths. Ficin treated cells tested in gel provided the most useful clue as 3/5 cells that reacted when untreated were nonreactive as ficin treated and one reaction was stronger, pointing toward at least one ficin sensitive antibody and one ficin enhanced antibody in addition to anti-M. Additional tube testing at IS, 37C and IAT revealed the presence of 4 alloantibodies: anti-M and anti-E reacting at IS, 37C, and IAT plus anti-Fya and anti-Jkb reacting at IAT. **Conclusion:** This case served as a reminder that antibodies are not always what they initially appear to be, as when multiple alloantibodies masquerade as a cold autoantibody, and that our assumptions can potentially misdirect a workup. It also demonstrates the value of flexibility in the use of multiple tools to pursue an answer during antibody identification studies.

Disclosure of Commercial Conflict of Interest

L. J. Sutor: Nothing to disclose; P. M. Wennersten: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. J. Sutor: Nothing to disclose; P. M. Wennersten: Nothing to disclose

SP312

Early Antibody Detection using Solid Phase Technology

A Stein[1] (astein2@nshs.edu), N M Nikolis[1], J Fischman[1], W Heaton[2], L E Logdberg[1], S Sanchez-Riffle[1], J Zenker[1]. *[1]Transfusion Medicine, North Shore University Hospital, Manhasset, NY, United States; [2]Transfusion Service, Hofstra North Shore-LIJ School of Medicine, Manhasset, NY, United States*

Background/Case Studies: This blood bank processes ~ 47,000 specimens per year and converted in June 2009 from manual gel technology (1992-2009) to automated solid phase methodology (ASPM). This was associated with a significant rise in inconclusive antibody panels. Any first time positive ASPM antibody screens were reflexed to manual gel, which became our confirmatory test and the final test of record. Negative confirmatory assay results were further identified as False Positive Antibody screen (FPA) for future reference. This report documents the long-term follow up of FPA patients to affirm patient safety. **Study Design/Methods:** Records of patients assigned FPA instruction during 2011-2012 were re-evaluated when patients returned for new testing. The new testing followed the standard algorithm. A control group with initial negative antibody screen (ABSCN) results was identified within the same time period. Both groups were evaluated for subsequent ABSCN results and antibody identification (ABID) follow-up, as required. **Results/Findings:** During 2011-2012, 495 patients were assigned an FPA instruction, out of a total of 93,000 specimens tested. 188 (17%) patients returned for new testing. Of these, 32 (17%) continued to demonstrate reactivity in solid phase and now were also ABSCN positive in gel testing. 150 patients (79%) were negative in solid phase. Six patients (3%) continued to have an FPA instruction. Our control group consisted of 915 solid phase ABSCN negative results. 397 (43%) of patients returned for new testing. Of these patients, 17 patients (4.3%) demonstrated reactivity in solid phase and Gel testing. 380 patients (95.7%) were negative, and 3 patients (0.76%) had new FPA instructions added. For ABID see Table. **Conclusion:** Notable differences include: a three-fold increased rate of identifying specific antibodies in the patients that previously had an FPA instruction (9.6% versus 3.5%);and a—9% rate of non-specific gel reactivity in the FPA instruction group, as opposed to 0% in the negative control group. Overall, FPA instruction is predictive of future identification of non-specific antibodies and also predictive of a higher incidence of future identifiable antibodies.

Disclosure of Commercial Conflict of Interest

J. Fischman: Nothing to disclose; W. Heaton: Immunetics Inc, Grants or Research Support, Haemonetics Inc, Grants or Research Support, Light Integra Inc., Grants or Research Support, Fresenius Inc., Travel Support or Honorarium, Fenwal Inc., Grants or Research Support; L. E. Logdberg: Nothing to disclose; N. M. Nikolis: Nothing to disclose; S. Sanchez-Riffle: Nothing to disclose; A. Stein: Nothing to disclose; J. Zenker: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Fischman: Nothing to disclose; W. Heaton: Immunetics Inc, Grants or Research Support, Haemonetics Inc., Grants or Research Support, Light Integra Inc., Grants or Research Support, Fresenius Inc., Travel Support or Honorarium, Fenwal Inc., Grants or Research Support; L. E. Logdberg: Nothing to disclose; N. M. Nikolis: Nothing to disclose; S. Sanchez-Riffle: Nothing to disclose; A. Stein: Nothing to disclose; J. Zenker: Nothing to disclose

Follow-up Antibody Identification Results

Antibody	Previous FPA w/ New ABID (n = 188)	Previous Neg ABSCN w/ New ABID (n = 397)
Non-specific	14	0
Anti-E	6	6
Anti-Jka	4	4
Anti-Jkb	4	0
Anti-C,K	1	0
Anti-K	0	2
Anti-C	0	1
Anti-c	1	0
Anti-V	1	0
Anti-Leb	1	0
Anti-M	0	1
Totals	32	14
Frequency of Specific Antibodies	9.6%	3.5%

SP313

Detection of Passive Anti-D Antibodies by Solid Phase Antibody Screening over Time Following Administration of Rh-IgG

K V Mikesell[1,2] (kmikesell@hmc.psu.edu), J W Davis[2], R E Domen[1,2], M R George[1,2]. *[1]Department of Pathology, Pennsylvania State, Hershey Medical Center, Hershey, PA, United States; [2]Blood Bank/Transfusion Medicine, Pennsylvania State, Hershey Medical Center, Hershey, PA, United States*

Background/Case Studies: Rhesus-immunoglobulin (Rh-IgG) is an IgG antibody used to prevent alloimmunization in Rh negative individuals exposed to Rh positive blood. This medication, however, can lead to confusing antibody screening panels. Little work has been done to demonstrate how quickly or how long passive anti-D antibodies remain present in antibody screens. **Study Design/Methods:** Using our pathology database, 165 pregnant patients were identified within one year's time with passive anti-D due to Rh-IgG administration. These patients received either intramuscular (IM) HyperRHO or Rhophylac with the vast majority (>80%) receiving Rhophylac. It was then recorded when each patient received Rh-IgG and had antibody screens performed by using electronic medical and laboratory records. Tube testing results due to its small sample size (five results) were then excluded. Only patients with at least one positive antibody screen were identified. **Results/Findings:** In total, 255 antibody screens were reviewed. Sixty-seven screens were performed within 1 day of Rh-IgG administration with the fastest positive result occurring at 84 minutes (Table). One-hundred and eighty-eight of the screens reviewed were performed 24 hours after Rh-IgG administration. Almost all screens were positive up to 54 days post Rh-IgG treatment with positive results present up to 134 days (Table). **Conclusion:** While most pharmacokinetic studies of Rh-IgG detect the antibodies by flow cytometry, these results are from solid phase antibody screening. A selection bias is present in this study because only patients with at least one positive antibody screen were identified. Regardless, the data show strong evidence that solid phase antibody screens will not be positive immediately after Rh-IgG administration. In fact, most screens will not be positive until several hours after treatment. It was also found that passive antibodies can be present in the antibody screen up to 19.1 weeks. Further work is needed to evaluate how different antibody screening methods compare to one another and how other factors like BMI, fetomaternal hemorrhage, type of Rh-IgG, or route of Rh-IgG administration affects antibody screens.

Disclosure of Commercial Conflict of Interest

J. W. Davis: Nothing to disclose; R. E. Domen: Nothing to disclose; M. R. George: Nothing to disclose; K. V. Mikesell: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. W. Davis: Nothing to disclose; R. E. Domen: Nothing to disclose; M. R. George: Immucor, Grants or Research Support; K. V. Mikesell: Nothing to disclose

Table 1

Hours From Rh-IgG Admin.	# of (-) Screens	# of (-) Screens	Days From Rh-IgG Admin.	# of (+) Screens	# of (-) Screens
<1	6	51	1-9	12	1*
1-2	1 (84 mm.)	6	10-19	18	0
2-3	0	1	20-29	8	0
3-4	0	1	30-39	3	0
4-5	0	0	40-49	7	0
5-6	0	0	50-59	12	1 (54 days)
6-7	0	3	60-69	12	1
7-8	1	0	70-79	38	5
8-9	0	0	80-89	26	1
9-10	1	0	90-99	13	4
10-11	0	0	100-109	1	0
11-12	0	0	110-119	2	0
12-13	0	0	120-129	2	3
13-14	1	0	130-139	1 (134 days)	1
14-15	0	1	>140	0	16

*It is unclear if this result could represent a false negative. The same patient had a positive antibody screen 72 days later.

SP314

Comparing Antibody Screening and Identification between the Immucor Galileo and NEO Instruments

S M Armstrong[1] (sarmstrong@mcvh-vcu.edu), N Bangs[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: As part of the instrument selection and validation process, the Immucor NEO was compared against the existing Immcuor Galileo for its ability to detect and identify unexpected antibodies. **Study Design/Methods:** The laboratory ran a total of 184 antibody screens and 32 identification panels on both the Galileo and NEO instruments. There were a total of 23 different technologists who each ran eight antibody screens. At least two of the eight antibody screens had to have been resulted as positive using the two-cell screen on the Galileo, the instrument of record. The antibody screens where then repeated on the NEO using the 3-cell screen. The 32 antibody identifications were run by 16 different technologists running two samples each in which the antibody screens had been resulted as positive and antibodies identified using the Galileo panels. The antibody identifications were repeated on the NEO using the same panel(s), including lot numbers, which had been run on the Galileo. **Results/Findings:** The NEO demonstrated a raw score Sensitivity rate of 95.5%, Specificity rate of 100%, and a concordance rate of 98.9% for antibody screens. These rates were due to two "false negative" results (positive on the Galileo and negative on the NEO). However, after investigation, it was determined that the original positive Galileo results were the errant results when subsequent testing could not reproduce the original positive results. With these two results removed from the calculations, the NEO was determined to have 100% Sensitivity, Specificity, and Concordance to the final reported antibody screen result. The antibody identifications detected the following antibodies: ant-D, -E, -C, -c, -M, -S, -K1, -Jka, an antibody of undetermined specificity, and an anti-D due to RhIg. The raw antibody identification results demonstrated a concordance rate of 93.3%. However, of the two non-concordant results between the Galileo and NEO, the NEO results matched the patients' historical record. The first patient had a history of an anti-Jka; the NEO identified the anti-Jka while the Galileo's panel was negative. The second patient had a history of an anti-K1 and antibody of undetermined specificity. The Galileo identified the anti-K1 only; the NEO identified the anti-K1 and antibody of undetermined specificity. **Conclusion:** The initial sensitivity, specificity, and concordance results for the NEO versus the Galileo, instrument of record, demonstrated acceptable results. However, when the results are compared not just to the Galileo results, but to the final reported results and historical records, the NEO outperformed the Galileo by reducing the false positive antibody screen rate and increasing the antibody identification rates. The NEO was found to be not only acceptable for use, but an even better instrument than the existing Galileo.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; N. Bangs: Nothing to disclose; K. Sanford: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; N. Bangs: Nothing to disclose; K. Sanford: Nothing to disclose

SP315

Implementation of Immucor NEO® Automated Platform for Routine Use in a Big Donor Testing Laboratory

R Person[1] (duarterosinha@bol.com.br), J G Muniz[1], C Arnoni[1], J Barreto[1], F Latini[1]. *[1]COLSAN, São Paulo, Brazil*

Background/Case Studies: Most donor testing laboratories use fully or partially automated testing platforms as a cost effective mean to perform massive donor blood group typing and antibody screening. This blood donor testing facility collects around 12,000 donors per month; therefore, a fully automated instrument is required to improve quality management, reduce manual steps, standardize testing, and the improve reviewing the results. Implementation of a new technology is a critical process that requires validation, staff perception, and rigorous analyses of the results. The efficiency and performance was evaluated of NEO® automated platform (Immucor) in a large blood bank from Brazil in terms of its functionality and throughput for ABO and D typing, antibody screening and identification, weak D testing, and Rh and K phenotyping. **Study Design/Methods:** From January 2013-April 2013, 11,900 samples were tested from the donor routine, including AB subtypes and weak D samples. ABO, Rh and K phenotyping were performed by direct hemagglutination and antibody screening and identification were performed with Capture-R Ready Screen on NEO® automated instrument.

Rh and K phenotyping were performed on 3450 samples. Reference methods included direct and indirect hemagglutination tests in gel cards (Biorad/DiaMed and Grifols). **Results/Findings:** Concordant results were obtained for ABO, Rh and K phenotyping. There were three RhD discrepancies; all samples typed primarily as D negative were typed as D positive in NEO® and molecular tests confirmed the positive results obtained in NEO®. There were 32 antibody screen discrepancies; positive results were obtained with Capture-R with NEO®, whereas completely negative results of reference methods were observed. NEO® did not detect 1 anti-Dia and 2 anti-M. ABO, Rh and K phenotyping sensitivity for NEO® was 100% while for antibody screening, sensitivity was 99.7%. **Conclusion:** NEO® instrument on capacity, throughput, turnaround times and sensitivity is shown to be an excellent alternative for big donor testing laboratories. The implementation of NEO® in our donor routine also improved the detection of weak D, reducing the risk of receptor alloimmunization. Although the data show an increase in the inconclusive antibody investigations, there is still a workload time saving following NEO® implementation.

Disclosure of Commercial Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Arnoni: Nothing to disclose; J. Barreto: No Answer; F. Latini: Nothing to disclose; J. G. Muniz: Nothing to disclose; R. Person: Nothing to disclose

SP316

Ficin-Treated Cells in ID-MTS Gel Test Identify Clinically Significant Antibodies which May Mimic Non-Specific Reactivity

B Hill[1] (benchill@uab.edu), C A Hanna[1], J Adamski[1], M B Marques[1].
[1]University of Alabama, Birmingham, Birmingham, AL, United States

Background/Case Studies: In December 2010 an antibody work-up was performed on a patient who, 6 months earlier, showed non-specific reactivity in the ID-Micro Typing System™ (ID-MTS) Gel Test. Although his plasma was again agglutinating a few reagent cells, all clinically significant alloantibodies had been ruled out with homozygous cells. It was elected to test the patient's sample with ficin-treated reagent cells in ID-MTS, recognizing that this proteolytic enzyme yields increased sensitivity to identify antibodies to the Rh, Kell, Kidd and Lewis antigens, while destroying antigens of the Duffy and MNS systems. Ficin-treated cells clearly demonstrated two clinically significant antibodies in the patient's plasma: anti-C and anti-e. Following this index case, a laboratory policy was instituted whereby all non-specific reactions identified in ID-MTS are followed by testing with ficin-treated cells. **Study Design/Methods:** The records of all positive antibody screens were reviewed by ID-MTS work-ups performed between January-March 2013. If a patient was tested more than once during this period, only the first work-up was included. **Results/Findings:** During the 3-month period, 501 antibody workups were performed by ID-MTS. After removal of repeat evaluations, the final data set included 352 cases from 252 females and 100 males with an average age of 46 $\pm$ 20.4 (range: 0-103) years. A total of 72 (20%) had non-specific reactivity in the panel and 67 were followed by a panel with ficin-treated cells. The five work-ups that did not include ficin either had a long history of unexplained reactivity, likely due to an alloantibody against a low frequency antigen (n = 3) or were not checked due to oversight (n = 2). Panels with ficin-treated cells allowed the identification of 20 alloantibodies: 17 were clinically significant: anti-E (n = 9), anti-D (n = 5), and one each of anti-c, anti-e, and anti-Jka. In addition, we identified three antibodies to the Lewis antigens. Of interest, 9/77 (12%) antibodies to E that were present in this data set required ficin for identification. Ficin did not contribute to 47 workups which were signed out as nonspecific reactivity. **Conclusion:** The findings show that one-third of antibody evaluations with non-specific reactivity in ID-MTS can be resolved using ficin-treated cells. The data illustrate the importance of this enzyme technique, since 17 patients (6% of the total) with clinically significant alloantibodies would have gone unrecognized without the use of ficin-treated cells. It is hoped this report encourages others who perform serologic workups in ID-MTS to incorporate ficin-treated panels as part of their protocol for evaluating non-specific reactions.

Disclosure of Commercial Conflict of Interest

J. Adamski: Nothing to disclose; C. A. Hanna: Nothing to disclose; B. Hill: Nothing to disclose; M. B. Marques: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Adamski: Nothing to disclose; C. A. Hanna: Nothing to disclose; B. Hill: Nothing to disclose; M. B. Marques: Nothing to disclose

SP317

Comparison of Jka/Jkb, Fya/Fyb, S/s phenotyping and Weak-D Identification using Erythrocyte Magnetization Technology (E.M.® Technology) and Conventional U-bottom Microplate Technique

C L Dinardo[1] (carlaluana@yahoo.com.br), P Loacono[2], G Abib[2], A Mendrone[1]. *[1]Immunohematology, Fundação Pró Sangue, São Paulo, Brazil; [2]Immunohematology, Diagast, Lille, France*

Background/Case Studies: Background: Erythrocyte Magnetization Technology has been validated for ABO and RhD/C/c/E/e phenotyping. In this technique, RBCs are covered with magnetic beads and incubated at room temperature with specific IgM antisera. Centrifugation step is eliminated by the application of a magnetic field, which attracts RBCs to the bottom of well and promotes the sensitive RBCs agglutination in case of a positive reaction. Phenotyping RBCs for other blood groups using the same methodology with the inclusion of a human antiglobulin (AHG) step raises concerns as it may be associated with non-specific reactions. The same applies to weak-D identification. The objective was to validate E.M.® Technology to determine Jka/Jkb, Fya/Fyb, S/s phenotype, and weak-D identification in which magnetic beads are linked to monoclonal anti-glycophorin A before they are added to RBCs and including an AHG step. Compare the results of E.M. Technology with those obtained from conventional U-bottom microplate technique **Study Design/Methods:** 148 samples were phenotyped for Fya, Fyb, Jka, Jkb, S and s antigens through the E.M.® Technology using specific monoclonal antisera. In parallel, 17 known variant RhD samples were tested with the same methodology using anti-D clones P3X35 and ESD1. With the new methodology of magnetization, RBCs were covered with magnetic beads which had been previously linked to anti-glycophorin A monoclonal antibody. RBCs were incubated at 37C after antisera addition. After incubation step, a magnetic field was applied so that RBCs moved towards anti-human globulin (AHG) layer which coated the bottom of reaction plate's wells. We considered a negative reaction if a clear dot was formed in the center of the wells and a positive reaction if a RBC layer was formed instead. Phenotype results were compared to previous conventional U-bottom microplate results, and 100% of accuracy was observed. All variant RhD were identified as positive. No false positive results were observed within 35 negative-control samples (rr phenotype). **Results/Findings:** Phenotype results were compared to previous conventional U-bottom microplate results, and 100% of accuracy was observed. All variant RhD were identified as positive. No false positive results were observed within 35 negative-control samples (rr phenotype). **Conclusion:** E.M.®Technology is reliable in determining Jka, Jkb, Fya, Fyb, S and s phenotype, as well as in identifying RhD weak samples.

Disclosure of Commercial Conflict of Interest

G. Abib: Nothing to disclose; C. L. Dinardo: Nothing to disclose; P. Loacono: No Answer; A. Mendrone: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Abib: Nothing to disclose; C. L. Dinardo: Nothing to disclose; P. Loacono: No Answer; A. Mendrone: Nothing to disclose

SP318

ABO and Rh Antigen Typing of Red Blood Cells (RBCs) Using a Novel Serology-Based Multiplexed Protein Microarray Platform (Project MosaiQ™)

J S Robb[1] (janine.robb@albabioscience.co.uk), L K Knowles[1], D C Robson[1]. *[1]Development, Quotient, Edinburgh, United Kingdom*

Background/Case Studies: Automated ABO and RhD typing are routinely performed on donor and patient samples as part of pre-transfusion testing, while manual testing is often required to do extended antigen typing and Direct Antiglobulin Testing (DAT). A platform that could provide ABO/Rh, extended blood type and DAT results in one comprehensive single-step test would offer significant efficiency savings in the provision of blood for patients. A new serology-based microarray technology platform is being developed, known internally as Project MosaiQ™. This platform has the capability to perform comprehensive blood group typing, antibody identification and DAT in parallel using one sample. An evaluation is reported here of the MosaiQ™ platform to perform ABO and Rh typing in a study of 195 donor samples, including Ax. **Study Design/Methods:** Specialized microarray printing equipment was used to spot antibodies to blood groups A, B, D, c, C, e and E, and buffers, onto wells of a glass substrate resulting in 144 discrete spots. The antibodies and buffers were bound to the substrate and then blocked before drying. Diluted test red cells were added to the microarray. The suspension of cells was incubated, before excess and unbound red cells were washed and aspirated from the well. The bound cells

were then fixed to the slide prior to drying. The presence of bound red cells was detected by the use of a scanner. All samples were tested in parallel by column agglutination and tube based comparator methods. **Results/Findings:** In one step, and using only 50 μL of sample, 100% correlation with comparator techniques was achieved with anti-A, -B, -D, -c, -C, and -E including Ax cells (n = 3). Further refinement is needed for optimal e antigen detection; this work is on-going. **Conclusion:** This study demonstrates that with on-going optimization the MosaiQ™ platform can provide comprehensive blood typing in a single step. Further development work is on-going to optimize, not only anti-e, but an extended range of typing antibodies (including antibodies to K, Fy, Jk, MNSs). Adaptations of this format have also successfully demonstrated that antibody identification, DAT and virology assays can be performed simultaneously. The MosaiQ™ platform may offer efficiencies over current serologic or molecular analysis systems.

Disclosure of Commercial Conflict of Interest

L. K. Knowles: Quotient, Grants or Research Support; J. S. Robb: Quotient, Grants or Research Support; D. C. Robson: Quotient, Grants or Research Support

Disclosure of Grants Conflict of Interest

L. K. Knowles: Quotient, Grants or Research Support; J. S. Robb: Quotient, Grants or Research Support; D. C. Robson: Quotient, Grants or Research Support

SP319

Evaluation of a Novel Serology-Based Platform for Irregular Blood Group Antibody Identification (Project MosaiQ™)

J S Robb[1] (janine.robb@albabioscience.co.uk), N K Renault[1], D C Robson[1]. *[1]Development, Quotient, Edinburgh, United Kingdom*

Background/Case Studies: Screening of donor and patient samples for irregular blood group antibodies is an essential part of pre-transfusion testing. Current automated systems initially run an antibody screening test. If positive an antibody identification investigation will subsequently be performed, often manually, before suitable blood can be selected for cross-matching. Identifying the antibody in the initial test would offer time and cost savings in the provision of blood for patients. A new antibody-antigen microarray technology in development, known internally as Project MosaiQ™, has the capability to combine screening and identification, using only 50 μL sample volume, in one step, while simultaneously performing full blood group typing. The technology has been developed to meet the sensitivity levels required for donor and patient testing and of current state-of-the-art systems, without unwanted positives. This work reports an evaluation of the technology to perform antibody identification on 230 samples. **Study Design/Methods:** Specialized microarray printing equipment was used to spot an identification panel of 10 red blood cells, in duplicate, onto wells of a glass substrate, resulting in 20 discrete spots. The antibody identification panel consisted of well-characterized donor red cells, expressing antigens required by international guidelines. The cells were bound to the substrate, then lysed and preserved. Test plasma plus diluent was added to the array and incubated at 37C, before washing away excess and unbound plasma. The presence of bound antibody was detected using a blend of conjugated anti-IgG/IgM followed by application of a development reagent, prior to antibody identification analysis. All samples were tested in parallel by column agglutination and tube based comparator methods. **Results/Findings:** In one pass, 100% correlation with comparator techniques was achieved. Of the 230 samples tested, 30 contained a blood group antibody and 200 were negative when tested by both the comparator and test platforms. The positive antibodies identified included examples of anti-D (n = 15), -C, -c, -E, -e, -K, -k, -Fy^a, -Fy^b, -C^w, -s, -S, -M and -P1. The MosaiQ™ method demonstrated identification of anti-D at 0.05 IU/mL. **Conclusion:** The study demonstrated that MosaiQ™ can be used for irregular blood group antibody identification in one step. The platform showed the required level of sensitivity against weak antibodies, whilst demonstrating no unwanted positives. Adaptations of this format have also successfully demonstrated that blood typing, direct antiglobulin testing and virology assays can be performed simultaneously. The MosaiQ™ platform offers an efficient antibody identification system which challenges current serologic and molecular systems.

Disclosure of Commercial Conflict of Interest

N. K. Renault: Quotient, Grants or Research Support; J. S. Robb: Quotient, Grants or Research Support; D. C. Robson: Quotient, Grants or Research Support

Disclosure of Grants Conflict of Interest

N. K. Renault: Quotient, Grants or Research Support; J. S. Robb: Quotient, Grants or Research Support; D. C. Robson: Quotient, Grants or Research Support

SP320

Direct Antiglobulin Testing Discrepancies due to Differences between Gel versus Tube Method

L McClannan[1] (mcclanl@labcorp.com), J McHarg[1], M E Brecher[1]. *[1]LabCorp., Burlington, NC, United States*

Background/Case Studies: Immunohematologic discrepancies are known to occur between testing methods. A retrospective review was performed of anti-human globulin direct antiglobulin test performed by the column agglutination method and additional testing for IgG and C3 by tube testing. **Study Design/Methods:** Specimens for testing were collected in a lavender top EDTA tube. The routine DAT was performed by the column agglutination method (gel) using the Anti-Human Globulin Anti-IgG, -C3d MTS™ card on the ORTHO ProVue® analyzer (Ortho Clinical Diagnostics, Raritan, NJ). Additional testing was performed in test tubes by anti-human globulin, anti-IgG and anti-C3b, -C3d antisera (Ortho Clinical Diagnostics). Data from October 17, 2012-December 28, 2012 were reviewed. **Results/Findings:** Of 674 DAT performed 126 (19%) were positive by the gel method and 78 of the 126 (62%) were positive with anti-AHG in tube. (Table). **Conclusion:** The data demonstrate that consistent reactions between the AHG gel method and the tube method do not occur until the gel reaction is 2+ or greater. At a 1+ reaction there is a 27% (18/66) correlation. Anti-IgG reactivity was not consistently reactive until the gel reaction was 3+ or greater. The gel method has demonstrated greater sensitivity in detection of antibody bound on to red blood cells. This data set may be useful in explaining apparent discrepancies when patients have multiple direct antiglobulin tests at different facilities.

Disclosure of Commercial Conflict of Interest

M. E. Brecher: Nothing to disclose; L. McClannan: Nothing to disclose; J. McHarg: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. E. Brecher: Nothing to disclose; L. McClannan: Nothing to disclose; J. McHarg: Nothing to disclose

AHG Gel rxn	Gel total	AHG neg	AHG pos	AHG w+	AHG 1+	AHG 2+	AHG 3+	AHG 4+
1+	66	48	18	6	10	2	0	0
2+	31	0	31	3	21	7	0	0
3+	26	0	26	0	3	11	11	1
4+	3	0	3	0	0	0	3	0
AHG Gel rxn	Gel total	IgG neg	IgG pos	IgG w+	IgG 1+	IgG 2+	IgG 3+	IgG 4+
1+	66	60	6	2	4	0	0	0
2+	31	8	23	1	18	4	0	0
3+	26	0	26	0	5	8	11	2
4+	3	0	3	0	0	0	3	0
AHG Gel rxn	Gel total	C3 neg	C3 pos	C3 w+	C3 1+	C3 2+	C3 3+	C3 4+
1+	66	55	11	3	7	1	0	0
2+	31	20	11	3	6	2	0	0
3+	26	17	9	3	6	0	0	0
4+	3	0	3	1	2	0	0	0

SP321

Comparing the Removal of IgG from Red Blood Cells (RBCs) by EDTA Glycine Acid (EGA) and Chloroquine Diphosphate (CDP) Measured by Flow Cytometry (FC), Tube and Gel Direct Antiglobulin Test (DAT) Methods

W A Beres[1] (wendy.beres@redcross.org), C M Beauchamp[1], P A Nickle[1], S Singh[1], S J Nance[1]. *[1]Technical Services, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: The effectiveness of removing IgG from RBCs to obtain DAT negative red blood cells (RBCs) can vary between elution methods (*Transfusion* 1997; 37:497-501). This was seen in the 2011-2013 AABB IRL proficiency sample which had a +^{w} direct antiglobulin test (DAT) requiring RBC treatment to remove RBC-bound IgG to perform the extended phenotype. 85% of IRLs used EDTA glycine acid (EGA) to treat to a negative DAT with 100% success while 21% of IRLs used chloroquine diphosphate (CDP) with 69% success. This abstract seeks to expand that dataset. The effectiveness of removing IgG from 21 IgG-DAT positive RBCs (+^{m} to 4+) by EGA and CDP treatment was measured by flow cytometry (FC) and DAT by Tube and Gel DAT. FC measurement of the efficiency of elution methods to remove RBC-bound IgG has not been previously reported. **Study Design/ Methods:** With IRB approval, RBCs from 21 IgG-DAT positive samples (20 patients, 1 donor) were split into 3 aliquots: untreated, EGA treated, and CDP treated. RBCs were EGA treated (Gamma EGATM Kit Immucor Inc., Norcross, GA) up to two times or CDP treated (Sigma-Aldrich, St. Louis, MO) up to four times; or until a negative Tube DAT was obtained. Samples were analyzed for RBC-bound IgG by Tube DAT (Immucor Inc.), Gel DAT (Ortho-Clinical, Raritan, NJ) and by FC (Becton Dickinson FACSCaliburTM San Jose, CA) with fluorescein isothiocyanate -labeled anti-human IgG (Life Technologies, Carlsbad, CA). FC normal ranges for RBC-bound IgG were determined by testing 10 group O allogeneic blood samples. **Results/ Findings:** After EGA elution of 21 IgG-DAT positive samples: 67% were negative for RBC-bound IgG by Tube DAT; 29% by Gel DAT; and 19% by FC. After CDP elution of the same samples: 14% were negative for RBC-bound IgG by Tube DAT; 43% by Gel DAT; and 0% by FC. The results show % of samples negative by each testing method (Table). **Conclusion:** When studying the effectiveness of EGA and CDP elution methods, there were somewhat surprising results in that the usefulness of each method in yielding negative DATs was dependent on the RBC-bound IgG detection method used. By Tube DAT, EGA treatment was more effective in removal of IgG, but by Gel DAT, CDP treatment was more effective. However, FC detected RBC-bound IgG in 81% of EGA and 100% of CDP treated RBCs indicating incomplete IgG removal in most samples tested. The study suggests that IRLs should use a method which is effective with the testing method intended for the treated RBCs.

Disclosure of Commercial Conflict of Interest

C. M. Beauchamp: Nothing to disclose; W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; P. A. Nickle: Nothing to disclose; S. Singh: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. M. Beauchamp: Nothing to disclose; W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; P. A. Nickle: Nothing to disclose; S. Singh: Nothing to disclose

Test Method	Elution Method	Initial Tube DAT Strengh						Percent Negative
		+^{m}	+^{w}	1+	2+	3+	4+	
Tube DAT	**EGA**	100%	67%	100%	100%	50%	33%	67%
	CDP	50%	17%	50%	0%	0%	0%	14%
Gel DAT	**EGA**	100%	50%	0%	50%	0%	0%	29%
	CDP	50%	17%	50%	100%	33%	67%	43%
FC	**EGA**	0%	33%	0%	0%	17%	33%	19%
	CDP	0%	0%	0%	0%	0%	0%	0%
Total Samples		2	6	2	2	6	3	

SP322

Use of Flow Cytometry (FC), Tube and Gel Direct Antiglobulin Test (DAT) to Evaluate levels of RBC-Bound IgG after Storage

W A Beres[1] (wendy.beres@redcross.org), C M Beauchamp[1], P A Nickle[1], S J Nance[1]. *[1]Technical Services, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: Frequently during a complex IRL evaluation, samples are tested over an extended time frame. It was noticed through testing that some samples yielded different Tube direct antiglobulin test (DAT) results over time. A review of samples tested after storage at 4C was performed to evaluate the impact of sample storage on DAT result. Semi-quantitative methods (Tube and Gel DAT) and a quantitative method using flow cytometry (FC) were performed. **Study Design/Methods:** In an IRB approved study, samples were tested initially by Tube DAT which baselined the DAT strength. After storage samples were divided into three aliquots and analyzed in parallel for RBC-bound IgG; by Tube IgG-DAT (Immucor Inc, Norcross, GA), Gel IgG-DAT (Ortho-Clinical, Raritan, NJ) and FC. The FC aliquot was washed using Dulbecco's phosphate buffered saline (PBS) and sensitized with fluorescein isothiocyanate (FITC)-labeled anti-human IgG (Life Technologies, Carlsbad, CA). After a 60 minute at 22C incubation the samples were washed with 0.6% bovine serum albumin (BSA) in phosphate buffered saline (PBS) and analyzed via FC. 50,000 RBCs were acquired from each sample by the Becton Dickinson FACSCaliburTM (San Jose, CA) FC for analysis. Samples were tested at varying sample ages. Normal ranges for RBC-bound IgG by FC were determined to be 0-5.0% using the mean %M2 ±3 SD of normal donor samples. **Results/Findings:** It was surprising that 8 of the 29 samples (28%) studied (+^{m} to 4+ DAT) were negative when retested by Tube IgG-DAT. However the Gel IgG-DAT remained moderately to strongly positive in 7/8 samples. All eight were positive by FC after storage with a range of 9.8% to 90.7% percent positive (Table). **Conclusion:** Of interest were the three initial 2+ Tube IgG-DAT samples that were negative when retested after storage by Tube DAT but had 3+ to 4+ Gel IgG-DAT and a FC RBC-bound IgG range of 57.3% to 90.7% percent positive. In general, in the eight aberrant samples, the initial Tube DAT was comparable to the Gel DAT and the FC interpretation at the end of the storage time but not the Tube DAT. Thus, in routine testing in IRLs, if the DAT or autocontrol is a critical test result on a stored sample, based on this small cohort, it might be best advised to perform a Gel DAT as 28% of samples were negative in Tube DAT after storage.

Disclosure of Commercial Conflict of Interest

C. M. Beauchamp: Nothing to disclose; W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; P. A. Nickle: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. M. Beauchamp: Nothing to disclose; W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose; P. A. Nickle: Nothing to disclose

Sample	Initial IgG-DAT Tube	Elapsed Time of Study (days)	Tube IgG-DAT	Gel IgG-DAT	FC, % Positive
1	+^{m}	8	-	-	9.8
2	+^{w}	10	-	2+	13.9
3	+^{w}	14	-	2+	25.2
4	+^{w}	18	-	3+	31.0
5	1+	5	-	3+	65.7
6	2+	4	-	3+	57.3
7	2+	11	-	3+	86.0
8	2+	14	-	4+	91.0

SP323

Stability of In Vitro Red Blood Cell (RBC)-Bound IgG, IgA, and IgM By Flow Cytometric (FC) Analysis

W A Beres[1] (wendy.beres@redcross.org), S J Nance[1]. *[1]Technical Services, American Red Cross, Philadelphia, PA, United States*

Background/Case Studies: Tannic acid (TA) induces passive adherence of immunoglobulins (Ig) to RBCs (Med Lab Sciences 1990;47;189) producing Ig-coated RBCs. RBCs coated with Ig are required for positive controls in FC studies. Storage of TA treated Ig coated RBCs has not been previously described. **Study Design/Methods:** Five sets of RBCs coated with Ig by TA treatment (Pools A-E) were analyzed by flow cytometry (FC) to study the effect of storage on levels of RBC-bound Ig. With IRB approval, RBCs from 10 group O allogeneic donors were washed using Dulbecco's phosphate buffered saline (PBS) and pooled. Pooled RBCs were TA (Sigma-Aldrich, St. Louis MO) treated with a 25ug/mL TA concentration in PBS (37C for 15 minutes) and coated with heated (56C for 60 minutes) human IgG, IgA or IgM whole molecules, (Chrompure, Jackson ImmunoResearch Laboratories, West Grove, PA) at 0.02 mg/mL in PBS. Aliquots of the pooled uncoated RBCs and Ig coated RBC were sensitized with fluorescein isothiocyanate (FITC)-labeled anti-human IgG, or -IgA or -IgM (Life Technologies, Carlsbad, CA; Jackson ImmunoResearch Laboratories). 50,000 RBC acquired from each sample by the Becton Dickinson FACSCaliburTM (San Jose, CA) FC were analyzed. Ig coated RBCs stored at 2-6C in RBC storage solution (Immucor Inc, Norcross, GA) were analyzed via FC for the % RBC-bound Ig initially, during and up to 37 days of storage. **Results/Findings:** FC reports the percent of RBCs with bound Ig as %M2. During storage the % RBCs bound with IgG decreased an average of 29.7%, the IgA decreased an average of 15.5%, and the IgM decreased an average of 24.6%, but were well above the normal range, except pool C on day 27 (11.2%) which recovered to 47.9% on day 31. Pools B, C, D and E demonstrated cross-reactivity with the IgM coated RBCs with anti-IgG 0 to 4 days after coating, but was not detectable by day 7. **Conclusion:** With storage, the % Ig coated RBCs decreased, but retained sufficient % coating in all but IgG coated pool C (day 27). The naïive RBC Pool E on day 35 had above normal levels of RBC-bound IgG, making it unusable for a negative control. Sample age at coating did not influence the level of coating, but Pool E, coated on day 20, was the only pool with cross-reactivity 4 days after coating. Cross-reactivity was observed on the initial day of IgM coated RBC tested with anti-IgG in Pools B, C and D, and not observed on A. From this study the optimal RBC coating and storage parameters would be: coat on days 8 to 17, test 4 days after coating, and use pool up to day 31. This study establishes data for storage of Ig coated RBCs prepared by TA treatment which has not been reported in the literature.

Disclosure of Commercial Conflict of Interest

W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. A. Beres: Nothing to disclose; S. J. Nance: Nothing to disclose

Pool	Pool Age When Coated (days)	Time of Study (days)	IgG Max %M2	IgG Min %M2	IgA Max %M2	IgA Min %M2	IgM Max %M2	IgM Min %M2
A	8	37	95.4	83.4	92.4	89.6	95.9	92.5
B	13	27	69.1	41.3	78.7	60.4	91.7	54.7
C	13	31	62.6	11.2	81.5	67.1	92.2	76.7
D	17	28	99.0	65.0	99.4	77.1	98.9	56.2
E	20	35	95.9	72.8	99.2	88.3	98.9	79.9

SP324

Teaching the Recognition of Hemolysis by Controlling Antibody Mediated In-Vitro Hemolysis with Kodecytes

E H Perry[1] (holly.perry@aut.ac.nz), S M Henry[1]. *[1]School of Applied Sciences, Aukland University of Technology University, Auckland, New Zealand*

Background/Case Studies: In-vitro hemolysis has long been recognised as a predictor of in-vivo hemolysis of incompatible red cells. The skill of recognising in-vitro hemolysis may be deteriorating, as the modern red cell diluents contain compounds that inhibit antibody induced hemolysis, and it is difficult to mimic in a teaching setting. A novel approach to teaching antibody-induced hemolysis is to prepare kodecytes with controlled levels of Galili antigen. Galili antigen is found in non-primate mammals and New World monkeys and Galili kodecytes can be prepared with FSL-Galili constructs and human red cells. As all human sera contain naturally occurring complement-activating antibodies to the α-gal epitope of the Galili antigen, Galili kodecytes will be incompatible with human sera and will hemolyse. **Study Design/Methods:** Galili kodecytes were prepared with FSL-GALILI(GALa3GALb4GLCNb)-SA1-L1 (www.kodebiotech.com) at concentrations ranging from 10μg/ml to 100μg/ml. Different examples of fresh human sera were incubated at 37C for 60 minutes with Galili kodecytes of different concentrations, and with unmodified red cell controls. Samples were graded by eye for agglutination and degree of hemolysis. Supernatants were also diluted in Drabkin's solution and absorbance was measured at 540nm. Galili kodecytes producing partial and complete hemolysis were tested with consenting medical laboratory scientists, including practising blood bankers. Scientists were supplied with one serum sample and three red cell samples (comprising of a 100μg/ml Galili kodecyte, 20μg/ml Galili kodecyte and one unmodified red cell), and asked to conduct a 60-minute, 37C saline crossmatch. Scientists graded and recorded the serologic reactions observed. **Results/Findings:** Galili kodecytes at higher concentrations (40μg/ml-100μg/ml) showed complete hemolysis with all serums. Galili kodecytes at lower concentrations (10μg/ml -20 μg/ml) showed degrees of partial hemolysis. Unmodified red cells showed no hemolysis. Visual observations were confirmed by spectrophotometry. Most scientists correctly identified partial and full haemolysis of the two Galili kodecytes. Surprisingly, a small number failed to note the haemolysis, and graded the reactions as negative. All scientists correctly graded the unmodified red cell as negative. **Conclusion:** Galili kodecytes can provide a valuable tool for training the recognition of antibody mediated in-vitro hemolysis. Partial or complete hemolysis can be easily controlled to produce the desired reaction. The observation that some medical laboratory scientists failed to recognise the significance of hemolysis supports the need for such a training tool.

Disclosure of Commercial Conflict of Interest

S. M. Henry: KODE Biotech, Stocks or Bonds; E. H. Perry: Kode Biotech, Ownership or Partnership

Disclosure of Grants Conflict of Interest

S. M. Henry: KODE Biotech, Other; E. H. Perry: Kode Biotech, Grants or Research Support

SP325

Evaluation of Cold Agglutinin Titers by Gel Method

P A Arndt[1] (patricia.arndt@redcross.org), B Williams[2], G Garratty[1]. *[1]American Red Cross Blood Services, Pomona, CA, United States; [2]Kaiser Foundation Hospital Antioch, Antioch, CA, United States*

Background/Case Studies: Serologic diagnostic work-ups for cold agglutinin syndrome (CAS) always include a cold agglutinin titer. Dilutions of patients' plasma are incubated with test RBCs at 4C and examined for direct agglutination. Patients with CAS have IgM cold autoagglutinins with 4C titers that are often > 1000; thermal amplitude is ≥ 30C. It has been suggested that buffered gel cards can be used for testing cold agglutinins. This study started as part of an SBB student project, which was later extended with further testing. **Study Design/Methods:** Stored plasma from patients with CAS (n = 14) or fresh plasma from blood donors (n = 10) had titrations performed at 4C by tube and buffered gel (MicroTyping System) methods. Tube method: 0.1mL plasma dilutions (doubling dilutions prepared in phosphate buffered saline) + 1 drop 4% RBCs, incubate 1 hour 4C, centrifuge 20 seconds at RT, incubate tubes 10 minutes in melting ice before reading. Gel method: 25 µL plasma dilutions + 50 µL 0.8% RBCs in Diluent 2, incubate 1 hour 4C, centrifuge 10 minutes at room temperature, read. **Results/Findings:** Titration results of patients' samples are shown in the table. Patients' 4C antibody titers varied from 256-32,768 by tube versus 8-8,192 by gel. One patient had equivalent 4C agglutinin titers by tube and gel, 3 titers were 2 dilutions higher by tube than gel, and 10 titers were 3-8 dilutions higher by tube than gel. Donors' 4C antibody titers ranged from 1-8 by tube; no cold antibodies were

detected by gel (data not shown). **Conclusion:** Ten of 14 (71%) 4C titers performed by the gel method were significantly weaker than by the tube method (3 dilutions or more). Four of 14 (29%) patients with CAS would have been classified as having normal cold agglutinin titers (<64) by the gel method. Using the gel method for cold agglutinin titrations cannot be recommended based on the findings here.

Disclosure of Commercial Conflict of Interest

P. A. Arndt: Nothing to disclose; G. Garratty: Nothing to disclose; B. Williams: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. A. Arndt: Nothing to disclose; G. Garratty: Nothing to disclose; B. Williams: Nothing to disclose

Results (Titers)

Patient #	Tube Method	Gel Method	# of dilutions difference
1	32,768	8192	2
2	32,768	512	6
3	16,384	2048	3
4	16,384	512	5
5	8192	512	4
6	4096	1024	2
7	2048	2048	0
8	2048	128	4
9	2048	8	8
10	1024	128	3
11	1024	32	5
12	1024	8	7
13	512	128	2
14	256	8	5

SP326

Correlation Between Warm Reactive Autoantibodies and the Monocyte Monolayer Assay (MMA)

Y M Kim[1] (ymkim3015@yahoo.com). [1]Blood Bank, Omega Diagnostics, Shreveport, LA, United States

Background/Case Studies: The monocyte monolayer assay (MMA) measures the adherence and phagocytosis of sensitized red cells with peripheral monocytes. The result of MMA therefore may reflect to some extent the in vivo destruction of sensitized red blood cells (RBCs). However, serological and clinical features of patients with warm reactive autoantibodies do not always show strict correlation with in vivo activity. To evaluate the correlation of warm reactive autoantibody between the MMA result and the clinical data may add value in distinguishing whether or not the autoantibodies are harmful. **Study Design/Methods:** Blood samples from 20 patients with warm reactive autoantibodies were investigated using the MMA. RBCs used in the in vitro sensitization phase of the MMA were selected based on the patient's RBC phenotype and the specificity of the antibody(ies) each patient possessed in addition to the autoantibody. Phenotypically matched antigen negative RBCs were from two different sources; screening cells from commercial reagents and segments from donor RBC units. The Monocyte Index (MI) values > 5% were considered positive while MI values ≤ 5% were considered negative. **Results/Findings:** Of the 20 patients, only one patient yielded a positive MMA result with an MI of 30%. All other 19 patients demonstrated a negative MI value (<5%). The direct antiglobulin test (DAT) was reactive for IgG in all patients with warm autoantibodies but none displayed C3 reactivity. The DAT strength ranged from microscopically reactive to a strong 3+. The DAT was microscopically positive in 5 patients, 1-2+ in 9 patients, and 3+ in 6 patients. There was no correlation between serologically detectable warm reactive autoantibodies and the MMA or DAT. The 20 patients with warm reactive autoantibodies exhibited hemoglobin (Hb) level between 4.0 and 14.0 g/dl. There were 10 patients who had an Hb level between 6.1 and 8.0 but only one patient demonstrated a clinical significant MI value. There was also no correlation found between the MMA results and the patient's Hb value. **Conclusion:** These findings were only a preliminary study; the final conclusion demands more comparative studies between the clinical and serologic features of patients with warm autoantibodies and the MMA tests. The mechanism of red cell destruction depends on many factors. Furthermore, there are many variables that exist within the MMA which may have an effect on the outcome. These variables should be taken into account when interpreting MMA results.

Disclosure of Commercial Conflict of Interest

Y. M. Kim: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. M. Kim: Nothing to disclose

SP327

Immunoprotein Analysis: A New Tool for the Diagnosis of Direct Antiglobulin Test (DAT) Negative Autoimmune Hemolytic Anemia (AIHA)

E Bloch[1,2] (evgenia.bloch@mail.utoronto.ca), D Sakac[1], J M Pendergrast[2,3], C M Cserti-Gazdewich[2,3], D R Branch[1,2]. *[1]Research & Development, Canadian Blood Services, Toronto, ON, Canada; [2]Laboratory Medicine/Pathobiology, University of Toronto, Toronto, ON, Canada; [3]Blood Transfusion Laboratory, University Health Network, Toronto, ON, Canada*

Background/Case Studies: Autoimmune hemolytic anemia (AIHA) is categorized by the direct antiglobulin test (DAT), with a positive DAT indicative of hemolysis that is immune-dependent. A positive DAT is not necessarily indicative of its severity, although results may critically influence diagnosis and treatment decisions. Due to the short- and long-term adverse effects of corticosteroids or the permanence of splenectomy, such first and second line tactics in the care of AIHA cannot be casually deployed in cases where an immune basis for the hemolytic anemia remains uncertain. DAT negative (-) AIHA, which is characterized by hemolysis without detectable immunoglobulins on patient RBCs, poses a particular diagnostic challenge. It is hypothesized here that Western immunoblotting would be a more sensitive and precise measure of IgG binding to suspected DAT(-) AIHA patient's RBCs than conventional DAT. **Study Design/Methods:** RBC samples were provided by the Canadian Blood Services and University Health Network from consenting volunteers and patients. DATs were performed manually by conventional or gel microtube technique, with indirect antiglobulin tests (IAT) performed on either of these platforms or by automation (solid phase microtiter plates). RBC membrane ghosts free of hemoglobin were made from fresh RBCs, resolved on SDS-PAGE and blotted with anti-human IgG antibodies conjugated to horseradish peroxidase and visualized by enhanced chemiluminescence. Blots were analyzed using densitometry, and normalized relative to purified IgG and for loading using beta-actin. **Results/Findings:** Densitometry analysis of 28 samples from normal donors were used to establish a relative quantity of IgG (RQIgG) normal range. All normal RBCs were found to have very low levels of IgG detectable on their cells with a mean RQIgG of 0.177± 0.155. Five DAT(-) hemolyzing patients were tested. All had a negative eluate. Two patients, one hemolyzing from pyruvate kinase deficency and the other unknown, had RQIgG in the normal range. Three DAT(-) patients were found to have RQIgG greater than the mean +1SD, and as a result were classified as immune-dependent DAT (-) AIHA. One of those patients, a 57-year-old male, opted to undergo splenectomy only after this demonstration of increased IgG on the RBCs as the first indication of an immune mechanism of hemolysis. Following a 20-year history of acquired hemolysis, he has remained in a steroid-free remission, currently 8 months post-splenectomy. **Conclusion:** Western blot analysis is more sensitive than conventional test tube DAT analysis. It is proposed that patients who have hemolysis who are DAT and eluate (-), where there is a question as to whether the hemolysis is immune-mediated can be tested by a reference laboratory using this proposed protocol to ensure correct classification of their disease, and appropriate treatment.

Disclosure of Commercial Conflict of Interest

E. Bloch: Nothing to disclose; D. R. Branch: Nothing to disclose; C. M. Cserti-Gazdewich: Nothing to disclose; J. M. Pendergrast: No Answer; D. Sakac: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Bloch: Nothing to disclose; D. R. Branch: Grifols, Grants or Research Support, ViroCarb, Stocks or Bonds; C. M. Cserti-Gazdewich: Nothing to disclose; J. M. Pendergrast: No Answer; D. Sakac: Nothing to disclose

Red Blood Cells: Kinship Testing

SP328

Sibship Index Variability among Multiple Sib Pairs: A Confounding Issue for the Determination of Indirect Relationships

E A Kopitke[1] (liz@ifi-test.com), L M Johns[1], K A Reich[1], P W Boonlayangoor[1]. *[1]Independent Forensics, Lombard, IL, United States*

Background/Case Studies: Kinship testing laboratories are increasingly being asked to provide evidence of relatedness in alleged indirect

relationships, or to distinguish between several possible relationships (eg full- and/or half-sibship). Such examinations do not enjoy the allele sharing requirements of direct inheritance (ie parent-child) and are heavily reliant on statistical analyses. While large family pedigrees can clarify such questions, many cases are limited to a pair of participants. Laboratory findings in these cases rely on the Combined Sibling Index (CSI) or Half-Sibling Index (CHSI), statistics for which multiple studies have shown difficulty in establishing conclusive cutoff values as to the alleged relationship, yet there are still no consensus guidelines for these tests. **Study Design/Methods:** Full- and half-sibling pairswere studied, using in-house methods and frequency tables, to inform the laboratory's reporting policies based on the range of CSI and CHSI values observed. While previous studies examined only single pairs of siblings from either real or simulated shared parents, our data encompass several relatively large families and demonstrate the wide range of statistics obtainable for CSI and CHSI from even a small pool of alleles. DNA profiles were developed from 76 Caucasian volunteers comprising 20 nuclear families using standard methods (buccal swabs, 15-STR multiplex, capillary electrophoresis). Full- and half-sibship likelihood ratios (LRs) were calculated for 140 pairs of full siblings using Microsoft Excel. For comparison, data from 17 laboratory members produced 136 unrelated pairs. **Results/Findings:** CSIs for full siblings ranged from $7.4x10^8$ to a low of 0.02 (median = 5142). For 17 of the 140 pairs (12%), CHSI exceeded CSI. Analysis of a family of 13 full siblings (78 pairs) yielded CSI values of $9.6x10^6$ to 0.02 (median = 2270) and 7 pairs where CHSI>CSI—an acute demonstration of the limitations of pairwise analysis with indirect relationships. Unrelated pairs produced a maximum CSI of 19—three pairs with CSI>1—and a low of 8.0×10^{-9} (median = 2.2×10^{-4}). Eleven of the 136 unrelated pairs (8%) produced a CHSI>1, with a total observed range of a convincing 28 down to 1.2×10^{-4} (median = 0.057). **Conclusion:** The overlap between CSI and CHSI and between related and unrelated pairs argues strongly against making conclusive exclusions of relationship in favor of objectively reporting LR data, whatever their numeric value. Thus, conclusions as to the validity of the alleged relationship are left to clients and government agencies who are ill equipped to evaluate probabilistic statements. Additional efforts exploring the effect on LRs of mutated alleles and the potential for counts of two-allele-sharing-loci (TASL) to contribute to relatedness conclusions are ongoing.

Disclosure of Commercial Conflict of Interest

P. W. Boonlayangoor: Nothing to disclose; L. M. Johns: Nothing to disclose; E. A. Kopitke: Nothing to disclose; K. A. Reich: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. W. Boonlayangoor: Nothing to disclose; L. M. Johns: Nothing to disclose; E. A. Kopitke: Nothing to disclose; K. A. Reich: Nothing to disclose

Transfusion Practice: Blood Component Utilization Review

SP329

Trends in Red Cell Transfusion: An Analysis of 38,265 Transfusions

N Shehata[1,2] (nshehata@mtsinai.on.ca), A Forster[3,4], N Lawrence[3], D M Rothwell[3], K Wilson[3,4]. *[1]Departments of Medicine and Laboratory Medicine and Pathobiology, Mount Sinai Hospital, Toronto, ON, Canada; [2]Canadian Blood Services, Toronto, ON, Canada; [3]Ottawa Health Research Institute, Ottawa, ON, Canada; [4]Medicine, University of Ottawa, Ottawa, ON, Canada*

Background/Case Studies: There is limited population-based data on red blood cell (RBCs)transfusion rates and trends in transfusion. **Study Design/Methods:** This was a retrospective study using administrative data. RBCs at three university affiliated hospitals were categorized according to disease subtype, and the trends were assessed in transfusion therapy over a 5-year interval. Transfusion episodes were divided by admitting service. The mean transfusions/service and the proportion of individuals who received a transfusion were identified. **Results/Findings:** 38,265 transfusions in 244,013 hospitalizations were analyzed. The general internal medicine service (GIM) (14%), hematology oncology (HEM/ONC) (14%) and the orthopedic services (14%) accounted for the highest proportion of RBCs utilized followed by cardiac surgery (9%), cardiology (8%), and general surgery (8%). The highest mean units transfused/service, aside from stem cell transplantation (SCT) (mean 5.3 units, range 1.6, 16.3) and HEM/ONC (mean 3.7, range 2.4, 6.6), occurred under the vascular surgery (mean 4.2, range 2.4, 7.57), cardiology (mean 4.1, range 2.3, 7.0) cardiac surgery (mean 4.04. range 1.2, 8.4) and the intensive care unit (ICU) (mean 3.1, range 1.4, 5.8) services. The majority of patients under the SCT service were transfused RBCs (67%) (Table). 20 to 40% of patients admitted to cardiac surgery, HEM/ONC, trauma, vascular surgery and the ICU received RBCs (Table). Over the 5 year interval analyzed, there was no evidence of a downward trend in RBCs for medical or surgical services. **Conclusion:** A large proportion of patients under the services described receive RBCs. The utilization of red cells has remained unchanged over the 5-year interval analyzed. Future studies are required to determine whether blood conservation methods can reduce these rates.

Disclosure of Commercial Conflict of Interest

A. Forster: Nothing to disclose; N. Lawrence: Nothing to disclose; D. M. Rothwell: Nothing to disclose; N. Shehata: Nothing to disclose; K. Wilson: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Forster: Nothing to disclose; N. Lawrence: Nothing to disclose; D. M. Rothwell: Nothing to disclose; N. Shehata: Nothing to disclose; K. Wilson: Nothing to disclose

TABLE: Proportion of Patients Transfused RBCs According to Admitting Service

	Total number of patients	Proportion transfused RBCs N (%)
Stem Cell Transplantation	889	597 (67.1)
Cardiac Surgery	7289	2563 (35.2)
Cardiology	27313	2211 (8.1)
Family Medicine	3997	402 (10.1)
General Medicine	30372	4095 (13.5)
General Surgery	24239	2477 (10.2)
Gynecology	8212	479 (5.8)
Gynecology/Oncology	3765	622 (16.5)
Hematology/Oncology	10667	4126 (38.7)
Intensive Care	5484	1570 (28.6)
Nephrology	4195	700 (16.7)
Neurosurgery	9676	626 (6.4)
Obstetrics	41895	482 (1.2)
Orthopedics	23298	3852 (16.5)
Radiation Oncology	2596	483 (1.7)
Thoracic Surgery	3641	392 (13.3)
Trauma	1064	267 (25.1)
Urology	9864	828 (8.4)
Vascular Surgery	6212	1522 (24.5)

SP330

Transfusion Epidemiology of Hospitalized Patients in the US: Data from an Integrated Health Care Delivery System

N Roubinian[1,2] (nareg.roubinian@ucsfmedctr.org), G J Escobar[2], B E Swain[2], M N Gardner[2], V Liu[2], D J Triulzi[3], J Gottschall[4], Y Wu[7], S Kleinman[6], E L Murphy[1,5]. *[1]Laboratory Medicine, University of California, San Francisco, San Francisco, CA, United States; [2]Division of Research, Kaiser Permanente Northern California, Oakland, CA, United States; [3]Institute for Transfusion Medicine, Pittsburgh, PA, United States; [4]BloodCenter of Wisconsin, Milwaukee, WI, United States; [5]Blood Systems Research Institute, San Francisco, CA, United States; [6]Kleinman Biomedical Research, Victoria, BC, Canada; [7]Yale University, New Haven, CT, United States*

Background/Case Studies: Few studies have reported the epidemiology of blood product transfusion in the US except for studies of individual medical centers or Medicare claims data. The objective here was to characterize transfusion events using clinical data available in an integrated healthcare delivery system. **Study Design/Methods:** A retrospective cohort study was conducted of transfusion events using comprehensive electronic medical record data between 2008-2011 from 21 hospitals in Kaiser Permanente Northern California (KPNC). Admission diagnoses were categorized using ICD-9 codes, and severity of illness and comorbidity scores were calculated according to previously published methods. Logistic regression was used to predict the likelihood of red blood cell (RBC) transfusion through hospitalization. **Results/Findings:** The overall annual RBC utilization rate in KPNC (29.4 RBC units/1000 members) was significantly lower than that (48.8 RBC units/1000 US population) reported in the 2009 US National Blood Collection and Utilization Survey. The number of patients and units transfused decreased from 2008-2011 for RBCs (-7% and -5%) and increased for platelet (+21% and +25%) and plasma (+31% and +32%) products. The cohort of adult non-obstetric inpatients included 275,874 patients and 444,969 hospitalizations. 24-hour and overall inpatient RBC transfusion rates were 7% and 14%, respectively. Inpatient RBC transfusion rates

ranged from 9% to 18% across the 21 hospitals. After correcting for patient demographics, comorbidities, diagnoses, and severity of illness in our logistic model, the observed-to-expected rate of RBC transfusion ranged between 0.75 and 1.25 across hospitals. From 2008-2011, inpatient pre-transfusion hemoglobin (8.2 to 8.0 g/dL) and INR (2.6 to 2.4) values decreased, and pre-transfusion platelet levels (37 to 47 x 109/L) increased. In transfused patients, the average admission pre-transfusion and post-transfusion hemoglobin values were 9.6, 8.1, and 9.9 respectively. Pre-transfusion hemoglobin was lower than average in patients with gastrointestinal bleeding (7.9), malignancy (7.7), and infection (7.9) and higher than average in patients admitted for orthopedic surgery (8.4) and cardiovascular conditions (8.4). **Conclusion:** The overall incidence of RBC transfusion within KPNC is lower than in national reports and decreased during the study period; however, platelet and plasma transfusion incidences increased. From 2008-2011, inpatient pre-transfusion levels for hemoglobin and INR decreased and increased for platelets. Pre-transfusion hemoglobin levels were higher than average in patients admitted for orthopedic surgery and cardiovascular conditions. Variation in RBC transfusion rates across hospitals within KPNC persisted after adjustment for covariates, suggesting residual variation in physician practice.

Disclosure of Commercial Conflict of Interest

G. J. Escobar: Nothing to disclose; M. N. Gardner: Nothing to disclose; J. Gottschall: Cerus, Other, Hemonetics, Other, Fenwal, Other, Macopharma, Other; S. Kleinman: Nothing to disclose; V. Liu: Nothing to disclose; E. L. Murphy: Nothing to disclose; N. Roubinian: Nothing to disclose; B. E. Swain: Nothing to disclose; D. J. Triulzi: Nothing to disclose; Y. Wu: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. J. Escobar: Novartis, Grants or Research Support; M. N. Gardner: Novartis, Grants or Research Support; J. Gottschall: Nothing to disclose; S. Kleinman: Nothing to disclose; V. Liu: Novartis, Grants or Research Support; E. L. Murphy: Nothing to disclose; N. Roubinian: Nothing to disclose; B. E. Swain: Nothing to disclose; D. J. Triulzi: Nothing to disclose; Y. Wu: Nothing to disclose

SP331

Group O Rh (D) Negative RBCs: Utilization by Recipient Blood Type at a Level 1 Trauma Medical Center

L Rich[1] (rich.lindsy@mayo.edu), K S Badjie[1], S Saathoff[1], J R Stubbs[1], C M van Buskirk[1]. *[1]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Group O, Rh(D) negative (O negative) RBCs are typically in short supply due to high demand because of their preferred "universal donor" characteristics for transfusion recipients in certain settings (emergencies, exchange/neonatal, transplantation etc). As a result, many transfusion services (TS) have guidelines or policies for triaging demands to mitigate recurrent inventory shortages. An analysis of the emergent release of O negative stock RBCs in the hospital emergency room and helipad in 2010 and 2011 showed the average distribution of recipients as follows: 96% of transfused males were >18 years of age and 52% of females were over the age of 55. Of those transfused females, 79% were Rh (D) positive (O positive). Based on this data, our TS modified the emergency release policy to allow group switching certain patient profiles to O positive RBCs upfront to conserve inventory. The objective of this study is to evaluate the impact of our new emergency release policy on O negative RBC inventory and utilization as part of our effectiveness assessment. **Study Design/Methods:** Two years of daily O negative RBC utilization by recipient blood type prior to policy modification was analyzed and compared to post implementation utilization using an excel spreadsheet formulated to capture daily usage and percent distribution by recipient blood type. Data was obtained from the Daily Transfused O negative report from our Laboratory Information System (LIS). Each day, the total number of O negative RBCs transfused, filtered by recipient blood type was transferred to the excel database and transfusion utilization was automatically adjusted accordingly. **Results/Findings:** In the two years prior to the emergency release policy update, total O negative RBC utilization averaged 12 units per day, with 60% of those units transfused to O negative recipients. Average utilization by recipients of other blood groups were as follows; 15% O positive, 11% A positive, 6% A negative, 4% B positive, 3% B positive, and 1% AB positive. Post emergency policy update data showed daily O negative RBC utilization averages 11 units per day, with 64% transfused to O negative recipients. While average utilization by recipients of other blood groups revealed the following distribution; 18% O positive, 8% A positive, 3% A negative, 4% B positive, and 3% B negative. **Conclusion:** The average number of O negative RBCs transfused daily showed a minor decrease, as compared to pre-implementation data. However, the percentage of transfusions to O negative recipients increased by an average of 4%, and no adverse hemolytic events were reported as a result of the upfront group switching. The change in emergent transfusion policy had a smaller impact on the overall pattern than anticipated, likely related to the fact that O negative products are still required to be transfused across blood groups in other situations, such as neonatal and transplant patients.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; L. Rich: Nothing to disclose; S. Saathoff: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; L. Rich: Nothing to disclose; S. Saathoff: Nothing to disclose; J. R. Stubbs: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

Transfusion Practice: Managing Alloimmunization in Chronic Transfusion

SP332

Transfusion Therapy in Thalassemia Intermedia: Ten Years of Experience as a Tertiary Care University Hospital

A Z Al Riyami[1] (arwa.alriyami@gmail.com), S A Al-Mahrooqi[1], S Al Hinai[1], S Al Hosni[1], N Karman[1], F Wasim[1], Z Jabeen[1], F Zia[1], A Al Madhani[2], S Daar[1]. *[1]Department of Hematology, Sultan Qaboos University Hospital, Muscat, Oman; [2]Department of Medicine, Sohar Hospital, Sohar, Oman*

Background/Case Studies: Thalassemia intermedia (TI) encompasses a wide clinical spectrum, ranging from mild anemia to a more severe phenotype presenting earlier in life. Many patients maintain their hemoglobin level without blood transfusions. However, there are occasional circumstances when transfusion therapy is needed in non transfusion dependent patients; including infections, pregnancy and preoperative settings. Transfusion is associated with its own complications including iron overload and alloimmunization. Alloimmunization appears to be common in thalassemic patients when they are started on blood transfusion after the age of 3 years. The transfusion practice and alloimmunization rate in transfusion independent TI patients in our centre is examined here. **Study Design/Methods:** A retrospective review of all transfusion independent TI patients attending this hospital over the last 10 years was performed. Patients' clinical details and transfusion indications were reviewed. Blood bank records were examined for the patient's blood group, red blood cell phenotype, detectable antibodies, and number of units transfused. **Results/Findings:** A total of 31 patients (10 males and 21 females), with a median age of 29 years (13-59) and median baseline hemoglobin of 9.1 g/dl (6.8-12.2) were reviewed. Blood transfusion was required in 23 patients at some stage of their followup. Indications for blood transfusion included pregnancy (21 pregnancies in 11 females), splenectomy (10 patients) and low hemoglobin level (8 patients). Iron overload was present in 22 patients. Among this cohort, the most common blood groups are O+ (12/31) followed by B+ (6/31) and A+ (5/31). Rh phenotype was available in 24 patients with the commonest being R1R1 (9/31), R1r (6/31), and R1R2 (4/31). There are 4 patients with multiple antibodies including anti-Jkb, anti-E, anti-c, anti-E, anti-K,anti-Jkb, anti-Kpa, anti-S and, anti-Fyb, all of which were developed prior to 2009, when our current policy of issuing phenotype matched blood for these patients was not possible due to limited blood inventory for extended cross match. Since 2009, total of 335 units were transfused with only one antibody developed after transfusing non phenotype matched red blood cells, due to the lack of the needed matched blood. Full red blood cell phenotype is currently available for 21 patients. **Conclusion:** The most common indication for blood transfusion in our non transfusion dependent TI patients was pregnancy. The strategy here of providing phenotype matched red blood cells has successfully reduced the rate of alloimmunization in these patients. Transfusion support of this group of patients necessitates the availability of needed expertise and blood bank facilities.

Disclosure of Commercial Conflict of Interest

S. Al Hinai: Nothing to disclose; S. Al Hosni: Nothing to disclose; A. Al Madhani: Nothing to disclose; A. Z. Al Riyami: Nothing to disclose; S. A. Al-Mahrooqi: Nothing to disclose; S. Daar: Nothing to disclose; Z. Jabeen: Nothing to disclose; N. Karman: Nothing to disclose; F. Wasim: Nothing to disclose; F. Zia: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Al Hinai: Nothing to disclose; S. Al Hosni: Nothing to disclose; A. Al Madhani: Nothing to disclose; A. Z. Al Riyami: Nothing to disclose; S. A. Al-Mahrooqi: Nothing to disclose; S. Daar: Nothing to disclose; Z. Jabeen: Nothing to disclose; N. Karman: Nothing to disclose; F. Wasim: Nothing to disclose; F. Zia: Nothing to disclose

SP333

Providing Extended Phenotyped Red Cells for Patients with Myeloproliferative Disorders

N Rubien[1] (nrubien@lifespan.org), L D Roberts[1,2], J D Sweeney[2,1].
[1]Transfusion Service, Rhode Island Hospital, Providence, RI, United States; [2]Transfusion Service, The Miriam Hospital, Providence, RI, United States

Background/Case Studies: Patients with hereditary anemias requiring chronic red cell transfusion are known to be at risk for alloimmunization and commonly receive extended phenotype matched red cells pre-emptively, mostly for Rhesus antigens and K-1. This practice is also employed in some acquired anemias, such as autoimmune anemias but has not generally been applied to other populations receiving recurrent transfusions such as patients with myelodysplastic anemias. A recent report indicates that such a practice may successfully prevent alloimmunization. Blood banks tend to be resistance to pre-emptive phenotyping for such patients partly because manually screening units with liquid reagent is time consuming and costly due to the cost of reagent and inventory management can present an additional challenge. However, availability of automated technology can abrogate the former objection. This study was conducted to capture the resource and cost savings associated with manual versus automated extended phenotype matching. **Study Design/Methods:** The predicate method for phenotyping red blood cells was the manual tube testing and the test method the automated ID-MTS gel technology. Extended phenotyping was performed for Rh antigens. Twenty donor samples from the Rhode Island Blood Center, and 19 patient samples from Rhode Island Hospital for the period between 2010-2011 were selected and phenotyped using both the predicate and test method. Associated labor and reagent costs were calculated for both methods. **Results/Findings:** 39/39 (100%) patient and donor samples showed concordance with serological findings, validating the approach. Two patient samples were omitted from the study due to recent transfusion with the observation of mixed field results on the gel card. For an average of 20 units screened per week, a net cost savings of $4,160 in technologist time and $4,891 in disposables would be realized annually using the automated versus the manual method. **Conclusion:** The results suggest that the automated ID-MTS technology is a practical method that can be implemented to perform extended red cell phenotyping and potentially allow the provision of extended phenotyped red cells to non-traditional patient populations requiring multiple transfusions.

Disclosure of Commercial Conflict of Interest

L. D. Roberts: Aeroscout, Other; N. Rubien: Johnson and Johnson, Other; J. D. Sweeney: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. D. Roberts: Nothing to disclose; N. Rubien: Nothing to disclose; J. D. Sweeney: Bio-Rad, Grants or Research Support

Transfusion Practice: Monitoring Blood Component Indication and Outcomes

SP334

A Program to Interdict Prescriptions for Fresh Frozen Plasma (FFP) Can Result in a Large Decrease in FFP Transfusion

J Mazer[2] (jmazer@lifespan.org), A Levinson[2], G Carino[2], J D Sweeney[1].
[1]Transfusion Service, The Miriam Hospital, Providence, RI, United States; [2]Critical Care Medicine, The Miriam Hospital, Providence, RI, United States

Background/Case Studies: Plasma tranfusions are commonly administered as an exchange fluid for therapeutic plasma exchange in patients with thrombotic thrombocytopenic purpura (TTP) or to correct clotting factor deficiencies in patients with coagulopathy. In the latter scenario, these transfuisons can be given prophylactically prior to an invasive procedure or in the management of patients with active bleeding. Despite decades of practice, there is limited or no data for any efficacy of plasma transfusion in these clinical situations and on-going uncertainty regarding dosage. Based on the absense of any evidence of efficacy and concerns regarding safety, this institution initiated a program to discourage prophylactic FFP transfusion when the INR was less than 2.0 (3.0 in the case of central line insertaion) and to encourage the use of intravenous vitamin K for warfarin reversal, even in bleeding patients who were hemodynamically stable in the absense of intracranial hemorrhage. **Study Design/Methods:** The program was initiated in 2009. All requests for FFP were screened by technologists . The prescribing physicians were contacted regarding appriopriatness (INR < 2.0) or the potential for use of intravenous vitamin K (any INR). This resulted in either cancellation of the order or dose adjustment in some cases, FFP use was tabulated per year over a 4-year period (2009-2012) and any FFP used as an exchange fluid in therapeutic apheresis separated from non-apheresis FFP. The acquisition cost of this FFP was calculated based on a unit price of $75. Total discharges were recorded for each year and the data represented as units FFP/1000 discharges to allow for inter-year comparison. **Results/Findings:** The data are shown in the table. A reduction of approximately 80% of non-apheresis FFP was achieved by year 4 with an associated acquisiton cost avoidance of $80,000. There were no reports or complaints of unexpected bleeding or morbidity reported to the blood bank in association with the cancellation of FFP for prophylactic procedures and warfarin reversal to an INR of approximately 2.0 was achieved typically 5-10 hours after IV administration. **Conclusion:** The data are consistent with the position that there is considerable unnecessary use of FFP and that changes in practice can be achieved by an intervention with the prescribing physician. There is no evidence that such an intervention results in unexpected bleeding.

Disclosure of Commercial Conflict of Interest

G. Carino: Nothing to disclose; A. Levinson: Nothing to disclose; J. Mazer: Nothing to disclose; J. D. Sweeney: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Carino: Nothing to disclose; A. Levinson: Nothing to disclose; J. Mazer: Nothing to disclose; J. D. Sweeney: Bio-Rad, Grants or Research Support

Year	Total # units Non Apheresis FFP	Units FFP transfused/1000 Discharges	Acquisition Cost
2009	1,502	94	105,140
2010	1,104	73	81,696
2011	722	49	53,428
2012	322	21	24,568

SP335

Examination of a Real-Time Clinical Decision Support System for Cryoprecipitate Orders

R A Collins[1] (collinsra2@upmc.edu), J H Waters[2], D J Triulzi[1,3], M H Yazer[1,3]. *[1]Pathology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [2]Anesthesiology, University of Pittsburgh Medical Center, Pittsburgh, PA, United States; [3]Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: Ensuring physician compliance with transfusion thresholds is a challenge. This study evaluated physician ordering practice after the implementation of real time clinical decision support software for cryoprecipitate (Cryo) transfusion orders. **Study Design/Methods:** Uniform Cryo ordering criteria was implemented at 12 hospitals in a regional healthcare system through the computerized physician order entry interface. The selectable ordering criteria included: Active Bleeding, Fibrinogen ≤ 100 mg/dl pre-procedure, and "Other". An on-screen alert appeared if the patient's most recent fibrinogen level exceeded the selected threshold for the second criterion, and it appeared for all orders placed under the "Other" indication regardless of the recipient's antecedent fibrinogen value. Orders placed in the operating room did not generate an alert. **Results/Findings:** There were 106 Cryo orders placed across the healthcare system in the 4 months following the implementation of the computerized ordering criteria (range: 23-32 orders per month), and 53 alerts were generated (50% of orders); 13% of the alerted orders were canceled. The numbers of orders and alerts have increased over time; however, the percentage of cancelled orders has been decreasing, with zero alerted orders cancelled over the most recent 2 months. Active Bleeding was the most frequently selected indication (45% of all orders), with an average antecedent fibrinogen level of 168 ± 72 mg/dl. The "Other" indication was the second most frequently selected criterion (35% of all orders) and was used frequently in patients who were being monitored with thromboelastography (TEG) or who had undefined "coagulopathy;" the mean antecedent fibrinogen level was 194 ±

108 mg/dl. Orders placed with the Fibrinogen ≤100 mg/dl indication were the least frequent (19% of all orders), but triggered alerts 63% of the time. These patients had an average antecedent fibrinogen level of 109 ± 22 mg/dl. The majority of Cryo transfusion orders were placed by attending physicians (63%). There was a trend towards attending physicians having a lower alert rate (39%) compared to resident physicians (59%, p = 0.06). Cryo transfusions were most frequently ordered by surgeons (53%). Hematology/Oncology physicians were the most likely to generate alerts (83%, p = 0.04), and were also the most likely to override them (p = 0.44). **Conclusion:** Ordering alerts provided by clinical decision support software can support clinical patient management and reduce non-evidence based transfusions. However, these data show that alerts alone are not sufficient to prevent a substantial proportion of inappropriate Cryo transfusions. Further interventions including education and "just-in-time" information will be necessary to further reduce the rate of non-compliant Cryo ordering.

Disclosure of Commercial Conflict of Interest

R. A. Collins: Nothing to disclose; D. J. Triulzi: Nothing to disclose; J. H. Waters: Nothing to disclose; M. H. Yazer: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. A. Collins: Nothing to disclose; D. J. Triulzi: Nothing to disclose; J. H. Waters: Haemonetics, Grants or Research Support, Coramed, Grants or Research Support; M. H. Yazer: Nothing to disclose

SP336

Effectiveness of a Real-Time Clinical Decision Support System for Computerized Physician Order Entry of Red Blood Cells (RBCs)

B K McWilliams[1] (bmac@zoominternet.net), J H Waters[2], D J Triulzi[1,2], M H Yazer[1,2]. *[1]The Institute for Transfusion Medicine, Pittsburgh, PA, United States; [2]University of Pittsburgh School of Medicine, Pittsburgh, PA, United States*

Background/Case Studies: The effect was investigated of implementing a requirement for prescribers to select an indication for red blood cell (RBC) transfusion when ordering RBCs using a computerized physician order entry (CPOE) system. **Study Design/Methods:** In a regional health care system consisting of 11 hospitals using a common CPOE, data on the number of RBC orders and alerts that were generated were collected over a 4-month period. During this period, an on-screen alert was triggered if the antecedent Hb was >8 g/dl. The CPOE was then modified to require the prescriber to select one of the following pre-populated indications for the RBC transfusion: Hb ≤ 10 Patient with Acute Cardiac Ischemia; Hb ≤ 8.0 Non-ICU Patient with Anemia, Hb ≤ 7.5 Stable ICU patient; Acute Bleeding with BP Instability; Surgical Blood Loss Anticipated; and Other. An on-screen alert was generated when the recipient's most recent Hb value exceeded the corresponding Hb value for the selected indication; the alerts were thus "adapted" to the selected indication for transfusion. For the Other category an alert was generated if the antecedent Hb was > 8.0 g/dl. Six months of data post-implementation of the ordering criteria was collected. **Results/Findings:** In the 4-month period before the ordering criteria were implemented, 69.7% of the RBC orders generated an alert; in the 6-month period thereafter the alert rate was significantly lower at 49.1% (p < 0.0001). The % of heeded alerts increased during the adaptive alert period (10.8% versus 13.5%, respectively, p < 0.0001). "Other" was the most commonly selected transfusion indication (29.8%), followed by Hgb ≤ 8.0 Non-ICU Patient with Anemia (28.6%), and Hgb ≤ 7.5 Stable ICU Patient (20.3%). The other four RBC indications had significantly lower frequencies, each ≤7% of the total number of alerts. Of the all the orders placed, the Other category had the highest percentage of patients whose antecedent Hb value was above the threshold value (79.6%), followed by Hb ≤ 7.5 Stable ICU Patient (34.8%) and Hb ≤ 8.0 Non-ICU Patient with Anemia (28.2%). There was also variation between the RBC indication categories with respect to mean antecedent Hb levels. Patients on whom RBCs were ordered using the Surgical Blood Loss Anticipated category had the highest average antecedent Hb of 9.63 +/- 2.34g/dl, which is in line with the fact that these units were ordered preemptively and not in response to anemia. The indications with the lowest mean recipient antecedent Hb values were Hb ≤ 8.0 Non-ICU Patient with Anemia (7.71 +/– 1.10 g/dl) and Hb ≤ 7.5 Stable ICU Patients (7.39 +/– 1.22 g/dl). **Conclusion:** Electronic interventions improve compliance with RBC guidelines but as implemented are not sufficient to completely curtail all non-evidence based orders.

Disclosure of Commercial Conflict of Interest

B. K. McWilliams: Nothing to disclose; D. J. Triulzi: Nothing to disclose; J. H. Waters: Nothing to disclose; M. H. Yazer: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. K. McWilliams: Nothing to disclose; D. J. Triulzi: Nothing to disclose; J. H. Waters: Haemonetics,Grants or Research Support, Coramed, Grants or Research Support; M. H. Yazer: Nothing to disclose

SP337

Plasma Product Transfusion Practices in Patients with Factor XII Deficiency

A M Svensson[1] (svenssa@ccf.org), S Bakdash[2]. *[1]Cleveland Clinic Fellowship in Blood Banking and Transfusion Medicine, Cleveland Clinic, Cleveland, OH, United States; [2]Transfusion Medicine, Cleveland Clinic, Cleveland, OH, United States*

Background/Case Studies: Factor XII (FXII) deficiency can prolong the activated partial thromboplastin time (aPTT), but is not a risk factor for bleeding. Transfusion of plasma products is not indicated to "correct" a prolonged aPTT when FXII deficiency is the only coagulation abnormality. Anecdotal observation suggested that patients with FXII deficiency may occasionally receive plasma transfusions with no apparent indication other than a prolonged aPTT. This study retrospectively evaluated plasma product transfusion practices in FXII deficient patients identified at our institution between January 1, 2008-December 31 2012. **Study Design/Methods:** Clot-based FXII assays are performed at this institution. After excluding reference samples, 292 patients with at least one FXII result below normal during the study period were identified. The subset of patients transfused any plasma product was selected for chart review. Patients with presumed FXII deficiency, defined as a below normal FXII result with no other apparent explanation such as liver disease, consumptive coagulopathy, lupus anticoagulant, etc, were identified by review of laboratory data and physician notes in the electronic medical record. Further evaluation was performed with respect to timing of laboratory diagnosis, indication for transfusion, and transfusion reactions (TRxNs). **Results/Findings:** Fifty-seven patients with low FXII received a plasma product transfusion. Of these, 18 had presumed FXII deficiency and 39 had a low FXII for other reasons. One of 18 patients with presumed FXII deficiency was transfused cryoprecipitate (cryo) only; the remaining 17 patients had 20 episodes of plasma transfusion, including one with both plasma and cryo. FXII results were available prior to transfusion in 11/20 episodes, but only specifically noted in the chart in two cases. By chart review, plasma transfusion was indicated in 9/20 episodes (warfarin reversal, acute bleeding, etc) and not indicated in 6/20. The indication for transfusion could not be determined with certainty on retrospective review of the remaining five episodes. The cryo-only transfusion was not indicated. A total of 22 units of plasma and a cryo pool were transfused with no readily apparent indication for transfusion. Two TRxNs were reported, both mild allergic. One occurred in a clinically indicated transfusion; the other was associated with a one-unit plasma transfusion to correct an elevated aPTT prior to evaluation of etiology. **Conclusion:** On retrospective chart review, 10 of 18 patients with presumed FXII deficiency received plasma products with no readily apparent indication, other than correction of a prolonged aPTT, in the 5-year period 2008-2012. One TRxN was reported in this group. Raising awareness of FXII deficiency as a possible cause of isolated prolonged aPTT, and emphasizing the importance of further evaluation of abnormal laboratory findings prior to transfusion, may help avoid unnecessary plasma product transfusion.

Disclosure of Commercial Conflict of Interest

S. Bakdash: Nothing to disclose; A. M. Svensson: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Bakdash: Nothing to disclose; A. M. Svensson: Nothing to disclose

SP338

Assessing Allogenic Perioperative Blood Transfusions for Total Hip Replacement Surgery Patients in the Saskatoon Health Region

R Schaan[1] (richard.schaan@usask.ca), C H Neveling[1], K L Dallas[2]. *[1]Anesthesia, University of Saskatchewan, Saskatoon, SK, Canada; [2]Laboratory Medicine & Pathology, University of Saskatchewan, Saskatoon, SK, Canada*

Background/Case Studies: Patients having elective total hip replacement (THR) surgery can have considerable blood loss and are therefore at risk of receiving allogenic blood transfusions. These transfusions are associated with potential risks to the patient and also pose a financial burden on the health care system. Preoperative subclinical anemia (PSCA) has been implicated as an independent risk factor for increased perioperative morbidity and

mortality and an increased risk of receiving transfusions. Some studies have suggested that appropriate preoperative diagnosis and treatment of anemia may result in less postoperative transfusions. However,locally it was largely unknown if PSCA patients were receiving more transfusions or were already taking anemia-related therapy. Therefore, the primary goal of this study was to assess whether patients with PSCA undergoing elective THR surgery here required more perioperative blood transfusions than non-anemic patients. A secondary goal was to assess how many patients were already receiving anemia-related therapy. **Study Design/Methods:** A retrospective chart review was performed for THR surgeries from 2011-2012. Ethics approval was granted by the Ethics Committee at the University of Saskatchewan. 181 charts were reviewed as determined by sample size calculations. Anemia cut-offs were <135 g/L for males and <110 g/L for females. It was noted if patients were taking potential anemia therapy preoperatively. **Results/Findings:** 172 patients met inclusion criteria. 11 had PSCA with 7 of those receiving postoperative blood transfusions (63.6%), while 24 of the 161 non-anemic patients received a transfusion (14.9%) ($p < 0.0001$). 50% of all patients and 4/11 (36%) PSCA patients were already taking some form of anemia therapy preoperatively; 3/4 patients were transfused. The average postoperative hemoglobin in those receiving transfusions was 82.4 g/L (range 68-101 g/L). Increasing age was determined to be a risk factor for transfusion ($p < 0.0001$). 10/31 transfused patients had a hemoglobin value <80 g/L, and only one was <70 g/L. Only 3/31 patients had documented symptoms for requiring a transfusion. One of the patients transfused developed a transfusion reaction. **Conclusion:** Patients with PSCA had higher rates of postoperative allogenic blood transfusion than non-anemic patients. Evidence suggests that transfusions were also age-related. Almost half of the PSCA patients were already taking anemia-related therapy preoperatively. The majority of the transfused patients had hemoglobin values >80 g/L and were asymptomatic, potentially placing these individuals at risk of transfusion related reactions with perhaps little benefit. The current postoperative transfusion-related practice for THR patients in our health region yields a transfusion related financial cost of >$200K annually.

Disclosure of Commercial Conflict of Interest

K. L. Dallas: Nothing to disclose; C. H. Neveling: Nothing to disclose; R. Schaan: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. L. Dallas: Nothing to disclose; C. H. Neveling: Nothing to disclose; R. Schaan: Nothing to disclose

SP339

Cost Tradeoffs of Low-, Medium-, and High Dose Platelet Transfusions

S Assmann[1] (SAssmann@neriscience.com), D J Triulzi[2], J McCullough[3], R W Harrison[1], S J Slichter[4]. *[1]New England Research Institutes, Watertown, MA, United States; [2]Institute for Transfusion Medicine, University of Pittsburgh, Pittsburgh, PA, United States; [3]Department of Laboratory Medicine & Pathology, University of Minnesota, Minneapolis, MN, United States; [4]Division of Research, Puget Sound Blood Center, Seattle, WA, United States*

Background/Case Studies: The Platelet Dose Study (PLADO) randomized thrombocytopenic patients undergoing stem cell transplant or chemotherapy to low dose (LD), medium dose (MD), or high dose (HD) prophylactic platelet transfusions. PLADO found no differences in bleeding outcomes between the three groups. Total platelets received during the study were lowest in LD patients and highest in HD patients, but LD patients required more transfusion episodes than the other two groups. Thus, there may be cost tradeoffs between the three dosing strategies. These cost tradeoffs may differ between patients supported by apheresis platelets and patients supported by whole blood derived (WBD) platelets. **Study Design/Methods:** Analyses were restricted to adults supported exclusively with either apheresis or WBD pooled platelets. Dose groups were compared pairwise on number of transfusion episodes, and on total number of split and whole apheresis units or WBD units provided. Each apheresis transfusion episode was assumed to cost $80 in labor. Each WBD transfusion episode was assumed to cost $100 in labor, including pooling. A variety of costs were assumed for whole and split apheresis units and for WBD units. Dose groups were compared pairwise on total cost of platelet support for each set of assumptions. Kruskal-Wallis p-values < 0.017 were considered statistically significant. **Results/Findings:** Dose groups differed in the number of transfusion episodes and in the total number of whole apheresis, split apheresis, and WBD units per patient. For hospitals that can split apheresis units, HD apheresis support was more costly than LD or MD, which did not differ significantly. For hospitals that do not split units, MD apheresis support was less costly than LD or HD, which did not differ significantly. HD WBD support was more costly than MD, while LD WBD support did not differ significantly from MD or HD. **Conclusion:** For WBD support, HD may be more costly than MD. For apheresis support, an LD strategy can reduce the number of apheresis collections needed, but cost tradeoffs depend heavily on cost assumptions.

Disclosure of Commercial Conflict of Interest

S. Assmann: Nothing to disclose; R. W. Harrison: Nothing to disclose; J. McCullough: Nothing to disclose; S. J. Slichter: Nothing to disclose; D. J. Triulzi: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Assmann: Nothing to disclose; R. W. Harrison: Nothing to disclose; J. McCullough: Nothing to disclose; S. J. Slichter: Terumo BCT, Grants or Research Support; D. J. Triulzi: Nothing to disclose

Dosing Strategy Comparisons

		Medians		Pairwise comparisons with P < 0.017
APHERESIS	LD (N = 187)	MD (N = 209)	HD (N = 235)	
Episodes	4	3	3	LD/MD, LD/HD
Whole Units (≥$3x10^{11}$ platelets)	0	3	4	LD/MD, MD/HD, LD/HD
Split Units (<$3x10^{11}$ platelets)	4	0	0	LD/MD, LD/HD
Total costs assuming $80/episode, $350/whole unit, $195/split unit	$1255	$1290	$1720	MD/HD, LD/HD
Total costs assuming $80/episode, $350/whole or split unit	$1720	$1290	$1910	LD/MD, MD/HD
Total costs assuming $80/episode, $625/whole unit, $332.50/split unit	$1943	$2115	$2820	MD/HD, LD/HD
Total costs assuming $80/episode, $625/whole or split unit	$2820	$2115	$3285	LD/MD, MD/HD
WBD	LD (N = 85)	MD (N = 75)	HD (N = 64)	
Episodes	5	2	2	LD/MD, LD/HD
Units	15	12	22	MD/HD, LD/HD
Total costs assuming $100/episode, $50/WBD unit	$1250	$900	$1300	MD/HD
Total costs assuming $100/episode, $75/WBD unit	$1625	$1200	$1850	MD/HD

Transfusion Practice: Transfusion Cases

SP340

Granulocyte Transfusions in an Outpatient Setting

F M Aung[1] (fmaung@mdanderson.org), B Lichtiger[1], V M Reddy[1], N Clancy[1], G Griffin[1], F Martinez[1], E J Freireich[2]. *[1]Laboratory Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX, United States; [2]Adult Leukemia Research Program, The University of Texas MD Anderson Cancer Center, Houston, TX, United States*

Background/Case Studies: Outpatient transfusions of blood components (red blood cells [RBCs], fresh frozen plasma [FFPs] and Platelets [PLTs]) are performed in many health care facilities and allow patients to schedule repeated infusions at convenient times for the patient as well as the treating facility. In this report, the experience is described of Granulocyte transfusions (GTs) to patients who continue to receive them as outpatients. This is for a specific patient population with hematologic malignancies, in different stages of their treatment. This may be the first report of patients receiving GTs in an outpatient setting. **Study Design/Methods:** Transfusion policies relating to donor selection and infusion of blood components were followed and met all institutional and regulatory requirements. All leukapheresis were performed at the MDACC Blood Donor Center. The donor recruitment, leukapheresis, and infusion processes was identical for both inpatients and outpatients and all leukapheresis products were irradiated prior to transfusion. This review was approved by the Institutional Review Board. **Results/Findings:** From January 2009-January 2012, 193 Granulocyte concentrates (GCs) were transfused to 38 patients (32 [84%] M: 6 F[16%]), median of 4 GCs (1-19) with a diagnosis of Acute Myeloid Leukemia/MDS in 27 (71%), ALL/CLL in 6 (16%), CML/CMML in 4 (11%) and Myeloma in 1(2%) patient. Twelve (31.5%) patients received a single GC, two (5.2%)/2 GCs, two (5.2%)/3 GCs, five (13.2%)/4 GCs, three (7.9%)/ 5 GCs, three (7.9%) patients/6 GCs, one (2.5%)/ 7 GCs, two (5.6%)/8 GCs, two (5.6%)/9 GCs, one (2.5%)/11 GCs, two (5.2%)/12 GCs, two (5.2%)/14 GCs and one (2.5%)/19 GCs. Each transfusion was infused over 2-4 hours with no adverse transfusion reactions observed or reported. All patients were pre-medicated with acetaminophen and benadryl. Upon completion of the transfusion and documentation of posttransfusion vital signs, the patients were discharged in stable condition. **Conclusion:** The outpatient transfusion center was implemented here in 1999, is staffed by fully trained health care personnel, and operates from 7 AM to 11 PM daily including weekends and holidays. The service has become a routine operational aspect of the overall outpatient support services. Of the institutional routine blood components (RBC's, FFPs and PLTs) 30% are administered in this outpatient setting. It is believed that Granulocyte transfusions in the proper clinical setting results in the desired clinical effect leading to cost savings from less usage of inpatient resources. One obvious advantage is the use of a centralized information system to track patients and their results and a dedicated team of blood donor and blood bank staff members. The utility, effectiveness and cost savings of Granulocyte transfusions in an outpatient setting has yet to be fully recognized by primary care teams serving oncology patients.

Disclosure of Commercial Conflict of Interest

F. M. Aung: Nothing to disclose; N. Clancy: No Answer; E. J. Freireich: Nothing to disclose; G. Griffin: Nothing to disclose; B. Lichtiger: Nothing to disclose; F. Martinez: Nothing to disclose; V. M. Reddy: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. M. Aung: Nothing to disclose; N. Clancy: No Answer; E. J. Freireich: Nothing to disclose; G. Griffin: Nothing to disclose; B. Lichtiger: Nothing to disclose; F. Martinez: Nothing to disclose; V. M. Reddy: Nothing to disclose

SP341

Comparison of Transfusion Requirements in High Risk Myelodysplastic Syndromes (HR-MDS) and Secondary Acute Myeloid Leukemia (sAML) Treated with Azacitidine or Intensive Chemotherapy

E Alonso Nogués[1] (ealonso@bst.cat), M Garcia Caro[2], B Xicoy[2], M Morgades[2], R Línio[1], L San Miguel[2], A Gallardo[1], M Prieto[1], J Grifols[1], J Ribera[2]. *[1]Banc de Sang i Teixits. Hospital Germans Trias i Pujol, Badalona, Spain; [2]Institut Català d'Oncologia and Josep Carreras Research Institut Hospital Germans Trias i Pujol, Badalona, Spain*

Background/Case Studies: The treatment goal in patients with high-risk myelodysplastic syndromes (HR-MDS) and secondary acute myeloid leukemia (sAML) is to modify the natural history of disease, to increase overall survival and to reduce the progression to acute leukemia in patients with HR-MDS. Several patients are not candidates for intensive chemotherapy and are treated with azacitidine (AZA), a drug with favorable toxicity profile compared with intensive chemotherapy. The aim of the study was to compare the transfusion requirements in HR-MDS and sAML treated with intensive chemotherapy or AZA in a single institution. **Study Design/Methods:** From August 2009-December 2012 28 patients diagnosed with HR-MDS/sAML were analyzed according to 2008 World Health Organization (WHO) classifications. Patients who underwent hematopoietic stem cell transplantation (allo-HSCT) were censored at the time of transplant. Patients with good prognostic cytogenetics were treated with intensive chemotherapy (ICE schedule: idarubicin, cytarabine and etoposide, one or two cycles) and those with poor cytogenetics were treated with AZA (six cycles). Patients not candidates for allo-HSCT were treated with maintenance schedule of AZA. Red blood cell (RBC) and platelet transfusion requirements were recorded during treatment in all patients. Demographic and clinical characteristics of patients were analyzed according to the Mann-Whitney test for quantitative variables and the chi-square or Fisher test, as appropriate, for qualitative variables. Finally, to compare the mean transfusion requirements the Mann-Whitney test was also used. **Results/Findings:** 29 patients were included. The median age was 65 (32-83). Twenty-six patients had ECOG score 0-1, and 2 an ECOG $\geq$ 2. HR-MDS patient stratification according to WHO classification was: RAEB-I 4 (13.8%), RAEB-II 10 (34.5%), secondary AML 8 (27.6%), RCMD 3 (10.3%), CMML-I 1(3.4%), CMML-II 1 (3.4%), MDS with isolated del(5q) 1 (3.4%), and RARS 1 (3.4%). Median (range) hemoglobin (Hb) and platelet levels in the whole series were 87 g/L (55-109) and 68 x 109/L (7-319), respectively. Thirteen patients (44.8%) were treated with ICE and 16 (55.2%) were treated with AZA receiving a median of 5 (range,1-5) cycles. Both treatment groups were comparable in terms of age, gender, ECOG score, WHO classification, hb, and platelet count. The mean (SD) of red blood cells transfusions received in the AZA group was 19.94 (SD = 17.68), whereas in the ICE group was 29.15 (SD = 16.95) ($p = 0.147$). The mean (SD) platelet transfusions (pools) received in the AZA group was 13.5 (SD = 17.24) whereas in the ICE group was 27.92 (SD = 22.18), being this difference statistically significant ($p = 0.033$). **Conclusion:** Treatments with ICE or AZA for patients with HR-MDS/sAML are associated with high need of transfusion support. Compared with intensive chemotherapy, treatment with AZA is associated with less platelet transfusions requirements.

Disclosure of Commercial Conflict of Interest

E. Alonso Nogués: Nothing to disclose; A. Gallardo: No Answer; M. Garcia Caro: Nothing to disclose; J. Grifols: Nothing to disclose; R. Línio: Nothing to disclose; M. Morgades: Nothing to disclose; M. Prieto: No Answer; J. Ribera: Nothing to disclose; L. San Miguel: Nothing to disclose; B. Xicoy: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Alonso Nogués: Nothing to disclose; A. Gallardo: No Answer; M. Garcia Caro: Nothing to disclose; J. Grifols: Nothing to disclose; R. Línio: Nothing to disclose; M. Morgades: Nothing to disclose; M. Prieto: No Answer; J. Ribera: Nothing to disclose; L. San Miguel: Nothing to disclose; B. Xicoy: Nothing to disclose

SP342

Estimation of Obstetrical Blood Loss to Improve Clinical Performance for Hemorrhage

M Villers[2] (villers@musc.edu), J E Squires[1], S Moncada[1], K A Stephenson[2], A Cooper-Smith[2], A Gaines-Garner[2]. *[1]Pathology and Laboratory Medicine, Medical University of South Carolina, Charleston, SC, United States; [2]Obstetrics and Gynecology, Medical University of South Carolina, Charleston, SC, United States*

Background/Case Studies: Obstetrical hemorrhage is a significant cause of maternal morbidity and mortality in the perinatal period. A recent national survey demonstrated a 26% increase in postpartum hemorrhage between 1994-2006 and a 92% increase in blood transfusion during delivery hospitalizations between 1997-2005. In spite of the fact that visual estimation of blood loss is notoriously imprecise, it continues to be the most commonly used method. To assess current performance in estimating obstetrical blood loss and to improve professional performance, an evaluation, and training experience was developed. **Study Design/Methods:** Hospital staff included in this evaluation included attending physicians and residents in: the obstetrics and gynecology department; the anesthesiology department; nursing staff in the labor and delivery departments; and medical students during their obstetrics clerkship. Units of red blood cells (RBCs) were obtained from the Transfusion Service and diluted to a physiologic concentration. Measured quantities of blood (ranging from 30 mL to 2000 mL) were displayed in a

variety of typical clinical settings including maternity pads, bed pan, vaginal drapes, Chux, etc. Staff members were asked to estimate the blood loss in each of these situations. Estimates were compared among the groups and by years of experience and percent error was calculated. **Results/Findings:** Participants included: obstetricians (23%); anesthesiologists (27%); nursing staff (36%); and medical students (14%). Fifty-six percent of participants had less than 5 years of experience in medicine, 16% had 5-10 years' experience, and 28% had over 10 years' experience. As a group, nursing staff performed the best with mean percent error of 1%. Medical students underestimated blood loss with a mean percent error of 16%. Obstetricians and anesthesiologists also underestimated blood loss by –20% and –25% respectively. Small amounts of blood were underestimated and large amounts of blood were overestimated. Longer experience in medicine did not improve visual estimates. **Conclusion:** Participants included Obstetricians (23%), Anesthesiologists (27%), nursing staff (36%), and medical students (14%). Fifty-six percent of participants had less than 5 years of experience in medicine, 16% had between 5 and 10 years' experience, and 28% had over 10 years. As a group, nursing staff performed the best with mean percent error of 1%. Medical students underestimated blood loss with a mean percent error of 16%. Obstetricians and Anesthesiologists also underestimated blood loss by –20% and –25% respectively. Small amounts of blood were underestimated and large amounts of blood were overestimated. Longer experience in medicine did not improve visual estimates.

Disclosure of Commercial Conflict of Interest

A. Cooper-Smith: Nothing to disclose; A. Gaines-Garner: Nothing to disclose; S. Moncada: Nothing to disclose; J. E. Squires: Nothing to disclose; K. A. Stephenson: Nothing to disclose; M. Villers: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Cooper-Smith: Nothing to disclose; A. Gaines-Garner: Nothing to disclose; S. Moncada: Nothing to disclose; J. E. Squires: Nothing to disclose; K. A. Stephenson: Nothing to disclose; M. Villers: Nothing to disclose

SP343

Post PAS 3 Launch Reaction Rates (RR) and Increments at North Shore University Hospital (NSUH)

R Galloway-Haskins[1,2] (rgallowayh@nshs.edu), W A Heaton[1,2], N M Nikolis[2], A Stein[2], M A Mulani[1,2], S Macwan[1]. *[1]Blood Component Research Laboratory, Feinstein Institute for Medical Research North Shore-LIJ Health System, Manhasset, NY, United States; [2]Blood Donor Transfusion Services, North Shore University Hospital, Manhasset, NY, United States*

Background/Case Studies: In 2012, a retrospective review study was carried out, evaluating single donor platelet (SDP) transfusion reactions and increments comparing PAS 3 (InterSol™, Fenwal) and Plasma apheresis platelets. PAS 3 was introduced as an equivalent product after an educational program. Transfusion triggers and SDP use per patient did not change before or during the study. **Study Design/Methods:** SDP requests were randomly assigned to PAS 3 or Plasma SDP based upon inventory. Blood center supply assumed equal PAS 3 and Plasma SDP availability. Staff members were in-serviced on reaction (Rxn) reporting, and type classification was monitored through transfusion (Tx) record and chart audit. Reactions (Rxns) were classified to CDC's National Healthcare Safety Networks Hemovigilance Protocol. Transplant (Tx) and Rxn data capture included corrected count increments (CCI), Tx indication, height/weight, and diagnosis. Patients were classified as Prophylactic—non-refractory without bleeding, sepsis, or consumption, or Therapeutic (all others). Refractory was defined as CCI < 5,000/m^3 > 2 times. **Results/Findings:** Out of 1377 Tx's (425 recipients), 32% were PAS 3, and 42% were categorized as Prophylactic. During the study, the average number of SDP Tx /recipient was consistent at 3.2/patient. Table 1 reports the increments for Prophylactic recipients, who had at least one CCI outcome of > 5,000/m^3. Comparing a RR of 0.23% (PAS 3) to 1.93% (plasma), the site observed a statistically significant reduction in reactions to Tx platelets. There was only one reaction with PAS 3 (categorized as nausea/dizziness); 78% of Plasma reactions being allergic. The frequency of excluded refractory recipients was significantly different, 27.15% and 15.23% for PAS and Plasma respectively. The difference in CCIs between the two groups were not significantly different ($p < 0.0674$.) **Conclusion:** With a reduction in reactions to PAS 3 platelets and a modest (statistically non-significant) reduction in increments, NSUH will monitor RR and CCI data further as PAS 3 platelets are implemented as a product equivalent to plasma SDP.

Disclosure of Commercial Conflict of Interest

R. Galloway-Haskins: Fenwal, Grants or Research Support; W. A. Heaton: Fenwal, Grants or Research Support, Immunetics, Grants or Research Support, Light Integra Inc., Grants or Research Support, Fresenius, Travel Support or Honorarium; S. Macwan: Fenwal, Grants or Research Support; M. A. Mulani: Fenwal, Grants or Research Support; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Galloway-Haskins: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, HemCon Medical Technologies Inc., Grants or Research Support, LightIntegra Technology, Grants or Research Support; W. A. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra Technology, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; S. Macwan: Fenwal, Grants or Research Support; M. A. Mulani: Immunetics,Grants or Research Support, LightIntegra Technology, Grants or Research Support, HemCon Medical Technologies, Inc., Grants or Research Support, Haemonetics, Grants or Research Support; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

TABLE 1

	Total Tx # (N = 1377)		Prophylactic* Tx (N = 584) Mean ± SD		
	PAS 3 Tx/Rxn RR	Plasma Tx/Rxn RR		PAS 3 (N = 161)	Plasma (N = 423)
April-May	146/0 0.00%	293/4 1.37%	Product Content (x10^{11})	3.96 ± 0.51	3.84 ± 0.49
June	134/1 0.75%	222/4 1.80%	Pre/Post Count (K/uL)	16.15 ± 1.37/31.75 ± 23.08	14.83 ± 10.66/33.35 ± 24.93
July-Aug	164/0 0.00%	418/10 2.37%	CCI (x1000/m^3)	7.03 ± 8.18	8.70 ± 10.01
Total	**444/1 0.23%**	**933/18 1.93%**	* CCI < 5000/m^3 > 2 times (refractory) and pre platelet count >50 K/ul were excluded		

SP344

Prospective Observational Study on the Efficacy of Excluding High Titer ABO Antibodies in Preventing Hemolysis due to Plasma Incompatible Plateletpheresis Transfusions

M K Scott[1] (MaScott@lifebridgehealth.org), S L Sheldon[1], W A Flegel[1]. *[1]Transfusion Medicine, NIH Clinical Center, Bethesda, MD, United States*

Background/Case Studies: In 2008 this facility implemented a policy for screening all plateletpheresis products (APLT) for high titer ABO antibodies. APLT with an isoagglutinin reacting at a dilution of 1:250 when tested with pooled A1 and B cells in buffered gel cards, are labeled as high titer, restricted to ABO identical or group O recipients, or washed prior to transfusion. ABO plasma incompatible APLT transfusions were evaluated with isoagglutinins below 1:250 for evidence of hemolysis in the recipient. **Study Design/Methods:** Data was collected for 2 months from non-group O recipients receiving an ABO plasma incompatible APLT with an isoagglutinin below the 1:250 dilution cutoff. Data included visual inspection for hemolysis of pre- and post-transfusion samples, direct antiglobulin test (DAT), total bilirubin and LDH to identify evidence of hemolysis. When DAT results were positive in the IgG gel card, elutions were tested against three group O screening cells and A1, A2 and B cells. If anti-A or anti-B was eluted from recipient RBC, serial titers were performed using EDTA plasma drawn at the time of APLT collection. Isoagglutinin titers were done in gel cards. **Results/Findings:** A total of 83 platelet transfusions in 23 different patients were evaluated. No hemolysis was observed visually in any sample, and clinical laboratory data were compared (Table). In 8/23 recipients DAT results were initially negative and remained negative after transfusion. In 9/23 recipients DAT results were negative pre-transfusion and positive post-transfusion, and in 6/23 recipients DAT results were positive pre- and post-transfusion. Of the nine recipients whose DATs transitioned from negative to positive, anti-A was eluted in six and eluates were negative in the remaining three recipients. Mean anti-A titers were 68 (buffered gel card) and 683 (anti-IgG gel card). **Conclusion:** The effectiveness of the 1:250 dilution screen in preventing hemolysis in recipients of ABO plasma incompatible APLT transfusions was

evaluated. There were no statistically significant results indicating intravascular hemolysis, although some recipients developed positive DAT, with anti-A eluted. An unexpected finding was that some anti-A titers in IgG gel cards were higher than in buffered gel cards. We concluded that our dilution cutoff was effective in avoiding clinically relevant hemolysis in this group of patients.

Disclosure of Commercial Conflict of Interest

W. A. Flegel: RHD Genotyping Patents, Other; M. K. Scott: Nothing to disclose; S. L. Sheldon: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. A. Flegel: Nothing to disclose; M. K. Scott: Nothing to disclose; S. L. Sheldon: Nothing to disclose

TABLE. Clinical laboratory results before and after transfusion

Mean ± SD			
Parameter	Before	After	P
LDH	240.58 ± 216.16	253.54 ± 214.90	0.054
Bilirubin	2.35 ± 3.96	2.46 ± 4.35	0.270

SP345

Peripartum Blood Transfusion in South Africa: Relative Contributions of Prenatal Anemia, Obstetric Hemorrhage, and Inter-Hospital Variation

J C Hull[1] (hulljc@iafrica.com), R Gangaram[2], S Fawcus[3], J Anthony[4], E Bloch[5], R Crookes[6], C Ingram[6], S M Ngcobo[6], J A Croxford[7], E L Murphy[8,5]. *[1]Chris Hani Baragwanath Hospital, Johannesburg, South Africa; [2]University of Kwa-Zulu Natal, Durban, South Africa; [3]Mowbray Maternity Hospital, Cape Town, South Africa; [4]University of Cape Town, Cape Town, South Africa; [5]Blood Systems Research Institute, San Francisco, CA, United States; [6]South African National Blood Service, Johannesburg/Durban, South Africa; [7]RTI International, Rockville, MD, United States; [8]University of California San Francisco, San Francisco, CA, United States*

Background/Case Studies: Peripartum blood transfusion is common in South Africa and poses a major demand on the blood supply. However there is little data on peripartum transfusion incidence nor contributing factors such as anemia, HIV infection, and inter-hospital variability. **Study Design/ Methods:** The incidence was measured of peripartum blood transfusion and associated clinical risk factors using a cross-sectional review of the medical records on women who delivered at four major hospitals in Soweto, Durban, and Cape Town, South Africa over a 4-month period (April-August 2012). Limited demographic and clinical data were collected on all peripartum obstetric patients; for the subset of patients who received a blood transfusion, detailed information was collected on blood transfusion, prenatal hemoglobin, obstetric diagnoses and management as well as HIV disease. Unadjusted incidence rates were calculated and logistic regression was used to estimate independent risk factors for transfusion. **Results/Findings:** 15,758 women were surveyed over the study period of whom 25.2% were HIV positive and 95.7% had received at least some antenatal care; 58.0% had normal vaginal deliveries, 40.4% had C sections and 189 (1.2%) delivered before arrival at hospital. Overall, 436 (2.8%) women were transfused with any blood product; transfusion incidence was 2.1%, 2.5%, 3.3% to 6.2% at Mowbray Maternity Hospital (MMH), Chris-Hani Baragwanath Hospital (CHB), King Edward VIII Hospital (KEH) and Groote Schuur Hospital (GSH), respectively. Mean prenatal (pre-transfusion) hemoglobin values were 11.4 (7.0), 11.5 (7.7), 10.5 (7.8), and 11.3 (7.5) g/dL at hospitals MMH, CHB, KEH, and GSH, respectively. Transfused women received a mean of 2.27 red cell, 0.29 plasma and 0.09 platelet units. In unadjusted analyses, women aged 20-24 years were less likely to be transfused compared to other age groups and transfusion incidence was positively associated with both parity and HIV infection. In this multiple logistic regression model, transfusion was independently associated with obstetric hemorrhage (OR = 94.38, 95%CI 71.16-125.19), prenatal hemoglobin (OR = 0.66 per g/dL, 95% CI 0.62-0.71 per g/dL), GSH (OR = 2.46, 95% CI 1.70-3.56 versus CHB), out of hospital birth (OR = 2.77, 95%CI 1.30-5.94), and HIV infection (OR = 1.41, 95% CI 1.08-1.84). C section was not associated with higher transfusion incidence. **Conclusion:** In the South African obstetrical setting, the incidence of blood transfusion is 10-fold higher than in the US and Europe. Low prenatal hemoglobin and, by inference, anemia appear to act synergistically with obstetric hemorrhage as the strongest predictors of transfusion. HIV infection is a common and independent risk factor for transfusion. Significantly higher transfusion incidence at GSH, a high-risk obstetric referral unit, suggests a role for inter-hospital variation, likely related to severity of illness.

Disclosure of Commercial Conflict of Interest

J. Anthony: No Answer; E. Bloch: Nothing to disclose; R. Crookes: No Answer; J. A. Croxford: Nothing to disclose; S. Fawcus: No Answer; R. Gangaram: Nothing to disclose; J. C. Hull: Nothing to disclose; C. Ingram: Nothing to disclose; E. L. Murphy: Nothing to disclose; S. M. Ngcobo: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Anthony: No Answer; E. Bloch: Nothing to disclose; R. Crookes: No Answer; J. A. Croxford: Nothing to disclose; S. Fawcus: No Answer; R. Gangaram: Nothing to disclose; J. C. Hull: Nothing to disclose; C. Ingram: Nothing to disclose; E. L. Murphy: Nothing to disclose; S. M. Ngcobo: Nothing to disclose

SP346

Do Rh(D) Negative Patients Who Receive Rh(D) Positive Platelets Pheresis Require Passive Immunization with Rh Immune Globulin?

A Simard[1] (amie.simard@umassmemorial.org), M Greene[1], J Ferschke[1], M Vauthrin[1], R Weinstein[1]. *[1]Transfusion Medicine, University of Massachusetts Memorial Medical Center, Worcester, MA, United States*

Background/Case Studies: Blood banks traditionally offer Rh Immune Globulin (RhIg) injection to Rh(D) negative patients who receive transfusion of Rh(D) positive platelets in order to prevent alloimmunization to Rh(D). Recent reports suggest that, given the low red cell content of platelets pheresis, and the relatively poor responder status of hospitalized patients who require platelet transfusion, these injections of RhIg may not be necessary. Having instituted a policy whereby Rh(D) positive platelets pheresis (but not RhIg) are issued to all patients, regardless of Rh(D) status (except for Rh(D) negative neonates who receive only Rh(D) negative platelets and Rh(D) negative women of childbearing age who do receive RhIg), all Rh(D) negative patients were followed for whom platelets pheresis were issued as part of quality assurance surveillance of this policy. Reported here are the results of our first 6 months of observation of these patients. **Study Design/ Methods:** All patients for whom this blood bank issues platelets pheresis are recorded in a prospective database. Data captured include gender, age, diagnosis, platelets pheresis transfused, and antibody screen results before and subsequent to platelet transfusion. Data collected between October 16, 2012-April 16, 2013 are reported here. Neonates and women of childbearing age are excluded from this analysis. Data for patients who received Rh(D) positive platelets pheresis but not RhIg, had not previously been alloimmunized to Rh(D) and from whom a subsequent blood specimen for type and screen had been received are presented. **Results/Findings:** 59 (41 male, 18 female) of 381 patients who received platelet transfusion during the observation interval were Rh(D) negative. Their median (range) age was 61 years (3-94). 30 of them received a total of 96 Rh(D) positive platelets pheresis and had a follow-up type and screen a median (range) of 28 (1-141) days after Rh(D) positive platelet transfusion. None of these 30 patients were subsequently found to have anti-Rh(D) alloantibodies. Seven were transfused platelets during treatment of a hematological malignancy, six during hemopoietic progenitor cell transplant, seven during treatment for advanced liver disease or liver transplant and ten for other diagnoses. Among the 29 patients excluded from this analysis, 7 received only Rh(D) negative platelets, 19 lacked a follow-up type and screen, 2 had been previously alloimmunized against Rh(D) and 1 had previously been passively alloimmunized with RhIg. **Conclusion:** Rh(D) negative hospitalized patients can receive Rh(D) positive platelets pheresis with a low risk of Rh(D) alloimmunization. Injection of RhIg is not necessary. This may not apply to neonates or women of childbearing potential.

Disclosure of Commercial Conflict of Interest

J. Ferschke: Nothing to disclose; M. Greene: Nothing to disclose; A. Simard: Nothing to disclose; M. Vauthrin: Nothing to disclose; R. Weinstein: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Ferschke: Nothing to disclose; M. Greene: Nothing to disclose; A. Simard: Nothing to disclose; M. Vauthrin: Nothing to disclose; R. Weinstein: Nothing to disclose

SP347

Decreased Awareness and Delayed Treatment of Patients with Hemoglobinopathy and Bone Marrow Necrosis can be Fatal

R Dasararaju[1] (r_dasararaju@yahoo.com), C A Hanna[1], M B Marques[1], J Adamski[1]. *[1]University of Alabama-Birmingham, Birmingham, AL, United States*

Background/Case Studies: Bone marrow necrosis (BMN) and embolization of fat and necrotic marrow causing acute multiorgan failure is a rare but potentially fatal complication of sickle cell β+ thalassemia (HgbSβ+), hemoglobin SC (HgbSC) and other compound heterozygous hemoglobinopathies. These patients are at greater risk of BMN compared to those who are homozygous for HgbS. BMN may be the first manifestation of hemoglobinopathy. Indeed, some patients are unaware that they have HgbS. This report aims to demonstrate that delayed recognition and treatment of BMN can result in permanent neurologic deficits or death. **Study Design/Methods:** The records of 10 patients were reviewed from the past 3 years, (including two diagnosed post-mortem), transferred to this institution with unrecognized BMN, and collected the following: history of hemoglobinopathy, initial laboratory data, evidence of multiorgan failure and, when applicable, type of treatment. **Results/Findings:** All patients presented to an emergency room with severe musculoskeletal pain. Within hours, all developed multiorgan failure, including altered mental status, respiratory compromise, and renal insufficiency. Six patients were transferred with a presumed diagnosis of thrombotic thrombocytopenic purpura (TTP) due to anemia and thrombocytopenia. However, the peripheral blood film showed an erythroblastic picture and the ADAMTS13 was normal. Five patients received multiple RBC transfusions (n = 1) or automated red cell exchange (n = 3) or both (n = 1) soon after hospital admission and made complete recovery within days to weeks. Those with delayed recognition of BMN either died (n = 3) or suffered permanent neurologic damage (n = 2) (Table 1). **Conclusion:** Increasing awareness of BMN in patients with compound heterozygous sickle disease, characterized by musculoskeletal pain and an erythroblastic response, is critical because delay in therapy can be catastrophic. Early establishment of the correct diagnosis and rapid intervention with red cell exchange or transfusion is essential to prevent permanent dysfunction or death.

Disclosure of Commercial Conflict of Interest

J. Adamski: Nothing to disclose; R. Dasararaju: Nothing to disclose; C. A. Hanna: Nothing to disclose; M. B. Marques: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Adamski: Nothing to disclose; R. Dasararaju: Nothing to disclose; C. A. Hanna: Nothing to disclose; M. B. Marques: Nothing to disclose

Table 1. Characteristics of patients with bone marrow necrosis

Patient	Hgb disease; Known at admit?	Hgb (g/dL)	Platelets ($x10^3$/uL)	BM biopsy	MTP or RCE	Outcome
1	HgbSE ; No	9.4	70	Necrosis	Neither	Neurologic impairment
2	HgbSβ+, No	10.9	50	Necrosis	Neither	Neurologic impairment
3	HgbSC; Yes	10.4	94	Necrosis	MT	Full Recovery
4	HgbSβ+, Yes	10.7	38	Not done	RCE	Full Recovery
5	HgbS trait, Yes	9.5	376	Necrosis	Neither	Death
6	HgbSC; Yes	8.8	58	Necrosis	Neither	Death
7	HgbSβ+, No	4.3	56	Necrosis	RCE	Full recovery
8	HgbSβ+, Yes	6.5	83	Not done	RCE	Full recovery
9	HgbSE ; No	8.4	32	Not done	MT, RCE	Delayed, partial recovery
10	HgbSC; Yes	7.8	63	Not done	Neither	Death

Hgb, hemoglobin; BM, Bone marrow; MT, multiple RBC transfusion (≥ 4 units); RCE, automated red cell exchange

Transfusion Practice: Transfusion Practice in Trauma/Emergent Settings

SP348

Transfusing a Plasma to Trauma Patients: Should We Worry About Anti-B?

C Tauscher[1] (Tauscher.Craig@mayo.edu), B Bendix[1], E K Jacob[1], J R Stubbs[1]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: In July 2009, this institution began stocking the emergency department and helipad refrigerators with type A, thawed plasma from male-only donors, rather than AB plasma due to inventory concerns. A Massive Blood Transfusion (MBT) protocol has since been implemented that requires six units of thawed plasma to be available for emergency transfusion. Type A plasma is most often used in this protocol. The anti-B titer range of the A plasma donors was determined to assess the potential risk of hemolysis in a B or AB individual receiving a "high-titer" plasma product. No clinical reports of hemolysis have been reported at our institution since the A thawed plasma program was implemented. **Study Design/Methods:** Anti-B titers were performed on 120 of our male, whole blood (WB)/fresh frozen plasma (FFP) donors. Serial dilutions of donor serum and saline were pipetted using a Tecan Freedom Evo. Commercially prepared (Bio-Rad) 3% A1 red blood cells were added to all tubes. The tubes were incubated for 30 minutes at 37C, washed x 3, and macroscopically read for agglutination at the Anti-human globulin (AHG) phase. The titer was determined to be the highest dilution reacting at a strength of 1+. **Results/Findings:** The median anti-B titer result of our 120 donors was 1:16. No donor had an anti-B titer result greater than 1:512 and 91.7% of our donors had a titer result of <1:64. (See Table) **Conclusion:** The serologic data supports the clinical experience that the use of A plasma from male donors in patients with unknown blood types allows for safe and timely provision of emergent transfusion therapy without untoward consequences or undue burden on a limited supply of AB plasma. This includes patients subsequently determined to be blood group B or AB.

Disclosure of Commercial Conflict of Interest

B. Bendix: Nothing to disclose; E. K. Jacob: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal, Other, Haemonetics, Other; C. Tauscher: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Bendix: Nothing to disclose; E. K. Jacob: Nothing to disclose; J. R. Stubbs: Nothing to disclose; C. Tauscher: Nothing to disclose

Table 1. Anti-B Titer Results for Type A Male Donors

1:2 or less	1:4	1:8	1:16	1:32	1:64	1:128	1:256	1:512
11	8	21	32	22	16	5	4	1
9.2%	6.7%	17.5%	26.7%	18.3%	13.3%	4.2%	3.3%	0.8%

n = 120

SP349

Development of a Pediatric Massive Transfusion Protocol (MTP)

N K Case[1] (ncalcaterra@mcvh-vcu.edu), A Stalley[1], A L Sutton[1], T VanBrackle[2], S Roseff[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States; [2]Laboratory Information Systems, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: This is a Level I trauma center and a regional referral center for the state of Virginia with 220 pediatric beds. Pediatric visits and admissions to the pediatric acute care units total 22,350 annually. In order to meet the medical needs of the growing pediatric population, its pediatric trauma program is being refined. Pediatric Trauma Surgery (PTS) met with Transfusion Medicine (TM) to discuss the creation of a pediatric massive transfusion protocol (PMTP). **Study Design/Methods:** The current adult MTP, which uses fixed ratios of red blood cells (RBCs): thawed plasma (TP): platelets (PLT): cryoprecipitate (Cryo), was used as a template in the development of the PMTP. The MTP was created as a care set in Cerner Millenium. TM and PTS developed a new template based on the weight of the patient. **Results/Findings:** The recommended weight-based care sets were created by TM and distributed for review and approval by PTS. Next, TM met with Laboratory Information Systems (LIS) personnel to discuss the

feasibility of building the PMTP in the LIS. Four weight categories were developed: 0-4 kg, 5-10 kg, 11-22 kg, and 23-50 kg. Pediatric patients weighing > 50 kg fall under our adult MTP. RBCs will be ≤ 7 days old for patients weighing < 5 kg and ≤ 14 days old for patients weighing 5-10 kg. Patients < 1 year of age will receive RBCs irradiated within the past 24 hours; however, the first group of RBCs transfused may be non-irradiated. The PMTP is activated when the patient's physician orders the "Massive Transfusion Protocol- Pediatric" care set in Cerner. The Cerner system automatically queries the patient's medical record for the most recent weight and selects the appropriate care set containing the blood product orders and release requests. In addition to releasing the products, a "Keep Ahead" order for RBCs is automatically placed upon activation. The second set triggers "Keep Ahead" orders for plasma and orders for PLTs and cryo in preparation for activation of the 3rd set. The protocol ends following the fourth set or after 24 hours, whichever occurs first. After 24 hours, the protocol can be restarted with set 1. PMTP care sets and associated blood products are shown in Table 1. **Conclusion:** Collaboration with ordering physicians is essential for meeting the needs of the patients. It is important to work with other departments in order to ensure all stakeholders can work together to overcome potential barriers.

Disclosure of Commercial Conflict of Interest

N. K. Case: Nothing to disclose; S. Roseff: No Answer; A. Stalley: Nothing to disclose; A. L. Sutton: Nothing to disclose; T. VanBrackle: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. K. Case: Nothing to disclose; S. Roseff: No Answer; A. Stalley: Nothing to disclose; A. L. Sutton: Nothing to disclose; T. VanBrackle: Nothing to disclose

Table 1

Care Set	0-4 kg	5-10 kg	11-22 kg	23-50 kg
1st	1 RBC	1 RBC	2 RBC	3 RBC
2nd	1 RBC 1 TP	1 RBC 1 TP	2 RBC 2 TP	3 RBC 2 TP
3rd & 4th	1 RBC 1 TP ½ dose[1] PLT 1 unit Cryo	1 RBC 1 TP 1 dose[1] PLT 1 unit Cryo	2 RBC 2 TP 1 dose[1] PLT 6 units Cryo	3 RBC 2 TP 1 dose[1] PLT 6 units Cryo

[1] One dose of PLT is either 4 pooled whole blood-derived PLTs or 1 Apheresis PLT

SP350

Defining Early Trauma-Induced Coagulopathy (ETIC): Is There a Role for the Clot Lifespan Analysis Model?

H Shafi[1] (heidishafi@hotmail.com), D Liou[2], S K Kitahara[1], E Ley[2], A Salim[2], D R Margulies[2], O Tcherniantchouk[1]. *[1]Department of Pathology and Laboratory Medicine, Cedars-Sinai Medical Center, Los Angeles, CA, United States; [2]Department of Surgery, Cedars-Sinai Medical Center, Los Angeles, CA, United States*

Background/Case Studies: Early trauma induced coagulopathy (ETIC) is defined as an early coagulopathy detected in trauma patients at time of presentation and is associated with increased mortality. ETIC has been contributed to hemodilution, depletion of coagulation factors and fibrinolysis. Prior studies attempting to define ETIC have used traditional coag testing parameters, PT and aPTT which are not designed for real time evaluation of coagulation. The goal of this study was to improve the understanding of the hemostatic changes responsible for ETIC by testing patients immediately upon arrival to the ED, prior to any blood transfusion, with thromboelastography (TEG 5000, Haemonetics). The clot lifespan analysis model was also evalutated (CLAM) of TEG to see if it allows for earlier identification of fibrinolysis. **Study Design/Methods:** Admission TEG was performed on a convenience sample of high-acuity trauma activations at a Level I trauma center. Patients receiving blood products prior to TEG were excluded. Patients with complete data at time of presentation were included (n = 47). Patients were stratified into three groups according to injury severity score (ISS): <16(MILD); 16-24(MOD); and >25(SEV). Clinical data and TEG parameters were compared. ETIC was defined as any abnormality on admission TEG before administration of any fluids and/or blood products. **Results/Findings:** 38 of 47 patients (81%) had abnormalities on TEG. There were no differences in TEG parameters among the 3 groups. A shortened R time was seen in all groups(MILD 4.1 versus MOD 3.4 versus SEV 4.1, p = 0.25); however, overall clot strength(G) was normal in all 3 groups. 3 of 38 patients showed hyperfibrinolysis (abnormal EPL/LY30): MILD 1, MOD 0, SEV 2; p = 0.33. 2 patients had primary fibrinolysis, both died on hospital day 1. A third patient had secondary fibrinolysis and discharged after 52 days. Of 44 patients with normal EPL/LY30, 15 had a time to maximum rate of lysis(TMRL) value generated using CLAM:18-49 min(mean 38 min). A TMRL was not generated/calculated by CLAM for 29 cases that had normal EPL/LY30; however, a subset of these cases may not have been evaluated for an adequate duration by TEG to calculate a TMRL value. **Conclusion:** 1) Abnormal admission TEG was seen in 81% of trauma patients irrespective of injury severity. 2) Within 1 hour of injury, transfusion naïve patients showed increased rate of thrombin generation (shortened R) with overall normal clot strength (G value); rapid-TEG may overlook this finding. 3) Both primary and secondary fibrinolysis can be seen in ETIC patients, therefore antifibrinolytic therapy would not be warranted in all trauma patients. TEG should be used to guide appropriate transfusion therapy. 4) CLAM is able to detect clot lysis in as early as 18 minutes even when EPL/LY30 is normal. Whether this is physiologic lysis or an indicator for risk of bleeding will require further follow-up with additional TEG and other clinical data.

Disclosure of Commercial Conflict of Interest

S. K. Kitahara: Nothing to disclose; E. Ley: Nothing to disclose; D. Liou: Nothing to disclose; D. R. Margulies: Nothing to disclose; A. Salim: Nothing to disclose; H. Shafi: Nothing to disclose; O. Tcherniantchouk: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. K. Kitahara: Nothing to disclose; E. Ley: Nothing to disclose; D. Liou: Nothing to disclose; D. R. Margulies: Nothing to disclose; A. Salim: Nothing to disclose; H. Shafi: Nothing to disclose; O. Tcherniantchouk: Nothing to disclose

SP351

Whole Blood and Transfusion of Incompatible Plasma Containing Products

J Olson[1] (jolson1@hmc.psu.edu), M R George[1], R E Domen[1]. *[1]Pennsylvania State University, Hershey Medical Center, Hershey, PA, United States*

Background/Case Studies: Component therapy has largely replaced the use of whole blood (WB). One of the few exceptions to component therapy has been in the military, where storage conditions, and the need to rapidly correct coagulopathy, favor the use of WB Lessons learned in military transfusion medicine have highlighted the utility of WB in trauma resuscitation. There has been recent interest in the use of fresh group O WB for acute trauma patients in the literature; however, consensus guidelines on the appropriateness of WB use are lacking. A survey was undertaken to determine if and when whole blood is used in the civilian realm. A related question is the titering of Anti-A and Anti-B in plasma containing products that are transfused to mismatched recipients. No standards exist and this study sets out to clarify what is current practice. **Study Design/Methods:** A four-question survey was sent out to 58 academic medical center transfusion medicine fellowship directors. This survey asked if their transfusion service transfused WB and, if so, in what circumstance(s). The second question asked if Anti-A, Anti-B titers were performed on group O plasma containing products for transfusion to mismatched recipients. The third and fourth questions asked what they consider a high-titer Anti-A, Anti-B result, and upon what specimen the titers would be performed. **Results/Findings:** Out of 58 possible responses, 23 were received (40% response rate). Six institutions indicated that they do transfuse whole blood. Of these, three were for autologous use only, 2 transfused whole blood only for pediatric or pediatric cardiac surgery use, and one did not specify the circumstances in which whole blood is used. No centers indicated use in a trauma setting. Five institutions reported that they do titer for Anti-A, anti-B and had reported high-titer cutoffs from 1:50 to 1:256. **Conclusion:** Given the recent wave of massive transfusion protocol literature with an emphasis on maintaining a ratio of pRBC to plasma units of approximately 1:1, the use of fresh WB may have some advantages in a trauma setting as thawing of plasma would not be necessary which could speed preparation time. Despite this possible advantage, none of the centers surveyed indicated its use in trauma.

Additionally, some pediatric cardiothoracic surgeons prefer fresh WB, although this remains controversial. The use of WB currently is not common in the civilian medical sphere; however it may become more common. Currently there are no uniform guidelines for transfusion of mismatched plasma containing products, or safety of cut-off values for Anti-A, Anti-B titers. These issues will need to be addressed if Group O WB is to see increased use in the future.

Disclosure of Commercial Conflict of Interest

R. E. Domen: Nothing to disclose; M. R. George: Nothing to disclose; J. Olson: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. E. Domen: Nothing to disclose; M. R. George: Immucor, Grants or Research Support; J. Olson: Immucor, Grants or Research Support

SP352

A Review of the Massive Transfusion Protocol (MTP) at a Single Institution

A Kaplan[1] (alesia@umn.edu), N D Zantek[1], S M Morgan[1], C S Cohn[1].
[1]Laboratory Medicine and Pathology, University of Minnesota Medical Center, Minneapolis, MN, United States

Background/Case Studies: Massive transfusion is defined as the transfusion of ≥10 red blood cell (RBC) units in 24 hours, a patient's blood volume in <24 hours, or more than 50% of the patient's estimated blood volume per hour. A massive transfusion protocol (MTP) was implemented to ensure timely and appropriate treatment when massive transfusions are needed and to utilize resources efficiently. The MTP may begin with two "emergency units" of O-negative RBCs and is followed by sets of 4 RBC, 4 plasma, and 1 apheresis platelet unit(s). Blood is switched to type specific as soon as possible. Per the protocol, it should take no more than 30 minutes for the first set to be available. Subsequent component sets should be available every 15 minutes thereafter. There has been occasional confusion among our clinicians concerning the difference between activating an MTP and requesting emergency units. This study was conducted to see if MTPs versus emergency units are ordered appropriately and to assess the timeliness of component delivery. **Study Design/Methods:** A retrospective review was performed on all initiated MTPs at a single institution from January 1, 2010-June 30, 2012. Data was collected on the following: time to first cooler, time to pick up, number of coolers ordered, blood components usage during each MTP, and whether emergency units were used. **Results/Findings:** A total of 117 MTPs were initiated. Activation of the MTP was the highest in the surgery unit (48/117). The longest time from initiating the MTP to the first component set was 25 minutes; the mean time for all first component sets was 10 minutes. The longest time between the first cooler available and a pick up by the requesting team was 26 minutes; the mean for all first component set pick-ups was 5 minutes. Emergency units were requested most often in MTPs when only one component set was used. Emergency unit orders were rare when the MTP extended to more than three component sets. Analysis of components returned to the blood bank indicated that the desired ratio of component transfusions was most often seen with smaller MTPs. When skewed, the ratio of components used had no identifiable relationship with the underlying diagnosis of the patient. **Conclusion:** This study confirms that blood was available without significant delay and within the protocol parameters during massive transfusion events. In some cases, the component set was available long before the patient's team requested the next set. There was an increase in emergency units used during smaller MTPs, suggesting that MTPs may be called when only emergency units are needed. The appropriate ratio of components is maintained for most MTPs; however, the ratio becomes more skewed during prolonged MTPs.

Disclosure of Commercial Conflict of Interest

C. S. Cohn: Fenwal, Grants or Research Support, Fenwal, Travel Support or Honorarium; A. Kaplan: Nothing to disclose; S. M. Morgan: Nothing to disclose; N. D. Zantek: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. S. Cohn: Fenwal, Travel Support or Honorarium, Fenwal, Grants or Research Support; A. Kaplan: Nothing to disclose; S. M. Morgan: Nothing to disclose; N. D. Zantek: Nothing to disclose

SP353

Are Uncrossmatched Red Cell Units an Acceptable Risk?

L Greebon[1,2] (Greebon@uthscsa.edu), S L Warner[2], G Meny[1,2].
[1]Pathology, University of Texas Health Science Center at San Antonio, San Antonio, TX, United States; [2]Pathology, University Health System Hospital, San Antonio, TX, United States

Background/Case Studies: Treatment of acutely hemorrhaging patients involves transfusion of group O red blood cells which are often uncrossmatched. Studies have sought to define the incidence of risks of transfusing uncrossmatched red cells. this is one experience at a level 1 trauma center in South Texas. **Study Design/Methods:** Demographic information on patients who received emergency release uncrossmatched O negative blood was obtained from emergency release transfusion (ERT) forms from January 2007-March 2013. The transfusion medicine laboratory information system and clinical electronic medical record system was then utilized to determine how many patients had an alloantibody (Ab) at the time of or subsequent to the ERT. Determination of unit compatibility was noted in the group of patients that had prior Abs. Suspected transfusion reactions reported to the blood bank were reviewed. **Results/Findings:** Of 1,661 patients, there were 42 patients (2.5%) identified with Abs either before or after ERT. One patient was excluded due to improper records. Of the 41 remaining patients, 33 had single Ab specificities and 8 had multiple Ab specificities. A total of 373 red cell units were transfused to the Ab patient population during the ERT episodes. Twenty two patients had a history of prior Ab and/or positive Ab screen determined at the time of ERT. The Ab specificities included anti-C, E, Fya, Fyb, K, Jka, S, M and Cw. Of these patients, 7 were issued a total of 39 incompatible units during ERT. Nineteen patients developed an Ab subsequent to the ERT. Ab specificities acquired post ERT were anti-C, c, E, M, and JKa. Subsequent antibody production was identified an average of 96 days after the ERT. Rh immunoglobulin was administered to six Rh negative patients who received Rh positive platelets during the ERT with subsequent passively acquired anti-D. There were four reported transfusion reactions. The other reported reactions were determined to be allergic (one), febrile non-hemolytic (one), and febrile episodes attributed to underlying medical disease (two). Noted within the transfusion medicine consult notes, there were two delayed hemolytic transfusion reactions in patients with prior antibodies that clinicians had not reported. **Conclusion:** Transfusion of uncrossmatched blood in an acutely bleeding patient is an acceptable therapy with minimal risk of Ab production and transfusion reaction. It is difficult to assess an accurate number of patients who developed Abs or the time frame in which they develop Abs due to lack of follow up, no subsequent antibody screens or long periods of time between patient follow up and subsequent antibody screens. Transfusion reactions may go unrecognized by clinicians and may be difficult to discern between a patient's underlying medical illness.

Disclosure of Commercial Conflict of Interest

L. Greebon: Nothing to disclose; G. Meny: Nothing to disclose; S. L. Warner: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Greebon: Nothing to disclose; G. Meny: Nothing to disclose; S. L. Warner: Nothing to disclose

SP354

What is the Right Direction for the Use of Optimal Bloods in the Extreme Massive Transfusion?

S Song[1] (song1221@yuhs.ac), S Choi[1], S Kim[1], H Kim[1]. *[1]Department of Laboratory Medicine, Severance Hospital, Seoul, Republic of Korea*

Background/Case Studies: Massive transfusion is defined as transfusion of 8-10 red blood cell (RBC) units to an adult patient in less than 24 hours or acute administration of 4-5 RBC units in 1 hour. In rare cases, extreme massive transfusion may have to be conducted during the management of massive hemorrhage. However, many urgent cases with extreme massive transfusion also have resulted in unsatisfactory outcomes. In other words, the amount of units of RBCs used in transfusion does not always reflect the outcome of survival of patients. The aim of this study is to determine whether extreme massive blood support is in fact the best method for patient survival. **Study Design/Methods:** A retrospective analysis was performed on the medical record data of 68 patients with extreme massive transfusion from January 1, 2008-December 31, 2012 at Yonsei University Medical Center, a university hospital of 2000 beds in Seoul, Korea. The definition of extreme massive transfusion is: over 30 RBC units within 24 hours or 50 RBC units within 72 hours. **Results/Findings:** The patients mortality rate was 68%, which is two times higher than 26% of the hospital data on massive

transfusion (Table). The median usage ratio of fresh frozen plasma (FFP)/RBC units and single donor platelet (SDP)/RBC units is 1.08(0.84-1.15) and 0.9(0.65-1.02), respectively. **Conclusion:** It is safe to conclude that extreme massive transfusion supports are empirically less effective and more costly. Although the decision of maintenance of continuous blood transfusion support should base on patient's condition, extreme massive transfusion is ineffective and must be studied in-depth with more evidence-based knowledge.

Disclosure of Commercial Conflict of Interest

S. Choi: No Answer; H. Kim: No Answer; S. Kim: Nothing to disclose; S. Song: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Choi: No Answer; H. Kim: No Answer; S. Kim: Nothing to disclose; S. Song: Nothing to disclose

TABLE 1. The mortality rate of patients with extreme massive transfusion

	Total	2008	2009	2010	2011	2012
Liver transplantation	10/19 (52.6%)	4/4 (100.0%)	4/6 (66.7%)	0/1 (0.0%)	0/5 (0.0%)	2/3 (66.7%)
Aortic dissection or aneurysm	7/10 (70.0%)	3/3 (100.0%)	1/1 (100.0%)	2/3 (66.7%)	0/1 (0.0%)	1/2 (50.0%)
Gastrointestinal bleeding	6/8 (75.0%)	0/1 (0.0%)	3/4 (75.0%)	1/1 (100.0%)	1/1 (100.0%)	1/1 (100.0%)
Cardiac surgery	5/7 (71.4%)	3/3 (100.0%)	1/1 (100.0%)	0/1 (0.0%)		1/2 (50.0%)
Trauma	2/3 (66.7%)		2/2 (100.0%)	0/1 (0.0%)		
Obstetric	1/1 (100.0%)		1/1 (100.0%)			
Other surgery	15/20 (75.0%)	2/3 (66.7%)	4/6 (66.7%)	6/6 (100.0%)	2/4 (50.0%)	1/1 (100.0%)
Total	46/48 (67.7%)	12/14 (85.7%)	16/21 (76.2%)	9/13 (69.2%)	3/11 (27.3%)	6/9 (66.7%)

SP355

Evolution of a Massive Transfusion Protocol (MTP)

F Shaw[1] (bldflo@msn.com). [1]Blood Bank, JPS Health Network, Fort Worth, TX, United States

Background/Case Studies: In 2000, this hospital decided to review the management of patients requiring transfusions to replace their blood volume in a short period of time following a trauma that involved a patient with lacerated jugular vein. It became apparent during the management of the patient that improved communication were needed between departments and the blood bank, and also protocols set for the quantity and type of blood products to issue and at what intervals. Since the initial meeting, the massive transfusion protocol (MTP) has frequently been revisited as new tests and advances in medicine have become available. **Study Design/Methods:** The main issues were identified as: proper sample type and labeling; proper cooler labeling; proper management of blood cooler; testing requirements for additional blood products; types and quantities of blood products; Group O negative RBC management; and switching patients to Group specific blood products. In 2003, the inclusion of CBC, PT/PTT and fibrinogen testing was included and the blood product mix was changed to reflect the lab results received. In 2012, the protocols were revisited again and included hemostasis testing in the MTPas well as removing the requirement for PT/PTT and fibrinogen. These tests were replaced by TEG (thromboelastograph) and INR.

Results/Findings:

Action 2013

Physician initiates blue card to obtain blood (patient may still be en route)

2003 4 O Neg Units RBC

2013 4 Group O Units RBC, 2 A FFP

MTP initiated: Cycle 1

2003 Batch 1: 10 RBC; Batch 2: 5 RBC, 4 FFP, 1 SDAPL (single donor apheresis platelet)

Batch 3: 5 RBC, 4 FFP

2013 Batch 1: 6 RBC, 1-1.25 L FFP, 1 SDAPL

Labs Drawn

2003 CBC, PT/PTT, Fibrinogen, Type and Cross

2013 TEG, CBC, i-STAT (Hgb), INR, Type and Cross

Cycle 2

2003 Batch 1: 5 RBC; Batch 2: 5 RBC, 4 FFP, 1 SDAPL; Batch 3: 5 RBC, 4 FFP

2013 Batch 2: 5 RBC, 0.8-1.0 L FFP, 1 SDAPL

Continuation

2003 Repeat labs after every 10 RBC are transfused and repeat Cycle 2 blood products.

2013 Batch 3: 5 RBC, 0.8-1.0 L FFP—LABS repeated, 1 SDAPL included based on TEG result; Batch 4: 5 RBC, 0.8-1.0 L FFP; Repeat until MTP is resolved.

Conclusion: Conclusion: By evaluating the MTP and utilizing the most up to date testing methodologies, patient outcomes have improved. The mortality rate of critically injured trauma patients at John Peter Smith Hospital is only 19.86% compared to that in the NTDB annual report of 28.05%. Blood utilization during traumas is also down and optimized by rapidly obtaining a blood type and switching patients to group specific blood products.

Disclosure of Commercial Conflict of Interest

F. Shaw: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Shaw: Nothing to disclose

Results

Action	2003	2013
Physician initiates blue card to obtain blood (patient may still be en route)	4 O Neg Units RBC	4 Group O Units RBC 2 A FFP
MTP initiated: Cycle 1	Batch 1: 10 RBC Batch 2: 5 RBC, 4 FFP, 1 SDAPL (single donor apheresis platelet) Batch 3: 5 RBC, 4 FFP	Batch 1: 6 RBC, 1-1.25 L FFP, 1 SDAPL
Labs Drawn	CBC, PT/PTT, Fibrinogen, Type and Cross	TEG, CBC, i-STAT (Hgb), INR, Type and Cross
Cycle 2	Batch 1: 5 RBC Batch 2: 5 RBC, 4 FFP, 1 SDAPL Batch 3: 5 RBC, 4 FFP	Batch 2: 5 RBC, 0.8-1.0 L FFP, 1 SDAPL
Continuation	Repeat labs after every 10 RBC are transfused and repeat Cycle 2 blood products.	Batch 3: 5 RBC, 0.8-1.0 L FFP—LABS repeated, 1 SDAPL included based on TEG result. Batch 4: 5 RBC, 0.8-1.0 L FFP Repeat until MTP is resolved.

SP356

Establishment and Refinement of the Massive Transfusion Protocol (MTP):A Cross-Departmental Approach

F Shaw[1] (bldflo@msn.com). [1]Blood Bank, JPS Health Network, Fort Worth, TX, United States

Background/Case Studies: John Peter Smith Hospital is a 537 bed, Level I trauma center in Fort Worth, TX. The Blood Bank has a laboratory in the main hospital as well as a satellite lab in the Pavilion where the ER and ORs are located. In 2012, it was decided to re-visit the Massive Transfusion Protocol (MTP) for multiple reasons. Inter-departmental communications needed to be addressed with the trauma, surgery, nursing, and the blood bank departments all on the same page in relation to expectations, requirements, and needs. Blood utilization needed to be improved and the length of time to switch patients to group specific blood products needed to be decreased. Finally and foremost, patient outcomes needed to be improved and their needs managed appropriately. **Study Design/Methods:** A number of cross-functional team meetings were held with representatives from every department with involvement with MTP patients. Blood product usage was analyzed during MTP and as well as the role of the thromboelastograph (TEG) result in defining a specific patient's needs. Critical blood products were established for managing patients as well as the volume most commonly required. The blood bank gave specific input into sample requirements for type and crossmatch as well as TEG testing. **Results/Findings:** The original plan included issuing 10 units of RBC, followed by 5 units of RBC, 4 units of FFP, and 1 apheresis platelet and was more aggressive than required. The first batch of blood products was therefore decreased to 6 units of RBC, 1-1.25 L of FFP, and 1 apheresis platelet. The blood products also triggered the trauma or surgery team to provide the following blood samples: TEG, CBC, i-STAT (for Hgb), INR, and a type and cross. Batch 2 would include the following blood products, without additional samples, including 5 units of RBC, 0.8-1.0 L of FFP and 1 apheresis platelet. If the MTP continues past Batch 2, it would be repeated except the apheresis platelet which would be dispensed pending a second TEG result. Additionally, a second CBC, INR, and Hgb would be run. Finally, Batch 4 would include 5 units of RBC and 0.8-1.0 L of FFP. If the MTP continues past Batch 4, Batch 3 and 4 repeat until the situation is resolved. **Conclusion:** By creating an MTP cross-functionally, each department understands the needs of the others that are ultimately supporting the patient. Creating a standardized approach to patient transfusion needs and laboratory testing requirements has allowed the MTP process to be streamlined and has also decreased the amount of blood products issued. This protocol ensures that the patient needs are met while ensuring the clinical integrity of the transfusion decisions and overall helping to manage the blood supply more effectively.

Disclosure of Commercial Conflict of Interest

F. Shaw: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. Shaw: Nothing to disclose

SP357

Experience with Anti-Fibrinolytic Treatment of Massively Bleeding Patients at a Large Trauma Center

Z Franco[1] (zfranco@yahoo.com), M Perez[1]. *[1]Pathology, Baylor College of Medicine, Houston, TX, United States*

Background/Case Studies: The Ben Taub General Hospital (BTGH) is one of the largest level I trauma centers in Texas. In 2009, a massive transfusion protocol (MTP) was put into place at BTGH. In July 2012 a policy of administering transexamic acid (TXA) to patients thought to be at risk of fibrinolysis began. This study shows the effects of the TXA policy on survival within the MTP group. **Study Design/Methods:** Chart review was performed on the 315 patients for whom an MTP was activated between September 2009-April 2013. Data collected included demographics, injury and treatment, neurological status, thromboelastography (TEG) data, and survival status. For this study, only 196 Code 3 trauma cases were included for which at least one TEG was available. Logistic regression of 30-day survival outcome with TXA policy as exposure was performed. **Results/Findings:** Of the 315 activated MTPs, 238 were for trauma, 127 penetrating (99 gunshot wounds and 28 stab wounds) and 111 blunt (96 motor vehicle accidents and 15 falls). Of the 196 cases with TEG data available, the survival rate before the TXA policy implementation was 58% compared to 75% afterwards with blunt trauma survival improving from 47% to 71% and penetrating trauma from 67% to 79%. The average initial fibrinolysis level for patients before the TXA policy was 10.4% and 3.6% afterwards. Additional summary statistics will be presented along with population characteristics before and after the policy change. Logistic regression of 30-day survival showed a crude odds ratio (OR) for TXA policy of 2.21 (1.14, 4.29) p = .02 with significant covariates as antifibrinolytic treatment: injury type, GCS, age, Rotterdam CT score, and initial level of fibrinolysis. Race was not significant except for a small number of Asians (4/13) with a lower survival rate. In a logistic regression of 30-day survival on TXA policy, OR was adjusted for age, Rotterdam CT score, and antifibrinolytic treatment, and was 2.58 (1.07, 6.28), p = 0.036. **Conclusion:** Despite being a retrospective study, this data supports the use of a antifibrinolytic therapy policy for severely bleeding trauma patients.

Disclosure of Commercial Conflict of Interest

Z. Franco: Nothing to disclose; M. Perez: Nothing to disclose

Disclosure of Grants Conflict of Interest

Z. Franco: Nothing to disclose; M. Perez: Nothing to disclose

SP358

A Systems Approach to a Massive Transfusion Protocol (MTP)

J M Faed[1] (jim.faed@otago.ac.nz). [1]Pathology, New Zealand Blood Service & University of Otago, Dunedin, New Zealand

Background/Case Studies: Recognition of the adverse effect from excessive haemodilution and development of massive transfusion protocols (MTPs) has improved clinical management of patients requiring urgent and massive transfusion. This step has ensured that a minimum standard of effectiveness can be assured in the support of affected patients. The MTP approach to rapidly evolving cases of transfusion has been focussed on the affected patient and immediate logistic issues but has not usually evaluated the wider clinical environment. Without additional modification, MTPs with fixed transfusion ratios have not always provided a timely response for platelet and fibrinogen replacement in individual patients. While these issues are implicit in developing a massive transfusion protocol, they can be submerged if the process is event and product focussed. **Study Design/Methods:** A systems approach was used to evaluate factors affecting emergency and urgent supply of large volumes of blood components. Discussion with Anaesthetists and Intensivists occurred and an MTP was developed. **Results/Findings:** Major process outputs identified were: appropriateness of the mix of blood components supplied; timeliness for components reaching the patient's bedside; timely availability of haemostatic and other relevant laboratory test results; and communication between key contributors, including availability of input from a Transfusion Medicine Specialist/ Haematologist (TMS/H). A potential process gap arising from timely and efficient access to haemostatic test results was identified. Process response was strengthened after expansion of the role for Blood Bank Scientists/Technologists to monitor haemostatic test results and maintain active liaison with the TMS/H (particularly when at remote locations) or directly with the Anaesthetist. An expanded role for Blood Bank Scientific/ Technical (S/T) staff requires that these staff can interpret the general significance of haemostatic test results, understand the clinical use of blood components/products, can identify components needed immediately, and anticipate impending needs for components such as platelets and cryoprecipitate/fibrinogen. Where direct access to haemostatic test results does not exist or is not used, the Blood Bank S/T staff operate in a relative information vacuum and an important link in the MTP response is weakened. **Conclusion:** An expanded role for the Blood Bank Scientist/Technologist has enhanced rapid and timely responsiveness for supporting massive transfusion protocols. It has provided greater job satisfaction for the staff and greater satisfaction from Anaesthetists. Where the Blood Bank response for massive transfusion requests does not have immediate and continuing Transfusion Specialist/Haematologist support a process redesign to involve Blood Bank S/T staff in an active liaison role can be an effective solution.

Disclosure of Commercial Conflict of Interest

J. M. Faed: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. M. Faed: Nothing to disclose

SP359

Standardization of Massive Transfusion Protocol (MTP)

L C Tse[1,2] (chiu.tse@advocatehealth.com), E A Halperin[3], C A Aronson[2,1]. *[1]Transfusion Services, Advocate Condell Medical Center, Libertyville, IL, United States; [2]Transfusion Services, ACL Laboratories, Rosemont, IL, United States; [3]Transfusion Safety, Advocate Health Care, Oak Brook, IL, United States*

Background/Case Studies: The purpose of a massive transfusion protocol (MTP) is to address the problem of exsanguinating coagulopathy, and to

provide a protocol to address the urgent need for blood products. The MTP alerts the clinical staff and Blood Bank (BB) to provide extraordinary support for the emergency. The MTP is intended to standardize laboratory monitoring and to aid clinicians in crisis situations. For this purposes, there was a hospital MTP policy written the previous year but no one in the Emergency Room (ER) or BB were familiar with the process which resulted in confusion and unacceptable delays. **Study Design/Methods:** The BB supervisor reviewed the hospital MTP at the level 1 trauma center and noted discrepancies and inconsistencies. A standard Hospital Information System (HIS) order set did not exist at this site so the ER nurses were placing separate orders for each blood product in the HIS or placing verbal orders with the BB. Since the ER nurses did not know how many units of RBCs, platelets, plasma or Cryoprecipitate (Cryo) to order, the confusion resulted in unacceptable delays. The hospital system MTP policies from the other four Level 1 trauma centers were reviewed for best practices. **Results/Findings:** These protocols were found to consider only blood products for transfusion but did not address other areas of the hospital that should also have standard responses to an MTP. One site protocol addressed the ordering of laboratory tests and delivery of pharmaceutical agents that have been shown to be helpful in the management of massive blood loss. Standardized processes that include all three of these aspects would improve delivery of care and decrease mortality in hemorrhaging patients. Blood product waste occurred due to over ordering and not following established MTP cycles. In 2012, there were 75 MTP activations for the five Level 1 trauma centers. Blood product wastage ranged from 4-8.5% mainly due to loss of Cryo, platelets and plasma. **Conclusion:** Based on the review of the five MTP policies, the necessity was determined for standardizing to one evidence-based MTP. This protocol needs to: 1) Describe clinical criteria to activate the protocol; 2) Define which care providers can activate an MTP; 3) Include evidence-based adjuvant therapies, including medications and laboratory studies; and 4) educate BB and clinical staff. An MTP HIS order set needs to be developed to automatically order the correct number of RBCs, platelets, plasma and Cryo. This order set also includes lab and pharmacy orders (ie tranexamic acid, CaCl, etc) to streamline the ordering process. The end results are to improve the care delivery processes for the hemorrhaging patients; emergent procedures, standard order sets, work flow, and reliability metrics.

Disclosure of Commercial Conflict of Interest

C. A. Aronson: Nothing to disclose; E. A. Halperin: Nothing to disclose; L. C. Tse: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. A. Aronson: Nothing to disclose; E. A. Halperin: Nothing to disclose; L. C. Tse: Nothing to disclose

SP360

1:1 Fresh Frozen Plasam : Red Blood Cell (FFP : RBC) Ratio Improves Patient Survival during Emergency Medical Air Helicopter Transport

K S Badjie[1] (badjie.karafa@mayo.edu), K S Berns[3], S C Bryant[2], L M Button[1], J Kreuter[1], C M van Buskirk[1]. *[1]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States; [2]Health Sciences Research, Mayo Clinic, Rochester, MN, United States; [3]Medical Transport, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Since the publication of a body of knowledge from military studies suggesting administering 1:1 fresh frozen plasma : red blood cell (FFP : RBC) ratio during massive transfusion resuscitation appeared to reduce mortality, many institutions have since developed guidelines towards achieving a 1:1 transfusion ratio. This helicopter service provides emergency medical air transportation within a 150-mile radius of three hospitals. Prior to 2008, the inflight blood products cooler (BPC) consisted of only RBCs. In 2009, Group A thawed plasma (TP) was added to the BPC. The objective of this study is to evaluate mortality rates of patients who received a 1:1 FFP: RBC ratio on board the helicopter en-route to the tertiary care hospital. **Study Design/Methods:** Retrospective analysis of in-flight blood transfusion was performed from February 2009-August 31, 2012. A total of 158 patients were reviewed and divided into two cohorts according to pre-hospital blood component resuscitation ratio—2:1 RBC: TP versus 1:1 TP: RBC, and mortality rates were compared. For adult patients, blood products are administered when indicated per our in-flight guidelines and transfused according to protocol. From February 2009-December 2011, the in-flight BCP consisted of two TP and four RBCs. However, the in-flight pre-hospital resuscitation protocol has since been modified from administering 2:2:2 RBC: TP: RBC in 2009 to 2:4 TP: RBC in 2010 (2:1), then to 3:3 TP: RBC (1:1) in January 2012, based on published reviews suggesting improved outcomes when plasma is administered first. With the new protocol TP is administered first or, concurrently with RBCs when indicated. Significance (defined as $p < 0.05$) was assessed using the Fisher exact test or Wilcoxon rank sum test, as appropriate, using SAS 9.3 software. **Results/Findings:** During the study period, 7087 flights were completed with an average transport time of 27 minutes to the tertiary hospital. There were a total of 79 patients in each cohort with comparable characteristics such as median age, transport location, reasons for transport, reasons for plasma and RBC transfusion, and transport time to the hospital. The first cohort from February 2009-December 2011 received a 2:1 RBC: TP, while the second cohort after January 2012 had a 1:1 TP: RBC. The median hospital length of stay in the 2:1 group was 7 days (range 1-40 days) versus 8.0 days (range 1-41 days) in the 1:1. The mortality rate for the 2:1 cohort (41.8% (33/79)) was significantly higher than 19.0% (15/79) for 1:1, ($p < 0.003$) **Conclusion:** A retrospective review of a small sample of pre-hospital 1:1 plasma to RBC transfusion during medical helicopter transport was associated with improved patient survival.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; K. S. Berns: Nothing to disclose; S. C. Bryant: Nothing to disclose; L. M. Button: Nothing to disclose; J. Kreuter: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; K. S. Berns: Nothing to disclose; S. C. Bryant: Nothing to disclose; L. M. Button: Nothing to disclose; J. Kreuter: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

Transfusion Practice: Transplantation and Surgery

SP361

Clinical Scoring System to Predict Blood Use in Orthotopic Liver Transplantation (BUILT)

C Liu[1,2] (cliu@path.wustl.edu), N Vachharajani[3], R Cooke[4], W C Chapman[3], B J Grossman[1]. *[1]Department of Pathology and Immunology, Washington University School of Medicine, Saint Louis, MO, United States; [2]Barnes-Jewish Hospital, Saint Louis, MO, United States; [3]Department of Surgery, Washington University School of Medicine, Saint Louis, MO, United States; [4]Department of Pathology, University of Maryland, Baltimore, MD, United States*

Background/Case Studies: Liver transplantation is frequently complicated by significant blood loss requiring blood transfusion. In addition to the intrinsic risks associated with blood transfusion, this treatment may not be applicable under certain medical or religious circumstances. A simple clinical score is developed here to estimate the probability of blood use in orthotropic liver transplantation (BUILT). **Study Design/Methods:** 605 patients were retrospectively evaluated who were undergoing primary liver transplantation at a single institution between 2002- 2011. This cohort was randomly split into a training (T) set ($n = 305$) and a validation (V) set ($n = 300$). Univariate and multivariate analyses were performed in the T set to develop a scoring system, which was validated in the V set. **Results/Findings:** In the T and V sets, 36 (12%) and 38 (13%) patients received liver transplantation without blood transfusion, respectively. Pretransplant diagnosis of hepatocellular carcinoma, hemoglobin level (<12 g/dL) and INR (≥2) independently predicted BUILT in the training set, which were used to construct a clinical score (BUILT score) ranging from 0 to 4. A score of two or above robustly predicted blood transfusion with a positive predictive value of 97% in both the T and V sets. Scores of 0-1, 2-3, and 4 correlated with lower, intermediate and higher risk of blood consumption intraoperatively. **Conclusion:** The BUILT score is a useful tool for gauging the probability of blood use in liver transplantation and for estimating the transfusion rate and amount of intraoperative blood transfusion.

Disclosure of Commercial Conflict of Interest

W. C. Chapman: Nothing to disclose; R. Cooke: No Answer; B. J. Grossman: Nothing to disclose; C. Liu: Nothing to disclose; N. Vachharajani: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. C. Chapman: Nothing to disclose; R. Cooke: No Answer; B. J. Grossman: Nothing to disclose; C. Liu: Nothing to disclose; N. Vachharajani: Nothing to disclose

TABLE 1. Components of the BUILT score

Variables	Score*
Pretransplant diagnosis of HCC	
Yes	0
No	+1
Hemoglobin	
≥12 g/dL	0
<12 g/dL	+1
INR	
<2	0
≥2	+2

* Scores of 0, 1-2, and 3-4 indicate lower, intermediate, and higher probabilities of blood use in liver transplantation, respectively

SP362

Intraoperative Blood Transfusion Need and Mortality during Liver Transplantation

A F Brown[1] (browna6@uthscsa.edu), G Meny[1]. *[1]Pathology, University of Texas HSC, San Antonio, TX, United States*

Background/Case Studies: Liver transplantation has been associated with major blood loss and the need for large-volume allogenic blood product transfusions. Improvements in surgical techniques and blood salvage have greatly decreased the amount of blood products transfused but blood loss has not been incorporated into the models for post-transplant survival (Am Coll Surg, 2013; 216(5): 902). The objective of this review was to assess the link between intraoperative red blood cell (RBC) and fresh frozen plasma (FFP) transfusion needs and perieoperative patient factors including model for endstage liver disease (MELD) score, total hospital stay and graft-related mortality at 6 months. **Study Design/Methods:** The records of all consecutive adult liver transplants from 2008-2012 at a tertiary care university hospital were retrospectively reviewed. Mean follow-up was 8 months. Patients were grouped by year to trend annual change. Additionally, univariate analysis of all data was used to correlate MELD score and preoperative international normalized ratio (INR) with intraoperative transfusion requirement, total hospital stay, and graft-related mortality at 6 months. **Results/Findings:** During the reviewed period, 227 adults received liver transplants with a median intraoperative RBC transfusion volume of 10.6 units and a plasma transfusion volume of 7.3 units. MELD score and preoperative INR were significantly correlated to graft-related mortality and length of hospitalization. Additionally, increasing MELD score and preoperative INR showed direct positive correlation with intraoperative RBC and FFP transfusion requirements. Annual trends showed a significant decreased in annual mortality over the observation period and a slight decrease of median intraoperative transfusions volumes. **Conclusion:** The overall assessment indicated that MELD and INR were predictive of intraoperative transfusion requirements. Lower MELD score and preoperative INR significantly correlated with increased postoperative survival and decreased total hospital stay. Technical factors including team experience may be important in dictating intraoperative transfusion needs and could be attributing to the discordance between decreasing annual 6-month mortality and intraoperative transfusion needs.

Disclosure of Commercial Conflict of Interest

A. F. Brown: Nothing to disclose; G. Meny: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. F. Brown: Nothing to disclose; G. Meny: Nothing to disclose

Year	N	Preoperative MELD[1](median)	Intraoperative FFP[2] (median)	Intraoperative RBC[3](median)	Total Hospital Stay (median)	Mortality (%)
2008	42	23.8	8.5	13	12	21.4
2009	46	26.9	7	10	15.5	17.4
2010	49	25.2	8	11	12	12.2
2011	47	19.9	6	8.5	9.5	10.6
2012	43	18.3	7	10.5	11	2.3
MELD Correlation (r^2)			0.68	0.73	0.85	0.73
Preoperative INR Correlation (r^2)			0.87	0.89	0.85	0.59
p-value			<0.01	<0.01	<0.01	<0.01

[1] MELD—model for end-stage liver disease, UNOS modified; [2] FFP-fresh frozen plasma unit; [3] RBC—red blood cell unit

SP363

The Factors Influence Transfusion in Patients with Allogeneic Peripheral Blood Stem Cells Transplantation : A Retrospective Study

Z Yuan[1] (zhuangyuan0215@126.com), P Jichun[1], W Deqing[1].
[1]Transfusion Department, The Chinese PLA General Hospital, Beijing, China

Background/Case Studies: Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an important means for treatment of malignant hematological diseases and refractory hematological diseases. Before transplantation, pretreatment (radiotherapy and chemotherapy) is the essential means to guarantee the success of allo-HSCT which can remove abnormal proliferation cells of the body but inhibit the immune function leading to serious bone marrow suppression and pancytopenia. Therefore, timely transfusion therapy in patients with allo-HSCT is an important support therapy, which directly affects the success rate of transplantation and the prognosis of patients. There are many factors affecting the amount of red blood cells (RBCs) and platelets (PLTs). To explore the influencing factors on transfusion during peri-transplantation, a retrospective study with patients undergoing allogeneic peripheral blood stem cell transplantation (allo-PBSCT) was performed. **Study Design/Methods:** Utilizing a clinic transfusion database explored by our department, 85 patients undergoing allo-PBSCT were enrolled at the Chinese PLA General Hospital between January 2009-December 2011. Multivariable logistic regression analysis was used to analyze the influencing factors on transfusion (more amounts of RBCs and PLTs infused) in the patients undergoing transplantation. **Results/Findings:** The results showed that in the patients during peri-transplantation the average RBC transfusion was 8.36 ± 8.24 U, and average PLT transfusion was 11.95 ± 8.1 U. Of 85 patients, 40% infused red blood cells no less than 8.5 U, and platelet recovery time after transplantation , narrow function status before pretreatment ($P < 0.01$), HLA matching and body surface area ($P < 0.05$) were associated with greater likelihood of more RBC infusion. 40% infused PLTs no less than 12 U, which was increasingly affected by platelet recovery time, HLA matching ($P < 0.01$), donor-recipient ABO incompatibility, and age ($P < 0.05$). The infusion of RBC and platelet during peri-transplantation was respectively 11.31 ± 14.16 U and 15.13 ± 10.01 U in major ABO incompatibility recipients, which was more than the compatibility recipients ($P < 0.05$). In patients whose narrow function status was poorest before pretreatment, the infusion of RBC was 13.22 ± 11.89 U, much higher than relative normal patients ($P < 0.05$). **Conclusion:** Platelet recovery time and HLA matching can greatly affect the amount of red blood cell and platelet transfusion; that donor-recipient ABO incompatibility can affect the amount of platelet transfusion; and that narrow function status before pretreatment can affect the amount of RBC transfusion in this study.

Disclosure of Commercial Conflict of Interest

W. Deqing: No Answer; P. Jichun: No Answer; Z. Yuan: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Deqing: No Answer; P. Jichun: No Answer; Z. Yuan: Nothing to disclose

Transfusion-Transmitted Infectious Diseases: Bacteria, Parasites, Prions

SP364

Detection of *Brucella* in Whole Blood Derived Platelet Unit : A Matter of Concern

F Anwar[1] (farzala@hotmail.com), M A Abdelaal[1], H Khalid[1], A M Osman[1]. *[1]Pathology & Laboratory Medicine, King Abdulaziz Medical City, Jeddah, Saudi Arabia*

Background/Case Studies: Bacterial contamination of platelets continues to be a hazard and the major infectious problem, despite the implementation of various detection methods. It is assumed that contamination of platelets occurs from the low concentration of skin bacteria at the time of phlebotomy and less common from contamination during processing or from asymptomatic donor bacteremia. *Brucellosis* is a major public health problem in many countries. Due to slow growth, *brucella bacteremia* in otherwise healthy donor may go undetected in the donated platelet unit with serious consequences to the recipient. A case of *brucella* detection is reported here in a platelet unit on day 3 by culture method; the unit, however, was issued after 24 hours incubation period cut-off. **Study Design/Methods:** Red blood cells (RBCs), fresh frozen plasma (FFP_ and platelet (PLT) were prepared from a 31-year-old whole blood (WB) regular donor with normal vital signs. BacT/ALERT microbiology culture system was used for bacterial contamination in platelet units. Platelet unit was issued after 24 hours incubation and was transfused. **Results/Findings:** Bacterial culture was found to be positive on day 3 of incubation. Patient physician was immediately notified as per facility policy. RBC and FFP from the same donation were quarantined. RBC and platelet unit was sent to microbiology section for culture and identification. *Brucella* species was identified in the platelet unit, however, no growth was found in the RBC unit. The donor was called-back for re-testing. Blood culture was found to be negative and *Brucella* antibody IgG titer was 200 and IgM was negative. Donor was permanently deferred as per the policy. **Conclusion:** This is the first reported case of *Brucella* detection in Saudi Arabia in the platelet concentrate from an apparently healthy blood donor. Human *Brucella* in Saudi Arabia and in other neighboring countries is high and is in the range of 1.6 %-2.6%. The serological evidence of exposure to brucella antigen in some areas is reported to as high as 19%. The main form of acquiring *brucellosis* is through ingestion of raw milk and milk products obtained mainly from infected goats and camels, a traditional custom fostered by the nomadic heritage and dietary habits of the people. Blood centers in the region need to be very cautious in blood donor selection and apply sensitive methods to detect bacterial contamination in platelet units.

Disclosure of Commercial Conflict of Interest

M. A. Abdelaal: Nothing to disclose; F. Anwar: Nothing to disclose; H. Khalid: No Answer; A. M. Osman: No Answer

Disclosure of Grants Conflict of Interest

M. A. Abdelaal: Nothing to disclose; F. Anwar: Nothing to disclose; H. Khalid: No Answer; A. M. Osman: No Answer

SP365

Comparison of Different Strategies for Platelet Bacteria Screening

M Stoermer[1] (melanie.stoermer@uk-koeln.de), S Radojska[1], V Petrescu-Jipa[1], N J Hos[2], B S Gathof[1]. *[1]Transfusion Medicine, University Hospital of Cologne, , Cologne, Germany; [2]Institute of Medical Microbiology, Immunology and Hygiene, University Hospital of Cologne, Cologne, Germany*

Background/Case Studies: Bacterial contamination of platelet concentrates (PCs) represents an ongoing risk in transfusion medicine although a variety of strategies have been developed for minimizing the risk of transfusion-transmitted sepsis. According to German regulations, the implementation of a bacterial detection system into routine screening may permit extending the shelf-life back to 5 days. Therefore the suitability of different screening technologies for the detection of bacteria in PCs at different time points during storage was evaluated. **Study Design/Methods:** Bacterial strains (*B. thuringiensis*, *S. pyogenes*, *K. pneumoniae*, *E. coli*, and *S. epidermidis*) were inoculated separately with 5-90 CFU/bag into freshly produced or expired PCs in replicates and stored under routine conditions. PCs were tested before and after contamination and during storage using the BactiFlow ALS (Biomerieux AES Chemunex), the ThromboLUX (LightIntegra Technology), platelet aggregometry according to Born or by using the Multiplate Analyzer (Roche Diagnostics), and aerobic and anaerobic classical and automated cultivation (Bactec, Beckton Dickinson). Isolated bacteria were identified by routine diagnostic tests and MALDI-TOF analysis. Sterile PCs served as negative controls. **Results/Findings:** Time to detection varied between bacterial species, sampling time and detection technology. With the exception of *S. epidermidis* who showed a prolonged growth, all strains were detectable by using the BactiFlow by sampling on day 2 in 92% (100% on day 3), and by using automated or classical culture in 100% on each sampling day. A decrease in the quality score of the ThromboLUX was significant for B. thuringiensis contaminated PCs already on day 2. Due to donor and PC individual properties (high score variations) no significant decrease could be observed using other species. A reduction in platelet aggregation in bacterially contaminated apheresis PCs could be observed. The level of reduction was dependent on donors and bacterial strain growth properties in the product. **Conclusion:** Late sampling in combination with the BactiFlow represents a good strategy for bacterial screening of PCs obtaining explicit results even for slow growing organisms. Measuring platelet aggregation would provide a low priced alternative method and would improve the quality and efficiency of platelet transfusion but the results are sometimes difficult to interpret. Moreover, a delay in detection of approximately 1 day for some bacterial species has to be considered due to the time between reaching high bacterial cell counts and inducing the detectable effect. The ThromboLUX should be further evaluated for bacterial detection by increasing the measurement time to allow the slow movement from micro-aggregates to be captured more accurately.

Disclosure of Commercial Conflict of Interest

B. S. Gathof: Nothing to disclose; N. J. Hos: Nothing to disclose; V. Petrescu-Jipa: Nothing to disclose; S. Radojska: Nothing to disclose; M. Stoermer: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. S. Gathof: AES Chemunex, Grants or Research Support, LightIntegra, Grants or Research Support, Roche, Grants or Research Support; N. J. Hos: Nothing to disclose; V. Petrescu-Jipa: Nothing to disclose; S. Radojska: Nothing to disclose; M. Stoermer: Nothing to disclose

SP366

A Novel, Sensitive and Rapid Point-Of-Issue Assay for the Universal Detection of Contaminated Platelet Concentrates (PCs)

D R Zweitzig[1] (dzweitzig@zeusscientific.com), N Riccardello[1], J Pester[2], R Jeanmonod[2], M Kopnitsky[1], S OHara[1]. *[1]R & D, ZEUS Scientific Inc., Branchburg, NJ, United States; [2]Emergency, St. Luke's University Health Network, Bethlehem, PA, United States*

Background/Case Studies: Serious health risks are associated with transfusion of bacterially contaminated platelet concentrates (PC). In an effort to help reduce these risks, rapid point-of-issue PC screening tests have recently been reported; however they suffer from various limitations. In this study, it was sought to determine the feasibility of detecting the presence of bacteria from spiked PC via the selective measurement of bacterially-derived DNA polymerase activity using a novel method called Enzymatic Template Generation and Amplification (ETGA). **Study Design/Methods:** ETGA background signal was characterized using 0.5 mL aliquots of whole blood (WB) (n = 4 non-leukoreduced units) and apheresis (n = 8 non-leukoreduced units, and n = 35 leukoreduced units) PC after these aliquots were subjected to a differential cell lysis procedure (DCLP). Next, the ETGA limit of detection (LOD) was assessed for *S. epidermidis*, *S. aureus*, *E. coli*, and *K. pneumoniae* using 6 independent apheresis PC units (n = 3 non-leukoreduced and n = 3 leukoreduced). Twenty two additional clinically relevant strains of bacteria were also spiked at two target levels (1000 and 100 CFU) into aliquots of apheresis PC and tested via ETGA. Finally, apheresis PC units were spiked at less than 1 CFU/mL with 7 clinically relevant bacterial species, and their growth was monitored via ETGA. **Results/Findings:** ETGA background signal derived from PC aliquots (totaling 329 sample preparations from 47 independent PC units) was effectively reduced to assay noise by the DCLP. The LOD experiments revealed an approximate sensitivity of 30-200 CFU/mL (average = 85 CFU/mL) for *S. epidermidis*, *S. aureus*, *E. coli*, and *K. pneumoniae*. An additional 22 clinically relevant strains of bacteria were also detected below 200 CFU/mL after spiking into apheresis PC aliquots. In addition, ETGA signals correlated well with the growth of seven bacterial species that were spiked in individual PC units at levels between 10-100 CFU/Unit. **Conclusion:** The feasibility of detecting bacterially contaminated PC by measurement of bacterially-derived DNA polymerase activity has been demonstrated here. These initial studies suggest that the ETGA-PC assay is 50-300-fold more sensitive than existing FDA-approved point-of-issue PC screening assays. With further

development, there is optimism that ETGA analysis of PC units could represent an improved approach towards rapid, sensitive, and universal point-of-issue PC screening.

Disclosure of Commercial Conflict of Interest

R. Jeanmonod: Nothing to disclose; M. Kopnitsky: Nothing to disclose; S. OHara: Nothing to disclose; J. Pester: Nothing to disclose; N. Riccardello: Nothing to disclose; D. R. Zweitzig: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Jeanmonod: Nothing to disclose; M. Kopnitsky: ZEUS Scientific, Inc., Ownership or Partnership; S. OHara: Nothing to disclose; J. Pester: Nothing to disclose; N. Riccardello: Zeus Scientific, Inc.,Other; D. R. Zweitzig: ZEUS Scientific,Other

SP367

Bacterial Detection in Platelet Concentrates (PCs): Evaluation of the BacTx® Test

P Morel[1] (pascal.morel@efs.sante.fr), M Deschaseaux[1], M Sauget[2], P Morata[1], C Naegelen[1], S Bégué[1], O Garraud[1], P Tiberghien[1], X Bertrand[2]. *[1]Etablissement Français du Sang (BFC, AuLo, Headquarters), Besançon- Saint Etienne-Paris, France; [2]Hygiene Laboratory, University Hôpital, Besançon, France*

Background/Case Studies: Despite preventive measures already in place and in the absence of a pathogen inactivation process, bacterial contamination remains the main risk of infection by transfusion in France with 2.4 deaths per million transfused Platelet Concentrates (PC). To improve transfusion safety, detection of contaminated PC is currently the choice in France. The advent of rapid tests able to detect contaminated PC at the time of delivery represents an attractive solution. An evaluation is presented here of the BacTx® test (Immunetics, MA, USA) conducted in France. **Study Design/Methods:** In the first series of tests, BacTx® was used to detect increasing concentrations of bacteria in PC. A defined bacterial inoculum was introduced into a sample. Three strains of the Institute Pasteur collection (CIP) (*S. epidermidis*, *K. oxytoca*, and *E. coli*) were investigated in apheresis PC (APC) and Pool of Buffy Coat (PPC) in 3 additive solutions used in France (PAS III, PAS III M, and Plasma) at 5 concentrations. Eight strains obtained from CIP or ATCC (*B. cereus*, *S. marcescens*, *S. aureus*, *S. dysgalactiae*, *S. enterica*, *P. aeruginosa*, *E. faecalis*, and *P. acnes*) were investigated in APC and in PPC, both in PAS III. A second series of tests consisted of the detection of bacteria in PC after growth. Inoculation of PC to achieve 10E2 CFU/mL (*S. aureus*, *E. coli*) or 10E3 CFU/mL (*B. cereus*) was performed, the PC were returned to normal storage conditions and testing with BacTx® was carried out every 3 hours on a sample taken from the PC. **Results/Findings:** BacTx® was easy to use, with short turnaround time: 13 min for preparation of 8 samples and results are available in 30 minutes. Detection with BacTx® was comparable for both PC types and the additive solutions had no significant effect on results. *S. epidermidis* was detected at 1.8E3 CFU/mL (Last Negative Concentration (LNC): 1.62E2); *K. oxytoca* at 4.9E3 (LNC: 3.0E3); *E coli* at 1.8E3 (LNC: 1.05E3). For the eight other strains the results showed detection between 3.5E3 (S. aureus) and 1.6E4 (B. cereus). After proliferation in APC and PPC in PAS III, BacTx® became positive for *S. aureus* 9 hours after spiking in APC (4.1E3) and PPC (2.4E3); for *E. coli*, 12 hours after spiking in APC (6.3E3) and PPC (5.0E3) ; for *B. cereus*, 27 hours after spiking in APC (1.66E4), and 48 hours in PPC (2.9E6, LNC: 2.2E2). No false positive results were observed during this study, although the number of tests on negative samples was low (n = 178). **Conclusion:** These results confirm the ability of the BacTx® test to detect bacterial contamination on the order of 2E3-2E4 CFU/mL regardless of the strain, the type of PC, or the additive solution used. It is believed that BacTx® can be implemented as a release test, to be performed just before the delivery of PC within 6 hours prior to transfusion. A feasibility study is underway.

Disclosure of Commercial Conflict of Interest

X. Bertrand: Nothing to disclose; S. Bégué: Immunetics, Other; M. Deschaseaux: Nothing to disclose; O. Garraud: Nothing to disclose; P. Morata: Nothing to disclose; P. Morel: Nothing to disclose; C. Naegelen: Nothing to disclose; M. Sauget: Nothing to disclose; P. Tiberghien: Bactx, Grants or Research Support

Disclosure of Grants Conflict of Interest

X. Bertrand: Nothing to disclose; S. Bégué: Nothing to disclose; M. Deschaseaux: Nothing to disclose; O. Garraud: Nothing to disclose; P. Morata: Nothing to disclose; P. Morel: Nothing to disclose; C. Naegelen: Nothing to disclose; M. Sauget: Nothing to disclose; P. Tiberghien: Nothing to disclose

SP368

Detection Limit and Time to Detection of Bacterial Contamination Using the Immunetics BacTx Test in Leukocyte Reduced Apheresis Platelet Units Inoculated with Representative Bacterial Species

W Heaton[1] (wheaton@nshs.edu), R Galloway-Haskins[1], C E Good[2], M R Jacobs[2]. *[1]Blood Component Research Laboratory, Feinstein Institute for Medical Research, Manhasset, NY, United States; [2]Pathology, Case Western Reserve University, Cleveland, OH, United States*

Background/Case Studies: Despite existing strategies, platelet bacterial contamination remains a problem. The BacTx Assay (Immunetics Inc., Boston, MA), which detects bacterial peptidoglycan, was evaluated. **Study Design/Methods:** In-date, leukocyte reduced apheresis platelet units were utilized. Level of detection (LoD) was determined on suspensions of eight aerobic and two anaerobic species in platelet aliquots. Time to detection (TtD) was determined on platelet units inoculated with bacterial suspensions of 2-5 CFU/mL, incubated in a platelet agitator at 22C, and tested after 48 hours, 72 hours, and 7 days. BacTx testing was performed in replicates of 10. LoD and TtD were defined as the point at which all 10 replicates were positive. **Results/Findings:** In LoD experiments, all 10 replicates were positive at bacterial titers ranging from 6.3e2 CFU/mL for S. epidermidis at site 2 to 7.6e4 CFU/mL for *E. coli* at site 1, with little variation in LoD between the two sites (Table). In TtD experiments, all 10 replicates of 6 of the 8 aerobic species were positive at 48 hours at both sites (Table). For S. epidermidis 0/10 replicates were positive at site 2 at 48 hours and 10/10 at 72 h. For P. aeruginosa 0/10 replicates were positive at both sites at 48 hours and 10/10 at 72 hours. All replicates at both sites were positive at 7 days. The two anaerobes failed to grow, with testing and cultures remaining negative throughout the 7-day study. **Conclusion:** The BacTx Assay reproducibly detected all 10 representative bacterial species at <10e5 CFU/mL at both sites in LoD experiments. In TtD experiments, the BacTx Assay reproducibly detected 7/8 aerobic bacterial species at site one at 48 hours after inoculation of low bacterial challenges, and 6/8 aerobic bacterial species at site two at 48 hours after inoculation; the remaining bacterial species were all detected at 72 hours, and all 8 aerobic bacterial species were detected at 7 days. These findings confirm the earlier findings with pooled, whole blood derived platelets, and suggest that the BacTx Assay will be a valuable test for detection of clinically relevant levels of bacterial contaminants in leukocyte reduced apheresis platelets close to time of use.

Disclosure of Commercial Conflict of Interest

R. Galloway-Haskins: Fenwal, Grants or Research Support; C. E. Good: Nothing to disclose; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; M. R. Jacobs: Immunetics, Grants or Research Support, Travel Support or Honorarium, Verax, Grants or Research Support, Travel Support or Honorarium, BioSense, Grants or Research Support

Disclosure of Grants Conflict of Interest

R. Galloway-Haskins: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, HemCon Medical Technologies, Grants or Research Support, LightIntegra Technology, Grants or Research Support; C. E. Good: Verax Biomedical, Grants or Research Support, Immunetics,Grants or Research Support; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; M. R. Jacobs: Immunetics, Grants or Research Support, BioSense, Grants or Research Support

Bacterial species	LoD: Lowest bacterial titer in CFU/mL at which all ten samples were positive by BacTx assay by test site		TtD: Time after inoculation at which all ten samples were positive by BacTx assay by test site	
	Site 1	Site 2	Site 1	Site 2
Bacillus cereus	1.4 x 10e3	1.9 x 10e3	48 h	48 h
Clostridium perfringens	9.4 x 10e2	4.8 x 10e3	NA	NA
Escherichia coli	7.6 x 10e4	3.3 x 10e4	48 h	48 h
Klebsiella oxytoca	1.6 x 10e4	7.5 x 10e3	48 h	48 h
Propionibacterium acnes	5.0 x 10e3	8.5 x 10e3	NA	NA
Pseudomonas aeruginosa	3.7 x 10e4	2.0 x 10e4	72 h	72 h
Serratia marcescens	4.2 x 10e3	5.3 x 10e3	48 h	48 h
Staphylococcus aureus	3.2 x 10e3	7.1 x 10e3	48 h	48 h
Staphylococcus epidermidis	1.3 x 10e3	6.3 x 10e2	48 h	48 h
Streptococcus agalactiae	4.5 x 10e3	3.3 x 10e3	48 h	72 h

NA, not applicable

SP369

BacT/ALERT Detection of Bacteria Grown in TRIMA-derived platelets stored in PAS compared to plasma

J Mundt[1] (janna.mundt@terumobct.com), L Rouse[1], D I Gilmour[1], A Razatos[1], S Marschner[1]. *[1]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Current practice in the US is to store platelets in 100% plasma; however, the use of platelet additive solutions (PAS) is gaining adoption. Terumo BCT has received FDA approval of Isoplate as a PAS solution for use with platelets collected on the Trima Accel System. This study compared the ability of the BacT/ALERT system (bioMerieux, France) to detect bacteria in platelet units suspended in plasma compared to those in PAS, to test whether a different storage medium has an effect on bacterial detection. **Study Design/Methods:** Paired platelet units stored in either 100% plasma or Isolyte S pH 7.4 (B. Braun, USA; the formulaic equivalent to Isoplate) with 35% plasma carryover, were split into 30 mL aliquots, each inoculated with one of eight strains of bacteria (~2 CFU/mL), while one aliquot served as a negative control. Three aerobic and three anaerobic BacT/ALERT bottles were each inoculated with 4.5 mL of sample from each 30 mL aliquot. The bottles were placed into the BacT/ALERT system and monitored for seven days. Eight organisms were evaluated in this study (Table) with a total of N = 4 replicates evaluated per strain. **Results/Findings:** For each BacT/Alert bottle that was inoculated with bacteria, a positive result occurred regardless of the media. The success rate (defined as Number of Positive Units/Total Number of Units) was 100% for all eight strains. The time to detection ranged from 9.9 hours (*B. cereus*, aerobic) to 22.9 hours (*S. epidermis*, anaerobic). The difference between PAS and plasma was not greater than 1.8 hours for anaerobic conditions and not greater than 3.7 hours under aerobic conditions. The difference between PAS and plasma was less than 1 hour for 6/8 strains tested. **Conclusion:** The BacT/Alert system is able to detect the organisms tested when PAS is used as a storage solution for platelets instead of plasma. The time to detection differed by at most 3.7 hours between PAS and plasma, thus indicating that the storage solution did not accelerate or delay detection.

Disclosure of Commercial Conflict of Interest

D. I. Gilmour: Terumo BCT, Grants or Research Support; S. Marschner: Employee of Terumo BCT, Other; J. Mundt: Terumo BCT, Ownership or Partnership; A. Razatos: Terumo BCT, Other; L. Rouse: Terumo BCT, Other

Disclosure of Grants Conflict of Interest

D. I. Gilmour: Nothing to disclose; S. Marschner: Employee of Terumo BCT, Other; J. Mundt: Terumo BCT, Ownership or Partnership; A. Razatos: Nothing to disclose; L. Rouse: US Deparment of Defense, Stocks or Bonds

TABLE 1. Results for different bacteria strains tested presented as difference between PAS and Plasma

Bacterial Species	Strain‡ ‡ ATCC = American Type Culture Collection †Clinical Isolate	Gram Stain	Type AE—aerobic FA—facultative anaerobe	Difference in Average Time to Detection (PAS—Plasma, in hours)	
				Anaerobic	Aerobic
Bacillus cereus	NI-0001†	+	AE/FA	0.00	0.18
Escherichia coli	ATCC #25922	–	FA	–0.04	–0.10
Staphylococcus aureus	ATCC #27217	+	FA	0.21	–2.88
Serratia marcescens	ATCC #43862	–	FA	–0.19	–0.07
Enterobacter cloacae	ATCC #29005	–	FA	0.04	–0.41
Klebsiella oxytoca	ATCC #43165	–	FA	–0.05	–0.22
Staphylococcus epidermidis	ATCC #12228	+	FA	0.95	0.42
Streptococcus mitis	ATCC #6249	+	FA	–1.83	–3.68

SP370

Performance Characteristics of Mass Spectrometry for the Rapid Identification of Cultured Bacterial Pathogens: Potential for Improving Laboratory Reporting in Patients Receiving Bacterially-Contaminated Units

J Weyhmiller[1] (weyhmiller.janice@marshfieldclinic.org), M Baldauf[1], B J Olson[1], T J Novicki[1], T R Fritsche[1,2]. *[1]Marshfield Clinic, Marshfield, WI, United States; [2]University of Wisconsin, La Crosse, WI, United States*

Background/Case Studies: Background: A transfusion-acquired bacterial infection may result in serious or life-threatening consequences for the recipient. For optimal patient management, clinicians require rapid pathogen identification and susceptibility results. The development and validation of proteomic profiling by MALDI-TOF mass spectrometry for the rapid, accurate and cost-effective identification of cultivable bacterial and fungal pathogens is changing the paradigm of clinical microbiology and may provide benefits for improving time to identification of pathogens recovered from blood products. When coupled with an automated instrument for antimicrobial susceptibility testing, such systems will likely improve accuracy and decrease time to reporting of critical laboratory results. **Study Design/Methods:** Bacterial isolates capable of contaminating blood products (351 total including 132 *Enterobacteriaceae*, 59 *Pseudomonas aeruginosa*, 48 other non-fermentative *bacilli*, 59 *Staphylococcus aureus* and 53 other *staphylococci*) were identified using an automated system (Vitek 2®, bioMerieux, Durham, NC) and results compared with those obtained on the MicroFlex LT mass spectrometer (Bruker Daltonics Inc., Billerica, Ma) using the MALDI Biotyper™ v3.0 software and library v3.31.0 according to the manufacturer's recommendations and validation algorithms. Consensus required agreement between both methods; discrepancies were adjudicated by 16S rDNA amplification and sequencing. **Results/Findings:** Correct genus-/species-level identifications for the automated method and mass spectrometry were, respectively, 98.0%/89.5% and 98.6%/93.2%. Species-level identification limitations (correct to genus-level only) for both methods included *Salmonella* and other *staphylococci*. Lack of species identifications (or misidentifications) occurred most commonly with *Enterobacteriaceae* (automated method and mass spectrometry, respectively, 18.2% and 18.2% excluding *Escherichia coli* and *Klebsiella*) and non-fermentative bacilli (29.2% and 12.5%). Bacterial identification by mass spectrometry may be performed directly upon observation of colonial growth and per test reagent cost for the automated method was >= 10X that of mass spectrometry. **Conclusion:** Pending FDA-approval, MALDI-TOF mass spectrometry offers a significant advance for the accurate, rapid, and cost-effective identification of bacterial pathogens, including those responsible for transfusion-acquired infections. Routine application of this technology in the clinical laboratory has the potential to more effectively guide therapy and decrease notification time to the blood product supplier.

Disclosure of Commercial Conflict of Interest

M. Baldauf: Nothing to disclose; T. R. Fritsche: bioMerieux, Grants or Research Support, Bruker Daltonics, Grants or Research Support; T. J. Novicki: No Answer; B. J. Olson: Nothing to disclose; J. Weyhmiller: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Baldauf: Nothing to disclose; T. R. Fritsche: Nothing to disclose; T. J. Novicki: No Answer; B. J. Olson: Nothing to disclose; J. Weyhmiller: Nothing to disclose

SP371

Dual False Positive Reactivity on PRISM Chagas Disease and PRISM Hepatitus C Virus (HCV)

S A Galel[1,2] (susangal@stanford.edu), A Layon[1,2], B A Marchlewicz[3]. *[1]Stanford Blood Center, Palo Alto, CA, United States; [2]Pathology, Stanford University, Stanford, CA, United States; [3]Blood Screening Technical Operations, Abbott Diagnostics, Abbott Park, IL, United States*

Background/Case Studies: The ABBOTT PRISM Chagas assay was implemented at this blood center in February 2011. Donors are tested on this assay if they were not previously tested by a licensed screen for *T cruzi* antibody (ie testing is performed on all first time donors and donors whose last donation was prior to March 2007). It was noticed that some donors were

false positive on both the PRISM Chagas and PRISM Hepatitus C virus (HCV) tests and investigated this further. **Study Design/Methods: A** retrospective review was performed of information system and donor records for *T cruzi* and HCV antibody reactive donations from February 2011 through March 2013. **Results/Findings:** Of 24,615 donations tested by PRISM Chagas, 19 (0.077%) were false positive for *T cruzi* antibody, ie repeat reactive on PRISM Chagas with negative radioimmunoprecipitation (RIPA) supplemental test results. In the same donor population, 32 donations (0.13%) were false positive on PRISM HCV, ie repeat reactive on PRISM HCV with a negative result on recombinant immunoblot or second licensed screen. Five donations were false positive on both assays, representing 26% of the Chagas false positive donations and 16% of the HCV false positives. Sample to cutoff (S/CO) ratios for these dual false positive donations are shown in the Table. Feedback from the manufacturer indicates that the dual reactivity could be caused by non-specific binding of human immunoglobulin to the reaction matrix which is then detected by the anti-human IgG component of each assay. Donor 2 was retested 1 year after the dual reactive donation: PRISM Chagas was nonreactive (S/CO 0.17); HCV remained reactive (S/CO 1.25). **Conclusion:** A high proportion (26%) of PRISM Chagas false positive donations are also false positive on PRISM HCV. Reactions may be related to human IgG trapped on the reaction surface that the acridinium labeled conjugate can then bind to; however, reactivity strength and persistence do not seem to be consistent between the two assays. This is consistent with the hypothesis that the reactivity is a result of non-specific binding due to levels and stickiness of subsets of antibodies. Blood samples drawn at different times from the same donor may contain differing concentrations and populations of IgG molecules.

Disclosure of Commercial Conflict of Interest

S. A. Galel: Nothing to disclose; A. Layon: Nothing to disclose; B. A. Marchlewicz: Abbott, Stocks or Bonds

Disclosure of Grants Conflict of Interest

S. A. Galel: Nothing to disclose; A. Layon: Nothing to disclose; B. A. Marchlewicz: Abbott, Stocks or Bonds

DONOR	CHAGAS S/CO	HCV S/CO
1	19.14	2.73
2	6.38	1.27
3	21.37	3.74
4	3.53	1.36
5	1.73	1.16

SP372

The New Automated TPHA Assay for Blood Donor's Screening:-Evaluation of Method

M Bohonek[1,2] (milos.bohonek@uvn.cz), H Zakoucka[2,1], H Kostalkova[1,2], M Pribylova[1,2], E Sladkova[1,2], J Lodinska[2,1]. *[1]Haematology, Biochemistry and Blood Transfusion, Central Military Hospital Prague, Prague, Czech Republic; [2]The National Institute of Public Health, Prague, Czech Republic*

Background/Case Studies: Screening for syphilis antibodies is a universal recommended test of blood donors. The preferred method for syphilis antibodies is to use a specific Treponemal assay. The new TPHA ScreenTM was launced by Immucor at the end of 2012, intended for use in the qualitative screening of blood donors for the detection of IgG and IgM antibodies to Treponema pallidum using the Immucor Galileo and/or NEO Automated Blood Bank System. This is an evaluation of the test using the Immucor Galileo instrument. **Study Design/Methods:** The 100 donor samples (both plasma and serum) and 124 reactive patients samples (100 samples with determined syphilis and 24 samples with excluded syphilis infection) sourced from the archives of the National Institute of Public Health (NIPH) were tested with Immucor TPHA ScreenTM using the Immucor Galileo Automated Blood Bank System. All samples tested by Immucor TPHA ScreenTM were tested twice: on Day 1 (D0) and the next Day (D1). Donor samples were tested in parallel using the Abbott Architekt Syphilis TP on the Architekt 2000i device on D0, as the current and reference method for donors. Samples from the NIPH were tested by these methods: VDRL (mikro Immutrep, Omega Diagnostics Ltd.), TP-PA (Serodia TP-PA, FujiRebio Inc.), TPHA (TPHA Immutrep, Omega Diagnostics Ltd.), FTA-ABS (in house method), IgM SPHA (in house method). **Results/Findings:** Results from the test group of donor samples performed by Immucor TPHA ScreenTM have the same interpretation as the results from the Abbott Architekt Syphilis TP. The results from plasma samples using Immucor TPHA ScreenTM have slightly higher agglutination values but a lower range of variation compared to results from serum samples, but without any impact on the interpretation (Table). The group of 100 syphilis positive samples showed good correlation

Table 1: Comparison results of blood donors syphilis screening tests made by Syphilis TP Abbott Architekt with TPHA Screen™

Test	TPHA Screen™ ImmucorGamma on Galileo Automated Blood Bank System.				Abbott Architekt Syphilis TP on Architekt 2000i
Sample	Plasma		Serum		Serum
Day of Measurement	D0	D1	D0	D1	D0
No of samples	100	100	100	100	100
Mean	7,856 ± 2,45	7,42 ± 2,69	5,421 ± 3,31	4,815 ± 2,88	0,0678 ± 0,01
CK (%)	31,191	36,257	60,958	59,769	59,969
Median	7,600	7,300	4,950	4,700	0,055
Upper quartile (percentil 0,75)	9,325	9,400	7,425	6,425	0,070
Lower quartile (percentil 0,25)	6,225	5,675	3,050	2,900	0,050
Interpretation	NEG	NEG	NEG	NEG	NEG

in 97 samples (the two discordant samples were from patients with old treated syphilis, and one from a 6-month-old child whose mother was syphilis positive; these samples also gave negative results in confirmatory TPHA test). There were three cases of false positive reactions in the group of 24 syphilis disease excluded samples which also agreed with other TPHA tests. **Conclusion:** Results produced by the Immucor TPHA ScreenTM test are fully comparable with current IgG specific tests and bring a new possibility for the automation of screening for syphilis antibodies in blood donors testing.

Disclosure of Commercial Conflict of Interest

M. Bohonek: Nothing to disclose; H. Kostalkova: No Answer; J. Lodinska: No Answer; M. Pribylova: No Answer; E. Sladkova: No Answer; H. Zakoucka: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Bohonek: Nothing to disclose; H. Kostalkova: No Answer; J. Lodinska: No Answer; M. Pribylova: No Answer; E. Sladkova: No Answer; H. Zakoucka: Nothing to disclose

SP373

Riboflavin and UV Illumination Treatment of Whole Blood Units (WBU) in a Ghanaian Blood Center

S Owusu-Ofori[1] (sowusu_ofori@hotmail.com), A Owusu-Ofori[2], G Freimanis[3], S D Keil[4], J Allain[3]. *[1]Transfusion Medicine Unit, Komfo Anokye Teaching Hospital, Kumasi, Ghana; [2]Microbiology, Komfo Anokye Teaching Hospital, Kumasi, Ghana; [3]Haematology, University of Cambridge, Cambridge, United Kingdom; [4]Research & Development, Terumo BCT, Denver, CO, United States*

Background/Case Studies: Plasmodium falciparum spiked into whole blood units (WBU) and treated with Riboflavin and UV illumination caused significant inhibition of long amplicon amplification and prevented parasite growth according to the level of energy applied. In preparation of a clinical trial of this methodology, the impact of the method on WBU collected from Ghanaian blood donors was examined. **Study Design/Methods:** 30 WBU were collected from random family or volunteer donors, treated with riboflavin and 80JRBC (15) or 120JRBC (15) illumination, and compared with 10 control WBU. Samples collected pre- and post-treatment were tested for hematocrit, hemoglobin (Hb) level, full cell count, plasmodium parasitemia level and species by polymerase chain reaction (PCR) and bacteremia by aerobic culture. **Results/Findings:** In a 10-day period, withdrawing 40 WBU did not affect supply to patients. WBU from 37 males and 3 female donors (median age 26) had median hematocrit of 41% and median Hb level of 12.5 g/dl. 20 WBU (50%) contained >10 plasmodium genome/ml but only three were detected by microscopy. Parasitemia mostly ranged between 10-100 gen/ml but 6 WBU contained >5000 copies/ml (5E3 to 5E5). Plasmodium species distribution was 70% P *falciparum*, 30% P *malariae* and 10% P *ovale* with 30% infected samples containing more than one species. No WBU contained cultivable bacteria. Riboflavin and illumination 80JRBC caused 0.78 ± 0.3 log inhibition of amplification of a 2316 nucleotide fragment of mitochondrial plasmodium gene. It did not affect hematocrit, RCC, or WBC counts but tended to moderately decrease platelet count from 153 ± 36 E3/µl to 135 ± 45 E3/µl. However, there was no significant difference in platelet count between controls, treated WBU pre- and post-80JRBC exposure (175.5 ± 44, 179.1 ± 63, and 170.9 ± 69 E3/µl, respectively) when corrected for dilution related to riboflavin addition. Further illumination to 120JRBC decreased the platelet count to 146.1 ± 41 E3/µl but did not modify amplification inhibition of plasmodium gene. Local personnel were trained in performing the pathogen reduction operation, collecting appropriate samples in sterile conditions, detecting and quantifying plasmodium genome, and

performing amplification inhibition assays with a real-time PCR instrument. **Conclusion:** This limited study demonstrated that a clinical trial of WBU in a west African, medium-size, blood center was feasible from both operational and scientific monitoring standpoints with local staff. Plasmodium amplification inhibition appeared a major biological endpoint to be correlated with clinical elements in treated recipients. Bacterial culture appeared less frequent than anticipated and a less attractive endpoint.

Disclosure of Commercial Conflict of Interest

J. Allain: Nothing to disclose; G. Freimanis: Nothing to disclose; S. D. Keil: Employee of Terumo BCT, Other; A. Owusu-Ofori: Terumo BCT, Grants or Research Support; S. Owusu-Ofori: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Allain: Terumo BCT, Grants or Research Support; G. Freimanis: Nothing to disclose; S. D. Keil: Employee of Terumo BCT, Other; A. Owusu-Ofori: Nothing to disclose; S. Owusu-Ofori: Terumo BCT, Grants or Research Support

SP374

Treatment of Whole Blood with Riboflavin and UV Light: Effectiveness against Clinically Relevant Bacterial Contamination

S Doane[1] (suzann.doane@terumobct.com), S D Keil[1], D I Gilmour[1], H Reddy[1], S Wilkinson[1], N Hovenga[1], R Goodrich[1]. *[1]Scientific, Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: Whole blood (WB) donations are most likely to become contaminated with bacteria during collection, and less often due to asymptomatic bacteremia or poor blood-handling techniques during processing. Units of WB are typically separated into components for transfusion. The incidence of contamination with bacteria is believed to be extremely low; however, if a WB unit is contaminated, then each component created from this unit may also contain bacteria. Red cell components have a reduced risk because many organisms either fail to grow or grow slowly at 1-6C. Platelet components have the highest risk because the 22C storage promotes bacterial growth. Bacteria can proliferate to high titers, especially in platelets prior to transfusion, resulting in complications including morbidity and mortality. The goal was to evaluate the prevention of bacterial growth in whole blood treated with the Mirasol System at a UV energy dose of 80 J/mL_{RBC}, as well as the components derived from Mirasol-treated whole blood. **Study Design/Methods:** WB was collected into either CPD or CPDA anticoagulant and was used for the experiments 24 to 30 hours post-collection. A panel of organisms identified in prior hemovigilance programs was selected for evaluation in this study and included the following species: *S. liquefaciens*, *S. marcescens*, *A. baumannii*, *Y. enterocolitica*, *B. cereus*, *E. coli*, and *K. pneumoniae*. Priority was placed on selecting bacterial strains that had been isolated from blood. Units of WB were spiked with clinically relevant bacteria loads of 10-100 CFU/product. Following Mirasol treatment, units were stored at 22C either as WB or as components (red blood cells, ?, platelet rich plasma [RBC, PPP, PRP]) separated from Mirasol-treated WB. Growth media was added to the RBC component (instead of additive solution) to ensure there would be no inhibition of bacteria growth. On day 7 post treatment, the Mirasol-treated units or components were sampled and screened for bacteria growth by BacT/ALERT. Positive and negative growth controls were also included to ensure the reliability of the results. **Results/Findings:** For reporting purposes, a treated WB unit was considered positive post treatment if any of its components were positive for bacterial growth. When WB products were inoculated with an initial bacterial load of 10-100 CFU per product, the following results were observed. **Conclusion:** Mirasol-treated WB units tested negative for bacterial growth in 96% to 100% of the units inoculated at clinically relevant levels for seven bacterial species of interest.

Disclosure of Commercial Conflict of Interest

S. Doane: Terumo BCT, Other; D. I. Gilmour: Terumo BCT, Grants or Research Support; R. Goodrich: Terumo BCT, Other; N. Hovenga: Terumo BCT, Grants or Research Support; S. D. Keil: Employee of Terumo BCT, Other; H. Reddy: Terumo BCT, Other; S. Wilkinson: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Doane: Nothing to disclose; D. I. Gilmour: Nothing to disclose; R. Goodrich: Terumo BCT, Other; N. Hovenga: Nothing to disclose; S. D. Keil: Employee of Terumo BCT, Other; H. Reddy: Nothing to disclose; S. Wilkinson: Nothing to disclose

	# pos/# tested
Y. enterocolitica	1/23
A. baumannii	0/4
B. cereus	0/3
E. coli	0/2
K. pneumoniae	0/3
S. marcescens	0/3
S. liquefaciens	0/7

SP375

Ultraviolet C (UVC)-Treatment of Residual Leukocytes in Platelet Concentrates Prevents x Graft Versus Host Disease (GVHD) Development in a Nod Scid Gamma (NSG) Mice Model

P Pohler[1] (pohler@bsd-nstob.de), M Mueller[2], D Schaudien[2], K Sewald[2], T H Müller[1], A Seltsam[1]. *[1]Research and Development, Blood Center of German Red Cross, Springe, Germany; [2]Immunology, Fraunhofer Institute of Toxicology and Experimental Medicine, Hannover, Germany*

Background/Case Studies: One of the deleterious effects of leukocytes present in blood components is the transfusion-associated graft versus host disease (TA-GVHD). It is mediated by lymphocytes in donated blood, which proliferate and mount a severe immunological response in susceptible patients. The THERAFLEX UV-Platelets system (MacoPharma) uses ultraviolet C (UVC) light for pathogen reduction. The previous in vitro studies have shown that UVC-irradiation inactivated T-cells by >5 log steps. In the present study, a xenogeneic mouse model was used to evaluate the efficacy of UVC to prevent GvHD in vivo. **Study Design/Methods:** Plasma-reduced PCs prepared from 5 buffy coats in additive solution SSP+ were spiked with 1 x 10E6/mL human peripheral blood mononuclear cells (PBMCs) and treated either with UVC (standard dose 0.2 J/cm2), γ-irradiation (30 Gy) or remained untreated (positive control). PBMCs were removed from PCs and transferred into nod scid gamma (NSD) mice (2.0-2.5 x 10E7 PBMCs/mouse). A total of 12 mice per study arm were treated with PBMCs. GVHD development was observed over a period of 61 days. GVHD were monitored by changes in external behaviour and phenotype, weight loss, survival days, and engraftment of human cells in different organs of the recipient mice by flow cytometry (FC) and immunohistological staining using human specific antibodies (eg CD45, CD3, CD4, CD8, and CD16/CD56). In addition, a comprehensive histopathological evaluation was performed on all mice. **Results/Findings:** All mice receiving untreated human PBMCs exhibited severe symptoms of GVHD with weight loss, ruffled fur, and reduced reactivity within 25 days after injection. High numbers of human T-cells were detected in various tissues like spleen, blood, and bone marrow, but also in the skin, lung, liver, and gastrointestinal tract. Major histopathological findings were human cell infiltration accompanied by a decreased cellularity of the bone marrow and a depletion of the red pulp of the spleen. In contrast, proliferation of human cell and development of GVHD was not detectable in animals inoculated with UVC-treated or gamma-irradiated PBMCs. Interestingly, human lymphocytes were observed in 2/12 mice in the gamma group, whereas in the UVC group no human lymphocytes were detectable. **Conclusion:** UVC irradiation effectively eliminates the proliferative capacity of PBMCs in PCs and inhibits the development of GVHD in a mouse model. These results indicate that the UVC-based THERAFLEX UV-Platelets system has the potential to substitute gamma irradiation for TA-GVHD prevention.

Disclosure of Commercial Conflict of Interest

M. Mueller: Nothing to disclose; T. H. Müller: Nothing to disclose; P. Pohler: Nothing to disclose; D. Schaudien: Nothing to disclose; A. Seltsam: Nothing to disclose; K. Sewald: No Answer

Disclosure of Grants Conflict of Interest

M. Mueller: Nothing to disclose; T. H. Müller: MacoPharma, Grants or Research Support; P. Pohler: MacoPharma, Ownership or Partnership; D. Schaudien: Nothing to disclose; A. Seltsam: MacoPharma, Grants or Research Support, Imusyn, Grants or Research Support; K. Sewald: No Answer

SP376

The Case of Leaky Blood Collection Devices: A "Quick and Dirty" Assessment to Evaluate and Mitigate the Risk

M Germain[1] (marc.germain@hema-quebec.qc.ca), G Delage[1], P Robillard[1]. *[1]Medical Affairs, Héma-Québec, Québec, QC, Canada*

Background/Case Studies: In November 2012, the staff started to report occasional leaks on a specific type of whole blood collection device. The leaks were always in the same location, on the Y-connector linking the

		Probability of bacterial proliferation				
		0,01	0,02	0,04	0,08	0,10
Probability of bacterial entry into the bag	0,005	79 760 198	39 880 099	19 940 050	9 970 025	7 976 020
	0,010	39 880 099	19 940 050	9 970 025	4 985 012	3 988 010
	0,020	19 940 050	9 970 025	4 985 012	2 492 506	1 994 005
	0,040	9 970 025	4 985 012	2 492 506	1 246 253	997 002
	0,080	4 985 012	2 492 506	1 246 253	623 127	498 501
	0,115	3 456 275	1 728 138	864 069	432 034	345 678

phlebotomy needle, the derivation pouch, and the entry line into the bag. This portion of the device is removed after the collection is completed, such that the leak can occur only during the collection phase. The manufacturer confirmed that the leaks were the result of a molding defect of the Y-connectors. Alternate collection sets were not readily available, so that it was necessary to determine the risk of bacterial contamination, if any, associated with using potentially defective devices. **Study Design/ Methods:** The overall risk was estimated by considering the following factors: the rate of observed leaks (0,00115); the probability that one or more bacteria would enter through the leak during collection (0,115); the probability that the bacteria would proliferate to potentially dangerous levels (0,01); the probability that a highly contaminated unit would cause a reaction in the recipient (0,435); and, finally; the probability of death resulting from that reaction (0,5). A sensitivity analysis was conducted in which two factors were varied for which there was little or no data to estimate the true probability. **Results/Findings:** The table shows the sensitivity analysis for the calculated risk of death, expressed as "one in n" units. The risk estimate for the baseline scenario is highlighted. **Conclusion:** Based on data from the literature, it was determined that products collected with potentially defective devices would present no additional risk if used within 14 days of collection, due to the lack of significant proliferation at 4C of most microbes within this timeframe. Beyond 14 days, the risk model predicted a risk of death of one in 3.5 million units in the baseline scenario, with considerable uncertainty as shown by the sensitivity analysis. This small level of risk is comparable to other risks for which mitigation strategies are currently applied; for example, nucleic acid testing to decrease the window period risk. Given the serious potential consequences of bacterial contamination, it was decided to mitigate the risk by systematically testing the individual collection sets for leaks prior to releasing the units through their full shelf life.

Disclosure of Commercial Conflict of Interest

G. Delage: Nothing to disclose; M. Germain: Nothing to disclose; P. Robillard: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Delage: Nothing to disclose; M. Germain: Nothing to disclose; P. Robillard: Nothing to disclose

SP377

Trypanosoma Cruzi Lineage-Specific Serologic Testing: Convergence of Parasitemia, High Antibody Titers, and Blood Donors at Greatest Risk for Transmitting Infection

M Nguyen[1] (Megan.Nguyen@redcross.org), R L Townsend[2], S L Stramer[2], T Bhattacharyya[3], A K Falconar[3], M A Miles[3], D A Leiby[1]. *[1]Transmissible Diseases, American Red Cross, Rockville, MD, United States; [2]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [3]Faculty of Infectious & Tropical Diseases, London School of Hygiene & Tropical Medicine, London, United Kingdom*

Background/Case Studies: Chagas disease is caused by *Trypanosoma cruzi*, (*T. Cruzi*) a genetically diverse parasite classified into the intra-species lineages TcI-VI that display disparate geographical distributions and ecologies throughout the Americas. In an earlier study, it was observed that blood donors infected with lineages TcII, V, and VI were significantly more likely to be parasitemic by hemoculture than donors infected with lineage TcI (45.8% versus 2.2%, respectively) and at a greater risk for transmitting infection by transfusion. *T. cruzi* lineage identification normally requires parasite isolation and subsequent molecular evaluation. However, an experimental assay based on trypomastigote small surface antigens (TSSA) detects *T. cruzi* lineage by serology, thereby offering a more direct approach. **Study Design/ Methods:** To investigate the ability of the TSSA assay to identify blood donors at greatest risk for transmitting infection, two panels of donor serum/ plasma were evaluated. Panel 1 (P1) was obtained from 23 parasitemic donors with molecularly characterized *T. cruzi* lineage (TcI, n = 4; TcII, n = 1; TcV, n = 16; TcVI, n = 2), while Panel 2 (P2) consisted of 43 seropositive and 7 seronegative from non-parasitemic donors (ie lineage not determined) with known country of birth (COB). Based on s/co values, samples associated with P1 (ie parasitemic donors) had higher antibody titers than those in P2. Samples were diluted 1:200, blinded, and tested using a lineage-specific peptide assay (devised by Bhattacharyya, Falconar and Miles—in preparation) to detect TcII/V/VI, III, IV and V/VI lineages. *T. cruzi* lysate was used as a control. **Results/Findings:** All 23 plasma samples in P1 reacted with the crude *T. cruzi* lysate confirming they were serologically positive for *T. cruzi*. When tested against the lineage-specific peptides, 7/23 (30%) showed reactivity with TcII/V/VI; all 7 had previously been typed as TcV (COBs; 4-Bolivia, 1-Argentina, 1-Brazil, 1-Chile). One of these 7 positives also reacted with the TcV/VI peptide. The 7 seronegative samples in the P2 were non-reactive; 7/43 (16%) confirmed positive samples were also non-reactive. The remaining 36 seropositives reacted with the *T. cruzi* lysate, but only 2 (5%) showed reactivity with the TcII/V/VI peptide (COBs; 2-Argentina). **Conclusion:** Previous studies suggested that donors infected with TcII, TcV and TcVI lineages were more likely to be parasitemic and transmit T. cruzi by blood transfusion. Evidence from the current study suggests that a serological assay may allow for rapid identification of these donors (ie those reactive with the TcII/V/VI antigen). Ability to detect specific lineages in donors is enhanced by higher antibody titers and demonstrable parasitemia. Studies are underway to increase assay sensitivity and further refine identification of donors infected with *T. cruzi* intra-specific lineages. Supported in part by EC FP7 contract 223034 (ChagasEpiNet).

Disclosure of Commercial Conflict of Interest

T. Bhattacharyya: Nothing to disclose; A. K. Falconar: Nothing to disclose; D. A. Leiby: Nothing to disclose; M. A. Miles: Nothing to disclose; M. Nguyen: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Bhattacharyya: Nothing to disclose; A. K. Falconar: Nothing to disclose; D. A. Leiby: Terumo BCT, Grants or Research Support; M. A. Miles: Nothing to disclose; M. Nguyen: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose

SP378

Demographic Profile of Blood Donors Deferred in Singapore due to Malaria Testing

R Alcantara[1] (ramirmd@yahoo.com), H Tan[1]. *[1]Blood Services Group, Health Sciences Authority, Singapore, Singapore*

Background/Case Studies: Transmission Transmitted Malaria (TTM) is one of the rare but serious complications of blood transfusion.The main risk for TTM comes from semi immune blood donors who are asymptomatic and have very low parasitemia in their blood. Singapore is a non-malaria endemic area but is surrounded by countries where malaria is endemic. Due to the frequent travel by its residents for business and leisure, and with the influx of foreign workers into Singapore, this is now of great concern. **Study Design/Methods:** The objective of the study is to determine the demographic profile of blood donors deferred due to malaria test results using polymerase chain reaction (PCR) and malaria antibody (EIA) testing. From September 1, 2012-February 28, 2013, there were 43,661 blood donors who donated blood in Singapore. Donors underwent pre donation medical screening and were asked if they previously stayed in a malaria endemic area for 6 consecutive months or recently traveled to a malaria endemic area. Donors were tested for malaria using PCR and Enzyme Immunoassay for the presence of antibodies against malaria. A total of 4247 donors were tested during the study period. **Results/Findings:** Only 68 (1.6%) were found to be repeatedly positive for the presence of malaria antibodies. None of the donors presented with detectable parasitemia by PCR and only 22 (32%) of the donors reported having a previous history of malaria infection. 26 (38%) of the donors were citizens and permanent residents while 42 (62%) were foreigners currently residing in Singapore. Other demographic parameters are in the Table. **Conclusion:** Well over half (62%) of the donors who were deferred are foreigners from a malaria endemic area. Analysis of the demographic profile of donors deferred due to malaria testing may identify critical areas (eg donor screening) in the donation process. Data collected may also be used in the formulation or revision of existing blood donation criteria and testing guidelines. In addition, blood centers can improve on their pre-donation education materials and strategies on preventing TTM by identifying donors who are more likely to be infected and

explaining more clearly the risks of malaria and the reason for the deferral.

Disclosure of Commercial Conflict of Interest

R. Alcantara: Nothing to disclose; H. Tan: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Alcantara: Nothing to disclose; H. Tan: Nothing to disclose

TABLE 1. Donor Demographics for Donors Deferred due to Malaria Testing

Gender		Number of donors with Travel to Malaria Endemic Area		EIA positive donors
Male	**62**	**India**	**2137**	**42**
Female	**6**	**Myanmar**	**276**	**22**
Residency Status		**Pakistan**	**1**	**1**
Foreigners	**42**			
Citizens/Permanent Resident	**26**	**Papua New Guinea**	**2**	**1**
Age Group		**Thailand**	**116**	**1**
18-30	**17**	**Indonesia**	**1101**	**0**
30-40	**32**	**Cambodia**	**157**	**0**
40-50	**15**	**Vietnam**	**45**	**0**
>50	**4**	**Philippines**	**55**	**0**
History of Malaria		**Malaysia**	**67**	
YES	**22**	**Others**	**289**	**0**
NO	**46**	**Not sure**	**1**	**1**

SP379

Serologic Evidence of Possible Transfusion-Transmission of *Babesia microti* in Maryland in 1987

P A Brunker[1] (pbrunke1@jhmi.edu), A M Thorp[2], L Tonnetti[2], K Nelson[3], D A Leiby[2], P M Ness[1]. *[1]Pathology, Div. of Transfusion Medicine, The Johns Hopkins School of Medicine, Baltimore, MD, United States; [2]Transmissible Diseases, American Red Cross, Rockville, MD, United States; [3]Epidemiology, The Johns Hopkins School of Public Health, Baltimore, MD, United States*

Background/Case Studies: First identified in a febrile patient on Nantucket Island in 1969, human infection by *Babesia microti* (*B. microti*)has progressively claimed a wider endemic area across the northeastern US seaboard. Babesiosis has been a reportable disease in Maryland since 2007. Since the first case of transfusion-transmitted babesiosis (TTB) in Boston in 1979, reports are increasing. In this study it was hoped to find if any transfusion-transmission or prevalent cases may have occurred as far south as Maryland earlier in the epidemic than previously documented. **Study Design/Methods:** An immunofluorescent assay (IFA) was performed for *B. microti* on serum from 2158 patients who underwent cardiac surgery at Johns Hopkins Hospital between 1985-1991 as part of the Frequency of Agents Communicated by Transfusion Study (FACTS). Samples collected 6 months post-operatively were screened and pre-transfusion samples were tested if the post-transfusion sample screened positive to determine if a case was seroincident or seroprevalent. IFA slides were produced in-house from *B. microti*-infected hamster blood. Briefly, samples were diluted 1:64 in PBS and 20 μL of the dilution was added to the slide and incubated at 37C for 30 minutes, washed in PBS for 10 minutes, rinsed, and air dried. 20 μL of FITC labeled goat anti-human IgG was added to each well and incubated again at 37C for 30 minutes followed by the same wash. Samples were analyzed at 40X on a fluorescent microscope and positives (≥1:64) were reanalyzed up to a dilution of 1:1024. **Results/Findings:** Six post-transfusion samples (0.28%) tested positive with a titre ≥1:64. Pre-transfusion aliquots were available for five. Three/five had both pre- and post-transfusion samples with the titres at or near the 1:64 cut-off; these were considered equivocal. One sample had both pre- and post-transfusion titres of 1:1024, consistent with *Babesia* exposure pre-transfusion. However, one sample demonstrated a rise in titre from <1:64 pre-transfusion to 1:128 6 months post-transfusion, raising the possibility of exposure to *B. microti* during the transfusion. **Conclusion:** Serological evidence of seroprevalent *B. microti* exposure was foundin a patient in Maryland in archived samples from 1988, which is earlier than previously documented in this state. Moreover, serum from one patient, a 42-year-old man who received two units of RBCs and one pool of cryoprecipitate during surgery, showed increased reactivity post-transfusion with a negative result pre-transfusion. This raises the possibility of a case of TTB in Maryland as early as 1987. The results suggest that *B. microti* may have been more widespread earlier than previously believed, and that TTB may have been clinically under-appreciated in Maryland as early as the late 1980s. For diseases like babesiosis that feature non-specific symptomatology, serological testing should be aggressively pursued to improve hemovigilance programs and improve detection of these often cryptic diagnoses.

Disclosure of Commercial Conflict of Interest

P. A. Brunker: Nothing to disclose; D. A. Leiby: Nothing to disclose; K. Nelson: Nothing to disclose; P. M. Ness: Nothing to disclose; A. M. Thorp: Nothing to disclose; L. Tonnetti: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. A. Brunker: Nothing to disclose; D. A. Leiby: Terumo BCT, Grants or Research Support; K. Nelson: Nothing to disclose; P. M. Ness: Nothing to disclose; A. M. Thorp: Terumo BCT, Grants or Research Support; L. Tonnetti: Terumo BCT, Grants or Research Support

SP380

Probable Transfusion-Transmission of Anaplasma phagocytophilum (AP) by leukoreduced platelets

R L Townsend[1] (Rebecca.Townsend@redcross.org), R D Calhoun[1], L B Fialkow[1], V P Berardi[2], S L Stramer[1]. *[1]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [2]IMUGEN, Norwood, MA, United States*

Background/Case Studies: Anaplasma phagocytophilum (AP) is a tick-borne bacterial (rickettsial) obligate intracellular pathogen that preferentially infects neutrophils and causes human granulocytic anaplasmosis; endemic areas are shared with Lyme and babesiosis as the same tick vector is involved. AP is also transfusion transmitted (TT): three cases involving nonleukoreduced (non-LRd) red blood cells (RBCs) (1998-2010; US-Minnesota (MN) and Slovenia; RBC age 8-30 days), two cases involving LRd RBCs (2010-2011; US-Wisconsin[WI] and RI; age not given-12 days) and one case with no implicated product (2007; US-WI; 25 LRd RBCs/apheresis platelets, 5 granulocytes, all irradiated) were published. Here we report the first probable case of TT-Anaplasma involving LRd platelets (US-Connecticut[CT]) is reported here. **Study Design/Methods:** A 41-year-old male, hospitalized for multiple gunshot wounds, received 13 LRd RBCs, 3 apheresis platelets and 9 FFP (25 total) in November 2012. Testing performed on the recipient by the hospital included: peripheral blood smears collected weekly (November 11-27) progressing from (-) to slightly and then predominantly (+); on December 3 (22 days post transfusion), PCR was (+) for AP. Recipient samples (November 25, November 27, and December 4) were sent to IMUGEN (Norwood, MA) for AP PCR and AP IgM/IgG ELISA; all were PCR (+) and IgM/IgG (-) (13, 15 and 23 days post transfusion). No recipient history is available for prior health, travel, or other risks for AP infection; the recipient died December 5 from trauma. All donors associated with the transfused (txd) components were investigated; 13 were excluded (9 FFP-since acellular, 4 RBCs-txd post AP (+) results). Ten of 12 donors provided follow-up (f/u) samples: all were PCR and ELISA (-); one RBC donor is pending. The 12th donor (platelets txd at day 5) provided 4 f/u samples. The donor is a 52-year-old-female from the Hudson Valley area of New York [NY]; reported endemic for AP). She was healthy at the time of donation but reported tick bites in September-October 2012 with no travel outside of NY. The donor remains asymptomatic and has received no treatment for AP from her physician due to resolution of PCR (+); see Table. The co-component implicated platelet was txd to a 78-yea-old male who died of causes unrelated to AP. **Results/Findings:** See Table **Conclusion:** This seventh case of probable TT-AP indicates that LRd platelets may also be infectious. Although genetic linkage studies were not done, a single antibody (+), PCR (+) donor was identified who had prior tick exposure from an endemic area of the US; the donor's elevated IgG levels (that continue to exceed the detectible range of the assay) even in the absence of IgM likely indicate recent infection.

Disclosure of Commercial Conflict of Interest

V. P. Berardi: Imugen, Ownership or Partnership; R. D. Calhoun: Nothing to disclose; L. B. Fialkow: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. P. Berardi: Nothing to disclose; R. D. Calhoun: Nothing to disclose; L. B. Fialkow: Nothing to disclose; S. L. Stramer: Nothing to disclose; R. L. Townsend: Nothing to disclose

Donor Follow-up Testing

Collect Date	Days Post Donation	PCR	ELISA Signal/cutoff (where ≥1=+) IgM	IgG
1/16/2013	70	Not Tested	<1	>7.8
2/20/2013	**105**	**Positive**	<1	>7.5
3/30/2013	143	Negative	<1	>8.1
4/15/2013	159	Negative	<1	>8.0

SP381

First Proof of PrP^{TSE} Presence in Blood-Circulating Exosomes from Transmissable Spongeform Encephalopathies (TSE)-Infected Mice by In Vitro Amplification of Misfolded Prion Protein

P Saa[1] (paula.saa@redcross.org), O Yakovleva[1], J de Castro[1], I A Vasilyeva[1], L Cervenakova[1]. *[1]Transmissible Diseases Department, American Red Cross Holland Laboratory, Rockville, MD, United States*

Background/Case Studies: Exosomes (membrane vesicles of 30-120 nm in diameter) were originally considered to be a cellular mechanism to eliminate unwanted proteins. Recent investigations however, have attributed new roles to these vesicles including intercellular communication and agent dissemination. Little is known about the distribution of prions (the agents implicated in transmissible spongiform encephalopathies (TSEs) in blood. Cellular prion protein, PrP^{C}, and its misfolded counterpart, PrP^{TSE}, have been identified in exosomes from various TSE models; and exosomes obtained from TSE-infected cell cultures have caused clinical disease in mice. Exosomes isolated from blood components contain PrP^{C}. This finding has raised the possibility that they may also contain PrP^{TSE} and serve as vehicles for the transfusion transmission of TSEs. In this study we isolated exosomes from plasma samples of mice infected with human TSE agents and demonstrated the presence of PrP^{TSE} by biochemical methods. These findings set the ground for the design of novel blood-based diagnostic tests for TSEs. **Study Design/Methods:** Exosomes were isolated with ExoQuick reagent (System Biosciences) from 250 µl of platelet poor plasma collected from clinically sick mice infected with human TSEs, and uninfected controls. Exosome pellets were mixed with uninfected mouse brain homogenates and amplified by saPMCA method (which allows the multiplication of minute amounts of PrP^{TSE} to levels detectable by standard biochemical assays). After proteinase K digestion, PrP^{TSE} was detected by Western blot using the anti-PrP monoclonal antibody 6D11. **Results/Findings:** Exosomal preparations were positive for exosomal markers Hsp70 and Flotillin by Western blotting. saPMCA allowed the specific detection of PrP^{TSE} in samples from TSE-infected animals. Different numbers of saPMCA rounds were necessary to amplify PrP^{TSE} to detectable levels in individual exosomal samples. These findings suggest that different quantities of PrP^{TSE} are present in plasma at the terminal stage. **Conclusion:** The biochemical detection of PrP^{TSE} in plasma exosomes, obtained from clinically sick animals, is reported here for the first time. These experiments provide an invaluable foundation for the development of new diagnostics and potential targets for TSEs treatment. Further characterization of PrP^{TSE}-containing microvesicles may lead to the identification of the cellular origin of prion-containing exosomes and the site(s) of prion replication in blood. Support. The study was partially supported by Fondation Alliance BioSecure, France.

Disclosure of Commercial Conflict of Interest

L. Cervenakova: Nothing to disclose; J. de Castro: No Answer; P. Saa: Nothing to disclose; I. A. Vasilyeva: Nothing to disclose; O. Yakovleva: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Cervenakova: Nothing to disclose; J. de Castro: No Answer; P. Saa: Nothing to disclose; I. A. Vasilyeva: Nothing to disclose; O. Yakovleva: Nothing to disclose

SP382

Prevalence of *T. Cruzi* Antibodies in Blood Donors from a Blood Bank Placed in a Non Endemic Area with High Migration Rates

G A Orjuela[1] (guillermo.orjuela@cruzrojacolombiana.org), J D Muñoz[1]. *[1]National Blood Bank, Colombian Red Cross, Bogotá, Colombia*

Background/Case Studies: Chagas disease, an infection caused by *Trypanosoma cruzi* (*T. Cruzi*), is endemic in several regions in Central and South America. Increasing migration rates have extended the number of cases to non-endemic regions such Europe and the US. Several mitigation strategies have been implemented in recent years for preventing transfusion-associated cases in these regions. In Colombia, serologic screening for *T. Cruzi* antibodies to all collected blood became mandatory in 1993. During the last 10 years, the capital city of Bogotá, a non endemic region, has experienced substantial migrations from the countryside caused by the internal social conflict. The prevalence of *T. cruzi* antibodies in blood donors from a reference blood bank placed in a non-endemic area (Bogotá, 8530 ft AMSL), during the last 8 years is reported here. **Study Design/Methods:** Results of screening tests for *T. cruzi* were retrospectively reviewed for all blood donations from January 2005-December 2012 at the National Blood Bank Colombian Red Cross in Bogotá, Colombia. Various ELISA-based assays were used for donor screening until January 2012, when a chemiluminescence method was introduced (ARCHITECT Chagas assay, Abbott GmbH & Co. KG). From 2005-2007 and 2009: ELISA cruzi test (bioMérieux, Mercy-l'Etoile, France). From 2008-2009: Chagatest (Wiener Laboratorios S.A.I.C. Argentina). From 2010-December 2011: ORTHO ELISA Test System (Johnson & Johnson, HighWycombe, UK). All reactive samples were confirmed by Indirect Immunofluorescence. **Results/Findings:** In total, 191,964 donations were tested during the time of the study, resulting in 645 samples being reactive. Only 280 samples were confirmed to be positive. The reactivity rate for *T. cruzi* was 0.37%, although, as the table shows, in 2007 and 2008 there was an unexpected rise in the detection rate, explained by transitory problems with the cut off values assigned during that time. The positivity rate was 0.14% (1 confirmed case every 686 units). **Conclusion:** The reactivity and positivity rates found in this bank are much higher than those reported recently in the US or Europe. Based on these data, Bogotá has comparable positive rates to other endemic places in the world.

Disclosure of Commercial Conflict of Interest

J. D. Muñoz: Nothing to disclose; G. A. Orjuela: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. D. Muñoz: Nothing to disclose; G. A. Orjuela: Nothing to disclose

Year	2005	2006	2007	2008	2009	2010	2011
Total Donors	22053	23862	23494	24270	25497	25970	21650
Reactive donors	65	60	167	175	80	43	55
Confirmed cases	32	35	45	35	36	31	35

Transfusion-Transmitted Infectious Diseases: Viruses

SP383

Eleven-Year Experience of Parental and Non-Parental Whole Blood Donor Infectious Disease Testing at a Pediatric Hospital Blood-Donor Center

N Lezama[1] (lezama.niara@gmail.com), V Criss[1], E Gourdine[1], C Colvin[3], N L Luban[1,2], E Wong[1,2]. *[1]Laboratory Medicine, Children's National Medical Center, Washington, DC, United States; [2]Pediatrics and Pathology, George Washington School of Medicine and Health Sciences, Washington, DC, United States; [3]Children's Research Institute, Children's National Medical Center, Washington, DC, United States*

Background/Case Studies: Directed donation is well known to be associated with increased infectious disease rates. However, little is known of the infectious disease risk of parental donation as a special subcategory of directed donations. It was hypothesized that: 1) parental donors would have higher infectious disease risk than homologous donors because of internal parental incentives to donate in order to help their child; and 2) have similar or greater infectious disease risk than non-parental directed donors because of internal pressure to donate. **Study Design/Methods: An** 11 year retrospective study (January 1997-December 2008) of whole blood donation at a pediatric hospital blood donor center was undertaken. Review of blood donor deferral records, including infectious disease testing positivity, was performed to obtain transfusion transmitted risk. Total yearly blood donations were stratified by donor relationship (parental, non-parental directed, and homologous) and first time versus repeat donation. Median (5th and 95th percentile) yearly incidence of positive infectious disease testing was determined for all groups. Mann-Whitney U test was used to compare median incidence rates between each group. **Results/Findings:** Over an 11-year period, a total of 23,865 first time and repeat donors attempted whole blood donation. Although there was a trend towards higher median yearly incidence of positive infectious disease testing among parental donors

compared to non-parental directed donors and homologous donors, the median yearly incidence rates between the donor groups (including first time and repeat donors) were not statistically significant. However, within the homologous donor group, first-time and repeat-donor median incidence rates were significantly different ($p < 0.001$). Among first-time donors, median incidence rates of non-parental directed donors were significantly lower than the median incidence rates of parental and homologous donors, ($p = 0.0262$ and $p = 0.0056$, respectively). Among repeat donors, there was no significant difference between the median incidence rates among the three groups. **Conclusion:** Overall, parental donors appear to be similar to homologous donors in terms of infectious disease risk. However, first–time, non-parental directed donors appear to have lower infectious disease risk than either first-time parental or homologous donors. This may reflect the recruitment methodology of prescreening both directed and parental donors. Alternatively, directed donors may have donated elsewhere and therefore may be a "preselected" population.

Disclosure of Commercial Conflict of Interest

C. Colvin: Nothing to disclose; V. Criss: Nothing to disclose; E. Gourdine: Nothing to disclose; N. Lezama: Nothing to disclose; N. L. Luban: Nothing to disclose; E. Wong: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Colvin: Nothing to disclose; V. Criss: Nothing to disclose; E. Gourdine: Nothing to disclose; N. Lezama: Nothing to disclose; N. L. Luban: Nothing to disclose; E. Wong: Nothing to disclose

Donor population	N	% (median)	5th Percentile	95th Percentile
First Time Parental	917	2.91	0.72	8.72
Repeat Parental	615	3.36	0.00	16.86
First Time Non-Parental Directed	2,338	1.09	0.16	5.60
Repeat Non-Parental	2,572	0.26	0.00	3.53
First Time Homologous	6,233	3.16	2.33	5.48
Repeat Homologous	11,190	0.29	0.04	0.16

SP384

Donor Management for NAT Non-Discriminated Reactive Test (NDR) Results

N Kiely[1] (nkiely@bloodsystems.org), M Etem[1], H Kamel[1]. *[1]Blood Systems, Scottsdale, AZ, United States*

Background/Case Studies: Non-discriminated reactive test results (NDR) using the Chiron Procleix HIV-1/HCV nucleic acid amplification test (NAT) assay resulted in the deferral of 445 blood donors between April 12, 1999-August 2, 2010. Each blood donation tested reactive(R) in the multiplex assay (HIV-1-HCV mTMA assay), non-reactive(NR) in discriminatory assays (HIV-1 and HCV dTMA assays), and NR in HIV-1/2 and HCV serological assays. **Study Design/Methods:** Records were evalutated of donors deferred for NAT NDR results. Those who successfully completed reentry were identified as well as the proportion of those who presented to donate following successful reentry. The number of subsequent donations collected were determined as well. **Results/Findings:** Of the 445 donors deferred for NAT NDR result, 116 were reentered (26%). Sixty one (53%) had successful subsequent donations, five donors attempted to donate but had unsuccessful phlebotomy and one donor deferred himself for another reason and could not donate. No donor who donated subsequently had a second R NAT test result. A total of 958 donations were gained from the reentered donors (an average of 15.7 donations per reentered donor). **Conclusion:** Based on the modest proportion of donors returning for reentry evaluation and the number of reentered donors with subsequent donations, there is a need to investigate why some donors do not present for sample collection when invited for reentry evaluation and why some people do not return to donate blood after requalification. These efforts represent good customer service to healthy donors with false reactive test results but there is a need to focus future strategy on what can be done to enrich the return and improve donation rate after reentry.

Disclosure of Commercial Conflict of Interest

M. Etem: Nothing to disclose; H. Kamel: Nothing to disclose; N. Kiely: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Etem: Nothing to disclose; H. Kamel: Nothing to disclose; N. Kiely: Nothing to disclose

SP385

Trend of Transfusion-Transmissible Infections (TTIs) among Blood Donors in Ethiopia

D G Burssa[1] (dburssag@gmail.com). [1]Ethiopian National Blood Transfusion Service, Addis Ababa, Ethiopia

Background/Case Studies: Transfusion transmitted infections create significant burden on health care system. Donor selection is of paramount importance because infected individuals serve as an asymptomatic reservoir and a potential source of transmission. The objective of this study is to determine the trend of transfusion transmitted infections among blood donors. **Study Design/Methods:** This is a retrospective study using data on donation testing for transfusion-transmissable infections (TTIs) from 2004-2011 at the national blood service donor database. During this period 286,570 donated bloods were screened for hepatitis B virus (HBV), hepatitis C virus (HCV), and human immunodeficiency virus (HIV) infections using enzyme-linked immunosorbent assays (ELISA); and positive results were discarded. **Results/Findings:** The prevalence of HBV, HCV, and HIV in 2004 was 5.6%, 5.4%, and 3.7% and decreased to 4.9%, 0.31%, and 0.68% in 2011, respectively. Generally the trend of prevalence of TTIs was sharp decline especially for HCV and HIV. The Prevalence of TTIs among Ethiopian blood donors was substantially lower than that in the general population. **Conclusion:** This study reflects the risk of TTIs has been greatly reduced in Ethiopia. The lower prevalence of TTIs in blood donors compared with the general population reflects the effectiveness of donor education and donor selection measures, but blood transfusion remains an ongoing risk factor for the spread of blood-borne infections, and further work and improvements are needed to strengthen both safety and availability of blood in Ethiopia.

Disclosure of Commercial Conflict of Interest

D. G. Burssa: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. G. Burssa: Nothing to disclose

SP386

Prioritizing Emerging Infectious Diseases(EIDs) for Transfusion Safety through Expert Elicitation

W Oei[1] (W.Oei@umcutrecht.nl), R Neslo[1], M Janssen[1]. *[1]Transfusion Technology Assessment, UMC Utrecht, Utrecht, Netherlands*

Background/Case Studies: Increasing identification of transfusion transmission of emerging infectious diseases (EIDs) raised the question which EIDs pose a risk for blood safety. A direct risk based prioritization is hindered by a lack of data as often uncertainty exists about some characteristics of EIDs. The aim of this study is therefore to obtain insight in how experts in the field of infectious disease control and blood transfusion prioritize risks from infectious diseases with various known and unknown disease characteristics. **Study Design/Methods:** A multiple ranking exercise is performed whereby experts rank sets of diseases with pre-specified characteristics: transfusion transmissibility; prevalence of infection; disease impact; and the asymptomatic phase of infection. For each of these characteristics six types of uncertainty are defined (with an uncertainty range presented as a percentage): unknown (1-99), likely (50-99), unlikely (1-50), very likely (75-99), possible (25-75), and very unlikely (1-25). Twenty-four hypothetical diseases were selected to be ranked in groups of four; two for low, moderate, and high risks each. These were disseminated in the form of an easy-to-use web-based form to experts from the Infectious Disease Monitor network (European Blood Alliance) and the ECDC Advisory Forum. In addition, experts were asked to rank eight existing diseases in two (overlapping) groups of five diseases as an external validation of the model. The perceived weights for each of the characteristics were obtained and validated using probabilistic inversion. **Results/Findings:** Sixteen out of thirty invited experts filled out the web-based form. Most experts had more difficulties ranking the hypothetical EIDs than ranking the existing EIDs. On the other hand, analyses of the responses show more agreement among experts in ranking the hypothetical EIDs than in ranking the existing EIDs. Furthermore, the rankings from the hypothetical EIDs show changes in preference of factors depending on the level of uncertainty and risk: disease impact was found to be most important for EIDs with medium to high levels of uncertainty; presence of asymptomatic phase was considered relevant for EIDs with low risk levels; and prevalence of infection is considered most relevant for EIDs with high risk levels. The rankings of the existing EIDs could be recovered quite well using the weights obtained from the ranking of the hypothetical diseases. **Conclusion:** The results from this study allows ranking of currently known emerging infections in accordance with the

revealed preferences of a group of experts in the field, and ranking of the risk of newly discovered EIDs according to the risk of emerging infections currently known. This provides a transparent and rational approach towards the interpretation of the risks from both existing and new EIDs.

Disclosure of Commercial Conflict of Interest

M. Janssen: Nothing to disclose; R. Neslo: Nothing to disclose; W. Oei: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Janssen: Sanquin Blood Supply Foundation, Other; R. Neslo: Nothing to disclose; W. Oei: Nothing to disclose

SP387

Sensitivity of Individual and Mini-Pool Nucleic Acid Testing (NAT) Assessed By Dilution of NAT Yield Samples

N Agarwal[1] (nuts_medico@rediffmail.com), K Chatterjee[1]. *[1]Transfusion Medicine, AIIMS, New Delhi, India*

Background/Case Studies: The replication rate of hepatitis B virus (HBV) is very low, and the viral load is also very low during the window phase. Depending on the sensitivity of the test, implementation of HBV nucleic acid testing (NAT) has the potential to reduce risk of infection to levels similar to those for human immunodeficiency virus (HIV) and hepatitus C virus (HCV). It has been debated in various studies whether pooling of samples results in decreased sensitivity of detection. Objective of this study was to investigate the effect of dilution on the sensitivity of tests with low viral load, serology negative, ID-NAT HBV reactive samples. **Study Design/Methods:** The study was performed on nine plasma samples which were exclusive Procleix Ultrio Plus HBV yields ie NAT Ultrio Plus Assay reactive but Serology and Procleix Ultrio non reactive during a Procleix Ultrio Plus pilot study1 conducted on blood donors for a period of 1 year, from September 2011-September 2012, at All India institute of Medical Sciences in New Delhi India. The samples were ID NAT tested by TMA technology using Procleix Ultrio Plus Assay for simultaneous detection of HIV-1, HCV-RNA, and HBV-DNA. Real Time PCR test for HBV viral load using COBAS Ampliprep and COBAS Taqman was performed on the nine samples at Dr.Lal's Path Lab, Rohini, Delhi (CAP & NABL accredited) to quantify the virus in each of the sample. These nine exclusive Ultrio Plus ID-NAT yield samples were diluted in 1:2, 1:4, 1:6, and and 1:8 dilutions using previously tested negative plasma and each dilution of every sample along with undiluted sample were retested in three replicates with Procleix Ultrio Plus Assay. **Results/Findings:** 8/9 samples showed presence of HBV DNA by Real Time PCR while 1 sample was not detected. This may be due to the low sensitivity of PCR kits as compared to Procleix Ultrio Plus kits. Among the eight samples which were detected by PCR, six had less than 20 IU/mL viral load, one sample had 810 IU/mL and another sample had 50 IU/mL viral load. All nine samples were further retested in neat and dilutions in triplicate and results observed are in Table below. **Conclusion:** In India, a high number of NAT yields are associated with low viral load especially with HBV and is apparent from the fact that out of 20 Ultrio Plus and Ultrio Yields for our initial Ultrio Plus pilot study, at least seven or 35% were low viral loads and were a mix of Window Period and Occult HBV cases. Individual Donor NAT is ideal methodology for NAT as dilution due to pooling may miss samples with low viral load.

Disclosure of Commercial Conflict of Interest

N. Agarwal: Nothing to disclose; K. Chatterjee: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Agarwal: Nothing to disclose; K. Chatterjee: Nothing to disclose

S/N	UP Yields	Viral Load Real Time PCR	Reactivity Neat samples	Reactivity 1:2 Dilutions	Reactivity 1:4 Dilutions	Reactivity 1:6 Dilutions	Reactivity 1:8 Dilutions
1	HBV	<20 IU/mL	3/3	2/3	1/3	1/3	2/3
2	HBV	<20 IU/mL	3/3	1/3	0/3	0/3	0/3
3	HBV	<20 IU/mL	3/3	2/3	1/3	0/3	0/3
4	HBV	<20 IU/mL	2/3	2/3	0/3	0/3	0/3
5	HBV	810 IU/mL	3/3	3/3	3/3	3/3	3/3
6	HBV	50 IU/mL	3/3	3/3	3/3	3/3	3/3
7	HBV	Not Detected	2/3	0/3	0/3	0/3	0/3
8	HBV	<20 IU/mL	2/3	2/3	1/3	0/3	0/3
9	HBV	<20 IU/mL	3/3	3/3	2/3	2/3	1/3

SP388

Evaluation of Procleix Ultrio Plus Nucleic Acid Test (NAT) Assay for Blood Donor Screening in a Tertiary Care Center in North India

R N Makroo[1] (makroo@apollohospitals.com), A Bhatia[1], M Chowdhry[1], B Arora[1], M[1]. *[1]Transfusion Medicine, Indraprastha Apollo Hospitals, New Delhi, India*

Background/Case Studies: The PROCLEIX ULTRIO PLUS Assay (Novartis Diagnostics, USA) is a new generation qualitative in vitro nucleic acid amplification test (NAT) used to screen for human immunodeficiency virus type 1 (HIV-1) RNA, hepatitis C virus (HCV) RNA, and hepatitis B virus (HBV) DNA in blood donors. India lies in the intermediate zone of HBV endemicity and the problem of window period or occult HBV infections are still a cause of concern. There is thus a growing need for a more sensitive test for detecting these infections especially HBV. This study was performed to compare the existing Procleix Ultrio assay with the new generation Procleix Ultrio Plus Nucleic Acid Test assays. **Study Design/Methods:** 10,302 donor samples were run in parallel for ID NAT using Procleix Ultrio and the Procleix Ultrio Plus assay. Simultaneously ELISA testing was performed on EVOLIS Walk away System (Bio-Rad) for HIV (Genscreen Ultra, 4th generation kits), HCV (Qualisa), HBsAg (Monolisa) and anti-HBc (IgG + IgM, Hepanostika). All initial NAT reactive samples were retested in triplicate and discriminatory assays for HIV 1, HCV, and HBV were performed. Reactive samples were confirmed using PCR. (Cobas Taqmam, Roche, USA). **Results/Findings:** In the 10, 302 samples tested during the study period, 15 cases were identified that were negative for HBsAg on ELISA but positive by initial NAT in either or both of two assays and subsequently HBV DNA was detected in the corresponding discriminatory assays. Eight of these were exclusive yields from the Ultrio Plus assay. All eight of these cases were detected as HBV in the discriminatory assays. However, only six out of these showed the presence of HBV DNA by PCR. Anti HBc antibodies were found in 7/8 cases indicating seropositive occult Hepatitis and one case was non reactive for anti HBc antibodies. The remaining seven cases were determined as HBV NAT yield both by the Procleix Ultrio as well as the Ultrio Plus assays, five were confirmed by PCR, and all were reactive for anti HBc antibodies by ELISA. No HCV or HIV 1 yields were detected during the study period by either of the two assays. **Conclusion:** With an overall yield rate of 1 in 687 and an exclusive yield rate of 1 in 1287, the Procleix Ultrio Plus assay proved to be highly sensitive in detecting occult HBV infections. Its incorporation in the blood screening protocol in place of Procleix Ultrio assay may go a long way in interdicting HBV infections, the incidence of which is still very high in India.

Disclosure of Commercial Conflict of Interest

M. : No Answer; B. Arora: Nothing to disclose; A. Bhatia: No Answer; M. Chowdhry: No Answer; R. N. Makroo: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. : No Answer; B. Arora: Nothing to disclose; A. Bhatia: No Answer; M. Chowdhry: No Answer; R. N. Makroo: Nothing to disclose

SP389

Organ Donor Screening using Nucleic Acid Testing (NAT)

S Jones[1] (scott.jones@qualtexlabs.org), K Leighton[1], L K Myers[1]. *[1]Lab Administration, QualTex Laboratories, San Antonio, TX, United States*

Background/Case Studies: Nucleic acid testing (NAT) for human immunodeficiency virus (HIV), hepatitis C virus (HCV) and hepatitis B virus (HBV) does increase the cost and time required for testing but decreases the window period between infection and detection. NAT assays are not required by the United Network for Organ Sharing (UNOS) to be performed by organ procurement organizations (OPOs) for organ donor screening; however over 80% of OPOs perform some level of NAT. The purpose of this study is to evaluate NAT results for organ donor screening from an OPO screening all donors using NAT. **Study Design/Methods:** All organ donors were screened using the PROCLEIX ULTRIO Assay on the PROCLEIX TIGRIS System. Individual samples were tested and discriminatory assays were run on ULTRIO positive samples. The percentage of ULTRIO positive samples and false positive samples was calculated. The turn around time from when samples were received to when results were sent to the customer was also calculated. The number of organs transplanted per donor was calculated for donors prior to NAT being implemented and after NAT implementation. **Results/Findings:** A total of 135 OPO donor samples were screened by NAT from 2011-May 2013. The rate of ULTRIO positive samples was 6.7% (9/135). The false positive rate was 0.0% (0/135). All nine ULTRIO positive samples were positive for HCV and negative for HIV and HBV using the discriminatory assays. The average turn-around time for this time period was

7.5 hours. The number of organs transplanted per donor did increase significantly (10%) after NAT was implemented for all donations. **Conclusion:** NAT was performed in a timely manner and false positive results did not affect the number of organs transplanted. Overall, performing NAT on all donors did not impede the transplantation process but there was a significant increase in the number of organs transplanted per donor.

Disclosure of Commercial Conflict of Interest

S. Jones: Nothing to disclose; K. Leighton: Nothing to disclose; L. K. Myers: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Jones: Nothing to disclose; K. Leighton: Nothing to disclose; L. K. Myers: Nothing to disclose

SP390

Case Report: A Repeat Blood Donor's Testing and Identification of Potential Public Health Risk

M H Fucci[1] (Mei-chien.Fucci@redcross.org), S L Stramer[2]. *[1]American Red Cross Biomedical Services, Southwest Region, Tulsa, OK, United States; [2]Scientific Support Office, American Red Cross Biomedical Services, Gaithersburg, MD, United States*

Background/Case Studies: Human immunodeficiency virus (HIV), hepatitis C virus (HCV), and hepatitis B virus (HBV) are known to cause acute infections that may be unrecognized, under-diagnosed, and under-reported. In previously reported transmissions of viral hepatitis in health care settings, investigation has depended upon chance observations of unsafe practices or clustering of symptomatic patients. Prompt recognition and investigation have been limited by sporadic transmissions, assumption of health care delivery as inherently safe, and limited resources; unfortunately, such limitations allow undetected, continued transmissions. In this report, a blood center helped identify a potential public health risk. **Study Design/Methods:** Retrospective review of a case file. **Results/Findings:** A repeat blood donor (BD), who had given regularly for over a decade, tested reactive for HIV/HCV by triplex NAT (Ultrio; Novartis) on an August 2012 donation. HIV RNA reactivity was unconfirmed on retesting (see Table). A May 2012 donation tested acceptably. With the help of a state epidemiologist, the blood center's medical director (MD) was able to reach BD for face-to-face counseling. BD shared results from the state and his physician, both confirming recent, acute HCV infection. He denied traditional HCV risk factors, but reported recent dental work between his May and August donations. His spouse tested HCV negative. Due to his NAT conversion in the absence of detectable anti-HCV during the 3-month interval between donations, and no traditional risk factors, other possible sources of risk warranted further investigation. The state visited the dental offices where BD had been during the exposure period. At one office, significant deviations from standard practices were cited. A patient, seen just before BD, was HCV-infected. At this time it is not known if that patient was the source. **Conclusion:** This is the second published report of a potential public health risk identified by the recent change in the HCV status of a frequent blood donor. Testing of frequent donors for HIV, HCV, and HBV can detect recently acquired infections. NAT converting or seroconverting donors should have a careful health history performed by the blood center MD. When traditional risk factors have been excluded, the MD should encourage those responsible for public health to perform appropriate follow-up, including inspection of health care facilities. Lapses in standard universal blood precautions may be detected during such an investigation, and blood centers can play a key role in identifying this potential public health risk.

Disclosure of Commercial Conflict of Interest

M. H. Fucci: Nothing to disclose; S. L. Stramer: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. H. Fucci: Nothing to disclose; S. L. Stramer: Nothing to disclose

Index Donation	HCV NAT S/CO (Novartis Ultrio)*	HIV NAT S/CO (Novartis Ultrio)*	Viral Load (copies/mL) [National Genetics Institute (NGI)]	Anti-HCV (Abbott PRISM)	Anti-HIV (Abbott PRISM)
Initial Results	21.59	1.53**	4.7 × 10^7	0.16	0.22
Retest of Plasma Unit	23.33	0.00	4.7 × 10^5	0.09	0.27

* Signal-to-cutoff ratio (>1.0 = reactive); ** Determined to be falsely reactive.

SP391

Prevalence of Human Immunodeficiency Virus (HIV), Hepatitus B Virus (HBV), and Hepatitis C Virus (HCV) Infections among Northeastern Thai Blood Donors: A Five-Year Analysis for Surveillance and Service Improvement

Y Urwijitaroon[1] (yupaurw@kku.ac.th), W Chaimanee[2]. *[1]Centre for Research and Development of Medical Diagnostic Laboratory, Faculty of Associated Medical Sciences, Khon Kaen University, Khon Kaen, Thailand; [2]Khon Kaen Red Cross Regional Blood Centre, National Blood Centre,Thai Red Cross Society, Khon Kaen, Thailand*

Background/Case Studies: This blood center provides donor blood processing for the upper northeastern blood banks on request. This study was performed to assess the seroprevalence of human immunodeficiency virus (HIV), hepatitis B virus (HBV), and hepatitis C virus (HCV) infectious markers found in blood donors from four provinces including Khon Kaen, Kalasin, Sakonnakhon, and Nakhonpanom for surveillance and service improvement. **Study Design/Methods:** A retrospective analysis of HIV, HBV, and HCV seropositivity detected in blood donors between January 2006-December 2010 was conducted. The prevalence in accordance with sex and age among first time and repeat donors were analyzed. **Results/Findings:** There were 310,323 units collected from 248,908 donors. Five years average prevalence of viral infectious markers among blood donors in Khon Kaen, Kalasin, Sakonakhon, and Nakhonpanom were as following: 0.16%, 0.11%, 0.12%, and 0.09% for HIV; 2.00%, 2.45%, 2.84%, and 2.78% for HBV; and 1.12%, 2.43%, 0.41%, and 0.44% for HCV, respectively. According to the annual prevalence, declining trends of HBV and HCV infections were observed. The prevalence of first time infected donor was significantly higher than repeat blood donors, especially in males. Significantly low prevalence was observed among the 17-20-year age group. Repeat donation following inconclusive (IC) testing results of all infectious markers was found in 854 IC donors. There were 465 IC donors who showed negative screening results on the consecutive donation, confirming false positive screening of the previous testing. Thirteen donors demonstrated positive infectious markers confirmed early seroconversion of the previous IC result. However, 376 IC donors still IC results for their consecutive donation testing. Repeat donation of infected donors was found in 1819 units of 1592 donors, of which 85% were collected at mobile sites. **Conclusion:** For improvement of transfusion service and cost effectiveness, active donor data management and improvement of IC infectious testing results guidelines should be considered for prevention of repeat donation from both infected and IC donors, to reduce waste and transfusion risk. On the other hand, for safe blood transfusion,the effort of recruiting and retaining new blood donors from the younger age group (17-20 years), especially female donors, should be encouraged to increase the number of safe donors for the future.

Disclosure of Commercial Conflict of Interest

W. Chaimanee: No Answer; Y. Urwijitaroon: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Chaimanee: No Answer; Y. Urwijitaroon: Nothing to disclose

SP392

Previous Human Immunodeficiency (HIV) Testing in Blood Donors in Brazil

R Bruhn[1] (rbruhn@bloodsystems.org), E C Sabino[2], E C Moreno[3], P F Blatyta[4], S M Carvalho[5], A C Proietti[3], D Sampaio[6], M Lopes[5], B Custer[1], T T Goncalez[1]. *[1]Epidemiology, Blood Systems Research Institute, San Francisco, CA, United States; [2]University of São Paulo, São Paulo, Brazil; [3]Fundação Hemominas/Hemocentro de Minas Gerais, Belo Horizonte, Brazil; [4]Fundação Pró-Sangue/Hemocentro de São Paulo, São Paulo, Brazil; [5]Hemocentro do Rio de Janeiro, Rio de Janeiro, Brazil; [6]Fundação Hemope/Hemocentro de Pernambuco, Recife, Brazil*

Background/Case Studies: More than 25% of a half million people living with HIV in Brazil are estimated to be unaware of their infection. Since previous studies indicated donors may use blood donation in lieu of testing, the objectives of this analysis were to assess, among donors with sexual risk

behavior in the 12 months before donation, the frequency of those who have previously been HIV tested and the demographic and behavioral factors associated with such testing. **Study Design/Methods:** Secondary analysis was performed using questionnaire data from an HIV case-control study of blood donors conducted at four blood centers in Brazil from January 2009-March 2011. The primary outcome was the response to the question: "Other than blood donation, have you ever been tested for HIV?" Using logistic regression analysis, demographic and disclosed risk behaviors were compared between donors with affirmative and negative responses to the question to assess factors associated with previous testing. Assessed sexual risk behaviors in the 12 months prior to donation included sex with any of the following: anonymous, one time, or acquaintance with condom, frequency of never or sometimes; HIV+; MSM; partner with potential job exposure; number of partners > center-specific deferral criteria; injection drug user; sex worker; transfusion recipient; hemophiliac; or inmate. **Results/Findings:** Of 341 HIV+ cases and 791 controls, 27% of controls and 40% of cases reported previous HIV testing. Per self-identified sexual orientation, 28% were hetero-, 51% were bi-, and 44% were homosexuals, and have been tested previously. After excluding 18 case donors who were aware of their HIV-positive status prior to donation, a subgroup of donors who acknowledged sexual risk (191 cases [42% tested] and 93 controls [37% tested]) was examined. Among cases with sexual risk behavior, increased odds of previous testing were associated with education >high school and ages 25-34 years compared to being older. Controls with sexual risk behavior were more likely to have been HIV tested if they were bisexual or homosexual or if they reported having sex with a sex worker. After adjusting for gender, race/ethnicity, education, sexual orientation, and blood center, cases with reported risk were more likely to have previously been HIV tested if they were female (AOR 3.5, $p = 0.03$) or educated beyond high school (AOR 5.5, $p = 0.009$). After similar adjustment, controls with reported risk were likely to have previously been tested for HIV if they were bisexual or homosexual (AOR 31.2, $p = 0.02$). **Conclusion:** While it is encouraging that previous testing was more frequent in donors with acknowledged sexual risk behavior, 60% still had not been tested for HIV outside of blood donation. Educating donors on the importance of HIV testing outside of blood donation, especially if they engage in higher risk behaviors (regardless of sexual orientation), could improve the safety of donated blood.

Disclosure of Commercial Conflict of Interest

P. F. Blatyta: Nothing to disclose; R. Bruhn: Nothing to disclose; S. M. Carvalho: No Answer; B. Custer: Novartis Vaccines and Diagnositics, Grants or Research Support; T. T. Goncalez: Nothing to disclose; M. Lopes: No Answer; E. C. Moreno: Nothing to disclose; A. C. Proietti: Nothing to disclose; E. C. Sabino: Nothing to disclose; D. Sampaio: No Answer

Disclosure of Grants Conflict of Interest

P. F. Blatyta: Nothing to disclose; R. Bruhn: Nothing to disclose; S. M. Carvalho: No Answer; B. Custer: Novartis Vaccines and Diagnostics, Grants or Research Support; T. T. Goncalez: Nothing to disclose; M. Lopes: No Answer; E. C. Moreno: Nothing to disclose; A. C. Proietti: Nothing to disclose; E. C. Sabino: Nothing to disclose; D. Sampaio: No Answer

SP393

The Use of Stimmunology and the SMARTube System to Further Reduce the Risk of Human Immunodeficiency Virus (HIV) Transmission

J L Jacobson[1] (jessica.jacobson@nyumc.org). [1]Pathology, Bellevue Hospital/New York University School of Medicine, New York City, NY, United States

Background/Case Studies: Screening blood, organ, and tissue donors for human immunodeficiency virus (HIV) is hindered by the long seronegative window period and the high rate of false-positive results. StimmunologyTM is a technology that enables the production of antibodies in a whole blood sample, in vivo. Such stimulation leads to an increase in HIV antibody levels in the blood sample, enabling the detection of HIV infection prior to seroconversion. The stimulation step acts upon HIV-primed lymphocytes in the blood sample. Therefore, increased antibody levels will be detected only in the presence of infection. This increase in antibody levels, which can be achieved within days of infection, can reduce the window period problem. Stimmunology has been evaluated in high risk populations and has been show to yield positive results earlier than nucleic acid testing (NAT) and testing of traditional plasma. The study was designed to evaluate if the SMARTubeTM (SMART Biotech Ltd., Israel), which is an application of Stimmunology, could be used to reduce the risk HIV transmission from low risk donors. **Study Design/Methods:** With IRB approval, a local blood center collected an unlabeled and unlinked eight ml heparin tube of blood from volunteer blood donors who had been cleared to donate allogeneic blood. The tubes were sent to a laboratory for processing and testing. One ml of the whole-blood sample was transferred to the SMARTube on the day after collection and was incubated for 3-5 days at 37C with 5% CO2. The antibodies secreted into the culture supernatant fluid due to Stimmunology and any antibodies present in the original sample are termed SMART-plasma. The SMART-plasma and traditional plasma were tested for HIV antibodies using the same assay. The SMART-plasma and traditional results were compared. **Results/Findings:** 719 donors were consented, and samples were collected in January 2012. SMART-plasma was made from 702 usable samples. 0/702 SMART-plasmas and 0/702 traditional plasmas were positive for HIV antibodies. The mean optical density (OD) of SMART-plasma was 0.069 (range 0.051 to 0.213), and mean OD of traditional plasma was 0.086 (range 0.049 to 0.365). The mean assay positive cut-off was 0.368 (range 0.412 to 0.347). **Conclusion:** Although no sample tested positively for HIV antibodies with either SMART-plasma or traditional plasma, the sample size was not large enough to assure that a positive sample would be identified. Stimmunology was effectively used to screen low risk blood donors and no discrepant results were identified. The required incubation time makes the SMARTube system, at present, not practical for screening volunteer blood donors. The SMARTube system however could be used to screen living-related organ donors and bone marrow/peripheral blood stem cell (PBSC) donors and thus could reduce the risk of HIV transmission in transplant patients.

Disclosure of Commercial Conflict of Interest

J. L. Jacobson: Fresenius, Other, Terumo BCT, Other, Genesis BPS, Other, SMART Bio Tech Ltd., Other

Disclosure of Grants Conflict of Interest

J. L. Jacobson: G & W Laboratories Inc., Grants or Research Support

SP394

The Experience of Hemotherapy and Hematology Center of Para-HEMOPA with Nucleic Acid Amplification Testing (NAT): First Detection of Human Immunodeficiency Virus (HIV) in Window Period

M K Palmeira[1] (mauriciopalmeira@gmail.com), R B Hermes[2], J R Lemos[4], L M Lamarao[5], K L de Almeida[5], K S Barile[5], M M de Lima[5], M R Ferreira[3], L C Maradei Pereira[6]. *[1]Laboraratory Coordination, Hemotherapy and Hematology Center of Para, Belem, Brazil; [2]Division of Transfusion Transmitted Diseases Laboratory, Hemotherapy and Hematology Center of Para, Belem, Brazil; [3]Technical Director, Hemotherapy and Hematology Center of Para, Belem, Brazil; [4]Division of Molecular Biology Laboratory, Hemotherapy and Hematology Center of Para, Belem, Brazil; [5]NAT Laboratory, Hemotherapy and Hematology Center of Para, Belem, Brazil; [6]Presidency, Hemotherapy and Hematology Center of Para, Belem, Brazil*

Background/Case Studies: In many countries, the nucleic acid amplification test (NAT) screening in potential blood donors has been implemented to reduce the risk of transfusion-transmitted viral infections. However, newly acquired infection may be undetectable in donors by serologic testing. The NAT for human immunodeficiency virus (HIV) and hepatitis C viruses (HCV) in blood banks was recommended by the Brazilian legislation in 2004, and a multiplex NAT, which allows the simultaneous detection, was started in June 2012 in Belen, Para. In order to prove that NAT shortens the window period, the first detection of the HIV infection with negative enzyme immunoassay result is described here. **Study Design/Methods:** Data collected during nine months since NAT implementation, from June 2012-May 2013. Donations were tested in six samples pool using commercial NAT HIV/HCV kit (Bio-Manguinhos/FIOCRUZ). Antibody screening kits for detection anti-HCV and HIV Ag/Ab combination (MUREX) were performed. **Results/Findings:** 60.321 blood samples were obtained from blood donors durin the HIV test period. Among these donors, 94 samples (0.15%) had NAT positive result: 74 HIV (0.12%), and 20 HCV (0.03%). All HCV-NAT positive samples were positive in serological screening, however, one positive pool were detected by HIV-NAT. The pooled samples were performed separately and one sample confirmed a positive test (29.03 CT).This sample was non-reactive in enzyme immunoassays, with negative confirmatory. The first detection during the HIV window period was a repeat donor, a 26-year-old female who made around 10 previous donations over a 3–year period, and the last two donations (June 2012 and November 2012) were also NAT negative. After call back, the donor returned in 30 days for a new sample collection that showed reactivity serological confirmed (values: 3,204 and 2,908/cutoff: 0.236). The first window period NAT detected occurred earlier than expected. Studies in other countries had an incidence estimate of undetected infection by serologic screening ranging in 1:600,000-1:3,100,000,

and Brazilian findings report 1:100,000 in the southern region. These findings suggest that NAT screening could significantly reduce the rate of undetected HIV and HCV in blood donors; however, the appropriate analysis should be performed in order to better investigate the transmission risk of window period cases in this area. **Conclusion:** NAT screening is feasible in blood banks services and it can become a huge tool for transfusional safety, so therefore its requirement is important to Brazilian transfusional laws.

Disclosure of Commercial Conflict of Interest

K. S. Barile: Nothing to disclose; K. L. de Almeida: Nothing to disclose; M. M. de Lima: Nothing to disclose; M. R. Ferreira: No Answer; R. B. Hermes: Nothing to disclose; L. M. Lamarao: Nothing to disclose; J. R. Lemos: No Answer; L. C. Maradei Pereira: No Answer; M. K. Palmeira: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Barile: Nothing to disclose; K. L. de Almeida: Nothing to disclose; M. M. de Lima: Nothing to disclose; M. R. Ferreira: No Answer; R. B. Hermes: Nothing to disclose; L. M. Lamarao: Nothing to disclose; J. R. Lemos: No Answer; L. C. Maradei Pereira: No Answer; M. K. Palmeira: Nothing to disclose

SP395

Prevalence of Hepatitis A Virus (HAV) and High Titer Parvovirus B19 in Recovered and Source Plasma Donations

S Jones[1] (scott.jones@qualtexlabs.org), K Leighton[1], J Chapa[1], D Sanchez-Watkins[1], L K Myers[1]. *[1]Lab Administration, QualTex Laboratories, San Antonio, TX, United States*

Background/Case Studies: Hepatitis A virus (HAV) is a single stranded non-enveloped RNA virus and Parvovirus B19 (B19) is a small non-enveloped DNA virus. Both viruses can be transmitted to susceptible individuals through blood transfusion and plasma-derived products. Plasma fractionators have screened solvent treated pooled and source plasma with molecular HAV and B19 assays as an additional safeguard. The purpose of this study was to determine the prevalence of HAV and high titer B19 in recovered and source plasma donors. **Study Design/Methods:** The prevalence of donations containing HAV and high titer B19 was determined by testing routine whole blood and source plasma donations from 2010-May 2013 using in-house molecular beacon assays. Master pool samples were created by an automated pipetting instrument used to aliquot a portion of 96 (HAV) or 288 (B19) individual samples into a single master pool sample. Nucleic acid was isolated using an automated pipetting instrument and a commercial genomic isolation kit. Specific primer sets and molecular beacons for B19 or HAV and the internal control were placed in the amplification reactions and potential amplicons were generated via PCR. Following amplification the epifluorescence was determined using a fluorometer. **Results/Findings:** The assays detected B19 DNA levels greater or equal to 347 IU/ml per master pool sample and greater or equal to 15 IU/ml per master pool sample for HAV. The rate of HAV in recovered plasma donations was 0.0% (0/100,360) and was 0.0005% (6/1,297,945) in source plasma donations. The rate of high titer B19 in recovered plasma donations was 0.02% (23/100,360) and was 0.03% (377/1,450,585) in source plasma donations. The majority (80%) of the high titer B19 donations for recovered and source plasma occurred during the months of February-July. **Conclusion:** In-house, semi-automated assays allowed for high throughput "in process" testing of recovered and source plasma donations for B19 and HAV. The rates of high titer B19 donations in recovered plasma versus source plasma are very similar. The rates of HAV donations in recovered plasma versus source plasma are very similar and significantly lower than the rates of high-titer B19 donations. The overall rates for positive donations for HAV and high titer B19 are similar to published reports.

Disclosure of Commercial Conflict of Interest

J. Chapa: No Answer; S. Jones: Nothing to disclose; K. Leighton: Nothing to disclose; L. K. Myers: Nothing to disclose; D. Sanchez-Watkins: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Chapa: No Answer; S. Jones: Nothing to disclose; K. Leighton: Nothing to disclose; L. K. Myers: Nothing to disclose; D. Sanchez-Watkins: Nothing to disclose

SP396

The Molecular Characterization of Hepatitus B Virus (HBV) Infection in the Hepatitis B Vaccinated Blood Donors from Shenzhen Area

W Zhu[1] (191252096@qq.com). *[1]ShenZhen Blood Center, ShenZhen, China*

Background/Case Studies: The purpose of this study is to investigate the molecular characteristics of hepatitus B virus (HBV) infection and the occurrence mechanism of HBsAg+/anti-HBs+ from the hepatitis B vaccinated blood donors in Shenzhen area. **Study Design/Methods:** A total of 58 HBsAg+/HBV DNA+ serum samples collected from 13504 donors were quantified by QPCR, and 35/58 samples amplified the Pre-S/S fragment successfully by the nested polymerase chain reactor (PCR) and determined the genotype. Through the follow-up of consulting,5/58 were confirmed as vaccinated blood donors with HBsAg+/anti-HBs+/anti-HBc+, and compared to consensus sequences of Pre-S/S of HBsAg+/anti-HBs-/anti-HBc+ strains with similar genotype. **Results/Findings:** The detection rate and infection rate of HBV in Shenzhen donors was 0.43% and 1.9%,respectively. Among 35/58 Pre-S/S positive samples, HBV genotypes B and C were dominant (42.6% and 57.4%, respectively). All five HBsAg+/anti-HBs+/anti-HBc+ samples were genotype B with low anti-HBs titre, ranging between 10.45-88.76 mIU/ml (median: 21.54 IU/ml). In the Pre-S/S, Pre-S2, S, MHR1, MHR2 and "a" determinant regions, the substitutions of amino acid mutations in HBsAg+/anti-HBs+ samples were significant popular than those HBsAg+/anti-HBs- samples($P < 0.05$), except Pre-S1 region. Most amino acid mutations occurred in major hydrophilic region 1 (MHR1) and Pre-S2 regions, rare mutations of N40S, P62L were found in MHR1. In major hydrophilic region 2 (MHR2) and Pre-S/S regions, no mutations were found in 2/5 and 1/5 samples, respectively. Besides, the popular mutation G145R was not found in "a" determinant region. **Conclusion:** The mutations in Pre-S/S region, especially in MHR1 and Pre-S2 regions, play an important role in the occurrence of HBsAg+/anti-HBs+ in vaccinated donors; more mutations occurred in HBV DNA of genotype B, which resulted in anti-HBs failed neutralizing with HBsAg under HBV vaccine pressure. There are no mutations in Pre-S/S regions. In some samples, the occurrence of HBsAg+/anti-HBs+ samples maybe related to other immune mechanisms.

Disclosure of Commercial Conflict of Interest

W. Zhu: Nothing to disclose

Disclosure of Grants Conflict of Interest

W. Zhu: Nothing to disclose

SP397

Summary of Hepatitus B Virus (HBV) Testing at a Blood Center

J Crane[1] (jecrane@uic.edu), K Burns[1], S Gordon[1], M Papari[1]. *[1]LifeSource Blood Center, Rosemont, IL, United States*

Background/Case Studies: Given the increased sensitivity of modern nucleic test methods in the detection of hepatitis B virus (HBV), debate is increasing regarding the possibility of removing surface antigen testing from the required testing. This report is a summary of testing data from donors as well as external clients at a blood center. **Study Design/Methods:** This is a retrospective study of the serologic and molecular results on samples from June 2011-February 2013. Serologic assays are performed using PRISM (Abbott Laboratories) methodology. Nucleic acid testing is performed using COBAS TaqScreen MPX (Roche Diagnostics) in mini-pools of six and COBAS Ampliscreen (Roche Diagnostics) for individual testing. Reported test results include HBV surface antigen (HBs), HBV core antibody (HBc), HBs neutralization confirmatory assay (HBn), multiplex (MPx) polymerase chain reaction (PCR), and confirmatory individual HBV nucleic acid testing (NATB). **Results/Findings:** In the timespan indicated above, 857 positive HBV serological tests were identified. Two were noted to question some results and were not used, yielding a total of 855 donors. Several clients perform testing outside the blood center, and may not perform all the listed tests or they may perform them but not transmit the results. The attached table demonstrates results of donors where all testing was readily available (N = 146). **Conclusion:** While the results are preliminary and require further study, some findings may be gleaned at this early stage. A small proportion of the donors were negative for HBs and positive for NATB. Conversely, a somewhat smaller number of donors tested positive for HBs, HBn, and HBc but were negative for MPx and NATB. The findings at this stage, though preliminary, are not supportive of removing surface antigen testing.

Disclosure of Commercial Conflict of Interest

K. Burns: Nothing to disclose; J. Crane: Nothing to disclose; S. Gordon: Nothing to disclose; M. Papari: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Burns: Nothing to disclose; J. Crane: Nothing to disclose; S. Gordon: Nothing to disclose; M. Papari: Nothing to disclose

Number With Pattern	HBs	HBn	HBc	MPx	NATB
31	pos	pos	pos	pos	pos
1	pos	pos	pos	pos	neg
4	pos	pos	neg	pos	pos
10	pos	pos	neg	neg	neg
8	pos	pos	pos	neg	neg
4	pos	neg	neg	pos	pos
63	pos	neg	neg	neg	neg
16	neg	N/A	pos	pos	pos
9	neg	N/A	pos	pos	neg

neg = negative, pos = positive, N/A = not applicable

SP398

Comparison of Seroprevalence to Human Immunodeficiency Virus (HIV), Hepatitis C (HCV) and Hepatitis B)HBV) in Tissue Donors and First-Time Blood Donors in North Texas

E Heck[1] (Ellen.Heck@UTSouthwestern.edu), V Winkelman[2], S Cyrus[2], D Drury[1], L J Sutor[3,1]. *[1]University of Texas, Southwestern, Dallas, TX, United States; [2]R&D Support, Creative Testing Solutions, Tempe, AZ, United States; [3]Carter BloodCare, Dallas, TX, United States*

Background/Case Studies: Characterization of tissue donor populations for risk factors by comprehensive screening and testing programs is essential to avoid transmission of blood-borne virus. The testing protocols for tissue donors are similar but not necessarily identical to those of the blood-banking industry. The health history and high risk screening, however, differs significantly in that it is not first-person information and therefore might indicate a potential for a different risk analysis. Therefore, this data examines the seroprevalence data for three major blood-borne viruses: hepatitis B (HBV), hepatitis C (HCV), and human immunodeficiency virus (HIV) in a university-based tissue bank and a large community blood center in the same geographic area. **Study Design/Methods:** Blood samples from approximately 70,000 first time blood donors per year and approximately 2800 tissue donors were sent to the same reference laboratory for testing over a four year period (2009-2012) for HBV, HV, and HIV. Testing was performed according to manufacturer's instructions for blood donors for HBsAg, anti-HBc, anti-HIV-1/2 Plus O, and nucleic acid amplification (NAT). The same assays were also used for tissue donors following manufacturer's instructions and FDA regulations. Some of these instructions and regulations differ from blood donor requirements as tissue donor samples represent cadaveric, post mortem, and some pre mortem samples. **Results/Findings:** The average reactive rate for first time blood donors from 2009-2012 ranged from a low of 0.02 % for NAT testing, to a high of 2.35 % for Hepatitis B core. In tissue donors the average low for combined NAT was 0.70%, with a similar high of 4.78 % for Hepatitis B core. **Conclusion:** Tissue donors had a higher rate of reactive viral markers than first time blood donors in the same geographic region. Factors accounting for this difference may relate to the inability to obtain a first-person health and risk factor history for tissue donors, although medical records are examined for elimination of high risk populations and health history of communicable disease.

Disclosure of Commercial Conflict of Interest

S. Cyrus: Nothing to disclose; D. Drury: Nothing to disclose; E. Heck: Nothing to disclose; L. J. Sutor: Nothing to disclose; V. Winkelman: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Cyrus: Nothing to disclose; D. Drury: Nothing to disclose; E. Heck: Nothing to disclose; L. J. Sutor: Nothing to disclose; V. Winkelman: Nothing to disclose

Average % Reactivity	HBsAg	HBc	HCV	NAT
First Time Donors	0.21	2.35	1.00	0.02
Tissue Donors	0.76	4.78	2.09	0.14 (HIV) 1.26 (HCV)

SP399

Comparison of Hepatitus C Virus (HCV) Immunoblot and Alternate HCV Screening

S Cyrus[1] (scyrus@mycts.org), R Spizman[1], P C Williamson[1], S Caglioti[1]. *[1]Creative Testing Solutions, Tempe, AZ, United States*

Background/Case Studies: A large regional donor testing laboratory has implemented alternate anti-hepatitus C virus (HCV) screening in place of a licensed confirmatory assay that is no longer available. These alternate screening test results are used for donor counseling and for determination of reinstatement eligibility by many US blood donor centers under an approved variance process. The discontinuation of the only FDA-approved confirmatory assay for anti-HCV (HCV immunoblot) has significantly impacted the donor counseling message provided to anti-HCV reactive donors. Comparison of available data was initiated to determine the degree of correlation between historical HCV immunoblot positive rates and alternate HCV reactive rates. **Study Design/Methods:** Data generated between January-October 2012 for HCV immunoblot was compared to data generated from November 2012-April 2013. A total of 2166 anti-HCV reactive donations were tested with the HCV immunoblot and a total of 1522 anti-HCV reactive donations were tested with the alternate anti-HCV screening assay. **Results/Findings:** A total of 1216 (56.1%) donations tested with HCV immunoblot were negative while 613 (28.3%) were indeterminate and 337 (15.6%) were positive as shown in the table below. The table also shows the alternate HCV screening test data in which 1089 (71.6%) were nonreactive and 433 (28.4%) were repeatedly reactive. These data demonstrate that a significantly higher number of anti-HCV reactive donations are found reactive with an alternate HCV test (28.4%) than were previously found positive by immunoblot (15.6%). This means that approximately 12.8% of anti-HCV reactive donations tested by an alternate assay would likely have been either immunoblot indeterminate or negative. **Conclusion:** The unavailability of a licensed confirmatory assay for donors found reactive for anti-HCV has led to difficulties with determination of the correct donor counseling message. The data presented here shows that up to 12.8% of donations found reactive with an alternate HCV screening assay would likely be immunoblot negative or indeterminate. The counseling message for these donors will indicate that HCV infection cannot be ruled out and may lead to unnecessary apprehension and additional medical follow-up that would not have been recommended if immunoblot results had been negative or indeterminate. Further analysis of data to determine the usefulness of signal/cutoff ratio in providing a better counseling message is indicated.

Disclosure of Commercial Conflict of Interest

S. Caglioti: Nothing to disclose; S. Cyrus: Nothing to disclose; R. Spizman: Nothing to disclose; P. C. Williamson: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Caglioti: Nothing to disclose; S. Cyrus: Nothing to disclose; R. Spizman: Nothing to disclose; P. C. Williamson: Nothing to disclose

HCV Immunoblot Results			Alternate HCV Results	
Indeterminate	Negative	Positive	Nonreactive	Reactive
613	1216	337	1089	433
28.30%	56.14%	15.56%	71.55%	28.45%

SP400

Photochemical Inactivation of Hepatitis C Virus (HCV) using the Mirasol® PRT System

A Bengrine[1] (abder.bengrine@u-picardie.fr), S D Keil[2], S Marschner[2], R Goodrich[2], G Duverlie[1]. *[1]Jules Verne University of Picardie and Biobanque de Picardie, Amiens, France; [2]Terumo BCT, Lakewood, CO, United States*

Background/Case Studies: A number of pathogens (viruses, bacteria, and parasites) can be transmitted through blood or blood products. Hepatitis C virus (HCV) is now mandatorily tested on blood donors worldwide due to the potential serious chronic clinical sequelae (fibrosis and hepatic-carcinoma) associated with its transmission. Despite testing, there remains a window period during which infectious units may test negative. Pathogen reduction technologies could offer a new strategy for dealing with window period viruses. In this study the efficacy of the Mirasol® PRT System for Platelets treated in PAS on the reduction of HCV was evaluated. **Study Design/Methods:** The virus used in this study is the JFH1 strain of HCV and it was produced in the Huh-7 cell culture model. This study used platelet depleted plasma (PDD) from five different donors in place of platelet units; previous work has shown PDD to yield comparable viral reduction results to treatment

performed with platelets. Due to the difficulty of obtaining large quantities of virus, all treatments were performed using a 250 μL volume in a 48-well plate. Each 48-well plate was treated with a scaled dose that was verified to be equivalent to the standard Mirasol PRT process. Additionally, bag material cut from a Mirasol PRT Illumination/Storage bag was placed on top of the 48-well plate during illumination so the that light was properly attenuated, thus mimicking the standard Mirasol PRT process. Each of the five donors was split into three treated units and one untreated unit. After treatment, the infectivity of treated and non-treated samples was measured by immunofluorescence 48hours after inoculation of Huh-7 cells with the samples using a monoclonal antibody directed against the glycoprotein E2 of HCV. Infectious titers before and after treatment were determined by counting the number of foci. **Results/Findings:** All three treated replicates for each of the five different donors contained no detectable foci and were at the limit of detection for the assay. The average pre-treatment titer for the five donor samples was 4.2 log FFU (foci forming units)/mL and the limit of detection for the assay was ≤ 0.1 log FFU/mL. The calculated reduction observed in this study was ≥ 4.1 log. **Conclusion:** The Mirasol PRT System for Platelets effectively reduced the infectivity of HCV to the limits of detection using an in vitro cell model. The ≥ 4.1 log reduction observed in this study could potentially eliminate window period transmission of HCV.

Disclosure of Commercial Conflict of Interest

A. Bengrine: Nothing to disclose; G. Duverlie: No Answer; R. Goodrich: Terumo BCT, Other; S. D. Keil: Employee of Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other

Disclosure of Grants Conflict of Interest

A. Bengrine: Nothing to disclose; G. Duverlie: No Answer; R. Goodrich: Terumo BCT, Other; S. D. Keil: Employee of Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other

	Log Reduction
Replicate #1	4,2
Replicate #2	4,1
Replicate #3	4,0
Replicate #4	4,2
Replicate #5	4,1
Replicate #6	4,0
Average Log reduction	4,1

SP401

The Associations between Human Leukocyte Antigens (HLA) and Chronic Hepatitus C Virus (HCV) Infection among Voluntary Chinese Blood Donors

X Rong[1] (joyjoy@126.com), J Huang[1], H Xiong[1], K Huang[1], L Lu[2], Y Fu[1]. *[1]Transfusion Institute, Guangzhou Blood Center, Guangzhou, China; [2]Center for Viral Oncology, University of Kansas Medical Center, Kansas City, KS, United States*

Background/Case Studies: The genetic variations of immunological importance have been shown in association with hepatitis C virus (HCV) clearance or persistence, of which the interactions between the human leukocyte antigens (HLA) and HCV are critical. However, inconsistent roles of some HLA alleles have been suggested among different ethnic groups while currently only limited data in this aspect are available from the Chinese. **Study Design/Methods:** The sequencing based typing was used to determine the polymorphism at the HLA-A, B, and DRB1 loci among 426 HCV-infected (HCV Ab+/RNA+) and 709 control (HCV Ab-/RNA-) donors. The allelic and genotypic distribution at the HLA-A, B, and DRB1 loci were compared between the two cohorts by using chi-square test and multivariate Logistic Regression analysis. False discovery rate (FDR) was used as a correction method. **Results/Findings:** 1. Distribution of the HLA-A, B and DRB1 alleles. At the HLA-A locus, 14 alleles each showed a frequency >1% and they accounted for 97.9% of the total HLA-A alleles. The first 3 alleles were HLA-A*1101 (31.1%), A*2402 (14.6%), and A*0207 (10.9%). 18 HLA-B alleles showed a frequency >1% and accounted for 86.7% of the total HLA-B. B*4001 (15.4%) and B*4601 (15.4%) showed the highest frequencies, followed by B*5801 (9.8%). 18 DRB1 alleles with frequency >1% accounted for 93.5% of the total HLA-DRB1; DRB1*0901 (13.6%), DRB1*1501 (11.4%), and DRB1*1202 (10.9%) were the most frequent. 2. Association of HLA alleles and genotypes with chronic HCV infection: four HLA-B alleles were found at the high-resolution level. The frequencies of B*13:02 (P = 0.009, OR = 1.515) and B*15:01 (P = 0.10, OR = 1.561) were higher in the HCV-infected than the control cohort, suggesting they were susceptibility genes of HCV. B*07:05 (P = 0.001, OR = 0.299), and B*15:02 (P = 0.016, OR = 0.778) were more frequent in the control cohort, which may act as a protective factor against HCV infection. Of note, B*1501 and B*1502 appeared to show opposite associations with HCV infection. The former seems to be a risk allele while the latter a protective one. At HLA-A, and DRB1 loci, the allelic frequency distributions were of no difference after adjustment for multiple comparisons. Similarly, no significant differences of genotypic distribution were found after adjustment for multiple comparisons between the two cohorts. **Conclusion:** Four novel HLA alleles were found in association with susceptibility/resistance to HCV infection. Notably, it is first reported here that B*15:01 and B*15:02 seemed to act inversely in the process of host anti HCV. It is also the first report regarding the association of HLA and HCV infection among the Han people of southern China.The current study contributed to the deeper understanding of the role of HLA on the host anti viral immunity.

Disclosure of Commercial Conflict of Interest

Y. Fu: No Answer; J. Huang: Nothing to disclose; K. Huang: Nothing to disclose; L. Lu: No Answer; X. Rong: Nothing to disclose; H. Xiong: Nothing to disclose

Disclosure of Grants Conflict of Interest

Y. Fu: No Answer; J. Huang: Nothing to disclose; K. Huang: Nothing to disclose; L. Lu: No Answer; X. Rong: Nothing to disclose; H. Xiong: Nothing to disclose

SP402

Research for Hepatitis C Virus (HCV) Infection in Chinese Blood Donors

B Dai[1,2] (danny-ning@163.com), G Xu[1,2], W Ding[1], Y Wu[1], F Zhu[1,2], H Lv[1,2]. *[1]Blood Center of Zhejiang Province, Hangzhou, China; [2]Key Laboratory of Blood Safety Research, Ministry of Health, Hangzhou, China*

Background/Case Studies: Hepatitis C virus (HCV) is a small, enveloped, single-stranded RNA virus, which leads to an infectious disease. The aim of this study was to determine the prevalence and viral loads of HCV among blood donors in the Blood Center of Zhejiang Province, China. **Study Design/Methods:** A test of antibodies against HCV (anti-HCV) was conducted twice using different ELISA reagents, and detection of HCV RNA was performed using TMA technique in the PROCLEIX® TIGRIS® system. All samples with HCV RNA positive were quantified for the viral loads by QT-PCR. **Results/Findings:** Among 127,169 blood donors, the prevalences of anti-HCV and HCV RNA were 0.24% (95% CI 0.26%-0.21%) and 0.02% (95% CI 0.03%-0.01%) , respectively from November 1, 2011-October 29, 2012. All HCV RNA carriers are the first time donors. The value of mean viral loads is 6.87 ± 0.71 (log10) IU/ml in the HCV RNA carriers, and the donors' gender and age are not associated with the measured HCV viral loads (Table 1). **Conclusion:** Low prevalence of HCV and high viral loads were identified in the blood donors, which may help to recruit blood donor and improve blood safety.

Disclosure of Commercial Conflict of Interest

B. Dai: Nothing to disclose; W. Ding: No Answer; H. Lv: No Answer; Y. Wu: No Answer; G. Xu: No Answer; F. Zhu: No Answer

Disclosure of Grants Conflict of Interest

B. Dai: The Science Foundation of Zhejiang Health Bureau (No. 2012KYA059) , Grants or Research Support, (No. 22013KYA053), Grants or Research Support, Science Foundation of Ministry of Health, China (No. WKJ2011-2-017), Grants or Research Support; W. Ding: No Answer; H. Lv: No Answer; Y. Wu: No Answer; G. Xu: No Answer; F. Zhu: No Answer

TABLE 1. Analysis of the viral loads of HCV

Factor		Viral loads(log10) IU/ml Mean ± SD	t-value	P-value
Gender	Male (n = 15)	6.93 ± 0.80	t = 0.42	p > 0.05
	Female (n = 9)	6.80 ± 0.57		
Age	18-35 (n = 13)	6.99 ± 0.65	t = 0.36	p > 0.05
	36-55 (n = 11)	6.75 ± 0.79		
Donor status	First-time donor (n = 24)	6.87 ± 0.71		

SP403

Searching for Non-Primate Hepacivirus (NHPV) Related Genomes in HCV-Reactive Donors with Indeterminate Blot Patterns

J E Levi[1,2] (dudilevi@usp.br), S N Cabral[1], A S Nishiya[2], S C Ferreira[2], C M Romano[3], M B Polite[1], R A Pereira[1], M A Mota[1], J M Kutner[1]. *[1]Blood Bank, Hospital Israelita Albert Einstein, São Paulo, Brazil; [2]Molecular Biology, Fundação Pró-Sangue, São Paulo, Brazil; [3]Virology, Tropical Medicine Institute, University of São Paulo, São Paulo, Brazil*

Background/Case Studies: Hepatitis C virus (HCV) belongs to the genus Hepacivirus, one of the four genera in the family Flaviviridae along two established genera (Flavivirus and Pestivirus,) and one proposed genus, Pegivirus. Among the Flaviviruses and Pestiviruses there are important human pathogens, known to be transmitted to humans by arthropods and bats, depicting the importance of the behavior of human blood sucking for cross-species transmission of infectious agents. Surprisingly, no primate homologue to HCV has ever been found, neither has any arthropod been demonstrated to transmit HCV to humans. The search for a zoonotic origin for HCV has been renewed recently when Kapoor and co-workers (PNAS 2011; 108:11608-11613) found a canine virus with a high homology to HCV, now named as Non-Primate Hepacivirus (NHPV). A variable proportion of anti-HCV reactive donors submitted to the immunoblot (IB) to confirm their HCV status, present indeterminate results, meaning that it was verified some reactivity, in general weak, to one or two antigenic fractions, that does not allow the assignment of a positive result but neither to discard it. The degree of homology between HCV and NPHV suggests that humans may be infected by NPHV or NPHV-like viruses. Maximum aminoacid similarity between NHPV and HCV is observed in the non-structural regions 3 and 5. Peptides representing both domains are present in IB assays, so we reasoned that blood donors reactive for anti-HCV in screening and further reacting to these antigenic fractions in IB may potentially harbor such viruses. This study searched for NPHV sequences in the plasma of blood donors found anti-HCV reactive but not confirmed by neither IB or RT-PCR analysis. **Study Design/Methods:** Fifty-seven plasma samples from blood donors found reactive for anti-HCV second or third generation assays and presenting immunoblot indeterminate results were submitted to five distinct PCR reactions under low-stringency conditions, employing primers targeting GBV-C 5′UTR and NS3, Flavivirus-genus NS5, and NHPV 5′UTR and NS3. **Results/Findings:** No amplification was obtained with all primer pairs tested except for three samples that amplified both 5′UTR and NS3 fragments from GBV-C. No other sample presented reproducible amplification of fragments within the expected size with any of the primer sets. **Conclusion:** The presence of any NHPV or Flavivirus related virus did not appear among well-selected blood donors with IB indeterminate patterns. Certainly, now-available metagenomics tools such as next-generation sequencing may provide or rule out definitely the existence of HCV-related viruses in immunoblot indeterminate samples.

Disclosure of Commercial Conflict of Interest

S. N. Cabral: No Answer; S. C. Ferreira: Nothing to disclose; J. M. Kutner: No Answer; J. E. Levi: Nothing to disclose; M. A. Mota: Nothing to disclose; A. S. Nishiya: Nothing to disclose; R. A. Pereira: Nothing to disclose; M. B. Polite: Nothing to disclose; C. M. Romano: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. N. Cabral: No Answer; S. C. Ferreira: Nothing to disclose; J. M. Kutner: No Answer; J. E. Levi: Novartis, Grants or Research Support; M. A. Mota: Nothing to disclose; A. S. Nishiya: Nothing to disclose; R. A. Pereira: Nothing to disclose; M. B. Polite: Nothing to disclose; C. M. Romano: Nothing to disclose

SP404

Case Report: A Case of Hepatitis E Virus (HEV) in a Blood Donor

A A Tendulkar[1] (anitatendulkar@gmail.com), S Shah[1], R Kelkar[2]. *[1]Transfusion Medicine, Tata Memorial Hospital, Mumbai, India; [2]Microbiology, Tata Memorial Hospital, Mumbai, India*

Background/Case Studies: Hepatitis E (HEV) is a liver disease caused by a non-enveloped, single stranded RNA hepatitis E virus. The virus is transmitted enterically, but parental transmissions via transfusions have been reported since 2004. The seroprevalence in blood donors reported from various countries ranges from 1% to 32%. **Study Design/Methods:** An apparently healthy, 27-year-old male donor passed all eligibility criteria tests and donated a triple blood bag at a tertiary cancer hospital. After 20 days, donor informed the blood bank that he had developed vomiting and jaundice 1 day post donation for which he had visited a physician. He was investigated by a local laboratory 1 day post donation for liver function tests which were elevated. (Bilirubin: total: 5.2, direct: 2.5, Indirect: 2.7), AST: 688, ALT: 680 (n = upto 40I U/L), Urine: bile pigment present, bile salt: present, Widal and Malaria test: negative, CBC: normal. **Results/Findings:** Donor contacted this blood bank and advised about an outbreak of similar symptoms in his residential community during the same period. Accordingly, he was investigated 20 days post donation for HAV and HEV in the hospital Microbiology laboratory. The HAV IgM antibody by chemiluminiscent microparticle immunoassay was nonreactive. The HEV IgM antibody by ELISA was positive. The liver profile had improved (total bilirubin: 2.68, Direct: 1.07, indirect: 1.61, AST: 47, ALT: 68, Total proteins: 8.2, S.Albumin:4.6, S.globulin: 3.6). The donor's packed cells had already been issued during this interim period of 20 days as all mandatory TTI tests were negative. However, the cryoprecipitate and Factor VIII deficient plasma were immediately discarded. The patient receiving the implicated packed cells was symptom free 35 days post transfusion. He could not be investigated for HEV as he was discharged from hospital and was a resident of another state. The incubation period following exposure to the HEV ranges from 3-8 weeks (mean~40 days). This donor was in incubation phase during blood donation and was detected as anti HEV IgM positive. The donor initiated call back was pivotal to this case. Even though the disease is self limiting and resolves within 4-6 weeks, a fulminant hepatitis may develop which can lead to death. **Conclusion:** HEV is being considered as a re-emerging infectious disease across the world. Routine screening for the virus is presently not performed in blood banks. Studies indicate that HEV seroprevalence exists in blood donors, especially in endemic areas. Screening donor blood for this virus may be worthwhile to safeguard patients as severity is higher in high risk patients. Being a referral cancer center, donors in this setup are representative of the nation's population. A pilot project for HEV RNA donor testing would help in understanding the incidence and facilitate in HEV screening policy decisions.

Disclosure of Commercial Conflict of Interest

R. Kelkar: Nothing to disclose; S. Shah: Nothing to disclose; A. A. Tendulkar: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Kelkar: Nothing to disclose; S. Shah: Nothing to disclose; A. A. Tendulkar: Humacount 30 TS, Grants or Research Support

SP405

Transfusion Transmission of Hepatitis E Virus (HEV) by Solvent-Detergent Inactivated Plasma

A Assal[1] (azzedine.assal@efs.sante.fr), Y Piquet[1], P Gallian[1], C Constant[1], P Tiberghien[1]. *[1]EFS (French Blood Transfusion Establishment), La Plaine Saint Denis, France*

Background/Case Studies: Hepatitis E Virus (HEV) belongs to hepeviridae family. This RNA virus is non-enveloped. Thus solvent detergent (SD) process used in France to inactivate pathogens in lots of 100 apheresis plasmas is ineffective on this virus. In 2012, two cases of HEV contamination (genotype 3) were identified in two recipients of SD-inactivated plasma. The first case (case A) was detected in a patient presenting elevated alanine aminotransferase (ALT) after treatment by plasma exchange. In the second case (case B), the patient who underwent a liver transplant and received SD-plasma showed high ALT level, 3 months after transplantation. The plasmas transfused to these patients were issued from two different lots A and B, respectively In both patients HEV RNA was detected and quantified by Real-Time (RT) polymerase chain reaction (PCR) and genotype determined by nucleotide sequencing. **Study Design/Methods:** Haemovigilance alerts from hospitals about recipient contaminations were notified to the SD-plasma production site. The corresponding contaminated lots (A and B) were then identified and screened positive for HEV ribonucleic acid (RNA). Donors' RNA positive plasmas (from A and B lots) were retrieved by testing crossing pools on the archive plates containing samples entering in the composition of the two contaminated lots. HEV RNA was extracted on Easy mag device (BioMérieux) and then amplified and detected by a RT PCR using HEV PCR realStar kit (Eurobio) on CFX instrument (Biorad). Sequencing of HEV RNA was performed in HEV positive donors, and then sequences compared to those isolated from patients. **Results/Findings:** RNA was detected in the two donor plasmas entering in the production of two different SD-plasma lots. The viral loads in the donors contaminated plasmas were 145,000 copies /mL and 3,060 copies/mL, respectively. The viral loads in corresponding lots A and B were respectively 541 and 62 copies/mL. In both cases the genotype was 3f and donor virus sequences were respectively identical to those isolated from patients. **Conclusion:** Recipient contamination by HEV has been evidenced for SD-plasma transfusion. The low viral load values in both lots can be explained by a dilution effect as well as the probable presence of anti-HEV neutralizing antibodies. Contamination by

HEV is often asymptomatic and it is probable that in number of cases transfusion of contaminated SD-plasma resulted in undiagnosed resolved infection. The protective role of anti-VHE antibodies in recipients, as well as in the pooled plasma lots, is being further investigated. Transfusion transmission of HEV is a matter of concern for the French Blood Transfusion Establishment (EFS). Hence HEV screening was implemented in December 2012 on pools of 96 samples, as it was already done for Parvovirus B19 and Hepatitis A virus. Infectivity of HEV-RNA containing SD-plasma lots and HEV transmission by other labile components are currently being investigated.

Disclosure of Commercial Conflict of Interest

A. Assal: Nothing to disclose; C. Constant: Nothing to disclose; P. Gallian: Nothing to disclose; Y. Piquet: Nothing to disclose; P. Tiberghien: Immunetics, Grants or Research Support

Disclosure of Grants Conflict of Interest

A. Assal: Nothing to disclose; C. Constant: Nothing to disclose; P. Gallian: Nothing to disclose; Y. Piquet: Nothing to disclose; P. Tiberghien: Nothing to disclose

SP406

Photochemical Inactivation of Hepatitis E Virus (HEV) in Platelet Samples (PCs) using Mirasol Pathogen Reduction Technology (PRT) system

T Owada[1] (t-owada@jrc.or.jp), M Kaneko[1], C Matsumoto[1], M Igarashi[1], S Uchida[1], M Satake[1], K Tadokoro[1]. *[1]Central Blood Institute, Japanese Red Cross Society, Tokyo, Japan*

Background/Case Studies: Hepatitis E Virus (HEV) had previously been considered to be transmitted perorally and to be particularly localized to developing regions, but has recently been found to be spreading throughout the world, including in industrialized nations. It has also been demonstrated that HEV can be transmitted via blood transfusion. Thus, the risk of HEV transmission via transfusion is now a major global concern. The Japanese Red Cross Society has become focused on this virus as a blood-borne infectious disease, and is the first manufacturer in the world of blood-related pharmaceuticals to introduce nucleic acid amplification test (NAT) screening for HEV tentatively. An assay system for the infectious titer (tissue culture infectious dose; [TCID]) of HEV cultured from human HEV-RNA-positive plasma and serum was recently successfully demonstrated. This methodology was applied to investigate the ability of the Mirasol PRT system (Terumo BCT, Lakewood, CO) to inactivate live HEV in contaminated PC. **Study Design/Methods:** PCs were spiked with cultured HEV genotype (G) 3 or G4 and treated with riboflavin and ultraviolet (UV) light using the Mirasol PRT system. PCs were examined before and after treatment for HEV load via infectivity titration (TCID) against A549, an HEV-infectible cell line. In addition, the quantitative real time reverse transcription-polymerase chain reaction (PCR) was performed to determine the load of HEV interacting with A549 cells based on a quantitative change in a continuous process of virus binding and entry, to clarify the effects of the Mirasol PRT system on the initial step of HEV infection. **Results/Findings:** A significant reduction in HEV G3 infectious titer of more than 3 log 10 was recorded after the treatment of PCs with the Mirasol PRT system. In addition, an inactivation of more than 2 log 10 was confirmed when HEV G4 was used. However, it was demonstrated that no alteration of the efficiency of HEV binding and entry into host cells had been observed. **Conclusion:** The current data indicated that the Mirasol PRT system efficiently inactivated live HEV in PCs, and could therefore potentially be used to lower the possibility of HEV transmission through the blood products. The limited load (infectious titer, RNA copies) of HEV obtained from in vitro cultivation precluded an evaluation of the robustness of the measured log reduction in HEV infectivity. Hence, a more reliable log reduction value might be obtained if a higher load of the virus could be applied. The inactivation mechanism expressed by this system may be attributed to an inhibition of HEV replication during the post-viral entry process. In summary, the Mirasol PRT system clearly reduced the risk of HEV transmission due to blood transfusion.

Disclosure of Commercial Conflict of Interest

M. Igarashi: No Answer; M. Kaneko: Nothing to disclose; C. Matsumoto: Nothing to disclose; T. Owada: Nothing to disclose; M. Satake: Nothing to disclose; K. Tadokoro: No Answer; S. Uchida: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Igarashi: No Answer; M. Kaneko: Nothing to disclose; C. Matsumoto: Nothing to disclose; T. Owada: Nothing to disclose; M. Satake: Nothing to disclose; K. Tadokoro: No Answer; S. Uchida: Nothing to disclose

SP407

Development of an Effective Prevention Method for T-cell Lymphotropic Virus Type I (HTLV-1) Infection Using HTLV-1 Sero-Positive Serum In Vitro

K Nojima[1] (ume@niid.go.jp), T Mizukami[1], R Sobata[2], C Matsumoto[2], M Kuramitsu[1], K Okuma[1], M Satake[2], K Tadokoro[2], K Yamaguchi[1], I Hamaguchi[1]. *[1]Department of Safety Research on Blood and Biological Products, National Institute of Infectious Diseases, Tokyo, Japan; [2]Central Blood Institute, Japanese Red Cross Society, Tokyo, Japan*

Background/Case Studies: Human T-lymphotropic Virus Type I (HTLV-1) is a human RNA retrovirus that causes adult T-cell leukemia. The route of viral transmission is from mother to infant via breast milk. Thus, many countries recommend HTLV-1 sero-positive mothers not to breastfeed. In a Japanese endemic area, Nagasaki, this recommendation was more effective in reducing the number of vertical infections from mother to infant. However, new prevention methods and adequate treatment options apart from preventing HTLV-1 sero-positive mothers from breastfeeding are required. To develop effective prevention methods, immunoglobulin (IgG) was isolated from HTLV-1 sero-positive healthy carriers serologically excluded from blood donation by the Japanese Red Cross, and tested for its ability to prevent cell-to-cell infection. **Study Design/Methods:** An in vitro assay was developed to screen the inhibitory effect of IgGs on HTLV-1 infection using HTLV-1 infected cell lines MT-2 and SLB-1. Secondly, 30 HTLV-1 positive serums were prepared from excluded blood donations by CLEIA (confirmed by immunofluorescence assay in Japan in 2008), according to pro-viral load (PVL) and immunoreactivity to HTLV-1 antigens (gp46, p24, pep180 [gp46 amino acids 180-204]) to assess the prevention of HTLV-1 infection in vitro. **Results/Findings:** An in vitro HTLV-1 infection system was established using HTLV-1 infected cell lines MT2 and SLB-1 with the immortalized T lymphocyte cell line, Jurkat. During co-culture of Mitomycin C treated MT-2 or SLB-1 cells with Jurkat cells, HTLV-1 was successfully transmitted to Jurkat cells. When SLB-1 cells were donors, syncytium formation typical of HTLV-1 infected cells was observed. Next, 30 HTLV-1 sero-positive serums were tested using the co-culture system, and it was demonstrated that HTLV-1 sero-positive serum effectively inhibited HTLV-1 infection from MT-2 or SLB-1 cells to the Jurkat cells ($P < 0.0001$). In addition, syncytium formation was also dramatically inhibited by HTLV-1 sero-positive serum treatment. These effects were more prominent when HTLV-1 sero-positive serum isolated from higher PVL carriers (PVL > 4%) was used. **Conclusion:** An assessment method was successfully developed for in vitro HTLV-1 infection using HTLV-1 infected cell lines. This method may be a powerful tool to screen anti-viral agents in vitro. In addition, it was found that HTLV-1 positive serum taken from asymptomatic donors with high PVL strongly inhibited HTLV-1 infection in the in vitro assay. Hyperimmune globulins are now being prepared from each donor's plasma to identify key epitopes that prevent HTLV-1 infection more effectively, both in vitro and in vivo using a humanized mouse model.

Disclosure of Commercial Conflict of Interest

I. Hamaguchi: Nothing to disclose; M. Kuramitsu: Nothing to disclose; C. Matsumoto: Nothing to disclose; T. Mizukami: Nothing to disclose; K. Nojima: Nothing to disclose; K. Okuma: Nothing to disclose; M. Satake: Nothing to disclose; R. Sobata: Nothing to disclose; K. Tadokoro: No Answer; K. Yamaguchi: Nothing to disclose

Disclosure of Grants Conflict of Interest

I. Hamaguchi: Nothing to disclose; M. Kuramitsu: Nothing to disclose; C. Matsumoto: Nothing to disclose; T. Mizukami: Nothing to disclose; K. Nojima: Nothing to disclose; K. Okuma: Nothing to disclose; M. Satake: Nothing to disclose; R. Sobata: Nothing to disclose; K. Tadokoro: No Answer; K. Yamaguchi: Nothing to disclose

SP408

Humana T-Lymphotropic Virus (HTLV)-Screening of blood donors: Follow-Up of Infected Donors for 16 Years

H P Fernandes[1] (heloisefernandes@gmail.com), A B Zangirolami[1], A M Azevedo[1], V A da Costa[1], C A Peterlini Pompeu[1], N S Gonçales[1], M M Magnus[1], V Castro[1], F Traina[1], M Barjas-Castro[1]. *[1]Hemocentro, Universidade Estadual de Campinas—UNICAMP, Campinas—SP, Brazil*

Background/Case Studies: Blood centers in Brazil began screening blood donors for HTLV antibodies in late 1993 and today these centers routinely screen for HTLV I/II. Infection occurs in certain geographic locations around the world and in Brazil the prevalence in blood donors ranges around 0.46%. The aim of this study was to evaluate the frequency of discarded blood donations over the past 16 years, due to positive HTLV I/II screening tests in

southeast Brazil; to determine the frequency of falsely positive donors and; finally, to follow up infected donors. **Study Design/Methods:** Blood donation data between January 1, 1997-March 31, 2013 were analyzed to determine the reactive HTLV I/II tests using enzyme immunoassay-EIA(Abbott) until July 2010, and after this date using HTLV I/II chemiluminescent immunoassay-CMIA(Abbott) till March 31, 2013. Donors with positive results in the first screening were considered ineligible for future donations and the blood components were discarded. These donors were recruited to collect a second sample to repeat the screening and to undergo a medical interview with clinical exams and questions regarding transfusions, pregnancies, breast feeding, sexual partner, drugs, and symptoms related to HTLV infection. To confirm, the anti-HTLVI/II that were repeatedly positive, "in house" PCR was carried out, and in some samples Western blot 2.4(MP Diagnostics) were also carried out. From 1997 to the end of 2002, PCR and hybridization for HTLV I were performed. From 2002 to March 31, 2013, PCR tests distinguished HTLVI and II. Donation samples that were reactive only by the first screening were considered false positive. Donors with positive confirmatory tests were invited for follow-up. **Results/Findings:** From 1997 to 2013 a total of 349,666 blood donations were tested for HTLVI/II(EIA or CMIA); 740 had positive results (0.21%) and all blood products were discarded. Only 192 (26%) returned to the service for collection of the second sample. Of these, 98 (51%) presented anti-HTLVI/II repeated reactive and also confirmatory tests. Ninety-four presented PCR/WB negatives and were considered false positive. Thirty-one people with positive confirmatory results were followed (mean age 45 years; female 17/male 14). The results showed that weakness of extremities and difficulty walking were a frequent donor complaint (35%), possibly representing the first manifestation of the virus, considering that the natural history of HTLV infection remains poorly defined. **Conclusion:** The results showed that although HTLV screening tests have contributed to prevent potential transmission of HTLV through blood transfusion, many blood products have been discarded and safe blood donors have been deferred unnecessarily. The improvement of HTLV tests and the introduction of additional diagnostic assays for screening donors in blood services are necessary for better results.

Acknowledgements: The authors are grateful to FAPESP and INCTS

Disclosure of Commercial Conflict of Interest

A. M. Azevedo: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; V. Castro: Nothing to disclose; V. A. da Costa: Nothing to disclose; H. P. Fernandes: Nothing to disclose; N. S. Gonçales: Nothing to disclose; M. M. Magnus: Nothing to disclose; C. A. Peterlini Pompeu: Nothing to disclose; F. Traina: Nothing to disclose; A. B. Zangirolami: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. M. Azevedo: Nothing to disclose; M. Barjas-Castro: Nothing to disclose; V. Castro: Nothing to disclose; V. A. da Costa: Nothing to disclose; H. P. Fernandes: Nothing to disclose; N. S. Gonçales: Nothing to disclose; M. M. Magnus: Nothing to disclose; C. A. Peterlini Pompeu: Nothing to disclose; F. Traina: Nothing to disclose; A. B. Zangirolami: Nothing to disclose

SP409

Investigational Dengue Testing Yields High Rates of Ribonucleic acid (RNA)-Positive Donors in Puerto Rico

S L Stramer[1] (susan.stramer@redcross.org), G A Foster[1], J Brodsky[2], J L Muñoz-Jordan[3], E Hunsperger[3], M C Lanteri[4], G Añez[5], M Rios[5], R Cory[6], J M Linnen[6]. *[1]Scientific Support Office, American Red Cross, Biomedical Services, Gaithersburg, MD, United States; [2]Quality Analytics, Inc., Riverwoods, IL, United States; [3]Centers for Disease Control and Prevention (CDC), San Juan, Puerto Rico; [4]Blood Systems Research Institute, San Francisco, CA, United States; [5]DETTD/OBRR/CBER/FDA, Bethesda, MD, United States; [6]Hologic-Gen Probe, San Diego, CA, United States*

Background/Case Studies: Dengue is a transfusion-transmissible emerging infectious disease caused by one of four related mosquito-borne viruses (DENV-1-4); no screening method is routinely available. Ag (NS1) or ribonucleic acid (RNA) screening is feasible, but Ag testing lacks sensitivity. Screening in Puerto Rico (PR), an endemic area, is being done on every Red Cross blood donation using investigational nucleic acid test (NAT). The yield, NAT performance characteristics, and outcome of acute infection are presented. **Study Design/Methods:** On Aug 13, 2012 individual-donation DENV-NAT (Hologic-Gen-Probe/Novartis) was initiated; TMA has 50% endpoint analytic sensitivities for DENV-1-4 of <5 copies/mL. Confirmatory testing is done by repeat TMA (x10) using primary and alternate primers/probes on an independent sample (Red Cross/Gen-Probe), antibody testing (IgM/IgG; CDC), and by type-specific, real-time RT-polymerase chain reaction (PCR) (CDC; FDA) and viral sequencing (FDA). NS1-Ag testing (Red Cross/Bio-Rad) is performed on TMA-reactive samples. All TMA-reactive donors are contacted to enroll in follow-up (f/u) for subsequent sampling through seroconversion, and asked to respond to a symptom questionnaire. **Results/Findings:** During 2012-2013 (Mar 11), over 17,000 suspected cases of clinical dengue infection were reported to CDC in PR as part of a passive DENV surveillance system of which 77% were DENV-1 and 23% DENV-4. From August 13, 2012-Mar 14, 2013, ~50,000 donations were DENV-TMA tested with 125 (0.25%) initial and 112 (0.22%) repeat reactive (see Table). To date, 114/125 (91.20%) TMA-reactive donors confirmed DENV-RNA (+) with 2-10 TMA-reactive replicates, quantifiable RNA (8.5 x 10∧6 to 1.25 x 10∧10 copies/mL; CDC) and/or seroconversion (76% DENV-1, 24% DENV-4); 15/114 (13.2%) were NS1 Ag reactive (9.7 x 10∧7-1.25 x 10∧10 copies/mL; CDC). Of 50 confirmed-(+) donors enrolled in f/u, 3 (10%) remained TMA-reactive for 8-11 days; 29 (58%) had DENV-related symptoms pre- or post-donation (most common: fever, joint pain, headache, and other body aches from −10 days pre- to +9 days post donation). One additional confirmed-(+) donor who lived in the Boise, Idaho (ID) reported developing symptoms 3 days post-donation while vacationing in PR; dengue fever was diagnosed by her physician. **Conclusion:** Using NAT, DENV-RNA confirmed positivity in donors were identified in a dengue-endemic area that corresponded in DENV type/distribution to the ongoing outbreak. DENV-RNA confirmed-(+) rates of 0.23% (of which ~10% were NS1-Ag reactive) were comparable to those observed from TMA retrospective screening in prior years (2005-2011: 0.03-0.31%). Over 50% of DENV acutely infected donors developed symptoms consistent with dengue fever.

Disclosure of Commercial Conflict of Interest

G. Añez: Nothing to disclose; J. Brodsky: No Answer; R. Cory: Nothing to disclose; G. A. Foster: Nothing to disclose; E. Hunsperger: No Answer; M. C. Lanteri: Nothing to disclose; J. M. Linnen: Hologic Gen-Probe, Stocks or Bonds; J. L. Muñoz-Jordan: No Answer; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Añez: Nothing to disclose; J. Brodsky: No Answer; R. Cory: Nothing to disclose; G. A. Foster: Nothing to disclose; E. Hunsperger: No Answer; M. C. Lanteri: Nothing to disclose; J. M. Linnen: Nothing to disclose; J. L. Muñoz-Jordan: No Answer; M. Rios: Nothing to disclose; S. L. Stramer: Nothing to disclose

	2012-2013 TMA Results					
#Tested	#Initial Reactive (%)	#Repeat Reactive (%)	#Confirmed (+) (%)	#NS1 Ag Rx (%)	Positive Predictive Value (%)	% Specificity (95% CI)
49,909	125 (0.25)	112 (0.22)	114 (0.23)	15 (13.2)	91.20	99.98 (99.96-99.99)

SP410

Photochemical Inactivation of La Crosse Virus (LCV) using the Mirasol® PRT System

S D Keil[1] (Shawn.keil@terumobct.com), S Marschner[1], E J Gosney[1], R Bowen[2]. *[1]Terumo BCT, Lakewood, CO, United States; [2]Colorado State University, Fort Collins, CO, United States*

Background/Case Studies: Beyond the well-studied and well-characterized pathogens that are known to be transmitted through blood or blood products, there lie lesser known organisms that have the potential to wreak havoc on blood supplies. One such organism is Severe Fever with Thrombocytopenia Syndrome (SFTS) virus, a member of the bunyavirus family. This virus was the causative agent in the 2011 outbreak in China, where a 12% mortality rate was observed. During this outbreak it was discovered that the virus was able to be transmitted person to person through blood contact, indicating the possibility this virus could be transmitted through a tainted blood product. In this study the efficacy of the Mirasol PRT System for Plasma was evaluated on the reduction of La Crosse virus (LCV), another member of the bunyavirus family. **Study Design/Methods:** A stock of LCV was prepared by inoculation of two 150 cm^2 flasks of Vero cells with ~10^6 plaque-forming units (PFU) of the H78 strain of virus. Virus was harvested three days later and was frozen at –80 C until it was needed. A total of three recovered plasma units were combined into one plasma pool. Each of three Mirasol Treatment bags were filled with 200 mL of pooled plasma, followed by 35 mL of riboflavin. Additionally, the units received a 13 mL aliquot of LCV. A 1.5-2.0 mL sample was removed from each unit both pre-treatment and post-treatment to measure the virus titer. Each of the resulting six samples (three pre-treatment and three post-treatment) were assayed in duplicate by plaque assay on Vero cells grown in 6-well plates. Serial 10-fold dilutions of each sample were prepared in BA-1 medium and 100 µL of those dilutions were inoculated onto confluent monolayers of cells drained of medium. The plates were rocked every 10-15 minutes for 45 minutes, after which a 2 mL overlay (MEM, 5% FBS, 0.5% agarose) was added to each well. The plates were placed into a 37 C CO_2 incubator. Two days later a second overlay, identical to the first, but containing 0.03% neutral red, was added to each well. Plaques were counted the following day. Virus titers were calculated based on plaque count and dilution. **Results/Findings:** All three replicates of treated samples contained no detectable plaques and were below the limit of detection for the assay. The average pre-treatment titer for the three replicates was 5.5 log PFU/mL and the limit of detection for the assay was ≤ 2.2 log PFU/mL. The calculated reduction observed in this study was ≥ 3.3 log. **Conclusion:** The Mirasol PRT System for Plasma effectively reduced the infectivity of LCV to the limits of detection using an in vitro cell culture model. The ≥ 3.3 log reduction of LCV observed in this study demonstrates that the Mirasol PRT System may provide protection against unscreened blood borne bunyviruses.

Disclosure of Commercial Conflict of Interest

R. Bowen: Terumo BCT, Grants or Research Support; E. J. Gosney: Employee of Terumo BCT, Other; S. D. Keil: Employee of Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other

Disclosure of Grants Conflict of Interest

R. Bowen: Terumo BCT, Grants or Research Support; E. J. Gosney: Employee of Terumo BCT, Other; S. D. Keil: Employee of Terumo BCT, Other; S. Marschner: Employee of Terumo BCT, Other

SP411

Syphilis Algorithm Solution: The World's First Automated NON-Treponemal Rapid Plasma Reagin (RPR) Analyzer

M D Hyde[1] (mhyde@arlingtonscientific.com), B Card[1], H Mody[1]. *[1]Arlington Scientific, Inc., Springville, UT, United States*

Background/Case Studies: In recent years, driven by economic reasons that include labor savings and data management, blood banks have begun using automated treponemal tests (CIA/EIA) to screen for syphilis; reverse sequence screening has been shown to complicate screening by producing discordant results that would not have resulted under the traditional algorithm. Although there is a strong need for automation, there are a number of important limitations to reverse sequence screening (eg higher cost, increased detection of reactive screening results, false reactives, increased total healthcare cost, and no standardized methodology) that will prevent the universal implementation of this testing algorithm. **Study Design/Methods:** Using enhanced imaging technology and mathematical algorithms to analyze images to differentiate and characterize flocculation and non-flocculation, new nontreponemal analyzers have been developmed to meet the blood banking needs of today. Independent evaluation has demonstrated

Sensitivity - 98.1%
Specificity - 99.4%

INDEPENDENT TESTING AT 5 SITES

ASiManager-AT™

Visual Results	Reactive	Nonreactive
Reactive	2178	43
Nonreactive	20	3274

the analyzer's function and digital interpretation software to be highly accurate, sensitive, and specific for detection of syphilis in serum specimens. **Results/Findings:** The increased reactive rate of treponemal assays has complicated the effort to maximize the available blood pool; it has reduced the number of donors, and damaged the reputation of blood banks. The implementation of this testing algorithm has created a substantial amount of turmoil, especially when the results of treponemal screening (eg EIA) and rapid plasma regain (RPR) are discordant (eg EIA reactive, RPR nonreactive). Reverse screening may also result in increased reporting, false results, time, retesting and workflow. **Conclusion:** For the first time blood screening laboratories can automate nontreponemal testing and drive additional workflow and efficiency gains due to new digital particle analyzers with a CDC recommended nontreponemal test. This new technology offers blood bank laboratories a choice of semi-automated and fully-automated analyzers that will aid in blood bank efficiencies and maximize the available blood donor pool and available blood.

Disclosure of Commercial Conflict of Interest

B. Card: Arlington Scientific, Inc., Ownership or Partnership; M. D. Hyde: Arlington Scientific, Inc., Other; H. Mody: Arlington Scientific, Inc., Other

Disclosure of Grants Conflict of Interest

B. Card: Arlington Scientific, Inc., Ownership or Partnership; M. D. Hyde: Arlington Scientific, Inc., Other; H. Mody: ASI, Other

SP412

Alteration of Individual Donor Nucleic Acid Amplification Testing Strategy in Response to the 2012 West Nile Virus Season

J P Giacoletti[1] (jgiacoletti@carterbloodcare.org), P Davenport[1], J Chiu[1], L J Sutor[1,2]. *[1]Carter BloodCare, Bedford, TX, United States; [2]University of Texas, Southwestern Medical Center, Dallas, TX, United States*

Background/Case Studies: After two years without West Nile virus (WNV) activity in donors, this large blood center experienced an unprecedentedly heavy WNV season in 2012. The usual testing strategy is to trigger to individual donor nucleic acid amplification testing (ID NAT) in a geographic zone defined by zip codes around the area of activity, as opposed to doing ID NAT in the entire collection area of 58 counties and two major urban areas. Traditionally, the trigger zones may range in size from three to hundreds of zip codes, and persist one week before reverting to minipool testing unless further "hits" of activity occur. **Study Design/Methods:** In 2012 the usual test strategy was followed at the beginning of the season, triggering on and off ID NAT as presumed viremic donors (PVD) were identified. Review of data from our donors as well as from the local health department showed early onset of WNV activity as well as much higher number of PVD cases than ever previously experienced. Mapping of PVD cases showed clustering in defined parts of the metropolitan areas. The medical staff made a decision to modify the testing strategy from previous years. The geographic areas with the highest WNV activity, encompassing 117 zip codes, were designated the "Permazone" (PZ) and donors living in those areas tested by ID NAT continuously from July 25-September 7, rather than having triggering on and off. Geographic areas outside the Permazone could trigger to ID NAT in the traditional way if needed throughout this period. **Results/Findings:** The 2012 season (June 5-October 22) yielded 72 PVD cases. Of these, 48 were identified by mini-pool NAT and 24 by ID NAT (13 within the PZ). Seven cases occurred in zip codes adjacent to the PZ, and 18 in other areas. PVDs

were 60% (43) male and 40% (29) female, and average age was 46. Repeat donors comprised 90% (65) cases and first time donors 10% (7). Signal to cut-off ratios (SC/O) were high (over 20.0) in 85% (61) of the cases, between 10.0 and 20.0 in 6% (4), and less than 10.0 in 9% (7) cases. No known transmissions by transfusion have been reported to our center. **Conclusion:** The exceptionally high number of WNV PVDs in 2012 caused the center to change the testing strategy midway through the season. This action reduced the complexity of workload in the high number of trigger events that were occurring and increased precautions in an area with focal high activity for WNV. Review of data from the 2012 PVDs showed an unexplained high occurrence in repeat donors and Caucasians relative to our baseline donor population.

Disclosure of Commercial Conflict of Interest

J. Chiu: Nothing to disclose; P. Davenport: Nothing to disclose; J. P. Giacoletti: Nothing to disclose; L. J. Sutor: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Chiu: Nothing to disclose; P. Davenport: Nothing to disclose; J. P. Giacoletti: Nothing to disclose; L. J. Sutor: Nothing to disclose

Ethnic Comparison of PVD to Other Populations (in Percentages)

	Caucasian	Hispanic	African American	Other/ Undisclosed
PVD 2012	86	11	3	
Blood center donor data base	61	10	8	21
Area general population	50	28	15	7

Administrative Abstracts

Quality Management

A1-030B

Best Tissue Ordering Practices: Custom Pediatric Ortho Order Sets

N McCarthy[1] (nanette.mccarthy@choa.org), C Josephson[2,1], Z M McCarthy[3], D B Robertson[1], E Meyer[2,1]. *[1]Blood and Tissue Services, Children's Healthcare of Atlanta, Atlanta, GA, United States; [2]Pathology and Lab Medicine, Emory University School of Medicine, Atlanta, GA, United States; [3]Medical College of Georgia, Georgia Regents University, Augusta, GA, United States*

Background/Case Studies: In 2007, the pediatric healthcare system here centralized human tissue allografts within the transfusion service. This allowed consistent assessment of allograft ordering practices. Defining single transducer and activator of transcription (STAT) as allografts issued within 20 minutes of receipt of the order, all allografts (bone, aortic/pulmonary valves, etc) were ordered STAT despite urgency of surgical procedure. This practice created delays and did not allow for optimal inventory management. Since orthopedic surgeries represent 79% of implanted allografts and are typically elective procedures, it was hypothesized that the creation of customized orthopedic order sets would not only establish the Best Tissue Ordering Practices but reduce STAT allograft orders by 70%, interruptions at the issue window by 50%, and allow inventory par adjustments. **Study Design/Methods:** A 7-month retrospective audit captured each orthopedic surgeon's allograft implantation history per surgery type to establish baseline data. Ordering priorities (STAT versus Routine), interruption events at the issue window, and the allograft ordered to implantation ratio, were also audited. Collaborating with orthopedic surgical services, custom preoperative allograft order sets were created for each surgeon by surgery type. The custom order sets were implemented in February 2012. To evaluate the effectiveness of the custom order sets, a 7-month follow-up audit was performed. **Results/Findings:** From January 2011-July 2011, 161 pediatric orthopedic surgeries were performed implanting 348 allografts. 100% of the orders were ordered STAT. The mean ordered to implantation ratio was 1.26. During this time frame, transporters made three trips to the blood bank issue window per surgery (obtain blood products, deliver STAT allograft order, obtain allografts). In the follow-up audit (June 2012-December 2012), after implementation of custom ortho order sets, 210 pediatric orthopedic surgeries were performed using 274 allografts: 8% were ordered STAT. Ortho order sets are now faxed, allowing surgical transporters to make a single stop at the issue window to obtain both blood and allograft products. This workflow change reduced interruptions at the issue window by 66% for orthopedic surgeries. Due to pediatric size ranges, different sizes of the same allograft are included in some order sets. This increased the number of allografts returned to inventory elevating the ordered to implantation ratio to an average of 2.4. **Conclusion:** The AABB supports centralization of the tissue service within the transfusion service. Improvements in patient safety were achieved by ensuring that blood components and allografts were available in OR prior to commencement of orthopedic surgeries. Establishing Best Tissue Ordering Practices minimized issuing delays by streamlining the workflow and allowed for optimizing inventory management.

Disclosure of Commercial Conflict of Interest

C. Josephson: Nothing to disclose; N. McCarthy: Nothing to disclose; Z. M. McCarthy: Nothing to disclose; E. Meyer: Nothing to disclose; D. B. Robertson: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Josephson: Immucor/BioArray, Grants or Research Support, Octapharma, Consulting or Board of Director Fees; N. McCarthy: Nothing to disclose; Z. M. McCarthy: Nothing to disclose; E. Meyer: Nothing to disclose; D. B. Robertson: Nothing to disclose

A2-030B

Clots in Jumbo Plasma Units: A Blood Center's Response

J Crane[1] (jecrane@uic.edu), L Petkovic[1], K L McKibben[1], L Gillard[1], J Szymanski[1], M K Bertman[1], L Sereseanu[2], S A Campbell Lee[2], M Papari[1]. *[1]LifeSource Blood Center, Rosemont, IL, United States; [2]Pathology, University of Illinois Hospital and Health Sciences System, Chicago, IL, United States*

Background/Case Studies: Prior to distribution, all blood products undergo visual inspection for irregularities. In September 2012, this blood center received multiple reports from a customer regarding "clots" in jumbo plasma units. This abstract documents the root cause analysis of a customer service issue. **Study Design/Methods:** As each report was received, all information pertaining to the unit was documented and reviewed for any inconsistencies. All units returned were cultured for possible bacterial contamination. Additionally, the customer was contacted and the thawing procedure was reviewed as a potential cause. Multiple units were thawed at the blood center to replicate the customer's procedure. Finally, the manufacturer of the kits used in collection was requested to analyze the composition of the "clots" as well as the citrate anticoagulant present in the units. **Results/Findings:** All the units were collected with the same apheresis system. Donation record review did not reveal any inconsistencies. Multiple phlebotomists drew the impacted units, ruling out a phlebotomy-related cause. Bacterial cultures were negative. The thawing procedure was repeated on separate units stored at the blood center and the "clots" were replicated. When multiple units were thawed together, it was noted the temperature of the water bath dropped below the recommended thawing temperature. When the units were placed in the refrigerator, "clots" were identified, specifically when the internal temperature of the donor bag (taken with a calibrated infrared thermometer reading from the center of each unit thawed) fell below 30 C. It was then suspected the "clots" were cryoprecipitate formed by mimicking the procedure for its isolation. The analysis of the "clots" by the manufacturer yielded the results shown in the table. It was also noted the precipitate dissolved when heated to 37 C and returned when incubated at 0-1 C for a minimum of 24 hours. These results are consistent with cryoprecipitate. **Conclusion:** As the issue was successfully replicated, and the "clots" identified as cryoprecipitate, it was concluded the thawing process was likely to blame. Not assuring proper water bath temperature during thaw, or confirming the product was not fully thawed prior to placement in the refrigerator, resulted in creating cryoprecipitate in the jumbo plasma units. A letter was sent to the customer describing the findings and reinforcing the need to maintain the water bath at 30-37 C with gentle agitation to ensure uniform temperature throughout the unit. A minimum thaw time of 75 minutes was suggested. This case provides an excellent example of a root cause analysis involving supplier and customer.

Disclosure of Commercial Conflict of Interest

M. K. Bertman: Nothing to disclose; S. A. Campbell Lee: Nothing to disclose; J. Crane: Nothing to disclose; L. Gillard: Nothing to disclose; K. L. McKibben: Nothing to disclose; M. Papari: Nothing to disclose; L. Petkovic: Nothing to disclose; L. Sereseanu: Nothing to disclose; J. Szymanski: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. K. Bertman: Nothing to disclose; S. A. Campbell Lee: Nothing to disclose; J. Crane: Nothing to disclose; L. Gillard: Nothing to disclose; K. L. McKibben: Nothing to disclose; M. Papari: Nothing to disclose; L. Petkovic: Nothing to disclose; L. Sereseanu: Nothing to disclose; J. Szymanski: Nothing to disclose

Fibrinogen	Elevated
Factor VII Activity	Normal
Factor VIII Activity	5 times normal
Citrate Concentration	Expected Range

A3-030B

Using an "Open House" Format for Qualifying the Blood Supplier as an Approved Vendor by the Hospital Transfusion Service

D Gilbert[1] (Denise.Gilbert@Oneblood.org), J B Smith[1], A B Prichard[1], R Reik[1], R R Gammon[1], G F Leparc[1]. *[1]Oneblood, Inc., St. Petersburg, FL, United States*

Background/Case Studies: This blood center (BC) supported a "Supplier Qualification Day" open house in several of the service regions for consignee hospital transfusion services (TS) to allow them to perform an onsite visit and get the information needed to satisfy the AABB Standards, Joint Commission and FDA requirements to qualify the blood center as an approved blood supplier. Invitations were sent to each hospital partner describing the event. A "Supplier Qualification Checklist" was developed in the event participants had not created one of their own. **Study Design/Methods:** "Stations" were organized to include the following rotations: 1) Sign in and basic overview from the moderator. Each participant was given a copy of the quality plan, a summary of external audits with dates and outcomes, a summary of proficiency testing for each laboratory, certificates of accreditation and licenses, a product insert from the collection container manufacturer, and information about testing performed on donor blood and the platforms used by our contracted facility. Each document was discussed and time for questions allotted as participants went through the chain of documents presented. 2) A kiosk provided access to "On line" standard operating procedures(SOPs) including a terminal and printer so TS could print SOPs they wanted to review. 3) A "Virtual Component Manufacturing" station: a blood collection container prop was used to show the TS the diversion pouch and leukocyte reduction filter as well as a "closed system" component prep simulation. 4) There were live tours of the donor room, reference lab, and manufacturing and distribution areas so the TS could observe the processes. Temperature monitoring records, computerized blood ordering displayed on monitors, training and competency testing records, bacterial testing records, and other QC records could be examined on site. 5). A working lunch, networking, and benchmarking opportunities were also provided. 6). Finally, the participant discussed the results with our QA department and both the BC and TS "signed off" on the joint audit. **Results/Findings:** Evaluations were given to each participant and they responded with 100% favorable responses, stating the program was productive and interesting. All members thought that the open house was well worth their time and the information presented was valuable. For those hospital partners that could not attend, materials were sent to them and follow-up provided either by phone or in person visit. Some attendees represented hospital networks and representatives were able to sign the paperwork for all members within their consortium. **Conclusion:** The supplier qualification day was well received by the TS participants. Another significant advantage of this approach is that multiple audits were done in one day, thus minimizing disruptions throughout the year for additional requests.

Disclosure of Commercial Conflict of Interest

R. R. Gammon: Nothing to disclose; D. Gilbert: Nothing to disclose; G. F. Leparc: Nothing to disclose; A. B. Prichard: Nothing to disclose; R. Reik: Nothing to disclose; J. B. Smith: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. R. Gammon: Nothing to disclose; D. Gilbert: Nothing to disclose; G. F. Leparc: Nothing to disclose; A. B. Prichard: Nothing to disclose; R. Reik: Nothing to disclose; J. B. Smith: Nothing to disclose

A4-030B

Stability of Protein Content and Factor Activity Of Fresh Frozen Plasma (FFP) and Cryoprecipitated AHF during One-Year of Storage At –30 C with Two Eight-Hour Temperature Excursions To –10 C

S J Wagner[1] (Steven.Wagner2@redcross.org), S Seetharaman[1], J Kurtz[1], G Moroff[1]. *[1]Blood Components Department, American Red Cross, Rockville, MD, United States*

Background/Case Studies: Some frozen blood units occasionally experience brief periods of warming to temperatures >-18 C due to freezer failure and result in discard of product, even if continuously frozen. This study follows a previously conducted small pilot study and was designed to determine if fresh frozen plasma (FFP) and cryoprecipitated AHF stored at –30 C for one year and subjected to two 8-hour temperature excursions to –10 C displayed any decrements in protein content or factor activities compared to those of identical units continuously stored at –30 C. **Study Design/Methods:** For each experiment, two ABO-Rh identical units of WB were collected in CPD from consenting donors. Plasma was prepared within 8 hours of collection; plasma units were pooled, mixed, and divided equally by weight, 5 mL tube samples (pre-CONTROL and pre-TEST) were removed for subsequent testing, and tube samples and plasma units were immediately frozen at –30 C. Some of the paired frozen FFP units were used to prepare cryoprecipitate (CRYO). One plasma (n = 13) or one cryoprecipitate (n = 20) unit (TEST) of each pair was transferred into a –10 C freezer for 8 hours, and then returned to the original –30 C freezer. Two temperature excursions on TEST units were performed separated by 1 month. The matched unit (CONTROL) was maintained continuously at –30 C. CONTROL and TEST FFP and CRYO were then held at –30 C until ≈1 year after collection, and analyzed for protein/factor activity. Given the eight plasma and two cyro activities measured, p values of <0.00625 and <0.025 were considered significant, respectively. **Results/Findings:** None of the factor activities and protein levels measured from FFP and CRYO units which underwent two 8-hour temperature excursions to –10 C differed significantly from those that were held continuously at –30 C after 1 year of storage. TEST cryo units contained 397 ± 66 mg of fibrinogen and 186 ± 55 U of F VIII : C; all met AABB standards. **Conclusion:** Key factor activities and protein levels in FFP and cryoprecipitate which underwent two 8-hour excursions to –10 C during –30 C storage for one year were indistinguishable from those of units continuously stored at –30 C.

Disclosure of Commercial Conflict of Interest

J. Kurtz: Nothing to disclose; G. Moroff: Nothing to disclose; S. Seetharaman: Nothing to disclose; S. J. Wagner: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Kurtz: Nothing to disclose; G. Moroff: Nothing to disclose; S. Seetharaman: Nothing to disclose; S. J. Wagner: Nothing to disclose

Factor levels in FFP and CRYO afer 1 year storage at –30 C with two 8 hour temperature excursions to –10 C

Factor activity	pre-CONTROL	pre-TEST	CONTROL unit	TEST unit	p value CONTROL vs TEST
FFP (n = 13)					
APPT (sec)	35.6 ± 2.7	37.4 ± 4.5	34.5 ± 3.8	36.3 ± 3.7	0.0100
PT (sec)	11.9 ± 0.4	11.9 ± 0.4	11.9 ± 0.4	12.0 ± 0.4	0.1432
AT III act. (%N)	101 ± 10	101 ± 17	92.5 ± 12.4	95.0 ± 10.3	0.0100
F V (U/mL)	0.905 ± 0.158	0.864 ± 0.140	1.038 ± 0.099	1.026 ± 0.101	0.3909
F VIII : C (U/mL)	0.923 ± 0.252	0.917 ± 0.282	1.037 ± 0.290	1.012 ± 0.283	0.4336
F XI (U/mL)	0.711 ± 0.184	0.685 ± 0.237	0.863 ± 0.103	0.832 ± 0.088	0.1496
Fibrinogen (mg/dL)	277 ± 70	276 ± 73	278 ± 54	276 ± 53	0.2178
Protein S act. (%N)	90.4 ± 17.0	93.0 ± 7.7	85.7 ± 18.1	85.4 ± 15.1	0.8740
	Plasma tube from units CRYO was prepared (n = 20)		CRYO unit (n = 20)		
FVIII : C (u/mL)	1.38 ± 0.32	1.36 ± 0.36	11.7 ± 3.7	12.4 ± 3.7	0.2654
Fibrinogen (mg/mL)	3.08 ± 0.31	3.10 ± 0.32	26.9 ± 6.4	26.5 ± 4.4	0.6687

A5-030B

Using LEAN DMAIC Framework to Redesign a Surgical Transfusion Ordering Practice at a Major Institution

K S Badjie[1] (badjie.karafa@mayo.edu), J C Rogers[3], P M Johnson[1], K L Bundy[1], C M van Buskirk[1], J R Stubbs[1], R R Cima[2]. *[1]Department of Laboratory Medicine and Pathology, Division of Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Department of Surgery, Mayo Clinic, Rochester, MN, United States; [3]Systems and Procedures Clinical Practice Support, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Regulatory and accrediting agencies allow Transfusion Services (TS) to develop safe practice guidelines for sample procurement and pre-transfusion testing for eligible patients scheduled for elective surgery prior to the surgery date to mitigate delays. At this institution, patients may have a pre-surgical sample (PSS) collected up to 56 days before their scheduled surgery date for pre-transfusion testing. A review of our PSS data revealed that an average of 17% of eligible patients did not have a type and screen completed prior to the day of surgery making the process both ineffective and inefficient in comparison to the national average of 5% according to a CAP Q-probe survey. As a quality improvement initiative in 2011, a multidisciplinary team was formed under the leadership of a surgeon and charged with appraising the current PSS process, developing an improved standardized process, educating stakeholders, and implementation. **Study Design/Methods:** The team followed the Define/Measure/Analyze/Improve/Control (DMAIC) framework in appraising the effectiveness and efficiency of the Current State (CS) process. In Define phase, the team used a project charter to identify the project scope, sponsor, champion, and stakeholders. In the Measure phase, baseline data was collected for the CS. Examples of data included PSS collection and type and screen (PSS TS) completion rate before surgery, number of patients without a PSS on the day of surgery (SDS), external benchmarking data, PSS ordering systems, and effectiveness of Standard Surgical Blood Ordering (SSBO) procedure. In Analyze phase, the team analyzed collected CS baseline data and found several problem areas including: multiple PSS ordering systems by different surgical specialties; ineffective patient instructions for PSS collection; unfamiliarity of ordering personnel of the valid system for ordering PSS; and significant variation in SSBO utilization. In Improve phase, data driven innovative solutions were implemented which entailed working across multiple disciplines to redesign a Future State (FS) map with a standardized and transformational best practice ordering system in November 2011. **Results/Findings:** First quarter 2012 post implementation results showed significant improvements including a 53% reduction in SDS; 13% increase in PSS TS before surgery; 26% reduction in total crossmatched RBCs; 24% reduction in crossmatched RBCs not issued, plus 43% reduction in RBCs wastage, but total RBCs issued versus returned did not change and represents a future opportunity to monitor and further Control the gains. **Conclusion:** The redesigned, transformational surgical blood ordering process eliminated the SSBO and has resulted in one of the most significant surgical practice changes here in the past 20-30 years, epitomizing teamwork and multidisciplinary collaboration. The gains realized were significant and resulted in the team being awarded Excellence Through Teamwork Award.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; K. L. Bundy: Nothing to disclose; R. R. Cima: Nothing to disclose; P. M. Johnson: Nothing to disclose; J. C. Rogers: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal, Other, Haemonetics, Other; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; K. L. Bundy: Nothing to disclose; R. R. Cima: Nothing to disclose; P. M. Johnson: Nothing to disclose; J. C. Rogers: Nothing to disclose; J. R. Stubbs: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

A6-030B

Developing a Current Good Manufacturing Practice (cGMP)-Compliant Cell Therapy Laboratory Culture through a Cycle of Education, Training, Auditing, and Improvement

W G Ward[1] (bward@saint-lukes.org), M C Monn[2]. *[1]Cellular Therapy Services, Saint Luke's Hospital, Kansas City, MO, United States; [2]Quality Resources, Saint Luke's Hospital, Kansas City, MO, United States*

Background/Case Studies: As the need for regulatory compliance in cell therapy laboratories has risen, so has the need for improved training in current Good Tissue Practices/current Good Manufacturing Practices (cGMP). Prior to 2005, the concern was maintaining compliance with accreditation standards. Then in 2005 the bar was raised and FDA tissue registration and inspections became a reality. Now, cGMP compliance is necessary for the cord blood Biologics License Application here, including the requirement for real cGMP training and related internal auditing. According to 21 CFR 211.25(a), "Training in current good manufacturing practice shall be conducted by qualified individuals on a continuing basis and with sufficient frequency to ensure that employees remain familiar with cGMP requirements applicable to them." The intent was to go beyond a perfunctory annual training, so a strategy was sought that could be woven into staff meetings that would address both the training and auditing requirements of the cGMP. **Study Design/Methods:** Guided by the performance improvement model (Plan, Design, Measure, Assess, Improve), a schedule was developed to create a cycle of training, auditing, and improvement for each of the ten cGMP Subparts and their cGTP counterparts. For example, training on 21 CFR Part 211 Subpart B, Organization and Personnel, was scheduled for a staff meeting: the Quality Management Supervisor taught the cGTP/cGMPs; related standard operating procedures (SOPs) were reviewed; methods for complying with the requirements were explained; and finally, related forms and records were reviewed. During the next month a quality audit of those SOPs and their accompanying records was performed by the Quality Resource Analyst. Audit results were discussed at the next meeting. **Results/Findings:** Through these independent quality audits, deficiencies in both SOPs and records have been brought to light. Corrections to SOPs and records have been made, but more importantly, the staff's understanding of the cGMPs and their application in the SOPs and records has increased. Although traversing the education/training/auditing cycle for all the cGMPs will take time, the investment is already yielding improvements in the practical daily demonstration of cGMP compliance, and movement toward a cGMP culture. **Conclusion:** A planned methodical approach that couples cGMP auditing with cGMP training improves the understanding of staff, achieves compliance with training/auditing requirements, and improves the overall cGMP posture of the facility. As all the cGMPs are addressed, it is assumed the increased understanding of the staff will lead to improved confidence and consistency, decreased errors, and the improved quality of our products and services.

Disclosure of Commercial Conflict of Interest

M. C. Monn: Nothing to disclose; W. G. Ward: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. C. Monn: Nothing to disclose; W. G. Ward: Nothing to disclose

cGTP/cGMP Training and Audit Schedule Snapshot

Training Date	cGMP	cGTP	SOPs	Audit Date
January	Organization & Personnel	Quality Program Personnel	Quality Management Plan Roles & Responsibilities	February
March	Buildings & Facilities	Facilities Environmental control & monitoring	Facilities & Environmental Management Surface Cleaning Non-viable Air Particle Counts	April

Collections

A7-030C

Feasibility of Constructing a Multicenter Blood Donation Database for Surveillance Purposes in the US

D M Nelson[1] (diane.nelson@redcross.org), E P Notari[1], S L Stramer[2], D E Krysztof[2], R Y Dodd[1], D A Kessler[3], L Milan-Benson[4], B H Shaz[3], Z Kaidarova[5], B Custer[5]. *[1]Transmissible Diseases, American Red Cross, Rockville, MD, United States; [2]Scientific Support Office, American Red Cross, Gaithersburg, MD, United States; [3]Special Donor Services, New York Blood Center, New York, NY, United States; [4]Operations Applications Services, New York Blood Center, Long Island City, NY, United States; [5]Epidemiology, Blood Systems Research Institute, San Francisco, CA, United States*

Background/Case Studies: US blood centers currently do not share a standardized data collection, testing, and reporting process. Thus, information cannot be easily integrated between centers. Such integration would improve the capabilities of surveillance efforts in the US. **Study Design/Methods:** Data from three large US blood collection agencies were used to develop a common database. Data regarding donor demographics with summary information for *T. cruzi*, West Nile Virus (WNV), and syphilis and detailed testing for human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV), and human T lymphotropic virus (HTLV) were selected for inclusion. The specific testing algorithms used by each blood center were elucidated and graphic algorithms representing process flow leading to classification of infection status were created. Some tests were common to all three centers but others, mainly confirmatory tests, were not mutual. A programming data dictionary defined unique field names, result options, and formats for every test and outcome included in the algorithms. Demographic information was defined by field name, variable, and format in a data dictionary. At each center, algorithms and data dictionaries guided creation of five standardized extracts; donor demographics and four supplemental virus extracts. Data from each set were linked through fields common to all five extracts. Processes for securely transferring extract data from the centers to a coordinating center were defined and tested. Extracts were analyzed for logic, completeness, and match to the algorithm and data dictionary at the coordinating center. Discrepancies were reported to the originating center for amendment and resubmission. **Results/Findings:** Donor demographic extracts consistent in format and structure were completed for all three centers for 2011 data. These extracts contain details for > 7.6 million donations, representing approximately 50% of the blood annually donated in the US. The processes for HIV, HBV, HCV and HTLV extraction, transmission and quality control are in place and vetted. Quality control of the initial 2011 virus extracts uncovered discrepancies in the reporting of some confirmatory tests. Data for 2012 will be added to the existing 2011 database. **Conclusion:** The 2-year database will allow standardized, nationally representative reporting of donor infectious disease marker testing. It is also designed as a tool for process improvement ranging from blood testing practices, to creating better donor questionnaires and educational programs to reduce risk. The creation of the database allows for an independent process for verifying true positives and false positive controls for risk factor assessment. Because of the large number of independent blood centers in the US, a true national database seems unlikely. However, by combining the data of three large US blood organizations, significant coverage can be achieved.

Disclosure of Commercial Conflict of Interest

B. Custer: Novartis, Grants or Research Support; R. Y. Dodd: Nothing to disclose; Z. Kaidarova: Nothing to disclose; D. A. Kessler: Nothing to disclose; D. E. Krysztof: Nothing to disclose; L. Milan-Benson: Nothing to disclose; D. M. Nelson: Nothing to disclose; E. P. Notari: Nothing to disclose; B. H. Shaz: Haemonetics, no financial relationship, Other, Fenwal, no financial relationship, Other; S. L. Stramer: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Custer: Novartis, Grants or Research Support; R. Y. Dodd: Nothing to disclose; Z. Kaidarova: Nothing to disclose; D. A. Kessler: Immunetics, Grants or Research Support; D. E. Krysztof: Nothing to disclose; L. Milan-Benson: Nothing to disclose; D. M. Nelson: Nothing to disclose; E. P. Notari: Nothing to disclose; B. H. Shaz: Nothing to disclose; S. L. Stramer: Nothing to disclose

A8-030C

Baby Boomers and the Different Generations of First-Time and Repeat Blood Donors: A Six-Year Analysis

M D Bravo[1] (mbravo@bloodsystems.org), C E Wilcox[2], P A Tomasulo[1], H Kamel[1]. *[1]Medical Affairs Department, Blood Systems, Inc., Scottsdale, AZ, United States; [2]BCD Operations, Blood Systems, Inc., Scottsdale, AZ, United States*

Background/Case Studies: Promoting a sustainable donor base is considered essential in maintaining an adequate blood supply. Blood banks rely on committed life-long donors and as donor population aging is inevitable, a shift in donor population base is expected. This study analyzed the proportion of first-time (FT) and repeat (RPT) donations across generations during a 6-year period. **Study Design/Methods:** Allogeneic donations from a network of blood centers were analyzed. Donation status was categorized as FT or RPT based on prior donation history. Generations were categorized as follows: Veterans (V), 1922-1945; Baby Boomers (B), 1946-1964; Generation X (X), 1965-1980; Generation Y (Y), 1981-1996, based on the western generation categories. Stratified rate calculations by year for each generation by donation status were performed. Chi-square tests were performed to compare 2007 (2008 for Y, based on acceptance of 16 year old in 2008) and 2012 rates across all groups. **Results/Findings:** During the 6 year period, there were approximately 4.7 million donations. There was an overall decrease in the proportion of older donors (V + B), 52.1-43.6%, while the proportion of donations from younger donors (X + Y) increased (47.9-56.4%) (Table). B RPT donors comprised the largest proportion of donations across all years (36-34.4%). RPT donations also increased among Y donors (13.3-22.8%). However, among donations from FT donors, there is a decrease in the older generations (V, 0.9-0.3% and B, 4-2%), as well as in younger donors (X, 5-2.9% and Y, 11.2-10.6%). All rate comparisons were highly statistically significant ($p < .0001$). **Conclusion:** The decreasing trend in donations from FT donors across all generations indicate a need to re-evaluate current collection and recruitment focus especially with the population shift in eligible donors. B donors contribute the majority of blood donations especially in RPT donations. As the B generation ages, health promotion will be valuable in prolonging their donation eligibility. At the same time, appealing to a broader population of younger donors to adopt the same engagement, promote a regular donation habit, and encourage altruism, as the older generations have shown, would be the continued challenge to blood bank organizations. The changing generations pose an opportunity for a holistic approach (with an understanding of the characteristics of these generations) in maintaining and recruiting younger and older donors which would be essential in sustaining an adequate blood supply for many years to come.

Disclosure of Commercial Conflict of Interest

M. D. Bravo: Nothing to disclose; H. Kamel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose; C. E. Wilcox: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. D. Bravo: Nothing to disclose; H. Kamel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose; C. E. Wilcox: Nothing to disclose

Year	Donations (N)	Generations									
		Veterans		Baby Boomers		Generation X		Generation Y		Overall	
		FT	RPT	FT	RPT	FT	RPT	FT	RPT	FT	RPT
2007	766,221	0.9	11.2	4.0	**36.0**	5.0	19.3	10.1	13.5	20.0	80.0
2008	828,195	0.7	10.2	3.6	**34.9**	4.7	19.5	11.2	15.3	20.2	79.8
2009	823,998	0.6	9.4	3.1	**34.0**	4.2	19.7	11.0	18.1	18.9	81.1
2010	783,910	0.4	8.8	2.3	**33.6**	3.3	19.9	10.6	21.0	16.7	83.3
2011	729,057	0.4	8.4	2.1	**33.7**	3.0	20.0	10.4	22.0	15.8	84.2
2012	719,170	0.3	7.9	2.0	**33.4**	2.9	20.1	10.6	22.8	15.8	84.2

Note: Chi-square test between 2007 and 2012 rates across all groups (except for Gen Y, comparison between 2008 (16 y/o acceptance) and 2012) are all statistically significant, p<.0001.

A9-030C

Prescreening with Noninvasive Hemoglobin(Hb) May Streamline the Blood Donation Process

P H Whitley[1] (Pam.Whitley@redcross.org), L Kline[2], S Graminske[3], M Corradino[4]. *[1]American Red Cross, Norfolk, VA, United States; [2]American Red Cross Holland Laboratory, Rockville, MD, United States; [3]BloodCenter of Wisconsin, Milwaukee, WI, United States; [4]Clinical Research, Masimo Corp., Irvine, CA, United States*

Background/Case Studies: Many blood donation centers spend significant time and resources screening prospective donors who are ultimately deferred due to low hemoglobin (Hb). In populations with high deferral rates,

such as high school and university students and certain ethnic groups, this issue is especially burdensome. A pain-free, quick, pre-screen assessment of Hb could prevent prospective donors likely to be deferred from having to proceed through additional testing, including an invasive blood test. This has the potential to improve efficiency and the donor experience while lowering overall cost. The aim of this study was to evaluate if noninvasive Hb measurement with Pronto-7 (SpHb) could be used to prescreen donors prior to an invasive test. **Study Design/Methods:** SpHb values (Pronto-7 sw ver 2.2.16, with rainbow 4D sensor, rev F, Masimo Corp.) and venous blood samples were collected from prospective blood donors at five blood donation facilities. Prospective donors likely to be near the qualifying Hb value of 12.5 g/dL were preferentially recruited to enrich the sample of subjects close to the cutoff. Venous samples were analyzed by cyanmethemoglobin assay (HiCN), the gold standard for hemoglobinometry, to serve as a reference. The percentage of donors appropriately deferred (sensitivity) and appropriately accepted (specificity) by SpHb were calculated. The effect of adjusting the qualification level of SpHb by −0.1 increments to optimize the sensitivity for detection of <12.5 g/dL was tested by retrospective analysis. **Results/Findings:** Data were collected from 429 subjects. Based on HiCN, 107 subjects would have been deferred (25%) and 322 accepted. SpHb sensitivity and specificity were 53% and 88%. Each 0.1 g/dL decrease in the qualification level of SpHb increased sensitivity by about 3% and decreased specificity by about 2%. A SpHb qualification level of 12.0 g/dL resulted in 73% sensitivity and 75% specificity for detection of <12.5 g/dL hemoglobin (Table). **Conclusion:** Adjusting the qualification level for SpHb improves the sensitivity for detecting deferrable donors and may be an option for centers who wish to adopt a quick pre-screen Hb assessment. Because SpHb testing is noninvasive, prospective donors likely to be deferred will avoid the finger stick required for an invasive capillary test. This has the potential to improve the donor experience and encourage future attempts to donate while streamlining donor screening.

Disclosure of Commercial Conflict of Interest

M. Corradino: Masimo, Other; S. Graminske: GenesisBPS, Grants or Research Support, MacoPharma, Grants or Research Support, Haemonetics, Grants or Research Support, Cerus, Grants or Research Support, Masimo, Grants or Research Support; L. Kline: Masimo, Grants or Research Support; P. H. Whitley: Massimo, Grants or Research Support, Citra, Grants or Research Support, Haemonetics, Grants or Research Support

Disclosure of Grants Conflict of Interest

M. Corradino: Masimo, Grants or Research Support; S. Graminske: Terumo BCT, Grants or Research Support, Fenwal, Grants or Research Support, Haemonetics , Grants or Research Support, Cerus, Grants or Research Support; L. Kline: Nothing to disclose; P. H. Whitley: Citra, Grants or Research Support, Haemonetics, Grants or Research Support

Donors Deferred and Accepted at Different SpHb Values

g/dL	Donors Deferred, n (%)	Donors Accepted, n (%)
SpHb = 12.4	64 (60)	280 (87)
SpHb = 12.3	70 (65)	270 (84)
SpHb = 12.2	73 (69)	264 (82)
SpHb = 12.1	77 (72)	254 (79)
SpHb = 12.0	78 (73)	242 (75)
SpHb = 11.5	90 (84)	199 (62)

Education and Training

A10-030D

Development of a Multiplayer,Virtual-Reality,Obstetric Hemorrhage Simulation Program

E L Lockhart[1] (lockh010@notes.duke.edu), T Allen[2], M Steele[2], A Bonifacio[2], E Hobbs[2], L R Brancazio[3], J Taekman[2]. *[1]Pathology, Duke University Medical Center, Durham, NC, United States; [2]Anesthesiology, Duke University Medical Center, Durham, NC, United States; [3]Obstetrics and Gynecology, Duke University Medical Center, Durham, NC, United States*

Background/Case Studies: Obstetric hemorrhage (OH) is a leading cause of maternal morbidity and mortality worldwide. Multidisciplinary initiatives are needed to improve outcomes in women experiencing OH. To this end, the Joint Commission recommends the use of clinical protocols for OH management. These protocols must be practiced by the team to ensure effectiveness, but mannequin-based simulations can be expensive and challenging to coordinate with all team members. In partnership with the Virtual Heroes division of Applied Research Associates, Inc., a web-based computer OH simulation was developed for the purpose of training and evaluating participants in use of an OH protocol as well as reinforcing teamwork and communication. **Study Design/Methods:** The Obstetric Hemorrhage Scenario (OHS) is an immersive multiplayer learning application built using the ILE@D platform and deployed via web plugins. Participant communication is accomplished via voice over IP (VoIP) and text chat windows. Participant move avatars through a virtual patient room and are able to interact with both the patient and objects in the room. Scenario content is constructed around an OH due to uterine atony in a primigravida. The instructor controls all patient physiology by either manual control or macro-based controls, and can adjust patient vitals and appearance based on participant therapeutic interventions. Participants are able to perform physical exam, administer fluids and medications, order labs, order and transfuse blood components, and view patient data in an electronic health record. Teamwork and communication behaviors of participants are observed and evaluated according to TeamSTEPPS principles. Knowledge learning objectives include: 1) early recognition of the signs of uterine atony; 2) appropriate use of uterotonics; 3) activation of the OH protocol; and 4) ordering laboratory studies and transfusing blood products according to the OH protocol and laboratory data. Key elements of transfusion practice observed include two-person bedside verification against the patient armband, ensuring ABO compatibility of blood products, and observation of blood product expiration times. **Results/Findings:** Simulation build is complete and currently undergoing multidisciplinary beta testing in both the US and Australia. **Conclusion:** The construction of a web-based virtual reality simulation has created a training tool that emphasizes team communication, evidence-based obstetric interventions, safe transfusion practices, and appropriate blood management for OH. This customizable, cost-effective tool allows for institution-wide training of team members in the OH protocol and provides a widely available opportunity for research into individual- and team-decision making during OH management without the need for geographic co-localization of participants.

Disclosure of Commercial Conflict of Interest

T. Allen: Nothing to disclose; A. Bonifacio: Nothing to disclose; L. R. Brancazio: Nothing to disclose; E. Hobbs: No Answer; E. L. Lockhart: Nothing to disclose; M. Steele: Nothing to disclose; J. Taekman: Applied Research Associates, Other

Disclosure of Grants Conflict of Interest

T. Allen: Nothing to disclose; A. Bonifacio: Nothing to disclose; L. R. Brancazio: Nothing to disclose; E. Hobbs: No Answer; E. L. Lockhart: Nothing to disclose; M. Steele: Nothing to disclose; J. Taekman: Pfizer, Grants or Research SupportAbbott,Grants or Research Support

A11-030D

Development of a Validated Exam to Assess Physician Transfusion Medicine Knowledge

R Haspel[1] (rhaspel@bidmc.harvard.edu), Y Lin[2], P B Fisher[3], A Ali[3], E R Parks[3]. *[1]Pathology, Beth Israel Deaconess Medical Center, Boston, MA, United States; [2]Clinical Pathology, Sunnybrook Health Sciences Center, Toronto, ON, Canada; [3]American Society for Clinical Pathology, Chicago, IL, United States*

Background/Case Studies: There is evidence that physicians lack adequate transfusion medicine knowledge. To design needs-based educational interventions to address this gap, a validated assessment tool is required. Previously published exams have not been created or validated using rigorous psychometric methods. **Study Design/Methods:** A modified Delphi method was used to achieve consensus regarding the essential knowldege and skills for physicians that transfuse blood products. To ensure content validity, members of an international organization of transfusion medicine experts (Biomedical Excellence for Safer Transfusion [BEST] Collaborative) participated in the exam design process. An exam, based on the most highly rated topics, was created and administered to individuals with a priori expected basic, intermediate and expert levels of transfusion medicine knowledge. Rasch analysis, a psychometric technique used in high stakes medical licensure and board testing, was used to determine exam accuracy and precision. **Results/Findings:** Thirty-six topics achieved ratings sufficient to be considered for inclusion in the exam (content validity index >0.8). A twenty-three question exam was administered to 49 individuals. Mean scores for individuals with expected basic, intermediate and expert knowledge were 42%, 62%, and 82% respectively ($p < .0001$). The exam achieved good fit with the Rasch model. **Conclusion:** A validated exam has now been

created to accurately assess transfusion medicine knowledge. This exam can be used to determine knowledge deficits and assist in the design of curricula to improve blood product utilization.

Disclosure of Commercial Conflict of Interest

A. Ali: No Answer; P. B. Fisher: Nothing to disclose; R. Haspel: Nothing to disclose; Y. Lin: Nothing to disclose; E. R. Parks: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Ali: No Answer; P. B. Fisher: Nothing to disclose; R. Haspel: Nothing to disclose; Y. Lin: Nothing to disclose; E. R. Parks: Nothing to disclose

A12-030D

Assessing Knowledge of Nurses on Transfusion using an Online Survey

M G Briggs[1] (margimargal@yahoo.com), S T Johnson[1], N L Fredrich[1], P E Kathleen[1]. *[1]SBB Education Program, BloodCenter of Wisconsin, Milwaukee, WI, United States*

Background/Case Studies: While nurses perform a majority of blood transfusions in US hospitals, their education in transfusion practice is limited to blood administration and transfusion reactions. Few published studies describe nurses' knowledge of transfusion medicine related topics. Using a confidential online survey the aim was to assess the current knowledge nurses have about transfusion practice and to identify gaps for future education. **Study Design/Methods:** An online survey was developed in the following areas where nurses are most involved with the transfusion process: 1) Sample collection and testing; 2) Blood product administration; 3) Components and compatibility; 4) Transfusion reactions; and 5) Blood supply risks. Demographic data was solicited to determine frequency of blood product administration, type of nursing license, years in practice, and primary patient care area employed. Survey consisted of 26 multiple-choice questions utilizing Checkbox® software. The target audience was nurses practicing in Wisconsin hospitals. Email addresses were obtained utilizing the 2011 state hospital directory. Distribution was via email letter with survey link to nurse educators and directors with instructions to distribute to their nursing staff. The survey was open for 4 weeks in 2012. Data was analyzed using SPSS software®. **Results/Findings:** 176 surveys were completed. In 24/176 surveys, the respondents had not transfused any blood products for 6 months or longer and were excluded. The remaining 152 surveys showed 96% of respondents were licensed as RNs with greater than 50% employed for more than 10 years. Frequency of blood administration varied. Survey results were scored with each question worth one point for a total of 25 points. A question on ABO compatibility of plasma was eliminated from the total (26) since only 7% of respondents answered it correctly. Median and mean for overall score and each category of the survey is shown in the Table. Nurses with more experience scored higher in sample collection and testing than those with less experience (2.9 versus 2.5). Nursing knowledge varied between type of nursing unit and frequency of blood administration. **Conclusion:** Nurses had an overall mean score of 17.4 out of 25. Knowledge in transfusion practice was wide and varied. Highest score was in blood administration with a mean score of 7.2 out of 9. Mean score of 2.7 out of 5 showed a knowledge gap in transfusion reactions. Survey results support the need for nursing education in blood transfusion practice.

Disclosure of Commercial Conflict of Interest

M. G. Briggs: Nothing to disclose; N. L. Fredrich: Nothing to disclose; S. T. Johnson: Nothing to disclose; P. E. Kathleen: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. G. Briggs: Nothing to disclose; N. L. Fredrich: Nothing to disclose; S. T. Johnson: Nothing to disclose; P. E. Kathleen: Nothing to disclose

Survey Scores (overall and 5 categories)

	Median	Mean	Standard Deviation	Variance
Overall Score (25)	18	17.4	2.286	5.227
Sample Collection & testing (4)	3	2.8	.917	.841
Blood Administration (9)	7	7.2	.969	.939
Blood components & compatibility (4)	3	2.9	.792	.628
Transfusion reactions (5)	3	2.7	1.031	1.063
Blood Supply risks (3)	2	1.9	.584	.341

A13-030D

Faster, Cheaper, More Personal: An Approach to Distance Learning for Hospital Customers (HCs)

M A Lieb[1] (mlieb@bloodsystems.org), B R Swanson[1], K Tellis[2], C Woods[2], J Sanders[2]. *[1]Quality Source, Blood Systems Inc, Scottsdale, AZ, United States; [2]Sales and Marketing, Blood Systems, Scottsdale, AZ, United States*

Background/Case Studies: Hospital staff (HS) are challenged to keep abreast of current industry trends and knowledge to maintain quality processes for compliance and patient care. Funding of off-site continuing education (CE) is limited, if available at all for HS. Human capital is also limited to support work processes, when HS are off-site for CE. A large blood center (BC) consulting division has implemented a refined approach to webinar (WEB) presentations for CE as a value-added benefit for the hospital customer (HC) educational needs. **Study Design/Methods:** The BC selected the web-based GoToWebinar to deliver the WEBs. The method allows the BC to schedule and deliver hot topics in an efficient and timely way. The WEBs cover regulatory/compliance topics, as well as clinically based, patient-focused applications. The WEBs are scheduled for 1 hour (10 AM PT-1PM ET) on a quarterly basis. HC in 20 states are invited by email which includes the topic and the registration link for the live presentation. An encore WEB is provided a month later for those who could not attend the first presentation. The presentation handouts are emailed to the registrants the evening before the presentation. The WEBs are recorded and available along with certificates of attendance (CA) for up to 6 months after the presentation on the BC's website. Each WEB can support a maximum of 500 registrants. During the WEB, there is an opportunity depending on topic, for participants to answer polling questions. Each WEB has a question and answer period at the end of the presentation. Participants have the capability to evaluate the WEB and print CA issued electronically after the WEB. **Results/Findings:** The WEBs began in July 2011. To date, 12 topics and 24 WEBs have been presented to 1645 registrants. WEB attendees include laboratory directors, supervisors, technologists, nurses, and physicians. An average of 187 people have clicked the registration link (range 88-295) and the average number of registrations is 137 (range 77-221). The WEB topics, "Adverse Reactions to Transfusion" and "Transfusion Administration-Getting the Right Blood to the Right Patient Every Time" were utilized for CE training for hospital transfusionists by some HC. The average number of questions generated during a presentation is 13 (range 2-36); the two WEBs with more regulatory focus, "Biological Product Deviation Reports" and "Competency Assessment-Is Your Program Competent?" generated the most questions, 54 and 45, respectively. **Conclusion:** The WEBs provide HC with an easy, timely, and pertinent educational opportunity to enhance their knowledge on blood industry topics provided in a cost-effective manner. It also allows for the BC to communicate current industry topics consistently across its entire customer base. Since WEB participation can be done individually or as a group, the WEBs facilitate discussion related to the HC's policies and procedures, while providing CE hours for the participants.

Disclosure of Commercial Conflict of Interest

M. A. Lieb: Nothing to disclose; J. Sanders: Nothing to disclose; B. R. Swanson: Nothing to disclose; K. Tellis: Nothing to disclose; C. Woods: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. A. Lieb: Nothing to disclose; J. Sanders: Nothing to disclose; B. R. Swanson: Nothing to disclose; K. Tellis: Nothing to disclose; C. Woods: Nothing to disclose

A14-030D

Focusing on Organizational Values to Improve Compliance/Customer Service

A Turner-Wilcox[1] (angelie.turner@redcross.org), D Haab[1], J Frieboes[1]. *[1]Collections, American Red Cross Blood Services, Pomona, CA, United States*

Background/Case Studies: Blood centers face the challenge of creating and reinforcing critical messaging which creates enthusiasm; recognizing, engaging, and developing staff; and cultivating a culture of compliance and promoting customer service. The American Red Cross Southern California Blood Region (SCR) employs over 470 frontline collection staff in a large geographical area. In June 2012 SCR recognized the need to develop a unique and creative program where strategies conducive to learning could be implemented every day to help achieve a higher level of performance. There was a need for a comprehensive program reaching staff daily, generating discussions about the organization's mission and that saving lives starts with happy donors. Creating and implementing a successful program

staff could identify with would strengthen the SCR. **Study Design/Methods:** A series of modules were designed for pre and post huddles using the Organization's Values called the 5 C's: Compassionate, Collaborative, Creative, Credible, and Committed. These 5 C's became the framework for supervisors to facilitate discussions at blood drives focusing on compliance/customer service. Staff was encouraged to share what each value meant to them as they tried to improve their compliance and customer service. During post huddle, staff was engaged to share what observations and outcomes they saw as they demonstrated this value in relation to compliance/customer service. Module orientation was delivered through interactive discussions with supervisors, followed by supervisor facilitation. Repetition and reinforcement was critical, with one of the values discussed for 2 weeks every day. Customer concern and compliance data was tracked during the 7-month implementation phase and compared to the same time the previous year. **Results/Findings:** Staff was engaged throughout their day and in huddles provided examples of an action where one of the 5 C's was modeled with positive results. Non-procedure type customer concerns in our largest districts decreased by 52%. A 2013 survey indicated 79% of our staff felt the focus with these modules was "effective in communicating priorities" and 92.5% felt quality efforts and compliance goals improved. The use of these modules in conjunction with procedural changes and effective corrective actions contributed to behavioral changes reducing total problems more than 25%. **Conclusion:** The use of these modules contributed to a significant impact promoting a culture of compliance and improved customer service. The models proved cost-effective, reaching our collection staff daily and generating pre and post huddle discussions that were engaging and promoted recognition and development. The curriculum design provided flexibility during non peak donor times.

Disclosure of Commercial Conflict of Interest

J. Frieboes: No Answer; D. Haab: Nothing to disclose; A. Turner-Wilcox: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Frieboes: No Answer; D. Haab: Nothing to disclose; A. Turner-Wilcox: Nothing to disclose

A15-030D

Implementation of Standards for Blood Transfusion in Zimbabwe

J J Parirewa[1] (jparirewa@bloodbank.co.zw), T Mapako[1], D A Mvere[1].
[1]Medical Services, National Blood Service Zimbabwe, Harare, Zimbabwe

Background/Case Studies: The Ministry of Health and Child Welfare (MoHCW) jointly with the National Blood Service Zimbabwe (NBSZ) developed and published standards for Blood Donation, Processing and Clinical Transfusion (Standards) in July 2011. The Standards are divided into four main areas: Quality Management Systems; Elements of a Blood Service; Blood Transfusion Activities; and Monitoring and Evaluation. The aim of this abstract is to share how the Standards have been implemented in hospitalsthrough PEPFAR support and highlight the key success factors to date. **Study Design/Methods:** Two workshops for nurses, doctors, and laboratory scientists aimed at disseminating the Standards for blood transfusion to key staff at health institutions were held. The workshops were also used to establish gaps between current practice in the area of blood transfusion in hospitals and desired standards. The baseline as well as areas that need improvement were established using a structured gap analysis questionnaire. The questionnaire focused on four main areas: policy and guidelines; documents and records; processes and control; and process improvement. Participants completed the questionnaire in groups according to the level of their institutions ie private, district, provincial, and central hospitals. Nineteen hospitals participated and 49 participants attended the training. **Results/Findings:** The baseline survey indicated that: There are few policies and procedures in hospitals throughout the country (range 14%-57%) ; There are documents and records in place in the hospitals (range 43%-67%), however they are not standardized; Process improvement needs to be established in all hospitals as these were rated 0% byall hospitals; The greatest strength was on process and controls where all hospitals managed to score at least 60%; District hospitals had the lowest overall average score (27)% and Ppivate hospitals had highest overall average score of 40%; Participants recommended a Training of Trainers as the way forward. **Conclusion:** The development of Standards for Blood Transfusion enabled a holistic approach in review of operations related to blood transfusion at health Institutions in Zimbabwe. In order to standardize documents in hospitals, a committee was chosen from the participants following a TOT to develop national materials for training, which will be adopted by all hospitals conducting blood transfusion in the country. Documents developed and piloted include two standard operating procedures on ordering and administration of blood/blood components, forms on adverse reactions, national blood bank requests, and hospital ordering and usage. District and provincial trainings are expected to be rolled out once national documents have been finalized. Establishment of Hospital Transfusion Committees after training will create a system for effective monitoring and evaluation in hospitals.

Disclosure of Commercial Conflict of Interest

T. Mapako: Nothing to disclose; D. A. Mvere: Nothing to disclose; J. J. Parirewa: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Mapako: Nothing to disclose; D. A. Mvere: Nothing to disclose; J. J. Parirewa: Nothing to disclose

Inventory Management, Storage, Distribution and Product Manufacturing

A16-030E

Improving Methods for Matching Single Donor Platelet (SDP) Production to True Demand and Reducing Outdate Waste

T Cummings[1] (tcummings@bloodcenters.org). [1]Blood Centers of the Pacific, San Francisco, CA, United States

Background/Case Studies: In an effort to reduce single donor platelet (SDP) outdates while maintaining adequate inventory in 2013 and beyond, the blood center has adopted a two-part strategy to help match the number of usable units in finished-goods inventory to anticipated customer usage by day of the week: 1) accurately predicting available inventory based on various production factors; and 2) understanding customer usage patterns by day. With 100% SDP consignment across 49 customers, the blood center retains complete responsibility for the transportation and management of SDPs (with >24 hours of shelf-life). Flexibility built into the platelet appointment schedule (to include available staff and machines) has allowed the blood center to routinely meet all customer orders while minimizing unplanned outside purchases. **Study Design/Methods:** SDP split rate, donor "no-show" and deferral rates, SDP laboratory losses, quantity not sufficient (QNS) collections rate, and current outdate rate were monitored at defined periodicities starting in January 2013 and continue on an on-going basis. These data were reviewed together on a weekly basis to help predict the number of products available by day from a given number of appointments. Net distribution averages were calculated over 4- and 10-week periods, and tied back to predicted appointment needs using a "Platelet Appointment Calculator" spreadsheet. Using the listed roduction factors, the calculator identified a specific number of appointments needed each day to fill predicted demand. In addition to monitoring total distributions, the blood center uses year-to-year data (2010-2013 current) to identify weekly trends and adjust the following week's appointments (during a weekly Production Meeting) to fit predicted fluctuations (eg temporary site closures, holidays, etc). **Results/Findings:** Platelet outdate rates have remained steady in the first 4 months of 2013, with rates of 8.6%, 7.4%, 8.4%, and 8.8% respectively. The center's overall outdate rate for 2012 was 10.5% (range: 4.4%-17.2) and imports averaged 105 per month (range: 16-293). 2013 YTD outdate rate is 8.3%, while imports average 23. **Conclusion:** Careful matching of production rates to a rolling current demand has helped the center achieve a more stable outdate rate over the first 4 months of 2013, while minimizing imports. Additional scaling of platelet collections to demand, and better monitoring of production factors impacting inventories as more data are available, will help continuously improve outdate rates.

Disclosure of Commercial Conflict of Interest

T. Cummings: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Cummings: Nothing to disclose

Platelet Imports and Outdates by Month

Month	Total SDP Produced	Total SDP Imported	Total SDP Outdates	Outdate Percentage
January 2013	2714	33	237	8.6
February 2013	2449	30	183	7.4
March 2013	2703	14	228	8.4
April 2013	2665	13	237	8.8

A17-030E

A Positive (A+) Outcome for Demand-Based Red Blood Cell (RBC) Collections

S Purcell[1] (spurcell@bloodsystems.org), T Ferber[1], J Falk[1], D Farnsworth[1], C Damm[1], M Nelson[1], L Asay[1], C Ellingson[1]. *[1]Blood Systems, Billings, MT, United States*

Background/Case Studies: An opportunity was identified in the blood center to improve the process of recruitment and collection of donors to optimize inventory levels by blood type. High A+ donor collections had resulted in variable inventory levels, causing excessive A+ expirations. Rather than suppression of A+ 2RBC collection, the center recognized the need to develop a method to control A+ inventory while maximizing 2RBC collections. **Study Design/Methods:** A cross functional team of department directors began meeting on a weekly basis to review data and make real time decisions based on product demand forecasts and operational efficiency. A spreadsheet was created to aid in tracking and decision making. The spreadsheet included data based on historical donation by blood type for each scheduled drive and projected procedure/product goal assuming a 25% 2RBC rate of total collections. Metric graphs and control charts were created and updated to assist with monitoring progress and making real time decisions. A pull system for A+ and O= recruitment and collections was created to align supply with demand. The center managed collection of A+ RBCs by maintaining a 10-12% A+ 2RBC rate with a cumulative 2RBC rate over 20%. Reports were provided to staff to aid in goal achievement. In addition, employee performance metrics included 2RBC rate and O= type specific goals. A+ donor lists were restricted for mobile blood drives based on current inventory levels and historical demand. The team evaluated the ratio of A+ to O= donors drawn at historical drives to determine whether to cancel blood drives or reduce marketing efforts. A subgroup production-planning team met weekly to carry out the objectives of the team and ensure communication to front line production staff. **Results/Findings:** The cumulative 2RBC rate increased from 18%-23% of total collections. The A+ 2 RBC rate was managed at 10-12% and the O= 2 RBC rate averaged a respective 36%. By aligning the A+ RBC collection level to customer demand, the center maintained an appropriate level of A+ RBCs while reducing waste due to expirations. The A+ RBC expiration rate prior to improvements was 2.71% and after improvements was decreased to 0.74%. The center realized an approximate 72% decrease in A+ RBC expirations for an annualized savings of approximately $60,000. **Conclusion:** The project success is attributed to the team's ability to proactively identify and mitigate concerning trends in real-time based on demand and to challenge goal targets. Efficiency has been increased by aligning A+ supply to demand, reducing A+ 2RBC expirations, and increasing 2RBC collections.

Disclosure of Commercial Conflict of Interest

L. Asay: Nothing to disclose; C. Damm: No Answer; C. Ellingson: Nothing to disclose; J. Falk: Nothing to disclose; D. Farnsworth: Nothing to disclose; T. Ferber: Nothing to disclose; M. Nelson: Nothing to disclose; S. Purcell: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Asay: Nothing to disclose; C. Damm: No Answer; C. Ellingson: Nothing to disclose; J. Falk: Nothing to disclose; D. Farnsworth: Nothing to disclose; T. Ferber: Nothing to disclose; M. Nelson: Nothing to disclose; S. Purcell: Nothing to disclose

A18-030E

The Optimization of Red Blood Cell (RBC) Conservation and Inventory Management in Rural Ontario

R D Scheuermann[1] (robert.scheuermann@hpha.ca), G Johnston[1,3], M Carlson[1,4], K Ellis[5,2], B McMahon[6,7]. *[1]InterHospital Laboratory Partnership, Huron Perth Healthcare Alliance, Stratford, ON, Canada; [2]Laboratory Services, South Huron Hospital, Exeter, ON, Canada; [3]Laboratory Services, Hanover & District Hospital, Hanover, ON, Canada; [4]Laboratory Services, Alexandra Marine & General Hospital, Goderich, ON, Canada; [5]Laboratory Services, Listowel Wingham Hospitals Alliance, Listowel, ON, Canada; [6]Laboratory Services, Groves Memorial Community Hospital, Fergus, ON, Canada; [7]Laboratory Services, North Wellington Health Corporation, Mt. Forest, ON, Canada*

Background/Case Studies: This initiative, through 2012, served to optimize red blood cell (RBC) conservation and inventory management to deliver a high level of blood transfusion support for 230,000 people over a large rural area of Ontario. An inventory redistribution system was optimized leveraging the strengths within the InterHospital Laboratory Partnership (IHLP) of twelve hospital laboratories and the support of Canadian Blood Services (CBS). This allows all of the rural hospitals to comfortably carry reduced and consistent inventories. An added advantage is it creates a large pool of blood products within a manageable geographic region for emergency preparedness. As dedicated stewards of Canada's finite blood resources, the IHLP was committed to markedly decrease the amount of outdated RBCs. The cost of outdated RBCs to the healthcare system for this region in 2011 was $197,500.00 equating to an 11.2% outdating rate. **Study Design/Methods:** A Project Charter detailed the following components: Problem Statement; Objectives and Goals; Scope; Metrics; Business Needs; DMAIC (Define, Measure, Analyze, Improve, Control (sustain); and Schedule. Other quality improvement methods were used during the project, such as Root Cause Analysis/Defect Management, LEAN® and Six Sigma® Methodologies. The RBC Inventory Management current state map and the Ishikawa Diagram allowed for the visualization and identification of possible causes for poor red cell inventory management. This initiative involved many different stakeholders; CBS and the Ontario Regional Blood Coordinating Network (ORBCoN) provided much of the collated data on blood inventories including transfused-to-total inventory and outdating ratios. **Results/Findings:** The results of the year-long project were remarkable. The hub Laboratory accepted near-to-expired red cell units from smaller partner hospital sites within a controlled manner. In the past year and half, efforts made to address inventory levels have resulted in a decrease of outdate rates from 20% in 2010/11, to 11% in 2011/12, to an outstanding low of <1% to date. Approximately $183,500.00 was saved to the Healthcare system in one calendar year 2012. **Conclusion:** The redistribution efforts of the IHLP's twelve hospital transfusion services optimized red blood cell inventory management in rural Ontario. Continuous Quality Improvement and Process Excellence are now hard-wired in the Quality Management System, which helps to ensure sustainability. Government stakeholder linkages, particularly with CBS and ORBCoN, continue to support blood transfusion services at all Ontario laboratories.

Disclosure of Commercial Conflict of Interest

M. Carlson: Nothing to disclose; K. Ellis: No Answer; G. Johnston: Nothing to disclose; B. McMahon: No Answer; R. D. Scheuermann: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Carlson: Nothing to disclose; K. Ellis: No Answer; G. Johnston: Nothing to disclose; B. McMahon: No Answer; R. D. Scheuermann: Nothing to disclose

The Optimization of Red Blood Cell Conservation and Inventory Management

TABLE 1. Initiative Metrics: Baseline, Goal, and Actuals

Metric	Baseline	Goal	Actual
% Outdated/Total # Units Received into Inventory	11.2%	<3.0%	0.9%
Cost (CAD$) of Outdated Red Cell Units/Year	$197,500.00	<$53,000.00	$14,000.00
% Transfused Red Cell Units/ Total # Red Cell Units Received into Inventory (Over processing metric)	49.5%	>50.0%	54.5%

A19-030E

Validation of Automated Screening for Duffy and Kidd Antigens on the Galileo

N Le[1] (nguyetbenkhoa@yahoo.com), W S Crews[1], R P Brown[1], J N Everetts[1], J M Gould[1], J H Herman[1]. *[1]Pathology, Transfusion Medicine, Thomas Jefferson University Hospital, Philadelphia, PA, United States*

Background/Case Studies: Providing phenotypically matched blood for sickle cell patients or antigen negative (AG-) blood for alloimmunized patients is costly, often delays transfusion, and requires additional technical time to screen for compatible blood products. In 2012, an AG- inventory was established by phenotyping red cell units already in inventory for C/c, E/e, and K using the Galileo (Immucor, Norcross, GA) and licensed reagents from Immucor. This proved to be both cost effective by reducing the number of AG- units ordered from blood suppliers, and more efficient by decreasing turnaround time (TAT) for product availability. To further reduce cost and improve TAT it was decided to expand antigen screening to the Duffy and Kidd blood groups. Unlike the phenotype methods for C/c, E/e, and K, antigen typing for Duffy and Kidd are not FDA approved as tests of record using the Galileo. It was decided to screen the general inventory for these antigens on the Galileo, and then reconfirm negative units by the standard

tube method. Method validation results for automated screening of Jk(a)/Jk(b) and Fy(a)/Fy(b) antigens are presented here. **Study Design/Methods:** Testing was performed using segments from each donor red blood cell (RBC) unit. For automated screening, the test reagents were diluted 1:11 by adding 1 part reagent antisera to 10 parts phosphate buffered saline. For tube testing, reagent antisera were used as provided following the manufacturer's insert. The same reagent lot number was used for comparative testing. Both antigen positive and negative samples were used. The validation for Jk(a)/Jk(b) was done by automated screening and by the standard tube method using Immucor reagents. Forty samples were tested, 19 of which were heterozygous. Because Ortho-Clinical Diagnostics (Raritan, NJ)was used for anti-Jk(a) and anti-Jk(b) by the standard tube method, due to faster incubation times, ten samples were also tested with Ortho reagents. The results were then compared to the manual and automated test results from Immucor. The validation for Fy(a)/Fy(b) was done by automated screening and by the standard tube method using Immucor reagents. A total of 40 samples, 21 of which were heterozygous, were tested. **Results/Findings:** There was complete correlation between the automated screening and the results from the manual tube testing. **Conclusion:** Jk(a)/Jk(b) and Fy(a)/Fy(b) automated screening using the Galileo was validated. This will allow implementation of screening general red cell inventory. Utilizing diluted reagent antisera and decreasing technical time of finding AG- units will lead to reduced cost and increased efficiency.

Disclosure of Commercial Conflict of Interest

R. P. Brown: Nothing to disclose; W. S. Crews: Nothing to disclose; J. N. Everetts: Nothing to disclose; J. M. Gould: Nothing to disclose; J. H. Herman: Nothing to disclose; N. Le: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. P. Brown: Nothing to disclose; W. S. Crews: Nothing to disclose; J. N. Everetts: Nothing to disclose; J. M. Gould: Nothing to disclose; J. H. Herman: Nothing to disclose; N. Le: Nothing to disclose

A20-030E

Multistate Provision of Phenotyped Red Blood Cells (RBCs)-Administrative Considerations

B A Lenes[1] (blenes@bloodsystems.org), T Glanzer[1], D F Fisher[1], D Figueroa[1], F Nizzi[1], K Land[1], P A Tomasulo[1]. *[1]Blood Systems, Scottsdale, AZ, United States*

Background/Case Studies: As part of a Patient Centric Blood Management Program at a large geographically and ethnically diverse multistate regional blood supplier, systems were created to increase antigen-typed red blood cells (RBC) availability to patients with antibodies such that the turn-around-time (TAT) is essentially the same as for patients without antibodies. The focus was to ensure the right phenotyped product(s) went to the right patient at the right time. **Study Design/Methods:** Data was collected and analyzed on all phenotyped RBC orders over a 12-month period involving over 550 healthcare facilities in 20 states. ABO/Rh, number of units requested, location of hospital, and the specific antigen negative requests were analyzed. A series of inventory control tools and procedures were implemented to help identify, isolate, and then pre-position antigen-typed RBC units to areas of expected need based on recent and longer term historical usage. Automated Rh (CEce) and K phenotyping were performed on approximately 300 donors per week. Selected phenotypes were further screened for Fy, Jk and MNS antigens. Overall, RBC antigen typing has been performed on greater than 350,000 donors. **Results/Findings:** 14,078 orders for 33,194 phenotyped RBC units were received out of 964,279 RBC collections, and out of 918,220 RBCs distributed, in a 12-month period. On average, 40% of collections had historical phenotypes with 9.7% of previously un-typed donors phenotyped per month. Data showed that delays in provision of products were more related to the geographic distance between collection and usage than to under-collection or mis-timing of collection across this large geographic footprint. Procedures and processes were implemented to isolate previously phenotyped products (to prevent unintended consignment), to provide real time tracking of orders and units across the system, and to pre-position fully phenotyped products in locations based on current and historical need for similar units. **Conclusion:** By creating new, and optimizing existing, processes, a multi-state blood center can efficiently meet alloimmunized RBC patient needs in a timely matter, while also leveraging its size to take advantage of centralization, geography-based ethnic diversity, low transportation cost, and customer-based historical analytics. This program demonstrates how the capabilities and resources of a large system of blood centers can be applied to solve local patient care issues by increasing the probability that the right RBC product(s) will be readily available to patients with antibodies, at the same time that it reduces overall customer cost.

Disclosure of Commercial Conflict of Interest

D. Figueroa: Nothing to disclose; D. F. Fisher: Nothing to disclose; T. Glanzer: Nothing to disclose; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; F. Nizzi: Mediware, Other, Blood Systems, Other; P. A. Tomasulo: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Figueroa: Nothing to disclose; D. F. Fisher: Nothing to disclose; T. Glanzer: Nothing to disclose; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; F. Nizzi: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

A21-030E

Phenotyping Versus Traditional Screening Method for Preparing Red Blood Cell (RBC) Units In Patients with High-Risk Pregnancies: A Decisional Analysis Approach

H P Pham[1] (hpp2103@columbia.edu), C Kim[2], J Schwartz[1]. *[1]Pathology and Cell Biology, Columbia University Medical Center and the New York-Presbyterian Hospital, New York, NY, United States; [2]Health Policy and Management, Columbia University Mailman School of Public Health, New York, NY, United States*

Background/Case Studies: The incidence of high-risk pregnancy, such as placenta accreta, is increasing due to multiple factors, including rising maternal age and rate of caesarean deliveries. These patients are at high risk of peripartum bleeding; thus, many are hospitalized approximately 21 days prior to their scheduled deliveries. Although it is rare for them to require transfusion prior to delivery, it is required to have at least one RBC unit ready throughout their hospitalization. Traditionally, this can be accomplished by performing a type and cross (T&C) every 72 hours. Alternatively, they could receive phenotype-matched RBC units if their RBC phenotypes are known at the time of admission. Each method has its own risks and benefits; thus, the aim of this analysis is to provide a rigorous assessment of the cost benefit between them. **Study Design/Methods:** A decisional analytic framework using TreeAge Pro 2012 was created to model and compare the cost benefits of the two methods of preparing RBC units for patients with high-risk pregnancies. A patient entered in the model can either undergo the traditional or alternative method. The traditional method is that she would undergo a T&C every 72 hours in order to have a RBC unit ready at all times. The alternative one is that she would be phenotyped at admission and have a phenotype-matched RBC unit ready at all times without the need of undergoing a T&C every 72 hours. If she has positive antibody screen at admission, then she can enter only the traditional method. All patients receive a T&C at the end of the 21st day in order to prepare RBC units for scheduled delivery. Probabilities of having positive screening (1%) as well as developing new alloantibodies (0.25%) using in the model were derived from the medical literature. Cost and time to prepare a RBC unit were derived from institutional experiences. **Results/Findings:** Monte Carlo simulation of 10000 trials was run for the model of the 21-day hospitalization of each patient. Assuming none of the patients required transfusion, the expected costs per RBC unit of the traditional method is significantly higher than the alternative method ($876.59 $\pm$ 36.01 versus $608.32 $\pm$ 216.59, p-value $<$ 0.01). Furthermore, for each RBC unit, 1.86 hours of technician working time are saved by the alternative method comparing to the traditional. After adjusting for both the technician-time saved and the costs, approximately $315.03 per RBC unit would be saved if the alternative method is employed. Sensitivity analysis showed that the model was only sensitive to the cost of the alternative method. Specifically, the alternative method should be used only if the total cost of a RBC unit was less than $857.67. **Conclusion:** The model demonstrates that preparing phenotype-matched RBC units at admissions is more cost-beneficial comparing to the traditional method of T&C every 72 hours if the cost for the matched unit is less than $857.67.

Disclosure of Commercial Conflict of Interest

C. Kim: Nothing to disclose; H. P. Pham: Nothing to disclose; J. Schwartz: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Kim: Nothing to disclose; H. P. Pham: Nothing to disclose; J. Schwartz: Nothing to disclose

Management, Finance and General Marketing

A22-030F

Looking for the Perfect Recruitment System

E P Gomez[1] (egomez@fbstib.org), M Vanrell[1], A Martorell[1], D Ruiz Alderton[1], A Galmes[1], M Sedeño[1], E Girona-Llobera[1]. *[1]Marketing, Fundació Banc de Sang i Teixits de les Illes Balears, Palma, Spain*

Background/Case Studies: The way blood donors are informed when and where they can give blood is based upon the place where they choose to donate through phone calls and SMS messages, duplicating effort and expenses. Since January 2012, donors have been classified into 5 groups here, according to the number of donations and the time that has elapsed since their last visit. Phone calls were made only to those people who have never donated or donated only once. Frequent donors received only SMS. **Study Design/Methods:** It was necessary to improve the way of donors were recruited. The goalss were to clean up the contact details of the donors in our database, reduce the recruitment costs, and maintain the number of blood donations. In 2012, 3030 donor satisfaction surveys were collected. 1,453 donors filled in a "comments" section that offered valuable information. In addition, 149 suggestions were received. Meetings were held with volunteers of donor associations to collect their opinions. The three sources of information were analyzed, and it was discovered there was a need for earlier contact, shorter waiting lines, and more precise information on where to donate. To debug the database, a Business Intelligence tool was used that detected that 10% of donor details were incorrect. It was also noticed that a great many donors had failed to respond to requests. **Results/Findings:** Donors are now informed twice before a blood drive is held (5-7 days before by phone, SMS, or email depending on the age of the donor, and all donors receive SMS reminder 24 hours prior). In the requests, donors are included who live near to the blood drive (by zip code). In email messages: 1) Donors who have more than one donation receive a link to download the medical questionnaire to fill in beforehand and reduce the waiting time; 2) A geolocation link is added to the location where the blood drive will take place. Finally, the recruitment database donors who had not returned the last 6 years or more are removed. This decision affected 35% of donors (19.306). During 2012, 100,609 phone calls were made; 217,467 SMS and 161,328 e-mail messages were sent. In 2012 44% was saved in the way donors are recruited compared with the year 2011. From January 1, 2013 donors who made their first donation over a year ago (and who have been contacted by phone during 2012) receive SMS only which results in a savings of 60% of the total cost of the recruitment budget compared to 2011. **Conclusion:** Analyzing the information offered by donors has helped turn the donor recruitment system more efficient and effective. The various modifications in the way donors are recruited has led to the right message, and to meet the needs of people who donate blood.

Disclosure of Commercial Conflict of Interest

A. Galmes: Nothing to disclose; E. Girona-Llobera: Nothing to disclose; E. P. Gomez: Nothing to disclose; A. Martorell: Nothing to disclose; D. Ruiz Alderton: Nothing to disclose; M. Sedeño: No Answer; M. Vanrell: No Answer

Disclosure of Grants Conflict of Interest

A. Galmes: Nothing to disclose; E. Girona-Llobera: Nothing to disclose; E. P. Gomez: Nothing to disclose; A. Martorell: Nothing to disclose; D. Ruiz Alderton: Nothing to disclose; M. Sedeño: No Answer; M. Vanrell: No Answer

Improvement in donor invitation to donate

	Year 2011	Year 2012	Forecast year 2013
Number of phone calls	201,428	100,609	57,142
Number of SMS messages	173,228	217,467	300,000
Total cost per year	153,126€	85,223€	61,000€

Efficiency is proved.

A23-030F

Aligning Tele-Recruitment Metrics with Organizational Goals Using LEAN Six Sigma Strategies

M Stefan[1] (mstefan@carterbloodcare.org), G Graham[1], J Biersmith[1]. *[1]Carter BloodCare, Bedford, TX, United States*

Background/Case Studies: What will a typical blood center look like in 10 years? It is hard to say for sure, but likely it will be complex, highly technical, and extremely efficient. With the changing dynamics in the industry today, it is imperative to find ways to operate more efficiently to improve productivity and reduce costs. This blood center adopted a LEAN Six Sigma program in 2005 and has achieved tremendous cost savings and improved performance in operational areas. More recently this program has been expanded into non-traditional departments including the tele-recruitment (TR) department. **Study Design/Methods:** Using the LEAN Six Sigma DMAIC method (define, measure, analyze, improve, and control) processes within (TR) were systematically reviewed, to challenge the status quo and identify opportunities for improvement with an aim to enhance efficiency and productivity. **Results/Findings:** It was found in our evaluation that the measuring tools for the processes in TR were measuring one particular metric (monthly collections), while disregarding others, in some cases chasing goals that may have been in conflict with the overall organizational goals. Additionally, the TR incentive plan (based on the core departmental metrics) was not fully aligned with the organizational goal to balance blood supply with local hospital usage and demand. TR staff was incentivized to book as many appointments as possible, regardless of usage and demand, which created excess inventory. Applying LEAN Six Sigma tools, the departmental metrics were completely redesigned and an incentive plan to fully align with corporate objectives was put in place. Additionally, individual report cards were developed for TR staff and standardized coaching, counseling, and retraining was implemented to ensure personal performance indicators were being met by staff. After implementing the realigned metrics and incentive plans, it was possible to increase the number of calls, improve the overall quality of donor calls, and decrease departmental absenteeism. Most importantly, by implementing metrics focused on system-wide level loading it is now easier to balance our productivity (collections) with current demand (inventory). **Conclusion:** Telerecruitment can play an integral part in the "LEANing" of the organization by finding innovative ways to operate more efficiently. Metrics and performance based measurement should be continuously reevaluated to ensure proper alignment with ever-changing corporate objectives. In TR, these metrics include CPH (calls per hour), CPR (calls per recruiter), Productive Call Hours, Appointments per Hour, Appointment Show Rates, unexcused absences, and Quality of Calls. Understanding how these measurements affect the departmental results was the key to increasing both departmental and individual performance.

Disclosure of Commercial Conflict of Interest

J. Biersmith: No Answer; G. Graham: Nothing to disclose; M. Stefan: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Biersmith: No Answer; G. Graham: Nothing to disclose; M. Stefan: Nothing to disclose

A24-030F

Designing and Implementing an Effective Monitoring and Evaluation System for Blood Safety: A Zimbabwean Experience

T Mapako[1,2] (tmapako@bloodbank.co.zw), D A Mvere[1]. *[1]Planning, Information and Research, National Blood Service Zimbabwe, Harare, Zimbabwe; [2]Unit of PharmacoEpidemiology & PharmacoEconomics (PE2), Department of Pharmacy, University of Groningen, Groningen, Netherlands*

Background/Case Studies: National Blood Service Zimbabwe (NBSZ) has the mandate to procure and distribute blood and blood products in Zimbabwe. Since 2006, NBSZ has been implementing a structured monitoring and evaluation system (M&E) at all blood safety levels. M&E is critical for blood safety as it informs key management functions of planning and implementation, resource management, communication, and advocacy. In 2010, funding was received for the program "Prevention of human immunodeficiency virus (HIV) transmission through strengthening blood safety in the Republic of Zimbabwe under PEPFAR". This 5-year program is addressing five goals including M&E strengthening. The aim of this abstract is to share the Zimbabwean M&E implementation experience. **Study Design/Methods:** In 2006, NBSZ developed a data management policy and trained all staff on this. Essential data forms for routine collection of M&E data were developed, validated, and implemented. The essential data forms included aspects on donor mobilisation, blood collection, processing, testing, distribution, and other key general areas. In 2011, through PEPFAR support, NBSZ received further M&E training from CDC, which was cascaded to all senior staff who participated in the development of the blood safety logic model. M&E data was gathered through the essential data forms and shared internally and externally with all key stakeholders. NBSZ also developed and resourced the necessary M&E structures as a separate independent department function. **Results/Findings:** Increased staff awareness enhanced their participation in M&E activities. The policy guidelines on M&E reinforced the NBSZ

commitment to accurate data management and its uses. By having a specific goal on M&E under the PEPFAR project, further attention could be given to M&E activities, which in turn makes it easier for the Service to better manage the blood safety program. The blood safety logic model was developed which included the intricate relationships within and between model components. M&E data was shared internally and externally with all key stakeholders including contribution to WHO Global Database on Blood Safety. M&E efforts have seen NBSZ enhancing its accountability and judgement, blood safety program improvement, and knowledge development. **Conclusion:** NBSZ experience has shown that it is possible for blood services in resource-constrained settings to develop and implement their M&E systems that will ensure timely availability of complete, accurate and quality data. This is achieved through various capacity building initiatives such as staff training and development of appropriate M&E framework. The first-ever developed blood safety logic model assisted in describing the blood safety programme in Zimbabwe. Designing and implementing an effective M&E system is a structured process that needs to be properly guided and not left to chance. This will address the glaring information gaps characterising most blood services in the developing world.

Disclosure of Commercial Conflict of Interest

T. Mapako: Nothing to disclose; D. A. Mvere: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Mapako: Nothing to disclose; D. A. Mvere: Nothing to disclose

A25-030F

Emergency Preparedness in the Modern Age of Blood Banking

D T Lo[1] (dlo@nyp.org), K M Crowley[1], M Cushing[2], C Goss[2], S T Avecilla[2]. *[1]Transfusion Medicine and Cellular Therapy, NewYork Presbyterian Hospital/WCMC, New York, NY, United States; [2]Pathology and Laboratory Medicine, Weill Cornell Medical College, New York, NY, United States*

Background/Case Studies: Incidents including but not limited to human, natural, technological, and hazardous material events that affect the environment of care within the hospital, and require a deviance from the day-to-day operations of the hospital, are considered emergency events (EE). More often than not, the blood bank (BB) is part of the hospital's multidisciplinary processes during an EE. Therefore, it is imperative that the BB has a collaborative emergency preparedness plan (EPP) to ensure that material, supplies and personnel are available to support patient care in the hospital and precious resources such as blood products (BP) are not wasted. **Study Design/Methods:** After "9/11" the EPP was reevaluated and external recommendation resources, such as the hospital EPP website, the *AABB Technical Manual*, and government emergency preparedness websites, were reviewed. **Results/Findings:** The EPP for the BB was drastically revised to include: 1)The method and terminology used by the hospital to notify employees on duty of an EE or an EE drill; 2)A BB contact person for all communications pertaining to the EE with the hospital's Command Center; 3)Defined responsibilities of the Medical Director; 4)A minimum BP inventory; 5)Current contact information of blood and reagent suppliers; 6)Assessment of the staffing situation and utilization of the established EPP Notification Tree to inform and to schedule off duty BB staff to ensure adequate staffing to support the BB operation; 7)An emergency phone that has an alternative telephone routing path; 8)A plan to assess and to complete the work in progress within the BB to prepare for the potential workload from the EE; 9)Alternative BP storage location(s) and method(s), such as preparation of coolers and ice packs; 10)Establishment of a mobile BP dispense station with appropriate authorization; 11)Guidelines for reviewing and canceling pending elective surgeries; 12)Descriptions of Emergency Department's processes for EE patient registration, specimen collection, and BP pickup; 13)Guidelines for BP requests with or without specimens or established blood types; 14)Equipment and power supply resource management; 15)Laboratory information system downtime and recovery procedures; 16)Provision of an alternative worksite for BB staff; 17)Provision of guidelines for transportation and possible bridge/tunnel restrictions; and 18)A drill plan and checklist for the EPP. **Conclusion:** This plan has served us well during all major snowstorms, the Blackout of 2004, Hurricane Irene 2011, and Super Storm Sandy 2012. However, challenges that remain include emergency staff accommodations, food and showers for the BB staff that remain onsite to support patient care during an EE, mass transit shutdowns, and alternative storage site and transportation for frozen products in the event of prolonged power outages.

Disclosure of Commercial Conflict of Interest

S. T. Avecilla: Nothing to disclose; K. M. Crowley: Nothing to disclose; M. Cushing: Nothing to disclose; C. Goss: Nothing to disclose; D. T. Lo: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. T. Avecilla: Nothing to disclose; K. M. Crowley: Nothing to disclose; M. Cushing: Nothing to disclose; C. Goss: Nothing to disclose; D. T. Lo: Nothing to disclose

A26-030F

Implementing In-House Infectious Disease Testing to Optimize Organ Allocation

K K Crissler[1] (kkc@cbckc.org), K M Coulter[1], S B McDonald[1]. *[1]Laboratories, Midwest Transplant Network, Westwood, KS, United States*

Background/Case Studies: With the recent consolidations of blood centers and the move to centralization of testing laboratories, locating FDA-licensed infectious disease laboratories willing to perform STAT testing is becoming a challenge for organ procurement organizations (OPOs). The limited number of large national testing laboratories may not have the desire, need, or capability to interrupt the scheduled blood donor testing runs to work with STAT organ donor samples. For OPOs, time is the major driver to successful procurement and transplantation. Shipping samples to a centralized laboratory capable of performing the testing is expensive, largely because of transportation costs. Even more damaging is the loss of potential transplantable organs due to time constraints.The push in blood management practices has resulted in having the local blood center making the decision, in October 2012, to outsource donor infectious disease testing. The blood center's decision directly impacted Midwest Transplant Network (MTN), the local OPO, to reevaluate how infectious disease testing would be performed. **Study Design/Methods:** Testing was sent to a national laboratory in January and February 2013. Concurrently, MTN reached out to the current FDA-licensed testing manufacturers to evaluate options of bringing testing in-house for approximately 350 donors a year. After review of instrumentation (size, throughput) and cost per test, two of the four vendors were selected that best fit the needs of MTN: Roche Diagnostics and Ortho Clinical Diagnostics. The following factors were taken into consideration: 1) Total turn-around-time (transportation, testing, review, and result reporting); 2) Cost (transportation and testing price-per-donor); and 3) Potential organs saved. **Results/Findings:** In-house testing was implemented in March 2013. 1) TAT: Sending testing to a national laboratory extended the TAT by two—three fold. 2) Cost 66% increased cost to bring testing in-house, when compared to the national laboratory. 3) Organs saved: 24% of all potential organ cases are donation after circulatory death. In order for organ transplantation to occur, serology testing needs to be completed prior to proceeding to OR. Decreased TAT with in-house testing allows for increased recovery and placement of livers and lungs. In-house testing allows for a 12% increase in organ allocation. **Conclusion:** Based on these findings, shipment of donor samples to a national testing laboratory was analyzed and determined not feasible. Optimization of organ allocation has a potential to save additional organs that were historically non-obtainable. Even though the cost of in-house infectious disease testing is higher than the national testing laboratory, the lives impacted by the OPO having total control over the donor screening process have proven to be invaluable.

Disclosure of Commercial Conflict of Interest

K. M. Coulter: Nothing to disclose; K. K. Crissler: Nothing to disclose; S. B. McDonald: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. M. Coulter: Nothing to disclose; K. K. Crissler: Nothing to disclose; S. B. McDonald: Nothing to disclose

A27-030F

Developing a Model to Capture Product Costs in Complex Blood Centers

J S Jacob[1] (jjacob@nybloodcenter.org), B McQuail[1], B H Shaz[1]. *[1]New York Blood Center, New York, NY, United States*

Background/Case Studies: Macroeconomic pressures and an inversion of the blood supply/demand ratio have created price deflation for blood products. Blood centers have sought to both understand and decrease costs in order to remain viable. One set of universal measures is the Cost Per Unit Collected/Distributed (CPUC/CPUD) for both red blood cells (RBCs) and

single donor platelets (SDPs). The variables, weights, and factors included in these benchmarks are not published and likely are not identical across institutions. It was sought to define a method of determining CPUC/CPUD using a detailed and precise approach that would both provide feedback on changes made in cost structures, and that was usable across institutions. **Study Design/Methods:** CPUC and CPUD models were built after: 1) detailed mapping of the collections, manufacturing and distribution processes, and identification of all activities that generate cost; 2) assignment of a % total effort for each step and for each product; and 3) allocation of all costs for each product including general & administrative and overhead costs. Each assumption was then tested repeatedly and the model was used to understand costs per department, functional area, and geography. **Results/Findings:** The previous model, and perhaps that of the current cost models of other centers, loads all costs on the RBC and extracts only the cost of producing and distributing other products. This leaves an inflated cost on the RBC and depresses the cost of other products. More importantly, this process invites aggregation of costs into fewer line items resulting in a less precise cost per product. This model, with nearly 800 input values, enables management to understand its costs more precisely across all products. Once the model's assumptions were populated, it was explored what would have to happen to relevant assumptions to get the RBC CPUC from an arbitrarily chosen value of $300 to $200 which reflects the postulated deflationary element in the national market in recent years. The model's sophistication enabled management to isolate particular variable(s) to determine the necessary effort to reach the target cost and therefore the average selling price to break even. **Conclusion:** Understanding one's cost structure is vital. Centers with more complex operations make identification of the true cost drivers a challenge, but a comprehensive review of the entire process and each of the cost drivers is essential for blood centers to precisely understand costs to remain competitive in a declining revenue environment.

Disclosure of Commercial Conflict of Interest

J. S. Jacob: Nothing to disclose; B. McQuail: Nothing to disclose; B. H. Shaz: Haemonetics, no financial relationship, Other, Fenwal, no financial relationship, Other

Disclosure of Grants Conflict of Interest

J. S. Jacob: Nothing to disclose; B. McQuail: Nothing to disclose; B. H. Shaz: Nothing to disclose

Blood Management Administrative

AP1

Reducing Platelet (PLT) and Red Blood Cell (RBC) Utilization in a 325-Bed Hospital

B C Sutton[1,3] (bsutton@sbmf.org), D R Roberts[2,9], M J Wright[1,4], R Pennoni[1,3], L Poeppelman[2,5], K Henke[7,5], T Belcher[9], A Mack[10,2], D T Mellin[9,8], A Haffner[2,6]. *[1]Pathology, South Bend Medical Foundation, South Bend, IN, United States; [2]Nursing, Elkhart General Hospital, Elkhart, IN, United States; [3]Laboratory, South Bend Medical Foundation, South Bend, IN, United States; [4]Blood Bank, South Bend Medical Foundation, South Bend, IN, United States; [5]Critical Care, Elkhart General Hospital, Elkhart, IN, United States; [6]Oncology, Elkhart General Hospital, Elkhart, IN, United States; [7]Education, Elkhart General Hospital, Elkhart, IN, United States; [8]Medicine, Elkhart General Hospital, Elkhart, IN, United States; [9]Administration, Elkhart General Hospital, Elkhart, IN, United States; [10]Performance Improvement, Elkhart General Hospital, Elkhart, IN, United States*

Background/Case Studies: Evolving data indicates that significant adverse events are associated with allogeneic blood transfusion, and that this procedure should be limited to those patients with true medical need. Also, cost associated with blood administration is substantial in any health system. Using a teamwork approach, Elkhart General Hospital (EGH), a full-service hospital in Elkhart, Indiana, initiated procedures that guided clinicians toward more conservative blood ordering practice and limited product wastage. **Study Design/Methods:** EGH Blood Utilization review indicated apheresis platelet usage was steadily increasing and that "two unit" transfusions often occurred without assessment of platelet increment between units. Following approval of the EGH Blood Bank Committee and education of staff, a physician order form was launched that automated a platelet count between units and expedited data resulting. Encouraging results from this project led to formation of two multidisciplinary teams. The "Plan for Every Platelet" Team represented largely laboratory and nursing staff. This team studied platelet purchasing and dispensing processes in an effort to reduce product outdating. The "Appropriate Utilization of Blood" Team included staff from high blood use areas such as oncology, critical care, and cardiac surgery. This team developed a physician order set for red blood cells (RBC) similar to that for platelets (PLT) and incorporated relevant educational information from recently updated institutional transfusion guidelines. Advisory Team Physicians included an internist and a transfusion medicine specialist. The RBC order set launched following physician and nursing education. **Results/Findings:** Apheresis platelet products transfused per year increased from 722 units in 2005 to 1079 in 2008. After introduction of a platelet order set, this decreased to an average of 543 units per year for the next 3 years. In 2010 35% of platelets expired prior to transfusion, and in 2011 31% of platelets expired. After adopting the new platelet distribution protocol in March 2012, outdated products dropped to 23% by 2013. After RBC physician order introduction, the template was utilized in 35.5% of RBC transfusions and hemoglobin/hematocrit assessment between units increased from 17.4% to 30% (by November 2012). "Single unit" RBC transfusions increased from about 21% of patients transfused in 2012 prior to order introduction, to an average of 34% in the last 6 months of 2012. **Conclusion:** 1) Providing useful information to clinicians at the time of blood product order and promptly following transfusion, encourages patient assessment and reduces inappropriate blood transfusion. 2) These changes will result in substantial savings in both blood acquisition and administration costs to the hospital system over time.

Disclosure of Commercial Conflict of Interest

T. Belcher: No Answer; A. Haffner: Nothing to disclose; K. Henke: Nothing to disclose; A. Mack: Nothing to disclose; D. T. Mellin: Nothing to disclose; R. Pennoni: Nothing to disclose; L. Poeppelman: Nothing to disclose; D. R. Roberts: Nothing to disclose; B. C. Sutton: Nothing to disclose; M. J. Wright: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Belcher: No Answer; A. Haffner: Nothing to disclose; K. Henke: Nothing to disclose; A. Mack: Nothing to disclose; D. T. Mellin: Nothing to disclose; R. Pennoni: Nothing to disclose; L. Poeppelman: Nothing to disclose; D. R. Roberts: Nothing to disclose; B. C. Sutton: Nothing to disclose; M. J. Wright: Nothing to disclose

AP2

Does Thromboelastogram Platelet Mapping (TEG) Reduce Transfusion in Heart Patients undergoing Bypass Surgery?

L LeeLo[1] (leelo278@gmail.com), L A Borowski[1], J Cruz[2], S F Gregurek[3]. *[1]Transfusion Service, Sunrise Hospital & Medical Center, Las Vegas, NV, United States; [2]Indiana Blood Center, Indianapolis, IN, United States; [3]Transfusion Medicine, LMC Pathology Services, Las Vegas, NV, United States*

Background/Case Studies: Thromboelastogram Platelet Mapping Assay (TEG) provides real-time coagulation assessment, enabling intraoperative data-driven transfusion decisions in coronary artery bypass graft (CABG) patients on antiplatelet medications. Some TEG users report a reduction in components transfused, with better component selection and associated cost reduction. This 638–bed, urban community-based facility acquired TEG since there is a large cardiac surgery service here. the impact of TEG was examined on component transfusion during coronary artery bypass surgery (CABG) to determine whether intraoperative transfusion and associated cost were reduced. **Study Design/Methods:** Thromboelastogram Instruments (Haemonetics, Braintree, MA) were utilized between April 2012-March 2013. The blood bank is in close proximity to surgery and performed TEG. Online tracing display enabled interpretation by the surgery team, who then decided if transfusion was indicated. The instrument manufacturer provided training, with additional hospital inservice. The number of packed cells, apheresis platelets, and thawed plasma were obtained for each CABG utilizing TEG (TEG-CABG). Similar data were obtained from billing for all CABG patients. Total cases and average transfusion rate by component were summarized. Statistical testing was performed using unpaired t-test. **Results/Findings:** (Table) Approximately 20% of CABG procedures utilized TEG. TEG-CABG was associated with a statistically significant, nearly two-fold increase in platelet transfusion compared to no TEG. Although not statistically significant, there were slight increases in number of packed cells, plasma, and total components transfused in TEG-CABG patients compared to no TEG. **Conclusion:** Rather than the expected reduction in transfused components for TEG-CABG, increased platelet transfusion was experienced, while red cells and plasma were unaffected. These findings may be directly driven by TEG testing, or may be due to underlying demographic differences in the study populations. Since plasma use was unchanged it does not appear that TEG testing corrected inappropriate component selection (plasma

transfusion when platelets were needed), but may indicate platelet undertransfusion. Given the higher cost of platelets, further study and detailed blood audit must be performed, and outcomes must be considered. The findings suggest that while TEG influences transfusion decisions, it may increase or decrease transfusion depending on the facility.

Disclosure of Commercial Conflict of Interest

L. A. Borowski: Nothing to disclose; J. Cruz: Nothing to disclose; S. F. Gregurek: Nothing to disclose; L. LeeLo: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. A. Borowski: Nothing to disclose; J. Cruz: Nothing to disclose; S. F. Gregurek: Nothing to disclose; L. LeeLo: Nothing to disclose

	Total Cases	Packed Cells*	Apheresis Platelets*	Plasma*	Total Transfused *
no TEG	199	1.5 (2.3)	0.4 (1.1)	0.6 (2.5)	2.5 (5)
TEG	49	1.9 (3.2)	0.8 (1.8)	0.7 (1.7)	3.4 (6.1)
p-value		0.16	0.03	0.39	0.14

* average units/patient (standard deviation)

AP3

Improving Appropriateness of Blood Utilization through Prospective Review of Requests for Blood Products: The Role of Pathology Residents as Consultants

H Zia[1,2] (ziahm@tmcmed.org), L J Haldiman[1,2], G Singh[1,2]. *[1]Pathology, University of Missouri-Kansas City School of Medicine, Kansas City, MO, United States; [2]Pathology, Truman Medical Centers, Kansas City, MO, United States*

Background/Case Studies: Transfusion of blood products constitutes an important part of the therapy for patients. While the use of these products is vital, blood transfusion is not without risks, in part due to the circumstances necessitating transfusion and in part due to the hazards associated with the transfusion itself. Blood products account for the second biggest expense in a pathology laboratory. Evidence is presented here that prospective review of orders for fresh frozen plasma (FFP) and platelets can result in significant decrease in utilization with the attendant reduction in risk to patients, reduction in cost, and an important dividend of preparing pathology residents for their role as clinical consultants. **Study Design/Methods:** Guidelines for the use of blood products were developed in collaboration with anesthesia, hematology/oncology, and emergency and trauma medicine to aid in the proper selection and use of blood and blood products, under the auspices of Transfusion Committee. If the patient met the criteria established in the guidelines, the unit(s) was (were) automatically approved. If the patient did not meet the criteria for the ordered blood component established in the guidelines, the on-call pathology resident was contacted. The resident electronically accessed the patient's medical record to determine if there were any extenuating circumstances. This often followed discussion with ordering clinicians to make sure appropriate evidence based and judicious blood usage. **Results/Findings:** There was an overall decrease in the number of units transfused for each blood product reviewed reaching nearly 40% for FFP usage. The number of discharges was also found to have decreased, however these did not decrease significantly enough to explain the 38.78% and 31.35% decreases in FFP and platelet utilization, respectively. Data normalized for discharges as number of products transfused per 1000 discharges per year showed a significant reduction. **Conclusion:** Prospective review of orders for blood products can result in significant improvement in utilization of blood products, thereby reducing risk to patients, avoiding costs to the healthcare system, and involving pathology residents in the prospective review, thereby providing an excellent opportunity for their training as consultants.

Disclosure of Commercial Conflict of Interest

L. J. Haldiman: Nothing to disclose; G. Singh: Nothing to disclose; H. Zia: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. J. Haldiman: Nothing to disclose; G. Singh: Nothing to disclose; H. Zia: Nothing to disclose

	RBC	RBC % Change	FFP	FFP Change (%)	PLTS	PLTS Change (%)	Cryo	Cryo Change (%)	Number of Discharges
2009	3181	—	900	—	319	—	81	—	13488
2010	3100	–2.6	875	–2.8	401	25.7	77	–5.0	14233
2011	3026	–2.4	612	–3.0	288	–28.2	75	–2.6	13007
2012	2872	–5.1	551	–1.0	219	23.9	73	–2.7	11740
Change from 2009-2012 (%)	–9.71		–38.78		–31.35		–9.88		–13.0

Table shows yearly blood component usage from 2009 through 2012 and any changes. It also shows number of discharges in the same time period as an indicator of the number of patients served. FFP- Fresh frozen plasma; PLTS- Platelets; Cryo- Cryoprecipitate.

AP4

Prospective Monitoring of Cryoprecipitate Resulting in Decreased Product Waste

E M Azzato[1] (elizabeth.azzato@uphs.upenn.edu), H Carpenter[1], M Miszczanczuk[1], L Dinh[1], F DeMuth[1], D Magee[1], N Aqui[1], D Siegel[1], B S Sachais[1]. *[1]Pathology and Laboratory Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA, United States*

Background/Case Studies: In 2010, the transfusion medicine service at the hospital noted increased wastage of cryoprecipitate due to returned units that could not be used before their 4-hour expiration window. An analysis of cryoprecipitate utilization revealed that the majority of wastage was due to returned products from operating rooms (ORs), with many returned orders being for pools of 20 units or more. Beginning in March 2011, clinical pathology residents began prospectively auditing cryoprecipitate orders, calling physicians for all orders greater than 10 units and orders from low usage locations. Residents would release 10 units and then, if necessary, allow a second order of 10 units to follow some time later; residents would only release pools of 20 units when the attending physician expressed acute clinical need. **Study Design/Methods:** All cryoprecipitate orders were evaluated in the year prior to intervention (March 1, 2010-February 28, 2011, baseline) and 2 years after intervention (March 1, 2011–March 31, 2013), looking at ordering service, location, status (transfused versus expired), transfusion date, and number of units ordered. The percentage of cryoprecipitate waste was calculated as expired units divided by total units ordered. Statistical significance was evaluated using chi-square test of proportions comparing percent cryoprecipitate waste post intervention to baseline waste. **Results/Findings:** During the year prior to intervention, 10,972 units of cryoprecipitate (mean 914 units/month, 23 units/patient) were dispensed. Of these, 9562 were transfused and 1410 were wasted (12.9% wastage). The majority of these orders were placed from ORs (5383 units of cryoprecipitate ordered) and/or by the cardiothoracic (CT) surgery service (6067 units of cryoprecipitate ordered). A high % of the wasted units had been dispensed to the ORs (82%) and/or CT surgery (72%), or had been dispensed in cryo-pools of 20 units or greater (62%). During the 2 years after the intervention, 19,609 units of cryoprecipitate (mean 784 units/month, 21 units/patient) were dispensed, with 1783 units destroyed. A significantly smaller proportion of cryoprecipitate units were wasted over this time period, compared with pre-intervention (9.1% versus 12.9%, $p < 0.0001$). Compared to pre-intervention, mean units ordered per patient and percentage of cryoprecipitate wastage in the ORs (17 versus 21 units/patient, $p < 0.0005$; 13.3% versus 21.4% wastage, $p < 0.0001$) and by CT surgery (18 versus 21 units/patient, $p < 0.002$; 12.4% versus 16.8% wastage, $p < 0.0001$) also decreased significantly. **Conclusion:** In this institution, blood bank physician intervention by prospective auditing and limiting orders of cryoprecipitate to pools of ten units resulted in a statistically significant decrease in the proportion of cryoprecipitate wastage.

Disclosure of Commercial Conflict of Interest

N. Aqui: No Answer; E. M. Azzato: Nothing to disclose; H. Carpenter: Nothing to disclose; F. DeMuth: Nothing to disclose; L. Dinh: Nothing to disclose; D. Magee: Nothing to disclose; M. Miszczanczuk: Nothing to disclose; B. S. Sachais: Nothing to disclose; D. Siegel: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Aqui: No Answer; E. M. Azzato: Nothing to disclose; H. Carpenter: Nothing to disclose; F. DeMuth: Nothing to disclose; L. Dinh: Nothing to disclose; D. Magee: Nothing to disclose; M. Miszczanczuk: Nothing to disclose; B. S. Sachais: Nothing to disclose; D. Siegel: Nothing to disclose

AP5

Transfusion Consent: Empowering the Patient

S K Cassidy[1] (sandy.cassidy@childrensmn.org), J Wenzel[1], M Dilley[1], M Livingston[1], K Mascotti[1]. *[1]Transfusion and Tissue Services, Children's Hospitals and Clinics of Minnesota, Minneapolis, MN, United States*

Background/Case Studies: Changes in healthcare are driven by technical innovation and increasingly engaged consumers. Access to information is key to empowering patients in making informed choices regarding care and impacts the patient experience. Transfusion of blood products is considered an invasive medical procedure with definable risks. Administration of blood components is governed by Joint Commission, College of American Pathologist and AABB, requiring informed consent to be part of the tranfusion process. Treating physicians are required to obtain consent for blood transfusions except in the case of a medical emergency. The consent process must include a discussion between the patient and physician. Barriers to the consent process include: 1) Physician lack of knowledge about transfusion medicine; 2) Patient recall of consent details; 2) Patient understanding of information; 3) Lack of patient information resources to address these barriers. This hospital created a video detailing the risks, benefits, and alternatives to transfusion. **Study Design/Methods:** 25% of transfusions here are ordered by physicians with little formal training in transfusion therapy. There are concerns regarding the consistency of information provided during the physician-patient discussion as well as the patient's understanding of the information. Patient/family surveys indicate a preference in technology-based systems to receive educational information over current paper models. Patient questions were addressed with the creation of an educational video capturing a provider dialogue of why a transfusion may be needed, the expected benefits, common risks, and altenatives to transfusion. The video concludes with a statement for the patient/family to continue the discussion with their treating provider. **Results/Findings:** Roll-out of the video included communication to clinical staff resulting in 425 views. Patient/families have generated 212 hits on the transfusion consent video since November 2012. During this time period 992 patients received transfusions indicating that one of five patients viewed the information. 84% viewed the video at least once. The next phase is to link the video to order sets and convert into other languages. **Conclusion:** The transfusion consent process begins at the time the physician determines that a patient's course of treatment may include transfusion therapy. Information as to why a transfusion maybe needed, and the risks and alternatives to transfusion, must be provided to the patient prior to all non-emergency transfusions. Readily available patient information is essential for the transfusion consent process. Improvements in educational material have empowered patients to make informed choices and to be more engaged in their healthcare decisions. It essential that patient educational material reinforce, not replace, the discussion between physician and patient. The physician remains ethically and legally obligated in the consent process.

Disclosure of Commercial Conflict of Interest

S. K. Cassidy: Nothing to disclose; M. Dilley: Nothing to disclose; M. Livingston: No Answer; K. Mascotti: No Answer; J. Wenzel: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. K. Cassidy: Nothing to disclose; M. Dilley: Nothing to disclose; M. Livingston: No Answer; K. Mascotti: No Answer; J. Wenzel: Nothing to disclose

AP6

Improving Blood Transfusion Practice by Educational Emphasis of the Blood Utilization Committee: The Experience of One Hospital

E Vrotsos[1] (elena.vrotsos@msmc.com), L H Howard[1,2], B Gonzalez[1], R C Goldszer[3]. *[1]Pathology, Mount Sinai Medical Center, Miami Beach, FL, United States; [2]Herbert Wertheim College of Medicine, Florida International University, Miami, FL, United States; [3]Internal Medicine, Mount Sinai Medical Center, Miami, FL, United States*

Background/Case Studies: Blood utilization reviews are expected to be conducted by the hospitals according to AABB standards and are required by the Code of Federal Regulations. In many hospitals blood utilization reviews are also part of the hospital bylaws. The Blood Utilization Committee usually provides an oversight of the blood usage. The structure of the committee can vary from hospital to hospital. Currently many hospitals accept cross-matched to transfused ratios (C : T) of less than two as a marker of good clinical practice. This ratio shows appropriate clinical judgment when requesting cross matches. Retrospective data from our cardiac unit blood utilization showed a C : T of 2.48 from October 2011-March 2012. **Study Design/Methods:** In October 2012, a new Blood Utilization Committee was organized. It consists of a cardiac surgeon, an anesthesiologist, the blood bank manager and the director, the institutional chief medical officer, a pathology resident, and a representative of the hospital process-improvement team. The meeting consists of an open format where all members have access to the data and participate freely in the discussion and decision making. Those discussions include reviews of transfusion protocols, blood usage, transfusion complications, and clarification of the type and screen concept. In December 2012 a new cardiac surgery protocol was established which aims at lowering C : T by ordering preoperative type and screen in the majority of non-emergent cardiac patients. **Results/Findings:** In this institution the C : T ratio was 2.48 at the beginning of the new protocol. This ratio was higher than the established goal of <2. In December 2012, after initiation of the new cardiac surgery protocol, the C : T decreased to 1.58. This decrease in C : T from 2.48 to 1.58 is statistically significant ($p < 0.05$). **Conclusion:** The implementation of a new Blood Utilization Committee with subsequent establishment of a new cardiac surgery protocol of preferentially ordering type and screens rather than cross matches has had a measurable effect. This effect includes an already evident decrease in numbers of cross matches, and is expected to also include a decrease in the numbers of units transfused.

Disclosure of Commercial Conflict of Interest

R. C. Goldszer: Nothing to disclose; B. Gonzalez: Nothing to disclose; L. H. Howard: Nothing to disclose; E. Vrotsos: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. C. Goldszer: Nothing to disclose; B. Gonzalez: Nothing to disclose; L. H. Howard: Nothing to disclose; E. Vrotsos: Nothing to disclose

AP7

First Year for Blood Management in a Midwest Hospital System

C A Aronson[1] (colleen.aronson@advocatehealth.com), E A Halperin[2], R J Fantus[3], L Roy[4], M Barrionuevo[5]. *[1]Transfusion Services, ACL Laboratories, Rosemont, IL, United States; [2]Transfusion Safety, Advocate Health Care, Oakbrook, IL, United States; [3]Trauma, Advocate Illinois Masonic Medical Center, Chicago, IL, United States; [4]Quality, Advocate Childrens Hospital, Oaklawn, IL, United States; [5]Heart and Vascular Service, Advocate Lutheran General Hospital, Park Ridge, IL, United States*

Background/Case Studies: Previous published reports have shown the advantages for a more restrictive Red Blood Cell (RBC) transfusion protocol including a decrease in complications and hospital stay. A large Midwest hospital system initiated a blood management program in the middle of 2011 to decrease the number of RBCs used for routine transfusions. This was implemented at eight hospitals, five of which are Level One trauma centers. **Study Design/Methods:** This was an administrative backed initiative that was aided by a consulting partner group. The project included education of clinical staff, physician champions, updated computer order sets and coordination of efforts by a Transfusion Safety Manager (TSM). The new RBC transfusion trigger for hemoglobin (Hb) was moved from the traditional Hb level of 10 to 7 g/dl. The traditional mentality that "if one is transfusing one unit of RBCs, one might as well give two," was also targeted for change. Slogans such as "7 is the new 10" and "1 is the new 2" were created to help remind and focus staff on the new triggers. A push was made to transfuse one unit of RBCs and then test the patient's hemoglobin prior to giving a second unit. The use of education events, department meeting presentations, FAQ sheets, as well as a transfusion web page helped get the word out to front line staff. A system wide transfusion safety committee was created with representatives from all sites with many areas of expertise (trauma, cardiac, transfusion service, pediatrics and oncology). This group led the charge both in practice and for education of peers. Efforts and ongoing monitoring were coordinated by the system TSM who was integral to keeping the project moving forward. New order sets were created to limit RBC orders for non-bleeding patients to one unit of RBCs per order. This also included a reminder built into the order set to test the hemoglobin post transfusion. **Results/Findings:** Not only was there a 20% decrease in the number of RBC transfusions from 2011 to 2012, but also a decrease in transfusion of other blood components. The average number of RBCs units per patient

decreased from 2.58 to 2.25. Actual blood cost savings for the hospital system was $5.6 million for the year. New data tracking is in process to look at "single unit transfusions" which is defined as an RBC transfusion with a hemoglobin check prior to transfusing the next unit. **Conclusion:** The system continues to monitor for an increase of single unit transfusions and is now moving to evaluate the use of other blood components. This incredible decrease in number or RBCs transfused was only possible due to the administrative backing and the physician champions that helped others understand that the new transfusion triggers were safe and effective for the patients.

Disclosure of Commercial Conflict of Interest

C. A. Aronson: Nothing to disclose; M. Barrionuevo: Nothing to disclose; R. J. Fantus: Nothing to disclose; E. A. Halperin: Nothing to disclose; L. Roy: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. A. Aronson: Nothing to disclose; M. Barrionuevo: Nothing to disclose; R. J. Fantus: Nothing to disclose; E. A. Halperin: Nothing to disclose; L. Roy: Nothing to disclose

Decrease in Units Transfused 2011 to 2012

	RBCs	PLATELETS	PLASMA	CRYO
# Units	10,200	1,200	1,050	980

AP8

Cardiac Surgery Blood Utilization

K S Gray[1] (gray.karen@mayo.edu), E A GIlman[1]. *[1]Transfusion Medicine, Mayo Clinic Hospital, Phoenix, AZ, United States*

Background/Case Studies: Due to the higher crossmatch to transfusion ratios (C : T) observed for the surgeons within the service of cardiothoracic surgery, a retrospective review was conducted of transfused blood products. The current pre-operative order for coronary artery bypass surgery (CABG), aortic valve replacement (AVR), and mitral valve replacement (MVR) is type and crossmatch for two units RBC. The current pre-operative order for redo CABG is type and crossmatch for four units of RBC. **Study Design/Methods:** In an effort to achieve more efficient blood utilization for the service of cardiothoracic surgery, a review of blood product use was conducted for all cases of CABG, AVR, and MVR between October 2012-March 2013. The case list was provided by the Dept. of Perioperative Services. There were a total of 94 cases. With the availability of the electronic crossmatch (EXM) through the SCC Soft system's SoftBank II v25.1, red blood cells can be provided promptly for patients meeting the standard EXM eligibility requirements. The transfusion history for each patient was reviewed in the SCC SoftBank module. The number of transfused red blood cells was noted for each case category during the study period. **Results/Findings:** The total number of cases with transfusion was 36 (36/94) = 38%. The number of cases with no transfusion was 58 (58/94) = 62%. Of the 36 patients/cases that were transfused, the number of patients who received < 2 units of RBC was 22 (22/36) = 61%. The number of patients who received > 3 units of RBC was 14 (14/36) = 39% or, if considering the figure out of the total number of cases, 15% (14/94). Of the five redo CABG cases, the number of transfused RBC ranged from 0-5. **Conclusion:** Taking into account the data from this audit and the more advanced technology available in the laboratory, a recommendation can be made for a reduction in the pre-operative standing order for most cardiothoracic surgical cases. The recommendation for change is as follows: Pre-op order for CABG, AVR, and MVR could be either a Type and Screen (TS) or TS with 1 unit RBC; Pre-op order for CABG redo could be TS with 2 units RBC crossmatched.

Disclosure of Commercial Conflict of Interest

E. A. GIlman: Nothing to disclose; K. S. Gray: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. A. GIlman: Nothing to disclose; K. S. Gray: Nothing to disclose

AP9

Cost Savings Associated with a Reduction in Predeposit Autologous Red Cell Transfusion

D Merhi[1] (dmerhi@lifespan.org), L D Roberts[1,2], J D Sweeney[2,1]. *[1]Transfusion Service, Rhode Island Hospital, Providence, RI, United States; [2]Transfusion Service, The Miriam Hospital, Providence, RI, United States*

Background/Case Studies: The practice of predeposit autologous donation (PAD) has been declining in the US over the past two decades. This is related to the perception that the allogeneic blood supply has improved in terms of safety. There has been residual resistance particularly in orthopedic surgery as the practice is perceived as safer by their older patient population. However, PAD carries risk of donation related morbidity as well as the higher overall rate of transfusion and on balance is difficult to justify medically. An additional aspect of PAD is that it is an expensive boutique product and eliminating or minimizing PAD can result in a large cost savings. **Study Design/Methods:** A program of education and engagement with the orthopedic department was initiated with the intention of decreasing the volume of PAD in late 2009. This program involved education on potential donation risks in addition to demonstrating that PAD resulted in a minimal reduction (if any) in allogeneic exposure. Reducing the transfusion threshold for post operative normovolemic anemia was emphasized. The data were re-inforced by periodic communications showing rates of transfusion (autologous and allogeneic) for individual surgeons. The cost savings aspect was not discussed with the orthopedic surgeons but was brought to the attention of the hospital administration which was very supportive. Annual data were recorded for both autologous and allogeneic units transfused and the acquisition costs of both product types. **Results/Findings:** Results are shown in the Table. After initial resistance, a steep decline in PAD is observed and by 2012 this approaches 90%. An increase in allogeneic red cell use is observed in 2011 but this is concurrent with the consolidation of two cardiac surgery programs to a single site, accounting for the majority of this increase. The net realized annual cost savings is at least $100,000 and does not include cost savings related to autologous unit processing, storage, or administration. **Conclusion:** There is resistance to changing the practice of PAD which can be at the surgeon or patient level. However, targeting surgeons alone can be successful in decreasing this practice but requires sustained effort. This reduction in PAD can be associated with a large cost savings.

Disclosure of Commercial Conflict of Interest

D. Merhi: Nothing to disclose; L. D. Roberts: Aeroscout, Other; J. D. Sweeney: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Merhi: Nothing to disclose; L. D. Roberts: Nothing to disclose; J. D. Sweeney: Bio-Rad, Grants or Research Support

Year	Autologous Units Received	Autologous Units Transfused	% of total Red Cell Transfusion	Wasted	% Reduction from baseline (2009)	Autologous Acquisition Cost	Allogeneic Transfusions	Allogeneic Acquisition Cost
2009	989	862	5.6%	127	—	$232,415	14,573	$3,328,425
2010	692	438	2.9%	254	49.2%	$162,620	14,445	$3,307,725
2011	221	173	1.1%	48	80%	$51,935	15,586	$3,564,225
2012	92	47	0.33%	45	95%	$21,620	14,788	$3,454,650

AP10

Regional Blood Center (BC) Based Patient Blood Management (PBM)

K Land[1] (kland@bloodsystems.org), P A Tomasulo[1], B A Lenes[1], M A Lieb[1], J Rataj[1], M Bassett[1], M Townsend[1], P Holt[1], F Nizzi[1]. *[1]Blood Systems, Scottsdale, AZ, United States*

Background/Case Studies: Over the past 30 years, the discipline of transfusion medicine has moved from serology to clinical practice, from physician led laboratory to quality controlled manufacturing under current good manufacturing practice (cGMP) and, most recently, from blood center (BC)-based blood product safety to hospital patient outcome and safety. The majority of the processes to achieve optimal transfusion practice are based within the hospital, where approximately 2/3 of the residual risk and 4/5 of the residual cost reside. Recognizing these paradigm shifts and the expected additional changes to healthcare, a large multistate regional blood supplier created a comprehensive, blood-center based patient blood management (PBM)

program designed to provide the tools and instruction for hospitals to effectively self deploy effective blood management. **Study Design/Methods:** Phone surveys of BCs known to provide sophisticated blood products and services were conducted. Literature was reviewed for methods and best practices in PBM, author experience documented, and customer focus groups performed. Information was collated and prioritized, strategies developed, and resources allocated. An affiliation with Strategic Healthcare Group for benchmarking tools as well as the hiring of physicians and medical technologists was undertaken. Program development ensued. **Results/Findings:** The BC-based PBM program provides active participation in the creation and running of hospital-based PBM programs and effectiveness teams and includes the following components: 1) 2-day, onsite assessment of transfusion service operational workflows and blood utilization protocols (Transfusion Dx.®); 2) benchmarking tools (BloodStat Online®) 3), series of analytical reports, educational modules, training programs, literature references, how-to guides, and tools to successfully transform hospital clinical transfusion practice (Optimum Tx™); 4) transdisciplinary support during the rollout of the hospital's PBM program; 5) targeted physician, nursing and technical staff education (Automatic communications and consultations for suboptimal or unique physician orders); 6) National Patient Antibody Registry (NPAR) 7) Transfusion Safety Officer program including an intensive 3-day training program, ongoing mentoring, and adjunct staffing opportunities; 8) charge-master review; 9) blood inventory process redesign; 10) LEAN performance improvement; 11) regulatory gap assessment; and 12) preparation for CAP, TJC, DNV, etc inspections. **Conclusion:** These BC programs are backed by a clinical services team of regionally deployed physicians, quality/transfusion specialists (including BC based TSOs), and other experts that are available to over 550 medical facilities in 20 states. The provision of this extensive list of PBM offerings from a regional, multistate supplier is unique. The effectiveness of these patient centric programs is currently being evaluated.

Disclosure of Commercial Conflict of Interest

M. Bassett: Nothing to disclose; P. Holt: No Answer; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; M. A. Lieb: Nothing to disclose; F. Nizzi: Mediware, Other, Blood Systems, Other; J. Rataj: No Answer; P. A. Tomasulo: Nothing to disclose; M. Townsend: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Bassett: Nothing to disclose; P. Holt: No Answer; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; M. A. Lieb: Nothing to disclose; F. Nizzi: Nothing to disclose; J. Rataj: No Answer; P. A. Tomasulo: Nothing to disclose; M. Townsend: Nothing to disclose

AP11

2012 Transfusion Safety and Patient Blood Management Survey

C Slapak[1] (cslapak@cbccts.org), K Gagliardi[2]. *[1]Community Blood Center/Community Tissue Services, Dayton, OH, United States; [2]Southwestern Ontario Regional Blood Coordinating Network, McMaster University, Hamilton, ON, Canada*

Background/Case Studies: In 2012, around 40 programs based on an informal network of transfusion safety (TS) and patient blood management (PBM) professionals existed in the US. In preparation for a TS/PBM workshop, an online survey was circulated to peers identified as transfusion safety officers, blood management coordinators, or similar roles across the US and Canada. **Study Design/Methods:** The survey of 49 questions was prepared in Survey Monkey™. Categories included professional background, location, size, scope, funding, education, and training. Many questions were multiple-choice with some yes/no closed-ended type. In some cases respondents selected all relevant responses from a list. The survey was circulated using informal networks in the US and the Transfusion mailing list originating in Canada with 240 subscribers. **Results/Findings:** There were 108 survey respondents: 35 (32%) from US, 67 (62%) from Canada and 6 (6%) from other countries including UK, India, Argentina, and Australia. Over 30 different program manager titles were provided; the most common variations included Transfusion Safety (50%), Transfusion (18%), Blood Management (16%), Quality (7%), and Hemovigilance (4%) as part of the title. As expected, the majority of programs identified are within a hospital/hospital group (74%). Facilitation of the remaining 26% of programs includes blood centers and provincial (Canada) offices. Professional backgrounds of program managers include 42% nursing (77% US, 24% Canada), 52% laboratory (9% US, 72% Canada), and 6% physicians or others. Almost 45% have been in their position for 5 years or less (US 74%, Canada 30%); 79% of the programs (US 54%, Canada 95%) have been in existence for over 5 years. Training is through professional experience (65%), networking (53%), and attending meetings (40%). Only 8% have used a formal TSO training program. When asked to identify specific TS and PBM activities performed, the following results were obtained (Table). **Conclusion:** A snapshot of formal, existing, TS/PBM programs was documented. Barriers to complete capture include the informality of networks between programs in North America. Many different job titles exist as well, although most managers agree on similar program elements. Training remains a challenge, with incumbents agreeing that professional experience and networking are factors in success. New directions include a national certification program, progress being limited by the number and scope of existing programs.

Disclosure of Commercial Conflict of Interest

K. Gagliardi: Nothing to disclose; C. Slapak: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Gagliardi: Nothing to disclose; C. Slapak: Nothing to disclose

TABLE 1: Common program activities

	Overall %	US %	Canada %
TRANSFUSION SAFETY			
Patient Identification	84	90	72
Blood administration	94	100	89
Appropriate orders	87	94	81
Transfusion reactions	88	97	83
PATIENT BLOOD MANAGEMENT			
Limit phlebotomy blood loss	34	45	20
Optimize hgb levels before OR	48	45	41
Minimize perioperative blood loss	63	71	52
Blood utilization review	91	97	85

AP12

Data-Driven, Pre-Transfusion Testing to Reduce Unnecessary Type and Screens

K E Finegan[1] (springdae@gmail.com), E A GIlman[1]. *[1]Blood Bank, Mayo Clinic Hospital, Phoenix, AZ, United States*

Background/Case Studies: The HOLD test is used here as an order to draw a properly labeled sample for pre-transfusion testing and to have a blood bank armband applied to the patient's wrist for possible future transfusion. There is no testing performed or billing charges associated with the HOLD test. The HOLD order is used to help manage outpatient Hematology/Oncology transfusion needs. When Hematology/Oncology patients have their blood drawn for labs, a HOLD order allows a blood bank sample to be drawn, but does not result in unnecessary testing or charges. The sample is kept in the blood bank until follow-up orders for transfusion are received for the patient, usually following the resulting of the hemoglobin (Hb) value. A study was done on the number of HOLD samples that were subsequently converted to a Type and Screen, and crossmatch for transfusion. **Study Design/Methods:** Computerized records were reviewed to determine the number of HOLD samples that were converted to a Type and Screen, and crossmatch for transfusion. Reports were generated from the Laboratory Information System (SCC Soft, Clearwater, FL) to indicate the number of HOLD tests that were drawn over the 6 month time frame of July 2012-December 2012. Review of the patient's medical record allowed us to determine how many HOLDs were converted to a Type and Screen, and crossmatch for transfusion. **Results/Findings:** During the study period, 1531 HOLD samples were drawn. Of those, 746 (48.7%) were converted to a Type and Screen, and crossmatch for transfusion. Therefore, there were 785 samples drawn that did not require ABO blood group typing and an antibody screen in the Blood Bank. The incidence of unexpected alloantibodies here is 10% of the patients. Use of the HOLD sample prevented numerous unnecessary antibody identifications (estimate of 79 avoided antibody identifications). This resulted in a savings of blood bank resources and technologist time. Patients also benefit from not being charged for blood bank testing when transfusion is not indicated. **Conclusion:** The HOLD test is an efficient way to prevent unnecessary pre-transfusion testing. The use of the HOLD test instead of a Type and Screen resulted in preventing the unnecessary testing of 785 samples.

Disclosure of Commercial Conflict of Interest

K. E. Finegan: Nothing to disclose; E. A. GIlman: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. E. Finegan: Nothing to disclose; E. A. GIlman: Nothing to disclose

AP13

Approaches to Improving Efficiency of Cell Saver Utilization

S Bakdash[1] (bakdass@ccf.org), J B Cywinski[2], D Tolich[3], R Lorenz[3,4], P I Figueroa[1]. *[1]Transfusion Medicine, Cleveland Clinic, Cleveland, OH, United States; [2]General Anesthesia, Cleveland Clinic, Cleveland, OH, United States; [3]Medical Operations, Cleveland Clinic, Cleveland, OH, United States; [4]Head and Neck Institute, Cleveland Clinic, Cleveland, OH, United States*

Background/Case Studies: Intraoperative autologous blood recovery may avoid many risks and costs associated with allogeneic transfusion; however, use of cell saver is not justified in every surgical case. Costs are greater for full set-ups, where blood is processed, than standby set-ups, where shed blood is collected and held. A minimum processing volume of 3 times the bowl size (50-250 cc) is typically required to return RBCs to the patient. The RBC content in 400 ml of cell saver blood is considered equivalent to 1 unit of allogeneic RBCs at this institution. Inefficient utilization of cell saver, defined as a yield <1 RBC equiv., may result in increased costs with no corresponding patient benefit. Evaluation of cell saver utilization, as part of a blood management initiative, identified variations in practice and opportunities for improvement. A comprehensive plan was developed and implemented, with reevaluation to assess for practice change and the need for further follow-up. **Study Design/Methods:** Cell saver utilization data (January 2009-October 2011) were analyzed by surgical subspecialty. A collaborative approach by the Blood Management, Transfusion Medicine, and Anesthesia departments resulted in meetings with representatives of surgical areas identified for improvement. Summary statistics and ideas for improvement were discussed and several educational interventions implemented, including staff meeting presentations, individual feedback and encouragement of active communication between anesthesia, surgery, and the cell saver operator prior to initiating processing. Data from Q1/Q2 2012 were used as a more current baseline for comparison to post-intervention results from Q1 2013, with Q2/Q3 2012 considered as a transition period. **Results/Findings:** Two targeted surgical areas demonstrated significant utilization improvement by different approaches. In Orthopedic/Joint Surgery, open discussion of utilization data in a staff meeting resulted in better understanding of volumes needed to process and costs associated with inefficient utilization. Subsequently, the proportion of cases yielding ≥1 RBC equivalent more than doubled (13% to 33%), with a corresponding drop in the proportion of cases yielding <1 RBC equivalent. (54% to 39%). A notable achievement in this group was decreasing conversion of standbys to full set-ups with 0 ml of returnable yield from 19% to 9%. Urological Surgery identified a physician champion and implemented a more personalized, peer-driven approach resulting in a 33% decrease in number of cases requesting cell saver and an increase in the proportion of cases yielding ≥1 RBC equiv. (33% to 42%). **Conclusion:** A collaborative approach to increasing awareness by surgeons of the costs of cell saver and their own utilization patterns, together with more active OR communication regarding minimum processing volumes required for meaningful RBC return, resulted in improved efficiency of cell-saver utilization.

Disclosure of Commercial Conflict of Interest

S. Bakdash: Nothing to disclose; J. B. Cywinski: Nothing to disclose; P. I. Figueroa: No Answer; R. Lorenz: Nothing to disclose; D. Tolich: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Bakdash: Nothing to disclose; J. B. Cywinski: Nothing to disclose; P. I. Figueroa: No Answer; R. Lorenz: Nothing to disclose; D. Tolich: Nothing to disclose

AP14

Cryoprecipitate Prospective Audit Program: Impact and Limitations

V Osegueda[1] (XVIOLETX@UCLA.EDU), A Ziman[1], R H Jeffery[1], D Goldfinger[1], Q Lu[1]. *[1]Pathology and Lab Medicine, University of California, Los Angeles Ronald Reagan Medical Center, Los Angeles, CA, United States*

Background/Case Studies: Cryoprecipitate usage has increased steadily at our university hospital in the last couple of years. A prospective audit program was implemented to reduce inappropriate transfusions. Data was collected to monitor program effectiveness and limitations. **Study Design/Methods:** Data on completed and cancelled orders of five-unit pre-pooled cryoprecipitate was collected from June 2012-February 2013. Due to operational issues, not all orders were audited. Orders from the emergency room, operating room, interventional radiology, and cardiac catheterization laboratory, as well as orders for patients with ventricular assist devices and patients on extracorporeal membrane oxygenation, were exempt from auditing. Orders from non-exempt areas were first reviewed by laboratory staff for a current fibrinogen. If there was no current fibrinogen or fibrinogen >150 mg/dL, orders were referred to on-call pathology resident for review. Upon clinician's agreement, unnecessary orders were cancelled. Additionally, transfusion data in 1 month (September 2012) was retrospectively reviewed to attain more detailed clinical information about cryoprecipitate recipients. **Results/Findings:** Between June 2012-February 2013, 1810 orders for five-unit pre-pooled cryoprecipitate were received, averaging 201 orders per month. A total of 799 (44%) orders were prospectively audited. Of those, 16 (2%) were cancelled. In September 2012, 219 pools of cryoprecipitate were transfused and 91 (42%) were transfused to patients with no current fibrinogen or fibrinogen >150 mg/dL. Among these 91 orders, 59 (65%) were from exempt service areas and 32 (35%) were from non-exempt service areas, mainly from the transplant surgical intensive care (10, 31%) and cardiothoracic intensive care units (9, 28%) (Table). **Conclusion:** The impact of the prospective audit program on reducing cryoprecipitate usage was limited given that less than half of orders were audited and a small percentage of those were determined to be inappropriate and cancelled. However, it provided one-on-one education to the clinical team especially to residents and fellows about indications for blood transfusion. A more comprehensive audit program to cover all service areas, and the institution of a dedicated transfusion safety officer may have a bigger impact on cryoprecipitate usage.

Disclosure of Commercial Conflict of Interest

D. Goldfinger: Nothing to disclose; R. H. Jeffery: Nothing to disclose; Q. Lu: Nothing to disclose; V. Osegueda: Nothing to disclose; A. Ziman: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Goldfinger: Nothing to disclose; R. H. Jeffery: Nothing to disclose; Q. Lu: Nothing to disclose; V. Osegueda: Nothing to disclose; A. Ziman: Nothing to disclose

List of non-exempt service areas with cryoprecipitate transfusions to patients with no fibrinogen or fibrinogen >150 mg/dL in September of 2012

Service Area	Pools of cryoprecipitate transfused	%
Transplant Surgical Intensive Care	10	31%
Cardiothoracic Intensive Care	9	28%
Coronary Care Intensive Care	5	16%
Medical Monitoring Unit	4	13%
Medical Intensive Care	2	6%
General Pediatrics	1	3%
Cardiac Observation Unit	1	3%
Total	32	100%

AP15

Transfusion Policy Snapshot Prior to Establishment of Blood Management Program

L Bonstein[1,3] (l_bonstein@rambam.health.gov.il), E J Dann[1,3], R Leiba[2], S Zarka[4]. *[1]Blood Bank, Rambam Health Care Campus, Haifa, Israel; [2]Statistics Unit, Rambam Health Care Campus, Haifa, Israel; [3]Bruce Rappaport Faculty of Medicine, Technion, Israel Institute of Technology, Haifa, Israel; [4]Israel Ministry of Health, Jerusalem, Israel*

Background/Case Studies: Appropriate blood use can reduce both transfusion-related adverse events and hospital costs. In 2012 the hospital administration of this 1000-bed, tertiary care academic center which serves over two million people, decided to design and implement a patient blood management (PBM) program. As a preliminary step, a study was conducted evaluating current transfusion policy at key departments aimed at locating major areas of intervention. **Study Design/Methods:** The transfusion practice of packed red blood cells units (PRBCu) was evaluated. All patients receiving PRBCu in five major departments (internal medicine, surgery, orthopedics, oncology, and hematology) from January-June 2012 were included in the study. Major bleeding events (≥4 units/ 24 hours) and emergency surgeries were excluded. Data on transfused patients, including demographics (age, sex), hemoglobin (Hb) level (g%) prior to transfusion, hospitalization period, surgery, and mortality rate, were collected. **Results/Findings:** A total of 6611 PRBCu transfused to 1836 patients were included in the analysis. Mean Hgb was 8.2 ± 1.3 g% for all percentiles and 9.7 for the 75% percentile. Average age of patients was 63 ± 17 yrs; 50 % of them were

males; 47% had surgery; mean hospitalization period was 23 ± 24 days; and 13% died during hospitalization. Most PRBCu were transfused at the department of hematology (27% of units/ 15% of patients); followed by internal medicine (24% of units/ 30% of patients); surgery (22% of units/27% of patients); oncology (19% of units/21% of patients); and orthopedics (8% of units/13% of patients). Mean Hgb values before transfusion were <8.0 g% at hematology and internal medicine wards and >8.8 g% at surgery and orthopedics services. **Conclusion:** Analysis of the data obtained revealed major differences in transfusion policy between the internal medicine and surgical services. There is a need for implementation of strict guidelines, mainly in surgical services, development of blood ordering software and computerized regulation as a tool for monitoring adherence to the updated transfusion policy. Such coordinated efforts can contribute to the establishment of effective blood management program.

Disclosure of Commercial Conflict of Interest

L. Bonstein: Nothing to disclose; E. J. Dann: No Answer; R. Leiba: No Answer; S. Zarka: Nothing to disclose

Disclosure of Grants Conflict of Interest

L. Bonstein: Nothing to disclose; E. J. Dann: No Answer; R. Leiba: No Answer; S. Zarka: Nothing to disclose

TABLE 1. Demographic and clinical data

	Total	Oncology	Hematology	Surgery	Orthopedics	Int. Med.
Units transfused	6611	1227	1802	1443	549	1596
Patients transfused	1836	385	279	504	247	559
Mean Hgb before transfusion	8.3 ± 1.5	8.4 ± 1.2	7.8 ± 0.9	9.2 ± 2.2	8.8 ± 1.4	7.6 ± 1.1
Age	63 ± 17	62 ± 14	56 ± 17	63 ± 18	69 ± 18	69 ± 14
Male sex (%)	50.0	46.2	50.5	60.3	34.4	51.7
Mortality (%)	13.2	10.1	8.2	7.3	1.6	24.9
Surgery (%)	46.9	17.9	36.2	76	95.5	14.3
Hospitalization period (median)[days]	23.3 ± 24.6 (15)	17.3 ± 16.9 (11)	28.9 ± 18.7 (27)	23.0 ± 26.4 (14)	15.9 ± 16.8 (9)	25.5 ± 28.9 (14)
Patients Fraction/ Units Fraction (%)		21.0/18.6	15.2/27.3	27.5/21.8	13.5/8.3	30.4/24.1
Units/patients	3.60	3.19	6.46	2.86	2.22	2.86

AP16

Reduction in Inappropriate Red Cell Transfusion through Prospective Computerized Order Auditing

K P Desrosiers[2] (Kevin.P.Desrosiers@hitchcock.org), S Yerrabothala[1], Z M Szczepiorkowski[1,2], N M Dunbar[1,2]. *[1]Pathology, Dartmouth-Hitchcock Medical Center, Lebanon, NH, United States; [2]Medicine, Dartmouth-Hitchcock Medical Center, Lebanon, NH, United States*

Background/Case Studies: The hospital transfusion policy was recently revised to recommend single unit red blood cell transfusion (RBC TXN) for non-bleeding patients when the hemoglobin (Hb) is <7 g/dL for most inpatients, <8 g/dL for acute coronary syndrome or <10 g/dL for early sepsis. To support compliance, our computerized provider order entry (CPOE) system was reconfigured to remove single click ordering for two unit RBC TXNs and provide prospective computerized order auditing based on the most recent hemoglobin level. When the patient Hb is ≥7 or has not been measured in the preceding 24 hours, a best practice alert (BPA) is triggered. The provider is then prompted to select from a limited menu of appropriate indications, provide a free text order justification or cancel the RBC TXN order. This study was undertaken to assess the impact of these changes on transfusion practice. **Study Design/Methods:** This study compared RBC TXNs in the pre-implementation period of October 2, 2011-March 31, 2012 (PRE) to the immediate post-implementation period of October 2, 2012-March 31, 2013 (POST). All RBC TXN events in adult inpatient medical, surgical and critical care areas (including the emergency, obstetric, and hematology-oncology units) were reviewed during both study periods. A "transfusion event" is defined as an order for TXN consisting of either one or two RBC units. The pre-transfusion Hb level was obtained for all RBC TXN events. One-unit RBC TXNs were classified as appropriate for HGB ≤ 8. **Results/Findings:** A total of 1800 RBC TXN events over 43,556 patient days were reviewed in the PRE study period and 1678 over 43,592 patient days in the POST. The proportion of two unit RBC TXNs decreased from 47% to 15% ($p < 0.0001$). The proportion of appropriate one unit RBC TXNs increased from 73% to 90% ($p < 0.0001$). Significant decreases were observed in pre-transfusion Hb triggers for both one and two unit RBC TXNs ($p < 0.0001$). There were also decreases in the total number of RBC units transfused/1000 patient days ($p < 0.0001$) and total transfusion events/1000 patient days ($p = 0.036$). **Conclusion:** Prospective computerized order auditing decreases red cell utilization and improves provider compliance with hospital transfusion policy. This may reduce unnecessary RBC TXNs (and associated risks) and result in direct cost savings to the institution.

Disclosure of Commercial Conflict of Interest

K. P. Desrosiers: Nothing to disclose; N. M. Dunbar: Nothing to disclose; Z. M. Szczepiorkowski: Nothing to disclose; S. Yerrabothala: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. P. Desrosiers: Nothing to disclose; N. M. Dunbar: Nothing to disclose; Z. M. Szczepiorkowski: Fenwal, Consulting or Board of Director Fees, Grants or Research Support, Terumo BCT, Grants or Research Support; S. Yerrabothala: Nothing to disclose

	PRE	POST
	N(%)	
Total Transfusion Events	1800 (100)	1678 (100)
One Unit Transfusions†	951 (53)	1431 (85)
Two Unit Transfusions†	849 (47)	247 (15)
One Unit Transfusions	951 (100)	1431 (100)
Appropriate (HGB < 8)†	696 (73)	1282 (90)
Pre-Transfusion HGB (g/dL)	Median (Mean. SD)	
One Unit Orders†	7.5 (7.6, 1.0)	6.9 (7.1, 0.84)
Two Unit Orders†	7.3 (7.4, 0.96)	6,8 (6.9, 1.1)
Total Patient Days	43,556	43.592
Total Units Transfused	2649	1925
Units Transfused/1000 patient days†	60.8	44.2
Transfusion Events/1000 patient‡	41.3	38.5

† $p < 0.0001$ ‡ $p = 0.036$

AP17

The Evolution of a Blood Center–Provided, Transfusion-Specialist Education Program

M Townsend[1] (mtownsend@bloodsystems.org), H Nguyen[1], M Lieb[1]. *[1]Blood Systems, Scottsdale, AZ, United States*

Background/Case Studies: In recent years, the interest in the implementation of blood management programs and monitoring the quality of transfusion medicine in hospitals has led to the emergence of hospital transfusion specialists such as Transfusion Safety Officers (TSOs). Results of recent informal global surveys (Streefker I, Deelen R., *TSOs in Hospitals, a New Dutch Profession*, 13th Annual IHN Symposium. Amsterdam, 2011) and US articles (C Slapack, 2012 unpublished) show that TSOs are trained on-the-job in addition to attending professional meetings (AABB, SABM) and workshops. In the US, most TSOs receive no official training, with only 7.8% identifying any kind of formal education. Currently there are three US blood centers (BC) offering some level of training, restricted to blood center employees functioning as resources to local hospitals. For the majority of hospitals, there are limited options for formal training programs available for TSO candidates. In 2012, one large BC identified the need for formal TSO training to serve hospital customers (HC) hiring TSOs to implement and direct Blood Safety and Management programs. The founders of this program sought to design a curriculum to fill knowledge/experience gaps in the array of candidates (RN, LVN, MT, Perfusionist) and to cover the theoretical and principal aspects of the job. With growing understanding of the role of the TSO, the faculty employed mechanisms to assist in the

development of the curriculum. **Study Design/Methods:** Following a 3-day program, students were asked to complete a detailed survey of each topic discussed to evaluate content, applicability, and presenter effectiveness. Students also participated in a monthly call to network and discuss challenges and successes experienced within their institutions. Information obtained through the surveys and calls were used to guide curriculum revisions. **Results/Findings:** The first iteration of the program consisted primarily of lectures with very little interaction. The program, now in its fifth revision, relies heavily on the use of case studies and workshops to make the training practical to applications in the hospital setting and now consists of topics including: Blood collection/ testing; Quality essentials; Transfusion indications/risks; Adverse event recognition; Blood banking essentials; Blood administration; Special Clinical Situations; Transfusion Alternatives; Blood Management; Soft skills; and Data management. **Conclusion:** Through a set of structured methods to identify expectations in transfusion management, a TSO training program has evolved to consist of standardized training topics and flexible adult learning methodologies. These follow-up techniques have also enabled the BC to understand the changing needs of hospitals in transfusion medicine.

Disclosure of Commercial Conflict of Interest

M. Lieb: No Answer; H. Nguyen: No Answer; M. Townsend: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Lieb: No Answer; H. Nguyen: No Answer; M. Townsend: Nothing to disclose

AP18

Tranexamic Acid (TA) Reduces Transfusions in Hip and Knee Arthroplasties without Increasing Thromboembolic Events

D Cole[1] (DCole@mtsinai.on.ca), J Freedman[2], N Shehata[1], E Goldszmidt[1], M Otremba[1], I M Fleming[1]. *[1]Transfusion Medicine, Mount Sinai Hospital, Toronto, ON, Canada; [2]St. Michael's Hospital, Toronto, ON, Canada*

Background/Case Studies: Antifibrinolytic agents (tranexamic acid [TA], aprotinin) have been shown in several studies to reduce blood loss during knee and hip arthroplasty. As part of a blood conservation program (ONTraC-Ontario Nurse Transfusion Coordinators) it was sought to determine whether TA protocol safely reduced transfusion rates in hip and knee revision arthroplasty. **Study Design/Methods:** This is a retrospective study of all patients undergoing hip and knee arthroplasty at a quaternary care centre. The proportion of patients transfused one year prior to TA implementation was compared with the year after using TA, to determine if there was a decrease in the proportion of patients requiring transfusion. Safety was assessed by comparing the rate of venous thromboembolism over the same time period. TA is given intraoperatively, prior to incision for patients undergoing hip arthroplasty and just after the release of the tourniquet for knee arthroplasty at a dose of 20 mg/kg. Patients are excluded from receiving TA if they have had a thromboembolic event ie myocardial infarction, stroke, deep venous thrombosis, pulmonary embolism within 1 year of surgery, if they are on lifelong anticoagulant, or have disseminated intravascular coordination (DIC) or sensitivity to TA. Patients are also excluded if they have cardiac disease, renal failure with serum creatinine >200 micromol/L and/or creatinine clearance <50 ml/min, and/or are dialysis dependant. **Results/Findings:** The Table describes the proportion of patients transfused and the mean units transfused prior to, and 1 year after the programmatic institution of TA. There was an absolute risk reduction of transfusion of 6% for hip arthroplasty and 12% for knee arthroplasty. The number of units per 100 patients operated on decreased from 33.9 to 29.7 for hip arthroplasty and from 28.3 to 8.7 for knee arthroplasty. A total of four thromboembolic events occurred after the use of TA was implemented, compared to a range of 0-4 events/year in the 5 years per approximately 600 patients yearly prior to the use of TA. **Conclusion:** TA results in a significant reduction in blood use without an increase in adverse thromboembolic events.

Disclosure of Commercial Conflict of Interest

D. Cole: Nothing to disclose; I. M. Fleming: Nothing to disclose; J. Freedman: Nothing to disclose; E. Goldszmidt: Nothing to disclose; M. Otremba: Nothing to disclose; N. Shehata: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Cole: Nothing to disclose; I. M. Fleming: Nothing to disclose; J. Freedman: Nothing to disclose; E. Goldszmidt: Nothing to disclose; M. Otremba: Nothing to disclose; N. Shehata: Nothing to disclose

	Pre Tranexamic Acid Program		Post Tranexamic Acid Program	
	Proportion Transfused	Mean Units Transfused	Proportion Transfused	Mean Units Transfused
Hip Arthroplasty n (%)	83/269 (31%)	1.1	66/266 (25%)	1.2
Knee Arthroplasty n (%)	68/264 (26%)	1.1	41/290 (14%)	0.62

AP19

Changing Blood Bank Policy for Managing Rh Negative Patients Who Receive Rh Positive Platelets: A Project Management Approach

M Greene[1] (mindy.greene@umassmemorial.org), A Simard[1], M Vauthrin[1], J Ferschke[1], R Weinstein[1]. *[1]Transfusion Medicine, University of Massachusetts Memorial Medical Center, Worcester, MA, United States*

Background/Case Studies: One approach to continuous quality improvement in a hospital transfusion service is to periodically audit the applicability of department policies as knowledge evolves in the field. One such policy required the issuing of Rh Immune Globulin (RhIg) to prevent Rh(D) alloimmunization of Rh negative patients who were transfused with Rh positive platelets. The policy was implemented in an era when blood banks issued both pooled platelet concentrate and platelets pheresis (PP), but PP only has been issued since August 2005. Given that PP contain less than 0.005 ml of red cells per unit, well below the immunizing dose of 0.03 ml, it was decided to reevaluate the policy. It was surmised that eliminating the routine administration of RhIg would save time, money, and resources while reducing patient risk that might be associated with the administration of RhIg. **Study Design/Methods:** Data were analyzed for a 1 year period to determine the impact of a potential policy change. During FY 2011, 89 doses of RhIg were administered to 56 Rh negative PP recipients to prevent Rh sensitization. 74 doses (83%) were administered in the inpatient setting. RhIg administration necessitated 373 antibody identifications to exclude alloantibodies on subsequent pretransfusion testing of 34 patients, resulting in longer wait times, in particular for same day outpatient transfusion. Estimated costs incurred for RhIg and antibody exclusion panels were $41,000. Based on the preliminary data,a project plan was developed with the aim of changing the policy to improve patient care and decrease costs. Planning steps included: 1) Evaluation of risks and inconvenience of RhIg administration to PP recipients; 2) Review of published evidence that would support or argue against a policy change; 3) Proposal to Transfusion Committee with supporting data and evidence regarding policy change; 4) Policy change implementation; and 5) Communication to medical and nursing staff regarding policy change. **Results/Findings:** The Transfusion Committee approved a revised policy whereby one dose (300 μg) of RhIg (IV) would be recommended only for female PP recipients age 50 years or younger at 4-week intervals while being transfused. During the first 6 months that the revised policy has been in force, there has been an estimated savings of $22,000 by avoiding unnecessary issuing of RhIg or antibody identification work-ups. **Conclusion:** The project management approach, whereby the data and the published evidence and evaluated risks and benefits regarding the proposed policy change were reviewed, resulted in ready acceptance of the proposal by the hospital Transfusion Committee. Amending the policy has improved the patient experience (shorter wait times for transfusion) and lowered costs.

Disclosure of Commercial Conflict of Interest

J. Ferschke: Nothing to disclose; M. Greene: Nothing to disclose; A. Simard: Nothing to disclose; M. Vauthrin: Nothing to disclose; R. Weinstein: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Ferschke: Nothing to disclose; M. Greene: Nothing to disclose; A. Simard: Nothing to disclose; M. Vauthrin: Nothing to disclose; R. Weinstein: Nothing to disclose

Collections

AP20

Targeting First-Time Donors to Apheresis

B Norman[1] (BNorman@carterbloodcare.org), M Stefan[1], D Lott[1]. *[1]Carter BloodCare, Bedford, TX, United States*

Background/Case Studies: Blood centers face enormous challenges with donor apathy, an aging donor base, an unsteady export market, and increasing pressures to reduce costs. Therefore, it is imperative to collect the right product, at the right time, from the right donor, to ensure inventory needs are appropriately met. With a first time donor ratio of ~ 30%, it is risky to use automation on mobile blood drives without knowing the donor's blood type. The result could be collecting two red blood cell units with a blood type not needed for inventory. With an unstable export market it is likely those units will end up being discarded which is costly and not good management of a precious resource. To better manage first time donor collections it was decided to pilot a blood typing card to solve three challenges: collecting the right product from the first time donor; increasing inventory of the right type product; and reducing waste by not discarding products such as AB+ RBCs. **Study Design/Methods:** The main objective was to identify first-time donor blood types prior to phlebotomy to determine the best collection procedure based on our inventory needs. The Eldon card testing methodology was chosen based upon cost, ease of use, and interpretation. Since this is not the test of record, an Eldon Card Reading Chart was developed which coded the testing results to ensure staff did not give blood typing results to donors (the test only identifies the optimal procedure and does not refer to blood type). This chart allowed staff to refer first time donors to the optimal procedure based on the results of the test. The goal was to decrease two RBC collections from A+, B+ and AB+ donors. Six high school blood drives were selected as this donor base provided sufficient numbers of eligible first time donors. Implementation was initiated by training a selected core staff of ABO Rh card typing screeners. The process flow consisted of an employee triaging donors as they present at the registration area. Direction to routine or Eldon card screening is based upon first time donation and meeting standard eligibility requirements. The donor chart was then color coded as a visual cue to assist staff in routing the donor to the appropriate procedure. **Results/Findings:** At the six drives, overall collections were effectively reduced of A+ double RBCs to 11% (previously as high as 29%), B+ to 14% (previously as high as 26%) and AB+ to 5% (previously as high as 32%). Collections were increased of O+ RBCs to 19% and O negative to 29%. Additionally, first time donors were targeted for platelet apheresis which had not been offered prior to this pre-donation blood typing. **Conclusion:** The Eldon card was effective in triaging first time donors to maximize specific component collection and provide more value from the mobile drives.

Disclosure of Commercial Conflict of Interest

D. Lott: Nothing to disclose; B. Norman: Macropharma,Ownership or Partnership; M. Stefan: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Lott: Nothing to disclose; B. Norman: Nothing to disclose; M. Stefan: Nothing to disclose

AP21

Universal Phlebotomy Action Plan for Decreasing Venipuncture Loss Rates Due To Unsuccessful Phlebotomies

T Gough[1] (tgough@carterbloodcare.org), N Coady[1]. *[1]Donor Center Collections, Carter Bloodcare, Bedford, TX, United States*

Background/Case Studies: Poor vein selection and phlebotomy technique is the primary cause of unsuccessful venipuncture procedures resulting in the loss of whole blood and apheresis units. This can be a particular problem for newer phlebotomists, but can also be exhibited in more tenured staff. To combat these, the blood center decided to create a consistent and progressive venipuncture action plan to improve the phlebotomy success rates of targeted staff. **Study Design/Methods:** After an initial period of monitoring and investigating unsuccessful venipunctures at three donor centers over an 18-month period, it was found that 5% of employees had a venipuncture loss rate >10%. Each donor center supervisor monitored loss rates, and implemented corrective action differently with varying degrees of success. A Universal Phlebotomy Action Plan (UPAP) was developed and utilized with employees that had an unsuccessful venipuncture rate greater than 10% in any 2 months of a preceding 3 month block. Loss rates were categorized as either apheresis or a combination of apheresis/whole blood venipunctures, and monitored. The UPAP consisted of five tiers: tier 1, the employee was given phlebotomy technique and vein selection instruction by their immediate supervisor; tier 2, the employee received the same instruction from their next-level supervisor; tier 3, a preceptor was assigned to work one-on-one with the phlebotomist at their home location for 2 months; tier 4, the phlebotomist worked with a preceptor at another location for 2 months; and tier 5, the phlebotomist was paired with a corporate trainer for 1 week. If improvements were not made at the initial tier level, the employee was moved to the next tier as applicable. To migrate out of the program, the employee was required to complete the current tier and reduce the loss rate <10% in both procedure categories. **Results/Findings:** During the study, ten phlebotomists entered tier 1, seven of those reached tier 2, and two of those seven eventually reached tier 3. None of the phlebotomists in this study reached tier 4 or 5. The number of products per apheresis kit at the three centers combined increased from 1.6 to 1.7. In addition, platelet quantity not sufficient rate decreased from 8.2% to 6.5%. Two of the phlebotomists were promoted after improving and maintaining acceptable phlebotomy loss rates. **Conclusion:** Implementation of the UPAP was effective in reducing the number of unsuccessful venipunctures, increasing product collection, and providing a consistent action plan for improving phlebotomy success rates. In addition, UPAP provided a mechanism to identify phlebotomists who could benefit from additional training, and also reduced supply wastage.

Disclosure of Commercial Conflict of Interest

N. Coady: No Answer; T. Gough: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Coady: No Answer; T. Gough: Nothing to disclose

AP22

The Benefits of Using a Sterile Connecting Device (SCD) to Recover Unsuccessful Apheresis Procedures

M T Wilson[1] (mwilson@carterbloodcare.org). [1]Collections, Carter BloodCare, Keller, TX, United States

Background/Case Studies: Historically, staff used a second apheresis kit to perform a phlebotomy if the initial venipuncture was unsuccessful or an infiltration occurred. Because of the cost of an apheresis kit, staff was reluctant to attempt a second phlebotomy and rarely did so. The result was countless lost opportunities to recover procedures of all types when the first phlebotomy attempt was unsuccessful. In addition to the unit losses on the day of donation, there was an inability to prevent platelet and plasma donors from being deferred for 56 days when red cells couldn't be returned in cases of infiltration. **Study Design/Methods:** With the goal of increasing product collection, decreasing deferrals and reducing cost, the blood center performed a project return on investment (ROI) to assess the feasibility of sterile connecting a second needle to the apheresis kit after an unsuccessful phlebotomy. Data was evaluated to determine the number of lost procedures and units due to unsuccessful venipuntures that possibly could be recovered using the welder. Data was retrieved from the blood center donor management and eligibility system. Once the implementation was deemed financially favorable from a cost saving and product inventory perspective, the sterile connecting devices (SCD)s and supplies were purchased for all 26 Donor Centers and SOPs were developed. Because this blood center is so large, the procedures were trained and implemented over several weeks from the end of November 2012 January, 2013. **Results/Findings:** 110 days post implementation, the restick procedure with the SCD needle was attempted 171 times on various types of apheresis procedures. The staff was successful in recovering 115 of these 171 procedures (63%), resulting in the collection of 109 platelets (PLTs), 91 red blood cells (RBCs), and 35 plasma units that would have been lost previously. Tracking these 115 donors over the next 8 weeks, it was calculated what the number of platelet and plasma units given would have been lost had the initial donation not been recovered 56 days before. These donations produced 78 platelet and 13 plasma units. In addition, the blood center saved $17,327 in kit cost by using only the needle and wafer instead of a new kit. **Conclusion:** Extrapolating the unit savings into an annual perspective, it is expected to recover 622 PLTt, 302 RBCs and 161 plasma units. In addition, $57,755 should be eliminated in kit cost, making this project extremely beneficial both financially and from an inventory perspective even after the cost of purchasing the SCDs for each center.

Disclosure of Commercial Conflict of Interest

M. T. Wilson: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. T. Wilson: Nothing to disclose

AP23

Who Orders Autologous Blood? A Retrospective Analysis of Preoperative Autologous Blood Utilization in a Medium-Sized Provider

M Almohammadi[1] (almohammadi77@gmail.com), N J McLaughlin[2], I Sadek[1], E Kahwash[1,2]. *[1]Dalhousie University, Halifax, NS, Canada; [2]Canadian Blood Services, Halifax, NS, Canada*

Background/Case Studies: The utilization of preoperative autologous blood donation (PABD) peaked in the 1990s in response to concern about the risk of transfusion-transmitted infection (TTI). Allogeneic transfusion is currently associated with a very low risk of TTI. Consequently, PABD has a limited set of clinical indications while being associated with higher cost than allogeneic transfusion. This study seeks to determine local trends in PABD utilization **Study Design/Methods:** A retrospective chart review was undertaken of all autologous donations from 2000-2011 at a medium-sized blood provider. Data reviewed included requesting physician name, physician specialty and date of graduation from medical school, requesting institution, number of autologous units requested and collected, and number of autologous and allogeneic transfusions received by the donor/patient. **Results/Findings:** 338 autologous donations occurred over the study period and a declining trend was observed. 626 autologous units were collected of which 188 (30%) were transfused. Requests for PABD were made by 99 individual physicians. Ten physicians accounted for 51% of all autologous requests. The average date of medical school graduation for this group was 1982. The most common procedures were gynecologic (53%) and orthopedic (35%). **Conclusion:** A declining trend in PABD utilization is observed over the study period. Residual demand appears to be physician-driven. The majority of residual PABD utilization is attributable to a small group of physicians that graduated from medical school prior to the development of highly sensitive tests for blood-borne pathogens. This suggests that the majority of PABD requests are being made without reference to recent utilization guidelines.

Disclosure of Commercial Conflict of Interest

M. Almohammadi: Nothing to disclose; E. Kahwash: Nothing to disclose; N. J. McLaughlin: Nothing to disclose; I. Sadek: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Almohammadi: Nothing to disclose; E. Kahwash: Nothing to disclose; N. J. McLaughlin: Nothing to disclose; I. Sadek: Nothing to disclose

AP24

The Impact of Using a Simple Blood Screen to Qualify First-Time Donors for Double Red Cell (DRC) Donation

J Ullmann-Bester[1] (Jane.UllmannBester@redcross.org). [1]American Red Cross, North Central Region, St Paul, MN, United States

Background/Case Studies: Double Red Cell (DRC) donation results in the collection of two units of leukoreduced (LR) red cells and no plasma. Collecting DRC from donors of the right type mix (RTM) provides significant benefit to the blood supply. RTM includes O Pos/O Neg/A Neg/B Neg and this blood center does not draw DRC from donors who are not of the RTM. Increased emphasis is placed on recruiting O Negative donors as DRC to maintain adequate inventories of "Universal Donor" blood. Use of a simple blood type screen during health history can help identify first-time RTM donors as DRC in a cost-effective manner. This blood type screen helped to increase RTM collections in 2012. **Study Design/Methods:** First-time blood donors who meet physical criteria (height, weight, hemoglobin [Hb]) are asked to donate DRC. Donors are tested using a standard Anti-A reagent, using blood from the same fingerstick performed to measure the Hb. The result of this simple slide test is not a test of record. If the test result is positive (agglutination), the donor is directed to donate whole blood. If the result is negative (no agglutination) the donor qualifies to donate DRC. Limitations of the Anti-A screen are that it does not exclude B Pos donors, and does exclude A Neg donors. **Results/Findings:** In 2012 in this blood center, 39,894 RTM units were drawn by automation. 1,042 first-time donors donated DRC. Taking into account limitations of the screen and errors inherent to a quick screen performed in a non-laboratory setting, 830 were ultimately confirmed to be O Pos/O Neg/A Neg/B Neg, including 158 O Neg donors. Costs of the Anti-A reagent and associated supplies required to perform the quick screen were minimal, and the Return on Investment was recovered when at least 22 additional units of the RTM were drawn (based on the average selling price for a unit of LR red cells in the U.S. as published in the 2009 DHHS National Blood Collection Utilization Survey). It is estimated, taking into account those procedures that did not result in a second unit collected, that 818 incremental units of RTM LR red cells were drawn, including 156 incremental units of O Negative. **Conclusion:** A quick screen with Anti-A reagent allows collection of additional units of RTM LR red cells. The additional RTM red cells collected from first-time donors can significantly help alleviate blood shortages, especially of type O blood. The screen enables first-time donors to donate DRC. The additional revenue generated also far outweighs the cost of reagent and supplies needed to qualify first-time donors to automation.

Disclosure of Commercial Conflict of Interest

J. Ullmann-Bester: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Ullmann-Bester: Nothing to disclose

AP25

Right Donor, Right Time: The Onsite Donor Conversion Process

J Douke[1] (janice.douke@bloodsource.org), C Judson[1]. *[1]BloodSource, Mather, CA, United States*

Background/Case Studies: On April 1, 2012 BloodSource implemented a whole blood (WB) to automation onsite donor conversion process. The WB onsite donor conversion process (WBODC) was implemented at 16 Donor Centers, 3 Regional Centers and 3 Mobile Operations locations. The WBODC process includes an invitation and education about the need for platelets, plasma and double red cell donations. The invitation starts with a local telephone or on-site Recruiter, with a follow-up invitation by the Registrars, Evaluators, and Phlebotomists. Donors expressing interest are able to donate immediately or book a future automated appointment. WBODC is designed to capture the right donor at the right time, increase equipment utilization, and provide a "friendly" first time automated donation. **Study Design/Methods:** WBODC replaced the previous new automated donor recruitment model; collect a CBC, invitation to donate by mail, and telephone recruitment. This old recruitment model included approximately 4000 CBCs annually, the number of new automated donors not increasing with an increase in the number of CBCs collected. The procedure offered for the WBODC donor is comprised of a single platelet, plasma (if male) and an RBC. A unique platelet target is used for this donor. This target is designed to lessen the chance of a high concentration, low concentration, or variable count platelet product that often results from using a default platelet count. This unique single platelet target is also intended to provide the novice automated donor with a friendlier experience by offering a shorter procedure. Evaluation of the adverse event rates for first time automated donors support this conclusion. The reaction rates for first time automated donors decreased with WBODC; October-December 2011 versus October-December 2012 showed a 3% decrease in mild syncopal reactions. Efficiencies gained by the use of donor conversion, eliminating the expense of CBCs and increased equipment utilization. Of the thousands of donors that consented to the CBC, approximately 8% made an appointment to donate; low return for a large expense. Equipment utilization improved as a by-product, with no additional expense. There is no need to increase the Donor Center appointments, hours, or staff to accommodate additional donors; the WBODC process is used primarily for automated appointment cancellations, donor no-shows, and open appointment slots. **Results/Findings:** See Table below for subsequent donations for WBODC April-October 2012. **Conclusion:** Onsite conversion is a more efficient process. Lab expense, paperwork, and manual work are eliminated. Novice donors are open to automation and younger donors are easier to convert. Reaction rates do not increase. All one has to do is just ask!

Disclosure of Commercial Conflict of Interest

J. Douke: Nothing to disclose; C. Judson: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Douke: Nothing to disclose; C. Judson: Nothing to disclose

Subsequent Donations for WBODC, April-October 2012

	# of conversions	Returned to WB	Not Eligible	Non-PLT Donation	Single PLT Plus	Double PLT Plus	Triple PLT Plus	Single PLT Only	No Subsequent Donation
Total	345	21	42	12	17	52	62	16	123
Percent	100%	6%	12%	3%	5%	15%	18%	5%	36%

AP26

Traveling Trimas at Bonfils Blood Centers

B Roberts[1] (bob_roberts@bonfils.org), A L Dirr[1], T Berry[1], C Propernick[1]. *[1]Community Donor Operations, Bonfils Blood Centers, Denver, CO, United States*

Background/Case Studies: In January 2013, discussions began about how this blood center (BBC) could match collections with the product demand of clients. The goal was to achieve self sufficiency and reduce the need to import product no later than Q1. After reviewing current collection strategies against client needs, it was decided to increase collections on specific days while decreasing on other days. BBC determined it needed to increase collections on Saturdays, Sundays, Mondays and decrease on Wednesdays, in order to be more closely aligned with client demand. The analytical data for client demand and optimizing operational days was provided by the BBC Manufacturing and Business Development Teams **Study Design/Methods:** With the goal defined to align collection strategies with client demand, the next step was to develop the methodology as to how BBC was going to achieve the appropriate increases and decreases. It was determined BBC would reach the goal using two methods. The first would be to adjust the days of operation of selected collection centers effective April 1, 2013 with one center being closed on Wednesdays and opened on Sundays. The second would be to transfer underutilized Trima machines to centers with high concentrations of platelet donors on Saturdays, Sundays, and Mondays. The transfer program began on March 1, 2013. **Results/Findings:** The traveling Trima program has been extremely effective. As a comparison BBC has included a table comparing the platelet collection data for March 2012 versus March 2013. Goals of becoming self sufficient have been met, with par levels being achieved and the alignment of collection strategies with client demand. **Conclusion:** The traveling Trima program has allowed BBC to achieve its primary goal of becoming self sufficient in the first month of operation.

Disclosure of Commercial Conflict of Interest

T. Berry: Nothing to disclose; A. L. Dirr: Nothing to disclose; C. Propernick: Nothing to disclose; B. Roberts: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Berry: Nothing to disclose; A. L. Dirr: Nothing to disclose; C. Propernick: Nothing to disclose; B. Roberts: Nothing to disclose

March 2012 versus March 2013 Platelet Collections

	Saturday	Sunday	Monday	Total
March 2012	59	68	112	239
March 2013	150	146	249	520

AP27

Apheresis Paperless Documentation Dawn M. Kaiser, Director, Community Donor Centers, LifeSource, Camille Lemaich, Donor Center Supervisor, LifeSource

D M Kaiser[1] (dkaiser@itxm.org), C Lemaich[1]. *[1]Donor Services, LifeSource Blood Services, Rosemont, IL, United States*

Background/Case Studies: This blood services company is striving for continued increase in efficiency; with that goal in mind the process for manual documentation was examined. In analysis, it was found that apheresis manual procedure records have the potential for documentation errors, omissions, and inconsistencies; these records also take up space and costs that could be used more productively. The data collected from donor screening as well as from the apheresis device needed to be captured, but streamlining was necessary. A complete solution would include: a fully integrated electronic donor record (EDR), workflow efficiencies, leveraging current technology investments, data network storage, wireless export capability, and monitoring target and actual split rate product yield. **Study Design/Methods:** LifeSource decided to implement DXT Relay (Fenwal, Inc.) with HealthCare ID to link the Amicus Separator (Fenwal, Inc.) to the rest of the EDR. This process took 6 months. DXT Relay software provides a flexible solution that supports paperless documentation and strengthens data decisions. To measure the workflow efficiency, time studies were conducted in centers with EDR versus those with paper records. The implementation process began with understanding current documentation needs and where the data was housed. Then we worked with the vendors to customize data fields (detailing the workflow plan). The team, consisting of quality, operations, and technical training, determined which fields were appropriate. Fenwal and HealthCare ID created customized fields that were identified and the process of updating our SOPs was begun, including apheresis operator training. **Results/Findings:** From beginning to end the project took 6 months and involved 10 staff members that evaluated and revised SOPS. The majority of time was spent understanding which fields were optimal to include in the record. Staff training was minimal as the only learning was to understand which field to scan during the procedure. Several time studies were performed, with and without EDR, and those studies showed that with EDR staff reduced time by an average of 3 minutes per procedure. From donor in bed to starting the procedure,the average time done manually: 17 minutes, 30 seconds. From donor in bed to starting the procedure, the average time with EDR: 14 minutes, 27 seconds. **Conclusion:** Using DXT Relay and HealthCare ID the goal of transitioning from a paper to an electronic record was accomplished with ease. The implementation was simple because both vendors were able to customize their products. Currently, we have successfully implemented the EDR in one donor center and we intend to push this throughout the rest of the blood center. The electronic record not only decreases time pre procedure but allows for enhanced monitoring, giving supervisors the ability to drill down on efficiency by instrument, staff, and site.

Disclosure of Commercial Conflict of Interest

D. M. Kaiser: Nothing to disclose; C. Lemaich: No Answer

Disclosure of Grants Conflict of Interest

D. M. Kaiser: Nothing to disclose; C. Lemaich: No Answer

AP28

Developing a Mobile Apheresis Platelet Program

B Norman[1] (BNorman@carterbloodcare.org), M Stefan[1], D Lott[1], B Wright[1]. *[1]Carter BloodCare, Bedford, TX, United States*

Background/Case Studies: With the changing dynamics in the industry, including the aging donor population and increased civic disengagement, the blood center recognized the need to increase the current pool of platelet donors. In this market, while red cell transfusions have been flat or declining over the past few years, there has been an increase in apheresis platelet transfusions here. This further enhanced the need for the center to increase the apheresis platelet donor base. **Study Design/Methods:** In reviewing the current donor base and collection data it was understood donor conversions were being maximized from whole blood to apheresis at the fixed donor centers. Additionally, the need was recognized to develop a mobile apheresis platelet program. This program would allow the introduction of the apheresis platelet technology to the current mobile donors and increase the apheresis platelet collections. After the initial mobile apheresis platelet donation, the donor would be recruited to the donor centers for routine apheresis collections. To begin the program, it was necessary to begin educating the donor coordinators about the need for apheresis platelets in the community. Highly productive drives were targeted where there was coordinator support and interest. The first few mobile platelet drives were at high schools and proved to be extremely effective. This success created interest and excitement among donor coordinators and staff which ultimately allowed the program.to grow. It has now grown to include six dedicated apheresis instruments, highly trained staff, specific core marketing materials, and two mobile platelet coaches. **Results/Findings:** May 2012-April 2013 Mobile Platelet Drive Statistics: 139 Mobile Platelet Blood Drives; 478 donors seen; Recruited 350 first time platelet donors; 600 total procedures, yielding 844 platelet products. **Conclusion:** By introducing a new technology and highlighting the need for apheresis platelet donations, this program increased the current donor base and provided an avenue to drive additional platelet donors to the fixed donor centers while reaching new donors through mobile collections.

Disclosure of Commercial Conflict of Interest

D. Lott: Nothing to disclose; B. Norman: MacroPharma, Ownership or Partnership; M. Stefan: Nothing to disclose; B. Wright: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Lott: Nothing to disclose; B. Norman: Nothing to disclose; M. Stefan: Nothing to disclose; B. Wright: Nothing to disclose

AP29

Make It Count: Trending Data That Can Effectively Make a Difference with Donor Reactions

W R Alaman[1] (walaman@carterbloodcare.org), V V Adams[1], T Jackson[1], A T Callies[1], G Paranjape[1]. *[1]Donor Advocate, Carter BloodCare, Bedford, TX, United States*

Background/Case Studies: Background: Typically, young donors have more adverse reactions. Minimizing reactions plays a vital role in maintaining

good donor relationships and promoting a healthy inventory of blood for the service area. An electronic data mining system Donor Reaction Tracker (ZREAC) was created which has been used for over 18 months. The system allows categorization and trend specific information regarding donor reactions in multiple categories. **Study Design/Methods:** Data from 2012 was taken from medical adverse reaction forms and examined , after being captured in the (ZREAC) program. Vasovagal reactions were studied by age, gender, and donation type of donors who fall into the following categories: 1) donors 16-21 years of age; and 2) donors 22 years of age and older who gave either Whole Blood (WB), Apheresis Platelets with concurrent products (plasma and red cells) and Double Red Cells (DRC). **Results/Findings:** See Table. **Conclusion:** Conclusion: The data show that young donors indeed have a higher rates of vasovagal reaction compared to older donors in all categories. Female donors in every category have almost twice the chance of having a reaction compared to their male counterparts. The reaction tracker program (ZREAC) has provided a timely and accurate means to retrieve data regarding our donors. Proactive initiatives need to be adopted to tackle adverse reactions, not just with the different age groups, but to tailor a message unique to each donation procedure type. This data should provide an opportunity to develop educational messages, fact sheets, and tutorials to build a diverse and productive donor management system at the blood center.

Disclosure of Commercial Conflict of Interest

V. V. Adams: Nothing to disclose; W. R. Alaman: Nothing to disclose; A. T. Callies: Nothing to disclose; T. Jackson: Nothing to disclose; G. Paranjape: Nothing to disclose

Disclosure of Grants Conflict of Interest

V. V. Adams: Nothing to disclose; W. R. Alaman: Nothing to disclose; A. T. Callies: Nothing to disclose; T. Jackson: Nothing to disclose; G. Paranjape: Nothing to disclose

Results

	Whole Blood		
	Donations	Reactions	%R
16-21 F	38406	6484	16.9
22+ F	100949	4923	4.9
16-21 M	24865	2031	8.2
22+ M	99228	2109	2.1
	Platelets		
	Donations	Reactions	%R
16-21 F	426	51	12.0
22+ F	9340	920	9.9
16-21 M	539	41	7.6
22+ M	16916	1061	6.3
	Double Red Cells		
	Donations	Reactions	%R
16-21 F	875	48	5.5
22+ F	2503	87	3.5
16-21 M	9132	340	3.7
22+ M	19247	302	1.6

AP30

The Mystery of the Missing Consent Forms of a 16 Year Old

F L Beatty[1] (fbeatty@itxm.org), L A Nelson[1]. *[1]Clinical Education and Training, The Institute for Transfusion Medicine, Pittsburgh, PA, United States*

Background/Case Studies: What is an effective approach to minimize the problems of losing the parental consent forms of a 16 year old? Teenagers are a growing and important donor group. Teenage donors roughly represent 13% of the donor population, with 25% presenting during mobile collections. Through specialized donor education specifically targeting this age group, 16 year olds are made aware and recognize their potential to save three lives. This organization has recognized unique aspects when dealing with 16 year old donors especially when complying with industry standards and legal requirements. The challenge experienced when processing the teenage donor population is that they do not typically have the proper identification. Often, the teenage donor is unable to present or verbalize required legal identifiers such as knowledge of their Social Security Number (SSN). The process requirements based on pre-established operational guidelines states that a current version of the parental consent form must be completed by a parent/guardian prior to donation date, which allows staff to confirm the donor's identification information for further processing, thus maintaining compliance. **Study Design/Methods:** A Root Cause Analysis team was formed to determine and mitigate the cause for the missing consents of teenagers. Using LEAN Six Sigma tools, the donation process for minors was reviewed. Upon review, it was determined that the process and SOP allowed room for error. The RCA team recommended revisions to the SOP, which consisted of establishing a pre-determined location of a green envelope, to hold consents once reviewed for completeness in each medical interview. The team performed observations to collect specific data related to the revised standard operating procedure. **Results/Findings:** Upon implementation, it was identified that the loss of parental consents decreased, but consents containing errors increased. Moreover, an additional set of challenges appeared. Initial findings proved that the green folder was not properly utilized at drives, which did not allow the Supervisor to review for errors and omissions on the consent forms. Recognizing specific trends, the team continued to monitor and identify specific errors utilizing the error management system. **Conclusion:** Our efforts were successful in solving the mystery of the missing consents. Revisions are currently in place to make changes to the SOP including further education and training for site Supervisors to enforce proper use of materials. The teenage donor continues to be a growing donor population that requires additional education and understanding of the importance of the consent. The parental consent being crucial to the donation process, it must meet all regulatory guidelines. With the growing number of teenage donors, it is important to stress pre-education for this donor group, ensuring availability and integrity of products to the customer.

Disclosure of Commercial Conflict of Interest

F. L. Beatty: Nothing to disclose; L. A. Nelson: Nothing to disclose

Disclosure of Grants Conflict of Interest

F. L. Beatty: Nothing to disclose; L. A. Nelson: Nothing to disclose

Education and Training

AP31

The Use of an Electronic Flashcard Program to Study the *AABB Blood Bank and Transfusion Service Standards*

M Friend[1] (Maribeth.Friend@uchealth.org). [1]Transfusion Service, University of Colorado Hospital-Memorial, Colorado Springs, CO, United States

Background/Case Studies: The transfusion service was looking for an innovative way to review the *AABB Blood Bank and Transfusion Standards* and to document competency of this review. **Study Design/Methods:** An electronic flashcard program was utilized to create flashcards, which matched pertinent terms from the *AABB Blood Bank and Transfusion Service* (BB/TS) *Standards*, 27th edition, Quality System Essentials (QSEs) to its definition or standard. The program is web-based. A set of flashcards was assigned each month. The team members accessed each set of flashcards by a user ID and password. The Transfusion Service focused on one QSE per month. Team members studied and reviewed using various games and learning tools from the online program. **Results/Findings:** In 2012, the transfusion service used this program to allow team members to study the terms, and then generate self-tests without limit, for self-evaluation. Once learners were able to achieve self-testing to 100% competency, the test was printed and submitted for documentation of learning the QSE of the month. **Conclusion:** The use of the online flashcards provided an easy access tool for team members to study the AABB BB/TS Standards. There is also an application available for mobile devices. By assigning one QSE per month and dividing the QSE, Process Control, into 2 months, ten QSEs were completed in 11 months to cover pertinent points of the 27th edition of AABB Standards. Participation was 100%. Compliance was greater than 99% and documented competency by the team was 100%. By December 2012, the format was used to add a set of flashcards to study "What's New in the 28th Edition." There are several comparable free flashcard tools available online, other than the one used in this case, that transfusion services or educational programs could use for similar learning and review. The program was successful enough that the Transfusion Service continues its use in 2013 with modification. Team members still study using the program and can self-test. This year, an added step is to take a closed book competency test by the end of the month that is pre-generated.

Disclosure of Commercial Conflict of Interest

M. Friend: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Friend: Nothing to disclose

AP32

Management/Development of Collection Staff Competency Assessments

A L Dirr[1] (Angela_Dirr@bonfils.org), A Burtchaell[1], D A Scherer[1].
[1]Community Donor Operations, Bonfils Blood Center, Denver, CO, United States

Background/Case Studies: The scope of the project was to design a competency assessment program for collection staff based on the underlying operating procedures in which they perform. Objectives included: providing an audit trail for aligning assessment with operating procedures; standardized assessment measurements; and simplicity of execution by end users. Additional key elements included time constraints for development, document management of records, and the ability replicate the process on a yearly basis with minimal modifications. **Study Design/Methods:** A total of 51 procedures were performed by a variety of collection staff. Grouping operating procedures by process and performance in field operation resulted in eight assessments being developed. A master spreadsheet outlining the competency assessment and correlating procedures was developed. The process or task was the driver for competency, not each individual operating procedure. This minimized documentation and confusion for supervisors in the execution phase. An Excel spreadsheet was developed to correlate staff with assessments. This served a dual role: to identify to supervisor's respective competencies needed for each staff member; and to track completion on the back end. With the basic framework identified, a project plan was developed for implementation. Standardized assessment templates were created to ensure uniformity, and assessment criteria was established. Assessment criteria included record review, direct observation, written test, or a combination of both based on the complexity of the process. Authors of the assessments completed competency on respective supervisors and training staff first. Supervisors and trainers were then deployed to complete assessments on remaining staff. Audits were conducted to ensure completion by deadline. Records were reviewed for accuracy and transferred to an offsite vendor for document management. **Results/Findings:** Implementation of the plan was completed by October 31, 2012. Developing the competency based on task was effective and narrowed the scope of document management from 51 procedures to 8 competencies. Subsequent assessment management time will be decreased each year since the framework has been established. Operating procedures which have been made obsolete or which address new sub processes are added to the existing competencies on a yearly basis. **Conclusion:** Tracking the completion of competency assessments for staff by task as the driver instead of operating procedures streamlined the overall management. The ability to demonstrate which procedures corresponded to which assessments was vital in terms of maintaining compliance. It has also been discovered that tailoring operating procedures based on how the process is performed in the live environment, how they are trained, and how they merge with competency assessment is vital when developing operating procedures.

Disclosure of Commercial Conflict of Interest

A. Burtchaell: Nothing to disclose; A. L. Dirr: Nothing to disclose; D. A. Scherer: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Burtchaell: Nothing to disclose; A. L. Dirr: Nothing to disclose; D. A. Scherer: Nothing to disclose

AP33

How Does a Transfusion Simulation Affect the Performance of Medical Students on Standard Assessments?

C Beavers[1] (charles.beavers@louisville.edu), J Wooldridge-Ware[1], M Ayyoubi[1]. *[1]Pathology and Laboratory Medicine, University of Louisville, Louisville, KY, United States*

Background/Case Studies: Medical simulation is becoming a widely used form of education, beyond what is experienced in didactic lectures. Transfusion medicine is often neglected in favor of other subjects, but becomes a substantial issue later in physicians' careers as blood product misuse is common. A remedy for this issue is suggested here by using a simulation case in conjunction with a didactic lecture. **Study Design/Methods:** A retrospective study was performed to gauge the effectiveness of a transfusion medicine based simulation and didactics in teaching basic principles of blood banking to medical students. A simulation was introduced into the sophomore course in addition to the usual lectures. This simulation was structured around an acute hemolytic transfusion reaction with complications and requires participants to assess vitals and laboratory values on an electronically controlled mannequin in order to make treatment decisions and perform interventions. Six years of exam results from 8009 medical students were gathered from transfusion based questions before the addition of the simulation, and three years after the simulation was instituted. Exam questions are changed from year to year, but reflected similar content and difficulty. An average of eight questions were asked per year. The average point-biserial correlation was used to measure the relationship between students' performance on the question with their overall test performance. These values range from −1.0 to +1.0. The more positive the correlational value, the more closely associated item performance is to overall test performance, suggesting higher validity for the question. **Results/Findings:** The average score of classes on blood banking questions from 2005-2009 from 4969 medical students was 65% and the average score for 3040 medical students from classes 2010-2012 was 80% (Table). The average point-biserial correlation of the 2010-2012 group was 0.224, and 0.201 for the 2005-2009 group, which suggests a strong correlation with student performance on transfusion questions compared to the overall exam. Overall, students scored more poorly on questions pertaining to transfusion transmitted diseases and product modifications. However, overall students did perform better on questions pertaining to ABO typing and RhIG administration. The average grade of students who answered questions correctly was 77% for 2005-2009 and 79% from 2010-2012. **Conclusion:** These data show that based upon similar class and question characteristics, the addition of a transfusion simulation aided in the learning of medical students on a standardized assessment. Although many variables are in place in the educational setting, a substantial difference of performance was noted of students who participated in the transfusion simulation.

Disclosure of Commercial Conflict of Interest

M. Ayyoubi: Nothing to disclose; C. Beavers: Nothing to disclose; J. Wooldridge-Ware: No Answer

Disclosure of Grants Conflict of Interest

M. Ayyoubi: Nothing to disclose; C. Beavers: Nothing to disclose; J. Wooldridge-Ware: No Answer

Class Averages on Transfusion Based Questions

2012	2011	2010	2009	2008	2007	2006	2005
70%	83%	85%	65%	62%	65%	72%	66%

AP34

A Successful Support Program for Cell Therapy: An Update from the Production Assistance for Cellular Therapies (PACT) Group

R Lindblad[1] (rlindblad@emmes.com), K Quinnan[1], D Wood[1]. *[1]The EMMES Corporation, Rockville, MD, United States*

Background/Case Studies: Cell therapy manufacturing continues to expand both in scope and complexity. Requirements to operate a cell manufacturing facility and produce a high quality current good manufacturing practice (cGMP) grade product are daunting. Those that are contemplating starting a new cell manufacturing facility or updating their current procedures often have little support to establish basic procedures including standard operating procedures (SOPs). The Production Assistance for Cellular Therapies (PACT) program provides SOPs to the cell therapy community via requests through the PACT website. This will describe how this program was built, maintained, and made successful. **Study Design/Methods:** PACT initially acquired SOPs from the manufacturing facilities. This was a small subset of all SOPs and included general facility processes. As the program has grown, additional SOPs have been added which include quality assurance, quality control, and regulatory affairs. The SOPs are watermarked as for informational purposes only and intended to be a reference for a facility to develop their own SOPs, or to compare their current SOPs to those of the PACT facilities. These SOPs are not intended to be used by a facility in their posted format. To maintained facility confidentiality the SOPs are de-identified. A disclaimer statement for each SOP is provided that states specific validation of the SOP is required per FDA regulations (21 CFR 1271 Subpart D—Current Good Tissue Practice, Section 1271.180 Procedures). Updated SOPs are provided by the PACT facilities and distribution is maintained by the PACT Coordinating Center. PACT maintains control of the SOP distribution process by requiring a requester to send an email request to pactinfo@pactgroup.net. In this way PACT could track both the demographics and the volume of requests received and fulfilled. The program was launched in 2007. The number of SOP categories available has increased from five initially to nine currently. A total of 30 SOPs are available across the nine categories. **Results/Findings:** Over 225 individual requests for SOPs have been received by PACT through April 2013. Overall, the most frequently requested SOP is environmental monitoring, followed in order of

frequency by facility cleaning, personnel training, and quality management. The country of origin is most frequently the US followed by Canada, Europe, the UK, and the Middle East. PACT has received consistent feedback as to the benefit of this program. **Conclusion:** PACT serves as a significant resource to the cell therapy community by providing template SOPs from its high quality manufacturing facilities. These serve as a reference to the cell therapy community with an ultimate goal to improve the operations and quality of all cell manufacturing. This project was supported by NHLBI contract #s HHSN268201000006C, HHSN268201000007C, HHSN268201000008C, HHSN268201000009C, HHSN268201000010C, and HHSN268201000011C.

Disclosure of Commercial Conflict of Interest

R. Lindblad: Nothing to disclose; K. Quinnan: Nothing to disclose; D. Wood: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Lindblad: Nothing to disclose; K. Quinnan: Nothing to disclose; D. Wood: Nothing to disclose

AP35

Implementation of a Process-Based Approach to Staff Training

P J Cappelli[1] (patricia.cappelli@bloodntissue.org), C Castaño[2], J Lowther[2]. *[1,2]Quality Assurance, South Texas Blood & Tissue Center, San Antonio, TX, United States*

Background/Case Studies: This center's training checklists were based on the Job Description. Since job descriptions include all tasks an employee can potentially perform, the training checklists were quite lengthy and often took months to complete, especially for highly complex positions. To meet production needs, new employees were often trained to perform only some of the tasks, and the training checklists were incomplete. The time to complete the training checklist for a Laboratory Technician took up to 6 months. While staff were competent in the tasks they were performing, unfinished sections were difficult to explain to outside assessors. Competency assessment forms were process based, but not detailed, requiring a free text explanation to describe any inadequacies and subsequent re-training. Documentation was inconsistent and made tracking and trending inadequacies quite challenging. Training and competency forms were controlled by the Education and Training Department, making the process of updating the forms when the SOP was revised inconsistent. Production increased rapidly and new staff needed to be trained and competent in one task in order to meet customer needs. **Study Design/Methods:** 300 job-based training checklists and 295 competency forms were reviewed for compliance with regulatory requirements and consistency with SOPs, ease of use, and timeliness of completion. New process based forms were designed to include SOP reading related to the specific procedure, skills and significant steps, and trainee acknowledgement regarding their ability to perform the activity. **Results/Findings:** A process based training checklists model has been created for each technical process, ie "Automated Immunoassay Analyzer." Responsibility for the management of forms was transferred to the Quality Assurance Department's document control process, and the forms are now reviewed concurrently with new and revised SOPs. Each significant step in the process is detailed in the training and competency forms. The training checklist can now be completed in 1 week and competency in 1 day. **Conclusion:** Process based training and competency forms facilitate employees' rapid transition into the workflow and quick response to increases in production. They help in assessing the potential of the a new employee more quickly, keep track of each employee's completed training, identify cross-training opportunities, and demonstrate training and competency for each technical process to outside assessors. The detailed forms ensure that all significant steps in the process are trained and assessed for regulatory compliance, allowing for easier identification of steps that may be problematic. Trends are identified and lead to improvements to the training program and the SOPs.

Disclosure of Commercial Conflict of Interest

P. J. Cappelli: Nothing to disclose; C. Castaño: Nothing to disclose; J. Lowther: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. J. Cappelli: Nothing to disclose; C. Castaño: Nothing to disclose; J. Lowther: Nothing to disclose

AP36

The Use of Cellular and Tablet Devices as a Teaching and Information Resource Tool

L M Tormos[1] (tormos@musc.edu), S Moncada[1], M E Clayton[1], J E Squires[1]. *[1]Pathology and Laboratory Medicine, Medical University of South Carolina, Charleston, SC, United States*

Background/Case Studies: Disseminating rapidly expanding medical and scientific information to large numbers of incoming student physicians and allied health professionals represents a unique challenge in academic medical institutions. Additionally, Transfusion Services provide these physicians with criteria (eg transfusion triggers) and transfusion ordering procedures that may differ from the hospitals in which they were originally trained; while for some, it may be their first exposure to blood product ordering. It was felt that the use of cellular ("smart") phone and tablet devices could supply timely and consistent Transfusion Service information in a convenient and easily adaptable method to incoming physicians as well as more senior medical, nursing, and perfusion staff. **Study Design/Methods:** Initially, a printed reference card was provided to all incoming physicians at orientation, but it was rarely used and often lost. In addition, controlling use of current editions became difficult. To avoid these problems, the information was transferred to an "app," a term loosely used to refer to a downloadable, non-modifiable interactive document. Included is critical information on all available blood products, transfusion indications, transfusion risks, reaction symptoms, and reaction reporting procedures. A QR code (matrix or two-dimensional bar code) was included to direct the smart phone or tablet browser to the appropriate website. **Results/Findings:** New physicians are targeted with the inclusion of the web address and QR code in the New Physician Orientation materials. Work area signage including the QR code and intranet websites has also been posted in targeted clinical areas. The Graduate Medical Education Office is the host of the main control document with redirected links also available through the pathology department website and the "Medical Toolbox" section of the university intranet. The benefits expected include: convenient access to transfusion service information, the ability to revise guidelines and policies, the elimination of outdated information in the hands of physicians, reduced printing costs, and potentially increased use of easily available information on blood product use and transfusion reaction identification. **Conclusion:** Rapid access to essential transfusion service information is critical to physicians and nursing staff. Ensuring this information reaches hundreds of new physicians-in-training increases the complexity of that task. With expanded use of smart phones and tablet devices in medical care, it seemed appropriate to utilize these devices to disseminate critical transfusion service information. Using this technology has expanded the availability of this information beyond new physicians to attending physicians, nurses and perfusionists, as well as residents and fellows.

Disclosure of Commercial Conflict of Interest

M. E. Clayton: Nothing to disclose; S. Moncada: Nothing to disclose; J. E. Squires: Nothing to disclose; L. M. Tormos: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. E. Clayton: Nothing to disclose; S. Moncada: Nothing to disclose; J. E. Squires: Nothing to disclose; L. M. Tormos: Nothing to disclose

AP37

Auditing Employee Records for Conversion to Electronic Data

B L Rose[1] (brose@mcvh-vcu.edu), A Hord[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: Records of competency and training for 34 employees at the Transfusion Medicine Service here proved cumbersome during assessments and annual evaluations. A comprehensive review of any employee record required multiple files from a mixture of paper and electronic sources. An Excel spreadsheet tracked annual requirements but failed to capture training for new tests or equipment. No master list of activities existed. The quality management team developed an action plan to gain better control of the process. Immediate goals included document reorganization, archiving papers of personnel no longer employed, and audit of current employee records. The long term goal is conversion of all records to electronic form and a simple mechanism for subsequent reviews. **Study Design/Methods:** The medical technologist trainer (MTT) and staff certified quality auditor (CQA) dismantled existing training and competency paper files. Records of those no longer employed joined their personnel file in the manager's office. While sorting documents, the CQA developed a master list of activities from 2007-2012. From this list, electronic audit forms were

created for Initial Training, On-going Training, and Annual Competency/ Education. Each audit form contained tables of expected tasks versus years. A key permitted notation for documents as present/complete, missing/ incomplete, or not required in table boxes. Missing or incomplete documentation was also noted on a Memo to Record coversheet for each employee file. A list of canned comments provided uniformity for notations. **Results/ Findings:** Employee records from 2007-2012 were audited along with Initial Training documents. Only four employees had fully complete records. 20 employees had incomplete initial training documents. In addition, 13 employees had no initial training records due to hire dates before 1998. Audit found 24 employees missing one or more on-going training and annual competency/education documents. The MTT addressed lapses in documentation with individual employees and noted actions on the Memo to Record. The manager could then review the Memo to Records and sign a Statement of Competency included on the form. **Conclusion:** Reorganization and auditing proved necessary for process control of employee records and conversion to a paperless system. MTT designed a comprehensive Excel spreadsheet to track on-going training, education, and competency tasks. Once sorted and reviewed, historic paper documents could be scanned and archived electronically. Subsequent training and competency documents could be scanned into each employee's electronic file and tracked on the new Excel spreadsheet. Annual review of employee record became electronically processed along with annual evaluations.

Disclosure of Commercial Conflict of Interest

A. Hord: Nothing to disclose; B. L. Rose: Nothing to disclose; K. Sanford: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Hord: Nothing to disclose; B. L. Rose: Nothing to disclose; K. Sanford: Nothing to disclose

AP38

Educating Residents as to the Relationship between Critical Results and Transfusion Practice

H Hong[1] (hong.hong@uhhospitals.org), K A Downes[1]. *[1]Pathology, University Hospitals Case Medical Center, Cleveland, OH, United States*

Background/Case Studies: The recent *AABB Standard 8.2* requires laboratories to monitor critical laboratory results pre and post transfusion. Integration and monitoring of such critical results may require coordination between the section of the laboratory that generates critical results and the transfusion service. Resident education in this aspect of transfusion practice may not be well defined. A resident education tool was designed for critical hemoglobin (Hb) and hematocrit (Hct) results and the response to red blood cell (RBC) transfusion using an existing interlaboratory communication system. **Study Design/Methods:** An interlaboratory system was developed here for the Hematology/Core laboratory to call the blood bank for critical Hb/Hct results in addition to notifying the clinical service as to the result. The blood bank staff assesses critical value calls received for the presence of blood component orders and type and screen test status which include: 1) Identification of a current type and screen sample; 2).Alloantibody history; and 3) If there is an antibody work-up under way and associated findings. A transfusion medicine resident (TMR) physician may be notified if: 1) no type and screen test is ordered; 2) no RBC components are ordered; and 3) if the patient has a known alloantibody history. The TMR physician then reviews the patient information and contacts the clinical team as needed. These critical Hb/Hct calls to blood bank were identified and monitored. These interlaboratory communication system data were used to develop a TMR teaching tool for critical Hb/Hct data and RBC transfusion increments. Resident teaching tool data elements included pre-transfusion (Pre-Txn) Hb/Hct values, transfusion date and time, number of RBC units transfused (tx'd), post-transfusion Hb/Hct values, predicted Hb/Hct increment, and actual Hb/Hct increment. **Results/Findings:** From 2011-2012 a total of 333 critical Hb/Hct blood bank notifications occurred. Of these, 165 (49.5%) of them were further reported to TMR physicians. The resident teaching tool developed is seen in the Table below. **Conclusion:** An interlaboratory communication system served as a systems based learning model for TMR physicians to interface with clinical services regarding critical Hb/Hct values. From this system a resident teaching tool was developed to assess resident understanding of predicted increments following RBC transfusion and for educating residents in critical result management in the laboratory and transfusion service.

Disclosure of Commercial Conflict of Interest

K. A. Downes: Nothing to disclose; H. Hong: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. A. Downes: Nothing to disclose; H. Hong: Nothing to disclose

Transfusion Medicine Resident Teaching Tool Critical Values

Case #	Date	Resident Name	Patient Name	Medical Record Number	Pre-Txn Hb/Hct Date/Time	Pre-Txn Hb/Hct g/dL; %	# RBC units tx'd	Post Txn Hb/Hct g/dL; %	Post-Txn Hb/Hct Date/time	Predicted Increment Hb/Hct g/dL; %	Actual Increment Hb/Hct g/dL; %
1											
2											

AP39

An Objective Method for Evaluating Reading and Grading Competency in the Immunohematology Reference Library (IRL)

J K Slayten[1] (jslayten@indianablood.org), K Evans[1], C A Kelly[1], S D Whelchel[1]. *[1]Immunohematology Reference Lab, Indiana Blood Center, Indianapolis, IN, United States*

Background/Case Studies: An important skill of blood bank testing is the ability to read and grade hemagglutination. Reading and grading is especially critical for Immunohematology Reference Laboratory (IRL) specialists. For this reason, the initial training and on-going competency of IRL specialists often includes the ability to detect not only serologic (microscopic, weak and 1–4+) reactivity but also mixed-field red cell agglutination (in direct antiglobulin test [DAT], autocontrol and antigen testing). Due to the subjective nature of hemagglutination reading and grading, it is a challenge to determine when the IRL specialist is considered competent. This lab created an objective approach to address this challenge. **Study Design/Methods:** A retrospective review was completed of serologic reading and grading exercises completed in the IRL from 2010-2012. Each reading and grading exercise was evaluated by the number of tubes tested (total) and the number of tubes which the staff members tested correctly (matching the key). The total number of tubes tested annually was compared to the total correct tube results. This data was then used to estimate the expected accuracy of the reading and grading exercises, verifying the ability of the IRL specialists to detect weak hemagglutination. The threshold or expectation for the reading and grading for the initial training and on-going competency for the IRL specialists was then established. **Results/Findings:** The Table summarizes the data from 2010-2012. The data demonstrated an average of 95% accuracy with a range from (92%-97%). The range of acceptable reading and grading results was determined to be 90%-100%. **Conclusion:** Prior to the definition of the threshold for reading and grading, IRL trainers or management would subjectively evaluate reading and grading competency of an IRL trainee or specialist. With the threshold set at 95% (range of 90%-100%), the IRL trainers and management now have an objective measure to demonstrate the accuracy of the IRL specialist's reading and grading skills.

Disclosure of Commercial Conflict of Interest

K. Evans: Nothing to disclose; C. A. Kelly: Nothing to disclose; J. K. Slayten: Nothing to disclose; S. D. Whelchel: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Evans: Nothing to disclose; C. A. Kelly: Nothing to disclose; J. K. Slayten: Nothing to disclose; S. D. Whelchel: Nothing to disclose

TABLE 1: 2010-2012 Reading and Grading Exercise Comparison

Year	Total	Total Correct	% Matching Key
2010	125	115	92
2011	275	283	97
2012	445	420	94

Overall Percentage = 95%

AP40

A Successful Web Seminar Program for Cell Therapy: An Update from the Production Assistance for Cellular Therapies (PACT) Group

R Lindblad[1] (rlindblad@emmes.com), D Styers[1], D Wood[1]. *[1]The EMMES Corporation, Rockville, MD, United States*

Background/Case Studies: Cell therapy manufacturing continues to expand both in scope and complexity. Requirements to operate a cell manufacturing facility and produce a high quality current good manufacturing practice (cGMP) grade product are daunting. The PACT program provides education outreach to the cell therapy community through the use of web seminars over the past 10 years. This describes how a PACT web seminar program was built, maintained, and made successful. **Study Design/Methods:** PACT initially acquired topics from the cell therapy community via customer surveys, cell therapy organizations, and at scientific meetings to develop its web seminar program. Several vendors were assessed to select the optimal service that would be required which included operator support, capacity, technical support, customer service, event recording, and software platform. Availability of rehearsal and preparation time in the service environment was an important factor. Presenters from outside the PACT program have been engaged to participate in these web seminars to encourage a broader viewpoint and leverage expertise and involvement of the cell therapy community. Metrics have been maintained to evaluate the topics presented, attendee number and demographics, and recommendations of future subject areas to be covered. **Results/Findings:** Over 20 web seminars have been conducted since PACT's inception in 2003. Topics have included facility management, equipment and process validation, cell product cryopreservation, regulatory requirements, adverse event reporting, standards, data management systems, cell therapy tourism, and many others. Vendor evaluation has been critical to make the program successful, including changing vendors when needed. Continuing education credit is provided to requesting attendees through a partnership with AABB. Involvement of non-PACT presenters creates challenges including available time and scheduling of both the event and rehearsal times in addition to obtaining presentations prior to the web seminar. Attendance initially averaged 75 over the first five web seminars and currently averages 240 over the last five web seminars. The majority of attendees are laboratory personnel but may also include laboratory directors, regulatory managers, and quality assurance directors. **Conclusion:** PACT serves as a significant educational resource to the cell therapy community by implementing web seminars to foster the advancement of cellular therapy. PACT has developed a structured well-defined program targeted to the cell therapy community. The PACT program is now in its tenth year of operation and continues to perform an ongoing assessment of current web seminar objectivesas we;; as evaluating customer feedback to anticipate the future needs of the cell therapy community to deliver a valuable program. This project was supported by NHLBI contract #s HHSN268201000006C, HHSN268201000007C, HHSN268201000008C, HHSN268201000009C, HHSN268201000010C, and HHSN268201000011C.

Disclosure of Commercial Conflict of Interest

R. Lindblad: Nothing to disclose; D. Styers: Nothing to disclose; D. Wood: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Lindblad: Nothing to disclose; D. Styers: Nothing to disclose; D. Wood: Nothing to disclose

AP41

Use of an Online Testing Service for Competency Testing

A Hord[1] (ahord@mcvh-vcu.edu), S M Armstrong[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: The Transfusion Medicine Department at this 780-bed health system performs routine transfusion service testing via manual and automated methods and performs the majority of our reference work in house. To meet the Clinical Laboratory Improvement Amendment (CLIA) requirements for judging the competency of medical technologists on routine and non-routine testing and processes, written tests must be developed and given to staff. In the past, hard copy written tests were administered to staff which were cumbersome to develop and grade. An easy and effective way was needed to administer these tests to all of our technologists across many shifts. **Study Design/Methods:** Last year a free online testing management system was implemented called Classmarker.com, to build and grade staff competency tests. The website provided template tools to create questions using different formats such as: multiple choice/response, true/false, free text, and essay. The website allows the user to administer 100 tests per month for free. This free version was used for our annual written competency test in 2012 to assess if the full service was cost effective to purchase. **Results/Findings:** The website allows administrators to assign each staff member a username and password and sends this information to users via email. This process maintains privacy when assigning and grading competency tests and notifies staff via email when a test is assigned to them. Creating test questions on the site is simple and archives test questions into categorized test banks for later use. As the administrator creates a test question a correct answer is entered, so as the employee completes the test it is graded and tests results are given to staff immediately for review. For additional costs, the site offers additional capabilities for test question creation and grading statistics. This version also loads images into the test questions, thereby creating more robust competency questions. The administrator can also review, grade, and question statistics, showing correct percentages for each test question, and also analyze test performance by cohorts (part time workers, full time workers). **Conclusion:** The staff feedback was positive. Creating, as well as grading, test questions was a great improvement for the test administrator. Staff appreciated the immediate feedback on competency tests rather than waiting for tests to be graded. Management uses the competency test results and statistics for process improvement opportunities. Based on the positive feedback, time saving aspects, and ease of use of the site, our facility decided to purchase a test pack from the site. The use of the online test creation service has allowed for better competency tests, less time used grading tests, and ways to identify opportunities for performance improvement.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; A. Hord: Nothing to disclose; K. Sanford: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; A. Hord: Nothing to disclose; K. Sanford: Nothing to disclose

AP42

Practical Guide for Safe Hospital Blood Transfusion

Y Fujii[1] (yfujii-ygc@umin.ac.jp), S Shimodaira[2], T Tasaki[3], T Asai[4], S Inaba[5]. *[1]Blood Transfusion, Yamaguchi University, Ube, Japan; [2]Transfusion Medicine, Cell Processing Center, Shinshu University Hospital, Matumoto, Japan; [3]Transfusion Medicine, Tokyo Jikei University Hospital, Tokyo, Japan; [4]Chiba Red Cross Blood Center, Funabashi, Japan; [5]Kanagawa Red Cross Blood Center, Atugi, Japan*

Background/Case Studies: The Japanese Red Cross Society (JRCS) has been collecting information on adverse reactions and infections due to blood and blood components since 1993. The most important task in hemovigilance by the JRCS is to improve the system for causal analysis. However, incidents are rarely reported voluntarily to the JRCS. The annual comprehensive surveys of blood transfusion by the Japan Society of Transfusion Medicine and Cell Therapy (JSTMCT) revealed that "identification error between patient and blood product" was the main cause of ABO-incompatible blood transfusion, but "phlebotomy error" also had serious consequences. In Japan, transfusions are conducted even in small hospitals. Knowledge of transfusion error is necessary for physicians, nurses, and medical laboratory technologists, whose activities include transfusion therapy. There is insufficient education for many trainees due to a lack of resources. A practical guide to prevent transfusion error in hospitals was created by the Study Group on Transfusion Reaction Monitoring Systems in Medical Institutions. **Study Design/Methods:** This practical guide was prepared according to the results of the annual comprehensive surveys of blood transfusion by JSTMCT conducted from January 1995-December 2009. The target readers are general clinicians, nurses, and medical laboratory technologists, but knowledge necessary for the chairperson of the blood transfusion therapy committee, manager of the blood transfusion department, and risk manager is also covered. Many figures and tables are used to aid understanding of the text. This guide was written in both English and Japanese. Blood transfusion therapy in Japan was compared to that in other countries to deepen understanding of it. Typical transfusion errors are illustrated graphically. Preparation of this guide was supported by a Health and Labour Science Research Grant for Research on Regulatory Science of Pharmaceuticals and Medical Devices. **Results/Findings:** This practical guide as a pdf file (51 pages) is now available at the website of the JSTMCT (http://www.jstmct.or.jp/jstmct/Document/Guideline/Ref22.pdf). The main contents were procedures of transfusion practice and transfusion-associated errors. The pathophysiology of acute hemolytic transfusion reactions resulting from ABO-incompatible and non-ABO blood groups incompatible blood transfusion was also described. Information technology for transfusion

practice and red blood cell testing for emergency transfusion were discussed. **Conclusion:** Educating hospital staff using this practical guide will promote a higher level of transfusion operations, which will subsequently lead to improved patient outcomes.

Disclosure of Commercial Conflict of Interest

T. Asai: Nothing to disclose; Y. Fujii: Nothing to disclose; S. Inaba: Nothing to disclose; S. Shimodaira: No Answer; T. Tasaki: No Answer

Disclosure of Grants Conflict of Interest

T. Asai: Nothing to disclose; Y. Fujii: Nothing to disclose; S. Inaba: Nothing to disclose; S. Shimodaira: No Answer; T. Tasaki: No Answer

AP43

Clinical Research Roadmap: A Resource for Cellular Therapy

J M Centanni[1] (jmcentanni@medicine.wisc.edu), R Wesselschmidt[2], S Ouseph[3], M Armant[4], R Draxler[5]. *[1]University of Wisconsin, Madison, WI, United States; [2]City of Hope, Duarte, CA, United States; [3]Baylor College of Medicine, Houston, TX, United States; [4]Center for Human Cell Therapy, Boston, MA, United States; [5]University of Minnesota, Minneapolis, MN, United States*

Background/Case Studies: The NHLBI Production Assistance for Cellular Therapies (PACT) Program's Cellular Therapy Clinical Research Roadmap was developed by PACT's cell processing facilities. This roadmap was developed as a resource for researchers new to the cellular therapy research field. It is intended to assist researchers in identifying critical areas that need to be considered when developing a cellular therapy intended for evaluation in human clinical studies under an Investigational New Drug (IND) application. **Study Design/Methods:** The Clinical Research Roadmap focuses on four key areas that include translational research, resource development, preclinical studies, and IND filing. The translational research phase bridges the gap between basic research and product development of a therapeutic candidate. A good understanding and management of a wide-range of resources is critical for product development to be cost effective yet able to meet projected milestones. Resource development includes the identification of a specialized study team and the ability to identify adequate funding. The study team is comprised of experts representing diverse, but complimentary, areas of expertise, who coordinate the planning of studies aimed at bringing the therapeutic candidate to the clinic. Preclinical studies are conducted to determine the value of the therapeutic candidate in treating a disease and are vitally important in evaluating safety and effectiveness. Development and implementation of a regulatory plan early in the cellular therapy development process is critical to the efficient use of resources and ultimately the success of getting a new cell therapy to the clinic. Filing an IND application with the US FDA is required to use these types of investigational products in human clinical studies. **Results/Findings:** During the translational research phase manufacturing needs must be critically evaluated to ensure sufficient data is obtained to warrant further development for clinical use. Technology transfer and product development experts are essential in transferring product manufacture from small-scale to large-scale methods required for clinical production. The NHLBI-funded PACT program provides support for -translational services and cell therapy product manufacturing. **Conclusion:** Timely evaluation and consideration of key resources are vital to successful translation of a cellular therapy product from bench top to bedside. This Clinical Research Roadmap developed by experienced investigators offers new investigators a strategic approach to product development with the benefit of rapid technology translation, a prompt regulatory approval process, and expedited transition into human clinical studies. This project was supported by NHLBI contract #s HHSN268201000006C, HHSN268201000007C, HHSN268201000008C, HHSN268201000009C, HHSN268201000010C, and HHSN268201000011C.

Disclosure of Commercial Conflict of Interest

M. Armant: Nothing to disclose; J. M. Centanni: Nothing to disclose; R. Draxler: Nothing to disclose; S. Ouseph: Nothing to disclose; R. Wesselschmidt: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Armant: Nothing to disclose; J. M. Centanni: Nothing to disclose; R. Draxler: Nothing to disclose; S. Ouseph: Nothing to disclose; R. Wesselschmidt: Nothing to disclose

AP44

A Multi-Faceted Approach to Transfusion Education for Nurses

M M Starkey[1] (mstarkey@mvrbc.org), J L Jackson[1]. *[1]Mississippi Valley Regional Blood Center, Davenport, IA, United States*

Background/Case Studies: Transfusion is the most common procedure performed in US hospitals, occurring in 11% of stays. Despite its frequency and inherent risk, education related to transfusion for nurses is lacking both in formal training programs and in clinical practice settings. This knowledge gap may result in patient harm, as well as regulatory and quality deficiencies. Barriers to obtaining this knowledge include limited time and lack of expert resources to develop programs, and scheduling conflicts with learners preventing attendance. To promote transfusion knowledge, this blood center offers continuing education to healthcare professionals at its hospitals. The programs are planned by a team consisting of a nurse, a specialist in blood banking, and a physician utilizing published best practice guidelines and highlighting essential patient safety information most useful to those transfusing blood and blood products. Continuing education credits are available at no charge to the attendee, following the program. A formal program evaluation is completed, providing feedback. In 2010, the list of course offerings at this blood center was expanded to include programs specifically designed to reach nurses in several learning formats. These nurse-focused programs were first offered to partner hospitals, and then were extended to local colleges of nursing. Presentation formats include: traditional classroom instruction taught by the blood center's Transfusion Safety Officer (nurse) or Client Support Coordinator (Specialist Blood Bank), computer-based learning (CPL), or designed as a competency evaluation for nurses. This study will evaluate the effect of changes to blood center education programs for nurses on attendance. **Study Design/Methods:** Following each presentation, information regarding the program is recorded in a database which contains: 1) program date; 2) attendee/hospital; 3) program title; 4) learning method; 5) number of attendees; and 6) evaluation ratings and comments. **Results/Findings:** Over two years, the choice of education programs grew from three to ten, and most were made available in both classroom and CPL formats. The number of programs given during this time doubled from 40 in 2010 to 81 in 2012. Similarly, the number of attendees increased to 1466 by 2012. CPL programs accounted for approximately 17% of the total number by 2012, despite persistent technical difficulties preventing full implementation. Evaluation feedback was consistently positive and provided valuable information for continuous improvement. **Conclusion:** Reaching today's busy nurses with transfusion knowledge is critical to patient safety but remains a challenging undertaking. Nurse attendance at education programs may be increased through thoughtful design of education programs on a variety of applicable topics in attractive, flexible formats to maximize interest, applicability to practice, and convenience.

Disclosure of Commercial Conflict of Interest

J. L. Jackson: Nothing to disclose; M. M. Starkey: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. L. Jackson: Nothing to disclose; M. M. Starkey: Nothing to disclose

AP45

Development of Resident Self-Tracking Tools for Measurement of Milestones in Transfusion Medicine Quality Assurance Education

H Hong[1] (hong.hong@uhhospitals.org), L Zhou[1], R W Maitta[1], K A Downes[1]. *[1]Pathology, University Hospitals Case Medical Center, Cleveland, OH, United States*

Background/Case Studies: Residency programs are essential in the maturation from medical student to independent physician along the lifelong continuum of medical education. In the past few years quality assurance (QA) has become a fundamental element of medical education for residency training programs. This QA renaissance has made it clear that some pathology residency programs may not have reorganized their curriculum in transfusion medicine (TM) to adapt to the new focus on QA didactics. This pathology resident training in TM-QA was evaluated in order to develop a tracking tool for individual residents to use during their rotation to measure individual QA educational milestones. **Study Design/Methods:** Current educational activities for pathology residents at this institution were reviewed over a 24-month period against quality plan elements, resident curriculum, and TM rotation resident activities to identify resident TM-QA activities. Accreditation requirements (AABB Standards/CAP requirements) were reviewed against resident participation in the TM service activities to identify areas of resident training and education in TM-QA. The objectives were to review current TM-QA activities, identify resident duties performed in these activities, and to define resident TM-QA exposure during training.

Educational instruction in TM-QA was divided into didactic and non-didactic. A mechanism was developed for residents to track TM-QA activities in order to document and measure reaching QA educational milestones. **Results/Findings:** Resident transfusion medicine didactic education included 60 hours of TM lectures out of 147 lecture hours in Clinical Pathology. Lectures topics with QA elements (50% of total hours) included Informed Consent, ASFA Indication Categories, Blood Utilization, Testing, Cost Containment, Adverse Reactions, Blood Administration, and ABO Incompatible hematopoietic progenitor cell (HPC). Non-didactic resident TM-QA experiential education included participation in QA Committees, Blood Administration Audits, Component calls, "Mock" laboratory inspections, TM/Apheresis/HPC Consultations, Inventory management, Monitoring critical values, and transfusion therapy. Nine forms were created for resident self-tracking of participation in TM-QA activities and progress toward meeting milestones in TM-QA. **Conclusion:** Pathology residents' TM education at this institution includes didactic and non-didactic learning opportunities in TM-QA with 50% of lecture hours having QA elements. TM-QA curricular activities permitted the development of an individualized resident self-tracking tool that captures participation in TM-QA activities as the resident moves toward milestones. Examining TM-QA resident training activities resulted in a process improvement for resident documentation and an opportunity for resident engagement in self-directed learning.

Disclosure of Commercial Conflict of Interest

K. A. Downes: Nothing to disclose; H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; L. Zhou: No Answer

Disclosure of Grants Conflict of Interest

K. A. Downes: Nothing to disclose; H. Hong: Nothing to disclose; R. W. Maitta: Nothing to disclose; L. Zhou: No Answer

AP46

Bridging the Gap: Providing a Web-Based Forum for Education and Communication in a Challenging Workplace Environment

K D Stasko[1] (karl_stasko@dfci.harvard.edu), H M Hansen[1]. *[1]Cell Manipulation Core Facility, Dana Farber Cancer Institute, Boston, MA, United States*

Background/Case Studies: This facility (CMCF) is approved to provide clinical stem cell transplant services to all Harvard affiliated hospitals. As the second largest stem cell processing center in the world, the CMCF processes over 1000 products per year. Providing in-service education to clinical team members is particularly complex given the everyday challenges of communicating technical details, allocating time for busy staff, and providing access to restricted good manufacturing practice (GMP) lab space. In addition to educating the clinical team, it is paramount to disseminate information as well as train internal department members within the CMCF. Projects and initiatives that involve administrative, quality, and clinical decision making are most effective when they are based on a solid understanding of laboratory practices. This in itself is a challenge, given the varied levels of knowledge and experience within the staff. Consequently, the fundamentals of clinical processing are often unfamiliar to CMCF staff members who aren't directly involved with the everyday workings of the lab. **Study Design/Methods:** In order to meet these challenges, a website has been developed which provides in-service training specifically centered on video and pictorial methods of communication. The website features a facility tour, as well as an overview of product types, all major processing steps, and standards for release. Additionally, links to the site are present on CMCF cell processing standard operating procedures. **Results/Findings:** Prior to the implementation of the in-service training website, eight staff members in the stem cell transplant laboratory had current working knowledge of cell processing procedures. Post implementation, all 40 internal departmental staff members, as well as the entire bone marrow transplant program, have access to the most up to date cell processing techniques and standards. The in-service training website slide show strategy allowed non technical staff to gain an understanding of individual lab processes in 5 minutes or less, in contrast to more cumbersome and time consuming standard operating procedures. **Conclusion:** In conclusion, the in-service training website proved efficacious as an efficient method to expand the knowledge base of new technologists, non-technical staff members within the CMCF, and external bone marrow transplant team members. By utilizing this simplified strategy, the gap that existed in the current training and education process has been bridged.

Disclosure of Commercial Conflict of Interest

H. M. Hansen: Nothing to disclose; K. D. Stasko: Nothing to disclose

Disclosure of Grants Conflict of Interest

H. M. Hansen: Nothing to disclose; K. D. Stasko: Nothing to disclose

AP47

Using Instructional Design Principles to Create Training for a Large Transfusion Service's Document Control Conversion

C A Bowler[1] (bowler.carrie@mayo.edu), D A Karger[1], E Bieri Patzke[1], J R Stubbs[1]. *[1]Division of Transfusion Medicine, Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Transfusion Medicine was moving from a paper based document control (DC) system to a web based system. In order to convert to the web based system, implementation of new standardized document templates for procedures, forms, and training modules was necessary. Additionally, a new DC review process was put into practice. Education of the large transfusion service regarding the number and degree of changes was critical to a successful implementation. The education group, along with DC subject matter experts, approached the creation of the ITE with principles learned from an instructional design model, ADDIE, which encompasses five phases: Analysis, Design, Development, Implementation, and Evaluation. **Study Design/Methods:** During **Analysis**, the ITE was clarified and potential barriers to learning were examined, such as space and time as 86 employees would need training. At the **Design** phase objectives, content, and assessment were linked together and a project timeline was created. Storyboarding allowed visualization of the ITE process flow and identified where content could be broken into chunks. At the **Development** phase the ITE material was created with 11 separate training events. Each training event identified the following: training audience, related documents and those that needed modification, trainers, and a specific training event order based on content and job title, noted to be most important. To efficiently utilize time in/out of class assignments were given. **Implementation** phase resulted in three separate ITE sessions with a total training time of 9 hours. The ITE was given on three consecutive days at 2-hour blocks of time; the remainder was performed as self-study or during work unit meetings. **Evaluation** consisted of formative and summative assessment. Trainees completed ITE written evaluations at the close of each training event. Passing criteria was 90%. Four weeks after DC implementation an anonymous survey was given. **Results/Findings:** Summative assessment provided evidence of a successfully articulated ITE. Approximately 95% (37/39) of survey respondents indicated they felt adequately prepared for the DC implementation. Written evaluations had a 97% (83/86) first time pass rate. The survey was broken into the following areas regarding what the respondents appreciated about the ITE: time frame/duration, content breakdown, class size, instructor/trainee ratio, and in/out of class assignments. ITE timeframe (17/39) and content breakdown (15/39) were indicated as the components that most likely aided trainees to be prepared. **Conclusion:** Applying the instructional design principals learned from ADDIE as well as collaboration between a core group of educators and DC subject matter experts resulted in an effective ITE. Not following the principles of ADDIE may have led to an inappropriate training event sequence and misidentification of content breakdown.

Disclosure of Commercial Conflict of Interest

E. Bieri Patzke: Nothing to disclose; C. A. Bowler: Nothing to disclose; D. A. Karger: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

E. Bieri Patzke: Nothing to disclose; C. A. Bowler: Nothing to disclose; D. A. Karger: Nothing to disclose; J. R. Stubbs: Nothing to disclose

Inventory Management

AP48

Evaluation of Validation Methods for Blood Product Transport Utilizing Pneumatic-Tube Systems

C C English[1] (cenglish75@hotmail.com). [1]Blood Bank, Central Arkansas Veterans Healthcare System, Little Rock, AR, United States

Background/Case Studies: Pneumatic tube systems have been widely used in hospitals for transport of blood products for many years in order to reduce the time involved in the blood product delivery process. Although these systems expedite blood product delivery time, the integrity, safety, and

effectiveness of blood products is at risk if there is not a process in place to validate pneumatic tube systems. **Study Design/Methods:** Data was collected by a self-developed survey form that was sent electronically to blood banks potentially using pneumatic tube systems for blood product delivery. Twenty-eight responses were received. The survey evaluated validation protocols to determine methods utilized to ensure the quality of products is unaffected and patient safety is maintained when pneumatic tube systems are used for transport of blood products. **Results/Findings:** Based on the survey, 26 of 28 respondents (93%) reported using pneumatic tube systems for blood product transport, and 24 (86%) performed validation studies prior to transportation of blood products. Two (7%) reported using pneumatic tube systems for blood product transport but did not perform validations on their pneumatic tube systems before using them for transport of blood products. Two (7%) did not use pneumatic tube systems for blood product transport. Data collected from 24 facilities that performed validation studies indicated 23 (96%) had methods to assess temperature, 24 (100%) had methods to assess product receipt by intended location, 24 (100%) assessed transit times, and 21 (88%) assessed visual integrity of the blood products. The study indicated validation studies did not emphasize testing for hemolysis, platelet destruction, factor analysis, or fibrinogen levels. Only 6 of 24 respondents (25%) performed laboratory testing to assess blood product quality and integrity. **Conclusion:** The study confirmed most facilities surveyed did perform validation of pneumatic tube systems used for blood product transport. Two facilities reported not validating. Previous literature emphasized laboratory testing to assess pneumatic tube systems and indicated pneumatic tube transport of blood products is safe and effective, but the potential for adverse effects on blood product integrity exists. Validation at the current time is focused on evaluating maintenance of temperature, product receipt by intended location, transit times, and visual integrity of the blood products. Laboratory testing of blood products may garner some interest as pneumatic tube systems change and become used for blood product transport more frequently. Validation methods used for pneumatic tube systems may be an area in need of further research to ensure blood products meet the safety requirements set forth by the FDA, CAP and AABB.

Disclosure of Commercial Conflict of Interest

C. C. English: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. C. English: Nothing to disclose

AP49

Information Technology in Blood Banks: A Step Toward Haemovigilance, from a Tertiary Care Hospital in Northern India

K Chatterjee[1] (kabita.chatterjee@gmail.com), N Agarwal[1]. *[1]Transfusion Medicine, All India Institute of Medical Sciences, New Delhi, India*

Background/Case Studies: A well-organized blood transfusion service (BTS), with quality systems in all areas, is a pre-requisite for the safe and effective use of blood and blood products. The major concern of blood transfusion service is to provide safest blood possible to the recipients and track any adverse consequences. Information and communication technology can greatly help BTS to improve their working efficiency and optimize the whole workflow. **Study Design/Methods:** Very few blood banks in India use blood bank software and rely mainly on manual method of documentation. Two years ago, this premier teaching and research institute decided to implement blood bank software (Biometrics,) that has a comprehensive approach to all the key aspects in blood banking to help monitor and improve standards in the blood bank. This is a report on that software. **Results/Findings:** Over 45,000 donations per year are performed here and in excess of one hundred thousand units for transfusion are issued. Traceability of all data from donation to transfusion is important and ICT plays a vital role here. The software has been running successfully since its implementation two years ago and is able to manage this volume of data very easily. The department introduced biometric (fingerprint) registration as well as photo capture for the purpose of unique identification of the donor that helps in tracking previous visits by the donor. The software automatically prevents the donor from proceeding ahead in case a donor returns before his/her eligibility date. Earlier, the incidence of paid/professional donors being caught before donations (on the basis of strict screening questions, identifying scars of prior donation at the phlebotomy site, or recognition by the staff from recent donation, etc) was around 2-3% but now it is reduced to 0.2-0.3 percent. A strict screening procedure is the cornerstone of blood safety programs. This is exemplified by the reduction in rate of TTI reactive donations (by nucleic acid amplification test [NAT]) that has come down from around 1.49 % in 2011 to 1.11 % in 2012, after implementation of Biometrics. All bags are identified with a bar-coded unit number that co-relates to samples that are tested for blood grouping, antibody screening, and TTI screening by ELISA and NAT. The software only accepts bags into stock after all the requisite infectious tests are screened as non-reactive and the grouping is confirmed. **Conclusion:** With the help of this software, all units that are issued to patients are traceable right from the purchase of bag, donor details, preparation details, testing history, storage information, and compatibility testing. The software also transfusion details to be keyed in and any adverse reactions suffered by the patients to be recorded. This enables vein-to-vein tracing which forms the basis for ensuring patient and donor safety.

Disclosure of Commercial Conflict of Interest

N. Agarwal: Nothing to disclose; K. Chatterjee: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Agarwal: Nothing to disclose; K. Chatterjee: Nothing to disclose

AP50

Platelet Inventory Management: Two Ways to be Wrong

S Hammel[1] (hammel.scott@mayo.edu). [1]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, Rochester, MN, United States

Background/Case Studies: One of the biggest challenges of blood product inventory management is volatility in platelet inventory. Balancing historic trends with current demands is an unforgiving balance; having too few disrupts patient care, and having too many leads to increased costs in waste. Knowing precisely what will be used today is beyond our ability to see, so we rely on examining the past. Two methods of predicting usage based on historical patterns involve looking at usage by date for the past 7 years and another method focuses on usage by day of the week for the past 4 weeks. **Study Design/Methods:** The 7-year (7 yr) method, employed by a large collection organization, predicts usage by averaging the previous 7 years. The 4-week (4 wk) method, employed by a hospital that collects its own products and purchases from a collection facility, predicts usage based on the prior 4 weeks, by day of the week. These methods were compared for 2012 and 2013 transfusion data. Of particular concern are the days that either method predicted more transfusions than actually occurred leading to an “overstock” situation. When fewer transfusions are predicted than the actual, one ends up short of platelets. The overall range, the biggest overstock minus the biggest shortage, of predictions produces a predictor of accuracy for the method. **Results/Findings:** 2012 transfusions totaled 14,140. The 7 yr method predicted 13,030 while the 4 wk method predicted 14,273. The 7 yr method had149 overstock days compared to 191 days using the 4 wk method, leading the 4 wk method to accumulate a larger overstock of platelets over the year, 901 and 1287, respectively. The range for the 7 yr method was 56.6 platelets versus the 4 wk method’s 52.8. 2013 year to date transfusions total 6224. The 7 yr method predicted 5929 and the 4 wk method predicted 6038. The 7 yr method has 50 overstock days with an average of 7.8 platelets whereas the 4 wk method has 51 overstock days with an average of 5.4 platelets. To date, the 7 yr method has 390 platelets in overstock while the 4 wk method has 276, with a range of 74.6 and 39.8, respectively. **Conclusion:** The first challenge is accumulating seven years of consistent data to establish patterns. At first glance the 4 wk method is closer to predicting actual transfusions. However, vacillations between overstock and shortage lead to instability in platelet inventory. Both facilities are geared toward providing the best care possible for patients, the real challenge lies in having enough for all patients at all times. The 7 yr method has a wider range but it does pick up known low usage days (holidays) better than the 4 wk method. The 4 wk method is always reactionary but is more in tune with current usage patterns. There isn’t one best method for all, but there may be a combination of methods that can work.

Disclosure of Commercial Conflict of Interest

S. Hammel: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Hammel: Nothing to disclose

AP51

Improved Cooler Design Reduces Wasted Blood Products on Medical Transport Flights

L M Button[1] (button.lisa@mayo.edu), K S Badjie[1], K S Berns[2], C M van Buskirk[1]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Medical Transport, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Provision of blood products during medical scene or inter-facility flights have been shown to improve patient outcome.

Validated blood storage coolers are an integral part of a successful aero-medical flight service as they enable blood products to accompany the medical team to the patient. This institution recently identified a new cooler design which improved length of time the blood can be away from the helipad refrigerator. Prior to May 2012, this institution's transfusion lab provided coolers to the medical transport team that were validated to maintain a temperature of 1-10 C for up to 4 hours. Laboratory management identified the need to implement coolers that maintained a temperature range of 1-6 C for a longer amount of time. The aim of this initiative was to reduce wasted RBCs and FFP due to improper temperature and to achieve the FDA and AABB standard temperature for blood storage conditions versus blood transport conditions. **Study Design/Methods:** The Transfusion Laboratory Event reporting database was used to identify red blood cells (RBCs) and fresh frozen plasma (FFP) units wasted due to improper storage during medical flight transport. Data was analyzed and it was noted that the majority of waste events were due to products remaining in the cooler for longer than the validated 4 hours resulting in an increase in temperature. A new cooler was identified and was subsequently validated by the institution to maintain a temperature of 1-6 C for at least 24 hours. The new cooler was implemented on May 29, 2012. **Results/Findings:** From June 2011-March 2012, the medical transport service conducted 644 flights carrying blood products. During this 10-month period, 0.78% (5/644) of flights resulted in the blood cooler being away from the lab for longer than 4 hours. Discard rate for the 10-month pre-implementation period was 54% (20/37) total transfused RBCs and 17% (10/58) total transfused FFP. From May 29, 2012-March 2013, there were 678 flights carrying blood products. In that time, 0.15% of flights resulted in the total time the products were in the cooler to be greater than the validated 24 hours (1/678). Upon return to the lab, the temperature of the blood products was measured. Three FFP units were 7 C and were discarded. Discard rate for the 10-month post-implementation period was 0% (0/18) for RBCs and 7% (3/42) for FFP. **Conclusion:** The implementation of a new cooler improved the discard rate for blood transported by the medical transport flight team by 100% for RBCs and 59% for FFP.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; K. S. Berns: Nothing to disclose; L. M. Button: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; K. S. Berns: Nothing to disclose; L. M. Button: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

AP52

Validation of a Medical Flight Blood Cooler

L M Button[1] (button.lisa@mayo.edu), E Girma[1], K S Badjie[1], K S Berns[2], A Barhoush[1], J R Stubbs[1]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Mayo Medical Transport, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: This institution began carrying coolers of blood on medical transport flights in 1988. The cooler was an off-the-shelf igloo cooler containing four red blood cells (RBC)s and no coolant. The cooler was stored in a refrigerator until time of use. It was validated to hold a temperature of 1-10 C for 4 hours once removed from the refrigerator. Event reports and discard data showed that a new cooler designed for transport of blood was necessary to reduce blood wasted due to improper storage conditions that allowed the blood to become warmer than 10 C. The institution studied the Credo Series 4-496 Shipper from MN Thermal Sciences, which consists of Thermal Isolation Chambers (TICs) containing Phase Change Material (PCM) as their cooling source. Vacuum Insulation Panels (VIPs) add an additional layer of insulation to the cooler design. All components are zipped in an outer shell for transport. **Study Design/Methods:** The study tested the manufacturer's claims for the cooler. This institution uses the cooler both inside and outside in Minnesota (MN) summer and winter conditions. This validation was designed to test the extreme conditions it could encounter as well as ambient conditions. The cooler was loaded with six units (max load) and placed on the roof of the hospital in January to expose it to MN Cold conditions. The environment ranged between −9.6 C to −0.2 C over the course of 8 hours. During the first 2 hours after loading, the cooler was opened three times for 30 seconds per instance to mimic likely use/access to the blood inside. After 4 hours, five units of blood were removed to leave the minimum load of one unit in place. This was repeated with MN Warm conditions by placing a cooler with the same load in a warm environment (external temperature of 29.6 C). For room temperature use (identified by flight crew as the most commonly encountered temperature in the helipad, helicopter, and outside facility emergency departments), the cooler was loaded with six units. The cooler was opened for 30 seconds, 3 different times. After 10 hours, 5 units were removed and the cooler was left in place for an additional 14 hours to bring total scenario time to 24 hours. **Results/Findings:** Even with opening the cooler several times and removing the majority of blood products, the unit of blood containing the TempTale that remained in the cooler showed that the internal temperature of the unit stayed in an acceptable storage range. The temperature range was 3.7 C-5.4 C during the MN Cold scenario; the temperature of the unit in the MN Warm scenario stayed between 3.2 C and 5.4 C. In room temperature conditions, the unit's temperature ranged between 2.8 C and 5.7 C. **Conclusion:** The coolers met validation criteria for extremes of temperature and length of usable time. Coolers were implemented with a validated time of 24 hours with the ability to be used at room temperature and outdoor summer and winter conditions in MN.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; A. Barhoush: Nothing to disclose; K. S. Berns: Nothing to disclose; L. M. Button: Nothing to disclose; E. Girma: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; A. Barhoush: Nothing to disclose; K. S. Berns: Nothing to disclose; L. M. Button: Nothing to disclose; E. Girma: Nothing to disclose; J. R. Stubbs: Nothing to disclose

AP53

Validation of an Operating Room Blood Cooler System

L M Button[1] (button.lisa@mayo.edu), E Girma[1], K S Badjie[1], A Barhoush[1], J R Stubbs[1], M J Brown[2]. *[1]Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [2]Anesthesiology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: This institution began providing coolers for blood storage in the Operating Room (OR) in 2000. The cooler was an off-the-shelf Igloo cooler, packed with two frozen blocks and an acrylic divider between the coolant and blood products to prevent the blood from becoming colder than 1 C. The coolers were validated to hold a temperature of 1-10 C for 8 hours. After reviewing event reports and discard data from the OR and identifying that "storage" versus "transport" temperature ranges must be maintained, it was determined that a new cooler designed specifically for blood storage was necessary to reduce blood wasted due to improper storage conditions. A multidisciplinary group was assembled to identify and validate the new cooler system. The Credo Series 4-824OR from MN Thermal Sciences was identified which consists of Thermal Isolation Chambers (TICs) containing Phase Change Material (PCM) as their cooling source. The TICs form a cube around the blood inside of an Igloo Maxcold Cooler. **Study Design/Methods:** The study was designed to test the manufacturer's claims for the cooler identified. The cooler is used only in room temperature conditions so the validation tested the performance of the cooler between 20-24 C. The TICs were frozen in the plasma freezer (average operating temperature −33 C) for a minimum of 12 hours prior to room temperature (RT) precondition phase. Manufacturer states RT TIC precondition phase should be at least 25 minutes or until surface frost disappears. Temperature monitoring device probes were inserted into one fake unit and set to record every 10 minutes. Coolers were loaded with six units of blood (from the refrigerator) and the scenario was run for at least 24 hours for each cooler. A second scenario was run where coolers were loaded with one unit of blood and was run for at least 24 hours. **Results/Findings:** It was found that the RT TIC precondition step produced the best phase change between 30-40 minutes (combination of freezer temperature and lab RT played role in length of precondition time). The coolers loaded with six units of blood maintained 1-6 C for at least 21 hours 50 minutes. Coolers loaded with one unit of blood maintained 1-6 C for at least 18 hours 20 minutes. If coolers were loaded with FFP removed directly from the thawer, the temperature did not stay 1-6 C for 18 hours. **Conclusion:** The coolers met acceptance criteria for maintaining an average load of blood (six units) and a minimum load of blood (one unit) for more than 18 hours in every trial run. It is imperative that both RBCs and FFP are loaded into the cooler at refrigerator temperature versus at just-thawed temperature for the FFP. The new cooler system extended the time a cooler can be in the OR from 8 hours-18 hours, tightened the temperature range from 1-10 C to 1-6 C and significantly reduced the number of wasted blood products for RBCs and FFP due to improper storage conditions.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; A. Barhoush: Nothing to disclose; M. J. Brown: Nothing to disclose; L. M. Button: Nothing to disclose; E. Girma:

Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; A. Barhoush: Nothing to disclose; M. J. Brown: Nothing to disclose; L. M. Button: Nothing to disclose; E. Girma: Nothing to disclose; J. R. Stubbs: Nothing to disclose

AP54

HLA Matched Platelet Donor Searches

R O Endres[1] (rendres@bloodsystems.org), J Julleis[1], M Marlowe[1], R Partel[1], C Mitchell[1], M Li[1], B A Lenes[1], K Land[1], F Nizzi[1], P A Tomasulo[1]. *[1]HLA Laboratory, Blood Systems Laboratories, Tempe, AZ, United States*

Background/Case Studies: Consistently providing HLA matched products to platelet refractory (PR) patients requires significant commitment and expense from the blood supplier. The number and the degree of match for the HLA matched products provided from a large regional blood supplier was reviewed to determine the relationship of donor pool size and the success of finding an acceptable donor. **Study Design/Methods:** Apheresis platelet donors were typed for HLA-A and -B alleles with the Luminex-based LABType kit. In donor searches, the A match grade was assigned to donors that have 4 antigens matched to the patient's 4 antigens. The B match grade was assigned to donors that have two or three matched antigens (due to homozygosity), with no mismatches. B1X and C match grades were assigned to donors with mismatched antigens. Searches were defined as successful if they identified at least one A or B match grade donor. Donor antigens that were not identical, but were within the same broad antigen group as a patient antigen, were considered matching. **Results/Findings:** Approximately 20% of donors who gave more than one platelet apheresis collection in 2012 were HLA typed. Between 2010-2013, the registry grew from 2437 to 5900; successful searches increased from 79% to 89%; and the average number of A or B match grade donors found per search rose from 7.8 to 16.2. When the registry contained 5404 donors, searches were repeated for all PR patients from 2003-2013. Of 1406 patients, 9% did not have any A or B match grade donors, 20% had fewer than five matched donors, and 71% had more than five donors, giving better assurance of product collection and delivery. Antigens in a broad antigen group are not truly serologically identical antigens. Therefore, donors with identical antigens are preferable to donors with broad antigen matches and should be distinguished in these searches. When this analysis was repeated with the criteria that the donor's antigens must be identical rather than broad antigen matched, 48% of patients had no A or B match grade donors, 17% had fewer than five matched donors and 35% had more than five donors. **Conclusion:** Given the current emphasis on healthcare cost control, the ideal number of HLA typed donors in a registry needs to be carefully assessed. There must be a balance between the cost of increasing the pool size and finding sufficient compatible donors to support PR patients. It was found that a pool of approximately 6000 active HLA-typed donors can identify more than five well-matched donors, about 71% of the time at a reasonable cost. Collaborative arrangements with other blood centers that maintain HLA typed donor pools serve as an appropriate backup for cases without donors in the local registry.

Disclosure of Commercial Conflict of Interest

R. O. Endres: Nothing to disclose; J. Julleis: Nothing to disclose; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; M. Li: Nothing to disclose; M. Marlowe: Nothing to disclose; C. Mitchell: No Answer; F. Nizzi: Mediware, Other, Blood Systems, Other; R. Partel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. O. Endres: Nothing to disclose; J. Julleis: Nothing to disclose; K. Land: Nothing to disclose; B. A. Lenes: Nothing to disclose; M. Li: Nothing to disclose; M. Marlowe: Nothing to disclose; C. Mitchell: No Answer; F. Nizzi: Nothing to disclose; R. Partel: Nothing to disclose; P. A. Tomasulo: Nothing to disclose

AP55

24 Hour Apheresis Plasma Implementation, PF24RT24

A L Dirr[1] (Angela_Dirr@bonfils.org), T Berry[2], C Larson[3]. *[1]CDO, Bonfils Blood Center, Denver, CO, United States; [2]Manufacturing, Bonfils Blood Center, Denver, CO, United States; [3]Facilities Management, Bonfils Blood Center, Denver, CO, United States*

Background/Case Studies: In October, 2012, it was learned that Terumo BCT obtained FDA approval for 24-hour Apheresis plasma. The clearance allowed blood centers 24 hours after collection before the plasma must be frozen. International Society of Blood Transfusion (ISBT) labels were released in December 2012. Once Terumo BCT released an updated circular of information, 24-hour Apheresis plasma was quickly implemented. **Study Design/Methods:** Collections led the project implementation. Impacted departments included: Client Development to ensure clients had appropriate ISBT labels to receive 24-hour Apheresis plasma; Manufacturing to update procedures to reflect the new freeze requirement; Regulated Software to update ISBT labels and product codes; Regulatory Affairs to ensure appropriate FDA notifications; Logistics to evaluate transportation costs; Accounting to establish billing procedures; and Business Development to prospect additional donor bases and blood drive opportunities to collect plasma in new markets. **Results/Findings:** The 24-hour Apheresis plasma was implemented on April 1, 2013. While awaiting the circular of information to be distributed, product label and Safetrace related code development was conducted to prepare the blood center for quick implementation. In addition to the 24-hour Apheresis plasma and opportunities for inventory management and improved product utilization, reduction of blood pick-ups was evaluated from every 2 hours to twice per day at fixed site locations. Historically, pick-ups were conducted by both in-house drivers and vendors. Boxes were not fully packed due to the frequency, and collection staff were pulled off the donor flow to prepare boxes for pick-up. Logistics and Collections developed a reduced pick-up schedule while maintaining sufficient cryoprecipitate product lines. Few hospital clients were not prepared to receive 24-hour Apheresis plasma, but became compliant quickly. **Conclusion:** While the ultimate goal of 24-hour Apheresis plasma implementation was to expand product line, an annual savings $55K in hard costs is projected, associated with the reduction in pick-ups and the reduction of packing materials now that shipping containers are packed full. Soft cost savings include the reallocation of in-house drivers to other business needs, and redirection of collection staff to donors and donor customer care. Next step is to evaluate growth potential to draw Apheresis plasma in rural locations with high concentrations of AB donors.

Disclosure of Commercial Conflict of Interest

T. Berry: Nothing to disclose; A. L. Dirr: Nothing to disclose; C. Larson: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Berry: Nothing to disclose; A. L. Dirr: Nothing to disclose; C. Larson: Nothing to disclose

AP56

Implementation of an Intra-Operative Blood Transport and Storage Initiative and its Effect on Reducing Red Blood Cell (RBC) and Plasma (FFP) Waste

L M Button[2] (button.lisa@mayo.edu), K S Badjie[2], J Guyer[1], S R Dhanorker[3], E J Brach[3], P M Johnson[2], J R Stubbs[2], M J Brown[1]. *[1]Department of Anesthesiology, Mayo Clinic, Rochester, MN, United States; [2]Department of Laboratory Medicine and Pathology, Division of Transfusion Medicine, Mayo Clinic, Rochester, MN, United States; [3]Department of Systems and Procedures, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Surgical patients remain one of the largest consumers of blood products in the US. The national waste rate for hospital-issued blood products ranges from 0-6%, with operating room-(OR) responsible waste representing up to 70% of total hospital blood product waste. A common reason for blood product waste is inadequate intraoperative storage. The effect of an intra-operative blood storage and transport initiative (BTSI) was reported on red blood cell (RBC) and plasma (FFP) waste. **Study Design/Methods:** The Transfusion Service database was used to quantify and categorize RBC and FFP units issued for intraoperative transfusion that were wasted over a 27-month period. Two cohorts were created: 1) Pre-implementation of BTSI: RBC and FFP waste January 1, 2011-May 31, 2012; and 2) Post-implementation of BTSI: RBC and FFP waste June 1, 2012-March 31, 2013. The BTSI replaced existing storage coolers (8-hour coolant lifespan with temperature range of 1-10 C) with a cooler that had a coolant lifespan of 18 hours and a temperature range of 1-6 C, and included an improved educational cooler placard and an alert mechanism in the electronic health record. Monthly median RBC and FFP waste were the primary outcomes. **Results/Findings:** An intra-operative BTSI significantly reduced median monthly RBC (1.3% versus 0.07%) and FFP (0.4% versus 0%) waste (Table). The majority of blood product waste was due to an unacceptable temperature of unused returned blood products. Waste due to reasons other than product temperature range was negligible. **Conclusion:** In this observational study, an intra-operative BTSI significantly reduced median monthly RBC and FFP waste.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; E. J. Brach: Nothing to disclose; M. J. Brown: Nothing to disclose; L. M. Button: Nothing to disclose; S. R. Dhanorker: Nothing to disclose; J. Guyer: Nothing to disclose; P. M. Johnson: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; E. J. Brach: Nothing to disclose; M. J. Brown: Nothing to disclose; L. M. Button: Nothing to disclose; S. R. Dhanorker: Nothing to disclose; J. Guyer: Nothing to disclose; P. M. Johnson: Nothing to disclose; J. R. Stubbs: Nothing to disclose

TABLE 1: Impact of a Blood Transport and Storage Initiative on Monthly Red Blood Cell and Thawed Plasma Waste

	Before Implementation n = 17 months	After Implementation n = 10 months	p value
Blood Product Issued#	1806 (l679-2014)	1547 (1481-1725)	<0.01
RBC	1365 (1277-1491)	1181 (1121-1263)	<0.01
FFP	487 (406-522)	396 (360-495)	0.07
RBC Wasted#	18 (12-24)	0 (0-3)	<0.01
Product temperature out of range	18 (12-23)	0 (0-5)	<0.01
Other	0 (0-1)	0 (0-0)	0.22
FFP Wasted#	3 (1-5)	0 (0-2)	<0.01
Pruduct temperature out of range	2 (1-4)	0 (0-2)	<0.01
Other	0 (0-1)	0 (0-0)	0.09

Product temperature out of range includes those products outside of the accepted temperature range: 1-10°C pre-implementation, 1-6°C post-implementation, and product stored in coolers whose coolant had expired without coolant replacement. "Other" includes units that were spiked, broken, or sent with a patient after their procedure had ended and never transfused. # = median value (25%-75% interquartile range)

AP57

Cross-Matched Platelets Unavailable? A Rapid Switch to a HLA-Matching Algorithm

J Duven[1] (duvenjennifer@gmail.com), S Shaikh[2], S M Morgan[1], N D Zantek[1], J Welbig[3], C S Cohn[1]. *[1]Laboratory Medicine and Pathology, University of Minnesota, Minneapolis, MN, United States; [2]Biomedical Services, American Red Cross, St. Paul, MN, United States; [3]Laboratory Medicine and Pathology, Fairview Health Services, St. Paul, MN, United States*

Background/Case Studies: Patients who have immune-mediated platelet refractoriness often require specially matched platelet units to maintain the minimally recommended platelet count of 10 × 109/L. This institution supports a large population of hematopoietic stem cell transplant (HSCT) patients who require multiple specially-matched platelet units every week. Compatible units can be identified by using a platelet cross-match assay or by matching the HLA-type of the recipient and donor. Cross-matched platelets have historically been sought due to the potential for identifying units currently in inventory. The HLA matching system may be less efficient due to the time needed to successfully identify and collect each individual matched donor. Because of the temporary unavailability of a commercially available cross-matching test, an HLA-matching protocol was rapidly implemented in order to provide platelet refractory recipients with good transfusion outcomes. **Study Design/Methods:** The HLA-type of all HSCT recipients is known and approximately 75% of the local American Red Cross (ARC) platelet donor population has been typed for their class I HLA. When an order for compatible platelets was received, an initial HLA-antibody screen was performed (Life Technologies, Carlsbad, CA). When positive, a single antigen bead assay (Luminex, Life Technologies) was used to identify the specificity of the HLA antibodies present. Simultaneously, the ARC searched their database for either HLA-matched units already in inventory or for potential donors. An algorithm was developed to determine the best available units and/or potential donors based on ABO-type, degree of HLA-match, presence and strength of HLA-antibodies, and patient engraftment status. One-hour corrected count increments (CCIs) were calculated to gauge the effectiveness of the transfusion. **Results/Findings:** A combination of HLA-matching and HLA antibody avoidance techniques allowed for rapidly obtaining a sufficient number of matched units. The protocol was developed, clinicians were educated, and the new workflow was implemented within 2 weeks. Since switching to this HLA matching and antibody avoidance algorithm, the platelet transfusion needs of patients have been supported with few delays. The CCI data show that units identified by HLA-matching and antibody avoidance techniques are equivalent to the cross-match units previously used. **Conclusion:** HLA matching and cross-matching are equivalent techniques for supplying compatible platelet units for refractory patients. A rapid switch can be made to an HLA-matching system to provide refractory patients with compatible platelet units.

Disclosure of Commercial Conflict of Interest

C. S. Cohn: Fenwal, Grants or Research Support, Travel Support or Honorarium; J. Duven: Nothing to disclose; S. M. Morgan: Nothing to disclose; S. Shaikh: Nothing to disclose; J. Welbig: Nothing to disclose; N. D. Zantek: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. S. Cohn: Fenwal, Travel Support or Honorarium, Grants or Research Support; J. Duven: Nothing to disclose; S. M. Morgan: Nothing to disclose; S. Shaikh: Nothing to disclose; J. Welbig: Nothing to disclose; N. D. Zantek: Nothing to disclose

AP58

Re-inventing the Inventory Management Paradigm: Comparison of Red Cell Inventory Flow in Different-Sized Hospital Transfusion Services Using a Novel Informatics-Driven Inventory Mapping Technique

C Cheng[3,2] (Calvino.Cheng@cdha.nshealth.ca), A Kumar-Misir[3], S Watson[3], J MacLeod[1], I Sadek[1,2]. *[1]Laboratory Medicine and Pathology, Capital District Health Authority, Halifax, NS, Canada; [2]Laboratory Medicine and Pathology, Dalhousie University, Halifax, NS, Canada; [3]Pathology Informatics Group, Capital District Health Authority, Halifax, NS, Canada*

Background/Case Studies: Current red cell (RBC) inventory management paradigms are based largely on models from the 1970s and 80s, and inventory performance is measured with quality indicators, such as discard/expiry rates. The limitation of these methodologies is that they are approximations and proxies for inferring the actual working of the inventory "black box", which the literature still has not fully characterized or understood. Presented here is a novel high-resolution inventory mapping technique which directly visualizes RBC unit flow inside the inventory "black box". This is used to study the red cell traffic in the inventory at different hospitals within the geographically distributed transfusion service to and discover areas for improving inventory practices. **Study Design/Methods:** RBC data during CY 2012 was queried from the laboratory information system (Cerner Millennium) for all sites at in the district here. This data contained unit and location attributes. Data validation, cleaning, and analysis was performed using commonly available database, spreadsheet, and data mining software. Ethics approval was not required by institutional policies. **Results/Findings:** There were 154,875 transactional states with 477 unique location and transfusion state specific activities involving 16,585 red cell units. There were 15,740 transactional variants in total, with most common variants and hospital types noted (Table). Qualitatively, the larger the hospital, the more complex the map was. There were also four main points of inventory receipt, with 88% of units at the tertiary hospital; smaller hospitals demonstrated delayed RBC recycling (mean 17.3days between available and transfer states); and hospitals without electronic crossmatching demonstrated crossmatching "loops" (pictures not shown). The main hospital also acted as an RBC distribution and reception point for smaller community hospitals. **Conclusion:** Using this novel technique, it was demonstrated that although RBC units travel though a common inventory receipt and processing pathway, subsequent transfusion transactions and practices are very

different at each hospital site. It was also demonstrated that it was possible to infer inventory practices at different hospitals, and determine areas where handling can be improved. Ultimately, in conjunction with quality/performance indicators, this technique can allow transfusion services to evaluate the effect of policy changes and enable true lab-to-lab benchmarking, ultimately enabling systemic cost savings and improving patient care.

Disclosure of Commercial Conflict of Interest

C. Cheng: Nothing to disclose; A. Kumar-Misir: Nothing to disclose; J. MacLeod: Nothing to disclose; I. Sadek: Nothing to disclose; S. Watson: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Cheng: Nothing to disclose; A. Kumar-Misir: Nothing to disclose; J. MacLeod: Nothing to disclose; I. Sadek: Nothing to disclose; S. Watson: Nothing to disclose

Most Common RBC Transactional Variant, processing times, and hospital size

Hospital Size and Type	Most Common Red Cell Inventory Transactional Variant (cases/total cases, %)	Inventory receipt and processing time (received>>>available)
All hospitals	receivedHI>unconfirmedHI>confirmedHI>availableHI>transferredVG> crossmatchedVG>issuedMDU>transfused MDU (735/15740, 4.7%)	n/a
Large, Tertiary/Emerg	receivedHI>unconfirmedHI>confirmedHI>availableHI>transferredVG> crossmatchedVG>issued8A>transfused8A (735/13881, 5.3%)	2.2 hours
Medium, Community	receivedDGBTS>unconfirmedDGBTS>confirmedDGBTS>availableDGBTS> >crossmatchedDGBTS>issuedDMPR>transfusedDMPR (245/1324, 18.5%)	47 minutes
Small, Community	receivedHCBTS>unconfirmedHCBTS>confirmedHCBTS>availableHCBTS> >crossmatchedHCBTS>issuedH2>transfusedH2 (101/395, 25.6%%)	77.3 minutes

AP59

Inventory Factors and Platelet Utilization at a Major Medical Institution

E A Douglas[1] (douglas.emily@mayo.edu), K S Badjie[1], M Budnick[1], K Schulte[1], J R Stubbs[1], C M van Buskirk[1]. *[1]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Every year, approximately 12,000 apheresis platelets are transfused here. The high demand for platelet transfusion therapy for the varied and complex patient population, in combination with the inventory challenges posed by the short storage period for platelets, has fostered a practice where the provisions of ABO and D compatible platelets for transfusion are the first choice but not an absolute requirement. Exceptions to the flexibility in the choice of platelets for transfusion are females of child bearing potential (≤54 years) and children (≤17 years) where the policy is to provide D compatible platelets when such products are available. Recent transfusion medicine publications have suggested that ABO-identical platelet transfusions are superior to non-identical transfusions from an efficacy and safety perspective (eg better post-transfusion platelet increments and prevention of potential hemolytic transfusion reactions). The objective of this study was to retrospectively evaluate the platelet transfusion practice here and measure how closely this complies with the ideal practice of ABO-identical platelets for all transfusion recipients. **Study Design/Methods:** A 2-year (2011-2012) retrospective review of platelet selection and transfusion trends by day was performed. Data was obtained by running and analyzing monthly transfused products crystal reports, and then calculating the percentage of patients who received ABO-identical platelets, ABO-compatible platelets, and ABO-incompatible platelets. **Results/Findings:** 71% of the platelets transfused daily are ABO-identical; 16% are ABO-compatible, and 13% are ABO-incompatible. Overall, patients are transfused with ABO-compatible platelets 87% of the time (Table). **Conclusion:** It is apparent from this analysis that with current inventory levels, the distribution of donor ABO types, and the current approach to inventory management, that this institution is unable to provide ABO identical platelets to all transfusion recipients. A transition to a total ABO identical approach for platelet transfusion recipients will require a detailed analysis of inventory adequacy, donor ABO distribution, impact of platelet outdate rates, and a post-implementation clinical and financial outcome analysis to assess the impact of such a change. In the meantime, there is an opportunity for specific patients and patient populations (eg hematology/oncology patients) to be designated with a special need of "transfuse ABO identical platelets only" as the feasibility analysis of a total transition to ABO identical platelets is conducted.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; M. Budnick: Nothing to disclose; E. A. Douglas: Nothing to disclose; K. Schulte: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; M. Budnick: Nothing to disclose; E. A. Douglas: Nothing to disclose; K. Schulte: Nothing to disclose; J. R. Stubbs: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

TABLE 1: Platelet Transfusions from Jan 2011-Dec 2012

	ABO-identical	ABO-compatible	ABO-incompatible	Total
Daily Average	25.5	5.7	4.6	35.8
Percentage	71%	16%	13%	

AP60

Cost-Effective Blood-Ordering Practices and Deliveries: A Hospital and Blood Center Working Together

K Yunker[1] (rkyunker@hotmail.com), P Nortmann[1], C Moorhead[1], A Sturgeon[2], T Renner[2]. *[1]Transfusion Services, St. Elizabeth Healthcare, Edgewood, KY, United States; [2]Hoxworth Blood Center, Cincinnati, OH, United States*

Background/Case Studies: As an integral part of the healthcare system, hospital transfusion services and blood centers have a responsibility to identify areas for improvement to continue providing patients with the safest blood products. Timely deliveries, correct product orders, managing staff and blood inventory levels, and meeting the financial responsibility of monitoring costs, were the expectations when a Kentucky hospital and community blood center took on a quality-improvement project to look at the costs associated with product ordering practices and deliveries. The initial data obtained identified several areas for improvement to meet the expectations. **Study Design/Methods:** A retrospective study of product deliveries was performed for a 3 month period (September-November 2012). Invoices from each delivery, the Laboratory Information System (LIS), and blood center's electronic ordering system were used to gather data for routine stock and special orders (platelets, irradiated products, reference units) and to determine the priority of an order that was placed and delivered, and total number of deliveries. As platelet orders were of special interest, the ordering location and transfusion requirements were identified. **Results/Findings:** During the 3-month period, there was an average of 85 Monday-Friday deliveries per month (range 78-96) and 15 weekend deliveries per month (range 9-24). This included stock, platelets, irradiated red blood cells, fresh frozen plasma, cryoprecipitate, and reference units. Other remarkable statistics included 1 day with 9 deliveries, and 20 high delivery days (those with 5 or more). The mean number of platelet orders was 87 (range 79-98). Average cost to the blood center for the delivery of blood components was just under $2000 per month. **Conclusion:** The 3-month study scrutinized physician and nurse ordering practices, inventory levels, duplicate orders, and delivery practices. An obvious reason for the high number of deliveries was the increased use of platelets, which were not kept in stock. Ideas to improve current practices included routine stocking of platelets, a later Friday stock delivery to accommodate expanded weekend Cancer Care hours, and standardization of orders in the electronic ordering system to improve efficiency. Estimated cost savings to the blood center would be around $1600 annually with the minor changes made thus far. Continuous improvements and monitoring will result in cost savings to the blood center and challenge other facilities in the region to review ordering practices, as resources and work are shared to better serve the community.

Disclosure of Commercial Conflict of Interest

C. Moorhead: Nothing to disclose; P. Nortmann: Nothing to disclose; T. Renner: Nothing to disclose; A. Sturgeon: Nothing to disclose; K. Yunker: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Moorhead: Nothing to disclose; P. Nortmann: Nothing to disclose; T. Renner: Nothing to disclose; A. Sturgeon: Nothing to disclose; K. Yunker: Nothing to disclose

Summary of monthly deliveries

MONTH (2012)	MONDAY-FRIDAY	AVG./DAY (M-F)	SATURDAY/SUNDAY
SEPTEMBER	81	4.05	24
OCTOBER	96	4.17	9
NOVEMBER	78	3.54	11

AP61

New Mexico Regional Blood Distribution Project

A Gonzales[1] (aimee.gonzales@tricore.org), D Schutt[2]. *[1]Commercial Blood Bank, Tricore Reference Laboratories, Albuquerque, NM, United States; [2]Hospital Services, United Blood Services New Mexico, Albuquerque, NM, United States*

Background/Case Studies: United Blood Services of New Mexico (UBSNM) and Tricore Reference Laboratories (TRL) jointly simplified inventory management to reduce costs while maintaining and improving services. A system was designed to reduce transportation costs, red blood cell (RBC) inventories, staff labor, reagents, and streamline testing in eight TRL hospital transfusion services (TS) throughout New Mexico. **Study Design/Methods:** In May 2010, two regional hospitals were chosen as pilot sites. The TRL commercial laboratory received all RBC inventory from the blood center, performing the confirmatory testing, and entering data into the laboratory information system inventory (LIS). Prior to this change, each hospital performed these activities on site. The commercial laboratory viewed the inventory electronically daily to determine the number of RBC units needed to be sent out. To reduce costs, inventory was delivered using established TRL courier routes instead of using blood center couriers which added expense. The hospital received the units into the LIS inventory and stored them onsite. Usage by the regional hospitals was tracked and deliveries adjusted to better fit transfusion needs. Regional hospitals were notified when units were approaching expiration and returned to the commercial laboratory. To maintain inventory levels, fresher units were delivered before the older units were returned. If there was a positive antibody screen, the patient sample was sent to the commercial laboratory to be evaluated and crossmatched RBCs sent to the hospital. To standardize billing and reduce costs, all RBCs were billed to the commercial laboratory starting in 2012 and costs allocated to hospitals based on usage. **Results/Findings:** Decreases were seen in the cost of reagents, labor, and transportation. In addition, testing protocols and procedures were standardized among eight hospitals. Approximately 1000 fewer RBC units were needed to maintain regional hospital inventory levels due to the management by a centralized location. Wastage was also decreased. Reagent cost and labor for confirmatory testing were eliminated for each hospital. Shipping costs were decreased by $40,000 by delivering inventory on established courier schedules. The move from paper to electronic tracking saved approximately 2000 staff hours between all participants. The electronic tracking also allowed real time tracking, better standardization, and fewer errors. Approximately $140,000 has been saved. **Conclusion:** Regional distribution was successful and is currently being used in eight hospitals. The project led to policy changes both at TRL and the blood center, as well as improved communications among employees. Patient care remains a top priority while inventory levels are better managed at a lower cost. RBC waste, labor, and reagent cost has decreased and electronic tracking has allowed for improved and more streamlined billing practices.

Disclosure of Commercial Conflict of Interest

A. Gonzales: Nothing to disclose; D. Schutt: No Answer

Disclosure of Grants Conflict of Interest

A. Gonzales: Nothing to disclose; D. Schutt: No Answer

AP62

Evaluation of Platelet Inventory Management Practices in a Large Pediatric Hospital Results in Opportunities for Improvement

V Bhoj[1] (vijay.bhoj@uphs.upenn.edu), C F Moultrie-Savage[2], D Sesok-Pizzini[2], D F Friedman[3]. *[1]Department of Pathology and Laboratory Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA, United States; [2]Department of Pathology and Laboratory Medicine, Children's Hospital of Philadelphia, Philadelphia, PA, United States; [3]Department of Pediatrics, Children's Hospital of Philadelphia, Philadelphia, PA, United States*

Background/Case Studies: Efficient platelet inventory management is challenging and complex chiefly because of short life-span and demand variability. This is reflected in reported discard rates of up to 20%. Efforts to minimize discards generally focus on ordering practices. This is an analysis of platelet inventory managementand pre- and post-corrective actions made as a result of findings at an academic pediatric hospital. **Study Design/Methods:** Analysis of platelet management practice included discussion with blood bank staff to understand their approaches to ordering and analysis of data regarding platelet transfusion, ordering, inventory, and discard patterns over two 6-month periods (October 2008-March 2009 and October 2012-March 2013) corresponding to pre- and post-corrective actions. Interventions included staff education, cancelling Saturday standing orders, discussion with clinicians, and limiting platelet reservations to 6 hours. **Results/Findings:** Analysis was performed separately for whole and aliquot units; results for whole units are presented. In total, 1111 and 2082 whole units were analyzed (transfused+discarded) in the pre- and post-periods, respectively. The discard rate changed from 5.2% to 1.2% after interventions were instituted in March 2012. Analysis of discards by ABO/Rh type revealed AB negative and B positive units were over-represented (10.2% and 22.9%) among discards pre-intervention. Discussion with staff suggested these units were likely over-held as "precious" and the desire to preserve them prompted additional ordering. In the post period, the corresponding values for AB negative and B positive units decreased to 6.3% and 15.6%, respectively. Analysis by day-of-discard and by day-of-receipt revealed higher discard rate on Mondays associated with higher discard rate on units ordered on Saturdays. At this institution, platelets are placed in "Assigned" status (ASN) when initially ordered by clinicians and may remain ASN indefinitely until transfusion, cancellation, or expiry. Analysis of duration of ASN by eventual status revealed a discard rate of 47% for those ASN >6 hours. Fewer than 50% were issued directly from ASN >6 hours. A small subset of clinical services were responsible for the majority of reservations >6 hours. Finally, the estimated cost of discarded whole units decreased from approximately $48,700 to $9,600 (normalized for total whole units). **Conclusion:** Platelets present a unique inventory management challenge. Analysis of platelet use patterns revealed opportunities for improvement in ordering, dispensing, and reserving practices. Substantially reduced waste may be attributed to changes implemented to address these areas, resulting in improved inventory management and financial savings.

Disclosure of Commercial Conflict of Interest

V. Bhoj: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; C. F. Moultrie-Savage: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose

Disclosure of Grants Conflict of Interest

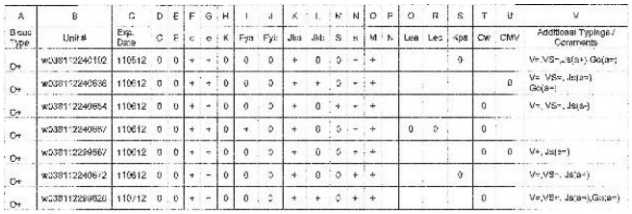
V. Bhoj: Nothing to disclose; D. F. Friedman: BioArray, Grants or Research Support; C. F. Moultrie-Savage: Nothing to disclose; D. Sesok-Pizzini: Nothing to disclose

AP63

Using Excel Spreadsheets and Logic Formulas to Manage Immunohematology Reference Lab (IRL) Inventory

A A Delk[2] (alexander.delk@oneblood.org), R R Gammon[1]. *[1]OneBlood, Inc., Orlando, FL, United States; [2]OneBlood, Inc., Lake Park, FL, United States*

Background/Case Studies: Immunohematology reference labs (IRLs) strive to maintain a diverse inventory of red cell units to meet the needs of alloimmunized patients. The IRL relied on the blood center information technology (IT) system to locate previously identified antigen negative units in the general inventory, but once in the IRL inventory, knowing exactly what was available could be a challenge. This IRL developed a method to track current liquid and frozen inventory by using spreadsheets and logic formulas to generate a dynamic list and overview of available units. **Study Design/Methods:** An Excel inventory workbook was created with separate worksheets for the following four groups that best met the needs of our IRL's



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
Blood Type	Unit #	Exp. Date	C	E	c	e	K	Fya	Fyb	Jka	Jkb	S	s	M	N	Lea	Leb	Kpa	Cw	CMV	Additional Typings / Comments
O+	W038112240102	110512	0	0	+	+	0	0	0	+	0	0	+	+				0			V+,VS+,Jsa(s+),Go(a=)
O+	W038112240698	110612	0	0	+	+	0	0	0	+	+	0	+	+						0	V+, VS+, Jsa(s+), Go(a+)
O+	W038112240854	110612	0	0	+	+	0	0	0	+	0	+	+	+					0		V+, VS+, Jsa(s+)
O+	W038112240857	110612	0	0	+	+	0	+	0	+	0	0	-	+	0	0			0		
O+	W038112239867	110612	0	0	+	+	0	0	0	+	0	0	+	+					0	0	V+, Jsa(s+)
O+	W038112240872	110612	0	0	+	+	0	0	0	+	0	0	+	+					0		V+,VS+, Jsa(s+)
O+	W038112239820	110712	0	0	+	+	0	0	0	+	+	0	+	+					0		V+,VS+, Jsa(s+),Go(a=)

O+R_0r

patients: O+ R_1R_1, O+ R_2R_2, O+ R_0r and O- rr. Each worksheet was labeled with ABO Rh-hr and the following column headers: Blood Type, Unit#, Exp. Date, C, E, c, e, K, Fy^a, Fy^b, Jk^a, Jk^b, S, s, M, N, Le^a, Le^b, Kp^a, Cw, CMV, and comments. Each row was an inventoried unit. Separate workbooks were created for liquid and frozen inventories. Each of the 4 groups was further divided into 30 specific profiles (each with an associated logic formula) within 4 antigen negative levels. Level 1 (6 profiles) was those units that were negative for at least one of the following: Fy^a, Fy^b, Jk^a, Jk^b, S, or s. Level 2 (12 profiles) were negative for at least 2 of the 6 antigens previously mentioned, 1 from each antithetical pair. Level 3 (8 profiles) was negative for at least 3 of the 6 antigens one from each antithetical pair. Level 4 was divided into 4 profiles: Fy(a-b-) Jk(b-) S-, Fy(a-b-) Jk(a-) S-, Fy(a-b-) Jk(b-) s, and Fy(a-b-) Jk(a-) s-. Levels 1-3 were calculated for all Rh. Level 4 was calculated only for rr and R_0r, as these phenotypes were most needed. When the logic workbook was opened, each unit was automatically simultaneously evaluated by 26(R_1R_1, R_2R_2) or 30(R_0r, rr) IF, AND logic formulas. Finally a basic SUM formula would provide the number of each specific profile for each of the four groups. Development time was 8 months and performed by Medical Technologists. **Results/Findings:** Once implemented, it took <5 minutes to open the spreadsheet and find the unit needed.(Table) **Conclusion:** Initial set-up time was substantial, but the ability to view current IRL inventory at-a-glance allowed for rapid unit identification, saved valuable technologist's time, and simplified inventory management. IRL inventory deficiencies or surpluses were immediately recognized and appropriate action from the blood center could be taken to balance the inventory.

Disclosure of Commercial Conflict of Interest

A. A. Delk: Nothing to disclose; R. R. Gammon: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. A. Delk: Nothing to disclose; R. R. Gammon: Nothing to disclose

AP64

Donor Supply Efficiency. . . . Just "Kitting"

G Schmitt[1] (Gschmitt@thebloodcenter.org). [1]Process Improvement, The Blood Center, New Orleans, LA, United States

Background/Case Studies: In 2011, a team at this organization was trained by an outside consultant, Patrick Maul, MT (ASCP), MBA, (Becton Dickinson-LEAN Six Sigma Black Belt) on LEAN methodologies and tools. During training, the team observed inefficiencies in distribution, storage, and use of collection supplies at donor centers and on mobile drives. **Study Design/Methods:** A kit contains every item needed to complete a process according to SOP. Over the past year, the team conducted several pilots to investigate whether kits would actually improve our processes and make them more efficient. Data was collected regarding the cost of kits, wastage, storage, and labor to determine whether they posed a cost savings to the organization. **Results/Findings:** Labor: The estimated yearly cost of labor and re-work is a minimum of $25,525. It was found that kits will save time and labor by reducing packing and set-up/breakdown time. In addition, fewer items will be received and stocked, resulting in faster and more accurate inventory counts. Storage: The estimated yearly cost of storage is $40,588. It was found that kits will require far less storage space than the supplies individually. Wastage: The estimated yearly wastage is a minimum of $21,605. It was found that kits will reduce waste by providing a standard location for supplies and reduce the chance that they will be found expired or out of temperature. It was also found that kits reduce clutter in donor collection areas. Control: An additional finding is increased control, as each item in a kit can be tracked by a master lot number. Used kits can be analyzed to investigate problems at individual donor sites or mobile drives. Cost Savings: Collected data was processed as a buy versus make decision and it was found that the purchase of kits actually project a cost savings. Yearly projected cost savings, with kits, is $11,446. **Conclusion:** The potential process improvements and cost savings were highly beneficial, so it was decided to implement the kits this year. In April 2013, all the collection centers and mobile drives began to use the kits. It is now planned to analyze the logistics of how the kits are distributed and used and what is left in the kits after use to further improve the process.

Disclosure of Commercial Conflict of Interest

G. Schmitt: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Schmitt: Nothing to disclose

Management, Finance and General Marketing

AP65

Strategic Assessment of New Projects

D A Dobler[1] (ddobler@indianablood.org). [1]Project Management, Indiana Blood Center, Indianapolis, IN, United States

Background/Case Studies: Organizations are constantly looking for ways to fulfill their missions in the most effective and cost conscious manner. Since new ideas exceed the resource and capital constraints of an organization, a formal process is needed to ensure new ideas (ie projects) are considered in a consistent and methodical manner by a group of people who understand the strategic interests of the organization and have budgetary, resource, and prioritization authority. **Study Design/Methods:** Project approval occurred in a decentralized manner with each department deciding whether a project was worth initiating. The availability of resources in other departments was given insufficient consideration. Critical feedback on project concepts was rarely sought, so project ideas were not fully developed before project work began. Not having a company-wide prioritized project list caused scheduling conflicts between project teams, since many departments had resources assigned to numerous projects. Once a project began there was no documentation that defined the project, which reinforced the concept that a project's scope, timeline, and budget could be changed as the project progressed. The project leader would inform all departments they assumed were affected by the project's existence, but sometimes there were additional departments affected by the project that were not identified upfront, which resulted in project rework or delays. **Results/Findings:** A project initiation process was developed to resolve the weaknesses identified above. This process included: 1) Creating a senior management steering committee to judge each project against the organization's strategic objectives, and budgetary and resource constraints. The steering committee is empowered to approve or reject projects, prioritize projects, or request that more information be gathered before a decision is made; 2) Developing a standard form ("project charter") by which project requests are made to the steering committee. The project charter includes background and objectives, scope, cutover schedule, justification, assumptions and risks, departments affected, cost, proposed project team members, and a section for documenting project authorization; 3) Creating a project management office (PMO) to review all documents for completeness before the documents are submitted to the steering committee, to monitor open requests, to inform all parties of steering committee decisions, and to ensure approved projects are seamlessly transitioned into an active project. The PMO is also responsible for the on-going existence and refinement of the project initiation process. **Conclusion:** The expected effect of these changes has been realized: fewer project requests, clearer project definition, a more synchronized organizational project portfolio, and a clearer understanding of the organization's priorities since each project has a numerical and distinct priority.

Disclosure of Commercial Conflict of Interest

D. A. Dobler: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. A. Dobler: Nothing to disclose

AP66

Why Don't Donors Come Back?

E P Gomez[1] (egomez@fbstib.org), D Ruiz Alderton[1], M Sedeño[1], M Sansaloni[1]. *[1]Marketing, Fundació Banc de Sang i Teixits de les Illes Balears, Palma, Spain*

Background/Case Studies: In January 2012, this regional blood bank (BB) improved the way donors were called to donate, into a more productive and

personalized system. Despite this improvement, there were still people who did not respond. In the BB quality system, one of the strategic processes is the measurement of customer satisfaction. **Study Design/Methods:** The objective was to detect how many donors do not return and find out the reasons in order to have a more effective recruitment donor system. The database was studied which selects people who had been called several times but had not returned. It was decided to survey donors who had not returned in the last 6 years or more. An on-line poll was sent to their email. 1448 surveys were sent. More than half the email addresses were erroneous or no longer active. It was estimated that approximately 700 emails were received. It was decided not to include these donors in future calls, and a new procedure was created so these donors would be withdrawn from the recruitment database. A survey.was sent **Results/Findings:** 125 surveys were collected (approx. 18% response). The reasons given for not coming back were: 57.1% for health reasons, 35.7% for lack of time (mostly mothers), 14% had moved, 4% for not knowing where to go to donate blood, and 3.1% had a bad experience at the last blood donation. 76.6% referred to wanting to continue to donate blood. We have withdrawn 19,306 people from the recruitment database, representing a saving of 35% of the cost on calls and sending SMS. About 5% of donors gave blood after receiving the survey. **Conclusion:** The withdrawal of 19,306 from the database used to recruit donors makes the recruitment program more efficient. Because of the messages sent, previous donors have returned. Health and lack of time are the main reasons for not attending the blood donation campaigns (35% of the survey respondents are women who have had children and who have less spare time). Comments in the survey helped to improve our recruitment system and the service given to blood donors.

Disclosure of Commercial Conflict of Interest

E. P. Gomez: Nothing to disclose; D. Ruiz Alderton: Nothing to disclose; M. Sansaloni: No Answer; M. Sedeño: No Answer

Disclosure of Grants Conflict of Interest

E. P. Gomez: Nothing to disclose; D. Ruiz Alderton: Nothing to disclose; M. Sansaloni: No Answer; M. Sedeño: No Answer

AP67

Improving Performance of Our Mobile Computer Aided Screening (CAS) system

S Goudar[1,2] (sgoudar@carterbloodcare.org), J M Centilli[2], D Nirode[2], C Bennett[2], M Kelkenberg[2], P Tran[2]. *[1]Corporate Services, Carter BloodCare, Bedford, TX, United States; [2]Information Services, Carter BloodCare, Bedford, TX, United States*

Background/Case Studies: In 2010, a Computer Aided Screening (CAS) system was implemented on the mobile drives to automate the donation for registration, eligibility, health history, mini-physical, and recording information about the collection. Users were often constrained by the lack of speed of the system. Many times during the drive, CAS would be abandoned due to slow performance, and the drive would be switched to manual process. CAS configuration requires a “master laptop” loaded with the donor database which functions as the main server at mobile blood drives. Master laptop has the capacity for 23 client laptops or tablets to connect wirelessly. Client laptops are used to perform actual donation process including screening and questionnaire. Because of performance constraints, CAS could be used only on smaller drives that did not exceed 4 wirelessly connected client laptops, and no more than 40 collected units. Since the CAS system offers substantial savings of decreased errors, it was important to identify and resolv performance issues. A series of tests was designed to identify root causes and establish the maximum throughput that could be expected from the CAS system. **Study Design/Methods:** In reviewing the CAS configuration, three potential areas for performance bottlenecks were identified; the master laptop itself, the wireless router, and the database. We first ran tests in a fully loaded production environment, using the actual collection staff to perform all CAS transactions, while IT staff monitored the performance of system components. The testing was designed to include registering a combination of new donors and existing donors, and performing all aspects of the registration process. During testing, the following was measured: the bandwidth used by the router, the resource utilization (CPU, hard disk, and network card) on the master laptop, and the database performance. **Results/Findings:** Testing ruled out the wireless router as a bottleneck, but there was a wait for transactions to complete on the master laptop. This latency was linked to the speed of master laptop hard disk. Additionally, it was identified that several SQL statements were not performing optimally. SQLs were analyzed and the need was identified for two additional indexes on the database. The indexeswere added, a solid-state hard disk was installed on the master laptop to minimize latency, and the testing scenarios were re-run with fully loaded 23 clients. 260 registrations were then completed in 2-hour time frame. This projects out to a maximum capacity of over 1,000 donors being registered in a single 8-9 hour drive. **Conclusion:** While operational constraints still prohibit automating every large blood drive, some of the technology was identified and removed as a bottleneck to operate CAS. When additional indexes and solid-state disk were added it resulted in the percentage of units being registered on CAS at mobile drives increasing from 60% to 72%.

Disclosure of Commercial Conflict of Interest

C. Bennett: Nothing to disclose; J. M. Centilli: Nothing to disclose; S. Goudar: Nothing to disclose; M. Kelkenberg: Nothing to disclose; D. Nirode: Nothing to disclose; P. Tran: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Bennett: Nothing to disclose; J. M. Centilli: Nothing to disclose; S. Goudar: Nothing to disclose; M. Kelkenberg: Nothing to disclose; D. Nirode: Nothing to disclose; P. Tran: Nothing to disclose

AP68

Peer-to-Peer Accountability: Developing the Ultimate Team Environment

S M Armstrong[1] (sarmstrong@mcvh-vcu.edu), K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: Enhancing patient safety beyond routine checks and balances requires accountability of staff members to each other and not just management. Peer-to-Peer Accountability (P2PA) is the ultimate indication that optimum teamwork has been achieved. Beginning January 2012, P2PA become the focus of this transfusion service, enhancing efficiency, teamwork, communication, and patient safety. **Study Design/Methods:** P2PA is a process whereby staff holds themselves accountable for processes, outcomes, and behaviors. It is important to note that if a highly effective team already exists prior to P2PA, assimilation will be successful. Implementation of the P2PA principles requires: 1) giving the staff the basic teamwork tools necessary to become a highly effective team; 2) training staff on P2PA; and 3) evaluating P2PA cultural integration. Team development tools include: teambuilding; conflict resolution; conducting crucial conversations; effective communication; and trust. These tools must be effectively integrated prior to implementing P2PA. With these tools already part of the culture, P2PA was introduced as a yearlong initiative to move beyond management-down accountability to peer-to-peer accountability. Information was distributed through the use of staff meetings, newsletters, and daily huddles to continuously discuss P2PA and its importance to the team and the patients. Through the various channels, staff was introduced into what P2PA is, and more importantly, what it is not. It was critically important that P2PA was not used as a mechanism for staff members to find fault with each other, but rather to understand that it is about each staff member looking out for each other and patients. P2PA also was used to address behavior issues (ie, tardiness, bad language, etc) by the staff who is enforcing group behavioral expectations. **Results/Findings:** Following implementation, It was studied to determine if P2PA was embraced and being used effectively by staff. Though empirical evidence was difficult to come by, the transfusion service survey information demonstrated that staff is more likely to speak up and question each other about work performance. Also, through observations, it was apparent that the staff began to ask each other if all steps in a process had been completed. This extended to observation of undesirable behaviors, when the staff took it upon themselves to let the offending employees know that their conduct was unacceptable. **Conclusion:** P2PA has been successful and the staff developed improved communication skills to discuss issues and problems, regulate inappropriate behaviors, and ask each other questions to ensure patient safety. By incorporating P2PA into the culture, the transfusion service began to change from a highly effective team, to a team with an even higher level of professionalism and effectiveness through self-regulation.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; K. Sanford: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; K. Sanford: Nothing to disclose

AP69

Transition Management: Putting Employees First During a Major Construction Project

S W Noone[1] (snoone@bloodsystems.org). [1]Administration, United Blood Services, California, Ventura, CA, United States

Background/Case Studies: In the fall of 2012, a major construction project was initiated at this facility to replace the existing component laboratory walk-in refrigerator and two walk-in freezers. Consideration of employee needs and concerns was a top priority for this project. Staff input and participation was crucial in maintaining positivity and morale during the construction phase. Several specific steps were taken before, during and after project implementation to help ensure a positive experience from employees' perspectives. **Study Design/Methods:** Pre-construction: Employees were informed of the reasons for undertaking the project at this time, and were consulted individually for their perspectives on building new walk-ins versus installing stand-alone storage units. It was made clear that their opinions mattered because they are the end-users, but that management would make the final decision. The construction company was carefully selected by the corporate Director of Facilities and Construction. During a pre-construction meeting with construction company representatives, it was emphasized that minimal staff disruption and respect for their workflow and needs were essential. A "kick-off" meeting was then held with staff to describe exactly what to expect during construction, including noise, safety concerns, how the construction area would be sectioned off, use of temporary storage units, disruption in workflow patterns, and timeframe to completion. Positive outcomes expected, including better functioning, safer storage units were stressed. All staff questions were answered. Construction phase: Open communication was maintained with the construction company throughout the project. Weekly conference calls were held with project leads to review progress and discuss any issues or concerns. Staff was asked regularly how things were going and if they were experiencing any problems. One subcontractor was removed from the job immediately by the construction company following a staff complaint regarding behavior. Post-Construction: Employees were invited to participate in a meeting with the construction crew after project completion to ensure all questions and concerns regarding the new walk-ins were addressed. Staff input was solicited for the Change Control File "Lessons Learned" report. **Results/Findings:** Despite several weeks of noise and disruption in the work area, only one complaint was voiced by staff working in the department, which was immediately addressed. Staff comments throughout the project were positive and morale remained high. **Conclusion:** Major construction projects in work areas can be a source of frustration, resulting in negativity and affecting employee morale. Involving employees in the project from start to finish, frequent and clear communication, acting immediately on employee concerns, and providing accurate information throughout the project, helped to ensure a positive experience for staff.

Disclosure of Commercial Conflict of Interest

S. W. Noone: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. W. Noone: Nothing to disclose

AP70

To Work or Not to Work, What Is the Challenge? An Explorative Qualitative Study of Sick Leave in Blood Banks

B Hermundstad[1] (brhe@broadpark.no), M Holth[2], J C Frich[3]. *[1]Department of Immunology and Transfusion Medicine, Akershus University Hospital, Lorenskog, Norway; [2]Department of Microbiology, Akershus University Hospital, Lorenskog, Norway; [3]Institute of Health and Society, University of Oslo, Oslo, Norway*

Background/Case Studies: Reducing sick leave is an important managerial challenge. Sick leave may be determined by individual or work-related factors. The individual employee's experience of the working environment is influenced by many different factors This is a study to explore blood bank employees' views about factors that influence work presence and sick leave in blood banks. **Study Design/Methods:** Two blood banks with different levels of sick leave were studied. In the first blood bank, sick leave varied from 1-4%, compared with 16-24 % in the second blood bank over the same 3-year period. Data were collectedby combining two qualitative methods: free dialogue meetings with many employees; and semi-structured interviews with eight employees. Karasek and Theorell's job demand-control-social support model has served as a framework for this study. **Results/Findings:** Collection of blood from donors is a standardized procedure. One group of employees reported that this work could be monotonous, while others found this task challenging and interesting. In the blood bank with high level of sick leave, employees reported that blood collection was less appreciated then other tasks in the blood bank. Employees said that this had a negative impact on those who worked in the blood collection department. Employees reported high and partly unclear job demands, and lack of work structure that caused an unpredictable working environment, and reduced a sense of control and autonomy. Lack of social support and feedback from the leaders of the department were reported by employees as factors that were associated with high levels of sick leave. In the blood bank with low level of sick leave figures, the management explicitly communicated that they valued all tasks. The leaders set clear demands, showed confidence in their employees' abilities to meet these demands, and appeared appreciative of their employees' efforts. Feedback was clear and timely and social interaction was good. The employees reported a high level of autonomy and responsibility. **Conclusion:** In this study, employees reported that high and unclear job demands, low sense of control and structure, and lack of social support were associated with high sick leave rates. High and explicit demands, a good sense of control, and social support was associated with low sick leave in blood banks. This study also suggests that high sick leave rates might result in less dedication among employees to their workplace, and less satisfactory interaction with blood donors. These findings are relevant for blood donor recruitment, work production, and quality in blood banks.

Disclosure of Commercial Conflict of Interest

J. C. Frich: Nothing to disclose; B. Hermundstad: Nothing to disclose; M. Holth: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. C. Frich: Nothing to disclose; B. Hermundstad: Nothing to disclose; M. Holth: Nothing to disclose

AP71

Implementation of Verax PGD Testing in a Transfusion Service

N M Nikolis[1] (nnikolis@nshs.edu), G Georgy[1], A Stein[1], R Galloway-Haskins[3], W Heaton[2], L E Logdberg[1]. *[1]Lab-Blood Bank, North Shore University Hospital, Manhasset, NY, United States; [2]Lab-Blood Bank, Hofstra North Shore-LIJ School of Medicine, Manhasset, NY, United States; [3]Research Laboratory, Feinstein Research Institute, Manhasset, NY, United States*

Background/Case Studies: In the Verax PGD multi-center study, ~1:3,000 platelet (SDP) units released as negative in BacT/Alert were found to be bacterially contaminated by PGD testing, yet no positives were identified here. The literature states that the first 24 hours of platelet transfusion associated infections (PTAIs) can be decisive in initiating sepsis treatment. This study reports an approach toward quickly identifying PTAIs, supporting the potential for rapid treatment, minimizing testing cost, and supporting staff competency. Verax PGD testing was implemented as part of our platelet transfusion workup in November 2012. **Study Design/Methods:** A team of 22 (16 full time equivalent [FTE]) were trained in the performance of the PGD test, utilizing a sample pouch from the SDP units. SOP preparation, equipment validation, training, and proficiency was established over a 4 month period prior to test launch. Testing was validated using a Zeptometrix Proficiency Panel, and ongoing proficiency maintained with enrollment in the CAP proficiency program. In addition, a procedure was developed to require collection of a valid (freshly stripped) sample on all SDP units at the time of release for transfusion. Release samples were held for 1 week post release. Reported below is the estimated start-up direct material and labor cost, as well as the annual running costs based on daily controls and testing of all transfusion reactions. Sample turnaround time averaged 2.5 hours from the time of report. **Results/Findings:** Between November 2012-April 2013, all retained samples tested negative. Start-up costs were $7,250 (21% labor, 46% test materials, and 33% other consumables) and the monthly running costs to maintain proficiency are $1,526. The total annual cost of for testing 50 reactions/year is $27,800. **Conclusion:** Operationally the time required for testing/shift is estimated to be 0.33 hour, and since testing implementation there have been no PGD positive platelet transfusion reactions. Published literature indicates that transfusion-associated sepsis develops after 1 in 25,000 platelet transfusions. It is likely to be 5 years before a positive unit is identified. Based on local medical policy input a decision upon universal testing will be made once the local bacterial positive unit rate has been identified. As strategy to minimize the risk of missing an unreported risk, this policy of testing all platelet reactions meets the need for evidentiary policy while offering timely treatment of transfusion associated sepsis.

Disclosure of Commercial Conflict of Interest

R. Galloway-Haskins: Fenwal, Grants or Research Support; G. Georgy: Nothing to disclose; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; L. E. Logdberg: Nothing to disclose; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Galloway-Haskins: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, HemCon Medical Technologies, Grants or Research Support, LightIntegra, Grants or Research Support; G. Georgy: Nothing to disclose; W. Heaton: Immunetics, Grants or Research Support, Haemonetics, Grants or Research Support, Light Integra, Grants or Research Support, Fresenius, Travel Support or Honorarium, Fenwal, Grants or Research Support; L. E. Logdberg: Nothing to disclose; N. M. Nikolis: Nothing to disclose; A. Stein: Nothing to disclose

Platelet Reactions Breakdown

Platelet Rxns	2010	2011	2012	2013 Q1
Transf'used patients	1567	1204	1209	356
Transf. Events	7127	5561	5741	1456
Reported Rxns	38	21	46	17
Febrile/Chills/ Flush	15	5	16	6
Hypotension	1	0	2	1
Allergic/Hives/ Urticaria	16	11	22	9
Other/Backpain/Nausea/SOB	6	5	6	1

AP72

Sustainability at SANBS: Harness the Power of Employee Retention for a Positive Impact on Blood Collection

I Padiachy[1] (Ivan.Padiachy@sanbs.org.za), S Wilson[1], C Fredericks[1], S Prithvi Raj[2], A S Mothokoa[1], V Reddy[2]. *[1]Human Resources, South African National Blood Service, Roodepoort, South Africa; [2]Operations, South African National Blood Service, Roodepoort, South Africa*

Background/Case Studies: SANBSThis blood service (SANBS) holds a core belief that people are critical to its success. While we empower employees, hold them accountable, and reward them appropriately, this may not be enough to ensure a competitive advantage and guarantee the retention of key skills. The retention of professional nurses presented SANBS with a challenge during 2008 when a threat emanated from state hospitals. State hospitals are this country's largest employer of professional nurses and had their own retention challenges. They embarked on a drive aimed at luring professional nurses internally and externally through attractive salary increases. The risk to SANBS was the failure to retain key staff which would cause skills depletion in donor centers with a resultant adverse impact on blood collection. Accordingly, SANBS implemented the following as part of a retention strategy for professional nurses: In April 2008 a 5% skills shortage increase on current salary was paid. The minimum payscale for professional nurses was adjusted upwards by 5% and a one-time "scarce skills retention bonus" was paid after service of 2 year. **Study Design/Methods:** This study involved a time series analysis from 2006-2012 of data extracted from the SANBS HR database pertaining to the recruitment and promotion of professional nurses. Analysed data was extracted from the SANBS Business Intelligence System regarding the impact of the retention strategies for professional nurses on blood collection. Further analysis showed the performance of blood collections to the national target. **Results/Findings:** Over the period 2006-2012 amidst this difficult and competitive environment, the study showed: a 25.19% increase in the professional nurse complement; an average of 7.69% appointment rate per annum; an average of 3.91% attrition rate per annum; an analysis of blood collection evidenced a positive impact of the retention strategies on the attainment of the annual blood collection targets, namely an average of 96.08% per annum of Actual against Target. **Conclusion:** The study demonstrated that recognizing and rewarding employees leads to higher retention rates of key staff, positively reinforces excellence in behaviour and performance, and sends a message to prospective employees that employees at SANBS are valued. Human value management and aligning employee interests to organizational goals and objectives can build long term value for donors, patients, and employees as well as contribute to sustainability.

Disclosure of Commercial Conflict of Interest

C. Fredericks: No Answer; A. S. Mothokoa: Nothing to disclose; I. Padiachy: Nothing to disclose; S. Prithvi Raj: Fenwal, Other; V. Reddy: Nothing to disclose; S. Wilson: No Answer

Disclosure of Grants Conflict of Interest

C. Fredericks: No Answer; A. S. Mothokoa: Nothing to disclose; I. Padiachy: Nothing to disclose; S. Prithvi Raj: Nothing to disclose; V. Reddy: Nothing to disclose; S. Wilson: No Answer

AP73

Biomedical Advanced Research and Development Authority (BARDA: Development of Medical Countermeasures for Chemical Agents, Radiation Exposure, and Radiation-Incident Injuries

B Tse[1] (brian.tse@hhs.gov), M Homer[1], R G Manning[1]. *[1]BARDA/ASPR/HHS, Washington, DC, United States*

Background/Case Studies: As part of its mission, the Biomedical Advanced Research and Development Authority (BARDA), within the Office of the Assistant Secretary for Preparedness and Response (ASPR) in the U.S. Department of Health and Human Services (HHS), provides an integrated, systematic approach to the development and purchase of the necessary drugs and diagnostic tools for chemical, radiological, and nuclear public-health medical emergencies. Using its advanced research and development authority, BARDA strengthens HHS efforts to bridge the valley of death funding gap that exists between the early stages of product development and the acquisition of approved or approvable medical countermeasures for the Strategic National Stockpile (SNS). BARDA advanced development programs support efforts to develop therapeutics to treat subsyndromes of acute radiation syndrome as well as decorporation agents. More recently, BARDA is beginning a new initiative to support development of blood products and blood product platform technologies, including but not limited to cellular therapies, plasma products, and platelet products, to bolster existing preparedness and response resources and tools. BARDA is also developing biodosimetry tools, both devices and assays. **Study Design/Methods:** Acute radiation syndrome (AES) is caused by exposure to high doses of damaging (ionizing) radiation. ARS includes injuries to multiple organs, hemorrhaging, infection, and suppression of the immune system's ability to fight organisms that cause infection. Because no products are licensed to treat any aspect of acute radiation syndrome, BARDA is supporting the development of products to treat bone marrow, gastrointestional, lung, and skin injury caused by radiation. BARDA is currently expanding its development portfolio to include mitigators or treatments of other radiological/nuclear incident-related medical treatment gaps, such as blood-derived products and skin substitutes for thermal burn injuries. **Results/Findings:** BARDA has re-issued a Long-range Broad Agency Announcement. The purpose is to identify innovative and promising medical countermeasure technologies for advanced development across several Chem/Bio/Rad/Nuc (CBRN) Research Areas. **Conclusion:** The BARDA mission is to ensure the availability of medical countermeasures to address public health emergencies. There are three major threat areas: 1) CBRN); 2) Pandemic Influenza; and 3) Emerging Infectious Diseases. BARDA maintains a comprehensive portfolio approach to development and acquisition of products and represents a unique niche in US Government biomedical research and development.

Disclosure of Commercial Conflict of Interest

M. Homer: Nothing to disclose; R. G. Manning: Nothing to disclose; B. Tse: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Homer: Nothing to disclose; R. G. Manning: Nothing to disclose; B. Tse: Nothing to disclose

AP74

The Operational and Managerial Challenges of Supporting a New Level II Trauma Facility

K M Unold[1] (kristy.unold@hcahealthcare.com), J Mason[1], J Nichols[1], S Renfro[1], M Cohen[1], R McVay[1]. *[1]Transfusion Services, Ocala Health, Ocala, FL, United States*

Background/Case Studies: Historically, the transfusion services here were contracted, not staffed in-house. Although faced with changes in reimbursement to hospitals and the increasingly demanding regulatory requirements which encouraged many hospitals to consider outsourcing the management of their transfusion services, the executive management of here made a different choice. It was decided to invest in the care of the community and to develop a level 2 trauma program; which included bringing the transfusion service in-house for cost savings based on economies of scale and leveraging the in-house transfusion service as an advantage. **Study Design/**

Methods: Becoming a Level 2 trauma center required reexamining the structure and strategic direction of the laboratories here. Hospital administration decided that, if the hospital assumed operational control of the transfusion services, meeting the requirements of a Level 2 trauma center would be more easily achieved than in the current two-party model. All parties agreed to realign the current two-party paradigm under one facility to ensure quality oversight and more cost effective production, and to deliver better quality services in a timely manner. The return on investment was projected to be obtained in less than one year. **Results/Findings:** During Q 1 2012, the contracted transfusion service achieved the STAT turn-around-time (TAT) goal of <45 minutes for an ABO type 28% of the time and 20.6% for the antibody screen. In the Q 1 operation, this transfusion service produced TATs for a STAT ABO type and antibody screen 63.1% and 48.9% of the time respectively. In addition to improved service delivery, becoming a Level 2 trauma center has impacted blood product usage. During the period between March April 2012, 1003 packed red blood cells (PRBCs), 124 platelet apheresis, and 85 plasma products were transfused. During this same period in 2013, 1140 PRBCs, 171 platelet apheresis, and 204 plasma products were transfused. This was an increase of 13.7% for PRBCs, 37.9% for platelet apheresis, and 140% for plasma products. **Conclusion:** The key to successful project management is to spend time in the planning cycle to ensure that the projects are aligned with the organization's goals, strategies, and key metrics; while ensuring that excellent patient care can be delivered. On December 8, 2012, after more than 40 years using a contracted services model, providing transfusion management services were begun in-house. This complete evolution in providing in-house blood banking and transfusion services occurred in just under ten weeks.

Disclosure of Commercial Conflict of Interest

M. Cohen: No Answer; J. Mason: Nothing to disclose; R. McVay: No Answer; J. Nichols: Nothing to disclose; S. Renfro: Nothing to disclose; K. M. Unold: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Cohen: No Answer; J. Mason: Nothing to disclose; R. McVay: No Answer; J. Nichols: Nothing to disclose; S. Renfro: Nothing to disclose; K. M. Unold: Nothing to disclose

AP75

Importance of a Focus Group Representing Hospital Partners When Implementing an Online Ordering System

T Sanders[1] (tcasper@cicbc.org). [1]Mississippi Valley Regional Blood Center, Springfield, IL, United States

Background/Case Studies: With increased regulatory requirements on supplier qualification, it is mandatory that a blood center achieve certain requirements as central to maintaining and securing hospital partnerships. An intuitive online ordering system is crucial to efficiently cross-check shipment accuracy and to deliver metrics documenting contract fulfillment. A focus group, representing the blood center's hospital partners can facilitate the selection of a system, communication to other hospital partners, and training requirements. **Study Design/Methods:** The online ordering system, BloodHub was implemented. The first version of the software was implemented without assistance of a focus group. A post implementation survey of hospital partners revealed satisfaction with the ease of ordering standard blood products: 98.3% rated the system was easy to use while 100% noted that the system was easy to use with job aides provided by the blood center. However, the same stellar results were not noted with other types of orders (ie antigen screen red blood cells, non-blood products, and blood center services). Because the submission process for all order types is similar, the scores were determined to be a result of the blood center's communication pertaining to the new system, including the training. Knowing enhancements to BloodHub were expected for the second version of the software, the blood center sought the feedback of a focus group anticipating increased satisfaction would be realized. **Results/Findings:** The group suggested communication improvements. With the first release, communication was to supervisors via email and contained only summary information with the assumption that it would be further communicated to staff. The group suggested two sets of communication for the second version: one with complete details emailed to the supervisor; and another with only essential information to the bench techs via fax. Also, the group was questioned about training sessions hosted by the blood center via audioconference over a 2-week time period, prior to the release of the first version of BloodHub. While the group was overwhelming satisfied with the sessions, it requested future sessions contain specific, detailed scenarios relating to the enhancements released, rather than a simple overview of the feature, and that training sessions span three weeks. **Conclusion:** Armed with the insight only an end user can provide, the blood center was able to implement an enhanced version of BloodHub with features most important to hospital partners, and in a manner that ensured applicable information was communicated to the appropriate staff and included an expanded training session. Increased hospital partner satisfaction with the implementation of subsequent versions of BloodHub was noted by the blood center. This satisfaction is a direct result of the focus group and is highly beneficial to maintaining and securing partnerships.

Disclosure of Commercial Conflict of Interest

T. Sanders: BloodHub, Travel Support or Honorarium

Disclosure of Grants Conflict of Interest

T. Sanders: Nothing to disclose

AP76

Bridging the Gap: The Success of Daily Transfusion Medicine Meetings

A L Sutton[1] (asutton@mcvh-vcu.edu), N K Case[1], K Sanford[1].
[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States

Background/Case Studies: In this academic institution, pathology residents rotate through the Transfusion Medicine (TM) laboratory here on a continual basis. It is the responsibility of both the pathology attending physicians and the TM medical laboratory scientists to teach and guide the residents through their TM rotations. The new director, appointed to TM in 2012, noted a communication gap between the laboratory, resident, and director activities. The director recognized the need to implement daily meetings that would serve as a communication bridge between the three parties. **Study Design/Methods:** The TM medical director established a set time for the daily meeting and assured that any time changes would be disseminated to all laboratory personnel and TM residents. The meetings are held in the director's office and attendance is required for all TM residents, the medical technologist in charge (MTIC) for the day, and TM supervisors. The format of the meetings is consistent and includes the following: status of the blood product inventories of both TM and the local blood supply service, scheduled liver transplant surgeries, information concerning patients who are either taking or expected to require a large amount of products, completed and in progress antibody workups, the status of all instruments and equipment, laboratory safety concerns, and scheduled apheresis patients. This time is also an opportunity for residents to ask questions and request clarification regarding TM laboratory testing and antibody workups. **Results/Findings:** The daily meetings have been held for 1 year now. The duration of each meeting is approximately 20-30 minutes. The advantages for all involved parties include: the medical director is always aware of the day to day operations of the TM laboratory; the pathology residents are not only aware of what is occurring in the TM lab, but they are also learning through the exchange of communication between all involved parties; and the TM lab personnel are able to communicate important TM information and are able to ask clarifying questions regarding difficult antibody workups and patients. In this TM laboratory, the MTIC position changes daily, and these meetings afford technologists an opportunity to sharpen their skills in preparing for meetings, and effectively communicating with other members of the healthcare team. **Conclusion:** The implementation of daily meetings is a testament of a new successful communication endeavor for TM. The meetings are an excellent method for exchanging important information and keeping all parties abreast of daily TM activities. This method of effective communication also contributes to maintaining a culture of safety for both TM and patients, and to promote excellent patient care in a complex academic medical center.

Disclosure of Commercial Conflict of Interest

N. K. Case: Nothing to disclose; K. Sanford: Nothing to disclose; A. L. Sutton: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. K. Case: Nothing to disclose; K. Sanford: Nothing to disclose; A. L. Sutton: Nothing to disclose

AP77

Further Development of Blood Services in Kyrgyzstan: Challenges and Strategies

K Sadyrbekov[1] (kubat_ks@mail.ru), B Karabaev[2], C T Smit Sibinga[3].
[1]Quality Management, Republican Blood Center, Bishkek, Kyrgyzstan; [2]Republican Blood Center, Bishkek, Kyrgyzstan; [3]Science, ID Consulting, Zuidhorn, Netherlands

Background/Case Studies: Socio-economic changes in Kyrgyzstan over the past quarter century have led to a significant decline in the performance

of the blood services, including total number of blood donors and volume of production. The increased incidence of TTI markers among the population, and a high proportion of family and primary donors, all are adding to the serious problem of supply of safe blood. **Study Design/Methods:** For the assessment of blood services, data from official statistical reports (1985-2012) were analyzed. To assess the infrastructure and sanitary conditions of the blood service organizations expert assessments were used. KAP-study was conducted in accordance with WHO guidelines. Microsoft Excel was used for statistical analysis of the data. **Results/Findings:** This country became independent in 1991 and has a population of 5.552 million. Currently there is a national blood service which includes 45 institutes, 5 blood centers, 1 area office, and 39 hospital transfusion departments. In 1985, the total number of donors was 178,232 with a donation rate of 36/1,000 population; by 2012—the number was 34,053 with a donation rate of 6/1,000 population and a decrease rate of 3 times over the first decade (during the collapse of the Soviet Union and early independence) to 5 times in 2012 **Conclusion:** The complicated social and economic situation in the country, the elimination of the old planning system for organizing blood donation, no donor promotion in the community, lack of trained professionals to promote, neglect of blood donors, as well as the lack of effective motivation of the population to voluntary non-paid donation, led to a significant decline in the operations of blood transfusion services. Inefficient organizational structure and functional subordination, poor physical infrastructure, inadequate legal framework, lack of qualified staff, and high turnover at present do hamper introduction of modern principles of accessibility, quality, and safety of transfusion services. To ensure availability, quality, and safety of blood transfusion, the following strategies need to be implemented here: 1) Improving organizational performance and functioning of the blood service; 2) Modernization of equipment and technologies and the introduction of modern methodologies; 3) Improving the legal framework and standardization of production processes in accordance with international standards; 4) Improving quality of training for all staff involved in the "vein to vein" transfusion process and enhance their scientific capacity; 5) Revival and development of voluntary non-remunerated blood donation at a national level; 6) Centralization of screening donated blood with the introduction of modern technologies and GLP; 7) Managing rational clinical use of blood components; 8) Implementation of an automated management information system at national level; and 9) Implementation of a system of monitoring and evaluation indicators in the blood service.

Disclosure of Commercial Conflict of Interest

B. Karabaev: Nothing to disclose; K. Sadyrbekov: Nothing to disclose; C. T. Smit Sibinga: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Karabaev: Nothing to disclose; K. Sadyrbekov: Nothing to disclose; C. T. Smit Sibinga: Nothing to disclose

AP78

Staffing to Workload in a Hospital-Based Immunohematology Reference Laboratory (IRL)

S Moldenhauer[1] (moldenhauer.sheila@mayo.edu), B Bendix[1], J R Stubbs[1], E K Jacob[1]. *[1]Department of Lab Medicine-Transfusion Medicine, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: Our Immunohematology Reference Laboratory (IRL) provides testing and services to support the care of patients at two large hospitals. In addition, reference work is performed for a variety of external clients. The lab wanted to determine if the current number of full time equivalent (FTE) employees aligned with current patient testing volumes, as well as to confirm that adequate staffing is in place for each shift based on workload volumes. Also, could idle time be decreased, and could workload leveling of non-stat tests be done? **Study Design/Methods:** A retrospective review of laboratory volumes, test type, time of availability for patient testing, and FTE time were analyzed based on test type averages. Current volumes and times of availability were collected for routine testing in which historical information was not available. Additionally the lab staff was asked to perform workload leveling of routine work and collect real time data as to when the work was being performed to determine the future state. Tools used to gather data included daily work tallies, timed test averages, employee schedules, and timekeeping reports. Minimum staffing levels were determined based on arrival of patient work into the lab and fixed time activities such as preventative maintenance (PM) and quality control (QC). Desired staffing levels were calculated to take in to account paid time off (PTO) used by staff throughout the year as well as by the addition of a 7/8th factor accounting for non-productive time. **Results/Findings:** The minimum FTE is 7.6 and our desired is 8.9. The lab's current FTE is 9.6. This FTE total is based on total volumes and FTE staffing for a one week period. Current FTE of 9.6 means thwere is 0.7 FTE available for added capacity. Based on when samples became available, it was determined that this lab was overstaffed on Thursday day shift and Monday evening shift, and understaffed on Tuesday day shift and Wednesday evening shift. This contradicted the assumption that Monday and Thursday were the lab's heavy workload days. As expected, there is a gap on most of the night shifts indicating being overstaffed. **Conclusion:** The Staffing to Workload analysis allowed confirmation of current FTE meets patient volume demands. The staffing schedule was altered to ensure there is coverage on the day and evening shifts that were understaffed to be better aligned with patient volume demands. No adjustments were made on the night shift as overstaffing is done purposefully to prevent technologist burnout in addition to the unpredictable test volume and a majority of the PM and QC duties performed on that shift. One year later it was possible to confirm that volumes and workload remained the same and staffing was adequate.

Disclosure of Commercial Conflict of Interest

B. Bendix: Nothing to disclose; E. K. Jacob: Nothing to disclose; S. Moldenhauer: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

B. Bendix: Nothing to disclose; E. K. Jacob: Nothing to disclose; S. Moldenhauer: Nothing to disclose; J. R. Stubbs: Nothing to disclose

AP79

Comprehensive Cost Analysis of Fresh Frozen Plasma (FFP) in an Urban Hospital Setting: An Observational and Decision Analysis

L Pizzi[1] (laura.pizzi@jefferson.edu), J K Karp[2], T Galanis[2], K M Prioli[1], J H Herman[2]. *[1]Jefferson School of Pharmacy, Philadelphia, PA, United States; [2]Thomas Jefferson University Hospital, Philadelphia, PA, United States*

Background/Case Studies: Population-adjusted plasma use is higher in the US versus other countries. Yet there are no published studies on its costs. The objective of this study is to report fresh frozen plasma (FFP) costs from the perspective of a large urban health system which both purchases plasma and produces its own. **Study Design/Methods:** Two scenarios were employed for the cost analysis: 1) FFP purchased; and 2) FFP produced. For the purchased scenario, FFP acquisition cost was based on blended rate from suppliers ($39.20); costs of thawing, labeling, and transfusion were collected from inpatient and process observations. For FFP produced, donor recruitment, donation and processing costs were collected via donor and process observations. These production costs were downward adjusted based on the relative market value of plasma versus red blood cells (RBCs) (15.1%). Plasma was assumed to be generated through whole blood donation as opposed to pheresis. Costs were computed per-unit and per-patient (assuming three units administered). Types of costs included were disposable equipment, supplies, donor giveaways, sample typing and testing, personnel time, and tasks performed. Supply costs were obtained from purchasing records. Personnel time was monetized by applying regional wage rates from the U.S. Bureau of Labor Statistics, plus 34.5% for fringe benefits. Since plasma-related adverse events (AEs) were not observable due to their infrequency, these costs were estimated using a decision analytic model informed by published literature on the rate and cost of specific events expected to prolong hospitalization (such as transfusion associated circulatory overload [TACO] and transfusion related acute lung injury [TRALI]). Fixed costs of durable (reusable) equipment were excluded from this analysis. **Results/Findings:** When FFP is purchased, per-patient total costs are $223.08; when produced, costs are $211.91 ($40.97/unit). Most production costs (61.5%) are attributable to supplies, with 38.5% due to staff time. Transfusion costs under both scenarios are $89.01/patient, mostly attributable to staff time (79.1%) for administration and monitoring. The additional cost of AEs based on decision analysis is estimated at $95.44/patient. **Conclusion:** FFP production and administration is complex and involves multiple processes, each of which requires human and material resources. Despite this, our production cost is competitive with typical purchase prices. Though transfusion requires staff time, it was found that it is not the largest cost driver relative to production and AE treatment. Major plasma-related AEs are uncommon, but costly.

Disclosure of Commercial Conflict of Interest

T. Galanis: Nothing to disclose; J. H. Herman: Nothing to disclose; J. K. Karp: Nothing to disclose; L. Pizzi: Nothing to disclose; K. M. Prioli: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Galanis: CSL Behring, Grants or Research Support; J. H. Herman: Nothing to disclose; J. K. Karp: CSL Behring, Travel Support or Honorarium; L. Pizzi: CSL Behring, Grants or Research Support; K. M. Prioli: CSL Behring, Grants or Research Support, Watson, Grants or Research Support

AP80

Employee Opinion Survey Improves Job Satisfaction and Reduces Turnover

N M Koblesky[1] (nkoblesky@rrvbc.org). [1]Administrative, Rock River Valley Blood Center, Rockford, IL, United States

Background/Case Studies: This blood center launched its first employee opinion survey in 2006 and has been conducting surveys every two years since. The purpose has been to gauge employee's satisfaction with particular emphasis on determining the overall climate of the organization, assessing which factors are contributing to employee satisfaction and which are detracting from their satisfaction. **Study Design/Methods:** The survey is outsourced and conducted by a professional employee engagement research firm. The survey is voluntary and the results are confidential. The survey results are placed into one of four factors that sustain employee involvement. The four factors are: Culture, Commitment, Communications, and Compensation. Following each survey, an executive summary and extensive analysis of the results is provided. This analysis provides industry and national benchmarks, along with areas where employees are most satisfied, and areas employees are least satisfied. This allows the senior management team to work on meaningful action plans to help sustain and improve overall levels of job satisfaction. Key to the survey's success is taking meaningful action following analysis of the results. Actions taken include but are not limited to: various recognition programs; improved communication methods ie monthly newsletters; cross departmental teams; quarterly CEO meetings; wellness initiatives ie health fair, MP screens, health club reimbursements, computer purchase and tuition reimbursement programs; improved supervisor accountability; employee training and team building; improved policies; updated job descriptions; and revising the performance management process. **Results/Findings:** See Table for job satisifaction and turnover results. Survey Cost: $4,775.00 every 2 years; COT (cost of turnover) for collection staff: $10,877. 18 per hire 2012 COT: $97,894.62 (9 collection positions filled); 2011 COT: $87,017.44 (8 collection positions filled) **Conclusion:** Conducting employee opinion surveys has proven to be a worthwhile investment. Taking meaningful action based on the survey results has gives the opportunity to increase job satisfaction and in effect reduce employee turnover. While turnover didn't decrease following the first survey, job satisfaction increased by 15 points. The long-term results reflect improvement in both job satisfaction and turnover. These efforts have a positive financial impact on the bottom-line and create a more positive, healthy, and productive work environment.

Disclosure of Commercial Conflict of Interest

N. M. Koblesky: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. M. Koblesky: Nothing to disclose

Employee Opinion Survey-Satisifaction and Turnover Results

Survey Year	Extremely/Very Satisfied	Participation Rate	Industry Norm	US Norm	Turnover Rate
2006	50%	70%	61%	60%	22%
2008	66%	80%	56%	56%	23%
2010	66%	90%	54%	57%	18%
2012	71%	96%	55%	57%	13%

Product Manufacturing

AP81

LEAN Laboratory Whole Blood (WB) Production

S Bassett[1] (sbassett@bloodsystems.org). [1]Blood Systems, Scottsdale, AZ, United States

Background/Case Studies: Separating whole blood (WB) into red blood cells (RBCs) and plasma involves the same process, but work staff performs it differently. A measurement system does not exist to assess how long it takes to process WB. **Study Design/Methods:** Six Sigma and LEAN Manufacturing tools were used to standardize the process of separating WB into RBCs and plasma. Six Sigma is a phased-analysis process known as DMAIC: Define, Measure, Analyze, Improve, and Control. LEAN manufacturing is a systematic approach for reducing production waste. Define: We first defined the problem with a project charter. The process was defined with a Supplier, Inputs, Process, Outputs, and Customer (SIPOC) diagram and process flow chart. Measure: The process was measured and it was determined what data to collect. A video was takenof staff performing the process and their total touch time was measured. A data collection form was used to manually track products. The data collection form included the start time of each process step. The process time was calculated at each step by using a Microsoft Excel Spreadsheet. Analyze: Data were analyzed with Sigma XL and Minitab software. The mean and standard deviation was calculated for each process step. The statistical t-test or Analysis of Variance (ANOVA) method was used to compare process times and look for significant differences. Improve: LEAN Manufacturing tools were used to improve the process. A team of staff to brainstormed process improvements, based on the identified process variations and root causes of waste. The recommended improvements were then categorized based on cost and ease of implementation. Control: The recommended improvements were implemented. Data were then collected post-implementation, using the same time collection form. Data collected before and after implementation were compared and the statistical t test was used to determine if the improvements made a significant difference in the process before and after improvements. **Results/Findings:** Total process time at each location was compared, as well as the time of each sub process by location. ANOVA revealed a statistically significant difference between locations overall ($p < 0.00001$). The mean total process time varied from one and a half, up to several hours. The standard deviation for total process time varied from about a half hour, up to several hours. ANOVA also revealed that each sub process differed by location significantly ($p < 0.00001$). The mean for each sub process time varied between locations, usually on the order of minutes, but sometimes on the order of hours. **Conclusion:** A team of staff from each location was assembled and trained on LEAN Manufacturing. Process time was reduced by improving the work area and balancing the production flow. The same standardized work was implemented at each location.

Disclosure of Commercial Conflict of Interest

S. Bassett: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Bassett: Nothing to disclose

AP82

Managing Growth in a Cellular GMP Facility

C Silva-Lentz[1] (clsilva@txch.org), A Gee[1]. *[1]CAGT, Baylor College of Medicine, Houston, TX, United States*

Background/Case Studies: Clinical uses of cellular therapies have expanded in the last few years. Combined with development of regulations by the FDA, and of standards by accrediting organizations this has stretched the resources of most cellular therapy facilities. As the field grows most facilities will face these challenges. **Study Design/Methods:** During the last 15 years the good manufacturing processes (GMP) here has grown from 2 rooms to 22 clean suites, and the number of product types has increased from 1 to >50. Here major issues are described that must be addressed to achieve expansion while operating within the regulations. **Results/Findings:** A primary obstacle to expansion is availability of space which can be solved only by the institution; however, when space is made available, it is important to determine the type of manufacturing environment required to meet FDA regulations. A misconception is that Class 10,000 clean rooms are required, resulting in over-engineering at considerable cost and high operational overhead. Two distinct environments were created here created here. The first, for vector manufacturing, was constructed with clean and dirty corridors and high level air handling. In contrast, the Cell Processing Facility (CPF) is a single corridor design to maximize space and accommodate the non-campaign style of manufacturing. CPF operations were developed to FDA GTP and GMP regulations for Phase 1 manufacturing, rather than the more restrictive European standards. This permits more flexibility and reduces costs. Suites that operate as part of either the CPF or the Vector Production Facility (VPF) were designed for the changing workload. Facility qualification was performed by the QC staff, although external companies were used for the HVAC system. Cleaning and environmental monitoring were redesigned for more efficient use of staff and to address different traffic patterns in the CPF and VPF. Manufacturing staff were not increased until justified by workload. This permitted controlled growth. Suites were opened gradually to allow for training of new staff. New changeover procedures

permitted production of more than one product /suite. The budget has increased during expansion and has been met by incorporating manufacturing, testing, and infrastructure costs into grant applications. Contract manufacturing is performed for the National Heart, Lung and Blood Institute's Production Assistance for Cellular Therapy (PACT) program and commercial companies. Although lucrative, this requires development of policies that detail whether space is leased to the company, and whether company staff performs manufacturing tasks, or control remains with the facility. **Conclusion:** This experience in managing growth may provide valuable information for other institutions anticipating expansion.

Disclosure of Commercial Conflict of Interest

A. Gee: Nothing to disclose; C. Silva-Lentz: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. Gee: Nothing to disclose; C. Silva-Lentz: Nothing to disclose

AP83

Switching Sterility Testing Methods to Reduce False Positive Rates and Associated Costs

J Powelson[1] (jpowelson@cbckc.org), B Anthis[1], A Hamel[1], J Comes[2], K B Peck[3]. *[1]Product Testing Lab, Community Blood Center of Kansas City, Kansas City, MO, United States; [2]Training & Development, Community Blood Center of Kansas City, Kansas City, MO, United States; [3]Administration, Community Blood Center of Kansas City, Kansas City, MO, United States*

Background/Case Studies: Testing was implemented for bacterial contamination of platelets using platform A. After experiencing equipment and IT interface malfunctions and specificity concerns, we switched to platform B. The experience was reviewed using platform A over a recent 2.5 year time span and it was found that 49 platelets and co-components were destroyed because of false positive sterility results. Each false positive resulted in product quarantine, hospital notification, investigation and follow up, and root cause events leading to increased direct and indirect costs to the center and hospitals. **Study Design/Methods:** After reviewing the data, it was determined that corrective action efforts made between 2010 and early 2012 reduced the false positive rates (Table). However, sporadic performance issues remained (not quantified in this report). In March 2012, it was decided to change the sterility testing methodology to platform B. Extensive validation case studies were performed to ensure sensitivity. Platform B was implemented July 9, 2012 without incident. **Results/Findings:** See Table. **Conclusion:** The sterility testing methodology was changed from A to B in July 2012. Product validation, QC, and proficiency testing indicate similar test sensitivity. Since changing to platform B in July 2012, the false positive rate decreased as did service issues resulting in direct and, more importantly, indirect time and cost savings.

Disclosure of Commercial Conflict of Interest

B. Anthis: Nothing to disclose; J. Comes: Nothing to disclose; A. Hamel: Nothing to disclose; K. B. Peck: Nothing to disclose; J. Powelson: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Anthis: Nothing to disclose; J. Comes: Nothing to disclose; A. Hamel: Nothing to disclose; K. B. Peck: Nothing to disclose; J. Powelson: Nothing to disclose

Summary of False Positive and associated component and co-component direct costs:

Date Range	False Positives	Number of Products Destroyed	Associated Cost	Corrective Actions
7/1/09 to 6/30/10	11	20	$9360	
7/1/10 to 6/30/11	11	22	$9312	Platform A: Process changes implemented
7/1/11 to 6/30/12	3	7	$2820	Platform A: Additional process changes implemented
7/1/12 to Current	1	1	$540	Changed Sterility Method from A to B July 2012

AP84

The Impact of Nucleated Red Blood Cells (RBC) on Percent Total Nucleated Cell Recovery Post-Thaw in Cord Blood (CB) Units: A Retrospective Study in 2013

A Lambert[1] (allambert@lifesouth.org), N Eckert[4], J Joye[2], J Merayo-Rodriguez[3]. *[1]Cellular Therapy, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States; [2]Quality Control Laboratory, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States; [3]Medical Office, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States; [4]Executive Office, LifeSouth Community Blood Centers, Inc., Gainesville, FL, United States*

Background/Case Studies: In October 2009, the FDA published *Guidance for Industry: Minimally Manipulated, Unrelated Allogeneic Placental/ Umbilical Cord Blood Intended for Hematopoietic Reconstitution for Specified Indications1* which provides guidelines for the completion and submission of the application for licensure for cord blood (CB). Detailed in this guidance are acceptable criteria for post-thaw total nucleated cell (TNC) recovery which is ≥70%. Nucleated red blood cells (nRBCs) do not have regenerative potential in hematopoietic transplantation, thus the loss of nRBCs during the thawing process would not impact recipient outcomes but may impact the required TNC percent recovery post-thaw2. **Study Design/ Methods:** Twenty CB units were thawed and washed for this study. The TNC percent recovery was analyzed for each of the 20 units. The TNC percent recovery was also calculated after removing the number of nRBCs from the initial TNC count. The percent recoveries were evaluated to determine if nRBCs impacted the % recovery on thawed and washed CBUs. **Results/ Findings:** Significance is shown in the average % recovery for TNC in which the nRBCs were removed from the cell count and the % recovery in which nRBCs were removed from the cell count. The average TNC percent recovery of 20 washed units including nRBCs was found to be 76.76% while the average TNC percent recovery for the same washed units removing nRBCs from the cell count was found to be 82.95%. Two CB units of significance were the two units having the highest and lowest post-processing nRBC %s. The raw data for these CB units are shown in the Table. Unit 1 had the highest % of nRBCs post-processing and did not meet the acceptable criteria of ≥70% TNC recovery before the nRBCs were taken into consideration. After the nRBCs were removed from the post-processing TNC, Unit 1 met the % recovery criteria. Unit 2 demonstrates that with smaller percentages of nRBCs post-processing, the % recovery of TNC is impacted very minimally. **Conclusion:** nRBCs should be taken into consideration when assessing the minimum criteria for TNC % recovery post-wash for stability studies in an institution. The loss of nRBCs during the thaw and wash procedure could significantly impact the resulting TNC % recovery depending on how high the concentration in the unit of nRBCs. Further study is recommended to assess the % recovery of nRBCs after the thaw and wash procedure to determine the exact amount of nRBC loss during the procedure.

1 (Administration, 2009)

2 (V Laroche, 2005)

Disclosure of Commercial Conflict of Interest

N. Eckert: Nothing to disclose; J. Joye: Nothing to disclose; A. Lambert: Nothing to disclose; J. Merayo-Rodriguez: Nothing to disclose

Disclosure of Grants Conflict of Interest

N. Eckert: Nothing to disclose; J. Joye: Nothing to disclose; A. Lambert: Nothing to disclose; J. Merayo-Rodriguez: Nothing to disclose

TABLE 2

	Post-Processing TNC (Absolute Count)	Post-Wash TNC (Absolute Count)	Uncorrected TNC % Recovery	nRBC Post-Processing %	Corrected TNC % Recovery
Unit 1	1.09E + 09	6.43E + 08	59.20%	28.8	83.15%
Unit 2	1.21E + 09	9.68E + 08	80.22%	2.61	82.37%

AP85

Barrier Improvement in Cryopreservation Bags with Nanotechnology Application

D Barocas[1] (dana.barocas@haemopharm.it), B Cadorin[1]. *[1]Haemopharm Healthcare, Milan, Italy*

Background/Case Studies: Nanotechnology is a powerful method for molecular reenginering of materials to impart unique surface properties without affecting the bulk properties in cryopreservation state, becoming an important tool also for the life sciences field. The method of processing and treating the plastic medical material destined for the production of cryopreservation bags with nanotechnology is aimed at giving additional protection to the bag's outer surface and to the bag's contents (blood, tissue, etc), as the treatment is destined, according to the preliminary evaluation, to give additional barrier features to the bag. The cryopreservation bag has a secondary protection (double bag, plastic material) and the contents are therefore protected, since the cryopreservation procedure foresees the conservation of the bags in liquid Nitrogen at minus 196 C for a very long time. Therefore, the nanotechnology treatment gives the bag an added value which consists of an additional barrier and therefore improved protection. **Study Design/Methods:** Bags have been treated with a nanotechnology method that involves the deposition of ions on the external surface that are able to interact with the molecules of the plastic material, creating a barrier. The ions derive from silica-like precursors, conferring to the bags the properties of a "glass-like" barrier and completely impeding the permeation of nitrogen (liquid or vapour) and other elements. These ions are totally non-reactive towards the inner contents of the bag since they remain on the external surface and do not pass through the plastic material. A few gas permeation tests (O2, CO2, N) have been performed in order to compare the permeability rate of standard cryopreservation bags (with secondary bag), and the same bags treated with the Nanotechnology. **Results/Findings:** Studies are still ongoing and conclusions at this point are positive, as up to this stage it has been found that nanotechnology treatment of the bag creates an additional safety barrier to its external surface. The silica-like effect confers to the bag a strong impermeability and completely isolates the contents of the bag from the outer elements of air, liquid and gas. This new feature creates an additional protection to the bag that already has a double bag system protection. Stronger barrier features have been confirmed after permeability tests were performed on the treated bags for a permeability rate of O2, CO2,N, in comparison to the parameters found on non-treated bags. **Conclusion:** Treatment of plastic medical bags with nanotechnology improves the material performance by creating a barrier of protection to the bag and an adequate isolation between the inner contents of the bag and the external atmosphere.

Disclosure of Commercial Conflict of Interest

D. Barocas: Haemopharm Healthcare, Ownership or Partnership, Siad Healthcare, Ownership or Partnership; B. Cadorin: Nothing to disclose

Disclosure of Grants Conflict of Interest

D. Barocas: Haemopharm Healthcare, Ownership or Partnership, Siad Healthcare, Ownership or Partnership; B. Cadorin: Nothing to disclose

AP86

Do It Yourself Trima Maintenance: Is it Worth It?

L H Parks[1] (lhp@cbckc.org), R Karre[1], T A Frame-Brown[2], K B Peck[3], J E Menitove[3]. *[1]BioMedical Services, Community Blood Center of Kansas City, Kansas City, MO, United States; [2]Donor Services Administration, Community Blood Center of Kansas City, Kansas City, MO, United States; [3]Administration, Community Blood Center of Kansas City, Kansas City, MO, United States*

Background/Case Studies: Manufacturer-trained technicians at this mid-size blood center maintain a complement of 23 Trima blood collection devices at five locations. Tracking parts costs for budgeting is standard practice. In 2010, the center began systematically collecting additional maintenance data, including internal labor costs, training, and machine down time in order to assess the total cost of the program. **Study Design/Methods:** Data were reviewed from 2010 to present. Actual parts costs and internal labor hours, at a nominal rate of $40 per hour, provide a direct comparison to manufacturer's contract services. Machine down time was calculated based on the value of products that could not be collected, eg the number of operating hours out of service/2 hour per PLT x cost of PLT unit. Although arbitrary, this quantifies the so-called "soft costs" incurred: rescheduling donor appointments, inconvenience to the donor, product loss during collection, and donor deferral due to blood spills. **Results/Findings:** See Table. Reduced annual cost: Median cost per Trima was $2494 for parts and labor ($1749 parts; ~18.8 hour technician labor @ $40/hr = $745). Including training (1 tech per year @ $8-9000), median annual cost is 51% of a standard maintenance contract (~$5900). Quicker response time and turn-around time: In-house technicians respond within a few hours and often provide same-day service. Although major repairs may be delayed due to replacement parts not kept in stock, most repairs are completed within 1-2 days. Interdepartmental collaboration: To reduce potential damage from preventable blood spills, technicians: 1) recommended a documented second person visual inspection before use; and 2) implemented a full machine evaluation after every blood spill. Scheduling efficiencies: Preventive maintenance visits can be scheduled after collection hours or when less disruptive; Measuring indirect soft costs facilitated program improvement: achieving savings here is an ongoing effort. **Conclusion:** The data demonstrate that blood centers using qualified in-house technicians to maintain Trima blood collection devices can save up to 51% ($69,596) of standard maintenance contracts and benefit from quicker response time and lower machine down time, which lowers "soft cost" expenses. Monitoring down time was found to be a useful tool to gauge program effectiveness.

Disclosure of Commercial Conflict of Interest

T. A. Frame-Brown: Nothing to disclose; R. Karre: Nothing to disclose; J. E. Menitove: Nothing to disclose; L. H. Parks: Nothing to disclose; K. B. Peck: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. A. Frame-Brown: Nothing to disclose; R. Karre: Nothing to disclose; J. E. Menitove: Nothing to disclose; L. H. Parks: Nothing to disclose; K. B. Peck: Nothing to disclose

Year	Parts	Labor	Training (1 tech/yr)	Parts, Labor & Training	PM Contract	Overall Savings
2010	S33,989	$17,908	S8,000	$59,897	$135,700	S75,803 / 56%
2011	S46,462	$16,350	S8,500	$71,312	$135,700	S64,388 / 47%
2012	S33,115	$23,757	S9,000	$65,872	$135,700	S69,823 / 51%
2013 (Jan-Apr)	S23,777	S4,840	S3,000	$31,617	S45,233	S13,616 / 30%
Median	S40,225	$17,129	S8,750	$66,104	$135,700	S69,596 / 51%

Quality Management

AP87

Alternative Method for Using Proficiency Samples for Bacterial Detection Testing

K Jones[1] (Keith.Jones@redcross.org), A Norwalt[1], J S Webb[1], M Curran[2], D K Gozelanczyk[3]. *[1]Manufacturing, Process Design, American Red Cross, Washington, DC, United States; [2]Quality Control Laboratory, American Red Cross, Philadelphia, PA, United States; [3]Quality Control Laboratory, American Red Cross, Baltimore, MD, United States*

Background/Case Studies: The Clinical Laboratory Improvement Amendment (CLIA) requires assessment of staff competency at least semiannually during the first year of employment and annually thereafter. Section (8)(v) requires assessment of test performance by testing previously analyzed specimens, internal blind testing samples, or external proficiency testing samples. For bacterial detection testing in blood centers, testing previously analyzed specimens presents unique challenges. For sampling previously negative products, there is greater chance for contamination due to multiple entries into the storage bag and the possibility of initially low concentrations of bacteria below the delectability threshold growing bacteria after day 5. For sampling previously positive products, there is concern regarding the use of live bacteria and that these products may produce negative cultures as a result of diminishing bacteria concentrations over time. The College of American Pathologists (CAP) samples have insufficient volume to be used by all staff in large blood centers. Because of these concerns and limitations, an alternative method to objectively assess aseptic techniques of sampling and inoculation utilized during the bacterial detection testing process was

developed using simulated negative products. **Study Design/Methods:** A set of three bags was prepared by sterile docking bags of sterile saline solution onto 300 mL transfer packs and transferring 75-100 mL of sterile saline solution into each. The three labeled transfer packs were managed the same as apheresis or pooled platelets and handed off to staff. The bacterial detection testing process was initiated by sampling the simulated platelets, loading the inoculated culture bottles into the bioMérieux® BacT Alert System, and monitoring for bacterial growth for five days. **Results/Findings:** The American Red Cross (ARC) Philadelphia and Baltimore facilities assessed 64 staff performing the bacterial detection testing process on a total of 192 simulated negative products. There were no false positives reported during this study. **Conclusion:** This method for assessing staff competency for the aseptic techniques of sampling and inoculation during the bacterial detection testing process has proven to be a successful alternative to using previously analyzed specimens and the bacterial detection CAP surveys.

Disclosure of Commercial Conflict of Interest

M. Curran: Nothing to disclose; D. K. Gozelanczyk: Nothing to disclose; K. Jones: Nothing to disclose; A. Norwalt: Nothing to disclose; J. S. Webb: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Curran: Nothing to disclose; D. K. Gozelanczyk: Nothing to disclose; K. Jones: Nothing to disclose; A. Norwalt: Nothing to disclose; J. S. Webb: Nothing to disclose

Results

ARC	Year	# of Staff Assessed	# of Simulated Negative Products Tested	# of False Positives
Philadelphia	2011	23	69	0
Philadelphia	2012	29	87	0
Baltimore	2012	12	36	0
Totals		64	192	0

AP88

Perceived Benefits and Barriers to Participation in the National Healthcare Safety Network (NHSN) Hemovigilance Program: A Survey of New England Region Blood Banks and Transfusion Services

L Uhl[1] (luhl@bidmc.harvard.edu), C Andrzejewski[2], P T Pisciotto[3], M Cumming[4], A Osinski[4], A DeMaria[4]. *[1]Pathology, Beth Israel Deaconess Medical Center, Boston, MA, United States; [2]Transfusion Medicine Services, Baystate Health System, Springfield, MA, United States; [3]American Red Cross Blood Services, Dedham, MA, United States; [4]Division of Epidemiology and Immunization, Massachusetts Department of Public Health, Boston, MA, United States*

Background/Case Studies: In February 2010, the first component of the US Biovigilance Network was launched, to conduct surveillance for adverse events related to transfusion. This Hemovigilance Module is part of the Centers for Disease Control and Prevention [CDC] National Healthcare Safety Network [NHSN], an internet-based surveillance program to monitor patient safety. As of April 2013, more than 140 institutions have enrolled in the NHSN hemovigilance program [NHSN-HV], yet less than 60 are actively reporting data into the system. Because an internet-based surveillance system offers a mechanism to report state-required, transfusion-related activity and adverse events, it was sought to examine current knowledge of NHSN and NHSN-HV, perceived barriers to participation, and perceived benefits of a centralized internet-based reporting system. **Study Design/Methods:** A web-based survey created using SurveyMonkey™ free online software was distributed to 212 New England institutions (ME, NH, VT, CT, and MA). The survey content included 28 questions designed to assess knowledge of and participation in NHSN-HV as well as to evaluate the level of interest in modification of the existing paper-based method of mandatory transfusion statistics reporting in MA. Responses were anonymously collected between November 26-December 29, 2012 and subsequently exported and analyzed using Microsoft Excel® and SAS® 9.3 software for collation and descriptive analysis. **Results/Findings:** The survey yielded a 25% (105 responses/425 requests) response rate representing community (CH), tertiary care (TH), and critical access (CA) hospitals. Overall awareness of NHSN-HV was high (80%), although higher among the TH versus CH (92% versus 77%). Only 9% of respondents indicated current participation in NHSN-HV. Responses revealed 18% were planning to participate, 53% were not sure, and 22% were not participating. Reasons for not participating included: labor intensity of use, perceived duplication of efforts, and uncertainty about how to initiate participation. Information security was not a concern among respondents. Interestingly, 53% of MA respondents indicated a willingness to participate in the NHSN-HV in lieu of the current state paper-based reporting mechanism. Reported perceived benefits included access to benchmarking of incidents as well as tracking of transfusion reactions and product wastage. **Conclusion:** Overall, there is high awareness of NHSN-HV in NE. Barriers to participation include perceived labor effort and duplication of current reporting programs for transfusion-related activities. However, MA survey respondents are in favor of reporting into NHSN-HV if it supplants the current state reporting process.

Disclosure of Commercial Conflict of Interest

C. Andrzejewski: Nothing to disclose; M. Cumming: Nothing to disclose; A. DeMaria: Nothing to disclose; A. Osinski: Nothing to disclose; P. T. Pisciotto: Nothing to disclose; L. Uhl: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Andrzejewski: Nothing to disclose; M. Cumming: Nothing to disclose; A. DeMaria: Nothing to disclose; A. Osinski: Nothing to disclose; P. T. Pisciotto: Nothing to disclose; L. Uhl: Nothing to disclose

AP89

Design of a Microsoft Access Application to Track Apheresis Platelet Statistical Sampling in the American Red Cross (ARC)

K Jones[1] (Keith.Jones@redcross.org), J S Webb[1], J Disciullo[2], T Sinski[2]. *[1]Information Delivery and Automation, American National Red Cross, Washington, DC, United States; [2]Manufacturing, Process Design, American National Red Cross, Washington, DC, United States*

Background/Case Studies: To facilitate apheresis platelet statistical sampling across the country, the American Red Cross (ARC) designed, developed, and validated a Microsoft (MS) Access application. There are 36 manufacturing facilities within the ARC that are grouped into 8 geographical Quality Control monitoring areas (QCMA) for the purpose of performing and monitoring of apheresis quality control. Each QCMA has access to the application. The application is an effective tool to track apheresis platelet statistical sampling in the American Red Cross. **Study Design/Methods:** The application monitors apheresis Quality Control (QC) sampling throughout the month, tracks test results and the apheresis instruments used, and generates summary reports. The basic functionality of the application includes: 1) menu-driven design allowing staff to enter test data for residual white blood cell counting and pH testing; 2) performing a second person review electronically; 3) entering deviation numbers related to specific test and sampling failures; and 4) generating reports, including reports that indicate if sampling criteria are not met or exceeded. User access and security levels were established to limit staff data entry capabilities; however, staff may view data entered by any user, including viewing any associated reports. **Results/Findings:** The application uses a single national database which permits facilities to manage their data and meet the sampling and testing requirements each month within the specific QCMA. Data is linked to each associated apheresis instrument allowing for a collection device summary report that lists the device identification, total tests for the device during report date range, and the date a sample from the device was last tested. This information is organized by test type, device type, and facility. A monthly summary report displays information for all of the manufacturing facilities within each QCMA. It provides a count of the number of samples that were 1) tested, 2) passed testing, and 3) failed testing. In addition, it provides the percentage pass rate for each manufacturing facility, as well as for the entire QCMA. **Conclusion:** The application provides the ARC with an effective tool to track apheresis platelet statistical sampling and quickly evaluate QC results and identify failures. The application also enhances the ARC's capability to manage workflow, collection device equipment, and staff resources. It provides enhanced visibility across facilities, allowing greater oversight and fostering standardization. The ARC has also been able to identify opportunities for process improvement and to enhance communications.

Disclosure of Commercial Conflict of Interest

J. Disciullo: Nothing to disclose; K. Jones: Nothing to disclose; T. Sinski: Nothing to disclose; J. S. Webb: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Disciullo: Nothing to disclose; K. Jones: Nothing to disclose; T. Sinski: Nothing to disclose; J. S. Webb: Nothing to disclose

AP90

The Blood Bank QuickView Page: Conducting a Phone Study to Assess Its Effectiveness

A L Sutton[1] (asutton@mcvh-vcu.edu), A Hord[1], S M Armstrong[1], K Sanford[1]. *[1]Pathology-Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: The Transfusion Medicine (TM) lab here has recently built and implemented a new functionality within the Hospital Information System (HIS). The application is called the Blood Bank QuickView page. This page grants access of all blood bank results and information on any patient to physicians and other members of the healthcare team. A phone study was conducted to assess the effectiveness of the QuickView Page. This study looked at the total number of phone calls as well as the nature of the phone call, ie could the question posed during the phone call be answered by accessing the QuickView page. **Study Design/Methods:** Prior to QuickView implementation, lab personnel were trained on the information available on the QuickView page and how to access it. Personnel on all three shifts participated in data collection for this study. When a phone call was answered, the person receiving the phone call was asked to document which nursing unit called and whether or not the phone call could be answered by the QuickView page. There was also a comment section for any additional notes that the caller deemed important. The staff was instructed to record only those phone calls from individuals with HIS access, as others such as patients are not eligible for this study. Data for the pre-QuickView phone study was collected for two weeks, followed by training of the nursing units. Once all nursing units were trained, QuickView went live. The post-QuickView phone study commenced one year after the QuickView go-live date. **Results/Findings:** During the 2 week pre-implementation data collection period, TM received 437 study eligible phone calls. Of the total phone calls, 248 could be answered by the QuickView page, or 57% of the total phone calls. During the 2 week post-implementation data collection period, a total of 211 study eligible phone calls were received, and of these, 51 phone calls or 24% of the total phone calls, could be answered by the QuickView page. These results indicate a 52% decrease in total phone calls to TM. **Conclusion:** The QuickView page has not only decreased the number of incoming phone calls to TM, but it has also made blood bank information easily accessible to the healthcare team. This is evident by the decrease in the number of phone calls that were received that could be answered by the QuickView page. Not only has the QuickView page contributed to patient safety by affording healthcare personnel quick access to accurate TM information, but the decrease in phone calls may result in a decrease in TM testing errors. Further studies will be conducted to assess the relationship between phone calls and testing errors. The success of the QuickView page has not only benefitted TM, but every individual who now has the capability to obtain information conveniently.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; A. Hord: Nothing to disclose; K. Sanford: Nothing to disclose; A. L. Sutton: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; A. Hord: Nothing to disclose; K. Sanford: Nothing to disclose; A. L. Sutton: Nothing to disclose

AP91

ABO/Rh Testing Mislabeled Specimens Reveals Additional WBITs

K Igo[1] (igok@ccf.org), C McGrath[1], P I Figueroa[1]. *[1]Transfusion Medicine, Cleveland Clinic, Cleveland, OH, United States*

Background/Case Studies: Wrong blood in tube errors (WBITs) are errors wherein the wrong person's name and identifying information is on a sample containing another individual's blood (misidentification). When one error in labeling a sample has been made, there is more likely to be another (eg WBIT). [Lumadue JA, Boyd JS, Ness PM. *Transfusion* 1997;37:1169-72.] **Study Design/Methods:** The Transfusion Service (TS) began forward ABO/Rh testing mislabeled samples in December 2009 in order to detect misidentification errors (WBITs) if also present. Mislabeled samples are reported immediately to the clinical service and in the hospital's error management system. A mislabeled sample is any sample whose label does not meet acceptability criteria: unlabeled, problems with the patient's name or medical record number, mismatch with the requisition, double labeled, missing phlebotomist's identity or draw date. WBIT errors are detected several ways: during routine testing, when a clinical service calls to notify the TS that a sample may have been incorrectly labeled, comparison to an independently drawn confirmatory sample, and by testing mislabeled samples. **Results/Findings:** Since the ABO/Rh testing policy was implemented, 364,569 samples have been submitted to the TS. Of these, 47 were found to be WBITs (including those derived from testing mislabeled samples), for an overall WBIT rate of 0.013%. In that same time period, 694 mislabeled samples were tested for ABO/Rh and yielded 29 WBITs, for a WBIT rate of 4.18% in mislabeled samples. Mislabeled samples had a WBIT rate more than 300 times greater than the overall rate. 29 WBITs were identified by testing mislabeled samples; 18 WBITs were identified by other methods (eg routine testing). Testing mislabeled samples revealed more WBIT errors than the other methods combined. **Conclusion:** This experience in implementing a policy of ABO/Rh testing all mislabeled samples demonstrates the importance of this testing, by revealing 29 WBITs that would otherwise have gone undetected. This finding has reinforced to the technical staff why it is critical that the TS enforce its labeling requirements and is now part of routine training. Had the TS not performed this testing, the clinical service would have been unaware of the magnitude of their labeling error and would also have been unaware of the need to investigate samples drawn at the same time in case of a reciprocal error.

Disclosure of Commercial Conflict of Interest

P. I. Figueroa: No Answer; K. Igo: Nothing to disclose; C. McGrath: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. I. Figueroa: No Answer; K. Igo: Nothing to disclose; C. McGrath: Nothing to disclose

AP92

Implementation of a Binomial Distribution Sampling Strategy for Quality Control (QC) Testing of Residual Leukocytes and pH in Apheresis Platelets

K Jones[1] (Keith.Jones@redcross.org), F Kugele[1], L Kline[2]. *[1]Manufacturing, Process Design, American Red Cross, Washington, DC, United States; [2]Holland Laboratory, American Red Cross, Rockville, MD, United States*

Background/Case Studies: FDA released the Guidance Document, *Collection of Platelets by Automated Methods* December, 2007, which outlines the process validation and quality monitoring for apheresis platelets. It includes an example of a binomial statistical sampling plan that provides a 95% confidence that greater than 95% of the apheresis platelets meet the criteria for leukoreduction and pH for Quality Control (QC) testing. The American Red Cross (ARC) implemented this process in all of its 36 blood centers throughout the United States. **Study Design/Methods:** The 36 blood centers were grouped into 8 geographical QC monitoring areas (QCMA) with a sampling strategy based on historical collections. For residual leukocytes (rWBC), a sampling size of 60 with no failures for each type of collection procedure (singles, doubles, or triples) was selected for a total of 180 procedures for each instrument type (Amicus or Trima) and for InterSol® apheresis platelets. For pH measurements, a sampling size of 60 using a combination of single, double, and triple products was selected for each QCMA. Should there be a product failure for either rWBC or pH, an investigation will be performed to identify the cause. After corrective actions are taken, if a process failure occurred or if the cause could not be determined, an additional sampling size of 91 is allocated and tested within the QCMA. No failures are allowed with a sample size of 91. A standardized national database was developed to monitor QC sampling throughout the month, track QC test results, track the apheresis equipment from which product was collected, and provide a monthly summary report. **Results/Findings:** After 3 months of monitoring under the new sampling plan (N = 5,151 for rWBC), there were two rWBC failures where the cause was not identified or was unclear, and testing of 91 additional samples was instituted. In both situations, there were zero failures, which indicate that the process was in control even when an isolated failure was identified. A total of 1637 products have been tested for pH with no failures. **Conclusion:** The apheresis platelets statistical sampling plan was implemented successfully within the organization. Continuous monitoring and a good communication network are essential to guarantee the success of this program.

Disclosure of Commercial Conflict of Interest

K. Jones: Nothing to disclose; L. Kline: MacoPharma, Grants or Research Support, Origen, Grants or Research Support, Masimo, Grants or Research Support; F. Kugele: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Jones: Nothing to disclose; L. Kline: Nothing to disclose; F. Kugele: Nothing to disclose

Binomial Sampling Plan/Per Month/QCMA

QC Test	Procedure Type	Amicus	Trima	Amicus InterSol	Total Tested All Devices	Fail*
rWBC counts per procedure type	Singles Doubles Triples	60 60 60	60 60 60	60 60 60	N/A	N/A
Total Tested	8 QCMA	2,982	1,808	361	5,151	2
pH at distribution or expiration	Mix of singles, doubles and triples	60	60	60	N/A	N/A
Total Tested	8 QCMA	951	566	120	1,637	0

* Tested additional 91 samples with no failures

AP93

Using Data to Increase Automated Blood Bank Analyzer Efficiency

E Carter[1] (ecarter@bloodsystems.org), K A Thompson[1], D Pientka[1].
[1]Blood Systems, Inc, Scottsdale, AZ, United States

Background/Case Studies: An automated blood bank analyzer was implemented for use in the centralized transfusion service in December 2011. The use of the instrumentation was not mandated by management; staff were allowed to optimize the process by choosing either the automated process or gel testing methodology for performing Type and Screen testing. Even with optional usage, it was believed that the analyzer's current use was optimized for cost reduction and efficiency. One year post implementation, a follow up analysis was initiated to evaluate the usage and benefits of the automated analyzer. **Study Design/Methods:** Various data sets were collected and analyzed. Time studies were performed for both manual and automated methods, including set up time, hands-on testing time, and time for quality control functions. Cost per test was calculated for both methodologies based on reagent costs. Technologists were surveyed as to usage, ease of use, and preferences. Testing volumes were collected from the Transfusion Services database for both methodologies and sorted by time of day and the day of the week to establish a clear profile regarding when and how each method was being utilized. **Results/Findings:** Once the instrument usage data was collated, several trends were noted. The analyzer's usage was optimized during weekday afternoons, when large batches were run, but it was rarely used during the early and late morning hours and never on the weekends. When comparisons were made using cost and hands-on time, the analyzer was found to require less hands-on technician time once it had been set up for the day (14 minutes for the gel screen and 9 minutes for the automated system), and the per test cost was lower. Survey results indicated that the technologists perceived the gel screen to be faster than the analyzer. This was found to not be the case when the walk-away capability of the analyzer was considered. Additionally, the technologists indicated a lack of interest to be trained on the newer technology. An overall savings of approximately $32,000 dollars in reagent cost and 139 hours of technician hands-on time was calculated annually if all of the tests performed using the gel methodology had been completed using the analyzer instrument. **Conclusion:** The perception that the automated analyzer instrument was being utilized to its full potential was found not to be supported when data was collected and analyzed. Areas of improved usage were identified by the day of the week and time of day to reap the benefits of the analyzer's capability and cost savings. Collected data can be used to change the perception of staff, and to support making changes to how and when instrumentation is used in the transfusion service to save money and technician time.

Disclosure of Commercial Conflict of Interest

E. Carter: Nothing to disclose; D. Pientka: Nothing to disclose; K. A. Thompson: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. Carter: Nothing to disclose; D. Pientka: Nothing to disclose; K. A. Thompson: Nothing to disclose

AP94

Promoting a Culture of Continuous Improvement in the Blood Bank Using LEAN and 5S Methodologies

D Fountain[1] (dfountain@partners.org), S R Boehler[1], R M Kaufman[1].
[1]Pathology, Brigham and Women's Hospital, Boston, MA, United States

Background/Case Studies: This is a process for developing a culture of continuous improvement in a high-volume hospital blood bank (BB) through the systematic adoption of LEAN principles. **Study Design/Methods:** Three interlinked paths were implemented in parallel to educate front line staff and promote the adoption of a culture of continuous improvement. Path 1) A Continuous Improvement Committee (CIC) was formed to provide high-level LEAN training. Upon completion, all BB staff (3-4 at a time) will have rotated through CIC for eight months. CIC was led by a BB technical director and clinical supervisor who were trained as LEAN Practitioners. Five training modules were created for new CIC members using the LEAN Orientation of Define, Measure, Analyze, Improve, and Sustain. CIC training included a focused process improvement (PI) project entitled "Irradiation Report Retention Goes Paperless." CIC members were surveyed for training effectiveness. Path 2) All staff underwent basic LEAN concept training led by the BB technical director. This included a focused PI project aimed at streamlining the attachment of pending orders to specimens. Path 3) Three teams of four staff each trained in 5S methodology, led by a senior technologist were formed. 5S implementation was done in three 8-week phases: Sort and Set in Order; Shine and Standardize; and Sustain. **Results/Findings:** This initiative successfully engaged all BB staff, and launched two PI projects, both ongoing currently. The CIC survey demonstrated 98% satisfaction with high-level LEAN training. The LEAN and 5S training sessions provided many opportunities to help strengthen working relationships between BB leadership and staff, while creating a foundation to initiate future PI projects. 5S implementation was enthusiastically driven by volunteer teams and allowed all staff to participate in decision-making about their work areas. A significant time commitment was required for the CIC and CIC leads (approximately 6 hours of prep time per module). The CIC required 2 hours/week for the PI project work. 5S teams needed scheduled time off the bench (2 hours/week). **Conclusion:** Implementation of the described interlinked paths has been successful in leading the BB towards a culture of continuous improvement. All staff had the opportunity to be educated in LEAN principles and to be directly involved in solving workflow problems. Leadership now has an established foundation for training staff in LEAN principles, a plan to move forward with completing process improvements, and adopting a culture of continuous improvement. Although requiring a significant time commitment, this initiative has proved to be an excellent investment of resources and has validated making the adoption of a culture of continuous improvement a high priority.

Disclosure of Commercial Conflict of Interest

S. R. Boehler: Nothing to disclose; D. Fountain: Nothing to disclose; R. M. Kaufman: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. R. Boehler: Nothing to disclose; D. Fountain: Nothing to disclose; R. M. Kaufman: Nothing to disclose

AP95

Improved Information Regarding Wrong Blood in Tube (WBIT) Errors: Impact on Error Rates

K Igo[1] (igok@ccf.org), C McGrath[1], P I Figueroa[1]. *[1]Cleveland Clinic, Cleveland, OH, United States*

Background/Case Studies: Wrong Blood in Tube (WBITs) are errors wherein the wrong person's name and identifying information is on a sample which contains another individual's blood. These errors are among the most dangerous errors that can be made in a laboratory and they are often "invisible," meaning that even careful examination of the label often cannot reveal the error. When one error in labeling a sample has been made, there is more likely to be another error (eg WBIT). [Lumadue JA, Boyd JS, Ness PM. *Transfusion* 1997;37:1169-72.] **Study Design/Methods:** At the end of 2009, the Transfusion Service (TS) began performing a forward ABO/Rh on all mislabeled specimens increasing the chance of detecting a WBIT error, thereby improving the accuracy of error reports so that appropriate corrective actions may be taken against the possibility of future errors. Also, in November 2009 a requirement for a confirmatory ABO/Rh on an independently drawn second sample ("CONABO") was instituted as well. Specimen collection and labeling errors are reported immediately to the clinical services by telephone and then in the hospital error management system, which mandates a follow-up by the appropriate clinical service. It was theorized that with better information being reported, there would be a decrease in the

number of WBIT errors made across the enterprise as floors making such errors would know to investigate the cause of the errors. Where trends were observed, education and retraining were provided. **Results/Findings:** Approximately 110,000 samples are drawn and sent to the TS annually. Prior to the implementation of testing mislabeled samples for their ABO/Rh and CONABO, the WBIT error rate was 0.009% and after the new processes were implemented, the error rate was 0.012%. There have been more WBITs documented as a result of the new processes (33); however, the number of WBIT errors detected annually has remained relatively stable over the past seven years. See Table. **Conclusion:** Had these policies not been implemented, thirty-three WBIT errors would have gone unnoticed, but informing the floors of these errors combined with ongoing education has not appreciably resulted in a reduction of the number of WBIT errors.

Disclosure of Commercial Conflict of Interest

P. I. Figueroa: No Answer; K. Igo: Nothing to disclose; C. McGrath: Nothing to disclose

Disclosure of Grants Conflict of Interest

P. I. Figueroa: No Answer; K. Igo: Nothing to disclose; C. McGrath: Nothing to disclose

WBIT Detection Methods

Year	Mislabels(3)	CONABO(4)	Other(5)	Total
2006(1)	—	—	6	6
2007	—	—	9	9
2008	—	—	12	12
2009	2	1	12	15
2010	10	0	4	14
2011	6	1	6	13
2012	8	2	4	14
2013(2)	3	0	1	4
Total	29	4	54	87

(1) April-December (2) January-March (3) Detected by testing mislabeled samples (4) Detected by testing second independent sample (5) Detected by other mechanisms

AP96

Use of a Secure Web Application for Documenting Transfusion Medicine Consults: Advantages, Disadvantages, and Areas for Improvement

H C Sullivan[1] (hcsulli@emory.edu), T M Schneider[1], L N Stuart[1], C E Hill[1], A M Winkler[1]. *[1]Pathology and Laboratory Medicine, Emory University, Atlanta, GA, United States*

Background/Case Studies: Transfusion Medicine consults encompass various types of clinical pathology calls which include but are not limited to blood product management, transfusion reactions, and requests for coagulation factor products. Until June 2012, all consults were documented in a handwritten log, which contained inherent problems including risk to patient privacy, difficulty tracing encounters, and inadequate transition of care between pathology residents. As a result, a secure, online database was created with data collection forms and reports using Research Electronic Data Capture (REDCap). The experience is presented here of transitioning from a handwritten log to this online format with attempts to discuss advantages, disadvantages, and areas for improvement. **Study Design/Methods:** The number of calls recorded in REDCap (RC) was tabulated and each encounter was categorized based on the type of call. To compare the handwritten log to the RC system and gauge the attitudes towards RC, an electronic survey was generated and distributed among pathology residents. **Results/Findings:** Since the implementation of RC 9 months ago, 1036 encounters have been documented. Transfusion medicine-related calls accounted for 75% of the encounters, while clinical chemistry, microbiology, and immunology calls constituted the remaining 25%. Twenty-eight residents (88%) and one fellow responded to the survey. Per responses, of the 15 residents who had previously used the handwritten log, 56% update calls and document primary care contact information more often using RC. Additionally, 31% of these residents admitted to having left the handwritten log in an unsecured area at some point in time. The Table contains a summary of additional responses. Overall, 94% of residents believe that RC is an improvement over the handwritten log. **Conclusion:** Adapting RC to fit our specifications, a secure and effective system has been created to record and manage clinical pathology calls. While the survey incited suggestions, which will be addressed, the pathology residents' experiences with RC are positive with general satisfaction and perceived superiority over the previous handwritten log.

Disclosure of Commercial Conflict of Interest

C. E. Hill: Nothing to disclose; T. M. Schneider: No Answer; L. N. Stuart: Nothing to disclose; H. C. Sullivan: REDCap, Other; A. M. Winkler: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. E. Hill: Qiagen, Other; T. M. Schneider: No Answer; L. N. Stuart: Nothing to disclose; H. C. Sullivan: Nothing to disclose; A. M. Winkler: Nothing to disclose

Summary of Resident REDCap Survey Responses

Do you think RC is . . .	% YES	% NO
User friendly	82	18
Cumbersome	31	69
Convenient to trace past encounters	72	28
A system that improves patient care	93	7
A system that creates additional work at	83	17
An efficient hand-off tool	86	14
Potentially beneficial to other pathology programs	90	10
Do you think the handwritten log was . . .		
User friendly	60	40
HIPPA* compliant	27	73
An efficient tool for hand-off	20	80
Able to supply sufficient patient data	33	67

* Health Insurance Portability and Accountability Act

AP97

Eliminating Manual Documentation of Red Blood Cell (RBC) Processing

J Bray[1] (jkb@cbckc.org), D Stanton[2], D Lewis[3], K Laughlin[4], S Royer[5], S Kelly[5], J Haggard[6], K B Peck[1]. *[1]Administration, Community Blood Center of Kansas City, Kansas City, MO, United States; [2]Product & Distribution Administration, Community Blood Center of Kansas City, Kansas City, MO, United States; [3]Donor Room Services, Community Blood Center of Kansas City, Kansas City, MO, United States; [4]Component Lab, Community Blood Center of Kansas City, Kansas City, MO, United States; [5]Quality Management, Community Blood Center of Kansas City, Kansas City, MO, United States; [6]Training & Development, Community Blood Center of Kansas City, Kansas City, MO, United States*

Background/Case Studies: Manual documentation errors associated with manufacturing blood products were tracked and it was found that these errors comprised 53.1% of manufacturing process errors at this blood center. In response, this study was conducted to determine the effect of utilizing LEAN principles and tools in reducing the quantity of manual documentation related to blood product manufacture. **Study Design/Methods:** Current state process maps were created for each blood product manufactured, and the current state map was evaluated to determine the number of forms and the number of non-value-added steps associated with each process. Forms and back-up processes used only during unexpected computer downtime were excluded. Future state process maps were designed focused on developing streamlined, electronic alternatives to currently used manual documentation forms. **Results/Findings:** 12 forms were identified, with 184 manual documentation error opportunities and 33 non-value-added process steps. In the new process, 2 forms are used, with 22 opportunities for manual documentation errors (an 83.3% reduction in forms and 88% reduction in manual documentation error opportunities). 14 non-value-added (NVA) process steps were removed from the processes (42.4%) (See Table). Three major categories of forms were identified: Whole Blood (WB) processing (one form); Cryoprecipitate processing (two forms); and special procedures (eight forms), with one form used during both WB processing and special procedures. It was found that the tracking form for slow filtration units discovered during WB processing had the largest number of error opportunities, with 30 per form. This form was completely eliminated. The two forms that are still used in the new processes are a communication log for special procedure requests, and a manual labeling verification for complex labeling processes. The communication log was not revised, due to the cross-departmental use of the form. When LEAN principles were applied to the revision of the manual labeling verification form, there was a 50% reduction in error opportunities (20 to 10), and an 80% reduction in the number of processes reference the form (5 to 1). **Conclusion:** Applying LEAN principles and tools to manual processes was found to significantly reduce redundancy and errors, and maximized implementation of electronic alternatives. This approach is recommended to be applied to manual process revision whenever feasible.

Disclosure of Commercial Conflict of Interest

J. Bray: Nothing to disclose; J. Haggard: Nothing to disclose; S. Kelly: Nothing to disclose; K. Laughlin: No Answer; D. Lewis: Nothing to disclose; K. B. Peck: Nothing to disclose; S. Royer: Nothing to disclose; D. Stanton: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Bray: Nothing to disclose; J. Haggard: Nothing to disclose; S. Kelly: Nothing to disclose; K. Laughlin: No Answer; D. Lewis: Nothing to disclose; K. B. Peck: Nothing to disclose; S. Royer: Nothing to disclose; D. Stanton: Nothing to disclose

	Total Before	Total After	% Reduction
Forms	12	2	83.3%
Error Opportunities	184	22	88.0%
NVA Process Steps	33	19	42.4%

AP98

Using LEAN Tools to Remodel a Flow Cytometry (FC) Testing Area in a Cellular Therapy Laboratory

A L Mohr[1] (mohr.amy@mayo.edu), S L Wittwer[1], E J Brach[1], J Glejf[1], M P Gustafson[1], M C Deeds[1], K L Bundy[1], D Gastineau[1]. *[1]Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: The Human Cellular Therapy Laboratory (HCTL) has an isolated 25.4 sq meters (273 sq ft) flow cytometry (FC) testing area able to accommodate two cytometers. This space was not originally designed for flow testing, resulting in an immovable and inefficient workspace. The HCTL needed a third FC and decided to remodel the testing area with the following goals in mind: 1) Arrange the space to accommodate three cytometers and two work benches. 2) Improve efficiency for one technologist but have workspace available for a second. 3) Use only mobile workbenches and structures to make the space adaptable for future changes. **Study Design/Methods:** Thirteen potential layouts were designed using paper models of the room and equipment. The decision for the final layout was based on the overall efficiency of the theoretical LEAN spaghetti diagram. The new layout consists of a primary workspace (which includes two cytometers and one bench) and a secondary workspace (which includes one cytometer and one bench). In the primary space, the two cytometers and workbench form a classic "U" pattern. In the secondary space the cytometer and workbench are parallel and face each other. The refrigerator and specimen drop off area are located in between the primary and secondary workspaces. **Results/Findings:** A survey was conducted 6 months after the remodel to gauge staff satisfaction with the new LEAN testing area. When asked how efficiency was affected after the remodel, 0% said efficiency had decreased, 18.2% said efficiency had stayed the same, and 81.8% said efficiency had increased. The pre- and post-remodel LEAN spaghetti diagrams were analyzed to identify any increases in efficiency. The "U" pattern of the primary workspace allowed the technologist to view both instruments at the same time and greatly reduced movement. A 326 liter (11.5 cu ft) refrigerator replaced a small under counter refrigerator, doubling the on hand storage space and eliminating untimely trips to the auxiliary storage area. These changes were the favorites of the technologists. The least favorite result of the remodel was the limited walkway space available when technologists are seated at the computer workstations. The workbenches were custom designed to be on wheels, making any future layout changes easy and inexpensive. Adjustable height benches were purchased to increase ergonomic functionality of the workspace. **Conclusion:** The three original goals were successfully met by the LEAN remodel. The post-remodel space is now conducive for FC testing and has improved overall efficiency. In addition, the mobile workbenches installed will make any future changes to the layout easy and inexpensive.

Disclosure of Commercial Conflict of Interest

E. J. Brach: Nothing to disclose; K. L. Bundy: Nothing to disclose; M. C. Deeds: Nothing to disclose; D. Gastineau: Nothing to disclose; J. Glejf: No Answer; M. P. Gustafson: Nothing to disclose; A. L. Mohr: Nothing to disclose; S. L. Wittwer: Nothing to disclose

Disclosure of Grants Conflict of Interest

E. J. Brach: Nothing to disclose; K. L. Bundy: Nothing to disclose; M. C. Deeds: Nothing to disclose; D. Gastineau: Nothing to disclose; J. Glejf: No Answer; M. P. Gustafson: Nothing to disclose; A. L. Mohr: Nothing to disclose; S. L. Wittwer: Nothing to disclose

AP99

Evaluation of Oral Calcium Carbonate Prophylaxis for Therapeutic Plasma Exchange during a Calcium Gluconate Shortage

J Parsons[1] (jeremy.parsons@ucdmc.ucdavis.edu), S Barnhard[1]. *[1]Pathology, University of California Davis, Sacramento, CA, United States*

Background/Case Studies: The apheresis service here uses prophylactic calcium gluconate (CG) to prevent hypocalcemia secondary to citrate during therapeutic plasma exchanges (TPE) and stem cell collections. Recently, a CG shortage led to rationing of the remaining supply by the apheresis service. Before the CG shortage, the standard practice was generally 3 g CG in 100 ml NS to be infused throughout the TPE. In addition, the nurse had a PRN order of 100-200 mg CG IV push if the patient exhibited any signs of hypocalcemia. During the shortage, the apheresis service utilized oral calcium carbonate (CaCO3) for pre-prophylaxis. In this abstract, the outcome of a trial switch is evaluated to oral CaCO3 administration for prophylaxis of hypocalcemia in a few patients to determine any potential cost savings to the hospital/service. **Study Design/Methods:** Electronic medical record review and consultation with hospital pharmacists were conducted. **Results/Findings:** A patient was selected who required once weekly TPE for chronic inflammatory demyelinating polyneuropathy (CIDP) to evaluate. She received more than 20 treatments prior to the shortage (3 g CG infusion) without any symptomatic episodes. Early in the shortage, an attempt was made to perform a TPE on this patient without prophylaxis to preserve CG for stem cell collection cases. During this procedure, the patient experienced perioral tingling, vomiting, and mild hypotension. Ionized calcium was low at 1.06 mmol/L (1.23 mmol/L pre-procedure). The patient's symptoms resolved with a 200 mg IV push of CG and 4 mg ondansetron. 500 mg oral CaCO3 was used instead prior to future procedures. She underwent two subsequent procedures with 1500 mg oral CaCO3 and exhibited no hypocalcemia symptoms. During the same time interval, the service provided 11 TPEs to a young man experiencing humoral kidney graft rejection. During four of these TPEs, he received 1500 mg oral CaCO3 in lieu of a CG IV infusion due to CG shortage and did not experience symptomatic hypocalcemia. A brief cost analysis revealed that the average TPE using these two different calcium prophylactic formulations cost $ 0.04 for 1500 mg CaCO3 and $5.71 for 3 g CG. **Conclusion:** An evaluation of two TPE patients that spanned a 2-month period during which the apheresis service experienced a CG shortage, reveals that 1500 mg CaCO3 given in an oral loading dose prior to the procedure can be adequate in preventing hypocalcemia symptoms in a patient who has exhibited such symptoms in the past when calcium prophylaxis was not given. Although the sample size is small, the preliminary findings and potential cost savings ($2.26 per year oral CaCO3 versus $297.13 per year IV CG in a once weekly CIDP patient) make the possibility of a permanent switch to oral calcium prophylaxis for TPE procedures worth pursuing further.

Disclosure of Commercial Conflict of Interest

S. Barnhard: Nothing to disclose; J. Parsons: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. Barnhard: Nothing to disclose; J. Parsons: Nothing to disclose

AP100

Development of an Audit Toolkit Utilizing Tracer Methodology which Incorporates AABB, CAP, and TJC Standards for Blood Centers and Hospital Blood Banks

S B Pelton[1] (sbpelton@gmail.com), W A Flegel[1], K M Byrne[1]. *[1]Department of Transfusion Medicine, National Institutes of Health, Bethesda, MD, United States*

Background/Case Studies: Blood centers and hospital blood banks can be accredited by three main organizations: AABB, College of American Pathologists (CAP), and The Joint Commission (TJC). TJC uses tracer methodology when performing audits, meaning that surveyors select a patient, review products/services received, and move through the organization to assess compliance with regulations. This approach has been used to perform internal audits of specific processes, such as blood administration, and can be expanded to trace products from donor selection to transfusion allowing review of critical steps. Internal audits are essential for detecting and correcting deficiencies before they are found by regulatory agencies. The purpose here was to develop an audit toolkit following tracer methodology encompassing the main requirements of three agencies. **Study Design/Methods:** A toolkit was created with three electronic documents using Excel 2007. Accreditation standards from AABB, CAP, and TJC were analyzed. A crosswalk linking the three sets of standards was developed using existing documents from the AABB national office. A checklist was constructed using

AABB Standards for Blood Banks and Transfusion Services, 28th ed. Any CAP and TJC requirements not part of AABB standards were added. A reference guide, which detailed requirements needed to complete the checklist, was developed. A request to review and use the toolkit was made to internal staff and external peers, asking them to evaluate the electronic platform, ease of use, and effectiveness of the toolkit. Using a standardized form, feedback was solicited from those volunteers who responded to the request. **Results/Findings:** A toolkit was developed using tracer methodology encompassing major requirements of three accrediting agencies. Electronic documents were reviewed and used by 13 volunteers. Five were current or former AABB assessors. The toolkit was reviewed by six volunteers. There were a total of eight product audits performed by seven volunteers. The products audited included 3 RBC, 2 APLT, 1 FFP, 1 CRYO, and 1 GRAN. Feedback was received from all volunteers. Most of the positive feedback and interestingly, most of the negative feedback, dealt with ease of use. Edits were made in response to comments. After completion and review of the eight product audits conducted, one noncompliance was detected. This finding was reported to appropriate supervisory staff. **Conclusion:** Internal audits can identify deficiencies in procedures and processes which may cause harm to patients and/or lead to poor outcomes during assessments by regulatory agencies. This audit toolkit, utilizing the tracer methodology, is an effective way to prepare for an upcoming assessment. In addition, it can serve as an educational and training tool.

Disclosure of Commercial Conflict of Interest

K. M. Byrne: Nothing to disclose; W. A. Flegel: RHD Genotyping Patents, Other; S. B. Pelton: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. M. Byrne: Nothing to disclose; W. A. Flegel: Nothing to disclose; S. B. Pelton: Nothing to disclose

AP101

Laboratory Information Service (LIS) Enhancement Requests: Development and Implementation of an Electronic Process Control System

A L Sutton[1] (asutton@mcvh-vcu.edu), S M Armstrong[1], A Stalley[1], C Mendoza[1], K Sanford[1]. *[1]Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: This Transfusion Medicine (TM) laboratory management promotes a culture of staff involvement in changes and improvements in all aspects of the laboratory, including suggestions to improve the Laboratory Information System (LIS). Typically, these requests were submitted via emails, screenshots, or even hand written notes. These requests, once received by the LIS Senior Technologist (ST), would be implemented, tested, and then approved by management and the Medical Director. After a large influx of suggestions were submitted, the ST and TM management recognized a need for developing a streamlined, electronic, process-control system for both tracking and updating the progress of suggestions submitted to LIS. **Study Design/Methods:** This new process involved construction of an electronic LIS enhancement request form that includes a description of the proposed change and rationale, necessity of staff training and procedural changes, testing protocol (completed by the ST), a method for tracking, and approval signatures. After the form was created, the staff was trained on the process of completing the form as well as the process of implementing LIS changes. The electronic form is easily accessible to every employee at any workstation. Once the form is submitted, the ST determines whether the enhancement is possible. The form is then taken through the approval process with TM management and the medical director. After approval, a tracking number is assigned to the form and all information is entered into a Microsoft® Access database. This database includes the tracking number, date received and the current status of the request ie waiting for approval, testing phase, and validation signatures. The ST updates the progress of the each request in the database. Lab personnel may access the database to stay informed of the progress of their submission. When all testing and training is completed, TM management and the medical director must approve all documentation before the new enhancement goes live. **Results/Findings:** The new process was implemented in October 2012 and has made the management of LIS requests much more manageable.TM staff is utilizing the new form and becoming acclimated with the process. Implementation of the new process has also increased the staff's awareness on the importance of process management and the proper steps that are taken to complete a new endeavor from beginning to end. **Conclusion:** Management and staff are now aware of the status of all LIS requests and can easily monitor the progress of each. The successful creation and use of the new LIS enhancement request and implementation process improved communication among lab personnel, LIS staff, and TM leadership while also allowing for easy tracking and management of requests.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; C. Mendoza: Nothing to disclose; K. Sanford: Nothing to disclose; A. Stalley: Nothing to disclose; A. L. Sutton: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; C. Mendoza: Nothing to disclose; K. Sanford: Nothing to disclose; A. Stalley: Nothing to disclose; A. L. Sutton: Nothing to disclose

AP102

Lessons Learned by the Bellevue Blood Bank Following Hurricane Sandy

J L Jacobson[1] (jessica.jacobson@nyumc.org). [1]Pathology, Bellevue Hospital/New York University School of Medicine, New York City, NY, United States

Background/Case Studies: The storm surge from Hurricane Sandy hit New York City on October 29-30, 2012. The blood bank transfuses approximately 8500 red blood cells (RBCs), 2000 single donor platelets (SDPs), and 2500 plasma units and performs roughly 60,000 ABORh types and 40,000 antibody screens annually. Prior to the arrival of the storm, Bellevue made preparations to shelter in place for 3 days. A transformer exploded knocking out electricity to southern Manhattan forcing the switch to backup generators. The fuel pumps were submerged. Fuel to power the generators was carried up the stairs by staff and the National Guard. On October 31, Administration decided to evacuate the patients. The final patient left on November 3. Although the blood bank never lost power, there were many unanticipated challenges. **Study Design/Methods:** Emergency operations were undertaken to enable continued service to our patients. Existing procedures covered manual testing and emergency issue of blood products, but did not adequately address how to return a 400+ blood product inventory to the blood supplier. **Results/Findings:** On October 31, NYBC arrived asking what it could do to help. NYBC sent people with shipping boxes, wet ice, and dry ice to package up most of our inventory. 285 RBCs, 6 SDPs, 169 frozen plasmas, and 13 CRYOs were appropriately boxed for shipping and returned so they could be used elsewhere. Without the computer system,the procedure could not be followed for recording the units' final disposition. Manually recording the ISBT unit numbers and product codes was not a viable option nor was photocopying the ISBT labels so that they could later be scanned. The decision was made to record a manual count of the products by type. NYBC scanned the units upon receipt and later provided an itemized list of all returned products. On November 3-4, after NYBC came to pickup the remaining liquid units, the lab was locked up. On November 15 after water and electricity had been restored, some staff were permitted to return to the lab. The disposition of products that had been transfused during the computer downtime was entered into HCLL by scanning the photocopy of the unit's ISBT label. The remaining products in the HCLL inventory were then compared to the NYBC document and the final dispositions were entered. The blood bank reopened and resumed limited services on November 19 and full services on December 10. **Conclusion:** Although emergency plans are part of every lab's policies and procedures, it may not be possible to plan for every type of contingency. Procedures need to include multiple options or have enough flexibility so that they can be followed even under extreme circumstances. Cooperation between hospital blood bank and blood supplier is essential and in this case enabled the blood products in our inventory to be returned so that they could be transfused to patients in need during the aftermath of Hurricane Sandy.

Disclosure of Commercial Conflict of Interest

J. L. Jacobson: Fresenius, Other, Terumo, Other, Genesis, Other, SMART Bio Tech Ltd., Other

Disclosure of Grants Conflict of Interest

J. L. Jacobson: G & W Laboratories, Grants or Research Support

AP103

Faster Blood Delivery in Surgical Emergencies by Pneumatic Tube Versus Cooler: A Pilot Audit of the Delivery Pathway

H Wang[1] (hwang37@partners.org), W Savage[2], R M Kaufman[2].
[1]Anesthesia, Brigham and Women's Hospital, Boston, MA, United States; [2]Pathology, Brigham and Women's Hospital, Boston, MA, United States

Background/Case Studies: Delivering red blood cell units (RBCs) rapidly to the operating room (OR) in bleeding emergencies is a critical responsibility for blood banks. Measuring the total turnaround time (TAT) for stat surgical blood orders is a key first step in optimizing the delivery system. **Study Design/Methods:** Over 10 days, stat RBC provision to the OR was monitored in real-time by study staff. Emergency intra- and pre-operative RBC orders were included; routine pre-operative orders were excluded. RBCs delivered by pneumatic tube or carried in a cooler by an OR courier were physically tracked from the blood bank to the OR, and arrival times were recorded. Other data were obtained from hospital electronic records. Anesthesiologists were surveyed on: 1) How quickly did the blood arrive? (0-5 scale; 0: much too slow; 5: ahead of schedule); and (2) How badly did the patient need the blood? (0: not needed; 5: life-threatening bleed). Linear regression was used to assess factors affecting TAT. **Results/Findings:** Stat blood delivery to the OR was evaluated in 14 surgical cases. In eight cases, RBCs (n = 22) were delivered by pneumatic tube; in six cases, RBCs (n = 23) were delivered by cooler. Key time intervals are shown in the Table. The median total TAT (ie time from receipt of the order in the BB until arrival of the unit in the OR) was 16 minutes for RBCs sent by pneumatic tube versus 24 minutes by cooler. This difference was partly due to unit preparation time within the blood bank. More time was required to fill larger blood orders (typically, 4-5 RBCs/cooler versus 2 RBCs/tube). Yet after controlling for the number of RBCs, delivery by pneumatic tube remained a significant predictor of faster TAT (–13 min., p = 0.002). Considerable time was spent waiting for couriers to arrive in the blood bank for cooler pickup (median of 5 minutes) A survey of anesthesiologist attitudes about blood delivery showed no significant relationship between TAT and satisfaction. Anesthesiologists were satisfied in the majority of cases evaluated. In one case when the TAT was judged to be unacceptably long, the units had been sent to the wrong OR tube station due to a miscommunication between the OR and blood bank. **Conclusion:** The complete pathway was audited for delivering RBCs to the OR in bleeding emergencies at a single hospital. Sending RBCs by pneumatic tube resulted in their arrival in the OR 33% faster than units carried in a cooler. This study demonstrates the feasibility of directly auditing stat blood delivery as a quality metric, and establishes a baseline for future improvements to the delivery system (eg remote electronic allocation of blood to the OR).

Disclosure of Commercial Conflict of Interest

R. M. Kaufman: Nothing to disclose; W. Savage: Nothing to disclose; H. Wang: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. M. Kaufman: Nothing to disclose; W. Savage: Nothing to disclose; H. Wang: Nothing to disclose

			Time INSIDE Blood Bank (min)			Time OUTSIDE Blood Bank (min)				
Delivery mode	Cases (n)	RBC units (n)	Order received—units ready	Units ready—units depart BB	Total time INSIDE BB	BB—tube station	Tube station—pickup	Pickup—OR arrival	Total time OUTSIDE BB	Total TAT to OR (min)
Tube	8	22	8 (5, 11)	1 (0, 2)	9 (5, 12)	3 (1, 5)	2 (1, 13)	0 (0, 7)	6 (4, 17)	16 (12, 24)
Cooler	6	23	12 (10, 17)	5 (3, 12)	16 (14, 31)	N/A	N/A	7 (6, 8)	7 (6, 8)	24 (20, 38)

All times are reported in minutes as median (10th percentile, 90th percentile).

AP104

Technology and Quality Control: Uncomplicated Functionality

J Shemenski[1] (jshemenski@mcvh-vcu.edu), K Sanford[1]. *[1]Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: When performing quality control (QC) on remotely located refrigerators within the medical center, it is difficult to access the temperature monitoring systems which are still located in the laboratory. Typically, temperature monitoring systems are physically located in the laboratory for ease of use by laboratory staff; however, a temperature monitoring system with software is used here which utilizes a centralized server via a web browser database that systemizes regulated QC of alarm checks for remotely located refrigerators. **Study Design/Methods:** The Transfusion Service here utilizes a temperature monitoring system which can access the program via a web browser. Using the web browser version of the temperature monitoring system along with a laptop computer or tablet enables the technologist performing QC to access temperature data during the alarm checks. The blood refrigerators are located in the operating room lab, neonatal ICU, pediatric ICU, emergency department trauma room, and the apheresis unit. When refrigerator alarm checks are performed, a technologist manually creates the alarm conditions and accesses the temperature monitoring system data simultaneously using the laboratory laptop. By using the web browser version of the temperature monitoring system, the tech can view the forced alarm in the system and answer the alert while performing QC in a remote location. The other required component to complete the alarm checks is to telephone the lab confirming the alarm is audible on site. The audible alarm confirmation is performed by lab staff utilizing the speaker phone to ensure that the tech in the remote location can hear the alarm. **Results/Findings:** Utilizing the web based temperature monitoring system along with a laptop for refrigerator alarm checks allows the tech in the remote site to manage the alarms, rather than staff back in the lab. This technology frees up any staff that normally would have to assist in the alarm check process. **Conclusion:** Using updated temperature monitoring technology that is now available creates a seamless QC check for remote refrigerator alarm checks. These newer approaches of using technology reduce the burden of performing alarm checks in remote locations, has reduced staff involvement to just one tech, and has decreased the interruption of the work flow in Transfusion Service.

Disclosure of Commercial Conflict of Interest

K. Sanford: Nothing to disclose; J. Shemenski: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. Sanford: Nothing to disclose; J. Shemenski: Nothing to disclose

AP105

Improving an Electronic Maintenance Documentation Form

S L Lamm[1] (slamm@mcvh-vcu.edu), J Shemenski[1], A Stalley[1], R M Bhavnagri[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: In October 2011 the system here switched to electronic documentation of daily, weekly, and monthly routine maintenance for each piece of critical equipment. There was also an electronic daily checklist to document maintenance, daily review, and other required daily quality control tasks. This allowed each shift's Medical Technologist in Charge (MTIC) to verify all required actions had been performed without opening individual equipment worksheets; however, this led to the inefficiency of documenting the same action in two places and required weekly and monthly maintenance not performed in a timely manner. **Study Design/Methods:** The Quality Assurance Coordinator (QAC) and Quality Control Senior (QCS) identified late weekly and monthly maintenance as a critical concern. Shift supervisors and daily MTICs identified missing documentation on the daily checklist, even though the daily maintenance had been performed, documented, and reviewed on the appropriate equipment electronic form. This was an opportunity for improvement. To address the late weekly and monthly maintenance, the template for all forms was switched from a year long, continuous weekly format to a discrete monthly format with separate tabs added for weekly and monthly routine maintenance. This allowed the QCS, QAC, and staff to determine if weekly or monthly had been performed without scrolling through multiple weeks. Using the linkage function of the spreadsheet software, the cells on the daily checklist worksheet were linked to the appropriate piece of critical equipment's daily maintenance and review signature cell. When the daily maintenance form was electronically signed the staff member's signature would automatically populate on the daily checklist. This remedied the missing documentation on the

daily checklist, as well as requiring only one entry of signatures. Finally, to allow an entire monthly overview of maintenance for the QAC and QCS, a worksheet was created linking all weekly and monthly maintenance signature cells for all pieces of critical equipment. As each task is documented, the date and signature auto populate the form, as well as generate a color change for easy visibility. A training presentation was given to all staff and the forms were validated in parallel with the existing forms for a 1-month period. **Results/Findings:** As part of the laboratory's continuous improvement program, the concerns of the QAC, QCS, supervisors, and MTIC about maintenance documentation were addressed. **Conclusion:** The implemented change contributed to overall laboratory efficiency for documenting QC. All weekly and monthly maintenance has been completed on time since the revised form's implementation in February 2013. The process will continue to be monitored.

Disclosure of Commercial Conflict of Interest

R. M. Bhavnagri: Nothing to disclose; S. L. Lamm: Nothing to disclose; K. Sanford: Nothing to disclose; J. Shemenski: Nothing to disclose; A. Stalley: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. M. Bhavnagri: Nothing to disclose; S. L. Lamm: Nothing to disclose; K. Sanford: Nothing to disclose; J. Shemenski: Nothing to disclose; A. Stalley: Nothing to disclose

AP106

The Passive Acquisition of Hemoglobin C Via Red Blood Cell (RBC) Exchange

C Di Loreto[1] (christina.diloreto@ucdmc.ucdavis.edi), L Gandy[1], L Freeman[1], C J Gresens[2], L Fernando[1], J Parsons[1]. *[1]Pathology, University of California Davis, Sacramento, CA, United States; [2]BloodSource, Sacramento, CA, United States*

Background/Case Studies: This apheresis service performs red blood cell (RBC) exchanges on patients with transfusion-dependent hemoglobin (Hgb) SS or sickle cell disease (SCD). A 10-year-old African American male with Hgb SS disease was inadvertently transfused an RBC unit derived from a donor with Hgb C trait during an RBC exchange. Hgb C disease is present in 0.02% of African Americans. Hgb C trait is present in 2-3% of this population. Donors with Hgb C trait typically exhibit no clinical symptoms and donor RBC units are not generally screened for the presence of Hgb C. With this in mind, the apheresis service performed a review to determine the frequency of this event and if any corrective action should be taken. **Study Design/Methods:** Using the electronic medical record and pre- and post-exchange high performance liquid chromatography (HPLC) Hgb analyses, the frequency of Hgb C trait transfusions in RBC exchanges was evaluated. **Results/Findings:** From January 1, 2011-March 31, 2013, the apheresis service performed 90 RBC exchange transfusions on 35 patients (range 1 to 22 treatments) using 658 RBC units (average 7.3 units per exchange, range 2-10 units). The only incident of Hgb C trait transfusion was the 10-year-old boy previously mentioned. The acquisition of temporary Hgb C occurred in 1.11% of procedures and in 0.15% of transfused units. HPLC Hgb analysis was performed on the patient during subsequent exchange transfusions and the Hgb C level, which began at 4.1%, eventually dissipated to 0% (Table). Segments of the six RBC units implicated in the exchange transfusion were submitted for HPLC Hgb analysis, and one Hgb C trait donor was identified. This donor was a 16-year-old African American male. **Conclusion:** The goal for a successful RBC exchange is to maximize the amount of normal hemoglobin present in the patient's post-exchange circulatory system. For this reason, pre-screening for Hgb S trait is performed on RBC units used for these procedures. No such screening occurs to prevent the use of RBCs with Hgb C. However, when such donors are identified, it is appropriate to prevent potentially disadvantageous RBC transfusions from occurring. The unit number from the RBC unit containing Hgb C was forwarded to the blood provider. The donor was counseled and removed from the RBC donor pool to alleviate the risk of any future RBC donations being transfused to at-risk patients, such as Hgb SS or SC patients or neonates receiving exchange transfusions. The donor will be given the option to continue donating platelets and/or plasma products.

Disclosure of Commercial Conflict of Interest

C. Di Loreto: Nothing to disclose; L. Fernando: Nothing to disclose; L. Freeman: Nothing to disclose; L. Gandy: Nothing to disclose; C. J. Gresens: Nothing to disclose; J. Parsons: Nothing to disclose

Disclosure of Grants Conflict of Interest

C. Di Loreto: Nothing to disclose; L. Fernando: Nothing to disclose; L. Freeman: Nothing to disclose; L. Gandy: Nothing to disclose; C. J. Gresens: Nothing to disclose; J. Parsons: Nothing to disclose

Serial pre- and post-red cell exchange Hgb Analyses in 10 year old boy with Hgb SS.

	3/1/2013 Pre	3/1/2013 Post	3/27/2013 Pre	3/28/2013 Post	4/24/2013 Pre
HGB A	47.3 (L)	79.6 (L)	65.8 (L)	89.9 (L)	74.9 (L)
HGB C	0.0	4.1 (H)	3.5 (H)	0.7 (H)	0.0
HGB S	45.7 (H)	13.3 (H)	27.3 (H)	6.8 (H)	22.1 (H)

AP107

Improved Outcomes in Bone Marrow Transplant (BMT) Patients through Standardization of Practices Related to the Use of Central Venous Catheters

E T Simpson[1] (esimpson@shc.org), A D Briggs[1], D Gibson[1], S Kendrick[1], J R Schriber[1]. *[1]Cancer Transplant Institute, at Virginia G. Piper Cancer Center, Scottsdale, AZ, United States*

Background/Case Studies: In 2011, the Centers for Disease Control and Prevention (CDC) estimated 248,000 bloodstream infections occurred in the US,and a large percentage of those were associated with the presence of central venous catheters (CVCs). Mortality rates were between 10-30% in patients who experienced central line infections. While CVCs are used in critical care settings for short periods of time, bone marrow transplant (BMT) patients maintain their lines for long periods of time, in both inpatient and outpatient settings. In addition, these patients are at greater risk of experiencing central line-associated blood stream infections (CLABSIs) due to the immunosuppressant effects of chemotherapy, decreased WBC counts, and diminished ability to fight infection. Standardization of practice related to central line insertion and care could lead to decreased numbers of infections in BMT patients. **Study Design/Methods:** A multi-disciplinary team reviewed practices of central line usage in the BMT population with emphasis placed on standardizing both insertion and nursing practices. Central venous catheter types were limited to two, based on patient need, with all insertions performed by properly trained Interventional Radiologists. Standardizing practice related to nursing processes included scheduled tubing and dressing changes, use of antimicrobial patches and moisture barriers, and established line flushing procedures for nursing staff, patients, and caregivers.The BMT Quality Management staff, working with Infection Control, developed a surveillance plan that meets current national reporting guidelines. Data was collected for all transplanted patients including: transplant type, catheter type, number of line days, length of stay, occurrence of positive blood cultures, and categorization utilizing CDC CLABSI criteria. **Results/Findings:** CLABSI rates were calculated by dividing the number of central line infections identified per 1000 line days. Initial results were obtained for 12 months from March 2009-February 2010. Final results were obtained for 12 months from March 2012 to February 2013. The average CLABSI rate obtained prior to implementing process improvement activities was 4.8. Following standardization of practices, the average CLABSI rate fell to 2.5, a 48% decrease. These results demonstrate a highly statistically significant improvement ($p = 0.02$). **Conclusion:** Data analysis revealed no CLABSIs attributed to line insertion. Limiting the number of catheters, developing a weekly schedule for dressing and line changes, and improving education for nurses, patients, and caregivers eliminated many opportunities for infection. Overall, standardization of practices resulted in a clinically significant decrease in CLABSI rates among BMT patients with potential decreases in length of stay, morbidity/mortality rates, and associated costs.

Disclosure of Commercial Conflict of Interest

A. D. Briggs: Nothing to disclose; D. Gibson: Nothing to disclose; S. Kendrick: Nothing to disclose; J. R. Schriber: No Answer; E. T. Simpson: Nothing to disclose

Disclosure of Grants Conflict of Interest

A. D. Briggs: Nothing to disclose; D. Gibson: Nothing to disclose; S. Kendrick: Nothing to disclose; J. R. Schriber: No Answer; E. T. Simpson: Nothing to disclose

AP108

Automated Environmental Monitoring in the Transfusion Service: A Case Study

J Syed[1] (jsyed@lifespan.org), L D Roberts[1]. *[1]Transfusion Service, Rhode Island Hospital, Providence, RI, United States*

Background/Case Studies: Environmental monitoring in the transfusion service can be problematic. Applying federal regulations of the Food and Drug Administration (FDA), the transfusion service follows strict criteria for blood/component storage. Good laboratory practice and standards from The Joint Commission (TJC), College of American Pathologists (CAP) and American Association of Blood Banks (AABB) mandate monitoring temperature and humidity. The traditional process of monitoring has been handwritten paper logs, which is cumbersome. Opportunities for tracking and trending equipment function can be delayed or overlooked. **Study Design/Methods:** A wireless temperature monitoring system was evaluated here. The system leverages existing WI-FI network. The system is FDA approved and EHA endorsed. Each tag is National Institute of Standards and Technology (NIST)-certified prior to receipt, allowing for easy placement and start-up monitoring. The tags are placed in equipment or the ambient environment. The software allows for customized temperature recording at intervals set by the user, from 1 second to hours. These reading are transferred via WI-FI to an on-line application, accessible from any computer with network capabilities. Any reading outside of the established threshold or target generates an alert event. Alerts are customizable and can be sent to the user's email, pager, or smart phone, enabling immediate reaction and corrective action recording in the on-line system. A broad range of scheduled or ad hoc reports can be generated for regulatory compliance. At a predefined interval, a Corrective Action Summary Report can be generated for review and sent to a network file or to user's email. Historical data can also be printed to evaluate the performance of equipment. **Results/Findings:** The automated temperature monitoring system is robust and does not require manual intervention to maintain accurate records of environmental conditions. It enables one to: 1) Improve compliance with regulatory standards through automated collection and logging environmental data; 2) Improve quality of care through constant monitoring of safe temperature and humidity ranges as well as analysis of logged data to identify trends; and 3) Reduce spoilage by shortening the time it takes to identify trends. Each probe costs $115, including NIST certification. It runs on regular watch batteries that cost $5 each and require replacement every 3 years. This comparison study indicated that the estimated time savings is 1 full day each week for labor, record keeping, review, and collation. Average annual cost for manual process is $5200-130 hours x average labor cost. The cost of probes for the lab was a onetime expense of $1725, which would be recouped in 3 months. **Conclusion:** By leveraging the technology currently available, this streamlined approach to environmental monitoring saves resources and provides a level of quality for transfusion service that surpasses the traditional processes used for years.

Disclosure of Commercial Conflict of Interest

L. D. Roberts: Aeroscout, Other; J. Syed: Aeroscout, Other

Disclosure of Grants Conflict of Interest

L. D. Roberts: Nothing to disclose; J. Syed: Nothing to disclose

AP109

Automation and Equipment Management: Computer and Interface Issues in a Small Immunohematology Laboratory

G M Liumbruno[1] (giancarlo@liumbruno.it), M Casiraghi[2], M Scrobogna[2]. *[1]Blood Bank & Transfusion Service, San Giovanni Calibita Fatebenefratelli Hospital, Rome, Italy; [2]Grifols Italia SpA, Ghezzano (Pise), Italy*

Background/Case Studies: Laboratory automation is necessary to simplify the operator work and eliminate every repetitive task that can be a potential source of error. The instruments able to automate the immunohematological pretransfusion tests can be interfaced with more than one laboratory information system (LIS). The aim of this study carried out in the Immunohematology Lab of an Italian Transfusion Center is to describe the implementation and management of an immunohematology system interfaced with two different LIS: one of the Clinical Pathology Unit (OpenLis, Engineering, Rome, Italy) and one of the Blood Bank & Transfusion Center (Emonet, Insiel S.p.A., Trieste, Italy). **Study Design/Methods:** A fully automatic analytical platform based on Grifols column agglutination technique with 8-column gel cards was chosen: it is composed of one main instrument Erytra® (Grifols Italia, Ghezzano [PI], Italy) and one WADiana ® 8XT (Grifols Italia) for back-up. The two aforementioned LIS are not integrated following Health Level Seven standard (HL7) which is helpful for the exchange of electronic (clinical, laboratory, and administrative) information. Therefore, the instrument connection was achieved via a computer program acting as "middleware", Nemo-DG Link (Inpeco SA/Grifols, Lugano, Switzerland). **Results/Findings:** Erytra® and WADiana® 8XT have been installed subsequently during a 6-month period, following a validation process. Both instruments were bidirectionally connected with Nemo DG Link, a web-based software acting as a middleware among the two instruments, and the two different LIS, OpenLis and Emonet. This allowed exporting the results to both the LIS and viceversa. In addition, the middleware has allowed, through user authentication, the access (from any PC connected to the LAN) to results (and images) obtained from both instruments, thus enabling the remote management and validation of tests. The middleware also allowed establishing some rerun and reflex testing rules with the conditional execution of customized second level testing based on first-level test results (eg D-weak on negative RhD samples or monospecific Coombs tests on samples with positive DAT). **Conclusion:** In laboratories with different LIS without HL7 standard that allows them to communicate, the use of connection software via web application (middleware) guarantees: 1) the interaction between the LIS despite possible differences in communication protocol, in architectures of local systems or in operating systems; 2) the exchange of electronic data/information, their concentration in a common interface, and the traceability of operators; 3) all the functionality necessary to telemedicine; and 4) compliance to the minimum required standards for institutional accreditation of transfusion facilities.

Disclosure of Commercial Conflict of Interest

M. Casiraghi: Nothing to disclose; G. M. Liumbruno: Nothing to disclose; M. Scrobogna: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Casiraghi: Nothing to disclose; G. M. Liumbruno: Nothing to disclose; M. Scrobogna: Nothing to disclose

AP110

Bring Out Your Dead: Utilizing the Social Security Death Index to Ease the Burden of Paper Blood-Bank Records

J M Gould[1] (joy.gould@jeffersonhospital.org), M Piacentino[1], J K Karp[1]. *[1]Pathology, Thomas Jefferson University Hospital, Methodist Division, Philadelphia, PA, United States*

Background/Case Studies: The Social Security Death Index (SSDI) is a death record database derived from the Social Security Administration (SSA) Death Master File (DMF). The SSDI and the DMF are composed of over 89 million records of decedents whose deaths were reported to the SSA. Unlike the DMF, the SSDI is available without cost from several genealogy websites. The SSDI was used to document the death of patients for whom paper blood-bank record files were retained on-site in our blood bank, to justify moving the decedents' paper record files to off-site storage. **Study Design/Methods:** Between January 25, 2013-March 22, 2013, 843 paper blood bank record files were reviewed. Patient information was extracted from each paper record file and entered into the SSDI to determine if the patient was known to be deceased. If the patient was found to be deceased, a copy of the SSDI entry was placed in the decedent's paper record file which was then transferred to off-site storage. **Results/Findings:** Of the 843 paper blood bank record files reviewed, 280 (33%) belonged to patients found to be deceased. Of these 280 paper record files, 251 (90%) documented serological investigations, 14 (5%) documented transfusion reactions, 10 (3%) documented the need for special blood products, and 5 (2%) documented both serological investigations and transfusion reactions. Of the 280 decedents, 9 (3%) died in the first weeks of 2013, 45 (16%) died in 2012, 45 (16%) died in 2011, 46 (16%) died in 2010, 48 (17%) died in 2009, 27 (10%) died in 2008, 28 (10%) died in 2007, 10 (4%) died in 2006, 8 (3%) died in 2005, 3 (1%) died in 2004, 5 (2%) died in 2003, 2 (1%) died in 2002, 2 (1%) died in 2001, 1 died in 2000, and 1 died in 1987. By report, the last review of the paper blood bank record files was performed in 2008, thus accounting for the smaller number of patients dying prior to that year. Transferring these 280 paper record files to off-site storage freed 19 (24%) of 78 inches of on-site blood bank shelf storage. **Conclusion:** Paper blood bank record files can be burdensome to laboratories, particularly those with limited on-site storage. The SSDI is a free, readily available online database that can be utilized to determine which paper record files belong to decedents. Decedent paper record files can then be retained in off-site storage as dictated by current regulatory requirements. The SSDI is particularly useful when patients die as outpatients or at other institutions and their death is not captured by internal electronic medical record systems. Based on the above, paper blood bank record files can be compared annually to the SSDI to minimize the number of paper record files occupying limited on-site storage.

Disclosure of Commercial Conflict of Interest

J. M. Gould: Nothing to disclose; J. K. Karp: Nothing to disclose; M. Piacentino: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. M. Gould: Nothing to disclose; J. K. Karp: CSL Behring, Travel Support or Honorarium; M. Piacentino: Nothing to disclose

AP111

An Audit of Stat Turnaround-Time (TAT) Compliance at a Large University Medical Center

J Mantini[1] (jessica.mantini@osumc.edu), M E Grevenow[2], T Bannerman[1], S V Rudmann[1], S Scrape[2]. *[1]Medical Laboratory Science Division, The Ohio State University, Columbus, OH, United States; [2]Department of Pathology, Wexner Medical Center at The Ohio State University, Columbus, OH, United States*

Background/Case Studies: Concerns were raised at a university medical center regarding compliance to a turnaround time (TAT) standard for stat type and crossmatch (XM) specimens. The proposed internal standard was for 90% of all stat XM results to be reported within 60 minutes. An in-depth audit was performed to assess the proposed standard with an evaluation of the compliance rate and work processes that may contribute to non-compliance. **Study Design/Methods:** Data collection occurred over a 3-month period. TAT was defined as the time from electronic receipt of the patient's sample to XM result entry. Each day a TAT report was generated for all stat XM orders. The following information was collected for each stat order: TAT, test ordered, shift involved in the testing of the specimen, and any serological complications. Specimens with TATs greater than 60 minutes were further evaluated to identify specific work flow and equipment-related contributors to the increased TAT. **Results/Findings:** Of the 2416 stat XM orders, only 52% were completed within 60 minutes The average TAT was 66 minutes. Of the 1160 stat XMs that exceeded 60 minutes, 48% had no obvious single reason for the delay. Of the remaining specimens, the following were noted as factors contributing to increased TAT: delayed ABO reconfirmation, antibody identifications, ABO discrepancies, weak reaction incubations, phone calls to floors, manual repeats, add-on tests, and instrument downtime. The largest contributors were ABO reconfirmation delays, instrument downtime, and serological complications. 53% of non-compliant specimens had delayed ABO reconfirmations greater than 15 minutes 21% of non-compliant TATs occurred during periods of instrument downtime. 6.4% of all stat XM specimens had serological complications. Only 18% of all stat XMs were associated with orders to transfuse, indicating a widespread overuse of the stat XM order. **Conclusion:** No simple process change could be identified that would result in conformance with the proposed goal of 90% compliance to a 60 minute TAT standard. It was determined that adjusting the TAT to 75 mins would be more appropriate. Specimens with serological complications could not reasonably be expected to be completed within a specific time frame. Their removal from future quality monitoring would provide more accurate quality data. These two changes would bring the compliance rate to at least 80%. Workflow factors to be addressed included ABO reconfirmation delays and equipment downtime. Physician education regarding the appropriate use of the stat XM order could potentially improve compliance by decreasing stat orders not associated with transfusion. There are currently no nationally accepted standards regarding blood bank TAT. By using appropriately updated standards, detailed auditing for identifiable contributing work flow factors, and physician education, internal TAT compliance goals can be met without sacrificing patient care.

Disclosure of Commercial Conflict of Interest

T. Bannerman: Nothing to disclose; M. E. Grevenow: Nothing to disclose; J. Mantini: Nothing to disclose; S. V. Rudmann: Nothing to disclose; S. Scrape: Nothing to disclose

Disclosure of Grants Conflict of Interest

T. Bannerman: Nothing to disclose; M. E. Grevenow: Nothing to disclose; J. Mantini: Nothing to disclose; S. V. Rudmann: Nothing to disclose; S. Scrape: Nothing to disclose

AP112

Emergency Blood Products: Review of the Verbal and Written Ordering Systems at a Major Medical Institution

E R Patterson[1] (patterson.emily@mayo.edu), K S Badjie[1], K Schulte[1], L M Copper[1], S L Coleman[1], M M Smith[1], L M Button[1], C M van Buskirk[1]. *[1]Laboratory Medicine and Pathology, Division of Transfusion Medicine, Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: The unpredictable nature of emergency medical care makes it challenging to obtain accurate documentation of emergency blood product usage. Yet this documentation is crucial for ongoing transfusion laboratory accreditation and investigation of poor patient outcome. Institutional policy here for emergency release (ER) of blood products states that products can be released after verbal order (VO), but that the verbal order must be followed by a written order (WO). This study serves as an internal review of the ER blood product ordering system. **Study Design/Methods:** A retrospective review was performed of patients who received ER blood products through the Safe Trace Tx ordering system during the year 2012. Patient electronic medical charts and order forms (VOs and WOs) were reviewed, collecting data regarding patient demographics, reason for transfusion, blood product utilization, and timing of orders. **Results/Findings:** During 2012, 180 patients received 586 ER blood products, including red blood cells (RBCs) (348 units), fresh frozen plasma (FFP) (226 units) and platelets (PLTs) (12 apheresis units). The average patient age was 59.4 years (range 18-98 years) with males out-numbering females 111:69. Reasons for emergency transfusions included trauma, gastrointestinal bleeding, cardiovascular disease, and procedural/surgical-induced bleeding. Patients were transfused in the emergency room (155), medical floor (23), and catheterization laboratory (2). Of WO copies that were stored in the transfusion lab (TL), 151 patients had a WO, 24 patients did not, and 6 patients had multiple orders that could not be definitively linked to their specific transfusion episode. Investigation of WO copies stored in the clinical folders revealed that only 67 patients had the same WO available for review. Almost half (45%) of the forms were missing more than one crucial piece of data, most commonly the informed consent box (85%), but also the location of transfusion (34%) and the time the order was placed (14%), among others. VOs were available for 28 out of 180 patients. Of the 7 patients with documented time for both VO and WO, the average time between VO and receipt of WO was 709 minutes (range 54 minutes to 1360 minutes). **Conclusion:** This study highlights documentation gaps in the current verbal and written ordering system for ER blood products. Although documentation from the TL is more complete than from the clinical folders, compliance for receiving written orders is still lower than expected. There is significant variability in how forms are filled out, with many partially completed orders. In an era that values transparency and ongoing process improvement, this data presents an opportunity for systems improvement.

Disclosure of Commercial Conflict of Interest

K. S. Badjie: Nothing to disclose; L. M. Button: Nothing to disclose; S. L. Coleman: Nothing to disclose; L. M. Copper: Nothing to disclose; E. R. Patterson: Nothing to disclose; K. Schulte: Nothing to disclose; M. M. Smith: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

Disclosure of Grants Conflict of Interest

K. S. Badjie: Nothing to disclose; L. M. Button: Nothing to disclose; S. L. Coleman: Nothing to disclose; L. M. Copper: Nothing to disclose; E. R. Patterson: Nothing to disclose; K. Schulte: Nothing to disclose; M. M. Smith: Nothing to disclose; C. M. van Buskirk: Nothing to disclose

AP113

Accuracy and Productivity for Three Residual White Blood Cell Counting Methods

L Kline[1] (linda.kline@redcross.org), G George[1], J S Webb[2], T E Keegan[2], K Jones[2]. *[1]American Red Cross, Holland Laboratory, Rockville, MD, United States; [2]Biomedical Headquarters, American Red Cross, Washington, DC, United States*

Background/Case Studies: Counting of residual white blood cells (rWBC) is performed for product and process quality control (QC) of leukoreduced (LR) whole blood or packed red cells (RBC) and apheresis or whole blood-derived platelet concentrates (PC). Three methods are approved: a subjective, microscopic, manual method (Nageotte); the Leucocount assay (BD Biosciences) with automated flow cytometry (FC); and automated ADAM-rWBC assay (ADAM) (NanoEnTek). These methods were evaluated for accuracy and staff productivity since accuracy and cost, including labor, are key considerations for method selection. **Study Design/Methods:** For accuracy studies LR RBC and PC (N = 12 each) were prepared from WB; 6 PC

were prepared with 65% InterSol platelet additive solution/35% plasma (PAS), 6 with plasma only. Aliquots of each product were spiked with autologous WBC at 10-100/μL. LR and spiked samples were tested for accuracy by a single investigator using Nageotte, FC and ADAM. Productivity was evaluated with 2-3 investigators/method using timing studies to determine the time required for daily tasks (startup, calibration, controls, shutdown; N = 2 repetitions/method) and batch testing of 5 samples (N = 6 repetitions/method). The time to test samples added to existing batches was calculated. For Nageotte, an informal survey of American Red Cross (ARC) QC labs provided data for maximum batch sizes and per-person limitations due to operator fatigue. **Results/Findings:** For RBC (N = 14) the mean rWBC values were 10.7, 12.1, and 11.4 cells/μL (Nageotte, FC and ADAM, respectively; $p < 0.05$ for Nageotte versus FC). For PC (PAS and plasma combined, N = 20) the mean rWBC values were 15.7, 17.4, and 17.1 cells/μL (Nageotte, FC and ADAM, respectively; $p < 0.05$ for Nageotte versus FC and ADAM). When results for either FC or ADAM were below assay linearity (<1 WBC/μL, N = 10 RBC and 4 PC) all results were excluded from calculations, but all three methods gave values <2 WBC/μL. Results of the timing studies are summarized in the Table. Although operator fatigue is variable and difficult to quantitate, survey responses were considered in establishing the maximum batch sizes used in calculations and the number of readings one operator can perform per day. **Conclusion:** All methods resulted in comparable WBC counts. However, productivity was not comparable, with the manual Nageotte method requiring 2-3 times more staff hands-on time than FC or ADAM to accomplish the same amount of testing.

Disclosure of Commercial Conflict of Interest

G. George: BD, Other, NanoEntek, Other; K. Jones: Nothing to disclose; T. E. Keegan: Haemonetics, Other, NanoEnTek, Other; L. Kline: MacoPharma, Grants or Research Support, Origen, Grants or Research Support, Masimo, Grants or Research Support; J. S. Webb: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. George: Nothing to disclose; K. Jones: Nothing to disclose; T. E. Keegan: Nothing to disclose; L. Kline: Nothing to disclose; J. S. Webb: Nothing to disclose

Timing Studies

	Nageotte RBC	Nageotte PC	Flow (RBC + PC)	ADAM (RBC + PC)
Daily (once/day) tasks	57 min		46 min	20 min
Batch of 5 samples	51 min	43 min	26 min	22 min
Time per sample added to batch, max batch size*	6 min, 6	4 min,8	2 min, 27	3 min, NA
Total time for Daily + 12 RBC + 18 PC/person	5 hr 12 min		2 hr 18 min	1 hr 57 min

* Max batch sizes defined by number that can be read within assay limits: Nageotte within 1 hour of charging chamber; Leucocount within 1 hour of staining; ADAM is a continuous process with no maximum batch size.

AP114

Blood-Product Recalls as a Measure of Quality of Blood-Supply Organizations

J McCullough[1] (mccul001@umn.edu), E Riley[1], B R Lindgren[2], S Pulkrabek[1], R Hall[3], W Riley[4]. *[1]Laboratory Medicine and Pathology, University of Minnesota, Minneapolis, MN, United States; [2]Biostatistics and Bioinformatics Core, Masonic Cancer Center, University of Minnesota, Minneapolis, MN, United States; [3]Law School, University of Minnesota, Minneapolis, MN, United States; [4]Health Policy and Administration; School of Public Health, University of Minnesota, Minneapolis, MN, United States*

Background/Case Studies: While there has been emphasis on good manufacturing practice (GMP), quality, and regulatory stringency during the last 25 years, there are no metrics that define quality in a blood supply organization. The blood recall process is intended to remove from use products that may be harmful, and thus could be a metric to define quality. **Study Design/Methods:** Recalls of all blood products were analyzed for 2010, categorizing the reason for the recall, the organizations producing the recalled products, the number of units of blood collected by those organizations, and the FDA district in which the blood was collected. **Results/Findings:** During 2012, there were 2468 recalls involving 8278 blood products. None of the recalls was considered to have a reasonable probability of causing serious adverse health consequences or death (FDA Class I). 79% presented a remote possibility of adverse health consequences (FDA Class II), and 21% were not likely to cause adverse health consequences (FDA Class III). Of the 10 categories of reasons for recalls, the most common were donation and donor qualification (73%) and finished product quality control (14%). There were 16.8 recalls/100,000 units of blood collected. Recalls represented an estimated 0.02% of the national blood supply. The class of recalls varied by month of the year, FDA district, units of blood collected per FDA district, and units of blood collected by the blood center. The number of recalls/100,000 units of blood, and the reason for recall varied by FDA district, and blood centers collecting smaller numbers of units had more recalls. **Conclusion:** The absence of Class I recalls suggest a high level of quality and safety in the US blood supply. There is no evidence that blood recalls could serve as an indication of the quality of a blood supply organization.

Disclosure of Commercial Conflict of Interest

R. Hall: No Answer; B. R. Lindgren: Nothing to disclose; J. McCullough: Nothing to disclose; S. Pulkrabek: Fenwal, Grants or Research Support; E. Riley: Nothing to disclose; W. Riley: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. Hall: No Answer; B. R. Lindgren: Nothing to disclose; J. McCullough: Nothing to disclose; S. Pulkrabek: Nothing to disclose; E. Riley: Nothing to disclose; W. Riley: Nothing to disclose

AP115

The Growing Number of Pre-Analytic, Analytic, and Post-Analytic Software Interfaces to the Transfusion Service: Quality Assurance at a Referral Medical Center for Maintaining the Integrity of the Information System

C Borek[1] (borek.clinton@marshfieldclinic.org), S Mowbray[1], J A Weyhmiller[1], M Baldauf[1]. *[1]Transfusion Services, Marshfield Clinic, Marshfield, WI, United States*

Background/Case Studies: When a transfusion service (TS) implements software, FDA requires end-user validation of computer systems and interfaces between systems. With the rapid growth in the electronic health record (EHR), computerized provider order/entry (CPOE), and interfaces to automation platforms, an initial software validation may prove insufficient to assure the ongoing integrity of the many information system (IS) interfaces. IS professionals from the interfacing parties may make changes that have unanticipated consequences and result in problems for the clinical laboratory and the transfusion service. **Study Design/Methods:** The FDA-approved commercial software ws reviewed for the transfusion services, the interfaces to the two automation platforms, and the interface to the laboratory onsite-developed information system (LIS), to determine the most critical points that require ongoing review. An abbreviated validation script was created that would challenge the process and uncover major problems arising from programming changes from within the laboratory. The pre-analytic portion, which includes orders for both tests and blood products, was more diverse and included interfaces to a hospital information system, an emergency room software package, and a multispecialty clinic electronic health record. These interfaces require ongoing testing of key fields, but are less extensive than the initial validation. The post-analytic portion was equally complex, involving both the dispensing of blood products and the reporting of test results to various sites and software packages. In conjunction with the laboratory quality manager and the IS representatives from the various entities, scripts were developed to assure that after an exhaustive initial validation, continued focused monitoring of the pre-analytic, analytic, and post-analytic facets of data transmission assures that these elements remain intact. **Results/Findings:** In addition to the complete validation for each software installation or major upgrade, an ongoing periodic verification has been added that challenges critical elements interfaced with the transfusion service. This uncovers problems that might arise as the interfaced parties perform database management, software changes, upgrades, and maintenance. The hope is to avoid errors, omissions, or data loss by taking proactive steps. **Conclusion:** The recent addition of periodic verifications of the critical areas in the TS interfaces should provide an early alert about any problems that may have arisen due to software changes from the interfacing parties. this audit will be included as part of the total quality management for the TS.

Disclosure of Commercial Conflict of Interest

M. Baldauf: Nothing to disclose; C. Borek: Nothing to disclose; S. Mowbray: No Answer; J. A. Weyhmiller: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Baldauf: Nothing to disclose; C. Borek: Nothing to disclose; S. Mowbray: No Answer; J. A. Weyhmiller: Nothing to disclose

AP116

Monitoring Quality Measures for Emergency Transfusion during Air or Ground Patient Transport

C Borek[1] (borek.clinton@marshfieldclinic.org), J A Weyhmiller[1], J J Redding[2], D Heegeman[2], M Baldauf[1]. *[1]Transfusion Services, Marshfield Clinic, Marshfield, WI, United States; [2]Emergency Medicine, Ministry St. Joseph's Hospital, Marshfield, WI, United States*

Background/Case Studies: The transfusion service (TS) provides red cells to accompany both ground and air patient transports between accident scenes/rural hospitals and regional referral medical centers. Because this service is not covered by existing internal auditing and review measures, the TS and the flight team (FT) developed a comprehensive system to monitor quality measures for all phases of the operation. **Study Design/Methods:** Members of the FT and the TS developed a plan to monitor quality measures for blood delivery, tracking, and return, as well as evaluating the compliance of the FT in transfusion administration, adverse reaction reporting, and completion of the documentation unique to the emergency transport situation. Other measures included the ongoing performance of the validated coolers and evaluation of the transportation services from the TS to the FT. **Results/Findings:** In the 18 months from the initiation of the service, the TS dispensed coolers 275 times. This allowed blood to accompany 872 air transports (156 air scene and 716 interfacility) and 5265 ground transports. There were 11 red cell transfusions in 8 patients occurring during 1-air scene and 7 air interfacility transports. No problems were identified related to transport of blood from the laboratory to the transport site. There was no time that the blood supply was insufficient to allow for two units of O neg cells to accompany both air and ground transport. There were three units of red cells that were wasted due to exceeding temperature limits. All necessary documentation related to transfusion administration was complete. The ongoing quality control for cooler performance did not reveal any problems. The TS had no lapses in the return of coolers, all acceptable unused units were checked and returned to inventory, and the disposition of all transfused units was documented in the patient's electronic health record. **Conclusion:** A joint quality program between the TS and FT allows us to monitor compliance to policies and procedures related to emergency transfusion for patients being transported by either ground or air. This provides documentation that transfusion outside of the hospital meets the quality standards similar to the inpatient setting.

Disclosure of Commercial Conflict of Interest

M. Baldauf: Nothing to disclose; C. Borek: Nothing to disclose; D. Heegeman: Nothing to disclose; J. J. Redding: Nothing to disclose; J. A. Weyhmiller: Nothing to disclose

Disclosure of Grants Conflict of Interest

M. Baldauf: Nothing to disclose; C. Borek: Nothing to disclose; D. Heegeman: Nothing to disclose; J. J. Redding: Nothing to disclose; J. A. Weyhmiller: Nothing to disclose

AP117

Improving Document Control for a Diverse Transfusion Service

E Bieri Patzke[1] (bieri.elise@mayo.edu), D A Karger[1], C A Bowler[1], J R Stubbs[1]. *[1]Mayo Clinic, Rochester, MN, United States*

Background/Case Studies: A process improvement opportunity was identified to decrease the full time equivalent (FTE) necessary to perform document control (DC) in Transfusion Medicine (TM) while additionally increasing the accessibility of web based electronic documents. DC is an essential component of the blood and tissue quality management program, and at this institution TM is responsible for the creation and archival of over 2100 process documents. The current DC process in TM is complex because of the number of process steps required to implement, manually archive, and annually review documents. Additional challenges are due to space constraints for paper copy archives, folder accessibility via a common server, and the time required to locate different file types. Furthermore, each electronic document type was based on the intended use instead of user accessibility. Successful conversion to electronic DC hinged on a robust process improvement plan that would allow for all documents to be accessible during the yearlong conversion process. Knowing this challenge, change management concepts were utilized to transition TM to the desired electronic DC state. Success would be evident with the use of a transparent communication plan to all key stakeholders in the process. **Study Design/Methods:** Monthly updates were communicated across TM during the conversion. A master index (MI) was utilized to capture document attributes, such as web location, review cycles, related documents, approval/review signatures, and additional current version information. TM focused the conversion to web-based documents on a staggered seven-week implementation plan for each work unit. While all documents were being updated into new document templates, a web page was designed to organize work unit documents, and a MI was populated. Hyperlinks were inserted into the body of the documents to create an avenue to connect related documents used within a single process. The seamless flow between process documents improved the accessibility and usability for the end user. The internal FTE was shifted to DC specialists to coordinate review cycles, electronic signatures, implementation, and electronic archival. **Results/Findings:** At baseline, the weekly mean for implementing the average 22 documents was 3.67 hours and annually utilized 1.06 internal TM FTE to manage the paper based DC process. Every seven weeks, a work unit transitioned from paper-based DC to the electronic environment. Currently, eight/nine work units have converted to the electronic DC process, and the personnel resources have been reduced to 0.12 FTE within TM. After the final conversion, the number of documents implemented, archived, and managed by TM will decrease to zero, which will result in 100% internal FTE saving. **Conclusion:** Web based electronic DC has improved document accessibility across TM while decreasing internal FTE resources required to manage the process.

Disclosure of Commercial Conflict of Interest

E. Bieri Patzke: Nothing to disclose; C. A. Bowler: Nothing to disclose; D. A. Karger: Nothing to disclose; J. R. Stubbs: Terumo BCT, Other, Fenwal Alyx, Other, Haemonetics, Other

Disclosure of Grants Conflict of Interest

E. Bieri Patzke: Nothing to disclose; C. A. Bowler: Nothing to disclose; D. A. Karger: Nothing to disclose; J. R. Stubbs: Nothing to disclose

AP118

Improving Staff Performance by Combining Metrics, Competition, and Recognition

R D Lucas[1] (rlucas@giveblood.org). [1]Donor Collections, Gulf Coast Regional Blood Center, Houston, TX, United States

Background/Case Studies: Our Donor Collections Quality Assurance Department was having a difficult time meeting the expected demand of at least seven Process Records (PR) a day, per employee. The issue resulted in an accumulation of process records in the database at the Supervisor/Department location, and poor performance standard ratings for staff. The expectations, accountability, and metrics to be successful were there. However, the output was not producing the expected result. Management decided to combine metrics, competition, and recognition into the incident management task and pilot the results to determine if the task could meet expectation for a consistent amount of time. **Study Design/Methods:** Management utilized a time study created by sitting with the staff, which illustrated how long it takes a routine incident to process through the Supervisor and Department Review of the Incident Management database. Management then averaged the amount of incidents that came through the Donor Collections department, by totaling each month's incidents over the last 12 months. Then by involving the individuals who would perform the task, an official competition was created which would display the totals for each month. The individual with the highest total would receive a certificate, department bulletin board recognition, and points that this organization provides for outstanding staff. The points totaled 1,500 a month. This recognition drove the competition. If the same individual won three consecutive months they would receive 5,000 points for that month. **Results/Findings:** The staff motivation for completing the process has improved. Also, individuals who performed consistently better than the others were acknowledged. The combination of metrics, competition, and recognition made staff that were not meeting expectation increase their work performance. This process increased the manager's opportunity to dedicate more time to review for consistent patterns of incidents. This allows for more in-depth incident evaluation of people, processes, and procedures. The results are fewer errors, and time for better examination of current and future projects and processes, ultimately permitting a more efficient output for staff. **Conclusion:** Feedback from direct reports is positive, and reflects the staff enjoyment of personal

recognition and competition. The finding indicates employees are excited that routine tasks, which can be tedious, are stimulating. The high performer exhibits pride in the recognition received for achieving the company goals. Further, the staff-management relationship becomes a win-win with effective and efficient processes that facilitate the organization's compliance and quality standards. At this time, the staff is waiting for the results to be displayed which will identify the outstanding performer for the month.

Disclosure of Commercial Conflict of Interest

R. D. Lucas: Nothing to disclose

Disclosure of Grants Conflict of Interest

R. D. Lucas: Nothing to disclose

AP119

Retrospective Audit on the Appropriateness of Red Cell Usage in a Tertiary Hospital in Singapore

S Lee[2] (shir_ying_lee@nuhs.edu.sg), S T Lim[1], G Che[1], J Goh[1], Y Lim[1], L Tan[1]. *[1]Department of Laboratory Medicine, Blood Transfusion Service, National University Hospital, Singapore, Singapore; [2]Department of Hematology-Oncology, National University Hospital, Singapore, Singapore*

Background/Case Studies: In Singapore, usage of blood has risen in recent years, which has not been matched by the supply of blood, owing to more stringent donor criteria and lack of donors during certain holiday periods. There is also a need to reduce unnecessary transfusion to enhance patient safety. A retrospective audit was conducted on the appropriateness of red cell usage here. It was hoped to identify practices in red cell usage which can be improved. **Study Design/Methods:** In a consecutive 8-day period in September 2011, all red cell requests from inpatient wards and intensive care units were reviewed. Requests from emergency room, operating theatres and outpatient clinics were excluded. A standardized algorithm based on Singapore Ministry of Health 2011 Guidelines was used to determine appropriateness of each request. Requests were appropriate if transfused for: 1) active bleeding; 2) hemoglobin (Hgb) <7 g/dL and ≤2 units requested or ≥3 units requested and post-transfusion Hgb did not exceed 11 g/dL; 3) Hgb <8 g/dL in the perioperative period; and 4) Hgb between 7 and 10 g/dL if any risk factor present. A risk factor was regarded as any of the following: symptoms of anemia, heart or lung disease, or aged above 65 years. Requests were inappropriate if transfused for: 1) Hgb >10 g/dL without active bleeding; 2) Hgb between 7 and 10 g/dL without any risk factor; and 3) Hgb ≥8 g/dL in the perioperative period without any risk factor. It was excessive if: 1) ≥2 units requested for Hgb ≥7 g/dL without risk factor and post-transfusion Hgb exceeded 10 g/dL; and 2) ≥3 units requested for Hgb <7 g/dL and post-transfusion Hgb exceeded 11 g/dL. **Results/Findings:** A total of 302 red cell requests were reviewed. Of these, 6% (18 cases) were deemed inappropriate. The Table summarizes the type of inappropriate indication. Common problems identified were: 1) transfusion for chronic compensated anemia even when alternatives were available; 2) transfusion of more than one unit when one may suffice; and 3) high hgb thresholds in asymptomatic, young patients without co-morbidities. **Conclusion:** The rate of inappropriate red cell transfusion was determined to be 6%, translating to approximately 67.5 units a month. The common problems identified will be used to educate physicians on ways to minimize unnecessary red cell use. This can help reduce cost for patients and release blood units for more appropriate usage.

Disclosure of Commercial Conflict of Interest

G. Che: Nothing to disclose; J. Goh: Nothing to disclose; S. Lee: Nothing to disclose; S. T. Lim: Nothing to disclose; Y. Lim: Nothing to disclose; L. Tan: Nothing to disclose

Disclosure of Grants Conflict of Interest

G. Che: Nothing to disclose; J. Goh: Nothing to disclose; S. Lee: Nothing to disclose; S. T. Lim: Nothing to disclose; Y. Lim: Nothing to disclose; L. Tan: Nothing to disclose

No of cases	Type of inappropriate indication
1	Hgb >10 g/dL without active bleeding
6	Hgb between 7 and 10 g/dL without any risk factor
6	Hgb ≥8 g/dL in the perioperative period without any risk factor
5	Excessive: ≥2 units requested for Hgb >7 g/dL without risk factor and post-transfusion Hgb exceeded 10 g/dL

AP120

Impact of Implementing CreaTech Solutions' Malaria Risk Maps for Assessing Donor Eligibility with Respect to Malaria Risk

J Mancilla[1] (jmancilla@donateblood.org), A Devine[2]. *[1]Quality Assurance, Central California Blood Center, Fresno, CA, United States; [2]Owner, CreaTech Solutions LLC, Bellevue, WA, United States*

Background/Case Studies: This blood supply is safeguarded from transfusion-transmitted malaria in part by following FDA screening guidelines for deferral of donors who have recently traveled to or resided in malaria endemic areas (malaria T/R screening). Information on areas with malaria is provided by the CDC through several resources including the Health Information for International Travel (Yellow Book), the CDC website, and the CDC Malaria Map Application. Even with these tools, uncertainty remains in correctly identifying geographic regions and applying risk definitions. In 2012 at this blood center (CCBC), 32% of all our Biological Product Deviation Reports (BPDRs) were related to malaria T/R screening, which are further classified as either "accidents" (due to previously undisclosed information) or "errors" (resulting from incorrect malaria risk assessment). In the summer of 2012 CCBC implemented Malaria Risk Maps (CTS MRM) (CreaTech Solutions LLC, Bellevue, WA) with the objective of reducing malaria T/R screening errors. The purpose of this study was to assess the impact of this implementation on the malaria T/R screening BPDR error rate. **Study Design/Methods:** In August 2012, CCBC implemented CTS MRM throughout our fixed and mobile collections operations. Malaria screening BPDRs in the 6 month period prior to implementation (pre-imp.) were compared to the 6 month period following implementation (post-imp.). Data for the implementation month August 2012 were omitted from analysis. **Results/Findings:** CCBC recorded 42 malaria T/R screening BPDRs, including 7 accidents and 35 errors (including 32 errors related to donor travel to Cancun, Mexico) during the pre-imp. period. The post-imp. period yielded 20 malaria T/R screening BPDRs, including 19 accidents and 1 error. Data are shown in the Table. **Conclusion:** After implementation of CTS MRM, a significant reduction in malaria T/R screening BPDR errors and overall rates were found compared to the previous 6 month period where the CDC Yellow Book and web-based resources were used. Subjective benefits of implementing CTS MRM included increased staff confidence and improved donor relations during travel discussions and a dramatic reduction in calls to our QA "Malaria Deferral Hotline" to the point where it was eventually discontinued. Taken together with the reduction in malaria T/R screening BPDR errors, these findings suggest that CTS MRM reduced staff uncertainty.

Disclosure of Commercial Conflict of Interest

A. Devine: CreaTech Solutions, LLC, Ownership or Partnership; J. Mancilla: CreaTech Solutions LLC, Other

Disclosure of Grants Conflict of Interest

A. Devine: Nothing to disclose; J. Mancilla: Nothing to disclose

TABLE 1: Comparison of Pre and Post-Implementation of CreaTech Solutions Malaria Risk Maps on BPDR Overall Occurrence and Rate

Period	Number of Donor Appearances	Number of Accidents	Accidents as a % of Appearances	Number of Errors	Errorrs as a % of Appearances	Total Number of Screening BPDRs	Total as a % of Appearances
Pre-implementation	45,408	7	0.015	35	0.077	42	0.092
Post-implementation	43,733	19	0.043	1	0.002	20	0.046

AP121

Improving an Electronic Reagent Quality Control Form

S L Lamm[1] (slamm@mcvh-vcu.edu), A Stalley[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: After converting to a spreadsheet-based electronic reagent control (ERC) form in September 2009, clerical error rates were reduced from 0.79% to 0.17% in an 8-month period. However, there were still clerical errors related to the manual entry of reagent information. **Study Design/Methods:** The Quality Assurance Coordinator (QAC) conducted a review of clerical errors from January 2011-January 2012 and identified the following critical concerns: incorrect reagent lot number entry; undocumented new lot reagents brought into use; and use of expired reagents. Staff feedback was solicited and visual confusion from all reagents being on a single spreadsheet was identified as the primary contributing factor to clerical errors. Therefore, the ERC was revised to reflect the reagent organization at the bench. Each rack type is represented by a separate tab in the ERC: reagents used for manual ABO/Rh and antiglobulin testing; reagents used for manual solid phase and direct antiglobulin testing; and critical supplies. The QAC's critical concerns were addressed by the following actions: *Incorrect reagent lot number entry*: 24 of 41 reagents had manufacturers' barcodes that include the lot number and expiration date. Using the ERC's concatenate function and existing barcode readers, both lot number and expiration date are now directly entered into the ERC. The remaining 17 reagents will continue to be manually entered. *Undocumented new lot reagents brought into use:* Previously, the lot numbers in the three manual ABO/Rh racks were visually compared, but only recorded once. The ERC was expanded to include the lot numbers for all three racks. Using the conditional formatting function, the three lot numbers for each reagent are compared and any differences highlighted, drawing attention to new lots. For single rack reagents and critical supplies, the lot number of the current day is compared with the lot number of the previous day and any differences highlighted. *Use of reagents past their expiration date*: A separate cell was created to count down the days until expiration for each reagent. This cell was formatted to turn bright purple when a reagent had 0-6 days until expiration, and turn red at expiration. A presentation was given to staff that would either perform or review the ERC, and the forms were validated in parallel with the existing forms for a 2-week period. **Results/Findings:** By soliciting end user's opinions and QAC concerns, the ERC form was improved as part of this laboratory's continuous improvement program. Staff satisfaction with the visual layout increased. **Conclusion:** Since implementation of the revised forms in January 2013 there have been no documented clerical errors. The process will continue to be monitored.

Disclosure of Commercial Conflict of Interest

S. L. Lamm: Nothing to disclose; K. Sanford: Nothing to disclose; A. Stalley: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. L. Lamm: Nothing to disclose; K. Sanford: Nothing to disclose; A. Stalley: Nothing to disclose

AP122

Reducing Health History Documentation Errors through Implementation of an Electronic Donor Health History Software Package

J Barba[1] (jsb@cbckc.org), D White[2], S Royer[2], T A Frame-Brown[3], A Geneser[3], K B Peck[4]. *[1]Information Technology, Community Blood Center of Kansas City, Kansas City, MO, United States; [2]Quality Management, Community Blood Center of Kansas City, Kansas City, MO, United States; [3]Donor Services Administration, Community Blood Center of Kansas City, Kansas City, MO, United States; [4]Administration, Community Blood Center of Kansas City, Kansas City, MO, United States*

Background/Case Studies: A continuous improvement initiative identified the need to reduce donor medical-history interview documentation errors. Electronic donor health history software was identified to assist. It was hypothesized that a modern, electronic method for completing health history documentation would significantly eliminate documentation errors. **Study Design/Methods:** A staggered deployment was begun of electronic donor health history software on January 4, 2012 and it was fully implemented at all neighborhood fixed site collection locations by March 7, 2012. Mobile drive units continued to utilize manual donor health history documentation (control group). Documentation errors were tracked and analyzed for both fixed site and mobile drives for the 6 months pre-implementation, as well as the 12 months post-implementation. The number of history errors were compared at each location per 1000 units collected pre- and post-implementation to determine if the error rate had changed. **Results/Findings:** Section 1 of the Table represents fixed site health history documentation errors: 76,875 units were collected from July 1, 2011-March 31, 2013; 351 of the health histories contained documentation errors. In the period up to full implementation of the electronic health history software on March 7, 2012, a mean of 5.4 health history documentation errors per 1000 units collected was observed. In the period after full implementation until March 31, 2013, a mean of 4.0 health history documentation errors was found, representing a drop of 1.4 errors per 1000 units collected ($\chi 2 = 4.2077$, $p = 0.04$). Section 2 of the Table represents mobile drive health history documentation errors: 137,584 units were collected during the specified timeframe; 1,103 of the health histories contained documentation errors. In the same period up to full implementation on March 7, 2012, a mean of 8.0 health history documentation errors per 1000 units collected was observed. In the same period after full implementation until March 31, 2013, a mean of 8.0 health history documentation errors was observed, representing no change from the prior period ($\chi 2 = 0.0003$, $p = 0.99$). **Conclusion:** Over a 12-month period, there was a significant ($p < 0.05$) drop in donor history documentation error rates at neighborhood fixed sites that utilized electronic donor health history software. Over the same time frame, there were no changes in donor history documentation error rates at mobile drives using paper donor history form questionnaires. It is concluded that deploying electronic donor history interview software effectively achieves lower health history documentation error rates in this case-control study.

Disclosure of Commercial Conflict of Interest

J. Barba: Nothing to disclose; T. A. Frame-Brown: Nothing to disclose; A. Geneser: Nothing to disclose; K. B. Peck: Nothing to disclose; S. Royer: Nothing to disclose; D. White: Nothing to disclose

Disclosure of Grants Conflict of Interest

J. Barba: Nothing to disclose; T. A. Frame-Brown: Nothing to disclose; A. Geneser: Nothing to disclose; K. B. Peck: Nothing to disclose; S. Royer: Nothing to disclose; D. White: Nothing to disclose

TABLE 1: Summary of documentation errors per 1000 units collected pre- and post-implementation

Section1 Fixed Sites	Pre-Implementation	Post-Implementation	Section 2 Mobiles	Pre-Implementation	Post-Implementation
Errors Observed	168	183	Errors Observed	422	681
Units Collected	30971	45904	Units Collected	52604	84980
Errors/ 1000 Units	5.4	4.0	Errors/ 1000 Units	8.0	8.0

AP123

Transfusion Audits: Looking Beyond the Obvious

R M Bhavnagri[1] (rbhavnagri@mcvh-vcu.edu), S M Armstrong[1], K Sanford[1]. *[1]Transfusion Medicine, Virginia Commonwealth University Medical Center, Richmond, VA, United States*

Background/Case Studies: Transfusion audits are mandatory requirements of regulatory agencies to monitor transfusion safety on a routine basis. The obvious reason for audit is to assess transfusion practices by nursing staff, and their adherence to written policies for providing safe transfusions. Looking beyond the obvious,it was decided to utilize the audits as an opportunity to improve communications, enhance education, and build better relationships and understanding between nursing and blood bank personnel. **Study Design/Methods:** Transfusion audits are rotated among every technologist in the transfusion service. This allows the blood bank staff an opportunity to form relationships with the nursing staff across the medical center on each shift. The audit process allows time for discussing issues and concerns with nursing staff while assuring safe transfusion practices. When

performing the audits, the blood bank staff is asked to engage and develop good working relationships with the nursing staff and solicit questions. This process is used to educate nursing staff on issues related to mislabeled samples, improper orders, atypical antibodies, sample requirements, product availability, and importance of patient's age, location, and diagnosis in setting up appropriate blood and blood products. The audits are also used to emphasize the importance of placing orders prior to surgery to eliminate possible delays in patient care, especially patients with atypical antibodies or special needs. Furthermore, blood bank's procedures for evaluating transfusion reaction workups are explained. Nurses are given the opportunity, and encouraged, to discuss their concerns with the blood bank technologist to create a better understanding and improved processes for overall safety and continuous quality improvement. **Results/Findings:** Improvement in relationships between nursing and the blood bank staff have been seen. This allows a more professional relationship and opens the doors for improved communication and collaboration on projects. These interactions are effective and well appreciated by all in creating environment that is conducive to better patient care. Insights gained during this process became a tool to design and improve nursing education and procedures. **Conclusion:** This mandatory process is normally overlooked as a tool to improve relationships. However, when used with a new focus, it can enhance communications, foster a learning environment, dispel myths, nurture good relationships, and form the basis for better understanding of each other's issues and concerns. It provides opportunities to improve teamwork while fulfilling the primary goal of ensuring safe transfusion practices.

Disclosure of Commercial Conflict of Interest

S. M. Armstrong: Immucor, Travel Support or Honorarium; R. M. Bhavnagri: Nothing to disclose; K. Sanford: Nothing to disclose

Disclosure of Grants Conflict of Interest

S. M. Armstrong: Nothing to disclose; R. M. Bhavnagri: Nothing to disclose; K. Sanford: Nothing to disclose

AP124

Laboratory Workflow Comparison Following the Implementation of a Fully Automated Immunoassay System

C Nare[1] (cnare@bbd.org), M Healy[1], R Roper[1], D Hoover[2], B Cavanaugh[2], B A Marchlewicz[2]. *[1]Blood Bank of Delmarva, Newark, DE, United States; [2]Abbott Laboratories, Abbott Park, IL, United States*

Background/Case Studies: The laboratory had been performing serology screening using a semi automated (SA) methodology for 10 years. As part of an initiative to improve overall laboratory efficiency, the effective use of technician resources, and to provide the opportunity for expansion of testing volume, the two currently available systems for serology screening were evaluated and the decision was made to switch to the fully automated infectious disease platform. This study evaluated laboratory workflow parameters pre- and post- implementation of the new system to assess if it achieved the expected benefits. **Study Design/Methods:** Workflow process steps and technician hands-on time were observed for the SA methodology (pre). Then, workflow process steps and technician hands-on time were observed 30 days after implementation of the fully automated system, to allow operators to gain familiarity with the analyzer (post). Approximately 400 samples were tested during process observation. This study was conducted to compare the number of user interface steps and the total user interaction time to complete certain functions, between the SA and the fully automated system. **Results/Findings:** Significant decreases in the total number of processing steps and the average amount of hands-on time were seen for the automated compared to the SA system. The SA process required a total of 23 process steps compared to 12 steps with the automated system. The SA process required a total hands-on time of 3 hours, which included processing steps for the data entry into an Excel spreadsheet and time for clean-up. Both of these steps occur after results are completed and have been transferred to the Laboratory Information Service (LIS). The fully automated system required only 1.7 hours of attended tech time and no need for manual data entry or review upon run completion. In addition, the SA method required the operator(s) to attend to the processing throughout the run, whereas the fully automated system enables blocks of time when the operator(s) were able to perform additional functions within the lab. **Conclusion:** The implementation of the fully automated infectious disease analyzer in the viral marker laboratory has resulted in significant labor savings and overall improvement in workflow. One FTE was reallocated to support the implementation of in-house nucleic acid amplification test (NAT) testing. There were also savings seen in the Quality and Documentation areas because of the reduction in manual data entry, and the review and verification processes. The fully automated system met the pre-implementation targets for overall workflow and efficiency.

Disclosure of Commercial Conflict of Interest

B. Cavanaugh: Nothing to disclose; M. Healy: Nothing to disclose; D. Hoover: Nothing to disclose; B. A. Marchlewicz: Abbott, Stocks or Bonds; C. Nare: Nothing to disclose; R. Roper: Nothing to disclose

Disclosure of Grants Conflict of Interest

B. Cavanaugh: Nothing to disclose; M. Healy: Nothing to disclose; D. Hoover: Nothing to disclose; B. A. Marchlewicz: Abbott, Stocks or Bonds; C. Nare: Nothing to disclose; R. Roper: Nothing to disclose